Supplement to: Root-n estimation of a parametric component in a panel NPIV model with a discrete instrument

Samuele Centorrino

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Monte Carlo: the limit law at the named design

Section numbers throughout this document refer to the article; this document's own headings carry no numbers. This section is the first of the article's four Monte Carlo experiments, the one run at the configuration the theory names. It adds no result: the text is the article's, moved here unchanged except for the cross-references a move breaks, and the floats are the ones it used to print. The section that follows carries the design validation underneath it. The design is the one the theory names, and naming it is also the concession. At $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$ the frontier holds, $\kappa\alpha = 0.256$ exceeding $\frac{1}{4}$, and the turn computed in Section 4 of the article falls at $n^* \approx 10^{58}$. The experiment therefore runs where the theorem is true and where its asymptotics are known in advance not to bind. Both laws draw a two-valued instrument, $\mathsf{L} = 2$. Theorem 29 is stated at general finite L and is asymptotic in n at fixed L, with no claim of uniformity in the cardinality, so nothing in the limit law itself turns on it; how large the remainder is at a given n is a different question, and the two sections that follow measure it rather than assume it away.

One hypothesis of Theorem 29 is imposed by none of the designs below, and whether it holds anyway is a question about the design rather than about the anchor. Every design builds its debiasing weight on the zero anchor, which Remark 52 admits with no further assumption, and setting the anchor to zero leaves $E[L_\star \mid \Delta Z]$ at whatever the design makes it. On the two laws run here it makes it zero at $\mathsf{L} = 2$ and not at $\mathsf{L} = 4$, which are the two cardinalities that check is run at on those laws. All of the solved representer's coefficient energy sits on odd q at the binary cardinality, where the antiderivative basis is odd in u, so $L_\star$ is odd in u, and ΔU is independent of $(\Delta Z, W)$ by construction and symmetric about zero, so the conditional mean vanishes by parity. Computed in population rather than estimated, the largest entrywise defect is 3.3×10^{-6} of $\|L_\star\|$ at $\mathsf{L} = 2$ against 0.191 at $\mathsf{L} = 4$ (Table S28). The binary arms therefore run inside the hypothesis in the propensity direction, which is the direction Proposition 20(iii) prices, and the four-cell panel of the cardinality arm does not.

That settles the N1 clause of (T-2) *and not* (T-2) *entire*, whose A2 and A2′ clauses nothing here certifies, and it settles it on the two laws of this section and of the cardinality arm; Design H of the next section runs the same anchor at $\mathsf{L} = 2$ and was not put through the population check. What the failure costs is measured in the cardinality arm. The multinomial designs of the last of these Monte Carlo sections carry that check at three cardinalities and fail it at all of them (Table S30). *That measurement is taken at the weight the engine solves for and not at* $\ell_\star$, which is the qualification that section states where it makes it, and it does not show that $\ell_\star$ fails N1 on those designs.

What we can measure is the size of the gap and the speed at which it closes.

We run two data-generating laws, DGP A and DGP B, the second harsher in its geometry, at a condition number of 27.7 against 3.6 (Table S6), and its linearisation residual larger at every n (Table S2, and drawn in Figure S2). The limit-law arm goes to $n = 181{,}019$ at 5,000 replications in every cell; the φ-step arm runs 36 cells at $n \in \{500, 2{,}000, 8{,}000\}$ and 300 replications, sweeping c_{14} over four decades. Monte Carlo error is fixed before any gap is read: the coverage standard error is 0.0031 at the 95% level and 0.0042 at the 90%, the standard-deviation-ratio standard error lies between 0.0099 and 0.0208, and precision does not vary with n. Everything below is produced by `scripts/simulations/sim_thm_rootn_run_all.R`.

The verdict is that the estimator matches its limit law at large n only.

Write c_δ for the constant in A5(ii)'s $O_P(\delta_n)$. At $c_\delta = 0.25$ the ratio of the simulated standard deviation to its asymptotic value fails a two-standard-error bar at every $n \leq 64{,}000$ in DGP A, at 16.90, 8.30, 5.83, 4.93, 3.09, 3.37 and 2.82 standard errors, and clears it only at $n = 181{,}019$ (Table S1). In DGP B the worst studentised quantile is still 2.92 standard errors off there (Figure S1, and Table S7).

At $c_\delta = 1$ nothing matches anywhere on the grid. The ratio is 1.232 at $n = 181{,}019$, which is 18.85 standard errors, and 90% coverage is 0.854, or 9.18 standard errors below nominal. Coverage at that level is below nominal in 8 of 8 cells of DGP A, at a sign-test p of 0.0078,

Table S1: Distance from the asymptotic value, in Monte Carlo standard error units

			Deviation from the asymptotic value, in MC SE units				
DGP	c_δ	n	$\hat{\text{sd}}/\text{sd}_\infty$	90% cov.	95% cov.	Worst quantile	Centring
A	0.25	125	16.90	-2.78	-3.73	2.68	0.60
A	0.25	354	8.30	-0.37	0.66	0.83	-0.05
A	0.25	1,000	5.83	-1.42	-1.73	1.55	-0.23
A	0.25	2,828	4.93	-1.24	-1.79	1.72	1.95
A	0.25	8,000	3.09	-1.82	-1.07	1.74	-1.92
A	0.25	22,627	3.37	-0.79	-1.13	1.73	-0.48
A	0.25	64,000	2.82	-1.11	-1.37	1.78	0.30
A	0.25	181,019	0.27	0.48	0.59	1.02	0.66
A	1.00	125	51.92	-3.86	-3.68	3.97	0.07
A	1.00	354	42.77	-3.16	-3.20	3.49	1.41
A	1.00	1,000	36.71	-5.81	-4.81	5.21	-0.44
A	1.00	2,828	33.03	-6.46	-6.16	5.16	0.54
A	1.00	8,000	27.09	-5.07	-5.11	4.61	0.10
A	1.00	22,627	24.61	-7.79	-8.12	6.73	-0.73
A	1.00	64,000	21.03	-8.12	-6.53	6.67	0.14
A	1.00	181,019	18.85	-9.18	-8.08	7.02	-0.12
B	0.25	125	21.19	-3.95	-2.98	3.21	0.85
B	0.25	354	11.47	-2.17	-1.90	3.20	-1.27
B	0.25	1,000	5.80	0.24	0.66	0.97	-0.21
B	0.25	2,828	4.71	-0.56	-0.82	0.61	-0.13
B	0.25	8,000	3.74	-0.74	0.87	1.73	0.12
B	0.25	22,627	2.22	-1.06	-1.55	1.11	-0.55
B	0.25	64,000	1.19	-1.29	-0.70	1.94	1.38
B	0.25	181,019	-0.77	0.67	0.07	2.92	1.76
B	1.00	125	49.99	-5.23	-4.86	4.37	1.26
B	1.00	354	41.16	-3.20	-2.81	3.63	-1.16
B	1.00	1,000	33.37	-1.82	-0.70	1.20	-0.52
B	1.00	2,828	27.94	-2.60	-3.09	3.57	-0.05
B	1.00	8,000	23.97	-3.95	-4.76	4.81	-0.02
B	1.00	22,627	21.02	-6.08	-4.30	4.33	-1.29
B	1.00	64,000	15.08	-3.66	-2.48	3.75	1.82
B	1.00	181,019	12.23	-4.79	-4.61	4.15	0.60

Notes: Every entry is a gap between a simulated quantity and its asymptotic value, divided by the Monte Carlo standard error of that gap. The entries are therefore counts of standard errors and not levels; the levels are in Table S2. $\hat{\text{sd}}/\text{sd}_\infty$ compares the simulated standard deviation of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ with $(\vartheta' V \vartheta)^{1/2}$, at the fixed unit direction $\vartheta = \frac{1}{2}(1,1,1,1)$. The two coverage columns compare the empirical coverage of nominal 90% and 95% intervals with those levels. "Worst quantile" is the largest absolute gap over the seven studentised quantiles $\{0.025, 0.05, 0.25, 0.5, 0.75, 0.95, 0.975\}$, which Table S7 prints for four of the eight sample sizes. "Centring" tests the mean of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ against zero. c_δ is the constant in the $O_P(\delta_n)$ of A5(ii), and both values satisfy the same hypotheses. 5,000 replications in every cell. An entry inside ± 2 is not distinguishable from zero at that replication count, which is not the same thing as a match. Source: `scripts/simulations/sim_thm_rootn_run_all.R`; table `paper/tables/sim/sim_thm_rootn_seunits.tex`.

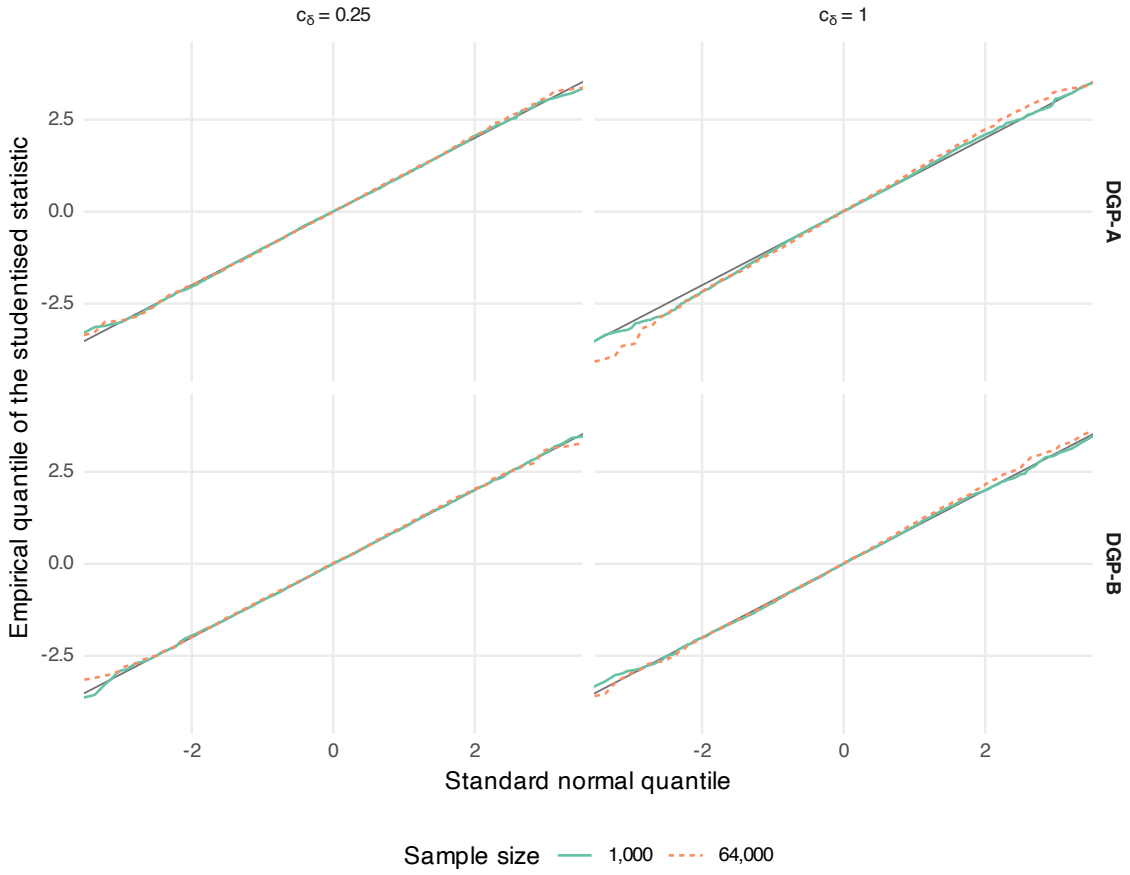


Figure S1: Studentised quantiles against the standard normal

Notes: Each curve plots the empirical quantiles of the studentised statistic $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)/(\vartheta'\hat{V}\vartheta)^{1/2}$, at $\vartheta = \frac{1}{2}(1, 1, 1, 1)$, against the quantiles of the standard normal. The straight grey line in each panel is the 45° line the limit law of Theorem 29 predicts, so a curve lying on it is the theorem's prediction and departure from it is read vertically. Columns are the two values of c_δ, the constant in the $O_P(\delta_n)$ of A5(ii); rows are the two data-generating laws. The solid curve is $n = 1{,}000$ and the dashed curve $n = 64{,}000$. The largest sample of the limit-law arm, $n = 181{,}019$, is not drawn, so the figure does not show the one cell at which the standard-deviation ratio clears its bar. Departure is in the tails and it is larger at $c_\delta = 1$ at both sample sizes. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_limitlaw.R`; figure `paper/figures/sim/sim_thm_rootn_qq.pdf`.

with every cell beyond three standard errors (Table S2).

Both configurations satisfy identical hypotheses, and we have no ground on which to set the failing one aside.

What separates them is the constant, quadratically. Dividing the $c_\delta = 1$ distortion by 16 reproduces the observed $c_\delta = 0.25$ ratio to within two standard errors at every $n \geq 22{,}627$. The distortion at the smaller constant is not absent. It is the same remainder, divided by sixteen.

Three statistics isolate the theorem itself, and none is in doubt. The linearisation residual, taken per replication so that nothing cancels across replications, decreases monotonically in all four core panels, from 1.015 to 0.319 in DGP A at $c_\delta = 0.25$ (Figure S2). The distance between the sample sandwich and its limit does the same, from 1.227 to 0.096. An ablation at exact nuisances gives 0.147, 0.052 and 0.019 at $n = 1{,}000$, 8,000 and 64,000, ratios of 2.83 and 2.74 against $\sqrt{8} = 2.828$, with coverage 0.951, 0.948 and 0.952 (Table S3).

So the gaps are the remainder and not the theorem. They track $\sqrt{n}\,\delta_n^2$, which on this grid rises in n and does not fall below one until $n \approx 10^{394}$ (Table S4; the ledger it is computed from is Table S9). What describes any feasible sample is the rate-floor corollary of Section 4 of the article.

The mean-zero injection of the structural error was an idealisation, and it is now measured (Table S8). At $c_\delta = 1$ and the largest n, interpolating to a fully deterministic error at the same $L^2(\pi)$ norm improves the standard-deviation ratio from 18.85 to 6.78 standard errors and worsens 95% coverage from -8.08 to -9.65. The centring moves across the whole table and not only there: under a mean-zero draw it stays within 1.92 standard errors of zero in all eight cells, and under a deterministic one it reaches 2.04 to 5.51 in the four $c_\delta = 1$ cells while staying within 1.6 in the four at $c_\delta = 0.25$. The idealisation was concealing a location distortion and not a scale one, and it was concealing it mostly at the larger constant.

Table S2: The limit law at levels, by data-generating law and by the constant in A5(ii)

		$\sqrt{n}\delta_n^2$		Coverage (MC SE)		Lin. resid.	$\|\hat{V} - V\|_\infty$
DGP	n	budget	$\hat{\text{sd}}/\text{sd}_\infty$ (MC SE)	90%	95%	mean (SE)	mean
Panel A: DGP-A, $c_\delta = 0.25$							
	125	0.73	1.203 (0.012)	0.888 (0.004)	0.937 (0.003)	1.015 (0.008)	1.227
	354	0.99	1.091 (0.011)	0.898 (0.004)	0.952 (0.003)	0.803 (0.005)	0.638
	1,000	1.27	1.062 (0.011)	0.894 (0.004)	0.944 (0.003)	0.654 (0.004)	0.416
	2,828	1.57	1.052 (0.011)	0.895 (0.004)	0.944 (0.003)	0.560 (0.003)	0.283
	8,000	1.88	1.032 (0.010)	0.892 (0.004)	0.947 (0.003)	0.477 (0.003)	0.208
	22,627	2.22	1.035 (0.010)	0.897 (0.004)	0.946 (0.003)	0.413 (0.003)	0.160
	64,000	2.56	1.029 (0.010)	0.895 (0.004)	0.946 (0.003)	0.359 (0.002)	0.124
	181,019	2.92	1.003 (0.010)	0.902 (0.004)	0.952 (0.003)	0.319 (0.002)	0.096
Panel B: DGP-A, $c_\delta = 1.00$							
	125	11.72	2.080 (0.021)	0.882 (0.005)	0.937 (0.003)	3.235 (0.024)	6.265
	354	15.82	1.748 (0.017)	0.886 (0.004)	0.939 (0.003)	2.805 (0.017)	3.466
	1,000	20.27	1.580 (0.016)	0.873 (0.005)	0.933 (0.004)	2.427 (0.014)	2.423
	2,828	25.06	1.493 (0.015)	0.869 (0.005)	0.927 (0.004)	2.139 (0.012)	1.722
	8,000	30.13	1.372 (0.014)	0.876 (0.005)	0.932 (0.004)	1.854 (0.011)	1.227
	22,627	35.45	1.327 (0.013)	0.862 (0.005)	0.919 (0.004)	1.619 (0.010)	0.885
	64,000	41.00	1.266 (0.013)	0.860 (0.005)	0.926 (0.004)	1.439 (0.009)	0.638
	181,019	46.74	1.232 (0.012)	0.854 (0.005)	0.919 (0.004)	1.261 (0.008)	0.476
Panel C: DGP-B, $c_\delta = 0.25$							
	125	0.73	1.269 (0.013)	0.882 (0.005)	0.940 (0.003)	1.761 (0.014)	3.296
	354	0.99	1.130 (0.011)	0.890 (0.004)	0.944 (0.003)	1.371 (0.009)	1.900
	1,000	1.27	1.062 (0.011)	0.901 (0.004)	0.952 (0.003)	1.107 (0.007)	1.234
	2,828	1.57	1.049 (0.010)	0.898 (0.004)	0.947 (0.003)	0.915 (0.006)	0.847
	8,000	1.88	1.039 (0.010)	0.897 (0.004)	0.953 (0.003)	0.767 (0.005)	0.612
	22,627	2.22	1.023 (0.010)	0.895 (0.004)	0.945 (0.003)	0.651 (0.004)	0.457
	64,000	2.56	1.012 (0.010)	0.894 (0.004)	0.948 (0.003)	0.535 (0.003)	0.352
	181,019	2.92	0.992 (0.010)	0.903 (0.004)	0.950 (0.003)	0.449 (0.003)	0.280
Panel D: DGP-B, $c_\delta = 1.00$							
	125	11.72	2.000 (0.020)	0.876 (0.005)	0.933 (0.004)	5.505 (0.037)	14.424
	354	15.82	1.700 (0.017)	0.886 (0.005)	0.941 (0.003)	4.800 (0.031)	10.423
	1,000	20.27	1.501 (0.015)	0.892 (0.004)	0.948 (0.003)	4.111 (0.027)	7.683
	2,828	25.06	1.388 (0.014)	0.888 (0.004)	0.940 (0.003)	3.486 (0.023)	5.629
	8,000	30.13	1.315 (0.013)	0.882 (0.005)	0.933 (0.004)	2.981 (0.019)	4.035
	22,627	35.45	1.266 (0.013)	0.871 (0.005)	0.935 (0.003)	2.566 (0.016)	2.844
	64,000	41.00	1.178 (0.012)	0.883 (0.005)	0.942 (0.003)	2.128 (0.013)	2.075
	181,019	46.74	1.139 (0.011)	0.878 (0.005)	0.934 (0.004)	1.780 (0.011)	1.522

Notes: The levels of the quantities Table S1 reports in standard error units. The budget column is $\sqrt{n}\,\delta_n^2$, the leading Step-4 product of the remainder accounting, at the panel's own c_δ; it rises in n across this grid rather than falling. $\hat{\text{sd}}/\text{sd}_\infty$ is the simulated standard deviation of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ divided by its asymptotic value, at $\vartheta = \frac{1}{2}(1,1,1,1)$. Coverage is of nominal 90% and 95% intervals. The Monte Carlo standard error in parentheses is evaluated at the *empirical* rate and not at the nominal one, which is why it moves across cells; at the nominal rates it is 0.0042 and 0.0031 throughout. "Lin. resid." is $E\|\sqrt{n}(\hat{\theta} - \theta_\dagger) - J^{-1}\sqrt{n}\,\mathbb{E}_n\psi_\star\|$, formed per replication and then averaged, so nothing cancels across replications; $\|\hat{V} - V\|_\infty$ is the distance between the sample sandwich and its limit. Panels A and B are the same law at two admissible values of c_δ, and so are Panels C and D. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_run_all.R`; table `paper/tables/sim/sim_thm_rootn_limitlaw.tex`.

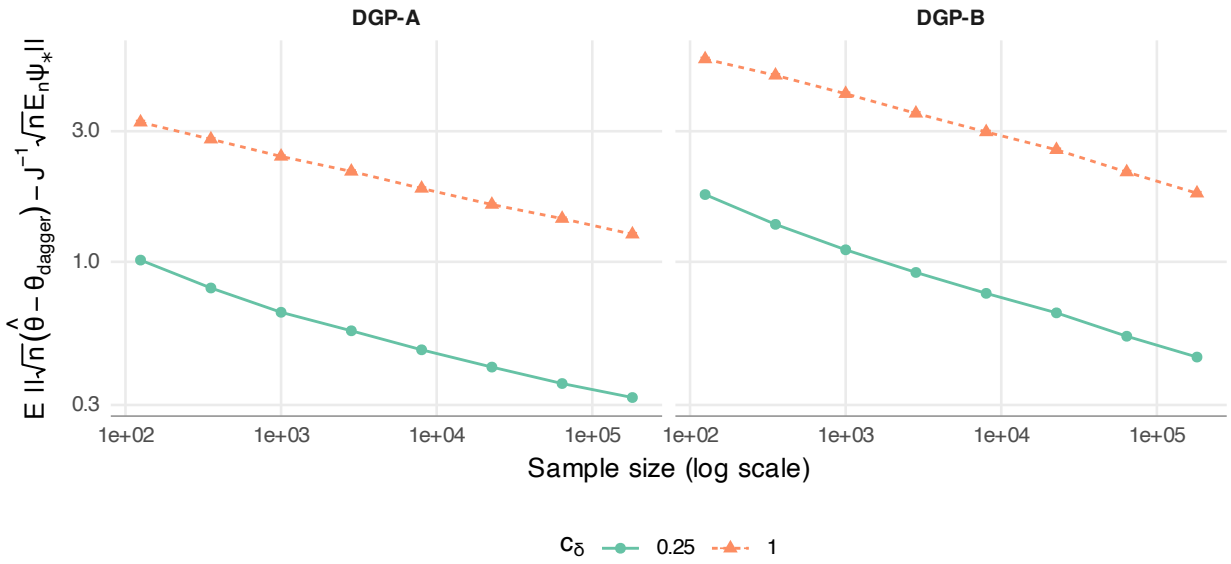


Figure S2: The linearisation residual

Notes: Each point is $E\|\sqrt{n}(\hat{\theta} - \theta_\dagger) - J^{-1}\sqrt{n}\,\mathbb{E}_n\psi_\star\|$, the quantity Theorem 29 sends to zero. It is formed per replication and then averaged, so no cancellation across replications can flatter it. Both axes are logarithmic, and a series falling to the right is the theorem's prediction. Panels are the two data-generating laws. The solid series with circles is $c_\delta = 0.25$ and the dashed series with triangles $c_\delta = 1$, where c_δ is the constant in the $O_P(\delta_n)$ of A5(ii). The residual decreases in n in all four series, which is the statement the figure carries; the figure says nothing about the *rate* at which it does so. The DGP-A levels are the "Lin. resid." column of Table S2. Grid: n from 125 to 181,019, 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_limitlaw.R`; figure `paper/figures/sim/sim_thm_rootn_linresid.pdf`.

Table S3: Exact nuisances, and the two departures from them

n	$\hat{\ell}$	h geometry	90%	95%	Lin. resid.	$\hat{\text{sd}}/\text{sd}_\infty$
1,000	$\hat{\ell} = \ell_\star$	rough	0.896 (0.004)	0.948 (0.003)	2.581	1.623
1,000	$\hat{\ell} = \ell_\star$	smooth	0.878 (0.005)	0.936 (0.003)	2.442	1.597
1,000	all exact	smooth	0.902 (0.004)	0.951 (0.003)	0.147	1.003
1,000	feasible	rough	0.894 (0.004)	0.945 (0.003)	2.589	1.640
8,000	$\hat{\ell} = \ell_\star$	rough	0.901 (0.004)	0.950 (0.003)	1.880	1.336
8,000	$\hat{\ell} = \ell_\star$	smooth	0.877 (0.005)	0.931 (0.004)	1.858	1.363
8,000	all exact	smooth	0.901 (0.004)	0.948 (0.003)	0.052	0.996
8,000	feasible	rough	0.889 (0.004)	0.950 (0.003)	1.885	1.359
64,000	$\hat{\ell} = \ell_\star$	rough	0.901 (0.004)	0.951 (0.003)	1.329	1.185
64,000	$\hat{\ell} = \ell_\star$	smooth	0.860 (0.005)	0.921 (0.004)	1.409	1.263
64,000	all exact	smooth	0.902 (0.004)	0.952 (0.003)	0.019	0.991
64,000	feasible	rough	0.894 (0.004)	0.946 (0.003)	1.337	1.203

Notes: DGP-A only. The "all exact" rows set every nuisance to its true value, which leaves the estimator's own second-order noise as the only gap to the oracle influence function; those are the rows the text reads. The $\hat{\ell} = \ell_\star$ rows supply the representer exactly while the remaining nuisances carry the error of the $c_\delta = 1$ configuration, which is the most favourable implementation the theory permits, since no construction of $\hat{\ell}$ is proposed anywhere in this paper. The "feasible" rows carry all three errors. "h geometry" places the injected structural error on the lowest available frequencies (smooth) or on the highest (rough), at the same $L^2(\pi)$ norm. The two coverage columns carry the Monte Carlo standard error in parentheses, evaluated at the empirical rate; "Lin. resid." and $\hat{\text{sd}}/\text{sd}_\infty$ are the quantities defined in the notes to Table S2. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_limitlaw.R`; table `paper/tables/sim/sim_thm_rootn_ablation.tex`.

Table S4: Where each Step-4 product falls below one, and where the leading one turns

Product	$\ln n$ at which it first falls below 1	$\log_{10} n$
$\sqrt{n}\,\delta_n^2$	908.2	394.4
$\sqrt{n}\,r_n\delta_n$	38.4	16.7
$\sqrt{n}\,\delta_n\epsilon_n$	8.1	3.5
$\sqrt{n}\,r_n\epsilon_n^L$	4.8	2.1
$\sqrt{n}\delta_n^2$ turns from increasing to decreasing at	133.3	57.9

Notes: The four products of the remainder accounting, extrapolated along the fitted log-log trend of the ledger in Table S9, at $c_\delta = 1$. A product below one is a remainder the $\sqrt{n}$ statement can absorb, so the first column is the $\ln n$ at which each first falls there and the second is the same point in $\log_{10} n$. The last row is where $\sqrt{n}\,\delta_n^2$ turns from increasing to decreasing, which is the turn $n^* \approx 10^{58}$ the section opens with. Both readings are extrapolations far outside the simulated grid, which ends at $n = 181{,}019$: they describe the design's own remainder budget and are not measurements. What the budget column depends on is the reading of A5(ii), and Table S10 prices that dependence. Source: `scripts/simulations/sim_thm_rootn_budget.R`; table `paper/tables/sim/sim_thm_rootn_crossings.tex`.

The φ-step arm measures the stopping rule of Section 3 of the article, and its finding concerns the criterion rather than the constant. Write $\hat{r}(m) := \hat{\mathcal{A}}_{\hat{\varphi}^{(0)}} v_m + \hat{T}_{h_x}(\hat{\varphi}^{(0)}, \theta)$ for the residual of the stage-one linear equation at Landweber pass m, measured in $L^2(\Pi)$, which is the quantity Definition 22(c) compares to its level; $\hat{r}(0)$ is that residual at the start $v_0 = 0$. Over the whole admissible budget the discrepancy criterion falls by 11 to 43 per cent, the ratio $\|\hat{r}(\bar{m}_n)\|/\|\hat{r}(0)\|$ running from 0.57 to 0.89 (Table S5, Panel B). A threshold therefore sits above the criterion at one step, or below it at the cap, unless it lands in a window between 0.06 and 0.27 decades wide, which is 1 to 7 per cent of the four decades swept.

The stop is then decided by c_{14} and not by the data. The cap is returned in every replication of all twelve cells at $\mathsf{c}_{14} \leq 10^{-3}$, one step is taken in every replication of all twelve at $\mathsf{c}_{14} \geq 10^{-1}$, and in exactly one of the 36 cells the interior branch is the largest of the three shares, at $n = 8{,}000$ under the zero preliminary and $\mathsf{c}_{14} = 10^{-2}$, where it takes 0.543 of the replications against 0.347 at the cap and 0.110 at one step. The interior branch carries an outright majority there and nowhere else, and the majority is a point estimate rather than a separation: its Monte Carlo standard error is 0.029, so 0.543 clears a half by 1.5 of them. That window moves with n, from $(0.022, 0.026)$ at $n = 500$ to $(0.0089, 0.016)$ at $n = 8{,}000$ under the zero preliminary, so no fixed c_{14} is stable in n (Table S5, Panel B; Figure S4 draws the sweep).

What the printed panels support is narrower than the sweep they summarise. The 36 cells are the whole of it and Panel B prints six, at $\mathsf{c}_{14} = 10^{-2}$; the frequencies are non-degenerate at two of the six swept values, 10^{-2} and 3×10^{-2}, and only the first is printed, the remaining four being the degenerate columns just described. Panel A reports the *median* stopped iteration, so a cell reading 1 or reading the cap does not by itself establish what every replication did; the branch frequencies that do are Panel B's.

Why the criterion barely moves is visible beside it: $\omega_n / \|\hat{\mathcal{A}}\|$ lies between 329,694 and 679,332, so one Landweber pass is numerically invisible and the cap exceeds $\hat{\delta}_n^{-2}$ by a factor of 7.32×10^8

to 2.78×10^9 (Table S5, Panel C).

Table S5: The discrepancy stop, its branches, and the Lepskii comparison

Panel A: median $\hat{m}^{(1)}$ by the unidentified c_{14}; $[q_{10}, q_{90}]$ in Panel B's branch frequencies

n	Prelim.	$\mathsf{c}_{14} = 10^{-4}$	10^{-3}	10^{-2}	3×10^{-2}	10^{-1}	1	mean $\bar{m}_n$
500	zero	3.95×10^{10}	3.94×10^{10}	3.91×10^{10}	1	1	1	3.96×10^{10}
500	rate	3.94×10^{10}	3.94×10^{10}	3.92×10^{10}	1	1	1	3.95×10^{10}
2,000	zero	1.85×10^{11}	1.85×10^{11}	1.82×10^{11}	1	1	1	1.85×10^{11}
2,000	rate	1.86×10^{11}	1.86×10^{11}	1.84×10^{11}	1	1	1	1.85×10^{11}
8,000	zero	8.76×10^{11}	8.76×10^{11}	3.97×10^{11}	1	1	1	8.76×10^{11}
8,000	rate	8.76×10^{11}	8.76×10^{11}	8.70×10^{11}	1	1	1	8.76×10^{11}

Panel B: which branch the rule takes at $\mathsf{c}_{14} = 10^{-2}$, and the residual range that decides it

n	Prelim.	Pr($\hat{m}^{(1)}$=1)	Pr(cap)	Pr(interior) (SE)	$\frac{\|\hat{r}(\bar{m}_n)\|}{\|\hat{r}(0)\|}$ (MC SE)	interior-stop c_{14} window
500	zero	0.053	0.923	0.023 (0.009)	0.8637 (0.0054)	$(0.0216, 0.0256)$
500	rate	0.027	0.973	0.000 (0.000)	0.8925 (0.0038)	$(0.0262, 0.0299)$
2,000	zero	0.053	0.750	0.197 (0.023)	0.7291 (0.0092)	$(0.0147, 0.0209)$
2,000	rate	0.080	0.853	0.067 (0.014)	0.8400 (0.0063)	$(0.0194, 0.0237)$
8,000	zero	0.110	0.347	0.543 (0.029)	0.5743 (0.0126)	$(0.0089, 0.0165)$
8,000	rate	0.120	0.780	0.100 (0.017)	0.7769 (0.0080)	$(0.0170, 0.0227)$

Panel C: the cap (36), its round-9 predecessor, and where the inflation comes from

n	$\bar{m}_n$ (MC SE)	$\hat{\delta}_n^{-2}$	Ratio	ω_n (MC SE)	$\|\hat{\mathcal{A}}\|_{\mathcal{S}_n}\|$ (MC SE)	$\omega_n / \|\hat{\mathcal{A}}\|$
500	3.95×10^{10} (1.45×10^{8})	54	7.32×10^{8}	27200.1 (49.71)	0.0825 (0.0009)	329,694
2,000	1.85×10^{11} (3.56×10^{8})	130	1.42×10^{9}	37777.9 (36.32)	0.0784 (0.0006)	481,922
8,000	8.76×10^{11} (7.59×10^{8})	315	2.78×10^{9}	52744.8 (22.86)	0.0776 (0.0004)	679,332

Panel C′: where the cap moved against the 2026-08-15 state, leg by leg. $\bar{m}_n \propto c_C\, \mathsf{c}_2^4$, so the column rescales without re-running

n	c_2	c_C	Total $\bar{m}_n$ ratio	Constants	Bandwidth h_x	Family Θ_I	h_u coupling	Reverse order
500	34.039	0.1551	344,503	89,741	3.47	1.00	1.108	1.00
2,000	34.039	0.1551	432,597	89,741	4.57	1.00	1.054	1.00
8,000	34.039	0.1551	556,276	89,741	6.03	1.00	1.027	1.00

Panel D: $\|\hat{\varphi} - \varphi_\dagger\|$ under Lepskii and under the discrepancy rule (DP), at the latter's oracle c_{14} *(an advantage Lepskii is not given) and at a fixed* c_{14}

n	Prelim.	$\hat{m}^{\mathrm{Lep}}$ (MC SE)	Lep. (SE)	oracle c_{14}	DP (SE)	Lep. at $3{\times}10^{-2}$ (SE)	DP at $3{\times}10^{-2}$ (SE)
500	zero	1.88×10^{9} (6.10×10^{7})	0.7023 (0.0062)	0.1	0.7071 (0.0000)	0.7110 (0.0070)	0.7986 (0.0101)
500	rate	2.03×10^{9} (5.90×10^{7})	0.4139 (0.0057)	1	0.3596 (0.0025)	0.4373 (0.0068)	0.5126 (0.0135)
2,000	zero	4.09×10^{9} (2.10×10^{8})	0.6736 (0.0032)	0.1	0.7071 (0.0000)	0.6743 (0.0037)	0.7441 (0.0069)
2,000	rate	6.30×10^{9} (2.91×10^{8})	0.3142 (0.0042)	0.1	0.2920 (0.0019)	0.3331 (0.0052)	0.3698 (0.0093)
8,000	zero	5.85×10^{9} (2.39×10^{8})	0.6643 (0.0017)	0.03	0.7033 (0.0039)	0.6643 (0.0017)	0.7033 (0.0039)
8,000	rate	1.61×10^{10} (1.20×10^{9})	0.2560 (0.0023)	0.1	0.2424 (0.0013)	0.2732 (0.0035)	0.2926 (0.0065)

Table S5, continued: *The discrepancy stop, its branches, and the Lepskii comparison.*

Notes: The φ-step arm, which is a separate experiment from the limit-law arm of Table S2. Its grid is $n \in \{500, 2{,}000, 8{,}000\}$ with two preliminaries and $\mathtt{c}_{14}$ swept over four decades, at 300 replications in every cell. "Prelim." is the initial condition, "rate" one that satisfies (P-a.2) and "zero" one that fails it. Panel A reports the median stopped iteration $\hat{m}^{(1)}$. Panel B reports the branch frequencies at $\mathtt{c}_{14} = 10^{-2}$, its last column the interval of $\mathtt{c}_{14}$ inside which the rule stops in the interior. Panel C reports the cap (36), the quantity $\hat{\delta}_n^{-2}$ it inflates, and the over-normalisation $\omega_n / \|\hat{\mathcal{A}}\|$ that produces it. Panel C′ decomposes the move in that cap against the configuration shipped through 2026-08-15, which used $h_x = n^{-0.20}$, $c_C = 4.0$ and $\mathtt{c}_2 = 0.8727$; its h_u leg is the residual between the three derived legs and the realised ratio, so it cannot absorb an arithmetic slip. Panel D compares a Lepskii-balanced stop with the discrepancy rule at the latter's oracle $\mathtt{c}_{14}$ in its fourth and sixth columns, and at the fixed $\mathtt{c}_{14} = 3 \times 10^{-2}$ in its last two, so that the fixed-threshold comparison is a difference of two adjacent columns inside one row. Where the oracle is 3×10^{-2}, at $n = 8{,}000$ under the zero preliminary, the two Lepskii columns coincide. Monte Carlo standard errors in parentheses. Source: `scripts/simulations/sim_thm_rootn_phistep.R`; table `paper/tables/sim/sim_thm_rootn_stopping.tex`; run object `quality_reports/sim/sim_thm_rootn_phistep.rds`.

A Lepskii-balanced stop was run alongside, and the comparison splits by preliminary. Given the discrepancy rule its own oracle c_{14}, an advantage no analyst can take because that constant is not identified, Lepskii is ahead in all three cells started from zero and behind in all three started from a rate-accurate preliminary. The margins are not all read the same way. In the two zero-preliminary cells at $n = 500$ and $n = 2{,}000$ the discrepancy arm returns $\|\varphi_\dagger\| = 0.7071$ in every replication, so the paired standard error of the difference is exactly the Lepskii arm's: at $n = 500$ the gap is 0.0048 against a standard error of 0.0062, which is 0.77 standard errors and is not a win, and at $n = 2{,}000$ it is 0.0335 against 0.0032, which is 10.6 on the unrounded gap and standard error and 10.5 on the four digits printed. In the remaining four cells both arms vary and their covariance is not stored, so we report the bound that holds whatever it is, $\mathrm{SE} \le \mathrm{SE}_{\mathrm{Lep}} + \mathrm{SE}_{\mathrm{DP}}$: the zero-preliminary gap at $n = 8{,}000$ is at least 6.9 standard errors, and the three rate-accurate cells go the other way by at least 6.6, 3.6 and 3.8. At a fixed $\mathsf{c}_{14} = 3 \times 10^{-2}$, read at that cell in both arms, Lepskii is ahead in all six by at least 1.9 to 6.9 on the same bound (Table S5, Panel D, columns 7 and 8, the two headed at 3×10^{-2}).

The win is unflattering where it happens. The oracle constant in every rate-preliminary cell is 10^{-1} or 1, values at which one step is taken in every replication, so the rule wins by returning the injected preliminary out of a budget of 10^{10} to 10^{11} steps. In every zero-preliminary cell the same prescription returns an error of exactly $\|\varphi_\dagger\| = 0.7071$ and loses. The oracle does not separate the two preliminaries: the prescriptions coincide at 10^{-1} at $n = 2{,}000$, sit at 10^{-1} against 1 at $n = 500$, which are both values at which one step is taken, and differ in kind only at $n = 8{,}000$, where the zero preliminary prefers 3×10^{-2}. One constant is therefore best under both preliminaries at the two smaller sample sizes while returning an estimate under one of them and nothing under the other, and nothing in the data chooses it in any case.

Four things this exercise does not do should be read with it. No constructed $\hat{\ell}$ is tested, because none exists to test. The realised operator is more ill-posed than H1 admits, its fitted

decay index running from 1.63 to 2.14 against the 0.75 of the design, and the injected error occupies six Fourier modes that the operator damps by a factor between 2.64 and 5.36, so no error in the damped directions is exhibited. Table S11 prints the index and the damping; the mode count is a constant of the injection basis, and the notes to that table read the damping off against the spectrum, at most a tenth of it in log terms. And no law violating an assumption is run, so a match is a non-rejection by a procedure shown able to reject only through the $-\hat{J}^{-1}$ counterfactual and the c_δ contrast (Table S12).

Two quantities are unavailable rather than reported. Both condition numbers sit at the double-precision floor and are printed as lower bounds, and at $n = 2{,}000$ the quadrature caps the Gram's rank at 780 of 2,000 (Table S11, Panel A).

The index measure is implemented as a quadrature and not as counting measure, and the distinction is not bookkeeping: counting measure overstates the cap inflation by a factor of 2.93, which is the node count divided by the volume of $\mathcal{T}$ and so is arithmetic on two constants of the configuration rather than a measurement. At the twelve-node grid the normalising constant is still 16 per cent below and the cap 29 per cent below their exact values, a convergent quadrature error we declare rather than absorb; that pair is read off a refinement run and appears in no table.

An applied reader should take three things from this section, all of them statements at $\mathsf{L} = 2$. The estimator runs, is correctly signed, and its remainder accounting is complete for nuisance errors of the priced size. Its normal approximation is accurate at $n = 181{,}019$ at one of two equally admissible constants, and at neither constant below $n = 64{,}000$. Nominal 90% intervals under-cover at 29 of the 32 cells, the three exceptions clearing nominal by less than one Monte Carlo standard error, and a panel of the size this readership has is described by the rate floor and not by the $\sqrt{n}$ statement. The cardinality scoping is not decoration: the section on cardinality below finds the first of the three failing at $\mathsf{L} = 4$ on this same design.

Monte Carlo: supplementary tables and figures

This section carries the design validation and the cell-level detail behind the experiment above. It adds no result. What follows is the validation of the two laws, the cell-level detail underneath two of the preceding section's summary tables, the ledger its extrapolations are computed from, the diagnostics of the φ-step arm, and the verification of the Jacobian. *The opening sentence of this paragraph named the article and not the section above until 2026-08-27*, when the experiment it validates moved here; a framing sentence that survives the thing it frames is the defect this project keeps finding, so it was rewritten rather than left. Every table is produced by `scripts/simulations/sim_thm_rootn_run_all.R` and the file it is read from is named in its notes.

Table S6: The two data-generating laws, and the conditions they were built to satisfy

DGP	A2 rel. resid.	$\max\|t\|$, $E\psi_\star$	$\max\|E[\Delta Z \Delta U]\|$	$\kappa(J)$	ϱ range	$E[\Delta U^4]$	$\sup\|\ell_\star\|$
A	4.2e-07	1.52	2.1e-03	3.6	[0.111, 0.896]	2.13	3.04
B	1.4e-06	1.60	2.7e-03	27.7	[0.039, 0.965]	3.98	3.09

Notes: Design validation, computed once per law and not per replication, so no Monte Carlo standard error attaches to it. "A2 rel. resid." is the relative residual of the balancing equation A2 at the constructed representer, and $\max|t|$ for $E\psi_\star$ and $\max|E[\Delta Z \Delta U]|$ test the two moment conditions the design must satisfy against zero. $\kappa(J)$ is the condition number of $J = E[\Delta Z \Delta Z']$ and is the sense in which DGP-B is the harsher geometry: 27.7 against 3.6. The ϱ range is the range of the propensity score, which stays inside $(0, 1)$ in both laws. $E[\Delta U^4]$ and $\sup|\ell_\star|$ record the moment and the sup-norm the asymptotics use. This table validates the design; it is not evidence about the estimator. Source: `scripts/simulations/sim_thm_rootn_dgp.R`; table `paper/tables/sim/sim_thm_rootn_dgpvalid.tex`.

Table S7: Studentised quantiles against the standard normal, cell by cell

DGP	c_δ	n	Quantile	Empirical (MC SE)	Asymptotic	Diff. (MC SE units)
A	0.25	1,000	0.025	-2.008 (0.038)	-1.960	-1.26
			0.050	-1.678 (0.030)	-1.645	-1.13
			0.250	-0.678 (0.019)	-0.674	-0.20
			0.500	0.008 (0.018)	0.000	0.48
			0.750	0.650 (0.019)	0.674	-1.27
			0.950	1.655 (0.030)	1.645	0.34

			0.975	2.018 (0.038)	1.960	1.55
A	1.00	1,000	0.025	-2.130 (0.038)	-1.960	-4.49
			0.050	-1.801 (0.030)	-1.645	-5.21
			0.250	-0.684 (0.019)	-0.674	-0.51
			0.500	0.020 (0.018)	0.000	1.11
			0.750	0.690 (0.019)	0.674	0.82
			0.950	1.759 (0.030)	1.645	3.83
			0.975	2.056 (0.038)	1.960	2.53
A	0.25	8,000	0.025	-2.026 (0.038)	-1.960	-1.74
			0.050	-1.696 (0.030)	-1.645	-1.70
			0.250	-0.701 (0.019)	-0.674	-1.38
			0.500	-0.014 (0.018)	0.000	-0.80
			0.750	0.653 (0.019)	0.674	-1.10
			0.950	1.649 (0.030)	1.645	0.12
			0.975	1.948 (0.038)	1.960	-0.31
A	1.00	8,000	0.025	-2.107 (0.038)	-1.960	-3.90
			0.050	-1.724 (0.030)	-1.645	-2.64
			0.250	-0.742 (0.019)	-0.674	-3.49
			0.500	0.008 (0.018)	0.000	0.44
			0.750	0.724 (0.019)	0.674	2.57
			0.950	1.783 (0.030)	1.645	4.61
			0.975	2.126 (0.038)	1.960	4.41
A	0.25	64,000	0.025	-1.960 (0.038)	-1.960	-0.01
			0.050	-1.670 (0.030)	-1.645	-0.83
			0.250	-0.682 (0.019)	-0.674	-0.38
			0.500	-0.004 (0.018)	0.000	-0.25
			0.750	0.691 (0.019)	0.674	0.86
			0.950	1.678 (0.030)	1.645	1.12
			0.975	2.027 (0.038)	1.960	1.78
A	1.00	64,000	0.025	-2.129 (0.038)	-1.960	-4.48
			0.050	-1.773 (0.030)	-1.645	-4.29

			0.250	-0.761 (0.019)	-0.674	-4.51
			0.500	0.018 (0.018)	0.000	1.00
			0.750	0.734 (0.019)	0.674	3.09
			0.950	1.844 (0.030)	1.645	6.67
			0.975	2.194 (0.038)	1.960	6.21
A	0.25	181,019	0.025	-1.946 (0.038)	-1.960	0.38
			0.050	-1.635 (0.030)	-1.645	0.32
			0.250	-0.663 (0.019)	-0.674	0.60
			0.500	0.008 (0.018)	0.000	0.47
			0.750	0.694 (0.019)	0.674	1.02
			0.950	1.638 (0.030)	1.645	-0.24
			0.975	1.930 (0.038)	1.960	-0.80
A	1.00	181,019	0.025	-2.170 (0.038)	-1.960	-5.57
			0.050	-1.855 (0.030)	-1.645	-7.02
			0.250	-0.776 (0.019)	-0.674	-5.29
			0.500	-0.008 (0.018)	0.000	-0.43
			0.750	0.777 (0.019)	0.674	5.31
			0.950	1.850 (0.030)	1.645	6.87
			0.975	2.189 (0.038)	1.960	6.07
B	0.25	1,000	0.025	-1.923 (0.038)	-1.960	0.97
			0.050	-1.630 (0.030)	-1.645	0.49
			0.250	-0.685 (0.019)	-0.674	-0.53
			0.500	-0.001 (0.018)	0.000	-0.05
			0.750	0.672 (0.019)	0.674	-0.15
			0.950	1.652 (0.030)	1.645	0.24
			0.975	1.973 (0.038)	1.960	0.35
B	1.00	1,000	0.025	-1.981 (0.038)	-1.960	-0.56
			0.050	-1.666 (0.030)	-1.645	-0.69
			0.250	-0.695 (0.019)	-0.674	-1.04
			0.500	0.002 (0.018)	0.000	0.11
			0.750	0.682 (0.019)	0.674	0.37

			0.950	1.681 (0.030)	1.645	1.20
			0.975	1.964 (0.038)	1.960	0.11
B	0.25	8,000	0.025	-1.920 (0.038)	-1.960	1.05
			0.050	-1.659 (0.030)	-1.645	-0.48
			0.250	-0.708 (0.019)	-0.674	-1.73
			0.500	0.012 (0.018)	0.000	0.67
			0.750	0.704 (0.019)	0.674	1.52
			0.950	1.659 (0.030)	1.645	0.46
			0.975	1.948 (0.038)	1.960	-0.32
B	1.00	8,000	0.025	-2.142 (0.038)	-1.960	-4.81
			0.050	-1.752 (0.030)	-1.645	-3.60
			0.250	-0.698 (0.019)	-0.674	-1.24
			0.500	0.002 (0.018)	0.000	0.14
			0.750	0.739 (0.019)	0.674	3.34
			0.950	1.723 (0.030)	1.645	2.61
			0.975	2.063 (0.038)	1.960	2.72
B	0.25	64,000	0.025	-1.946 (0.038)	-1.960	0.38
			0.050	-1.646 (0.030)	-1.645	-0.04
			0.250	-0.653 (0.019)	-0.674	1.12
			0.500	0.019 (0.018)	0.000	1.09
			0.750	0.708 (0.019)	0.674	1.74
			0.950	1.703 (0.030)	1.645	1.94
			0.975	2.019 (0.038)	1.960	1.57
B	1.00	64,000	0.025	-1.964 (0.038)	-1.960	-0.11
			0.050	-1.660 (0.030)	-1.645	-0.50
			0.250	-0.676 (0.019)	-0.674	-0.09
			0.500	0.020 (0.018)	0.000	1.13
			0.750	0.730 (0.019)	0.674	2.88
			0.950	1.757 (0.030)	1.645	3.75
			0.975	2.092 (0.038)	1.960	3.49
B	0.25	181,019	0.025	-1.940 (0.038)	-1.960	0.52

			0.050	-1.629 (0.030)	-1.645	0.51
			0.250	-0.618 (0.019)	-0.674	2.92
			0.500	0.034 (0.018)	0.000	1.90
			0.750	0.678 (0.019)	0.674	0.19
			0.950	1.640 (0.030)	1.645	-0.16
			0.975	1.983 (0.038)	1.960	0.60
B	1.00	181,019	0.025	-2.117 (0.038)	-1.960	-4.15
			0.050	-1.758 (0.030)	-1.645	-3.78
			0.250	-0.693 (0.019)	-0.674	-0.94
			0.500	0.019 (0.018)	0.000	1.07
			0.750	0.723 (0.019)	0.674	2.50
			0.950	1.725 (0.030)	1.645	2.67
			0.975	2.074 (0.038)	1.960	3.01

Notes: The quantile-by-quantile detail behind the "worst quantile" column of Table S1, and the numbers Figure S1 draws for two of these cells. "Empirical" is the sample quantile of the studentised statistic $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)/(\vartheta'\hat{V}\vartheta)^{1/2}$ at $\vartheta = \frac{1}{2}(1,1,1,1)$, with its Monte Carlo standard error in parentheses; that standard error is the Bahadur expression $\sqrt{q(1-q)/B}/\phi(\Phi^{-1}(q))$, evaluated at the density of the *asymptotic* law, which is the comparison target, and it therefore does not vary with n. "Asymptotic" is the corresponding standard normal quantile and the last column is the gap between them in those units. Four of the limit-law arm's eight sample sizes are printed. 5,000 replications in every cell. The column head does not repeat on continuation pages. Source: `scripts/simulations/sim_thm_rootn_limitlaw.R`; table `paper/tables/sim/sim_thm_rootn_quantiles.tex`.

Table S8: The mean-zero injection of the structural error, relaxed

n	Bias frac.	Coverage (MC SE) 90%	Coverage (MC SE) 95%	$\hat{\text{sd}}/\text{sd}_\infty$ (MC SE)	Lin. resid. mean (SE)	Centring (MC SE units)
$c_\delta = 0.25$						
1,000	0.0	0.894 (0.004)	0.944 (0.003)	1.062 (0.011)	0.654 (0.004)	-0.23
1,000	0.5	0.899 (0.004)	0.948 (0.003)	1.042 (0.010)	0.652 (0.004)	-0.51
1,000	1.0	0.897 (0.004)	0.949 (0.003)	0.967 (0.010)	0.615 (0.004)	1.54
8,000	0.0	0.892 (0.004)	0.947 (0.003)	1.032 (0.010)	0.477 (0.003)	-1.92
8,000	0.5	0.896 (0.004)	0.948 (0.003)	1.012 (0.010)	0.480 (0.003)	-1.72
8,000	1.0	0.898 (0.004)	0.950 (0.003)	0.960 (0.010)	0.444 (0.003)	-1.07
64,000	0.0	0.895 (0.004)	0.946 (0.003)	1.029 (0.010)	0.359 (0.002)	0.30
64,000	0.5	0.896 (0.004)	0.949 (0.003)	1.017 (0.010)	0.358 (0.002)	0.16
64,000	1.0	0.897 (0.004)	0.953 (0.003)	0.972 (0.010)	0.330 (0.002)	0.38
181,019	0.0	0.902 (0.004)	0.952 (0.003)	1.003 (0.010)	0.319 (0.002)	0.66
181,019	0.5	0.902 (0.004)	0.949 (0.003)	1.003 (0.010)	0.317 (0.002)	0.96
181,019	1.0	0.895 (0.004)	0.947 (0.003)	0.981 (0.010)	0.289 (0.002)	1.21
$c_\delta = 1.00$						
1,000	0.0	0.873 (0.005)	0.933 (0.004)	1.580 (0.016)	2.427 (0.014)	-0.44
1,000	0.5	0.873 (0.005)	0.933 (0.004)	1.540 (0.015)	2.459 (0.014)	0.16
1,000	1.0	0.864 (0.005)	0.927 (0.004)	1.293 (0.013)	2.298 (0.013)	5.51
8,000	0.0	0.876 (0.005)	0.932 (0.004)	1.372 (0.014)	1.854 (0.011)	0.10
8,000	0.5	0.871 (0.005)	0.926 (0.004)	1.353 (0.014)	1.873 (0.011)	-0.64
8,000	1.0	0.870 (0.005)	0.926 (0.004)	1.123 (0.011)	1.722 (0.010)	2.04
64,000	0.0	0.860 (0.005)	0.926 (0.004)	1.266 (0.013)	1.439 (0.009)	0.14
64,000	0.5	0.861 (0.005)	0.923 (0.004)	1.236 (0.012)	1.438 (0.009)	0.73
64,000	1.0	0.865 (0.005)	0.928 (0.004)	1.059 (0.011)	1.321 (0.008)	2.14
181,019	0.0	0.854 (0.005)	0.919 (0.004)	1.232 (0.012)	1.261 (0.008)	-0.12
181,019	0.5	0.853 (0.005)	0.918 (0.004)	1.211 (0.012)	1.250 (0.008)	1.55
181,019	1.0	0.849 (0.005)	0.911 (0.004)	1.073 (0.011)	1.151 (0.007)	3.34

Notes: DGP-A. "Bias frac." interpolates the injected structural error from a mean-zero draw (0.0) to a fully deterministic one (1.0) at a fixed $L^2(\pi)$ norm, so the rows of a block differ in the location of the injection and in nothing else. The 0.0 rows repeat the corresponding cells of Table S2. Columns are those of that table, with "centring" added in Monte Carlo standard error units: it tests the mean of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ against zero and is the column the idealisation was concealing. Coverage standard errors in parentheses are evaluated at the empirical rate. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_limitlaw.R`; table `paper/tables/sim/sim_thm_rootn_bias.tex`.

Table S9: The four Step-4 products of the remainder accounting

			Step-4 products, scaled by $\sqrt{n}$				
n	$\ln n$	δ_n	δ_n^2	$r_n\delta_n$	$\delta_n\epsilon_n$	$r_n\epsilon_n^L$	$\delta_n/n^{-1/4}$
Panel A: the Monte Carlo grid							
125	4.83	1.024	11.72	2.44	1.303	0.272	3.42
354	5.87	0.917	15.82	2.64	1.230	0.205	3.98
1,000	6.91	0.801	20.27	2.78	1.131	0.155	4.50
2,828	7.95	0.686	25.06	2.87	1.021	0.117	5.01
8,000	8.99	0.580	30.13	2.93	0.910	0.088	5.49
22,627	10.03	0.485	35.45	2.95	0.801	0.067	5.95
64,000	11.07	0.403	41.00	2.95	0.700	0.050	6.40
181,019	12.11	0.331	46.74	2.93	0.607	0.038	6.84
Panel B: extended log-log grid, to the point where each product turns over							
125	4.83	1.024	11.72	2.44	1.303	0.272	3.42
248	5.51	0.955	14.37	2.58	1.259	0.226	3.79
541	6.29	0.870	17.60	2.70	1.191	0.183	4.19
1,318	7.18	0.770	21.51	2.81	1.103	0.144	4.64
3,644	8.20	0.660	26.27	2.89	0.994	0.109	5.13
11,638	9.36	0.545	32.02	2.94	0.870	0.080	5.66
43,814	10.69	0.431	38.95	2.95	0.736	0.056	6.24
199,008	12.20	0.326	47.28	2.93	0.599	0.037	6.88
$10^{6.0}$	13.93	0.233	57.24	2.85	0.467	0.023	7.57
$10^{6.9}$	15.90	0.156	69.09	2.73	0.346	0.014	8.31
$10^{7.9}$	18.15	0.097	83.12	2.56	0.242	0.007	9.12
$10^{9.0}$	20.72	0.056	99.62	2.34	0.158	0.004	9.98

Notes: The ledger Table S4 extrapolates. Every product is scaled by $\sqrt{n}$, so the column headed δ_n^2 is $\sqrt{n}\,\delta_n^2$ and likewise for the other three; a product below one is a remainder the $\sqrt{n}$ statement can absorb. δ_n itself is in the third column, unscaled, and the last column measures it against the $n^{-1/4}$ the frontier asks it to beat. Panel A is the Monte Carlo grid. Panel B extends the log-log grid to the point at which each product turns over; $\sqrt{n}\,\delta_n^2$ has not turned at $n = 10^9$ and rises throughout. These are deterministic functions of the design at $c_\delta = 1$, not simulated quantities, so no Monte Carlo standard error attaches to them. Figure S3 draws this table and extends it past the point where Panel B stops, which is where the leading product turns. Source: `scripts/simulations/sim_thm_rootn_budget.R`; table `paper/tables/sim/sim_thm_rootn_budget.tex`.

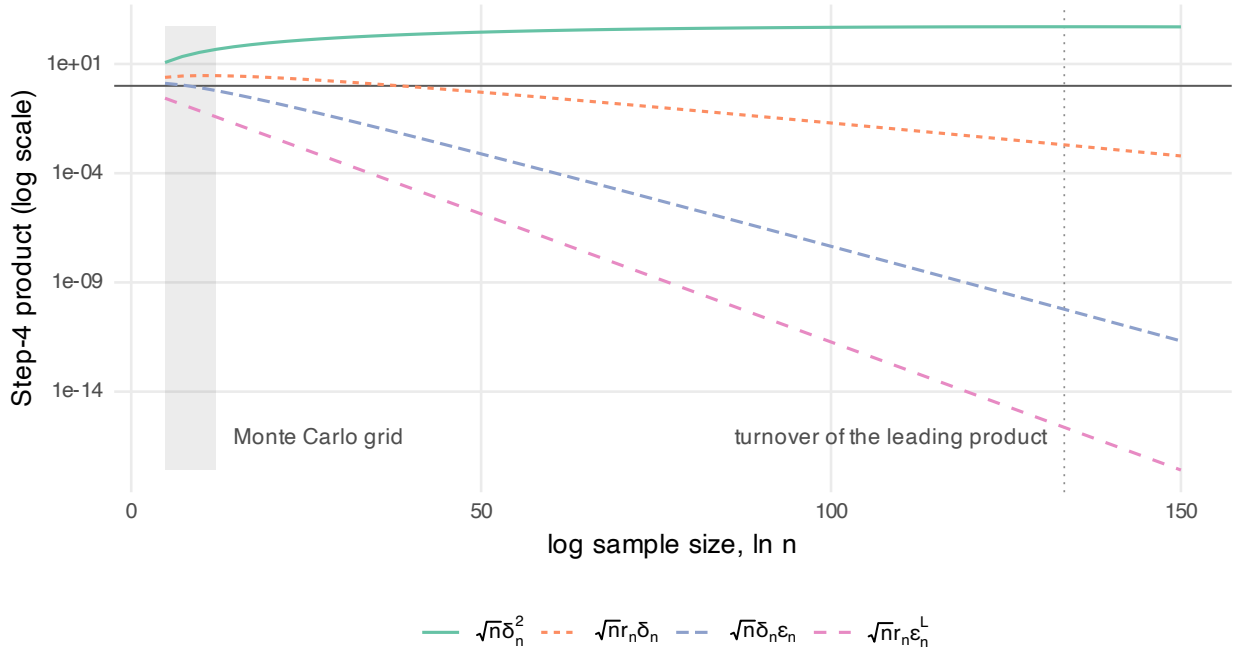


Figure S3: The four Step-4 products, and where the leading one turns

Notes: Table S9 drawn, and extended. The horizontal axis is $\ln n$ and the vertical axis is the product on a log scale, so a curve crossing the solid horizontal line at one is a remainder passing into the range the $\sqrt{n}$ statement can absorb. The shaded band at the left, labelled "Monte Carlo grid", is the simulated range $\ln n \in [4.83, 12.11]$; everything to its right is extrapolation. The dotted vertical line at $\ln n = 133$, labelled "turnover of the leading product", is where $\sqrt{n}\,\delta_n^2$ turns from increasing to decreasing. That curve, the solid one, is above one over the whole range drawn and does not cross back until $\ln n = 908$, which is off this figure. Source: `scripts/simulations/sim_thm_rootn_budget.R`; figure `paper/figures/sim/sim_thm_rootn_budget.pdf`.

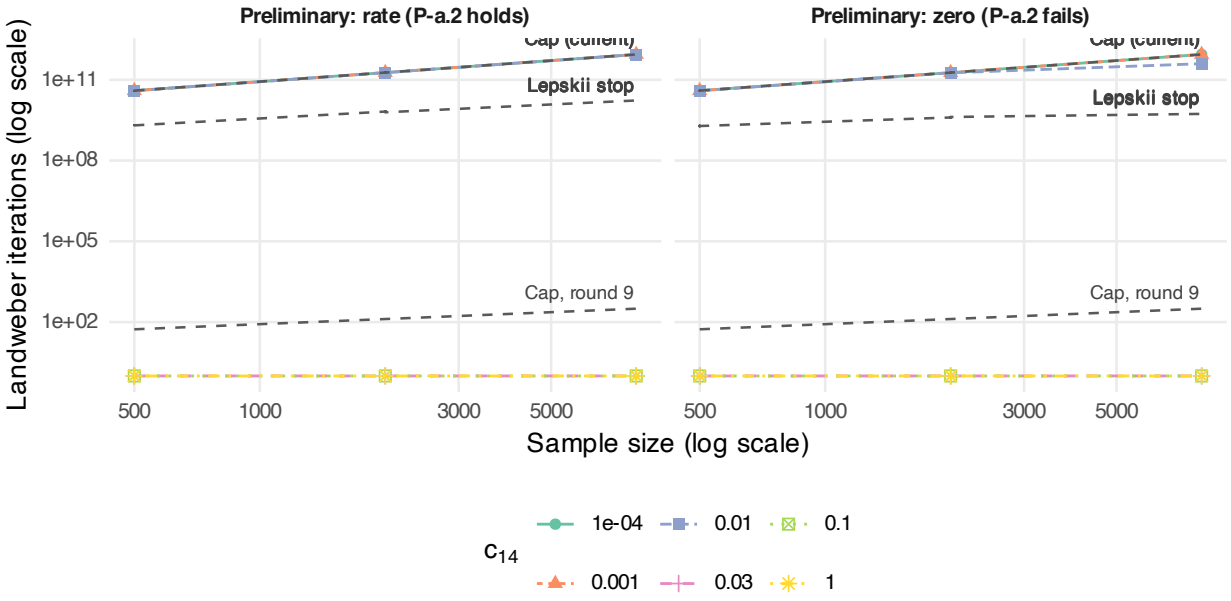


Figure S4: The stopped iteration across four decades of c_{14}

Notes: Panel A of Table S5 drawn. Both axes are logarithmic. Each coloured series is one value of the tuning constant c_{14} and plots the median stopped iteration $\hat{m}^{(1)}$ against n; panels are the two preliminaries. The three dashed reference lines are the cap of (36), annotated *Cap (current)*; the Lepskii stop; and the cap as it stood at round 9 of the theory, annotated *Cap, round 9*. The series separate into two groups: $\mathsf{c}_{14} \leq 10^{-2}$ tracks the cap and $\mathsf{c}_{14} \geq 0.03$ sits at one iteration. Exactly one of the eighteen cells lies between them, $\mathsf{c}_{14} = 10^{-2}$ at $n = 8{,}000$ under the zero preliminary, which stops at a median 3.97×10^{11} against a cap of 8.76×10^{11}, both of which Panel A of Table S5 prints. That separation, and not the level of either group, is what the figure shows. The interior-stop window that lies between the two groups is too narrow to be visible at this scale and is reported numerically in Panel B of Table S5. $n \in \{500, 2{,}000, 8{,}000\}$, 300 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_phistep.R`; figure `paper/figures/sim/sim_thm_rootn_stopping.pdf`.

Table S10: The budget under three readings of A5(ii)

	$\sqrt{n}\,\delta_n^2$ under three readings of A5(ii)			Ratio to amended	
n	Log-free (pre-disp. 10)	Amended	Double (forbidden)	Log-free	Double
125	0.944	11.72	145.5	0.081	12.4
354	0.932	15.82	268.5	0.059	17.0
1,000	0.920	20.27	446.6	0.045	22.0
2,828	0.909	25.06	690.7	0.036	27.6
8,000	0.898	30.13	1011.0	0.030	33.6
22,627	0.887	35.45	1417.4	0.025	40.0
64,000	0.876	41.00	1919.4	0.021	46.8
181,019	0.865	46.74	2526.7	0.019	54.1

Notes: What the leading product $\sqrt{n}\,\delta_n^2$ of Table S9 depends on, priced. "Amended" is the reading this paper uses and is the column reproduced in Table S9 and in the budget column of Table S2 at $c_\delta = 1$. "Log-free" drops the $\ln n$ factor from δ_n and "double" squares it; the latter is ruled out and is printed so that the direction and the size of the sensitivity are both visible. The last two columns are the ratios of the two alternatives to the amended reading. The choice of reading moves the product by between one and two orders of magnitude, which is larger than anything the simulation measures. Deterministic functions of the design; no Monte Carlo standard error attaches. Source: `scripts/simulations/sim_thm_rootn_budget.R`; table `paper/tables/sim/sim_thm_rootn_logreadings.tex`.

Table S11: Conditioning, ill-posedness, and the geometry of the injection

Panel A: two condition numbers, and they are two different objects (≥: at the double-precision floor)

n	Prelim.	L_x	κ, compression (53)	κ, Gram (112)	n_{m}	Gram s.v. computed
500	zero	6	≥4.5e+15	≥2.0e+31	500	500 of 500
500	rate	6	≥4.5e+15	≥2.0e+31	500	500 of 500
2,000	zero	6	≥4.5e+15	≥2.0e+31	2,000	780 of 2,000
2,000	rate	6	≥4.5e+15	≥2.0e+31	2,000	780 of 2,000
8,000	zero	9	≥4.5e+15	—	8,000	—
8,000	rate	9	≥4.5e+15	—	8,000	—

Panel B: ill-posedness of the realised design, against H1's $\sigma_j \asymp j^{-a}$ with $a = 3/4$

n	Prelim.	Power fit: $\hat{a}$ (R^2)	Exponential fit: $\hat{c}$ (R^2)	Pr(exp. wins)
500	zero	1.63 (0.944)	0.66 (0.966)	0.68 (0.03)
500	rate	1.67 (0.964)	0.66 (0.943)	0.34 (0.03)
2,000	zero	2.02 (0.943)	0.83 (0.975)	0.81 (0.02)
2,000	rate	2.05 (0.973)	0.80 (0.919)	0.14 (0.02)
8,000	zero	2.14 (0.972)	0.61 (0.946)	0.12 (0.02)
8,000	rate	2.05 (0.982)	0.56 (0.881)	0.00 (0.00)

Panel B′: L_x against (54), at a literal constant-1 reading of (P-c) and (P-d)

n	L_x used	(P-c) lower	(P-d) upper	Window	Upper/lower
500	6	2.11	1.51	empty	0.715
2,000	6	2.49	2.03	empty	0.815
8,000	9	2.94	2.83	empty	0.962

Panel C: does the φ-step attain δ_n? $\|\hat{\varphi} - \varphi_\dagger\|_{L^2(\pi)}/\delta_n$ at $\mathsf{c}_{14} = 1$

n	δ_n	Prelim. "rate": measured (MC SE)	Prelim. "zero": $\|\varphi_\dagger\|/\delta_n$, arithmetic
500	0.879	0.409 (0.0028)	0.805
2,000	0.724	0.405 (0.0026)	0.977
8,000	0.580	0.419 (0.0021)	1.218

Panel D: can the limit-law arm's injected h reach the directions this operator damps?

n	Prelim.	Sieve-captured	Resolvable dim.	Energy c.o.m. index	Energy in damped half	σ_1/σ at c.o.m.
500	zero	0.783	5.0	3.00	0.595	2.64
500	rate	0.783	5.0	3.02	0.605	2.96
2,000	zero	0.783	5.0	3.00	0.593	2.93
2,000	rate	0.783	5.0	3.04	0.608	3.54
8,000	zero	0.973	8.0	4.28	0.462	4.26
8,000	rate	0.973	8.0	4.36	0.482	5.36

Notes: Diagnostics of the φ-step arm: $n \in \{500, 2{,}000, 8{,}000\}$, two preliminaries, 300 replications. Panel A reports two distinct condition numbers, the compression of (53) and the Gram of (112). Both sit at the double-precision floor, so both are printed as lower bounds and neither is a measurement; at $n = 2{,}000$ the quadrature caps the Gram's rank at 780 of 2,000 and at $n = 8{,}000$ it is not computed. Panel B fits the realised spectrum against H1's $\sigma_j \asymp j^{-a}$ with $a = 3/4$: the fitted power index runs from 1.63 to 2.14, so the realised operator is more ill-posed than H1 admits, and the fit runs on five to eight points, so the functional form is not identified and the exponential column is reported beside it rather than chosen against it. Panel B′ checks L_x against (54) at a literal constant-1 reading of (P-c) and (P-d), where the admissible window is empty at all three sample sizes. Panel C asks whether the φ-step attains δ_n. Panel D measures whether the limit-law arm's injected h reaches the directions the operator damps: it occupies six Fourier modes spanning three frequencies, its energy centre of mass sits at index 3.00 to 4.36, and the damping there is a factor of 2.64 to 5.36, which is at most a tenth of the spectrum in log terms. No error in the damped directions is therefore exhibited. Monte Carlo standard errors in parentheses where the quantity is simulated. Source: `scripts/simulations/sim_thm_rootn_phistep.R`; table `paper/tables/sim/sim_thm_rootn_condition.tex`.

Table S12: Verification of the Jacobian, its sign, and the anchor

Panel A: finite-difference check of $\partial_\theta \bar{\psi}_n = -\hat{J}$ *of* (46)

DGP	θ	Step	Max rel. gap to $-\hat{J}$	Min rel. gap to $+\hat{J}$	Sign agreement
A	$\theta_\dagger$	1e-03	6.3e-07	2.000	1.000
A	$\theta_\dagger$	1e-04	6.3e-07	2.000	1.000
A	$\theta_\dagger$	1e-05	6.3e-07	2.000	1.000
A	displaced	1e-03	6.3e-07	2.000	1.000
A	displaced	1e-04	6.3e-07	2.000	1.000
A	displaced	1e-05	6.3e-07	2.000	1.000
B	$\theta_\dagger$	1e-03	1.0e-06	2.000	1.000
B	$\theta_\dagger$	1e-04	1.0e-06	2.000	1.000
B	$\theta_\dagger$	1e-05	1.0e-06	2.000	1.000
B	displaced	1e-03	1.1e-06	2.000	1.000
B	displaced	1e-04	1.1e-06	2.000	1.000
B	displaced	1e-05	1.1e-06	2.000	1.000

Panel B: direction of the one-step, $\|\hat{\theta} - \theta_\dagger\| / \|\hat{\theta}^{(0)} - \theta_\dagger\|$

DGP	Displacement	$+\hat{J}^{-1}$ (MC SE)	$-\hat{J}^{-1}$ (MC SE)
A	0.15	0.2947 (0.0080)	2.027 (0.012)
A	0.59	0.0799 (0.0023)	2.000 (0.004)
B	0.15	0.5599 (0.0186)	2.052 (0.024)
B	0.59	0.1347 (0.0043)	2.008 (0.006)

Panel C: the second summand of J, $E[\widetilde{W}\ell_\star \Delta Z']$, *measured directly*

DGP	$\max \lvert J \rvert$	$\max \lvert E[\widetilde{W}\ell_\star \Delta Z'] \rvert$	MC SE	$\max \lvert t \rvert$	$\|J - J'\|_\infty$
A	1.122	2.19e-03	8.93e-04	2.46	1.06e-03
B	1.010	9.21e-04	8.40e-04	1.10	9.22e-04

Panel D: the asymptotic sd of $\hat{\theta}_2$ *under two admissible anchors*

DGP	Zero anchor	N1 anchor	N1 / zero	$\|\hat{b}\| / \|L_\star\|$ (rms)	Zero / no debias.
A	1.0055	1.0055	1.0000	1.2e-02	1.042
B	2.0810	2.0810	1.0000	1.2e-02	1.032

Notes: Verification of the implementation rather than evidence about the estimator, which is why it is here and not in the article. Panel A checks $\partial_\theta \bar{\psi}_n = -\hat{J}$ of (46) by finite differences at three step sizes, at the truth and at a displaced θ: the relative gap to $-\hat{J}$ is at most 1.1×10^{-6} while the gap to $+\hat{J}$ is 2, so the sign is identified and not merely consistent. Panel B is the counterfactual the article names: a one-step in the $+\hat{J}^{-1}$ direction contracts the error and one in the $-\hat{J}^{-1}$ direction doubles it, which is the one mechanism by which this design has been shown able to reject. Panel C measures the second summand of J directly and finds it indistinguishable from zero, consistent with $J = E[\Delta Z \Delta Z']$ exactly. Panel D reports the asymptotic standard deviation of $\hat{\theta}_2$ under the two admissible anchors, which coincide here, so this design cannot detect anchor sensitivity. Monte Carlo standard errors in parentheses. Source: `scripts/simulations/sim_thm_rootn_jacobian.R`; table `paper/tables/sim/sim_thm_rootn_jacobian.tex`.

Monte Carlo: the design with frontier headroom

This section carries the second Monte Carlo experiment of the article, the one run at Design H, together with its supplementary material. It adds no result. The experiment opens the section, in the article's own text and with its own two floats: the remainder budget both designs realise, with the contrast between them, and the outcome against the pre-committed criterion. What follows them is the configuration behind those two floats, the validation of the two laws drawn from it, the ledger the budget is read off, and the cell-level detail underneath the summary. *Through 2026-08-27 this paragraph said the article printed the two floats*, which it did until that pass moved them here. Every table is produced by `scripts/simulations/sim_thm_rootn_headroom.R` and the file it is read from is named in its notes.

Everything above is about one design, and we chose that design because the theory names it. At $(\mu, s, a, r, d_Z) = (2, 3, \frac{3}{4}, \frac{8}{3}, 3)$ the frontier holds at $\kappa\alpha = 0.2560$ against its threshold of $\frac{1}{4}$, a margin of 0.0060, and the leading Step-4 product accordingly climbs from 11.72 to 46.74 across the grid at $c_\delta = 1$ (Table S2, Panel B). What the experiment measured was therefore the remainder, in a design where the remainder was always going to dominate. Whether the limit law binds anywhere a reader can reach is a different question, and nothing above answers it.

We answer it by rerunning the same estimator at a configuration with room to spare. Design H is $(\mu, s, a, r, d_Z, p_K) = (5, 4, \frac{2}{5}, \frac{21}{8}, 1, 8)$ at $p = 2$, which gives $\kappa\alpha = 0.3818$ against the same threshold and a frontier margin of 0.1318, more than twenty times the shipped one. Neither the estimator nor the replication path is touched; what changes is the configuration and the two laws drawn from it, DGP-HA and DGP-HB. The configuration clears the ten conditions the shipped one clears, which is what Table S14 sets side by side; both laws are validated at the same moment conditions (Table S15), and the operator each realises is recorded beside

them (Table S16).

The headroom is bought from the smoothness index μ and not from the propensity score. Design H runs $r = 2.625$ against the shipped 2.667, so its propensity is no smoother than the one already in hand, and the reason is A5(i). That condition sets $\hat{\delta}_n$ to the larger of a propensity branch and a parametric branch of order $n^{-1/2}\sqrt{\ln n}$, so raising α pushes the propensity branch under a floor, below which α stops mattering. The floor binds at every cell of Design H and at no cell of the shipped design, which the ledger records in its own branch column (Table S17).

What the experiment turns on is the realised budget. At $c_\delta = 0.25$ it runs from 0.635 at $n = 125$ up to 0.721 at $n = 1{,}000$ and then down to 0.397, so it is below one at every cell of both laws and falling across the part of the grid where the samples are large. At $c_\delta = 0.35$ it crosses below one at $n = 64{,}000$ and reaches 0.778. The shipped arm at the smaller of its two constants runs from 0.73 to 2.92 and rises throughout (Table S2, Panels A and C; Figure S5 draws both designs). That contrast is the experiment: a remainder present but below budget, which is a stronger test than the exact-nuisance ablation of Table S3 and one the shipped grid could not run.

We fixed the decision rule before the run and we state it in the form it was written. Nominal coverage within two Monte Carlo standard errors, at cells whose realised budget is below one, confirms the limit law at a feasible n and scopes the verdict above to the shipped design; coverage off nominal at those cells exhausts the remainder explanation and implicates the theorem, the variance estimator or the implementation; and no admissible law with a budget below one at a feasible n would itself have been the finding, with the binding constraint named.

The criterion is met at 21 of the 23 cells the table prints, and both of the misses are at the smallest sample size on the grid (Table S13). At $n = 125$ coverage is 0.940 in both laws and 2.98 standard errors below nominal in each, the two laws agreeing to every digit printed. At

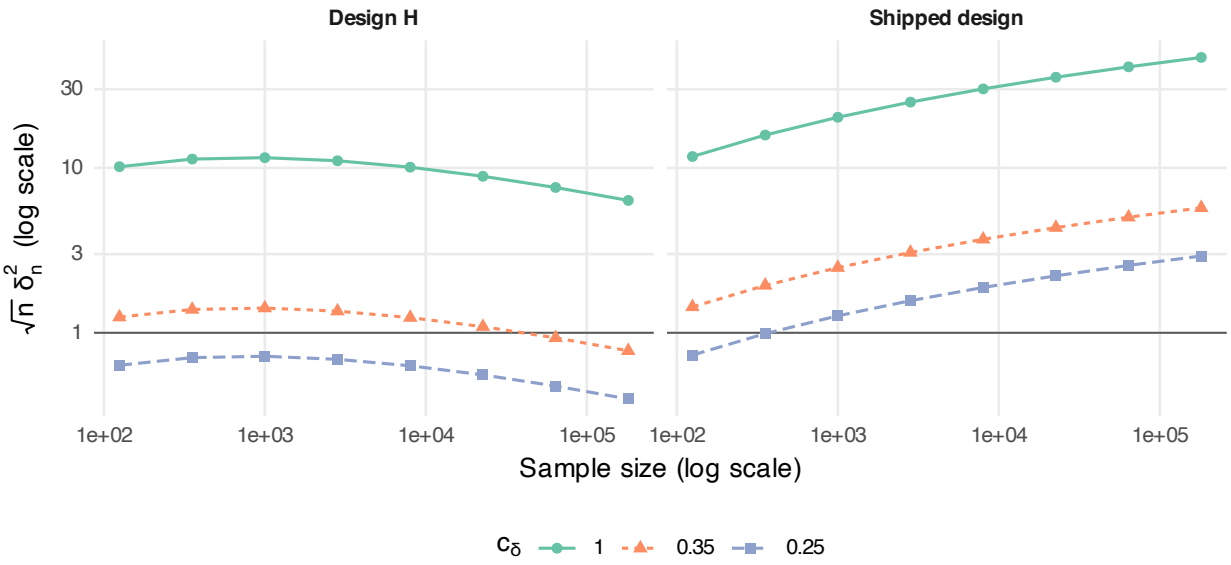


Figure S5: The remainder budget at the two designs

Notes: The leading Step-4 product $\sqrt{n}\,\delta_n^2$ at the two configurations. It is a deterministic function of the design, so no Monte Carlo standard error attaches to any point drawn. Both axes are logarithmic. The horizontal line at one separates remainders the $\sqrt{n}$ statement can absorb from those it cannot, so a series below that line at a given n marks a cell in which A5(ii) holds at the realised size of the nuisance error. Panels are the two configurations: Design H, $(\mu, s, a, r, d_Z, p_K) = (5, 4, \frac{2}{5}, \frac{21}{8}, 1, 8)$ at $p = 2$, and the design of the preceding pages, $(2, 3, \frac{3}{4}, \frac{8}{3}, 3, 8)$ at $p = 4$. The three series are the three values of c_δ, the constant in the $O_P(\delta_n)$ of A5(ii), which scales the product quadratically. Design H turns over at $n = 1{,}000$ and falls thereafter at all three constants, while the shipped design rises at all three across the whole range; that difference in direction, and not the level of either, is what the figure carries. The levels are Table S17 for Design H and Table S2 for the shipped design. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; figure `paper/figures/sim/sim_thm_rootn_headroom_budget.pdf`.

every $n \geq 354$ coverage is inside the bar in both laws, at a budget below one. The outcome is ordered by the sample size and not by the law: on the criterion fixed above, and with the remainder present and priced, the estimator matches its limit law from $n = 354$ upward and at $n = 125$ it does not. That threshold is a threshold on 95% coverage and on nothing else; the standard-deviation ratio reaches the same bar at a larger sample size, which we give below.

We read the experiment as confirming Theorem 29 at a feasible sample size, and the qualification the section opened with is carried into that reading: Design H runs the same zero anchor and was not put through the population check that settles N1 on the two laws of the preceding pages, so whether this confirmation sits inside the N1 clause of (T-2) is not established. The two misses are recorded and not set aside, and the reason they do not overturn that reading is a property of the budget rather than a judgement about them: below one is not the same as small, and at $n = 125$ the budget falls below one through $\sqrt{n}$ rather than through δ_n. We give that argument, and the statistic that orders the misses across both designs and both constants, in the closing paragraphs of this section rather than here.

The measure the shipped verdict failed on is the ratio of the simulated standard deviation to its asymptotic value, and it is where the two designs separate. At matched sample size and matched constant $c_\delta = 0.25$ the shipped design sits 5.83, 4.93, 3.09 and 3.37 standard errors from its target at $n = 1{,}000$ through $n = 22{,}627$ (Table S1), while Design H sits at 2.12, 1.74, 0.61 and 0.18 (Table S19); over those four cells the budget runs from 1.27 to 2.22 and rises in the first case (Table S2, Panel A), and from 0.721 to 0.556 and falls in the second (Table S17). Design H clears the two-standard-error bar on that ratio in both laws at that constant from $n = 8{,}000$ upward, at 0.61 and 0.51, and misses it in at least one law at every smaller sample size, at 6.35 and 7.35 as late as $n = 354$. The sample size at which this design's limit law becomes accurate is therefore $n = 354$ on coverage and $n = 8{,}000$ on scale, and we carry both numbers rather than the smaller one alone. Both designs clear the bar at $n = 181{,}019$ and

Table S13: The headroom experiment against the criterion fixed before it ran

				Pre-committed criterion: 95% coverage			Whole battery	
DGP	c_δ	n	Budget	Coverage (MC SE)	Gap (SE)	Verdict	Worst $\lvert z\rvert$	Verdict
Panel A: every core cell with realised budget below one								
HA	0.35	64,000	0.932	0.948 (0.003)	-0.57	within 2 SE	1.73	within 2 SE
HA	0.35	181,019	0.778	0.944 (0.003)	-1.85	within 2 SE	2.58	outside 2 SE
HA	0.25	125	0.635	0.940 (0.003)	-2.98	outside 2 SE	11.50	outside 2 SE
HA	0.25	354	0.707	0.948 (0.003)	-0.64	within 2 SE	6.35	outside 2 SE
HA	0.25	1,000	0.721	0.950 (0.003)	0.13	within 2 SE	2.12	outside 2 SE
HA	0.25	2,828	0.690	0.946 (0.003)	-1.37	within 2 SE	1.79	within 2 SE
HA	0.25	8,000	0.631	0.948 (0.003)	-0.64	within 2 SE	1.89	within 2 SE
HA	0.25	22,627	0.556	0.953 (0.003)	1.00	within 2 SE	1.00	within 2 SE
HA	0.25	64,000	0.475	0.950 (0.003)	0.00	within 2 SE	3.28	outside 2 SE
HA	0.25	181,019	0.397	0.947 (0.003)	-0.82	within 2 SE	1.01	within 2 SE
HB	0.35	64,000	0.932	0.950 (0.003)	0.13	within 2 SE	1.77	within 2 SE
HB	0.35	181,019	0.778	0.953 (0.003)	1.00	within 2 SE	1.53	within 2 SE
HB	0.25	125	0.635	0.940 (0.003)	-2.98	outside 2 SE	17.26	outside 2 SE
HB	0.25	354	0.707	0.947 (0.003)	-1.07	within 2 SE	7.35	outside 2 SE
HB	0.25	1,000	0.721	0.947 (0.003)	-0.89	within 2 SE	3.74	outside 2 SE
HB	0.25	2,828	0.690	0.949 (0.003)	-0.19	within 2 SE	2.10	outside 2 SE
HB	0.25	8,000	0.631	0.951 (0.003)	0.46	within 2 SE	1.15	within 2 SE
HB	0.25	22,627	0.556	0.949 (0.003)	-0.45	within 2 SE	1.02	within 2 SE
HB	0.25	64,000	0.475	0.955 (0.003)	1.85	within 2 SE	2.19	outside 2 SE
HB	0.25	181,019	0.397	0.952 (0.003)	0.59	within 2 SE	0.86	within 2 SE
Panel B: zero-remainder anchor (nuisances exact)								
HA	—	181,019	0.000	0.949 (0.003)	-0.45	within 2 SE	—	—
HA	—	22,627	0.000	0.951 (0.003)	0.20	within 2 SE	—	—
HA	—	1,000	0.000	0.950 (0.003)	-0.06	within 2 SE	—	—

$B = 5,000$ at every cell; coverage MC SE ≈ 0.003 at nominal 95%. Bar fixed before the run. Worst $\lvert z\rvert$ is the largest of five measures. Cells: 51 of 51.

Notes: Design H, $(\mu, s, a, r, d_Z, p_K) = (5, 4, \frac{2}{5}, \frac{21}{8}, 1, 8)$ at $p = 2$, and its two data-generating laws DGP-HA and DGP-HB. Panel A lists every core cell whose realised budget is below one; the cells whose budget is above one are not listed here and are in Table S19. Panel B is the zero-remainder anchor, which sets every nuisance to its true value and is the Design H counterpart of Table S3. "Budget" is $\sqrt{n}\,\delta_n^2$ at the cell's own c_δ, the constant in the $O_P(\delta_n)$ of A5(ii), and is deterministic. "Coverage" is the empirical coverage of a nominal 95% interval, with its Monte Carlo standard error in parentheses evaluated at the empirical rate; "Gap" is the distance from 0.95 in those units, and the verdict beside it applies the two-standard-error bar that was fixed in advance. "Worst $\lvert z\rvert$" is the largest of five measures at that cell, namely the standard-deviation ratio, coverage at 90% and at 95%, the worst studentised quantile and the centring; the verdict beside it applies the same bar to that maximum, which is a stricter test than the pre-committed one and is printed beside it rather than in place of it. The five measures are defined in the notes to Table S1 and their levels are in Table S18. 5,000 replications in every cell, and 51 of 51 cells completed. An entry inside ± 2 is not distinguishable from zero at that replication count, which is not the same thing as a match. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; table `paper/tables/sim/sim_thm_rootn_headroom_verdict.tex`.

both miss it at $n = 125$, and the second of those needs saying.

We read the two failures as a defect in the criterion and not as evidence against Theorem 29. The budget $\sqrt{n}\,\delta_n^2$ is a product, and at $n = 125$ it falls below one through $\sqrt{n}$ and not through δ_n: the realised δ_n there is 0.2384 at $c_\delta = 0.25$ (Table S17) against a $\|\varphi_\dagger\|$ of 0.7071, so the nuisance error is about a third of the object being estimated and no premise of the theorem is close to holding. The statistic that does order the failures is the realised linearisation residual, and it orders them across both designs and both constants. It is 1.015 at the shipped design's $n = 125$ cell (Table S2, Panel A) and 0.611 and 1.029 at Design H's two (Table S18); all three have a budget below one and all three miss. It is 0.265 at Design H's $n = 181{,}019$ cell at $c_\delta = 1$, whose budget is 6.350, and that cell misses nothing.

That last cell is the converse case and it carries as much of the point as the failures do. Its budget is more than six times the bar and the largest of its five measures is 1.63 standard errors (Table S19). A budget above one does not predict a miss and a budget below one does not predict a match; what predicts both is the size of the remainder the estimator realises.

Eight cells inside the passing range depart on the wider battery, and we flag them rather than absorb them: all eight carry an "outside 2 SE" whole-battery verdict at $n \geq 354$ in Table S13, at five of them the largest of the five measures is the standard-deviation ratio of the preceding paragraphs, and at the other three it is the worst studentised quantile, at 2.58, 3.28 and 2.19. The largest of the three needs its own reading. At DGP-HA, $c_\delta = 0.25$ and $n = 64{,}000$, the largest quantile gap is 3.28 standard errors while 95% coverage is exactly nominal at $z = 0.00$, the standard-deviation ratio is 0.22 and the centring is 2.49. That gap sits at the median while the four tail quantiles are inside a quarter of a standard error, a reading taken from the run object and not from Table S20, which prints eight cells and not this one. So the two measures that move at that cell are location measures and the two that do not are scale measures, and the cells on either side of it, at $n = 22{,}627$ and $n = 181{,}019$, are clean on all five. Table S19 reports five measures at each of 48 cells, and a two-standard-error bar is

expected to be crossed about once in twenty comparisons. We report the cells; we do not claim they are nothing.

Two of the caveats above carry over unchanged. The realised operator is again more ill-posed than H1 admits, its fitted power index running to 8.252 and 8.516 against the 0.400 of the design (Table S16), and no law violating an assumption is run here either, so a match remains a non-rejection. The studentised quantiles at $n = 1{,}000$ and $n = 181{,}019$, in both laws and at both of the extreme constants, are drawn in Figure S6; the largest quantile gap among those eight cells is 2.15 standard errors. Table S20 prints the same statistic cell by cell at $n = 22{,}627$ and $n = 181{,}019$, where the largest is 2.29.

We separate what this settles from what it does not. It settles that the verdict of the preceding pages is a statement about the shipped design and not about Theorem 29: at a frontier margin of 0.1318 the pre-committed criterion is met from $n = 354$ upward, and at the matched constant $c_\delta = 0.25$ the standard-deviation ratio clears its bar from $n = 8{,}000$, while the same estimator, run from the same code, does not clear that ratio until $n = 181{,}019$ when the margin is 0.0060. It does not settle that the limit law binds at every admissible constant. Design H clears the budget nowhere on this grid at $c_\delta = 1$, and the parametric branch of A5(i) binds at each of its cells, so what puts the budget below one here is the constant and not the configuration; c_δ is not identified, and the two values remain as equally admissible as they were above. Of the three readings we offered an applied reader above, the second is therefore a statement about the shipped design, and so is the under-coverage the third opens with: at every cell of Panel A with $n \geq 354$ Design H's nominal 90% coverage is inside the same two-standard-error bar in both laws, and it sits above nominal at four of them: DGP-HA's $n = 1{,}000$ at $+0.81$, and DGP-HB's $n = 2{,}828$, $n = 64{,}000$ and $n = 181{,}019$ at $+0.38$, $+1.86$ and $+0.43$, with DGP-HB's $n = 8{,}000$ exactly at nominal (Table S19). None of the four clears a two-standard-error bar. The first stands as written at $\mathsf{L} = 2$, which is the cardinality both designs run; the section on cardinality below is where that qualifier is earned.

So does the third's closing clause, which turns on whether a reader's own configuration clears (7) and not on the coverage record of any one design.

Table S14: The two configurations, and the conditions each was built to satisfy

		Design H		Shipped design		
Condition	Source	Threshold	Realised	Threshold	Realised	Both pass
Propensity frontier $\kappa\alpha > 1/4$	A5(iii), eq:frontier	0.250	0.382	0.250	0.256	yes
$r > d_Z(2\mu+1)/(2(2\mu-1))$	A5(iii), eq:frontier	0.611	2.625	2.500	2.667	yes
Linearisation remainder $\mu > 5/6$	(P-f), eq:muhalf	0.833	5.000	0.833	2.000	yes
Mild ill-posedness $s > a + 1/2$	H1, plan sec.2.6 scope	0.900	4.000	1.250	3.000	yes
L_x window, rate route	(P-c)$\wedge$(P-d), eq:lxwindow	2.953	4.000	1.000	3.000	yes
L_x window, both routes	A8 lower end $\wedge$ (P-d)	3.125	4.000	1.389	3.000	yes
h_x window	eq:hxwindow, (P-g)	1.497	4.000	1.227	3.000	yes
Kernel order $p_K \geq s - 1/2$	lem:unifop Step 4	3.500	8.000	2.500	8.000	yes
Spline degree $> s - 1/2$	A8 basis clause	3.500	4.000	2.500	4.000	yes
$1 \leq d_Z \leq p - 1$	A4(ii)	1.000	1.000	3.000	3.000	yes

Design H: $(\mu, s, a, r, d_Z, p_K) = (5, 4, 2/5, 21/8, 1, 8)$, $p = 2$; $\kappa\alpha = 0.3818$, frontier margin 0.1318.
Shipped: $(\mu, s, a, r, d_Z, p_K) = (2, 3, 3/4, 8/3, 3, 8)$, $p = 4$; $\kappa\alpha = 0.2560$, frontier margin 0.0060.

Notes: Design validation, computed from the configuration alone and not simulated, so no Monte Carlo standard error attaches to any entry. Each row is one condition the theory imposes, with the source of the condition beside it; "Threshold" is the value the condition requires at that design and "Realised" the value the design supplies, so a row passes when the realised value is on the admissible side of its own threshold. The thresholds differ across the two designs because most of them depend on d_Z, μ and s. The leading row is the frontier: $\kappa\alpha = 0.3818$ at Design H against 0.2560 at the shipped design, both above $\frac{1}{4}$, and the margins are 0.1318 and 0.0060. The second row is the reason the headroom is not bought from the propensity score: Design H runs $r = 2.625$ against the shipped 2.667, so the propensity is no smoother. This table validates the two configurations; it is not evidence about the estimator. Source: `scripts/simulations/sim_thm_rootn_headroom_dgp.R`; table `paper/tables/sim/sim_thm_rootn_headroom_design.tex`.

Table S15: The two headroom laws, and the conditions they were built to satisfy

DGP	A2 rel. resid.	$\max\lvert t\rvert$, $E\psi_\star$	$\max\lvert E[\Delta Z \Delta U]\rvert$	$\kappa(J)$	ϱ range	$E[\Delta U^4]$	$\sup\lvert \ell_\star\rvert$
HA	4.0e-07	0.70	1.4e-03	2.4	[0.119, 0.891]	2.10	2.83
HB	1.9e-06	0.11	3.0e-04	23.2	[0.031, 0.970]	3.90	5.12

Validation draw 4×10^5; pilot for J, Σ 4×10^6. Monte Carlo replications per cell $B = 5,000$.

Notes: The Design H counterpart of Table S6, computed once per law and not per replication, so no Monte Carlo standard error attaches to it. "A2 rel. resid." is the relative residual of the balancing equation A2 at the constructed representer, and $\max|t|$ for $E\psi_\star$ and $\max|E[\Delta Z \Delta U]|$ test the two moment conditions the design must satisfy against zero. $\kappa(J)$ is the condition number of $J = E[\Delta Z \Delta Z']$ and is the sense in which DGP-HB is the harsher geometry: 23.2 against 2.4. The ϱ range is the range of the propensity score, which stays inside $(0, 1)$ in both laws. $E[\Delta U^4]$ and $\sup|\ell_\star|$ record the moment and the sup-norm the asymptotics use. The validation draw and the pilot draw for J and Σ are larger than any Monte Carlo cell and are stated in the last row. This table validates the design; it is not evidence about the estimator. Source: `scripts/simulations/sim_thm_rootn_headroom_dgp.R`; table `paper/tables/sim/sim_thm_rootn_headroom_dgpvalid.tex`.

Table S16: The operator the two headroom laws realise

		Power law $\sigma_j \propto j^{-a}$		Exponential		$\varphi_\dagger$ energy	$\mathsf{R}_*(\mu)$	
DGP	Resolvable	$\hat{a}$	R^2	rate	R^2	index at $1 - 10^{-6}$	$\mu = 2$	$\mu = 5$
HA	13	8.252	0.852	1.751	0.993	16	9.0e+34	6.9e+90
HB	14	8.516	0.848	1.707	0.995	18	6.3e+38	2.1e+100

Population operator on the residual index, not the sieve compression of (P-a.3); R_* over the resolvable spectrum is a lower bound.

Notes: The Design H counterpart of Panel B of Table S11, computed on the population operator and not per replication. "Resolvable" is the number of singular values the quadrature resolves, 13 and 14, which is also the number of points each fit runs on. The power-law columns fit the realised spectrum against H1's $\sigma_j \asymp j^{-a}$: the fitted index is 8.252 and 8.516 against the $a = 0.400$ of the design, so the realised operator is more ill-posed than H1 admits, exactly as it is at the shipped design. The exponential columns fit a geometric decay instead and carry the higher R^2 in both laws, 0.993 and 0.995 against 0.852 and 0.848; on this many points the functional form is not identified, so the two fits are printed beside one another rather than one chosen against the other. "$\varphi_\dagger$ energy" is the index below which the structural function holds all but 10^{-6} of its $L^2(\pi)$ energy, and $\mathsf{R}_*(\mu)$ is the source-condition constant at two values of μ, computed over the resolvable spectrum and therefore a lower bound. Source: `scripts/simulations/sim_thm_rootn_headroom_dgp.R`; table `paper/tables/sim/sim_thm_rootn_headroom_spectrum.tex`.

Table S17: The remainder ledger at Design H, with the shipped design for reference

		A5(i)		$\sqrt{n}\,\delta_n^2$				Budget
n	$\ln n$	branch	δ_n	A5(i) as written	r_n only	$\sqrt{n} r_n \delta_n$	$\delta_n / n^{-1/4}$	< 1?
$c_\delta = 1.00$								
125	4.83	parametric	0.9535	10.164	4.903	1.403	3.19	no
354	5.87	parametric	0.7752	11.308	5.314	1.240	3.36	no
1,000	6.91	parametric	0.6038	11.530	5.434	1.049	3.40	no
2,828	7.95	parametric	0.4557	11.045	5.331	0.861	3.32	no
8,000	8.99	parametric	0.3359	10.094	5.068	0.689	3.18	no
22,627	10.03	parametric	0.2431	8.892	4.702	0.542	2.98	no
64,000	11.07	parametric	0.1734	7.606	4.277	0.420	2.76	no
181,019	12.11	parametric	0.1222	6.350	3.828	0.322	2.52	no
$c_\delta = 0.35$								
125	4.83	parametric	0.3337	1.245	0.601	0.491	1.12	no
354	5.87	parametric	0.2713	1.385	0.651	0.434	1.18	no
1,000	6.91	parametric	0.2113	1.412	0.666	0.367	1.19	no
2,828	7.95	parametric	0.1595	1.353	0.653	0.301	1.16	no
8,000	8.99	parametric	0.1176	1.237	0.621	0.241	1.11	no
22,627	10.03	parametric	0.0851	1.089	0.576	0.190	1.04	no
64,000	11.07	parametric	0.0607	0.932	0.524	0.147	0.97	**yes**
181,019	12.11	parametric	0.0428	0.778	0.469	0.113	0.88	**yes**
$c_\delta = 0.25$								
125	4.83	parametric	0.2384	0.635	0.306	0.351	0.80	**yes**
354	5.87	parametric	0.1938	0.707	0.332	0.310	0.84	**yes**
1,000	6.91	parametric	0.1510	0.721	0.340	0.262	0.85	**yes**
2,828	7.95	parametric	0.1139	0.690	0.333	0.215	0.83	**yes**
8,000	8.99	parametric	0.0840	0.631	0.317	0.172	0.79	**yes**
22,627	10.03	parametric	0.0608	0.556	0.294	0.136	0.75	**yes**
64,000	11.07	parametric	0.0433	0.475	0.267	0.105	0.69	**yes**
181,019	12.11	parametric	0.0305	0.397	0.239	0.080	0.63	**yes**
Shipped design at $c_\delta = 1$, for reference								
125	4.83	propensity	1.0238	11.720	11.720	2.442	3.42	no
354	5.87	propensity	0.9169	15.818	15.818	2.637	3.98	no
1,000	6.91	propensity	0.8007	20.274	20.274	2.776	4.50	no
2,828	7.95	propensity	0.6864	25.057	25.057	2.870	5.01	no
8,000	8.99	propensity	0.5804	30.127	30.127	2.926	5.49	no
22,627	10.03	propensity	0.4855	35.450	35.450	2.951	5.95	no
64,000	11.07	propensity	0.4026	40.997	40.997	2.951	6.40	no
181,019	12.11	propensity	0.3315	46.744	46.744	2.930	6.84	no

Notes: The levels behind Figure S5, and the source of the budget column of Table S13. Deterministic functions of the configuration and the constant, not simulated quantities, so no Monte Carlo standard error attaches to them. "A5(i) branch" names which of the two branches of A5(i) attains $\hat{\delta}_n$ at that cell: the parametric branch, of order $n^{-1/2}\sqrt{\ln n}$, at every Design H cell, and the propensity branch at every shipped cell. "A5(i) as written" is $\sqrt{n}\,\delta_n^2$ at the maximum the condition takes and "r_n only" the same product computed from the propensity branch alone, so the gap between the two columns is what the parametric floor costs; the two coincide at every shipped row, which is what it means for the propensity branch to attain the maximum there. $\sqrt{n} r_n \delta_n$ is the second Step-4 product and $\delta_n / n^{-1/4}$ measures δ_n against the rate A5(ii) asks it to beat. The last column marks the cells at which the leading product is below one, and those are the cells Table S13 lists. The first three panels are Design H at the three values of c_δ, which scales the product quadratically; the last is the shipped design at $c_\delta = 1$, printed so that the two designs can be read on one page. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; table `paper/tables/sim/sim_thm_rootn_headroom_budget.tex`.

Table S18: The limit law at Design H, at levels, by law and by the constant in A5(ii)

	$\sqrt{n}\delta_n^2$		Coverage (MC SE)		Worst quantile	Lin. resid.
n	budget	$\hat{\text{sd}}/\text{sd}_\infty$ (MC SE)	90%	95%	(MC SE units)	mean
Panel A: DGP-HA, $c_\delta = 1.00$, $B = 5,000$						
125	10.164	1.760 (0.018)	0.886 (0.004)	0.938 (0.003)	3.19	2.054
354	11.308	1.526 (0.015)	0.880 (0.005)	0.934 (0.004)	4.00	1.608
1,000	11.530	1.322 (0.013)	0.888 (0.004)	0.944 (0.003)	2.00	1.261
2,828	11.045	1.220 (0.012)	0.887 (0.004)	0.936 (0.003)	3.29	0.952
8,000	10.094	1.101 (0.011)	0.896 (0.004)	0.948 (0.003)	0.92	0.709
22,627	8.892	1.081 (0.011)	0.889 (0.004)	0.945 (0.003)	2.29	0.521
64,000	7.606	1.045 (0.010)	0.891 (0.004)	0.947 (0.003)	1.46	0.375
181,019	6.350	1.006 (0.010)	0.901 (0.004)	0.955 (0.003)	1.25	0.265
Panel B: DGP-HA, $c_\delta = 0.35$, $B = 5,000$						
125	1.245	1.191 (0.012)	0.888 (0.004)	0.941 (0.003)	2.15	0.780
354	1.385	1.088 (0.011)	0.897 (0.004)	0.945 (0.003)	1.99	0.599
1,000	1.412	1.050 (0.011)	0.897 (0.004)	0.948 (0.003)	1.17	0.461
2,828	1.353	1.031 (0.010)	0.901 (0.004)	0.950 (0.003)	1.10	0.343
8,000	1.237	1.034 (0.010)	0.891 (0.004)	0.945 (0.003)	1.88	0.250
22,627	1.089	0.996 (0.010)	0.903 (0.004)	0.952 (0.003)	1.26	0.184
64,000	**0.932**	1.012 (0.010)	0.892 (0.004)	0.948 (0.003)	1.23	0.133
181,019	**0.778**	1.023 (0.010)	0.892 (0.004)	0.944 (0.003)	2.58	0.093
Panel C: DGP-HA, $c_\delta = 0.25$, $B = 5,000$						
125	**0.635**	1.130 (0.011)	0.884 (0.005)	0.940 (0.003)	3.06	0.611
354	**0.707**	1.068 (0.011)	0.893 (0.004)	0.948 (0.003)	2.46	0.452
1,000	**0.721**	1.022 (0.010)	0.903 (0.004)	0.950 (0.003)	1.94	0.338
2,828	**0.690**	1.018 (0.010)	0.896 (0.004)	0.946 (0.003)	1.79	0.250
8,000	**0.631**	1.006 (0.010)	0.896 (0.004)	0.948 (0.003)	1.89	0.185
22,627	**0.556**	1.002 (0.010)	0.897 (0.004)	0.953 (0.003)	0.76	0.133
64,000	**0.475**	1.002 (0.010)	0.900 (0.004)	0.950 (0.003)	3.28	0.094

181,019	**0.397**	1.002 (0.010)	0.899 (0.004)	0.947 (0.003)	1.01	0.067
Panel D: DGP-HB, $c_\delta = 1.00$, $B = 5,000$						
125	10.164	1.843 (0.018)	0.883 (0.005)	0.939 (0.003)	3.19	3.256
354	11.308	1.502 (0.015)	0.892 (0.004)	0.941 (0.003)	2.56	2.478
1,000	11.530	1.287 (0.013)	0.896 (0.004)	0.949 (0.003)	2.15	1.917
2,828	11.045	1.193 (0.012)	0.883 (0.005)	0.937 (0.003)	2.52	1.440
8,000	10.094	1.092 (0.011)	0.895 (0.004)	0.948 (0.003)	1.98	1.046
22,627	8.892	1.051 (0.011)	0.903 (0.004)	0.948 (0.003)	1.19	0.755
64,000	7.606	1.015 (0.010)	0.906 (0.004)	0.952 (0.003)	1.22	0.537
181,019	6.350	1.013 (0.010)	0.899 (0.004)	0.950 (0.003)	1.60	0.386
Panel E: DGP-HB, $c_\delta = 0.35$, $B = 5,000$						
125	1.245	1.251 (0.013)	0.882 (0.005)	0.938 (0.003)	3.32	1.298
354	1.385	1.100 (0.011)	0.897 (0.004)	0.948 (0.003)	1.29	0.919
1,000	1.412	1.065 (0.011)	0.896 (0.004)	0.944 (0.003)	1.30	0.685
2,828	1.353	1.027 (0.010)	0.894 (0.004)	0.948 (0.003)	1.44	0.521
8,000	1.237	1.025 (0.010)	0.895 (0.004)	0.949 (0.003)	2.73	0.379
22,627	1.089	1.010 (0.010)	0.896 (0.004)	0.948 (0.003)	1.54	0.277
64,000	**0.932**	0.989 (0.010)	0.900 (0.004)	0.950 (0.003)	1.77	0.196
181,019	**0.778**	1.002 (0.010)	0.900 (0.004)	0.953 (0.003)	1.53	0.137
Panel F: DGP-HB, $c_\delta = 0.25$, $B = 5,000$						
125	**0.635**	1.209 (0.012)	0.888 (0.004)	0.940 (0.003)	3.37	1.029
354	**0.707**	1.079 (0.011)	0.898 (0.004)	0.947 (0.003)	2.31	0.709
1,000	**0.721**	1.039 (0.010)	0.898 (0.004)	0.947 (0.003)	1.37	0.521
2,828	**0.690**	1.021 (0.010)	0.902 (0.004)	0.949 (0.003)	0.86	0.381
8,000	**0.631**	1.005 (0.010)	0.900 (0.004)	0.951 (0.003)	1.15	0.274
22,627	**0.556**	1.009 (0.010)	0.899 (0.004)	0.949 (0.003)	1.02	0.200
64,000	**0.475**	0.988 (0.010)	0.908 (0.004)	0.955 (0.003)	2.19	0.141
181,019	**0.397**	0.997 (0.010)	0.902 (0.004)	0.952 (0.003)	0.86	0.099

Notes: The levels of the quantities Table S19 reports in standard error units, and the source of the linearisation residuals the article reads at $n = 125$ and at $n = 181{,}019$. The budget column is $\sqrt{n}\,\delta_n^2$ at the panel's

own c_δ, set in bold where it is below one. $\hat{\mathrm{sd}}/\mathrm{sd}_\infty$ is the simulated standard deviation of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ divided by its asymptotic value $(\vartheta' V \vartheta)^{1/2}$, with its Monte Carlo standard error in parentheses. Coverage is of nominal 90% and 95% intervals, with the Monte Carlo standard error in parentheses evaluated at the *empirical* rate and not at the nominal one, which is why it moves across cells. "Worst quantile" is the largest absolute gap over the seven studentised quantiles $\{0.025, 0.05, 0.25, 0.5, 0.75, 0.95, 0.975\}$, in Monte Carlo standard error units. "Lin. resid." is $E\|\sqrt{n}(\hat{\theta} - \theta_\dagger) - J^{-1}\sqrt{n}\,\mathbb{E}_n\psi_\star\|$, formed per replication and then averaged, so nothing cancels across replications. Panels A to C are DGP-HA at the three constants and Panels D to F are DGP-HB at the same three; all six are the same estimator at the same configuration, $(\mu, s, a, r, d_Z, p_K) = (5, 4, \frac{2}{5}, \frac{21}{8}, 1, 8)$ at $p = 2$. 5,000 replications in every cell. The column head does not repeat on continuation pages. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; table `paper/tables/sim/sim_thm_rootn_headroom_limitlaw.tex`.

Table S19: Distance from the asymptotic value at Design H, in Monte Carlo standard error units

				Deviation from the asymptotic value, in MC SE units				
DGP	c_δ	n	Budget	$\hat{\text{sd}}/\text{sd}_\infty$	90% cov.	95% cov.	Worst quantile	Centring
HA	1.00	125	10.164	43.17	-3.07	-3.52	3.19	-1.59
HA	1.00	354	11.308	34.47	-4.27	-4.66	4.00	0.51
HA	1.00	1,000	11.530	24.34	-2.73	-1.73	2.00	-0.46
HA	1.00	2,828	11.045	18.04	-2.99	-4.10	3.29	0.02
HA	1.00	8,000	10.094	9.14	-0.97	-0.64	0.92	-0.16
HA	1.00	22,627	8.892	7.51	-2.39	-1.55	2.29	-0.78
HA	1.00	64,000	7.606	4.31	-1.95	-0.82	1.46	-0.47
HA	1.00	181,019	6.350	0.63	0.24	1.63	1.25	0.63
HA	0.35	125	1.245	16.01	-2.65	-2.81	2.15	0.48
HA	0.35	354	1.385	8.09	-0.74	-1.67	1.99	-0.32
HA	0.35	1,000	1.412	4.80	-0.79	-0.51	1.17	0.86
HA	0.35	2,828	1.353	2.98	0.19	-0.13	1.10	1.01
HA	0.35	8,000	1.237	3.27	-2.00	-1.55	1.88	-0.94
HA	0.35	22,627	1.089	-0.44	0.77	0.73	1.26	-0.74
HA	0.35	64,000	0.932	1.20	-1.73	-0.57	1.23	-0.31
HA	0.35	181,019	0.778	2.25	-1.91	-1.85	2.58	0.57
HA	0.25	125	0.635	11.50	-3.45	-2.98	3.06	1.64
HA	0.25	354	0.707	6.35	-1.65	-0.64	2.46	0.78
HA	0.25	1,000	0.721	2.12	0.81	0.13	1.94	0.76
HA	0.25	2,828	0.690	1.74	-0.88	-1.37	1.79	0.85
HA	0.25	8,000	0.631	0.61	-0.97	-0.64	1.89	-0.35
HA	0.25	22,627	0.556	0.18	-0.61	1.00	0.76	0.84
HA	0.25	64,000	0.475	0.22	-0.05	0.00	3.28	2.49
HA	0.25	181,019	0.397	0.16	-0.19	-0.82	1.01	0.05
HB	1.00	125	10.164	45.74	-3.78	-3.20	3.19	0.26
HB	1.00	354	11.308	33.44	-1.87	-2.65	2.56	-0.85
HB	1.00	1,000	11.530	22.32	-0.88	-0.32	2.15	1.33

HB	1.00	2,828	11.045	16.18	-3.70	-3.68	2.52	0.86
HB	1.00	8,000	10.094	8.42	-1.15	-0.64	1.98	-0.10
HB	1.00	22,627	8.892	4.89	0.77	-0.76	1.19	-0.62
HB	1.00	64,000	7.606	1.50	1.45	0.66	1.22	-0.03
HB	1.00	181,019	6.350	1.24	-0.23	0.07	1.60	1.11
HB	0.35	125	1.245	20.05	-3.90	-3.47	3.32	1.67
HB	0.35	354	1.385	9.06	-0.65	-0.57	1.29	-0.57
HB	0.35	1,000	1.412	6.09	-0.97	-1.85	1.30	0.80
HB	0.35	2,828	1.353	2.66	-1.33	-0.51	1.44	-0.15
HB	0.35	8,000	1.237	2.40	-1.15	-0.45	2.73	2.00
HB	0.35	22,627	1.089	1.01	-0.97	-0.70	1.54	1.20
HB	0.35	64,000	0.932	-1.10	-0.09	0.13	1.77	0.18
HB	0.35	181,019	0.778	0.18	-0.09	1.00	1.53	0.56
HB	0.25	125	0.635	17.26	-2.60	-2.98	3.37	1.86
HB	0.25	354	0.707	7.35	-0.56	-1.07	2.31	-1.89
HB	0.25	1,000	0.721	3.74	-0.37	-0.89	1.37	0.46
HB	0.25	2,828	0.690	2.10	0.38	-0.19	0.86	-0.22
HB	0.25	8,000	0.631	0.51	0.00	0.46	1.15	0.38
HB	0.25	22,627	0.556	0.94	-0.33	-0.45	1.02	-0.21
HB	0.25	64,000	0.475	-1.22	1.86	1.85	2.19	0.12
HB	0.25	181,019	0.397	-0.28	0.43	0.59	0.86	-0.10

$B = 5,000$ at every cell. Positive means the empirical quantity exceeds its asymptotic target.

Notes: Every entry in the last five columns is a gap between a simulated quantity and its asymptotic value, divided by the Monte Carlo standard error of that gap, so the entries are counts of standard errors and not levels; the levels are in Table S18. Table S13 summarises this table; it is the only place all 48 core cells are reported, including the 28 whose budget is above one and which Table S13 does not list. The five measures are those defined in the notes to Table S1: $\hat{\text{sd}}/\text{sd}_\infty$ compares the simulated standard deviation of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ with $(\vartheta' V \vartheta)^{1/2}$; the two coverage columns compare empirical coverage of nominal 90% and 95% intervals with those levels; "worst quantile" is the largest absolute gap over the seven studentised quantiles; and "centring" tests the mean of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)$ against zero. "Budget" is $\sqrt{n}\,\delta_n^2$ at the cell's own c_δ, and all three values of c_δ satisfy the same hypotheses. 5,000 replications in every cell. An entry

inside ± 2 is not distinguishable from zero at that replication count, which is not the same thing as a match; with five measures at each of 48 cells, some entries outside that bar are expected. The column head does not repeat on continuation pages. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; table `paper/tables/sim/sim_thm_rootn_headroom_seunits.tex`.

Table S20: Studentised quantiles against the standard normal at Design H, cell by cell

DGP	c_δ	n	Quantile	Empirical (MC SE)	Asymptotic	Diff. (MC SE units)
HA	0.25	181,019	0.025	-1.984 (0.038)	-1.960	-0.65
			0.050	-1.643 (0.030)	-1.645	0.05
			0.250	-0.668 (0.019)	-0.674	0.34
			0.500	0.000 (0.018)	0.000	0.01
			0.750	0.656 (0.019)	0.674	-0.96
			0.950	1.653 (0.030)	1.645	0.26
			0.975	1.998 (0.038)	1.960	1.01
HA	0.25	22,627	0.025	-1.946 (0.038)	-1.960	0.37
			0.050	-1.657 (0.030)	-1.645	-0.41
			0.250	-0.661 (0.019)	-0.674	0.70
			0.500	0.014 (0.018)	0.000	0.76
			0.750	0.683 (0.019)	0.674	0.43
			0.950	1.656 (0.030)	1.645	0.37
			0.975	1.943 (0.038)	1.960	-0.46
HA	1.00	181,019	0.025	-1.930 (0.038)	-1.960	0.81
			0.050	-1.633 (0.030)	-1.645	0.39
			0.250	-0.660 (0.019)	-0.674	0.74
			0.500	0.000 (0.018)	0.000	0.03
			0.750	0.698 (0.019)	0.674	1.22
			0.950	1.651 (0.030)	1.645	0.20
			0.975	1.913 (0.038)	1.960	-1.25
HA	1.00	22,627	0.025	-1.976 (0.038)	-1.960	-0.41
			0.050	-1.703 (0.030)	-1.645	-1.94
			0.250	-0.719 (0.019)	-0.674	-2.29
			0.500	-0.005 (0.018)	0.000	-0.31
			0.750	0.673 (0.019)	0.674	-0.06
			0.950	1.683 (0.030)	1.645	1.27
			0.975	2.011 (0.038)	1.960	1.36

HB	0.25	181,019	0.025	-1.955 (0.038)	-1.960	0.12
			0.050	-1.625 (0.030)	-1.645	0.67
			0.250	-0.691 (0.019)	-0.674	-0.86
			0.500	-0.003 (0.018)	0.000	-0.15
			0.750	0.674 (0.019)	0.674	-0.02
			0.950	1.635 (0.030)	1.645	-0.32
			0.975	1.944 (0.038)	1.960	-0.43
HB	0.25	22,627	0.025	-1.979 (0.038)	-1.960	-0.50
			0.050	-1.657 (0.030)	-1.645	-0.42
			0.250	-0.688 (0.019)	-0.674	-0.70
			0.500	-0.018 (0.018)	0.000	-1.02
			0.750	0.690 (0.019)	0.674	0.82
			0.950	1.640 (0.030)	1.645	-0.15
			0.975	1.958 (0.038)	1.960	-0.05
HB	1.00	181,019	0.025	-1.938 (0.038)	-1.960	0.59
			0.050	-1.610 (0.030)	-1.645	1.18
			0.250	-0.666 (0.019)	-0.674	0.45
			0.500	0.028 (0.018)	0.000	1.60
			0.750	0.667 (0.019)	0.674	-0.39
			0.950	1.665 (0.030)	1.645	0.67
			0.975	1.987 (0.038)	1.960	0.72
HB	1.00	22,627	0.025	-1.983 (0.038)	-1.960	-0.62
			0.050	-1.644 (0.030)	-1.645	0.03
			0.250	-0.696 (0.019)	-0.674	-1.13
			0.500	-0.002 (0.018)	0.000	-0.10
			0.750	0.681 (0.019)	0.674	0.35
			0.950	1.609 (0.030)	1.645	-1.19
			0.975	1.972 (0.038)	1.960	0.33

$B = 5,000$. Studentised statistic in the mixed direction; asymptotic law $N(0,1)$.

Notes: The quantile-by-quantile detail behind the "worst quantile" column of Table S19. *It is not the*

cell set Figure S6 draws: the figure is at n = 1,000 and n = 181,019 and this table is at n = 22,627 and n = 181,019, so only the larger sample size is common to both, and their largest gaps differ accordingly at 2.15 and 2.29. "Empirical" is the sample quantile of the studentised statistic $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)/(\vartheta'\hat{V}\vartheta)^{1/2}$, with its Monte Carlo standard error in parentheses; that standard error is the Bahadur expression $\sqrt{q(1-q)/B}/\phi(\Phi^{-1}(q))$, evaluated at the density of the *asymptotic* law, which is the comparison target, and it therefore does not vary with n. "Asymptotic" is the corresponding standard normal quantile and the last column is the gap between them in those units. Eight of the 48 core cells are printed, at n = 22,627 and n = 181,019 in each of the two laws and at each of the two extreme constants; the cell at n = 64,000 the article discusses is not among them. 5,000 replications in every cell. The column head does not repeat on continuation pages. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; table `paper/tables/sim/sim_thm_rootn_headroom_quantiles.tex`.

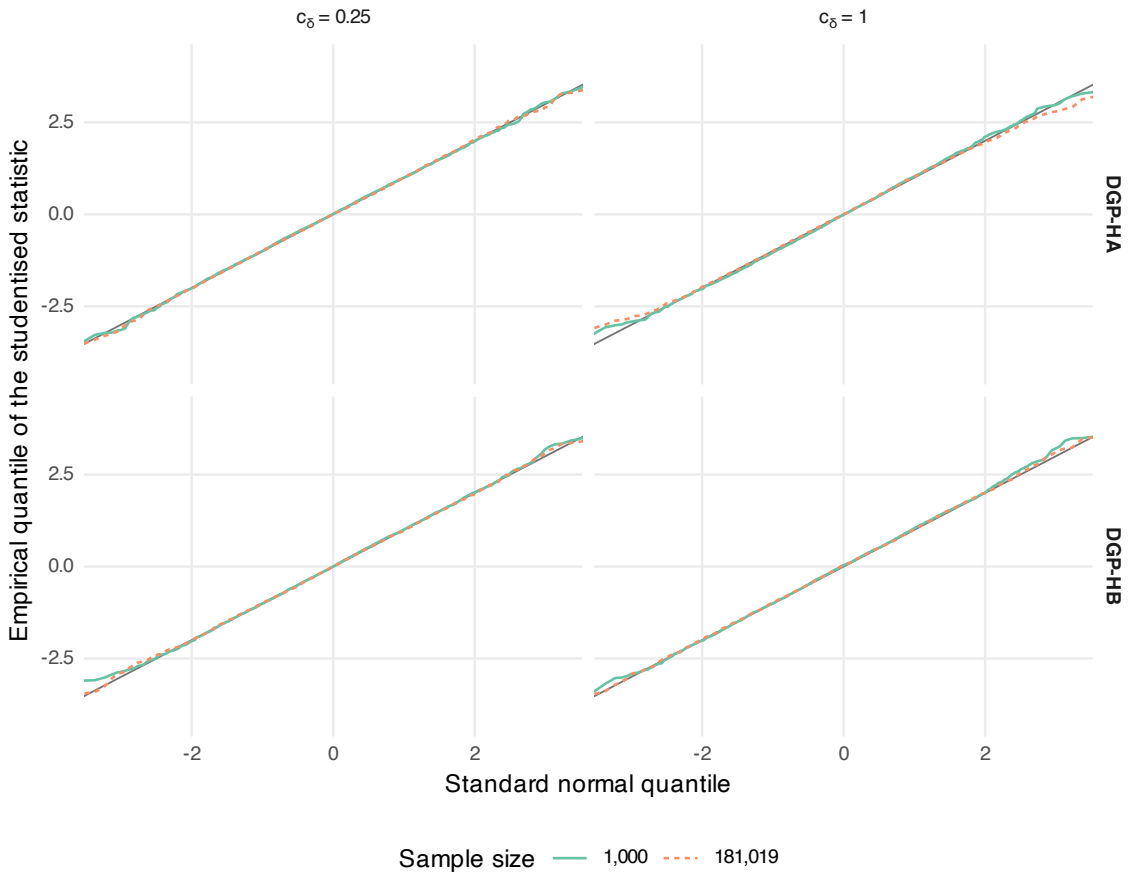


Figure S6: Studentised quantiles against the standard normal at Design H

Notes: The Design H counterpart of Figure S1. *It is not Table S20 drawn*: that table is at $n = 22{,}627$ and $n = 181{,}019$ while this figure is at $n = 1{,}000$ and $n = 181{,}019$, so only the larger sample size is common to the two and their largest gaps differ accordingly. Each curve plots the empirical quantiles of the studentised statistic $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)/(\vartheta'\hat{V}\vartheta)^{1/2}$ against the quantiles of the standard normal. The straight grey line in each panel is the 45° line the article's limit law predicts, so a curve lying on it is that law's prediction and departure from it is read vertically. Columns are the two extreme values of c_δ, the constant in the $O_P(\delta_n)$ of A5(ii); rows are the two data-generating laws. The solid curve is $n = 1{,}000$ and the dashed curve $n = 181{,}019$. The curves lie on the line in all four panels and at both sample sizes, including the $c_\delta = 1$ column, where the realised budget is between 6.350 and 11.530; the statistic is studentised, so a scale error common to $\hat{V}$ and to the sampling distribution does not appear here and is read from Table S19 instead. The largest quantile gap among the eight cells drawn is 2.15 standard errors. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_headroom.R`; figure `paper/figures/sim/sim_thm_rootn_headroom_qq.pdf`.

Monte Carlo: the instrument's cardinality

This section is the third of the article's four Monte Carlo experiments, which varies the instrument's cardinality on the design of the two sections above and holds everything else fixed. It adds no result: the text is the article's, moved here unchanged except for the cross-references a move breaks. Everything above runs at $\mathsf{L} = 2$, and we now vary the one thing the two designs held fixed. Theorem 29 is stated at general finite L, so a reader entitled to the four-cell instrument of Section 6 of the article is entitled to ask whether the limit law is as accurate there. This arm runs the shipped design's own configuration, $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$ at $p = 4$ and $d_Z = 3$, at $\mathsf{L} \in \{2, 4\}$ and $n \in \{1{,}000, 8{,}000, 64{,}000\}$, with 5,000 replications in every cell.

One factor varies. The three rate constants, the injection regime, the n grid, the replication count and the overlap floor are identical across the two panels, and the leading Step-4 product $\sqrt{n}\,\delta_n^2$ is 1.27, 1.88 and 2.56 in both, so nothing the remainder accounting prices differs between them (Table S21). The binary panel is the estimator of the preceding pages and not a re-implementation of it: thirty-six objects are bit-identical against the pre-cardinality sources, from the configuration symbols through the population objects to a whole replication (Table S25). That identity holds at the object level and at one replication and not at a cell mean: the cardinality arm draws its own stream, so its Panel A and Panel A of Table S2 are distinct 5,000-replication realisations of one design rather than the same numbers printed twice. At the three sample sizes they share they differ by at most 1.6 Monte Carlo standard errors of the difference on every statistic that carries one, worst at the standard-deviation ratio at $n = 1{,}000$.

The two panels do not agree. At $\mathsf{L} = 2$ the linearisation residual that (58) sends to zero falls $0.657 \to 0.474 \to 0.363$ and 95% coverage is 0.946, 0.950 and 0.948. At $\mathsf{L} = 4$ the same residual *rises*, $0.856 \to 0.909 \to 1.230$, and coverage decays $0.941 \to 0.925 \to 0.895$.

At $n = 64{,}000$ the worst of the five measures is 2.20 standard errors at $\mathsf{L} = 2$ and 18.91 at $\mathsf{L} = 4$, and the four-cell entry grows monotonically across the grid while the binary one falls (Table S22). The studentised quantiles carry the same shape: at $\mathsf{L} = 4$ and $n = 64{,}000$ the 0.95 quantile is 2.042 against the normal's 1.645, a gap of 13.29 standard errors, while no quantile at $\mathsf{L} = 2$ departs by more than 2.20 anywhere on the grid (Table S23; Figure S7 draws them).

We read this as Proposition 20(iii) confirmed rather than Theorem 29 refuted, and the reason is that the two panels differ in the condition the proposition names and in nothing else the remainder accounting prices. By Corollary 21 N1 is $p(\mathsf{L} - 1)$ scalar conditions and not p, which at the $p = 4$ of this design is four conditions at $\mathsf{L} = 2$ and twelve at $\mathsf{L} = 4$. The design satisfies the four and violates the twelve. Computed in population rather than estimated, the largest entrywise defect $\max_{j,l} \|E[L_\star \mid \Delta Z]_{jl}\|$ is 8.23×10^{-6} against a $\|L_\star\|$ of 2.4613 at $\mathsf{L} = 2$, and 0.7293 against 3.8227 at $\mathsf{L} = 4$, so the ratio moves from 3.3×10^{-6} to 0.191 (Table S28).

The identity at $\mathsf{L} = 2$ is structural and not a small number. All of the solved representer's coefficient energy sits on odd q there, where A_q is odd in u, so $L_\star$ is odd in u; ΔU is independent of $(\Delta Z, W)$ by construction and symmetric about zero; and the conditional mean vanishes by parity. The share of that energy sitting on odd q is 1.000000 at $\mathsf{L} = 2$ against 0.707 at $\mathsf{L} = 4$, so the mechanism is measured and not inferred from the defect it produces, and the residual 3.3×10^{-6} is the antiderivative grid's own parity error rather than a live entry. DGP-B repeats the pattern at both cardinalities, so this is a property of the construction and not of one law.

Proposition 20(iii) makes N1 necessary *and sufficient* for the moment's ϱ-derivative to vanish, and the two panels exhibit both directions. Under N1 the first-order propensity term $\sqrt{n}\,\mathbb{E}_n[L_\star(\hat{\varrho} - \varrho)]$ is a centred average of quantities of size r_n and decays at $n^{-\alpha} = n^{-0.320}$; when N1 fails, that term carries a non-vanishing mean and grows at $n^{1/2-\alpha} = n^{+0.180}$. Both

references are read off the design rather than fitted. Measured, the term decays at $n^{-0.302}$ where the condition holds, which is 1.4 Monte Carlo standard errors from the orthogonality reference, and grows at $n^{+0.206}$ where it fails, which is 2.0 from the failure reference, at a standard error of 0.013 on each exponent (Panel E of Table S28). *The condition holds where the proposition requires the term to decay and fails where it requires the term to grow.* At $\mathsf{L} = 2$ the term decays at the predicted rate. At $\mathsf{L} = 4$ it grows, which is the direction N1's failure forces and the one a centred average cannot produce; the $+0.206$ sits two standard errors above the failure reference rather than on it, and we do not read it as that reference recovered. Under N1 failure T_2 carries a growing mean term beside a still-live centred one, whose size Panel E prices at $\Lambda\tau_n = 0.1813$ against a total of 0.8225 at $n = 8{,}000$, so a slope over a single leg is a weighted average of two exponents and bounds the mean term's exponent from below rather than measuring it.

Those are not the exponents Table S27 implies, and the difference is a confound in that arm's measurement code, which we removed rather than absorbed. The arm forms its propensity residual against $\Delta U - \Delta h$ and evaluates $L_\star$ at that residual, so the T_2 it accumulates carries the $\sqrt{n}\,\|h\|r_n$ cross-product the theorem prices separately: a contamination free of L, carrying a non-zero conditional mean, negligible beside a live defect and dominant where the defect is absent. Its levels fall $0.337 \to 0.270$ at $\mathsf{L} = 2$ and rise $0.569 \to 0.843$ at $\mathsf{L} = 4$, which are slopes of -0.108 and $+0.189$ over the leg, and those are the figures that table prints. Re-running the T_2 accumulation alone with h switched off, at 1,500 replications on a separate stream whose h-on rows reproduce the shipped ones to within 2.1 standard errors of the difference, moves $\mathsf{L} = 2$ from -0.100 to -0.302 and $\mathsf{L} = 4$ from $+0.154$ to $+0.206$. The binary arm sits 17 standard errors from the orthogonality reference with h on and 1.4 from it with h off, so removing the confound dissolves the binary anomaly. It does not sharpen the four-cell one: $+0.154$ and $+0.206$ are both 0.026 from the failure reference, so the estimate crosses that reference rather than closing on it, and it crosses to the side the preceding paragraph reads as a bound.

Each exponent is still a slope over one leg, $n = 1{,}000$ to $8{,}000$, and one leg shows no curvature. What has changed is where the legs land: both now sit on their own reference rather than between the two, and the discrepancy a third sample size was going to adjudicate has been answered instead by switching off the term that produced it. A third point would buy a check on curvature and we did not run one, for the reason the closing paragraphs of this section give.

The ablation localises the failure to one nuisance and exonerates the other twice. Two arms switch on exactly one estimated nuisance each, at $n \in \{1{,}000, 8{,}000\}$ and 5,000 replications. The $\hat{\varrho}$ arm reaches 12.10 standard errors at $\mathsf{L} = 4$ and rises, while converging at $\mathsf{L} = 2$ at 2.97 and falling; its residual at four cells is 0.874 and 0.913, which reproduces the full arm's 0.856 and 0.909 to within 2.2 and 0.4 per cent, both computed on the unrounded values and not on the four digits printed, so switching that one nuisance on reproduces the whole failure. The $\hat{\ell}$ arm's residual decays at -0.171 at $\mathsf{L} = 2$ and -0.172 at $\mathsf{L} = 4$, the same exponent to a thousandth, which is what "the cardinality is not the problem in this channel" looks like (Table S26). A5(iv) is exonerated a second time by a measurement and not by an arm: the norm the moment consumes divided by the norm A5(iv) prices is 0.463 at $\mathsf{L} = 2$ against 0.420 at $\mathsf{L} = 4$, so the Frobenius bound is *conservative* at the higher cardinality rather than slack there, and the measurement moves the wrong way for any account that blames it.

One entry of that ablation does not fall and we flag it rather than absorb it. The $\hat{\ell}$ arm's worst of five at $\mathsf{L} = 4$ goes only from 6.49 to 5.03, against 6.29 to 1.75 at $\mathsf{L} = 2$, and at $n = 8{,}000$ the statistic carrying it is the centring and not the scale. The residual (58) names decays there at the binary arm's own exponent, so the locating claim stands; what does not stand is a reading of that arm as clean on every measure.

The design remains admissible at both cardinalities and the deterioration is not a boundary artefact. Overlap clears its floor at four cells, at $\lambda_{\min}(\Omega) = 0.0273$ against $\varepsilon_{\varrho}/\mathsf{L} = 0.0250$; the Jacobian's condition number is 3.63 against 3.62; and the balancing residual A2 asks for is

3.6×10^{-5} against 4.2×10^{-7}, which the remainder accounting prices at 1.9×10^{-4} against a leading product of 2.56 (Table S24). What does move with the cardinality is the budget (R-a) prices, $\sup |\ell_\star|$ rising from 3.04 to 7.56.

Two limits. The four-cell grid stops at $n = 64{,}000$ and the ablation at $n = 8{,}000$, in both cases on cost: one replication at $\mathsf{L} = 4$ and $n = 64{,}000$ takes 2.20 seconds, so a cell of 5,000 of them is three hours of processor time and a little under twenty-three minutes on the eight workers the run distributes it over, and we would rather carry two well-separated points at 5,000 replications than three at a thinned count. And the arm injects its nuisances at fixed rate constants, so it prices a remainder of a declared size and establishes nothing about whether any estimator attains that size.

What this changes above is the first of the three readings, and only its last clause. The remainder accounting is complete for nuisance errors of the priced size at $\mathsf{L} = 2$ and is not at $\mathsf{L} = 4$, because the term it completes is the one N1 deletes at the first cardinality and leaves standing at the second. Section 6 of the article runs at $\mathsf{L} = 4$ and reports what the same defect measures on its own data, where at $n = 983$ it is not resolvable from zero.

Table S21: The limit law at two instrument cardinalities, on the design of Table S2

	$\sqrt{n}\delta_n^2$		Coverage (MC SE)		Lin. resid.	$\|\hat{V} - V\|_\infty$
n	budget	$\hat{\text{sd}}/\text{sd}_\infty$ (MC SE)	90%	95%	mean (SE)	mean
Panel A: $\mathsf{L} = 2$						
1,000	1.27	1.087 (0.011)	0.888 (0.004)	0.946 (0.003)	0.657 (0.004)	0.412
8,000	1.88	1.027 (0.010)	0.899 (0.004)	0.950 (0.003)	0.474 (0.003)	0.211
64,000	2.56	1.019 (0.010)	0.895 (0.004)	0.948 (0.003)	0.363 (0.002)	0.123
Panel B: $\mathsf{L} = 4$						
1,000	1.27	1.097 (0.011)	0.885 (0.005)	0.941 (0.003)	0.856 (0.007)	0.669
8,000	1.88	1.129 (0.011)	0.865 (0.005)	0.925 (0.004)	0.909 (0.008)	0.318
64,000	2.56	1.233 (0.012)	0.821 (0.005)	0.895 (0.004)	1.230 (0.012)	0.174

Notes: DGP-A at the frontier-clearing configuration $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$, $p = 4$, $d_Z = 3$, which is the design of Table S2 Panel A. *The two panels differ in the instrument's cardinality and in nothing else the theorem prices*: $c_\delta = 0.25$, $c_\epsilon = c_r = 1$, the smooth h geometry, the n grid, the overlap floor $\varepsilon_\varrho = 0.10$ and the injected-nuisance regime are common, and the three rate constants are those of the three injected nuisance errors, c_δ the constant in the $O_P(\delta_n)$ of A5(ii) on $\hat{\varphi} - \varphi_\dagger$, c_ϵ the constant in the ϵ_n of A5(iv) on $\hat{\ell} - \ell_\star$, and c_r the constant in the r_n of A4(iii) on $\hat{\varrho} - \varrho$; each pins a size and none pins an exponent. The budget column, which is $\sqrt{n}\,\delta_n^2$, is therefore identical row by row across the two panels. What L forces is three propensity indices instead of one and four first-stage shifts instead of two, generated by a rule that returns the binary design of the preceding pages at $\mathsf{L} = 2$; Table S25 is the check that it does. The columns are those of Table S2 and are defined in its notes. 5,000 replications in every cell. *"Lin. resid." is the statistic* (58) *sends to zero*, formed per replication and then averaged, so nothing cancels across replications; it falls in Panel A and rises in Panel B. Source: `scripts/simulations/sim_thm_rootn_cardinality.R`; table `paper/tables/sim/sim_thm_rootn_cardinality.tex`.

Table S22: The same six cells in Monte Carlo standard error units

		Deviation from the asymptotic value, in MC SE units				
L	n	$\hat{\text{sd}}/\text{sd}_\infty$	90% cov.	95% cov.	Worst quantile	Centring
2	1,000	7.98	-2.65	-1.37	1.56	0.08
2	8,000	2.58	-0.14	0.00	1.55	-1.33
2	64,000	1.91	-1.24	-0.70	2.20	-0.85
4	1,000	8.86	-3.41	-2.76	2.85	2.67
4	8,000	11.39	-7.24	-6.71	6.84	0.79
4	64,000	18.91	-14.51	-12.72	13.29	2.76

Notes: The five measures of Table S1, defined in its notes, at the six cells of Table S21. Every entry is a gap between a simulated quantity and its asymptotic value divided by the Monte Carlo standard error of that gap, so the entries are counts of standard errors and not levels. The "worst of five" the text quotes is the largest absolute entry in a row: 2.20 at $\mathsf{L} = 2$ and 18.91 at $\mathsf{L} = 4$, both at $n = 64{,}000$. An entry inside ± 2 is not distinguishable from zero at 5,000 replications, which is not the same thing as a match. Source: `scripts/simulations/sim_thm_rootn_cardinality.R`; table `paper/tables/sim/sim_thm_rootn_cardinality_seunits.tex`.

Table S23: Studentised quantiles at the two cardinalities

L	n	Quantile	Empirical (MC SE)	Asymptotic	Diff. (MC SE units)
2	1,000	0.025	-1.998 (0.038)	-1.960	-1.01
		0.050	-1.691 (0.030)	-1.645	-1.54
		0.250	-0.673 (0.019)	-0.674	0.09
		0.500	0.008 (0.018)	0.000	0.45
		0.750	0.679 (0.019)	0.674	0.23
		0.950	1.692 (0.030)	1.645	1.56
		0.975	1.994 (0.038)	1.960	0.91
2	8,000	0.025	-1.935 (0.038)	-1.960	0.65
		0.050	-1.657 (0.030)	-1.645	-0.42
		0.250	-0.701 (0.019)	-0.674	-1.36
		0.500	-0.028 (0.018)	0.000	-1.55
		0.750	0.653 (0.019)	0.674	-1.14
		0.950	1.634 (0.030)	1.645	-0.35
		0.975	1.963 (0.038)	1.960	0.09
2	64,000	0.025	-2.018 (0.038)	-1.960	-1.54
		0.050	-1.710 (0.030)	-1.645	-2.20
		0.250	-0.698 (0.019)	-0.674	-1.23
		0.500	-0.001 (0.018)	0.000	-0.06
		0.750	0.677 (0.019)	0.674	0.11
		0.950	1.639 (0.030)	1.645	-0.20
		0.975	1.932 (0.038)	1.960	-0.75
4	1,000	0.025	-2.047 (0.038)	-1.960	-2.31
		0.050	-1.727 (0.030)	-1.645	-2.74
		0.250	-0.698 (0.019)	-0.674	-1.23
		0.500	0.014 (0.018)	0.000	0.78
		0.750	0.714 (0.019)	0.674	2.04
		0.950	1.730 (0.030)	1.645	2.85
		0.975	2.029 (0.038)	1.960	1.82
4	8,000	0.025	-2.181 (0.038)	-1.960	-5.85
		0.050	-1.849 (0.030)	-1.645	-6.84
		0.250	-0.754 (0.019)	-0.674	-4.10
		0.500	0.023 (0.018)	0.000	1.27
		0.750	0.743 (0.019)	0.674	3.54
		0.950	1.801 (0.030)	1.645	5.24
		0.975	2.146 (0.038)	1.960	4.94
4	64,000	0.025	-2.334 (0.038)	-1.960	-9.91
		0.050	-1.951 (0.030)	-1.645	-10.25
		0.250	-0.803 (0.019)	-0.674	-6.66
		0.500	0.040 (0.018)	0.000	2.23
		0.750	0.872 (0.019)	0.674	10.26
		0.950	2.042 (0.030)	1.645	13.29
		0.975	2.407 (0.038)	1.960	11.84

Notes: The seven studentised quantiles $\{0.025, 0.05, 0.25, 0.5, 0.75, 0.95, 0.975\}$ of $\vartheta'\sqrt{n}(\hat{\theta} - \theta_\dagger)/(\vartheta'\hat{V}\vartheta)^{1/2}$ at the six cells of Table S21, against the quantiles of the standard normal that Theorem 29 predicts, with the gap divided by its own Monte Carlo standard error in the last column. *The departure at* $\mathsf{L} = 4$ *is two-sided and grows in* n: both tails move outward and the median moves less than either, which is a scale distortion and not a location one. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_cardinality.R`; table `paper/tables/sim/sim_thm_rootn_cardinality_quantiles.tex`.

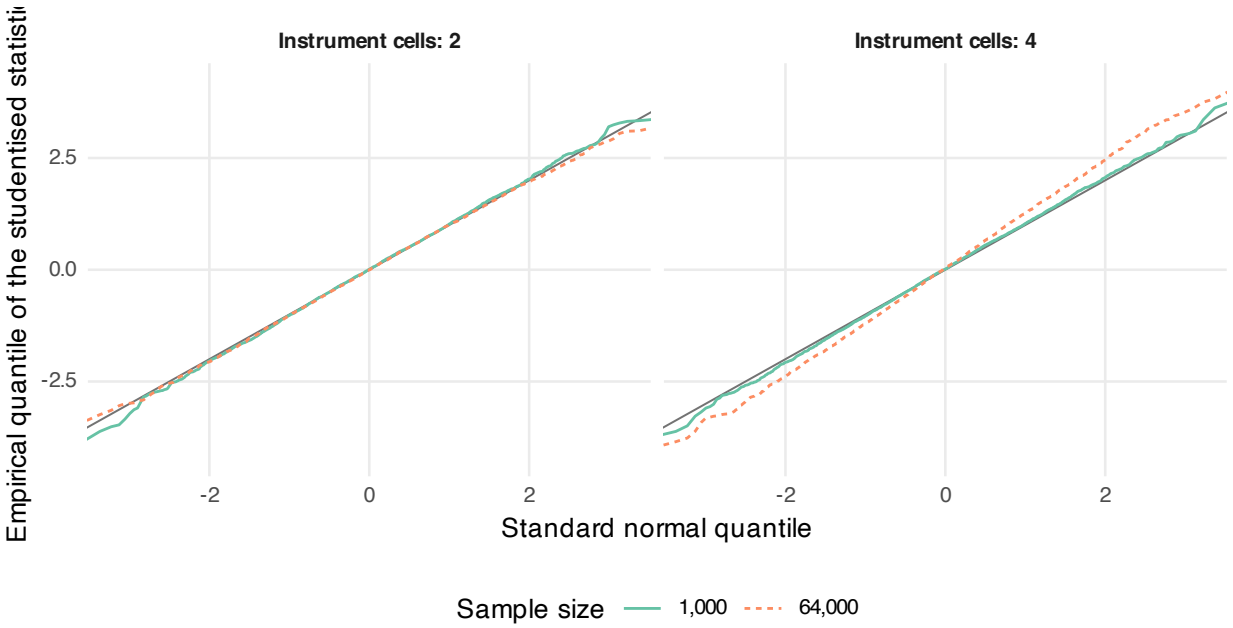


Figure S7: Studentised quantiles against the standard normal, by cardinality

Notes: The quantiles of Table S23 drawn against the standard normal. The straight grey line in each panel is the 45° line the limit law of Theorem 29 predicts, so a curve lying on it is the theorem's prediction and departure from it is read vertically. Panels are the two instrument cardinalities on one design. The solid curve is $n = 1{,}000$ and the dashed curve $n = 64{,}000$, the ends of the grid; $n = 8{,}000$ is not drawn. *In the binary panel both curves lie on the line and neither departs from it by more than* 2.20 *standard errors anywhere; in the four-cell panel the larger sample lies farther from it than the smaller at all seven quantiles Table S23 prints.* That difference in direction, and not the level of either, is what the figure carries. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_cardinality.R`; figure `paper/figures/sim/sim_thm_rootn_cardinality_qq.pdf`.

Table S24: The design at each cardinality, and what the estimator inherits

	$\mathsf{L}=2$	$\mathsf{L}=4$
The design at this cardinality		
Non-base contrasts, $\mathsf{L}-1$	1	3
Columns in the population balancing solve	12	72
Realised cell shares	0.422/0.578	0.213/0.313/0.231/0.243
A2: the balancing equation, and what its failure costs		
Relative residual $\|\nu_{g_\star}\|/\|\nu_{\Delta Z}\|$	4.2e-07	3.6e-05
Priced, $\sqrt{n}\,\|\nu_{g_\star}\|\,c_\delta\delta_n$	6.4e-06	1.9e-04
Against the leading remainder $\sqrt{n}\,c_\delta^2\delta_n^2$	2.56	2.56
A4(i): overlap, realised against the bound it must clear		
$\min_l \varrho_l$ (all cells, base included)	0.1042	0.1001
Floor ε_ϱ	0.10	0.10
$\lambda_{\min}(\Omega)$	0.0933	0.0273
Floor $\varepsilon_\varrho/\mathsf{L}$	0.0500	0.0250
What the estimator inherits		
$\kappa(J)$	3.62	3.63
$\sup\|\ell_\star\|$, the (R-a) budget	3.04	7.56
sd_∞ of the mixed direction	0.992	1.110
$\max_m \|t\|$ on $E[\psi_\star]$	1.52	1.62

Notes: The design validity of Table S21's two panels, computed from the data-generating law's closed forms rather than from the replications. The first block counts what the cardinality forces. The second reports how nearly A2's balancing equation holds, as the relative residual $\|\nu_{g_\star}\|/\|\nu_{\Delta Z}\|$ and as the quantity the remainder accounting prices it at, $\sqrt{n}\,\|\nu_{g_\star}\|\,c_\delta\delta_n$, which is to be read against the leading product $\sqrt{n}\,c_\delta^2\delta_n^2$ on the row below it. The third reports A4(i) at the realised propensity, against the floors the assumption sets: ε_ϱ for the cell probabilities and $\varepsilon_\varrho/\mathsf{L}$ for $\lambda_{\min}(\Omega)$, both of which are cleared. *The floor on* $\lambda_{\min}(\Omega)$ *falls with* L *by construction*, so clearing it at four cells is a weaker statement than clearing it at two. The last block reports what the estimator inherits from the design, including the $\sup|\ell_\star|$ that (R-a) budgets. Source: `scripts/simulations/sim_thm_rootn_cardinality.R`; table `paper/tables/sim/sim_thm_rootn_cardinality_design.tex`.

Table S25: The binary panel against the pre-cardinality sources

Layer	Checks	Bit-identical	Objects compared
config	5	5	CFG, every rate and bandwidth symbol on the eight-point grid
population	12	12	$\ell_\star$, J, Σ, perturbation norms, design validity
replication	9	9	Δ-draws, injected h, $\hat{\ell}$, $\hat{\varrho}$, `one_rep_full`
headroom	6	6	$d_Z = 1$ specs, which declare no cardinality at all
closed form	2	2	binary closed forms against the general algorithms
provenance	2	2	the cardinality arm's replication against the limit-law arm's
Total	36	36	

Notes: Adding a cardinality parameter to a data-generating law can change the binary case without anyone noticing, so the binary case is compared object by object against the sources that predate the parameter. Each row is a layer of the construction and "Bit-identical" counts the comparisons returning exact equality, not equality to a tolerance. The replication layer compares a whole replication and not only its summary, so a difference anywhere inside the estimator would appear there. *All thirty-six comparisons pass*, which is what licenses reading Panel A of Table S21 as the estimator of the preceding pages rather than as a re-implementation of it. Source: `scripts/simulations/sim_thm_rootn_l2identity.R`; table `paper/tables/sim/sim_thm_rootn_l2identity.tex`.

Table S26: Which nuisance carries the divergence

			Coverage (MC SE)		Lin. resid.	Worst of
L	n	$\hat{\mathrm{sd}}/\mathrm{sd}_\infty$ (MC SE)	90%	95%	mean (SE)	five (MC SE)
Panel A: $\hat{\ell}$ only ($c_\epsilon = 1$, $c_r = 0$)						
2	1,000	1.067 (0.011)	0.894 (0.004)	0.943 (0.003)	0.593 (0.003)	6.29
2	8,000	1.018 (0.010)	0.904 (0.004)	0.949 (0.003)	0.416 (0.002)	1.75
4	1,000	1.069 (0.011)	0.888 (0.004)	0.941 (0.003)	0.663 (0.004)	6.49
4	8,000	1.022 (0.010)	0.898 (0.004)	0.946 (0.003)	0.464 (0.003)	5.03
Panel B: $\hat{\varrho}$ only ($c_\epsilon = 0$, $c_r = 1$)						
2	1,000	1.063 (0.011)	0.901 (0.004)	0.954 (0.003)	0.661 (0.004)	5.95
2	8,000	1.031 (0.010)	0.899 (0.004)	0.951 (0.003)	0.474 (0.003)	2.97
4	1,000	1.076 (0.011)	0.898 (0.004)	0.945 (0.003)	0.874 (0.007)	7.08
4	8,000	1.138 (0.011)	0.863 (0.005)	0.923 (0.004)	0.913 (0.008)	12.10

Notes: The design of Table S21 primitive for primitive, at $n \in \{1{,}000, 8{,}000\}$ and 5,000 replications in every cell, with exactly one estimated nuisance switched on per panel: Panel A supplies the true propensity and injects the representer error, isolating A5(iv); Panel B supplies the true representer and injects the propensity error, isolating A4(iii). The structural error h is injected in both, at $c_\delta = 0.25$. The panel headers switch the arms by setting a rate constant to zero: c_ϵ and c_r are the constants in the ϵ_n of A5(iv) and the r_n of A4(iii), defined in the notes to Table S21, and a zero suppresses that nuisance error rather than rescaling it. $\mathsf{L} = 2$ *is the control and not a courtesy*: an arm that diverges at $\mathsf{L} = 4$ without its binary counterpart says that a nuisance is expensive, not that it is expensive *at* $\mathsf{L} = 4$. The columns are those of Table S21; "worst of five" is the largest absolute entry over the five measures of Table S22 at that cell. The grid stops at $n = 8{,}000$ on cost, which makes a flat arm here weaker evidence than a diverging one; both findings the text draws are in the diverging direction, so the caveat binds on neither. Source: `scripts/simulations/sim_thm_rootn_ablsplit.R`; table `paper/tables/sim/sim_thm_rootn_ablsplit.tex`.

Table S27: The instrument block, term by term, and the two norms A5(iv) separates

			Instrument-block terms, $\times\sqrt{n}$			Norm ratio, consumed/priced		
Arm	L	n	$\|T_1\|$ (MC SE)	$\|T_2\|$ (MC SE)	$\|T_3\|$	$\hat{\ell}-\ell_\star$	$\hat{L}-L_\star$	Lin. resid.
$\hat{\ell}$ only	2	1,000	0.020 (0.0001)	—	—	0.463	0.462	0.593
$\hat{\ell}$ only	2	8,000	0.012 (0.0001)	—	—	0.463	0.462	0.416
$\hat{\ell}$ only	4	1,000	0.014 (0.0001)	—	—	0.420	0.420	0.663
$\hat{\ell}$ only	4	8,000	0.006 (0.0000)	—	—	0.419	0.419	0.464
$\hat{\varrho}$ only	2	1,000	—	0.337 (0.0036)	—	—	—	0.661
$\hat{\varrho}$ only	2	8,000	—	0.270 (0.0029)	—	—	—	0.474
$\hat{\varrho}$ only	4	1,000	—	0.569 (0.0060)	—	—	—	0.874
$\hat{\varrho}$ only	4	8,000	—	0.843 (0.0089)	—	—	—	0.913

Notes: The cells of Table S26, decomposed. Writing $\widehat{\widetilde{W}}_l = \widetilde{W}_l - (\hat{\varrho}_l - \varrho_l)$ gives $\hat{L}\widehat{\widetilde{W}} - L_\star\widetilde{W} = (\hat{L}-L_\star)\widetilde{W} - L_\star(\hat{\varrho}-\varrho) - (\hat{L}-L_\star)(\hat{\varrho}-\varrho)$, and the three terms are T_1, T_2 and T_3, each reported scaled by $\sqrt{n}$. The arms separate them exactly, so an em-dash marks a term that is zero by construction in that arm and not a term that was not computed. *T_2 is first order in r_n and survives only because orthogonality in the propensity direction kills its expectation*, which by Proposition 20(iii) requires N1. Read against the two reference exponents, which are not fitted but read off the design: under orthogonality T_2 is a centred average and decays at $n^{-\alpha} = n^{-0.320}$; under outright failure it carries a non-vanishing mean and grows at $n^{1/2-\alpha} = n^{+0.180}$. *The T_2 column carries a confound, and the text reads its exponents from another table in consequence*: this arm evaluates $L_\star$ at $\Delta U - \Delta h$ rather than at ΔU, so the entries below also carry the $\sqrt{n}\,\|h\|r_n$ cross-product the theorem prices separately. The slopes they imply over the single leg $n = 1{,}000 \to 8{,}000$, computed from the two unrounded cell means, are -0.108 at $\mathsf{L} = 2$ and $+0.189$ at $\mathsf{L} = 4$. Panel E of Table S28 re-runs the same accumulation with h switched off and returns -0.302 and $+0.206$, and those are the figures the text reads against the two references. The last two numeric columns divide the norm the moment consumes, $\|\sum_l(\hat{\ell}-\ell_\star).{}_l\widetilde{W}_l\|_{L^2(P)}$, by the Frobenius norm A5(iv) prices; a ratio below one is a conservative bound. 5,000 replications in every cell. Source: `scripts/simulations/sim_thm_rootn_ablsplit.R`; table `paper/tables/sim/sim_thm_rootn_ablsplit_channel.tex`.

Table S28: Does the zero anchor impose N1? The defect, computed rather than estimated

Design	L	$\max_{j,l}\|b_{jl}\|$	$\max_{j,l}\|L_{\star,jl}\|$	ratio	$\|b\|_F$	$\|L_\star\|_F$	ratio
Panel A: the N1 defect, computed							
DGP-A	2	8.23×10^{-6}	2.4613	3.34×10^{-6}	9.86×10^{-6}	2.9477	3.34×10^{-6}
DGP-A	4	0.7293	3.8227	0.1908	1.0359	5.5730	0.1859
DGP-B	2	4.18×10^{-5}	2.4964	1.67×10^{-5}	4.92×10^{-5}	2.9424	1.67×10^{-5}
DGP-B	4	0.4295	3.4628	0.1240	0.5779	4.3852	0.1318

Panel B: contracted, per unit $\|\hat{\varrho}-\varrho\|_{L^2}$

Design	L	n	Γ_{op}	Γ_{real}	Γ trunc.	Λ	$\Gamma_{\text{real}}/\Lambda$
DGP-A	2	1,000	9.86×10^{-6}	5.19×10^{-6}	5.33×10^{-6}	2.9681	1.75×10^{-6}
DGP-A	2	8,000	9.86×10^{-6}	5.19×10^{-6}	5.75×10^{-6}	2.9681	1.75×10^{-6}
DGP-A	4	1,000	1.0349	0.3258	0.3120	3.2161	0.1013
DGP-A	4	8,000	1.0349	0.3258	0.2920	3.2161	0.1013
DGP-B	2	1,000	4.92×10^{-5}	2.79×10^{-5}	2.76×10^{-5}	2.9506	9.46×10^{-6}
DGP-B	2	8,000	4.92×10^{-5}	2.79×10^{-5}	2.71×10^{-5}	2.9506	9.46×10^{-6}
DGP-B	4	1,000	0.5778	0.1814	0.1833	2.5518	0.0711
DGP-B	4	8,000	0.5778	0.1814	0.1856	2.5518	0.0711

Panel C: mechanism and confirmation

Design	L	odd-q share	bal. resid.	sampled max $\|b_{jl}\|$	MC SE	binned $\|\hat{b}\|/\|L_\star\|$	binned floor
DGP-A	2	1.000000	4.15×10^{-7}	0.0037	0.0017	0.0110	0.0115
DGP-A	4	0.707117	3.69×10^{-5}	0.7281	0.0022	–	–
DGP-B	2	1.000000	1.36×10^{-6}	0.0015	0.0019	0.0122	0.0115
DGP-B	4	0.722202	3.64×10^{-5}	0.4264	0.0024	–	–

Panel E: the shipped T_2 *statistic, with and without the* φ*-error (DGP-A, sampled)*

L	h	$n=1{,}000$	(MC SE)	$n=8{,}000$	(MC SE)	β	$\Lambda\tau_n$ at 8,000
2	on	0.3417	0.0065	0.2776	0.0053	−0.100	0.1673
4	on	0.5950	0.0109	0.8200	0.0155	+0.154	0.1813
2	off	0.2448	0.0047	0.1308	0.0026	−0.302	0.1673
4	off	0.5358	0.0100	0.8225	0.0154	+0.206	0.1813

Notes: N1 at $L_\star$ asks that the defect $E[L_\star \mid \Delta Z]$, written b in the column headers and not to be confused with the anchor $b_\star$, vanish; by Corollary 21 that is $p(\mathsf{L}-1)$ scalar conditions. On the designs of Table S21 it is a closed form rather than a quantity to be estimated: ΔU is independent of $(\Delta Z, W)$ and symmetric about zero, and $L_\star$ separates as $\sum_{r,q} c_{m,(l,r,q)} A_q(u)\eta_r(z)$ with A_q the zero-anchored antiderivative of the u-basis. *Panels A to C therefore carry no sampling error*, and they measure the defect of the shipped $L_\star$, on the shipped antiderivatives and clamp. Panel A reports the entrywise maximum over the $p(\mathsf{L}-1)$ pairs, because N1 is a matrix identity and a Frobenius aggregate can hide one live entry; the aggregate and $\|L_\star\|$ in both conventions are printed beside it. Panel B contracts b against the propensity error, which is what ψ and $\hat{J}$ see: $\Gamma_{\text{op}} = \sup_{\|r\|=1}\|E[br]\|$ is the sharp constant, Γ_{real} its root mean square over the ablation arm's own direction law, "Γ trunc." the same under projection onto the overlap set, and $\Lambda = (E\|L_\star r\|^2)^{1/2}/\|r\|$ the centred-piece constant they are read against. Panel C carries the mechanism and one sampled check: the share of the representer's coefficient energy on odd q, A2's relative balancing residual, the defect re-estimated by sampling with its Monte Carlo standard error, and a binned estimate of $\|b\|/\|L_\star\|$ beside the noise floor $1/\sqrt{7{,}500}$ that binning carries when the truth is exactly zero. *That binned estimate resolves nothing at* $\mathsf{L}=2$ *and is printed against its own floor so that it cannot be read as though it did. The printed panels run A, B, C, E and there is no Panel D*: the producing script computes one, an independent sampled confirmation of Panels A and B by least squares of $L_\star$ on the propensity basis $(1,\varrho)$ at $n=2\times10^6$, and its output is the last four columns of Panel C rather than a panel of its own. The letters are the script's and are left as it assigns them. Panel E re-runs the ablation arm's first-order propensity term $\|T_2\| = \sqrt{n}\,\|\mathbb{E}_n[L_\star(\hat{\varrho}-\varrho)]\|$ on a separate stream at 1,500 replications, with the φ-error h on and off; β is the slope over the leg, at a Monte Carlo standard error of 0.013 in every row, and $\Lambda\tau_n$ is the size a purely centred T_2 would carry at $n=8{,}000$. The h-on rows reproduce Table S27 to within 2.1 standard errors of the difference, worst at $\mathsf{L}=4$ and $n=1{,}000$, which is what makes the h-off rows comparable to it. Source: `scripts/simulations/sim_thm_rootn_n1defect.R`; table `paper/tables/sim/sim_thm_rootn_n1defect.tex`.

Monte Carlo: the shipped engine on three designs

This section is the fourth of the article's four Monte Carlo experiments, the one the article still prints. Three of its floats stay there: the balanced design at three cardinalities is Table 1 of the article, the misspecified propensity is Table 2 of the article, and the record of when the stopping rule fires is Table 3 of the article. What follows is the rest of the experiment: the argument that the three suites do not merge, the three blocks set side by side, N1 measured at the weight the engine solves for, the skewed design, the interval's scale, and the paired cardinality contrast, together with the article's full text for the experiment. It adds no result. Both experiments above run the theory's own sieve construction against population objects computed in closed form, and both draw their nuisances as the truth plus an error of controlled size rather than estimating them. Section 6 runs neither. A third suite closes that gap: a multinomial design estimated by the shipped engine, the same code Section 6 estimates with, with every nuisance fitted from the sample and the instrument read at $\mathsf{L} \in \{2, 3, 4\}$ off one draw.

One departure of that engine from the theory it implements has since been repaired, and we state the repair here with what it is worth measured rather than asserted. The measurements of this paragraph are printed in Panels B and D of Table S48. C3(ii) asks the boundary kernel for vanishing truncated moments at every order $0 \le j \le p_K - 1$, and the family Lemma 66 constructs delivers all of them; the engine used to renormalise to unit mass at each design point, which realises the $j = 0$ member alone, and it now runs that family at $p_K = 8$, which is the kernel Section 6 estimates with. Both configurations were run, and each lands on the exponent its own construction predicts. The renormalised kernel converges to 1.003, the degraded h_x^1 that enforcing the zeroth moment and nothing higher buys; the family converges to 7.984, which is $h_x^{p_K} = h_x^8$, because $\varphi_\dagger$ is C^∞ on this design and (15) kills the orders $j = 1, \dots, 7$. The guarantee (172) carries is $h_x^{s-1/2} = h_x^{2.5}$, and the two exponents bracket it: the first fails it and the second clears it.

Two things moved together, and the second is a bandwidth. The x-bandwidth is now $h_x = n^{-0.40}$, the midpoint of the window $(0.360, 0.440)$, and the boundary layer stopped swallowing the design. Within three bandwidths of an endpoint sat every pooled design point at every n up to 2,828 before, and 99.41 down to 51.93 per cent at the four sample sizes above that; after, the share is 48.55 per cent at $n = 500$ and 4.22 at $n = 181{,}019$ (Table S48, Panel A). The same measurement returns 884 of the 1,966 pooled design points of Section 6's panel, which the notes to Table S47 record, and that is 44.96 per cent. The bias that enters one hypothesis, (P-g), read as an upper bound on h_x, is where the two changes are visible together. Its realised ratio used to *climb* across the grid, from 3.13 at $n = 125$ to 8.09 at $n = 181{,}019$, where the hypothesis asks for $O(1)$; it now falls to a plateau of about 9.6×10^{-7}, moving by -0.0406 in log-log terms over the last four sample sizes at a level spread of 13.4 per cent. *We report a plateau and not a fitted rate.* A log-log line through the whole grid returns -0.813, and it is the wrong model for a curve that falls by a factor of 328 and then flattens; its regression standard error is 175 times the Monte Carlo one, which is the diagnostic that says so; that ratio is taken on the unrounded standard errors, which the panel's three digits round to 173. Under the renormalised arm the window (P-g) left against C5′(ii) was non-empty but narrow, clearing by 0.061 at the frontier-clearing $\alpha = 0.32$ and by 0.006 at $\alpha = 3/8$, which Remark 125 already declares an illustration rather than an admissible design.

The cap moves with the repair, and the move corrects two constants rather than adding a fresh distortion. In the old configuration c_C was carrying a compact-support half-width retired with the kernel it belonged to, and c_2 was the interior kernel's maximum where C3(ii) asks for the sup of the envelope. *Correcting them is not the same as making the cap tight*: (36) still exceeds $\hat{\delta}_n^{-2}$ by 7.32×10^8 to 2.78×10^9 through the over-normalisation $\omega_n / \|\hat{\mathcal{A}}\|$ that Panel C prices, and nothing below touches that. Correcting both, at $n = 500$ the cap (36) moves by a factor of 344,503, of which the constants carry 89,741, the bandwidth 3.47 and the kernel family exactly 1.00. The constants leg is 0.1551×34.039^4 against the

retired 4.0×0.8727^4; the bandwidth enters (36) as h_x^{-1} and nothing else, so its leg is the ratio 0.2885/0.0833 of the two bandwidths as the exponent moves from $n^{-0.20}$ to $n^{-0.40}$, taken on the unrounded bandwidths and not on the four digits printed, which give 3.46. What the three derived legs leave over is the realised h_u coupling, 1.108 at $n = 500$ and 1.054 and 1.027 at $n = 2{,}000$ and 8,000, and it is reported as a residual so that it cannot absorb an arithmetic slip. *The family contributes exactly* 1.00. Θ_I enters (36) through c_C and c_2 alone, so the boundary family is free in the cap and is paid for in the bias, which is where it should be paid. The post-repair values, at $n = 500$, 2,000 and 8,000, are $\omega_n = 27{,}200.1$, 37,777.9 and 52,744.8, and $\bar{m}_n = 3.95 \times 10^{10}$, 1.85×10^{11} and 8.76×10^{11}, at $\mathsf{c}_2 = 34.039$ and $c_C = 0.1551$ (Table S5, Panel C′).

We have priced the two exponents against the x-smoothing bias and against (P-g) and nowhere else: neither is traced through the sieve entropy bound, through A5(vi′), or through Theorem 29's reading of δ_n off A5(ii), and if either reaches any of those three the accounting just given is incomplete. The gap was between the theory and the implementation and never inside the theory, which is why C3(ii) is stated at every order and why closing it cost a kernel and not a hypothesis.

It also cost draws, and we price that here rather than in the notes to the tables that pay it. A kernel that kills the truncated moments of every order $1 \le j \le p_K - 1$ cannot be non-negative, so $\hat{\pi}$ under the boundary family is a higher-order density estimate and can take a non-positive value at a design point. The adjoint divides by $\hat{\pi}$, so the engine refuses to build the operator and the replication is dropped rather than returned degraded. *What fails is the sample reference density and not a hypothesis.* The adjoint of Section 3 of the article carries the population π, and the article states no condition on the estimate that stands in for it, so we name the object rather than an assumption tag. It is the same failure that leaves the estimator infeasible in the μg/m^3 coordinate of Section 6, where Panel C of Table S43 prints it in those words; it is reached there through a rescaled coordinate and here through a

small design set.

The mechanism is that design set together with the cross-fitting geometry, and neither of the two alone. Every loss across the four arms of this suite falls where the operator is built on a cross-fitting fold of 160, 320 or 640 points, which is $\lceil 0.4n \rceil$ at $n = 400$, 800 and 1,600, and the count falls as the fold grows. No operator built on a whole draw has tripped the guard, neither at the 400 points and more the uncross-fitted arms hand it nor at the 160 of 60 fresh balanced draws taken at the smallest fold size. *The two counts are on different scales and the second is not a threshold in the first*: what separates the arms that lose draws is that a cross-fitted operator builds $\hat{\pi}$ on a subsample while the support it is evaluated over still comes from the whole draw. The multinomial φ-step arm, which is not the 36-cell arm of the limit-law section above, builds its operator on the whole draw and keeps all 24 of its paths in each of its twenty-seven cells, which is what the bracketed denominators of Table 3 of the article record; its plug-in companion, also on the whole draw, keeps all 504 of its replications, a count printed in no table and read off `quality_reports/sim/sim_multinomial_phiplugin.rds`; the arms that partition into $\mathsf{F} = 5$ folds, of which the tables below are one, hand the operator 160 points at $n = 400$ and lose between none and eight draws per cell, most of them there. What is lost does not depend on the cardinality, which is what the three tables below record when their headers print the count as equal at all L, and what leaves the paired contrast defined. The realised counts are stated with the precision statement they govern below.

The three suites do not merge into a single cardinality contrast, and we say so before setting them beside each other. They differ in the design, in the estimator and in the nuisance regime, and their sample-size grids do not intersect at any point: the first two run 125 to 181,019 and the third runs 400, 800 and 1,600. A table whose columns read $\mathsf{L} = 2$ and $\mathsf{L} = 4$ but whose rows come from two of them would attribute three differences to the cardinality. Every table in this section therefore obeys one rule, carried in the panel headers rather than in the notes so that a reader who reads only the body still cannot merge two blocks by accident: *one row*

is one design, one estimator and one nuisance regime, and the cardinality varies within a row and never across one (Table S29).

One precision statement governs the third suite at rates near nominal, and it does not govern the rest of it. The suite was run at 120 replications and returns between 112 and 120 usable ones, for the reason given above. The count is a property of the cell and not of the cardinality: it is identical across $\mathsf{L} = 2$, 3 and 4 in all nine cells, which is what leaves the paired contrast defined, and it is lowest at the misspecified design at $n = 400$. At a coverage rate of 0.95 the Monte Carlo standard error is about 0.02, and two of them resolve to ± 0.040 at the fullest cell and ± 0.041 at the thinnest, so a gap smaller than that is not evidence of anything. *That is the smallest resolution in the suite, and it is calibrated at one point.* The standard error grows as coverage moves toward a half, and at the misspecified constant, where coverage runs from 0.277 to 0.795, the standard errors the table prints are 0.038 to 0.046; two of those are ± 0.076 to ± 0.092, about twice the figure just quoted. Every reading below is therefore priced at the standard error its own cell prints. The paired contrasts are the sharper instrument, because the three cardinalities run on the same draw and the same partition, so the difference carries its own standard error rather than the sum of two (Table S33), and the claims below rest on those.

What the dropped draws could have done is bounded and no longer merely disclosed. A cell drawn at $B = 120$ and used at b has its coverage inside $[\hat{p}\, b/B,\ \hat{p}\, b/B + (B - b)/B]$ whatever the missing draws would have returned, and each of the three tables now carries that bound in its own header rather than in its notes: at most 0.024 on the balanced design, 0.032 on the skewed one and 0.063 on the misspecified one; at $n = 1{,}600$ it is at most 0.008, which is under the smallest coverage standard error the three tables print at that sample size, 0.014. *That comparison is against the coverage standard errors of Panel C and against no other column*: Panel A's bias standard errors at the same sample size run down to 0.0053, and the bound does not govern a bias. Readings below taken at $n = 400$ and $n = 800$ carry the larger

figure rather than that one. The selection is real and its direction is known, a draw surviving when the nonparametric stage behaved, so the printed rates are the optimistic reading of that interval.

One hypothesis of Theorem 29 had been computed on no multinomial design, and computing it here returns a failure. N1 asks that $E[L(\Delta U, \Delta Z) \mid \Delta Z]$ vanish, which Corollary 21 types as $p(\mathsf{L} - 1)$ scalar conditions, and it is the clause the preceding section found breaking at $\mathsf{L} = 4$ on the design the theory names. Measured on the weight the engine solves for, and reported as the entrywise defect over $\max_{j,l} \|L_{jl}\|$, which is that section's own statistic to the digit, it runs from 0.531 to 0.720 across the nine design-by-cardinality cells at $n_{\text{solve}} = 640$ (Table S30, Panel A). The weight is solved at the true propensity and the true residual, so what is measured is the weight class and not the two nuisance errors (T-3) and (T-4) price separately; on the misspecified design that means its own true propensity, and its row is a third design rather than a statement about the misspecification. Block I returns 3.3×10^{-6} at $\mathsf{L} = 2$ and 0.191 at $\mathsf{L} = 4$ against the same denominator. *N1 fails on all three designs at all three cardinalities*, and the spread across L within a design is at most 0.088 where the level is 0.53 to 0.72.

We give three certificates for that reading, because a negative result about a quantity whose null value is zero is worth more measured against a floor than asserted. The computation is closed form rather than estimated: the design draws ΔU conditionally on $\Delta Z = z$ as a normal whose mean and whose variance both move with z, so the inner expectation is a normal integral taken at machine precision and the routine reads no z-sample. Redrawn with that dependence switched off, which leaves ΔU independent of ΔZ, the same code returns 5.4×10^{-15} to 7.6×10^{-14} over its 72 draws, and the smallest live ratio anywhere in the arm, 0.3145, sits 12.6 orders of magnitude above that floor. Nor does the defect shrink: across the three solve folds the arm realises, $n_{\text{solve}} \in \{160, 320, 640\}$, the log-log slope runs -0.133 to $+0.064$, where a sampling error would fall and a restriction on the weight class does not

(Panel B). And an independent quadrature route agrees with the closed form to at most 1.7×10^{-4} at every printed cell, against quantities of size 23 to 60, and to 1.4×10^{-5} in relative terms at the worst of the 324 live solve draws (Panel C).

This is not a defect in Theorem 29 and we do not report it as one. It sits between the theory and the implementation, and it is the same species as the C3(ii) gap the opening of this section prices. `solve_balance_weight` builds ℓ in a function class over the residual alone, so Λ carries no z argument, and `crossfit_once` anchors it by subtracting a constant. The conditional laws of ΔU given ΔZ therefore form a location-scale family with a continuum of means, and a bounded function of u alone orthogonal to every one of them is zero almost everywhere, so the zero weight is the only member of the implemented class that satisfies N1 on this design. The supremum of the same ratio over centred weights that are polynomials of degree at most six in the residual is a property of the design and of no weight, and it reads 0.8615 at the two well-specified designs and 0.9021 at the third, collapsing to 8.94×10^{-8} once the conditional dependence is switched off (Panel C). The re-anchoring of Section 4 of the article would restore N1 at the same ℓ, and Remark 52 admits it with no further assumption; the engine does not build it. Two things this does not establish. It says nothing about Section 6's data, where the defect is estimated on a panel whose conditional law is unknown and the population argument just given has no purchase, and it does not show that the theory's own $\ell_\star$ fails N1 here.

At the two well-specified designs the cardinality does very little, and the measurement above says why rather than leaving it unattributed: there is no cardinality-varying N1 mechanism on this suite to produce a cardinality-varying divergence, the defect being live at every cell and of about one size. The bias at $n = 1{,}600$ is inside two of its own Monte Carlo standard errors at all nine cells of the balanced design and at eight of the nine skewed ones, the exception being the skewed constant at four cells, at -0.0138 (0.0053). The quantity Theorem 29 makes constant in n, $\sqrt{n}$ times the standard deviation, falls across the grid at sixteen of the eighteen

cells of those two designs and is larger at $n = 1{,}600$ than at $n = 400$ at two, both of them the balanced constant, by 19.19 and 5.35 per cent. *Neither of those two moves is resolved from flat.* Panel B now prints a Monte Carlo standard error for that quantity, computed from the fourth moment rather than from the Gaussian formula, the realised kurtosis reaching 10.8 over the suite's 81 cells; against it the larger of the two is 1.50 standard errors and the smaller is 0.45. The paired differences at $n = 1{,}600$ are inside 2.4 of their own standard errors at every coefficient and both cardinalities, the largest being the balanced constant at -0.0339 (0.0144) and the skewed z_2 at -0.0152 (0.0076). The first of those is 2.35 standard errors and stands clear of the rest; the second is 2.00 on the printed digits, which the print cannot separate from two either way; none of the remaining ten reaches 1.2.

Coverage at $n = 1{,}600$ sits between 0.866 and 0.975 at all nine balanced cells and between 0.867 and 0.958 at all nine skewed ones (Table 1 of the article, Table S31). *That range is true and it conceals the shape of its own low end*, which is one-sided and does not shrink. Seven of those eighteen cells sit more than two Monte Carlo standard errors below nominal, and they are not scattered: three are the balanced constant, at 0.882, 0.866 and 0.866 down the three cardinalities, and four are on the skewed design, its constant at two cells and z_2 at all three. On the favourable design the constant at four cells runs 0.863, 0.842 and 0.866 across the three sample sizes, which is flat and 2.7 standard errors under nominal at the largest of them; skewed z_2 at four cells moves *away* from nominal as n grows, at 0.948, 0.915 and 0.867.

What kind of failure that is, the interval answers and the limit law does not, and the statistic that answers it was computed all along and printed in no table. The mean estimated standard error over the simulated standard deviation is below one in 77 of the 81 cells of this suite, below 0.85 in 42 and below 0.75 in 18 (Table S32); the balanced constant sits between 0.705 and 0.805 and does not improve across a grid spanning a factor of four in n. Take the coverage a centred normal carrying that ratio would deliver and subtract the realised coverage: the difference is inside ± 0.070 at all 54 well-specified cells, at a root mean square of 0.024. *The*

entire well-specified coverage shortfall is the interval's scale and not the limit law. On the misspecified design the same statistic runs -0.028 to $+0.033$ on the slopes and $+0.030$ to $+0.511$ on the constant, which is the bias a centred approximation sets aside by construction. A live N1 defect predicts this shape: by Proposition 20(iii) the moment's ϱ-derivative no longer vanishes, so $\hat{\varrho}$ enters at first order, and where the propensity is correctly specified it converges at $\sqrt{n}$ and leaves $\hat{\theta}$ asymptotically normal while the sandwich omits a term and is wrong by a factor that does not shrink. *We report that as consistent with the measurement and not as demonstrated by it*, the variance correction itself not being computed here.

The third design is the one that breaks, and it breaks in the propensity. It misspecifies $\hat{\varrho}$, which is the nuisance A4(iii) prices and the one the cardinality section above isolated. Its constant's bias does not fall in n at four cells, running -0.3040, -0.3242 and -0.3830, and $\sqrt{n}$ times its standard deviation *rises*, 6.031, 6.960, 8.605, where the theorem makes it constant. That last move is 3.40 of the standard error Panel B now prints, so unlike the two at the balanced design it *is* resolved from flat. Coverage of the constant at $n = 1{,}600$ is 0.697 at two cells, 0.294 at three and 0.277 at four, and the paired bias difference against the binary arm is -0.2287 (0.0226) and -0.2765 (0.0241), which are 10.1 and 11.5 standard errors and are not a resolution artefact (Table 2 of the article). The two slope coefficients are untouched at $n = 1{,}600$, covering between 0.891 and 0.950 at every cardinality. The damage is therefore confined to the coordinate the propensity enters, and it grows with the number of cells at the two larger sample sizes: the $n = 1{,}600$ coverages just quoted fall down the three cardinalities, those at $n = 800$ run 0.771, 0.525, 0.424, and the bias grows in step at both. *At $n = 400$ it does not.* Coverage there runs 0.795, 0.661, 0.732 and the bias -0.1813, -0.3266, -0.3040, so four cells show less damage than three on both measures. Both reversals are inside the bound $\mathrm{SE} \leq \mathrm{SE}_3 + \mathrm{SE}_4$, at 0.82 and 0.38 of it, and $n = 400$ under this design is the thinnest cell in the suite, at 112 usable replications. The direction is the one the injected-nuisance ablation found, reached here through estimated nuisances and through the code Section 6 runs on.

One competing explanation for that rising standard deviation is removed by an arm that already existed and had not been read. The φ-step sweep runs the discrepancy rule at $\tau_{\mathrm{DP}} = 0$, where it never fires and the returned iterate *is* the cap, so a sweep over the cap is a fixed-m variant: at $n = 1{,}600$ and $\mathsf{L} = 4$, moving $\bar{m}$ from 15 to 100 raises the misspecified design's integrated squared error from 0.195 to 0.488, at Monte Carlo standard errors of 0.010 and 0.067, while it lowers the balanced design's from 0.091 to 0.036 and the skewed design's from 0.118 to 0.051. Those six figures are read off the $(\tau_{\mathrm{DP}}, \bar{m})$ sweep that `scripts/simulations/sim_multinomial_phistep.R` writes to `quality_reports/sim/sim_multinomial_phistep_sweep.rds`, at $\tau_{\mathrm{DP}} = 0$ and $\bar{m} \in \{15, 100\}$; the sweep is a run object and not a float, which is why they carry a path here and not a cross-reference. *The cap is therefore not an independent cause*, iterating longer helping where the propensity is right and hurting where it is wrong. The same separation is not available on the θ side, which would need that arm re-run at a fixed budget, and we have not run it.

The stopping rule fails in the same place and in the same direction. At the two well-specified designs the discrepancy rule selects an iterate inside its budget in most replications at every cardinality, at 1.000, 0.917 and 0.875 on the balanced design at $n = 1{,}600$ and 1.000, 0.792 and 0.667 on the skewed one. At the misspecified design it selects one in 0.875 of replications at two cells, 0.250 at three and *none at all* at four (Table 3 of the article). A rule that never fires returns the cap, which is the configuration Section 6's bootstrap runs under and says so.

Three readings this section does not support. It does not establish that the shipped engine and the abstract construction agree, because no cell of the third suite shares a sample size with either of the first two, and the merged tables print the grids rather than interpolate between them. It does not establish that the cardinality is harmless at the well-specified designs, because 112 to 120 replications cannot resolve a coverage gap below about ± 0.04 and the paired contrasts bound the *bias* difference rather than the coverage one. And it does

not establish that the N1 defect is what the standard-error ratio measures: the correction that Proposition 20(iii) implies is not computed, and the two quantities are set beside each other rather than fitted to one another.

Table S29: The three simulation suites at the cardinality all three share

Design and nuisance regime	n	B	$\mathsf{L} = 2$	$\mathsf{L} = 3$	$\mathsf{L} = 4$
Block I — named configuration, abstract operator, nuisances injected at controlled rates $(\mu, s, a, r, d_Z, p_K) = (2, 3, \frac{3}{4}, \frac{8}{3}, 3, 8)$, $p = 4$; $\mathsf{L} = 2$ in this block					
DGP A, injected $c_\delta = 0.25$	181,019	5,000	0.952 (0.003)	—	—
DGP A, injected $c_\delta = 1.00$	181,019	5,000	0.919 (0.004)	—	—
DGP B, injected $c_\delta = 0.25$	181,019	5,000	0.950 (0.003)	—	—
DGP B, injected $c_\delta = 1.00$	181,019	5,000	0.934 (0.004)	—	—
Block II — headroom configuration, abstract operator, nuisances injected at controlled rates $(\mu, s, a, r, d_Z, p_K) = (5, 4, \frac{2}{5}, \frac{21}{8}, 1, 8)$, $p = 2$; $\mathsf{L} = 2$ in this block					
DGP-HA, injected $c_\delta = 0.25$	181,019	5,000	0.947 (0.003)	—	—
DGP-HA, injected $c_\delta = 0.35$	181,019	5,000	0.944 (0.003)	—	—
DGP-HA, injected $c_\delta = 1.00$	181,019	5,000	0.955 (0.003)	—	—
DGP-HB, injected $c_\delta = 0.25$	181,019	5,000	0.952 (0.003)	—	—
DGP-HB, injected $c_\delta = 0.35$	181,019	5,000	0.953 (0.003)	—	—
DGP-HB, injected $c_\delta = 1.00$	181,019	5,000	0.950 (0.003)	—	—
Block III — multinomial design, shipped engine, nuisances estimated `sim_multinomial_dgp.R` with `lf_panel.R`; $\mathsf{L} \in \{2, 3, 4\}$ off one draw					
Balanced, Constant, estimated	1,600	119	0.882 (0.030)	0.866 (0.031)	0.866 (0.031)
Balanced, z_1, estimated	1,600	119	0.975 (0.014)	0.916 (0.025)	0.958 (0.018)
Balanced, z_2, estimated	1,600	119	0.916 (0.025)	0.975 (0.014)	0.958 (0.018)
Skewed, Constant, estimated	1,600	120	0.883 (0.029)	0.925 (0.024)	0.933 (0.023)
Skewed, z_1, estimated	1,600	120	0.958 (0.018)	0.950 (0.020)	0.950 (0.020)
Skewed, z_2, estimated	1,600	120	0.875 (0.030)	0.867 (0.031)	0.867 (0.031)
Misspecified $\hat{\varrho}$, Constant, estimated	1,600	119	0.697 (0.042)	0.294 (0.042)	0.277 (0.041)
Misspecified $\hat{\varrho}$, z_1, estimated	1,600	119	0.950 (0.020)	0.941 (0.022)	0.933 (0.023)
Misspecified $\hat{\varrho}$, z_2, estimated	1,600	119	0.899 (0.028)	0.891 (0.029)	0.908 (0.027)

Notes: Coverage of a nominal 95% interval with its Monte Carlo standard error in parentheses, at the largest sample size each suite reaches. *One row is one design, one estimator and one nuisance regime*; the cardinality varies across the columns of a row and never across rows. Block I is the configuration of Table S2 and Block II that of Table S13, both run against population objects computed in closed form inside the data-generating law, with nuisances injected at controlled rates. Block III is a multinomial design estimated by the shipped engine, with every nuisance fitted from the sample. *Blocks I and II are binary for different reasons*: the headroom design has no cardinality parameter at all, while the named design does have one and is binary here only because the arms this table reads are the binary ones, which is what Table S21 runs at four cells. Their $\mathsf{L} = 3$ and $\mathsf{L} = 4$ cells are printed as em-dashes rather than left blank. *The B column is not decoration*: at the 119 and 120 it prints for Block III a coverage rate resolves to about ± 0.04 and at $B = 5{,}000$ to about ± 0.006, so the blocks are not comparable digit for digit. Block III was run at 120 replications everywhere and returns fewer in the cells where the boundary family drove $\hat{\pi}$ non-positive and the operator could not be built; the column prints the count used and not the count requested. The sample-size grids do not intersect at any point. This table is generated and not transcribed. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_designs.tex`.

Table S30: N1 at the weight the engine solves for, on three multinomial designs

Block III — multinomial design, weight from `solve_balance_weight`
the N1 functional is exact; the weight is the estimator's own

Panel A: the N1 defect at $n_{\text{solve}} = 640$*, i.e.* $n = 1{,}600$*.*
Monte Carlo standard errors over 12 *solve draws.*

Design	L	$\max_{j,l} \|b_{jl}\|$	$\max_{j,l} \|L_{jl}\|$	entrywise ratio	$\|b\|_F/\|L\|_F$	Block I, DGP-A
Balanced	2	3.561	5.555	0.641 (0.012)	0.632 (0.011)	3.34×10^{-6}
	3	3.020	5.342	0.558 (0.032)	0.545 (0.024)	—
	4	4.846	7.431	0.646 (0.018)	0.589 (0.015)	0.191
Skewed	2	1.341	2.494	0.531 (0.030)	0.530 (0.021)	3.34×10^{-6}
	3	2.267	3.851	0.591 (0.020)	0.562 (0.014)	—
	4	2.945	4.945	0.588 (0.021)	0.550 (0.013)	0.191
Misspecified $\hat{\varrho}$	2	5.048	6.982	0.720 (0.010)	0.714 (0.010)	3.34×10^{-6}
	3	5.762	8.452	0.675 (0.020)	0.645 (0.020)	—
	4	7.241	10.593	0.688 (0.012)	0.649 (0.007)	0.191

Panel B: the entrywise ratio against the solve fold, $\lceil 0.4n \rceil$*.*
A restriction on the weight class does not shrink in n*.*

Design	L	$n_{\text{solve}} = 160$	$n_{\text{solve}} = 320$	$n_{\text{solve}} = 640$	log-log slope	worst j
Balanced	2	0.613	0.606	0.641	0.032	const. [1.00]
	3	0.633	0.599	0.558	-0.091	const. [1.00]
	4	0.620	0.604	0.646	0.029	const. [1.00]
Skewed	2	0.639	0.600	0.531	-0.133	z_1 [0.42]
	3	0.621	0.604	0.591	-0.036	const. [0.67]
	4	0.593	0.651	0.588	-0.006	const. [0.50]
Misspecified $\hat{\varrho}$	2	0.659	0.675	0.720	0.064	const. [1.00]
	3	0.620	0.649	0.675	0.061	const. [1.00]
	4	0.708	0.666	0.688	-0.021	const. [1.00]

Panel C: the M3-off control, the design ceiling and the quadrature certificates.

Design	L	control ratio	$R_{\max}$	$R_{\max}$ control	cert. $\|\Delta E[L]\|$	scale
Balanced	2	1.63×10^{-14}	0.8615	8.94×10^{-8}	5.70×10^{-6}	25.98
	3	2.07×10^{-14}			1.24×10^{-5}	46.44
	4	2.40×10^{-14}			7.78×10^{-6}	55.26
Skewed	2	2.63×10^{-14}	0.8615	8.94×10^{-8}	3.45×10^{-6}	25.90
	3	2.71×10^{-14}			4.18×10^{-6}	23.27
	4	2.27×10^{-14}			5.38×10^{-6}	29.12
Misspecified $\hat{\varrho}$	2	—	0.9021	8.94×10^{-8}	8.57×10^{-5}	40.94
	3	—			9.41×10^{-5}	59.90
	4	—			1.67×10^{-4}	57.33

Notes: The defect $b = E[L \mid \Delta Z]$ that N1 asks to vanish, L being the anchored antiderivative the estimator builds; by Corollary 21 that is $p(\mathsf{L} - 1)$ scalar conditions. *It is a closed form here and not an estimate*, and what carries sampling error is the weight, which `solve_balance_weight` solves on $n_{\text{solve}} = \lceil 0.4n \rceil$ points at each design's own true propensity and true residual; every standard error printed is therefore over the 12 solve draws, and Panel A's 640 is the $n = 1{,}600$ of Table 1 of the article. Panel A takes the entrywise maximum over the $p(\mathsf{L} - 1)$ pairs, a Frobenius aggregate being able to hide one live entry, and closes with Block I's figure from Table S28 at the two cardinalities that arm runs. Panel B repeats the ratio over the three folds and names the coefficient attaining the maximum at the median draw; *its bracket is the share of draws at which the constant attains it*, not the share for the coefficient named. Panel C carries the three certificates the text reads: the control, which switches the conditional dependence off and returns this script's numerical floor, run at the largest fold and at the two well-specified designs only; $R_{\max}$, the supremum of the same ratio over centred weights polynomial of degree at most six in the residual, which is a property of the design and of no weight; and the largest gap between the closed form and an independent quadrature rule over a cell's draws, beside the largest scale compared. Source: `scripts/simulations/sim_multinomial_n1defect.R`; table `paper/tables/sim/sim_multinomial_n1defect.tex`.

Table S31: The shipped engine at three cardinalities: the skewed design

Block III — multinomial design, shipped engine, nuisances estimated, *Skewed design*
`sim_multinomial_dgp.R` with `lf_panel.R`; $\mathsf{L} \in \{2, 3, 4\}$ off one draw
B used of 120 drawn: 116 (n=400), 117 (n=800), 120 (n=1,600); equal at all L
Dropped draws move any coverage by at most 0.032; worst 0.948 → [0.917, 0.950]

Design	n	Coefficient	$\mathsf{L} = 2$	$\mathsf{L} = 3$	$\mathsf{L} = 4$
Panel A: bias, with its Monte Carlo standard error. Truth (0, 1.0, −0.5).					
Skewed	400	Constant	-0.0220 (0.0199)	-0.0134 (0.0143)	-0.0127 (0.0127)
		z_1	-0.0049 (0.0214)	-0.0140 (0.0182)	-0.0108 (0.0153)
		z_2	0.0064 (0.0191)	0.0268 (0.0150)	0.0026 (0.0148)
	800	Constant	-0.0075 (0.0126)	0.0016 (0.0105)	-0.0180 (0.0087)
		z_1	0.0108 (0.0132)	0.0018 (0.0106)	-0.0166 (0.0091)
		z_2	0.0119 (0.0142)	0.0073 (0.0099)	0.0050 (0.0093)
	1,600	Constant	-0.0123 (0.0080)	-0.0108 (0.0061)	-0.0138 (0.0053)
		z_1	-0.0089 (0.0074)	-0.0058 (0.0061)	-0.0073 (0.0059)
		z_2	0.0045 (0.0091)	-0.0108 (0.0070)	-0.0042 (0.0072)
Panel B: $\sqrt{n}\times$ *s.d., which the theorem makes constant in n, with its s.e.*					
Skewed	400	Constant	4.280 (0.427)	3.082 (0.230)	2.736 (0.190)
		z_1	4.612 (0.277)	3.916 (0.258)	3.290 (0.188)
		z_2	4.120 (0.247)	3.234 (0.236)	3.180 (0.238)
	800	Constant	3.858 (0.313)	3.223 (0.251)	2.656 (0.164)
		z_1	4.031 (0.325)	3.257 (0.247)	2.786 (0.191)
		z_2	4.333 (0.373)	3.017 (0.165)	2.838 (0.173)
	1,600	Constant	3.525 (0.312)	2.654 (0.190)	2.344 (0.179)
		z_1	3.249 (0.248)	2.665 (0.164)	2.607 (0.156)
		z_2	3.981 (0.367)	3.067 (0.178)	3.156 (0.198)
Panel C: coverage of the nominal 95% *interval, with its Monte Carlo standard error.*					
Skewed	400	Constant	0.897 (0.028)	0.914 (0.026)	0.931 (0.024)
		z_1	0.862 (0.032)	0.879 (0.030)	0.888 (0.029)
		z_2	0.905 (0.027)	0.931 (0.024)	0.948 (0.021)
	800	Constant	0.880 (0.030)	0.872 (0.031)	0.897 (0.028)
		z_1	0.906 (0.027)	0.906 (0.027)	0.915 (0.026)
		z_2	0.863 (0.032)	0.897 (0.028)	0.915 (0.026)
	1,600	Constant	0.883 (0.029)	0.925 (0.024)	0.933 (0.023)
		z_1	0.958 (0.018)	0.950 (0.020)	0.950 (0.020)
		z_2	0.875 (0.030)	0.867 (0.031)	0.867 (0.031)

Notes: As Table 1 of the article, at the skewed design, which moves the distribution of the differenced controls and leaves the propensity correctly specified. Panels, resolution and the absence of a standard-deviation ratio are as there. *The replication count is not*, and it is read off this table's own header row rather than repeated here. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_rootn_skewed.tex`.

Table S32: The interval's scale, and the coverage shortfall it accounts for

Block III — multinomial design, shipped engine, nuisances estimated
the interval's scale, not its centre: `lf_panel.R`, L off one draw

			Mean s.e. over Monte Carlo s.d.			Implied − realised coverage		
Design	n	Coefficient	L = 2	L = 3	L = 4	L = 2	L = 3	L = 4
Balanced	400	Constant	0.776	0.805	0.718	0.017	-0.004	-0.023
		z_1	0.635	0.874	1.000	-0.068	-0.053	-0.007
		z_2	0.625	0.827	0.848	-0.024	0.032	-0.037
	800	Constant	0.776	0.783	0.739	-0.070	-0.033	0.011
		z_1	1.017	0.928	0.930	-0.005	0.014	-0.010
		z_2	0.991	0.970	0.916	-0.027	-0.007	0.002
	1,600	Constant	0.762	0.705	0.758	-0.018	-0.033	-0.003
		z_1	0.986	0.868	0.981	-0.028	-0.005	-0.012
		z_2	0.893	0.963	1.016	0.004	-0.034	-0.005
Skewed	400	Constant	0.739	0.883	0.903	-0.044	0.003	-0.008
		z_1	0.782	0.743	0.821	0.013	-0.025	0.005
		z_2	0.853	0.912	0.876	0.000	-0.005	-0.034
	800	Constant	0.741	0.755	0.838	-0.027	-0.011	0.002
		z_1	0.834	0.826	0.897	-0.008	-0.011	0.007
		z_2	0.734	0.876	0.868	-0.014	0.016	-0.003
	1,600	Constant	0.803	0.888	0.901	0.001	-0.007	-0.011
		z_1	0.901	0.953	0.915	-0.036	-0.012	-0.023
		z_2	0.749	0.842	0.758	-0.017	0.035	-0.004
Misspecified $\hat{\varrho}$	400	Constant	0.691	0.714	0.737	0.030	0.178	0.119
		z_1	0.841	1.002	0.989	-0.028	0.013	0.019
		z_2	0.821	0.923	0.919	-0.009	-0.017	-0.018
	800	Constant	0.787	0.703	0.657	0.106	0.306	0.378
		z_1	0.822	0.894	0.847	-0.014	-0.004	0.005
		z_2	0.783	0.763	0.893	-0.015	-0.007	-0.004
	1,600	Constant	0.661	0.662	0.562	0.108	0.511	0.452
		z_1	0.930	0.912	0.921	-0.018	-0.015	-0.004
		z_2	0.856	0.903	0.927	0.007	0.033	0.023

Notes: Block III of Table S29, on the cells of Table 1 of the article and its two companions and at their replication counts, which those tables carry in their own header rows. The left block is the mean estimated standard error over the simulated standard deviation, so a value below one is an interval narrower than the sampling distribution it is meant to cover. The right block subtracts the realised coverage of those tables' Panel C from the coverage a centred normal carrying that ratio would deliver, $2\Phi(1.96\,\text{ratio}) - 1$: a value near zero says the shortfall is the interval's scale, and a value away from zero says it is something a centred normal does not carry, which on this suite is the bias in the coordinate the propensity enters. *The right block is a decomposition and not a second measurement*, being a function of the left block and of a coverage already printed, and it adds no replication. No population variance exists for this block, so the denominator is the simulated standard deviation and never an asymptotic one. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_seratio.tex`.

Table S33: The paired cardinality contrast, on the same draws

Block III — multinomial design, shipped engine, nuisances estimated
`sim_multinomial_dgp.R` with `lf_panel.R`; $\mathsf{L} \in \{2, 3, 4\}$ off one draw

			s.d. ratio against $\mathsf{L} = 2$		Paired bias difference (s.e.)	
Design	n	Coefficient	$\mathsf{L} = 3$	$\mathsf{L} = 4$	$\mathsf{L} = 3 - \mathsf{L} = 2$	$\mathsf{L} = 4 - \mathsf{L} = 2$
Balanced	400	Constant	0.808	0.751	-0.1053 (0.0428)	-0.1678 (0.0420)
		z_1	0.619	0.473	-0.0235 (0.0316)	-0.0231 (0.0280)
		z_2	0.651	0.561	-0.0100 (0.0261)	-0.0250 (0.0244)
	800	Constant	0.983	0.889	-0.0380 (0.0246)	-0.0503 (0.0240)
		z_1	0.950	0.871	-0.0247 (0.0111)	-0.0251 (0.0106)
		z_2	0.894	0.855	0.0009 (0.0112)	0.0015 (0.0113)
	1,600	Constant	1.181	0.970	0.0005 (0.0175)	-0.0339 (0.0144)
		z_1	1.059	0.867	-0.0030 (0.0073)	-0.0069 (0.0064)
		z_2	0.793	0.695	-0.0042 (0.0073)	-0.0032 (0.0071)
Skewed	400	Constant	0.720	0.639	0.0087 (0.0205)	0.0093 (0.0207)
		z_1	0.849	0.713	-0.0091 (0.0179)	-0.0059 (0.0209)
		z_2	0.785	0.772	0.0204 (0.0188)	-0.0038 (0.0197)
	800	Constant	0.836	0.689	0.0092 (0.0118)	-0.0105 (0.0124)
		z_1	0.808	0.691	-0.0090 (0.0124)	-0.0274 (0.0121)
		z_2	0.696	0.655	-0.0046 (0.0122)	-0.0069 (0.0118)
	1,600	Constant	0.753	0.665	0.0014 (0.0083)	-0.0015 (0.0084)
		z_1	0.820	0.802	0.0030 (0.0071)	0.0015 (0.0073)
		z_2	0.771	0.793	-0.0152 (0.0076)	-0.0087 (0.0079)
Misspecified $\hat{\varrho}$	400	Constant	0.891	0.802	-0.1452 (0.0414)	-0.1227 (0.0401)
		z_1	0.752	0.690	0.0406 (0.0226)	0.0422 (0.0218)
		z_2	0.775	0.729	0.0117 (0.0230)	0.0345 (0.0235)
	800	Constant	1.003	1.034	-0.1366 (0.0303)	-0.1692 (0.0314)
		z_1	0.771	0.767	0.0120 (0.0149)	0.0222 (0.0132)
		z_2	0.873	0.715	-0.0019 (0.0151)	-0.0071 (0.0134)
	1,600	Constant	1.114	1.207	-0.2287 (0.0226)	-0.2765 (0.0241)
		z_1	0.907	0.840	0.0120 (0.0078)	0.0139 (0.0080)
		z_2	0.846	0.778	-0.0138 (0.0075)	-0.0089 (0.0076)

Notes: Block III only, because pairing across blocks is not defined: the three suites draw different laws and share no sample size. *Every comparison here is within a replication*, the three cardinalities being read off one draw and one cross-fitting partition, so the standard error in parentheses is that of the difference and not the sum of two independent ones. The first two numeric columns are the simulated standard deviation at $\mathsf{L} = 3$ and at $\mathsf{L} = 4$ divided by the binary one at the same cell, so a value below one is a tighter sampling distribution at the higher cardinality; the last two are the paired bias difference against the binary arm. No asymptotic variance is available for this block, so the ratio is against the binary arm and never against a population value. Between 112 and 120 replications per cell, and identical across the three cardinalities within a cell, which is what leaves the pairing defined; the counts are given design by design in the header rows of Table 1 of the article, Table S31 and Table 2 of the article. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_precision.tex`.

The empirical application: supplementary material

This section carries the material for Section 6 of the article that the article names but does not print. It adds no result. The article's three tables and its figure carry the section's findings; what follows is the construction of the estimation sample, the variables it holds, the balance and overlap the design realises, the parametric component along the control-set ladder, the diagnostics of the estimator that produced the article's structural function, the tuning arms, and the three floats that moved here from Section 6 in the second OBES cut: the iteration-cap contrast, the N1 defect over the class of constant anchors, and $\hat{\varphi}$ at eleven percentiles with its bootstrap envelope. Every table but one is produced by `scripts/application/app_tsp_run_all.R`; the exception is Table S48, a Monte Carlo float placed here beside the boundary material it explains, which comes from `scripts/simulations/sim_thm_rootn_bdrylayer.R` and is reached by neither run-all driver. The file each table is read from is named in its notes. The sample is the Chay–Greenstone county panel, $T = 2$, first differenced, $n = 983$ throughout except where a row says otherwise.

Table S34: Construction of the estimation sample

Step	Dropped	Remaining
Raw rows in data_tsp.txt	–	989
Drop column ExpPol (NaN in 989 of 989 rows)	0	989
Drop counties missing the local-fiscal block	6	983
Analysis sample	–	983

Notes: Every row dropped from the raw file, in the order the drops are applied, with the count remaining after each. `ExpPol` is per-capita expenditure on pollution abatement; it is `NaN` in all 989 rows, so it is an empty column rather than a column with gaps, and the count of usable candidate controls is 22 and not 23. The six counties missing the local-fiscal block are three New York City boroughs and three Virginia independent cities, which is the documented County and City Data Book reporting pattern and not item non-response. The whole ladder of the article runs on these 983 counties so that movement along it is attributable to d_Z alone; the full 989 is a robustness arm in Table S42. Source: `scripts/application/app_tsp_common.R`; table `paper/tables/app_tsp_sample.tex`.

Table S35: Descriptive statistics

Variable	Mean	SD	p10	Median	p90
ΔY: change in log property value	0.254	0.172	0.055	0.236	0.480
X_1: mean TSP 1970 (μg/m^3)	66.018	27.809	37.895	62.749	96.845
X_2: mean TSP 1980 (μg/m^3)	56.281	16.640	36.925	54.953	76.497
ΔX: change in TSP (μg/m^3)	-9.738	23.322	-32.183	-7.471	10.865
W: 1975 nonattainment (imputed)	0.303	0.460	0.000	0.000	1.000
ΔZ_1: per capita income	0.177	0.086	0.075	0.173	0.289
ΔZ_2: population	0.147	0.166	-0.023	0.121	0.343
ΔZ_3: unemployment rate	0.023	0.020	-0.002	0.021	0.047
ΔZ_4: manufacturing employment	-0.023	0.041	-0.071	-0.024	0.027
ΔZ_5: high school	0.143	0.033	0.105	0.141	0.183

Notes: $n = 983$ counties. ΔY is the 1970–1980 change in log county median property value; X_1 and X_2 are mean total suspended particulates in μg/m^3 in the two periods and ΔX their difference; W is the imputed 1975 nonattainment designation, so its mean is the share of designated counties. The five ΔZ rows are the core control set, rung 1 of the ladder, each entered as a 1970–1980 difference. Percentiles are of the county distribution. Source: `scripts/application/app_tsp_01_diagnostics.R`; table `paper/tables/app_tsp_sumstats.tex`.

Table S36: Balance of the twenty-two differenced covariates on the instrument

Differenced covariate	Diff.	Robust SE	Clustered SE	Norm. diff.
Core (rung 1)				
PCIncome	-0.0045	(0.0057)	[0.0108]	-0.053
Population	-0.0129	(0.0122)	[0.0228]	-0.078
UnempRate	0.0003	(0.0015)	[0.0025]	0.017
EmpManufacturing	-0.0013	(0.0028)	[0.0048]	-0.032
HighSchool	-0.0033	(0.0022)	[0.0033]	-0.101
Demographic (rung 2)				
College	0.0024	(0.0017)	[0.0030]	0.098
PopDens	-0.0088	(0.0139)	[0.0246]	-0.042
Urban	-0.0013	(0.0053)	[0.0053]	-0.016
Poverty	0.0140	(0.0024)	[0.0042]	0.378
White	-0.0165	(0.0031)	[0.0047]	-0.376
Housing and fiscal (rung 3)				
Female	-0.0010	(0.0005)	[0.0005]	-0.121
Senior	-0.0003	(0.0009)	[0.0018]	-0.023
VacancyRate	0.0006	(0.0023)	[0.0021]	0.019
OwnerOccRate	0.0072	(0.0036)	[0.0060]	0.125
UnitsWOPlumbing	0.0259	(0.0037)	[0.0067]	0.419
PCGovRev	0.0028	(0.0018)	[0.0020]	0.089
PCPropTax	-0.0020	(0.0005)	[0.0008]	-0.253
PCGenExpLocalGov	0.0024	(0.0017)	[0.0018]	0.080
ExpEduc	0.0110	(0.0061)	[0.0077]	0.119
ExpHighways	0.0032	(0.0022)	[0.0029]	0.098
ExpWelf	-0.0094	(0.0030)	[0.0051]	-0.220
ExpHosp	-0.0022	(0.0041)	[0.0044]	-0.035
Imbalanced at 5% (robust / clustered)				7 / 4 of 22
Joint F (robust)				6.85 $[p = 0.0000]$
Joint F (clustered)				10.50 $[p = 0.0000]$

Notes: $n = 983$. Each row regresses one differenced covariate on W alone and reports the coefficient, its robust standard error in parentheses, its state-clustered standard error in square brackets, and the normalised difference, the difference in means divided by the square root of the average of the two within-group variances. Rows are grouped by the ladder rung at which the covariate first enters. The last three rows count the covariates whose difference is significant at 5 per cent under each variance, and test all twenty-two jointly. This table is not a randomisation check: W is not randomly assigned and A1 does not require it to be independent of ΔZ. What it establishes is that W and ΔZ are dependent, which matters because the estimator centres at $W - \hat{\varrho}(\Delta Z)$: the dependence this table measures is exactly what $\hat{\varrho}$ has to capture, and a design in which W and ΔZ were independent would make the conditional and unconditional centrings coincide. Source: `scripts/application/app_tsp_01_diagnostics.R`; table `paper/tables/app_tsp_balance.tex`.

Table S37: Overlap of the propensity score, condition A4(i)

Conditioning set	Min	p1	p99	Max	Share $\notin [.05, .95]$	Share $\notin [.10, .90]$	Pseudo-R^2
$d_Z = 0$ (intercept only)	0.303	0.303	0.303	0.303	0.000	0.000	0.000
$d_Z = 5$	0.196	0.230	0.362	0.408	0.000	0.000	0.003
$d_Z = 10$	0.026	0.062	0.682	0.800	0.005	0.044	0.065
$d_Z = 22$	0.001	0.030	0.757	0.901	0.025	0.099	0.112
Baseline TSP only	0.013	0.034	0.940	1.000	0.039	0.172	0.192
Baseline TSP + rung 1	0.012	0.031	0.946	1.000	0.044	0.178	0.195

Notes: $n = 983$. Each row fits a logit of W on the stated conditioning set and summarises the fitted $\hat{\varrho}(\Delta Z) = \widehat{P}(W = 1 \mid \Delta Z)$ across counties: its minimum, its first and ninety-ninth percentiles, its maximum, the share of counties outside $[0.05, 0.95]$ and outside $[0.10, 0.90]$, and McFadden's pseudo-R^2. A4(i) asks $\hat{\varrho}$ to stay away from zero and one, so a row fails when the extremes approach them. *Since 2026-08-07 this table is no longer a diagnostic*: $\hat{\varrho}$ is an estimated nuisance inside the moment, truncated into $[0.03, 0.97]$, so A4(i) is a maintained condition on an object the estimator uses rather than a property it merely reports. The truncation binds on roughly one per cent of the sample at the widest control set, where the first percentile of $\hat{\varrho}$ is 0.030. The last two rows condition on baseline particulates, which is the variable the designation rule thresholds; both drive the maximum to exactly one, so overlap fails there. Source: `scripts/application/app_tsp_01_diagnostics.R`; table `paper/tables/app_tsp_overlap.tex`.

Table S38: The parametric component along the control-set ladder

	(1) $d_Z = 0$	(2) $d_Z = 5$	(3) $d_Z = 10$	(4) $d_Z = 22$
Constant	0.2277	-0.0041	-0.0314	-0.0484
	(12.9675)	(0.0400)	(0.0597)	(0.0550)
	[12.9676]	[0.0624]	[0.0762]	[0.0710]
Per capita income	–	0.5732***	0.4662*	0.6735***
	–	(0.1409)	(0.2356)	(0.1479)
	–	[0.1947]	[0.2769]	[0.1857]
Population	–	0.3411***	0.2752***	0.2041***
	–	(0.0665)	(0.0619)	(0.0436)
	–	[0.0788]	[0.0656]	[0.0524]
Unemployment rate	–	-1.0178*	-1.1425*	-0.9645**
	–	(0.4577)	(0.5754)	(0.4638)
	–	[0.5937]	[0.6833]	[0.4882]
Manufacturing employment	–	0.0404	0.1412	-0.1325
	–	(0.2270)	(0.2716)	(0.2096)
	–	[0.2383]	[0.2883]	[0.1988]
High school	–	0.8684**	0.9460**	0.7853**
	–	(0.2872)	(0.4058)	(0.3219)
	–	[0.3616]	[0.4502]	[0.3787]
College	–	–	0.3914	0.2657
	–	–	(0.4350)	(0.3826)
	–	–	[0.4985]	[0.4303]
Population density	–	–	0.0946**	0.0768***
	–	–	(0.0390)	(0.0239)
	–	–	[0.0386]	[0.0295]
Urban share	–	–	-0.0967	-0.1174
	–	–	(0.1305)	(0.0899)
	–	–	[0.1341]	[0.0907]
Poverty rate	–	–	-0.3698	-0.5758
	–	–	(0.4252)	(0.4168)
	–	–	[0.4743]	[0.4682]
White share	–	–	-0.4633	-0.6145*
	–	–	(0.2488)	(0.1963)
	–	–	[0.3903]	[0.3183]
Controls in ΔZ	none	core	core + demographic	all usable
Coefficients estimated (p)	1	6	11	23
Observations	983	983	983	983

Notes: $\hat{\theta}$ from the cross-fitted one-step of (47), at $\mathsf{F} = 5$ folds and 10 partitions, $n = 983$ in every column. The instrument here is the binary coarsening, at a truncation level $\varepsilon_\varrho = 0.03$ and an iteration cap $\bar{m} = 15$. The article reads this table against Table 6 of the article, which runs a different cross-fitting stream at 0.01 and 60; those are the three factors that separate the two. ordered by how far each block of controls is from being an outcome of house prices: an intercept alone, five core hedonic demand shifters, those plus five demographic variables, and all twenty-two usable controls, whose per-capita property-tax revenue is mechanically a function of the dependent variable. Rows are the coefficients the first three rungs share; the housing and fiscal coefficients of column (4) are estimated but not printed. Robust standard errors in parentheses,

state-clustered standard errors in square brackets, and stars computed from the clustered standard error, which is the one A0's failure at Air Quality Control Region level licenses. Column (1) is a reference point and not an estimate: its standard error is two failed cross-fitting partitions, as Table S40 shows. The object of the article's Section 6 is the structural function and not θ, so this table is a sensitivity axis rather than a headline. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: `scripts/application/app_tsp_02_ladder.R`; table `paper/tables/app_tsp_ladder.tex`.

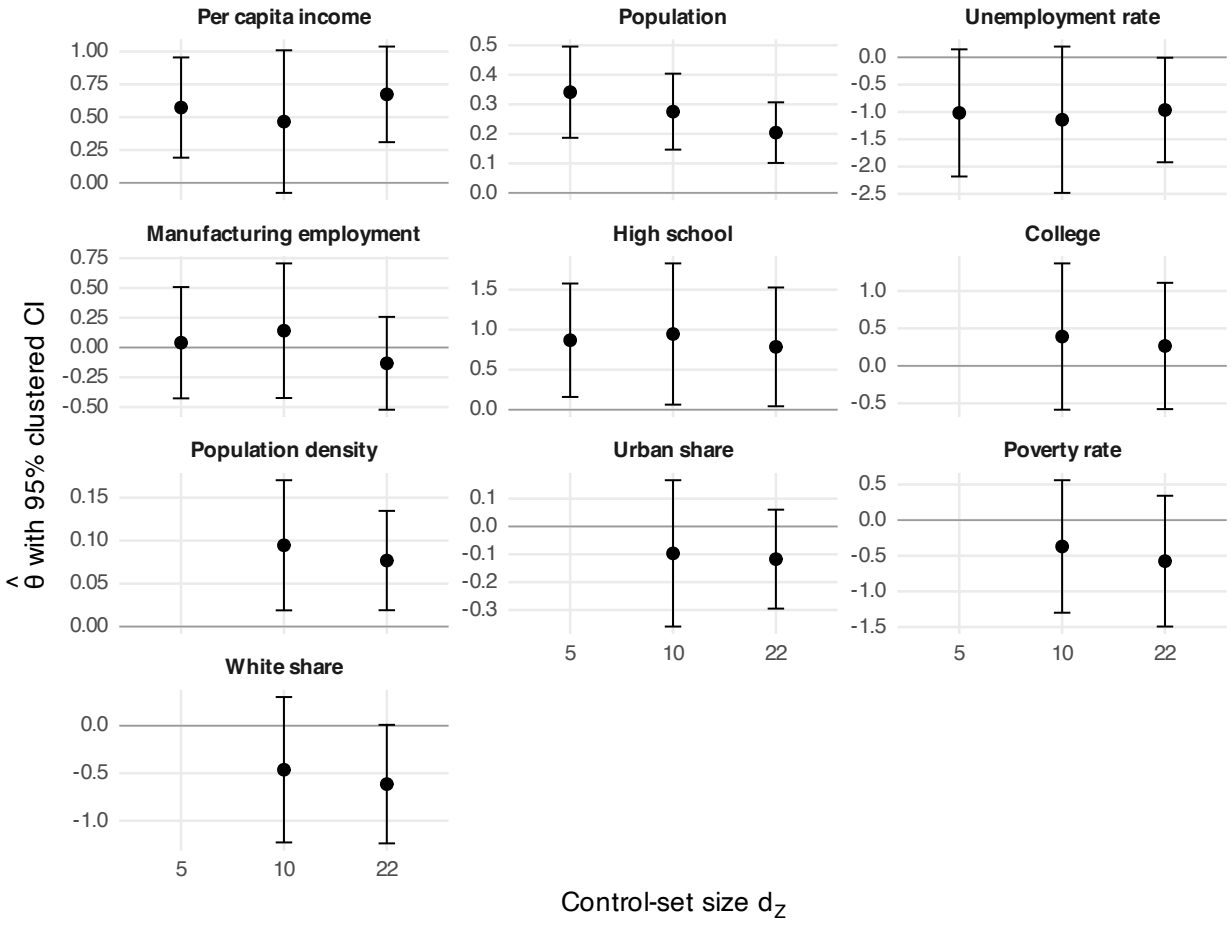


Figure S8: The parametric component along the control-set ladder

Notes: Table S38 drawn. Each panel is one coefficient of $\hat{\theta}$ and each point is its estimate at one rung of the ladder, with a 95 per cent interval built from the state-clustered standard error; the horizontal line at zero is drawn in every panel, so a coefficient whose interval crosses it is not distinguishable from zero at that rung. The horizontal axis is the number d_Z of differenced controls, and the intercept-only rung is not drawn because it estimates no slope. Panels have independent vertical scales, so the figure carries the movement of each coefficient along the ladder and not the relative size of one coefficient against another; the levels are in Table S38. What it shows is that population keeps its sign and decays monotonically, that per capita income keeps its sign and does not, and that manufacturing employment is indistinguishable from zero at all three rungs and changes sign at the last. $n = 983$ at every point. Source: `scripts/application/app_tsp_04_figures.R`; figure `paper/figures/app_tsp_ladder.pdf`.

Table S39: The iteration cap, held fixed and then varied

	$\varepsilon_{\varrho} = 0.0100$		
	$\mathsf{L} = 4$ $\bar{m} = 60$	$\mathsf{L} = 2$ $\bar{m} = 60$	$\mathsf{L} = 2$ $\bar{m} = 15$
Constant	-0.0025	-0.0156	-0.0071
	(0.0560)	(0.0568)	(0.0599)
Per capita income	0.7112	0.6057	0.6504
	(0.2027)	(0.1862)	(0.2119)
Population	0.3102	0.3433	0.3480
	(0.0754)	(0.0838)	(0.0868)
Unemployment rate	-0.9905	-1.1148	-1.1095
	(0.5922)	(0.6238)	(0.6511)
Manufacturing employment	0.3918	0.4198	0.2864
	(0.2398)	(0.2751)	(0.2547)
High school	0.8743	0.9008	0.7078
	(0.3188)	(0.3396)	(0.3520)
Truncation binds (%)	0.997	0.000	0.000
Mean rel. balance	0.0012	0.0013	0.0013

Notes: The same panel, control set, estimator, cross-fitting stream and truncation level as Table 6 of the article, whose two columns are repeated here as the first two so that the two contrasts can be read off one table. *Each adjacent pair varies one factor.* Columns one and two differ in the instrument's cardinality at a common Landweber cap $\bar{m} = 60$; columns two and three differ in the cap alone, at $\bar{m} = 60$ against the $\bar{m} = 15$ that the binary cardinality would itself have set. Clustered standard errors in parentheses, clustered on state; no significance stars are computed. The two contrasts are opposite in sign at five of the six coordinates, population being the exception, so a design that moves both at once nets them against each other rather than adding them. Source: `scripts/application/app_tsp_08_localise.R`; table `paper/tables/app_tsp_localise_theta_tied.tex`.

Table S40: What the estimator did: stopping, conditioning, component scale, and the variance decomposition

	$d_Z = 0$	$d_Z = 5$	$d_Z = 10$	$d_Z = 22$
Panel A: Landweber–Fridman step, over $K \times B$ fold solves				
Fold solves	50	50	50	50
Median stopping index $\hat{m}$	15	15	15	15
Range of $\hat{m}$	[15, 15]	[15, 15]	[15, 15]	[15, 15]
Solves stopping at $\hat{m} = 1$	0	0	0	0
Solves on a discrepancy path that rises throughout	0	0	2	7
Solves reaching the cap $\bar{m} = 15$	50	50	50	50
Panel B: moment and Jacobian				
Relative balancing residual	1.0e-03	1.3e-03	1.5e-03	1.5e-03
Median $\kappa(\hat{J})$	1	3972	4544	23852
Max $\kappa(\hat{J})$	1	4658	5163	29752
$\sigma_{\min}(\hat{J}) \,/\, \sigma_{\min}(E_n[\Delta Z \Delta Z'])$	1.042	0.991	1.011	0.950
Panel C: scale of the fitted components, and stability across the B partitions				
Median $\mathrm{sd}(\Delta\hat{\varphi}) / \mathrm{sd}(\Delta Y)$	1.19	0.14	0.12	0.08
Median $\mathrm{sd}(\Delta\hat{U}) / \mathrm{sd}(\Delta Y)$	1.58	0.76	0.76	0.69
Max $\mathrm{sd}(\Delta\hat{\varphi}) / \mathrm{sd}(\Delta Y)$	1382.4	0.2	0.1	0.1
sd / 1.4826×MAD of $\hat{\theta}_b$	791.46	1.35	1.25	1.22
Partitions > 10 robust scales from the median	2	0	0	1
Panel D: variance decomposition and rate budget				
Sampling share of SE (median)	0.003	0.976	0.963	0.977
Cross-fitting share of SE (median)	1.000	0.218	0.269	0.215
δ_n at $(\mu, r) = (2, 8/3)$	0.644	1.129	1.796	2.736
$\sqrt{n}\,\delta_n^2$	13.0	40.0	101.1	234.6

Notes: Diagnostics of the fits reported in Table S38, over the $K \times B = 5 \times 10 = 50$ fold solves at each rung, $n = 983$. Panel A reports the realised Landweber–Fridman stopping index $\hat{m}$ of Definition 22: a solve stopping at $\hat{m} = 1$ returned its initial condition unchanged, and one reaching the cap $\bar{m} = 15$ was still descending when it was stopped. The shipped artifact writes that index $\hat{m}$ here and $\hat{m}_\varphi$ in Table S43; they are the same object, realised at a θ-step fold solve in this table and at a bootstrap replication of the φ-step in that one. Panel B reports the relative residual of the balancing solve for the representer, the condition number of the moment's Jacobian $\hat{J}$ of (46), and $\sigma_{\min}(\hat{J})$ measured against $\sigma_{\min}$ of the design second-moment matrix; the last is within four per cent of one at every rung, so the debiasing term does not degrade identification and nothing here is a near-singular-Jacobian problem. At $d_Z = 0$ the Jacobian is a scalar and its condition number is one by construction, which is why $\sigma_{\min}$ is reported beside it. Panel C compares the standard deviation of the fitted structural difference $\Delta\hat{\varphi}$ and of the fitted residual $\Delta\hat{U}$ with that of the outcome they decompose, so an entry above one is a component larger than the quantity it explains, and reports the dispersion of the B partition estimates against a robust scale. Panel D splits the reported standard error into the part that is sampling variation and the part that is spread across cross-fitting partitions, and repeats the frontier quantities of Table S49. Figure S9 traces the quantity Panel A's stopping index is selected from, which is where the difficulty of locating that stop is visible. Source: `scripts/application/app_tsp_02_ladder.R` and `app_tsp_common.R`; table `paper/tables/app_tsp_estimator_diag.tex`.

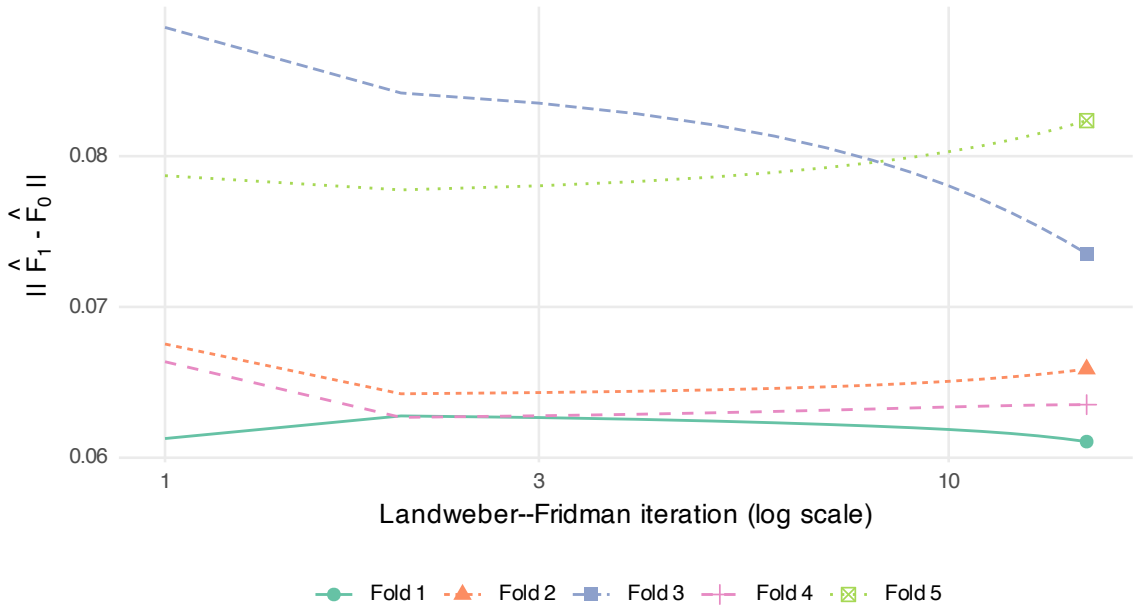


Figure S9: Landweber–Fridman discrepancy paths, five fold solves

Notes: The quantity the stopping rule of Definition 22 thresholds, traced along the iteration for the five cross-fitting folds of one partition at the five-control set, $n = 983$. The horizontal axis is the iteration index on a log scale and the vertical axis is the norm of the discrepancy, the difference between the two periods' fitted transforms; the plotted marker on each series is where that series stopped. A series whose marker sits at the right-hand edge ran to the cap $\bar{m} = 15$ and was still descending. The figure is drawn to show why the stop is hard to locate here rather than to compare folds: three of the five paths are close to flat over the first decade of iterations, so a threshold that is slightly too high stops at once and one slightly too low runs to the cap, which is the behaviour Table S40 counts. Fold identities carry no meaning across panels of other figures. Source: `scripts/application/app_tsp_03_robustness.R` and `app_tsp_04_figures.R`; figure `paper/figures/app_tsp_lf_path.pdf`.

Table S41: Is N1 imposed? The defect over the class of constant anchors

	Conditional mean fitted on a quadratic basis in ΔZ, rank 21					
Statistic	Plug-in	Null floor	Debiased	95% upper	Upper/$\|\widehat{L}\|$	p
$\mathsf{L} = 4$, *four trigger cells:* $\varepsilon_\varrho = 0.010$, $\bar{m} = 60$, *headline partition stream*						
Entrywise $\max_{j,l}$	0.5647	0.6158	0.2522	0.4021	0.137	0.206
before mean removal	3.3031	3.2958	3.2439	3.2719	0.686	0.198
Contracted, ϖ_w only	0.2457	0.2615	0.1196	0.1887	0.142	0.151
Contracted, $\varpi_w \widetilde{W}$	0.0920	0.0920	0.0252	0.0620	0.129	0.279
$\mathsf{L} = 2$, *binary coarsening:* $\varepsilon_\varrho = 0.010$, $\bar{m} = 60$, *same partition stream*						
Entrywise $\max_{j,l}$	0.1546	0.1943	0.0245	0.1053	0.110	0.394
before mean removal	0.4633	0.4876	0.4273	0.4471	0.375	0.400
Contracted, ϖ_w only	0.1546	0.1943	0.0245	0.1053	0.110	0.394
Contracted, $\varpi_w \widetilde{W}$	0.0794	0.0884	0.0128	0.0561	0.111	0.468
$\mathsf{L} = 2$, *binary coarsening:* $\varepsilon_\varrho = 0.030$, $\bar{m} = 15$, *its own partition stream*						
Entrywise $\max_{j,l}$	0.1893	0.2219	0.0156	0.1328	0.105	0.455
before mean removal	0.4597	0.4781	0.4031	0.4307	0.353	0.419
Contracted, ϖ_w only	0.1893	0.2219	0.0156	0.1328	0.105	0.455
Contracted, $\varpi_w \widetilde{W}$	0.0836	0.1057	0.0040	0.0607	0.104	0.585

Notes: Chay–Greenstone county panel, $n = 983$, at the headline control set $d_Z = 5$, measured on the shipped cross-fitted $\hat{L}^{(-k)}$ by replaying the estimation streams of Table 6 of the article rather than on a reconstruction of it. Every entry is a norm of $\widehat{E}[\cdot \mid \Delta Z]$ in L^2, the conditional mean being fitted on a quadratic basis in ΔZ of rank 21, which is the basis that spans the estimator's own propensity index. "Entrywise" is $\max_{j,l}$ over the $p(\mathsf{L}-1)$ conditions Corollary 21 shows N1 to be, eighteen of them at four cells; "Contracted, ϖ_w only" is the single weighted contraction over l, which is the weaker condition; "Contracted, $\varpi_w \widetilde{W}$" is the term that enters ψ, which is conditionally centred under A1 *whatever the anchor*, and is printed so that an orthogonal-looking moment cannot be read as evidence for N1. "Raw" is the projection norm itself, which at rank 21 is dominated by overfitting and is therefore reported only against "Null floor", the 95th percentile of the same statistic under 300 permutations of the observation index in state blocks, which break the dependence between $\hat{L}$ and ΔZ the projection is measuring; "Debiased" is the HC2-debiased point estimate and "95% upper" its one-sided upper bound, and those two are the columns to read. The last column is a permutation p on the largest of the entries, at the median over the cross-fitting partitions; the draw permutes the index commonly across cells, so the reference carries the cross-cell dependence the observed statistic has and the test is multiplicity-correct by construction. *The reported defect is minimal over the whole class of constant anchors*, because a projection on a basis containing an intercept preserves sample means, so no u-free, ΔZ-free constant lowers it and only a genuinely ΔZ-measurable anchor can. *Nothing is re-anchored and no estimator is changed*; the $\mathsf{L} = 2$ blocks are the paired arm of Table 6 of the article and the binary ladder of Table S38, *which differ from each other in three factors and not in one*, the cross-fitting stream, the truncation level ($\varepsilon_\varrho = 0.01$ against 0.03) and the iteration cap ($\bar{m} = 60$ against 15), so the two are not a repeated measurement of the binary defect. Source: `scripts/application/app_tsp_10_n1_anchor.R`; table `paper/tables/app_tsp_n1_anchor.tex`.

Table S42: Fifteen tuning arms and two comparison estimators

	Constant		Population		High school		Median	Fold solves	
Arm	Estimate	(SE)	Estimate	(SE)	Estimate	(SE)	$\hat{m}$	at $\hat{m} = 1$	at the cap
Baseline	-0.0006	(0.0656)	0.3673	(0.0745)	0.8564	(0.3212)	15	0/30	30/30
X in μg/m^3 or min–max	h_x matched; $\hat{\pi} \leq 0$: 2 of 31 rungs feasible, $[1.410, 1.485]$; rank 23 of 31						–	–	–
Step $c = 1.0$	-0.0160	(0.0617)	0.3713	(0.0771)	0.9672	(0.3655)	15	0/30	30/30
Step $c = 0.5$	0.0070	(0.0677)	0.3717	(0.0911)	0.8926	(0.3651)	15	0/30	30/30
Step $c = 0.1$	0.0445	(0.0779)	0.3485	(0.0872)	1.0073	(0.4005)	15	0/30	30/30
Start: Robinson	0.0086	(0.0674)	0.3061	(0.0830)	1.0056	(0.3429)	15	0/30	30/30
Start: cubic	-0.0042	(0.0752)	0.3484	(0.1026)	0.8822	(0.4857)	15	0/30	30/30
Folds $K = 2$	0.0029	(0.0616)	0.3780	(0.0866)	0.8888	(0.3812)	15	0/12	12/12
Folds $K = 10$	-0.0074	(0.0627)	0.3165	(0.0946)	1.0481	(0.4026)	15	0/60	60/60
Ridge $\rho = 10^{-4}$	0.0046	(0.0670)	0.3253	(0.0858)	1.0629	(0.3614)	15	0/30	30/30
Ridge $\rho = 10^{-2}$	-0.0136	(0.0621)	0.3223	(0.0818)	0.9926	(0.3316)	15	0/30	30/30
Rule: $\tau_{\mathrm{DP}} = 0.7$, $\bar{m} = 2000$	-0.0277	(0.0595)	0.3624	(0.0810)	0.9546	(0.3270)	14	4/20	0/20
Budget $\bar{m} = 30$	0.0013	(0.0612)	0.3733	(0.0927)	0.9430	(0.3672)	30	0/30	30/30
Budget $\bar{m} = 60$	-0.0113	(0.0575)	0.3715	(0.0989)	0.9075	(0.3516)	60	0/30	30/30
Budget $\bar{m} = 8$	0.0180	(0.0631)	0.3946	(0.0965)	0.8758	(0.4018)	8	0/30	30/30
All 989 counties	-0.0284	(0.0549)	0.3688	(0.0887)	1.0020	(0.3280)	15	0/30	30/30
Comparison estimators									
Linear IV (2SLS)	-0.2955	(0.3150)	0.5840	(0.2700)	0.2258	(0.7909)	–	–	–
Alternating LF (draft estimator)	-0.0075	–	0.3270	–	0.8822	–	15	–	–

Notes: This table's baseline and the $d_Z = 5$ column of Table S38 are the same specification, and what separates their numbers is the cross-fitting draw. Both are the five-control set on the same 983 counties, at the same initial condition, step, stopping rule $\tau_{\mathrm{DP}} = 0$, budget $\bar{m} = 15$ and five folds; they differ in how many partitions are averaged over, ten in the ladder against six here, and in the seed, which is why the baseline reports 30 fold solves against the ladder's 50. *The gaps between the two tables are therefore a stability check and not a sensitivity. The two tables differ in which coefficients they print and in which standard error.* This table prints a fixed set, the constant with population and the high-school share, pinned rather than reselected because the high-school share holds the second slot over per capita income by 0.0024 in $|t|$, a margin of 0.0917 per cent the cross-fitting draw alone can overturn; and it prints the state-clustered standard error alone, where the ladder prints the independent one in parentheses and the clustered one in brackets, so a figure here is read against the ladder's bracketed column. Per capita income, which this table does not print, is 0.5363 (0.2013) against the ladder's 0.5732 [0.1947]; in units of that column the three gaps are a third of a standard error at population, a fifth at per capita income and a thirtieth at the high-school share. Table S40 Panel D puts the cross-fitting share of the standard error at 0.218. Every

arm is the five-control set on the 983-county sample except "All 989 counties", which restores the six dropped in Table S34, and each varies one tuning constant from the baseline and nothing else. The coefficients of $\hat{\theta}$ that are not printed are estimated in every arm. "Median $\hat{m}$" is the realised Landweber–Fridman stopping index across that arm's fold solves; the last two columns count the solves that returned their initial condition and those that reached the cap, so an arm that looks stable while those counts are high is stable for the wrong reason. *The X-coordinate row returns no estimates*: outside the pooled-rank coordinate $\hat{\pi}$ is not bounded away from zero, and the adjoint divides by it. What fails is the sample reference density: C1(ii) bounds the population π away from zero and is untouched here. That row prints what the bandwidth ladder found, and the three items in it read as follows. Of 31 rungs only 2 leave $\hat{\pi} > 0$; those two span $h_x \in [1.410, 1.485]$ in $\mu\mathrm{g/m}^3$, an interval 5.1 per cent wide about its own midpoint and a factor of 5.00 below the 7.42 that the rule of thumb returns, which is the bandwidth every other arm runs at; and "rank 23 of 31" is the same ladder in the pooled-rank coordinate, which is feasible at 23 rungs where this one is feasible at 2. Even on those two rungs $\hat{\pi}$ is negative on the dense grid in 120 of 120 builds. What fails is the lower endpoint and not the 428.1 outlier: the guard minimum sits between the 0.25th and 0.38th percentile of X, whose smallest value is 0.0 and next 7.84, and trimming the largest counties raises the failures from 5 of 120 builds at $k \leq 2$ to 18 at $k = 5$ and 10. One row carries both coordinates because min–max is an exact affine image of $\mu\mathrm{g/m}^3$ to a relative 7.07×10^{-14}; the rows they printed separately differed through the L'Ecuyer stream each drew, and so through the cross-fitting partition. Panel C of Table S43 reports the same failure for $\hat{\varphi}$. The comparison block reports just-identified linear instrumental variables on the same sample and the draft alternating estimator, for which no standard error is computed. Source: `scripts/application/app_tsp_03_robustness.R`; table `paper/tables/app_tsp_robustness.tex`; run object `quality_reports/application/app_tsp_robustness.rds`.

Table S43: The estimated hedonic price function and its bootstrap variability

Pctile of X	TSP	$\hat{\varphi}(x) - \hat{\varphi}(x_{0.5})$	Bootstrap median	Bootstrap s.d.	Bootstrap envelope 5th	Bootstrap envelope 95th	Counties within $\pm 5\,\mu g/m^3$ 1970	Counties within $\pm 5\,\mu g/m^3$ 1980
Panel A: $\hat{\varphi}$ at deciles of the pooled X distribution								
5	31.4	0.0175	0.0149	0.0195	-0.0175	0.0436	61	67
10	37.5	-0.0012	-0.0020	0.0177	-0.0310	0.0242	94	124
20	44.9	-0.0151	-0.0173	0.0154	-0.0435	0.0073	138	206
30	49.6	-0.0169	-0.0200	0.0133	-0.0426	-0.0004	170	272
40	53.6	-0.0075	-0.0086	0.0080	-0.0232	0.0019	184	280
50 (anchor)	58.0	0.0000	0.0001	0.0017	-0.0016	0.0033	187	253
60	62.8	0.0083	0.0094	0.0084	-0.0023	0.0242	178	210
70	68.1	0.0150	0.0168	0.0132	-0.0010	0.0411	168	154
80	75.4	0.0089	0.0127	0.0168	-0.0097	0.0421	135	96
90	87.7	0.0005	0.0045	0.0207	-0.0273	0.0352	76	28
95	100.2	0.0090	0.0131	0.0233	-0.0261	0.0506	46	9

Panel B: bootstrap design and convergence	
Replications drawn	99
Replications with a finite curve	99
Failures: error / non-finite	0 / 0
$\hat{m}_\varphi$: median [min, max]	60 [60, 60]
Retained with $\hat{m}_\varphi = 1$	0
Retained with $\hat{m}_\varphi = \bar{m}$	99
S.d. of the anchor across resamples	0.63
Share at $\hat{m}_\varphi = 1$: resample of n / subsample of 621	0 of 99 / 0 of 24
θ re-estimated as well ($B = 1$): replications	25, 25 finite
θ re-estimated: band width on the same draws	0.0388 vs. 0.0412
θ-step fold solves at the cap (median of 5)	5

Panel C: the same normalised curve recomputed in one other X coordinate						
Coordinate	Max $\lvert\Delta\rvert$	RMS Δ	Max/SD	RMS/SD	Steps up	Band width, 25 paired draws
Pooled empirical CDF (baseline)	0.000	0.000	0.00	0.00	32/59	0.041
X in $\mu g/m^3$ or min–max	Infeasible: $\hat{\pi}$ not bounded away from zero					

Notes: Chay–Greenstone county panel, $n = 983$, at the headline control set $d_Z = 5$, with the instrument entered as its four trigger cells. Panel A evaluates $\hat{\varphi}(x) - \hat{\varphi}(x_{0.5})$ at eleven percentiles of the pooled distribution of $\{X_1, X_2\}$, normalised at its median $x_{0.5} = 58.0\,\mu g/m^3$; the "TSP" column is the particulate level in $\mu g/m^3$ at that percentile, and the last two columns count the counties within $\pm 5\,\mu g/m^3$ of it in each period, so a row supported by few counties is visible as such. The bootstrap columns summarise 99 replications that resample counties with replacement and re-run the regularisation in each, holding $\hat{\theta}$ fixed; the envelope is the 5th and 95th percentile of the replications at that level. *The envelope illustrates variability and is not a confidence set. No coverage is claimed for it and none is computed*, which follows Centorrino, Fève, and Florens (2025) for the same estimator on the same equation. Panel B records the design and what it returned: $\hat{m}_\varphi$ is the realised Landweber–Fridman stopping index and $\bar{m} = 60$ the cap, which is set from the instrument's cardinality at 15, 30 and 60 for $\mathsf{L} = 2$, 3 and 4. *Every solve reported here terminates at that cap*, because the discrepancy rule runs at $\tau_{\mathrm{DP}} = 0$ and the patience equals the cap, so the rows recording $\hat{m}_\varphi$ describe the budget and not a selected index. The two rows counting replications at $\hat{m}_\varphi = 1$, where a replication would have returned its initial condition unchanged, are zero for that reason and not for a favourable one, in the subsample comparison as in the resample. "S.d. of the anchor across resamples" is the standard deviation across replications of the weighted pooled median each curve is re-anchored at, in $\mu g/m^3$ and not in log points. *The cap is a calibration constant and not the $\bar{m}_n$ of* (36), which Panel A of Table S5 measures at 3.95×10^{10} to 8.76×10^{11}, a factor of 7.32×10^8 to 2.78×10^9 above the $\hat{\delta}_n^{-2}$ it is meant to dominate; the two are different objects and only the first bounds the runs reported here. Panel B's last block re-estimates θ inside each of 25 draws rather than holding $\hat{\theta}$ fixed, and the resulting band on those same draws is narrower, 0.0388 against 0.0412; the block's last row counts the θ-step fold solves that reach the cap, at the median over the five folds. The multiplier arm's counterpart to Panel A, which the text compares this envelope with at the ends of the support, is Table S45, and Table S44 carries the two schemes' summary statistics side by side. Panel C recomputes the whole normalised curve in one other strictly increasing coordinate of X, $\mu g/m^3$, and returns nothing: $\hat{\pi}$ is not bounded away from zero there, so the estimator is infeasible in that coordinate, and a min–max rescaling shares the row as an exact affine image of it. *No hypothesis of this article is violated by that*: what fails is the sample reference density the engine builds, and the conditions here are stated on the population π. The baseline row is filled in pointwise over the 60-point grid: "Max $\lvert\Delta\rvert$" and "RMS Δ" are in log points, the two "/SD" columns divide them by the bootstrap standard deviation at the same point, "Steps up" counts the grid steps on which the curve rises, and the last column is the band width on 25 paired draws. A strictly increasing reparameterisation leaves $\varphi_\dagger \circ T^{-1}$ structural and $\theta_\dagger$ unchanged, so what fails is the estimator at this sample and not the model; the magnitudes reported in Panel A and drawn in Figure 1 of the article are those of the pooled-rank coordinate. Source: `scripts/application/app_tsp_05_phi_band.R`; table `paper/tables/app_tsp_phi.tex`.

Table S44: Five resampling schemes for the variability band, and what each one perturbs

	Full sample	Pairs (C0)	Multiplier (M)	Subsample, no replacement	
	point estimate	with replacement	Exp(1) weights	$n(1-e^{-1})$	$n/2$
Panel A: what each scheme perturbs, and the stopping index it produces					
Counties entering a replication	983	983	983	621	492
Distinct counties, i.e. distinct X values	983	621 (exp.)	983	621	492
Kish effective sample size	983	492	492	621	492
Replications drawn	–	99	99	24	24
Replications with a finite curve	–	99	99	24	24
Failures: error / non-finite	–	0 / 0	0 / 0	0 / 0	0 / 0
$\hat{m}_\varphi$: median	60	60	60	60	60
$\hat{m}_\varphi$: [min, max]	–	[60, 60]	[60, 60]	[60, 60]	[60, 60]
Retained with $\hat{m}_\varphi = 1$	–	0	0	0	0
Retained with $\hat{m}_\varphi = \bar{m}$	–	99	99	24	24
Stopped before the cap, $\hat{m}_\varphi < \bar{m}$ (%)	–	0.0	0.0	0.0	0.0
Panel B: the band					
Band width, median over the grid	–	0.046	0.043	–	–
Band width, first 25 replications	–	0.041	0.036	–	–
Deciles where 5–95 excludes zero (of 11)	–	1	1	–	–
Range of the curve	0.035	0.038	0.039	–	–
Range as a share of the point estimate (%)	100.0	108.1	111.0	–	–
Panel C: the multiplier design's own side-arm					
Coordinate map rebuilt: replications	–	–	25	–	–
Coordinate map rebuilt: at $\hat{m}_\varphi = 1$ / at $\bar{m}$	–	–	0 / 25	–	–
Coordinate map rebuilt: band width, 25 paired draws	–	–	0.035 vs. 0.036	–	–
Weight s.d. (population value 1): median	–	–	1.000	–	–

Notes: All five schemes estimate the same object on the same 983-county sample, hold θ at $\hat{\theta}$, re-select the stopping index in every replication, and re-anchor each curve at its own pooled median. They differ only in how the sample is perturbed. The *Kish effective sample size* is $(\sum_i w_i)^2 / \sum_i w_i^2$; multinomial and Exp(1) weights both give $n/2$, so the pairs and multiplier columns are matched on information and differ only in duplication. The two subsample columns draw distinct counties without replacement at the sizes named, so they vary information while holding duplication at zero. Three rows record the stopping index. *Retained with* $\hat{m}_\varphi = 1$ counts the replications that returned the initial condition, where the Landweber step never fired; *Retained with* $\hat{m}_\varphi = \bar{m}$ counts those that ran to the cap; and *Stopped before the cap* is the complementary share. Under this configuration the first is zero and the second is every replication in all four columns, for the reason Table S43 gives, so those rows describe the budget and not a selection. The bootstrap columns draw 99 replications and the subsample columns 24, so a difference of one replication is worth 1.0 and 4.2 percentage points respectively. *None of these bands is a confidence set and no coverage is computed for any of them.* Source: `scripts/application/app_tsp_06_phi_mult.R` and `app_tsp_05_phi_band.R`; table `paper/tables/app_tsp_phi_boot_compare.tex`.

Table S45: The variability band under multiplier weights, and the design that produced it

					Bootstrap envelope		Counties within $\pm 5\,\mu g/m^3$	
Pctile of X	TSP	$\hat{\varphi}(x) - \hat{\varphi}(x_{0.5})$	Bootstrap median	Bootstrap s.d.	5th	95th	1970	1980
Panel A: $\hat{\varphi}$ at deciles of the pooled X distribution								
5	31.4	0.0175	0.0148	0.0198	-0.0183	0.0474	61	67
10	37.5	-0.0012	-0.0044	0.0176	-0.0316	0.0221	94	124
20	44.9	-0.0151	-0.0187	0.0151	-0.0417	0.0026	138	206
30	49.6	-0.0169	-0.0201	0.0127	-0.0412	-0.0016	170	272
40	53.6	-0.0075	-0.0086	0.0081	-0.0229	0.0023	184	280
50 (anchor)	58.0	0.0000	0.0000	0.0017	-0.0024	0.0031	187	253
60	62.8	0.0083	0.0092	0.0086	-0.0025	0.0231	178	210
70	68.1	0.0150	0.0157	0.0142	-0.0026	0.0418	168	154
80	75.4	0.0089	0.0119	0.0179	-0.0161	0.0416	135	96
90	87.7	0.0005	0.0035	0.0217	-0.0343	0.0379	76	28
95	100.2	0.0090	0.0103	0.0241	-0.0313	0.0508	46	9
Panel B: multiplier design and convergence								
Replications drawn	99							
Replications with a finite curve	99							
Failures: error / non-finite	0 / 0							
$\hat{m}_\varphi$: median [min, max]	60 [60, 60]							
Retained with $\hat{m}_\varphi = 1$	0							
Retained with $\hat{m}_\varphi = \bar{m}$	99							
S.d. of the anchor across replications	0.71							
Weight s.d. (population value 1): median	1.000							
Coordinate map rebuilt ($B = 25$): replications	25, 25 finite							
Coordinate map rebuilt: band width on the same draws	0.0352 vs. 0.0356							

Notes: The multiplier arm of Table S44, at the eleven percentiles Table S43 reports for the pairs arm. Chay–Greenstone county panel, $n = 983$, headline control set $d_Z = 5$, instrument entered as its four trigger cells, $\hat{\theta}$ held at its full-sample value. Panel A's columns are those of Table S43 and are formed the same way: $\hat{\varphi}(x) - \hat{\varphi}(x_{0.5})$ at percentiles of the pooled distribution of $\{X_1, X_2\}$, re-anchored in each replication at that replication's own weighted pooled median, with the envelope the 5th and 95th percentile of the replications at that level. *The envelope illustrates variability and is not a confidence set. No coverage is claimed for it and none is computed*, exactly as for the pairs envelope. What differs between the two is the scheme: one weight per county, w_i i.i.d. Exp(1) normalised to mean one and shared by both of that county's periods, against a resample of counties with replacement. Exp(1) matches the first two moments of the multinomial weights the resample induces, so the two perturb the sample equally hard and differ in what they perturb, mass rather than support; all 983 counties and all 983 distinct values of X are present in every replication here. The coordinate map is the one further departure. It is held at its full-sample value, which a with-replacement resample cannot do, so the second-stage

grid is identical across replications and differences between curves are differences in the estimate rather than in where it was evaluated. What that conditioning omits is measured rather than asserted: the last two rows of Panel B rebuild the map from the weighted empirical CDF on the same first 25 draws, and the band on those draws is 0.0352 with the map rebuilt against 0.0356 with it held, so holding it overstates the width by 0.0004 there rather than understating it. *No verdict is reported, and none is available.* The rule fixed before the run judged this scheme against the pairs scheme on the stopping-index distribution alone, on the share of replications at $\hat{m}_{\varphi} = 1$, the share at the cap, and the median index, because a band can be narrow either because the procedure is stable or because it collapsed to its initial condition and only that distribution separates the two. At $\tau_{\mathrm{DP}} = 0$ the discrepancy branch never fires and the patience equals the cap, so every solve that returns returns $\hat{m}_{\varphi} = \bar{m} = 60$; the index is a constant, all three criteria are then satisfied by construction for any data in either arm, and they discriminate nothing. The script records no verdict rather than one read off an unfailable test, and the three rows of Panel B that report the stopping index carry no information under this configuration, which is what their $[\,60, 60\,]$ range says. *Neither envelope is uniformly the narrower.* Against the pairs envelope of Table S43 this one is wider at the 5th percentile, at the 70th and at the 80th, 90th and 95th, and narrower from the 10th to the 30th and at the 60th; at the 40th and the 50th it is wider by less than 0.0006, taken on the unrounded envelope bounds and not on the four digits printed, which give exactly 0.0006 at the 50th. It is therefore the wider of the two at seven of the eleven levels and the narrower at four, a split that is the same on the printed digits and on the unrounded bounds, so a single summary of the two widths does not carry to the level at which it is read, and Section 6 of the article reads that pattern off this table. Table S44 reports each band's median width over the 60-point grid and the two schemes' other outcomes side by side. The remaining rows of Panel B record the draw: 99 replications, all 99 returning a finite curve, no errors and no non-finite curves; "S.d. of the anchor across replications" is the standard deviation across replications of the weighted pooled median each curve is re-anchored at, in μg/m^3; and "Weight s.d." checks the draw itself against the value Exp(1) implies. Source: `scripts/application/app_tsp_06_phi_mult.R`; table `paper/tables/app_tsp_phi_mult.tex`.

Table S46: The operator's squared norm in three coordinates, and what the default Landweber step does in each

Coordinate	$\|A\|^2$	Bound $2/\|A\|^2$	Step 0.5÷ bound	Iterations run	max $\|\Delta\hat{\varphi}\|$
Pooled empirical CDF (C2, baseline)	2.57	0.78	0.641	200	0.104
μg/m^3	6.4×10^4	3.1×10^{-5}	1.6×10^4	101	0.646
Min–max rescaling to $[0, 1]$	0.347	5.77	0.0867	200	0.402

Notes: $\|A\|^2$ is the squared norm of the composite map $q \mapsto \mathfrak{w} \cdot \{h_q(X_2) - h_q(X_1)\}$, $h'_q = q$, obtained by the same 30-step power iteration the estimator itself runs to set its relaxation; the reported values are stable to six significant figures across four starting seeds. It is the *squared* norm, which is the quantity (35) bounds: the iteration converges for a step below $2/\|A\|^2$, the third column. The fourth column divides the default step 0.5 by that bound, so a value above one is a step the convergence condition forbids, which is the μg/m^3 row and no other. The last column reports what the default step produces in each coordinate, namely the largest fitted $|\Delta\hat{\varphi}|$ over the sample, in log points of property value. *Only the $\mu g/m^3$ coordinate violates the bound*, and it is the only one whose fitted curve leaves an economically interpretable range. The "Iterations run" column is printed so that the last column can be read as a comparison: all three coordinates are still iterating when the count runs out, so what separates them is the magnitude and not the stopping. That count is a diagnostic choice and not the application's cap, which C4 sets from the instrument's cardinality at 15, 30 and 60. Source: `scripts/application/app_tsp_07_opnorm.R`; table `paper/tables/app_tsp_opnorm.tex`.

Table S47: Truncated moments of the shipped boundary kernel, at every order C3(ii) asks for

	Worst over design points			At $\underline{x} + kh_x$	
	Interior	Within $3h_x$	At the endpoint	$k = 1$	$k = 3$
Panel A: C3's boundary family at $p_K = 8$, the shipped kernel					
$j = 0$	1.3×10^{-15}	2.0×10^{-13}	0.0000	3.6×10^{-15}	6.7×10^{-16}
$j = 1$	8.9×10^{-16}	4.5×10^{-13}	2.3×10^{-13}	1.8×10^{-15}	0.0000
$j = 2$	3.6×10^{-15}	1.2×10^{-12}	4.5×10^{-13}	3.2×10^{-14}	1.8×10^{-15}
$j = 3$	5.3×10^{-15}	2.0×10^{-12}	2.3×10^{-13}	5.7×10^{-14}	0.0000
$j = 4$	3.6×10^{-14}	6.4×10^{-12}	5.0×10^{-12}	2.8×10^{-14}	1.8×10^{-14}
$j = 5$	4.3×10^{-14}	1.8×10^{-11}	5.5×10^{-12}	2.3×10^{-13}	1.4×10^{-14}
$j = 6$	2.5×10^{-13}	8.7×10^{-11}	5.8×10^{-11}	4.5×10^{-13}	2.8×10^{-14}
$j = 7$	4.5×10^{-13}	2.6×10^{-10}	2.3×10^{-10}	3.6×10^{-12}	0.0000
Panel B: the renormalised kernel it replaces ($j = 0$ enforced, order 2)					
$j = 0$	0.0000	0.0000	0.0000	0.0000	0.0000
$j = 1$	0.0044	0.7979	0.7979	0.2876	0.0044
$j = 2$	1.0000	1.0000	1.0000	0.7124	0.9867
$j = 3$	0.0487	1.5958	1.5958	0.8628	0.0488
$j = 4$	3.0000	3.0000	3.0000	1.8496	2.8402
$j = 5$	0.5534	6.3831	6.3831	3.7388	0.5547
$j = 6$	15.0000	15.0000	15.0000	8.9604	13.1228
$j = 7$	6.5517	38.2985	38.2985	22.7204	6.5636

Notes: $\int_I \varsigma^j \Theta_I(\varsigma)\, d\varsigma - \delta_{j0}$, the departure from (15), for $j = 0, \ldots, p_K - 1$ at $p_K = 8$, over all 1,966 design points of the 983-county panel with both periods pooled, on the support of the pooled-rank coordinate. "Within $3h_x$" is the boundary layer, 884 of those points; "at the endpoint" is the two ends of the support; the last two columns read the same quantity at the fixed offsets $\underline{x} + kh_x$. *C3(ii) asks the truncated moments to vanish at the boundary as well as in the interior, and this table is that check rather than an assertion of it.* Panel A is the family of Lemma 66 at $p_K = 8$, which is the kernel the estimates in this paper are computed with; its worst departure anywhere, boundary included, is 2.6×10^{-10}, at $j = 7$. Panel B is the renormalisation it replaces, which enforced $j = 0$ and nothing above it, and that one failed in two directions at once: at $j = 7$ it departs by 38.30 both in the layer and at the endpoint, and $j = 2$ never comes within 0.7055 of vanishing at *any* design point, interior ones included. The second failure is the one an interior-only check would have missed. Source: `scripts/application/app_tsp_09_boundary.R`; table `paper/tables/app_tsp_bdrymom.tex`; run object `quality_reports/application/app_tsp_boundary.rds`.

Table S48: The boundary layer, the x-smoothing bias, and whether (P-g) holds on the φ-step grid

	Through 2026-08-15 (D11)			From 2026-08-16 (D11a/b)		
Panel A: is the boundary layer populated? Pooled design $\{X_{1i}\} \cup \{X_{2i}\}$, DGP-A						
n	h_x	$\lvert I \rvert$	Within $3h_x$ (%)	h_x	$\lvert I \rvert$	Within $3h_x$ (%)
125	0.3807	2.63	100.00	0.1450	6.90	86.52
354	0.3092	3.23	100.00	0.0956	10.46	56.07
500	0.2885	3.47	100.00	0.0833	12.01	48.55
1,000	0.2512	3.98	100.00	0.0631	15.85	36.36
2,000	0.2187	4.57	100.00	0.0478	20.91	27.25
2,828	0.2040	4.90	100.00	0.0416	24.02	23.59
8,000	0.1657	6.03	99.41	0.0275	36.41	15.33
22,627	0.1346	7.43	80.11	0.0181	55.19	9.97
64,000	0.1093	9.15	64.51	0.0120	83.65	6.49
181,019	0.0888	11.26	51.93	0.0079	126.79	4.22
Panel B: $\int_I \varsigma^j \Theta_I - \delta_{j0}$, worst over design points, each configuration on its own quadrature						
	D11, 241 trap.	C3, 961 fixed	C3, $N \propto n^{0.40}$	N	Nh_x	
n=500, j=0	2.22×10^{-16}	3.30×10^{-7}	3.30×10^{-7}	961	80.0	
n=500, j=1	0.2530	3.19×10^{-7}	3.19×10^{-7}	961	80.0	
n=500, j=2	0.0395	1.22×10^{-7}	1.22×10^{-7}	961	80.0	
n=500, j=3	0.3297	7.49×10^{-8}	7.49×10^{-8}	961	80.0	
n=500, j=4	0.4763	1.22×10^{-7}	1.22×10^{-7}	961	80.0	
n=500, j=5	2.4416	2.02×10^{-7}	2.02×10^{-7}	961	80.0	
n=500, j=6	9.6273	3.47×10^{-7}	3.47×10^{-7}	961	80.0	
n=500, j=7	33.4443	6.17×10^{-7}	6.17×10^{-7}	961	80.0	
n=2,000, j=0	2.22×10^{-16}	3.03×10^{-6}	3.28×10^{-7}	1,675	80.1	
n=2,000, j=1	0.2509	2.93×10^{-6}	3.17×10^{-7}	1,675	80.1	
n=2,000, j=2	0.1613	1.12×10^{-6}	1.21×10^{-7}	1,675	80.1	
n=2,000, j=3	0.1098	6.89×10^{-7}	7.45×10^{-8}	1,675	80.1	
n=2,000, j=4	1.1185	1.12×10^{-6}	1.21×10^{-7}	1,675	80.1	
n=2,000, j=5	1.4069	1.86×10^{-6}	2.01×10^{-7}	1,675	80.1	
n=2,000, j=6	5.5053	3.18×10^{-6}	3.44×10^{-7}	1,675	80.1	
n=2,000, j=7	19.1120	5.68×10^{-6}	6.14×10^{-7}	1,675	80.1	
n=8,000, j=0	2.22×10^{-16}	2.79×10^{-5}	3.28×10^{-7}	2,915	80.1	
n=8,000, j=1	0.2493	2.69×10^{-5}	3.17×10^{-7}	2,915	80.1	
n=8,000, j=2	0.1138	1.02×10^{-5}	1.21×10^{-7}	2,915	80.1	
n=8,000, j=3	0.2804	6.33×10^{-6}	7.45×10^{-8}	2,915	80.1	
n=8,000, j=4	0.2956	1.03×10^{-5}	1.21×10^{-7}	2,915	80.1	
n=8,000, j=5	2.5047	1.70×10^{-5}	2.01×10^{-7}	2,915	80.1	
n=8,000, j=6	3.6977	2.92×10^{-5}	3.45×10^{-7}	2,915	80.1	
n=8,000, j=7	23.2493	5.22×10^{-5}	6.14×10^{-7}	2,915	80.1	

Panel C: (172) realised, and (P-g) measured rather than assumed. The middle column is the superseded fixed-N pass

n	h_x	$\sup/(h_u\hat{\delta}_n)$ (SE)	h_x	$\sup_{\mathcal{X}}\lvert\varphi_{h_x}-\varphi_\dagger\rvert$	fixed N: $\sup/(h_u\hat{\delta}_n)$	$N\propto n^{0.40}$: same
125	0.3807	3.13 (0.048)	0.1450	7.08×10^{-5}	3.00×10^{-4} (0.0000)	2.95×10^{-4} (0.0000)
354	0.3092	3.73 (0.020)	0.0956	1.51×10^{-5}	9.17×10^{-5} (0.0000)	9.23×10^{-5} (0.0000)
500	0.2885	3.87 (0.027)	0.0833	7.36×10^{-6}	5.06×10^{-5} (0.0000)	5.05×10^{-5} (0.0000)
1,000	0.2512	4.14 (0.023)	0.0631	1.26×10^{-6}	1.11×10^{-5} (0.0000)	1.09×10^{-5} (0.0000)
2,000	0.2187	4.45 (0.014)	0.0478	1.63×10^{-7}	9.99×10^{-6} (0.0000)	1.84×10^{-6} (0.0000)
2,828	0.2040	4.59 (0.016)	0.0416	8.45×10^{-8}	1.71×10^{-5} (0.0000)	1.08×10^{-6} (0.0000)
8,000	0.1657	5.19 (0.007)	0.0275	5.47×10^{-8}	8.62×10^{-5} (0.0000)	1.02×10^{-6} (0.0000)
22,627	0.1346	5.97 (0.007)	0.0181	3.61×10^{-8}	4.37×10^{-4} (0.0000)	9.78×10^{-7} (0.0000)
64,000	0.1093	6.93 (0.004)	0.0120	2.39×10^{-8}	0.0022 (0.0000)	9.35×10^{-7} (0.0000)
181,019	0.0888	8.09 (0.010)	0.0079	1.57×10^{-8}	0.0111 (0.0000)	9.00×10^{-7} (0.0000)

Panel D: the two exponents, by configuration (D11 / fixed N / grown N). (P-g) holds only if the first row is ≤ 0

$d\log(\text{ratio})/d\log n$, full grid	0.125 (0.001 MC, 0.003 reg.)	0.647 (0.001 MC, 0.258 reg.)	−0.813 (0.001 MC, 0.173 reg.)
restricted to $n\in\{500,2000,8000\}$	0.107 (0.003 MC, 0.003 reg.)	0.192 (0.002 MC, 0.787 reg.)	−1.407 (0.002 MC, 0.567 reg.)
minimum of the ratio at $n=$ (interior ⇒ a U remains)	125 (edge)	2,000 (interior)	181,019 (edge)
$d\log\sup\lvert\varphi_{h_x}-\varphi_\dagger\rvert/d\log h_x$, full sweep	1.085	5.690	
over $h_x\leq 0.09$, and the tightest local slope	1.013 / 1.003	7.542 / 7.984	

Panel E: what the repair costs (36), leg by leg, from the constants

n	Total	Constants	Bandwidth	Family	h_u coupling	Reverse-order cost
500	3.458e+05	8.974e+04	3.466	1.00	1.112	1.00
2,000	4.317e+05	8.974e+04	4.573	1.00	1.052	1.00
8,000	5.602e+05	8.974e+04	6.034	1.00	1.034	1.00

Notes: The two hypotheses the φ-step arm makes about its x-smoother, measured on DGP-A's pooled design $\{X_{1i}\}\cup\{X_{2i}\}$ at ten sample sizes from 125 to 181,019. The table sets the configuration shipped through 2026-08-15, which smoothed at $h_x=n^{-0.20}$ with a kernel renormalised to unit mass at each design point, beside the current one, at $h_x=n^{-0.40}$ with the family of Lemma 66 at $p_K=8$; "D11" in the panel heads is the former and "C3" the latter. The spanning header dates the two blocks of Panel A, below which each panel names its own configurations, and no header repeats on a continuation page. Panel A counts the design points within $3h_x$ of an endpoint, the layer C3(ii) governs. The old bandwidth put every pooled point inside it up to $n=2{,}828$ and 51.93 per cent of them at $n=181{,}019$, so the interior kernel was never the operative one anywhere on the arm's grid; the new one holds 48.55 per cent at $n=500$, against the 884 of 1,966 pooled points, 44.96 per cent, that Section 6 of the article realises on the county panel (Table S47, notes). Panel B is the departure from (15) at each order j, worst over design points, each configuration on its own quadrature; the third column grows the node count as $N\propto n^{0.40}$ and is the rule the current arm uses. Panel C reports (172) realised and the ratio (P-g) requires to be $O(1)$, $\sup_{\mathcal{X}}\lvert\varphi_{h_x}-\varphi_\dagger\rvert/(h_u\hat{\delta}_n)$, at 3 to 25 draws per n, and Panel D fits that ratio's slope in $\log n$ against two standard errors; the slope is compatible with $O(1)$ only if it is at or below zero by both. *Section 5 of the article claims the arm now runs on-hypothesis; this is the evidence, and the claim is about the slope and not the level.* Under the old configuration the ratio climbs monotonically from 3.13 to 8.09 at a slope of +0.125 whose regression standard error is 0.003, so the climb is not a Monte Carlo artifact and no re-reading of the constants removes it. Under the current one the slope is −0.813 at a regression standard error of 0.173 against a Monte Carlo one of 0.001, which is the diagnostic saying one line is the wrong model for a curve that falls and then flattens; the article reports the plateau, 9.6×10^{-7} averaged over the last four sample sizes, rather than the fitted rate.

Panel E decomposes the move in the cap (36) into the legs of $\bar{m}_n\propto c_C\mathsf{c}_2^4h_x^{-1}h_u^{-2}$ at $n=500$, 2,000 and 8,000. *It is an independent recomputation of Panel C′ of Table S5, not a second assertion of it, and the two reach the h_u leg by different routes.* Panel C′ takes the total from the φ-step arm's realised $\bar{m}_n$ over 300 replications and recovers the h_u leg as the residual against the three derived legs; Panel E measures h_u on this arm's own draws, 25 at each of the three sizes, and multiplies the four legs out. The derived legs agree exactly, the constants at 8.974×10^4, the bandwidth at $n^{0.20}$ and the family at 1.00; the h_u leg, the only quantity the two routes compute differently, agrees to

within 0.68 per cent, and the totals inherit that and agree to within 0.71 per cent, 3.458×10^5 here against the 344,503 printed there at $n = 500$. Two routes agreeing to that tolerance check the arithmetic and the realised $\bar{m}_n$ in a way neither table does alone, which is why both are printed. Monte Carlo standard errors in parentheses. Source: `scripts/simulations/sim_thm_rootn_bdrylayer.R`; table `paper/tables/sim/sim_thm_rootn_bdrylayer.tex`; run object `quality_reports/sim/sim_thm_rootn_bdrylayer.rds`.

Table S49: The frontier at each rung of the control-set ladder

d_Z	r required to clear the frontier			$\kappa\alpha$	δ_n	$\sqrt{n}\,\delta_n^2$	n for budget < 1	Verdict
	$\mu = 1$	$\mu = 2$	$\mu = 3$					
0	0.00	0.00	0.00	0.400	0.644	13.0	2.1×10^{11}	inside
5	7.50	4.17	3.50	0.206	1.129	40.0	never	outside
10	15.00	8.33	7.00	0.139	1.796	101.1	never	outside
22	33.00	18.33	15.40	0.078	2.736	234.6	never	outside

Notes: Chay–Greenstone county panel, $n = 983$, first differenced. Rows are the four rungs of the control-set ladder of Table S38, indexed by the number d_Z of differenced controls entered: an intercept alone, five core hedonic demand shifters, those plus five demographic variables, and all twenty-two usable controls. The first three columns are the frontier read in the primitives of the article's (7), $r > d_Z(2\mu + 1)/\{2(2\mu - 1)\}$ with r the Hölder smoothness of the propensity score and μ the source index, evaluated at three values of μ; they are properties of the condition and of d_Z alone and involve no estimate. The remaining columns are evaluated at $(\mu, r) = (2, \frac{8}{3})$, which is the frontier-clearing configuration of the article's Section 5: $\kappa\alpha$ with $\kappa = 2\mu/(2\mu + 1)$ and $\alpha = r/(2r + d_Z)$, against the $\frac{1}{4}$ the frontier requires; δ_n at this n; the leading Step-4 product $\sqrt{n}\,\delta_n^2$; and the sample size at which that product would fall below one. The last two of those are the two rows Panel D of Table S40 repeats. *"Verdict" reports the frontier and nothing else.* It is "inside" at the intercept-only rung because any $r > 0$ clears a threshold of zero there, and that rung is nevertheless unavailable: A4(ii) requires $d_Z \geq 1$, and at $d_Z = 0$ the propensity exponent is $\alpha = \frac{1}{2}$ exactly, at which the sieve-entropy condition admits no integer sieve dimension and Theorem 29 is vacuous there. So the frontier is met at no control set the assumption block also admits, which is the statement Sections 6.4 and 7 of the article read off this table. At the three rungs that do satisfy A4(ii) the budget never falls below one at any n, because $\frac{1}{2} - 2\kappa\alpha > 0$ there and the product grows. Every entry is a deterministic function of the design and of n; no Monte Carlo standard error attaches to any of them, and nothing here is estimated from the outcome. Source: `scripts/application/app_tsp_01_diagnostics.R`; table `paper/tables/app_tsp_feasibility.tex`; run object `quality_reports/application/app_tsp_diagnostics.rds`.

Assumptions

The instrument, its contrasts, and the range reference measure. The instrument W takes finitely many values, $W \in \{0, 1, \ldots, \mathsf{L}-1\}$, with $\mathsf{L} \geq 2$ an integer that is *fixed and known to the analyst. Sans-serif* L *is the number of instrument cells and nothing else*: italic L remains the debiasing weight of Definition 19, L_x and L_u remain the sieve dimensions, and $L^2(\cdot)$ remain the function spaces. Write, for $l \in \{0, 1, \ldots, \mathsf{L}-1\}$, $\varrho_l(z) := P(W = l \mid \Delta Z = z)$, and collect the *non-base* cells into

$$\varrho \;:=\; \big(\varrho_1, \ldots, \varrho_{\mathsf{L}-1}\big)', \qquad \widetilde{W}_l \;:=\; \mathbb{1}\{W = l\} - \varrho_l(\Delta Z), \qquad \widetilde{W} \;:=\; \big(\widetilde{W}_1, \ldots, \widetilde{W}_{\mathsf{L}-1}\big)' \in \mathbb{R}^{\mathsf{L}-1}, \tag{1}$$

so that $\varrho_0 = 1 - \mathbf{1}'\varrho$ with $\mathbf{1}$ the vector of ones in $\mathbb{R}^{\mathsf{L}-1}$. Cell 0 is the base cell. The *conditional contrast covariance* is

$$\Omega(z) \;:=\; \mathrm{Var}\left(\widetilde{W} \mid \Delta Z = z\right) \;=\; \mathrm{diag}\left(\varrho(z)\right) - \varrho(z)\varrho(z)', \qquad z \in \mathbb{R}^p. \tag{2}$$

At $\mathsf{L} = 2$ these are the scalar objects: $\varrho = \varrho_1 = P(W = 1 \mid \Delta Z)$ is the propensity, $\widetilde{W} = W - \varrho(\Delta Z)$, and $\Omega = \varrho(1 - \varrho)$.

The contrasts span, and nothing in the W direction is approximated: every measurable $a(W, \Delta Z)$ with $E[|a| \mid \Delta Z] < \infty$ and $E[a \mid \Delta Z] = 0$ almost surely is $a = c(\Delta Z)'\widetilde{W}$ for a ΔZ-measurable c, uniquely so at a $\Delta Z = z$ at which every cell is non-null, where the space has dimension exactly $\mathsf{L} - 1$ and (1) is a basis for it. *The spanning statement needs no overlap condition*; at a z with null cells the dimension is one less than the number of non-null cells. Consequently the Bierens device is needed only in the ΔZ direction, where it always was, and B1 is unchanged apart from B1(iv.d). The derivation, the exact statement at a degenerate z, and the fact that neither depends on which cell is taken as the base are in the Discussion paragraph *A1*.

Type declaration, stated once and binding on every statement in this corpus. The debiasing weight and the objects built from it carry one column per contrast: L, $\ell = \partial_u L$, the anchor b and their starred and hatted variants are $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued, one *row* per coordinate of θ and one *column* per contrast, and the range-space representer ℓ^{Π} carries the contrast index as a third argument, $\ell^{\Pi}(u,t,l)$. *Norms carried by these objects are Frobenius*; matrix norms elsewhere in this corpus are spectral unless the site says otherwise, and the two are named at every site at which both occur. The moment function is $\psi = \Delta Z\,\Delta U + L(\Delta U, \Delta Z)\,\widetilde{W}$ of (38), the contrast standing to the *right* of the $p \times (\mathsf{L}-1)$ weight. *At* $\mathsf{L} = 2$ *each of these is the printed object*: a $p \times 1$ matrix is a vector in $\mathbb{R}^p$, the Frobenius norm is the Euclidean norm, $\ell^{\Pi}(u,t,1) = \ell^{\Pi}(u,t)$, and $L\widetilde{W} = \widetilde{W}L$ because $\widetilde{W}$ is then a scalar.

The estimator carries the **raw** contrast (1), on both sides: the operator equation is written at $\widetilde{W}$ and so is ψ, while Ω enters as the conditional covariance (2) and as the prefactor (62), and *never as a weight*. The normalised contrast $\widetilde{W}^{\Omega} := \Omega(\Delta Z)^{-1}\widetilde{W}$, what it makes explicit, and the reason the paper does **not** estimate at it are Remark 35.

The range reference measure gains a third factor: $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ with Π_{w} a probability measure on $\{1,\dots,\mathsf{L}-1\}$, masses $\varpi_{\mathsf{w}}(l)$, required strictly positive by B1(iv.d). *At* $\mathsf{L} = 2$, Π_{w} *is the point mass at* 1 *and* $\Pi = \Pi_u \otimes \Pi_t$, under the canonical identification of $\mathbb{R}\times\mathcal{T}\times\{1\}$ with $\mathbb{R}\times\mathcal{T}$; c_{Π} of B1(iv.c) is unchanged at every L, Π_{w} being a probability measure and contributing the factor 1.

How a citation of this block is read. *On a result's hypothesis line* — inside a "Suppose … hold" — a citation is a **carrier**: naming a tag carries every condition under it, and naming a clause carries that clause and no other. *Inside a definition* a citation is a **pointer** when it merely locates the object it names, as Definition 5's "$\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ as in Assumption B1(iv)" does, and imposes nothing; it is **not** a pointer when it is constitutive, as A2b's "admits a version satisfying A2′(i.a)–(i.b)" and Definition 2's use of the constant A4(i) declares are. *And several conditions below are written in two halves*: a **declarative**

half, which introduces a symbol, a type or a carrier set and is *standing* — in force at every statement in this corpus, named or not, since without it the statement has no objects to be about, so it is never listed as a hypothesis — and a **clause**, which restricts, and which is carried on the hypothesis line of every result whose proof reads it. A0, A4(i), A8, B1(iii) and B1(iv) are split that way. Where the split falls, why it replaced an earlier test, and the register of the three constants the analyst chooses are in the Discussion of the assumptions, under the heading *The three registers*.

Assumption (A0: Sampling)**.** $\{(Y_{i1}, Y_{i2}, X_{i1}, X_{i2}, Z_{i1}, Z_{i2}, W_i)\}_{i=1}^{n}$ are independent and identically distributed draws from a distribution P on $\mathbb{R}^2 \times \mathcal{X}^2 \times \mathbb{R}^{2p} \times \{0, 1, \ldots, \mathsf{L}-1\}$, where $\mathcal{X} \subseteq \mathbb{R}$ is the carrier of the regressor, $\underline{x} := \inf \mathcal{X}$ and $\bar{x} := \sup \mathcal{X}$ — *that much is A0's* ***declaration*** — and $\mathsf{L} \geq 2$ is a finite integer that does not depend on n and is known to the analyst. **The condition on** $\mathcal{X}$ is that it is a *compact interval*, so that $\mathcal{X} = [\underline{x}, \bar{x}]$ with $\underline{x} < \bar{x}$; *that is a clause and is carried by the results that read it.*

Assumption (A1: Conditional exogeneity)**.** $\Delta U \perp\!\!\!\perp W \mid \Delta Z$ and $E[\Delta U \mid \Delta Z] = 0$.

Note on A1. A1 is a statement about σ-fields and carries no reference to the cardinality of W's support. Its representation is (1), and that representation is exact: restriction (A) of Definition 7 written at the $\mathsf{L}-1$ contrasts is *equivalent* to $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, at every ΔZ, at every L, and under **no** overlap condition — *A4(i) is not a hypothesis of that equivalence and must not be listed as one.* The equivalence, the base-cell invariance, and the sense in which extra cells are *surplus* rather than necessary are in the Discussion of the assumptions, under the heading *A1 (conditional exogeneity).*

Assumption (A1b: Identification)**.**

(i) *Normalisation.* $\varphi_\dagger \in \mathcal{V}$, that is, $\int_{\mathcal{X}} \varphi_\dagger \, d\pi = 0$.

(ii) *Measure-separability.* Every function of X_1 that is almost surely equal to a function of X_2 is almost surely equal to a constant.

(iii) *Global identification of* $\varphi_\dagger$. $\mathcal{V}_{CI} = \{\varphi_\dagger\}$, where $\mathcal{V}_{CI} := \{\varphi \in \mathcal{V} : \ \Delta Y - \Delta\varphi \perp\!\!\!\perp W \mid \Delta Z\}$.

(iv) *Identification of* $\theta_\dagger$. $E[\Delta Z\, \Delta Z']$ is nonsingular, and $E\|\Delta Z\|^2 < \infty$, $E[\Delta U^2] < \infty$, $E\|\Delta Z\, \Delta U\|^2 < \infty$.

Assumption (A1c: First-order local identification)**.** The linearisation is injective: $\ker \mathcal{A}_{\text{cond}} = \{0\}$, where $\mathcal{A}_{\text{cond}} : \mathcal{V} \to L^2(\Pi)$ is the Fréchet derivative of Proposition 8 evaluated at $(\varphi_\dagger, \theta_\dagger)$.

Assumption (B1: Bierens family and range space)**.**

(i) *Transform.* $\Phi : \mathbb{R}^p \to \mathbb{R}^p$ is Borel, one-to-one, and bounded: $c_\Phi := \sup_z \|\Phi(z)\|_\infty < \infty$. The maintained choice is Φ componentwise arctan, for which $c_\Phi = \pi/2$. *The value of* c_Φ *is not pinned by this block and no result reads it*: only $c_\Phi < \infty$ is consumed, at Lemma 39 (compactness of the image cube) and through $\bar{w}$ at (ii). c_Φ *is not separately identified from the scale of* $\mathcal{T}$ — see Lemma 85(vii) — *and enters the programme only through* c_Π *of* (iv.c).

(ii) *Weight.* $w = \exp$. Write $\bar{w}(t) := e^{c_\Phi \|t\|_1}$, so that $|w(t'\Phi(z))| \le \bar{w}(t)$ for every z.

(iii) *Index set.* $\mathcal{T} \subset \mathbb{R}^p$ has strictly positive Lebesgue measure.

(iv) *Range reference measure.* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$, where

(iv.a) Π_u is a finite measure on $\mathbb{R}$ that is absolutely continuous with respect to Lebesgue measure, with a density ϖ_u that is bounded, strictly positive at every point of $\mathbb{R}$, and continuously differentiable with $\|\varpi_u'\|_\infty < \infty$; in particular Π_u has full topological support, $\Pi_u(O) > 0$ for every nonempty open $O \subseteq \mathbb{R}$;

(iv.b) Π_t is a finite measure on $\mathcal{T}$ for which there exists a Borel set $\mathcal{T}_0 \subseteq \mathcal{T}$ such that $\text{Leb}|_{\mathcal{T}_0}$ and $\Pi_t|_{\mathcal{T}_0}$ are *not mutually singular*;

(iv.c) $c_\Pi^2 := \Pi_u(\mathbb{R}) \int_{\mathcal{T}} \bar{w}(t)^2\, \Pi_t(dt) < \infty$;

(iv.d) Π_{w} is a *probability* measure on $\{1, \ldots, \mathsf{L}-1\}$ whose masses $\varpi_{\mathsf{w}}(l) := \Pi_{\mathsf{w}}(\{l\})$ are strictly positive at every $l \in \{1, \ldots, \mathsf{L}-1\}$.

(v) *Correction measure.* γ is a finite signed measure on $\mathcal{T}$ with $\int_{\mathcal{T}} \bar{w}(t)\,|\gamma|(dt) < \infty$; and, where $\partial_u L$ is required, $|\partial_u \lambda(u,t)| \le \Lambda_1(t)$ for all u, with $\int_{\mathcal{T}} \Lambda_1(t)\,\bar{w}(t)\,|\gamma|(dt) < \infty$.

Assumption (A2: Source condition for θ)**.** Each coordinate of ν/π lies in $\mathcal{R}(\mathcal{A}^*_{\text{cond}})$, the range of the adjoint of $\mathcal{A}_{\text{cond}} : \mathcal{V} \to L^2(\Pi)$, where $\nu := \nu_{\Delta Z}$ and $\nu_g(x) := E[g \mid X_2 = x]f_{X_2}(x) - E[g \mid X_1 = x]f_{X_1}(x)$ as in (20). Equivalently (Picard): $\nu/\pi \perp \ker \mathcal{A}_{\text{cond}}$ in $L^2(\pi)$ and $\sum_{j\ge 1} \sigma_j(\mathcal{A}_{\text{cond}})^{-2}\,\langle \nu/\pi,\, e_j\rangle^2_{L^2(\pi)} < \infty$.

Assumption (A2′: Source condition with usable representer)**.** A2 holds, and the selected solution $\ell^{\Pi}_\star \in \ell^{\Pi}_\dagger + \ker \mathcal{A}^*_{\text{cond}}$ of $\mathcal{A}^*_{\text{cond}}\ell^{\Pi} = -\nu/\pi$, where $\ell^{\Pi}_\dagger$ denotes any one particular solution, so that the displayed set is the whole solution coset and $\ell^{\Pi}_\star$ is a selection from it, *admits a version* satisfying

(i) *Range-space regularity.*

(i.a) $\Lambda^{(0)}_\star := \max_{1\le l\le \mathsf{L}-1} \int_{\mathcal{T}} \left(\sup_{u\in\mathbb{R}} \left\| \ell^{\Pi}_\star(u,t,l) \right\| \right) \bar{w}(t)\,\Pi_t(dt) < \infty$;

(i.b) for every $l \in \{1,\dots,\mathsf{L}-1\}$ and Π_t-almost every t the map $u \mapsto \ell^{\Pi}_\star(u,t,l)$ is continuously differentiable, and $\Lambda^{(1)}_\star := \max_{1\le l\le \mathsf{L}-1} \int_{\mathcal{T}} \left(\sup_{u\in\mathbb{R}} \left\| \partial_u \ell^{\Pi}_\star(u,t,l) \right\| \right) \bar{w}(t)\,\Pi_t(dt) < \infty$;

and $\ell_\star := \mathcal{W}\ell^{\Pi}_\star$ is defined *pointwise* from that version, through (68), columnwise in l;

(iii) *Anchor.* The debiasing weight attached to $\ell_\star$ is $L_\star := L^{b_\star}_{\ell_\star}$ of Definition 19 for a measurable $b_\star : \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$ with $E\left\| b_\star(\Delta Z) \right\|^2 < \infty$, the norm being Frobenius.

The two budgets aggregate the $p(\mathsf{L}-1)$ *slicewise quantities in* ***two different ways***: the $\|\cdot\|$ inside each supremum is the **Euclidean** norm of the vector $\ell^{\Pi}_\star(u,t,l) \in \mathbb{R}^p$, and the aggregation over the contrast index l is by the **maximum**. *The asymmetry is deliberate*, and why each half is what it is is in the Discussion of the assumptions, under *A2′*. At $p = 1$ the Euclidean norm on $\mathbb{R}$ is the modulus, and at $\mathsf{L} = 2$ the maximum is over a one-element set.

Assumption (A2b: Admissible representer (a *property*, not maintained))**.** A function $\ell^{\Pi} \in L^2(\Pi)$ is an *admissible representer* if it admits a version satisfying A2′(i.a)–(i.b) with ℓ^{Π}

in place of $\ell^{\Pi}_{\star}$, that is, with

$$\Lambda^{(0)} := \max_{1\le l\le \mathsf{L}-1} \int_{\mathcal{T}} \Big(\sup_{u\in\mathbb{R}} \big\| \ell^{\Pi}(u,t,l) \big\| \Big) \bar{w}(t)\, \Pi_t(dt) < \infty,$$
$$\Lambda^{(1)} := \max_{1\le l\le \mathsf{L}-1} \int_{\mathcal{T}} \Big(\sup_{u\in\mathbb{R}} \big\| \partial_u \ell^{\Pi}(u,t,l) \big\| \Big) \bar{w}(t)\, \Pi_t(dt) < \infty,$$

the norms and the aggregation over l being A2′(i.a)–(i.b)'s, and $u \mapsto \ell^{\Pi}(u,t,l)$ continuously differentiable for every $l \in \{1,\dots,\mathsf{L}-1\}$ and Π_t-almost every t, and if $\ell := \mathcal{W}\ell^{\Pi}$ is defined *pointwise* from that version through (68), columnwise in l. *A2b is NOT MAINTAINED*: it is a hypothesis a result may impose on a representer it quantifies over. Under A2′(i) the selected solution $\ell^{\Pi}_{\star}$ satisfies A2b, with $\Lambda^{(0)} = \Lambda^{(0)}_{\star}$ and $\Lambda^{(1)} = \Lambda^{(1)}_{\star}$; that implication is the *only* relation between the two tags.

Assumption (A2c: Source condition of index μ'' on the representer (a *property*, not maintained))**.** Fix $\mu'' > \frac{3}{2}$. For $g \in L^2(\mathbb{R}, du)$ write $(\mathcal{F}g)(\xi) := \int_{\mathbb{R}} g(u) e^{-\mathrm{i}u\xi} du$ for the Fourier transform in u, and let

$$H^{\mu''}(\mathbb{R}) \;:=\; \Big\{ g \in L^2(\mathbb{R}, du) :\; \big\|g\big\|^2_{H^{\mu''}(\mathbb{R})} := \tfrac{1}{2\pi} \int_{\mathbb{R}} \big(1+\xi^2\big)^{\mu''} \big|(\mathcal{F}g)(\xi)\big|^2 d\xi < \infty \Big\} \tag{3}$$

be the u-Sobolev scale of order μ'' on $\mathbb{R}$, *taken with respect to Lebesgue measure du and not with respect to* Π_u. For an $\mathbb{R}^p$-valued g on $\mathbb{R}$ write $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ for the same scale with $|\cdot|$ replaced inside (3) by the Euclidean norm $\|z\| := \big(\sum_{j=1}^p |z_j|^2\big)^{1/2}$ on $\mathbb{C}^p$, which is where $(\mathcal{F}g)(\xi)$ takes its values, so that $\|g\|^2_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} = \sum_{j=1}^p \|g_j\|^2_{H^{\mu''}(\mathbb{R})}$ and $g \in H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ if and only if every coordinate lies in $H^{\mu''}(\mathbb{R})$; *at $p=1$ they are the same object.* A function $\ell^{\Pi} \in L^2(\Pi)$ *satisfies the source condition of index μ''* if it admits a jointly measurable representative, *fix one; everything below is read at it*, whose $\mathsf{L}-1$ slices satisfy: there is a Borel set $\mathcal{T}_* \subseteq \mathcal{T}$, *one set, serving every l*, with $\Pi_t(\mathcal{T}\setminus\mathcal{T}_*) = 0$ such that the slice $\ell^{\Pi}(\cdot,t,l)$

lies in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ for *every* $t \in \mathcal{T}_*$ and every $l \in \{1,\dots,\mathsf{L}-1\}$, and

$$\Lambda^{(\mu'')} \;:=\; \max_{1\le l\le \mathsf{L}-1} \int_{\mathcal{T}_*} \left\| \ell^{\Pi}(\cdot,t,l) \right\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} \bar{w}(t)\,\Pi_t(dt) \;<\; \infty, \tag{4}$$

the budget being aggregated as in A2′(i).

Notes on A2c. That it is a property and not a maintained condition, whose sole role is to imply A2′(i) at the starred representer, and the $\bar{w}$-weighted geometry it is stated in, are Remark 32.

Assumption (H1: Hilbert-scale link)**.** $\mathcal{A}_{\mathrm{cond}} : \mathcal{V} \to L^2(\Pi)$, read into the *whole* three-factor range space of Definition 5, not contrast by contrast, has infinite rank and its singular values satisfy $\sigma_j(\mathcal{A}_{\mathrm{cond}}) \asymp j^{-a}$ for some $a > 0$; the source set for $\varphi_\dagger$ is a ball of the *classical* Sobolev space $H^s(\mathcal{X})$, for non-integer s, the interpolation space between the neighbouring integer orders, **with no boundary condition imposed at either endpoint of** $\mathcal{X}$; and the operator is *matched* to that scale in the sense of Engl, Hanke, and Neubauer (2000, §8.5), which is here the following stated comparison: there is a finite c, not depending on ζ, such that every $\zeta \in \mathcal{V} \cap H^s(\mathcal{X})$ admits

$$\zeta \;=\; \left(\mathcal{A}^*_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}}\right)^{s/(2a)} h \qquad \text{with} \quad \|h\|_{L^2(\pi)} \;\le\; \mathsf{c}\,\|\zeta\|_{H^s}. \tag{5}$$

Notes on H1. That a is an exponent of the **stacked** operator and not of any one contrast, that it does not depend on the range weights Π_{w}, what a per-contrast reading of it does and does not license at $\mathsf{L} \ge 3$, and why $\mathsf{L} = 2$ reduces with no exception, are the supplement's Discussion paragraph *H1.* The disposition of the two numbers the glyph μ names is Remark 86.

Assumption (A4: Overlap and propensity regularity)**.**

(i) *Overlap.* **Fix** $\varepsilon_\varrho \in \left(0, \frac{1}{\mathsf{L}}\right]$, once and for all in this block — *that much is A4(i)'s* ***declaration****, and it is what Definition 2 builds* (11) *from.* **The condition** is that

$\varrho_l(\Delta Z) \geq \varepsilon_\varrho$ almost surely, *for every* $l \in \{0, 1, \ldots, \mathsf{L}-1\}$: *every cell of the instrument is reached, at every value of the controls, with probability bounded away from zero*, which rules out a design in which some cell empties out over part of the support of ΔZ. Consequently

$$\lambda_{\min}\Big(\Omega(\Delta Z)\Big) \;\geq\; \varepsilon_\Omega \;:=\; \frac{\varepsilon_\varrho}{\mathsf{L}} \;>\; 0 \qquad \text{almost surely.} \tag{6}$$

(ii) *Smoothness.* Each of $\varrho_1, \ldots, \varrho_{\mathsf{L}-1}$ belongs to a Hölder ball of order $r > 0$, the *same* r for every l, on the support of the non-degenerate coordinates of ΔZ, whose number is d_Z with $1 \leq d_Z \leq p-1$.

(iii) Estimation, and the measurability that makes the display a probability statement. $\hat{\varrho} = (\hat{\varrho}_1, \ldots, \hat{\varrho}_{\mathsf{L}-1})'$ is computed on an auxiliary fold; its *own pointwise function* $(\omega, z) \mapsto \hat{\varrho}(\omega)(z)$, the one Definition 2 evaluates at each ΔZ_i, is **jointly measurable**, with respect to the product of the σ-algebra generated by the auxiliary fold with $\mathfrak{B}(\mathbb{R}^p)$; and

$$\big\|\hat{\varrho} - \varrho\big\|_{L^2} \;:=\; \Big(\int \big\|\hat{\varrho}(z) - \varrho(z)\big\|^2 P_{\Delta Z}(dz)\Big)^{1/2} = O_P\Big(n^{-r/(2r+d_Z)}\Big) =: O_P(r_n),$$

the inner $\|\cdot\|$ being Euclidean on $\mathbb{R}^{\mathsf{L}-1}$ and the integral running over z alone, so that the left side is a function of the auxiliary fold.

Assumption (N1: Conditional centring of a weight)**.** A debiasing weight L *satisfies N1* if $E\Big[L(\Delta U, \Delta Z) \mid \Delta Z\Big] = 0$ almost surely, *the zero on the right being the zero matrix of* $\mathbb{R}^{p\times(\mathsf{L}-1)}$: N1 is $p(\mathsf{L}-1)$ scalar conditions, one per pair (j, l), and it is **not** a condition on any contraction of the $\mathsf{L}-1$ columns. For a fixed $\mathsf{v} \in \mathbb{R}^{\mathsf{L}-1}$ with $\mathsf{v} \neq 0$ the contracted condition $E\Big[L \mid \Delta Z\Big]\,\mathsf{v} = 0$ is p conditions; it is implied by N1 and implies N1 *only* at $\mathsf{L} = 2$, where the columns number one and the two statements are the same p conditions. Corollary 21 makes the gap explicit and prices it. N1 is a property that a weight may or may not have; it is not maintained, and a result that needs it says so.

Assumption (N1′: Conditional centring of a direction)**.** A perturbation direction D in the weight *satisfies N1′* if $E\big[D(\Delta U, \Delta Z) \mid \Delta Z\big] = 0$ almost surely.

Assumption (A5: Nuisance rates)**.** Let $h := \hat{\varphi} - \varphi_\dagger$, let $\hat{\delta}_n$, δ_n and ϵ_n be the rates the clauses below define, and let r_n be the propensity rate of A4(iii).

(i) Operator noise, at a declared level with two branches. The level is the *deterministic* sequence

$$\hat{\delta}_n \;:=\; \max\Big\{\hat{\delta}_n^{(T)}, \hat{\delta}_n^{(\varrho)}\Big\}, \qquad \hat{\delta}_n^{(T)} \;:=\; n^{-1/2}\sqrt{\ln n}, \qquad \hat{\delta}_n^{(\varrho)} \;:=\; r_n,$$

so that $\hat{\delta}_n \geq n^{-1/2}\sqrt{\ln n}$ at every n; and the operator error is $O_P(\hat{\delta}_n)$.

(ii) Structural rate, with the logarithm of (56) carried **in the symbol**. $\|h\|_{L^2(\pi)} = O_P(\delta_n)$ with

$$\delta_n \;:=\; \Big(\hat{\delta}_n \ln n\Big)^{2\mu/(2\mu+1)},$$

and $\delta_n = o_P(n^{-1/4})$. *The first half is* ***assumed*** *and is not delivered by Proposition 28*: $h = \hat{\varphi} - \varphi_\dagger$ is formed at the estimator Definition 22(b) instructs, while that proposition concludes at the infeasible reading $\hat{\varphi}^\pi$, and the passage between them is Remark 89(7), two of whose five parts are open. The second half is a statement about the deterministic δ_n alone and *is* delivered, under (7).

(iii) *Propensity frontier.* Writing $\hat{\delta}_n = n^{-r/(2r+d_Z)}$, which is (i)'s maximum for all n large enough, A4(ii)'s $d_Z \geq 1$ giving $r/(2r+d_Z) < \frac{1}{2}$, condition (ii) holds if and only if

$$\frac{2\mu}{2\mu+1} \cdot \frac{r}{2r+d_Z} \;>\; \frac{1}{4} \qquad\Longleftrightarrow\qquad r \;>\; \frac{d_Z\,(2\mu+1)}{2\,(2\mu-1)}, \tag{7}$$

which in particular requires $\mu > \frac{1}{2}$, whatever the value of r.

(iv) *Debiasing weight, on the derivative.* The estimator $\hat{\ell}$ of the representer, a random element of $L^2(P_{\Delta U,\Delta Z})$ measurable with respect to the σ-algebra generated by the

auxiliary fold and read at one fixed *jointly* measurable representative, satisfies

$$\left\|\hat{\ell} - \ell_\star\right\|_{L^2(P)} = O_P(\epsilon_n), \qquad \delta_n\,\epsilon_n = o_P\Big(n^{-1/2}\Big).$$

(v) *Plug-in.* $\sqrt{\hat{m}}\,n^{-1/2} = o_P(\delta_n)$.

(vi) Sobolev stability of the first-stage error, at the final iterate. $\|h\|_{H^s} = O_P(1)$, with s as in H1.

(vi′) Sobolev stability along the linearisation path, as a w.p.$\to 1$ envelope at a **fixed** radius. With $\hat{\varphi}^{(0)}, \ldots, \hat{\varphi}^{(\bar{k})}$ the stages of Definition 22 and $\bar{k}$ finite, there is one finite R, not depending on n, such that

$$P\Big(\max_{0\le k\le \bar{k}} \left\|\hat{\varphi}^{(k)} - \varphi_\dagger\right\|_{H^s} \;\le\; \mathsf{R}\Big) \;\longrightarrow\; 1.$$

Assumption (A8: Sieve window)**.** The balancing sieve $b_1, \ldots, b_{L_x}$ is a *B-spline* basis of *degree exceeding* $s - \frac{1}{2}$ on a quasi-uniform mesh over $\mathcal{X}$ *with simple interior knots*, with

$$n^{1/(2s)} \;\ll\; L_x \;\ll\; n^{1/2},$$

and $\beta_1, \ldots, \beta_{L_u}$ is a basis on the residual range with $L_u \le L_x$. *Consequently the sieve is contained in the Sobolev space the source conditions are written in*:

$$\mathcal{S}_n \;=\; \operatorname{span}(b_1, \ldots, b_{L_x}) \cap \mathcal{V} \;\subset\; H^s(\mathcal{X}), \tag{8}$$

so $\|\zeta\|_{H^s} < \infty$ at every $\zeta \in \mathcal{S}_n$, with $\mathcal{S}_n$ the sieve of Definition 22 and $H^s(\mathcal{X})$ the *classical* Sobolev space of H1, *the one that carries no boundary condition*, which is what makes the containment available to a basis that satisfies none.

Definition 1 (The integrated kernel)**.** Let C_{h_u} be the kernel of C3 and, for $I \in \mathfrak{I}$, define

$$\mathsf{C}_{h_u}(v; I) \;:=\; \int_{-\infty}^{v/h_u} \Theta_I(\varsigma)\, d\varsigma \;=\; \int_{-\infty}^{v/h_u} C_{h_u}\Big(h_u\varsigma, I\Big)\, d\varsigma \;=\; \frac{1}{h_u}\int_{-\infty}^{v} C_{h_u}(\varsigma, I)\, d\varsigma, \qquad v \in \mathbb{R}, \tag{9}$$

with Θ_I of (14), and write $\mathsf{C}_{h_u}(v) := \mathsf{C}_{h_u}(v; \mathbb{R})$. The same definition with h_x in place of h_u defines C_{h_x}. *The boundary index is carried only in the x-direction*: in the residual direction the argument u ranges over all of $\mathbb{R}$ and there is no boundary, so (12) uses $\mathsf{C}_{h_u}(\cdot) = \mathsf{C}_{h_u}(\cdot; \mathbb{R})$ throughout — *in u the interval is always* $\mathbb{R}$; in the x-direction $\mathcal{X}$ is a compact interval by A0 and the index is used. Its properties are Lemma 67.

Definition 2 (The sample operator $\hat{T}$)**.** Adopt the sample-splitting convention of Lemma 60: $\{1, \ldots, n\} = \mathcal{I}_a \cup \mathcal{I}_m$, $n_{\mathrm{a}} := \#\mathcal{I}_a$, $n_{\mathrm{m}} := \#\mathcal{I}_m$, the two folds independent by A0, with $\hat{\varrho}$ computed on $\mathcal{I}_a$ as A4(iii) requires. *The split is* ***balanced***: n_{a} and n_{m} are non-random with

$$n_{\mathrm{m}} \asymp n \qquad (\text{equivalently, } \liminf_n n_{\mathrm{m}}/n > 0). \tag{10}$$

Construction convention, and it costs nothing: $\hat{\varrho}$ is taken to have values in the ε_ϱ-interior of the simplex,

$$\Big\{q \in \mathbb{R}^{\mathsf{L}-1} :\; q_l \geq \varepsilon_\varrho \;\text{ for every } l \in \{1, \ldots, \mathsf{L}-1\},\;\; \textstyle\sum_{l=1}^{\mathsf{L}-1} q_l \leq 1 - \varepsilon_\varrho\Big\}, \tag{11}$$

with ε_ϱ the overlap constant of A4(i); if a first-stage estimator does not, replace it by its metric projection onto that set, which is well defined because (11) is closed, convex and, since A4(i) *declares* $\varepsilon_\varrho \leq 1/\mathsf{L}$, non-empty — *the declaration alone and not the condition.* Writing $\widehat{\widetilde{W}}_i := \Big(\mathbb{1}\{W_i = 1\} - \hat{\varrho}_1(\Delta Z_i), \ldots, \mathbb{1}\{W_i = \mathsf{L}-1\} - \hat{\varrho}_{\mathsf{L}-1}(\Delta Z_i)\Big)'$, the *sample operator* is

$$\hat{T}(\varphi, \theta)(u, t, l) \;:=\; \frac{1}{n_{\mathrm{m}}} \sum_{i \in \mathcal{I}_m} \widehat{\widetilde{W}}_{i,l}\, \mathsf{C}_{h_u}\Big(u - \Delta U_i(\theta, \varphi)\Big)\, w\Big(t'\Phi(\Delta Z_i)\Big), \qquad (u, t, l) \in \mathbb{R}\times\mathcal{T}\times\{1, \ldots, \mathsf{L}-1\}, \tag{12}$$

with $\Delta U_i(\theta, \varphi)$ the trial residual of (19). Its x-smoothed variant $\hat{T}_{h_x}$, which is what Defini-

tion 22 iterates on and what $\hat{\mathcal{A}}_\varphi$ differentiates, is (97).

Notes on Definition 2. What the balanced split (10) costs and where it is consumed, what the truncation buys, the scope note for a smaller known lower bound, and which assumption names (12) are Remark 33.

Definition 3 (The smoothed sample linearisation $\hat{\mathcal{A}}$)**.** For $\varphi \in \mathcal{V}$ let $\mathcal{A}_\varphi : \mathcal{V} \to L^2(\Pi)$ be the Fréchet derivative of $T(\cdot, \theta)$ at φ, so that $\mathcal{A}_{\text{cond}} = \mathcal{A}_{\varphi_\dagger}$, with kernel $\mathsf{k}(u, t, l; x) = \nu_{g_{u,t,l}}(x)/\pi(x)$ as in Proposition 9, where $g_{u,t,l}$ is that proposition's $g_{u,t}$ read at the l-th contrast. Write $\hat{\mathcal{A}}_\varphi$ for the *kernel-smoothed sample analogue* of $\mathcal{A}_\varphi$: the integral operator whose kernel is obtained from k by replacing the indicator $\mathbb{1}\{\Delta U(\theta, \varphi) \le u\}$ inside $g_{u,t,l}$ by its C_{h_u}-smoothed version, and each of the two products $E[\cdot \mid X_j = x] f_{X_j}(x)$ constituting $\nu_{g_{u,t,l}}$ by its C_{h_x} kernel estimate. Write $\hat{\mathcal{A}}^*_\varphi$ for its adjoint in the $L^2(\pi)$–$L^2(\Pi)$ pairing, and

$$\hat{\mathcal{A}}_{\text{cond}} := \hat{\mathcal{A}}_{\varphi_\dagger}, \qquad \hat{\mathcal{A}}^*_{\text{cond}} := \hat{\mathcal{A}}^*_{\varphi_\dagger},$$

the running Landweber–Fridman iterate being written $\hat{\mathcal{A}}_{\hat{\varphi}_m}$ wherever the base point matters.

Assumption (A7: Stopping feasibility, at the amended threshold)**.** The level used by the discrepancy principle dominates the true operator noise level with probability approaching one: for some $\tau_{\text{DP}} \ge 1$,

$$P\Big(\big\| \hat{T}(\varphi_\dagger, \theta) \big\|_{L^2(\Pi)} \le \tau_{\text{DP}}\, \hat{\delta}_n \, \ln n \Big) \longrightarrow 1, \tag{13}$$

with $\hat{T}$ as in (12). *Any divergent factor serves in place of* $\ln n$; $\ln n$ is fixed here so that one symbol, not a family, propagates downstream.

Assumption (C1: Densities)**.** Write $x = (x_1, x_2) \in \mathcal{X}^2$.

(i) The conditional density $f_{\Delta Y, X_1, X_2 | \Delta Z, W}$ and the joint density $f_{\Delta Y, X_1, X_2 | \Delta Z}$ are p_K times continuously differentiable and uniformly bounded above.

(ii) f_{X_1,X_2}, f_{X_1}, f_{X_2} and π are uniformly bounded above and below away from 0 on $\mathcal{X}^2$ and $\mathcal{X}$ respectively; consequently $\bar{c}_\pi := \max_{j\in\{1,2\}} \|f_{X_j}/\pi\|_\infty < \infty$.

Assumption (C2′: Smoothness of the residual law)**.**

(i) The conditional density $f_{\Delta Y|X_1,X_2,\Delta Z,W}(\cdot \mid \cdot)$ exists, is $p_K - 1$ times continuously differentiable in its first argument, and admits envelopes $\bar{f}_k(x_1, x_2, z, \mathsf{w}) := \sup_u \left|\partial_u^k f_{\Delta Y|X_1,X_2,\Delta Z,W}(u \mid x_1, x_2, z, \mathsf{w})\right|$ with $E[\bar{f}_k] < \infty$ for $k = 0, 1, \ldots, p_K - 1$ and $\|\bar{f}_0\|_\infty + \|\bar{f}_1\|_\infty < \infty$.

(ii) For every $t \in \mathcal{T}$ the map $u \mapsto \mathcal{M}(u,t)$ is p_K times continuously differentiable, with $p_K = 8$ and

$$\max_{0\le k\le p_K} \sup_{t\in\mathcal{T}} \bar{w}(t)^{-1} \sup_{u\in\mathbb{R}} \left|\partial_u^k \mathcal{M}(u,t)\right| \;<\; \infty.$$

Assumption (C3: Kernel, and the defining property of the boundary adjustment)**.** Fix $p_K = 8$. *Write $\mathfrak{I}$ for the set of closed intervals $I \subseteq [-\infty, +\infty]$ with $\inf I \in [-\infty, 0]$, $\sup I \in [0, +\infty]$ and $\sup I - \inf I \geq 2$ — half-lines and $\mathbb{R}$ included — and, for a bandwidth $h > 0$ and $I \in \mathfrak{I}$,*

$$\Theta_I(\varsigma) \;:=\; C_h\Big(h\varsigma,\, I\Big), \qquad \varsigma \in \mathbb{R}, \tag{14}$$

for the kernel read as a function of its scaled argument. The smoothing kernel used in the residual direction is a univariate generalized kernel C_{h_u} with:

(i) Order p_K at the interior index. Writing $\Theta_1 := \Theta_{\mathbb{R}}$ — *the subscript* 1 *is an abbreviation for the untruncated index $I = \mathbb{R}$, and it is the **digit** that is not an interval and so not a member of $\mathfrak{I}$; the index it abbreviates, $I = \mathbb{R}$, is a member,* the notation being retained because it is the interior kernel every bound below names —

$$\int_{\mathbb{R}} \Theta_1(\varsigma)\, d\varsigma \;=\; 1, \qquad \int_{\mathbb{R}} \varsigma^j\, \Theta_1(\varsigma)\, d\varsigma \;=\; 0 \quad (1 \le j \le p_K - 1), \qquad \int_{\mathbb{R}} |\varsigma|^{p_K} \left|\Theta_1(\varsigma)\right| d\varsigma \;<\; \infty.$$

(ii) *Boundary adjustment.* There is a measurable **envelope** $\bar{\Theta} : \mathbb{R} \to [0, \infty)$ with $\|\bar{\Theta}\|_\infty <$

∞, $\int_{\mathbb{R}} \bar{\Theta} < \infty$ and $\int_{\mathbb{R}} |\varsigma|^{p_K} \bar{\Theta} < \infty$, such that for every $I \in \mathfrak{I}$ the function Θ_I vanishes off I, satisfies $|\Theta_I| \le \bar{\Theta}$, and

$$\int_I \varsigma^j \, \Theta_I(\varsigma) \, d\varsigma \;=\; \delta_{j0} \qquad (0 \le j \le p_K - 1), \tag{15}$$

with δ_{j0} the Kronecker symbol. Write

$$\mathsf{c}_2 \;:=\; \|\bar{\Theta}\|_\infty, \qquad c_C \;:=\; \frac{\|\bar{\Theta}\|_{L^1}}{2\,\mathsf{c}_2}, \qquad \text{so that} \qquad \int_{\mathbb{R}} \bar{\Theta} \;=\; 2\,c_C\,\mathsf{c}_2. \tag{16}$$

(iii) *Regularity at the interior index.* Θ_1 is symmetric and Lipschitz, with Lipschitz constant c_4.

The same three conditions with the same p_K apply to the kernel C_{h_x} used in the x-direction inside $\hat{\mathcal{A}}_{\text{cond}}$ and $\hat{\mathcal{A}}^*_{\text{cond}}$, both as defined in Definition 3. *C3 is not vacuous at* $p_K = 8$: Lemma 66 exhibits a family satisfying (i)–(iii) whose interior member is *exactly* the order-p_K Gaussian-based kernel the application computes.

Notes on C3. What (14) is for, and how clause (ii)'s boundary index and defining property are read, are Remark 34. *Clause* (ii) *asks for* ***an*** *envelope, so* (16) *returns a different pair for each admissible* $\bar{\Theta}$; every result below holds at whichever pair the analyst can verify, and none is sharper for a larger one. Lemma 66 discharges (ii) at the **smallest** envelope the family admits, the pointwise supremum $\sup_{I \in \mathfrak{I}} |\Theta_I|$, and *its* constants at $p_K = 8$ are the ones reported here: $\mathsf{c}_2 = 34.0$ and $c_C = 0.155$, so that $2c_C\mathsf{c}_2 = \|\bar{\Theta}\|_{L^1} = 10.56$. *Neither constant enters a rate*; they enter the relaxation parameter (35), with c_2 **squared**, and hence the cap (36), where the combination is $c_C\mathsf{c}_2^4$. Which further bounds they enter, how a looser envelope moves each, and which of three admissible envelopes a given reading should use are in the Discussion of the assumptions, under *C3*.

Assumption (C5′: Smoothing parameters)**.**

(i) *Residual direction, operator component.* $p_K = 8$ and

$$h_u \asymp \mathrm{sd}(\Delta U)\,\hat{\delta}_n^{1/p_K} = \mathrm{sd}(\Delta U)\,n^{-\alpha/8}, \qquad \alpha := \frac{r}{2r + d_Z} \in \left(0, \tfrac{1}{2}\right),$$

(ii) *x-direction, derivative and adjoint components.* $h_x \le (\bar{x} - \underline{x})/2$ at every n, $h_x \to 0$, and

$$\left(n\, h_x\, h_u\right)^{-1} \ln n \longrightarrow 0.$$

(iii) *Propensity.* $\hat{\varrho}$ satisfies A4(iii) with r obeying the frontier (7).

(iv) *No further smoothing.* No bandwidth is used in t, in ΔZ, or in W.

Notes on C5′. What clause (i)'s display buys, and clause (ii)'s stronger inherited form together with the window on s it leaves against (P-g), are Remark 31.

Definition 4 (Model and differencing)**.** The structural model is

$$Y_{ij} \;=\; \varphi_\dagger(X_{ij}) \;+\; Z_{ij}'\theta_\dagger \;+\; \eta_i \;+\; U_{ij}, \qquad j \in \{1,2\}, \tag{17}$$

with $X_{ij} \in \mathcal{X} \subset \mathbb{R}$ scalar and continuously distributed, $Z_{ij} \in \mathbb{R}^p$, $W_i \in \{0,1,\ldots,\mathsf{L}-1\}$ with $\mathsf{L} \geq 2$ fixed and known, and η_i an individual effect that may depend on (X,Z,W). Differencing across the two periods,

$$\Delta Y_i \;=\; \Delta\varphi_{\dagger i} \;+\; \Delta Z_i'\theta_\dagger \;+\; \Delta U_i, \qquad \Delta\varphi_i := \varphi(X_{i2}) - \varphi(X_{i1}), \tag{18}$$

and, for a trial pair (φ,θ), the residual at trial values is

$$\Delta U_i(\theta,\varphi) \;:=\; \Delta Y_i - \Delta\varphi_i - \Delta Z_i'\theta, \qquad \Delta U_i \;=\; \Delta U_i(\theta_\dagger,\varphi_\dagger). \tag{19}$$

The subscript i is suppressed whenever a statement concerns the population distribution P.

Definition 5 (Spaces, and the domain of the operator)**.** Let π be the reference density on $\mathcal{X}$ of Assumption C1, namely the *pooled design density* of the regressor across the two periods, $\pi := \frac{1}{2}\big(f_{X_1} + f_{X_2}\big)$. *It is an* ***unknown*** *feature of P and not a measure the analyst selects*: the estimator of Definition 22 divides by an *estimate* $\hat\pi$ of it. *It is fixed at P once and is not re-indexed by a restriction of the sample*, which is the reading at which Remark 18(2)'s sub-population transfer is stated. What that declaration costs, what it delivers for free and the one thing it does not settle are Remark 104(4). Set

$$\mathcal{V} \;:=\; \Big\{\varphi \in L^2(\mathcal{X},\pi) :\; \textstyle\int_{\mathcal{X}} \varphi\, d\pi = 0\Big\}, \qquad L^2(\Pi) \;:=\; L^2\Big(\mathbb{R}\times\mathcal{T}\times\{1,\ldots,\mathsf{L}-1\},\; \Pi\Big),$$

with $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ as in Assumption B1(iv), the base carrying $\mathfrak{B}(\mathbb{R})\otimes\mathfrak{B}(\mathcal{T})\otimes 2^{\{1,\ldots,\mathsf{L}-1\}}$, *the third factor is a finite set with its power set*, which is what makes B1(iv.d)'s $\varpi_{\mathsf{w}}(l) := \Pi_{\mathsf{w}}(\{l\})$ well posed and what makes Π-measurability equivalent to measurability of the $\mathsf{L}-1$ sections.

The elements of $L^2(\Pi)$ are *scalar* functions of (u,t,l); *and, for an integer* $q \geq 1$, $L^2(\Pi;\mathbb{R}^q)$ *denotes the same space with* $\mathbb{R}^q$*-valued elements*, normed by $\left(\int \|\cdot\|^2 d\Pi\right)^{1/2}$ with $\|\cdot\|$ the Euclidean norm on $\mathbb{R}^q$, so that $L^2(\Pi;\mathbb{R}^1) = L^2(\Pi)$. The *range* of the operator is the scalar one, which is why the pairing below is written on scalars; the *representer* ℓ^{Π} of A2′, A2b and A2c is $\mathbb{R}^p$-valued and lives in this space at $q = p$, with $\mathcal{A}^*_{\text{cond}}$ read coordinatewise, as Definition 19 spells out. At $p = 1$ the two spellings are one symbol. Inner products are $\langle h, h' \rangle_{L^2(\pi)} = \int hh' \, \pi \, dx$ and

$$\begin{aligned} \langle \Xi, \Xi' \rangle_{L^2(\Pi)} &= \int \Xi \, \Xi' \, d\Pi \\ &= \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \Xi(u,t,l) \, \Xi'(u,t,l) \, \Pi_u(du) \, \Pi_t(dt), \end{aligned}$$

where h, h' denote elements of the domain and Ξ, Ξ' elements of the range. This pairing is where the masses $\varpi_{\mathsf{w}}(l)$ of B1(iv.d) act, and every adjoint taken in it carries them. (68) *displays* them because $\mathcal{W}$ absorbs what the pairing *supplies*; they are not applied twice. Write $P_{\mathcal{V}}$ for the $L^2(\pi)$-orthogonal projection onto $\mathcal{V}$. *At* $\mathsf{L} = 2$ *the third factor is the one-point set* $\{1\}$ *and* Π_{w} *the point mass at it*, under the canonical identification of $\mathbb{R} \times \mathcal{T} \times \{1\}$ with $\mathbb{R} \times \mathcal{T}$. The period-difference operator is $\Delta : \mathcal{V} \to L^2(P)$, $\Delta h = h(X_2) - h(X_1)$. For an integrable random variable g write

$$\nu_g(x) \;:=\; E[g \mid X_2 = x] \, f_{X_2}(x) \;-\; E[g \mid X_1 = x] \, f_{X_1}(x), \qquad \nu \;:=\; \nu_{\Delta Z} \in \mathbb{R}^p, \tag{20}$$

the latter understood coordinatewise.

Every operator in this paper has domain $\mathcal{V}$*, with one declared exception named below*, and the decomposition $L^2(\pi) = \mathcal{V} \oplus \{\text{constants}\}$ is orthogonal.

The exception is the second-order form operator $\mathcal{B}$ *of* (74)*, which is read on all of* $L^2(\pi)$. No other object in the paper leaves $\mathcal{V}$.

Notes on Definition 5. Which letters are the bound variables on each side, why the restriction

to $\mathcal{V}$ costs nothing, what governs the sharp constant of the form on the declared exception, and what the declaration of π as an unknown feature of P costs are Remark 104.

Definition 6 (The operator equation)**.** Let $\Delta Y_\theta := \Delta Y - \Delta Z'\theta$, so that $\Delta U(\theta,\varphi) = \Delta Y_\theta - \Delta\varphi$. For $(\varphi,\theta) \in \mathcal{V}\times\mathbb{R}^p$ define the *Bierens moment family*

$$\mathcal{M}(u,t,l;\varphi,\theta) \;:=\; E\Big[\widetilde{W}_l\,\mathbb{1}\big\{\Delta Y_\theta - \Delta\varphi \le u\big\}\,w\big(t'\Phi(\Delta Z)\big)\Big], \qquad (u,t,l)\in\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}, \tag{21}$$

with $\widetilde{W}_l = \mathbb{1}\{W=l\} - \varrho_l(\Delta Z)$ the l-th contrast of (1), and let

$$T:\ \mathcal{V}\times\mathbb{R}^p \longrightarrow L^2(\Pi), \qquad T(\varphi,\theta) \;:=\; \Big[(u,t,l)\mapsto \mathcal{M}(u,t,l;\varphi,\theta)\Big], \tag{22}$$

the contrast index being an argument of $\mathcal{M}$ and not a coordinate of its value, so that $T(\varphi,\theta)$ is a scalar function on the base set of Π fixed in Definition 5. The *operator equation of the model* is

$$T(\varphi,\theta) \;=\; 0 \quad \text{in } L^2(\Pi). \tag{23}$$

We abbreviate $\mathcal{M}(u,t,l) := \mathcal{M}(u,t,l;\varphi_\dagger,\theta_\dagger)$ and $T(\varphi) := T(\varphi,\theta_\dagger)$ when no ambiguity arises. At $\mathsf{L}=2$ the index takes the single value 1 and $\widetilde{W}_1 = W - \varrho(\Delta Z)$.

Definition 7 (Admissible sets)**.** Restrictions (A) and (B) of plan §2.1 are

$$\text{(A)}\quad E\Big[\widetilde{W}\,\varsigma(\Delta U)\,\Big|\,\Delta Z\Big] = 0 \text{ a.s. for every bounded measurable } \varsigma, \qquad \text{(B)}\quad E\Big[\Delta Z\,\Delta U\Big] = 0.$$

Define

$$\mathcal{I} \;:=\; \Big\{(\varphi,\theta)\in\mathcal{V}\times\mathbb{R}^p:\ \Delta Y - \Delta\varphi - \Delta Z'\theta \;\perp\!\!\!\perp\; W \,\Big|\, \Delta Z\Big\}, \qquad \mathcal{V}_{CI} \;:=\; \big\{\varphi\in\mathcal{V}:\ (\varphi,0)\in\mathcal{I}\big\}.$$

Conditional independence $\xi \perp\!\!\!\perp W \mid \Delta Z$ is understood in the factorisation sense, $P(\xi\in B, W=j\mid\Delta Z) = P(\xi\in B\mid\Delta Z)\,P(W=j\mid\Delta Z)$ a.s. for all Borel B and $j\in$

$\{0, 1, \ldots, \mathsf{L}-1\}$. Two features of this form are used below, and at $\mathsf{L} \geq 3$ they are features of two *different* sets of values of ΔZ. *First*, at a $\Delta Z = z$ at which every W-cell is non-null it is equivalent to the conditional-expectation form $P(\xi \in B \mid \Delta Z, W) = P(\xi \in B \mid \Delta Z)$ a.s., and where some cell is null it is the form that behaves correctly, the factorisation holding at a null cell with both sides zero whatever version is taken there. *Second*, it holds automatically, whatever the conditional law of ξ, exactly where W is *conditionally degenerate*, that is, where $\max_{0 \leq l \leq \mathsf{L}-1} \varrho_l(z) = 1$. Both facts are used in the proof of Lemma 11. *Degeneracy is that condition, and a null cell is not by itself degeneracy*: at $\mathsf{L} \geq 3$ a z may have null cells and two or more surviving ones, and there the restriction is not vacuous, it still forces the surviving cells to share one conditional law of ξ. *At* $\mathsf{L} = 2$ *the two conditions coincide*, one null cell leaving one survivor.

Proposition 8 (Fréchet derivative in φ, with a Lipschitz derivative map)**.** *T is Fréchet differentiable in φ with a quadratic remainder, and the derivative map is Lipschitz. Suppose B1(i), (ii), (iv.c), (iv.d), C1(ii) and C2′(i) hold;* B1(iv) is cited by clause*, its factorisation $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ being a declaration and not a hypothesis. Then, for every $\theta \in \mathbb{R}^p$ and every $\varphi \in \mathcal{V}$, the linear operator $\mathcal{A}_\varphi : \mathcal{V} \to L^2(\Pi)$ defined, at each $(u, t, l) \in \mathbb{R} \times \mathcal{T} \times \{1, \ldots, \mathsf{L}-1\}$, by*

$$\mathcal{A}_\varphi h(u,t,l) \;=\; E\Big[\widetilde{W}_l\, w\big(t'\Phi(\Delta Z)\big)\, f_{\Delta Y_\theta - \Delta\varphi \mid X_1, X_2, \Delta Z, W}(u)\, \big\{h(X_2) - h(X_1)\big\}\Big] \tag{24}$$

is bounded, with $\|\mathcal{A}_\varphi\| \leq 2\,\bar{c}_\pi^{1/2} \|\bar{f}_0\|_\infty c_\Pi$, and:

(i) Fréchet differentiability with a quadratic remainder: *for all $h \in \mathcal{V}$,*

$$\big\|T(\varphi + h, \theta) - T(\varphi, \theta) - \mathcal{A}_\varphi h\big\|_{L^2(\Pi)} \;\leq\; C_{\mathcal{A}}\, \|h\|^2_{L^2(\pi)}, \qquad C_{\mathcal{A}} := 2\,\bar{c}_\pi\, \|\bar{f}_1\|_\infty\, c_\Pi; \tag{25}$$

(ii) Lipschitz derivative map: *for all $h \in \mathcal{V}$,*

$$\big\|\mathcal{A}_{\varphi+h} - \mathcal{A}_\varphi\big\| \;\leq\; 2\, C_{\mathcal{A}}\, \|h\|_{L^2(\pi)}. \tag{26}$$

At the truth we write $\mathcal{A}_{\text{cond}} := \mathcal{A}_{\varphi_\dagger}$ *with* $\theta = \theta_\dagger$*, in which case the conditional density in* (24) *is* $f_{\Delta U|X_1,X_2,\Delta Z,W}$*. At* $\mathsf{L} = 2$ *the index takes the single value* 1 *and* $\widetilde{W}_1 = W - \varrho(\Delta Z)$*.*

Proposition 9 (Compactness)**.** $\mathcal{A}_{\text{cond}}$ *is Hilbert–Schmidt, hence compact. Suppose B1(i), (ii), (iv.c), (iv.d), C1(ii) and C2′(i) hold;* B1(iv) is cited by clause*, its factorisation being a declaration and not a hypothesis. Then* $\mathcal{A}_{\text{cond}}$ *is the integral operator on* $\mathcal{V}$ *with kernel*

$$\mathsf{k}(u,t,l;x) \;=\; \frac{\nu_{g_{u,t,l}}(x)}{\pi(x)}, \qquad g_{u,t,l} \;:=\; \widetilde{W}_l\, w\Big(t'\Phi(\Delta Z)\Big)\, f_{\Delta U|X_1,X_2,\Delta Z,W}(u), \tag{27}$$

and $\|\mathcal{A}_{\text{cond}}\|_{\text{HS}} \le 2\,\bar{c}_\pi\,\|\bar{f}_0\|_\infty\, c_\Pi < \infty$*. Moreover* $\mathsf{k}(u,t,l;\cdot) \in \mathcal{V}$ *for every* $(u,t,l) \in \mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$*, so* $\mathcal{A}^*_{\text{cond}}$ *maps* $L^2(\Pi)$ *into* $\mathcal{V}$*; and* $\mathcal{A}_{\text{cond}}$ *admits a singular system* $(\sigma_j; e_j, \tilde{e}_j)_{j\in\mathcal{J}}$*,* $\mathcal{J} \subseteq \mathbb{N}$*, with* $\{e_j\}_{j\in\mathcal{J}}$ *an orthonormal basis of* $(\ker \mathcal{A}_{\text{cond}})^\perp \cap \mathcal{V}$ *and, when* $\mathcal{J}$ *is infinite,* $\sigma_j \downarrow 0$*.* At $\mathsf{L} = 2$ the index takes the single value 1 and $\mathsf{k}(u,t,1;x) = \mathsf{k}(u,t;x)$*, the Hilbert–Schmidt constant included,* Π_{w} *being a probability measure at every* L*.*

Lemma 10 (L1: shift invariance)**.** *Restriction (A) is invariant to the value of* θ*, so it does not identify* $\theta_\dagger$*. The set* $\mathcal{I}$ *of Definition 7 satisfies, with no assumption beyond that definition,*

$$\mathcal{I} \;=\; \mathcal{V}_{CI} \times \mathbb{R}^p; \tag{28}$$

and if in addition A1 holds, then $\varphi_\dagger \in \mathcal{V}_{CI}$*, so* $(\varphi_\dagger, \theta) \in \mathcal{I}$ *for every* $\theta \in \mathbb{R}^p$*.*

Lemma 11 (Bierens equivalence for the panel restriction)**.** *The* ΔZ*-conditional restriction (A) is equivalent to the vanishing of the unconditional family* $\{\mathcal{M}(u,t,l)\}$*, so no conditional distribution given* ΔZ *ever has to be estimated. Suppose B1(i)–(iv) hold. Then the following are equivalent:*

(a) $\Delta U \perp\!\!\!\perp W \mid \Delta Z$*;*

(b) $E[\widetilde{W}\,\varsigma(\Delta U) \mid \Delta Z] = 0$ *a.s. for every bounded measurable* ς*, that is, restriction* (A)*;*

(c) $\mathcal{M}(u,t,l) = 0$ *for every* $(u,t,l) \in \mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$*;*

(d) $T(\varphi_\dagger, \theta_\dagger) = 0$ *in* $L^2(\Pi)$.

Note on Lemma 11. Clause (b) is the $\mathbb{R}^{\mathsf{L}-1}$-valued identity (1) types it as, hence the $\mathsf{L}-1$ scalar restrictions $E[\widetilde{W}_{l\varsigma}(\Delta U) \mid \Delta Z] = 0$; that (a) and (b) are *equivalent* at every L, and under **no** overlap condition, is established in the Discussion of the assumptions, under the heading *A1 (conditional exogeneity)*, and is cited and not reproved here.

Corollary 12 (to Lemma 10 and Lemma 11)**.** *The operator equation has the same solution set in* φ *at every* θ*, and that set is* $\mathcal{V}_{CI}$*. Suppose B1(i)–(iv) hold. Then for every* $\theta \in \mathbb{R}^p$*,*

$$\left\{\varphi \in \mathcal{V} : \; T(\varphi, \theta) = 0 \text{ in } L^2(\Pi)\right\} \;=\; \mathcal{V}_{CI},$$

and under A1b(iii) this set equals $\{\varphi_\dagger\}$*.*

Proposition 13 (R2: when a weight yields a φ-free moment)**.** *A moment is free of the structural function exactly when the two periods' g-tilted marginals of the endogenous regressor coincide. Suppose C1(ii) holds and let* g *be a random variable satisfying*

$$E|g| < \infty, \qquad E\Big|g\, h(X_j)\Big| < \infty \;\; (j = 1, 2) \text{ for every } h \in \mathcal{V}, \qquad E\Big|g\, \Delta U(\theta, 0)\Big| < \infty. \quad (29)$$

Then the map $\varphi \mapsto E\Big[g\, \Delta U(\theta, \varphi)\Big]$ *is constant on* $\mathcal{V}$*, for one and hence every* $\theta \in \mathbb{R}^p$*, if and only if*

$$E[g \mid X_2 = x]\, f_{X_2}(x) \;=\; E[g \mid X_1 = x]\, f_{X_1}(x) \qquad \text{for almost every } x \in \mathcal{X}, \qquad (30)$$

that is, if and only if $\nu_g \equiv 0$*.*

Corollary 14 (to Proposition 13: the debiasing weight must vary with the residual)**.** *If the debiasing weight does not depend on the residual, the debiased moment is not merely first-order insensitive to the structural function but exactly free of it. Suppose A1, A1b(i), A1b(iv) and C1(ii) hold, let* L *be differentiable in* u at every (u, z) *with* $\partial_u L(u, z) = c(z)$ *for*

some Borel $c : \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$*, and write*

$$c(z)\,\widetilde{W} \;:=\; \sum_{l=1}^{\mathsf{L}-1} \widetilde{W}_l\; c_{\cdot l}(z) \;\in\; \mathbb{R}^p$$

for the Euclidean contraction of the $\mathsf{L}-1$ *columns of* c *against the contrast vector. Suppose* $g := \Delta Z + c(\Delta Z)\widetilde{W}$ *satisfies* (29) *coordinatewise, and suppose* L *satisfies N1. Then* $L(u,z) = c(z)\,u$*, the moment function is* $\psi = g\,\Delta U(\theta,\varphi)$ *with* g *measurable with respect to* $\sigma(W,\Delta Z)$*, and the following are equivalent:*

(a) $E[\psi]$ *is first-order insensitive to* φ *at* $\varphi_\dagger$*, i.e.* $\nu_{\Delta Z + c\widetilde{W}} \equiv 0$*;*

(b) $\varphi \mapsto E[\psi]$ *is constant on* $\mathcal{V}$*;*

(c) $\nu_{c\widetilde{W}} = -\nu$ for the c of the hypothesis*; equivalently, membership* $-\nu \in \mathcal{N}_0 := \{\nu_{\tilde{c}(\Delta Z)\widetilde{W}} : \tilde{c} \in \mathcal{C}_0\}$ *is realised* at that c*, where* $\mathcal{C}_0$ *is the set of* $\tilde{c}$ *for which* $\Delta Z + \tilde{c}(\Delta Z)\widetilde{W}$ *satisfies* (29) *coordinatewise.*

Moreover $\varphi \mapsto E[\psi]$ *is affine on* $\mathcal{V}$*, so every term of order two or higher in its expansion vanishes identically* whether or not (a)–(c) hold*; what* (a)–(c) *add is that the first-order term vanishes as well. Contrapositively, if* $-\nu \notin \mathcal{N}_0$*, then every* $\ell_\star$ *satisfying* $\nu_{\Delta Z + \ell_\star \widetilde{W}} \equiv 0$ *has* $\partial_u \ell_\star \not\equiv 0$*. At* $\mathsf{L} = 2$ *there is one column,* $c(z) \in \mathbb{R}^p$*,* $\widetilde{W}$ *is the scalar* $W - \varrho(\Delta Z)$*, and* $c\widetilde{W} = \widetilde{W}c$*.*

Proposition 15 (Characterisation of the kernel of the linearisation)**.** *A direction is annihilated by the linearised operator exactly when the instrument fails to move its period difference, given the residual and the controls. Suppose B1(i)–(iv), C1(ii) and C2′(i) hold, and let* $h \in \mathcal{V}$*. Then*

(i) the following are equivalent: (a) $\mathcal{A}_{\mathrm{cond}} h = 0$ *in* $L^2(\Pi)$*;* (b) $E\big[\widetilde{W}\,\{h(X_2) - h(X_1)\}\,\big|\,\Delta U, \Delta Z\big] = 0$ *almost surely;*

(ii) if in addition A1 and A4(i) hold, (b) *is equivalent to* (c) $E\big[h(X_2) - h(X_1) \mid \Delta U, \Delta Z, W =$

$l\big] = E\big[h(X_2) - h(X_1) \mid \Delta U, \Delta Z\big]$ *for every* $l \in \{0, 1, \ldots, \mathsf{L}-1\}$, *almost surely on* $\{\varrho_l(\Delta Z) > 0\}$.

Notes on Proposition 15. Clause (b) is the $\mathbb{R}^{\mathsf{L}-1}$-valued identity (1) types it as, hence the $\mathsf{L}-1$ scalar restrictions $E[\widetilde{W}_l\{h(X_2)-h(X_1)\} \mid \Delta U, \Delta Z] = 0$. *Clause* (c) *compares each cell against the expectation given* $(\Delta U, \Delta Z)$ *alone, and not the cells against one another*, which is the comparison the companion treatment of discrete instruments, Centorrino, Fève, and Florens (2025), is stated at; it names no base cell and is well posed where a cell is null, on which it says nothing, exactly as the factorisation form of Definition 7 does. *The two readings coincide at every* L: under A1, $P(W = l \mid \Delta U, \Delta Z) = \varrho_l(\Delta Z)$, so $E[h(X_2) - h(X_1) \mid \Delta U, \Delta Z]$ is the ϱ-weighted average of the cell expectations over the non-null cells, and all cells agreeing with one another and all cells agreeing with that average are the same condition. The quantifier over l runs over the fixed set $\{0, \ldots, \mathsf{L}-1\}$ and the qualification is on the **event** $\{\varrho_l(\Delta Z) > 0\}$, so no index set depends on the data; *under* (ii)*'s own hypotheses that qualification is vacuous*, A4(i) making each of those events almost sure, and it is carried only so that (c) remains a statement if A4(i) is relaxed.

Definition 16 (Per-subset restriction sets)**.** For a set of instrument cells $\mathsf{S} \subseteq \{0, 1, \ldots, \mathsf{L}-1\}$ put

$$\mathcal{V}_{\mathsf{S}} := \Big\{\varphi \in \mathcal{V} : \varrho_k(\Delta Z)\, P\big(\Delta Y - \Delta\varphi \in B,\ W = l \mid \Delta Z\big) = \varrho_l(\Delta Z)\, P\big(\Delta Y - \Delta\varphi \in B,\ W = k \mid \Delta Z\big)\Big\},$$

the identity holding, *for each Borel* B *and each pair* $l, k \in \mathsf{S}$, almost surely; by right-continuity in the endpoint of a half-line and a π–λ argument a single null set serves every B and every pair, so the stronger simultaneous reading defines the same set. Here $\varrho_0 = 1 - \mathbf{1}'\varrho$ as in (1). *The restriction is written in the factorisation sense of Definition 7 and in cross-multiplied form*, so that no conditional law at a null cell is ever named: where every cell of S is non-null it says that $\Delta Y - \Delta\varphi$ has *one* conditional law given ΔZ across the cells of S, and where a cell of S is null it says nothing about that cell. *The set is unchanged if* $\Delta Y - \Delta\varphi$ *is replaced*

by $\Delta U(\theta,\varphi)$ *of* (19) *at any* $\theta\in\mathbb{R}^p$, the two differing by a ΔZ-measurable shift; this is the invariance Lemma 10 records for $\mathcal{I}$, read at a set of cells. Call S *consistent* if $\mathcal{V}_{\mathsf{S}}\neq\emptyset$; and, for a consistent S, call

$$\Big\{l\in\{0,1,\dots,\mathsf{L}-1\}\setminus\mathsf{S}:\ \mathcal{V}_{\mathsf{S}\cup\{l\}}=\emptyset\Big\}$$

the set of cells *in conflict with* S. *Sans-serif* S *is a set of instrument cells and nothing else*, on the pattern of L: the subscript of $\mathcal{V}_{\mathsf{S}}$ is always a set of cells and never an integer, which separates it from the source-condition ball $\mathcal{V}_n$ of (120) and from $\mathcal{V}_{CI}$, whose subscript is a word.

Proposition 17 (Localisation of a violation to a set of cells)**.** *Extra instrument cells localise a failure of conditional exogeneity to a set of cells, and do so without any identification hypothesis on any subset of cells. Suppose B1(i)–(iv) hold, and read* $\mathcal{V}_{\mathsf{S}}$, consistent *and* in conflict with *as in Definition 16. Then*

(i) Structure of the family. $\mathcal{V}_{\mathsf{S}'}\subseteq\mathcal{V}_{\mathsf{S}}$ *whenever* $\mathsf{S}\subseteq\mathsf{S}'$; $\mathcal{V}_{\mathsf{S}}=\mathcal{V}$ *whenever* $\#\mathsf{S}\leq 1$; *and*

$$\mathcal{V}_{\{0,1,\dots,\mathsf{L}-1\}}\ =\ \mathcal{V}_{CI}\ =\ \Big\{\varphi\in\mathcal{V}:\ T(\varphi,\theta)=0\ \text{in}\ L^2(\Pi)\Big\}\qquad \text{for every}\ \theta\in\mathbb{R}^p.$$

B1(i)–(iv) is consumed by the second equality of this clause, through Corollary 12, and by nothing else in this proposition, *so localisation does not rest on the Bierens apparatus.*

(ii) Joint consistency. *If in addition A1 holds, then* $\varphi_\dagger\in\mathcal{V}_{\mathsf{S}}$ *for every* S, *so every set of cells is consistent.* This clause uses no identification hypothesis*: A1b(iii) is a hypothesis of no part of this proposition. Contrapositively, if some* S *is not consistent then A1 fails.*

(iii) The conflict set, and what it cannot be freed from. Consistency is inherited by subsets, and every S *with* $\#\mathsf{S}\leq 1$ *is consistent. The cells in conflict with* S *are among those in conflict with* S' *whenever* $\mathsf{S}\subseteq\mathsf{S}'$ *are both consistent; at a maximal consistent* $\mathsf{S}^\star$ *they are exactly* $\{0,1,\dots,\mathsf{L}-1\}\setminus\mathsf{S}^\star$*; and some consistent* S *has a non-empty conflict set if and*

only if $\mathcal{V}_{CI} = \emptyset$. *Moreover there is exactly one maximal consistent set of cells if and only if* $\mathcal{V}_{CI} \neq \emptyset$, *and no cell is in conflict with* every *consistent* S. *So wherever there is a violation to attribute, the attribution is relative to a reference set of cells. What* P *determines is the* family *of maximal consistent sets, which is a functional of* P *and is identified; what it does not determine is which member of that family is the reference set, and any rule for choosing one is a convention.*

(iv) Localisation. *Suppose* $\mathcal{V}_{CI} = \emptyset$ *while* $\mathcal{V}_{\mathsf{S}_0} \neq \emptyset$ *for some* S_0 *with* $\#\mathsf{S}_0 \geq 2$. *Then there is a maximal consistent* $\mathsf{S}^\star \supseteq \mathsf{S}_0$, *necessarily with* $\#\mathsf{S}^\star \geq 2$, *and the cells in conflict with it form a non-empty* proper *subset of* $\{0, 1, \ldots, \mathsf{L}-1\}$ *omitting at least two cells. The hypothesis of this clause is unsatisfiable at* $\mathsf{L} = 2$.

At $\mathsf{L} = 2$ clauses (i)–(iii) are true and say nothing beyond Corollary 12*: under a violation the maximal consistent sets are* $\{0\}$ *and* $\{1\}$, *the cells in conflict with them are* $\{1\}$ *and* $\{0\}$, *so the verdict is that the restriction fails and it names no channel.* Clause (iv) is **vacuous** at $\mathsf{L} = 2$, *its hypothesis being unsatisfiable there, so the* $\mathsf{L} = 2$ *reading of this proposition is a vacuity and not the verbatim reduction the statements around it carry.*

Remark 18 (The slices of T, the sub-population reading, and the statistic).

(1) The slices of T carry joint consistency and not localisation. Write $\mathcal{V}_{CI}(g(W))$ for the set of Definition 7 computed at the instrument $g(W)$ in place of W, so that $\mathcal{V}_{CI} = \mathcal{V}_{CI}(W)$. Under B1(i)–(iii), for each $l \in \{1, \ldots, \mathsf{L}-1\}$ the set of $\varphi \in \mathcal{V}$ at which the l-th slice of $T(\cdot, \theta_\dagger)$ vanishes at every $(u, t) \in \mathbb{R} \times \mathcal{T}$ is exactly $\mathcal{V}_{CI}(\mathbb{1}\{W = l\})$, the solution set at the *binary coarsening* that separates cell l from the rest; and the intersection of those $\mathsf{L}-1$ sets is $\mathcal{V}_{CI}$. *So the slices carry Proposition 17*(ii) *exactly*, and what they do not carry is localisation: a coarsening imposes fewer restrictions, $\mathcal{V}_{CI}(W) \subseteq \mathcal{V}_{CI}(g(W))$, and slice l compares cell l against all remaining cells *pooled*, so one deviant cell moves the pooled benchmark and with it every slice. *The same two readings meet at Proposition 15*(ii)(c), *where they coincide*, there the quantifier runs over all L cells at once, and the Notes

on that proposition show cell-against-average and cell-against-cell are then the same condition. They part company only at a *proper* subset of cells, which is the object Definition 16 supplies and the slices do not.

(2) Sub-population reading, and exactly what transfers with it. Under A4(i), for $\#\mathsf{S} \geq 2$ the set $\mathcal{V}_{\mathsf{S}}$ is $\mathcal{V}_{CI}$ of Definition 7 computed under the law $Q := P(\cdot \mid W \in \mathsf{S})$, the same restriction, on the sub-population on which the instrument takes its values in S. *No new operator and no normalisation enters*, and the propensities it induces are $\varrho_l/\varrho_{\mathsf{S}}$ with $\varrho_{\mathsf{S}} := \sum_{k\in\mathsf{S}} \varrho_k$. Whether the maintained conditions survive the passage to Q is a separate question from the set identity, and it is answered here rather than assumed. A0 survives: $\{i : W_i \in \mathsf{S}\}$ is an i.i.d. sample from Q. A4(i) survives with the *same* ε_ϱ, since $\varrho_l/\varrho_{\mathsf{S}} \geq \varrho_l \geq \varepsilon_\varrho$ and $\varepsilon_\varrho < 1/\mathsf{L} \leq 1/\#\mathsf{S}$. B1 survives unchanged, being a set of choices made by the analyst and not a condition on P. C1(ii)'s *upper* bounds survive with $\bar{c}_\pi$ inflated to $\bar{c}_\pi/P(W \in \mathsf{S})$. *Two things do not survive, and neither is repaired here*: C1(ii)'s *lower* bounds **on the X-densities** f_{X_1}, f_{X_2} and f_{X_1,X_2}, because A4(i) bounds ϱ_l given ΔZ and says nothing about $P(W \in \mathsf{S} \mid X_j)$, which is what a density of X_j under Q is bounded below by; and the source conditions A2, A2′ and H1, which are conditions on the linearisation the law induces, and at Q that is a different operator. *Only the second of the two binds anything*: those three lower bounds are consumed by no result in this paper, whereas A2, A2′ and H1 carry the whole rate theory, which therefore has no counterpart at Q until it is posed there. *The narrowing to the X-densities is deliberate*: C1(ii) also bounds π below, and *that* bound is consumed — at Proposition 70(i), at (35) and hence at (36), at Lemma 78 and at Lemma 47(i). *Nothing is claimed here about π.*

(3) The statistic is apparatus and is subordinate. Proposition 17 is a statement about population sets. The corresponding procedure is a J-type one, the letter is used as a name and not as a symbol, italic J being the Jacobian of (40): for a candidate reference

set S, solve for φ on the sub-population of clause (2), evaluate at that solution the criterion of the cells outside S, and read a value large relative to its own noise level as evidence that the cell is in conflict with S. *Its construction, its rate under the null and its limit law are not established here*, and clause (iii) of Proposition 17 bounds what any such procedure can deliver: the output is a family of maximal consistent sets of cells, and a selection among them, the largest, or the one containing a designated reference cell, is a convention and not an identified object. *And if* S *is itself chosen from the data, a conflict set read off it is a post-selection object*, whose coverage is not the coverage of the same set read off a reference set fixed in advance; nothing here prices that.

(4) *Base invariance.* $\mathcal{V}_{\mathsf{S}}$ is symmetric in the cells of S and names no base cell, so the family, the consistent sets and the conflict sets are all invariant to which cell (1) omits. This is the invariance established in the Discussion of the assumptions, under the heading *A1 (conditional exogeneity)*, and it is cited and not reproved here.

(5) *What is* ***not*** *claimed.* Neither Proposition 17 nor this remark asserts that extra cells make A1 testable or that they make the instrument stronger. A1 is testable at $\mathsf{L} = 2$, the restriction is a continuum of equations against one unknown function, and splitting an instrument creates no variation. The separation of testability, instrument strength and localisation is argued in that same Discussion paragraph.

(6) *Two constructions establish that the hypothesis of Proposition 17*(iv) *is not empty*, and they are in the online supplement, under the heading *Two constructions for the hypothesis of Proposition 17*(iv). The first is a witness at $\mathsf{L} = 3$ with an *irrelevant* instrument: W independent of (X_1, X_2), so there is no first stage, $\mathcal{V}_{\{0,1\}} = \mathcal{V}$ precisely because nothing is identified, and $\mathcal{V}_{CI} = \emptyset$ all the same — which isolates the fact that emptiness of $\mathcal{V}_{CI}$ is settled at the law and not by an identification hypothesis. *The companion treatment of discrete instruments says what that regime is*: Centorrino, Fève, and Florens (2025, Remark 2) records that when the residual is independent of the

regressor and the instrument, the relevance condition of that paper reverts to ordinary completeness, which a purely discrete instrument cannot satisfy, and the structural function is then only *partially* identified. The second is a worked example at $\mathsf{L} = 3$ with a **relevant** instrument — relevant in all three senses the question has — in which $\varphi_\dagger \in \mathcal{V}_{\{0,1\}}$ while $\mathcal{V}_{\{0,2\}} = \emptyset$, so that $\{0,1\}$ is a maximal consistent set and the cells in conflict with it are exactly $\{2\}$; the same construction at $\mathsf{L} = 4$ puts them at exactly $\{2,3\}$, which is the form the application's chronic-against-transient hypothesis takes. *The hypothesis of clause* (iv) *is maintained nowhere in this paper*: it is the configuration the procedure of clause (3) exists to detect, and its standing is that of the conditional-relevance condition A1b(iii), the conditional analogue of Centorrino, Fève, and Florens (2025, Assumption 2.2) — a condition on the law whose plausibility is discussed and whose scope is bounded, not a theorem. An example illustrates such a hypothesis and is not asked to exhaust it.

Definition 19 (Representers, weights, and the anchor)**.** Let $\ell^\Pi \in L^2(\Pi; \mathbb{R}^p)$ and $\ell := \mathcal{W}\ell^\Pi$ with $\mathcal{W}$ as in (68), applied coordinatewise in the p coordinates of θ and *columnwise in* l: column l of the $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued ℓ is generated by the slice $\ell^\Pi(\cdot,\cdot,l)$ and by no other slice, so the $\mathsf{L}-1$ contrasts carry $\mathsf{L}-1$ *separately choosable* generators — separately choosable, not linearly independent: nothing prevents two slices coinciding, and localisation needs only the former. *This is the reading A2′(i) and A2b fix, not a further restriction.*

(a) ℓ^Π is an *admissible representer* if it satisfies **A2b**. *That tag, and not A2′(i), is what this clause names*: A2′(i) is a *maintained assertion* about $\ell^\Pi_\star$; A2b is a *property* a candidate representer may or may not have, and it is the property every result quantifying over representers imposes. Under B1(i), (ii), (iv.a) and (iv.d), Corollary 49, itself stated at a generic admissible representer, applies to ℓ entry by entry: each entry is bounded and continuously differentiable in u, uniformly in z, with budgets $\Lambda^{(0)}, \Lambda^{(1)}$ aggregated as A2b aggregates them, which therefore bound the Euclidean norm of a whole column

and hence *a fortiori* every entry.

(b) For a Borel $b : \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$, the *weight generated by ℓ and anchored at b* is

$$L_\ell^b(u,z) \;:=\; \int_0^u \ell(v,z)\,dv \;+\; b(z), \qquad (u,z)\in\mathbb{R}\times\mathbb{R}^p, \tag{31}$$

the integral taken entrywise and well defined because each entry of $\ell(\cdot,z)$ is continuous and bounded. It is the unique function that is differentiable in u at *every* point with $\partial_u L_\ell^b = \ell$ and $L_\ell^b(0,\cdot) = b$. The weight is *admissible* if $E\|b(\Delta Z)\|^2 < \infty$, the norm being Frobenius.

(c) ℓ^Π is an *admissible selection* if it is an admissible representer and solves $\mathcal{A}^*_{\text{cond}}\ell^\Pi = -\nu/\pi$.

Under A2 and A2′(i), $\ell^\Pi_\star$ is an admissible selection, A2 gives that it solves the adjoint equation, A2′(i) that it satisfies A2b, with $\ell_\star = \mathcal{W}\ell^\Pi_\star$, and $L_\star := L^{b_\star}_{\ell_\star}$ is the admissible weight fixed by A2′(iii). Write $\zeta_\star := (\varphi_\dagger, L_\star, \varrho)$ for the nuisance triple at the truth.

Proposition 20 (Neyman orthogonality)**.** *The debiased moment is orthogonal in all three nuisances: in the structural function exactly when the debiasing weight solves the adjoint equation, and in the weight and the propensity exactly, for every admissible direction. Suppose A1b(i), A1b(iv), B1(i), (ii), (iv), C1(ii) and C2′(i) hold. Then*

(i) Structural function, at an arbitrary admissible representer. Let ℓ^Π be an admissible representer in the sense of A2b, $\ell := \mathcal{W}\ell^\Pi$ with $\mathcal{W}$ as in (68), and L an admissible weight generated by ℓ in the sense of Definition 19. Write $\ell\widetilde{W} := \sum_{l=1}^{\mathsf{L}-1}\widetilde{W}_l\,\ell_{\cdot l}(\Delta U,\Delta Z)\in\mathbb{R}^p$ for the contraction of the $\mathsf{L}-1$ columns of ℓ against the contrast vector. For $h\in\mathcal{V}$ define

$$\begin{aligned}\Psi_L(\tau_1) \;&:=\; E\Big[\psi\Big(\theta_\dagger,(\varphi_\dagger+\tau_1 h,\,L,\,\varrho)\Big)\Big]\\ &=\; E\Big[\Delta Z\,(\Delta U-\tau_1\Delta h)+L\Big(\Delta U-\tau_1\Delta h,\Delta Z\Big)\,\widetilde{W}\Big],\qquad \tau_1\in\mathbb{R},\end{aligned} \tag{32}$$

which is finite for every $\tau_1\in\mathbb{R}$. *At $L = L_\star$, $\Psi_{L_\star}(\tau_1) = \Psi(\tau_1,0,0)$ in the notation*

of Lemma 56, so the two agree wherever both are defined. Then for every $h \in \mathcal{V}$, $\partial_{\tau_1}\Psi_L(0) = -\langle \nu_{\Delta Z+\ell\widetilde{W}}/\pi,\, h\rangle_{L^2(\pi)}$, and this vanishes for every $h \in \mathcal{V}$ if and only if

$$\nu_{\Delta Z+\ell\widetilde{W}} \equiv 0 \qquad \Longleftrightarrow \qquad \mathcal{A}^{*}_{\mathrm{cond}}\ell^{\Pi} = -\nu/\pi. \tag{33}$$

The two sides of (33) are one condition written before and after routing through $\mathcal{W}$: *$\ell\widetilde{W}$ is the combination $\widetilde{W}'(\mathcal{W}\ell^{\Pi})$ of (69), read coordinatewise in the p coordinates of θ, so the masses $\varpi_{\mathsf{w}}(l)$ of B1(iv.d) enter only inside $\mathcal{W}$ and ψ carries the raw contrast on both sides.* In particular, under A2′ the left-hand condition holds at $\ell = \ell_\star$, *and* $\partial_{\tau_1}\Psi(0) = 0$ *for every* $h \in \mathcal{V}$.

(ii) Debiasing weight, exactly. *Assume in addition A1, A2, A2′(i) and A2′(iii). For every Borel $D : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$ with $E[\|D\| \mid \Delta Z] < \infty$ almost surely and $E\|D\widetilde{W}\| < \infty$, and every $\tau_2 \in \mathbb{R}$, $E[\psi(\theta_\dagger, (\varphi_\dagger, L_\star + \tau_2 D, \varrho))] = E[\psi(\theta_\dagger, \zeta_\star)]$.*

(iii) Propensity: exact orthogonality is equivalent to N1. Assume in addition A2, A2′(i), A2′(iii) and A4(i), and do not assume *N1, nor A1. Then, for every bounded Borel $\mathsf{r} : \mathbb{R}^p \to \mathbb{R}^{\mathsf{L}-1}$ with $\varrho+\tau_3\mathsf{r}$ taking values in the simplex $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \geq 0$ for every l, $\mathbf{1}'q \leq 1\}$ for all small τ_3, and every such τ_3,*

$$E\Big[\psi(\theta_\dagger, (\varphi_\dagger, L_\star, \varrho + \tau_3\mathsf{r}))\Big] = E\Big[\psi(\theta_\dagger, \zeta_\star)\Big] \qquad \Longleftrightarrow \qquad E\Big[L_\star(\Delta U, \Delta Z) \mid \Delta Z\Big] = 0 \text{ a.s.},$$

the right-hand condition being exactly N1, both sides being well defined.

At $\mathsf{L} = 2$, *L, ℓ and D have one column, r is scalar, the three products against $\widetilde{W}$ commute, and the simplex of clause* (iii) *is the interval* $[0, 1]$.

Notes on Proposition 20. Why A1 is not in the global list, why N1 is assumed in no part, and what the finiteness of (32) and the well-posedness of clause (iii)'s two sides rest on are Remark 59.

Corollary 21 (to Proposition 20(iii): the arity of N1, and what a contracted centring leaves unpriced)**.** *N1 is a condition on a matrix and therefore $p(\mathsf{L}-1)$ scalar conditions, not p: centring one contraction of the columns is strictly weaker at every $\mathsf{L} \geq 3$, it fails inside the class of weights this paper admits, and what it leaves uncentred enters the propensity direction at the full propensity rate. Suppose A1, A1b(iv), A2, A2′(i), A2′(iii), A4(i) and B1(i), (ii), (iv.a), (iv.d) hold, and fix $\mathsf{v} \in \mathbb{R}^{\mathsf{L}-1}$ with $\mathsf{v} \neq 0$.* Do not assume *N1.* The four B1 clauses are the $\|\ell_\star\|_\infty$ bridge*: the finiteness* (34) *requires runs through the sup-norm bound on $\ell_\star$, which is reached from A2′(i) under exactly those four. They are a proper part of Proposition 20's own line, so they add no restriction on P that this corollary's parent does not already carry. Then*

(i) Arity, and the failure of the converse inside the admissible class. *N1 at $L_\star$, namely $E\big[L_\star(\Delta U, \Delta Z) \mid \Delta Z\big] = 0$ almost surely, is $p(\mathsf{L}-1)$ scalar conditions, one per entry $(j,l) \in \{1,\dots,p\} \times \{1,\dots,\mathsf{L}-1\}$, the zero being the zero matrix of $\mathbb{R}^{p\times(\mathsf{L}-1)}$. It implies the p contracted conditions $E\big[L_\star \mid \Delta Z\big]\,\mathsf{v} = 0$ almost surely.* The converse fails at every $\mathsf{L} \geq 3$, and it fails at a weight the paper's own hypotheses admit. *Let $\mathsf{v}^\perp \in \mathbb{R}^{\mathsf{L}-1}$ satisfy $\mathsf{v}^\perp \neq 0$ and $(\mathsf{v}^\perp)'\mathsf{v} = 0$, which is possible if and only if $\mathsf{L}-1 \geq 2$, let b_{N1} be the N1 anchor of Remark 52 at $\ell_\star$, and let $\mathsf{f} : \mathbb{R}^p \to \mathbb{R}^p$ be Borel with $E\|\mathsf{f}(\Delta Z)\|^2 < \infty$ and $P\big(\mathsf{f}(\Delta Z) \neq 0\big) > 0$. Then $L := L^b_{\ell_\star}$ with $b(z) := b_{\mathrm{N1}}(z) + \mathsf{f}(z)\,(\mathsf{v}^\perp)'$, a Borel map $\mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$* — the **outer** product of the $\mathbb{R}^p$-valued $\mathsf{f}(z)$ with the $\mathbb{R}^{\mathsf{L}-1}$-valued $\mathsf{v}^\perp$, and not f evaluated at $\mathsf{v}^\perp$, which is not even typed — *is, in the sense of Definition 19*(b)*, an admissible weight with $\partial_u L = \ell_\star$, it satisfies the contracted condition at v, and it violates N1 on an event of positive probability.*

(ii) What the gap costs at the population propensity. *For every bounded Borel $\mathsf{r} : \mathbb{R}^p \to \mathbb{R}^{\mathsf{L}-1}$ admissible in the sense of Proposition 20*(iii)*,*

$$E\big[L_\star(\Delta U, \Delta Z)\,\mathsf{r}(\Delta Z)\big] \;=\; E\Big[E\big[L_\star \mid \Delta Z\big]\,\mathsf{r}(\Delta Z)\Big],$$

the $\sigma(\Delta Z)$-measurable factor standing on the right, *so the derivative in the propensity reads the* whole *matrix and not its contraction against* v. *If* $P\big(E[L_\star \mid \Delta Z] \neq 0\big) > 0$ *there is such an* r *making it non-zero, and it may be taken of the entrywise form* $\mathsf{r} = \varepsilon_\varrho \operatorname{sign}\Big(\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big) e_l$ *with* $e_l \in \mathbb{R}^{\mathsf{L}-1}$ *the* l*-th standard basis vector, at any entry* (j,l) *on which* $E[L_\star \mid \Delta Z]$ *is non-zero with positive probability.*

(iii) What it costs at the estimated propensity, and why it does not vanish. *For every bounded Borel* r *as in* (ii),

$$\Big\| E\big[L_\star(\Delta U, \Delta Z)\, \mathsf{r}(\Delta Z)\big] \Big\| \;\leq\; \Big(E\big\|E[L_\star \mid \Delta Z]\big\|^2\Big)^{1/2} \Big(E\|\mathsf{r}(\Delta Z)\|^2\Big)^{1/2}, \tag{34}$$

the first norm Frobenius and the second Euclidean. If in addition $\liminf_n \Big(E\|\mathsf{r}\|^2\Big)^{-1/2} \Big\|E[L_\star \mathsf{r}]\Big\| > 0$ *along a sequence of such directions, the left side is of* exact *order* $\Big(E\|\mathsf{r}\|^2\Big)^{1/2}$. *Read at* $\mathsf{r} = \hat{\varrho} - \varrho$ — conditionally on $\sigma(\mathcal{I}_a)$, on which $\hat{\varrho}$ is frozen, so that the clause is applied at a **bounded Borel** r as it is stated and not at a random one*; boundedness holds because* (11) *confines both* $\hat{\varrho}$ *and* ϱ *to the* ε_ϱ*-interior simplex, and the expectation in* (34) *is then over a* $(\Delta U, \Delta Z)$ *draw independent of* $\mathcal{I}_a$*, by Definition 2's sample split — A4(iii) makes that order* $r_n = n^{-\alpha}$ *with* $\alpha = r/(2r + d_Z)$*, so after the* $\sqrt{n}$ *scaling of Theorem 29 the contribution is of order* $n^{1/2-\alpha}$, which diverges*: A4(ii) gives* $r > 0$ *and* $d_Z \geq 1$*, so* $\alpha = r/(2r + d_Z) < \frac{1}{2}$*, the range C5′(i) also records.*

At $\mathsf{L} = 2$ the whole statement collapses to the printed one*: there is one column,* v *is a non-zero scalar, no* $\mathsf{v}^\perp \neq 0$ *exists, so clause* (i)*'s witness cannot be built, and N1 and the contracted condition are the same* p *conditions.*

Definition 22 (The estimator: iterated linearisation with sieve Landweber–Fridman)**.** Let $\hat{\varphi}^{(0)}$ be a preliminary estimator which is a *measurable random element of* $\mathcal{V}$ *with respect to the sample it is computed on* **and which lies in** $\mathcal{S}_n$, the sieve fixed at the foot of this paragraph; and let $\hat{\varepsilon}_0 \to 0$ be a *known* deterministic sequence with $P\big(\|\hat{\varphi}^{(0)} - \varphi_\dagger\|_{L^2(\pi)} \le \hat{\varepsilon}_0\big) \to 1$ and $P\big(\|\hat{\varphi}^{(0)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}\big) \to 1$. Let $\mathcal{S}_n := \operatorname{span}(b_1, \dots, b_{L_x}) \cap \mathcal{V}$ with $b_1, \dots, b_{L_x}$ the B-spline basis of A8. Why the second display is a statement about a finite quantity, and why the sieve-membership clause is not idle, are in the supplement's discussion of the estimator.

Fix the reference density's estimate, and floor it. The adjoint (101), through which every step of (b) below is taken, carries $1/\pi(x)$ in front of the whole expression, and π is an unknown feature of P by Definition 5; the operator the analyst forms is (101) at an estimate $\hat{\pi}$. **Fix** $\varepsilon_\pi \in (0, 1]$ before the data are seen. **The construction convention** is that $\hat{\pi}$ takes values in $[\varepsilon_\pi, \varepsilon_\pi^{-1}]$ at every $x \in \mathcal{X}$; a density estimator that does not is replaced by the pointwise clipping $(\hat{\pi} \vee \varepsilon_\pi) \wedge \varepsilon_\pi^{-1}$, its metric projection in $L^2(\mathcal{X}, dx)$ onto *the set of functions with values in* $[\varepsilon_\pi, \varepsilon_\pi^{-1}]$ *Lebesgue-almost everywhere on* $\mathcal{X}$. What the floor buys, what it costs, and the one thing it does **not** establish — that running the iteration at $\hat{\pi}$ rather than at π leaves (56) in place, an **open item** stated as one at Remark 89(7) — are in the supplement's discussion of the estimator.

Normalise the scale, ***in the pairing the iteration runs in****.* Let

$$\omega_n \;:=\; 2\,(2c_C)^{1/2}\,\mathsf{c}_2^2\,c_\Pi\,\max\Big\{\|1/\pi\|_\infty^{1/2},\ \varepsilon_\pi^{-1/2}\Big\}\,h_u^{-1}h_x^{-1/2} \tag{35}$$

where c_2 and c_C are the two kernel-envelope constants (16) of C3(ii) fixes, and c_Π is the range-measure constant of B1(iv.c). This is the bound of Proposition 70(i) and (i′), deterministic and sure at every $\varphi \in \mathcal{V}$, read at the two costs *the maximum pays*: $\|1/\pi\|_\infty^{1/2}$ in $L^2(\pi)$ and $\varepsilon_\pi^{-1/2}$ in $L^2(\hat{\pi})$. (35) *does not move with* L. Run the iteration below on $\omega_n^{-1}\hat{\mathcal{A}}_\varphi$ and $\omega_n^{-1}\hat{T}_{h_x}$, so that $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\| \le 1$ surely *as an operator out of* $L^2(\pi)$ ***and*** *out of* $L^2(\hat{\pi})$. By Lemma 85 this is a choice of relaxation parameter: it changes neither the iterates nor the stopping index

and leaves the rate (56) unchanged, while inflating by ω_n^2 the **bound** (F-f) supplies on that index. The cap is sized accordingly:

$$\bar{m}_n \;\geq\; \omega_n^2\,\hat{\delta}_n^{-2}. \tag{36}$$

The stopping levels are computed before the data are touched. Fix $\mathsf{c}_{14}, \mathsf{c}_{15} < \infty$ and $\kappa := 2\mu/(2\mu+1)$, and define the deterministic *stagewise noise schedule*

$$\hat{\delta}_{n,k}^{\mathrm{eff}} \;:=\; \mathsf{c}_{14}\Big(\hat{\delta}_n \ln n \;+\; h_u^{-2}\,\hat{\varepsilon}_{k-1}^2\Big), \qquad \hat{\varepsilon}_k \;:=\; \mathsf{c}_{15}\big(\hat{\delta}_{n,k}^{\mathrm{eff}}\big)^{\kappa}, \qquad k \geq 1. \tag{37}$$

c_{14} and c_{15} must be *chosen*, because they are not identified by the data. They must dominate two analytic constants, (176)'s c_{16} and (178)'s c_{17}, which Proposition 28 Steps 2 and 3 consume and neither of which is known to the analyst. What each is built from, which way the error bites, and why the radius the second carries is the *spectral pre-image* radius R_* of (P-a.3) and **not** R, are in the supplement's discussion of the estimator. For $k = 1, \ldots, \bar{k}$:

(a) form the linear equation in $v \in \mathcal{S}_n$, $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v = -\hat{T}_{h_x}\big(\hat{\varphi}^{(k-1)}, \theta\big)$;

(b) run Landweber–Fridman on it from $v_0 = 0$, at the relaxation (35), $v_{m+1} = v_m - \omega_n^{-2}\,\hat{\mathcal{A}}^*_{\hat{\varphi}^{(k-1)}}\big[\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v_m + \hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta)\big]$, projected onto $\mathcal{S}_n$, which by Lemma 85(i) *is* the unrelaxed iteration for the normalised pair, *the adjoint being formed at the floored* $\hat{\pi}$ *fixed above and not at* π;

(c) stop at

$$\hat{m}^{(k)} \;:=\; \min\Big\{m \leq \bar{m}_n :\; \big\|\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v_m + \hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta)\big\|_{L^2(\Pi)} \;\leq\; \tau_{\mathrm{DP}}\,\hat{\delta}_{n,k}^{\mathrm{eff}}\Big\},$$

the discrepancy principle *at the stage-k level* (37), *with the convention that* $\hat{m}^{(k)} := \bar{m}_n$ *when the set is empty*;

(d) set $\hat{\varphi}^{(k)} := \hat{\varphi}^{(k-1)} + v_{\hat{m}^{(k)}}$.

Put $\hat{\varphi} := \hat{\varphi}^{(\bar{k})}$.

Two readings of clause (b)*, and they are two different random functions.* As written, (b) forms the adjoint at the floored $\hat{\pi}$: that is the **feasible** reading, the one an analyst can run, and $\hat{\varphi}$ names its output — here and in every other file. Write $\hat{\varphi}^{\pi}$ for the output of the *same* recipe with two ingredients replaced: $\hat{\mathcal{A}}^{*}_{\hat{\varphi}^{(k-1)}}$ read as (101) at the *population* π, and the projection in (b) the $L^2(\pi)$-orthogonal projection onto $\mathcal{S}_n$. That is the **infeasible** reading: no analyst can run it, π being an unknown feature of P by Definition 5. *The stage superscripts* $\hat{\varphi}^{(0)}, \ldots, \hat{\varphi}^{(\bar{k})}$ *are* ***reading-relative***, denoting the stages of whichever reading the ambient statement fixes; so $\hat{\varphi} = \hat{\varphi}^{(\bar{k})}$ read here and $\hat{\varphi}^{\pi} = \hat{\varphi}^{(\bar{k})}$ read there, and no third family of symbols is introduced. (56) is a statement about $\hat{\varphi}^{\pi}$ and Definition 24(b) plugs in $\hat{\varphi}$; what is settled and what is open about the passage from the first to the second is Remark 89(7), in five parts of which two are open, and the convention's scope and typography are in the supplement's discussion of the estimator.

Write $\mathcal{I}_*$ for the fold the iteration is run on, $\mathcal{I}_{-k}$ under Definition 24(b), $\mathcal{I}_m$ under Definition 68, and read every "measurable" below at $\sigma(\mathcal{I}_*)$.

The output is a measurable random element, and that is **verified** rather than required elsewhere — *verified at the* ***fixed*** *sieve* $\mathcal{S}_n$*, which is what the infeasible reading projects onto, and* ***pending*** *at the feasible one, whose sieve is random.* Induction on $k \leq \bar{k}$, from a $\sigma(\mathcal{I}_*)$-measurable $\hat{\varphi}^{(0)}$, gives the claim for $\hat{\varphi}^{\pi} = \hat{\varphi}^{(\bar{k})}$, and *that is not the whole of what Definition 24*(c′.1) *asks.* The verification, the two features of (c) it consumes, the three stated results that consume it, and what it does **not** reach at the feasible, random sieve are in the supplement's discussion of the estimator. So are the two load-bearing features of (c) — the level is stage-dependent, and it is deterministic and supplied from outside — together with the status of Bakushinskii's veto, and the price the schedule charges, which is a *rate* $\hat{\varepsilon}_0$ for the preliminary and not mere consistency and which hypothesis (P-a.2) of Proposition 28 carries.

Every Landweber iteration in this construction is applied to a ***linear*** *equation*; the nonlinearity

is carried by the outer index k and by nothing else. That is the entire difference from the route Remark 88 declines, and it is what makes a cone condition unnecessary.

Definition 23 (The debiased moment function, and the two matrices)**.** For a nuisance triple $\zeta = (\varphi, L, \varrho)$ with $\varphi \in \mathcal{V}$, $L : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$ Borel and $\varrho : \mathbb{R}^p \to \{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \geq 0 \text{ for every } l,\ \mathbf{1}'q \leq 1\}$ Borel, write $\widetilde{W}_l(\varrho) := \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ for $l = 1, \ldots, \mathsf{L}-1$ and $\widetilde{W}(\varrho) := \big(\widetilde{W}_1(\varrho), \ldots, \widetilde{W}_{\mathsf{L}-1}(\varrho)\big)'$, the contrast vector (1) formed at the trial propensity, and define the $\mathbb{R}^p$-valued *debiased moment function*

$$\psi(\theta, \zeta) \;:=\; \Delta Z\, \Delta U(\theta, \varphi) \;+\; L\Big(\Delta U(\theta, \varphi),\, \Delta Z\Big)\, \widetilde{W}(\varrho), \tag{38}$$

with $\Delta U(\theta, \varphi)$ as in (19). Write $\widetilde{W} := \widetilde{W}(\varrho)$ at the true propensity, which is (1) itself, $\zeta_\star := (\varphi_\dagger, L_\star, \varrho)$ as in Definition 19, and

$$\psi_\star \;:=\; \psi(\theta_\dagger, \zeta_\star) \;=\; \Delta Z\, \Delta U + L_\star(\Delta U, \Delta Z)\, \widetilde{W}. \tag{39}$$

Define

$$J \;:=\; E\Big[\Delta Z\, \Delta Z'\Big] \;+\; E\Big[\ell_\star(\Delta U, \Delta Z)\, \widetilde{W}\, \Delta Z'\Big], \qquad \Sigma \;:=\; E\Big[\psi_\star \psi_\star'\Big]. \tag{40}$$

Every product above is read at the type declaration of the instrument block: L, $L_\star$ and $\ell_\star$ carry one column per contrast, $\widetilde{W}(\varrho)$ is a column of $\mathsf{L}-1$ contrasts, so $L\widetilde{W}(\varrho)$ is $\mathbb{R}^p$-valued and the second summand of (40) is $(p \times 1)(1 \times p)$, the shape J must have. *At* $\mathsf{L} = 2$ *every display above is the printed one*: the simplex is the interval $[0, 1]$, $\mathbb{1}\{W = 1\} = W$ makes $\widetilde{W}(\varrho) = W - \varrho(\Delta Z)$ a scalar, L, $L_\star$ and $\ell_\star$ have one column, and $L\widetilde{W} = \widetilde{W}L$, $L_\star\widetilde{W} = \widetilde{W}L_\star$ and $\ell_\star\widetilde{W} = \widetilde{W}\ell_\star$ for that reason.

Notes on Definition 23. Why the second summand of (40) carries $\ell_\star$ and not $L_\star$ and is nevertheless the zero matrix, why J is not "a different matrix" from $E[\Delta Z \Delta Z']$, and how the typing of $E[\ell_\star \widetilde{W} \Delta Z']$ transfers to $\hat{J}$ are Remark 103.

Definition 24 (Cross-fitting, and the fixed anchor of the φ-step)**.** Fix $\mathsf{F} \geq 2$, not depending on n, and a partition of $\{1, \ldots, n\}$ into folds $\mathcal{I}_1, \ldots, \mathcal{I}_\mathsf{F}$ with $n_k := |\mathcal{I}_k|$ and $\lfloor n/\mathsf{F} \rfloor \leq n_k \leq \lceil n/\mathsf{F} \rceil$, chosen independently of the data. Write $\mathcal{I}_{-k} := \{1, \ldots, n\} \setminus \mathcal{I}_k$, O_i for the i-th observation vector of A0, and

$$\mathbb{E}_n[\varsigma] := \frac{1}{n} \sum_{i=1}^{n} \varsigma(O_i), \qquad \mathbb{E}_{n,k}[\varsigma] := \frac{1}{n_k} \sum_{i \in \mathcal{I}_k} \varsigma(O_i).$$

At $\mathsf{F} = 2$ this is the split of Definition 2, with $(\mathcal{I}_k, \mathcal{I}_{-k}) = (\mathcal{I}_m, \mathcal{I}_a)$.

Convention (P8)*, re-declared here.* (P8)'s *conditioning convention*, "every expectation and every O_P is conditional on the auxiliary fold", was stated **for the φ-step**, at $\sigma(\mathcal{I}_a)$, and is therefore **re-declared** rather than inherited: throughout the θ-step it is read at $\sigma(\mathcal{I}_{-k})$, with $\mathcal{I}_k$ in place of $\mathcal{I}_m$. A0 supplies the independence of $\mathcal{I}_k$ from $\mathcal{I}_{-k}$, which with clause (c′) below is what (P8)'s *freezing lemma* consumes. Which of Lemma 115(i)–(iii) reaches which site — and in the θ-step it is **never the transfer rule** — is in the supplement's discussion of the estimator.

For each k the nuisances are computed on $\mathcal{I}_{-k}$:

(a) *Propensity.* $\hat{\varrho}^{(-k)} = (\hat{\varrho}_1^{(-k)}, \ldots, \hat{\varrho}_{\mathsf{L}-1}^{(-k)})'$ satisfies A4(iii) with n replaced by n_{-k} and is truncated to the ε_ϱ-interior simplex (11) as in Definition 2, so that the fold contrast vector

$$\widehat{\widetilde{W}}^{(-k)} := \Big(\mathbb{1}\{W = 1\} - \hat{\varrho}_1^{(-k)}(\Delta Z),\ \ldots,\ \mathbb{1}\{W = \mathsf{L} - 1\} - \hat{\varrho}_{\mathsf{L}-1}^{(-k)}(\Delta Z)\Big)',$$

written $\widehat{\widetilde{W}}_i^{(-k)}$ when evaluated at observation i and $\widehat{\widetilde{W}}_{i,l}^{(-k)}$ for its l-th entry there, on the indexing of Definition 2, satisfies $\varepsilon_\varrho \leq \big|\widehat{\widetilde{W}}_{i,l}^{(-k)}\big| \leq 1$ surely *at every* $l \in \{1, \ldots, \mathsf{L}-1\}$, this is Remark 33(2) read at the fold, and it is the **per-cell** bound and **not** a bound on $\|\widehat{\widetilde{W}}_i^{(-k)}\|$, which (183) **derives** from it rather than assumes. And $\mathsf{r}_k := \hat{\varrho}^{(-k)} - \varrho$ satisfies $\|\mathsf{r}_k\|_\infty \leq 1 - 2\varepsilon_\varrho < 1$ **by A4(i)**, the supremum taken over z *and* over the $\mathsf{L} - 1$

components. Why A4(i) is named, and why this display is **not** (183) — a sup-norm and a Euclidean statement about the same object, only the first of which reads an overlap condition — are in the supplement's discussion of the estimator.

(b) *Structural function.* $\hat{\varphi}^{(-k)}$ is the estimator of Definition 22, run on $\mathcal{I}_{-k}$ **at a fixed anchor** $\theta_\circ \in \mathbb{R}^p$ **chosen before the data are seen**, and $h_k := \hat{\varphi}^{(-k)} - \varphi_\dagger$. The anchor is what removes the circularity, and it is licensed by Corollary 12 and Lemma 10 under A1b(iii). *What is not θ-free is the conditioning of the inverse problem*, so Proposition 28's hypotheses (P-a)–(P-g), (55) and A7 in the form (13) are read *at* $\theta_\circ$, and a, μ, R_* and $\hat{\delta}_n$ are $\theta_\circ$-specific.

(c) *Debiasing weight.* $\hat{L}^{(-k)}$ is an admissible weight in the sense of Definition 19(b), $\hat{L}^{(-k)} = L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$, generated by an estimator $\hat{\ell}^{(-k)}$ of $\ell_\star$ that is Borel, bounded, and continuous in u at every (u, z); $D_k := \hat{L}^{(-k)} - L_\star$. Hypothesis (R-a) of Theorem 29 strengthens that continuity to continuous differentiability, which (c′.2) uses and no other clause here does, so the sup-norm convention (42) is **not** unconditional. What the three properties buy is in the supplement's discussion of the estimator.

(c′) Measurability of the five estimated nuisances, and the reading of their sup-norms. Clause (c) states a per-realisation "Borel" and nothing more, without which the hypotheses of (R-a) and the displays (43) are not well-posed probability statements; *and the right container is not the same for all five.* (c′.1) is a requirement on the estimators, with **one** exception: for $\hat{\varphi}^{(-k)}$ the property is **verified** rather than required, at the measurability paragraph of Definition 22, and *that verification is discharged at the fixed sieve $\mathcal{S}_n$ and* ***pending*** *at the feasible, random one.*

(c′.1) Measurability, and it is two different statements, because three of the five are only ever integrated and two are evaluated pointwise at arguments their container does not control.

The three integrated nuisances, as random elements of L^2 at a fixed representative.

Each of $\hat{\varrho}^{(-k)}$, $\hat{b}^{(-k)}$ and $\hat{\varphi}^{(-k)}$ is a $\sigma(\mathcal{I}_{-k})$**-measurable random element** of, respectively, $L^2(P_{\Delta Z})$, $L^2(P_{\Delta Z})$ and $\mathcal{V}$, *a measurable map from the sample space into the space and not a family of functions indexed by the sample point*; *fix a jointly measurable representative for each, once, as A2c does for* $\ell^{\Pi}_{\star}$. *For the third this is a verification and not a requirement*, supplied by Definition 22's measurability paragraph, which is written at the fixed $\mathcal{S}_n$ — the sieve the *infeasible* reading $\hat{\varphi}^{\pi}$ projects onto, while clause (b) names the *feasible* output, for which **both** halves of the present requirement — measurability, by Remark 89(7a), and membership in $\mathcal{V}$, by Remark 89(7c) — are *pending*, and are printed as such rather than discharged by being carried as a hypothesis. Why, and the two grounds on which the choice of representative is immaterial for these three, are in the supplement's discussion of the estimator.

For $\hat{\varrho}^{(-k)}$ the choice is moreover **pinned**, and that is what clause (a)'s *sure* bound rests on: A4(iii), read at the fold by clause (a), requires product measurability of $\hat{\varrho}$'s *own pointwise function*, the (41) shape, so for that object no equivalence class is formed and no representative is chosen at all.

The two weight objects, for which the L^2 container is not merely unnecessary but wrong. What is required of $\hat{\ell}^{(-k)}$, in addition to clause (c), is that its *own* pointwise function, at every realisation and every (u, z), be product measurable:

$$(\omega, u, z) \longmapsto \hat{\ell}^{(-k)}(\omega)(u, z) \quad \text{is} \quad \sigma(\mathcal{I}_{-k}) \otimes \mathfrak{B}\Big(\mathbb{R} \times \mathbb{R}^p\Big)\text{-measurable}, \tag{41}$$

and $\hat{L}^{(-k)} = L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$ then *inherits* the same property from (31), *so* (41) *is imposed on one object and not on two*; under (R-a) the derivative inherits it too, which (c′.2) needs for its second envelope. *What is retained from the* L^2 *reading is membership*: $\hat{\ell}^{(-k)}$ and $\hat{L}^{(-k)}$ lie in $L^2(P_{\Delta U, \Delta Z})$ almost surely, the second *derived and not required*, and that is what makes the third display of (43) finite. (41) is also what (P8)'s

freezing lemma consumes for the $\hat{\ell}^{(-k)}$ *ingredient* of its composite. Why the L^2 reading fails for these two and not for the three above, both inheritances, and why the third display has a truth value, are in the supplement's discussion of the estimator.

(c′.2) The sup-norms are essential suprema in z and genuine suprema in u. For the two envelopes (R-a) prices,

$$\left\|\hat{\ell}^{(-k)}\right\|_{\infty} := \operatorname{ess\,sup}_z \left\{ \sup_{u\in\mathbb{R}} \left\|\hat{\ell}^{(-k)}(u,z)\right\| \right\}, \qquad \text{the essential supremum taken under } P_{\Delta Z}, \tag{42}$$

and likewise for $\|\partial_u \hat{\ell}^{(-k)}\|_{\infty}$. Both halves of that reading are needed and neither can be interchanged with the other. Hence $\omega \mapsto \|\hat{\ell}^{(-k)}\|_{\infty}$, and likewise $\omega \mapsto \|\partial_u \hat{\ell}^{(-k)}\|_{\infty}$, is $\sigma(\mathcal{I}_{-k})$-measurable as a map into $[0,+\infty]$, which is what (R-a) needs to price both envelopes as $O_P(1)$. The measurability of the inner supremum and of the **outer** essential supremum, and why the essential supremum must be taken in z and **not** in (u,z), are in the supplement's discussion of the estimator.

(d) The fold reading of the maintained nuisance-rate clauses, declared once. A4(iii) is read at each fold in (a); the **four** clauses of A5 that price an *estimated* nuisance are read the same way, and every later citation of them is a citation of the corresponding display here:

$$\max_{k\le \mathsf{F}} \left\|h_k\right\|_{L^2(\pi)} = O_P(\delta_n), \qquad \max_{k\le \mathsf{F}} \left\|h_k\right\|_{H^s} = O_P(1), \qquad \max_{k\le \mathsf{F}} \left\|\hat{\ell}^{(-k)} - \ell_\star\right\|_{L^2(P)} = O_P(\epsilon_n), \tag{43}$$

being A5(ii)'s first half, A5(vi), and A5(iv)'s first half respectively, *the third in force precisely when (R-e.1) of Theorem 29 is the branch taken, since under (R-e.2) no rate for $\hat{\ell}^{(-k)}$ is assumed at all. The second display is finite at every realisation only* ***once*** $\hat{\varphi}^{(0)}$ ***is in*** $\mathcal{S}_n$*, a premise stated here because this is a* ***definition*** *and inherits no hypothesis list*: Definition 22(d) puts the path in $\mathcal{S}_n$ if and only if its *starting point* is

there, (8) places $\mathcal{S}_n$ in H^s and H1 places $\varphi_\dagger$ there. *Every result that reads* (43) *supplies the premise*, (P-a.1) being in group (T-5) of Theorem 29. *The indexing is not cosmetic and it is not free either*: A5(ii), (iv) and (vi) are stated at a **single** estimated nuisance, and for the first two the cost is nil, Theorem 29 Step 0 *deriving* them from (T-5) with F fixed and $n_{-k} \asymp n$. *For the third it is a strengthening of A5(iv) in form*, of the same order for fixed F but a strengthening nonetheless, and it is therefore carried explicitly by hypothesis (R-e.1) of Theorem 29 rather than read into A5(iv). *The fourth clause is A5(vi′)*, which prices the *stagewise* Sobolev stability of the φ-step, $P(\max_{j\le\bar{k}} \|\hat{\varphi}^{(j)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}) \to 1$ — $\bar{k}$ a *stage* index and not a fold index — and is read at each fold complement exactly as the other three are. What that costs, and why the radius R must nonetheless serve all F fold complements at one value, are in the supplement's discussion of the estimator.

Put $\hat{\zeta}^{(-k)} := (\hat{\varphi}^{(-k)}, \hat{L}^{(-k)}, \hat{\varrho}^{(-k)})$ and, for $\theta \in \mathbb{R}^p$,

$$\bar{\psi}_n(\theta) \;:=\; \frac{1}{n}\sum_{k=1}^{\mathsf{F}} \sum_{i\in\mathcal{I}_k} \psi\Big(\theta, \hat{\zeta}^{(-k)}\Big)(O_i), \qquad \hat{\psi}^{(-k)} := \psi\Big(\theta_\dagger, \hat{\zeta}^{(-k)}\Big). \tag{44}$$

Since F *is fixed,* $n_k/n \to \mathsf{F}^{-1}$ *and* $n_k \asymp n$; the constant F is carried explicitly in every bound rather than absorbed.

Definition 25 (The preliminary and the one-step estimator)**.** With the notation of Definition 24 define

$$\hat{\theta}^{(0)} \;:=\; \Big(\mathbb{E}_n\big[\Delta Z \Delta Z'\big]\Big)^{-1} \frac{1}{n}\sum_{k=1}^{\mathsf{F}} \sum_{i\in\mathcal{I}_k} \Delta Z_i \Big(\Delta Y_i - \Delta\hat{\varphi}_i^{(-k)}\Big), \tag{45}$$

on the event that $\mathbb{E}_n[\Delta Z \Delta Z']$ is nonsingular, and $\hat{\theta}^{(0)} := 0$ otherwise. Define the *cross-fitted Jacobian*, for $\theta \in \mathbb{R}^p$,

$$\hat{J}(\theta) \;:=\; \frac{1}{n}\sum_{k=1}^{\mathsf{F}} \sum_{i\in\mathcal{I}_k} \Big[\Delta Z_i \Delta Z_i' + \hat{\ell}^{(-k)}\Big(\Delta U_i(\theta, \hat{\varphi}^{(-k)}), \Delta Z_i\Big)\, \widehat{\widetilde{W}}_i^{(-k)}\, \Delta Z_i'\Big], \tag{46}$$

so that $\partial_\theta \bar{\psi}_n(\theta) = -\hat{J}(\theta)$ identically, and put $\hat{J} := \hat{J}(\hat{\theta}^{(0)})$. *The order of the three factors is what the identity forces*: by (38) the θ-derivative of $\hat{L}^{(-k)}(\Delta U_i(\theta,\cdot), \Delta Z_i)\, \widehat{\widetilde{W}}_i^{(-k)}$ is $-\hat{\ell}^{(-k)}(\cdot)\, \widehat{\widetilde{W}}_i^{(-k)}\, \Delta Z_i'$, which is $(p \times (\mathsf{L}-1))((\mathsf{L}-1)\times 1)(1\times p)$. At $\mathsf{L}=2$ the three factors commute in the first two positions. The *one-step estimator* is

$$\hat{\theta} \;:=\; \hat{\theta}^{(0)} \;+\; \hat{J}^{-1}\, \bar{\psi}_n\Big(\hat{\theta}^{(0)}\Big) \tag{47}$$

on the event $\{\hat{J}$ nonsingular$\}$, and $\hat{\theta} := \hat{\theta}^{(0)}$ otherwise. The variance estimator is

$$\hat{\Sigma} \;:=\; \frac{1}{n}\sum_{k=1}^{\mathsf{F}} \sum_{i\in\mathcal{I}_k} \psi\Big(\hat{\theta}, \hat{\zeta}^{(-k)}\Big)(O_i)\; \psi\Big(\hat{\theta}, \hat{\zeta}^{(-k)}\Big)(O_i)', \qquad \hat{V} \;:=\; \hat{J}^{-1}\hat{\Sigma}\,\hat{J}^{-1\prime}, \tag{48}$$

the last **on the event $\{\hat{J}$ nonsingular$\}$, with $\hat{V} := 0$ otherwise**.

Remark 26 (How Definition 25's construction is read). Three clauses, each of which stood inside the definition. None of it is construction, and the estimator Definition 25 defines is unchanged by its removal.

(1) *That identity uses **one** derivative and no more*: $\partial_u \hat{L}^{(-k)} = \hat{\ell}^{(-k)}$, which Definition 19(b) supplies at every point under the three properties Definition 24(c) requires of $\hat{\ell}^{(-k)}$, together with $\theta \mapsto \Delta U_i(\theta,\cdot)$ being affine by (19) and the sum being finite. No second derivative is claimed here. $\partial_u^2 \hat{L}^{(-k)} = \partial_u \hat{\ell}^{(-k)}$ is used at Theorem 29 Step 1, for Taylor's theorem in θ, and it is hypothesised *there*, at (R-a), *a definition should not carry a differentiability claim that a theorem assumes. That step is not the only reader of the envelope $\|\partial_u \hat{\ell}^{(-k)}\|_\infty$, and no claim of exhaustiveness is made here*: Lemma 93 Step 4 and Lemma 95 Step 5 read it too, on the right of an inequality, as (R-a)'s own reading records. The draft this sentence replaces said "only" and named one of three, while that reading named two of the same three; the set is one and the two claims covered disjoint halves of it.

(2) *The fallback is not optional.* $\hat{\theta}^{(0)}$ is given a value off $\{\mathbb{E}_n[\Delta Z \Delta Z']$ nonsingular$\}$ and

$\hat{\theta}$ is given one off $\{\hat{J}$ nonsingular$\}$; $\hat{V}$ inverts the *same* $\hat{J}$ and needs one too. That matters because Theorem 29's **second** conclusion, $\hat{V} \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$, and the studentised statement that follows it are statements about $\hat{V}$ at every n, and a convergence in probability is not well posed for a quantity that is undefined on an event of positive probability. *The choice of value is immaterial and the zero matrix is chosen because it is the one value that needs no further object*: by Lemma 95(i), $P(\hat{J}$ nonsingular$) \to 1$, so the event on which the fallback is active has probability tending to zero and no conclusion of Lemma 95(ii) or of Theorem 29 changes. *In a finite sample the fallback is not innocuous and should not be reported as a variance estimate*: $\hat{V} = 0$ makes the studentised statistic undefined, which is the honest report on that event and is the reason the value is stated rather than left to the reader.

(3) One step and not a root, and the ground is the **sample** criterion and not the population one. The realised map $\theta \mapsto \bar{\psi}_n(\theta)$ is *not* affine: by (38) θ enters $\hat{L}^{(-k)}$ through its first argument, and $\widehat{\widetilde{W}}^{(-k)}$ is not conditionally centred, so $\bar{\psi}_n(\theta) = 0$ need not have a unique root at any n. *The ground is* ***not*** *that no global rank condition is available*: by Lemma 91(iv) and (125) the population moment $\theta \mapsto E[\psi(\theta, \zeta_\star)]$ is affine with gradient $-E[\Delta Z \Delta Z']$, so a global rank condition *is* available, it is A1b(iv), and $\theta_\dagger$ is its unique root on all of $\mathbb{R}^p$, which says nothing about the sample criterion at estimated nuisances. *What the block supplies is a preliminary with a rate* (Lemma 92), and a one-step from a preliminary with a rate is exactly the construction that converts a rank condition on the population Jacobian into a limit distribution without solving anything; the cross-fitted form is that of Chernozhukov et al. (2018), and the two classical anchors for the one-step itself are Newey and McFadden (1994, §3.4) and Vaart (1998, Ch. 5). *The Jacobian is evaluated at* $\hat{\theta}^{(0)}$ *and not at* $\hat{\theta}$, which is what makes (47) a closed-form estimator and what makes the algebra of Theorem 29 Step 1 exact rather than asymptotic.

Three objects in what follows are constructed in the supplement and are recalled here so that the statement can be read where it is printed. $\mathcal{B}_{\mathrm{sym}} := \frac{1}{2}(\mathcal{B} + \mathcal{B}^*)$ is the symmetric part of $\mathcal{B}$; $q := E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2] \in \mathbb{R}^p$, the $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued $\partial_u \ell_\star$ contracted against the contrast vector $\widetilde{W}$ of (1) and then conditioned; and, for an integrable g, $\sigma_g(x) := E[g \mid X_2 = x] f_{X_2}(x) + E[g \mid X_1 = x] f_{X_1}(x)$, taken coordinatewise for a vector-valued g, so that σ_q is that object at $g = q$ and is $\mathbb{R}^p$-valued. The multiplier is written out as $\vartheta'\sigma_q/\pi$ throughout and is given no abbreviation, because $\mathcal{B}_{\mathrm{mult}}$ is multiplication *by* it and an abbreviation would collapse the distinction. At $\mathsf{L} = 2$ the contraction is a product of scalars in the second factor, $\partial_u \ell_\star \widetilde{W} = \widetilde{W} \partial_u \ell_\star$, and q, σ_q and $\vartheta'\sigma_q/\pi$ are the objects earlier rounds printed.

Proposition 27 (The second-order remainder on the ball: the sharp constant, and the exact δ^2 order under O-16(a))**.** *The supremum of the second-order remainder over a ball of radius δ in the structural direction is exactly $\frac{1}{2}\delta^2$ times the norm of the compressed symmetrised form, and it is bounded below by $\frac{1}{2}\delta^2 \|\vartheta'\sigma_q/\pi\|_\infty$; so, provided the form does not degenerate, no bound of the form $o(\delta^2)$, and in particular no bound in the weak norm $\|\mathcal{A}_{\mathrm{cond}} \cdot\|_{L^2(\Pi)}$, holds uniformly on that ball. Suppose the hypotheses of Lemma 55 hold, and for clause* (iv) *in addition* ***C2′(i)****. Fix $\vartheta \in \mathbb{R}^p$ with $\|\vartheta\| = 1$, and let the* second-order remainder in the structural direction *at $h \in \mathcal{V}$ be*

$$\tfrac{1}{2}\,\vartheta'\partial^2_{\tau_1\tau_1}\Psi(0) \;=\; \tfrac{1}{2}\,E\Big[\vartheta'\,\partial_u\ell_\star\,\widetilde{W}\,(\Delta h)^2\Big] \;=\; \tfrac{1}{2}\,\Big\langle \mathcal{B}h, h\Big\rangle_{L^2(\pi)}, \tag{49}$$

which is Lemma 56(i)*, read at the admissible direction pair $(D, \mathsf{r}) = (0, 0)$, contracted in ϑ and transferred to $L^2(\pi)$ by Lemma 55*(iii)*.* Two non-degeneracy hypotheses are carried by name below, and this display is where they are defined*: at the fixed ϑ above,*

$$\text{O-16(a)}: \quad \exists\, h \in \mathcal{V} \;\; \textit{with} \;\; \vartheta' E\Big[q\,(\Delta h)^2\Big] \neq 0; \qquad\qquad \text{O-16(b)}: \quad \vartheta'\sigma_q \not\equiv 0.$$

Both are hypotheses and neither is a consequence of O-11*, which is Remark 65*(3)–(4)*;* that

(b) implies (a), and that the converse fails, are claims of clause (iii) below and of Remark 65(3) respectively, not background assumed here. *Then, for every $\delta > 0$:*

(i) The sharp constant.

$$\sup_{h\in\mathcal{V},\ \|h\|_{L^2(\pi)}\le\delta} \tfrac{1}{2}\left|\left\langle \mathcal{B}h, h\right\rangle_{L^2(\pi)}\right| \;=\; \tfrac{1}{2}\,\delta^2\left\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\right\|, \tag{50}$$

a supremum that need not be attained.

(ii) Bracketing. *The constant in* (50) *obeys*

$$\left\|\vartheta'\sigma_q/\pi\right\|_\infty \;\le\; \left\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\right\| \;\le\; \left\|\mathcal{B}_{\mathrm{sym}}\right\| \;\le\; \left\|\mathcal{B}\right\|, \tag{51}$$

where the first inequality holds under O-16(b) *and the outer two hold always.*

(iii) Positivity, which is a hypothesis and is stated as one. Under ***O-16*****(a)** *the constant in* (50) *is strictly positive, so* $\sup_{\|h\|\le\delta}\frac{1}{2}|\langle\mathcal{B}h,h\rangle| \asymp \delta^2$ *with a constant free of* δ*, and no bound of the form* $o(\delta^2)$ *as* $\delta\downarrow 0$ *holds uniformly over* $\{h\in\mathcal{V}:\|h\|_{L^2(\pi)}\le\delta\}$*. Under* ***O-16*****(b)** *the constant is at least* $\|\vartheta'\sigma_q/\pi\|_\infty > 0$*, and O-16*(b) *implies O-16*(a).

(iv) No weak-norm bound. *Under O-16*(b) *there is no finite* λ *with*

$$\tfrac{1}{2}\left|\left\langle \mathcal{B}h, h\right\rangle_{L^2(\pi)}\right| \;\le\; \lambda\left\|\mathcal{A}_{\mathrm{cond}}h\right\|^2_{L^2(\Pi)} \qquad \text{for all } h\in\mathcal{V}; \tag{52}$$

nor with $\|\mathcal{A}_{\mathrm{cond}}h\|^2$ *replaced by* $\|h\|_{L^2(\pi)}\,\|\mathcal{A}^*_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}}h\|_{L^2(\pi)}$.

Notes on Proposition 27. How its hypothesis list is read is Remark 64; what it claims and the four things it does not are Remark 65.

Proposition 28 (The rate for the φ-step)**.** *Suppose A0, A1, A1b(iii)–(iv), A4(ii)–(iii), A5(vi′), B1(i)–(v), C1(ii), C2′, C3 and C5′(i) hold. Suppose H1 holds,* for the Sobolev scale H^s that Definition 80, A5(vi′), (P-c) and (P-g) are written in, and for A2's Picard form; the

identity $\mu = s/(2a)$ is **not** assumed and is **not** used, μ being primitive at (P-a.3). *Suppose A8 holds in its* basis *clause, $b_1, \ldots, b_{L_x}$ is a B-spline basis of degree exceeding $s - \frac{1}{2}$ on a quasi-uniform mesh over $\mathcal{X}$, and in its* upper end *$L_x \ll n^{1/2}$, but **not** in its lower end $L_x \gg n^{1/(2s)}$; and suppose $p_K \geq s - \frac{1}{2}$, which with $p_K = 8$ caps $s \leq \frac{17}{2}$ and therefore, against (54)'s floor,* restricts $\alpha < \frac{68}{145}$. *And suppose*

(P-a) (preliminary, and the **spectral** source condition at the compression) *Three clauses, all three* envelopes holding with probability tending to one *rather than O_P statements, Remark 87 says why the distinction is load-bearing.*

(P-a.1) (preliminary, H^s, and $\sigma(\mathcal{I}_*)$-measurable) *$\hat{\varphi}^{(0)}$ is a $\sigma(\mathcal{I}_*)$-measurable random element of $\mathcal{V}$, in the sense and with the fold $\mathcal{I}_*$ of Definition 22; $\hat{\varphi}^{(0)} \in \mathcal{S}_n$; and $P\big(\|\varphi_\dagger - \hat{\varphi}^{(0)}\|_{H^s} \leq \mathsf{R}\big) \to 1$, with R the* same *deterministic radius as in Definition 80, A5(vi′), (P-c) and (P-g).* A5(vi′), (P-c) and (P-g) are named here as instances and this clause maintains no list*: Definition 80 carries the enumeration — seven sites, this clause itself the third of them — and this points at it.* The distinction is not clerical here, because this clause is a **hypothesis** *and a reader must act on it: a radius chosen to clear only the sites named above need not clear (121) or Lemma 81 Step 4, which are the remaining two, and the existential is over **one** R meeting all seven at once.*

(P-a.2) (preliminary, rate) *$P\big(\|\varphi_\dagger - \hat{\varphi}^{(0)}\|_{L^2(\pi)} \leq \hat{\varepsilon}_0\big) \to 1$ for a known deterministic $\hat{\varepsilon}_0 = O(n^{-\varsigma_0})$ whose exponent clears the repelling fixed point of the stagewise recursion,*

$$\varsigma_0 \;>\; \frac{\kappa\,\alpha}{4\,(2\kappa - 1)}, \qquad \kappa = \frac{2\mu}{2\mu + 1}, \quad \alpha = \frac{r}{2r + d_Z}$$

(P-a.3) (source condition, stated **spectrally** and **at the compression**) *There are $\mu > 0$ and a finite R_*, neither depending on n, such that with*

$$\hat{\mathcal{P}}_{\varphi}^{\mathcal{S}} \;:=\; P_{\mathcal{S}_n}\,\hat{\mathcal{A}}_{\varphi}^{*}\hat{\mathcal{A}}_{\varphi}\big|_{\mathcal{S}_n} \;:\; \mathcal{S}_n \to \mathcal{S}_n \tag{53}$$

the compression *of* $\hat{\mathcal{A}}_{\varphi}^{*}\hat{\mathcal{A}}_{\varphi}$ *to* $\mathcal{S}_n$,

$$P\Big(\ \bar{v}_{\dagger}^{(k)} = \big(\hat{\mathcal{P}}_{\hat{\varphi}^{(k-1)}}^{\mathcal{S}}\big)^{\mu}\varsigma^{(k)} \quad \textit{with} \quad \big\|\varsigma^{(k)}\big\|_{L^2(\pi)} \le \mathsf{R}_{*} \quad \textit{for every } k \le \bar{k}\ \Big) \ \longrightarrow\ 1,$$

where $\bar{v}_{\dagger}^{(k)} := P_{\mathcal{S}_n}\big(\varphi_{\dagger} - \hat{\varphi}^{(k-1)}\big)$ *and* $\bar{k}$ *is the stage count of Definition 22,* which the proof's own construction puts at $\bar{k} \ge 2$, so this clause is **never** a one-stage condition and is always read at a $\bar{v}_{\dagger}^{(k)}$ formed at a realised iterate as well as at one formed at the preliminary.

(P-b) (no hypothesis). *There is* ***no*** *hypothesis (P-b): the list runs* (P-a), (P-c)–(P-g)*;*

(P-c) (sieve approximation, a **lower** end for L_x) $\mathsf{R}\,L_x^{-s} = O\big(h_u\,\hat{\delta}_n\big)$*;*

(P-d) (uniformity, an *upper* end for L_x) (123)*, so that Lemma 81 clause (b) is available.* (P-c) and (P-d) are the two ends of one window and must be read together*: with* $\hat{\delta}_n = n^{-\alpha}$ *and* $h_u \asymp \hat{\delta}_n^{1/8}$ *they are* $L_x \gg n^{9\alpha/(8s)}$ *and* $L_x = o\big(n^{1-2\alpha}/\ln n\big)$*, so the window is non-empty if and only if*

$$s \ >\ \frac{9\alpha}{8\,(1-2\alpha)}; \tag{54}$$

(P-e) (stopping) *A7 in the amended form* (13)*, with* $\tau_{\mathrm{DP}} > 2$*;*

(P-f) (the linearisation remainder is absorbed) $\delta_n^2 = O\big(h_u^2\,\hat{\delta}_n \ln n\big)$*, a hypothesis in which a bandwidth appears explicitly, as (P-c) and (P-g) also do, and no claim of exhaustiveness is made in naming it so;*

(P-g) (x-smoothing bias) $\mathsf{R}\,h_x^{s-1/2} = O\big(h_u\,\hat{\delta}_n\big)$*, an* upper *bound on* h_x*, which C5′ does not carry;*

and suppose finally

$$\mu \ >\ \tfrac{5}{6}. \tag{55}$$

Then there is a finite $\bar{k} \ge 2$*, not depending on* n *and computable from* $(\mu, \varsigma_0, \alpha)$*, such that*

*the **infeasible** reading $\hat{\varphi}^{\pi}$ of Definition 22 — clause* (b) *of that definition with the adjoint formed at the population π and the projection $L^2(\pi)$-orthogonal onto $\mathcal{S}_n$, which is the object every step of the proof below runs, and **not** the estimator $\hat{\varphi}$ that clause instructs, whose adjoint is formed at the floored $\hat{\pi}$ — satisfies*

$$\left\|\hat{\varphi}^{\pi} - \varphi_{\dagger}\right\|_{L^2(\pi)} = O_P(\delta_n), \qquad \delta_n := \left(\hat{\delta}_n \ln n\right)^{2\mu/(2\mu+1)}, \tag{56}$$

and $\delta_n = o_P(n^{-1/4})$ whenever (7) *holds.* What is **not** claimed is that $\hat{\varphi}$ satisfies (56)*, and no hypothesis of this proposition reaches that passage: it is the open item Remark 89*(7)*, whose parts* (7c) *and* (7d) *are open.* In particular this proposition does **not** discharge A5(ii)*, which is stated at $\hat{\varphi}$; Theorem 29 carries that clause on its own line, at* (T-4)*, and carries it as a hypothesis.*

Notes on Proposition 28. What its hypotheses carry, where the threshold (55) comes from and why (P-a.3) is the load-bearing one are Remark 90; why (P-a.3) is spectral and at the compression is Remark 86; why (P-a)'s clauses are w.p.→ 1 envelopes is Remark 87; the nonlinear route it declines is Remark 88; and what it does *not* claim is Remark 89. *The last of those is not a pointer to fine print*: the conclusion above is scoped to the infeasible reading $\hat{\varphi}^{\pi}$, and Remark 89(7) is where the passage to the estimator Definition 24(b) actually plugs in is divided into five parts — two settled outright, one settled with an undischarged rider, two open.

Theorem 29 (Asymptotic normality of the cross-fitted one-step estimator)**.** *The one-step estimator is $\sqrt{n}$-consistent and asymptotically normal with the sandwich variance, under mild ill-posedness and a propensity smooth enough to put the structural rate strictly inside $n^{-1/4}$. Suppose*

(T-1) (sampling, model, identification) *A0; A1; A1b(i), A1b(iii), A1b(iv);*

(T-2) (the representer and the weight) *A2; A2′(i.a), A2′(i.b), A2′(iii); N1 at $L_\star$,* which is

$p(\mathsf{L}-1)$ scalar conditions and **not** a condition on a contraction of the $\mathsf{L}-1$ columns *(Corollary 21),* and which, together with (R-a), pins the anchor A2′(iii) leaves free*:* (R-a) *forces* $\hat{b}^{(-k)} \to b_\star$*, so* $b_\star$ *must be the N1 anchor of Remark 52 and the construction must target it;*

(T-3) (index set, densities, propensity) *B1(i), (ii), (iv); C1(ii); C2′(i); A4(i), A4(ii), A4(iii);*

(T-4) (rates) *A5(i) and A5(ii) in the form (137); A5(iii); and A5(vi) with* $s > \frac{1}{2}$*, read at each fold as in Definition 24(d).* A5(ii) is **assumed** here and is not delivered by (T-5)*: it is stated at* $\hat{\varphi}$*, the estimator Definition 24*(b) *plugs in, while Proposition 28 concludes at the infeasible* $\hat{\varphi}^\pi$*, and the passage between them is Remark 89*(7)*, two of whose five parts are open.* This theorem is therefore conditional on that clause, and says so on its own hypothesis line rather than in a remark*;*

(T-5) (the φ-step) *the hypotheses of Proposition 28, read at the anchor* $\theta_\circ$ *of Definition 24(b), H1; A5(vi′); A8's basis and upper-end clauses;* $p_K \geq s - \frac{1}{2}$*, which against (54)'s floor gives* $\alpha < \frac{68}{145}$*; (P-a.1)–(P-a.3), (P-c)–(P-g); and (55),* $\mu > \frac{5}{6}$ *strictly, so that (56) holds on each* $\mathcal{I}_{-k}$ — at the **infeasible** reading $\hat{\varphi}^\pi$ of Definition 22 that (T-4) names, and not at the $\hat{\varphi}^{(-k)}$ of Definition 24(b)*; so this group supplies A5(i) and the fold reading of A5(vi′) and does* ***not*** *supply A5(ii);*

and suppose in addition

(R-a) (the weight is built as an anchored antiderivative, with a bounded derivative) *for each* k*,* $\hat{L}^{(-k)} = L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$ *in the sense of Definition 19(b), with* $\hat{\ell}^{(-k)}$ *continuously differentiable in* u *at every point and bounded,*

$$\max_{k\leq \mathsf{F}} \left\{ \left\| \hat{\ell}^{(-k)} \right\|_\infty + \left\| \partial_u \hat{\ell}^{(-k)} \right\|_\infty \right\} = O_P(1), \qquad \max_{k\leq \mathsf{F}} E \left\| \hat{b}^{(-k)}(\Delta Z) \right\|^4 = O_P(1),$$

and $\max_{k\leq \mathsf{F}} \|\hat{L}^{(-k)} - L_\star\|_{L^2(P)} = o_P(1)$*;*

(R-b) (fourth moments, and exactly two of them) $E\|\Delta Z\|^4 < \infty$ *and* $E[\Delta U^4] < \infty$*;*

(R-c) (the $\varrho \times L$ cross term) ***either*** (c.1) $D_k = \hat{L}^{(-k)} - L_\star$ *satisfies N1′ for every* k, ***or*** (c.2) $\max_k \|D_k\|_{L^2(P)} = O_P(\epsilon_n^L)$ *with* $r_n\, \epsilon_n^L = o(n^{-1/2})$;

(R-d) (no hypothesis). *There is* ***no*** *hypothesis (R-d): the list runs (R-a), (R-b), (R-c), (R-e), on the* (P-b) *pattern above;*

(R-e) (the first-order bias) ***either*** (e.1) *A5(iv) read at each fold, i.e. the third display of* (43) *together with* $\delta_n \epsilon_n = o_P(n^{-1/2})$, ***or*** (e.2) *the hypotheses of Lemma 60*(iii)*, the sieve solve sets term* (II) *of* (78) *to zero, A8 holds with a B-spline basis of degree exceeding* $s - \frac{1}{2}$, *A5(vi) holds and* (79) *holds, together with* $\max_k \|\hat{\ell}^{(-k)} - \ell_\star\|_{L^2(P)} = o_P(1)$, ***and*** *with*

$$s \;>\; \frac{1}{2\,(1-2\alpha)}. \tag{57}$$

Then, with $\hat{\theta}$ *of* (47) *and* J, Σ *of* (40),

$$\begin{aligned} \sqrt{n}\left(\hat{\theta} - \theta_\dagger\right) \;&=\; J^{-1}\,\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_\star(O_i) \;+\; o_P(1) \\ &\xrightarrow{d}\; \textit{the centred Gaussian law on } \mathbb{R}^p \textit{ with covariance matrix } J^{-1}\Sigma J^{-1\prime}, \end{aligned} \tag{58}$$

and $\hat{V} \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$, *so that for* ***every*** $\vartheta \in \mathbb{R}^p$ *with* $\|\vartheta\| = 1$,

$$\sqrt{n}\,\vartheta'\left(\hat{\theta} - \theta_\dagger\right) \Big/ \left(\vartheta'\hat{V}\vartheta\right)^{1/2} \;\xrightarrow{d}\; \textit{the standard Gaussian law on } \mathbb{R},$$

the ratio being read on $\{\vartheta'\hat{V}\vartheta > 0\}$, *an event of probability tending to one.*

Notes on Theorem 29. What each hypothesis group does and does not carry is Remark 100; how the conclusion is read is Remark 101; what the theorem covers and the six things it does not are Remark 98; the clause-by-clause usage audit is Remark 99; the logarithm in (56) is Remark 97.

Corollary 30 (T2′: the rate floor, without (7))**.** *The same decomposition delivers a rate*

whether or not the frontier holds; on the sieve branch, and only there, every exponent in that rate is a function of the primitives. *Suppose (T-1)–(T-3), (T-5), (R-a), (R-b) hold, together with A5(i), A5(vi) with* $s > \frac{1}{2}$, $\delta_n \to 0$, $\epsilon_n \to 0$, $r_n \to 0$, *and (R-c.2), (R-e.1) with the two products left* unrestricted. *Then*

$$\left\|\hat{\theta} - \theta_\dagger\right\| \;=\; O_P\Big(n^{-1/2} \;+\; \delta_n\epsilon_n \;+\; \delta_n^2 \;+\; r_n\delta_n \;+\; r_n\epsilon_n^L\Big). \tag{59}$$

Under (R-e.2) in place of (R-e.1) the term $\delta_n\epsilon_n$ *is replaced by* $O_P(n^{-1/2}\delta_n) + O_P(\mathsf{R}\, L_x^{-s})$, *and no rate for* $\hat{\ell}$ *enters* (59) *at all. Finally,* (58) *is the special case of* (59) *in which every term after the first is* $o(n^{-1/2})$.

Notes on Corollary 30. How much of (59) is in primitives, and which branch buys which answer, is Remark 102.

Discussion of the assumptions

This section carries the material the article does not print: for each numbered condition of the assumption block, the interpretation, the provenance, the record of what it does and does not restrict, and – where a condition was retired – the ground on which it was retired. The statements themselves are printed above, verbatim and unchanged. Headings below name the condition; the text under each is reproduced verbatim from the consolidated block.

The three registers: how a citation of the assumption block is read. A tag is named in more places than three; the three below are the three that have produced mis-citations and been repaired against. *No claim of exhaustiveness is made, and that is deliberate* — the two positions left out are named at the end of this paragraph, and one of them is where the under-charging rule the assumption block enforces actually lives. *On a result's hypothesis line* — inside a "Suppose . . . hold" — a citation is a **carrier**: naming a tag carries every condition under it, and naming a clause carries that clause and no other. *Inside a definition* a citation is a **pointer** *when it locates*: Definition 5's "$\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ as in Assumption B1(iv)" names the object it uses and imposes nothing, so a result whose proof reads B1(iv.d) must carry that clause on its own line and cannot borrow it from the definition. *It is* ***not*** *a pointer when it is constitutive*, and two definitions are: A2b *is* "admits a version satisfying A2′(i.a)–(i.b)", the citation being its whole content, and Definition 2 builds (11) out of the constant A4(i) declares. *Ask what survives deleting the citation*: a pointer's object is defined elsewhere and survives; a constitutive citation's object does not exist without it. *The third register is neither.* Several conditions have a **declarative half**: the part that introduces a symbol, a type, a carrier set or a construction, as against the part that restricts it. *A citation of a declarative half is a* ***declaration*** *and carries no hypothesis*, because a declarative half is **standing**: it is in force at every statement in this corpus, named or not, since without it the statement has no objects to be about. It is therefore written *outside* the "Suppose . . . hold"

list, and never with the word *clause*.

Where the split falls, so that it is checkable and not a matter of taste. Cut the fragment at the point where it stops **introducing** and starts **restricting**. What introduces is the declaration; the test that what remains is a **clause** is that *there is a spelling of the assumption block, with the declaration in force, under which every symbol used anywhere in the corpus still has a referent and the remainder is false.* B1(iii) is the model and is already written this way: $\mathcal{T} \subseteq \mathbb{R}^p$ introduces the index set, $\mathrm{Leb}(\mathcal{T}) > 0$ restricts it and can fail with $\mathcal{T}$ still named, so the first is a declaration and the second a clause. B1(iv)'s **stem** $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ is all introduction and leaves no remainder — it is what makes Definition 5's three-factor pairing, the masses $\varpi_{\mathsf{w}}(l)$ and the third argument of $\ell^{\Pi}(u, t, l)$ writable at all — so it is a declaration entire, while B1(iv.a) and B1(iv.d) are clauses: negate either and Proposition 36 is still a well-formed claim, with a wrong constant. H1's $H^s(\mathcal{X})$, the *classical* scale read for non-integer s as the interpolation space and carrying no boundary condition, is a declaration on the same ground, while H1's rank, its decay and (5) are clauses.

Three fragments are split that were not split before, and in each the remainder is a clause and is carried by the results that read it. **A0** introduces $\mathcal{X} \subseteq \mathbb{R}$ as the carrier of the regressor, with $\underline{x} := \inf \mathcal{X}$ and $\bar{x} := \sup \mathcal{X}$ — *spelled that way from 2026-08-26, because as endpoints of an interval the two symbols lose their referents exactly when the restriction is negated, which is the trap A4(i) fell into* — and *restricts* it to a compact interval: a fat Cantor $\mathcal{X}$ leaves every symbol named and the restriction false, so the restriction is a clause, and Lemma 67 accordingly carries it. **A4(i)** *fixes* $\varepsilon_\varrho \in (0, 1/\mathsf{L}]$ — which is what Definition 2 builds (11) from, and all that definition reads of A4(i), (11) being non-empty on $\varepsilon_\varrho \leq 1/\mathsf{L}$ alone — and *imposes* $\varrho_l(\Delta Z) \geq \varepsilon_\varrho$, which is the overlap condition and is a clause at each of the results carrying it. *The upper end of that range is* ***weak*** *from 2026-08-27 and was strict before; why, and why the change is forced by the paragraph after next rather than by taste, is recorded there.* **A8** names $b_1, \ldots, b_{L_x}$ and $\beta_1, \ldots, \beta_{L_u}$ as the two bases, which is what Definition 22 builds $\mathcal{S}_n$ from, and

requires of them a B-spline of degree exceeding $s - \frac{1}{2}$ on a quasi-uniform simple-knot mesh inside $n^{1/(2s)} \ll L_x \ll n^{1/2}$: restrictive, refutable at A8's own $s = 1.7$ witness, and correctly carried on Lemma 81's line.

One family needs the rule at a finer grain, because there the split falls ***inside*** *a numeric range: an interval declared for a constant the* ***analyst*** *chooses.* Three such constants are printed in this corpus — A4(i)'s ε_ϱ, Definition 22's ε_π and A7's τ_{DP} — in three different registers, which is a defect unless one rule decides all three, and *is* a defect until the rule is printed. *The rule, and it is the split-first test applied to a range rather than to a sentence.* A declared range is a declaration exactly as far as its **referent boundary**: the widest range on which every symbol used anywhere in this corpus still has a referent. Whatever the printed range cuts off *strictly inside* that boundary restricts nothing but the analyst's own hand, is therefore a **clause** by the test above — the witness being any spelling in the gap — and must either sit where τ_{DP}'s bound sits, on a hypothesis line, or be struck. *The three then come out in three registers because the rule puts them there, not because three conventions are in use.* (a) ε_ϱ. (11) is non-empty *if and only if* $\varepsilon_\varrho \le 1/\mathsf{L}$, the witness at equality being the single point $(1/\mathsf{L}, \dots, 1/\mathsf{L})$; so the referent boundary is the **weak** inequality and the strict one the assumption block printed through 2026-08-27 lay inside it, at the one spelling $\varepsilon_\varrho = 1/\mathsf{L}$. It is **struck** rather than relocated, on the block's retirement discipline: it excludes no law — a P meeting overlap at some constant meets it at half that constant — and no step in this corpus consumes it. *That last is a reading and not a census, and it is the half of this ruling a later pass should re-measure first.* (b) ε_π. Definition 22's projection target is non-empty *if and only if* $\varepsilon_\pi \le 1$, the witness at equality being the constant function 1; so there the printed range **is** the referent boundary, the remainder is empty, and there is nothing to charge. *That is why that definition prints a declaration and no clause, and the asymmetry with* (a) *is the rule working rather than two conventions: one range was strict and one is weak because one boundary is attained inside the model and the other is not.* (c) τ_{DP}. No referent anywhere depends on its value — Definition 22(c) is total at every $\tau_{\mathrm{DP}} > 0$

by its emptiness convention — so the referent boundary is all of $(0, \infty)$ and the *whole* of $\tau_{\mathrm{DP}} > 2$ is a clause. It is carried on a hypothesis line, Proposition 28(P-e), correctly and already. *The rule is stated for analyst-chosen constants and for no others*; a numeric range on a *functional of* P is a restriction on P and was never in question.

Why the split, and not the negation test it replaces. Through 2026-08-25 the printed discriminant read: negate the fragment, and if statements stop parsing it is a declaration. *That test is refuted inside this corpus.* A4(i) was written existentially, so negating it left ε_ϱ with no referent and stopped (11), $\widehat{\widehat{W}}$, (12) and the results built on them from parsing — among them Proposition 70, Proposition 71, Proposition 74, Lemma 77, Lemma 78, Lemma 81, Proposition 79, Proposition 28 and the conditions grouped as (T-5), and *no claim of exhaustiveness is made about that list.* The old test therefore ruled *overlap* a declaration carrying no hypothesis, while the assumption block carries it as a *condition* on the hypothesis line of every result whose proof reads it — the lower half of (95) among them — correctly. *And the verdict turned on an unstated reading convention*: read ungenerously at A0 it gave "declaration", read generously at A8's basis clause — which also *introduces* the b_j that Definition 22 needs — it gave "carrier", and the two readings stood three lines apart. Splitting first removes both defects, because it is the *remainder* that is tested and the remainder always has its symbols.

The word ***clause*** *covers a restrictive fragment whether or not it is numbered.* A8's basis clause is one and is correctly carried on Lemma 81's line; B1(iv)'s stem is not one and is carried nowhere. *And a result whose proof opens only some clauses under a tag cites those clauses:* the stem is not among them, because the stem was never optional, and a *block* citation instead would charge the result with clauses its proof never opens — the ground on which Corollary 49 is cited by clause, stated once here instead of six times at the sites.

The two positions this paragraph leaves out, and what each does. At a proof step: naming a tag inside a proof records where the condition is consumed and charges nothing by itself — the hypothesis line does the charging — so a clause consumed at a step and named on no

line is *under-charged*, and that is the defect repaired at eight results on 2026-08-24 and at both steps of Lemma 78. This register is not optional to state: it is the one in which the under-charging rule is defined. *Inside another assumption*: A2′ opens "A2 holds, and …", so a result citing A2′(i.a) carries A2 transitively, and a census run over hypothesis lines alone does not see the edge.

A1 (conditional exogeneity). A1's statement is cardinality-free, and this paragraph is the reason the passage from a binary to an L-category instrument is a change of representation and not of content. A1 says $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, a statement about σ-fields; nothing in it refers to how many values W takes. What referred to the cardinality, in every round before the L-category one, was the single scalar contrast $\widetilde{W} = W - \varrho(\Delta Z)$ through which the restriction was carried. Three facts fix the general representation, and all three are established here rather than assumed.

First: the contrasts span the conditionally centred functions, exactly. Fix z at which $\varrho_l(z) > 0$ for every $l \in \{0, \dots, \mathsf{L}-1\}$, which A4(i) supplies for almost every z. Let a be any measurable function of $(W, \Delta Z)$ with $E[|a| \mid \Delta Z = z] < \infty$ and $E[a \mid \Delta Z = z] = 0$, and write $a_l(z) := a(l, z)$, a finite list of numbers at each z, since W takes finitely many values, so that the centring is $\sum_{l=0}^{\mathsf{L}-1} a_l(z)\varrho_l(z) = 0$. Put $c_l(z) := a_l(z) - a_0(z)$ for $l = 1, \dots, \mathsf{L}-1$. Then

$$\sum_{l=1}^{\mathsf{L}-1} c_l(z)\,\mathbb{1}\{W = l\} \;=\; a(W,z) - a_0(z), \qquad \sum_{l=1}^{\mathsf{L}-1} c_l(z)\,\varrho_l(z) \;=\; \Big(\sum_{l\geq 1} a_l\varrho_l\Big) - a_0\Big(1-\varrho_0\Big) \;=\; -a_0(z),$$

the second using the centring in the form $\sum_{l\geq 1} a_l\varrho_l = -a_0\varrho_0$. Subtracting, $c(z)'\widetilde{W} = a(W,z)$. So every conditionally centred function of $(W, \Delta Z)$ is a ΔZ-measurable linear combination of the $\mathsf{L}-1$ contrasts, and the family is exhaustive with nothing approximated. The representation is unique: if $c(z)'\widetilde{W} = 0$ almost surely given $\Delta Z = z$ then $0 = \mathrm{Var}(c'\widetilde{W} \mid \Delta Z = z) = c(z)'\Omega(z)c(z)$, and $\Omega(z) \succ 0$ forces $c(z) = 0$, $\Omega(z) \succ 0$ *being supplied by* ***the standing hypothesis of this paragraph****, that every cell is non-null at z, through the*

determinant identity $\det\Omega = \prod_l \varrho_l$ *below, and* ***not*** *by* (6), which is A4(i)'s global consequence and is not assumed here. The space therefore has dimension exactly $\mathsf{L} - 1$ at such a z and (1) is a basis for it.

At a degenerate z the count is different, and it is stated here because the notation block promises it and because a reader who stopped at the display above would conclude that spanning needs A4(i). *It does not.* Let $A_0 := \{l \in \{0, \ldots, \mathsf{L}-1\} : \varrho_l(z) > 0\}$ be the non-null cells. The contrasts indexed by null cells are identically zero at z, and the argument above, re-run with the sums restricted to A_0, shows that the conditionally centred functions of $(W, \Delta Z)$ at z are exactly the ΔZ-measurable combinations of $\{\widetilde{W}_l : l \in A_0,\, l \geq 1\}$, of dimension $\#A_0 - 1$ when the base cell is non-null, and $\#A_0 - 1$ again when it is not, the constraint $\sum_{l \in A_0} a_l \varrho_l = 0$ removing one dimension from an $\#A_0$-dimensional space in both cases. *So the spanning statement holds at every z, with the dimension being one less than the number of non-null cells*; A4(i) makes $\#A_0 = \mathsf{L}$ everywhere and removes the case distinction, but it is not a hypothesis of the statement. *This is why no Bierens-type device is needed in the W direction*: a Bierens family exists to span an infinite-dimensional space of test functions by a countable or continuum-indexed family, and here the space is finite-dimensional and a basis is available in closed form. The Bierens family remains where it was, over ΔZ, and B1 is unchanged apart from B1(iv.d).

Second: nothing above depends on which cell is the base. The derivation used no property of cell 0 beyond its being one of the cells, so re-running it with base k produces a second basis of the *same* space; the two are related by an invertible ΔZ-measurable linear map, and every statement of the form "the restriction holds" or "the span is exhausted" is therefore invariant to the choice. *Objects that are not invariant are flagged where they occur, and the distinction is finer than it looks*: the individual contrast $\widetilde{W}_l = \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ does not mention the base cell at all, but *which* $\mathsf{L} - 1$ of the L contrasts are collected into the vector does, the base cell is the omitted one, and Ω, being the covariance of that collection, moves with it. The

quadratic form $\widetilde{W}'\Omega^{-1}\widetilde{W}$ does not move: by (61) it equals $(1-\varrho_W)/\varrho_W$, which names no cell but the realised one. See the paragraph *A4*.

Third: restriction (A) *at the* $\mathsf{L}-1$ *contrasts is equivalent to A1, not merely implied by it,* ***and the equivalence uses no overlap condition.*** For bounded measurable ς the computation of (62) with $\varsigma(\Delta U)$ in place of $\mathbb{1}\{\Delta U \le u\}$ gives $E[\widetilde{W}\varsigma(\Delta U) \mid \Delta Z = z] = \Omega(z)\, d_\varsigma(z)$ with $d_{\varsigma,l}(z) = E[\varsigma(\Delta U) \mid W = l, \Delta Z = z] - E[\varsigma(\Delta U) \mid W = 0, \Delta Z = z]$, read at any versions of the conditional expectations, those at null cells being arbitrary. *Fix* z *and put* $A := \{l \ge 1 : \varrho_l(z) > 0\}$; $\Omega(z)$ vanishes off $A \times A$ and its A-block is $\Omega_A = \operatorname{diag}(\varrho_A) - \varrho_A\varrho_A'$ with $\det \Omega_A = \varrho_0 \prod_{l\in A} \varrho_l$. If $\varrho_0 > 0$ then Ω_A is nonsingular and $\Omega d_\varsigma = 0$ forces $d_{\varsigma,A} = 0$; if $\varrho_0 = 0$ then $\ker \Omega_A = \operatorname{span}\{\mathbf{1}_A\}$, from $\varrho_l(c_l - \varrho_A' c) = 0$ with $\varrho_l > 0$ on A, and $d_{\varsigma,A}$ is constant on A. *In both cases the restriction holds for every bounded* ς *if and only if the conditional law of* ΔU *given* $(W, \Delta Z = z)$ *is the same across all non-null cells*, which is $\Delta U \perp\!\!\!\perp W \mid \Delta Z = z$. *No condition on* ϱ *was used, and A4(i) is* ***not*** *a hypothesis of this equivalence*: what A4(i) is needed for is (64), the normalisation, which is a different matter and is taken up in the paragraph *A4*. *At* $\mathsf{L} = 2$ *this is the equivalence the block has used since round 1*, with the scalar $\varrho(1-\varrho)$ in place of Ω, and the two cases collapse to the single observation that a null cell makes W conditionally degenerate.

What extra cells buy, and what they do not. They do not buy identification, and no sentence in this corpus may suggest that they do. Identification of $\varphi_\dagger$ is A1b(iii), and A1b(iii) at the full L-category W is *implied by* A1b(iii) at any coarsening of W *together with A1*. The two halves are separate and only the first is about coarsening. *Inclusion*: a coarsening $g(W)$ imposes fewer restrictions, so $\mathcal{V}_{CI}(W) \subseteq \mathcal{V}_{CI}(g(W))$, and $\mathcal{V}_{CI}(g(W)) = \{\varphi_\dagger\}$ gives $\mathcal{V}_{CI}(W) \subseteq \{\varphi_\dagger\}$, *at most* one element. *Non-emptiness*: $\varphi_\dagger \in \mathcal{V}_{CI}(W)$ is exactly $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, which is A1 and is not delivered by the coarsening. *With both,* $\mathcal{V}_{CI}(W) = \{\varphi_\dagger\}$. A first draft of this paragraph gave the inclusion alone and called it the implication, which would leave $\mathcal{V}_{CI}(W)$ possibly empty. *A finer instrument therefore makes the identification hypothesis weaker,*

never stronger, A1 being maintained throughout, and identification already held at $\mathsf{L} = 2$.

What the extra cells **do** buy is **localisation**, and three concepts have to be kept apart to see it.

First, testability, which is **present at** $\mathsf{L} = 2$ and is not what $\mathsf{L} > 2$ adds. At $\mathsf{L} = 2$ the restriction is $d_1(u \mid z) = 0$ *for every* u *at almost every* z, a continuum of equations against one unknown function. This is a conditional-**independence** restriction with a continuously distributed conditioning variable, not the mean-independence case in which a binary instrument supplies two moments and exhausts them. A1 is therefore testable at $\mathsf{L} = 2$, and the paper's own application demonstrates it: the Section 6 probe rejects A1 at $\mathsf{L} = 2$. A statement that more cells make A1 testable is false and must not be written.

Second, instrument strength, which is a property of the data and not of the model. How hard W moves X bears on precision. *Splitting an instrument creates no variation*, and strength within a sub-population can be examined at $\mathsf{L} = 2$ by subsetting. This is not what $\mathsf{L} > 2$ adds either.

Third, and this is the claim, **localisation**. At $\mathsf{L} = 2$ the restriction is the single object $d_1(u \mid z) = 0$, so a violation concentrated in one channel of the instrument and its absence in another are *averaged into one verdict*: the test rejects, and cannot say through which channel. At $\mathsf{L} \geq 3$ the restriction is $d_l(u \mid z) = 0$ *cell by cell*, so a violation can be *attributed* to a cell. *That question cannot be posed at* $\mathsf{L} = 2$ *at all*, and posing it is what the finer instrument is for.

The formal apparatus is joint consistency, and it is apparatus rather than motivation. Under A1 each of the $\mathsf{L} - 1$ contrast directions carries a valid restriction on the *same* $\varphi_\dagger$, each a whole family, indexed by the test function and by the Bierens index t, so the testable content is that there exists a single $\varphi \in \mathcal{V}$ at which all $\mathsf{L} - 1$ systems hold, and a J-type statistic for that hypothesis is the vehicle. *This route needs no identification from any subset*, which is why it is the one taken; and its per-cell decomposition is what delivers localisation. *A route a first draft took is withdrawn*: it pinned $\varphi_\dagger$ from a subset of the directions and

called the remainder testable, which presumes identification from a *singleton* contrast, and contrast l alone is the binary coarsening $\mathbb{1}\{W = l\}$, so identification from it is A1b(iii) at that coarsening, *stronger* than A1b(iii) at the full instrument and not implied by it.

The application is what the split exists for, and it makes the three concepts concrete. The Section 6 instrument divides into a *chronic-level* channel, the annual-mean-only and both-triggers cells, and a *transient* channel, the bad-day-only cell. *The hypothesis the split is built to test is that A1 fails through the chronic channel*, where W is close to a function of X_1, the endogenous regressor's own initial value, *and not through the transient one.* At $\mathsf{L} = 2$ that hypothesis has no expression. *The proposition that states the joint-consistency restriction and the statistic that localises a violation to a cell are not in this dispatch*; they are recorded as owed, and Phase 2 builds them off this paragraph.

And consistency of every cell set is necessary for A1, not sufficient, which governs how a non-rejection may be reported and is stated here because it is stated nowhere else. Proposition 17(ii) runs in one direction only: A1 implies that every cell set is consistent, and its contrapositive is that a conflict refutes A1. The converse is never asserted, and it is false in general. The reason is structural rather than statistical, and survives at infinite sample size. What the cell system interrogates is whether $\mathcal{V}_{CI}$ is *empty*; what A1 is about is whether $\varphi_\dagger$ *lies in* $\mathcal{V}_{CI}$. *Those are different questions.* At a law where A1 fails at the truth but some other $\varphi_1 \in \mathcal{V}$ satisfies the conditional-independence restriction, every cell set is satisfied *at* φ_1, every conflict set is empty, the diagnostic reports nothing, and the estimator converges to $\varphi_1 \neq \varphi_\dagger$. Nothing in this block excludes such a law: A1b(iii) would also fail there, it is the clause that says $\mathcal{V}_{CI} = \{\varphi_\dagger\}$, and the point is precisely that *the diagnostic cannot see the failure of the clause it would need to see. So a cell showing no conflict is* ***not exonerated***: the procedure detects $\mathcal{V}_{CI} = \varnothing$ and nothing more, which is not an interpretation but Proposition 17(iii)'s own equivalence, that some consistent S has a non-empty conflict set *if and only if* $\mathcal{V}_{CI} = \emptyset$, so a violation reconciled by some other $\varphi \in \mathcal{V}$

is invisible to it, and the false-negative rate of the attribution is unpriced by anything in this paper. *Section 6 will attribute a non-rejection to a channel and must report it in exactly those terms*, as the absence of a detected conflict, not as evidence for A1 in the un-flagged cell.

A1b (identification). *(The corresponding statement for a generic $\varphi \in \mathcal{V}$ is Definition 5 and not an assumption; only membership of the truth is content.)* Mode of consumption, stated once and binding on every citation of this clause: A1b(i) is consumed for **well-posedness** and never for a bound. What it buys is that the truth, and hence the path $\varphi_\dagger + \tau_1 h$ for $h \in \mathcal{V}$, lies in the parameter space. A result whose conclusion concerns a generic $h \in \mathcal{V}$, and whose proof needs only $\int h \, d\pi = 0$, consumes Definition 5 and must cite Definition 5; citing A1b(i) there charges the result with a hypothesis it does not use, because the two statements are typographically identical and semantically disjoint. The genuine consumers are Lemma 56 (Step 1), Proposition 20(i), Corollary 14 (clause (a), at $\varphi_\dagger$), Lemma 93 and, transitively, Lemma 55 and Theorem 29. *Lemma 93's use is the archetype*: A1b(i) is what puts $h_k = \hat{\varphi}^{(-k)} - \varphi_\dagger$ in $\mathcal{V}$, hence what makes the balancing pairing $\langle \nu_{g_k}/\pi, h_k \rangle_{L^2(\pi)}$ of (130) the object Lemma 45(iii) computes and the object A2 asserts a range condition about, a *well-posedness* use and not a bound. Theorem 29 carries A1b(i) in group (T-1) for the same use, inherited. *Lemma 92 Step 4, Lemma 94 Step 3 and Lemma 95 Steps 3 and 6′ are **not** consumers*, although each applies Lemma 45(i) at h_k: that clause is stated for every $h \in L^2(\mathcal{X}, \pi)$, which is all its proof establishes.

A1d (RETIRED: instrument completeness given the residual), moved entire.

Assumption (A1d: Instrument completeness given the residual)**. ~~RETIRED, 2026-08-05.~~** It is unsatisfiable in this model. The condition read: for every $\Lambda \in L^2(P_{X_1,X_2})$, $E[\widetilde{W}\,\Lambda(X_1, X_2) \mid \Delta U, \Delta Z] = 0$ a.s. implies $\Lambda(X_1, X_2) = 0$ a.s. ~~[retired]~~. *Take* $\Lambda \equiv 1$, which lies in $L^2(P_{X_1,X_2})$ because P is a probability measure. Since $\widetilde{W}_l = \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ by

(1) and A1 gives $\Delta U \perp\!\!\!\perp W \mid \Delta Z$,

$$E\big[\widetilde{W}_l \,\big|\, \Delta U, \Delta Z\big] \;=\; E\big[\mathbb{1}\{W = l\} \,\big|\, \Delta Z\big] - \varrho_l(\Delta Z) \;=\; 0 \quad \text{a.s., for every } l,$$

so the antecedent holds while $\Lambda \neq 0$. Hence *A1 implies the negation of this condition*, and so does A1b(iii), which delivers A1 at $\varphi_\dagger$. More generally, $\widetilde{W}$ is centred given ΔZ, so $\Lambda \mapsto E[\widetilde{W}\Lambda \mid \Delta U, \Delta Z]$ annihilates every Λ agreeing a.s. with a $\sigma(\Delta U, \Delta Z)$-measurable function: no completeness condition on that operator over a class containing the constants can hold here. That is what separates this design from the conditional-mean one of Fève and Florens (2014), whose operator $g \mapsto E[g \mid W]$ does not annihilate constants, *their hypothesis transplants as a contradiction, not as an excess.* Lemma 45(v) is the unconditional shadow of the display above and was in this corpus throughout.

The status of A1d, written with A1d as its subject. **A1d** is not maintained, is not available for use, and is a hypothesis of no result in this programme; the tag is kept rather than reused, on the A3 and B1(vi) precedent, so that a stale citation lands on the reason rather than on a gap. *Its consumer count is zero*: the corollary that derived A1c from it, `cor:a1cprim`, was withdrawn with it and no other statement takes it as a hypothesis; see Remark 44, which carries the withdrawal and says so in terms. *And A1d is **retired**, on A6's ground and not on A3's*: it is *false* in this model rather than *proved* or merely unconsumed, which is the distinction the five-clause list of unconsumed-but-live conditions below turns on. *This block therefore has **five** retirements and not four*, on **three** grounds — A3 and A2′(ii) because they are proved, A6 and A1d because they are false, B1(vi) because it is about a different estimator — and the census below counts five.

B1 (Bierens family and range space).

(vi) ~~**RETIRED.**~~ It describes an estimator this block does not implement. The clause read: *the family used on the φ-side is a finite set $\mathcal{T}_n \subset \mathcal{T}$ with $\#\mathcal{T}_n \to \infty$ and $\overline{\bigcup_n \mathcal{T}_n} \supseteq \mathcal{T}$*

~~[retired]~~. It is not maintained, is not available for use, and is a hypothesis of no result in this programme. The clause number is kept rather than reused, on the A3 and (P-b) precedent, so that a stale citation lands somewhere. That second claim was false for four dispatches, and what made it false is recorded here rather than repaired silently. A retirement is complete only when the clause is struck, when no hypothesis list carries it *including through a block citation of the parent tag*, and when nothing describes it in the present tense as a live condition. B1(vi) failed the second test at Proposition 28, whose list read "B1" in full and now reads B1(i)–(v); and it failed the third at three compiled sites, the Discussion of this block, Remark 43 in `results.tex`, and that remark's rendering in `paper/sections/theorems/lem-bierens.tex`, each of which called it an identification restriction in the present tense. All four are closed. *A clause that is struck in one file and live in three is worse than a clause nobody struck*, because the strike is what stops the next reader from checking. Why it is retired, and this one is retired for the third of the three reasons, neither "it is a theorem" (A3) nor "it is false" (A6) but "it is about a different estimator". The **growing finite family** $\mathcal{T}_n$, a finite subset of $\mathcal{T}$ with $\#\mathcal{T}_n \to \infty$, is instantiated **nowhere in this project**, in the compiled text or in the simulation code. The estimator does not use a growing finite grid: Definition 2 integrates against Π_t over *all* of $\mathcal{T}$, and every operator norm in Parts F–H is an $L^2(\Pi)$ norm over the whole index set. The clause is a leftover from a Domínguez–Lobato-style design with a growing finite family, which this block considered and did not adopt. What it was for, and where that work is actually done. Its heading claimed identification of $\varphi_\dagger$. Identification of $\varphi_\dagger$ in this block is **A1b(iii)**, assumed outright, and A1b(iii)'s own comment records that no primitive route to it is offered. A growing family would be one ingredient of such a route, *and the counterexample-hunter's sub-probe shows it would be a genuine ingredient and not decoration*: at a *fixed finite* grid $\Pi_t = \sum_{k \le K} \delta_{t_k}$ with ΔZ supported on $K+1$ atoms, the $K \times (K+1)$ matrix $[e^{t_k \Phi(z_j)}]$ has a null vector, and perturbing along it makes the Bierens family hold while the conditional

restriction fails. So the clause is correctly labelled an identification condition; what it is not is a condition of this estimator. It is recorded here rather than deleted so that a later dispatch reinstating the growing-grid design finds the statement and the witness together. *Note also that (vi) was never sufficient for A1b(iii)*: it constrains the index family and says nothing about $\mathcal{V}_{CI}$.

A2′ (source condition with usable representer). *Why the two budgets aggregate in two different ways.* **Euclidean over the coordinates of** θ because every consumer of $\Lambda_\star^{(0)}$ and $\Lambda_\star^{(1)}$ reads a Euclidean or Frobenius norm of the assembled object — Corollary 49's two sup-norms and everything downstream of them — so a coordinatewise maximum would leave every one of those constants short by $\sqrt{p}$. **Maximal over** l because (68) treats the contrast index diagonally, each slice generating its own column and no other, so no downstream bound ever sums across l.

(ii) ~~**RETIRED.**~~ It is a consequence of (i). The clause read $E[\widetilde{W}^2\|\ell_\star\|^2] < \infty$ ~~[retired]~~ and is not maintained, is not available for use, and is a hypothesis of no result in this programme. The clause number is kept rather than reused, on the A3 precedent and for the same reason (P-b) is kept: a stale citation must land somewhere. Why it is retired, and it is retired because it is a **theorem**, exactly as A3 was, not because it is false and not because it is too strong. Under (i) in full, together with B1(i), (ii), (iv.a), (iv.d), Corollary 49 gives $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty \Lambda_\star^{(0)} < \infty$, and $\|\widetilde{W}\| \le \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$, in the Euclidean norm on $\mathbb{R}^{\mathsf{L}-1}$, is (P1b), free from (1), W taking finitely many values and $\varrho_l \in [0,1]$; hence $E[\widetilde{W}^2\|\ell_\star\|^2] \le 2\|\ell_\star\|_\infty^2 < \infty$ with no further hypothesis. *The factor* 2 *is the square of that bound and is therefore* (P1b)*'s constant and not* (P1)*'s*: (P1) is the *per-cell* $|\widetilde{W}_l| \le 1$ and does not bound the Euclidean norm of the vector, and a derivation routed through it here would produce the constant Corollary 50's proof records as **false** at $\mathsf{L} \ge 3$. *The bound is written out rather than called by its symbol*, that symbol being declared at Corollary 50, which this file precedes in reading

order. *This tag was* (P1) *through 2026-08-24, at both sites of this retirement ground*, while Lemma 91's closing paragraph, which runs the identical derivation, printed "the bound on $\|\widetilde{W}\|$ is the Euclidean one and **not** (P1)"; *soundness was never at stake, but a retirement ground is a derivation and must name every clause it uses.* The factor is 1 at $\mathsf{L} = 2$; it is a constant, it is uniform in L, and it changes nothing about the retirement. *The attribution is (i)* ***in full*** *and not (i.a) alone*: the only route to $\|\ell_\star\|_\infty < \infty$ in this block is Corollary 49, that corollary is stated at A2b, and A2b requires (i.a) *and* (i.b). *Every tag in that derivation is already carried by every list that could have carried (ii)*, so the retirement changes no conclusion anywhere. *The clause has* ***no consumer***, as Lemma 60, Lemma 55, Lemma 56's own hypothesis line — "A2′ cited by clause; A2′(ii) is consumed by no part of this lemma" — and Lemma 91's closing paragraph each state independently. *Four attestations, and the number is not maintained*: the draft this sentence replaces said three and named three, and the fourth had been printing on its own hypothesis line throughout. What survives, and it is the half the clause was really carrying. At Step 8 (`prp:effbound`), where the variance is computed and where tier T3 relaxes (i), $\|\ell_\star\|_\infty$ is no longer available and $E[\widetilde{W}^2\|\ell_\star\|^2] < \infty$ becomes a genuine restriction, it is exactly the Khan–Tamer irregularity boundary, since holding the conditional laws given W fixed and driving any one cell probability to zero makes $\|\Omega^{-1}\| \geq \max_{0\leq l\leq \mathsf{L}-1} \varrho_l^{-1}$ diverge by (61), read at the diagonal entry $e_l'\Omega^{-1}e_l = \varrho_l^{-1} + \varrho_0^{-1}$, and with it $E[\widetilde{W}^2\|\ell_\star\|^2]$. *At* $\mathsf{L} = 2$ *that is the printed computation*, $\varrho \equiv \varepsilon$ giving $E[\widetilde{W}^2\|\ell_\star\|^2] \asymp \varepsilon^{-1}$; *at* $\mathsf{L} \geq 3$ *it is* ***one*** *cell that has to be thin and not all of them*, which is the reading the application needs, its thin cell being a single trigger category. *That is a* ***forward*** *obligation and not a present consumer*, and it is recorded here rather than carried as a maintained clause: Step 8 must state the condition it needs, at the tier where it needs it, and under T3 it will not be this clause because (i) will not be available to make it redundant. *Flagged for Step 8*, on the model of Remark 58;

Status of clause (i), stated here so that the tag names exactly one object. A2′(i) is a *maintained assertion about $\ell^{\Pi}_{\star}$, and about $\ell^{\Pi}_{\star}$ alone.* The same two conditions read at a generic $\ell^{\Pi} \in L^2(\Pi)$ are not an assumption but a *property* that a candidate representer may or may not have; that property is tagged **A2b** below and is what Definition 19(a) names *admissible representer. A result that needs the property at a generic representer cites A2b and never A2′(i)*, exactly as a result needing the centring property of a generic weight cites N1 and never a maintained clause. This is the treatment N1/N1′ received in round 7 and it is applied here for the same reason: one tag, one object.

A2c (source condition on the representer). Two things in (4) are established here rather than assumed, because neither is automatic and the second cannot be had from the transform.

(m.1) The exceptional set is a Borel set by construction, and stating it that way is a choice. The weaker phrasing "for Π_t-almost every t the slice lies in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$" delivers the displayed form at no cost: it means that the set of t failing the slice condition is contained in some $\mathcal{N} \in \mathfrak{B}(\mathcal{T})$ with $\Pi_t(\mathcal{N}) = 0$, and $\mathcal{T}_* := \mathcal{T} \setminus \mathcal{N}$ is then Borel, carries full Π_t-measure, and contains *only* points at which the slice condition holds. *Nothing anywhere requires $\mathcal{T}_*$ to be the largest such set*, which need not be Borel. What the choice buys is that $\mathbb{R} \times \mathcal{T}_*$ belongs to $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathcal{T})$, the property Lemma 61's Step 2 needs when it asserts joint measurability there, and the property an "almost every" quantifier does not by itself supply. With L cells the construction is run once per slice and the sets are then intersected. *What A2c asks for is* ***one*** *$\mathcal{T}_*$ serving every l*, its slice condition being membership of the **vector** slice $\ell^{\Pi}(\cdot, t, l)$ in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$, so that it has $\mathsf{L}-1$ slices and not more, *while the construction below runs at the $p(\mathsf{L}-1)$* ***coordinate*** *slices and delivers that requirement rather than a stronger one*: by the identity A2c prints when it extends (3) to $\mathbb{R}^p$-valued g, membership of the vector slice in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ *is* membership of all p coordinates in $H^{\mu''}(\mathbb{R})$, so the finer index is bookkeeping and not a further demand. There are $p(\mathsf{L}-1)$ coordinate slices $\ell^{\Pi}_j(\cdot, t, l)$; take $\mathcal{N}_{j,l}$ for each

and set $\mathcal{N} := \bigcup_{j,l} \mathcal{N}_{j,l}$, a *finite* union, hence Borel and Π_t-null, and $\mathcal{T}_* := \mathcal{T} \setminus \mathcal{N}$ then carries full Π_t-measure and contains only points at which *every* coordinate slice condition holds, equivalently, by that same identity, only points at which $\ell^{\Pi}(\cdot, t, l) \in H^{\mu''}(\mathbb{R}; \mathbb{R}^p)$ for every l, which is A2c's quantifier. *Finiteness of* L *is what makes this free*: a countable union would serve too, but a union over an index set of unbounded cardinality would not, which is one more place where A0's "L finite and fixed in n" is doing work rather than housekeeping. *A per-slice* $\mathcal{T}_{*,j,l}$ *would not serve* Lemma 61's Step 2, which needs one product set on which all the slices are simultaneously well behaved.

(m.2) The integrand is measurable, by pairing against one fixed countable family of functions of u. There are real $\varkappa_1, \varkappa_2, \ldots \in L^2(\mathbb{R}, du)$ with

$$\big\|g\big\|_{H^{\mu''}(\mathbb{R})} \;=\; \sup_{k \geq 1} \left| \int_{\mathbb{R}} g(u)\, \varkappa_k(u)\, du \right| \qquad \text{for every real } g \in H^{\mu''}(\mathbb{R}). \tag{60}$$

Construction, with no negative-index space and no completion. Call a function of ξ *Hermitian* when its value at $-\xi$ is the complex conjugate of its value at ξ, which is what transforms of real functions are. The Hermitian elements of $L^2(\mathbb{R}, (2\pi)^{-1} d\xi)$ form a closed real-linear subspace of a separable space, hence a separable real Hilbert space; take a countable subset of its closed unit ball, dense there, and replace each member by its truncations to $\{|\xi| \leq R,\ |\cdot| \leq R\}$ over rational R, truncation preserves Hermitian symmetry, because the truncation set is symmetric, does not increase the norm, and converges in L^2, so the subset may be taken to consist of *bounded, compactly supported* Hermitian functions and to remain dense in the ball. Let $\varkappa_k$ be the inverse Fourier transform of $(1 + \xi^2)^{\mu''/2}$ times the k-th member: that product is bounded with compact support, hence in $L^1 \cap L^2$, so $\varkappa_k$ is a bounded continuous element of $L^2(\mathbb{R}, du)$, and it is real because the product is Hermitian. For real $g \in H^{\mu''}(\mathbb{R})$, Plancherel in the normalisation of (3) gives

$$\int_{\mathbb{R}} g\, \varkappa_k\, du \;=\; \frac{1}{2\pi} \int_{\mathbb{R}} \big[(1 + \xi^2)^{\mu''/2} (\mathcal{F} g)(\xi)\big]\, \overline{\big[(1 + \xi^2)^{-\mu''/2} (\mathcal{F} \varkappa_k)(\xi)\big]}\, d\xi,$$

whose second factor is the k-th member of the dense family and whose first is a Hermitian element of $L^2(\mathbb{R},(2\pi)^{-1}d\xi)$ of norm exactly $\|g\|_{H^{\mu''}(\mathbb{R})}$, by (3). The supremum over k of the modulus of a continuous linear functional over a dense subset of the closed unit ball is the functional's norm, which is (60).

The remaining step is run at a fixed coordinate slice (j,l) and then aggregated exactly as (4) aggregates: Euclidean over the p coordinates, maximal over the $\mathsf{L}-1$ contrasts. *The second stage is the only place the* L*-category form changes anything here*; the first is present at every L, $\mathsf{L}=2$ included. At each fixed k and each (j,l) the map $t\mapsto\int_{\mathbb{R}}\ell_j^{\Pi}(u,t,l)\varkappa_k(u)\,du$ is Π_t-measurable on $\mathcal{T}_*$: the product $(u,t)\mapsto\ell_j^{\Pi}(u,t,l)\varkappa_k(u)$ is $\mathfrak{B}(\mathbb{R})\otimes\mathfrak{B}(\mathcal{T})$-measurable, *the contrast index l ranges over a finite set, so a jointly measurable representative of ℓ^{Π} on $\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$, with that set given its discrete σ-algebra, is exactly a finite list of jointly measurable functions of (u,t), one per (j,l)*, Tonelli on the σ-finite product $(\mathbb{R},du)\otimes(\mathcal{T},\Pi_t)$ makes the u-integrals of its positive and negative parts measurable in t, and both are finite at each $t\in\mathcal{T}_*$ because $\ell_j^{\Pi}(\cdot,t,l)$ and $\varkappa_k$ are both in $L^2(\mathbb{R},du)$. A countable supremum of measurable functions is measurable, so $t\mapsto\|\ell_j^{\Pi}(\cdot,t,l)\|_{H^{\mu''}(\mathbb{R})}$ is Π_t-measurable on $\mathcal{T}_*$ for each (j,l). One aggregation then remains, and it is the one over j: the integrand of (4) is the **vector** norm, which the preceding sentence does not by itself deliver. It closes on the identity A2c prints when it extends (3) to $\mathbb{R}^p$-valued g, read at $g=\ell^{\Pi}(\cdot,t,l)$:

$$\left\|\ell^{\Pi}(\cdot,t,l)\right\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} \;=\; \left(\sum_{j=1}^{p}\left\|\ell_j^{\Pi}(\cdot,t,l)\right\|^2_{H^{\mu''}(\mathbb{R})}\right)^{1/2}, \qquad t\in\mathcal{T}_*,$$

a *finite* sum of squares of Π_t-measurable functions followed by $z\mapsto z^{1/2}$, which is continuous on $[0,\infty)$ and hence Borel; so $t\mapsto\|\ell^{\Pi}(\cdot,t,l)\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}$ is Π_t-measurable on $\mathcal{T}_*$, now for each l. Every summand is finite there, A2c placing the slice in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ at every $t\in\mathcal{T}_*$ and that membership being, by the same identity, membership of all p coordinates. *The step is one line because p is a fixed finite dimension, exactly as* (m.1)*'s intersection was free because* L *is finite*; nothing here is a Fubini statement in j. Each of the $\mathsf{L}-1$ integrals in (4), *one per*

*contrast l, and **not** one per (j,l): the p coordinates are aggregated inside the integrand, by the Euclidean norm, so only the contrast index survives as an index of integrals*, is therefore well defined in $[0,\infty]$, and so is their maximum, *a maximum over a finite index set requiring no further argument.* That is the quantity of which A2c asserts finiteness. *This is the same countable-supremum device Lemma 61 uses at its Step 3 and Definition 53 at (73), applied one level up*, and it is used *instead* of a Fubini statement about $(\xi,t)\mapsto(\mathcal{F}\ell^{\Pi}(\cdot,t))(\xi)$, which is not available, for the reason Lemma 61's Step 2 prints: on $L^2(\mathbb{R},du)$ the transform is an L^2-limit and not an integral. (60) pairs the slice against a family fixed *once*, in the variable u in which the slice is jointly measurable, so no such statement is needed.

H1 (Hilbert-scale link). *Two notational points. First*, bare sans-serif c is this project's *generic finite positive constant*: bound by the quantifier of the clause in which it appears, free to take a different value in every such clause, and never carried from one result to another. This clause binds it, in the words "there is a finite c, not depending on ζ", and no site downstream reads its value. *Second, the pre-image is written h and **not** ς, h* being the *domain*-side bound variable Remark 104(1) declares, the counterpart of Ξ,Ξ' on the range side, and the pre-image being exactly that: a generic element of $L^2(\pi)$, existentially bound per ζ inside this clause. *The object is the spectral source pre-image*, the same species as in (P-a.3) and Lemma 85(iii), read here at the *population* operator on all of $\mathcal{V}$. *H^s is the **classical** Sobolev space and **not** a scale generated by $\mathcal{D}$*: for a Hilbert scale generated by a differential operator on a bounded interval the second scale step imports the generator's boundary conditions, so the space the generator produces at order s' coincides with $H^{s'}(\mathcal{X})$ only for $s'<\frac{3}{2}$ (Neumann-type) or $s'<\frac{1}{2}$ (Dirichlet-type), *s' throughout this assumption and A8's derivation denoting a **generic** order of the scale, bound where it appears and used nowhere else in this project*, while (54) forces $s>\frac{27}{16}$ at $\alpha=\frac{3}{8}$, and it is (54) and not (63) that binds there, the latter asking only $s>1.4818$; at every $\alpha<\approx 0.359$ the two exchange roles and (63) is the higher floor. *A B-spline on a quasi-uniform mesh satisfies no boundary*

condition, so under the literal generated reading (8) would be **false** at every admissible s and the four conditions it underwrites , A5(vi), A5(vi′), Proposition 28(P-a.1) and (43)'s second display, would again be events on the *empty* set, which is the failure the amendment of A8 closed and which the generated reading reopened verbatim. *What held it together was a directive and not a condition*: one sentence in a scope note in `proofs.tex` instructing that this scale be read on the same scale as (162), i.e. classically, in a different file, and cited at neither this assumption, Definition 80, nor (8). *It is settled here instead, in the clause, once*: the space is the classical one, and the comparison the generator was there to produce is (5), assumed outright. *Nothing downstream loses anything*: only the single index s is ever used, no site reads a second member of the scale, and (5) is exactly what the generated construction was invoked to deliver. *Consequently the source index in A2 is*

$$\mu_{\mathrm{H1}} \;:=\; \frac{s}{2a}.$$

The subscripted name is used **only** where the two indices must be told apart, which is the disposition Remark 86(5) makes binding. Bare μ in this project is (P-a.3)'s *primitive* index at the finite-rank sieve compression of the *realised* operator at the anchor $\theta_\circ$; μ_{H1} is the index of A2 at the *population* operator $\mathcal{A}_{\mathrm{cond}}$ on all of $\mathcal{V}$ at $\theta_\dagger$, *read into the whole three-factor range space and not contrast by contrast, so that a is the decay exponent of the* ***stacked*** *operator at every* L, and s and a in the display are objects at $\theta_\dagger$ as well, against the $\theta_\circ$-specific a of Definition 24(b). *They are equal only under the interpolation proviso this file states and declines*, and the sites that computed one and read it into a condition on the other are qualified in place: the two daggered lines of the smoothness-budget table, the sentence deducing $\mu > \frac{5}{6}$ from $s > a + \frac{1}{2}$, Remark 98(1), and clause (2) of Remark 125, which Remark 98(1) cites while qualifying its own copy and which carried no proviso of its own until it was given one. *No claim of exhaustiveness is made, and that is deliberate*: the sentence this one replaces said "the three sites", then named four things, and the fourth site

was unqualified while the sentence asserted that all of them were. ***The same identity is not asserted for Proposition 28.*** Proposition 28's hypothesis (P-a.3) states a **spectral** source condition, at the sieve compression of the *realised* operator, with μ and its radius R_* *primitive*. What H1 supplies to that route is the *scale* H^s, which Definition 80, A5(vi′), (P-a.1), (P-c) and (P-g) are written in, and nothing more; the passage from a Hilbert-scale source condition plus a link condition to a spectral one is operator interpolation at exponent 2μ, licensed only at $2\mu \leq 1$, and Proposition 28 lives at $2\mu > \frac{5}{3}$. Remark 86 carries the full disposition, including the commutativity proviso under which the identity above *would* extend to the sample operator, and what that proviso costs.

What a is an exponent **of**, at $\mathsf{L} \geq 3$. H1 is a hypothesis about the **stacked** operator, $\mathcal{A}_{\text{cond}}$ read into $L^2(\Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}})$ entire, as Definition 5 fixes it, and not about any single contrast. *At* $\mathsf{L} \geq 3$ *that is a genuinely different operator from the binary one*, and what makes it different is the **range enlargement**: the domain $\mathcal{V}$, the scale H^s and (5)'s form are untouched, and the range gains a third factor. *It is not the normalisation*, which was reversed, the operator carries the raw contrast (1) on both sides at every L, as the paragraph *A4* sets out.

And a does not depend on Π_{w}, which is worth saying because Π_{w} is the analyst's choice. B1(iv.d) puts strictly positive masses on a finite index set, so any two admissible Π_{w}, Π'_{w} give *equivalent* range norms,

$$\min_l \frac{\varpi'_{\mathsf{w}}(l)}{\varpi_{\mathsf{w}}(l)} \, \|\Xi\|^2_{L^2(\Pi)} \;\leq\; \|\Xi\|^2_{L^2(\Pi')} \;\leq\; \max_l \frac{\varpi'_{\mathsf{w}}(l)}{\varpi_{\mathsf{w}}(l)} \, \|\Xi\|^2_{L^2(\Pi)} \qquad \text{for every } \Xi,$$

with both ratios finite and strictly positive because the index set is finite and B1(iv.d)'s masses are; and the *domain* norm is untouched, Π_{w} appearing only on the range side. Singular values of a compact operator are its approximation numbers, infima of operator-norm distances to operators of rank below j, and an operator norm inherits those same two constants, so the singular values under the two weightings differ by at most their square roots *at every* j *alike. The constants are* j*-free, so the exponent is identical and only the constant in the* $\asymp$ *moves*;

and $\max_l \varpi_{\mathsf{w}}(l)^{-1/2}$, the same factor that prices the passage between the $L^2(\Pi)$ norm and the per-contrast norms, is what bounds the move. So the choice of range weights is not a choice of ill-posedness.

One reading of a in terms of the contrasts is available, and it is conditional. Write $\mathcal{A}^{(l)}h := \mathcal{A}_{\mathrm{cond}}h(\cdot,\cdot,l)$, so that $\mathcal{A}^{(l)} : \mathcal{V} \to L^2(\Pi_u \otimes \Pi_t)$ is the l-th contrast operator, Hilbert–Schmidt by Proposition 9 read at fixed l, its kernel being $\mathsf{k}(\cdot,\cdot,l;\cdot)$ of (27), and let $\mathcal{A}^{(l)*}\mathcal{A}^{(l)} \succeq 0$ be its Gram operator on $\mathcal{V}$, the adjoint taken in the *unweighted* range pairing. Since $\|\mathcal{A}_{\mathrm{cond}}h\|^2_{L^2(\Pi)} = \sum_l \varpi_{\mathsf{w}}(l)\,\|\mathcal{A}^{(l)}h\|^2$ by Definition 5's pairing, and since two bounded self-adjoint operators with the same quadratic form are equal,

$$\mathcal{A}^{*}_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}} \;=\; \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l)\; \mathcal{A}^{(l)*}\mathcal{A}^{(l)} \qquad \text{on } \mathcal{V}.$$

Two comparisons follow, both from the max–min characterisation of the eigenvalues of a compact positive operator. *Discarding all but one summand* leaves that sum $\succeq \varpi_{\mathsf{w}}(l_0)\,\mathcal{A}^{(l_0)*}\mathcal{A}^{(l_0)}$ in the Löwner order for each l_0, and the max–min is monotone in that order, so $\sigma_j(\mathcal{A}_{\mathrm{cond}}) \geq \max_l \varpi_{\mathsf{w}}(l)^{1/2}\,\sigma_j(\mathcal{A}^{(l)})$ at every j. *In the other direction*, Weyl's inequality for compact positive operators, the $(i+i'-1)$-th eigenvalue of a sum of two is at most the i-th of the first plus the i'-th of the second, which is the same max–min evaluated on the intersection of two finite-codimensional subspaces , iterated over the $\mathsf{L}-1$ summands gives

$$\lambda_{\sum_l j_l-(\mathsf{L}-2)}\Big(\mathcal{A}^{*}_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}}\Big) \;\leq\; \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l)\,\sigma_{j_l}\Big(\mathcal{A}^{(l)}\Big)^2 \qquad \text{for any } j_1,\dots,j_{\mathsf{L}-1} \geq 1.$$

Taking $j_l = \lceil j/(\mathsf{L}-1)\rceil$, for which $\sum_l j_l - (\mathsf{L}-2) \leq j$ at every $j \geq 1$, and using that λ is non-increasing in its index,

$$\sigma_j(\mathcal{A}_{\mathrm{cond}})^2 \;\leq\; \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l)\,\sigma_{\lceil j/(\mathsf{L}-1)\rceil}\Big(\mathcal{A}^{(l)}\Big)^2.$$

So if the contrast operators have polynomial exponents, $\sigma_j(\mathcal{A}^{(l)}) \asymp j^{-a_l}$, the two bounds pinch

and

$$\sigma_j(\mathcal{A}_{\mathrm{cond}}) \asymp j^{-a}, \qquad a = \min_{1\le l\le \mathsf{L}-1} a_l,$$

the constants depending on L and on ϖ_{w} and *not* on j. *Read the lower bound at any l attaining the minimum* and it is $\gtrsim j^{-\min_l a_l}$; *read the upper bound through* $j^{-2a_l} \le j^{-2\min_l a_l}$ *for* $j \ge 1$, and, since $\lceil j/(\mathsf{L}-1)\rceil \asymp_{\mathsf{L}} j$, it is $\lesssim j^{-\min_l a_l}$ as well. *At* $\mathsf{L} = 2$ *every step is an identity and not a special case*: there is one contrast, $\Pi_{\mathsf{w}} = \delta_1$, $\mathcal{A}^{(1)} = \mathcal{A}_{\mathrm{cond}}$, both comparisons hold with constant 1 and with $j_1 = j$ and offset $\mathsf{L} - 2 = 0$, and $a = a_1$ is the binary a of every earlier round.

Two things do **not** follow, and each is a way the display above invites being over-read. First, the implication runs one way. Per-contrast polynomial decay is **sufficient** for H1's decay clause and is not necessary, a design in which one contrast operator has finite rank and another decays polynomially satisfies that clause and has no a_l at the first contrast, so H1 stays a hypothesis about the stacked operator, assumed and not derived. And it reaches only the decay clause. (5) compares H^s with the powers of the *summed* Gram operator $\sum_l \varpi_{\mathsf{w}}(l)\, \mathcal{A}^{(l)*}\mathcal{A}^{(l)}$; per-contrast matchings do not aggregate to it, and nothing above supplies it. Second, $\min_l a_l$ is not a comparison with "the binary a" unless the binarisation is named. $\mathcal{A}^{(l)}$ is contrast l against the *base* cell in the L-category parameterisation. Merging cells, $W \ge 1$ against $W = 0$, say, produces the contrast $\sum_{l\ge 1} \widetilde{W}_l$, hence by linearity of (21) in the contrast the moment $\sum_{l\ge 1} \mathcal{M}(\cdot,\cdot,l)$ and the linearisation $\sum_{l\ge 1} \mathcal{A}^{(l)}$, which is a **sum** of contrast operators and is none of them; a sum admits cancellation that no hypothesis on its summands excludes, so no exponent transfers to it in either direction. *And merging is not a free re-parameterisation in any case*, for the reason the *Discussion* paragraph gives at its coarsening warning: it adopts the *stronger* identification hypothesis, since $\mathcal{V}_{CI}(W) \subseteq \mathcal{V}_{CI}(g(W))$ for every coarsening g, and it forfeits the localisation. A reader who takes $a = \min_l a_l$ to mean "the exponent improves on the binary design" is reading more than this gives. *What it does license is the comparison between the full stack and a sub-collection of its own contrasts,*

inside one fixed L*-category design*: $\varpi_{\mathsf{w}}(l) > 0$ makes $\ker \mathcal{A}_{\text{cond}} = \bigcap_l \ker \mathcal{A}^{(l)}$, so the full stack has the weakly smaller kernel and the weakly larger rank, and H1's *infinite-rank* clause is to that extent weakly easier at the stack than at any one contrast. *The* ***level*** *of* σ_j *carries no such monotonicity*: Π_{w} is a probability measure, so $\|\mathcal{A}_{\text{cond}} h\|^2$ is an *average* over contrasts and not a sum, and adding a contrast that carries little signal can lower it.

A3 (RETIRED: local rank), moved entire.

Assumption (A3: RETIRED, local rank, now Lemma 91(iv))**.** ~~**RETIRED.**~~ This assumption is not maintained, it is not available for use, and no result in this programme takes it as a hypothesis. It is retired because it is a **theorem**, not because it is false, as A6 was, and not because it is too strong. It read: *suppose A2 holds, so that* $\ell_\star$ *exists; then*

$$J \;:=\; E[\Delta Z\, \Delta Z'] \;+\; E\Big[\widetilde{W}\, \ell_\star(\Delta U, \Delta Z)\, \Delta Z'\Big] \quad \textit{is nonsingular.} \qquad \text{~~[retired]~~}$$

Why it is retired. Lemma 91(iv) proves that under A1's *first* clause the second summand is the zero matrix, $\ell_\star(\Delta U, \Delta Z)\Delta Z'$ is $\sigma(\Delta U, \Delta Z)$-measurable and $E[\widetilde{W} \mid \Delta U, \Delta Z] = 0$, so $J = E[\Delta Z \Delta Z']$ whenever J is defined at all, and nonsingularity of that matrix is A1b(iv). The proof consumes A1, A1b(iv), A2′(i), A2′(iii) and B1(i), (ii), (iv.a), (iv.d), *A1 for the vanishing and A1b(iv) for the nonsingularity; A2′(i), A2′(iii) and the three B1 clauses only to make the second summand a finite matrix in the first place, through Corollary 49. Every one of those tags was already carried by every list that carried A3*, so striking the tag is a strict weakening of four hypothesis lists with no change to any conclusion: Lemma 95(i) and (i′), Theorem 29's group (T-2), and Corollary 30 through (T-2).

What survives, and it is the half the conditional statement was invented to carry. A3 was stated conditional on A2 because $\ell_\star$ exists only under A2, so the display above is a well-formed object only under A2 and "A3 holds while A2 fails" had no truth value. *That remains exactly true of the display* and Remark 98(4) still says so, in the sharper form that what fails under

¬A2 is the *existence* of $\ell_\star$, $L_\star$ and $\psi_\star$ and not a rank condition. It also remains true that A3 was never what identifies $\theta_\dagger$: that is A1b(iv). What is withdrawn is the further claim, made at A1b(iv)'s comment and at three sites in `results.tex`, that the two are **different** matrices. They are the same matrix wherever both are defined.

*What is **not** conceded.* No conclusion weakens, no rate moves, and no other tag changes status. In particular A2′(i.b) is **not** freed, see Lemma 91(iv.1), and A2 is not freed, being consumed at Lemma 93, (131), for the first-order bias, which is where the source condition does its work. *And the counterexample probe is redirected as it always was*: the object to probe is A2.

A4 (overlap and propensity regularity). *The word "consequently" in A4(i) is a derivation, and it is printed here rather than asserted*: (6) is what every statement that divides by Ω points at. Write $\varrho = \varrho(z)$ and $\Omega = \Omega(z)$ throughout, and recall $\varrho_0 = 1 - \mathbf{1}'\varrho$.

The inverse in closed form, and it is where everything else comes from. For any ϱ with $\varrho_l > 0$ at every $l \in \{0, \dots, \mathsf{L}-1\}$,

$$\Omega^{-1} \;=\; \operatorname{diag}(\varrho)^{-1} + \varrho_0^{-1}\mathbf{1}\mathbf{1}', \qquad \text{and} \qquad \left(\Omega^{-1}\widetilde{W}\right)_l \;=\; \frac{\mathbb{1}\{W=l\}}{\varrho_l} - \frac{\mathbb{1}\{W=0\}}{\varrho_0}. \tag{61}$$

A notational convention, stated here because this is the identity map's first site in compiled order: the identity map, on any space and in any dimension, is written Id *throughout this project and never I, italic I being C3's boundary index* (14) *— a closed interval, not an operator.* For the first identity, multiply out: $\big(\operatorname{diag}(\varrho) - \varrho\varrho'\big)\big(\operatorname{diag}(\varrho)^{-1} + \varrho_0^{-1}\mathbf{1}\mathbf{1}'\big) = \mathrm{Id} + \varrho_0^{-1}\varrho\mathbf{1}' - \varrho\mathbf{1}' - \varrho_0^{-1}(1-\varrho_0)\varrho\mathbf{1}' = \mathrm{Id} + \varrho\mathbf{1}'\big\{\varrho_0^{-1} - 1 - \varrho_0^{-1} + 1\big\} = \mathrm{Id}$, using $\operatorname{diag}(\varrho)\mathbf{1} = \varrho$, $\varrho'\operatorname{diag}(\varrho)^{-1} = \mathbf{1}'$ and $\varrho'\mathbf{1} = 1-\varrho_0$; every one of the three needs $\varrho_l > 0$ at some point, and $\varrho_0 > 0$ is what makes the second summand finite. For the second identity, $\widetilde{W}_l/\varrho_l = \mathbb{1}\{W=l\}/\varrho_l - 1$ and $\mathbf{1}'\widetilde{W} = (1 - \mathbb{1}\{W=0\}) - (1-\varrho_0) = \varrho_0 - \mathbb{1}\{W=0\}$, so the two added terms are $\mathbb{1}\{W=l\}/\varrho_l - 1$ and $1 - \mathbb{1}\{W=0\}/\varrho_0$ and the ± 1 cancel.

The floor. Matrix norms in this paragraph are spectral, so that $\lambda_{\min}(\Omega) = \|\Omega^{-1}\|^{-1}$ for the symmetric positive definite Ω. From (61) and the triangle inequality, $\|\Omega^{-1}\| \le \max_{l\ge 1} \varrho_l^{-1} + \varrho_0^{-1}\|\mathbf{1}\mathbf{1}'\| = \max_{l\ge 1} \varrho_l^{-1} + (\mathsf{L}-1)\varrho_0^{-1}$, since the only nonzero eigenvalue of $\mathbf{1}\mathbf{1}'$ on $\mathbb{R}^{\mathsf{L}-1}$ is $\mathbf{1}'\mathbf{1} = \mathsf{L}-1$. Under **A4(i)'s condition** — the sentence beginning "The condition is"; this pointer read "A4(i)'s *first sentence*" through 2026-08-25, and A4(i)'s first sentence is now its *declaration*, which fixes ε_ϱ and asserts no inequality — each of the L cell probabilities is at least ε_ϱ, so $\|\Omega^{-1}\| \le \mathsf{L}/\varepsilon_\varrho$ and $\lambda_{\min}(\Omega) = \|\Omega^{-1}\|^{-1} \ge \varepsilon_\varrho/\mathsf{L}$, which is (6). *The bound is valid and not sharp*: at $\mathsf{L} = 2$ it asserts $\varrho(1-\varrho) \ge \varepsilon_\varrho/2$ where the exact minimum over $[\varepsilon_\varrho, 1-\varepsilon_\varrho]$ is $\varepsilon_\varrho(1-\varepsilon_\varrho)$. Nothing in this corpus is stated *at* the value of the floor, only under it, so the slack costs nothing; *a dispatch that begins to track the constant should re-derive it rather than sharpen this one in place. The base cell is in A4(i)'s quantifier for this reason and not for symmetry*: $\varrho_0 \ge \varepsilon_\varrho$ is what bounds the second summand, and if $\varrho_0 \to 0$ then $\|\Omega^{-1}\| \to \infty$ however far the non-base cells stay from zero. *A version of A4(i) quantified only over* $l \in \{1, \dots, \mathsf{L}-1\}$ *would therefore not deliver* (6), and at $\mathsf{L} = 2$ it would be the one-sided condition $\varrho \ge \varepsilon_\varrho$, which is strictly weaker than the two-sided interval every earlier round assumed.

The prefactor, which is what A4(i) cancels. Write $F_m(u \mid z) := P(\Delta U \le u \mid W = m, \Delta Z = z)$ and $d(u \mid z) := \big(F_1 - F_0, \dots, F_{\mathsf{L}-1} - F_0\big)'(u \mid z)$. Then, at $\sigma(\Delta Z)$ and only there,

$$E\Big[\widetilde{W}\, \mathbb{1}\{\Delta U \le u\} \,\Big|\, \Delta Z = z\Big] \;=\; \Omega(z)\, d(u \mid z). \tag{62}$$

Component l of the left side is $\varrho_l F_l - \varrho_l \sum_m \varrho_m F_m = \varrho_l\big(F_l - \sum_m \varrho_m F_m\big)$; writing $F_l - F_m = d_l - d_m$ with $d_0 := 0$ and using $\sum_m \varrho_m = 1$ gives $F_l - \sum_m \varrho_m F_m = d_l - \varrho' d$, so the vector is $\mathrm{diag}(\varrho)d - \varrho\varrho' d = \Omega d$. At $\mathsf{L} = 2$ this is $\varrho(z)(1-\varrho(z))\{F_1(u \mid z) - F_0(u \mid z)\}$, the display the block has carried since round 1. Under (6) the map $\Omega(z)$ is invertible and the restriction can be *divided through. That division is what makes the reading of Remark 35 available; it is* ***not*** *performed by the estimator*, whose operator and moment both carry the raw contrast.

What that division is **not** needed for, and a claim withdrawn here. It is not what makes Lemma 11 an equivalence. A first version of this paragraph said it was, and located the failure on $\{z : \det \Omega(z) = 0\}$; *both halves are wrong at* $\mathsf{L} \geq 3$. Take $\mathsf{L} = 3$ and $\varrho(z) = (q, 0)'$ with $q \in (0, 1)$: cell 2 is null, $\det \Omega(z) = 0$, and yet W is not conditionally degenerate, $\widetilde{W} \neq 0$, and the restriction still forces $F_1 = F_0$. The equivalence $\Omega d_\varsigma = 0 \Leftrightarrow$ conditional independence holds at *every* z and under *no* overlap condition, by the two-case argument given in the paragraph *A1* above, Ω_A nonsingular when $\varrho_0 > 0$, $\ker \Omega_A = \operatorname{span}\{\mathbf{1}_A\}$ when $\varrho_0 = 0$, and in both cases the non-null cells are forced to share one conditional law. *The only set on which the restriction is vacuous is where* W *is conditionally degenerate*, that is, where $\Omega(z) = 0$, equivalently where $\max_{0 \leq l \leq \mathsf{L}-1} \varrho_l(z) = 1$. *This is the* L*-category reading of* $\mathcal{Z}_0$ *and it is* ***not*** *a new definition*: $\mathcal{Z}_0$ is defined at (149) in exactly this form, which recovers the binary $\{z : \varrho(z)(1 - \varrho(z)) = 0\}$ verbatim at $\mathsf{L} = 2$. *As a condition on the propensity vector, degeneracy is strictly stronger than singularity of* Ω *once* $\mathsf{L} \geq 3$, $\varrho = (q, 0)'$ is singular and not degenerate, *though for a particular data-generating process the two* z*-sets may of course coincide*, and nothing here asserts that they do not. *Lemma 11 states its equivalence under B1(i)–(iv) alone and its clauses (a) and (b) are correct as stated at every* L; what does not generalise is the binary *ground* recorded beside it, that the scalar prefactor vanishes on the degenerate region. A dispatch that generalises the printed reason instead of the printed result will import an overlap hypothesis (a)⇔(b) neither has nor needs.

The statement is unqualified, and a qualification an earlier draft attached to it is withdrawn. That draft said Lemma 11's clauses naming $\mathcal{M}$ and T acquire A4(i), because those objects would not exist at a normalised contrast. The estimator carries the raw contrast, so $\mathcal{M}$ and T are defined everywhere and bounded, and no clause of Lemma 11 needs an overlap condition. *The draft's reason was also wrong on its own terms*: $\Omega^{-1}\widetilde{W}$ is not "non-integrable" where a cell vanishes, it is *undefined* there, Ω having no inverse, and where it is defined it is integrable with a ϱ-free bound, as Remark 35(4) computes. *Undefinedness and non-integrability are different failures* and the corpus must not conflate them.

The determinant, since it was used above and its usual derivation presumes what fails there. For ϱ in the open region where every $\varrho_l > 0$, the matrix determinant lemma applied to (61) gives $\det \Omega = \det \operatorname{diag}(\varrho)\,(1 - \varrho' \operatorname{diag}(\varrho)^{-1}\varrho) = \left(\prod_{l\geq 1} \varrho_l\right)\varrho_0$. *Both sides are polynomials in* $(\varrho_1, \ldots, \varrho_{\mathsf{L}-1})$, they agree on a non-empty open subset of $\mathbb{R}^{\mathsf{L}-1}$, and two polynomials agreeing on a non-empty open set agree identically; hence $\det \Omega = \prod_{l=0}^{\mathsf{L}-1} \varrho_l$ *everywhere*, including at the boundary where Ω is singular and the lemma does not apply. *That is exactly where the identity is wanted*, which is why the continuation step is printed rather than the lemma cited alone.

If the prefactor is inverted at all, which the estimator does not do, but which the reading of Remark 35 does, the inverse is Ω^{-1}, and that is settled by an identity rather than by a preference. The scalar $\varrho(1-\varrho)$ admits exactly one componentwise generalisation, the diagonal matrix with entries $\varrho_l(1-\varrho_l)$, and it is wrong. (62) says the map carrying the cell-versus-base contrasts d into the observable moment is Ω, whose off-diagonal entries $-\varrho_l\varrho_m$ are the negative cross-cell correlation that a multinomial has and a collection of separate binaries does not. Dividing component l by $\varrho_l(1-\varrho_l)$ therefore does not return d_l, and what it returns is not a contrast between cell l and the base cell at all. Two further facts make Ω^{-1} the right inverse to take. By (61), $\Omega^{-1}\widetilde{W}$ is the vector of *inverse-probability* contrasts of each cell against the base, so Ω^{-1} is the only inverse under which the contrast vector is the Horvitz–Thompson object the binary design already was, and at $\mathsf{L} = 2$ it reads $(W-\varrho)/\{\varrho(1-\varrho)\} = \mathbb{1}\{W=1\}/\varrho - \mathbb{1}\{W=0\}/(1-\varrho)$, the identity a binary design writes down without comment. And the quadratic form is base-free: $\widetilde{W}'\Omega^{-1}\widetilde{W} = (1-\varrho_W)/\varrho_W$, where ϱ_W is the conditional probability of the *realised* cell, a quantity that names no base cell, is symmetric in all L cells, and satisfies $E[\widetilde{W}'\Omega^{-1}\widetilde{W} \mid \Delta Z] = \mathsf{L} - 1$. The componentwise weighting has neither property. So **which** inverse to take, if one is taken at all, is settled by an identity and not by a preference.

Whether to take it is a second question, and the answer is **no**: the estimator carries the

raw contrast (1) on both sides. The operator equation and ψ of (38) are both written at $\widetilde{W}$, and Ω^{-1} appears nowhere in either. The reading it supports is retained, as Remark 35, and is interpretation rather than specification.

What normalising would have changed, so that the trade is legible. By the tower property with (62) the two candidate forward residuals are

$$T(u,t,\cdot) \;=\; E\Big[\Omega(\Delta Z)\, d(u \mid \Delta Z)\, w(t'\Phi(\Delta Z))\Big], \qquad T^{\Omega}(u,t,\cdot) \;=\; E\Big[d(u \mid \Delta Z)\, w(t'\Phi(\Delta Z))\Big],$$

which differ in where Ω sits relative to the *outer* integration over ΔZ, a difference no choice of norm on the range space can undo. The gain would be that the conditional forward residual is the cell difference d itself rather than a matrix mixture of it.

The price is that T^{Ω} has a different linearisation, and the whole smoothness budget is stated at the one the paper has. A2's range condition, A2′, A1c and H1, the last through $\sigma_j(\mathcal{A}_{\text{cond}}) \asymp j^{-a}$, the exponent of the *stacked* operator on the whole three-factor range space at every L, are all stated at $\mathcal{A}_{\text{cond}}$, and $\mu_{\text{H1}} = s/(2a)$, the two daggered lines of the smoothness-budget table below and the comparison to the cross-sectional companion sit downstream of a. (7) *does not*, its μ being (P-a.3)'s primitive index, and no other line of that table imposes a condition involving a; *that narrows what a re-derivation would have to cover and does not remove the objection*, since the scale H^s that Definition 80, A5(vi′), (P-c) and (P-g) are written in, and A2's Picard form, would both have to be re-established at the normalised operator. *A4's note (c) already records that the two operators must not be assumed to share range or kernel*, and the paragraph on Khan–Tamer irregularity scopes range-invariance to a *constant* ϱ, a z-varying Ω being exactly what that scoping excludes. *An interpretational gain does not buy a re-derivation of the ill-posedness budget*, and the author declines the trade. *Nothing else in the dispatch depends on it*: the L-category instrument, the contrast vector, Ω as covariance and prefactor, A4(i), (11) and (6) all stand.

One consequence of the reversal is worth stating positively. At $\mathsf{L} = 2$ the operator, ψ and

every assumption clause reduce to the printed binary objects *verbatim, with no exceptions*, and the binary estimator is unchanged. Under the normalisation they did not: (12) would have carried a factor the pre-dispatch display did not have.

And one argument made along the way is false and is recorded because it is the first repair a re-inventor will reach for. It ran: Ω is $\sigma(\Delta Z)$-measurable, so $L\widetilde{W} = (L\Omega)\widetilde{W}^{\Omega}$, and $L \mapsto L\Omega$ is a bijection of Definition 19's admissible class onto itself, whence a normalised operator could be paired with a ψ printed at $\widetilde{W}$ without loss. *Four bounds were checked and each of the four is correct, the norms being spectral on Ω and Frobenius on L, ℓ and b, as the notation block fixes, which is why the two are named here rather than left to context*, $L_{\ell}^{b}\Omega = L_{\ell\Omega}^{b\Omega}$ since Ω does not depend on u; $\|\Omega\| \le 1$, because for a unit c, $c'\Omega c = \mathrm{Var}(c'\widetilde{W} \mid \Delta Z) \le E[(c'\widetilde{W} + c'\varrho)^2 \mid \Delta Z] = \sum_l c_l^2 \varrho_l \le 1$, so $\|\ell\Omega\|_\infty \le \|\ell\|_\infty$ and $E\|b\Omega\|^2 \le E\|b\|^2$, *a Frobenius norm is submultiplicative against a spectral one,* $\|AB\|_F \le \|A\|_F \|B\|$; the inverse map with $\|\Omega^{-1}\| \le \mathsf{L}/\varepsilon_\varrho$; and $E[L\Omega \mid \Delta Z] = E[L \mid \Delta Z]\Omega$, so N1 transfers. They are jointly insufficient, because Definition 19 does not define its admissible class by bounds. Its clause (a) binds $\ell := \mathcal{W}\ell^{\Pi}$, so membership requires the generator to lie in $\mathcal{R}(\mathcal{W})$; the z-sections of $\mathcal{W}$ are Laplace transforms against $w = \exp$ on a compact index set under B1(i)–(ii) and (iv.c), hence *real-analytic* in z, while A4(ii) makes ϱ, and therefore Ω, only Hölder of order r. *Right-multiplication by a non-analytic $\Omega(z)$ leaves $\mathcal{R}(\mathcal{W})$*, and the failure occurs already at $\mathsf{L} = 2$ with $p = 1$ and a ϱ that is Hölder but non-analytic at a point. So the two parameterisations do not have the same admissible weight class, and no bijection is claimed anywhere in this corpus. *The four bound-checks are left on record rather than deleted*, because the lesson is that verifying the bounds of a class is not verifying membership of it when the class is defined by a range condition, *and because the same analytic-versus-Hölder mechanism is what makes the author's ground for keeping the raw contrast bite*: a normalised $\mathcal{A}_{\mathrm{cond}}$ cannot be assumed to inherit the range and decay statements A2, A2′, A1c and H1 are made at.

Nothing in the assumption block changes on account of any of this, because the estimator

was not re-written. The bounds the block does use are the free ones, $\|\widetilde{W}\|_\infty \le 1$ and $\|\widetilde{W}\|_2 \le \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$, uniform in L and free of ϱ; A4(i) remains idle in Proposition 36, Proposition 8 and Proposition 9; $\hat{T}$ remains affine in $\hat{\varrho}$, so the propensity branch of the operator error keeps its exactly-zero second-order term; and no clause of Lemma 11 needs an overlap condition.

The lower bound $d_Z \ge 1$ is load-bearing. At $d_Z = 0$ the exponent $\alpha = r/(2r + d_Z)$ is $\frac{1}{2}$ *exactly*, and:

- One site genuinely breaks, and it is the φ-step. (P-d)'s sieve-entropy condition is $L_x \ln n = o(n\hat{\delta}_n^2)$. At $\alpha = \frac{1}{2}$, A5(i)'s $\hat{\delta}_n = \max\{n^{-1/2}\sqrt{\ln n}, r_n\}$ is $n^{-1/2}\sqrt{\ln n}$, *an equality, and it is the declared floor of clause* (i) *that supplies it, not a two-sided reading of an* O_P, so $n\hat{\delta}_n^2 = \ln n$ exactly and the condition reads $L_x = o(1)$, **no integer** $L_x \ge 1$ **satisfies it**. Equivalently, the window (54) is empty at every s, its right-hand side $9\alpha/\big(8(1-2\alpha)\big)$ diverging as $\alpha \uparrow \frac{1}{2}$. Proposition 28 therefore has no admissible sieve dimension, group (T-5) of Theorem 29 is unsatisfiable, and the theorem is *vacuous* at $d_Z = 0$ rather than false.
- *Three further sites print "*$\alpha < \frac{1}{2}$*" as a* ***reason*** *and are false as printed, although each conclusion happens to survive at the boundary*: Lemma 92's statement, which writes $\alpha = r/(2r + d_Z) < \frac{1}{2}$; that lemma's Step 6, whose collapse argument gives $\kappa\alpha < \frac{1}{2}$ from "$\kappa < 1$ and $\alpha < \frac{1}{2}$", true at $\alpha = \frac{1}{2}$ from $\kappa < 1$ alone; and Remark 125's "impossible because $\alpha < \frac{1}{2}$ always". *A true conclusion resting on a false printed reason is the harder defect to find*, which is why they are listed.

$d_Z = 0$ *is not a pathology and it is not being excluded as one*: it says ϱ has no non-degenerate argument, hence is a constant, hence is estimable at the parametric rate, a genuinely easier propensity problem. *What fails there is not the propensity but the* φ*-step's entropy budget*, because $\hat{\delta}_n$ then sits at $n^{-1/2}$ and leaves no room between the noise level and the sieve dimension. *A block that assumes a strict inequality should say which primitive delivers it*,

and this is that primitive.

The upper bound $d_Z \leq p-1$ is a **modelling** restriction, and read with A1b(iv) it is sharper than it looks. It is stated on one line and, unlike its companion above, has been discussed nowhere; the asymmetry is not warranted, because the two bounds do different work. The lower bound protects a rate. *The upper bound restricts the specification of* ΔZ, as follows.

- d_Z counts the *non-degenerate* coordinates of ΔZ, degenerate meaning almost surely equal to a constant, which is what makes a coordinate carry no information for the propensity regression, so $d_Z \leq p-1$ says that *at least one* coordinate of ΔZ is a.s. constant.
- *That constant cannot be zero.* If coordinate j is a.s. 0 then row and column j of $E[\Delta Z \Delta Z']$ vanish and the matrix is singular, contradicting A1b(iv).
- And there cannot be two of them. If coordinates $j \neq k$ are a.s. the non-zero constants c_j and c_k, then rows j and k of $E[\Delta Z \Delta Z']$ are $c_j E[\Delta Z]'$ and $c_k E[\Delta Z]'$, proportional, and the matrix is singular again.

So A4(ii) and A1b(iv) together force $d_Z = p-1$ ***exactly****, and force* ΔZ *to contain exactly one constant coordinate, non-zero:* ***the differenced equation must carry a period effect.*** That is a restriction on how the analyst writes the model and not a regularity condition, and it is substantive: a specification of ΔZ in which every coordinate varies is *excluded* by A4(ii), and one in which two do not is excluded by A1b(iv). *What is* ***not*** *claimed is that the period effect is non-zero in the population*, the coefficient on the constant coordinate is an ordinary component of $\theta_\dagger$ and may be zero; what is forced is that the coordinate is *present*, so that ΔZ spans an intercept in differences. *Nor is* $d_Z = p-1$ *a strengthening of anything assumed*: it is a consequence of two clauses the block already maintains, which is precisely why it had to be derived somewhere rather than left for a reader to notice.

The σ*-algebra of the fold is written out and given no symbol here*, because Definition 2,

which names it $\sigma(\mathcal{I}_a)$ by adopting Lemma 60's convention, comes several hundred lines later in this file, and a maintained clause must not name an undefined object; this is the write-it-out precedent C12/C24/C78, applied for the same reason A7's own statement forced Definition 2 into Part V. *The joint-measurability half is load-bearing*: two things this clause is asked to support are not well posed without it. *First, its own display.* "O_P" is a statement about the distribution of $\omega \mapsto \|\hat{\varrho} - \varrho\|_{L^2}$, which is an *integral of a random function*; joint measurability makes it a measurable function of the sample, by Tonelli's theorem applied to the non-negative jointly measurable integrand $(\omega, z) \mapsto \|\hat{\varrho}(\omega)(z) - \varrho(z)\|^2$ against the law of ΔZ. Per-realisation Borel measurability of $z \mapsto \hat{\varrho}(\omega)(z)$ does not supply it. *The integrand is a sum of* $\mathsf{L} - 1$ *squares and is jointly measurable exactly when each component is*, so the clause is read componentwise and the argument is the one it always was; *and the two norms in the display are different objects*, the inner $\|\cdot\|$ is Euclidean on $\mathbb{R}^{\mathsf{L}-1}$ and the outer $\|\cdot\|_{L^2}$ integrates it against the law of ΔZ, which is what A4(iii)'s ":=" spells out. *Second, Lemma 77 Step 4*, which freezes $\hat{\varrho}$ on the auxiliary fold and applies (P8)'s freezing lemma. That lemma's hypothesis is joint measurability of the *composite*, and the composite there is $\|\hat{\varrho}(\omega)(\cdot) - \varrho(\cdot)\|$ read at the main fold's ΔZ-draws, which the clause above supplies by composition with the Borel coordinate maps. *The shape is Definition 24*(c′.1)*'s, one Part later*: a requirement on the estimator's own pointwise function and **not** a container declaration. That matters and is not a stylistic preference, no equivalence class is formed and no representative is chosen, so Definition 2's truncation, being a pointwise operation, continues to make $\varepsilon_\varrho \leq \left|\widehat{\widetilde{W}}_{i,l}\right| \leq 1$ hold *surely* at every i and every l, and Definition 24(a)'s reading of that bound at the folds is exact rather than almost sure. *And it costs nothing about P*: it is a construction requirement on an estimator, satisfied by any $\hat{\varrho}$ given by a formula jointly measurable in the auxiliary sample and the evaluation point, which every kernel and every series construction is, by inspection of the formula. *The Carathéodory route Definition 24*(c′.1) *records for* (41) *is available here too but is* ***not*** *the same statement*: it delivers joint measurability from continuity in one argument and measurability in the others, so here it would run on continuity of $z \mapsto \hat{\varrho}(\omega)(z)$

at each realisation, true of a kernel or series construction and not of a measurable selection, and in any case a sufficient condition and not the requirement, which is the display above.

A5 (nuisance rates). *Clause* (i) ***declares*** *its two branches and* ***assumes*** *one thing, and the two must not be run together.* What the clause assumes about the data-generating process is that the operator error is $O_P(\hat{\delta}_n)$, the last equality of (116), which Lemma 77 delivers rather than requires. What it declares is the level itself: $\hat{\delta}_n$ is the noise level *fed to* the discrepancy principle, and the stagewise schedule is computed from it before the data are seen, so it is a deterministic sequence and the two superscripts are component tags naming which summand of the maximum is meant, not estimators. *A deterministic sequence does not carry an* O_P, and this block's own preamble already says the clauses *define* these rates.

Nothing about the data-generating process is strengthened by the repair, and the floor is the thing gained. The $O_P(\hat{\delta}_n)$ half is upper-only, as it was, and is still discharged by Lemma 77; every "A5(i) is discharged, not assumed" downstream survives verbatim. What changes is that

$$\hat{\delta}_n \;\geq\; \hat{\delta}_n^{(T)} \;=\; n^{-1/2}\sqrt{\ln n}$$

is now **arithmetic**, a property of a maximum whose branches are named, where before it was a two-sided reading of an O_P. *Lemma 77*(i) *cannot supply it and is no longer cited for it*: that clause is a Chebyshev tail bound and bounds the empirical component from above. The floor is what Proposition 79(i) runs Chebyshev against, and it is the reason a statement proved at $\frac{1}{2}\mathsf{c}\, n^{-1/2}\sqrt{\ln n}$ may be printed at $\mathsf{c}\,\hat{\delta}_n$.

The clause's closing sentence is struck, and what replaces it is a quantifier. It read "in every configuration of interest $\hat{\delta}_n = \hat{\delta}_n^{(\varrho)} = O_P(r_n)$", an identity with no "eventually" in it, and *false as printed* over an admissible range. The propensity branch attains the maximum exactly when $n^{1-2\alpha} > \ln n$, so the crossover sits near $n = 5{,}500$ at $\alpha = \frac{3}{8}$ and *diverges* as $\alpha \uparrow \frac{1}{2}$; at $n = 2{,}000$, the sample size Lemma 85 prints an inflation factor at, the *parametric* branch is

the maximum, and the shipped headroom ledger carries a branch column that records which branch binds cell by cell. *What survives is an eventual statement and it is a consequence of A4(ii), not a configuration label*: $d_Z \geq 1$ gives $\alpha = r/(2r + d_Z) < \frac{1}{2}$ strictly, hence for each admissible configuration there is a finite $n_\star$ with $\hat{\delta}_n = \hat{\delta}_n^{(\varrho)} = r_n$ at every $n \geq n_\star$. *That is precisely the licence clause* (iii) *needs* in order to write $\hat{\delta}_n = n^{-r/(2r+d_Z)}$ inside an asymptotic condition, and clause (iii) now names it.

It is **not** the licence the two sites that consumed the struck sentence as strict dominance needed, and neither of them cites it any longer. Remark 82(2) needs $n\hat{\delta}_n^2$ to diverge *polynomially*, which the floor does not give, the floor gives $n\hat{\delta}_n^2 \geq \ln n$, and it now carries $\alpha < \frac{1}{2}$-with-a-margin as its own scope condition on Step 6. Remark 125(2) needs only $\hat{\delta}_n \gg n^{-1/2}$, which the floor delivers outright, in *every* configuration and not merely where the propensity branch binds; that site is strengthened by the repair rather than weakened.

The factor $\ln n$ *is not optional*: Proposition 28 delivers (56) at $(\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)}$, and (P-f), Remark 125(0) and Proposition 79(iv) all read that log-inflated δ_n. A log-free reading of this clause is off by $(\ln n)^{2\mu/(2\mu+1)}$. *The* ***rate*** *is what that proposition delivers and the* ***object*** *is not the object of this clause*, a distinction that costs nothing here and is load-bearing two paragraphs over: (56) is stated at the infeasible reading $\hat{\varphi}^\pi$ of Definition 22 from 2026-08-27, while this clause is stated at $\hat{\varphi}$, so the clause's first half is **assumed** and not discharged — and δ_n is the same deterministic sequence either way, which is all this paragraph needs. Remark 97 carries the full disposition, including which of the four rate products Theorem 29 needs are affected. In one line: the second half of this clause and clause (iii) are **unchanged**, because (7) is a strict inequality between polynomial exponents and its strictness absorbs the logarithm; clause (iv) becomes **stronger** by the factor $(\ln n)^{2\mu/(2\mu+1)}$, and that is the only place the logarithm bites; clause (v) becomes **weaker**, its right-hand side being inflated; and (55) does not move, having been computed at the inflated δ_n already.

Clause (iv)*'s fold and measurability are fixed by Lemma 60's sample-splitting convention,*

which also names what that measurability is of. *The naming is not decoration.* The first display is an O_P statement about $\omega \mapsto \|\hat{\ell} - \ell_\star\|_{L^2(P)}$, an integral of a random function; joint measurability of the fixed representative makes it a measurable function of the sample by Tonelli's theorem, and *membership plus an unspecified "measurability" does not. Nothing here is strengthened*: the joint measurability is scoped to what Lemma 60 Steps 0 and 2 use, which is strictly less than Definition 24(c′.1)'s (41), because Lemma 60 evaluates $\hat{\ell}$ at an undistorted $(\Delta U, \Delta Z)$ draw. *In Part I it is* (41) *that governs and not this convention*: what Definition 24(d) reads at the folds is this clause's *display*, and the measurability of $\hat{\ell}^{(-k)}$ is Definition 24(c′.1)'s own. What this clause does and does not say about $\hat{L}$. It prices $\hat{\ell}$, and nothing else. The identity $\|\partial_u(\hat{L} - L_\star)\|_{L^2(P)} = \|\hat{\ell} - \ell_\star\|_{L^2(P)}$ is **an assertion, not notation**, and is *not* made here: it holds if and only if $\hat{L}$ is differentiable in u at every point with $\partial_u \hat{L} = \hat{\ell}$, that is, if and only if $\hat{L} = L_{\hat{\ell}}^{\hat{b}}$ in the sense of Definition 19(b), which is exactly the construction obligation Remark 57 records as owed by Step 6 or 7 and *undischarged in this block*. Lemma 60(i)–(ii) consume the clause in the form printed above, at $\hat{\ell}$ itself, and need nothing further; Lemma 56(iv) consumes it at $\|\partial_u D\|_{L^2(P)}$ with $D = \hat{L} - L_\star$, and may therefore be instantiated at the estimated direction only *after* that obligation is discharged.

The clause is a statement about a finite quantity, and what makes it one is (8) *together with H1*: $h = \hat{\varphi} - \varphi_\dagger$ with $\hat{\varphi} \in \mathcal{S}_n$ at the configuration Lemma 83 states, so $\hat{\varphi} \in H^s$ by (8) and $\varphi_\dagger \in H^s$ by H1's source set; without the first of those $\|h\|_{H^s} = +\infty$ surely and the $O_P(1)$ statement would hold on the *empty* event rather than with probability tending to one.

Clause (vi′)*'s display is a probability statement about a finite quantity only because* ***both*** *memberships hold, and both are needed.* Every stage $\hat{\varphi}^{(0)}, \ldots, \hat{\varphi}^{(\bar{k})}$ lies in $\mathcal{S}_n$ once $\hat{\varphi}^{(0)}$ does, by Definition 22(b),(d), and hence in H^s by (8), *while* $\varphi_\dagger \in H^s$ *by H1's source set.* The display being a norm of a *difference*, either membership failing makes it $+\infty$ at every realisation and the event **empty** rather than merely improbable.

That same R *serves at every site Definition 80 enumerates, and* ***that*** *is where the enumeration*

is maintained: Definition 80 carries the list, and this clause, Lemma 83 and Remark 87 point at it and maintain no copy. *"No second copy is maintained" is what this sentence used to say and it was false*: Definition 24's clause-(d) reading carries a three-entry instance list, which now declares itself an instance list and points here — the very failure the archaeology below condemns, committed by the sentence recording it. *The quantifier is an existential over one radius meeting all of those demands at once*, which is what Lemma 83 consumes and what Remark 87 explains; an $O_P(1)$ reading does not serve, because Definition 80's ball must be deterministic and an $O_P(1)$ bound is a family indexed by tolerance rather than a number.

A8 (sieve window). *The word "consequently" is a derivation, and it is printed here rather than asserted*: ten live citations rest on it. Write p_b for the degree, so $p_b > s - \frac{1}{2}$; and write s' for a *generic* order of the Sobolev scale, bound in this derivation and in H1's boundary-condition paragraph and used nowhere else. A polynomial spline of degree p_b on a mesh with simple interior knots lies in $C^{p_b-1}(\mathcal{X})$ (Schumaker, 2007); its p_bth derivative is constant on each mesh cell, hence a finite step function; and a step function on a bounded interval lies in $H^{s'}$ for every $s' < \frac{1}{2}$, and, unless it is constant, i.e. unless the spline is a single polynomial, in no $H^{s'}$ with $s' \geq \frac{1}{2}$. Hence $\mathcal{S}_n \subset H^{s'}(\mathcal{X})$ for every $s' < p_b + \frac{1}{2}$, and the exponent $p_b + \frac{1}{2}$ cannot be raised uniformly over the span. Since $p_b > s - \frac{1}{2}$ gives $s < p_b + \frac{1}{2}$, this is (8). Three ingredients of that derivation are load-bearing and each is now stated rather than read in. The interior knots must be *simple*: at an interior knot of multiplicity two the spline is only C^{p_b-2} there, its (p_b-1)st derivative already jumps, and membership drops by a full order to $s' < p_b - \frac{1}{2}$, so the containment fails at the stated degree. The space must be the *classical* one, for the reason H1 now records. And $p_b > s - \frac{1}{2}$ is *strict*, $s' = p_b + \frac{1}{2}$ being exactly where membership fails.

the sample operator $\hat{T}$. Why the truncation is in the maintained block and not in a proof. Without it, $\hat{\varrho}$ taking the value of the realised dummy vector $\left(\mathbb{1}\{W=1\},\ldots,\mathbb{1}\{W=\mathsf{L}-1\}\right)'$ has positive probability, hence so does $\widehat{\widetilde{W}}_i = 0$ in $\mathbb{R}^{\mathsf{L}-1}$; on the event that this holds at every

$i \in \mathcal{I}_m$ the sample operator is identically zero and every statement about it is vacuous. A4(i) constrains ϱ and says nothing about $\hat{\varrho}$, so the degeneracy cannot be excluded by assumption, it is excluded by construction, which is where it belongs.

And the truncation set is the ε_ϱ-interior of the simplex (11) and not a box, for a second and separate reason. On (11) the implied base cell $\hat{\varrho}_0 = 1 - \sum_{l \geq 1} \hat{\varrho}_l$ is itself at least ε_ϱ, so $\hat{\varrho}$ is a probability vector bounded away from every face; *on the box* $[\varepsilon_\varrho, 1 - \varepsilon_\varrho]^{\mathsf{L}-1}$ *it need not be one at all*, the implied base cell being possibly negative, in which case $\hat{\varrho}_0$ names nothing, the *estimated contrast covariance*

$$\hat{\Omega}_i \;:=\; \operatorname{diag}\left(\hat{\varrho}(\Delta Z_i)\right) - \hat{\varrho}(\Delta Z_i)\hat{\varrho}(\Delta Z_i)'$$

need not be positive semi-definite, and the overlap diagnostic $\min_l \hat{\varrho}_l$ the application reports is not what it is taken for. $\hat{\Omega}_i$ *is declared here, at its first use, and not in Definition 2*: the estimator does not evaluate it, (12) carries the raw contrast, so a declaration inside the definition would name an object that definition never uses. *It is used by the diagnostics and by the two clauses below*, which is why it is declared at all. *Be exact about what does and does not need this*: the two surely-bounds of Remark 33(2) hold on the box as well, since each uses only $\hat{\varrho}_l \in [\varepsilon_\varrho, 1 - \varepsilon_\varrho]$. *What needs* (11) *is that the fitted object be a probability vector*, which is a requirement on the first stage rather than a bound any proof invokes. At $\mathsf{L} = 2$ the box and (11) coincide, which is why no earlier round had to distinguish them.

the smoothed sample linearisation. This fixes a reading, and the reading is substantive rather than notational. $\hat{\mathcal{A}}_\varphi$ is a **separately smoothed operator** and *not* the Fréchet derivative of $\hat{T}$ in φ; the hat marks a sample analogue of $\mathcal{A}_\varphi$, not a derivative of $\hat{T}$. *The bandwidth is what forces the choice*: C3 attaches C_{h_x} to $\hat{\mathcal{A}}_{\mathrm{cond}}$ and $\hat{\mathcal{A}}^*_{\mathrm{cond}}$, and a derivative of $\hat{T}$ would carry no bandwidth of its own. *And the rejected reading is now known to be unavailable rather than merely different*: by Proposition 70(v), $\hat{T}(\cdot, \theta)$ is nowhere Fréchet

differentiable on $\mathcal{V}$, so "the derivative of $\hat{T}$" names no operator.

Forward obligation, DISCHARGED at Definition 69 and Proposition 70, with one negative answer. This definition names a construction; it does not perform one. Step 6 was to exhibit $\hat{\mathcal{A}}_\varphi$ explicitly, in particular the smoothed indicator and the two kernel estimates, and to verify the two conditions then maintained for the object exhibited. (100) exhibits it; A7 is verified in the amended form (13) by Proposition 79; and *the tangential cone condition A6 is not verified but refuted* (Proposition 71), which is why the block below carries it as retired. Recorded in Remark 58 as obligation (O4).

A6 (RETIRED: tangential cone condition), moved entire.

Assumption (A6: RETIRED, tangential cone condition for the *random* operator)**.** ~~**RETIRED.**~~ This assumption is not maintained, it is not available for use, and no result in this programme takes it as a hypothesis. It read: *there exist* $\eta_{\mathrm{TCC}} < \frac{1}{2}$*, not depending on* n*, and a ball* $\mathcal{V}_n \subset \mathcal{V}$ *around* $\varphi_\dagger$ *containing the Landweber–Fridman path, such that, with* $\hat{\mathcal{A}}_{\tilde{\varphi}}$ *the smoothed sample linearisation of Definition 3,*

$$P\Big(\big\|\hat{T}(\varphi,\theta)-\hat{T}(\tilde{\varphi},\theta)-\hat{\mathcal{A}}_{\tilde{\varphi}}(\varphi-\tilde{\varphi})\big\|_{L^2(\Pi)} \;\le\; \eta_{\mathrm{TCC}}\,\big\|\hat{T}(\varphi,\theta)-\hat{T}(\tilde{\varphi},\theta)\big\|_{L^2(\Pi)}\ \forall\,\varphi,\tilde{\varphi}\in\mathcal{V}_n\Big)\;\longrightarrow\;1. \qquad \text{~~[retired]~~}$$

Why it is retired, and it is not a matter of strength. Proposition 71 proves that the displayed event has probability zero, at every n, at every $\eta_{\mathrm{TCC}} < 1$, at every ball of positive $L^2(\pi)$-radius, and **for every bounded operator** $\mathcal{V} \to L^2(\Pi)$ **put in place of** $\hat{\mathcal{A}}_{\tilde{\varphi}}$. An assumption whose event is null cannot be maintained and cannot be repaired by lowering a constant or shrinking a ball. *What it should have said is Definition 73 (A6b)*, which pairs $\hat{\mathcal{A}}_\varphi$ with the operator it actually differentiates, $\hat{T}_{h_x}$; A6b is a *property*, on the A2b/A2c model, never asserted and consumed by no theorem. *Scope of the refutation, stated because it is narrower than the retirement*: Proposition 71 is a statement about $L^2(\pi)$-balls in $\mathcal{V}$. On the spline sieve $\mathcal{S}_n$ the spike it uses does not exist, so a cone condition *restricted to* $\mathcal{V}_n \cap \mathcal{S}_n$ is untouched by

it, and would still not deliver the inherited route, for the independent reason recorded at Remark 72(3). *Nothing downstream is affected*: Proposition 28 uses neither A6 nor A6b.

C2′ (smoothness of the residual law). Two readings of that display are fixed here, **no letters are attached to them**, so that nothing below can be cited as a sub-clause of C2′(i), because without them neither $E[\bar{f}_k]$ nor $\|\bar{f}_k\|_\infty$ is a well-defined quantity.

First, the version, and why each $\bar{f}_k$ is measurable. The density is read at a version that is **jointly Borel** in $(u, x_1, x_2, z, \mathsf{w})$ and, at *every* conditioning point rather than almost every one, $p_K - 1$ times continuously differentiable in u, which is what "continuously differentiable in its first argument" is read to mean above. *The supplier is one Part away and is now cited rather than left implicit*: (P6a) of Definition 109 fixes exactly such a version for the conditional law of $\Delta Y_\theta - \Delta\varphi$ given $\mathcal{F}_{XZW} = \sigma(X_1, X_2, \Delta Z, W)$, for every ω, at $(\theta, \varphi) = (0, 0)$ that law *is* the one this clause states, and every other is a translate of it by the shift identity of (P6), and the Doob–Dynkin lemma transports an $\mathcal{F}_{XZW}$-measurable function of ω to a Borel function of the conditioning *value*, which is the argument $\bar{f}_k$ carries. Those two supply the version **one variable at a time**, and the declaration above is a **joint** one, so a third step is named rather than elided. (P6a) gives joint measurability in (ω, u); Doob–Dynkin runs at each *fixed* u and returns, for that u, a Borel function of the conditioning value; neither, nor the two together, is joint Borel measurability in $(u, x_1, x_2, z, \mathsf{w})$. *What delivers it is the rationals-plus-continuity step this paragraph already performs below for the envelope*: at every conditioning point the map $u \mapsto \partial_u^k f$ is continuous, and a function that is Borel in one argument at each fixed value of the other and continuous in the other at each fixed value of the first is jointly Borel, the Carathéodory criterion, reached here by taking u along the rationals and passing to the limit, which is the same device Definition 24(c′.1) runs for (41) and the proof of Proposition 8 runs for G_φ. *Nothing that consumes this clause is affected*: every site reads $\bar{f}_k$ or $\|\bar{f}_k\|_\infty$, and the sentences below build both from the version fixed above at each separate conditioning point. *The shortfall was in the* ***declaration*** *and not in the*

consumption, and it is repaired here because a declaration a reader cannot derive from its own stated suppliers is a gap wherever it sits. *It is also the version the corpus already runs*: the Carathéodory argument opening the proof of Proposition 8 establishes joint measurability of G_φ and of its first two u-derivatives G'_φ and G''_φ, which *are* $\partial_u^k f$ at $k = 0$ and $k = 1$, from measurability in ω and continuity in u, and says there that it is used silently below. *Extending it to* $k \leq p_K - 1$ *is the same argument and no more*: each $\partial_u^k f$ is the pointwise limit of the difference quotients of $\partial_u^{k-1} f$ in u taken along the reciprocals of the positive integers, and a pointwise limit of countably many jointly Borel functions is jointly Borel, the device Definition 24(c′.1) uses for $\partial_u \hat{\ell}^{(-k)}$. *Then the envelope is a countable supremum*: continuity in u makes $\sup_{u\in\mathbb{R}} = \sup_{u\in\mathbb{Q}}$ at every conditioning point, so each $\bar{f}_k$ is Borel as a map into $[0, +\infty]$, $E[\bar{f}_k]$ is well defined in $[0, +\infty]$, and the display above is the requirement that it be finite. *Without this the display is an expectation of a function not known to be measurable*, the ground on which Lemma 61 Step 3 refuses the same elision, and the same species as the defect repaired at A4(iii).

Second, the outer norm is an **essential** supremum, and the inner one is genuine. $\|\bar{f}_k\|_\infty :=$ ess sup $\bar{f}_k$ under the law of $(X_1, X_2, \Delta Z, W)$, the supremum in u inside $\bar{f}_k$ remaining a genuine supremum, *the same split Definition 24*(c′.2) *settles for* (42)*, and settled the same way and for the same reason. The essential reading is what every consumer needs and is the weaker of the two*: in the proof of Proposition 8 at Steps 3–5, in that of Proposition 9, in that of Lemma 46 and at Lemma 78 Step 4, $\|\bar{f}_0\|_\infty$ and $\|\bar{f}_1\|_\infty$ appear only on the right of an inequality *inside* an expectation or a conditional expectation, so an almost-sure bound is exact there and a sure bound would be surplus strength. *One thing does* ***not*** *carry over from* (42): there the outer essential supremum needed a measurability argument of its own because the object inside it was random; here $\bar{f}_k$ is a *deterministic* function and $\|\bar{f}_k\|_\infty$ is a number, so no measurability question arises for the outer norm and none is settled. What had to be settled was only which of the two readings it is.

C3 (kernel and boundary adjustment). *The normalisation* $\int \Theta_1 = 1$ *is printed rather than left inside the words "of order* p_K*"*: it is consumed at Lemma 67(i), (iii) and (iv).

At $I = \mathbb{R}$ *clause (ii)* ***is*** *clause (i)*, so the interior kernel is Θ_1 and there is no second convention. *At a point near the* ***right*** *endpoint of a compact interval no separate clause and no reflection convention are needed*: the realised interval I is then truncated on the right rather than the left, Θ_I is the member of the family that interval indexes, and (15) holds at it by hypothesis. *The reflection is a* ***theorem*** *rather than a convention* — $\Theta_{-I}(\varsigma) = \Theta_I(-\varsigma)$, Lemma 66(f) — *and, being a theorem, it is not needed*: the interval already records which side is truncated. *The case a scalar index could not express, and which the old Lemma 67(iv) excluded by hypothesis, is now covered by the same clause*: both endpoints close, i.e. I bounded on both sides.

The envelope pair, and how it moves. A smaller envelope returns a smaller $\left(\mathsf{c}_2, \|\bar{\Theta}\|_{L^1}\right)$, which is the monotone object — c_C is a ratio and is **not** monotone under pointwise domination. Beyond the relaxation parameter (35) and the cap (36), the two constants enter $\mathsf{c}_3 := 1 \vee 2c_C\mathsf{c}_2$ of Lemma 67(i), increasingly; and they enter the certified radius of Proposition 74(ii) *reciprocally*, so that a loose envelope does not merely inflate a printed number there — it shrinks a sufficient condition towards vacuity. *This list is Lemma 66's, at its note to clause* (g)*, and it is offered as exhaustive neither there nor here*: the sentence it replaces read "both enter three bounds", named three, and was short by c_3. *The Bernstein exponent* c_{18} *of Proposition 28 Step 1 is* ***not*** *a second such case, and an earlier draft made it one*: it is a ratio of a mean bound to a range bound, both quadratic in $\|\bar{\Theta}\|_{L^1}$, so the envelope cancels and it is invariant.

Why the envelope is taken minimal and not in the closed form that proves it finite. The explicit majorant inside Lemma 66(d) is admissible too, and returns $\mathsf{c}_2 = 4.50 \times 10^8$ with $c_C = 1.82$ — the same mathematics at a cap larger by 3.6×10^{29} and a certified radius smaller by 4.5×10^7. *There is a third envelope between the two, and it is the only one whose pair is* ***proved*** *rather than measured*: Lemma 66(g) exhibits a closed-form admissible envelope

with $\mathsf{c}_2 = 2^{1/4} e^{1/2} p_K^2 = 125.5$ and $c_C = \sqrt{\pi}\big(1 + \operatorname{erf}(2^{-3/2})\big) = 2.451$, so $2 c_C \mathsf{c}_2 = 615.2$ — six orders inside the majorant on the product the downstream displays read, *their number is deliberately not printed*, no criterion for membership having been stated — and a factor 58 outside the minimal envelope. *Which pair a given reading should use is decided by whether the inequality it feeds must be* ***proved*** *or may be* ***measured***, and that is (g)'s subject.

C4′ (residual truncation), moved entire.

Assumption (C4′: Residual truncation)**.** Let ℓ_n be a real sequence that is either bounded or diverges slowly to ∞ with n. The marginal density $f_{\Delta U}$ exists, is continuous, and is strictly positive on $[-\ell_n, \ell_n]$ for every n; set $\kappa_n := \inf_{|u| \le \ell_n} f_{\Delta U}(u)$, so that $\kappa_n > 0$. If ℓ_n is bounded then $\inf_n \kappa_n > 0$.

Discussion. A0 and A1 are the sampling and exogeneity primitives; A1 is the draft's own conditional-exogeneity primitive and is the weaker of the two candidate exogeneity conditions, since it permits the instrument to be assigned on characteristics that also enter ΔZ. A1b(iii) rules out a second structural function generating the same conditional law of the residual given $(W, \Delta Z)$; A1b(iv) is what identifies $\theta_\dagger$, and it involves no source condition and no debiasing weight; A1c is the linearisation of A1b(iii) and governs the local behaviour of the regularised inverse. B1 is a set of conditions on objects the analyst *chooses*, not on the data-generating process, with the exception of B1(iii)–(iv), which restrict how coarse an index family may be used. *B1(vi) is* ***retired*** *and is not part of that set*: it was an identification restriction, it is not a condition of this estimator, and it is not available as a hypothesis, see the struck clause and the reason recorded with it. A2 is the Severini–Tripathi condition and is the binding assumption for $\sqrt{n}$ *through this moment function*: in the linear model its necessity for $\sqrt{n}$-estimability is known (Severini and Tripathi, 2006), but in the present nonlinear problem necessity remains conjectural, and two claims must be kept apart, necessity for this moment, and necessity for $\sqrt{n}$-estimability by *some* estimator. A2′ adds what is needed for the second-order term to be a finite number. A3 is retired, and it never ruled out "a design in which the debiased moment has no local information about $\theta_\dagger$". By Lemma 91(iv) the debiasing term contributes nothing to the θ-Jacobian at all, $E[\psi(\theta, \zeta_\star)]$ is affine with gradient $-E[\Delta Z \Delta Z']$, so *all* of the moment's information about $\theta_\dagger$ comes from restriction (B), which is what Lemma 10 and Remark 38 said in words from Step 1 onward and what A1b(iv) prices. *The debiasing term buys orthogonality in φ, not information about θ*, and (127) shows it is paid for in variance. A4 rules out a design in which the instrument is degenerate on part of the support of the controls. The application's 689/300 split is not itself a difficulty: at $\varrho \approx 0.303$ the factor $\varrho(1-\varrho) \approx 0.211$ costs $0.25/0.211 \approx 1.18$ relative to a balanced design, about 18% in variance and about 9% in standard error. A4(i) binds only if some ϱ_l approaches 0 somewhere on the support of ΔZ, which is a question about the conditional propensity and not about the marginal split. *At* $\mathsf{L} \geq 3$ *that sentence is*

the one to read twice, because the marginal split can look comfortable while a single cell is thin: the binding quantity is $\min_{0\le l\le \mathsf{L}-1}\varrho_l(\Delta Z)$ and, through (61), one thin cell alone sends $\|\Omega^{-1}\|$ to infinity whatever the others do. *The realised* $\min_l\hat{\varrho}_l$ *and* $\lambda_{\min}(\hat{\Omega})$ *are therefore reportable diagnostics and not optional ones*, and a design whose thinnest cell will not support a conditional multinomial fit against d_Z controls should be coarsened by merging cells rather than pushed through, but *merging is not free, and its cost runs in the direction opposite to the one a reader expects.* Since $\mathcal{V}_{CI}(W)\subseteq\mathcal{V}_{CI}(g(W))$ for any coarsening g, identification at the merged instrument *implies* identification at the full one: merging therefore **adopts the stronger** of the two hypotheses, and it forfeits the localisation the separated cells were carrying. Both are to be stated whenever a design is coarsened, and neither is a formality. N1 is a *property* of a debiasing weight, not a normalisation built into it: it can always be arranged by re-anchoring, at no cost to $\partial_u L$ or to $E[\psi]$, and Proposition 20(iii) shows that it is *necessary and sufficient* for exact orthogonality in the propensity. Stating it as a property rather than as a construction is what leaves that equivalence with content. *At* $\mathsf{L}\ge 3$ *the count is the thing to carry away, because a construction can satisfy it at two cells and fail it at four without any display changing*: N1 is $p(\mathsf{L}-1)$ scalar conditions and the re-anchoring that arranges it is $p(\mathsf{L}-1)$ conditional expectations, computed slice by slice, so centring *one contraction* of the $\mathsf{L}-1$ columns discharges N1 at $\mathsf{L}=2$ and leaves $p(\mathsf{L}-2)$ of it undischarged above, at a cost of r_n before the $\sqrt{n}$ scaling and $n^{1/2-\alpha}$ after it (Corollary 21). *The realised* $\max_{j,l}\big\|E\big[\hat{L}_{jl}^{(-k)}\mid\Delta Z\big]\big\|_{L^2(P)}$, *the largest of the* $p(\mathsf{L}-1)$ *entrywise centring residuals of the fitted weight — the object* (O2) *writes* $E[\hat{L}-L_\star\mid\Delta Z]$, *read entry by entry and estimated by a regression on* ΔZ *— is therefore a reportable diagnostic too*, on exactly the footing of the $\min_l\hat{\varrho}_l$ and $\lambda_{\min}(\hat{\Omega})$ named just above: all three are quantities a design either produces or does not, and none of them is visible in a $\mathsf{L}=2$ rehearsal of the same estimator. A5 collects what the regularised first step must deliver; clauses (i)–(iii) are discharged, not assumed, by Proposition 28, and clause (iii) is the paper's real feasibility frontier. *Clauses (vi) and (vi′) are not discharged*: they are unproven inputs, and from Step 6 onward they are consumed on

both routes, (vi) by Lemma 60(iii) on the balancing route, (vi′) by Lemma 83 on the rate route, which needs the Sobolev bound at every stage and not only at the final iterate. *A7 is the condition under which a discrepancy-principle stopping rule is a theorem rather than a heuristic when the noise sits in the operator, and it is now the only one*: A6 is retired, having been refuted for the operator it named (Proposition 71), and the stopping theorem Proposition 28 runs on A7 in the amended form (13) together with the stage-adaptive schedule the estimator computes, with *no* cone condition anywhere. A8 fixes the sieve window and, with it, the basis; from Step 6 onward its *basis clause* and its *upper end* are consumed by the φ-step as well as by the balancing solve, through Lemma 84(iii) and Lemma 81(3d), *but its lower end is not*, and Proposition 28 cites A8 by clause for that reason; (P-c) is the rate route's own lower end. C1–C5 are regularity conditions for uniform consistency of the kernel components, carried over from the draft's appendix and restated for the panel operator.

Two clauses outside Part VI in the same position: A1b(ii) and A1c. Both are, as of the withdrawal recorded at Remark 44, clauses no theorem rests on; both are maintained and printed; and neither is retired. They are collected here because the withdrawal that orphaned them is recorded in the *results* file while every comparable status note in this programme sits in the assumption block, so a reader checking the block alone would not meet either.

- **A1b(ii)** (measure separability) reached results only through Lemma 45(ii), and that clause's only consumer was the withdrawn corollary. It is *not* retired, it is a substantive and checkable restriction on the joint law of (X_1, X_2), and it is the natural hypothesis of any future primitive for A1c, but nothing in the present programme consumes it. Lemma 45(ii) carries the same note at its own site.
- **A1c** (the linearisation of A1b(iii), governing the local behaviour of the regularised inverse) *is a hypothesis of no result in this paper*, and its non-declaration occurrences are *mentions* and not consumptions — the lists pairing it with A2, A2′ and H1 as

statements posed at $\mathcal{A}_{\text{cond}}$, of which there are **three** and not two; the Discussion sentence naming what it is; Remark 44, which withdraws its primitive; Remark 104, which names it as the completeness condition alongside the singular system; and the closing line of Proposition 9's proof, which records that the e_j span a proper subspace "unless A1c holds". *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces counted three and the count is at least six. It is *not* retired, and the distinction is the same one the five Part VI clauses turn on: A1c is available as a hypothesis and is a real restriction, it simply has no consumer. *What it lacks is a primitive*: Remark 44 withdraws the only one offered, because that route's hypothesis set is empty in this model, and records that A1b(iii) is therefore assumed and not derived. That is the status note, and this is where a reader of the assumption block will look for it.

Five clauses of Part VI that no theorem rests on, and the status of each. These five are collected here because a maintained clause with no theorem behind it is exactly what a reader cannot check by grepping, and because the block already does this for A5(v) (Remark 98(4)), that is the model, and the treatment below is uniform with it. *None of the five is retired, and the distinction matters*: this block has **five** retirements resting on three grounds, A3 and A2′(ii) because they are *proved*, A6 **and A1d** because they are *false*, B1(vi) because it is *about a different estimator*, and none of the five below meets any of the three. of the five is instead:

– **C1(i)** has exactly one consumer in the corpus, Step 2 of Lemma 78, where it supplies the p_K derivatives of the two products $E[\breve{g}_{u,t} \mid X_j = \cdot]f_{X_j}$, *the breve marks the* ***infeasible*** *version of a hatted object, the one built at the true ϱ rather than at $\hat{\varrho}$; it is declared at Lemma 77 Step 1, which is where the two stems that carry it are defined, and this is its first occurrence in compiled order*, behind that step's $O(h_x^{p_K})$ x-smoothing bias. *That consumer is itself consumed by nothing*, Lemma 78 opens "Consumers: none" and

is retained for the two reasons its reading (iv) gives, so C1(i) is a hypothesis of one lemma and, transitively, of no theorem. It is kept because that lemma's proof needs it. *Proposition 28 does* ***not*** *consume C1(i).*

– **C4′.** ℓ_n and κ_n occur nowhere in this project outside C4′ itself, and no object anywhere in these files is truncated to $[-\ell_n, \ell_n]$. The clause prices a residual truncation the estimator does not perform. *It is inert*: no bound stated anywhere degrades if it is deleted. It is nevertheless not retired, because *inert* and *unavailable* are different registers, a truncated variant of the residual direction would need exactly this clause, and Definition 73 is carried in this block on the same footing, being "never asserted and consumed by no theorem".

– **C5′(ii).** No hypothesis list consumes it, *and that statement is true only from this dispatch*: Lemma 78, Proposition 28 and Remark 99(3) block-cited C5′ and therefore carried this clause, on the register rule stated with the retirements above, and all three now cite C5′(i). Lemma 78 reading (i) shows that the one operator-error bound stated at this pair of bandwidths needs only $(nh_xh_u)^{-1}\ln n \to 0$, which is what the clause now states. *What it does do is fix the lower end of the admissible h_x window*, and two printed computations turn on that end: Remark 72's third reading, which collects C5′(ii) with A8, (123), (P-c) and Lemma 81 Step 4 and concludes that h_xL_x is free in both directions, and Lemma 85's cap inflation, which reads $\bar{m}_n$ across the h_x window that C5′(ii) and (P-g) leave. Both are computed at the strengthened form of the clause and at $s = 3$, where (63) holds with room, so both stand. So C5′(ii) is consumed *arithmetically* and not as a hypothesis, and the two readings must be kept apart: an auditor grepping hypothesis lists will find nothing, and deleting the clause would nonetheless leave two displays without a lower limit. *And a lower bound stated more strongly than any proof needs is* ***not*** *free here*, which is the correction (63) records: it must intersect the upper bound (P-g) imports, and the strengthened form and (P-g)

have no common solution at any $s \le \frac{1}{2} + 18\alpha/(8 - 3\alpha)$, a band that is non-empty at every $\alpha \in (\frac{1}{4}, \approx 0.359)$, and (7) forces $\alpha > \frac{1}{4}$.

– **C5′(iii)** is a *cross-reference* and not a restriction. It repeats A4(iii) and (7), and every result that needs the propensity rate cites those two directly. Nothing cites C5′(iii) and nothing should; it is here so that the smoothing-parameter block can be read on its own.

– **C5′(iv)** is a *statement of scope* about the estimator, not a restriction on P. It is discharged by construction, Definition 68 and Definition 69 introduce h_u and h_x and no bandwidth in t, in ΔZ or in W, so there is nothing for a proof to cite it for.

Why a status sentence and not a retirement, applied to all five alike. A retirement is a claim that the clause is unavailable as a hypothesis, and for C1(i) and C5′(ii) that claim would be false, the first is a live hypothesis of Lemma 78 and the second is what puts the lower end on the h_x window the two printed computations above read. Retiring the remaining three alone would file clauses of one species under two registers and leave a reader inferring the difference. The uniform remedy is the one A5(v) already carries: state the standing, in one place, in the printed text.

The smoothness budget, in one place. Four smoothness parameters appear above and they do not compete for the same budget once H1 is imposed:

Condition	In terms of (s, a, r)	Route	Source
Mild ill-posedness†	$s > a + \frac{1}{2}$	both	plan §2.6 scope
$\mu_{\mathrm{H1}} > \frac{1}{2}$†	$s > a$	both	implied by the line above, with room
Sieve window non-empty	$s > 1$ (splines); $s > \frac{3}{2}$ (polynomials)	*balancing only*	A8
Sieve conditioning ($L_x \ll n^{1/2}$)	,	both	A8
Propensity frontier	$\frac{2\mu}{2\mu+1} \cdot \frac{r}{2r+d_Z} > \frac{1}{4}$	both	A5(iii)
Kernel bias	$u \mapsto \mathcal{M}(u,t)$ in C^{p_K} uniformly in t, $p_K = 8$	both	C2′(ii)
Representer	source index $\mu'' > \frac{3}{2}$ for $\ell^{\Pi}_{\star}$, independent of μ	both	A2c, Lemma 61
Linearisation remainder	$\mu > \frac{5}{6}$ *strict*, at C5′(i) as amended	*rate only*	(P-f), Remark 125
Sieve entropy	$L_x \ln n = o\big(n\hat{\delta}_n^2\big)$	*rate only*	(P-d), (123)
x-smoothing bias	$\mathsf{R}\, h_x^{s-1/2} = O\big(h_u \hat{\delta}_n\big)$ (an *upper* end for h_x; the h_u is what Step 4 charges in operator norm)	*rate only*	(P-g), Lemma 81 Step 4
Sobolev stability, stage-wise	$P\big(\max_{k \le \bar{k}} \|\hat{\varphi}^{(k)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}\big) \to 1$	*both*	A5(vi′)
L_x *window, rate route*	$s > \dfrac{9\alpha}{8(1-2\alpha)}$	*rate only*	(P-c) ∧ (P-d), (54)
L_x *window, both routes at once*	$s > \dfrac{1}{2(1-2\alpha)}$	*both, jointly*	A8 lower end ∧ (P-d)
Kernel order vs. Sobolev order	$p_K \ge s - \frac{1}{2}$, i.e. $s \le \frac{17}{2}$ at $p_K = 8$; against (54) this restricts $\alpha < \frac{68}{145}$, i.e. $9r < 68\, d_Z$	*rate only*	Lemma 81 Step 4, gates (P-g), (N1)
Source index, now primitive	$\mu > \frac{5}{6}$ at (P-a.3), **not** derived from $s/(2a)$	*rate only*	(P-a.3), Remark 86

†These two lines carry $\mu_{\mathrm{H1}} = s/(2a)$, H1's index at the **population** operator $\mathcal{A}_{\mathrm{cond}}$ on all of $\mathcal{V}$ at $\theta_\dagger$, **not** the primitive μ of (P-a.3), which is what every other μ in this table and

in (7) means. They are *interpretations* and transfer to the primitive index only under the commutativity proviso Remark 86(4) states and declines. Remark 86(5) carries the binding reading.

The table is read at the **stacked** operator, and that is what makes it exact at every L rather than merely safe. a is the exponent H1 assumes for $\mathcal{A}_{\text{cond}}$ on the whole three-factor range space, so at $\mathsf{L} \geq 3$ the a in the two daggered lines is already the right object and *no line of this table needs re-deriving*; what that operator is, and what it is not, is the Discussion paragraph *H1*. Those two lines are also the only ones that condition on a at all. Every other row restricts some combination of s, μ, μ'', L_x, h_x, p_K, α and r, and none of them a; and μ has been *primitive* at (P-a.3) since the identity was struck, so (7), (55) and the $\mu > \frac{5}{6}$ line do not move with a in either direction, and the last row names $s/(2a)$ only to deny that μ is derived from it. *The operative content of this table is therefore free of* L, and the flagship configuration $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$ below is a statement about four numbers that does not care how many contrasts produced $a = \frac{3}{4}$. *There is one reading a reader can be misled by*, and naming it is the whole of what $\mathsf{L} \geq 3$ changes here: someone who imports an exponent measured on a *single* contrast, a_l for cell l against the base, and reads $s > a + \frac{1}{2}$ and $s > a$ at that a_l, is reading them *conservatively*, under per-contrast polynomial decay the stacked exponent is $\min_l a_l \leq a_l$, and both lines weakly relax as a falls. *That is a statement about the reading and not about the table*; it is conditional on per-contrast exponents existing at all, which H1 does not assume; and it says nothing about a *merged* binary design, whose operator is a *sum* of contrast operators and none of them. The subsumption argument below inherits exactly this and nothing more: it turns on $a \leq \frac{3}{4}$, which a smaller a makes weakly easier to meet, and it already runs under the proviso Remark 86(4) declines.

Two readings of the table are needed rather than one. *The "Route" column is not decoration:* by Lemma 60 the first-order bias is controlled by *either* A5(ii)+A5(iv) *or* A8+(79)+A5(vi), not by both, so the $s > 1$ line is charged on the balancing route only. On the rate route

the s budget faces the first two lines, the frontier, the representer condition and *the last two rows of the table*, and *nothing from A8's lower end*, which is why Proposition 28 cites A8 by clause, taking its basis and its upper end and *not* its lower end. The upper end of A8, which is the conditioning constraint on the sieve solve, is charged on both routes and carries no s requirement. The trade-off between μ and r in the frontier should be displayed rather than hidden: at $(\mu, d_Z) = (1, 3)$ the frontier reads $r > 4.5$, at $(1, 5)$ it reads $r > 7.5$, and at $(2, 3)$ it reads $r > 2.5$.

The representer line is the one that changed at Step 4, and the change is a discharge rather than a budget. Through round 8 it read "source index μ'' for $\ell^{\Pi}_{\star}$" and pointed at A2′(i), a *primitive*, with no route to it. It now points at A2c and at Lemma 61: the index is pinned at $\mu'' > \frac{3}{2}$, which is the one-dimensional C^1-embedding threshold and not a free choice, and the line records a *sufficient condition* rather than an assumption. It still competes for no part of the s budget, because μ'' indexes smoothness of $\ell^{\Pi}_{\star}$ in the residual and s indexes smoothness of $\varphi_\dagger$ in x.

The four lines added at Step 6 should be read as one addition and three bookkeeping entries. *The addition is* $\mu > \frac{5}{6}$, and it *subsumes* the $\mu_{\mathrm{H1}} > \frac{1}{2}$ line, which is therefore now slack rather than binding: under H1 and the scope condition $s > a + \frac{1}{2}$ one has $\mu_{\mathrm{H1}} > \frac{1}{2} + \frac{1}{4a}$, so $\mu > \frac{5}{6}$ is *implied* whenever $a \le \frac{3}{4}$, *under the interpolation proviso of Remark 86*(4)*, without which the computed* μ_{H1} *is not the* μ *of* (55); see the dagger note above, and is a genuine extra restriction otherwise. *It is strict*, and the strictness is not decoration: at the *boundary* configuration $(\mu, d_Z, r) = (1, 3, 4.5)$ the underlying condition (P-f) holds in powers of n with no margin and *fails* once the logarithms A7's amended threshold introduces are counted , see Remark 125, clauses (1) and (5). *The three bookkeeping entries* are the sieve-entropy line (which replaces the stronger Sobolev-entropy condition (121), not consumed by any result), the x-smoothing line (which is the block's *only* upper bound on h_x and is currently carried as a proposition hypothesis rather than as a clause of C5′), and the stagewise Sobolev line (which is A5(vi′),

an input and not a discharge). *None of those four competes with s against the frontier*; they compete with μ, with L_x and with h_x respectively.

Two further lines **do** compete with s. (P-c) and (P-d) are the two ends of one window for L_x, and reading them together at $h_u \asymp \hat{\delta}_n^{1/p_K}$ gives $n^{9\alpha/(8s)} \ll L_x = o(n^{1-2\alpha}/\ln n)$, which is non-empty if and only if (54), $s > 9\alpha/\big(8(1-2\alpha)\big)$. That is $s > \frac{27}{16} = 1.6875$ at $\alpha = \frac{3}{8}$ and $s > 1$ at $\alpha = 0.32$: satisfiable, but at $\alpha = \frac{3}{8}$ it consumes most of the available L_x margin, the admissible window there being only $\big(n^{0.2411}, n^{0.25}/\ln n\big)$. *And if A8's* ***lower*** *end is carried as well , which it must be by any configuration on which both the rate route and the balancing route are to be available, the requirement is instead $s > \big(2(1-2\alpha)\big)^{-1}$, i.e. $s > 2$ at $\alpha = \frac{3}{8}$*: at $\alpha < \frac{4}{9}$ the two thresholds differ by $(4-9\alpha)/\big(8(1-2\alpha)\big) > 0$ and A8's is strictly the larger. *The practical consequence, and it is the reason the line is recorded rather than absorbed*: the boundary configuration $(1, 3, 4.5)$ was priced at $s > 1.75$ in earlier rounds, which clears (54) and does *not* clear A8 in full. It is therefore not an admissible design for a paper that keeps both routes open, independently of the fact that it already fails (P-f) by $(\ln n)^{1/3}$. The frontier-clearing configuration $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$ clears every line in this table. *Read that configuration as* ***four*** *numbers and not as three plus a derivation*: $\mu = 2$ is (P-a.3)'s primitive index, assumed at the compression of the realised operator at $\theta_\circ$, while $(s, a) = (3, \frac{3}{4})$ fix the scale and the population decay at $\theta_\dagger$; that $s/(2a) = 2$ *coincides* with the assumed μ is a property of the numbers chosen and is not an identity this block can invoke, the bridge being the proviso Remark 86(4) declines. *The coincidence is deliberate and is worth keeping*, a configuration on which the two indices agree is the one a reader will find least surprising, but it is not what makes the configuration admissible. *Mechanised at* `scripts/symbolic/specs/phirate-sieve-joint-window.yaml`.

Two lines were added in round 12, and the second changes how the whole table is read. *The first is a budget line that binds*: $p_K \geq s - \frac{1}{2}$, which Lemma 81 Step 4's Taylor expansion consumes and which no earlier round stated. At $p_K = 8$ it caps $s \leq \frac{17}{2}$, and *against* (54)*'s*

floor that cap empties the hypothesis set at every $\alpha \geq \frac{68}{145}$, as Remark 90(2) works out with an admissible witness; it is a condition, it gates (P-g) and (N1), and it was previously supplied by an appeal to $H^s \hookrightarrow C^{0,s-1/2}$ that is **vacuous** for every $s > \frac{3}{2}$ and hence at every admissible configuration. The rate $\mathsf{R}h_x^{s-1/2}$ survives; its justification did not.

The second is that μ is no longer a function of (s, a) on the rate route. (P-a.3) states the source condition **spectrally**, at the compression (53), with μ and R_* primitive; the identity $\mu = s/(2a)$ is *struck* from Proposition 28's hypothesis line and re-appears in Remark 86(4) as an interpretation under an explicitly stated interpolation proviso. Three consequences for reading the table. (i) The frontier line and the $\mu > \frac{5}{6}$ line are now conditions on a *primitive* μ, so they no longer translate automatically into conditions on s, the entry "$\mu = s/(2a)$" has been removed from the frontier row. (ii) *H1 is not retired and the rest of the table is unchanged*: H1 still fixes the scale H^s that Definition 80, A5(vi′), (P-c) and (P-g) are written in, still supplies A2's Picard form, and still delivers the mild-ill-posedness and $\mu > \frac{1}{2}$ lines *as an interpretation*. (iii) *The honest reading of the collapse*: H1's matching is a statement about $\mathcal{A}_{\text{cond}}$ on all of $\mathcal{V}$, while (P-a.3) is about a finite-rank, sieve-compressed operator at a random base point, and Lemma 84(iii.a) records that a link condition *cannot* hold on $\mathcal{V}$. So H1 never delivered $\mu = s/(2a)$ for the operator (P-a) is about, and round 12 makes visible an assumption the block was already carrying rather than adding one. See Remark 86, all four clauses.

Remark 31 (How the smoothing-parameter block C5′ is read). Two clauses, both of which stood inside the statement of C5′. Neither is a condition, and the block restricts exactly the same bandwidth sequences without them.

(1) *What clause* (i)*'s display buys.* It is chosen so that $h_u \to 0$ and the smoothing bias $h_u^{p_K}$ is $\asymp \hat{\delta}_n$. No lower bound on h_u is required for this component.

(2) *Clause* (ii)*'s displayed form, the stronger inherited form, and the window they leave.* The displayed form is the one Lemma 78 consumes. *The stronger inherited form*

$\left(nh_x^2h_u^3\right)^{-1}\ln n\to 0$ *remains available, and is what the two arithmetic readings recorded elsewhere in this Discussion are computed at, but it is imposed* ***only*** *where it is compatible with an upper bound on* h_x, because clause (ii) is a *lower* bound and the block carries no upper bound at all: the upper bound is Proposition 28's hypothesis (P-g), $\mathsf{R}h_x^{s-1/2}=O\left(h_u\hat{\delta}_n\right)$. At the tuning of clause (i), $h_u\asymp\hat{\delta}_n^{1/p_K}=n^{-\alpha/8}$, $\hat{\delta}_n=n^{-\alpha}$, $p_K=8$, so that $h_u\hat{\delta}_n=n^{-9\alpha/8}$, the stronger form reads $h_x\gg n^{-(1-3\alpha/8)/2}$ and (P-g) reads $h_x=O\left(n^{-9\alpha/(8(s-1/2))}\right)$, so the two leave a non-empty window *if and only if*

$$s\;>\;\tfrac{1}{2}\;+\;\frac{18\,\alpha}{8-3\alpha},\tag{63}$$

whereas the form displayed in the statement reads $h_x\gg n^{-(1-\alpha/8)}$ and is compatible with (P-g) as soon as $s>\frac{1}{2}+9\alpha/(8-\alpha)$, *the* ***displayed-form threshold****, which is what the block actually imposes*, hence, since $\alpha<\frac{1}{2}$ by A4(ii), at *every* $s>\frac{11}{10}$, uniformly in α. And the strengthening costs this block no configuration, which is worth stating because it is not obvious from the thresholds alone. Remark 125(5)'s two running configurations are the frontier-clearing $\alpha=0.32$ and the boundary $\alpha=\frac{3}{8}$. At those, the displayed-form threshold reads $s>0.8750$ and $s>0.9427$, against $s>0.8333$ and $s>0.8934$ before, and (63) reads $s>1.3182$ and $s>1.4818$, against $s>1.2273$ and $s>1.3727$. *The frontier-clearing configuration fixes* $s=3$ *and clears both, at* $+2.13$ *against the displayed-form threshold and* $+1.68$ *against* (63); the boundary configuration fixes only (μ,d_Z,r) and leaves s free, so it is quoted at the same $s=3$, where it clears at $+2.06$ and $+1.52$, and it still clears at $s=2$, the value below which Remark 125(5) records its sieve window as empty, with margins $+1.06$ and $+0.52$. So no conclusion of this corpus is lost to the correction, and the reader who wants the one-line version has it: at $s=3$ nothing in this block moves. *One printed witness did have to move with it*, Remark 100(5)'s, which is chosen to sit *below* (57) and had to be raised to stay *above* (63), and it is the only one.

Remark 32 (What A2c is, and what it is for). One clause, which stood inside the statement of A2c. It is a status note and not a condition, and A2c restricts exactly the same representers without it.

(1) *A2c is NOT MAINTAINED.* It is a hypothesis Lemma 61 imposes on a representer it quantifies over, on the model of A1d, whose sole consumer was the withdrawn corollary of Remark 44. Its only role is to *imply* A2′(i) at $\ell^{\Pi} = \ell^{\Pi}_{\star}$, where the budget is written $\Lambda^{(\mu'')}_{\star}$, and it is never asserted. *A2c is stated in the same $\bar{w}$-weighted $L^1(\Pi_t)$ geometry as A2′(i.a)–(i.b) themselves*, which is what makes Lemma 61(iii) a one-line integration of a slicewise inequality rather than an interchange of a supremum with an integral.

Remark 33 (How Definition 2's two construction conventions are read). Four clauses, each of which stood inside the definition. None of it is construction: (10), the truncation, the definition of $\widehat{\widetilde{W}}_i$ and (12) are unchanged by its removal.

(1) What the balanced split costs, and where it is consumed. *This is a construction convention and it costs nothing, the analyst chooses the split, and $n_{\mathrm{m}} = \lceil n/2 \rceil$ serves, but it is* ***used***: every variance bound for the φ-step is denominated in n_{m} and every rate in n, so the two must be tied. It is consumed at Lemma 77 Step 5, Proposition 79(i) and Proposition 28 Step 1 (N2.3), each of which converts an $n_{\mathrm{m}}^{-1/2}$ or n_{m}/L_x into the corresponding statement in n. Without (10) no bound for the φ-step is false, but nothing is stated in n either.

(2) Three things follow from the truncation and all three are used. (a) The truncation can only decrease $\|\hat{\varrho} - \varrho\|_{L^2}$, because $\varrho(\Delta Z)$ lies in (11) almost surely by A4(i) and the metric projection onto a closed convex set containing the target is a pointwise contraction; so A4(iii) is preserved. *Convexity is what the argument uses, not one-dimensionality*: at $\mathsf{L} = 2$ the set (11) is the interval $[\varepsilon_\varrho, 1 - \varepsilon_\varrho]$ and the sentence is the one earlier rounds printed. (b) With $\widehat{\widetilde{W}}_{i,l} := \mathbb{1}\{W_i = l\} - \hat{\varrho}_l(\Delta Z_i)$ as Definition 2 defines it, we have $\varepsilon_\varrho \le \left|\widehat{\widetilde{W}}_{i,l}\right| \le 1$ *surely*, for every i and every l, because $\mathbb{1}\{W_i = l\} \in \{0, 1\}$ while

$\hat{\varrho}_l(\Delta Z_i) \in [\varepsilon_\varrho, 1-\varepsilon_\varrho]$: the two cases $W_i = l$ and $W_i \neq l$ give $1-\hat{\varrho}_l \geq \varepsilon_\varrho$ and $\hat{\varrho}_l \geq \varepsilon_\varrho$ respectively. *This is the degeneracy guard*: a row and a column of (112) vanish only if $\widehat{\widetilde{W}}_i$ vanishes as a vector, which the bound excludes at every l. (c) *$\hat{\varrho}$ is a probability vector*, its implied base cell $\hat{\varrho}_0 = 1 - \sum_{l\geq 1} \hat{\varrho}_l$ being at least ε_ϱ, which is what (11) buys over the box $[\varepsilon_\varrho, 1-\varepsilon_\varrho]^{\mathsf{L}-1}$, on which the implied base cell may be negative. *Clauses* (a) *and* (b) *would hold on the box too*, each using only $\hat{\varrho}_l \in [\varepsilon_\varrho, 1-\varepsilon_\varrho]$; what needs (11) is that $\hat{\varrho}$ estimate a probability vector at all, hence that $\hat{\varrho}_0$ and $\hat{\Omega}$ mean anything and that the reported $\min_l \hat{\varrho}_l$ be the overlap diagnostic it is taken for. *A consequence with no consumer in the estimator as adopted*, recorded because the application reports it and a normalised variant would need it: $\lambda_{\min}(\hat{\Omega}_i) \geq \varepsilon_\Omega$ surely, by (6) read at $\hat{\varrho}(\Delta Z_i)$. *At $\mathsf{L} = 2$ the box and (11) coincide and clause* (c) *says nothing new*, which is why no earlier round had to distinguish them.

(3) *If only a smaller lower bound $\varepsilon \in (0, \varepsilon_\varrho]$ is known to the analyst, truncating at ε serves equally and every constant below carries ε in place of ε_ϱ*, provided $\varepsilon < 1/\mathsf{L}$, without which (11) is empty at ε and there is nothing to project onto. At $\mathsf{L} = 2$ that proviso is $\varepsilon < \frac{1}{2}$ and is implied by $\varepsilon \leq \varepsilon_\varrho$; at $\mathsf{L} \geq 3$ it is also implied, since A4(i) already puts $\varepsilon_\varrho < 1/\mathsf{L}$. It is stated because a reader who reads clause (3) alone, as a licence to pick any smaller ε, must not read it as a licence to pick a larger one.

(4) (12) is the object A7 names.

Remark 34 (How the kernel block C3 is read). Two clauses, both of which stood inside the statement of C3. Neither is a condition, and the block restricts exactly the same kernels without them.

(1) *What (14) is for.* Θ_I is the object every bound below C3 is stated for, and the object (9) integrates.

(2) *How clause* (ii) *is read.* The family is indexed by the **realised truncation interval** $I \in \mathfrak{I}$ — the interval, in scaled coordinates, over which the kernel is actually integrated

at the evaluation point. That index encodes both *how far* the nearer endpoint is, in bandwidths, and *which* endpoint it is; the scalar boundary index of earlier rounds, and the reflection convention it needed, are absorbed into it. Support is no longer a condition: what (ii) requires is the **envelope** $\bar{\Theta}$, and the defining property (15) is a **truncated** moment condition, as before.

Remark 35 (The inverse-probability reading of the contrast, and why the estimator does not use it). This remark is interpretation. Nothing in the assumption block, in ψ or in the operator is stated at the object it introduces, and the paper does **not** estimate at it.

(1) *The normalised contrast.* Wherever every cell is non-null, so that $\Omega(\Delta Z)$ is invertible, set

$$\widetilde{W}^{\Omega} := \Omega(\Delta Z)^{-1}\widetilde{W}, \qquad \text{so that} \qquad \widetilde{W}^{\Omega}_{l} = \frac{\mathbb{1}\{W=l\}}{\varrho_l(\Delta Z)} - \frac{\mathbb{1}\{W=0\}}{\varrho_0(\Delta Z)}, \tag{64}$$

the second equality being (61). *This is the vector of inverse-probability (Horvitz–Thompson) contrasts of each cell against the base*, and at $\mathsf{L}=2$ it is $(W-\varrho)/\{\varrho(1-\varrho)\} = \mathbb{1}\{W=1\}/\varrho - \mathbb{1}\{W=0\}/(1-\varrho)$, the identity a binary design writes down without comment.

(2) *What it is for.* By (62) the raw contrast has conditional forward residual $\Omega(z)d(u \mid z)$, a matrix mixture of the cell-versus-base differences, while $E[\widetilde{W}^{\Omega}\mathbb{1}\{\Delta U \le u\} \mid \Delta Z = z] = d(u \mid z)$ is the difference itself. *So (64) is the reading under which the restriction is most directly legible*, and the quadratic form $\widetilde{W}'\Omega^{-1}\widetilde{W} = (1-\varrho_W)/\varrho_W$ is base-cell-free where the componentwise weighting $\mathrm{diag}(\varrho_l(1-\varrho_l))^{-1}$ is not.

(3) Two moments, and they are different objects. Conditionally on ΔZ, at a point where every cell is non-null,

$$E\left[\widetilde{W}'\Omega^{-1}\widetilde{W}\,\middle|\,\Delta Z\right] = \mathrm{tr}(\Omega^{-1}\Omega) = \mathsf{L}-1, \qquad E\left[\left\|\Omega^{-1}\widetilde{W}\right\|^2\,\middle|\,\Delta Z\right] = \mathrm{tr}\left(\Omega^{-1}\right) = \sum_{l\ge 1}\frac{1}{\varrho_l} + \frac{\mathsf{L}-1}{\varrho_0}.$$

The first is bounded by $\mathsf{L}-1$ whatever the cell probabilities are; the second diverges

as any cell probability tends to zero. At $\mathsf{L} = 4$ with a thin non-base cell at 0.05 and the other three at $0.31\overline{6}$ they are 3.00 and 35.79, against $\varepsilon_\varrho^{-1} = 20$: *the second is not controlled by the level bound*, exceeding it here by a factor of 1.79. Under A4(i) it is at most $2(\mathsf{L}-1)/\varepsilon_\varrho$, so the *worst case* is a factor $2(\mathsf{L}-1)$, 6 at $\mathsf{L} = 4$, which the configuration above does not attain.

(4) *And the first absolute moment is free of ϱ altogether*: $E[|\widetilde{W}_l^{\Omega}| \mid \Delta Z] = \varrho_l/\varrho_l + \varrho_0/\varrho_0 = 2$, exactly, at every l and every cell configuration. This is the standard inverse-probability pattern, the mean is untouched and the variance diverges, and it is the reason (3)'s two displays must not be substituted for one another.

(5) Where Ω is singular, $\widetilde{W}^{\Omega}$ is **undefined**, not non-integrable. Ω has no inverse there and (64) names nothing; where it is defined, clause (4) shows it is integrable with a ϱ-free bound. The two failures are different and the corpus must not conflate them.

(6) Why the estimator does not use any of this. Normalising would replace $\mathcal{A}_{\text{cond}}$ by a different operator, and A2's range condition, A2′, A1c and H1, the last through $\sigma_j(\mathcal{A}_{\text{cond}}) \asymp j^{-a}$, are all stated at $\mathcal{A}_{\text{cond}}$, with $\mu_{\text{H1}} = s/(2a)$ and the smoothness-budget table's two daggered lines downstream of a, (7) is *not*, its μ being primitive at (P-a.3). *A4's note (c) already records that the two operators must not be assumed to share range or kernel*, and the mechanism is the same one that defeats the weight-class argument in the paragraph *A4*: $\mathcal{W}$'s z-sections are real-analytic while Ω is only Hölder. *The gain would be interpretational and the price is the whole smoothness-and-ill-posedness budget*, so the paper keeps the raw contrast and keeps the reading here.

(7) *What is **not** claimed.* Ω^{-1} is not the efficient weight: under A1 the conditional variance of the indicator moment is $\Omega(z)F(u \mid z)$, so the efficient weight differs from Ω^{-1} by a scalar depending on both u and z, and no efficiency calculation for this estimator exists in this corpus. *Nor is any conditioning improvement claimed*: Ω^{-1} up-weights thin cells, which is where the application's own design sits.

Discussion of the estimator

This section carries the material the article does not print about the two definitions that construct the estimator: for the φ-step of Definition 22 and the cross-fitting scheme of Definition 24, the measurability verifications, the grounds on which each nuisance's container is chosen, the re-declarations of conventions stated elsewhere, and the record of what each clause does and does not rule out. The constructions themselves are printed above, verbatim and unchanged. Headings below name what the text under them discusses.

The φ-step (Definition 22)

The measurability requirement on the preliminary, and where it is stated The opening of Definition 22 declares the preliminary a measurable random element of $\mathcal{V}$ at $\sigma(\mathcal{I}_*)$, with $\mathcal{I}_*$ the fold fixed there: *this is a requirement on the preliminary and is stated* ***here****, where the preliminary is declared, rather than only inside the verification that consumes it*; (P-a.1) of Proposition 28 carries the same requirement, and it is of the kind read off a formula, which is why it costs nothing to impose.

The sieve-membership clause, and why the H^s bound is an envelope *The sieve-membership clause is* ***imposed here***, and (P-a.1) of Proposition 28, this opening's companion, carries it in terms. *The clause is not idle*: every iterate after the zeroth lies in $\mathcal{S}_n$ by construction, so without it the preliminary is the one element of the path that need not, and the H^s display above is then a statement about a quantity no clause makes finite; *envelopes, not O_P statements, and Remark 87 is why*: the constants $\mathsf{c}_{14}, \mathsf{c}_{15}$ below are printed before the data are seen, and an O_P hypothesis supplies no number for them to dominate. *What makes the display finite once the clause is in force are two containments and neither is this definition's own*: (8), which is A8's basis clause read as the containment it delivers, places $\mathcal{S}_n$ in H^s, and H1's source set places $\varphi_\dagger$ there.

The letter ω The letter is ω because Lemma 85(i) shows the normalisation **is** a Landweber relaxation parameter, which is what ω names in the regularisation literature.

What in the schedule is computable, and from what *Every quantity in* (37) *is* ***computable*** *by the analyst, from four inputs and two constants, and the two constants are not free*: $\hat{\delta}_n$ and h_u are already inputs to the procedure, κ is a function of μ, and $\hat{\varepsilon}_0$ is the declared rate of the preliminary, those are the four; c_{14} and c_{15} are the two. *The scale* (35) *and the cap* (36) *are a different matter, and the next paragraph is where they are treated*: both read $\|1/\pi\|_\infty$, which no analyst knows.

The reference density is unknown, what the floor buys, and the one thing it does not Definition 5 declares π an unknown feature of P — the pooled design density $\frac{1}{2}(f_{X_1} + f_{X_2})$ — and Definition 22 floors an estimate $\hat{\pi}$ of it at ε_π. Three things follow, and the order matters, because the third is an open item and the first two are what make it a narrow one.

(i) What the floor buys, and it is not only well-posedness — nor, as this paragraph read through 2026-08-26, only computability. $\hat{\pi} \geq \varepsilon_\pi$ gives $\|1/\hat{\pi}\|_\infty \leq \varepsilon_\pi^{-1}$, so (101) formed at $\hat{\pi}$ is defined at every design point. *The same floor also fixes the relaxation*, through $\|h\|_{L^2(\mathcal{X},dx)} \leq \varepsilon_\pi^{-1/2}\|h\|_{L^2(\hat{\pi})}$: that is the entry $\varepsilon_\pi^{-1/2}$ in the maximum (35) now carries, and by Proposition 70(i′) it is what makes $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi\| \leq 1$ true in the $L^2(\hat{\pi})$ pairing the iteration actually runs in — not a substitute for an unknown $\|1/\pi\|_\infty^{1/2}$ but the constant (F-f) requires there. *Through 2026-08-26 this paragraph offered exactly that number and filed it under computability alone*, which is why the entry was absent from (35) and Remark 89(7) enumerated four parts rather than five; the disposition is (7e). *Computability then follows, and it now follows* ***on a range*** *rather than after a substitution*: at $\varepsilon_\pi \leq \|1/\pi\|_\infty^{-1}$ — implied by the threshold clause (ii) names — the maximum in (35) *is* $\varepsilon_\pi^{-1/2}$, leaving c_C, c_2 and c_Π, which C3(ii) and B1(iv.c) fix and the analyst therefore knows, and h_u, h_x, which the analyst chooses; (36) then

follows, being $\omega_n^2 \hat{\delta}_n^{-2}$ at that ω_n. *Above the threshold the maximum reads* $\|1/\pi\|_\infty^{1/2}$ *and the two displays are population displays and are* ***not*** *computable*, and that is worth saying beside the paragraph above, which reports (37) computable and is about a different set of quantities. *Nor are those the only displays carrying the norm*: Remark 18(2) names Proposition 70(i), (35), (36), Lemma 78 and Lemma 47(i), and the Cauchy–Schwarz step of the measurability verification below is a further one. *No claim of exhaustiveness is made about either list.*

(ii) What the floor costs, and where the cost is charged. The clipping is the metric projection in $L^2(\mathcal{X}, dx)$ onto an order interval, and that set contains π itself whenever $\varepsilon_\pi \le \min\{\|1/\pi\|_\infty^{-1}, \|\pi\|_\infty^{-1}\}$; at such an ε_π the projection can only decrease $\|\hat{\pi} - \pi\|_{L^2(\mathcal{X},dx)}$, which is Definition 2's accounting for (11) read one object over. C1(ii) makes that threshold positive and names no value for it, so ε_π joins c_{14}, c_{15} and τ_{DP} as a constant the theory does not pin; above the threshold the clipping is a bias this paper does not price. *The population condition is untouched, and the direction of that statement is the point*: C1(ii) bounds π below on $\mathcal{X}$ and says nothing whatever about an estimate of it, so a $\hat{\pi}$ that goes negative violates **no** condition the paper states — it is not a failure of C1(ii) and cannot be one. The convention in Definition 22 is *additional* and *sample-level*, and C1(ii) is neither weakened nor re-spelled to accommodate it. *The event is not hypothetical*: at the kernel order the estimator runs, $\hat{\pi}$ is a higher-order density estimate, higher-order kernels are sign-changing, and the Monte Carlo of Section 5 loses replications to exactly this failure when the operator is built on a small enough fold.

(iii) The open item. What none of this establishes is that the substitution of $\hat{\pi}$ for π *inside* the iteration leaves (56) in place — *which is why Proposition 28 states that display at* $\hat{\varphi}^{\pi}$*, the infeasible reading, from 2026-08-27, and does not state it at* $\hat{\varphi}$. Remark 89(7) states that item in full, in five parts. *Two are settled outright*: the two norms are equivalent with deterministic constants, so the norm (56) is read in is not the difficulty; and, as of 2026-08-27, the normalisation (35) is admissible in *both* pairings, so (F-f)'s $\|\mathcal{L}\| \le 1$ holds whichever the

iteration is read in — that last is (7e), and it was not among the parts this corpus enumerated before that date. *A third is settled* ***with a rider****, and the rider bears on the verification below*: the feasible recursion is a genuine Landweber iteration on $L^2(\hat{\pi})$, but its sieve is a **random** subspace, because $\mathcal{S}_n$ is cut out by π, whereas the four steps below are written at a fixed one; *so those four steps verify the measurability of* $\hat{\varphi}^{\pi}$ *and leave the measurability of* $\hat{\varphi}$ *pending*, which is what Definition 24(c′.1) now prints and what Lemma 94 consumes. *Two parts are simply open.* The level of the iterate is pinned by $\hat{\pi}$ and not by π, which leaves a constant component of the error bounded by $\|\hat{\pi} - \pi\|_{L^2(\mathcal{X},dx)}$ — that part is *priced*, and what it needs is a rate condition on $\hat{\pi}$ of the form A4(iii) imposes on $\hat{\varrho}$, which this paper does not impose. And $\hat{\mathcal{A}}^*_{\hat{\pi}} = (\pi/\hat{\pi})\hat{\mathcal{A}}^*_{\varphi}$ perturbs the operator multiplicatively while (P-a.3) is a condition on the unperturbed one — that part is not priced at all.

Why the clipping is well defined, and why the printed range charges no clause

The projection target of Definition 22 — the set of functions with values in $[\varepsilon_\pi, \varepsilon_\pi^{-1}]$ Lebesgue-almost everywhere on $\mathcal{X}$, *written out rather than named, and carrying no bound variable, on the write-it-out precedent* — is closed, convex and, $\mathcal{X}$ having finite Lebesgue measure — which is A0's *compactness* and not its interval, the constant function lying in $L^2(\mathcal{X}, dx)$ on every $\mathcal{X}$ of finite measure — and $\varepsilon_\pi \leq 1$, **non-empty**: the constant function ε_π lies in it. *And* $\varepsilon_\pi \leq 1$ *is exactly what non-emptiness needs and nothing more*, the target being empty at every $\varepsilon_\pi > 1$ and the singleton $\{1\}$ at $\varepsilon_\pi = 1$, *which is why the range that definition prints is a* ***declaration*** *entire and charges no clause*: it is the referent boundary, in the sense the assumption block's paragraph on citation registers rules for analyst-chosen constants, and A4(i)'s ε_ϱ and A7's τ_{DP} are decided by that same rule and land elsewhere. *This is Definition 2's device for* (11)*, applied to a second first-stage nuisance that has to be held off a boundary* — there the boundary of the simplex, at which the contrast degenerates; here zero, at which the adjoint is undefined — and the two are read alike: a declaration fixes the constant, non-emptiness rests on the declaration alone, and the property that the projection

cannot increase the error is charged where it is used and not in the definition.

Why (35) carries a maximum, and why it does not move with L (35) *does not move with* L*, and the reason is which bound on the contrast it reads*: the estimate behind it is taken at each contrast index separately, so it consumes the **per-cell** bound $|\widehat{\widetilde{W}}_{i,l}| \le 1$ of Remark 33(2) and not the Euclidean bound on the vector $\widehat{\widetilde{W}}_i$; Π_{w} is a probability measure by B1(iv.d), so integrating out the third factor of Π contributes the factor 1 and c_Π of B1(iv.c) is unchanged. *The maximum is what makes the normalisation admissible in* ***both*** *of the two pairings the construction puts in play, and carrying only one of the two entries would be a defect and not a saving.* Clause (b) of Definition 22 forms the adjoint at $\hat\pi$, so the iteration it runs is a Landweber iteration on $L^2(\hat\pi)$, while the analysis of Proposition 28 runs it on $L^2(\pi)$; (F-f)'s hypothesis $\|\mathcal{L}\| \le 1$ is a statement about the operator norm *out of the domain Hilbert space whose inner product defines* $\mathcal{L}^*$, so it must be checked in whichever of the two is in force, and Proposition 70(i′) supplies both from one bound, at the two conversion costs the paragraph on the reference density names above. *On the range that paragraph recommends* — ε_π at or below $\min\{\|1/\pi\|_\infty^{-1}, \|\pi\|_\infty^{-1}\}$, the range at which the clipping costs nothing — the maximum **is** $\varepsilon_\pi^{-1/2}$, so there the same number is the stability constant and the computable one; the two recommendations coincide rather than compete. *The bound is in fact the unrestricted one and the restriction is free*, an operator norm out of a subspace never exceeding the norm out of the whole space — *which is what lets the same* ω_n *serve the feasible reading, where the subspace is not* $\mathcal{S}_n$ *but the* $\hat\pi$*-centred one Remark 89*(7a) *names. And the realised index and the* a priori *bound on it are two objects, so Lemma 85(ii) and (v) do not collide*: (ii) says $\hat m^{(k)}$ itself does not move with the scale, (v) says the bound does, and a reader should not have to reconcile two clauses named in one breath.

R against R_*, and the two directions of a mis-set level *Neither of the two constants Definition 22's schedule must dominate is known to the analyst, and they are built from different things*: c_{16}, the constant of (176), from the kernel constants and the design; $\mathsf{c}_{17} =$

$2c_\mu \mathsf{R}_*^{1/(2\mu+1)}$, the constant of (178), from (F-f)'s c_μ — the constant in the Landweber–Fridman bound under a discrepancy stop, which depends only on the source index μ and on A7's τ_{DP}, and on nothing about the operator or the sample — and from the *spectral pre-image* radius R_* of (P-a.3). The radius there is R_* and **not** R, and the two must not be the same symbol, this is the same species of defect as c_{14} versus c_{16} and it sits on the same inequality. R is an H^s *ball radius*: it is what Definition 80 measures, what A5(vi′) and (P-a.1) bound, and what (P-c), (P-g) and (121) price, so it carries units of Sobolev norm and enters *approximation* statements. R_* is a *spectral pre-image norm* in $L^2(\pi)$: it is (P-a.3)'s bound on $\varsigma^{(k)}$, it enters only through (F-f), and it is what c_{15} must dominate. They are not comparable in either direction without the interpolation step Remark 86(1) shows is unavailable. The two directions of error are not symmetric. Chosen too large, the level is conservative: the rule stops earlier, the rate in (56) is unchanged in order and the constant degrades. Chosen too small, the implemented level sits below the true data error at some stage, (L3) fails, and with it (F-f): the discrepancy guarantee is void and the estimator is whatever the cap (36) returns. *This is the one place where the procedure carries a tuning constant the theory does not pin, and it is said here rather than left to be discovered.*

The two readings of clause (b): the scope of the convention, and its typography

Definition 22 declares the feasible $\hat{\varphi}$ and the infeasible $\hat{\varphi}^{\pi}$ and makes the stage superscripts reading-relative: clauses (a)–(d) there and Definition 24 fix the feasible reading, Proposition 28 and its proof the infeasible one at every step. *The convention is safe only while no single statement quantifies over stages of* ***both*** *readings at once*; no statement in this corpus is known to, *and that is a reading and not a census* — the two places that name both readings in one paragraph, that definition's measurability discussion and Theorem 29 Step 0, each write the feasible object as $\hat{\varphi}$ or $\hat{\varphi}^{(-k)}$ and the other as $\hat{\varphi}^{\pi}$, with no stage superscript in play. *The superscript is a* ***pairing*** *mark and not a stage index*: every stage index in that definition is parenthesised, $\hat{\varphi}^{(k)}$, while an unparenthesised superscript naming the measure an object is

formed in is this project's $\ell^{\Pi}_{\star}$ convention. *One thing the two readings* ***share****, and it is not an accident*: (35) carries the maximum of the two conversion costs and is therefore admissible in both pairings, so a single relaxation and a single cap serve both — which is Remark 89(7e), and is why the distinction drawn there does not reopen it.

The output is a measurable random element: why inspection of the formula will not settle it *And the discharge available for the other nuisances is not available here*: A4(iii) and Definition 24(a) settle $\hat{\varrho}$ "by inspection of the formula", which every kernel and every series construction passes; inspection of the formula settles nothing for a Landweber–Fridman iterate stopped by a discrepancy rule, *because the stopping index of* (c) *is random. The requirement is one Definition 24*(c′.1) *needs of each* $\hat{\varphi}^{(-k)}$*, and clause* (b) *of that definition names* ***this*** *estimator as the object, so the property cannot be imposed there as a construction requirement — one cannot require a property of an object one has already built*, which is Remark 58's test read on an estimator rather than on an assumption. *Two features of* (c) *are what the verification below consumes*: the cap (36) is deterministic, so the family of stopping events is **finite** and no limit is taken, and the emptiness convention makes $\hat{m}^{(k)}$ **total**. *Three stated results consume the verification*, each a probability statement about an output of one of the two readings and none of them well posed without it: (56); Lemma 83's bound on $\max_{j\le\bar{k}}$ with probability tending to one; and (P-a.3)'s event, which is defined through the realised operator at $\hat{\varphi}^{(k-1)}$.

The verification, in four steps *One index warning, because both readings occur in this paragraph*: the superscript in $\hat{\varphi}^{(-k)}$ and $\mathcal{I}_{-k}$ just above is Definition 24's *fold* index; every k from here to the end of the verification is this definition's *stage* index, and the two are one glyph apart.

– *The iterates.* Suppose $\hat{\varphi}^{(k-1)}$ is a $\sigma(\mathcal{I}_*)$-measurable random element of $\mathcal{V}$. The finitely many numbers $\Delta_{h_x}\hat{\varphi}^{(k-1)}_i$, $i \in \mathcal{I}_*$, are then random variables, being values of the *bounded*

linear functionals (96) at a measurable random element, *and the step from a random function to a random number is the* ***Carathéodory*** *criterion and not the evaluation of a fixed functional, because the functional* (96) *is itself* ***random***, its kernel carrying X_{i1}, X_{i2} and I_{i1}, I_{i2}, which lie in $\mathfrak{I}$ at every n by C5′(ii) — *the per-n form of that clause and not its limit*, since measurability is asserted at each n. "Value of a bounded linear functional at a measurable random element" gives measurability only for a *fixed* functional; what is available, and what is needed, is the superposition argument: the map $F_i(\omega, \varphi) := \Delta_{h_x}\varphi_i(\omega)$ is $\sigma(\mathcal{I}_*)$-measurable in ω at each fixed φ, because X_{i1}, X_{i2} are random variables and $(x, x^*) \mapsto C_{h_x}\Big(x - x^*, I(x^*)\Big) = \Theta_{I(x^*)}\Big((x - x^*)/h_x\Big)$ is *jointly continuous on $\mathcal{X} \times \mathcal{X}$*, hence jointly Borel there. *The route is* ***not*** *continuity of $I \mapsto \Theta_I(\varsigma)$, and that map is* ***not*** *continuous*: Lemma 66's proof of (d) records that it jumps at each I whose endpoint crosses ς, and (94) moves both endpoints of $I(x^*)$ affinely with x^*, so this is exactly the configuration that clause names. *Nor is it* (f) *read as a statement about* Θ: (f) carries the **coefficient** map $I \mapsto \mathsf{G}(I)^{-1}e_0$ and nothing more. *What removes the jump is that the two arguments move together, and the collapse is exact rather than approximate*: at $\varsigma = (x - x^*)/h_x$ and $I = I(x^*)$ of (94), $\varsigma \in I(x^*)$ holds if and only if $\underline{x} \le x \le \bar{x}$, because $h_x > 0$, so the indicator of (89) is $\mathbb{1}\{x \in \mathcal{X}\}$ — *free of* x^*, and identically 1 on the domain of integration of (96). On that domain the kernel is $\Big(\sum_k (\mathsf{G}(I(x^*))^{-1}e_0)_k\, \varsigma^k\Big)e^{-\varsigma^2/2}$ with no indicator left in it, and this is jointly continuous in (x, x^*): $x^* \mapsto I(x^*)$ is continuous into $\mathfrak{I}$ for the topology (90), both endpoints being affine in x^* and tanh continuous; $I \mapsto \mathsf{G}(I)^{-1}e_0$ is continuous by Lemma 66(f); and polynomial evaluation is jointly continuous in coefficient vector and argument. So the dx-integral in (96) is measurable in ω by Tonelli *applied to positive and negative parts, and the qualification is not decoration here*: the integrand of (96) is a difference of two boundary-corrected kernels of order p_K, which (15) makes *sign-changing* by construction, against a φ that integrates to zero against π, so Tonelli in its nonnegative form does not apply to it as written. The split is legitimate because the integrand is absolutely

integrable at every sample point: $|C_{h_x}| \le \mathsf{c}_2$ by C3(ii) and (P9), $\mathcal{X}$ is a bounded interval by A0, and $\int_{\mathcal{X}} |\varphi|\, dx \le |\mathcal{X}|^{1/2} \|1/\pi\|_\infty^{1/2} \|\varphi\|_{L^2(\pi)} < \infty$ by C1(ii) and the Cauchy–Schwarz inequality, so each of the four one-signed integrals is finite and their signed combination is well defined and measurable. *This is the treatment the same paragraph gives the* Π*-pairing below*; it is also A2c(m.2)'s and Definition 24(c′.1)'s, on the other side of the pairing. The integral is, moreover, $\|\cdot\|_{L^2(\pi)}$-continuous in φ at each fixed ω, which is exactly the boundedness of (96) that Definition 68 records and which holds at every sample point. Since $\mathcal{V}$ is a subspace of $L^2(\mathcal{X}, \pi)$ with $\mathcal{X}$ compact, it is separable, so the Carathéodory criterion makes F_i jointly $\sigma(\mathcal{I}_*) \otimes \mathfrak{B}(\mathcal{V})$-measurable, and F_i then composes with the measurable map $\omega \mapsto \big(\omega, \hat{\varphi}^{(k-1)}(\omega)\big)$. *The device is this paper's own and is used at three further sites*: it is what Definition 24(c′.1) names for (41), what the opening of the proof of Proposition 8 runs for G_φ and its u-derivatives, and what C2′(i) runs for the envelopes $\bar{f}_k$; and $\hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta)$, $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}$ and $\hat{\mathcal{A}}^*_{\hat{\varphi}^{(k-1)}}$ depend on $\hat{\varphi}^{(k-1)}$ through those numbers *alone*, by (97), (98) and (100), the adjoint being determined by the operator and the two inner products of Definition 5. Restricted to $\mathcal{S}_n$, which is a *fixed* finite-dimensional space, the iteration (b) is therefore an affine recursion on a Euclidean space of that dimension whose coefficients are random variables, started at the constant $v_0 = 0$; so each v_m, $m = 0, 1, \ldots, \bar{m}_n$, is a $\sigma(\mathcal{I}_*)$-measurable random element of $\mathcal{S}_n$, being a finite composition of measurable maps. That the coefficients are random variables is **Tonelli** and not composition, and the distinction is the same one the previous link turns on. *The coordinates are those of a basis of* $\mathcal{S}_n$ *and* ***not*** *of* $\{b_1, \ldots, b_{L_x}\}$, and two facts force the distinction: $\mathcal{S}_n = \operatorname{span}(b_1, \ldots, b_{L_x}) \cap \mathcal{V}$, and the linear functional $\zeta \mapsto \int_{\mathcal{X}} \zeta\, d\pi$ is *not* identically zero on $\operatorname{span}(b_1, \ldots, b_{L_x})$: A8's b_j are B-splines, hence *nonnegative* and not identically zero, and π is bounded below away from zero by C1(ii), so $\int b_j\, d\pi > 0$ for every j, no b_j lies in $\mathcal{V}$, $\{b_j\}$ spans a strictly larger space, and $\dim \mathcal{S}_n = L_x - 1$. *Both consequences matter*: the coordinate map written in the b_j is not a coordinate map on $\mathcal{S}_n$, and $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} b_j$ evaluates the operator

off the domain Definition 5 declares for it. *Neither touches the conclusion*, and that is why the repair is a rewording: fix once any basis of $\mathcal{S}_n$, every element of it lies in $\mathcal{V}$, so the operator is applied only where it is defined, and every element is a finite linear combination of $b_1, \dots, b_{L_x}$, and read the recursion in its coordinates. Its matrix is that of $\hat{\mathcal{A}}^{*}_{\hat{\varphi}^{(k-1)}} \hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}|_{\mathcal{S}_n}$, whose entries are the $L^2(\Pi)$ pairings of the images under $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}$ of two basis elements, and its constant vector carries the $L^2(\Pi)$ pairings of those images with $\hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta)$, each an *integral over* (u, t, l) *against* Π *of a random integrand*, (101) writing the adjoint as exactly such an integral. *A finite composition of measurable maps does not reach an integral*; what does is joint measurability in (ω, u, t, l) followed by Tonelli's theorem. The joint measurability is again Carathéodory, *run at each contrast index separately*: by (100) and (98), at a fixed element ζ of $\mathcal{V}$ and a fixed $l \in \{1, \dots, \mathsf{L}-1\}$ the maps $(u,t) \mapsto \hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}\zeta(u,t,l)$ and $(u,t) \mapsto \hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta)(u,t,l)$ are continuous at every sample point, C_{h_u} and C_{h_u} being continuous in u by Lemma 67 and $t \mapsto w(t'\Phi(\Delta Z_i))$ continuous, while each is $\sigma(\mathcal{I}_*)$-measurable in ω at fixed (u,t) by the paragraph above, so both are $\sigma(\mathcal{I}_*) \otimes \mathfrak{B}(\mathbb{R} \times \mathcal{T})$-measurable at that l. *Measurability in* (ω, u, t, l) *then costs nothing, and the reason is that the third factor of* Π *carries a* ***finite*** *set*: $\{1, \dots, \mathsf{L}-1\}$ with its power set, so each map is the finite disjoint union over l of the pieces just shown measurable, and no further argument is needed. Each factor lies in $L^2(\Pi)$, by Proposition 70(i) for the operator and, for $\hat{T}_{h_x}$, by $\mathsf{c}_3 c_\Pi < \infty$, a bound that reads the *per-cell* $|\widehat{\widetilde{W}}_{i,l}| \le 1$ of Remark 33(2) at each l against a Π_{w} that B1(iv.d) makes a *probability* measure, and so does not move with L, so the products are Π-integrable by the Cauchy–Schwarz inequality, and Tonelli applied to positive and negative parts then makes each pairing a $\sigma(\mathcal{I}_*)$-measurable function of ω. This is the inheritance Definition 24(c′.1) performs for $\omega \mapsto \|\hat{\ell}^{(-k)} - \ell_\star\|_{L^2(P)}$, read on Π instead of P. At $\mathsf{L} = 2$ the third factor of Π is the one-point set $\{1\}$ and Π_{w} the point mass at it, so every display in this bullet is the one earlier rounds printed.

– *The stopping index.* τ_{DP} and $\hat{\delta}^{\mathrm{eff}}_{n,k}$ are deterministic by (37), and $\|\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v_m +$

$\hat{T}_{h_x}(\hat{\varphi}^{(k-1)},\theta)\|_{L^2(\Pi)}$ is a random variable for each m, *by the previous step* ***and*** ***Tonelli***, *the two being needed and neither sufficing alone*, since an $L^2(\Pi)$ norm is a Π-integral of a random function and not a value of one: the previous step makes $(\omega,u,t,l)\mapsto\big(\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}v_m+\hat{T}_{h_x}(\hat{\varphi}^{(k-1)},\theta)\big)(u,t,l)$ jointly measurable, v_m being $\sigma(\mathcal{I}_*)$-measurable with values in the fixed finite-dimensional $\mathcal{S}_n$, and Tonelli then makes $\omega\mapsto\int(\cdot)^2\,d\Pi$ measurable, the square root being continuous. So, for each $m\le\bar{m}_n$, the event that *that* norm is at most $\tau_{\mathrm{DP}}\hat{\delta}^{\mathrm{eff}}_{n,k}$ is measurable, *it carries no letter*, every short candidate being claimed, and $\{\hat{m}^{(k)}=m\}$ is that event intersected with the complements of the same events at the indices $m'<m$: a *finite* intersection of measurable sets. *Two further features of* (c), *not the two the paragraph after this one names, which are about the* level, *are what this step consumes*: the cap $\bar{m}_n$ of (36) is *deterministic*, so the family is **finite** and no limit is taken; and the emptiness convention makes $\hat{m}^{(k)}$ **total**, so no sample point is left without a value and $\{\hat{m}^{(k)}=\bar{m}_n\}$ absorbs the residual event. A random cap or a partial index would break the argument here, and **not** only here: the next bullet's finite sum of indicators needs the same two features, and so does (L4) of Proposition 28's Step 2, which compares the stopping index to $\bar{m}_n$ and then applies the emptiness convention on the complement. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces read "and nowhere else" and was short by its own next bullet.

– *The output.* $v_{\hat{m}^{(k)}}=\sum_{m\le\bar{m}_n}\mathbb{1}\{\hat{m}^{(k)}=m\}\,v_m$ is a *finite* sum of indicators of measurable events against measurable random elements, hence a $\sigma(\mathcal{I}_*)$-measurable random element of $\mathcal{S}_n$; and $\hat{\varphi}^{(k)}$ of (d) is one of $\mathcal{V}$, $\mathcal{V}$ being a linear space and $\mathcal{S}_n\subset\mathcal{V}$ (Definition 5). Induction on $k\le\bar{k}$, from a $\sigma(\mathcal{I}_*)$-measurable $\hat{\varphi}^{(0)}$, *which is a requirement on the preliminary, and* ***is*** *of the kind read off a formula; it is* ***stated*** *in this definition's opening and in (P-a.1) of Proposition 28, and is cited rather than imposed here*, gives the claim for $\hat{\varphi}=\hat{\varphi}^{(\bar{k})}$.

– The H^s norm of the same object, which the paragraph owes because Lemma 83 and A5(vi′) are H^s statements. Both take probabilities of $\max_{j\le\bar{k}} \|\hat{\varphi}^{(j)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}$, so $\omega \mapsto \|\hat{\varphi}^{(j)} - \varphi_\dagger\|_{H^s}$ must be a random variable as well, and measurability into the $L^2(\pi)$ container does not by itself give it. *It follows from the three steps above at the configuration Lemma 83 states*, where $\hat{\varphi}^{(0)} \in \mathcal{S}_n$ and hence the whole path lies in $\mathcal{S}_n$: since $\mathcal{S}_n \subset H^s$ by (8), $\|\cdot\|_{H^s}$ is a norm on a *fixed* finite-dimensional space, hence a continuous, and so measurable, function of the coordinates, which the steps above make random variables. *The membership is A8's business and it is now* ***stated*** *there rather than flagged here*: nothing in this definition places the B-spline basis in H^s, and (8) is A8's basis clause read as the containment it delivers. (162) *is* ***not*** *the right pointer, and the error would be one of direction rather than of precision*: (162) states how well splines *approximate* H^s functions, and what is needed is that splines *lie in* H^s. *Approximation power does not give membership, and the two conditions are genuinely different inequalities on the same degree*, a basis of degree d approximates H^s at the rate (162) states as soon as $s \le d+1$, while $\mathcal{S}_n \subset H^s$ requires $d > s - \frac{1}{2}$; on the window $s - 1 \le d \le s - \frac{1}{2}$ the first holds and the second fails, and the witness $s = 1.7$, $d = 1$ sits in it. A8's basis clause is the right home, as this bullet's own first sentence already said, and it is stated there in *degree*, the one reading of "order" on which the spline conventions agree.

This is a discharged verification and **not** a fifth entry in Remark 58's inventory, and the distinction is that remark's own. That inventory carries obligations whose *discharge site is a later step*, something a stated result's truth depends on and no present result establishes. Here the discharge is performed in the same definition that constructs the object, so nothing is owed forward. *That is not the same as having no consumer, and the draft this sentence replaces said it was*: Definition 24(c′.1) consumes exactly this measurability when it fixes the jointly measurable representative of $\hat{\varphi}^{(-k)}$, which is what supplies the composite at Lemma 92 Steps 4 and 4′ — the second precedent named just below is therefore also a consumer. The

precedents are A4(iii)'s joint-measurability half and Definition 24(c′.1)'s Tonelli inheritance, neither of which is inventoried either, for the same reason.

What the verification does not reach, and why *pending* is the honest register The four steps above are written at the **fixed** $\mathcal{S}_n$, and by Remark 89(7a) the *feasible* clause (b) of Definition 22 projects instead onto $\operatorname{span}(b_1, \ldots, b_{L_x}) \cap \{h : \int_{\mathcal{X}} h \, d\hat{\pi} = 0\}$, a **random** finite-dimensional subspace. *Two things follow for* $\hat{\varphi}$ *and neither is established anywhere in this corpus. Measurability*: the induction above evaluates a projection onto a fixed subspace, and at a random one the step is a superposition argument — the Carathéodory device the same verification already runs for the random functionals (96) — *which is not run for a random basis anywhere here. Membership*: the feasible iterate has zero $\hat{\pi}$-mean while $\mathcal{V}$ is cut out by zero π-mean, so $\hat{\varphi} \in \mathcal{V}$ is not automatic; that is Remark 89(7c), open and priced. *Both are properties Definition 24*(c′.1) *needs of* $\hat{\varphi}^{(-k)}$ *and not of* $\hat{\varphi}^{\pi}$, and they have a consumer of their own that no rate reaches: Lemma 94 kills the empirical-process term off $\sigma(\mathcal{I}_{-k})$-measurability, not off any rate, so this gap is not (56)'s gap wearing a second hat — the rate condition on $\hat{\pi}$ that (7c) prices leaves the measurability half exactly where it stands, *and does not close the membership half either, a rate making the* π*-mean small and not zero. Neither is discharged by being carried as a hypothesis*, on Remark 58's standard for (O2) and (O3): one cannot require a property of an object one has already built, so the honest register is *pending*, and it is printed as one. *The repair available for membership is a container weakening, reading Definition 24*(c′.1)*'s third entry at* $L^2(\pi)$ *rather than at* $\mathcal{V}$*, on the Lemma 45*(i) *precedent; it is named and not made, because it must be checked at every consumer of that clause.*

The price the schedule charges, which is separate from both features of (c) The schedule requires a *rate* $\hat{\varepsilon}_0$ for the preliminary and not mere consistency. *That price is not new, only newly visible*: the claim that $\bar{k}$ is finite and does not grow with n already required $\hat{\varepsilon}_0$ to be polynomially small. Hypothesis (P-a.2) of Proposition 28 carries it.

The two load-bearing features of the stopping rule, and the status of Bakushinskii's veto Two features of (c) are load-bearing and one of them is a correction. First, the level is stage-dependent, and it must be. The data error of the stage-k equation (a) is not $\hat{\delta}_n$ but $\hat{\delta}_n \ln n$ *plus* the linearisation remainder $h_u^{-2}\varepsilon_{k-1}^2$, where $\varepsilon_{k-1} = \|\varphi_\dagger - \hat{\varphi}^{(k-1)}\|_{L^2(\pi)}$, see (174). A fixed level $\tau_{\mathrm{DP}}\hat{\delta}_n \ln n$ sits *below* the true data error in exactly the regime the recursion must pass through, namely $h_u^{-2}\varepsilon_{k-1}^2 \geq \hat{\delta}_n \ln n$; there the classical discrepancy guarantee does not apply and the rule either over-fits the noise or fails to stop before $\bar{m}_n$. Second, the level is deterministic and supplied from outside, and the reason has two parts of very different status. The first is this paper's own and is sharp; the second is classical and cited, and is weaker than it looks.

(2a) The realised residual cannot supply the level, and the reason is what ill-posedness is. The natural data-driven repair is to replace $h_u^{-2}\hat{\varepsilon}_{k-1}^2$ by $h_u^{-2}\hat{r}_{k-1}^2$ with $\hat{r}_{k-1}$ the residual realised at stage $k-1$. *It cannot work.* The quantity the level must dominate is $h_u^{-2}\varepsilon_{k-1}^2$ with $\varepsilon_{k-1} = \|\varphi_\dagger - \hat{\varphi}^{(k-1)}\|_{L^2(\pi)}$, and a residual bounds ε_{k-1} from below only through a lower bound on $\|\hat{\mathcal{A}}_{\varphi_\circ} v\|$ in terms of $\|v\|$. That $\|\hat{\mathcal{A}} v_\dagger\| \ll \|v_\dagger\|$ is available is not an accident of this construction; it is the definition of ill-posedness. So $\hat{r}_{k-1}$ can be arbitrarily smaller than ε_{k-1}, and a threshold built from it sits below the data error for exactly the reason the fixed threshold did. This observation is the paper's own and it is the sharp part of the argument.

(2b) The general statement is Bakushinskii's veto, it is cited, and it is weaker than it looks in a statistical setting. That *no* rule depending on the data alone, and not on the noise level , can make a regularisation method convergent for a genuinely ill-posed problem is Bakushinskii (1984); see Engl, Hanke, and Neubauer (2000, §3.3). Two scope statements, because the veto is routinely over-read. It is a *worst-case, deterministic* statement: convergence is required uniformly over the admissible data, and the counterexample it turns on is a worst case. *In the statistical setting this paper inhabits it does not rule out data-driven selection*: Lepskii-type balancing over the regularisation index is purely data-driven and attains the optimal rate up

to a logarithmic factor (Lepskii, 1990), and Section 5 runs such a stop alongside the schedule. *So the schedule is preferred here, not forced here*, and the reasons are three and local. The number of stages $\bar{k}$ is finite and $\bar{k}_a$ is deterministic, so there is no adaptation problem across stages to solve. A deterministic level is what makes the rule Definition 22(c) implements and the rule Proposition 28 Step 2 analyses coincide *by construction*. And a Lepskii stop over m would need a variance proxy at each candidate index that the operator-noise model of this paper does not supply without a further bound. A balancing stop is therefore a live alternative rather than a foreclosed one, and were it to work here the requirement that (P-a.2) deliver a *rate* would soften materially.

The θ-step (Definition 24)

Convention (P8): the transfer rule and its two companions (P8)'s *transfer rule* and its two companions, the O_P companion and the o_P companion, are Lemma 115(i)–(iii), stated at a *generic* σ-field, written $\mathcal{G}$ there and instantiated here, and they apply with no extension.

The conditioning σ-field is written out and given no symbol The licence is the same either way, since the convention is a reading of a proof and not a hypothesis; the sentence is not, and this definition is where the re-declaration lives. *That σ-field is written out at every site and is given no symbol*: the three natural candidates are all claimed, $\mathcal{G}$ is Lemma 41's closed span, $\mathcal{E}$ is the best-approximation error of (162), and $\mathcal{F}$ is the Fourier transform of (3), and Lemma 60 and (P8) already write $\sigma(\mathcal{I}_a)$ out, so the convention is inherited rather than invented.

How each conditional bound of the θ-step is produced, and which companion converts it Each conditional bound below is produced by conditional Markov or conditional Chebyshev, hence holds at *every* threshold with a $\sigma(\mathcal{I}_{-k})$-measurable numerator, and is converted by the O_P *companion* when the numerator is unconditionally $O_P(1)$ after division by the stated rate (Lemma 92 Step 4, Lemma 95 Step 2) and by the o_P *companion* when

it is unconditionally $o_P(1)$ (Lemma 94 Step 6, Lemma 95 Step 8). *The transfer rule needs the conditional probability to vanish at a* ***single*** *moving threshold, which neither inequality delivers*; that is the observation (P8) already recorded for the o_P case, now applied to the O_P case. What supplies the tightness of the numerators is A4(iii), A5(ii), (iv) and hypothesis (R-a) of Theorem 29.

Clause (a): the Euclidean norm the file does not use, and why A4(i) is named *No statement of Definition 24's own file uses the Euclidean norm* $\|\widehat{\widetilde{W}}_i^{(-k)}\|$*; several proofs in this supplement need exactly it, and they* ***derive*** *it from the per-cell bound rather than assume it*: (183) is that derivation, and Lemma 94, Lemma 95 and Theorem 29 are its consumers. So that clause scopes the absence to its own file and does not say the object is unwanted. *A4(i) is named on the second bound because the* ***constant*** *rests on it and nothing else on that line does*: (11) puts $\hat{\varrho}_l^{(-k)}$ in $[\varepsilon_\varrho, 1-\varepsilon_\varrho]$ by construction, and it is A4(i), read at *every* $l \in \{0, \dots, \mathsf{L}-1\}$, the base cell included, that puts ϱ_l in the same interval; without it ϱ_l ranges over $[0,1]$ and the constant is $1-\varepsilon_\varrho$, not $1-2\varepsilon_\varrho$. *And that display is* ***not*** (183): this one is *Euclidean*, carries the constant declared there, and is derived from the truncation alone with A4(i) expressly *off* its list. They are a sup-norm and a Euclidean statement about the same object, and only the first reads an overlap condition, which is what makes them confusable and is why the difference is printed. *At* $\mathsf{L}=2$, (11) is the interval $[\varepsilon_\varrho, 1-\varepsilon_\varrho]$, $\mathbb{1}\{W=1\}=W$ makes $\widehat{\widetilde{W}}^{(-k)} = W - \hat{\varrho}^{(-k)}(\Delta Z)$ a scalar, and the cell subscript is vacuous.

Clause (b): what licenses the anchor, and what it is not By Corollary 12 and Lemma 10 the population solution set of (23) in φ is $\mathcal{V}_{CI}$ at *every* θ, so under A1b(iii) the φ-step targets $\varphi_\dagger$ at $\theta_\circ$ as well as at $\theta_\dagger$; the population solution set is θ-free and the operator is not, which is why the $\theta_\circ$-specific reading of Proposition 28's hypotheses is stated in the clause. This is Remark 38(iv), used and not extended.

What clause (c)'s three properties buy The three properties clause (c) requires of $\hat{\ell}^{(-k)}$ are the minimum Definition 19(b) needs to be well posed, they are what make $\int_0^u \hat{\ell}^{(-k)}(v,z)dv$ exist and make $\partial_u \hat{L}^{(-k)} = \hat{\ell}^{(-k)}$ hold at every point, *and they are stated here rather than inherited*, so that this definition is well posed on its own terms.

Clause (c′) re-declares nothing: what Lemma 60 already states *This clause re-declares nothing*: Lemma 60's sample-splitting convention states $\sigma(\mathcal{I}_a)$-measurability of $\hat{\ell}$ as a *random element* of $L^2(P_{\Delta U,\Delta Z})$ together with a fixed *jointly measurable representative*, which is strictly weaker than (41) below and is all that site needs, it evaluating $\hat{\ell}$ at an *undistorted* $(\Delta U, \Delta Z)$ draw.

Why the carve-out for $\hat{\varphi}^{(-k)}$ is forced For $\hat{\varphi}^{(-k)}$ the property of clause (c′.1) is verified rather than required; *and the exception is forced, not stylistic*, since one cannot require a property of an object one has already built, Remark 58's test, read on an estimator.

Clause (c′.1): what supplies $\hat{\varphi}^{(-k)}$'s measurability, and the choice of representative *A random element of an L^2 space over a separable metric space admits a jointly measurable representative*, and the representative clause (c′.1) fixes is measurable with respect to the product of the sample σ-algebra with $\mathfrak{B}(\mathbb{R}^p)$ for $\hat{\varrho}^{(-k)}$ and $\hat{b}^{(-k)}$ and with $\mathfrak{B}(\mathcal{X})$ for $\hat{\varphi}^{(-k)}$. Clause (c′.1) records that the ground which settles the other two, A4(iii)'s "by inspection of the formula", is unavailable for $\hat{\varphi}^{(-k)}$: no formula is inspected when an index is chosen by a discrepancy rule. *What supplies it instead is the measurability paragraph of Definition 22*, which builds the property from the deterministic cap (36), the emptiness convention of Definition 22(c), and the finiteness of the resulting family: the stopping index is a hitting index of finitely many measurable events, the iterates are finite compositions of measurable maps, and the stopped iterate is a finite indicator-weighted sum of them. *The superscripts are not written out here*, because Definition 22's index is a *stage* index and this clause's is a *fold* index, and the two are one glyph apart. *The other two remain requirements, as printed*,

and $\hat{b}^{(-k)}$'s is read off its formula on the $\hat{\varrho}$ model. *The sample σ-algebra is written out and given no symbol*, because calligraphic $\mathcal{F}$ is A2c's Fourier transform and $\mathcal{F}_{XZW}$ never appears without its subscript; the same convention applies to $\sigma(\mathcal{I}_{-k})$ itself. For these three the choice of representative is immaterial, and that is the test a container has to pass, but the test is passed on **two different grounds**. For $\hat{\varrho}^{(-k)}$ and $\hat{b}^{(-k)}$ the evaluation is at ΔZ_i, whose law *is* the one the container is built on, so identity of laws does it. *For $\hat{\varphi}^{(-k)}$ it is not*: its container is $\mathcal{V} \subset L^2(\mathcal{X}, \pi)$, whose null sets are the π-null ones, while its evaluations are at X_{i1} and X_{i2}, whose laws are f_{X_1} and f_{X_2}. *What transfers the null sets there is **domination**, and it has a tag*: C1(ii) gives $\bar{c}_\pi = \max_{j\in\{1,2\}} \|f_{X_j}/\pi\|_\infty < \infty$, so every π-null set is P_{X_j}-null for $j \in \{1, 2\}$; *"the law its container is built on" is the wrong ground for the third object.* Either way any two representatives agree at all of the finitely many evaluation points outside a P-null set, the union of the finitely many exceptional sets being P-null.

Clause (c′.1): the inheritance is Tonelli on positive and negative parts *The inheritance is the **measurability half** of Fubini–Tonelli and it is taken on positive and negative parts, entrywise*: (31)'s integrand is $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued and of arbitrary sign, and $\int_0^u = -\int_u^0$ when $u < 0$. For each entry (j, l) with $j \le p$ and $l \le \mathsf{L} - 1$, and each sign part, the two maps

$$(\omega, u, z) \longmapsto \int_{\mathbb{R}} \mathbb{1}\{0 \le v \le u\} \left(\hat{\ell}_{jl}^{(-k)}\right)^{\pm}(\omega)(v, z)\, dv, \qquad (\omega, u, z) \longmapsto \int_{\mathbb{R}} \mathbb{1}\{u \le v \le 0\} \left(\hat{\ell}_{jl}^{(-k)}\right)^{\pm}(\omega)(v, z)\, dv$$

are product measurable by Tonelli's theorem, their integrands being non-negative and $\sigma(\mathcal{I}_{-k}) \otimes \mathfrak{B}(\mathbb{R} \times \mathbb{R} \times \mathbb{R}^p)$-measurable by (41); all four are *finite* at every point because clause (c) makes $\hat{\ell}^{(-k)}$ bounded at each realisation, so their signed combination is well defined and product measurable. *This is the device A2c*(m.2) *already owns*, on the other side of the pairing, where the same positive-and-negative-part split carries Tonelli across a signed u-integral. *The inheritance runs together with the representative that clause fixes for $\hat{b}^{(-k)}$*, which is why (41) is imposed on $\hat{\ell}^{(-k)}$ and not on $\hat{L}^{(-k)}$ as well.

And the third display of (43) *is a statement with a truth value*: by (41) and Tonelli's theorem $\omega \mapsto \|\hat{\ell}^{(-k)} - \ell_\star\|_{L^2(P)}$ is $\sigma(\mathcal{I}_{-k})$-measurable, and likewise every conditional expectation and every O_P statement about these objects is well posed.

Clause (c′.1): why the derivative inherits product measurability under (R-a) *Under (R-a) the derivative inherits it too, and* (c′.2) *needs that for its second envelope*: $\partial_u \hat{\ell}^{(-k)}$ exists at every point under (R-a) and is the pointwise limit of the difference quotients of $\hat{\ell}^{(-k)}$ in u taken along the reciprocals of the positive integers, *the increment is written out and carries no letter*, on the same ground as (42)'s countable dense set; each such quotient is product measurable by (41), and a pointwise limit of countably many product-measurable functions is product measurable. *A representative of an L^2 class supplies nothing of the kind*, having no derivative in general. No equivalence class is formed and no representative is chosen, so there is nothing for a choice of representative to disturb, and clause (c)'s continuity in u is untouched.

Clause (c′.1): why the L^2 reading fails for the two weight objects *Why the L^2 reading fails here and not three lines above*: these two are evaluated at a *shifted* first argument, at $(\Delta U - \lambda \Delta h_k, \Delta Z)$ in Lemma 93 Steps 1 and 4–5, at $(\Delta U_i(\theta, \hat{\varphi}^{(-k)}), \Delta Z_i)$ in (46) and Lemma 95 Steps 2 and 9, whose law (c′.2) below argues, and needs, *not* to be dominated by $P_{\Delta U, \Delta Z}$; on an undominated set two representatives of one class disagree, so $\hat{J}$ of (46), $\Psi_k^{(1)}$ of Lemma 93, and hence $\hat{\theta}$ of (47) and $\hat{\Sigma}$ of (48), would depend on which was chosen and *the estimator would not be a function of the object the clause declares.*

Clause (c′.1): the derivation of $\hat{L}^{(-k)}$'s membership The membership of $\hat{\ell}^{(-k)}$ in $L^2(P_{\Delta U, \Delta Z})$ almost surely is Lemma 60 Step 0's third point, and it is what makes the third display of (43) finite. The membership of $\hat{L}^{(-k)}$ in $L^2(P_{\Delta U, \Delta Z})$ is derived and not required: by (31), $\|\hat{L}^{(-k)}(u, z)\| \le \|\hat{b}^{(-k)}(z)\| + \|\hat{\ell}^{(-k)}\|_\infty |u|$, so $E\|\hat{L}^{(-k)}(\Delta U, \Delta Z)\|^2 \le 2E\|\hat{b}^{(-k)}(\Delta Z)\|^2 + 2\|\hat{\ell}^{(-k)}\|_\infty^2 E[\Delta U^2] < \infty$ almost surely under Definition 19(b)'s admissibility, (R-a) and A1b(iv).

Both statements are about the function and neither is a container declaration.

Clause (c′.1): the seven Part-I sites of the freezing lemma, and the direction the proofs use *Equation* (41) *is what* (P8)*'s freezing lemma consumes for the* $\hat{\ell}^{(-k)}$ ingredient *of its composite, at six of its seven* ***Part-I*** *sites*, that lemma being stated at a composite of the form $(\omega, e) \mapsto f\big(\hat{\ell}^{(-k)}(\omega), e\big)$. *The seventh is Lemma 92 Steps 4 and 4′, where* $\hat{\ell}^{(-k)}$ *does not appear at all and the frozen object is* h_k: what supplies the composite there is the jointly measurable representative fixed above for $\hat{\varphi}^{(-k)}$, so both halves of this clause are load-bearing and neither covers the other. *Seven is the* ***Part-I*** *count and not the count of the lemma's sites, which is nine*, and the qualifier is load-bearing because the other two are supplied by something else: Lemma 77 Step 4 sits in Part H, is stated before Definition 24 exists and does not adopt it, and takes its composite from A4(iii); Lemma 60 Steps 0 and 2 sit in Part F and take theirs from that lemma's own sample-splitting convention, which is deliberately weaker than (41). The catalogue is Remark 120, which carries all nine; this paragraph points into it and maintains no count of its own. The composite is **not** product measurable "precisely when" (41) holds, in either direction. Not sufficient: the composite is product measurable only once f is itself jointly Borel in its two arguments, and at the sites where the evaluation point is built from $\hat{\varphi}^{(-k)}$ only together with that object's own clause. Not necessary: $f \equiv 0$ is product measurable whatever $\hat{\ell}^{(-k)}$ is. *What is true is the direction the proofs use*, and it is the one this clause is for: (41) is what this definition *supplies*, and it is what the per-realisation "Borel" of clause (c) does not. *The file owns the device by which* (41) *is verified, and it is recorded rather than imposed*: Lemma 93 Step 1 obtains joint measurability in (λ, ω) from "continuous in λ, measurable in ω" by the Carathéodory criterion, and the same criterion gives (41) whenever, for each fixed u, the map $(\omega, z) \mapsto \hat{\ell}^{(-k)}(\omega)(u, z)$ is $\sigma(\mathcal{I}_{-k}) \otimes \mathfrak{B}(\mathbb{R}^p)$-measurable, since (c) already makes $u \mapsto \hat{\ell}^{(-k)}(u, z)$ continuous at every (ω, z). *That is a sufficient condition and not the requirement*; the requirement is (41), which is what the proofs use.

Clause (c′.2): the inner supremum, and why the essential supremum is taken in z The inner supremum in (42) is over an uncountable index but is attained on a countable dense set, because clause (c) makes $u \mapsto \hat{\ell}^{(-k)}(u, z)$ continuous at every z and (R-a) makes $u \mapsto \partial_u \hat{\ell}^{(-k)}(u, z)$ continuous as well, so $(\omega, z) \mapsto \sup_u \|\cdot\|$ is jointly measurable by (c′.1). *That is the one clause of Definition 24 which consumes (R-a)'s strengthening of* (c)*'s continuity to continuous differentiability, and it is why* (42) *is* ***not*** *an unconditional convention. And the essential supremum must be taken in z and not in (u, z)*, because every consumer of (42) evaluates the envelope at a *shifted* first argument, whose law need not be dominated by $P_{\Delta U, \Delta Z}$, while its *second* argument is in every case a draw of ΔZ. *The shift takes the two shapes named above, and the list that follows is* ***illustrative*** *and not a complete enumeration of the consumers*: beyond Lemma 93 Steps 1 and 4–5 and Lemma 95 Steps 2 and 9, Lemma 94 Steps 3 and 5 read the first shape, and Lemma 95 Steps 3 and 5 and Theorem 29 Step 1 read the second.

Clause (c′.2): measurability of the outer essential supremum The **outer** essential supremum is measurable as well, and this does **not** follow from joint measurability without an argument, it is not a countable supremum. The argument is the tool of the previous paragraph, once more. At each fixed real level, the map $\omega \mapsto P_{\Delta Z}\Big\{z : \sup_u \|\hat{\ell}^{(-k)}(\omega)(u, z)\| > \text{that level}\Big\}$ is $\sigma(\mathcal{I}_{-k})$-measurable by Tonelli's theorem applied to the non-negative jointly measurable indicator; the levels at which that probability vanishes are exactly the levels *at or above* the essential supremum, since it vanishes at the essential supremum itself, the exceptional set there being the union of the countably many exceptional sets one reciprocal-of-an-integer above it, and at no smaller level, by the definition of the essential supremum as an infimum; and therefore, for every real level, the event that the essential supremum does not exceed it is the intersection, over the *countably many rational levels strictly above it*, of the events that the probability vanishes, a countable intersection of measurable sets, the *strict* inequality being what makes those exceptional sets exhaust the one at the level itself.

Clause (c′.2): why the envelope's measurability is load-bearing *The levels are written out and carry no letter*, on the same ground as the countable dense set above. *This is load-bearing and not hygiene*: (R-a) prices both envelopes as $O_P(1)$, which is a statement about the distribution of these two maps, and (P8)'s O_P companion requires its *bounding numerator* to be measurable with respect to the conditioning σ-field, and at Lemma 95 Step 2 that numerator is $\|\hat{\ell}^{(-k)}\|_\infty^2 E\|\Delta Z\|^2$, whose only random factor is the envelope. *No symbol of the companion's is imported into compiled text*; the numerator is named by what it is.

Clause (c′.2): a genuine supremum over z would not be measurable *A genuine supremum over z, by contrast, would not be a measurable function of ω*, since no clause makes $z \mapsto \hat{\ell}^{(-k)}(u, z)$ continuous.

Clause (c′.2): where the exceptional set falls, and why $\ell_\star$ needs no convention *The exceptional set is therefore P-null where the envelope is used*: for each i, $\sup_u \|\hat{\ell}^{(-k)}(u, \Delta Z_i)\| \leq \|\hat{\ell}^{(-k)}\|_\infty$ almost surely, and the union over the finitely many $i \leq n$ of the exceptional sets is still P-null, so the bound holds simultaneously at all n observations almost surely. *For the deterministic $\ell_\star$ no convention is needed and none is imposed*: A2′(i)'s closing clause defines $\ell_\star$ pointwise and Corollary 49 bounds it at *every* (u, z), so $\|\ell_\star\|_\infty$ is a genuine supremum and the two readings agree.

Clause (d): what the second display would say without the sieve premise The second display of (43) is finite at every realisation only under (8): without the first the display is an $O_P(1)$ statement about a quantity equal to $+\infty$ surely, which is a condition on the *empty* event and not a rate.

Clause (d): what the fold reading of A5(vi′) costs This costs nothing and is nonetheless not without consequence. It costs nothing because F is fixed, so the fold reading is the intersection of F events each of probability tending to one; it is not without consequence

because the radius R must then serve all F fold complements *at one value*, R being a constant of Definition 80 and not a fold-indexed object, and R also appears in (P-c), (P-g), (121) and elsewhere, so a fold-indexed radius is not available. *Those three are named as instances and this paragraph maintains no list*: Definition 80 carries the enumeration and this points at it. The schedule constants carry R_*, which that definition rules a **different** radius, not comparable to R in either direction. (T-5) already reads A5(vi′) at each fold *operationally*, through "(56) holds on each $\mathcal{I}_{-k}$"; what was missing is that this enumeration, which declares itself the place where the reading lives, did not list it. *That operational reading survives the 2026-08-27 scoping of* (56) *and is worth saying so*: what (T-5) delivers at each fold is the display at $\hat{\varphi}^{\pi}$, and A5(vi′) is a hypothesis *consumed* inside Proposition 28 at Lemma 83(c) rather than a conclusion drawn from it, so the fold reading it needs is of the hypothesis and is untouched by which reading of Definition 22(b) the conclusion names. *What does* ***not*** *survive is the parallel reading for A5(ii)*, and that clause is carried at (T-4).

The parts of (P8) do not have the same standing Convention (P8) is re-declared at Definition 24, and the parts of (P8) do not have the same standing.

How clause (c′)'s two sub-clauses stand to clause (c) Two clauses follow; neither is available from clause (c) alone, and (c′.1) is a requirement on the estimators and not a consequence of anything above it, *with* ***one*** *exception, marked at the clause itself, and it is the object clause* (b) *has already fully constructed.*

Clause (a)'s truncation is a pointwise operation on the very function it names For $\hat{\varrho}^{(-k)}$ no representative is chosen at all; Definition 2's truncation is then a pointwise operation on the very function clause (a) names.

Why the L^2 container is nonetheless retained for $\hat{\varrho}^{(-k)}$ The L^2 container is retained for $\hat{\varrho}^{(-k)}$ only because A4(iii)'s *rate* is an L^2 statement; it is not what any pointwise bound

rests on.

Why the fold count F is carried explicitly and not absorbed The constant F is carried explicitly in every bound rather than absorbed, because Proposition 96 quantifies over partitions and a suppressed F would hide the one place a partition enters an order.

Proposition 36 (T is well defined and bounded)**.** *The moment family* (21) *defines an element of* $L^2(\Pi)$ *at every trial pair. Suppose B1(i), (ii), (iv.c) and (iv.d) hold.* **B1(iv) is cited by clause, and the factorisation is not among the clauses***:* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ *is B1(iv)'s* declaration*, standing at every statement in this corpus by the assumption block's* ***three registers*** *paragraph, and it is what lets Step 2's iterated integral be written at all. What the proof* opens *is* (iv.c) *and* (iv.d) *and nothing else — Step 2 reads "by B1(iv.c) and B1(iv.d)" in terms — so a block citation would charge this proposition with* (iv.a) *and* (iv.b)*, which no step reads.* Through 2026-08-24 this line was the block*, and before that day it was* (iv.c) *alone. Then for every* $(\varphi, \theta) \in \mathcal{V} \times \mathbb{R}^p$ *the function* $\mathcal{M}(\cdot, \cdot; \varphi, \theta)$ *is Borel measurable and*

$$\left\| T(\varphi, \theta) \right\|_{L^2(\Pi)} \;\leq\; c_\Pi \;<\; \infty,$$

with c_Π *from Assumption B1(iv.c), so that* T *maps* $\mathcal{V} \times \mathbb{R}^p$ *into a ball of* $L^2(\Pi)$ *of radius not depending on* (φ, θ)*.*

Remark 37 (ν versus ν/π). Suppose in addition A1b(iv) holds, so that $\nu = \nu_{\Delta Z}$ is well defined. Then $\int_{\mathcal{X}} (\nu/\pi)\, \pi \, dx = \int_{\mathcal{X}} \nu \, dx = 0$, whereas $\int_{\mathcal{X}} \nu \, \pi \, dx = E[\Delta Z \{\pi(X_2) - \pi(X_1)\}]$, which does not vanish at a generic design — π is a feature of P by Definition 5 and not a shape the analyst may choose to make this term zero. Since $\mathcal{R}(\mathcal{A}^*_{\text{cond}}) \subseteq \mathcal{V}$ by Proposition 9 and $\mathcal{V} = \{\zeta : \int \zeta \, \pi \, dx = 0\}$, the source condition A2 can only be imposed on ν/π, never on ν.

Remark 38 (R1). Three things follow from Lemma 10, and one does not.

(i) What it establishes. Restriction (A) carries no first-order information about $\theta_\dagger$: the set of pairs it admits is a product set, so no functional of the conditional law of the residual given $(W, \Delta Z)$ can distinguish $\theta_\dagger$ from any other θ.

(ii) What it does not establish. It does not say that $\theta_\dagger$ is unidentified. $\theta_\dagger$ is identified by (B)

alone: under A1 and A1b(iv),

$$\theta_\dagger \;=\; \Big(E[\Delta Z \Delta Z']\Big)^{-1} E\Big[\Delta Z\,(\Delta Y - \Delta\varphi_\dagger)\Big],$$

the inverse existing by A1b(iv) and the expectations being finite by the moment clauses of that same assumption. This route uses neither A2 nor the retired A3. A3 asserted nonsingularity of $J = E[\Delta Z \Delta Z'] + E[\ell_\star \widetilde{W} \Delta Z']$ and was itself conditional on A2; routing identification of $\theta_\dagger$ through it would make identification contingent on a source condition, which it is not. By Lemma 91(iv) the second summand is zero under A1, so J is $E[\Delta Z \Delta Z']$ wherever it is defined, and A3 is retired at A3 as a consequence of A1 and A1b(iv). *What must not be routed through A2 is the existence of* $\ell_\star$, and the display above needs none of it. Nor does Lemma 10 say that (A) is useless. A Neyman correction is mean-zero at the truth *by construction*, that is what makes it a correction rather than a new restriction, so "carries no information about θ" is a property of every Riesz representer in the debiased-machine-learning construction (Chernozhukov et al., 2018), not a defect peculiar to this model.

(iii) What it costs. Over-identification. Under an *unconditional* independence assumption one would have $\Delta Y - \Delta\varphi_\dagger - \Delta Z'\theta \not\perp\!\!\!\perp W$ for $\theta \neq \theta_\dagger$ whenever $\Delta Z \not\perp\!\!\!\perp W$, so (A) would carry information about θ and deliver a testable restriction. Under A1 as stated, $\theta_\dagger$ is *exactly* identified by (B) and (A) is a pure debiasing device: *(A) yields no over-identifying restriction on* θ, which is precisely what Lemma 10 proves. It does not follow that (A) is untestable. Solvability of the operator equation remains testable in the usual nonparametric-instrumental-variables sense, $\mathcal{V}_{CI} = \emptyset$ is an observable event, witnessed for instance by $\Delta Y = W\chi + e$ with $\chi \neq 0$ and $e \perp\!\!\!\perp W$, where no φ makes the residual conditionally independent of W. The claim to make in the paper is about θ, not about (A).

(iv) A population consequence, and a conditional algorithmic one. By Corollary 12 the solution set of (23) in φ is $\mathcal{V}_{CI}$ at every θ, so *as a population statement* the first stage may be posed at any fixed θ, including $\theta = 0$, that is, on raw ΔY. Two qualifications are required

before this says anything about the estimator. First, "targets $\varphi_\dagger$" needs $\mathcal{V}_{CI} = \{\varphi_\dagger\}$, i.e. A1b(iii); without it the first stage targets a set. Second, consistency of the *regularised sample* solve is not delivered by Corollary 12: it is delivered by Proposition 28, whose hypotheses, A7 in its amended form (13), the spectral source condition (P-a.3), and the bandwidth condition (P-f), are conditions on $\hat{T}(\cdot, \theta)$ and $\hat{\mathcal{A}}$, not on T, and must hold at the running value of θ — *and it is delivered at the* ***infeasible*** *reading* $\hat{\varphi}^\pi$ *of Definition 22, not at the estimator that definition instructs*, which is a third qualification of the same kind and is Remark 89(7). *No tangential cone condition is among them*: A6 is retired (A6, Proposition 71). What is θ-free is the population solution set. What is emphatically not θ-free is the conditioning of the inverse problem: $\mathcal{A}_\varphi$ in (24) evaluates the conditional density at a θ-dependent point, so the singular values, the degree of ill-posedness a and the source index μ are all θ-specific.

Lemma 39 (Bierens core)**.** *A bounded, one-to-one transform and the exponential weight convert a conditional moment restriction into an unconditional family indexed by* any *set of positive Lebesgue measure. Suppose B1(i) and (ii) hold, let* $\mathcal{T}' \subseteq \mathbb{R}^p$ *be Borel with* $\mathrm{Leb}(\mathcal{T}') > 0$*, let* V *be a random vector in* $\mathbb{R}^p$*, and let* ε *be a random variable with* $E|\varepsilon| < \infty$*. Then*

$$E\big[\varepsilon \mid V\big] = 0 \ \text{a.s.} \qquad \Longleftrightarrow \qquad E\big[\varepsilon\, e^{t'\Phi(V)}\big] = 0 \ \ \text{for every } t \in \mathcal{T}'.$$

B1(iii) is not a hypothesis of this lemma, and this is a restatement and not a change of content*: the proof reads of the index set only that its Lebesgue measure is positive, and that is now* quantified in the statement *instead of asserted about the proof. B1(iii) splits under the assumption block's* ***three registers*** *paragraph:* $\mathcal{T} \subseteq \mathbb{R}^p$ *is its* declaration *and is standing, while* $\mathrm{Leb}(\mathcal{T}) > 0$ *is its* clause*, and that clause is exactly the instance* $\mathcal{T}' = \mathcal{T}$*.* So every result that instantiates this lemma at the maintained index set carries B1(iii) on its own line — *Lemma 11, Corollary 12, Proposition 15, Proposition 17 and Lemma 41 all do, and none of their lists moves* — while Lemma 48, which instantiates it at $\mathcal{T}_1 = \mathcal{T} \cap \operatorname{supp} \Pi_t$ and **derives** $\mathrm{Leb}(\mathcal{T}_1) > 0$ from B1(iv.b), correctly does not. *That omission was previously*

legible only from a comment and from one sentence inside a proof; it is now a consequence of this statement. And the two sites that substituted an index set into a **proof** — Step 5 of Lemma 11's and Step 4 of Lemma 48's — now instantiate a **statement**, *which is the move Lemma 95*(i′) *was created to make available.* Every set this lemma is instantiated at in this corpus is Borel*: $\mathcal{T}$ by Definition 5, which equips it with $\mathfrak{B}(\mathcal{T})$, and by B1(iv.b), which quantifies over Borel subsets of it; $\mathcal{T}_0 \setminus N_{\mathcal{T}}$ because $\mathcal{T}_0$ is Borel by B1(iv.b) and $N_{\mathcal{T}}$ is a finite union of Π_t-null sets supplied by Fubini on Definition 5's* uncompleted *product σ-field, hence Borel; and $\mathcal{T}_1$ as the intersection of a Borel set with a closed one.* $\mathcal{T}'$ is bound by this lemma's own quantifier and is carried to no other result.

Lemma 40 (L-3: quantifier management in u)**.** *A family of conditional restrictions indexed by a real number reduces to a countable family, with a single exceptional null set. Fix $(\varphi, \theta) \in \mathcal{V} \times \mathbb{R}^p$. Then a single regular conditional distribution of $(\Delta U(\theta,\varphi), W)$ given ΔZ supplies* three *objects at once: a version of the conditional distribution functions $F_j^{(\theta,\varphi)}(u \mid z) := P\big(\Delta U(\theta,\varphi) \le u \mid W = j, \Delta Z = z\big)$, $j \in \{0,\dots,\mathsf{L}-1\}$, that is a distribution function in u for every z; a version $\varrho_j(\cdot)$ of $P(W = j \mid \Delta Z = \cdot)$ at each of those same j; and, for each $l \in \{1,\dots,\mathsf{L}-1\}$, a version of the left-hand side of (65), by which is meant one function of (u,z), jointly measurable and right-continuous in u at* every *z, agreeing at each fixed u almost surely with the conditional expectation that left-hand side names.* No symbol is introduced for the third*, the corpus's short capitals being claimed, Λ is the integrand of convention* (P4c) *in* ***proofs.tex*** *and $\mathcal{M}$ is the unconditional moment of Definition 6, so it is called "the left-hand side, read at the version fixed here". Writing $\bar{F}^{(\theta,\varphi)}(u \mid z) := \sum_{j=0}^{\mathsf{L}-1} \varrho_j(z) F_j^{(\theta,\varphi)}(u \mid z)$ for the W-free conditional distribution function, with $\varrho_0 := 1 - \mathbf{1}'\varrho$,*

$$E\Big[\widetilde{W}_l \, \mathbb{1}\{\Delta U(\theta,\varphi) \le u\} \mid \Delta Z = z\Big] \;=\; \varrho_l(z)\Big\{F_l^{(\theta,\varphi)}(u \mid z) - \bar{F}^{(\theta,\varphi)}(u \mid z)\Big\} \tag{65}$$

for every $u \in \mathbb{R}$, every $l \in \{1,\dots,\mathsf{L}-1\}$ and ***every*** *z, both sides being read at the versions fixed here.* That the second quantifier is "every z" and not "a.e. z" is the whole content of

the lemma*: a null set allowed to move with u says nothing at uncountably many u, the union of uncountably many null sets not being null, and it is the regular version, one kernel, read at all u simultaneously, that removes the dependence.* The freedom in F_j at a null cell is untouched*: an arbitrary distribution function there is multiplied by $\varrho_j(z) = 0$ on both sides, so* (65) *holds at every (u, z) for every such choice. Consequently, if the left-hand side of* (65)*, read at that version, vanishes for every u in a countable dense subset of $\mathbb{R}$, every l, and a.e. z, then it vanishes for every $u \in \mathbb{R}$, every l, and every z outside a single null set not depending on u or on l,* a finite union of the $\mathsf{L} - 1$ per-cell null sets, each itself a countable union over the dense set, which is why the index costs nothing here. At $\mathsf{L} = 2$ the display is the printed one*: l takes the single value 1, $\varrho_0 = 1 - \varrho_1$, so $\bar{F} = (1 - \varrho_1)F_0 + \varrho_1 F_1$ and $\varrho_1\{F_1 - \bar{F}\} = \varrho_1(1 - \varrho_1)\{F_1 - F_0\}$.* **Consumers: none**, as of 2026-08-09*, on the model of Lemma 45*(ii)*. Its only proof-step consumer was Lemma 11 Step 3, which has been lifted onto the cell-factorisation route of Remark 18*(1)*, a route that does not pass through this lemma, and Corollary 12 now inherits from that route rather than from here. The lemma is kept because it is correct, because* (**??**)*'s clause sends a reader here for what the identity does and does not deliver at $\sigma(\Delta Z)$, and because a reader following a citation should land on a statement rather than on a gap; it is* ***not*** *available as an inherited step, since nothing inherits it.* A4(i) is **not** among its hypotheses, and that is a consequence of the proof and not an omission*: the extension from a countable set of u to all of $\mathbb{R}$ propagates a zero along a right-continuous function and* divides by no prefactor*, so no lower bound on ϱ_l is needed and no case split on the degenerate region is made.*

Lemma 41 (L-2: density of the closed span)**.** *The debiasing weights representable through the Bierens family are dense, so the choice of weighting measure is immaterial on the θ-side. Suppose B1(i)–(iii) and (v) hold, and let*

$$\mathcal{G} \;:=\; \overline{\operatorname{span}}\Big\{(u,z) \mapsto \varsigma(u)\, w\big(t'\Phi(z)\big) \;:\; \varsigma \in C_b(\mathbb{R}),\; t \in \mathcal{T}\Big\}$$

with closure taken in $L^2(P_{\Delta U,\Delta Z})$. *Then*

(i) $\mathcal{G} = L^2(P_{\Delta U,\Delta Z})$;

(ii) the set $\mathcal{G}_\gamma := \Big\{L : L(u,z) = \int_{\mathcal{T}} \lambda(u,t) w(t'\Phi(z))\,\gamma(dt),\ \gamma$ *finite signed satisfying B1(v),* λ *bounded*$\Big\}$ *is dense in* $L^2(P_{\Delta U,\Delta Z})$;

(iii) $\overline{\{L - E[L \mid \Delta Z] : L \in \mathcal{G}_\gamma\}} = \{L \in L^2(P_{\Delta U,\Delta Z}) : E[L \mid \Delta Z] = 0\}$, *the subspace of weights* satisfying *N1.*

Remark 42 (Why the weight is the exponential, and what a general analytic weight would cost). B1(ii) fixes $w = \exp$, and the restriction is substantive rather than expositional.

The textbook condition is not enough. Suppose w were required only to be real-analytic and non-polynomial, as in the usual statement of the device. Take $V = \pm 1$ with probability $\frac{1}{2}$ each, let Φ be the maintained componentwise arctan, an *odd* function, so $\Phi(-1) = -\Phi(1)$, and set $\varepsilon := V$. Then $E|\varepsilon| < \infty$ and $E[\varepsilon \mid V] = V \neq 0$, yet for $w = \cos$ and *every* t,

$$E\Big[\varepsilon\, \cos\Big(t\Phi(V)\Big)\Big] \;=\; \tfrac{1}{2}\cos\Big(t\,\Phi(1)\Big) - \tfrac{1}{2}\cos\Big(-t\,\Phi(1)\Big) \;=\; 0.$$

B1(i)–(iii) hold and the converse direction of Lemma 39, read at $\mathcal{T}' = \mathcal{T}$, fails. The mechanism is that an odd Φ against an even w exposes only the even part of the image measure, so Step 4 of the proof recovers the even moments and no others. The failure is not confined to the abstract lemma: with ΔZ scalar and two-valued at $\pm c$, $\mathcal{M}(u,t,l) = \cos\Big(t\,\Phi(c)\Big)\,E[\widetilde{W}_l \mathbb{1}\{\Delta U \le u\}]$, so (c) of Lemma 11 holds whenever the *unconditional* restriction does, while the ΔZ-conditional restriction can fail through opposite-signed violations that cancel across the two values of ΔZ.

The correct general statement, and what it would cost. The repair is not to strengthen Φ but to add an intercept. Index the family by (t_0, t) over a subset of $\mathbb{R}^{1+p}$ of positive Lebesgue measure and use $\{w(t_0 + t'\Phi(z))\}$. For w real-analytic and non-polynomial, each derivative $w^{(k)}$ is analytic and not identically zero, otherwise w is a polynomial, so its zero set is discrete and

hence countable; therefore all but countably many $c \in \mathbb{R}$ satisfy $w^{(k)}(c) \neq 0$ for every $k \geq 0$, and the argument runs at such a c. This is the Stinchcombe–White family and the result is Stinchcombe and White (1998, Thm. 3.1); it is load-bearing here, not a "see also". Adopting it would change the range space to $L^2(\mathbb{R} \times \mathcal{T}^+, \Pi)$ with $\mathcal{T}^+ \subseteq \mathbb{R}^{1+p}$, and would require the envelope bookkeeping of B1(iv)–(v) to be redone with $\bar{w}(t_0, t) = \sup_{|v| \leq |t_0| + c_\Phi \|t\|_1} |w(v)|$ throughout Proposition 36, Proposition 8 and Proposition 9.

Why it is not adopted. $w = \exp$ is the maintained choice of the estimator, it is what Bierens (1990, Thm. 1) characterises, so that the citation is exact rather than approximate, and the additional generality is consumed nowhere in the programme. The general family is recorded here so that the statement is on the record without the range space paying for it.

Remark 43 (On $\mathcal{T}$, on γ, and on what compactness is for). It is worth recording what the identification layer costs, because the statement of Lemma 11 does not show it. The equivalence requires no moment condition whatsoever on ΔY, ΔU or ΔZ, and no Lebesgue density for ΔU or ΔZ. ΔZ may be discrete, ΔU may have atoms, and neither need have a finite mean. That is exactly what boundedness of Φ in B1(i) purchases: it makes $\varepsilon\, w(t'\Phi(\Delta Z))$ integrable for the bounded $\varepsilon = \widetilde{W}_l \mathbb{1}\{\Delta U \leq u\}$, *scalar at each contrast index, which is the arity* (27) *and Lemma 39 are read at*, with no condition on the law of ΔZ, and it makes the image measure of Lemma 39 compactly supported, hence determined by its moments. The smoothness and moment conditions of C1, C2′ and A1b(iv) enter only when the *operator* is differentiated or the *parameter* is identified, never in the equivalence itself.

Compactness of $\mathcal{T}$ is *sufficient but not necessary* and should not be assumed. What the proofs use is B1(iv.c) and B1(v), which for $w = \exp$ read $\Pi_u(\mathbb{R}) \int e^{2c_\Phi \|t\|_1} \Pi_t(dt) < \infty$ and $\int e^{c_\Phi \|t\|_1} |\gamma|(dt) < \infty$; a compact $\mathcal{T}$ carrying a finite Π_t and a finite γ satisfies both, but so do many unbounded index sets. Plan §18.3 introduced compactness in order to license differentiation under the t-integral. That is a specific interchange with a specific dominating function, and B1(v) now supplies it: differentiating $L(u, z) = \int_{\mathcal{T}} \lambda(u, t) w(t'\Phi(z)) \gamma(dt)$ in u

under the integral sign is licensed by the bound $|\partial_u \lambda(u,t)\, w(t'\Phi(z))| \le \Lambda_1(t)\bar{w}(t)$, valid for all (u,z), together with $\int_{\mathcal{T}} \Lambda_1 \bar{w}\, d|\gamma| < \infty$.

Two clauses of B1(iv) are minimal and should not be strengthened back. Step 5 of the proof of Lemma 11 consumes only (iv.a), full topological support of Π_u, which is what makes the zero set of $\mathcal{M}(\cdot,t,l)$ dense in $\mathbb{R}$, and (iv.b), non-mutual-singularity of $\mathrm{Leb}|_{\mathcal{T}_0}$ and $\Pi_t|_{\mathcal{T}_0}$, which is exactly the statement that no Π_t-null set can swallow $\mathcal{T}_0$. Neither absolute continuity nor equivalence is used in either factor.

Two things about (iv.b) are worth stating precisely, because it is easy to describe it wrongly. First, by the Lebesgue decomposition it is equivalent to Π_t having a nonzero absolutely continuous part, so it excludes *every* Lebesgue-singular Π_t and not merely the atomic ones: a Cantor measure is non-atomic, has uncountable support, and is excluded. Second, it is sufficient and not necessary. What the argument really needs is an analytic-uniqueness condition on $\operatorname{supp}\Pi_t$: by Step 2 of the proof of Lemma 39, $t \mapsto \mathcal{M}(u,t,l)$ is real-analytic on all of $\mathbb{R}^p$ at each fixed (u,l), so it suffices that its zero set cannot contain $\operatorname{supp}\Pi_t$ up to a Π_t-null set unless it vanishes identically. When $p=1$ that holds for any Π_t whose support has an accumulation point, a Cantor measure included, since a Cantor set is perfect, and Lemma 11 survives such a Π_t. When $p \ge 2$ the requirement is that $\operatorname{supp}\Pi_t$ not lie inside a proper real-analytic variety, which positive Lebesgue measure guarantees and which many Lebesgue-null sets also satisfy, but which perfectness alone does not: a hyperplane is perfect and is the zero set of a coordinate. (iv.b) is retained as the clean sufficient version.

What (iv.b) does exclude in every dimension is a finite grid. *The growing-family condition B1(vi) is* ***retired***, because the clause describes an estimator this programme does not implement: Definition 68 integrates against Π_t over all of $\mathcal{T}$ and never over a finite subfamily. Identification of $\varphi_\dagger$ here is A1b(iii), assumed outright. The exclusion (iv.b) performs is recorded here so that it remains visible to a reader who reinstates the growing-grid design.

The correction measure plays two different roles on the two sides of the construction. *On the*

estimation side the statement stands as written: by Lemma 41 the estimator *solves for* L inside a dense class rather than integrating a fixed weighting measure against a continuum, so no t-integral is ever computed in forming $\hat{L}$, and no weighting measure has to be chosen. This is also the surviving argument against a continuum-GMM formulation: that would require choosing a weighting measure on $\mathcal{T}$ and regularising the inverse of its covariance operator. On the population side the integral representation is not optional. The representer selected by the adjoint equation is an element of $L^2(\Pi)$, and what makes it evaluable at $(\Delta U, \Delta Z)$ is exactly the induction map $\mathcal{W}$ of Lemma 48(ii), which is an integral against Π_t. The correction measure of B1(v) is therefore not a free object *on that side*: Lemma 48(ii) identifies it with Π_t, and *at* $\gamma = \Pi_t$*, and only there*, the first half of B1(v) is implied by B1(iv.c) rather than assumed, since $\int \bar{w}\, d\Pi_t \leq (\Pi_t(\mathcal{T}) \int \bar{w}^2 d\Pi_t)^{1/2} < \infty$ by the Cauchy–Schwarz inequality. That implication must not be widened. Lemma 41(ii) quantifies over *every* finite signed γ satisfying B1(v), and its proof constructs an *atomic* $\gamma \neq \Pi_t$ for which the first half is verified directly; so the first half is live on the estimation side and is retained for that reason, while the second half is live here, in this remark, and is declined by Corollary 49 for the reason recorded there. What reconciles the two sides is Lemma 48(ii)'s density claim: the sieve and the Bierens family parameterise the same closure, so the estimator may search the larger class without losing the population object.

Finally, Lemma 11 is stated at the equivalence, not at a strict inclusion. The claim that $\mathcal{R}(\mathcal{A}^*_{\text{cond}})$ strictly contains the range of the unconditional adjoint is not proved here and is not claimed: with W taking finitely many values the two moment classes are not nested, since, under A1, $\{g : E[g \mid \Delta U, \Delta Z] = 0\} = \{\widetilde{W}'\ell(\Delta U, \Delta Z)\}$ is an equality of sets, *the* $\mathsf{L} - 1$ *contrasts spanning the conditionally centred functions of* W*, which is the spanning statement accompanying* (1)*, and at* $\mathsf{L} = 2$ *the single contrast doing it* , and the comparison across the two modelling assumptions is heuristic. The conditional route is defended on economic grounds and on internal consistency with the maintained exogeneity assumption, not on a weaker source condition.

Remark 44 (The withdrawn primitive for A1c). A corollary stood in Section 2, deriving A1c from measure separability together with an instrument completeness condition A1d, on the model of Fève and Florens (2014, Prop. 2.2). It is withdrawn, because its hypothesis set is empty in this model. $\widetilde{W}$ is conditionally centred given ΔZ, so under A1 the operator $\Lambda \mapsto E[\widetilde{W}\Lambda \mid \Delta U, \Delta Z]$ annihilates the constants; the completeness condition therefore fails at $\Lambda \equiv 1$ at every law A1 admits, and *a fortiori* at every law A1b(iii) admits. Any completeness hypothesis posed on that operator over a class containing the constants is unsatisfiable here, which is the structural difference from the conditional-mean design, where the corresponding operator $g \mapsto E[g \mid W]$ does not annihilate constants. *The transplant fails as a contradiction and not as an excess*, and the version stated there was too strong in a way that made it empty rather than merely demanding. *No result is affected*: A1c is a hypothesis of no result in this paper, and the withdrawn corollary was A1d's only consumer. What the withdrawal costs is the paper's only offered primitive route to first-order local identification; A1b(iii) is assumed and not derived. *Nor would the linearised condition have delivered a neighbourhood of uniqueness had it been reachable*: $\mathcal{A}_{\text{cond}}$ is compact by Proposition 9, so it is not bounded below, and by Proposition 8(i) a competitor at $\varphi_\dagger + \varepsilon e_j$ is admissible once $\varepsilon \gtrsim \sigma_j / C_{\mathcal{A}}$, a threshold that shrinks to zero along the singular system. First-order injectivity therefore separates competitors at no fixed radius.

Two constructions for the hypothesis of Proposition 17(iv). *A first witness that the hypothesis of Proposition 17*(iv) *is not empty as a matter of set theory. It is* ***not*** *a witness for this paper, and the worked example below is*: the instrument here is *irrelevant*, and $\mathcal{V}_{\{0,1\}}$ is all of $\mathcal{V}$ precisely because nothing is identified. It is retained because it is a true statement about the mathematics and because what it isolates, that emptiness of $\mathcal{V}_{CI}$ is settled at the law and not by an identification hypothesis, is what the worked example builds on. Take $\theta_\dagger = 0$ and any $\varphi_\dagger \in \mathcal{V}$; let (X_1, X_2) be continuously distributed and independent of $(W, \Delta U, \Delta Z)$; let W be independent of ΔZ with $\varrho_0 = \varrho_1 = \varrho_2 = \frac{1}{3}$, so that A4(i) holds with $\varepsilon_\varrho = \frac{1}{4} < \frac{1}{3}$; let the

conditional law of ΔU given $(W, \Delta Z)$ depend on W alone, being $N(\mu_l, 1)$ on $\{W = l\}$ with $\mu_0 = \mu_1 = 0$ and $\mu_2 = 1$; and let $\Delta Y = \Delta\varphi_\dagger + \Delta U$, which is (18) at $\theta_\dagger = 0$. Then $\mathcal{V}_{\{0,1\}} = \mathcal{V}$ and $\mathcal{V}_{CI} = \emptyset$, so Proposition 17(iv) applies at $\mathsf{S}_0 = \{0, 1\}$: the maximal consistent sets are $\{0, 1\}$ and $\{2\}$, and the cells in conflict with $\{0, 1\}$ are exactly $\{2\}$. *No identification on any subset is used, and none is available*: $\mathcal{V}_{\{0,1\}}$ is all of $\mathcal{V}$. The same construction at $\mathsf{L} = 4$, with the four cells equally likely, $\varepsilon_\varrho = \frac{1}{5}$, $\mu_0 = \mu_1 = 0$ and $\mu_2 \neq \mu_3$ both non-zero, puts $\{2, 3\}$ in conflict with $\{0, 1\}$; *the reading of that shape against the application belongs to the worked example and not here*, an irrelevant instrument having no application to read it against. *One feature of the construction places it, and is stated rather than hidden*: there W is independent of (X_1, X_2), so the instrument is irrelevant, there is no first stage, and $\mathcal{V}_{\{0,1\}} = \mathcal{V}$. *The companion treatment of discrete instruments says what that regime is*: Centorrino, Fève, and Florens (2025, Remark 2) records that when the residual is independent of the regressor and the instrument, the relevance condition of that paper reverts to ordinary completeness, which a purely discrete instrument cannot satisfy, and the structural function is then only *partially* identified. This construction sits there, which is why the reading against the application is carried by the worked example and not by it.

A worked example at $\mathsf{L} = 3$, with a **relevant** instrument, in which the hypothesis of Proposition 17(iv) holds and the attribution it delivers is the one the application is after. It illustrates that hypothesis; it does not exhaust it, and nothing in this paper asks it to. The hypothesis of clause (iv) is not maintained anywhere here, it is the configuration the procedure of Remark 18(3) exists to detect, and its standing is that of the conditional-relevance condition A1b(iii), which is the conditional analogue of Centorrino, Fève, and Florens (2025, Assumption 2.2): a condition on the law whose plausibility is discussed and whose scope is bounded, not a theorem. Take $\mathcal{X} = [0, 1]$ and $\pi \equiv 1$ on it; $\mathsf{L} = 3$; $\theta_\dagger = 0$ and $\varphi_\dagger$ any C^∞ element of $\mathcal{V}$; ΔZ bounded and centred with $E[\Delta Z \Delta Z']$ nonsingular and independent of $(W, X_1, X_2, \Delta U)$; and W independent of ΔZ with $\varrho_l \equiv \frac{1}{3}$, so that A4(i) holds

at $\varepsilon_\varrho = \frac{1}{4} < \frac{1}{3}$. Let the conditional density of (X_1, X_2) given $(W = l, \Delta Z)$ be

$$f^{(0)} = f^{(2)} \equiv 1 \ \text{ on } [0,1]^2, \qquad f^{(1)}(x_1, x_2) = 1 + \gamma\,(x_2 - x_1), \quad \gamma \in (0,1),$$

let ΔU be independent of $(X_1, X_2, \Delta Z)$ given W with $\Delta U \mid W = l \sim N(\mu_l, 1)$, $\mu_0 = \mu_1 = -\frac{1}{3}$ and $\mu_2 = \frac{2}{3}$, and set $\Delta Y = \Delta\varphi_\dagger + \Delta U$. Then $\varphi_\dagger \in \mathcal{V}_{\{0,1\}}$ and $\mathcal{V}_{\{0,2\}} = \emptyset$, so $\mathcal{V}_{CI} = \emptyset$; Proposition 17(iv) applies at $\mathsf{S}_0 = \{0,1\}$, which is a maximal consistent set, and the cells in conflict with it are exactly $\{2\}$. The instrument is relevant, in all three senses the question has. It moves the period difference of the endogenous regressor: $E[X_2 - X_1 \mid W = 1] = \gamma/6$ against $E[X_2 - X_1 \mid W = 0] = E[X_2 - X_1 \mid W = 2] = 0$. The linearisation is not the zero map: at $h(x) = x - \frac{1}{2} \in \mathcal{V}$, $\mathcal{A}_{\text{cond}} h(u,t,1) = \frac{\gamma}{27}\, E\Big[w(t'\Phi(\Delta Z))\Big]\, \phi\Big(u + \frac{1}{3}\Big) \neq 0$ at every (u,t), with ϕ the standard normal density. And the restriction carried by the reference set has bite: $\mathcal{V}_{\{0,1\}}$ is a *proper* subset of $\mathcal{V}$, since $\varphi_\dagger - h \notin \mathcal{V}_{\{0,1\}}$ at that same h. That last is exactly what fails in the witness above, and it is the difference between a witness and an example. The maintained conditions are verified and not assumed. A0 holds, $\mathcal{X}$ being a compact interval; A1's centring $E[\Delta U \mid \Delta Z] = 0$ holds, and A1's conditional independence holds on $\{0,1\}$ and fails only through cell 2; A1b(i) holds by choice of $\varphi_\dagger$, A1b(ii) because the joint density of (X_1, X_2) is bounded below on $[0,1]^2$, and A1b(iv) by the choice of ΔZ; A4(i) as above; B1 is met by the maintained Φ, w, $\mathcal{T}$ and Π, none of which is a functional of P; C1(i) because $\varphi_\dagger$ and the cell densities are C^∞, and C1(ii) because $f_{X_1,X_2} = 1 + \frac{\gamma}{3}(x_2 - x_1) \in [1 - \frac{\gamma}{3}, 1 + \frac{\gamma}{3}]$ with marginals in $[1 - \frac{\gamma}{6}, 1 + \frac{\gamma}{6}]$ and $\bar{c}_\pi = 1 + \frac{\gamma}{6}$. A1b(iii) is neither claimed nor used. The mechanism is this paper's own, and naming it is what makes the construction portable. Write $\delta_l := f^{(l)}_{X_2} - f^{(l)}_{X_1}$ for the period-difference tilt of cell l, so that $E[\Delta h \mid W = l] = \int h\, \delta_l$, the object (30) is stated at, read at a cell. What empties $\mathcal{V}_{\{0,2\}}$ is $\delta_0 = \delta_2$ together with $\mu_0 \neq \mu_2$: the structural function can separate the two cells' means only through $\int h\,(\delta_0 - \delta_2) = 0$, while ΔU separates them by $\mu_2 - \mu_0 = 1$. *Equality of the two cells' joint laws is not needed, only equality of their tilts*, so cells 0 and 2 may carry different conditional laws of (X_1, X_2).

At $\mathsf{L} = 4$ the same construction with $f^{(0)} = f^{(2)} = f^{(3)} \equiv 1$, $f^{(1)}$ as above, $\varrho_l \equiv \frac{1}{4}$, $\varepsilon_\varrho = \frac{1}{5}$, $\mu_0 = \mu_1 = -\frac{3}{4}$, $\mu_2 = \frac{1}{2}$ and $\mu_3 = 1$ puts the cells in conflict with $\{0, 1\}$ at exactly $\{2, 3\}$, which is the form the application's chronic-against-transient hypothesis takes. What the example covers, and what it does not. It has $\delta_0 = \delta_2$: the instrument moves the tilt between the reference set and cell 1, and not between the base cell and the violating cell. When $\delta_0 \neq \delta_2$ the mean argument is unavailable, some $h \in \mathcal{V}$ matches the two cells' means, and an obstruction at a higher cumulant would be needed, equality of the two laws forcing equality of every cumulant of order at least two together with a mean gap of $\mu_2 - \mu_0$; nothing above settles that case either way. *That is a scope statement about the example and not a gap in Proposition 17*, whose clause (iv) is a hypothesis and not a claim: an example illustrates such a hypothesis and is not asked to exhaust it, which is the role Centorrino, Fève, and Florens (2025, Examples 1 and 2) play for the relevance condition of that paper.

Lemma 45 (L-1: the difference and sum functionals)**.** *The period-difference operator is bounded with kernel the constants, and the two functionals it generates differ in exactly one respect: the difference functional integrates to zero and the sum functional does not. Suppose C1(ii) holds and let g be a random variable with $E|g\,h(X_1)| < \infty$ and $E|g\,h(X_2)| < \infty$ for the h named in each clause. Then*

(i) $\|\Delta h\|_{L^2(P)} \leq 2\,\bar{c}_\pi^{1/2}\|h\|_{L^2(\pi)}$ *for every $h \in L^2(\mathcal{X}, \pi)$,* the whole ambient space and not only $\mathcal{V}$*, so Δ extends from the domain Definition 5 gives it to a bounded operator $L^2(\mathcal{X}, \pi) \to L^2(P)$ with the same norm bound.* The quantifier is stated at $L^2(\mathcal{X}, \pi)$ because that is what the proof establishes and because three later results consume it there*: the proof of* (i) *uses $(a-b)^2 \leq 2a^2 + 2b^2$ and C1(ii) and* ***never*** *$\int h\,d\pi = 0$, and C1(ii) is also what makes Δh well defined for a π-equivalence class, since $f_{X_j} \leq \bar{c}_\pi \pi$ makes $P_{X_j} \ll \pi$ for $j = 1, 2$. Lemma 92 Step 4, Lemma 94 Step 3 and Lemma 95 Steps 3 and 6′ all apply this bound at $h_k = \hat{\varphi}^{(-k)} - \varphi_\dagger$, whose membership of $\mathcal{V}$ would otherwise need A1b(i);* the wider quantifier removes that need at all three at once,

which is strictly preferable to adding A1b(i) to three hypothesis lists. Clause (ii) *is a different matter and genuinely needs* $h \in \mathcal{V}$*;*

(ii) **Consumers: none**, as of 2026-08-05. *Its only consumer was the corollary withdrawn at Remark 44, and with it A1b(ii), which enters no other result, became a clause no theorem rests on. The clause is kept because it is the statement of a fact about* Δ *that the surrounding lemma's other clauses are read against, and because a reader who follows a stale citation should land on it rather than on a gap; it is not available as an inherited step, since nothing inherits it.* The content: $\Delta h = 0$ *almost surely if and only if* h *is almost surely constant, provided A1b(ii) holds; consequently, under A1b(ii) alone,* $\ker \Delta \cap \mathcal{V} = \{0\}$, the passage from "constant" to "zero" is $\int h \, d\pi = 0$ for $h \in \mathcal{V}$, which is Definition 5 and not an assumption; A1b(i) is not needed, since it asserts $\varphi_\dagger \in \mathcal{V}$ and h here is generic;

(iii) $E[g\,\Delta h] = \int_{\mathcal{X}} \nu_g(x)\, h(x)\, dx = \langle \nu_g/\pi,\, h\rangle_{L^2(\pi)}$*, for each* h *satisfying the integrability proviso above, in particular for every* $h \in \mathcal{V}$ *and for* $h \equiv 1$*;*

(iv) $\int_{\mathcal{X}} \nu_g \, dx = 0$ *and* $\int_{\mathcal{X}} \sigma_g \, dx = 2\, E[g]$*, where*

$$\sigma_g(x) \;:=\; E[g \mid X_2 = x]\, f_{X_2}(x) \;+\; E[g \mid X_1 = x]\, f_{X_1}(x); \tag{66}$$

(v) if in addition A1 holds and $g = \Xi(\Delta U, \Delta Z)\, \widetilde{W}$ *for an integrable* $\mathbb{R}^{p\times(\mathsf{L}-1)}$*-valued* Ξ*,* the direction to the **left** of the contrast, which is the only order that multiplies, and the type is Definition 53's*, then* $E[g] = 0$*, hence* $\int_{\mathcal{X}} \sigma_g \, dx = 0$ *as well.* The conclusion is coordinatewise in the p rows*, so it holds verbatim for an array-valued* Ξ *whose last index is the one contracted against* $\widetilde{W}$*; that is how Remark 103*(2) *reads it. At* $\mathsf{L} = 2$*,* $\widetilde{W}$ *is a scalar,* Ξ *is* $\mathbb{R}^p$*-valued and* $\Xi\widetilde{W} = \widetilde{W}\Xi$ *is the printed object.*

Lemma 46 (L-4: interchange in the adjoint direction)**.** *The* (u, t, l)*-integration defining the adjoint may be taken inside the expectation. Suppose B1(i), (ii), (iv), C1(ii) and C2′(i) hold,*

and recall from Definition 4 that P is a probability measure, hence σ-finite. **B1(iv) is cited as a block because this proof opens all four of its clauses**, *which is what the block register requires: the interchange **is** this lemma's conclusion and not a constant inside it, it is Fubini for $\Pi \otimes P$, and that needs Π σ-finite — which it is because B1(iv) makes Π a product of* three *finite measures, Π_u and Π_t by* (iv.a) *and* (iv.b) *and the finitely supported Π_{w} by* (iv.d). *Drop any one of the three and there is no interchange left to state.* **It is not the only line that carries the block, and through 2026-08-25 this statement said it was**: *Lemma 48 carries it on the same ground, its own proof opening all four clauses at named steps, and so does Proposition 71; while Lemma 55 carries it by inheritance, its proof naming no clause of B1 at all and invoking Lemma 48*(ii) *and Proposition 20, which carry the block between them.* No claim of exhaustiveness is made, and that is deliberate: *the assumption block's own audit of B1(iv) block-form lines names, besides the three above, Lemma 11 and Proposition 15, which consume* (iv.c) at the statement *rather than at a step, and Corollary 12, Proposition 17, Lemma 93 and Proposition 20, which carry it by inheritance as Lemma 55 does, and the group* (T-3) *of the main theorem.* A sentence whose whole subject is a miscount is the last place to leave a bare count standing, *which is why the hedge is attached here and not left to the audit.* Through 2026-08-23 this line read "(iv.c)" alone, *so the clause the conclusion rests on was named in the proof and nowhere a reader of the statement would meet it. Then for every $\ell \in L^2(\Pi)$ and every $h \in \mathcal{V}$ the function $(u,t,l,\omega) \mapsto g_{u,t,l}(\omega)\,\Delta h(\omega)\,\ell(u,t,l)$ is $\Pi \otimes P$-integrable, with*

$$\begin{aligned}\sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} E\Big[\big|g_{u,t,l}\,\Delta h\big|\Big]\,\big|\ell(u,t,l)\big|\,\Pi_u(du)\,\Pi_t(dt) \\ \le\; 2\,\bar{c}_\pi^{1/2}\,\|\bar{f}_0\|_\infty\,c_\Pi\,\|h\|_{L^2(\pi)}\,\|\ell\|_{L^2(\Pi)} \;<\; \infty,\end{aligned} \tag{67}$$

and consequently, with $\int(\cdot)\,d\Pi$ abbreviating the same three-factor integration, $\int E[g_{u,t,l}\Delta h]\,\ell(u,t,l)\,d\Pi = E\Big[\Delta h \int g_{u,t,l}\,\ell(u,t,l)\,d\Pi\Big]$. The constant does not move with L: *the envelope reads the **per-cell** bound $|\widetilde{W}_l| \le 1$ inside $g_{u,t,l}$, and the Π-integration of $\bar{w}$ against $|\ell|$ is bounded by*

$c_\Pi \|\ell\|_{L^2(\Pi)}$ *by B1(iv.c) and the Cauchy–Schwarz inequality in* Π, Π_{w} *being a probability measure. At* $\mathsf{L} = 2$ *the sum has one term with* $\varpi_{\mathsf{w}}(1) = 1$ *and every display is the printed one.*

Lemma 47 (L-5: levels versus derivatives)**.** *The orthogonality condition and the source condition are the same in the two parameterisations; the remainder bound and the rate condition are not, and levels is the weaker of the two.* "The source condition" in that headline means **A2**, a plain range condition, and clause (iii) states exactly that. It does **not** mean the source condition *at index* $\mu_{\mathrm{H1}} = s/(2a)$, which is **not** parameterisation-invariant*: composing with* $\mathcal{D}_0^{-1}$ *raises the degree of ill-posedness from* a *to* $a+1$ *and lowers the index from* $s/(2a)$ *to* $(s-1)/(2(a+1))$*, a loss of* $\frac{8}{3}$ *units of* s *uniformly in* a*. Remark 72*(4) *is where that computation is used and is this lemma's principal consumer. Suppose C1(ii) and* ***A0's compactness clause*** *hold, and let* $\mathcal{D}_0^{-1} : L^2(\pi) \to \mathcal{V}$ *be* $(\mathcal{D}_0^{-1}h)(x) := \int_{[\underline{x},x]\cap\mathcal{X}} h(v)\,dv$*, with* $\underline{x} := \inf \mathcal{X}$*, followed by centring in* $L^2(\pi)$*.* A0 is cited **by clause**, and what this lemma reads is the compactness and **not** the interval*:* $|\mathcal{X}| < \infty$ *in the constant of* (i)*, and* $\underline{x}$ *finite.* Through 2026-08-25 this line said the opposite and defended it*, calling the compact interval a* ***declaration*** *in the sense of the assumption block's register paragraph and arguing that negating it does not make the lemma false but stops it parsing, "on a fat Cantor* $\mathcal{X}$ *there is no interval* $[\underline{x}, x]$ *to integrate over".* Clause (ii) below refutes that four lines down in this same statement*:* (ii) *already writes* $\int_{\{x\geq v\}\cap\mathcal{X}}$*, the intersection form, which parses on any measurable* $\mathcal{X}$*, and* (i) *is now spelled the same way — one intersection symbol, and nothing else moves.* On a fat Cantor $\mathcal{X}$ the lemma is then **true***: the Cauchy–Schwarz bound of* (i) *is untouched; injectivity holds because the uncentred integral* $F(x) := \int_{[\underline{x},x]\cap\mathcal{X}} h\,dv$ *is absolutely continuous with* $F' = h$ *almost everywhere on* $\mathcal{X}$*, so that at almost every point of a level set of* F *the derivative vanishes, and* F *constant* π*-a.e. — hence Lebesgue-a.e.,* π *being bounded below by C1(ii) — forces* $h = 0$ *a.e. on* $\mathcal{X}$*;* (ii) *and* (iv) *are unchanged in statement and in proof; and* (iii) *is unchanged in statement while* its proof is not, and that is recorded here rather than left to the proof. Through 2026-08-26 this sentence read "(ii)–(iv) are unchanged", which is true of the three statements and false of one of the three proofs*:* (iii)*'s density step*

asserted that $\mathcal{R}(\mathcal{D}_0^{-1})$ contains every centred C^1 function on $\mathcal{X}$, an interval argument that fails at $\zeta(x) = x$ on a fat Cantor $\mathcal{X}$ — a density-point argument forces $h' = 1$ a.e. on $\mathcal{X}$, so the primitive is $x \mapsto |\mathcal{X} \cap [\underline{x}, x]|$, which differs from x by a function strictly decreasing across every complementary gap and therefore constant on no set meeting both sides of one. The clause survives because injectivity of $(\mathcal{D}_0^{-1})^*$ on $\mathcal{V}$ is clause (ii) itself — *read at $\nu_g = \zeta\pi$, after extending the tail function continuously from $\mathcal{X}$ to $\mathbb{R}$ — so density becomes a conclusion of the proof rather than a step in it, and neither step reads connectedness.* The other live consequence is the register*: the interval is a **clause** and not a declaration, on the split the assumption block now prints, so a result that reads it carries it — Lemma 67 does — and this lemma, which does not read it, does not.* A1b(i) is **not** a hypothesis here*: all four clauses concern generic h, h', ν_g, the codomain $\mathcal{V}$ is Definition 5, and $\varphi_\dagger$ does not occur in the statement or in any step of the proof. Then*

(i) $\mathcal{D}_0^{-1}$ is bounded, with $\|\mathcal{D}_0^{-1}\| \le 2\,(\,|\mathcal{X}|\,\|1/\pi\|_\infty)^{1/2}$, and injective, while $\mathcal{D}$ is unbounded on $L^2(\pi)$;

(ii) for integrable ν_g, writing $\mathcal{N}_g(v) := \int_{\{x \ge v\} \cap \mathcal{X}} \nu_g(x)\,dx$, one has $\mathcal{N}_g \equiv 0$ if and only if $\nu_g \equiv 0$ almost everywhere on $\mathcal{X}$;

(iii) A2 holds for $\mathcal{A}_{\mathrm{cond}}$ if and only if the corresponding source condition holds for $\mathcal{A}_{\mathrm{cond}}\mathcal{D}_0^{-1}$;

(iv) $\|h\|_{L^2(\pi)} \le \|\mathcal{D}_0^{-1}\|\,\|h'\|_{L^2(\pi)}$ whenever $h = \mathcal{D}_0^{-1}h'$, so a rate in the derivative norm implies the same rate in the level norm, while the converse fails.

Lemma 48 (Closed form of the adjoint)**.** *The adjoint of the linearisation is the difference functional of the debiasing weight, divided by the reference density, once the weight is read through the Bierens family. Suppose B1(i), (ii), (iv) hold together with C1(ii) and C2′(i). Then, with k and $g_{u,t,l}$ as in (27):*

(i) Kernel form. *For every* $\ell \in L^2(\Pi)$,

$$\big(\mathcal{A}^{*}_{\mathrm{cond}}\ell\big)(x) \;=\; \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \mathsf{k}(u,t,l;x)\,\ell(u,t,l)\,\Pi_u(du)\,\Pi_t(dt),$$

and $\mathcal{A}^{*}_{\mathrm{cond}}\ell \in \mathcal{V}$. The integration is over all three factors of Π, the third being the weighted sum over the contrast index*; this is Definition 5's pairing read on the left-hand argument, and it is where the masses* $\varpi_{\mathsf{w}}(l)$ *act that* (68) *displays. At* $\mathsf{L}=2$ *the third factor is the one-point set* $\{1\}$, $\varpi_{\mathsf{w}}(1)=1$, $\mathsf{k}(u,t,1;x)=\mathsf{k}(u,t;x)$, *and the display is the two-factor integral earlier rounds printed.*

(ii) The induction map. *The operator*

$$\big(\mathcal{W}\ell\big)_{l}(u,z) \;:=\; \varpi_{\mathsf{w}}(l)\,\varpi_u(u) \int_{\mathcal{T}} \ell(u,t,l)\,w\Big(t'\Phi(z)\Big)\,\Pi_t(dt), \qquad (u,z)\in\mathbb{R}\times\mathbb{R}^p, \quad l=1,\dots,\mathsf{L}-1, \tag{68}$$

is a bounded linear map from $L^2(\Pi)$ *into* $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$ *with* $\|\mathcal{W}\| \le \Big(\|\bar f_0\|_\infty \,\|\varpi_u\|_\infty \int_{\mathcal{T}} \bar w^2\, d\Pi_t\Big)^{1/2}$, *its range is dense in* $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$, *and* $\ker\mathcal{W} \subseteq \ker\mathcal{A}^{*}_{\mathrm{cond}}$. *Three readings of this clause are fixed here, because each changes with* L *and none is visible at* $\mathsf{L}=2$. First, the codomain norm *is* $\|\Xi\| := (E\|\Xi(\Delta U,\Delta Z)\|^2)^{1/2}$ *with the Euclidean norm on* $\mathbb{R}^{\mathsf{L}-1}$ *inside. Second, the constant is the printed one at every* L, *and R1 is what makes that true.* (68) *acts* diagonally *in* l, *slice* l *of* ℓ *generates component* l *of* $\mathcal{W}\ell$ *and no other, so the passage from the slicewise estimate to the vector norm costs the factor* $\big(\max_l \varpi_{\mathsf{w}}(l)\big)^{1/2} \le 1$, Π_{w} *being a probability measure by B1(iv.d);* without the $\varpi_{\mathsf{w}}(l)$ that R1 puts in (68) the same passage would cost $\big(\min_l \varpi_{\mathsf{w}}(l)\big)^{-1/2} \ge 1$ instead, *the domain norm carrying the masses and the codomain not, so the printed constant would be too small.* Third, density holds in the $\mathbb{R}^{\mathsf{L}-1}$-valued space and not merely componentwise*: the diagonal action makes* $\mathcal{R}(\mathcal{W})$ *the algebraic sum of* $\mathsf{L}-1$ *subspaces, one dense in each component by the scalar argument, and a finite sum of dense subspaces of the corresponding factors is dense in the product.* At

$\mathsf{L}=2$ all three readings are the printed ones*:* $\varpi_{\mathsf{w}}(1)=1$*, the codomain is scalar, and the constant does not move.*

(iii) Closed form. *For every* $\ell\in L^2(\Pi)$*,*

$$\mathcal{A}^{*}_{\mathrm{cond}}\ell \;=\; \frac{\nu_{\widetilde{W}'(\mathcal{W}\ell)}}{\pi}, \tag{69}$$

where $\widetilde{W}'(\mathcal{W}\ell)$ *abbreviates the Euclidean inner product* $\sum_{l=1}^{\mathsf{L}-1}\widetilde{W}_l\left(\mathcal{W}\ell\right)_l(\Delta U,\Delta Z)$*; consequently* $\mathcal{A}^{*}_{\mathrm{cond}}\ell=-\nu/\pi$ *if and only if* $\nu_{\Delta Z+\widetilde{W}'(\mathcal{W}\ell)}\equiv 0$ *almost everywhere on* $\mathcal{X}$*.*

Corollary 49 (to Lemma 48: the representer is differentiable, with explicit bounds)**.** *Regularity of a range-space representer transfers to its observable-space image, and the transfer is what makes the sup-norm requirements of A2′ well posed. Stated at a generic admissible representer, with* $\ell^{\Pi}_{\star}$ *as the specialisation. Suppose B1(i), (ii), (iv.a) and (iv.d) hold and let* $\ell^{\Pi}\in L^2(\Pi)$ *be an* admissible representer *in the sense of A2b, with budgets* $\Lambda^{(0)},\Lambda^{(1)}$ *and with* $\ell:=\mathcal{W}\ell^{\Pi}$ *defined pointwise from the version A2b fixes. Then* ℓ *is bounded, is continuously differentiable in* u *with*

$$\partial_u\ell_l(u,z)=\varpi_{\mathsf{w}}(l)\Big\{\varpi_u'(u)\int_{\mathcal{T}}\ell^{\Pi}(u,t,l)\,w(t'\Phi(z))\,\Pi_t(dt)+\varpi_u(u)\int_{\mathcal{T}}\partial_u\ell^{\Pi}(u,t,l)\,w(t'\Phi(z))\,\Pi_t(dt)\Big\},\qquad l=1,. \tag{70}$$

and satisfies

$$\big\|\ell\big\|_{\infty}\le\|\varpi_u\|_{\infty}\,\Lambda^{(0)},\qquad \big\|\partial_u\ell\big\|_{\infty}\le\|\varpi_u'\|_{\infty}\,\Lambda^{(0)}+\|\varpi_u\|_{\infty}\,\Lambda^{(1)},$$

where $\|\ell\|_{\infty}:=\sup_{(u,z)}\|\ell(u,z)\|$ *and* $\|\partial_u\ell\|_{\infty}$ *likewise, the norm being* Frobenius *on the* $p\times(\mathsf{L}-1)$ *array,* which is the norm every consumer of these two symbols reads*:* Remark 59(3)*,* Remark 52*,* Lemma 55(iii) *and* (76)*. Neither constant moves with* L *or with* p*, and the two directions are paid for by different things. The* p*-direction is paid for by A2b, whose budgets are* ***Euclidean over the*** p ***coordinates of*** θ*, so that column* l *of* ℓ*, the* $\mathbb{R}^p$*-valued* ℓ_l *of*

(68), *obeys* $\|\ell_l(u,z)\| \le \varpi_{\mathsf{w}}(l)\|\varpi_u\|_\infty \Lambda^{(0)}$ *directly, by the triangle inequality for the* Π_t*-integral of a vector-valued integrand. The* L*-direction is paid for by R1: assembling the columns gives* $\|\ell(u,z)\|^2 \le \|\varpi_u\|_\infty^2 (\Lambda^{(0)})^2 \sum_l \varpi_{\mathsf{w}}(l)^2$*, and* $\sum_l \varpi_{\mathsf{w}}(l)^2 \le \big(\sum_l \varpi_{\mathsf{w}}(l)\big)^2 = 1$ *because* (68) *carries* $\varpi_{\mathsf{w}}(l)$ *in column* l *and* B1(iv.d), which is on the hypothesis line above, makes Π_{w} a probability measure*, so the* $\sqrt{\mathsf{L}-1}$ *a mass-free induction map would have cost is paid for, and it is paid for exactly.* **B1(iv.d) is assumed here; it is no longer also asserted**. *Through 2026-08-23 the clause stood at this sentence alone, on the* conclusion *side, so a reader who acted on the hypothesis line had not assumed what the constant needs, while a reader of the whole statement was told it holds without ever being asked for it. It is cited* by clause *and not through the whole of B1(iv), because no step of the proof reads* (iv.b) *or* (iv.c) *and a block citation would charge this corollary — and everything downstream of it — for two clauses it never opens.* That is the assumption block's **three registers** rule and is no longer this corollary's own defence*: the factorisation* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ *is B1(iv)'s* declaration*, standing at every statement here, so citing the two clauses this proof opens leaves nothing unnamed.* This corollary's line was the model for the six repaired on 2026-08-24 *and is itself unchanged.* And because it is a hypothesis **here** it is a hypothesis of every result that runs a bound through this corollary*, which is why Lemma 56, Lemma 91, Lemma 95*(i′) *and Remark 57 carry* (iv.d) *on their own lines from the same date.* No conclusion anywhere moves*:* (iv.d) *costs nothing about* P*, and at* $\mathsf{L}=2$ *it is forced.* Specialisation. *Under A2′(i.a)–(i.b) the selected solution* $\ell^{\Pi}_\star$ *is an admissible representer with* $\Lambda^{(0)} = \Lambda^{(0)}_\star$, $\Lambda^{(1)} = \Lambda^{(1)}_\star$*, so the corollary applies at* $\ell = \ell_\star := \mathcal{W}\ell^{\Pi}_\star$ *and yields* $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty \Lambda^{(0)}_\star$ *and* $\|\partial_u \ell_\star\|_\infty \le \|\varpi_u'\|_\infty \Lambda^{(0)}_\star + \|\varpi_u\|_\infty \Lambda^{(1)}_\star$.

Corollary 50 (to Lemma 48: how far A2 can exceed what the moment needs)**.** *The source condition and the orthogonality equation the estimator targets can differ only inside the non-closedness of the range of the adjoint. Suppose the hypotheses of Lemma 48 hold together with A1b(iv), and define the* tilted difference map $\Upsilon : L^2(P_{\Delta U,\Delta Z}; \mathbb{R}^{\mathsf{L}-1}) \to \mathcal{V}$ *by* $\Upsilon\Xi := \nu_{\widetilde{W}'\Xi}/\pi$*, where* $\widetilde{W}'\Xi = \sum_l \widetilde{W}_l \Xi_l$ *is the Euclidean inner product of* (69)*,* Υ's domain is Lemma 48(ii)'s codomain, which is what makes the factorisation below a composition of maps that meet.

Then Υ *is a bounded linear operator with*

$$\big\|\Upsilon\big\| \;\le\; 2\,c_{\widetilde{W}}\,\bar{c}_{\pi}^{1/2}, \qquad c_{\widetilde{W}} \;:=\; \min\Big\{\sqrt{2},\;\sqrt{\mathsf{L}-1}\Big\},$$

$\mathcal{A}^{*}_{\mathrm{cond}} = \Upsilon \circ \mathcal{W}$, *and*

$$\mathcal{R}\big(\mathcal{A}^{*}_{\mathrm{cond}}\big) \;\subseteq\; \mathcal{R}(\Upsilon) \;\subseteq\; \overline{\mathcal{R}\big(\mathcal{A}^{*}_{\mathrm{cond}}\big)}. \tag{71}$$

$c_{\widetilde{W}}$ *is a surely-valid bound on* $\|\widetilde{W}\|$*, it is* ***not*** *a consequence of* (P1)*, and it is the constant every contraction of a* $\widetilde{W}$ *against a vector in this file carries.* It is italic and symbol-subscripted, joining c_{Φ}, c_{Π}, c_{C}, c_{μ} and $c_{\mu''}$*: in the c-family every symbol-subscripted member is italic and every sans-serif member,* c_0 *through* c_{17}*, is integer-subscripted, so the font separates the two without a further rule. By* (1)*,* $\widetilde{W} = e - \varrho$ *is the difference of two points of the simplex* $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \ge 0,\ \mathbf{1}'q \le 1\}$*, e a vertex and* ϱ *an interior or boundary point, so* $\|\widetilde{W}\|$ *is at most that simplex's diameter, which is* $\sqrt{2}$ *at every* $\mathsf{L} \ge 3$ *and* 1 *at* $\mathsf{L} = 2$*; and separately* $\|\widetilde{W}\|^2 = \sum_l \widetilde{W}_l^2 \le \mathsf{L} - 1$*, each coordinate lying in* $[-1, 1]$*. The minimum of the two is* $c_{\widetilde{W}}$*, and it equals* 1 *at* $\mathsf{L} = 2$ *and* $\sqrt{2}$ *at every* $\mathsf{L} \ge 3$*,* so the constant saturates at $\mathsf{L} = 3$ and does not grow with the number of cells. (P1) does not deliver it*:* (P1) *is the* per-cell *statement* $|\widetilde{W}_l| = |\mathbb{1}\{W = l\} - \varrho_l(\Delta Z)| \le 1$*, one scalar bound at each of the* $\mathsf{L} - 1$ *contrast indices, and, at* $\mathsf{L} = 2$*, the single bound* $|W - \varrho(\Delta Z)| \le 1$ *earlier rounds printed; the Euclidean bound on the vector is a different statement and is not* 1. No sharpness is claimed for the bound on $\|\Upsilon\|$*: it is a pointwise Cauchy–Schwarz bound, and* $\|\widetilde{W}\|$ *approaches* $\sqrt{2}$ *only where one cell probability approaches* 1*, an event whose probability the bound does not use. At* $\mathsf{L} = 2$*,* $c_{\widetilde{W}} = 1$ *and the display reads* $\|\Upsilon\| \le 2\bar{c}_{\pi}^{1/2}$*, the constant earlier rounds printed.*

Remark 51 (Consequences of Lemma 48, and one independent confirmation). Four things follow, three of which later results consume.

(i) The representer is evaluable at the data, and that is now a theorem. A solution of the adjoint equation is an element of $L^2(\Pi)$, a function of (u, t) with t the Bierens index, while

A2′(i), whose closing sentence fixes $\ell_\star := \mathcal{W}\ell_\star^\Pi$ pointwise, and (40) evaluate $\ell_\star(\Delta U, \Delta Z)$ with z a value of ΔZ. Part (ii) supplies the map between the two and part (iii) shows it commutes with the adjoint equation, so the object A2′(i) and (40) speak about is constructed rather than posited. *And* (68) *is exactly why the evaluation is well posed*: $\ell_\star = \mathcal{W}\ell_\star^\Pi$ is a Borel map of (u, z) and of nothing else, in particular not of W, which is the measurability Lemma 91(iv) consumes. *The construction consumes exactly one thing beyond the identification block*: absolute continuity of Π_u in B1(iv.a), a restriction on a reference measure the analyst chooses.

(ii) The correction measure is not free on the population side, the map (68) integrating against Π_t; Remark 43 carries the disposition of B1(v)'s two halves.

(iii) What the construction does not give, and what it gives anyway. $\mathcal{W}$ has dense range but need not be surjective, so a selection criterion posed over all of $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$ may be unreachable; and $\mathcal{W}$ need not be injective, though $\ker \mathcal{W} \subseteq \ker \mathcal{A}^*_{\text{cond}}$ means no direction relevant to the selection is lost. It does not follow that the gap between A2 and the equation the estimator targets is uncontrolled. By Corollary 50, $\mathcal{R}(\Upsilon)$, the set of ν/π for which the primitive orthogonality equation is solvable in $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$, is squeezed between $\mathcal{R}(\mathcal{A}^*_{\text{cond}})$ and its closure, *with no characterisation of* $\mathcal{R}(\mathcal{W})$ *required*. So A2 can exceed what the moment needs only within $\overline{\mathcal{R}(\mathcal{A}^*_{\text{cond}})} \setminus \mathcal{R}(\mathcal{A}^*_{\text{cond}})$. That set is nonempty under H1: $\mathcal{A}^*_{\text{cond}}$ is compact, being the adjoint of a compact operator, and a compact operator has closed range if and only if that range is finite-dimensional, which H1's infinite-rank clause excludes. *Injectivity is neither available nor needed for this*, $\mathcal{A}^*_{\text{cond}}$ may well have a kernel, since $\ker \mathcal{W} \subseteq \ker \mathcal{A}^*_{\text{cond}}$ two sentences above. The gap is therefore exactly where the ill-posedness lives.

(iv′) The derivative of the representer. The sup-norm requirements on $\ell_\star$ remain assumptions, but they are stated on $\ell_\star^\Pi$, in A2′(i.a)–(i.b), and Corollary 49 derives from them both (70) and the two bounds. *The ordering matters*: the C^1 clause of B1(iv.a) alone controls only the first term of (70), so $\|\partial_u \ell_\star\|_\infty < \infty$ cannot be assumed directly without assuming the

existence of a derivative nothing else in the block delivers. (70) is what makes the choice of Π_u a design lever on q and on σ_q; it is a *conditional* identity whose hypothesis is A2′(i.b), and A2′(i.b) is a primitive awaiting L-21, not a theorem.

(iv) The ν/π referent is confirmed independently, and in the same form. *Notation:* Fève and Florens (2014) write the endogenous regressors as Z_1, Z_2 with generic value z, the instrument as W with generic value w, and a generic element of the range space as ψ; the translation to this paper is $Z_t \mapsto X_t$, $z \mapsto x$, and their ψ is our ℓ, the letter ψ being reserved here for the moment function. For the linear conditional-mean panel operator $K\varphi = E[\varphi(Z_2) - \varphi(Z_1) \mid W]$, their Proposition 2.1 gives, *verbatim in their notation*,

$$\big(K^*\psi\big)(z) \;=\; \int \psi(w)\, \frac{f_{Z_2,W}(z,w) - f_{Z_1,W}(z,w)}{\pi(z)}\, dw,$$

followed immediately by the statement that $K^*\psi$ satisfies the normalising constraint $\int \big[(K^*\psi)(z)\big]\pi(z)\, dz = 0$. Translated, the numerator is $\nu_{\ell(W)}$ and the display is $\nu_{\ell(W)}/\pi$. *One difference of setting must be flagged rather than absorbed:* their dw integral ranges over a general instrument space, whereas W takes the L values $\{0, \ldots, \mathsf{L}-1\}$ here, so the counterpart in this paper is the finite sum $\nu_{\ell(W)}(x) = \sum_{j=0}^{\mathsf{L}-1} \ell(j)\{f_{X_2,W}(x,j) - f_{X_1,W}(x,j)\}$; the difference-of-densities structure is what carries over, not the integral. At $\mathsf{L} = 2$ this is the two-point sum over $j \in \{0,1\}$ earlier rounds printed.

Both features of (69) *therefore appear in print in the linear case:* the division by the reference density, with the same letter π, which the draft takes from them, and the zero π-mean, which is membership of $\mathcal{V}$ and which Proposition 9 Step 2 proves here from $\int \nu_g\, dx = 0$ at $h \equiv 1$. This matters because the referent was contested: an earlier version of this project imposed the source condition on ν rather than on ν/π, and the correction was forced internally by a Picard membership test. The π^{-1} is not an artefact of the $L^2(\pi)$ convention adopted here; it is the shape the adjoint takes in a first-differenced panel. What transfers is that shape and nothing more: their K is a fixed linear conditional-expectation operator, while $\mathcal{A}_{\text{cond}}$ is a

Fréchet derivative, and the induction map $\mathcal{W}$ in (69) has no counterpart there because their restriction is already unconditional in the residual.

Remark 52 (The two anchors of interest, and why the anchor is priced at all). Two anchors are canonical and both are admissible with no further assumption. The *zero anchor* $b \equiv 0$ is trivially admissible. The *N1 anchor* $b(z) = -E\Big[\int_0^{\Delta U} \ell(v, z)\, dv \;\Big|\; \Delta Z = z\Big]$ produces the unique weight generated by ℓ that satisfies N1, and it is admissible whenever A1b(iv) holds, since $\|b(z)\| \le \|\ell\|_\infty E[|\Delta U| \mid \Delta Z = z]$ and conditional Jensen give $E\|b(\Delta Z)\|^2 \le \|\ell\|_\infty^2 E[\Delta U^2] < \infty$. So N1 can always be arranged, at no cost to $\partial_u L$ and, by (P3) and A1, none to $E[\psi]$; it does not hold automatically, and Proposition 20(iii) is the statement of exactly what its failure costs.

"Arranged" is $p(\mathsf{L}-1)$ conditional expectations, and the arithmetic of that count is where a construction goes wrong. The N1 anchor is a $p \times (\mathsf{L}-1)$ matrix computed *slice by slice*: entry (j, l) is $-E\Big[\int_0^{\Delta U} \ell_{jl}(v, \Delta Z)\, dv \mid \Delta Z\Big]$, and the $\mathsf{L}-1$ columns are generated by $\mathsf{L}-1$ separately choosable slices of ℓ^{Π} per Definition 19. A construction that centres only a *contraction* of the columns against a fixed $\mathsf{v} \in \mathbb{R}^{\mathsf{L}-1}$ therefore discharges p of the conditions and leaves $p(\mathsf{L}-2)$ uncentred, which is zero at $\mathsf{L} = 2$ and strictly positive at every $\mathsf{L} \ge 3$; Corollary 21 exhibits such a weight *inside* the admissible class and prices what it leaves. *Nor is targeting this anchor optional once Theorem 29 is read whole*: (T-2) assumes N1 at $L_\star$ and (R-a) forces $\hat{b}^{(-k)} \to b_\star$, so the two together select this anchor and no other, and a construction free to choose b is free only before (T-2) is imposed.

The anchor is priced in A2′(iii) because it is a *free design object*: the weight is determined by the pair (ℓ, b), the adjoint equation constrains only ℓ, and an analyst may choose a b with $E\|b(\Delta Z)\|^2 = \infty$. *What A2′(iii) buys, and nothing more, is that* $E\|L_\star(\Delta U - \tau_1 \Delta h, \Delta Z)\| < \infty$ *for every* τ_1, *that* $E[\|L_\star\| \mid \Delta Z] < \infty$ *almost surely, and that* $E\big\|L_\star \widetilde{W}\big\|^2 < \infty$, *the third quantity is the squared Euclidean norm of the* $\mathbb{R}^p$*-valued product and* ***not*** $E[\widetilde{W}^2 \|L_\star\|^2]$, *which is not a well-formed expression once* $\widetilde{W}$ *is a vector; at* $\mathsf{L} = 2$ *the two are the same number*

, the first because $\|L_\star(u,z)\| \le \|b_\star(z)\| + \|\ell_\star\|_\infty |u|$ by (31) with all norms Frobenius, the second by conditional Jensen, the third by $\|L_\star \widetilde{W}\| \le c_{\widetilde{W}} \|L_\star\|$ and $(\varsigma + \varsigma')^2 \le 2\varsigma^2 + 2\varsigma'^2$ applied to that same bound, together with A1b(iv). *No shift restriction is needed on the $L_\star$ term for this reason*, which is why the shift proviso of Definition 53 is imposed on the direction and not here.

Definition 53 (Admissible perturbation directions)**.** Fix $\bar\tau > 0$ and $h \in \mathcal{V}$, and for a Borel $\Xi : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}^{p \times (\mathsf{L}-1)}$ write

$$\big\|\Xi\big\|_{0,h} := \sup_{|\tau_1| \le \bar\tau} \Big(E \big\| \Xi(\Delta U - \tau_1 \Delta h, \Delta Z) \big\|^2 \Big)^{1/2} \in [0, \infty], \tag{72}$$

the norm inside the expectation being Frobenius, as the type declaration of (1) fixes for every object carrying one column per contrast. A pair (D, r) is an *admissible direction pair at* $(h, \bar\tau)$ if

(a) $\mathsf{r} : \mathbb{R}^p \to \mathbb{R}^{\mathsf{L}-1}$ is Borel and bounded, and $\varrho + \tau_3 \mathsf{r}$ takes values in the simplex $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \ge 0 \text{ for every } l,\ \mathbf{1}'q \le 1\}$ for all small τ_3; *the range condition is the* ***simplex*** *and not the box* $[0,1]^{\mathsf{L}-1}$, which would permit $\mathbf{1}'(\varrho + \tau_3 \mathsf{r}) > 1$ and hence a negative base cell, so that the perturbed vector would not be a propensity at all. It is Proposition 20(iii)'s condition verbatim;

(b) $D : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}^{p \times (\mathsf{L}-1)}$ is Borel with $\|D\|_{0,h} < \infty$.

It is *differentiable* if in addition D is differentiable in u at every point, $\partial_u D$ is continuous in u, and

$$\big\|D\big\|_{1,h} := \Big(E \Big[\sup_{|s| \le \bar\tau} \big\| \partial_u D(\Delta U - s\Delta h, \Delta Z) \big\|^2 \Big] \Big)^{1/2} < \infty. \tag{73}$$

Taking $\tau_1 = 0$ in (72) gives $\|D\|_{L^2(P)} \le \|D\|_{0,h}$, and taking $s = 0$ inside the supremum in (73) gives $\|\partial_u D\|_{L^2(P)} \le \|D\|_{1,h}$; *so the unshifted $L^2(P)$ bounds are implied and are not assumed separately.* Continuity of $\partial_u D$ in u makes the supremum in (73) equal to the supremum over rational s, hence measurable.

The two value types are forced by what the directions multiply. D is a direction in the debiasing weight L, which is $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued by the type declaration of (1), and it enters ψ only as $D\widetilde{W}\in\mathbb{R}^p$; r is a direction in the propensity vector $\varrho\in\mathbb{R}^{\mathsf{L}-1}$ and enters only as $\varrho+\tau_3\mathsf{r}$. *At* $\mathsf{L}=2$ *both are the printed objects*: a $p\times 1$ matrix is a vector in $\mathbb{R}^p$, the Frobenius norm is the Euclidean norm, r is scalar, and the simplex of clause (a) is the interval $[0,1]$. Two pairs a reader will want are admissible. $(0,0)$ is admissible at every $(h,\bar\tau)$ and is differentiable there, which is what Remark 64(2) reads Lemma 56 at; and $\mathsf{r}=\hat\varrho-\varrho$ satisfies clause (a)'s range condition at every $\tau_3\in[0,1]$ by convexity, both ϱ and $\hat\varrho$ taking values in (11) under A4(i) and Definition 2.

Both glyphs suppress a dependence, and the suppression is safe for one reason only, which is worth printing. $\|\cdot\|_{0,h}$ and $\|\cdot\|_{1,h}$ depend on the *pair* $(h,\bar\tau)$, both are suprema over $|\tau_1|\le\bar\tau$ or $|s|\le\bar\tau$, while the subscript carries only h. The $\bar\tau$ is suppressed because every statement that uses either seminorm fixes $\bar\tau$ at its head, exactly as Definition 53 does; *and* because the dependence is **monotone**: enlarging $\bar\tau$ enlarges the index set of both suprema, so

$$\bar\tau\le\bar\tau' \quad\Longrightarrow\quad \|\Xi\|_{0,h}\big|_{\bar\tau}\le\|\Xi\|_{0,h}\big|_{\bar\tau'},\qquad \|D\|_{1,h}\big|_{\bar\tau}\le\|D\|_{1,h}\big|_{\bar\tau'}.$$

The monotonicity is what forbids one false reading, and it is recorded here rather than left to be inferred: in any display that quantifies over $\bar\tau$ with $\bar\tau$ appearing on one side only, the seminorm side is *also* increasing in $\bar\tau$, so the bound cannot be optimised by sending $\bar\tau\downarrow 0$. See Remark 57, where the quantifier "for every $\bar\tau>0$" stands in front of exactly such a display.

Remark 54 (The two seminorms of Definition 53: why the suprema sit on opposite sides, and why the shift-uniform form is not implied). The two clauses of Definition 53 are not stated in the same form, and the asymmetry is forced.

(72) *may take the supremum outside*, because the derivatives it licenses are *exact* rather than limits: ψ is affine in each of τ_2,τ_3 *separately*, with one bilinear term $-\tau_2\tau_3\,D(\Delta U-\tau_1\Delta h,\Delta Z)\,\mathsf{r}(\Delta Z)$, *the direction to the left of the propensity direction, which is the only order*

that multiplies, D being $p \times (\mathsf{L}-1)$ and r being $(\mathsf{L}-1)\times 1$ by Definition 53, and the product lying in $\mathbb{R}^p$ as ψ does. *It is bi-affine, not jointly affine, and the distinction is not pedantic*: read literally, joint affineness would give $\partial^2_{\tau_2\tau_3}\Psi \equiv 0$, which is exactly what Lemma 56(iii) denies. So $\partial_{\tau_2}\Psi$, $\partial_{\tau_3}\Psi$ and $\partial^2_{\tau_2\tau_3}\Psi$ are obtained by splitting a finite sum of finite expectations, the bilinear term included, since it too is one finite expectation with a coefficient, with no interchange of derivative and integral anywhere. All that is needed is that Ψ be finite at each τ separately, and $\sup_{|\tau_1|\le\bar\tau} E[\,\cdot\,] < \infty$ delivers exactly that.

(73) *may not.* The mixed partial $\partial^2_{\tau_1\tau_2}\Psi(0)$ differentiates in τ_1, which *is* a limit, and dominated convergence needs an integrable envelope valid *simultaneously* for all τ_1 in a neighbourhood, that is, $E[\sup_{|s|\le\bar\tau}\|\partial_u D(\Delta U - s\Delta h, \Delta Z)\|^2] < \infty$, with the supremum *inside*. A sup-outside condition gives $L^2(P)$-boundedness of the family $\{\partial_u D(\cdot - s\Delta h, \cdot)\}_{|s|\le\bar\tau}$ without uniform integrability of its supremum, and $\sup_s \|\cdot\|_{L^2} < \infty$ does not imply $\|\sup_s \cdot\|_{L^2} < \infty$. This is precisely the quantifier that Leibniz's rule consumes, and it is the one A2′(i.b) is stated in. Since $\sup_s E[\,\cdot\,] \le E[\sup_s \cdot\,]$, (73) implies the sup-outside version, so nothing is lost by stating the stronger form where it is needed and the weaker form where it is not.

Why the shift-uniform seminorm at all, and what the alternative would have cost. Definition 53(b) is what makes $\Psi(\tau)$ a real number off $\tau_1 = 0$, and it is not implied by anything stated at the true residual. The two conditions are genuinely different, and a one-line example separates them. Let ΔZ be degenerate, $\Delta U \sim \mathrm{Unif}[0,1]$, let $h \in \mathcal{V}$ be such that $P(\Delta h = -1) = \frac{1}{2}$ and $\Delta h = 0$ elsewhere, and take $D(u,z) = \mathbb{1}\{u > 1\}(u-1)^{-1} e_{11}$, with e_{11} the $p \times (\mathsf{L}-1)$ matrix whose $(1,1)$ entry is 1 and whose other entries are 0, one entry carries the example and the Frobenius norm of D is the modulus of that entry, so nothing below depends on p or on L; at $\mathsf{L} = 2$ and $p = 1$, $e_{11} = 1$ and D is the scalar function earlier rounds printed. Then $D(\Delta U, \Delta Z) = 0$ almost surely, so $E[D \mid \Delta Z] = 0$ and $\|D\|_{L^2(P)} = 0$; but on $\{\Delta h = -1\}$ the argument at $\tau_1 = 1$ is $\Delta U + 1$ and $E|D(\Delta U + 1, \Delta Z)| = \frac{1}{2}\int_0^1 u^{-1} du = \infty$, so $\|D\|_{0,h} = \infty$ for every $\bar\tau \ge 1$. *With such a D the moment $\Psi(\tau)$ is not merely large at*

$\tau_1 = 1$*: it is undefined*, its positive and negative parts both being infinite whenever $\widetilde{W}$ is non-degenerate. A statement of the form "$\partial^2_{\tau_2\tau_2}\Psi \equiv 0$ identically in τ" would then be ill-posed rather than false.

The alternative repair was rejected and it is worth saying why. One could instead require a uniformly bounded density ratio, $\sup_{|\tau_1|\le\bar\tau} \left\| dP_{(\Delta U-\tau_1\Delta h,\Delta Z)}/dP_{(\Delta U,\Delta Z)} \right\|_\infty < \infty$, which implies $\|\Xi\|_{0,h} \le C\|\Xi\|_{L^2(P)}$ for *every* Ξ at once. That is strictly stronger and it fails in the most ordinary designs: if ΔU has compact support and $\Delta h \not\equiv 0$, the shifted law charges points the true law does not. And the failure is sharper than unboundedness of a ratio: it is failure of absolute continuity, so there is no Radon–Nikodym derivative to bound. In the display above the law of $(\Delta U - \tau_1\Delta h, \Delta Z)$ at $\tau_1 = 1$ assigns mass $P(A) = \frac{1}{2}$ to $\{u > 1\}$, which is a $P_{(\Delta U,\Delta Z)}$-null set; hence $P_{(\Delta U-\Delta h,\Delta Z)} \not\ll P_{(\Delta U,\Delta Z)}$ and the quotient in the display is not merely large, it *does not exist*. The rejected condition is therefore not a stronger assumption one might be willing to pay for, in this design it cannot be stated at all. Definition 53(b) is a condition on the *direction class*, where the other direction conditions already live, and it restricts the law of the data not at all.

Lemma 55 (L-16: the solution set and the selection)**.** *Every admissible solution of the adjoint equation yields a valid moment, so the selection is a free design choice; and the functional it should be chosen to control is the multiplier of the second-order form, not the sup-norm of the representer. Suppose A1, A1b(i), A1b(iv), A2, A2′(i), A2′(iii), B1(i), (ii), (iv) and C1(ii) hold,* A2′ is cited by clause and not as a block: A2′(ii) is the variance clause, is consumed by no part of this lemma, and is a Step-8 condition, so listing A2′ whole would over-state the hypotheses*, and let* admissible representer *(A2b),* admissible weight *and* admissible selection *be as in Definition 19. Then*

(i) the set of ℓ^Π *solving* $\mathcal{A}^*_{\mathrm{cond}}\ell^\Pi = -\nu/\pi$ *is the affine set* $\ell^\Pi_\dagger + \ker \mathcal{A}^*_{\mathrm{cond}}$*, and its image under* $\mathcal{W}$ *is the affine set* $\ell_\dagger + \mathcal{W}(\ker \mathcal{A}^*_{\mathrm{cond}})$ *with* $\ell_\dagger := \mathcal{W}\ell^\Pi_\dagger$*;*

(ii) every admissible *selection, paired with* any *admissible anchor, gives a moment satis-*

fying Proposition 20(i)–(iii); in particular validity does not select. The restriction to admissible members is not decoration: $\mathcal{W}(\ell_\dagger^\Pi + \ker \mathcal{A}^*_{\mathrm{cond}})$ *contains members with no sup-norm bound, for which Proposition 20(i)'s differentiation under the expectation is unlicensed;*

(iii) fix $\vartheta \in \mathbb{R}^p$ *with* $\|\vartheta\| = 1$ *and write* $\mathcal{S}\ell := \sigma_{\partial_u \ell \widetilde{W}}$ *for the map into the sum functional of* (66)*, so that* $\sigma_q = \mathcal{S}\ell_\star$, the contrast standing to the right of the $p \times (\mathsf{L}-1)$ derivative, which is the only order that multiplies and is the order ψ itself carries at (38). *The second-order form of Lemma 56(i), contracted in the direction* ϑ*, depends on the selection through the* pair $\big(\vartheta'\sigma_q,\ \vartheta'\mathcal{K}\big)$*;* read on $L^2(\pi)$, which is Definition 5's one declared exception*, it is* $\langle \mathcal{B}h, h\rangle_{L^2(\pi)}$ *with*

$$\mathcal{B} \;=\; \big(\vartheta'\sigma_q/\pi\big)\cdot \;-\; 2\,\vartheta'\tilde{\mathcal{K}}, \qquad \tilde{\mathcal{K}}h := (\mathcal{K}h)/\pi, \tag{74}$$

of which $\vartheta'\sigma_q/\pi$, not $\vartheta'\sigma_q$*, is the multiplication part;* $\|\vartheta'\sigma_q/\pi\|_\infty \le 2\, c_{\widetilde{W}}\, \bar{c}_\pi\, \|\partial_u \ell_\star\|_\infty < \infty$*, so* $\mathcal{B}$ *is bounded; and* $\int_{\mathcal{X}} \sigma_q\, dx = 0$ *for every admissible selection.* The constant carries $c_{\widetilde{W}} = \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$ of Corollary 50 because the bound contracts the $p \times (\mathsf{L}-1)$ matrix $\partial_u \ell_\star$ against the vector $\widetilde{W}$*, and* $\|\partial_u \ell_\star\|_\infty$ *is Corollary 49's Frobenius sup-norm; at* $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ *and the display is the one earlier rounds printed.*

Lemma 56 (L-6: second-order expansion in the three nuisances)**.** *The moment is affine in two of its three nuisances and quadratic in the third, so of the six second-order terms three vanish identically and three are products of two estimation errors. Suppose A1b(i), A1b(iv), B1(i), (ii), (iv.a), (iv.d), C1(ii), A2, A2′(i) and A2′(iii) hold,* A2′ cited by clause; A2′(ii) is consumed by no part of this lemma. *Fix* $\bar{\tau} > 0$ *and* $h \in \mathcal{V}$*, let* (D, r) *be an admissible direction pair at* $(h, \bar{\tau})$ *in the sense of Definition 53, and put*

$$\zeta_\tau \;:=\; \big(\varphi_\dagger + \tau_1 h,\ L_\star + \tau_2 D,\ \varrho + \tau_3 \mathsf{r}\big), \qquad \Psi(\tau) \;:=\; E\big[\psi(\theta_\dagger, \zeta_\tau)\big], \qquad \tau \in [-\bar{\tau}, \bar{\tau}] \times \mathbb{R}^2.$$

Then Ψ *is finite on* $[-\bar\tau,\bar\tau]\times\mathbb{R}^2$ *and*

(i) $\partial^2_{\tau_1\tau_1}\Psi(0) = E\big[\partial_u\ell_\star\,\widetilde W\,(\Delta h)^2\big] = \int_{\mathcal X}\sigma_q\,h^2\,dx - 2\big\langle\mathcal K h,h\big\rangle_{L^2(dx)}$, *where* $q(x_1,x_2) = E[\partial_u\ell_\star\,\widetilde W \mid X_1=x_1,X_2=x_2]\in\mathbb{R}^p$, *in the jointly measurable version fixed in the proof, and* $\mathcal K$ *is the integral operator with kernel* $q\,f_{X_1,X_2}$*;* the contrast stands to the right of $\partial_u\ell_\star$ in both displays, which is the only order that multiplies and is the order ψ carries; $(\Delta h)^2$ is a scalar and commutes*;*

(ii) $\partial^2_{\tau_2\tau_2}\Psi\equiv 0$ *and* $\partial^2_{\tau_3\tau_3}\Psi\equiv 0$ identically on $[-\bar\tau,\bar\tau]\times\mathbb{R}^2$*, not merely at* $\tau=0$*;*

(iii) if D *satisfies N1′ then* $\partial^2_{\tau_2\tau_3}\Psi(0)=0$*; away from* $\tau_1=0$ *the cross-partial is* $-E[D(\Delta U-\tau_1\Delta h,\Delta Z)\,\mathsf r(\Delta Z)]$*, which need not vanish and is bounded in Euclidean norm by* $\|\mathsf r\|_\infty\big\|D(\cdot-\tau_1\Delta h,\cdot)\big\|_{L^2(P)}\le\|\mathsf r\|_\infty\|D\|_{0,h}$ *, the norm of the* shifted *direction, which is not controlled by* $\|D\|_{L^2(P)}$*, and* $\|\mathsf r\|_\infty:=\sup_z\|\mathsf r(z)\|$ *the sup of the Euclidean norm of the* $\mathbb{R}^{\mathsf L-1}$*-valued direction.* No factor $c_{\widetilde W}$ appears here*: the product* $D\mathsf r$ *contracts* D *against* $\mathsf r$ *and not against* $\widetilde W$*, and* $\|D\mathsf r\|\le\|D\|\,\|\mathsf r\|$ *with the Frobenius norm on* D*;*

(iv) if the pair is differentiable in the sense of Definition 53, then, with the mixed partials taken in the order $\partial_{\tau_1}\big[\partial_{\tau_2}\Psi\big]$ *and* $\partial_{\tau_1}\big[\partial_{\tau_3}\Psi\big]$*,* the reverse order returns the same two values, verified by direct computation in the proof, so no symmetry theorem is invoked*,* $\partial^2_{\tau_1\tau_2}\Psi(0)=-E\big[\partial_u D\,\widetilde W\,\Delta h\big]$ *and* $\partial^2_{\tau_1\tau_3}\Psi(0)=E\big[\ell_\star\,\mathsf r(\Delta Z)\,\Delta h\big]$*, bounded in Euclidean norm by*

$$\big\|\partial^2_{\tau_1\tau_2}\Psi(0)\big\| \;\le\; 2\,c_{\widetilde W}\,\bar c_\pi^{1/2}\,\big\|\partial_u D\big\|_{L^2(P)}\,\|h\|_{L^2(\pi)} \;\le\; 2\,c_{\widetilde W}\,\bar c_\pi^{1/2}\,\|D\|_{1,h}\,\|h\|_{L^2(\pi)} \tag{75}$$

and, respectively, by

$$\big\|\partial^2_{\tau_1\tau_3}\Psi(0)\big\| \;\le\; 2\bar c_\pi^{1/2}\,\big\|\ell_\star\big\|_\infty\,\big\|\mathsf r\big\|_{L^2(P)}\,\|h\|_{L^2(\pi)} \;\le\; 2\bar c_\pi^{1/2}\,\big\|\ell_\star\big\|_\infty\,\big\|\mathsf r\big\|_\infty\,\|h\|_{L^2(\pi)}, \tag{76}$$

where $\|\mathsf r\|_{L^2(P)}:=\Big(E\big\|\mathsf r(\Delta Z)\big\|^2\Big)^{1/2}$*,* the second moment of the **norm** of the $\mathbb{R}^{\mathsf L-1}$-valued direction, which is what A4(iii) prices, and not the scalar second moment, which is

not a norm on a vector*, in each case a product of the two direction magnitudes.* (75) carries $c_{\widetilde{W}}$ and (76) does not, and the asymmetry is exactly which vector the matrix is contracted against*: $\partial_u D\,\widetilde{W}$ contracts against the contrast, costing $\|\widetilde{W}\| \le c_{\widetilde{W}}$, while $\ell_\star \mathsf{r}$ contracts against r, whose norm is already the quantity being priced. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$, r is scalar, and both displays are the ones earlier rounds printed.*

In each of (75) *and* (76) *the first inequality is the operative one and the second is a stated weakening, and in both the operative norm is one the assumption block prices. $\|\partial_u D\|_{L^2(P)}$ is what the proof delivers and what A5(iv) prices, at $D = \hat{L} - L_\star$, as ϵ_n; $\|D\|_{1,h}$ is priced by nothing in the assumption block and appears here only because it is the envelope that licenses the interchange producing the identity. $\|\mathsf{r}\|_{L^2(P)}$ is what A4(iii) prices, at $\mathsf{r} = \hat{\varrho} - \varrho$, as r_n; $\|\mathsf{r}\|_\infty$ is priced by nothing, no clause of the block bounds $\|\hat{\varrho} - \varrho\|_\infty$. The sharper L^2 form costs one application of the Cauchy–Schwarz inequality against the bounded factor $\ell_\star$, and r is bounded by Definition 53(a), so* (76) *is a strengthening at no cost in hypotheses. Remark 57 states what must be established before either bound is instantiated at $D = \hat{L} - L_\star$, and gives a sufficient condition under which it is.*

Remark 57 (Membership of the estimated direction in Definition 53, and why it is a forward obligation rather than an assumption)*. This remark does* ***not*** *import Lemma 56's hypothesis block, and in particular not its direction-pair hypothesis*, that membership is exactly what (77) exists to establish, so importing it would make the remark assume what it delivers. What the displays below use, exhaustively: A1b(iv); C1(ii), through Lemma 45(i); B1(i), (ii), (iv.a), (iv.d) together with A2′(i.a)–(i.b), through Corollary 49; and A2′(iii), through Definition 19.

Lemma 56(iii)–(iv) are stated at a *fixed* admissible direction pair (D, r) at $(h, \bar{\tau})$. Step 6 or 7 will instantiate them at $D = \hat{L} - L_\star$, $\mathsf{r} = \hat{\varrho} - \varrho$ and $h = \hat{\varphi} - \varphi_\dagger$, all three random, and *that instantiation requires membership, which is a separate statement from any rate*: clause (iii) needs $\|\hat{L} - L_\star\|_{0,h} < \infty$, clause (iv) needs in addition that $\hat{L} - L_\star$ be differentiable in u with continuous derivative and $\|\hat{L} - L_\star\|_{1,h} < \infty$, and both need $\hat{\varrho} - \varrho$ bounded and satisfying

Definition 53(a)'s range condition, *which it does, at every* $\tau_3 \in [0,1]$*, by convexity of* (11)*, so the propensity half of the pair costs nothing.* Neither seminorm is assumed anywhere, and neither is discharged before Step 6.

It is, however, implied by the construction of $\hat{L}$, and that is why it is recorded here rather than assumed. Suppose $\hat{L} = L_{\hat{\ell}}^{\hat{b}}$ is an admissible weight in the sense of Definition 19(b), generated by an $\hat{\ell}$ that is bounded and continuous in u. Then $D := \hat{L} - L_\star$ is differentiable in u at every point with $\partial_u D = \hat{\ell} - \ell_\star$ continuous in u, and for every $h \in \mathcal{V}$ and every $\bar{\tau} > 0$

$$\big\|D\big\|_{1,h} \;\le\; \big\|\hat{\ell}-\ell_\star\big\|_\infty, \qquad \big\|D\big\|_{0,h}^2 \;\le\; 2E\big\|\hat{b}(\Delta Z)-b_\star(\Delta Z)\big\|^2+4\big\|\hat{\ell}-\ell_\star\big\|_\infty^2\Big\{E\big[\Delta U^2\big]+4\bar{\tau}^2\bar{c}_\pi\|h\|_{L^2(\pi)}^2\Big\}, \tag{77}$$

both finite, $\|\ell_\star\|_\infty < \infty$ by Corollary 49 under A2′(i), so $\|\hat{\ell} - \ell_\star\|_\infty < \infty$. The first holds because the sup-norm bound on $\partial_u D$ is uniform over *all* arguments, so the supremum inside (73) is dominated by the constant $\|\hat{\ell} - \ell_\star\|_\infty$ before any expectation is taken, this is the one place where a sup-inside condition costs nothing, and it is why the asymmetry of Remark 54 is not a practical burden. The second follows from $\|D(u,z)\| \le \|\hat{b}(z) - b_\star(z)\| + \|\hat{\ell} - \ell_\star\|_\infty |u|$ via (31), $(\varsigma + \varsigma')^2 \le 2\varsigma^2 + 2\varsigma'^2$ applied twice, A1b(iv), and $\|\Delta h\|_{L^2(P)} \le 2\bar{c}_\pi^{1/2}\|h\|_{L^2(\pi)}$ from Lemma 45(i); the first term is finite because both weights are admissible, $E\|\hat{b}(\Delta Z)\|^2 < \infty$ by hypothesis and $E\|b_\star(\Delta Z)\|^2 < \infty$ by A2′(iii).

One reading of (77) must be blocked explicitly, because the notation invites it. The display is quantified over *every* $\bar{\tau} > 0$; its right-hand side carries $\bar{\tau}^2$ and its left-hand side carries no $\bar{\tau}$ at all. Read naively one would minimise over $\bar{\tau}$, send $\bar{\tau} \downarrow 0$, and conclude a $\bar{\tau}$-free bound $\|D\|_{0,h}^2 \le 2E\|\hat{b} - b_\star\|^2 + 4\|\hat{\ell} - \ell_\star\|_\infty^2 E[\Delta U^2]$. *That inference is false.* The seminorm $\|D\|_{0,h}$ suppresses a dependence on $\bar{\tau}$ in its glyph but not in its value, it is a supremum over $|\tau_1| \le \bar{\tau}$, and by the monotonicity recorded in Definition 53 the left-hand side is *also* nondecreasing in $\bar{\tau}$. The display is therefore a family of bounds indexed by $\bar{\tau}$, matched term for term, and neither side may be optimised in isolation. *Step 6 or 7 carries this display forward at the* $\bar{\tau}$ *its own statement fixes, and at no other.*

Boundedness of $\hat{\ell}$ is needed *per realisation only*; no uniformity in n is used, because finiteness at each n is all that clauses (iii)–(iv) consume and the rate is A5(iv)'s separate job.

What Step 6 or 7 owes is therefore a verification, not a hypothesis: that its $\hat{L}$ is an admissible weight in the sense of Definition 19(b) with $\|\hat{\ell}\|_\infty < \infty$ and $E\|\hat{b}(\Delta Z)\|^2 < \infty$. *It is not a new clause of A5*, for the reasons Remark 58 gives, and it is inventoried there with the other forward obligations.

Remark 58 (The open verification obligations, in one place). The block carries four obligations that are *verifications* rather than assumptions. (O1) is discharged, (O4) is discharged, and (O2) and (O3) **remain open**. *Neither open obligation is discharged by being carried as a hypothesis*, since a hypothesis of a theorem is not a verification; what each has is a **named consumer that carries it explicitly**, which is what this remark's test asks of an open obligation. (O3) is carried by hypothesis (R-a) of Theorem 29, which asserts the construction $\hat{L}^{(-k)} = L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$ with a bounded derivative and so implies Definition 53 membership through (77). (O2) is carried by (R-c), which is a *disjunction*: (R-c.1) assumes N1′ at every D_k, and (R-c.2) instead **prices its failure** at $r_n \epsilon_n^L$, where ϵ_n^L is an assumed $L^2(P)$ rate for $\|D_k\|$ and (134) is the exact cost. Pricing a failure is the honest third option and it is neither a discharge nor an evasion.

The test used, stated once. A forward obligation is a verification and not an assumption **so long as no stated result's truth depends on it**: every present result carries the condition as an explicit hypothesis, so that a reader can see what is owed and by whom. *The moment a stated assumption cannot be written down without the obligation, the obligation has already become an assumption*, which is why A5(iv)'s first equality is printed as a conditional consequence rather than as part of the clause.

(O1) A2′(i.a)–(i.b) in sup-inside form, on $\ell^{\Pi}_{\star}$. **DISCHARGED** at Lemma 61, which derives both clauses from the source condition A2c on $\ell^{\Pi}_{\star}$ at index $\mu'' > \frac{3}{2}$, in the sup-inside form and with the explicit budget (85). *A2′(i) does not thereby cease to be an assumption*,

an assumption with a proved sufficient condition is still an assumption, *but it ceases to be an unverifiable primitive*, A2c being one condition on one object, stated in the Hilbert-scale language H1 already uses and reportable through (3).

(O2) N1′ at the estimated direction, $E[\hat{L} - L_\star \mid \Delta Z] = 0$. **Where the condition lives now:** the tagged property N1′, stated as a hypothesis of Lemma 56(iii), *and, as of Step 7, hypothesis (R-c) of Theorem 29.* **Discharge site:** Step 6 or 7, by constraining the estimator so that $\hat{L}$ is centred; failing that, the leading cross term does not vanish and enters at order $\|\mathsf{r}\|_\infty \|D\|_{0,h}$. **Status: STILL OPEN, and now priced rather than only placed.** Step 7 does not construct a centred $\hat{L}$, so the obligation stands; what it adds is (R-c.2), under which the failure costs exactly $r_n \epsilon_n^L$ by (134), and (R-c.1), under which $\Psi_k^{(3)} = 0$. *The cheap discharge, re-anchoring $\hat{L}$ at the N1 anchor of Remark 52, requires an estimate of $E[\int_0^{\Delta U} \hat{\ell} \mid \Delta Z]$ and is therefore a further nuisance*, which is why it is not taken **for the direction** D_k. *That declining is scoped to D_k and does not extend to $L_\star$*, and the two are not interchangeable here. N1′ at D_k is optional *because it is priced*: (R-c.2) makes its failure cost exactly $r_n \epsilon_n^L$ by (134), a product of two estimation errors. N1 at $L_\star$ is not optional and is not priced: it is hypothesis (T-2) of Theorem 29, and **nothing anywhere prices its failure**, which by Corollary 21(iii) enters at r_n *undamped* — no second small factor multiplies it, so after $\sqrt{n}$ scaling it diverges. *A construction that targets a non-centred anchor therefore does not fail* (O2)*; it fails* (T-2), and reading the decline above as covering both is the error Corollary 21 exists to block.

(O3) Membership of $\hat{L} - L_\star$ in Definition 53, via $\hat{L} = L_{\hat{\ell}}^{\hat{b}}$. **Where the condition lives, three homes, and the third is of a different kind:** Remark 57, untagged, with the discharge already written at (77) under one property of the estimator; hypothesis (R-a) of Theorem 29; and Definition 24(c), which asserts the construction $\hat{L}^{(-k)} = L_{\hat{\ell}^{(-k)}}^{\hat{b}^{(-k)}}$ *together with* the three properties of $\hat{\ell}^{(-k)}$, Borel, bounded, u-continuous, that make

Definition 19(b) well posed. *A remark can be declined and a theorem hypothesis can be refused, but a* ***definition*** *is adopted*: every Part-I result that adopts Definition 24 carries (O3) whether or not it lists (R-a), and Definition 25's Jacobian identity $\partial_\theta \bar{\psi}_n(\theta) = -\hat{J}(\theta)$ consumes it directly through $\partial_u \hat{L}^{(-k)} = \hat{\ell}^{(-k)}$. **Discharge site:** Step 6 or 7, which must verify that its $\hat{L}$ is an admissible weight generated by a bounded, u-continuous $\hat{\ell}$ with $E\|\hat{b}(\Delta Z)\|^2 < \infty$. **Status: STILL OPEN, and carried as an explicit hypothesis by a stated theorem.** (R-a) of Theorem 29 asserts exactly that verification, at each fold and with $\|\partial_u \hat{\ell}^{(-k)}\|_\infty$ added, so (77) applies and Definition 53 membership follows; *assuming the construction is not performing it. The obligation is in addition presupposed by A5(iv)*, whose identity $\|\partial_u(\hat{L} - L_\star)\|_{L^2(P)} = \|\hat{\ell} - \ell_\star\|_{L^2(P)}$ cannot be written without it; A5(iv) therefore prices $\hat{\ell}$ alone and states the identity as a consequence conditional on this obligation.

(O4) *Construction of* $\hat{\mathcal{A}}_\varphi$. **DISCHARGED, with one negative answer and one amendment.** The operator is exhibited at (100) and characterised by Proposition 70; $\hat{T}$ is exhibited at (12); A7 is verified in the amended form (13) (Proposition 79); and *A6 is refuted, not verified* (Proposition 71) and has been retired. *A discharge with a negative answer is a discharge*: the obligation was to verify, and the verification has an outcome.

No new clause of A5, and no new maintained assumption, is created for any of (O1)–(O4). In particular A2c is not maintained: it is a property tag with exactly one consumer, and promoting it would convert a discharged obligation into a new assumption, the trade this remark's test forbids. The ground is that every clause of A5 constrains an object the maintained block *introduces*, while (O2)–(O4) are constructive statements about objects the block does not construct.

Remark 59 (The hypothesis split of Proposition 20, and what makes its displays well posed). (1) *A1 is* ***not*** *in the global list*: it is consumed by part (ii) alone, where it is assumed and cited, while parts (i) and (iii) use it nowhere. The split follows the precedent

already in this file at Lemma 10, and its practical point is the same: Lemma 60, which inherits "the hypotheses of Proposition 20(i)", would otherwise inherit A1 as surplus and consume it in no step.

(2) *N1 is not assumed in any part*: (i) and (ii) do not use it and (iii) characterises it. *And part (i) does not assume A2 either*, being stated at an arbitrary admissible representer.

(3) Why (32) is finite for every $\tau_1 \in \mathbb{R}$. The second term because $\|L(u,z)\widetilde{W}\| \le c_{\widetilde{W}}\{\|b(z)\| + \|\ell\|_\infty |u|\}$, by (31) and $\|L\widetilde{W}\| \le \|L\|\,\|\widetilde{W}\|$ with the Frobenius norm on L and $\|\widetilde{W}\| \le c_{\widetilde{W}}$ from Corollary 50, *the vector bound and not* (P1), *which is the* ***per-cell*** *statement* $|\widetilde{W}_l| \le 1$ *and does not bound the Euclidean norm; at* $\mathsf{L} = 2$ *the two coincide and* $c_{\widetilde{W}} = 1$, $E\|b(\Delta Z)\|^2 < \infty$ by admissibility, $\|\ell\|_\infty < \infty$ by Corollary 49 under A2b , *the tag is A2b and not A2′(i), because* ℓ^{Π} *there is a generic admissible representer and not* $\ell^{\Pi}_\star$, and $E|\Delta U|, E|\Delta h| < \infty$ by A1b(iv) and Lemma 45(i); the first because $E\|\Delta Z \Delta U\| < \infty$ and $E\|\Delta Z \Delta h\| \le \|\Delta Z\|_{L^2(P)} \|\Delta h\|_{L^2(P)} < \infty$ by A1b(iv), the Cauchy–Schwarz inequality and Lemma 45(i).

(4) *Proposition 20(i) defines* Ψ_L *and does not import it*: no radius $\bar{\tau}$, no direction pair (D, r) and no clause of A2′ enters (32), which is what makes the hypothesis reduction of that clause readable off its own statement.

(5) *Part (ii) is the one part that consumes A1*, through the conditional independence that makes $E\big[D\,\widetilde{W} \mid \Delta Z\big] = E\big[D \mid \Delta Z\big]\,E\big[\widetilde{W} \mid \Delta Z\big] = 0$, a $p \times (\mathsf{L}-1)$ matrix against the zero vector of $\mathbb{R}^{\mathsf{L}-1}$, hence 0 in $\mathbb{R}^p$. *The product exists only in this order*, D being $p \times (\mathsf{L}-1)$ and $\widetilde{W}$ being $(\mathsf{L}-1) \times 1$ by Definition 53 and (1); *what A1 supplies is unchanged*, namely conditional independence of $D(\Delta U, \Delta Z)$ and W given ΔZ, which is what lets the conditional expectation of the product factor at any value type of D. At $\mathsf{L} = 2$, $\widetilde{W}$ is a scalar and the display is the one earlier rounds printed.

(6) *Both sides of part (iii)'s equivalence are well defined*, because A2′(iii) with Corollary 49 and A1b(iv) give $E[\|L_\star\| \mid \Delta Z] < \infty$ almost surely.

Lemma 60 (L-7: the balancing bound on the first-order bias)**.** *The first-order bias in the structural function is negligible under either of two independent conditions: a product rate for the debiasing weight, or a sieve balancing constraint with an approximation condition. Suppose the hypotheses of Proposition 20(i) hold together with A2, A2′(i) and A2′(iii),* A2′ cited by clause; A2′(ii) is consumed by no part of this lemma*, and adopt the following* sample-splitting convention*: the index set* $\{1,\dots,n\}$ *is partitioned into an auxiliary fold* $\mathcal{I}_a$ *and a main fold* $\mathcal{I}_m$*, independent by A0;* $\hat{\ell}$ *is a* $\sigma(\mathcal{I}_a)$*-measurable* ***random element*** *of* $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{p\times(\mathsf{L}-1)})$*, and one* ***jointly measurable*** *representative of it is fixed, measurable with respect to* $\sigma(\mathcal{I}_a)\otimes\mathfrak{B}(\mathbb{R}\times\mathbb{R}^p)$*, which a random element of an* L^2 *space over a separable metric space admits, exactly as Definition 24*(c′.1) *fixes one for the three integrated nuisances of Part I.* Membership and joint measurability are two statements and both are needed here*: Steps 0 and 2 freeze* $\hat{\ell}$ *inside* ν_g *and Step 2's identity* $\nu_g=\nu_{(\hat{\ell}-\ell_\star)\widetilde{W}}$ is (P8)*'s freezing lemma at* $\mathcal{G}=\sigma(\mathcal{I}_a)$*, whose hypothesis is joint measurability of the* composite*, and a measurable map into* L^2 *supplies nothing of the kind on its own.* What this site needs is strictly **less** than Part I needs, and the clause is scoped to it and not imported*: here* $\hat{\ell}$ *is evaluated only at an* ***undistorted*** $(\Delta U,\Delta Z)$ *draw, whose law is the one the container is built on, so any two representatives agree outside a* P*-null set and the choice is immaterial;* (41)*, which Definition 24*(c′.1) *imposes because Part I evaluates* $\hat{\ell}$ *at* shifted *first arguments whose law need not be dominated by* $P_{\Delta U,\Delta Z}$*, is* ***not*** *required here and is* ***not*** *imposed here ,* and here $\hat{\ell}$ is a free estimator of $\ell_\star$, not the u-derivative of anything: nothing in this lemma assumes that $\hat{\ell}=\partial_u\hat{L}$ for a weight $\hat{L}$, and A5(iv) prices $\hat{\ell}$ in exactly this sense. The relation $\hat{L}=L^{\hat{b}}_{\hat{\ell}}$ is a construction constraint, and it is owed by Step 6 or 7 (Remark 58), not supplied here*, ;* $\hat{\varphi}$ *and* $\hat{\nu}_g$ *are computed on* $\mathcal{I}_m$*; and every* ν*-object and every* O_P *statement below is conditional on* $\sigma(\mathcal{I}_a)$*, so that* $\hat{\ell}$ *is frozen wherever it appears inside a conditional expectation. Put* $g:=\Delta Z+\hat{\ell}\,\widetilde{W}\in\mathbb{R}^p$*,* the contrast to the right of the $p\times(\mathsf{L}-1)$ estimate, the order (38) carries and the only one that multiplies*, and* $h:=\hat{\varphi}-\varphi_\dagger$*, let* $\hat{\nu}_g$ *be an estimator of* ν_g *computed on* $\mathcal{I}_m$*, and let* P_{L_x} *denote the* $L^2(\pi)$*-orthogonal projection onto* $\operatorname{span}(b_1,\dots,b_{L_x})$.

Then

$$\left\|\langle \nu_g/\pi,\, h\rangle_{L^2(\pi)}\right\| \;\le\; \underbrace{\left\|(\nu_g-\hat\nu_g)/\pi\right\|\,\|h\|}_{(I)} \;+\; \underbrace{\left\|\langle \hat\nu_g/\pi,\, P_{L_x}h\rangle\right\|}_{(II)} \;+\; \underbrace{\left\|\hat\nu_g/\pi\right\|\,\left\|(\mathrm{Id}-P_{L_x})h\right\|}_{(III)}, \tag{78}$$

the bars on the left-hand side and on (II) being the **Euclidean** norm on $\mathbb{R}^p$ and **not** a modulus, $g \in \mathbb{R}^p$ *here, so* (20) *makes* ν_g *and* $\hat\nu_g$ *coordinatewise* $\mathbb{R}^p$*-valued and both pairings are vectors, scalar only at* $p=1$*, while every norm on a function is* $L^2(\pi)$*, that of an* $\mathbb{R}^p$*-valued function being the aggregate* $\left(\int \|\cdot\|^2 \pi\, dx\right)^{1/2}$ *at which clause* (i) *is proved, and*

(i) $\|\nu_g/\pi\|_{L^2(\pi)} \le 2\, c_{\widetilde W}\, \bar c_\pi^{1/2}\, \|\hat\ell - \ell_\star\|_{L^2(P)}$*, with* $c_{\widetilde W} = \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$ *of Corollary 50 and* $\|\cdot\|_{L^2(P)}$ *the Frobenius* L^2 *norm of the* $p \times (\mathsf{L}-1)$ *difference.* The constant is Corollary 50's and it is $c_{\widetilde W}$ and not 1, because Step 2's identity puts ν_g at $g = (\hat\ell - \ell_\star)\widetilde W$ and the passage $\|(\hat\ell-\ell_\star)\widetilde W\| \le \|\hat\ell-\ell_\star\|\,\|\widetilde W\|$ reads the Euclidean bound on the contrast vector and **not** (P1)*, which is the per-cell statement* $|\widetilde W_l| \le 1$*;* at $\mathsf{L}=2$ the two agree, $c_{\widetilde W}=1$, and the display is $2\bar c_\pi^{1/2}\|\hat\ell-\ell_\star\|_{L^2(P)}$ as earlier rounds printed. The factor is a constant free of n*, so clauses* (ii) *and* (iii) *are unaffected: neither reads the value of the constant, only that it is finite;*

(ii) (rate route) *under A5(ii) and A5(iv) alone, the left-hand side is* $o_P(n^{-1/2})$ with no condition on L_x, no balancing constraint, no appeal to (79) below, and no appeal to A5(vi)*;*

(iii) (balancing route) *suppose the sieve solve sets* (II) $=0$*, that A8 holds with a B-spline basis of degree exceeding* $s-\frac12$*, that A5(vi) holds, and that*

$$\left\|(\hat\nu_g-\nu_g)/\pi\right\|_{L^2(\pi)} = O_P\!\left(n^{-1/2}\right) \qquad \textit{and} \qquad \left\|\hat\nu_g/\pi\right\|_{L^2(\pi)} = O_P(1). \tag{79}$$

Then the left-hand side is $o_P(n^{-1/2})$ with no rate for $\hat\ell$*.*

Lemma 61 (L-21: the source condition delivers A2′(i), in sup-inside form)**.** *A source condition*

of index exceeding the one-dimensional C^1-embedding threshold on the selected solution implies both sup-norm clauses of A2′(i), with the supremum inside the t-integral and with explicit budgets; neither B1 nor any Picard criterion supplies them. Suppose $\ell^{\Pi}_{\star}$ satisfies ***A2c*** *at some $\mu'' > \frac{3}{2}$, with budget $\Lambda^{(\mu'')}_{\star}$ as in (4), and set*

$$c_{\mu''} \;:=\; \Big(\frac{2}{\pi}\int_{\mathbb{R}} \big(1+\xi^2\big)^{1-\mu''}\, d\xi\Big)^{1/2} \;<\; \infty, \tag{80}$$

together with the two constants the level *and the* derivative *carry separately,*

$$c^{(0)}_{\mu''} \;:=\; \Big(\frac{1}{2\pi}\int_{\mathbb{R}} \big(1+\xi^2\big)^{-\mu''} d\xi\Big)^{1/2}, \qquad c^{(1)}_{\mu''} \;:=\; \Big(\frac{1}{2\pi}\int_{\mathbb{R}} \xi^2\big(1+\xi^2\big)^{-\mu''} d\xi\Big)^{1/2}. \tag{81}$$

The three constants are one identity apart, and the separation costs nothing, neither in p, nor in L*, nor in the admissible range of μ''. The two integrands of (81) add to $(1+\xi^2)^{1-\mu''}$, so*

$$\big(c^{(0)}_{\mu''}\big)^2+\big(c^{(1)}_{\mu''}\big)^2 \;=\; \tfrac{1}{4}\,c^2_{\mu''}, \qquad \textit{hence} \qquad c^{(0)}_{\mu''}+c^{(1)}_{\mu''} \;\le\; \sqrt{2}\,\Big(\big(c^{(0)}_{\mu''}\big)^2+\big(c^{(1)}_{\mu''}\big)^2\Big)^{1/2} \;=\; \frac{c_{\mu''}}{\sqrt{2}} \;<\; c_{\mu''}, \tag{82}$$

the inequality being the Cauchy–Schwarz inequality on two terms, with equality exactly when $c^{(0)}_{\mu''} = c^{(1)}_{\mu''}$. Both split constants are available in closed form against $c_{\mu''}$, which settles that equality case and makes them computable without a quadrature. One integration by parts, whose boundary term $\big[\xi(1+\xi^2)^{1-\mu''}\big]$ vanishes at $\pm\infty$ precisely *because $\mu'' > \frac{3}{2}$, gives $\int_{\mathbb{R}}(1+\xi^2)^{1-\mu''}d\xi = 2(\mu''-1)\int_{\mathbb{R}}\xi^2(1+\xi^2)^{-\mu''}d\xi$, whence*

$$c^{(1)}_{\mu''} \;=\; \frac{c_{\mu''}}{\sqrt{8(\mu''-1)}}, \qquad c^{(0)}_{\mu''} \;=\; c_{\mu''}\sqrt{\frac{2\mu''-3}{8(\mu''-1)}}, \qquad \mu'' > \tfrac{3}{2}.$$

The two therefore agree exactly at $\mu'' = 2$, where both are $c_2/\sqrt{8} = \frac{1}{2}$, so (82) is tight at that one index and strict at every other. And (81) imposes no new restriction on μ''*: $c^{(0)}_{\mu''}$ is finite for $\mu'' > \frac{1}{2}$ and $c^{(1)}_{\mu''}$ for $\mu'' > \frac{3}{2}$, and the second threshold is the one $c_{\mu''}$ already carries, (80) diverging at $\mu'' = \frac{3}{2}$ for the same reason (81)'s second integral does. A2c's index*

range is untouched. What else enters clauses (i)–(iii)*, exactly, and it is not nothing. Not C1 and not A2, A2c does not presuppose that* $\ell_\star^\Pi$ *solves anything. But* ***B1(iv.a)–(iv.b)*** *enter* substantively*, at two points and in two different ways:* finiteness *of* Π_u *and* Π_t *makes the* ***two****-factor product* $\Pi_u \otimes \Pi_t$ *a product of* σ*-finite measures, which is what supplies a product-measurable representative and licenses Tonelli; and* absolute continuity *of* Π_u *with respect to Lebesgue measure is what turns a Lebesgue-null modification of a slice into a* Π*-null modification, so that the function clause* (i) *constructs is a* version*. The third factor of* Π *is reached by* ***neither*** *clause, and the route that leaves it outside is the more general one. It enters only through Definition 5's declaration that* $\{1, \ldots, \mathsf{L}-1\}$ *carries its power set, which makes* Π*-measurability equivalent to measurability of the* $\mathsf{L}-1$ *sections; the measure theory therefore runs on* $\Pi_u \otimes \Pi_t$ *and the sections are assembled at the end, with the zero function at any* l *of zero* Π_{w}*-mass, where the section is unconstrained, a case B1(iv.d) happens to exclude but that the argument never needs excluded.* So **B1(iv.d) does not enter clauses (i)–(iii)***, and that is a consequence of the route and not an oversight: the three-factor alternative,* Π *itself a product of finite measures, would consume it, B1(iv.d) being the only clause that makes* Π_{w} *a finite measure.* The scope qualifier is load-bearing and is new on 2026-08-24*: the clause* ***does*** *enter the concluding sentence, through Corollary 49, whose hypothesis line carries it from that date. This paragraph printed the claim unqualified while that line was silent, and unqualified it would now be false of the lemma taken whole — the reading a reader of a printed* exact *accounting is entitled to. In clauses* (i)–(iii) *B1(ii) enters only in that it* defines $\bar{w}$*, the symbol A2c is written in, its nonnegativity being part of that definition; and B1(i), (ii), (iv.a), (iv.d) are what the concluding sentence consumes,* but it consumes them in **two** ways, and only one of the two is through Corollary 49*. B1(ii)'s envelope* $|w(t'\Phi(z))| \leq \bar{w}(t)$ *dominates* (68)*'s integrand* directly*, which is what makes* $\ell_\star := \mathcal{W}\ell_\star^\Pi$ *defined pointwise, A2b's last clause, and hence what entitles Corollary 49, stated at an admissible representer, to be applied at all; the corollary then supplies the two sup-norm bounds under B1(i), (ii), (iv.a) and (iv.d) jointly. B1(i) and B1(iv.d) enter there and nowhere else in this lemma; B1(iv.a) enters there*

and, for the different reason above, at the two earlier points. The contrast index is carried ***explicitly*** *in every clause below, and it is quantified differently in different ones, which is why the two indices cannot be described by one rule. The p coordinates of θ are carried* inside *a Euclidean norm, the geometry A2c and* (4) *are stated in, and one that no display below ever opens. The $\mathsf{L}-1$ contrasts are carried as a* free *index: clauses* (i) *and* (ii)*, and the measurability half of* (iii)*, hold at* every l*, and* (i)*'s joint measurability is asserted on the three-fold base $\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$, so it is one statement about all of them and not $\mathsf{L}-1$ statements, whereas the three budgets $\Lambda_\star^{(0)}$, $\Lambda_\star^{(1)}$ and $\Lambda_\star^{(\mu'')}$ that* (85) *relates are* maxima *over l of single-contrast integrals, by A2′(i.a)–(i.b) and* (4)*.* One Borel set $\mathcal{T}_*$ serves every l, as A2c requires*, which is what lets the version be constructed by one argument rather than by $\mathsf{L}-1$ of them.* At $\mathsf{L}=2$ the index is single-valued and at $p=1$ the Euclidean norm is the modulus, so every display below reduces to the one earlier rounds printed, with **no** exception*: under those two identifications, the third argument l dropped, $\|\cdot\|$ read as $|\cdot|$ and $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ as $H^{\mu''}(\mathbb{R})$,* (84)*, and* (85) *read at its* outer *members, are those displays character for character.* And the two sharper inner members of (85) are sharper at **every** L, $\mathsf{L}=2$ included*, they are an improvement in the constant and not an L-dependence, so nothing here is an $\mathsf{L}=2$ exception in either direction. Then, at that μ'':*

(i) The version. $\ell_\star^{\Pi}$ *admits a jointly measurable version, given slicewise by the* unique continuous representative *of the slice at every $t\in\mathcal{T}_*$, the Borel full-Π_t-measure set A2c attaches to its representative, so the version is determined there and not merely selected, and by 0 off $\mathcal{T}_*$; for every $t\in\mathcal{T}_*$ and every l the slice $u\mapsto\ell_\star^{\Pi}(u,t,l)$ is continuously differentiable on all of $\mathbb{R}$ with $\ell_\star^{\Pi}(\cdot,t,l)$ and $\partial_u\ell_\star^{\Pi}(\cdot,t,l)$ bounded and vanishing as $|u|\to\infty$; and both $(u,t,l)\mapsto\ell_\star^{\Pi}(u,t,l)$ and $(u,t,l)\mapsto\partial_u\ell_\star^{\Pi}(u,t,l)$ are jointly measurable on $\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$.*

(ii) The embedding, slicewise, and it holds ***separately*** *in the level and in the derivative.*

For every $t \in \mathcal{T}_*$ and every l,

$$\begin{aligned} \sup_{u\in\mathbb{R}} \left\|\ell_\star^{\Pi}(u,t,l)\right\| &\le c_{\mu''}^{(0)} \left\|\ell_\star^{\Pi}(\cdot,t,l)\right\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}, \\ \sup_{u\in\mathbb{R}} \left\|\partial_u \ell_\star^{\Pi}(u,t,l)\right\| &\le c_{\mu''}^{(1)} \left\|\ell_\star^{\Pi}(\cdot,t,l)\right\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}, \end{aligned} \tag{83}$$

and therefore, adding the two and applying (82)*,*

$$\sup_{u\in\mathbb{R}} \left\|\ell_\star^{\Pi}(u,t,l)\right\| + \sup_{u\in\mathbb{R}} \left\|\partial_u \ell_\star^{\Pi}(u,t,l)\right\| \le c_{\mu''} \left\|\ell_\star^{\Pi}(\cdot,t,l)\right\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}. \tag{84}$$

(83) is the load-bearing form and (84) is its corollary*, not the reverse: clause* (iii) *takes a maximum over l in each of the two budgets* before *adding them, and only a bound that has already been separated survives that order of operations. The two are $\sqrt{2}$ apart,* (83) *summed gives the constant $c_{\mu''}^{(0)} + c_{\mu''}^{(1)} \le c_{\mu''}/\sqrt{2}$, and* (84) *is printed at the weaker $c_{\mu''}$ because that is the constant every consumer downstream reads.* Neither constant is inflated by p*: Step 1 of the proof is* written *at the Euclidean norm throughout, at $\mathbb{R}^p$-valued arity and with no coordinate index fixed, the same Fourier inversion bound, the same Cauchy–Schwarz inequality, now applied once to the level and once to the derivative rather than once to their sum, so* (80) *and* (81) *serve the vector-valued scale verbatim. The sum over the p coordinates sits inside the definition of $\|\cdot\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}$, which is where Plancherel deposits it, and* ***no step of that argument ever forms one****.*

(iii) A2′(i.a)–(i.b), in sup-inside form. *Both maps $t \mapsto \sup_u \|\ell_\star^{\Pi}(u,t,l)\|$ and $t \mapsto \sup_u \|\partial_u \ell_\star^{\Pi}(u,t,l)\|$ are Π_t-measurable at every l, and*

$$\Lambda_\star^{(0)} + \Lambda_\star^{(1)} \le \left(c_{\mu''}^{(0)} + c_{\mu''}^{(1)}\right)\Lambda_\star^{(\mu'')} \le \frac{c_{\mu''}}{\sqrt{2}}\,\Lambda_\star^{(\mu'')} \le c_{\mu''}\,\Lambda_\star^{(\mu'')} < \infty, \tag{85}$$

so A2′(i.a) and A2′(i.b) hold at that version, with the supremum inside *the Π_t-integral in both.* (85) *is stated at the* ***sum****, it holds at every* L *and every p, and read at its*

outer members, left-hand side against $c_{\mu''}\Lambda_\star^{(\mu'')} < \infty$*, it is the display earlier rounds printed, character for character. What is new are the two inner members, and they are a* strengthening *by a factor* $\sqrt{2}$ *that is available at every* L*,* $\mathsf{L} = 2$ *included: nothing in* (85) *depends on* L*, so the binary case reduces with no exception. The one thing* L *does change is the* ***route****, and it is worth saying which step fails, because it is the step a reader will try. All three budgets are maxima over* l *of* $\bar{w}$*-weighted* Π_t*-integrals, write* $\Lambda_l^{(0)}$*,* $\Lambda_l^{(1)}$ *and* $\Lambda_l^{(\mu'')}$ *for the integral at a single contrast* l*, so that* $\Lambda_\star^{(0)} = \max_l \Lambda_l^{(0)}$ *and likewise for the other two, by A2′(i.a)–(i.b) and* (4)*, and the maximum does not distribute over a sum: integrating* (84) *at a fixed* l *and then maximising delivers* $\max_l\{\Lambda_l^{(0)} + \Lambda_l^{(1)}\} \le c_{\mu''}\Lambda_\star^{(\mu'')}$*, which at* $\mathsf{L} \ge 3$ *is* weaker *than* (85)*, because* $\Lambda_\star^{(0)}$ *and* $\Lambda_\star^{(1)}$ *may be attained at different contrasts and* $\max_l a_l + \max_l b_l \ge \max_l\{a_l + b_l\}$*. Chaining that route over the two maximising contrasts recovers only the factor* 2. That is a defect of the combined bound (84) and not of the statement*: the separated bound* (83) *maximises over* l *in each budget* first*, giving* $\Lambda_\star^{(0)} \le c_{\mu''}^{(0)}\Lambda_\star^{(\mu'')}$ *and* $\Lambda_\star^{(1)} \le c_{\mu''}^{(1)}\Lambda_\star^{(\mu'')}$ *separately, and the sum of those two is* (85)*. At* $\mathsf{L} = 2$ *the two routes coincide and the naive one is not wrong, merely* $\sqrt{2}$ *short. And the representer that would attain the factor* 2 *does not exist, so that route was not merely wasteful but empty. Attaining it asks for one contrast saturating* (84) *in the level and another saturating it in the derivative; saturation at the first leaves nothing on the right of* (84) *for its derivative, so it needs* $\Lambda_l^{(1)} = 0$ *at a contrast with* $\Lambda_l^{(0)} > 0$*. That is unrealisable.* $\Lambda_l^{(1)} = 0$ *integrates a nonnegative function to zero, so* $\sup_u \|\partial_u \ell_\star^{\Pi}(u,t,l)\| = 0$ *for* Π_t*-almost every* t *with* $\bar{w}(t) > 0$*; each such slice is then constant in* u*, hence identically zero by the vanishing at* $|u| \to \infty$ *that clause* (i) *supplies; and the* t *with* $\bar{w}(t) = 0$ *contribute nothing to either budget. Hence* $\Lambda_l^{(0)} = 0$ *too.* The factor 2 is therefore not attained, and it is not the supremum either, (85) *bounding the sum by* $c_{\mu''}/\sqrt{2}$ *throughout.* Every consumer downstream reads the outer form, at $c_{\mu''}$*, the concluding sentence below, and Corollary 49 through it, so the strengthening propagates nowhere and nothing outside this lemma retypes.* The p-direction costs

nothing at all, (84) *and* (83) *being stated at the Euclidean norm that* (4) *is also stated at.*

Consequently Corollary 49 applies at $\ell^{\Pi} = \ell^{\Pi}_{\star}$ under B1(i), (ii), (iv.a), (iv.d), *delivering* $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty \, c_{\mu''} \Lambda_\star^{(\mu'')}$ *and* $\|\partial_u \ell_\star\|_\infty \le \big(\|\varpi'_u\|_\infty + \|\varpi_u\|_\infty\big) c_{\mu''} \Lambda_\star^{(\mu'')}$. The target is A2′(i) and never A2b*: A2b is a predicate over generic representers, is never asserted, and therefore has nothing to discharge.*

Remark 62 (Why A2c, and what the registered route would have cost). Four things about Lemma 61 that a reader will ask, in the order they arise.

(a) Why the sup-inside form is the target, and why it is met rather than approximated. A sup-*outside* form does not dominate: $\sup_u \int (\cdot)\, d\Pi_t < \infty$ gives $L^1(\Pi_t)$-boundedness in u without uniform integrability, and Leibniz's rule, which is what (70) consumes, needs $\int \sup_u (\cdot)\, d\Pi_t < \infty$. A2c meets the sup-inside target because it is stated in the *same* $\bar{w}$-weighted $L^1(\Pi_t)$ geometry: each half of (83) is a bound between two functions of t, holding pointwise in t, so (85) is *two* integrations of *two* pointwise inequalities, followed by a maximum over the finite contrast index in each and one addition, and no supremum is ever moved across an integral at any of those steps. The asymmetry of Remark 54 is not reproduced here, because there is no interchange to license.

(b) The obvious alternative route, and why it is three steps short. One might instead require $\ell_\star \in \mathcal{R}\Big((\mathcal{A}_{\text{cond}} \mathcal{A}^*_{\text{cond}})^{\mu''}\Big)$ followed by an embedding into C^1. *First*, that names $\ell_\star$, the observable-space object, whereas A2′(i) is stated on the range-space object $\ell^{\Pi}_{\star}$; the bridge between them is Corollary 49, which *consumes* A2′(i) and so cannot be used to reach it. *Second*, and decisively, membership in $\mathcal{R}((\mathcal{A}_{\text{cond}} \mathcal{A}^*_{\text{cond}})^{\mu''})$ is a statement about the singular system of $\mathcal{A}_{\text{cond}}$ and implies *nothing* about $\sup_u$ or about u-differentiability unless that scale is *matched* to a u-smoothness scale. H1 supplies exactly such a matching on the domain side; there is no analogue on the range side, and supplying one would be a *second* Hilbert-scale link, an assumption about how the left singular functions $\tilde{e}_j$ oscillate in u, on an object the

analyst does not choose. That is strictly larger than A2c and buys nothing A2c does not. *Third*, such a range condition says nothing about the Π_t-integration, which is half of what (i.a) and (i.b) assert.

(c) What A2c costs beyond A2, and it is not only smoothness. Membership $\ell^{\Pi}_{\star} \in L^2(\Pi)$ gives, at every l and for Π_t-almost every t, that $\ell^{\Pi}_{\star}(\cdot, t, l) \in L^2(\Pi_u; \mathbb{R}^p) = L^2(\varpi_u\, du; \mathbb{R}^p)$, *the* $\mathsf{L}-1$ *slices being separately in* $L^2(\Pi_u)$ *because* $\varpi_{\mathsf{w}}(l) > 0$ *by B1(iv.d), so no slice is free.* B1(iv.a) makes ϖ_u bounded and strictly positive *pointwise*, not bounded below, so $L^2(\mathbb{R}, du) \subseteq L^2(\Pi_u)$ and *not* conversely: a slice may lie in $L^2(\Pi_u)$ with tails too fat to lie in $L^2(\mathbb{R}, du)$. A2c therefore imposes a *tail-decay* condition in $|u|$ on top of smoothness in u. *This is stated rather than hidden*, it is the same species of restriction as C4′'s residual truncation, and it is the mirror image of the asymmetry recorded at A2 for π, a one-sided density bound gives one containment and not the other. Two sufficient conditions, in increasing strength: an $L^2(\Pi_t)$-form,

$$\Lambda^{(\mu'')}_{\star} \;\le\; \Big(\int_{\mathcal{T}} \bar{w}^2 \, d\Pi_t \Big)^{1/2} \max_{l \le \mathsf{L}-1} \Big(\int_{\mathcal{T}_*} \big\| \ell^{\Pi}_{\star}(\cdot, t, l) \big\|^2_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} \, \Pi_t(dt) \Big)^{1/2}$$

by the Cauchy–Schwarz inequality, the first factor being finite by B1(iv.c), *it is* $\int \bar{w}^2 d\Pi_t$ *and not* c_{Π}*, which carries an extra* $\Pi_u(\mathbb{R})$ *that this inequality does not produce*; and the uniform form $\max_l \sup_{t \in \mathcal{T}_*} \|\ell^{\Pi}_{\star}(\cdot, t, l)\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} < \infty$, which with B1(v)'s $\int \bar{w}\, d\Pi_t < \infty$ implies both. The weakest of the three is the one A2c states. *The maximum is over a finite index set and introduces no* L*-dependent factor*, the aggregation being the same maximum (4) takes; and the p coordinates sit inside the norm at every one of the three, so none of them carries a $\sqrt{p}$ either.

(d) The integer case needs no transform, and the two constants differ. *Both routes are written at an* $\mathbb{R}^p$*-valued* g*, at the Euclidean norm throughout*, and neither constant acquires a factor in p: the identity $\|g\|^2_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} = \sum_j \|g_j\|^2_{H^{\mu''}(\mathbb{R})}$ is A2c's *definition* of the vector scale and not a step of either argument, the transform route forms no coordinatewise sum at

all, and the elementary route forms only the one that defines $\|g\|^2$. At $\mu''=2$, $H^2(\mathbb{R})$ is $\{g : g, g', g'' \in L^2(\mathbb{R}, du)\}$ with

$$\|g\|^2_{H^2(\mathbb{R})} \;=\; \|g\|^2_{L^2} + 2\|g'\|^2_{L^2} + \|g''\|^2_{L^2}, \tag{86}$$

which is read at the vector scale by reading each L^2 *norm as* $\left(\int \|\cdot\|^2 du\right)^{1/2}$, that being A2c's coordinate identity at $\mu''=2$; and (84) holds with $c_{\mu''}$ replaced by 2 by the elementary argument of Step 1′ of the proof, no Fourier transform, no complex numbers. *The factor* 2 *on the middle term is not decoration and is the one thing an applied reader will get wrong*: (3)'s weight at $\mu''=2$ is $(1+\xi^2)^2 = 1+2\xi^2+\xi^4$, and Plancherel in that same normalisation sends 1, ξ^2, ξ^4 to $\|g\|^2_{L^2}$, $\|g'\|^2_{L^2}$, $\|g''\|^2_{L^2}$, so the cross term of the square survives as a 2. *The set* $\{g : g, g', g'' \in L^2\}$ *is right either way*, the two quadratic forms are equivalent, with ratio in $[1,2]$, so printing the sum without the 2 names the right space and the wrong norm. *And the elementary route's* 2 *is a weaker constant than* (80) *gives at the same index*: $\int_{\mathbb{R}}(1+\xi^2)^{-1}d\xi = \pi$, so $c_2 = \sqrt{2} < 2$. Both are correct and the difference is the price of the transform-free route; the sharper one is the one (80) defines, and (85)'s last member is stated with it. The separated constants sharpen the comparison once more, and the arithmetic is worth seeing at one index. At $\mu''=2$, $\int_{\mathbb{R}}(1+\xi^2)^{-2}d\xi = \int_{\mathbb{R}}\xi^2(1+\xi^2)^{-2}d\xi = \frac{\pi}{2}$, so $c_2^{(0)} = c_2^{(1)} = \frac{1}{2}$ and $c_2^{(0)} + c_2^{(1)} = 1$; (82) is tight at that index, the two integrals being equal there. Step 1′ gives the same two constants elementarily at the value 1 each, hence 2 for the sum. A reader who prefers to check A2c the elementary way therefore loses the range $\mu'' \in (\frac{3}{2}, 2)$ and a constant: a factor $\sqrt{2}$ if the comparison is against (80)'s combined bound, and a factor 2 in each budget separately if it is against (81). *And the threshold is binding, not a margin*: at $\mu'' = \frac{3}{2}$ the integral in (80) diverges, the integrand behaving like $|\xi|^{-1}$, so $c_{\mu''}$ is undefined there and Step 1 has no dominating function. The extra 1 over the L^∞ threshold $\mu'' > \frac{1}{2}$ is exactly what clause (i.b), the *derivative* clause, costs over clause (i.a), *and* (81) *is where that statement becomes an arithmetic one rather than an interpretation*: $c_{\mu''}^{(0)}$ is finite

for $\mu'' > \frac{1}{2}$ and $c^{(1)}_{\mu''}$ only for $\mu'' > \frac{3}{2}$, and it is the second that binds in $c_{\mu''}$, whose integrand is the sum of the other two. *So separating the constants exhibits a threshold* (80) *already carried and imposes no new one*, and A2c's index range is exactly what it was. *And* μ'' *is independent of* μ: $\mu = s/(2a)$ indexes smoothness of $\varphi_\dagger$ in x; μ'' indexes smoothness of $\ell^{\Pi}_\star$ in u. Nothing links them, which is why the smoothness-budget table lists them on separate lines and why the representer line competes for no part of the s budget.

What remains an assumption. A2c is a condition on $\ell^{\Pi}_\star$, an object the model and the selection determine jointly; Lemma 61 does not derive it from anything more primitive and does not claim to. What it buys is that *one* condition, on *one* object, in a language the block already uses, replaces two sup-norm primitives with no route to them, and that the condition is *reportable*: by (3) it is a weighted summability condition on the u-frequency content of the slices, the same kind of object as A2's Picard sum.

Lemma 63 (L-20: structure of the second-order form)**.** *The symmetrised second-order form is a bounded multiplication operator perturbed by a compact one; its essential spectrum is the essential range of the multiplier, it is compact exactly when that multiplier vanishes, and it is indefinite whenever it does not. Suppose the hypotheses of Lemma 55 hold* together with *C2′(i),* which Lemma 55 does not carry, and which is consumed by clause (ii) alone*, transitively through Proposition 9, for the compactness of* $\mathcal{A}_{\mathrm{cond}}$ *that delivers* $\|\mathcal{A}_{\mathrm{cond}} h_k\| \to 0$*; clauses* (i) *and* (iii)–(v) *need it not. Fix* $\vartheta \in \mathbb{R}^p$ *with* $\|\vartheta\| = 1$*, and let* $\mathcal{B}$*,* $\mathcal{B}_{\mathrm{mult}}$ *and* $\mathcal{B}_{\mathrm{sym}} := \frac{1}{2}(\mathcal{B} + \mathcal{B}^*)$ *be the operators on* $L^2(\pi)$ *of* (74)*,* (158) *and Lemma 55(iii).* The multiplier is written out as $\vartheta'\sigma_q/\pi$ throughout and is given no abbreviation*:* σ_q *is the multiplier,* $\mathcal{B}_{\mathrm{mult}}$ *is multiplication* by $\vartheta'\sigma_q/\pi$*, and an abbreviation would collapse the distinction. Then*

(i) Decomposition and boundedness. $\mathcal{B} = \mathcal{B}_{\mathrm{mult}} - 2\,\vartheta'\tilde{\mathcal{K}}$ *and* $\mathcal{B}_{\mathrm{sym}} = \mathcal{B}_{\mathrm{mult}} - \vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$*;* $\mathcal{B}_{\mathrm{mult}}$ *is self-adjoint with* $\|\mathcal{B}_{\mathrm{mult}}\| = \|\vartheta'\sigma_q/\pi\|_\infty \le 2\, c_{\widetilde{W}}\, \bar{c}_\pi\, \|\partial_u \ell_\star\|_\infty < \infty$, Lemma 55(iii)'s bound, carrying $c_{\widetilde{W}}$ for the contraction of $\partial_u \ell_\star$ against $\widetilde{W}$ and equal to the printed constant at $\mathsf{L} = 2$*, the essential supremum being with respect to* $\pi\,dx$ *and hence, by*

C1(ii), with respect to Lebesgue measure on $\mathcal{X}$; $\vartheta'(\tilde{\mathcal{K}}+\tilde{\mathcal{K}}^*)$ is compact and self-adjoint; and $\mathcal{B}$ is bounded but not *self-adjoint.*

(ii) The singular sequence, and it carries (iii)–(v) *and Proposition 27*(ii),(iv). *For every $\varepsilon>0$ and every Borel $\mathcal{X}_0\subseteq\mathcal{X}$ of positive Lebesgue measure on which $\vartheta'\sigma_q/\pi$ is of one sign with $|\vartheta'\sigma_q/\pi|\geq \mathsf{c}_0-\varepsilon$, where $\mathsf{c}_0:=\operatorname{ess\,sup}_{\mathcal{X}_0}|\vartheta'\sigma_q/\pi|$, there is a sequence $(h_k)_{k\geq1}\subset\mathcal{V}$ with $\|h_k\|_{L^2(\pi)}=1$, $h_k\rightharpoonup 0$ weakly in $L^2(\pi)$, $\|\mathcal{A}_{\mathrm{cond}}h_k\|_{L^2(\Pi)}\to 0$, and*

$$\lim_{k\to\infty}\left\langle \mathcal{B}_{\mathrm{sym}}h_k,h_k\right\rangle_{L^2(\pi)} \;=\; \lim_{k\to\infty}\int_{\mathcal{X}_0}\left(\vartheta'\sigma_q/\pi\right)h_k^2\,\pi\,dx, \tag{87}$$

a limit whose modulus is at least $\mathsf{c}_0-\varepsilon$ and whose sign is that of $\vartheta'\sigma_q$ on $\mathcal{X}_0$, along a subsequence. Such an $\mathcal{X}_0$ exists whenever $\vartheta'\sigma_q\not\equiv 0$, and the proof constructs one with $\mathsf{c}_0\geq\|\vartheta'\sigma_q/\pi\|_\infty-\varepsilon$, by taking the near-supremum set first and splitting it by sign second, an order that cannot be reversed.

(iii) Essential spectrum. *$\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}})=\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{mult}})=\operatorname{ess\,ran}(\vartheta'\sigma_q/\pi)$, and the same holds for the compression $P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}$; consequently*

$$\left\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\right\| \;\geq\; \left\|\vartheta'\sigma_q/\pi\right\|_\infty. \tag{88}$$

(iv) Compactness dichotomy. *$\mathcal{B}_{\mathrm{sym}}$ is compact if and only if $\vartheta'\sigma_q=0$ almost everywhere; and the same dichotomy holds for the compression. In particular $\mathcal{B}_{\mathrm{sym}}$ is non-compact whenever $\vartheta'\sigma_q\not\equiv 0$.*

(v) Indefiniteness. *$\int_{\mathcal{X}}\sigma_q\,dx=0$, so if $\vartheta'\sigma_q\not\equiv 0$ then $\vartheta'\sigma_q/\pi$ is strictly positive on a set of positive measure and strictly negative on another; hence $\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}})$ contains points of both signs, and neither $\mathcal{B}_{\mathrm{sym}}$ nor its compression is positive or negative semidefinite.*

Remark 64 (How the hypothesis list of Proposition 27 is read). (1) Lemma 55 is cited by clause and not as a block. Clause (iv) in addition requires **C2′(i)**, which it consumes

transitively through Lemma 63(ii) and Proposition 9, for the compactness of $\mathcal{A}_{\mathrm{cond}}$ that delivers $\|\mathcal{A}_{\mathrm{cond}} h_k\| \to 0$; *clauses* (i)–(iii) *need it not*, and Step 5 of the proof runs the implication O-16(b)⇒O-16(a) through (88) and the numerical radius precisely so that they need not.

(2) *Lemma 56 is read at the admissible direction pair* $(D, \mathsf{r}) = (0, 0)$, which is admissible at every $(h, \bar{\tau})$ by Definition 53 and is differentiable there, so that $\Psi(\cdot, 0, 0) = \Psi_{L_\star}$ by (32) and (49) is half the second τ_1-derivative of the moment along the *structural direction alone*; no condition on a direction in L or in ϱ is inherited.

(3) *The remainder carries no symbol of its own*; italic Q names the regular conditional distributions of (P6b).

(4) *Clause* (i)*'s sharp constant carries no hypothesis on* q *beyond Lemma 55's, and clause* (ii) *brackets that constant without pinning it*: the left-hand member of (51) is a lower bound and is not the sharp constant; the right-hand members are upper bounds and are not the sharp constant either.

Remark 65 (What Proposition 27 claims, and the four things it does not). The quantifier is a supremum over a ball, and the whole result lives there. (50) is a statement about $\sup_{\|h\| \le \delta}$, hence about the *best bound valid uniformly on the ball*. Four readings must be blocked, and each has a version that is true.

(1) It is not a claim about the estimator. Proposition 27 does *not* say that the realised remainder at $h = \hat{\varphi} - \varphi_\dagger$ fails to be $o_P(\|h\|^2_{L^2(\pi)})$. A regularisation path confined to directions on which the form happens to be small could do better, and nothing here excludes that. *What is true:* the direct bound $\frac{1}{2}|\langle \mathcal{B} h, h\rangle| \le \frac{1}{2}\|P_{\mathcal{V}} \mathcal{B}_{\mathrm{sym}} P_{\mathcal{V}}|_{\mathcal{V}}\| \, \|h\|^2$ is *order-sharp*, so A5(ii)'s $\delta_n = o_P(n^{-1/4})$ cannot be relaxed *by finding a better bound on the remainder*. Relaxing it requires either a different moment, a selection with a smaller σ_q and $\mathcal{K}$, which is lever 1, or a restriction on the *direction* of the first-stage error, which is an extra hypothesis and not a sharper argument. The two are different claims and the paper should print both.

(2) The lower bound is not the sharp constant, and the identity $\sup_{\|h\|\le\delta} \frac{1}{2}|\langle \mathcal{B}h, h\rangle| = \frac{1}{2}\delta^2\|\mathcal{B}_{\mathrm{sym}}\|$ holds on the wrong ball. It is exact when the supremum runs over the ball of $L^2(\pi)$ and is an *upper bound* when it runs over the ball of $\mathcal{V}$, which is the operative one, since $h = \hat{\varphi} - \varphi_\dagger$ lies in $\mathcal{V}$ by Definition 5. The exact constant on $\mathcal{V}$ is the compression's norm, Definition 5's one declared exception is what forces the distinction, and (51) brackets it. *No downstream consequence changes*, because what Theorem 29 consumes is the lower bound. *And* $\frac{1}{2}\|\vartheta'\sigma_q/\pi\|_\infty$ *must not be printed as the sharp constant*: it is a lower bound, delivered by the multiplication part through Lemma 63(iii).

(3) $\sigma_q \equiv 0$ does not make the remainder vanish, and this is what separates O-16(a) from O-16(b). Lemma 55(iii)'s witness exhibits a $q \not\equiv 0$ with $\sigma_q \equiv 0$ whose form is nonzero: σ_q is a marginalisation and the form sees the *symmetric part* of $q\, f_{X_1,X_2}$, which a marginalisation does not determine. Consequently, when $\sigma_q \equiv 0$ but O-16(a) holds, clauses (i) and (iii) survive *with the same* δ^2 *order*, while three things change and they change in three different ways: (88) becomes *vacuous*, its right-hand side being zero; the non-compactness of Lemma 63(iv) *fails*, the form then being compact; and the weak-norm refutation (iv) becomes *unavailable*, *which is not the same as a weak-norm bound existing*. Whether one exists is then open, and it is what the Douglas route of the strategy memo §3.6 would have to settle. *So lever 1's target* $\sigma_q \equiv 0$ *removes the essential spectrum and does not remove the* δ^2 *order*, and SG-15, which sweeps $\|\vartheta'\sigma_q/\pi\|_\infty$, minimises a certified *lower bound* on the sharp constant and not the sharp constant. *The objective is not even monotone in the quantity of interest, and that is stronger than saying it is a lower bound*: the compact part $\vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ is *unconstrained* by σ_q, so a selection that lowers $\|\vartheta'\sigma_q/\pi\|_\infty$ may *raise* $\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\|$, and driving the sweep to zero makes (88) vacuous while leaving the sharp constant free. *Any claim that SG-15 improves the constant therefore requires the compression's norm computed directly*, which (51) brackets but does not deliver. The target that removes the remainder outright is $q \equiv 0$, or a cancellation in the symmetric part of $q\, f_{X_1,X_2}$; probe CP-12 is aimed at exactly this, and it is the only probe that could reopen the scope. *And* $q \not\equiv 0$ *alone is not enough for O-16(a)*

either: if $q\, f_{X_1,X_2}$ is antisymmetric the form vanishes identically, since $(\Delta h)^2$ is symmetric in (x_1, x_2), which is why (a) is stated as "the form does not vanish" and not as "$q \not\equiv 0$".

(4) O-16 is a hypothesis and not a consequence of O-11. O-11 delivers $\partial_u \ell_\star \not\equiv 0$: a debiasing weight must vary with the residual. It does *not* deliver $q \not\equiv 0$, because $q = E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2]$ can vanish by cancellation inside the conditional expectation, still less $\sigma_q \not\equiv 0$, which is a second marginalisation. Both O-16 clauses are therefore carried as explicit hypotheses of Proposition 27(iii)–(iv) and are asserted nowhere. The chain reads: O-11 says the quadratic term must *exist*; O-16 says it does not degenerate; Proposition 27 says that if it does not degenerate then its order cannot be improved.

Lemma 66 (The boundary family exists, and its interior member is the shipped kernel)**.** *Let $p_K \geq 1$ and let $\mathfrak{I}$ be as in C3. For $I \in \mathfrak{I}$ put*

$$\mathsf{G}(I) \;:=\; \Big[\int_I \varsigma^{j+k} e^{-\varsigma^2/2}\, d\varsigma\Big]_{j,k=0}^{p_K-1}, \qquad \Theta_I(\varsigma) \;:=\; \Big(\sum_{k=0}^{p_K-1} \big(\mathsf{G}(I)^{-1} e_0\big)_k\, \varsigma^k\Big) e^{-\varsigma^2/2}\, \mathbb{1}\{\varsigma \in I\}, \tag{89}$$

e_0 the first standard basis vector of $\mathbb{R}^{p_K}$. Fix the topology on $\mathfrak{I}$ here, because (d), (f) and (g) are cited from other files and a continuity claim without a topology is not a claim*: $\mathfrak{I}$ carries* ***endpointwise convergence in the two-point compactification****, the topology induced by*

$$\mathfrak{I} \;\ni\; I \;\longmapsto\; \big(\tanh \inf I,\ \tanh \sup I\big) \;\in\; [-1,0] \times [0,1]. \tag{90}$$

(c) *below shows this map to be injective with closed image, so that $\mathfrak{I}$ is compact and metrisable; "continuous on $\mathfrak{I}$" below means continuous for (90).* tanh carries no content*: it is one homeomorphism $[-\infty, +\infty] \to [-1, 1]$ and any other returns the same topology, which is why the clauses below are stated in I and not in coordinates. Then:*

(a) $\mathsf{G}(I) \succ 0$ for every $I \in \mathfrak{I}$, so (89) is well posed.

(b) Θ_I vanishes off I and $\int_I \varsigma^j \Theta_I(\varsigma)\, d\varsigma = \delta_{j0}$ for $0 \leq j \leq p_K - 1$; that is, (15) holds.

(c) The map (90) *is injective with closed image, so* $\mathfrak{I}$ *is compact and metrisable for it; and* $\underline{g} := \inf_{I\in\mathfrak{I}} \lambda_{\min}\big(\mathsf{G}(I)\big) > 0$.

(d) Put

$$\bar\Theta(\varsigma) \;:=\; \sup_{I\in\mathfrak{I}} \big|\Theta_I(\varsigma)\big|, \qquad \textit{for which} \qquad \bar\Theta(\varsigma) \;\le\; \underline{g}^{-1}\Big(\sum_{k=0}^{p_K-1}\varsigma^{2k}\Big)^{1/2} e^{-\varsigma^2/2}. \tag{91}$$

Then $|\Theta_I| \le \bar\Theta$ *for every* $I \in \mathfrak{I}$ *by construction, and* $\bar\Theta$ *is continuous — hence measurable — bounded and integrable, with* $\int_{\mathbb{R}} |\varsigma|^m \bar\Theta < \infty$ *for every* $m \ge 0$. *Hence C3(ii) holds, with* c_2 *and* c_C *the constants* (16) *derives from this* $\bar\Theta$. *Moreover* $\bar\Theta$ *is* ***minimal****: every* $\bar\Theta'$ *meeting C3(ii) satisfies* $\bar\Theta' \ge \bar\Theta$ *pointwise, hence* $\|\bar\Theta'\|_\infty \ge \mathsf{c}_2$ *and* $\|\bar\Theta'\|_{L^1} \ge 2c_C\mathsf{c}_2$ — no admissible envelope returns a smaller pair*, so* (35) *and* (114) *are at their sharpest here.* The second half of (91) is the only role the explicit majorant plays **in any proof***: a fixed polynomial times a Gaussian, it is bounded, integrable, of finite moment of every order* and vanishing at infinity*, and* $\bar\Theta$ *inherits all four from it — the fourth is not decoration, it is what* (g) *turns into attainment.* So $\underline{g}$ reaches every **consumed** constant only through the finiteness and the decay it certifies*: no bound, no rate and no constant that any result below reads is computed from it.* It does have a second role, that role is expository and not inferential, and the numbers it carries there are printed at three places — the sentence this one replaces denied that, and was false at all three. *The majorant is the* ***comparison*** *envelope, the closed form an analyst writes down first, against which C3's note and the note below weigh the minimal one.* (16) *evaluated at it returns the pair* $(4.50\times10^{8},\ 1.82)$*, printed in C3's note;* of that pair the first entry is computed from $\underline{g} = 6.59\times10^{-8}$ and the second is **not**, $\underline{g}^{-1}$ *cancelling in the ratio that defines* c_C*. The note below prints a third number,* 1.97×10^{6}*: the same sup-norm formed with the realised family's infimum* 1.51×10^{-5} *in place of* $\underline{g}$. All three are quoted and consumed nowhere*, and the comparison is the whole of their work — it is what makes "take the envelope minimal" a ruling with a stated price rather than a preference,*

and (g) *now supplies the third term of that comparison, a* proved *pair between the two.*

(e) At $p_K = 8$, $\Theta_{\mathbb{R}}$ *is* exactly *the order-*8 *Gaussian-based kernel*

$$\Theta_1(\varsigma) \;=\; \frac{105 - 105\varsigma^2 + 21\varsigma^4 - \varsigma^6}{48\sqrt{2\pi}}\, e^{-\varsigma^2/2},$$

which is symmetric, C^∞ *and Lipschitz; so C3(i) and C3(iii) hold at the same family, and the interior member is the kernel the application computes.*

(f) $\Theta_{-I}(\varsigma) = \Theta_I(-\varsigma)$, *and* $I \mapsto \mathsf{G}(I)^{-1}e_0$ *is continuous on* $\mathfrak{I}$ *for the topology* (90). It is the **coefficient** map that is continuous and **not** $I \mapsto \Theta_I(\varsigma)$, *which jumps at every* I *with an endpoint at* ς*; the two are one indicator apart, and every consumer of this clause must dispose of that indicator before invoking it.* Both do, and by different devices*:* (d)*'s proof restricts the supremum to* $\{I : \varsigma \in I\}$, *where the indicator is identically* 1*; the measurability paragraph of Definition 22 tracks* ς *and* I together *along* (94), *where the indicator collapses to* $\mathbb{1}\{x \in \mathcal{X}\}$.

(g) A closed-form admissible envelope, two-sided bounds on (16), and attainment. *For every* $I \in \mathfrak{I}$ *and every* $\varsigma \in \mathbb{R}$,

$$\left|\Theta_I(\varsigma)\right| \;\le\; 2^{1/4}\, p_K^2\, e^{1/4}\, \exp\!\left(\frac{\left(|\varsigma| + 2^{-1/2}\right)^2}{4} - \frac{\varsigma^2}{2}\right), \tag{92}$$

whose right-hand side is bounded, integrable, of finite moment of every order and hence itself an admissible envelope for C3(ii), with (16) *returning at it the* exactly computed *pair* $\mathsf{c}_2 = 2^{1/4}e^{1/2}p_K^2$ *and* $c_C = \sqrt{\pi}\big(1 + \operatorname{erf}(2^{-3/2})\big)$, *the second free of* p_K. *Consequently, at the minimal envelope of* (d),

$$\tfrac{1}{2}\, p_K^2 \;\le\; \mathsf{c}_2 \;\le\; 2^{1/4}e^{1/2}p_K^2, \qquad \|\bar{\Theta}\|_{L^1} = 2c_C\mathsf{c}_2 \;\le\; 2^{5/4}\sqrt{\pi}\, e^{1/2}\big(1+\operatorname{erf}(2^{-3/2})\big)\, p_K^2, \tag{93}$$

and hence $c_C \le 2^{5/4}\sqrt{\pi}e^{1/2}\big(1 + \operatorname{erf}(2^{-3/2})\big)$, *also free of* p_K. *Moreover the supremum*

defining $\|\bar\Theta\|_\infty$ *is* ***attained****: there exist* $\varsigma^*\in\mathbb{R}$ *and* $I^*\in\mathfrak{I}$ *with* $\varsigma^*\in I^*$ *and* $|\Theta_{I^*}(\varsigma^*)|=\|\bar\Theta\|_\infty=\mathsf{c}_2$*. Finally, along the slice* $\varsigma=0$ *the supremum is carried by the* shortest *admissible intervals:* $\bar\Theta(0)=\max\{\Theta_I(0):I\in\mathfrak{I},\ |I|=2\}$.

Note on (g)*: which of its bounds to substitute, into what, and in which direction each consumer reads the pair. The two bounds of* (93) *are the ones to substitute and the standalone bound on* c_C *is* ***not***: that bound is formed with the *lower* half of (93) in its denominator, so pairing it with the upper half double-counts. *Five printed quantities read the pair* $(\mathsf{c}_2,\|\bar\Theta\|_{L^1})$*, the list is given in full because the directions are not all the same.* It is not offered as exhaustive, and no criterion is given for membership*: readers of a single component exist too — Lemma 67*(ii)*'s* $\|\partial_v\mathsf{C}_{h_u}\|_\infty\le\mathsf{c}_2h_u^{-1}$ *is one, increasing in* c_2 *and so safe under the same substitution — and two drafts of this sentence were short by a site while claiming not to be.* What is asserted is only the ruling below, and it holds for any reader whatever, monotone in the pair or invariant in it*: substituting the upper bounds of* (93) *cannot falsify a conclusion.* **Three read it increasingly**: (35) is $\|\bar\Theta\|_{L^1}^{1/2}\mathsf{c}_2^{3/2}$ up to L-free factors, (36) is its square, and c_3 of Lemma 67(i) is $1\vee\|\bar\Theta\|_{L^1}$. **One reads it reciprocally**: (114) carries $\big(\mathsf{c}_2\|\bar\Theta\|_{L^1}\big)^{-1/2}$, since $(2c_C)^{1/2}\mathsf{c}_2=\big(\mathsf{c}_2\|\bar\Theta\|_{L^1}\big)^{1/2}$. **And one does not read it at all**: c_{18}, of Proposition 28's proof, is *invariant* in the pair. It is a Bernstein exponent, a ratio of a mean bound to a range bound — $\mathsf{K}=4\bar c_\pi\|\bar\Theta\|_{L^1}^2\|\pi\|_\infty\|1/\pi\|_\infty$ over $(4c_C\mathsf{c}_2\mathsf{c}_{11})^2=4\|\bar\Theta\|_{L^1}^2\mathsf{c}_{11}^2$ — and both bounds are quadratic in the same Δ_{h_x} operator norm, so the envelope cancels exactly and $\mathsf{c}_{18}=\frac{3}{8}\bar c_\pi\|\pi\|_\infty\|1/\pi\|_\infty\big/\mathsf{c}_{11}^2$. *So substituting the* ***upper*** *bounds* (93) *is safe throughout, and for one reason*: it overstates the first three, which are read as upper bounds; it understates the fourth, which is a certified curvature radius, and a smaller radius is a weaker statement of the same form; and it moves the fifth not at all. Substituting (93) therefore leaves every conclusion true.

Note on the constants, and read every quantifier — including the epistemic one. At $p_K=8$, numerically: $\mathsf{c}_2=\|\bar\Theta\|_\infty=34.0$ and $\|\bar\Theta\|_{L^1}=10.56$, hence $c_C=\|\bar\Theta\|_{L^1}/(2\mathsf{c}_2)=0.155$, the

supremum in (91) being taken over *all* of $\mathfrak{I}$. *These three are* ***located*** *values and not proved ones, and the distinction is not pedantic here, because the direction of a search error and the direction the constants must err in are opposite.* By (g) the supremum *is* attained, so there is a maximiser to locate; a search reports its value *at the points searched*, which is a **lower** bound on a supremum. *Every consumer needs an* ***upper*** *one, and in both directions of use*: (35), (36) and c_3 read the pair increasingly, so understating it understates ω_n and Definition 22's $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\| \le 1$ — which is Proposition 70(i) and (i′) read at ω_n, in each of the two domain pairings, and holds *surely* — ceases to be implied; (114) reads it reciprocally, so understating it overstates a certified radius. *Overstating is safe in both directions of use; understating is unsafe in both, and a number below* $\|\bar{\Theta}\|_\infty$ *is the sup-norm of no admissible envelope at all.* c_{18} *is* ***not*** *a third case and an earlier draft of this sentence made it one*: it is a ratio of a mean bound to a range bound, both quadratic in $\|\bar{\Theta}\|_{L^1}$, so the envelope cancels and it is invariant — the note to (g) above computes it. *What is proved, and what it costs.* (g) brackets the located values: $32 \le \mathsf{c}_2 \le 125.5$ and $\|\bar{\Theta}\|_{L^1} \le 615.2$, both in closed form, so 34.0 sits 6.4% above a proved floor and a factor 3.7 below a proved ceiling, and the independently confirmed ratio 1.06373 to the Legendre value $\sum_{j=0}^{7}(2j+1)/2 = 32$ is **that floor**, not a coincidence. *A reading that requires its inequality* ***proved*** *rather than* ***measured*** *runs at* (93), and pays a factor 54.0 in (35), 2.92×10^3 in (36) and 58.3 in c_3 — against 6.0×10^{14} and 3.6×10^{29} at the explicit majorant. *Two parts of the location remain unproved and are named rather than assumed*: that the maximiser sits at an *endpoint* placement, and that $\varsigma^* = 0$. (g)'s Step 8 proves that, *given* $\varsigma^* = 0$, the interval has $|I| = 2$; the placement is then a one-parameter maximisation over a compact interval, provable term-by-term under the Lebesgue weight from $|\mathrm{Leg}_j| \le 1$ and **not** under the Gaussian one, whose orthonormal family is not uniformly bounded. *And* $\varsigma^* = 0$ *cannot be reached by monotonicity*: $I \mapsto |\Theta_I(\varsigma)|$ is **not** antitone at fixed $\varsigma \neq 0$, since $|\Theta_I|$ has zeros that a larger I moves. Over the intervals the application realises, for which $|I(x^*)| = |\mathcal{X}|/h_x \in [10.1, 12.4]$, the same two norms are 17.4 and 6.22. *The uniform pair is what is quoted, and the realised pair is not*, because (35)

is a constant of a theorem while $\mathfrak{I}$ is what the theorem quantifies over: h_x is free in every statement below, Lemma 67(iv) admits every $h_x \le (\bar{x} - \underline{x})/2$, and a number measured at one bandwidth is a function of that bandwidth. The uniformity costs a factor of 3.6 in (35) and 13 in (36), and that is the entire price of quoting a constant rather than a measurement.

The Gram spectrum, reported as an infimum because that is what (c) *defines.* Every eigenvalue here is of (89) as printed, with the bare $e^{-\varsigma^2/2}$; a $\sqrt{2\pi}$-normalised Gram divides each by $\sqrt{2\pi}$, and an earlier draft of this note mixed the two conventions. $\underline{g} = 6.59 \times 10^{-8}$ over the grid searched, attained at $I = [-2, 0]$, the shortest admissible interval; $\lambda_{\min}(\mathsf{G}(\mathbb{R})) = 2.718 \times 10^{-1}$; and over half-lines $\lambda_{\min}(\mathsf{G}([-\infty, 0])) = 1.51 \times 10^{-5}$. *The application does* ***not*** *escape the half-line value, and an earlier draft of this note said it did.* A design point at an endpoint of $\mathcal{X}$ realises $I(x^*) = [0, |\mathcal{X}|/h_x]$ by (94), a truncated half-line, and $\mathsf{G}([0,\infty]) - \mathsf{G}([0,b]) = \mathsf{G}([b,\infty]) \succeq 0$ gives $\lambda_{\min}(\mathsf{G}([0,b])) \le \lambda_{\min}(\mathsf{G}([0,\infty])) = 1.51 \times 10^{-5}$ at every right endpoint b, by (f) for the reflection. So 1.51×10^{-5} — not the 2.57×10^{-1} that draft quoted — is the infimum over the realised family, and it is attained *exactly* at every bandwidth the application runs; 2.57×10^{-1} is a correct value at the *symmetric* realised interval, i.e. at an interior design point, and is the family's **maximum**. *The* $\mathsf{c}_2 = 115.4$ *derived from it is withdrawn*: $\|\cdot\|_\infty$ of the majorant divided by $\lambda_{\min}$ is an envelope bound only at the infimum, which returns 1.97×10^6 there. *Nothing above depends on the repair*, c_2 and c_C being read off $\bar{\Theta}$ directly and never through $\underline{g}$, which is why (d) is written the way it is. *These are searched values and not proved bounds* — (c) proves the infimum positive; the numbers here locate it.

Lemma 67 (Properties of the integrated kernel)**.** *Suppose C3 and* ***A0's compact-interval clause*** *hold.* A0 moves **onto** this list on 2026-08-26, and the ground is the assumption block's own split*: the interval is what A0* restricts $\mathcal{X}$ *to and not what it introduces, a fat Cantor* $\mathcal{X}$ *leaves every symbol named and the restriction false, so it is a* ***clause****, and the result that reads it carries it.* Clause (iv) is where it is read, and it reads it twice*: once for* (94) *to be an element of* $\mathfrak{I}$ *at all, and once for the substitution* $x = x^* + h_x\varsigma$ *to carry the* $\mathcal{X}$*-integral onto a*

single *interval.* Through 2026-08-25 this line said the opposite — *that the compact interval is a* ***declaration*** *standing at every statement, named outside the "Suppose … hold" list and not a hypothesis* — and cited two sites as precedent, neither of which supports it*: Lemma 47, whose defence is retyped in this same pass because that lemma does* ***not*** *read the interval, and Lemma 81, which* ***does*** *carry A0, as a block and first on its own hypothesis line, so it was never an instance of the idiom.* And through 2026-08-23 this was the one site that used the interval silently*, which is the reading that survives all of it. Then, with* $\mathsf{c}_2 = \|\bar{\Theta}\|_\infty$ *and* c_C *of* (16)*, and* Θ_I *of* (14)*:*

(i) Tails and normalisation. *For every* $\epsilon > 0$ *there is a finite* $c_\Theta(\epsilon)$ *with* $\int_{|\varsigma|>c_\Theta(\epsilon)} \bar{\Theta} \le \epsilon$*, and then* $|\mathsf{C}_{h_u}(v)| \le \epsilon$ *for* $v \le -c_\Theta(\epsilon)h_u$ *and* $|\mathsf{C}_{h_u}(v) - 1| \le \epsilon$ *for* $v \ge c_\Theta(\epsilon)h_u$*; and* $|\mathsf{C}_{h_u}| \le \mathsf{c}_3$ *with* $\mathsf{c}_3 := 1 \vee 2c_C\mathsf{c}_2$*, uniformly in* h_u*.* Nonnegativity is **not** claimed and is not available*: at* $p_K = 8$ *the kernel* Θ_1 *must take negative values, so* C_{h_u} *does too. Every consumer uses only the two-sided bound.*

(ii) Derivative, and a second-order bound without a second derivative. C_{h_u} *is continuously differentiable with* $\partial_v \mathsf{C}_{h_u}(v) = h_u^{-1} C_{h_u}(v, \mathbb{R})$ *and* $\|\partial_v \mathsf{C}_{h_u}\|_\infty \le \mathsf{c}_2 h_u^{-1}$*;* $\partial_v \mathsf{C}_{h_u}$ *is Lipschitz with constant* $\mathsf{c}_4 h_u^{-2}$*,* c_4 *being the Lipschitz constant of* $C_{h_u}(h_u\cdot, \mathbb{R})$ *furnished by C3(iii); and consequently, for all* $v, v' \in \mathbb{R}$*,*

$$\left|\mathsf{C}_{h_u}(v') - \mathsf{C}_{h_u}(v) - \partial_v \mathsf{C}_{h_u}(v)\,(v' - v)\right| \;\le\; \tfrac{1}{2}\,\mathsf{c}_4\,h_u^{-2}\,(v' - v)^2.$$

No second derivative of C_{h_u} is assumed or used*; C3(iii) gives a Lipschitz kernel and the display above is what a Lipschitz derivative buys.* The increment carries no symbol*: it is written out as* $v' - v$*, in the pair this lemma's own proof already uses for the two arguments of* $\partial_v \mathsf{C}_{h_u}$*.*

(iii) Order, for a function of ***bounded variation****, which is what its consumers supply. Let* $G : \mathbb{R} \to \mathbb{R}$ *be of bounded variation with* $G(-\infty) = 0$*, let* dG *denote the associated finite signed Borel measure, so that* $G(u) = dG\big((-\infty, u]\big)$ *and* $|dG|(\mathbb{R}) < \infty$*, and suppose in*

addition that G is p_K times continuously differentiable with $\|G^{(p_K)}\|_\infty < \infty$. Then, for every u,

$$\left| \int_{\mathbb{R}} \mathsf{C}_{h_u}(u-a)\, dG(a) - G(u) \right| \;\le\; \mathsf{c}_5\, \|G^{(p_K)}\|_\infty\, h_u^{p_K}, \qquad \mathsf{c}_5 := \frac{1}{p_K!} \int_{\mathbb{R}} |\varsigma|^{p_K} \left|\Theta_1(\varsigma)\right| d\varsigma,$$

c_5 *finite and uniform in h_u by C3(i). The two hypotheses on G are not decoration. A version asking only that G map into $[0,1]$ and be smooth is* false*: at $G \equiv \frac{1}{2}$ it reads $|0 - \frac{1}{2}| \le 0$, because $dG = 0$ while $G(u) = \frac{1}{2}$.* What the identity needs is that G be recovered from dG, *which is $G(-\infty) = 0$ together with bounded variation, and* that is precisely what its consumers have*: Lemma 77 Step 3 applies it to $\mathcal{G}_{t,l} = \mathcal{M}(\cdot, t, l)$, a difference of two nondecreasing functions each vanishing at $-\infty$ and of total variation at most $\bar{w}(t)$* , at every L, the total variation of the signed measure $E[\widetilde{W}_l\, \delta_{\Delta U}\, w(t'\Phi(\Delta Z))]$ being at most $E[|\widetilde{W}_l|\, w] \le \bar{w}(t)$ by the **per-cell** bound $|\widetilde{W}_l| \le 1$, so the constant does not move*; and Lemma 78 Step 2 applies its argument to conditional-mean products that are likewise differences of finite measures' distribution functions; and Lemma 81 Step 2 re-runs Lemma 77 Step 3's computation verbatim at $\mathcal{M}(\cdot, t, l; \varphi, \theta_\dagger)$ and carries this clause's c_5 into its own aggregation, so it is a third. None of them needs $G \ge 0$ and none needs $\int dG = 1$.* No claim of exhaustiveness is made, and that is deliberate*: the draft this clause replaces said "the two consumers" twice and was short by that site both times.*

(iv) Exact x-normalisation, and the property of the boundary adjustment it rests on is now ***stated****. Suppose $h_x \le (\bar{x} - \underline{x})/2$. For every $x^* \in \mathcal{X}$ put*

$$I(x^*) \;:=\; \left[\frac{\underline{x} - x^*}{h_x},\; \frac{\bar{x} - x^*}{h_x} \right] \;\in\; \mathfrak{I}, \tag{94}$$

the realised truncation interval *at x^* in scaled coordinates, which lies in $\mathfrak{I}$ because $\underline{x} \le x^* \le \bar{x}$ gives $\inf I(x^*) \le 0 \le \sup I(x^*)$ and $h_x \le (\bar{x} - \underline{x})/2$ gives $\sup I(x^*) -$*

$\inf I(x^*) = (\bar{x} - \underline{x})/h_x \geq 2$. *Then,* for every $x^* \in \mathcal{X}$, one case,

$$h_x^{-1} \int_{\mathcal{X}} C_{h_x}\big(x - x^*,\, I(x^*)\big)\, dx \;=\; 1.$$

The clause of C3 this consumes is (15) at $j = 0$. There are no longer three cases and no reflection convention*: the interval index records both how far the nearer endpoint is and which endpoint it is, so the interior point, the two boundary points and the point close to* ***both*** *endpoints — which a scalar index could not name at all, and which the round-12 form of this clause excluded by hypothesis — are one clause at four values of* I. Load-bearing*: Proposition 70(iii) rests entirely on this clause, and without it* $\int \hat{\nu}_{u,t,l}\, dx \neq 0$ *at* every *design point — the defect being the tail* $\int_{\mathbb{R}\setminus I(x^*)} \Theta_1$*, which is* $O(1)$ *only where the nearer endpoint is within* $O(h_x)$ *of* x^* *and tends to* 0 *with* h_x *at every interior point, by integrability of* $\bar{\Theta}$ *— so that clause's conclusion fails:* $\hat{\mathcal{A}}^*_\varphi \Xi \notin \mathcal{V}$ *for some* $\Xi \in L^2(\Pi)$*, read at Proposition 70*(iii)*'s own bound variable, and (L2) of Proposition 28 Step 2 loses the projection identity it runs on.* The width is written $O(h_x)$ and not $c_C h_x$*: under the envelope* c_C *is the ratio* (16) *of two norms of* $\bar{\Theta}$ *and is not a length.*

Definition 68 (The x-smoothed sample operator)**.** Throughout Part H, $\hat{T}$ is (12) of Definition 2, under that definition's sample-splitting and truncation conventions: $\{1, \ldots, n\} = \mathcal{I}_a \cup \mathcal{I}_m$ with $n_{\mathrm{a}} := \#\mathcal{I}_a$, $n_{\mathrm{m}} := \#\mathcal{I}_m$, $\hat{\varrho}$ computed on $\mathcal{I}_a$ and taking values in the ε_ϱ-interior simplex (11), and $\widehat{\widetilde{W}}_{i,l} := \mathbb{1}\{W_i = l\} - \hat{\varrho}_l(\Delta Z_i)$, $\widehat{\widetilde{W}}_i := (\widehat{\widetilde{W}}_{i,1}, \ldots, \widehat{\widetilde{W}}_{i,\mathsf{L}-1})'$, satisfying

$$\varepsilon_\varrho \;\leq\; \big|\widehat{\widetilde{W}}_{i,l}\big| \;\leq\; 1 \qquad \text{surely, for every } i \text{ and every } l \in \{1, \ldots, \mathsf{L}-1\}, \tag{95}$$

the upper bound as in (P1) *read at the* l*-th contrast* and the lower bound by A4(i) and the truncation. *Both bounds are* ***per cell*** *and neither is a statement about* $\|\widehat{\widetilde{W}}_i\|$: (11) puts $\hat{\varrho}_l \in [\varepsilon_\varrho, 1 - \varepsilon_\varrho]$ for every l, the lower bound by the first constraint, the upper because

$\hat{\varrho}_l \le \mathbf{1}'\hat{\varrho} \le 1-\varepsilon_\varrho$, so $\mathbb{1}\{W_i = l\} - \hat{\varrho}_l$ lies in $[\varepsilon_\varrho, 1-\varepsilon_\varrho]$ in modulus at both values of the indicator. At $\mathsf{L} = 2$ there is one contrast, $\mathbb{1}\{W_i = 1\} = W_i$, and (95) is the display earlier rounds printed. (95) *is used at four places and each would fail without it*: Proposition 70(v), Proposition 71 and Proposition 76(ii) all need a design point at which the sample operator responds, and Proposition 74(iii) needs $\mathbf{G}_n$ to have no zero row. *All four therefore carry A4(i) on their own hypothesis lines, and each of the four does*; the free *upper* half of (95) is a different matter and is read, without A4(i), at Proposition 70(i) and at Lemma 78 Step 1, whose own hypothesis line carries A4(iii) and not A4(i) and whose $\sum_l \varpi_{\mathsf{w}}(l)\widehat{\widetilde{W}}_{1,l}^{\,2} \le 1$ is that upper half read with B1(iv.d). *Those two are named as readers of the upper half and no claim of exhaustiveness is made about it*: the draft this sentence replaces named one.

For $\varphi \in \mathcal{V}$ and $i \in \mathcal{I}_m$ let the *smoothed period difference* be

$$\Delta_{h_x}\varphi_i \;:=\; \frac{1}{h_x}\int_{\mathcal{X}} \Big[C_{h_x}\big(x - X_{i2}, I_{i2}\big) - C_{h_x}\big(x - X_{i1}, I_{i1}\big)\Big]\varphi(x)\,dx, \tag{96}$$

with $I_{ij} := I(X_{ij})$ of (94), which lies in $\mathfrak{I}$ *at every* n by C5′(ii), and write $\Delta_{h_x}\varphi := (\Delta_{h_x}\varphi_i)_{i\in\mathcal{I}_m} \in \mathbb{R}^{n_{\mathrm{m}}}$. The *$x$-smoothed sample operator* is (12) with $\Delta\varphi_i$ replaced by $\Delta_{h_x}\varphi_i$:

$$\hat{T}_{h_x}(\varphi,\theta)(u,t,l) \;:=\; \frac{1}{n_{\mathrm{m}}}\sum_{i\in\mathcal{I}_m} \widehat{\widetilde{W}}_{i,l}\, \mathsf{C}_{h_u}\big(u - \Delta Y_{i,\theta} + \Delta_{h_x}\varphi_i\big)\, w\big(t'\Phi(\Delta Z_i)\big), \qquad \Delta Y_{i,\theta} := \Delta Y_i - \Delta Z_i'\theta, \tag{97}$$

at each $(u,t,l) \in \mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$, *which is* (12)*'s arity read argument for argument*, $\hat{T}_{h_x}$ differing from $\hat{T}$ in the period difference alone. *The two agree to the accuracy of the x-smoothing and differ in one respect that decides everything below*: $\varphi \mapsto \hat{T}(\varphi,\theta)$ depends on φ through the *point evaluations* $\varphi(X_{ij})$, while $\varphi \mapsto \hat{T}_{h_x}(\varphi,\theta)$ depends on it through the *bounded linear functionals* (96).

Definition 69 (The smoothed sample linearisation, exhibited)**.** Fix θ and $\varphi \in \mathcal{V}$, and put,

for $i \in \mathcal{I}_m$,

$$\hat{g}_{u,t,l,i}(\varphi,\theta) \;:=\; \widehat{\widetilde{W}}_{i,l}\; w\Big(t'\Phi(\Delta Z_i)\Big)\;\frac{1}{h_u}\,C_{h_u}\Big(u-\Delta Y_{i,\theta}+\Delta_{h_x}\varphi_i,\;1\Big), \tag{98}$$

the sample counterpart of $g_{u,t,l}$ of (27) with the conditional density replaced by the derivative of the integrated kernel. *The contrast index enters through* $\widehat{\widetilde{W}}_{i,l}$ *and through nothing else*, the kernel and the Bierens factor being l-free; that is what makes every bound below a per-cell bound. *The residual inside* (98) *carries the* ***smoothed*** *difference* $\Delta_{h_x}\varphi_i$ *of* (96)*, not the raw* $\Delta\varphi_i$*, and the choice is not cosmetic*: with the raw difference every clause of Proposition 70 except (iv) is unchanged, and (iv) acquires an additional *first-order* term, which is exactly the discrepancy Proposition 71 exploits. Define the *kernel-smoothed sample difference functional*

$$\hat{\nu}_{u,t,l}(x) \;:=\; \frac{1}{n_{\mathrm{m}}h_x}\sum_{i\in\mathcal{I}_m}\hat{g}_{u,t,l,i}(\varphi,\theta)\Big[C_{h_x}\Big(x-X_{i2},I_{i2}\Big)-C_{h_x}\Big(x-X_{i1},I_{i1}\Big)\Big], \tag{99}$$

which replaces each of the two products $E[\,\cdot\mid X_j=x]f_{X_j}(x)$ constituting $\nu_{g_{u,t,l}}$ by its C_{h_x} kernel estimate, exactly as Definition 3 prescribes. Then

$$\begin{aligned}\hat{\mathcal{A}}_{\varphi}h(u,t,l) \;&:=\; \int_{\mathcal{X}}\frac{\hat{\nu}_{u,t,l}(x)}{\pi(x)}\,h(x)\,\pi(x)\,dx \;=\; \int_{\mathcal{X}}\hat{\nu}_{u,t,l}(x)\,h(x)\,dx\\ &=\; \frac{1}{n_{\mathrm{m}}}\sum_{i\in\mathcal{I}_m}\hat{g}_{u,t,l,i}(\varphi,\theta)\,\Delta_{h_x}h_i,\end{aligned} \tag{100}$$

with kernel $\hat{\mathsf{k}}_{\varphi}(u,t,l;x) := \hat{\nu}_{u,t,l}(x)/\pi(x)$ in the $L^2(\pi)$ pairing, and

$$\hat{\mathcal{A}}^{*}_{\varphi}\Xi(x) \;:=\; \frac{1}{\pi(x)}\sum_{l=1}^{\mathsf{L}-1}\varpi_{\mathsf{w}}(l)\int_{\mathbb{R}\times\mathcal{T}}\hat{\nu}_{u,t,l}(x)\,\Xi(u,t,l)\,\Pi_u(du)\,\Pi_t(dt), \qquad \Xi\in L^2(\Pi). \tag{101}$$

Where the first three arguments are suppressed, $\hat{g}_{\cdot,\cdot,\cdot,i}(\varphi,\theta)$ denotes the element $(u,t,l)\mapsto\hat{g}_{u,t,l,i}(\varphi,\theta)$ of $L^2(\Pi)$; the third dot is the contrast index, and at $\mathsf{L}=2$ it stands for the single value $l=1$, so the glyph is the two-dot one earlier rounds printed. *The sum over* l *against the masses* $\varpi_{\mathsf{w}}(l)$ *is Definition 5's pairing and is what makes* (101) *the adjoint of* (100); it is the

sample counterpart of Lemma 48(i), and it is the third factor of Π written out rather than folded into the measure. *At* $\mathsf{L} = 2$ *there is one term,* $\varpi_{\mathsf{w}}(1) = 1$*, and* (98)*,* (99)*,* (100) *and* (101) *are the displays earlier rounds printed* under the canonical identification of $\mathbb{R} \times \mathcal{T} \times \{1\}$ with $\mathbb{R} \times \mathcal{T}$. *The generic element of* $L^2(\Pi)$ *is* Ξ*, as fixed in Remark 104*(1)*, and not* α, which is **binding** as the propensity rate exponent $r/(2r + d_Z)$ in (P-a.2), (P-f), C5′(i), A5(iii), (7), (54) and (181). As in Definition 3, $\hat{\mathcal{A}}_{\mathrm{cond}} := \hat{\mathcal{A}}_{\varphi_\dagger}$ and $\hat{\mathcal{A}}^*_{\mathrm{cond}} := \hat{\mathcal{A}}^*_{\varphi_\dagger}$.

Two features of (100) are worth naming because they are not accidents. First, the reference density cancels: the operator is $h \mapsto \int \hat{\nu}_{u,t,l} h \, dx$ and is therefore *computable without knowing* π, which the population identity $\langle \nu_g/\pi, h \rangle_{L^2(\pi)} = \int \nu_g h \, dx$ of (P5) already predicted; only its *kernel* with respect to $\pi \, dx$ carries π^{-1}, which is the same bookkeeping Remark 37 records for A2. *That is a statement about* (100) *and it does not extend to* (101)*, which is the operator the estimator actually iterates*: the adjoint carries $1/\pi(x)$ in front of the whole expression, π is an unknown feature of P by Definition 5, and *every* Landweber step of Definition 22(b) is taken through $\hat{\mathcal{A}}^*_\varphi$. So the forward operator is free of π, the iteration is not, and the analyst forms (101) at the floored estimate $\hat{\pi}$ that Definition 22 fixes. *The cancellation above is true and, read by itself, misleading*, which is why the qualification is printed here, four lines from the display it qualifies, rather than left to a reader who has just been told that π need not be known. Second, (100) depends on h through $\Delta_{h_x} h$ and through nothing else, which is what makes it a bounded operator and is the whole content of Proposition 70.

Proposition 70 (What $\hat{\mathcal{A}}_\varphi$ is, and what it is not)**.** *Suppose A0, B1(i), (ii), (iv.a), (iv.b), (iv.c), (iv.d), C1(ii) and C3 hold, and let* $\hat{\mathcal{A}}_\varphi$ *be* (100)*; for clause (v) suppose in addition A4(i), so that* (95) *holds. Then, conditionally on the sample, and for every* $\varphi \in \mathcal{V}$ *and* $\theta \in \mathbb{R}^p$*:*

(i) Boundedness. $\hat{\mathcal{A}}_\varphi : \mathcal{V} \to L^2(\Pi)$ *is a bounded linear operator with* $\|\hat{\mathcal{A}}_\varphi\| \le 2\,(2c_C)^{1/2}\,\mathsf{c}_2^2\,c_\Pi\,\|1/\pi\|_\infty^{1/2}\,h_u^{-1}h_x^{-1/2}$. *This constant does not move with* L*, and the derivation is per cell, which is what* (35) *asserts of it. At each fixed* l *the estimate of*

$|\hat{\mathcal{A}}_\varphi h(u,t,l)|$ *reads the contrast only through* $|\widehat{\widetilde{W}}_{i,l}| \le 1$ *of* (95), the per-cell bound, not the Euclidean bound $\|\widehat{\widetilde{W}}_i\| \le c_{\widetilde{W}}$, which would not reproduce this constant; and the **free upper half** of that display and not its lower half, the upper bound being (P1) read at the l-th contrast, which holds at every ϱ. **So this clause reads no overlap condition**, and A4(i), which the hypothesis line above scopes to clause (v) alone, is correctly absent from it, *and the resulting bound on* $\int_{\mathbb{R}\times\mathcal{T}} |\hat{\mathcal{A}}_\varphi h(u,t,l)|^2 \Pi_u(du)\Pi_t(dt)$ *is the same at every* l*, the kernel and the Bierens factor in* (98) *being* l*-free. Summing over* l *against the masses therefore contributes* $\sum_l \varpi_{\mathsf{w}}(l) = 1$*,* Π_{w} *being a probability measure by B1(iv.d), and* c_Π *of B1(iv.c) is unchanged for the same reason. At* $\mathsf{L} = 2$ *the sum has one term and the derivation is the printed one.*

(i′) The same estimate, in the two pairings the estimator puts in play, and neither implies the other. *The bound behind* (i) *is primitively a bound out of* $L^2(\mathcal{X}, dx)$*, and* as such it reads no condition on π*:*

$$\left\|\hat{\mathcal{A}}_\varphi h\right\|_{L^2(\Pi)} \;\le\; 2\,(2c_C)^{1/2}\,\mathsf{c}_2^2\,c_\Pi\,h_u^{-1}h_x^{-1/2}\,\|h\|_{L^2(\mathcal{X},dx)} \qquad \text{for every } h \in L^2(\mathcal{X},dx). \tag{102}$$

Clause (i) *is* (102) *read in* $L^2(\pi)$*, at the conversion cost* $\|1/\pi\|_\infty^{1/2}$ *that C1(ii) makes finite.* For any measurable weight $\hat{\pi}$ on $\mathcal{X}$ with $\hat{\pi} \ge \varepsilon_\pi > 0$ Lebesgue-almost everywhere — *in particular the floored estimate Definition 22 fixes, which is* ***not*** *required to be a probability density, only the pointwise floor being used — the same estimate read in* $L^2(\hat{\pi})$ *costs* $\varepsilon_\pi^{-1/2}$ *in place of* $\|1/\pi\|_\infty^{1/2}$*:*

$$\left\|\hat{\mathcal{A}}_\varphi\right\|_{L^2(\hat{\pi})\to L^2(\Pi)} \;\le\; 2\,(2c_C)^{1/2}\,\mathsf{c}_2^2\,c_\Pi\,\varepsilon_\pi^{-1/2}\,h_u^{-1}h_x^{-1/2}. \tag{103}$$

The two readings do not imply one another, and the discrepancy is unbounded in ε_π*: at the constant weight* $\hat{\pi} \equiv \varepsilon_\pi$*, which is the very function Definition 22's non-emptiness argument exhibits and which the construction convention there admits,* $\|h\|^2_{L^2(\hat{\pi})} =$

$\varepsilon_\pi \|h\|^2_{L^2(\mathcal{X},dx)}$ *exactly, whence*

$$\left\|\hat{\mathcal{A}}_\varphi\right\|_{L^2(\hat{\pi})\to L^2(\Pi)} \;\geq\; \left(\varepsilon_\pi \,\|1/\pi\|_\infty\right)^{-1/2} \left\|\hat{\mathcal{A}}_\varphi\right\|_{L^2(\pi)\to L^2(\Pi)}, \tag{104}$$

and the right-hand side diverges as $\varepsilon_\pi \downarrow 0$ *at a fixed sample on which* $\hat{\mathcal{A}}_\varphi \neq 0$. So a scale calibrated to one of the two pairings is not a scale for the other*, which is why* (35) *carries the* ***maximum*** *of the two conversion costs and why Definition 22 can then assert* $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi\| \leq 1$ *surely in both.* This clause is stated here and not inside Definition 22 on Remark 58's test*: a definition may* fix *a constant, but the inequality that constant has to satisfy is a claim and belongs to a result.* And it consumes **less** than (i) does*: C1(ii) is read by neither* (102) *nor* (103)*, and only by* (104)*, which compares the two.*

(ii) Finite rank. $\operatorname{rank}\hat{\mathcal{A}}_\varphi \leq n_{\mathrm{m}}$*, so* $\hat{\mathcal{A}}_\varphi$ *is Hilbert–Schmidt, and* $\ker\hat{\mathcal{A}}_\varphi$ *has infinite dimension.* The rank bound is n_{m} and not $n_{\mathrm{m}}(\mathsf{L}-1)$*: by* (100) *the operator factors through the* n_{m} *functionals* $h \mapsto \Delta_{h_x} h_i$*, which do not depend on* l*, so the contrast index enlarges the range space and not the rank.*

(iii) The adjoint maps into $\mathcal{V}$. $\int_{\mathcal{X}} \hat{\nu}_{u,t,l}(x)\,dx = 0$ *for every* (u,t,l)*, hence* $\hat{\mathcal{A}}^*_\varphi \Xi \in \mathcal{V}$ *for every* $\Xi \in L^2(\Pi)$*, and* (101) *is the adjoint of* (100) *in the* $L^2(\pi)$*–*$L^2(\Pi)$ *pairing.* This is what C3(ii)'s boundary index is for*, through Lemma 67(iv); without it the identity fails at* every *design point, by an amount that is* $O(1)$ *only within* $O(h_x)$ *of an endpoint of* $\mathcal{X}$ *and vanishes with* h_x *elsewhere, and* $\hat{\mathcal{A}}^*_\varphi$ *leaves the parameter space.*

(iv) It is the exact Fréchet derivative of $\hat{T}_{h_x}$*. The map* $\varphi \mapsto \hat{T}_{h_x}(\varphi,\theta)$ *is twice Fréchet differentiable on* $\mathcal{V}$ *with* $\partial_\varphi \hat{T}_{h_x}(\varphi,\theta) = \hat{\mathcal{A}}_\varphi$ *and, for all* $\varphi, \varphi+v \in \mathcal{V}$*,*

$$\left\|\hat{T}_{h_x}(\varphi+v,\theta) - \hat{T}_{h_x}(\varphi,\theta) - \hat{\mathcal{A}}_\varphi v\right\|_{L^2(\Pi)} \;\leq\; \frac{\mathsf{c}_4\, c_\Pi}{2\,h_u^2} \cdot \frac{\|\Delta_{h_x} v\|_2^2}{n_{\mathrm{m}}}. \tag{105}$$

(105) *is sharper than the closest antecedent, and the sharpening is exactly what generates*

(P-f). Centorrino, Fève, and Florens (2025, Prop. 4.2(a)) establish the same quadratic-remainder property for their sample operator with a constant written only as "$\hat{M} < \infty$"; (105) exhibits that constant, and its bandwidth factor is h_u^{-2}. An unnamed finite constant and a constant diverging like h_u^{-2} support different theorems*: the second is what (176) carries into the stagewise noise level and hence what forces (P-f) and, through it, (55). A treatment that leaves the constant unnamed cannot see that condition.*

(v) It is not, and cannot be, a derivative of $\hat{T}$, the reading Definition 3 rejects is unavailable, not merely different. The Gateaux derivative of $\hat{T}(\cdot,\theta)$ at φ in the direction v exists for every $v \in \mathcal{V}$ and equals

$$\mathrm{d}\hat{T}_\varphi[v](u,t,l) \;=\; \frac{1}{n_{\mathrm{m}}} \sum_{i \in \mathcal{I}_m} \widehat{\widehat{W}}_{i,l}\, w\Big(t'\Phi(\Delta Z_i)\Big)\, \frac{1}{h_u} C_{h_u}\Big(u - \Delta U_i(\theta,\varphi), \mathbb{R}\Big)\, \Delta v_i, \qquad \Delta v_i = v(X_{i2}) - v(X_{i1}) \tag{106}$$

, the raw *residual and the* raw *difference, which is what distinguishes it from (100), but $v \mapsto \mathrm{d}\hat{T}_\varphi[v]$ is* unbounded on $\mathcal{V}$*: almost surely, the only exception event being a tie among the $2n$ points X_{ij}, which is null because X is continuously distributed by Definition 4, there is a sequence $v_k \in \mathcal{V}$ with $\|v_k\|_{L^2(\pi)} \to 0$ and $\|\mathrm{d}\hat{T}_\varphi[v_k]\|_{L^2(\Pi)} \geq \mathsf{c} > 0$. Consequently $\hat{T}(\cdot,\theta)$ is* nowhere Fréchet differentiable *on $\mathcal{V}$, and no bounded operator is its derivative.* No event involving $\hat{\varrho}$ is excluded here, *because (95) holds surely.*

Proposition 71 (A6 is false, and not only for the operator it names)**.** *Suppose A0, A4(i), A4(iii), B1(i), (ii), (iv), C1(ii), C3 and C5′(i) hold, and, for clause (iii) only, C2′(i), and adopt Definition 2, in particular its fold split and its truncation, so that (95) holds surely. Let $\hat{T}$ be (12) and let $\eta_{\mathrm{TCC}} < 1$. Then, on the event that the $2n$ points $X_{11}, X_{12}, \ldots, X_{n2}$ are pairwise distinct, an event of probability one, because X is continuously distributed by Definition 4, and for all n large enough that $h_u < 1/\big(2c_\Theta(\epsilon)\big)$ for some $\epsilon < \frac{1}{4}$, with c_Θ the envelope-tail radius of Lemma 67(i):*

(i) No bounded operator serves. *For* every $\tilde{\varphi} \in \mathcal{V}$*, every $\bar{\rho} > 0$, and* every bounded linear

operator $\mathcal{Q}:\mathcal{V}\to L^2(\Pi)$ *there exists* $\varphi\in\mathcal{V}$ *with* $\|\varphi-\tilde\varphi\|_{L^2(\pi)}\le\bar\rho$ *and*

$$\left\|\hat T(\varphi,\theta)-\hat T(\tilde\varphi,\theta)-\mathcal{Q}(\varphi-\tilde\varphi)\right\|_{L^2(\Pi)}\;>\;\eta_{\mathrm{TCC}}\left\|\hat T(\varphi,\theta)-\hat T(\tilde\varphi,\theta)\right\|_{L^2(\Pi)}. \tag{107}$$

The candidate linearisation is $\mathcal{Q}$ and not $\mathcal{L}$, *which in Part H names the* one specific *normalised compression* $\omega_n^{-1}\hat{\mathcal{A}}_{\varphi_\circ}|_{\mathcal{S}_n}$ *of Definition 22 and Lemma 85, an operator that is bounded by construction, whereas the whole point of clause (iii) below is a candidate that is* not. $\mathcal{Q}$ *carries no properties beyond those the statement gives it.*

(ii) In particular A6 is false. *Taking* $\mathcal{Q}=\hat{\mathcal{A}}_{\tilde\varphi}$ *of* (100), *which is bounded by Proposition 70(i), the event in A6 has probability* 0, *for every* n *large enough, every* $\eta_{\mathrm{TCC}}<1$, *and every ball of positive* $L^2(\pi)$*-radius.*

(iii) And the one unbounded candidate does not serve either. Let $\mathcal{Q}$ *be the Gateaux derivative* $\mathrm{d}\hat T_{\tilde\varphi}[\cdot]$ *of* (106), *the reading Definition 3 rejects, which by Proposition 70(v) is unbounded and so is* not *covered by (i). Fix* $i^*:=\min\mathcal{I}_m$. *Then* (107) *holds at the same* φ, *for all* n *large enough, at every* $\tilde\varphi$ *for which*

$$\liminf_{h\downarrow0}\;\frac{1}{h}\int_{\mathbb{R}}C_h\Big(u-\Delta U_{i^*}(\theta,\tilde\varphi),1\Big)^2\,\Pi_u(du)\;>\;0. \tag{108}$$

At each fixed $\tilde\varphi$, (108) holds with probability one*: under C2′(i) the law of* $\Delta U_{i^*}(\theta,\tilde\varphi)$ *is absolutely continuous, Lebesgue-almost every point of* $\mathbb{R}$ *is a Lebesgue point of* ϖ_u, *and at such a point the displayed limit equals* $\varpi_u(\Delta U_{i^*})\int\Theta_1^2>0$ *by B1(iv.a),* Θ_1 *being the interior rescaled kernel* (14) *of C3, whose* ***envelope*** *bound C3(ii) makes free of* h — C3(ii) posits an envelope and states no support clause.

A6 is not a strong assumption; it is a false one, and the falsity is a property of $\hat T$ *alone.* Three qualifiers scope this proposition and all three are load-bearing*: it is about* $\hat T$ ***on*** $\mathcal{V}$, ***in the levels parameterisation***, *and* ***off the sieve***. *The sieve qualifier is Remark 72*(3)*;* the parameterisation qualifier is Remark 72(4) and is the sharper of the two, *because in a*

space of derivatives *point evaluation is bounded, Proposition 70(v)'s mechanism fails, and the proposition dissolves, at a cost in the source index that* (55) *will not pay.*

Remark 72 (Why it fails, what the failure is not, and where it stops). (1) The mechanism, in one sentence. A tangential cone condition asks a *second-order* quantity to be dominated by a *first-order* one. Pairing $\hat{T}$, which sees φ through point evaluations, with $\hat{\mathcal{A}}_{\tilde{\varphi}}$, which sees it through the smoothed functionals (96), leaves a *first-order* discrepancy $\mathrm{d}\hat{T}_{\tilde{\varphi}}[v] - \hat{\mathcal{A}}_{\tilde{\varphi}} v$ on the left-hand side, and by Proposition 70(v) that discrepancy is unbounded relative to $\|v\|_{L^2(\pi)}$ while the right-hand side is not bounded below by anything. The two readings Definition 3 separates are not interchangeable, and C3's bandwidth forces the separately-smoothed one; under that reading A6 pairs two objects whose difference is first order. The defect is therefore *in the pairing*, not in the constant η_{TCC} and not in the radius of $\mathcal{V}_n$, which is why Proposition 71 is stated at every $\eta_{\mathrm{TCC}} < 1$ and every $\bar{\rho} > 0$.

(2) The refutation does not depend on the construction, and that matters. Steps 1–4 of the proof use nothing about $\hat{\mathcal{A}}_{\tilde{\varphi}}$ beyond boundedness: a spike of vanishing $L^2(\pi)$-norm is annihilated by *any* bounded operator, while $\hat{T}$ moves by a fixed nonzero amount. *So the obstruction is not that Definition 69 chose the wrong linearisation*; no choice of bounded linearisation whatsoever satisfies A6, and Proposition 71(iii) closes the one unbounded candidate. This is why the disposition is retirement rather than re-specification, and it is what immunises the refutation against the deviation Definition 69 itself flags, that the residual inside (98) carries $\Delta_{h_x}\varphi_i$ and not $\Delta\varphi_i$. That deviation changes which bounded operator is used; Proposition 71(i) quantifies over all of them.

(3) Where the refutation stops: the sieve, and it stops there completely. Proposition 71 is a statement about $L^2(\pi)$-balls in $\mathcal{V}$, and Definition 22 does not run there, it runs on $\mathcal{V}_n \cap \mathcal{S}_n$. On the sieve the spike is not damped; it is not representable. On A8's quasi-uniform mesh the narrowest feature of $\mathcal{S}_n$ has width $\asymp L_x^{-1}$ while the design spacing is $\asymp n^{-1}$, so every nonzero element of $\mathcal{S}_n$ has, in the support of its narrowest feature, of order $n/L_x \gg n^{1/2}$

design points; A8's upper end $L_x \ll n^{1/2}$ is what forces the margin. The inverse inequality $\|\zeta\|_\infty \le \mathsf{c}_{11} L_x^{1/2} \|\zeta\|_{L^2(\pi)}$ of Lemma 81(3d) is the same fact read as a norm bound. So a cone condition restricted to the sieve is untouched by Proposition 71, and it is also unproved. The first-order limit below is taken in A6, the condition on the raw $\hat{T}$, and **not** in (111). Put $\varphi = \tilde{\varphi} + cv$ with $v \in \mathcal{S}_n$ and let $c \downarrow 0$. Since $\hat{T}(\cdot, \theta)$ is Gateaux differentiable in every direction (Proposition 70(v)) and $\hat{\mathcal{A}}_{\tilde{\varphi}}$ is linear, A6's inequality divided by c passes to the limit and becomes $\|\mathrm{d}\hat{T}_{\tilde{\varphi}}[v] - \hat{\mathcal{A}}_{\tilde{\varphi}} v\|_{L^2(\Pi)} \le \eta_{\mathrm{TCC}} \|\mathrm{d}\hat{T}_{\tilde{\varphi}}[v]\|_{L^2(\Pi)}$, which at the base point $\tilde{\varphi} = 0$, where the raw and the smoothed period differences of the base point coincide, both being zero, so that (106) and (100) carry the *same* $\hat{g}_{\cdot,\cdot,\cdot,i}$, is

$$\sup_{v \in \mathcal{S}_n \setminus \{0\}} \frac{\left\| \sum_{i \in \mathcal{I}_m} \hat{g}_{\cdot,\cdot,\cdot,i}(0,\theta) \left(\Delta v_i - \Delta_{h_x} v_i \right) \right\|_{L^2(\Pi)}}{\left\| \sum_{i \in \mathcal{I}_m} \hat{g}_{\cdot,\cdot,\cdot,i}(0,\theta)\, \Delta v_i \right\|_{L^2(\Pi)}} \;\le\; \eta_{\mathrm{TCC}}. \tag{109}$$

Why not (111), and the answer is that there the same limit is empty. By Proposition 70(iv) the operator $\hat{\mathcal{A}}_{\tilde{\varphi}}$ is the **exact** Fréchet derivative of $\hat{T}_{h_x}$, with remainder $O(c^2)$ by (105); (111)'s right-hand side is $O(c)$; so dividing by c and letting $c \downarrow 0$ gives $0 \le \frac{\eta}{1+\eta} \|\hat{\mathcal{A}}_{\tilde{\varphi}} v\|$, which is true of every pair and says nothing. *The pairing defect A6 has and A6b does not is precisely what survives the limit*, and it is why the display must be read off A6. *Two scope statements.* **(a) The base point.** (109) is written at $\tilde{\varphi} = 0$. At a general base point the two displays run on *different* kernels, (106) evaluates C_{h_u} at the raw residual $u - \Delta Y_{i,\theta} + \Delta\tilde{\varphi}_i$ and (98) at the smoothed one $u - \Delta Y_{i,\theta} + \Delta_{h_x}\tilde{\varphi}_i$, and the two arguments differ by $|\Delta\tilde{\varphi}_i - \Delta_{h_x}\tilde{\varphi}_i| \le \mathsf{c}_{12} \mathsf{R}\, h_x^{s-1/2}$, which is (172), *whose justification is the* $\lfloor s - \frac{1}{2} \rfloor$*-th order Taylor expansion of Lemma 81 Step 4 and not the vacuous embedding* $H^s \hookrightarrow C^{0,s-1/2}$, and which therefore also inherits the hypothesis $p_K \ge s - \frac{1}{2}$, and which is $O\left(h_u \hat{\delta}_n\right)$ under (P-g). Since $h_u^{-1}\Theta_1(\cdot/h_u)$ is Lipschitz at scale h_u, the resulting relative displacement of the two families is $O(\hat{\delta}_n)$, so (109) is the exact limit at $\tilde{\varphi} = 0$ and the limit up to that factor elsewhere. **(b) "Only if", not "if and only if".** (109) is *necessary* for a cone condition on $\mathcal{V}_n \cap \mathcal{S}_n$, being its first-order shadow. It is not shown sufficient: recovering A6 at finite c from its first-order limit needs the second-order

remainder (105) to be dominated by $\|\sum_i \hat{g}_{\cdot,\cdot,\cdot,i}\Delta v_i\|$ uniformly on the sieve, which is a *lower* bound on that quantity in terms of $\|v\|_{L^2(\pi)}$, i.e. a bound on $\lambda_{\min}(\mathbf{G}_n(\cdot))$, the same object Proposition 74(iii) declines to supply and Q-3 records as open. The paragraph below declines it too, so the display is stated as an "only if". Two things should be read off (109), and the second is why it is not pursued. First, it is a *first-order* condition and not a curvature condition, the quadratic remainder scales away, so "A6 on the sieve" is not the tangential cone condition restricted but the statement that *x-smoothing acts as a near-identity on $\mathcal{S}_n$*. Second, (109) *turns on the product $h_x L_x$, which no assumption in this block constrains*: C5′ places no upper bound on h_x (see (P-g)) and A8 no joint restriction with it, and the block's admissible window at the frontier-clearing configuration $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$ leaves $h_x L_x$ free over a range containing both $n^{-0.21} \to 0$ and $n^{0.19} \to \infty$, *both verdicts*. Establishing it would cost *a new joint budget line $h_x L_x = O(\eta_{\mathrm{TCC}})$ plus* a lower bound on $\lambda_{\min}(\mathbf{G}_n)$, which is the same object Proposition 74(iii) declines to bound and Q-3 records as open. Neither is charged here, because Proposition 28 needs no cone condition at all.

(4) Where Proposition 71 stops, second reading: the **parameterisation**. And this is the one that matters for the programme, because the parameterisation that dissolves the proposition is exactly the one that cannot feed Theorem 29. Clause (3) records that the refutation is a statement about $\hat{T}$ on $\mathcal{V}$ and says nothing about $\mathcal{V}_n \cap \mathcal{S}_n$. There is a second, sharper limit of the same kind, and it is a limit on the *space* rather than on the sieve. Proposition 70(v) refutes Fréchet differentiability of $\hat{T}$ because $\hat{T}$ sees φ through point evaluations and point evaluation is unbounded on $L^2(\pi)$. Centorrino, Fève, and Florens (2025) face the same construction and *do* assume differentiability of their sample operator, legitimately, because their parameter space is a space of **derivatives**, in which point evaluation *is* bounded: for φ with $\varphi(\underline{x}) = 0$, $|\varphi(x)| = |\int_{\underline{x}}^{x} \varphi'| \le |\mathcal{X}|^{1/2}\|\varphi'\|_{L^2}$ by the Cauchy–Schwarz inequality. So the mechanism of Proposition 70(v), and with it Proposition 71, dissolves under the derivative parameterisation. *And it cannot be adopted*: by Lemma 47 the range condition A2 is the same in the two parameterisations, but the *source index* is not, composing with $\mathcal{D}^{-1}$ raises the

degree of ill-posedness from a to $a+1$ and lowers the index from $s/(2a)$ to $(s-1)/(2(a+1))$, which at the frontier-clearing configuration $(s,a)=(3,\frac{3}{4})$ is $\mu=0.571$ and **fails** (55). *The scope statement is therefore:* Proposition 71 is a statement about $\hat{T}$ *on* $\mathcal{V}$*, in the levels parameterisation, off the sieve*; each of the three qualifiers can be removed, and removing any one of them costs something the programme needs, the sieve costs $h_x L_x$ and $\lambda_{\min}(\mathbf{G}_n)$ (clause (3)), and the parameterisation costs $\frac{8}{3}$ units of s, uniformly in a, which (55) will not pay.

(5) One sentence about Proposition 76's reach, because it is narrower than its title suggests. Range invariance *implies* the relative-ill-posedness condition along the path that Centorrino, Fève, and Florens (2025, Assumption (18)) run their nonlinear rate on, and the implication does **not** reverse. *So Proposition 76 spends the range-invariance route and leaves the weaker condition open*: a tangential-cone-free treatment built on (18) rather than on A6 is not closed by anything in this file. It would still need a lower bound on $\lambda_{\min}(\mathbf{G}_n)$ (Q-3), which is why it is recorded as the most promising unexplored route in the neighbourhood and not as an omission.

Definition 73 (A6b: restricted curvature (a *property*, not maintained))**.** For $\varphi_\circ \in \mathcal{V}$ and $\rho > 0$ write

$$\mathcal{V}(\varphi_\circ, \rho) \;:=\; \Big\{\varphi \in \mathcal{V} :\; \|\varphi - \varphi_\circ\|_{L^2(\pi)} \le \rho\Big\} \tag{110}$$

for the closed $L^2(\pi)$-ball in $\mathcal{V}$. *The two arguments are the separation*: bare $\mathcal{V}_n$ is reserved, throughout Part H, for the *source-condition* ball (120), which is a different object and is not a ball of $L^2(\pi)$ at all.

Let $\eta \in (0, \frac{1}{2})$. The pair $(\hat{T}_{h_x}, \hat{\mathcal{A}})$ *satisfies A6b at* $(\eta, \varphi_\circ, \rho)$ if

$$\big\|\hat{T}_{h_x}(\varphi,\theta) - \hat{T}_{h_x}(\tilde{\varphi},\theta) - \hat{\mathcal{A}}_{\tilde{\varphi}}(\varphi-\tilde{\varphi})\big\|_{L^2(\Pi)} \;\le\; \frac{\eta}{1+\eta}\,\big\|\hat{\mathcal{A}}_{\tilde{\varphi}}(\varphi-\tilde{\varphi})\big\|_{L^2(\Pi)} \qquad \text{for all } \varphi, \tilde{\varphi} \in \mathcal{V}(\varphi_\circ, \rho). \tag{111}$$

Note the quantifier: (111) ranges over *all pairs* in the ball, so the base point $\tilde{\varphi}$ at which $\hat{\mathcal{A}}$ is evaluated varies too. That is what Proposition 74(ii) must deliver, and it is why its

hypothesis is a bound on $\lambda_{\min}(\mathbf{G}_n(\cdot))$ *uniform over base points* and not at one point.

A6b is NOT MAINTAINED, on the A2c model: it is a property that a pair may or may not have, it has exactly one consumer (Proposition 74), it is never asserted, and *no theorem in this programme uses it*, Proposition 28 takes a route that needs no cone condition at all. It is stated so that the paper can say what A6 should have said, and so that the simulation grid has a testable object (probe CP-8).

Proposition 74 (A6b implies a tangential cone condition, and holds on an explicit ball)**.** *Suppose the hypotheses of Proposition 70 hold, A4(i) so that (95) holds, fix θ and $\varphi_\circ \in \mathcal{V}$, and for $\varphi \in \mathcal{V}$ let*

$$\mathbf{G}_n(\varphi) \;:=\; \Big(\big\langle \hat{g}_{\cdot,\cdot,\cdot,i}(\varphi,\theta),\, \hat{g}_{\cdot,\cdot,\cdot,k}(\varphi,\theta)\big\rangle_{L^2(\Pi)}\Big)_{i,k\in\mathcal{I}_m} \tag{112}$$

be the Gram matrix of the n_{m} functions (98) in $L^2(\Pi)$, at base point φ. Then:

(i) A6b delivers A6 for the consistent pair. If $(\hat{T}_{h_x}, \hat{\mathcal{A}})$ satisfies A6b at $(\eta, \varphi_\circ, \rho)$ then A6 holds verbatim on $\mathcal{V}(\varphi_\circ, \rho)$ with $\hat{T}$ replaced by $\hat{T}_{h_x}$ and $\eta_{\mathrm{TCC}} = \eta$, deterministically on the same event; and, on that event, for all $\varphi, \tilde{\varphi} \in \mathcal{V}(\varphi_\circ, \rho)$,

$$(1-\eta)\big\|\hat{T}_{h_x}(\varphi,\theta) - \hat{T}_{h_x}(\tilde{\varphi},\theta)\big\| \;\le\; \big\|\hat{\mathcal{A}}_{\tilde{\varphi}}(\varphi-\tilde{\varphi})\big\| \;\le\; (1+\eta)\big\|\hat{T}_{h_x}(\varphi,\theta) - \hat{T}_{h_x}(\tilde{\varphi},\theta)\big\|.$$

(ii) A sufficient condition, with an explicit radius. Fix $\rho_0 > 0$ and suppose

$$\underline{\lambda}_n \;:=\; \inf_{\varphi\in\mathcal{V}(\varphi_\circ,\rho_0)} \lambda_{\min}\Big(\mathbf{G}_n(\varphi)\Big) \;>\; 0, \tag{113}$$

and let

$$\bar{\rho}_n \;:=\; \frac{\eta}{1+\eta}\cdot\frac{h_u^2\, h_x^{1/2}}{(2c_C)^{1/2}\,\mathsf{c}_2\,\mathsf{c}_4\,c_\Pi\,\|1/\pi\|_\infty^{1/2}}\cdot\frac{\underline{\lambda}_n^{1/2}}{n_{\mathrm{m}}^{1/2}}. \tag{114}$$

Then $(\hat{T}_{h_x}, \hat{\mathcal{A}})$ satisfies A6b at $\big(\eta, \varphi_\circ, \frac{1}{2}\min\{\bar{\rho}_n, \rho_0\}\big)$. The halving is not cosmetic*:* (111)

quantifies over pairs, so two points of a ball of radius $\frac{1}{2}\bar{\rho}_n$ *are at distance at most* $\bar{\rho}_n$*, which is what* (114) *governs; and the base point of* $\hat{\mathcal{A}}_{\tilde{\varphi}}$ *moves with the pair, which is what* (113)*'s infimum pays for. A bound at the single base point* $\varphi_\circ$ *does* not *suffice, and rounds before this one proved* (111) *only there.*

(iii) And the radius is governed by the smallest singular value, which is where the ill-posedness sits. At each φ*,* $n_{\mathrm{m}}^{-1}\lambda_{\min}(\mathbf{G}_n(\varphi))^{1/2}$ *is exactly the smallest ratio* $\|\hat{\mathcal{A}}_\varphi v\|_{L^2(\Pi)}\big/\|\Delta_{h_x} v\|_2$*, i.e. the injectivity modulus of* $\hat{\mathcal{A}}_\varphi$ *read in the* Δ_{h_x} *coordinates, so* (114) *shrinks with the ill-posedness of the empirical linearisation.* No conclusion of this file rests on $\bar{\rho}_n$ being bounded away from zero, and none is claimed*; whether it is, is a design question about the conditioning of* $\mathbf{G}_n$*, the same species of question A8's upper end answers for the balancing solve, and it is the object probe CP-8 should report,* at pairs not anchored at the centre. One degeneracy is excluded by construction and would otherwise be common*: if* $\widehat{\widetilde{W}}_{i,l} = 0$ *for some* i *and* every l *then* $\hat{g}_{\cdot,\cdot,\cdot,i} \equiv 0$*, row and column* i *of* $\mathbf{G}_n(\varphi)$ *vanish and* $\lambda_{\min} = 0$ *exactly, so* (114) *returns* $\bar{\rho}_n = 0$ *and (ii) is vacuous.* (95) *rules this out* surely*, and* a fortiori at $\mathsf{L} \geq 3$*: it bounds every component away from zero, while the degeneracy needs all of them to vanish at once. It does not rule out near-collinearity of the* $\hat{g}_{\cdot,\cdot,\cdot,i}$*, which is the residual content of Q-3.*

Remark 75 (The disposition of A6 and A7). *The operator is exhibited* at (100), with $\hat{T}$ at (12). Proposition 70 establishes that the exhibited object is bounded, finite rank, adjointable into $\mathcal{V}$, and is the exact Fréchet derivative of the x-smoothed operator $\hat{T}_{h_x}$, so Definition 3's reading is *realised*, and its rejected alternative is proved unavailable rather than merely declined.

A6 is not verified; it is refuted, by Proposition 71. Three consequences.

(1) A6 has left the maintained block. An assumption whose event has probability zero cannot be maintained. It is not repairable by lowering η_{TCC}, by shrinking the ball, or by re-specifying $\hat{\mathcal{A}}$: Proposition 71 quantifies over all three. The retired statement is kept, struck, at A6. *Its scope is recorded with it*: refuted on $L^2(\pi)$-balls, untouched on

$\mathcal{V}_n \cap \mathcal{S}_n$, where the question is (109) and is open, see Remark 72(3).

(2) *What A6 should have said is A6b*, Definition 73, which pairs $\hat{\mathcal{A}}$ with the operator it actually differentiates. Proposition 74 proves A6b non-vacuous and gives an explicit radius. A6b is a *property*, not maintained, exactly as A2b and A2c are.

(3) *Nothing downstream needs either.* Proposition 28 is proved without A6 and without A6b, by a route that treats the nonlinearity as an addition to the noise level rather than controlling it by a cone condition; Remark 88 explains why that route is available here and why the inherited one is not. The range-invariance route is spent at Proposition 76: it fails, and it fails for a reason that is proved rather than reported.

A7 is verified in an amended form, by Proposition 79: at a fixed τ_{DP} for the component of the noise generated by the smoothed indicator and the empirical average, and *not* at a fixed τ_{DP} for the component generated by $\hat{\varrho}$, because A4(iii) is an O_P statement and an O_P statement carries no fixed-constant exceedance bound. The amendment (13) does not cost nothing. It is free against (7), whose strictness absorbs the extra $(\ln n)^{2\mu/(2\mu+1)}$ (Proposition 79(iv)); it is *not* free against (P-f), which is an exact domination in which both sides now carry logarithms, and it is what turns (P-f)'s thresholds strict, see Remark 125.

Proposition 76 (The range-invariance route, spent)**.** *Range invariance in the sense of Hanke, Neubauer, and Scherzer (1995, (2.6)–(2.7)) would require a family of bounded operators* $\mathcal{O}(\varphi, \tilde{\varphi})$ *on* $L^2(\Pi)$*, with* $\|\mathcal{O}(\varphi, \tilde{\varphi}) - \mathrm{Id}\| \le \mathsf{c}\|\varphi - \tilde{\varphi}\|_{L^2(\pi)}$*, such that* $\hat{\mathcal{A}}_\varphi = \mathcal{O}(\varphi, \tilde{\varphi})\, \hat{\mathcal{A}}_{\tilde{\varphi}}$ *for all* $\varphi, \tilde{\varphi}$ *in a ball* $\mathcal{V}(\tilde{\varphi}, \rho)$ *of* (110)*. The intertwining family is written* $\mathcal{O}$*, and neither* Ω *nor* $\mathcal{R}$*. Calligraphic* $\mathcal{R}(\cdot)$ *is the* range *throughout this project, it appears in the next clause four lines below, and sans-serif* R *is already the source radius of* (120)*.* And Ω is the conditional contrast covariance (2), an $\mathbb{R}^{(\mathsf{L}-1)\times(\mathsf{L}-1)}$ matrix, which (62) carries as a prefactor and Remark 35 inverts*: that is the only object the symbol names anywhere in this corpus, so no disambiguation rule attaches to it, and the one that formerly did, the two families having been separated by type and not by arity, both being written bare at some sites, is discharged*

by the renaming performed here. $\mathcal{O}$ is free: it occurs at no site in the corpus outside this proposition and its proof. It is not Landau's O, *which this project writes italic and never calligraphic.* Wherever it **denotes** a member of the family it carries its two arguments*; the bare occurrences in this paragraph name the symbol rather than apply it, and the only bare* applications *are inside the proof, where the opening line binds $\mathcal{O} := \mathcal{O}(\varphi, \varphi')$ once and the abbreviation runs to the end of that proof and no further. Suppose the hypotheses of Proposition 70, A4(i) so that (95) holds, and that*

$$\rho \;>\; \mathsf{c}_9\, \mathsf{D}_n\, h_x^{1/2}, \qquad \mathsf{c}_9 := 2\Big(\|\pi\|_\infty / \mathsf{c}_{13}\Big)^{1/2}, \tag{115}$$

with c_{13} the uniform lower bound (168) on $\int_I \Theta_I^2$, where $\mathsf{D}_n := \max_{i,k\in\mathcal{I}_m} \big|(\Delta Y_{i,\theta} - \Delta_{h_x}\tilde{\varphi}_i) - (\Delta Y_{k,\theta} - \Delta_{h_x}\tilde{\varphi}_k)\big|$ is the spread of the n_{m} kernel centres, an $O_P(1)$ quantity. Then, almost surely for all n large enough:

(i) range invariance forces $\mathcal{R}(\hat{\mathcal{A}}_\varphi) = \mathcal{R}(\hat{\mathcal{A}}_{\varphi'})$ for every $\varphi, \varphi' \in \mathcal{V}(\tilde{\varphi}, \rho)$, first for pairs with $\mathsf{c}\|\varphi - \varphi'\|_{L^2(\pi)} < 1$, and then for all *pairs, by chaining along the segment joining them, which (110) contains;*

(ii) and that equality fails, provided *the n_{m} design weights $\mathsf{d}_i(x) := h_x^{-1}[C_{h_x}(x - X_{i2}, I_{i2}) - C_{h_x}(x - X_{i1}, I_{i1})]$, $i \in \mathcal{I}_m$, are linearly independent in $L^2(\mathcal{X}, dx)$: there are then $\varphi, \varphi' \in \mathcal{V}(\tilde{\varphi}, \rho)$ with $\mathcal{R}(\hat{\mathcal{A}}_\varphi) \neq \mathcal{R}(\hat{\mathcal{A}}_{\varphi'})$.*

Hence no such family exists, for any $\mathsf{c} < \infty$, on any ball whose radius does not shrink faster than $h_x^{1/2}$. The mechanism is that $\hat{\mathcal{A}}_\varphi$'s range is spanned by kernels translated *in u by $\Delta_{h_x}\varphi_i$, and a sufficiently displaced translate cannot lie in their span.* Displacement, and not disjointness of supports, is what the proof uses*: the translates are linearly independent at distinct centres because the interior kernel's Fourier transform is strictly positive at every real frequency, so the displacement D_n has only to* ***separate*** *the displaced centre from the others and not to clear their whole spread.*

The independence proviso in (ii), and why it is stated rather than read off distinctness of the design. What (ii) needs is that $\hat{g}_{\cdot,\cdot,\cdot,i^*}$ *lie in the* range *of* $\hat{\mathcal{A}}_\varphi$ *and not merely in the span of the* $\hat{g}_{\cdot,\cdot,\cdot,i}$*; by* (100) *that is joint surjectivity of the* n_{m} *functionals* $v \mapsto \Delta_{h_x} v_i = \int_{\mathcal{X}} \mathsf{d}_i v \, dx$ *onto* $\mathbb{R}^{n_{\mathrm{m}}}$*, which is equivalent to linear independence of* $\{\mathsf{d}_i\}$ *in* $L^2(\mathcal{X}, dx)$ *and is unaffected by the constraint* $\int v \, d\pi = 0$ *defining* $\mathcal{V}$*, since* $\int \mathsf{d}_i \, dx = 0$ *makes every functional constant-free.* Distinctness of the $2n$ points X_{ij} is **not** the reason*: the minimum design gap is* $O_P(n^{-2})$*, far below* h_x*, so the bumps overlap heavily and the family is nowhere near a disjointly supported one. Where distinctness genuinely enters is one step further in, and* the route through a common kernel shape is no longer available at all — *a structural fact, and worth stating as one.* Exactness of the $j = 0$ truncated moment at **every** design point is incompatible with a design-point-invariant kernel shape whenever the kernel is not compactly supported. *Under a compactly supported kernel the incompatibility is confined to a boundary layer: at design points further than the support half-width from* $\partial\mathcal{X}$ *every* $\Theta_{I_{ij}}$ *is the one interior kernel, each* d_i *is a difference of two translates of the single function* $x \mapsto h_x^{-1} C_{h_x}(x, \mathbb{R})$*, its image under the Fourier transform* $\mathcal{F}$ *of* (3) *is entire of exponential type and not identically zero, so its zero set is discrete, and* $\sum_i c_i \mathsf{d}_i = 0$ *forces* $\sum_i c_i \left(e^{-\mathrm{i}\xi X_{i2}} - e^{-\mathrm{i}\xi X_{i1}} \right) = 0$ *for every* ξ*, whence* $c \equiv 0$ *by linear independence of the characters* $\xi \mapsto e^{-\mathrm{i}\xi x}$ *at the* $2n$ distinct *frequencies* X_{ij}*. Under C3 as it now stands,* $I(x^*)$ *moves with* x^* *at* every *design point — there is no interior region on which it is constant — so the shape* $\Theta_{I_{ij}}$ *varies everywhere, the convolution structure is lost everywhere, and this sufficient condition is unavailable.* That is the one genuine loss of the envelope route, and it is recorded rather than absorbed. *Three things keep the cost at zero. It is a* qualitative *nonsingularity and not a lower bound on* $\lambda_{\min}$*, so it does not reintroduce the conditioning input Q-3 records as unavailable and Proposition 74(iii) declines to supply. It is a condition on clause (ii) only; clause (i) and the population paragraph below use nothing of the kind. And* no result in this programme consumes Proposition 76, *Proposition 28 needs no cone condition, so it needs no route to one, which is why the proviso is printed as a proviso rather than promoted to a hypothesis of the block. Probe CP-8's grid can report it directly.*

The step this replaces, and why*: an earlier draft produced a direction annihilating* $n_{\mathrm{m}} - 1$ *of the functionals* (96) *and read* (165)*, an* upper *bound on* $|\Delta_{h_x} v_i|$ *given* $\|v\|$*, backwards to bound that direction's norm. The norm of such a direction is governed by the conditioning of* $\{\mathsf{d}_i\}$*, i.e. by the* $\lambda_{\min}$ *information Q-3 records as unavailable. The construction below uses* one *explicit direction,* d_{i^*}/π*, whose norm and whose functional value are both computed directly, and needs no conditioning input at all. The same argument applies verbatim to the population pair* $(T, \mathcal{A}_\varphi)$*, where by* (24) *and the shift identity of (P6) the base point enters* $\mathcal{A}_\varphi$ *as an argument shift of the conditional density.*

Lemma 77 (L-18: the operator error at the fixed truth)**.** *Suppose A0, A1, A4(iii), B1(i), (ii), (iv.c), (iv.d), C2′(ii), C3 and C5′(i) hold, and adopt Definition 68.* **B1(iv) is cited by clause, and this line has now moved twice***: it read "*(iv.c)*" alone, which was short by* (iv.d) *— cited at the* $\sum_l \varpi_{\mathsf{w}}(l) = 1$ *aggregations of Steps 2 and 4 — and the first 2026-08-24 repair promoted it to the whole block, which was long by* (iv.a) *and* (iv.b)*, neither of which any step opens.* The two clauses named here are exactly the two the proof opens*, at Steps 0, 2, 3 and 4 for* (iv.c) *and Steps 2 and 4 for* (iv.d)*; and the factorisation* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ *is B1(iv)'s* declaration *and carries nothing, on the assumption block's* ***three registers*** *paragraph.* Proposition 79 is repaired by this one edit and must not be repaired a second time*: its own line is "the hypotheses of Lemma 77", so it inherited both the gap and the over-charge, and inherits this fix.* **C2′ is cited by clause.** *The single C2′ citation in the proof is at Step 3, to (ii), for* $\mathfrak{m}_{p_K}$*; no step reads (i)'s envelopes.* ***A1b(iv) is not a hypothesis****, here, or at Lemma 78, Lemma 81 and Lemma 94. The reason is B1(i)'s own:* Φ *is* bounded*, so* $|w(t'\Phi(z))| \le \bar{w}(t)$ *uniformly in* z *and every* ΔZ*-integral in Part H is an integral of a bounded summand, with no moment condition whatsoever on* ΔZ*. What A1b(iv) buys is nonsingularity of* $E[\Delta Z \Delta Z']$ *and the* θ*-side moments, and Part H inverts no such matrix; its consumers lie in Parts F and I — Lemma 91 Steps 1 and 3–6, which is clauses* (ii)–(v) *and the reading* (iv.4)*; Lemma 92 Steps 1 and 3; Lemma 93 Steps 0 and 2; Lemma 95 Steps 1, 2, 3, 5 and 6; and Theorem 29 Step 1 — all of which retain it.* The list is illustrative and no

claim of exhaustiveness is made*: the draft this sentence replaces named three of them and was short by at least five.* At Lemma 94 the ground is different and is recorded there*: its own hypothesis (R-b) gives* $E\|\Delta Z\|^4 < \infty$ *and* $E[\Delta U^4] < \infty$*, hence the second moments a fortiori. Then* $T(\varphi_\dagger, \theta_\dagger) = 0$ *and*

$$\left\|\hat{T}(\varphi_\dagger, \theta_\dagger)\right\|_{L^2(\Pi)} = \underbrace{O_P\left(n^{-1/2}\right)}_{\text{empirical}} + \underbrace{O\left(h_u^{p_K}\right)}_{\text{smoothing bias}} + \underbrace{O_P(r_n)}_{\text{propensity}} = O_P\left(\hat{\delta}_n\right), \tag{116}$$

with $\hat{\delta}_n$ *as in A5(i). Moreover the three components are separately controlled:*

(i) Empirical, with no logarithm and with a fixed-constant tail. Writing $\bar{T}$ *for* (12) *with* ϱ *in place of* $\hat{\varrho}$ *and for every* $\mathsf{s} > 0$

$$E\left\|\bar{T}(\varphi_\dagger, \theta_\dagger) - E\bar{T}(\varphi_\dagger, \theta_\dagger)\right\|^2_{L^2(\Pi)} \le \frac{\mathsf{c}_3^2 c_\Pi^2}{n_{\mathrm{m}}}, \qquad P\left(\left\|\bar{T} - E\bar{T}\right\|_{L^2(\Pi)} \ge \mathsf{s}\right) \le \frac{\mathsf{c}_3^2 c_\Pi^2}{n_{\mathrm{m}}\,\mathsf{s}^2}.$$

The $L^2(\Pi)$ norm needs no uniformity in (u, t) and hence no logarithm*, because* $L^2(\Pi)$ *is a Hilbert space and the summands are bounded in* norm*; the logarithm A5(i) carries is what turns the second display into a* fixed-constant *statement at* $\mathsf{s} = \mathsf{c}\, n^{-1/2}\sqrt{\ln n}$, for every fixed $\mathsf{c} > 0$*, where the right-hand side is then* $O(1/\ln n) \to 0$*, and that is where it earns its keep. No exponential inequality and no Hilbert-space concentration result is used; Chebyshev suffices, which matters because no such result is pinnable in this project's bibliography.* Sans-serif c is a generic finite positive constant *and is* ***not*** *A7's discrepancy constant, which never appears without its subscript* τ_{DP}*; Proposition 79(i)'s proof runs the identical Chebyshev step at* $\mathsf{s} = \frac{1}{2}\mathsf{c}\, n^{-1/2}\sqrt{\ln n}$. Bare sans-serif s is a *tail level: a positive number bound by the quantifier of the clause in which it appears, "for every* $\mathsf{s} > 0$*" here, and likewise at Lemma 123*(i)*–(ii), free to take a different value in every such clause and never carried from one result to another. It is separated by* ***font*** *from the two binding italic s's: the Sobolev order s of H1 and the generic order s′ of A8.*

(ii) Smoothing bias. $\|E\bar{T}(\varphi_\dagger,\theta_\dagger) - T(\varphi_\dagger,\theta_\dagger)\|_{L^2(\Pi)} \le \mathsf{c}_5\, c_\Pi\, \mathfrak{m}_{p_K}\, h_u^{p_K}$, *where* $\mathfrak{m}_{p_K} := \max_{k\le p_K}\max_{l\le \mathsf{L}-1}\sup_t \bar{w}(t)^{-1}\sup_u|\partial_u^k\mathcal{M}(u,t,l)| < \infty$ *by C2′(ii); this is* $O(\hat{\delta}_n)$ *under C5′(i) as amended, and* $O(n^{-1/2})$ *under the unamended bandwidth* $h_u \asymp n^{-1/16}$. Only the first is needed*: (116) is a statement at* $\hat{\delta}_n$*, and matching the bias to* $n^{-1/2}$ *rather than to* $\hat{\delta}_n$ *is what C5′(i)'s amendment corrects.*

(iii) Propensity. $\|\hat{T}(\varphi_\dagger,\theta_\dagger) - \bar{T}(\varphi_\dagger,\theta_\dagger)\|_{L^2(\Pi)} \le \mathsf{c}_3 c_\Pi\, n_{\mathrm{m}}^{-1}\sum_{i\in\mathcal{I}_m}\left\|\hat{\varrho}(\Delta Z_i) - \varrho(\Delta Z_i)\right\| = O_P(r_n)$, the norm being Euclidean on $\mathbb{R}^{\mathsf{L}-1}$ and the passage from the slicewise bound to it costing nothing, since $\max_l|\cdot| \le \|\cdot\|$ and $\sum_l \varpi_{\mathsf{w}}(l) = 1$; at $\mathsf{L} = 2$ it is the modulus earlier rounds printed.

Scope, per the memo's own correction to L-18*: this is a statement at the* fixed *truth. It is not sufficient for Proposition 28, which evaluates the residual along a data-dependent path; that is Lemma 81.*

Lemma 78 (The operator-estimation rate, and the re-derivation C5′(ii) asks for)**.** ***Consumers: none.*** *This lemma, together with (117) and (119), is consumed by no result in this file. It is retained for two reasons, both recorded at reading (iv): it is the re-derivation C5′(ii) asks for, and it is the price-tag on the route Step 0 of Proposition 28 declines.* ***And it is the sole consumer of C1(i) in the corpus****, Step 2 below, at two lines and nowhere else, so C1(i) is a hypothesis of this lemma and, transitively, of no theorem. C1 is cited here* in full*: (i) at Step 2 for the x-smoothing bias, (ii) at the Hilbert–Schmidt reduction that opens the proof, for* $\|1/\pi\|_\infty$*. Suppose A0, A4(iii), B1(i), (ii), (iv.a), (iv.c), (iv.d), C1, C2′, C3 and C5′(i) hold.* **Three clauses of B1(iv) and not four, and not the block***:* (iv.b) *is opened nowhere below, and the factorisation* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ *is B1(iv)'s* declaration*, which carries nothing, on the assumption block's* ***three registers*** *paragraph.* (iv.c) is on this line although a draft of the 2026-08-24 pass counted only (iv.a) and (iv.d)*: Step 1's variance bound and Step 2's bias bound both integrate* $\bar{w}^2$ *against* Π_t*, and the stated order* $O\big((nh_xh_u)^{-1}\big)$ *needs that integral finite, which is* (iv.c) *and only* (iv.c)*. The clause was* consumed *at two steps*

and named *at neither until that day;* Step 1 was repaired then and **Step 2 was not**, and named nothing until 2026-08-26. Both steps now name it, *and the sentence this replaces said the repair was complete when it covered one of the two consuming steps — the same shape of defect, one level down, as the under-charging it was written to record.* **B1(iv.a) is on that line for a reason, and neither promotion retires the reason**: *Step 1 of the proof writes* $\Pi_u(du) = \varpi_u(u)\,du$ *and reads* $\|\varpi_u\|_\infty < \infty$, *which is B1(iv.a) and only B1(iv.a), it is what turns the* Π_u*-integral of* $C_{h_u}^2$ *into a du-integral and produces the* h_u^{-1} *of* (117)*'s variance term.* The same clause reaches the step a second time through (P7), *whose own hypothesis line is B1(iv.a) and C2′(i).* A1b(iv) is **not** a hypothesis, *on the ground recorded at Lemma 77: no step below reads a moment of* ΔZ, *B1(i)'s boundedness of* Φ *making every* ΔZ*-integral here an integral of a bounded summand.* C5′ is cited **by clause**, and the narrowing is not cosmetic: *what this lemma consumes in the x-direction is the proviso it states for itself below,* $(nh_xh_u)^{-1}\ln n \to 0$, *while C5′(ii)'s stronger form is a strictly higher floor for* h_x, *and* a floor stated higher than any proof needs stops being free the moment it must meet a ceiling, *which is* (63) *at C5′. Clauses (iii) and (iv) are a* cross-reference *and a* statement of scope, *and nothing here reads them. Define*

$$\gamma_n \;:=\; \underbrace{\big(n\,h_x\,h_u\big)^{-1/2}\sqrt{\ln n}}_{\text{variance}} \;+\; \underbrace{h_x^{p_K}+h_u^{p_K}}_{\text{smoothing bias}} \;+\; \underbrace{r_n\,\big(h_x\,h_u\big)^{-1/2}}_{\text{propensity}}. \tag{117}$$

The propensity term carries the bandwidth factor and the first term is therefore never binding, *since* $r_n \geq n^{-1/2}$ *by A4(iii) and* (7). The factor $(h_xh_u)^{-1/2}$ is a triangle-inequality bound and is very likely improvable: *the mean of that term is a* ν*-functional of an integrable function and is* $O(r_n)$ *with no bandwidth at all, and only its fluctuation carries the factor, which a sup-norm rate for* $\hat{\varrho}$ *would remove, but* nothing in the block prices $\|\hat{\varrho}-\varrho\|_\infty$, *as the glossary's* r_n *row records, so the crude bound is carried. This is recorded, not claimed away. Then, with*

$\mathcal{V}_n$ *the source-condition ball* (120) *below and* $\mathcal{S}_n$ *the spline sieve of A8,*

$$\sup_{\varphi\in\mathcal{V}_n\cap\mathcal{S}_n}\big\|\hat{\mathcal{A}}_\varphi-\mathcal{A}_\varphi\big\|_{\mathcal{V}\to L^2(\Pi)}\;=\;O_P(\gamma_n),\tag{118}$$

provided $\big(n\,h_x\,h_u\big)^{-1}\ln n\to 0$, $\mathsf{R}\,h_x^{s-1/2}=o\big(h_u\,\gamma_n\big)$, *which aligns the base points,* $\hat{\mathcal{A}}_\varphi$ *evaluating the residual at* $\Delta_{h_x}\varphi$ *and* $\mathcal{A}_\varphi$ *at* $\Delta\varphi$, and the factor h_u is there for the reason Lemma 81's own proviso carries one*: the alignment is charged in* ***operator*** *norm and therefore at* $h_u^{-1}\mathsf{R}\,h_x^{s-1/2}$, *which is* $o(\gamma_n)$ *exactly when the displayed condition holds, and*

$$L_x\,\ln n\;=\;o\big(n\,h_x\,h_u\,\gamma_n^2\big).\tag{119}$$

That the supremum in (118) is a random variable at all is Lemma 81(3c)'s separability paragraph, read at the subset $\mathcal{V}_n\cap\mathcal{S}_n$*; Step 4 of the proof below does the reading, and the claim is stated once, there, rather than repeated here.* **The supremum in** (118) **is nested, and that paragraph covers the outer one only***:* $\|\hat{\mathcal{A}}_\varphi-\mathcal{A}_\varphi\|_{\mathcal{V}\to L^2(\Pi)}$ *is itself a supremum over the* uncountable *unit ball of* $\mathcal{V}$*, so at each fixed* φ *it is not a random variable on the strength of the definition alone.* The ingredient is present and was uncited*:* $\mathcal{V}$ *is a subspace of* $L^2(\mathcal{X},\pi)$ *with* $\mathcal{X}$ *compact, hence separable, and* $\zeta\mapsto\|(\hat{\mathcal{A}}_\varphi-\mathcal{A}_\varphi)\zeta\|_{L^2(\Pi)}$ *is continuous in* $\|\cdot\|_{L^2(\pi)}$ *at every sample point, both operators being bounded there, Proposition 70(i) for the first, Proposition 9 for the second, so the inner supremum equals the supremum over a fixed countable dense subset of the unit ball, and each of those is a random variable because* $(\hat{\mathcal{A}}_\varphi-\mathcal{A}_\varphi)\zeta$ *is jointly measurable in* (ω,u,t) *by Definition 22's measurability verification, first step and* $L^2(\Pi)$ *is separable. The two suprema are then countable one after the other, and* (118) *is a statement about a random variable.* This is a formality with a bounded blast radius, and it is stated rather than left*: this lemma declares* ***Consumers: none****, so no result inherits the gap, but a display that is not defined is not a display. The supremum is taken over* $\mathcal{V}_n\cap\mathcal{S}_n$ *and not over* $\mathcal{V}_n$*, and the restriction removes a budget line. On the sieve the covering numbers are elementary, the volumetric bound of Lemma 81(3d), so* (119) *is a*

condition on L_x and rests on no classical entropy theorem; over all of $\mathcal{V}_n$ the chaining step would consume the imported *sup-norm entropy of a Sobolev ball, item (F-e) of the citation note, and would in addition require $\mathsf{R}\, h_x^{s-1/2} = o\big(h_u\, \gamma_n\big)$ to align the two operators' base points.* The restriction costs nothing even prospectively*: every iterate of Definition 22(b) lies in $\mathcal{S}_n$ by construction, so a later result that did consume this lemma would evaluate it only there.* No result consumes it now*, and reading (iv) below says why the lemma is nonetheless retained. Three readings, and the third is the deliverable.*

(i) The exponents in (117) are one *on h_x and* one *on h_u, not two and three. C5′(ii)'s $(nh_x^2h_u^3)^{-1}\ln n \to 0$ is inherited from a density-*derivative *estimator in the cross-sectional companion; $\hat{\mathcal{A}}_\varphi$ estimates a density-weighted conditional mean and carries one bandwidth in each direction.* C5′(ii) **is** the condition used here as of this dispatch*: it previously stated the $h_x^2h_u^3$ pairing and now states the h_xh_u one, retaining the stronger form only where (63) makes it compatible with (P-g).* The ground formerly given for declining the relaxation, that the stronger clause implies the weaker one, "so nothing proved is lost", was an invalid inference and is struck*: C5′(ii) is a* lower *bound on h_x, and a lower bound stated above what any proof needs is free only until it has to meet an upper bound.*

(ii) γ_n is not *a component of $\hat{\delta}_n$. It would multiply $\|\varphi_\dagger - \varphi\|_{L^2(\pi)}$ wherever it appeared, entering an effective noise level at $\gamma_n\delta_n$ and not at γ_n.*

(iii) The $\sqrt{\ln n}$ in (117) buys uniformity over $\varphi \in \mathcal{V}_n \cap \mathcal{S}_n$ and nothing else; at a fixed φ it may be deleted, by the argument of Lemma 77(i).

And a fourth reading, which is a scoping statement rather than a property of the lemma. ***Proposition 28 does not consume this lemma.*** *Its proof applies the linear regularisation theory* conditionally on the sample*, at the realised operator $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}|_{\mathcal{S}_n}$, and the spectral source hypothesis (P-a.3) is stated for the compression (53) of* that *operator and not for $\mathcal{A}_{\hat{\varphi}^{(k-1)}}$. Under that reading no transfer of a source condition from $\mathcal{A}$ to $\hat{\mathcal{A}}$ is needed, which is fortunate, since Lemma 84(ii) shows the only general device for such a transfer is unavailable at the*

source indices this programme lives at. So (118) is retained for two other reasons and neither is a rate*: it is the re-derivation C5′(ii) asks for, per reading (i), and it is what a reader needs in order to see that the conditional-on-sample reading is a choice and not an evasion, the operator error is $O_P(\gamma_n)$ and is priced, it is simply not the channel through which the rate is proved.*

Proposition 79 (A7, verified in the amended form, and the one place the unamended form does not close)**.** *Suppose the hypotheses of Lemma 77. Then:*

(i) The T-component closes at a fixed constant. There is a finite c, not depending on n, with

$$P\Big(\big\|\bar{T}(\varphi_\dagger,\theta_\dagger)\big\|_{L^2(\Pi)} > \mathsf{c}\,\hat{\delta}_n\Big) \longrightarrow 0;$$

$\mathsf{c} \geq 4\mathsf{c}_3 c_\Pi + \mathsf{c}_5 c_\Pi \mathfrak{m}_{p_K}$ serves. The statement is at $\hat{\delta}_n$ and not at $\hat{\delta}_n^{(T)} = n^{-1/2}\sqrt{\ln n}$, and the reason is C5′(i)*: under the amended bandwidth the smoothing bias is $\asymp \hat{\delta}_n$, which is larger than $\hat{\delta}_n^{(T)}$ whenever the propensity component binds. Under the unamended bandwidth $h_u \asymp n^{-1/16}$ the sharper conclusion, with $\hat{\delta}_n^{(T)}$ in place of $\hat{\delta}_n$, holds for every fixed $\mathsf{c} \geq 4\mathsf{c}_3 c_\Pi$. Only the displayed version is consumed.*

(ii) The ϱ-component does not close at a fixed constant. Under A4(iii) alone, $\|\hat{T}(\varphi_\dagger,\theta_\dagger) - \bar{T}(\varphi_\dagger,\theta_\dagger)\| = O_P(r_n)$ and no more; in particular there is no finite c for which $P(\|\hat{T} - \bar{T}\| > \mathsf{c}\,r_n) \to 0$ follows. An O_P hypothesis cannot produce a fixed-constant exceedance conclusion, and A7 unamended was a fixed-constant statement.

(iii) Two repairs, either sufficient. *The unamended A7, the display of (ii) at threshold $\tau_{\mathrm{DP}}\hat{\delta}_n$, holds if A4(iii) is strengthened to: there is a finite c with $P(\|\hat{\varrho} - \varrho\|_{L^2} > \mathsf{c}\,r_n) \to 0$. Alternatively, and this is the route the block takes, (13) holds under A4(iii) exactly as stated, for every $\tau_{\mathrm{DP}} \geq 1$; and any divergent factor serves in place of $\ln n$.*

(iv) What the amendment costs, and it is not nothing. The substitution is at the level of $\hat{\delta}_n$, and A5(ii) is the record of its effect*: replacing $\hat{\delta}_n$ by $\hat{\delta}_n \ln n$ in the discrepancy threshold*

is exactly why (137) *sets* $\delta_n := (\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)}$ *rather than* $\hat{\delta}_n^{2\mu/(2\mu+1)}$*, so* δ_n *below already carries the logarithm and is not inflated a second time here. Against* (7) *the substitution is free. Since* (7) *is a* strict *inequality,* $\delta_n = o_P(n^{-1/4})$ *survives it at every* (μ, r, d_Z) *satisfying it: if* $\frac{2\mu}{2\mu+1} \cdot \frac{r}{2r+d_Z} = \frac{1}{4} + \varepsilon$ *with* $\varepsilon > 0$ *then* $\delta_n = (\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)} = n^{-1/4-\varepsilon}(\ln n)^{2\mu/(2\mu+1)} = o(n^{-1/4})$*. The strictness of the frontier is what pays for it. Against* (P-f) *it is not free, and* (P-f) *is the other condition* δ_n *enters.* (P-f) *is an* exact domination*,* $\delta_n^2 = O(h_u^2 \hat{\delta}_n \ln n)$*, in which the left side now carries* $(\ln n)^{4\mu/(2\mu+1)}$ *and the right side* $(\ln n)^1$*; and* $\frac{4\mu}{2\mu+1} > 1$ *precisely when* $\mu > \frac{1}{2}$*, which is everywhere in this block.* So the logarithms never cancel in (P-f) and its thresholds must be strict*: at a configuration where* (P-f) *holds with equality in powers of* n *it fails by a power of* $\ln n$*. Remark 125 computes both thresholds and exhibits the* boundary *configuration at which this is not hypothetical; clause (5) there names the block's two running configurations and says which is a design and which is an illustration.*

Definition 80 (The source-condition ball)**.** Let s and R be as in H1 and A5(vi′) and set

$$\mathcal{V}_n := \Big\{ \varphi \in \mathcal{V} : \ \|\varphi - \varphi_\dagger\|_{L^2(\pi)} \le \bar{\rho}, \quad \|\varphi - \varphi_\dagger\|_{H^s} \le \mathsf{R} \Big\}, \tag{120}$$

with $\bar{\rho} > 0$ and $\mathsf{R} < \infty$ fixed and not depending on n, and H^s the *classical* Sobolev space $H^s(\mathcal{X})$ of order s, carrying no boundary condition, as in H1. *It is* ***not*** *a* $\mathcal{D}$*-generated scale*: the two spaces differ at every $s > \frac{3}{2}$, hence at every s this block admits, and the norm $\|\varphi - \varphi_\dagger\|_{H^s}$ in (120) is the classical one. Take R large enough that $\|\varphi_\dagger\|_{H^s} \le \mathsf{R}$ as well, which H1's source set permits and which Lemma 81 Step 4 uses. $\mathcal{V}_n$ *is deterministic*, as L-19 requires, and it is a *source-condition* ball rather than an $L^2(\pi)$ ball: the second constraint is exactly A5(vi′)'s H^s envelope made into a set.

R is an H^s radius and it is not the source radius (F-f) is applied at. The latter is R_*, the *spectral pre-image* norm of (P-a.3), which enters $\mathsf{c}_{17} = 2c_\mu \mathsf{R}_*^{1/(2\mu+1)}$ and hence c_{15} of (37). The two are not comparable in either direction without the operator interpolation Remark 86(1)

shows is unavailable at $2\mu > 1$. R *is the radius that is* ***not*** *free*: it is shared by this definition, A5(vi′), (P-a.1), (P-c), (P-g), (121) and Lemma 81 Step 4, so "fixed and large" tightens two conditions and inflates none, the hypothesis is an existential over *one* R meeting seven demands, and each of the seven is listed rather than left to be assembled. ***This is where the list is maintained***: A5(vi′), Lemma 83 and Remark 87 point here and maintain no copy, and Definition 24's clause-(d) reading names three of the seven as instances and says in terms that it maintains none, *as does* (P-a.1)*, which names four and is itself the third of the seven. No stronger claim is made, and that is deliberate*: the draft this sentence read "the corpus's only one" while that partial copy stood unmarked. *The subscript* n *is conventional*; nothing in (120) varies with n, and the containment statement that does vary is Lemma 83.

One symbol, one object, from here on. Bare $\mathcal{V}_n$ means (120) and nothing else in Part H. An $L^2(\pi)$-ball is written $\mathcal{V}(\varphi_\circ, \rho)$ per (110), with its centre and radius shown; the two are separated by the presence of arguments.

Lemma 81 (L-19: uniform operator error over the source-condition ball)**.** *Suppose A0, A4(iii), B1(i), (ii), (iv.c), (iv.d), C1(ii), C2′, C3, C5′(i),* ***A8's basis clause*** *and H1 hold, adopt Definition 68, suppose that* $s > \frac{1}{2}$*, so that on A0's compact interval the one-dimensional Sobolev embedding* $H^s(\mathcal{X}) \hookrightarrow C(\mathcal{X})$ *is available, suppose that* $p_K \geq s - \frac{1}{2}$*, and that*

$$\left(\frac{\mathsf{R}}{h_u\,\hat{\delta}_n}\right)^{1/s} \;=\; o\big(n\,\hat{\delta}_n^2\big). \tag{121}$$

Then, with $\mathcal{V}_n$ *as in* (120)*,*

$$\sup_{\varphi\in\mathcal{V}_n} \big\|\hat{T}(\varphi,\theta_\dagger) - T(\varphi,\theta_\dagger)\big\|_{L^2(\Pi)} \;=\; O_P\big(\hat{\delta}_n\big), \tag{122}$$

and the same bound holds with $\hat{T}$ *replaced by* $\hat{T}_{h_x}$ *provided in addition* $\mathsf{R}\,h_x^{s-1/2} = O\big(h_u\,\hat{\delta}_n\big)$, whose factor h_u is not decoration and is what Step 4 actually charges*: that step bounds the displacement of the two operator families in* ***operator norm*** *by* $\mathsf{c}_2 c_\Pi h_u^{-1} \max_i \big|\Delta_{h_x}\varphi_i - \Delta\varphi_i\big|$,

the h_u^{-1} *being Lemma 67*(ii)*'s* $\|\partial_v C_{h_u}\|_\infty \le \mathsf{c}_2 h_u^{-1}$*, the same* h_u^{-1} *that forces the net radius at* (3c) *and the* h_u *inside* (121)*, so with* (172) *the displacement is* $\mathsf{c}\, h_u^{-1} \mathsf{R}\, h_x^{s-1/2}$*, and it is* that product *and not* $\mathsf{R}\, h_x^{s-1/2}$ *alone that has to be* $O(\hat{\delta}_n)$. This is the same h_u (P-c) carries*, and the proviso is Proposition 28's (P-g).* **B1(iv) is cited by clause and A8's is cited as a clause, and the two are different registers.** *A8's basis clause is a* condition *on the sieve — it can fail, and A8's own note exhibits* $s = 1.7$ *with linear splines — so it is a carrier and belongs on the line. B1(iv)'s factorisation is a* declaration*, standing at every statement here and carried by nobody; what this proof opens is* (iv.d) *at Steps 2 and 3(3a) and* (iv.c) *at Steps 3(3a) and 4, and nothing opens* (iv.a) *or* (iv.b). Through 2026-08-24 this line was the block. Furthermore*, if the supremum is restricted to* $\mathcal{V}_n \cap \mathcal{S}_n$ *with* $\mathcal{S}_n := \operatorname{span}(b_1, \ldots, b_{L_x}) \cap \mathcal{V}$ *the spline sieve of A8, then* (122) *holds under*

$$L_x \ln n \;=\; o\Big(n\, \hat{\delta}_n^2\Big) \tag{123}$$

in place of (121)*, and that is the version Definition 22 and Proposition 28 consume;* (121) *is consumed by nothing.* (121) *is written in the primitive variables deliberately. Reading it at a bandwidth requires choosing one, and that choice is C5′(i)'s; Remark 82 unpacks the reading there.*

What makes (122) *a statement about the distribution of a random variable, and where it is established. The supremum in* (122) *is over an* uncountable *index set, so it is not a random variable on the strength of the definition alone.* It is one, and the proof establishes it at (3c)*:* $\mathcal{V}_n$ *is* $\|\cdot\|_\infty$*-**separable**, A0's* $\mathcal{X}$ *is a compact interval and* $s > \frac{1}{2}$*, so* $\mathcal{V}_n \subset C(\mathcal{X})$*, and the process is sup-norm Lipschitz at every sample point outside one fixed* φ*-free null set, so the supremum agrees almost surely with a countable one. That paragraph consumes three things this hypothesis list carries **because** it consumes them. **C1(ii)** is load-bearing rather than incidental: it makes* π *strictly positive on* $\mathcal{X}$*, so the continuous representative of an* $L^2(\pi)$ *class is* unique *and* $\|\cdot\|_\infty$ *is well defined on* $\mathcal{V}_n$*, without it one class can carry two continuous*

representatives of different sup-norms and the norm the entire ϵ_0-net runs in is not a function of the class. ***H1*** *enters through Definition 80, which chooses* R *with* $\|\varphi_\dagger\|_{H^s} \le \mathsf{R}$ *"which H1's source set permits", and hence gives* $\|\varphi\|_{H^s} \le 2\mathsf{R}$ *on* $\mathcal{V}_n$*. And the passage from that bound to continuity is the* ***Sobolev embedding*** $H^s(\mathcal{X}) \hookrightarrow C(\mathcal{X})$ *at* $s > \frac{1}{2}$*, named in the hypothesis list above.* **Definition 68 is adopted rather than presumed***: the conclusion* (122) *is a statement about* $\hat{T}$*, which is that definition's object under that definition's sample-splitting and truncation conventions, and (3c) reads its record that* $\varphi \mapsto \hat{T}_{h_x}(\varphi)$ *depends on* φ *only through the bounded linear functionals* (96)*.* **A8's basis clause is on the line** *because clause (b) is proved at (3d) from the B-spline inverse inequality on a quasi-uniform mesh, which carries the entire sieve entropy bound and hence* (123)*, the version Proposition 28 consumes.* **A1b(iv) is not a hypothesis***, on the ground recorded at Lemma 77: nothing in this proof reads a moment of* ΔZ*.* And (95), cited twice at (3c), is **not** on this line*: that display is two-sided and its lower half rests on A4(i), which neither appeal reads; both are narrowed at the site to free upper bounds,* (P1) *for the population weight and* $\varrho, \hat{\varrho} \in [0,1]$ *for the propensity difference, which is why Definition 68 is adopted above and A4(i) is not assumed.* The claim is made **once**, there, and referred to at the three other suprema over $\mathcal{V}_n$ in this block*: Step 1's and Step 4's in the same proof, and the one in* (118)*.* Neither an outer-probability convention nor a Suslin hypothesis is invoked, and none is needed*: the separability of the index set and the pathwise sup-norm continuity of the process are both available, the second from clause (3a) of the proof.*

What is used, and what is not. The proof of clause (a) uses a ***bracketing*** *entropy bound for the source-condition ball* (120) *in the sup-norm,* $\log N_{[]}(\epsilon, \mathcal{V}_n, \|\cdot\|_\infty) \le \mathsf{c}_6(\mathsf{R}/\epsilon)^{1/s}$*, valid for* $s > \frac{1}{2}$*, item (F-e) of the citation note and* not proved here*, together with a bounded-difference inequality at each bracket centre, which* is *proved here, as Lemma 123(i). The pin for (F-e) is Birman and Solomjak (1980, Ch. 2); the obvious alternative is not merely weaker but unavailable. Routing through a Hölder-ball entropy bound at exponent* $s - \frac{1}{2}$ *would yield* $\epsilon^{-1/(s-1/2)}$ *rather than* $\epsilon^{-1/s}$ *and would therefore move* (121)*'s threshold in* s *up by* exactly

$\frac{1}{2}$, *at every configuration and at every bandwidth, the right-hand side* $n\hat{\delta}_n^2$ *being untouched: at Remark 125*(5)*'s* ***boundary*** *configuration* $(\mu, d_Z, r) = (1, 3, 4.5)$*, equivalently* $\alpha = \frac{3}{8}$*, from* $s > \frac{27}{16}$*, Remark 82*(1)*'s reading at the amended* $h_u \asymp \hat{\delta}_n^{1/8}$*, to* $s > \frac{35}{16}$, but that route needs the embedding $H^s \hookrightarrow C^{0,s-1/2}$, which is vacuous for every $s > \frac{3}{2}$*, since* $C^{0,\beta}$ *on an interval contains only constants at* $\beta > 1$*, and* (54) *already requires* $s > \frac{27}{16}$. The correct Hölder route runs through $H^s \hookrightarrow C^{k,\beta}$ with $k = \lfloor s - \frac{1}{2} \rfloor$*, whose sup-norm entropy is governed by* k *and* β *separately and does* not *reduce to a single Hölder exponent, so Birman–Solomjak is not merely the pin taken but the only one available. The proof does* not *use a VC-subgraph bound, and it does* not *use stochastic equicontinuity of the* $\sqrt{n}$*-scaled empirical process at an estimated point.* The VC route registered in the strategy memo §3.4 is not available here*: the class* $\{\widetilde{W}_l \, \mathsf{C}_{h_u}(u - \Delta Y_\theta + \Delta\varphi) \, w(t'\Phi(\Delta Z)) : \varphi \in \mathcal{V}_n\}$ *is indexed by an infinite-dimensional* φ *entering through a smooth transformation, and it is* not *a VC-subgraph class of indicator-weighted functions, the indicator has been smoothed, and even before smoothing the index is a function and not a halfspace.* Smoothness of the index set, not combinatorial dimension, is what supplies the entropy*, and that is precisely why the ball must be a source-condition ball and not an* $L^2(\pi)$ *ball: the* $L^2(\pi)$ *ball has infinite sup-norm entropy at every radius.* On the sieve the entropy is elementary and needs no classical citation*:* $\mathcal{V}_n \cap \mathcal{S}_n$ *is a bounded subset of a space of dimension* at most L_x*, exactly* $L_x - 1$*, by the count Definition 22's measurability verification, first step performs, and the bound below is stated at* L_x *because it is an upper bound and nothing here is sharp in the dimension, so a volumetric bound gives* $\log N(\epsilon) \leq L_x \log(\mathsf{c}\,\mathsf{R}\, L_x^{1/2}/\epsilon)$. The two versions therefore differ in what they rest on as well as in what they require*, and the one consumed is the elementary one.*

Remark 82 (What Lemma 81 costs, and which clause charges it). (121) is a restriction linking s, r and d_Z, and it should be read three ways. Write $\hat{\delta}_n = n^{-\alpha}$ with $\alpha = r/(2r + d_Z) \in (0, \frac{1}{2})$ and ignore the fixed R. Since $\epsilon_0 \asymp h_u \hat{\delta}_n$, (121) reads $\left(h_u \hat{\delta}_n\right)^{-1/s} = o(n^{1-2\alpha})$, and the bandwidth enters through h_u alone.

(0) Which clause charges it, and the answer is neither Proposition 28 nor Definition 22. (121) is clause (a)'s condition; what the estimator and the rate consume is clause (b) under (123), which (P-d) names. *So* (121) *is charged to nothing*, and readings (1)–(2) below are a discussion of a clause no result consumes, kept because a referee will ask what the Sobolev-ball route would have cost.

(1) It is satisfiable, and it is not the binding line. At the amended bandwidth $h_u \asymp \hat{\delta}_n^{1/8}$ it reads $\frac{9}{8}\alpha/s < 1 - 2\alpha$, i.e. $s > \frac{27}{16} = 1.6875$ at $\alpha = \frac{3}{8}$, *the same number as* (54)*'s, and not a coincidence*: both conditions compare a resolution level against $n\hat{\delta}_n^2$ and both resolutions are $h_u\hat{\delta}_n$, the bracket radius $\epsilon_0 \asymp h_u\hat{\delta}_n$ here and the sieve approximation level $\mathsf{R}L_x^{-s} = O(h_u\hat{\delta}_n)$ in (P-c). They are two readings of one budget and only the second is charged. *It also runs the wrong way in* r, the left-hand side increasing in α while $n^{1-2\alpha}$ decreases, so *a smoother propensity makes Lemma 81 harder* and the block's two constraints on r pull in opposite directions.

(2) It excludes the configuration in which the propensity component does not bind. As $\alpha \uparrow \frac{1}{2}$ the right-hand side $n\hat{\delta}_n^2 = n^{1-2\alpha}$ of (121) ceases to diverge and no finite s suffices. *So Lemma 81 requires* $r_n \gg n^{-1/2}$ *strictly*, i.e. requires $\alpha < \frac{1}{2}$ with a *polynomial* margin. *A5(i) does not supply it and is not cited for it*: that clause's floor gives $n\hat{\delta}_n^2 \geq \ln n$, a *logarithmic* divergence, and (121) needs a polynomial one, the same distinction A4(ii)'s $d_Z \geq 1$ note draws at $\alpha = \frac{1}{2}$, where $L_x \ln n = o(\ln n)$ admits no integer $L_x \geq 1$. That is a scope condition on Step 6 and it is stated rather than assumed away: with a parametric propensity the uniform bound at rate $\hat{\delta}_n$ is *not* available by this route, and the $L^2(\Pi)$-norm rate would have to be relaxed to $\hat{\delta}_n$ times a slowly divergent factor, which, by Proposition 79(iv), the strictness of (7) again absorbs.

Lemma 83 (Containment of the path)**.** *Let* $\hat{\varphi}^{(0)} \in \mathcal{S}_n$ *satisfy (P-a.2),* $P(\|\hat{\varphi}^{(0)} - \varphi_\dagger\|_{L^2(\pi)} \leq \hat{\varepsilon}_0) \to 1$ *with* $\hat{\varepsilon}_0 \to 0$*, let* ***A5(vi)*** *hold* in its envelope form at the fixed radius R of Definition 80*,* $P\big(\max_{k \leq \bar{k}} \|\hat{\varphi}^{(k)} - \varphi_\dagger\|_{H^s} \leq \mathsf{R}\big) \to 1$*, and let* $\hat{\varphi}^{(k)}$ *be the stages of Definition 22,*

each stopped at an index $\hat{m}^{(k)} \le \bar{m}_n$. Then, with that same R *and with $\bar{\rho}$ fixed and large enough, $P\big(\hat{\varphi}^{(k)} \in \mathcal{V}_n \cap \mathcal{S}_n$ for all $k \le \bar{k}\big) \to 1$.*

The radius is ***not*** *chosen here, and that is the point of stating A5(vi′) as an envelope. Definition 80 requires $\mathcal{V}_n$ to be* deterministic*, so the H^s constraint must be a fixed number; but* R *also appears at the other sites Definition 80 enumerates, in two of which it is an* upper *bound that "fixed and large" tightens.* So the hypothesis is an existential over one R satisfying all of them simultaneously*, which is a statement about the logical structure of the hypothesis and not about its size. The enumeration is Definition 80's and is not repeated here.* An $O_P(1)$ reading of A5(vi′) does not serve*: $O_P(1)$ is a family of bounds indexed by tolerance, and a deterministic ball cannot be defined from a family.*

Three things about this lemma, and the third is a finding. (a) *The $L^2(\pi)$ half is the stagewise bound of Proposition 28 Step 4 and is genuinely proved; the within-stage Landweber path satisfies $\|v_m\|_{L^2(\pi)} \le m \sup_{\mathcal{V}_n} \|\hat{T}_{h_x}\| \le 2m\,\mathsf{c}_3 c_\Pi$ under the normalisation of Definition 22, which bounds the path but by $\bar{m}_n$ and not by $\bar{\rho}$; it is the DP stopping rule, not the Landweber algebra, that keeps the* stopped *iterate inside the ball. The memo's registered mechanism, "$\varphi_m - \varphi_0 \in \mathcal{R}(\hat{\mathcal{A}}^*)$ with norm growing at most polynomially in the cap", gives the first bound and not the second, and the distinction matters because $\mathcal{V}_n$'s radius is fixed while $\bar{m}_n$ diverges.* (b) *The sieve constraint $\hat{\varphi}^{(k)} \in \mathcal{S}_n$ is by construction.* (c) ***The H^s half is A5(vi′), assumed and not proved, and it is assumed as a w.p.$\to 1$ envelope at the fixed*** R***.*** *A5(vi) prices $\|\hat{\varphi} - \varphi_\dagger\|_{H^s}$ at the* final *iterate and is consumed by Lemma 60*(iii) *alone, on the balancing route. What is needed here is the bound* at every stage*, because Lemma 81's ball is a source-condition ball and every $\hat{\varphi}^{(k)}$ must lie in it;* that is a different statement, not a different route for the same one*, and it is A5(vi′). Since $\bar{k}$ is finite and does not grow with n, A5(vi′) is a finite union of $O_P(1)$ statements and not a uniformity requirement, which is the only reason it is affordable. Deriving the H^s bound instead of assuming it is possible on the sieve, Bernstein's inequality for splines gives $\|\zeta\|_{H^s} \le \mathsf{c}_7 L_x^s \|\zeta\|_{L^2(\pi)}$ on $\mathcal{S}_n$, but at the cost*

of $L_x^s \delta_n = O(1)$, which is a genuine extra restriction and is not *imposed; A5(vi′) is cheaper and is what the block now carries.*

Lemma 84 (Source-condition transfer: what is free, what is unavailable, and what is merely unproved)**.** *Let $\mathcal{A}, \hat{\mathcal{A}} : \mathcal{V} \to L^2(\Pi)$ be bounded with $\|\hat{\mathcal{A}} - \mathcal{A}\| \le \gamma$, and let $v \in \mathcal{V}$.*

(i) $\mu \le \frac{1}{2}$: transfer is free. *If $v = (\mathcal{A}^*\mathcal{A})^\mu \varsigma$ with $\mu \in (0, \frac{1}{2}]$ and $\|\varsigma\| \le \mathsf{R}_*$, then there is $\tilde{\varsigma}$ with*

$$\|\tilde{\varsigma}\| \;\le\; \mathsf{c}_8 \mathsf{R}_* \qquad \text{and} \qquad \left\| v - (\hat{\mathcal{A}}^*\hat{\mathcal{A}})^\mu \tilde{\varsigma} \right\|_{L^2(\pi)} \;\le\; \mathsf{c}_8\, \mathsf{R}_*\, \gamma^{2\mu}.$$

Here $\mathsf{c}_8 := 2^{2\mu}$, the constant clause (ii.b) *also carries. The proof of* (i) *in fact delivers the smaller 2^μ; the symbol is written at the larger value so that* one letter carries one number *across the two clauses, and* (i) *holds* a fortiori *because $2^\mu < 2^{2\mu}$ at every $\mu > 0$.* The radius here is the **spectral pre-image** norm R_* throughout this lemma, never the H^s radius R of Definition 80, *clause (iii) is the one place an H^s radius appears, and it says R for that reason.* The proof is Heinz's inequality*, operator monotonicity of $z \mapsto z^{2\mu}$ on $[0, \infty)$, valid for $2\mu \le 1$, and $\mu \le \frac{1}{2}$ is exactly its range of validity; Engl, Hanke, and Neubauer (2000, Prop. 8.21), which Engl, Hanke, and Neubauer name the inequality of Heinz, state it for a Hilbert-scale generator and hence under a* strict *positivity that $\mathcal{A}^*\mathcal{A}$ does not have, so the passage to a merely positive bounded operator is the standard shift: apply it to $\mathcal{A}^*\mathcal{A} + \epsilon\,\mathrm{Id}$ and let $\epsilon \downarrow 0$.*

(ii) $\mu > \frac{1}{2}$*: the device of (i) is unavailable, and the failure is one of non-commutativity, not of size. Two statements, and* ***the second is a limitation on the first.***

(ii.a) The tool fails. *For every $2\mu > 1$ there are $\mathcal{Q}_1, \mathcal{Q}_2$ on a two-dimensional space with $0 \preceq \mathcal{Q}_1 \preceq \mathcal{Q}_2$ and $\mathcal{Q}_1^{2\mu} \not\preceq \mathcal{Q}_2^{2\mu}$; at $\mu = 1$, $\mathcal{Q}_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $\mathcal{Q}_2 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ serve.* The two clauses of that sentence are proved by two different arguments*: the general $2\mu > 1$ case by Löwner's divided-difference criterion for matrix monotonicity of order two (Bhatia, 1997, Ch. V), the displayed pair by inspection. So step (i)'s*

passage from $\mathcal{A}^*\mathcal{A} \preceq 2\hat{\mathcal{A}}^*\hat{\mathcal{A}} + 2\gamma^2 \,\mathrm{Id}$ *to the same inequality raised to the power* 2μ *is* not available *above* $\mu = \frac{1}{2}$*, and no bound in* $\|\hat{\mathcal{A}} - \mathcal{A}\|$ *alone is obtainable by that route.*

(ii.b) But no counterexample to the conclusion *is claimed, and none can be diagonal. If* $\mathcal{A}$ *and* $\hat{\mathcal{A}}$ *share a singular system, transfer holds at* every $\mu > 0$*, with* $\mathsf{c}_8 = 2^{2\mu}$ *and residual* $(2\gamma)^{2\mu}\mathsf{R}_*$*: split the index set at* $\sigma_j \geq 2\gamma$*, where* $\hat{\sigma}_j \geq \sigma_j/2$ *bounds* $\tilde{\varsigma}$*, and below it, where* $|v_j| \leq (2\gamma)^{2\mu}|\varsigma_j|$ *bounds the residual.* So the obstruction is that the perturbation may rotate the singular system, not that it is large, *and whether transfer itself fails above* $\frac{1}{2}$ *is left open here.* ***What is established is the unavailability of the device, and that is all this programme uses****: (P-a.3) is stated spectrally on the compression of* $\hat{\mathcal{A}}$ *directly, so no transfer and no interpolation is performed anywhere in Part H.*

(iii) The Hilbert-scale route: what it needs, and why it is ***not*** *the route taken. A Hilbert-scale source condition,* $\|v\|_{H^s} \leq \mathsf{R}$*, is a statement about* v *alone and transfers trivially.* But it is not the hypothesis of (F-f)*, and it delivers one only through the* link *condition* $\|\hat{\mathcal{A}}\zeta\|_{L^2(\Pi)} \asymp \|\zeta\|_{H^{-a}}$ together with an operator-interpolation step at exponent 2μ*. Two facts about that step, and they are the reason Proposition 28 does not take this route.*

(iii.a) The link condition cannot hold on $\mathcal{V}$*: the left side vanishes on the infinite-dimensional* $\ker\hat{\mathcal{A}}$ *of Proposition 70(ii) and the right side vanishes at no nonzero* ζ*, so it must be imposed on the sieve* $\mathrm{span}(b_1, \ldots, b_{L_x})$ *of A8. Restricting removes the obstruction whenever* $L_x \leq \mathrm{rank}\,\hat{\mathcal{A}}_\varphi$*, for which A8's* upper end $L_x \ll n^{1/2} \leq n_{\mathrm{m}}$ *is sufficient in generic position,* and "generic position" is an L_x-dimensional injectivity statement, hence **Q-4**'s object and not Q-3's*, Q-3 being the* n_{m}*-dimensional Gram-matrix question Proposition 74(iii) and Proposition 76 decline.*

(iii.b) And the interpolation step is unavailable in the regime this block inhabits. Read in the Löwner order, the link condition says that the compression (53) *and the*

compressed scale operator each dominate a fixed positive multiple of the other. *What the interpolation step needs is the corresponding comparison of their* -2μ *powers; inverting the two-sided inequality delivers that comparison at the power* -1*, and* raising a Löwner inequality to a power preserves it only for exponents in $[0, 1]$*, by operator monotonicity of the power map, available at* $2\mu \leq 1$ *by (i)'s Heinz step and* ***refuted at every exponent above*** 1 ***by (ii.a)****.* (55) *forces* $2\mu > \frac{5}{3}$. The two operators are named rather than symbolised*: the compression and the compressed scale operator are the objects the clause is about, and the Löwner statement is more readable said than written.* So the pair (Hilbert-scale source, link) does not imply the spectral condition (F-f) consumes, at any configuration Proposition 28 admits*, which is why (P-a.3) states the spectral condition directly and why this clause is now a statement about a route* declined *rather than about the route taken. Remark 86 carries the full disposition, including the commutativity proviso under which (iii.a)+(iii.b)* would *deliver* $\mu = s/(2a)$*, and its price.*

Lemma 85 (The normalisation is a relaxation parameter, and (F-f)'s bound is invariant under it)**.** *Let* $\mathcal{L}$ *be a bounded linear operator from a Hilbert space into* $L^2(\Pi)$*, let* $\bar{v}_\dagger$ *be in its domain, let* $y^\delta \in L^2(\Pi)$ *satisfy* $\|y^\delta - \mathcal{L}\bar{v}_\dagger\| \leq \delta$*, let* $\omega_* > 0$*, and put* $\mathcal{L}_{\omega_*} := \omega_*^{-1}\mathcal{L}$ *and* $y^\delta_{\omega_*} := \omega_*^{-1}y^\delta$*.* ω_* is the generic scale*; the single value the estimator uses is* ω_n*, fixed at* (35) *below.* The star is not decoration. *Bare* ω *is the* ***sample point*** *of the probability space throughout this project, it is the argument of the regular versions fixed at (P6), of (P7a)'s "for a.e.* ω*", and of the joint-measurability statements in Lemma 46 and Proposition 9, and a generic positive scalar and a sample point cannot share a bare glyph. The three live objects are separated by form: bare* ω *the sample point,* ω_n *the estimator's normalising factor,* ω_* *this lemma's generic scale. Then:*

(i) Same iterates, different relaxation. *The Landweber iteration* $v_{m+1} = v_m + \mathcal{L}^*_{\omega_*}(y^\delta_{\omega_*} - \mathcal{L}_{\omega_*}v_m)$ *from* $v_0 = 0$ *is identical, term by term, to the relaxed iteration* $v_{m+1} = v_m +$

$\omega_*^{-2}\mathcal{L}^*(y^\delta - \mathcal{L}v_m)$ *for the unscaled pair. In particular the normalisation is a choice of relaxation parameter and not a change of estimator.*

(ii) Same stopping index. *For every* m, $\|\mathcal{L}_{\omega_*}v_m - y^\delta_{\omega_*}\| = \omega_*^{-1}\|\mathcal{L}v_m - y^\delta\|$, *so the discrepancy rule at level* $\tau_{\mathrm{DP}}\omega_*^{-1}\delta$ *for the scaled pair stops at exactly the index at which the rule at level* $\tau_{\mathrm{DP}}\delta$ *stops for the unscaled pair.*

(iii) The source pair transforms, and in opposite directions. If $\bar{v}_\dagger = (\mathcal{L}^*\mathcal{L})^\mu\varsigma$ *with* $\|\varsigma\| \le \mathsf{R}_*$, ς *being the* spectral *source pre-image and* R_* *its norm bound, as in Lemma 84 and (P-a.3), and* ***not*** *the* H^s *radius* R *of Definition 80, then* $\bar{v}_\dagger = (\mathcal{L}^*_{\omega_*}\mathcal{L}_{\omega_*})^\mu\left(\omega_*^{2\mu}\varsigma\right)$, *so the source radius transforms as* $\mathsf{R}_* \mapsto \omega_*^{2\mu}\mathsf{R}_*$ *while the data-error bound transforms as* $\delta \mapsto \omega_*^{-1}\delta$.

(iv) Hence (F-f)'s error bound is exactly invariant*: for every* $\omega_* > 0$ *and every* $\mu > 0$,

$$\left(\omega_*^{2\mu}\mathsf{R}_*\right)^{1/(2\mu+1)}\left(\omega_*^{-1}\delta\right)^{2\mu/(2\mu+1)} \;=\; \mathsf{R}_*^{1/(2\mu+1)}\,\delta^{2\mu/(2\mu+1)}. \tag{124}$$

(v) The stopping-index bound is not invariant, and that is the whole cost. (F-f)*'s bound on* $\hat{m}$ *transforms as* $(\mathsf{R}_*/\delta)^{2/(2\mu+1)} \mapsto \omega_*^2(\mathsf{R}_*/\delta)^{2/(2\mu+1)}$, *so the iteration cap must absorb a factor* ω_*^2.

(vi) The ***reference measure****'s normalisation is a second free scale, it is* ***not*** *covered by* (i)–(v), *and* (36) *is proportional to it. Fix* $c_* > 0$ *and replace* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ *of Definition 5 by* $\Pi_u \otimes (c_*\,\Pi_t) \otimes \Pi_{\mathsf{w}}$, the **second** factor alone is scaled, the other two are untouched, *with every other primitive of the model held fixed; the replacement satisfies B1(iv) whenever* Π *does. Then every* $L^2(\Pi)$ *norm is multiplied by* $c_*^{1/2}$, *hence so are* c_Π *and* ω_n *of* (35), *while* $\hat{\delta}_n$ *of A5(i),* $\max\{n^{-1/2}\sqrt{\ln n}, r_n\}$, *a functional of* P *and* n *with no* Π *in it, is unchanged. So the right-hand side of* (36) *is multiplied by* c_* exactly. The three free scales of this lemma are notated alike*:* ω_* *scales the operator,* c_* *scales the reference measure,* a_* *scales the index.*

(vii) A third free scale: the index scale, and it is why c_Φ carries no number. *Fix* $a_* > 0$ *and replace* $(\Phi, \mathcal{T}, \Pi_t)$ *by* $(a_*\Phi,\ a_*^{-1}\mathcal{T},\ \Pi_t \circ (a_* \cdot))$*, every other primitive held fixed.* The subscript is not decoration*: bare a is the degree of ill-posedness, live as* $\mu = s/(2a)$ *in the clause that closes the preceding lemma; the starred form joins* ω_* *and* c_* *as this lemma's free scales. The family* $\{w(t'\Phi(z)) : t \in \mathcal{T}\}$ *is* ***unchanged as a set of functions****,* $\bar{w}$ *is unchanged at corresponding indices,* $c_\Phi \mapsto a_* c_\Phi$ *and* c_Π *is* ***invariant****; so* ω_n *and* (36) *do not move.* Hence normalising c_Φ is a change of units in the index and pins nothing. *What is* ***not*** *covered: holding* Π_t *fixed while changing* c_Φ *is a* different family*, not a change of units, and it moves* c_Π *— exponentially in* $\dim \mathcal{T}$ *through* $\int e^{2c_\Phi \|t\|_1} \Pi_t(dt)$*.*

Accordingly, and this is a standing convention rather than a clause of the assumption block*,* Π *is normalised to a* ***probability measure****,* $\Pi_u(\mathbb{R})\,\Pi_t(\mathcal{T}) = 1$*, throughout this project.* B1(iv) leaves the normalisation entirely free*: (iv.a) constrains the shape of* ϖ_u*, (iv.b) is scale-invariant, and (iv.c) bounds* c_Π *without fixing* $\Pi(\mathbb{R} \times \mathcal{T})$ *— and, separately, without fixing the* ***dispersion*** *of* Π_t *— so* c_* *above ranges over* $(0, \infty)$ *inside the maintained block.* Fixing it is a choice of units and not a restriction on P*:* Π *is an object the analyst chooses, on the B1 reading.* No display in this project needs rewriting*, each being stated in* c_Π*, in* $\asymp$ *or in* O_P*;* the one display whose numerical content moves is (36). The convention pins the **mass** scale c_* and no more*: the dispersion of* Π_t *is left free by* (iv.b) *and* (iv.c)*, and it enters* c_Π *exponentially through* $\int e^{2c_\Phi \|t\|_1} \Pi_t(dt)$*.* So $\bar{m}_n$ remains a number times a free constant, and any instantiation of (36) must state the Π_t it is computed at. *Pinning the mass is what removes the* c_* *ambiguity of* (vi) *and nothing else. The inflation figures computed below, and any simulation that reads them, inherit that constant.* That is not hypothetical*: a Monte Carlo run of this estimator at an unnormalised* Π_t *reported a cap inflation about* 2.9 *times the correct one, and the discrepancy is exactly* c_**, not an error in* (v)*. Why* (i)–(v) *do not already cover it, since they look as though they should. They rescale the* operator *by* ω_* *at a* fixed Π*, and clause (ii) preserves the stopping index precisely* ***because*** *the discrepancy threshold rescales with the residual, both carrying* ω_*^{-1}*. Under a change of* Π *that symmetry breaks:*

the residual is an $L^2(\Pi)$ *norm and moves by* $c^{1/2}$*, while the threshold* $\tau_{\mathrm{DP}}\hat{\delta}^{\mathrm{eff}}_{n,k}$ *of* (37) *is built from* $\hat{\delta}_n$*,* h_u *and* $\hat{\varepsilon}_{k-1}$ *and does not move at all. One rescaling is a relaxation parameter; the other is a change of units in which the noise level is denominated, and only the first is invariant. Why this is a lemma and not a sentence. Definition 22 normalises so that* $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\| \le 1$*, which is (L1) of Proposition 28 Step 2 and is a hypothesis of (F-f).* That bound is **pairing-specific** and this lemma is not*: (F-f) reads the operator norm out of the domain Hilbert space whose inner product defines* $\mathcal{L}^*$*, so the constant in* (35) *has to cover the pairing the iteration runs in —* $L^2(\hat{\pi})$ *for the feasible recursion,* $L^2(\pi)$ *for the analysed one, and* (35) *carries the maximum of the two conversion costs by Proposition 70*(i′). Clauses (i)–(vii) below are indifferent to which is in force*, each being stated for an abstract* $\mathcal{L}$ *and an arbitrary* $\omega_* > 0$*; what is* not *indifferent is the single number* ω_n *they are instantiated at, and that is fixed in Definition 22 and not here.* The normalisation does not cost nothing, because the only norm bound available in this file diverges*: Proposition 70(i) gives* $\|\hat{\mathcal{A}}_\varphi\| \le \mathsf{c}\, h_u^{-1}h_x^{-1/2}$*, so the required* ω_n *diverges, and a divergent rescaling moves* $\|\mathcal{L}\|$*, the data error* and *the source radius at once. Clauses (iii)–(iv) are what make the movement cancel: the exponents* $1/(2\mu+1)$ *on the radius and* $2\mu/(2\mu+1)$ *on the noise are exactly the pair for which it does, which is why* (178) *must be written with* R_* *shown and not absorbed into an* O_P*. And the cancellation does not depend on the direction of the movement. If the unscaled norm were instead* $o(1)$*, which is plausible, since near-orthogonal* $\hat{g}_{\cdot,\cdot,\cdot,i}$ *would give* $\|\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\| \asymp (n_{\mathrm{m}}h_u)^{-1/2}$*, then* $\omega_n \to 0$ *would be admissible and (iv) would still hold verbatim, clause (iv) being stated for* every $\omega_* > 0$*. Nothing in Part H therefore needs* $\|\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\| \asymp 1$*, and that is the whole of what this lemma buys.*

What the file does and does not know about the unscaled norm. Only the upper *bound of Proposition 70(i) is available. A matching* lower *bound on* $\mathcal{S}_n$ *is the statement* $\|\hat{\mathcal{A}}_\varphi\zeta\|_{L^2(\Pi)} \gtrsim \|\zeta\|$ *for* $\zeta \in \mathcal{S}_n$*, which is* one side of the link condition*, and the link condition on the sieve at a random base point is* ***Q-4****'s object, not Q-3's.* Q-3 is a different question and is classified as not a theory obligation*: it asks for* $\lambda_{\min}(\mathbf{G}_n(\cdot))$ *on the* n_{m}*-dimensional Gram matrix of* (112)*,*

which governs Proposition 74(iii)'s radius and Proposition 76's proviso and which those two results correctly decline. Two objects, two dimensions, two open items*: the sieve-restricted injectivity modulus is L_x-dimensional and is assumed, at (P-a.3); the Gram matrix is n_{m}-dimensional and is declined. Lemma 84(iii)'s "generic position" is the same L_x-dimensional object and carries the same redirect.*

What is not free is (v), and it is priced here rather than left as a symbol. Definition 22 pays for (v) by sizing the cap at $\bar{m}_n \geq \omega_n^2 \hat{\delta}_n^{-2}$ rather than at $\hat{\delta}_n^{-2}$, and the inflation is polynomial and not negligible. *With $\omega_n \asymp h_u^{-1} h_x^{-1/2}$ from (35), $\omega_n^2 \asymp h_u^{-2} h_x^{-1}$.* That $\asymp$ survived the 2026-08-27 amendment of (35) and the figures below with it, and the reason is which kind of quantity moved*: the display now carries $\max\{\|1/\pi\|_\infty^{1/2}, \varepsilon_\pi^{-1/2}\}$ where it carried $\|1/\pi\|_\infty^{1/2}$, and both entries are fixed positive numbers free of n — the first finite by C1(ii), the second a constant Definition 22 fixes before the data are seen — so every* exponent *in this paragraph is unchanged and the* constant *multiplying $\bar{m}_n$ rises by at most $(\varepsilon_\pi \|1/\pi\|_\infty)^{-1}$, and by nothing at all at $\varepsilon_\pi \geq \|1/\pi\|_\infty^{-1}$.* An instantiation must therefore now state the ε_π it is computed at as well as the Π_t*, on exactly the reading clause* (vi) *imposes for c_*. At the frontier-clearing configuration $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$ of Remark 125(5), where $h_u \asymp n^{-0.04}$ and C5′(ii) ∧ (P-g) leave $h_x \in (n^{-0.44}, n^{-0.144})$:*

$$\omega_n^2 \asymp n^{0.08} h_x^{-1} \in \left(n^{0.224}, n^{0.52}\right), \qquad \bar{m}_n \geq \omega_n^2 \hat{\delta}_n^{-2} \in \left(n^{0.864}, n^{1.16}\right) \quad \textit{against} \quad \hat{\delta}_n^{-2} = n^{0.64}.$$

At $n = 2000$ that is an inflation of the iteration budget by a factor between about 5 and about 50, 130 *Landweber steps become between* 710 *and* 6750, and the factor is decreasing in h_x, so the upper end of the h_x window is also the cheap end of the cap. *The cap is a computational budget and not a rate condition, so nothing in (56) moves with it; but it is a real cost of the normalisation, a reader tuning the estimator will meet it, and it is stated here rather than left to be discovered. The alternative to paying it is a lower bound on $\|\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\|$, which is Q-4 again.*

Remark 86 (Why (P-a.3) is spectral and at the compression, what H1 still does, and what a derived μ would cost). Read this before Remark 89. Four clauses: why the Hilbert-scale pair does not deliver (F-f)'s hypothesis, what (P-a.3) asserts instead, why that is a **weakening**, and the alternative, priced.

(1) The Hilbert-scale pair does not deliver what (F-f) consumes. The pair is (a) a *Hilbert-scale* source condition, $\|\bar{v}_\dagger^{(k)}\|_{H^s} \le \mathsf{R}$, and (b) the *link* condition of Lemma 84(iii), $\|\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}\zeta\|_{L^2(\Pi)} \asymp \|\zeta\|_{H^{-a}}$ on $\mathcal{S}_n$; what (F-f) consumes is the **spectral** condition $\bar{v}_\dagger^{(k)} = (\mathcal{L}^*\mathcal{L})^\mu\varsigma$ with $\|\varsigma\| \le \mathsf{R}_*$. The step from (a) and (b) to that is Hilbert-scale operator interpolation, and it is available only at $2\mu \le 1$. Read in the Löwner order, (b) says that the compression (53) and the compressed scale operator *each dominate a fixed positive multiple of the other*; what the step needs is the corresponding comparison of their -2μ powers. Inverting the two-sided inequality gives that comparison at the power -1, and *raising it to a power preserves it only for exponents in* $[0, 1]$, by operator monotonicity of the power map , Heinz's range, and Engl, Hanke, and Neubauer (2000, §8.5)'s interpolation range, which is the pin H1 itself names. Here the exponent is 2μ, and (55) forces $2\mu > \frac{5}{3}$: *no configuration this block admits escapes*, and Lemma 84(ii.a) is a proof *in this file* that the device fails at every exponent above 1. Finite-dimensionality of $\mathcal{S}_n$ does not rescue it. On $\mathcal{S}_n$ range membership is vacuous, every positive-definite operator there is surjective, so all the content of the step is the *uniform-in-n pre-image norm bound*, and that bound *is* the interpolation inequality. Finite rank converts the failure from qualitative to quantitative; it does not remove it.

(2) (P-a.3) states the hypothesis where the theorem consumes it. It asserts the spectral condition directly, at the operator Definition 22 actually regularises, named as the compression (53). Two consequences. *Step 2 (L5) becomes two lines*, no interpolation, no Heinz range, no Löwner, and Lemma 84 is used nowhere on the rate route, its clause (ii.a) surviving as the reason the *inherited nonlinear* theorem is unavailable (Remark 88(3)) and its clause (iii) as the reason a Hilbert-scale route *would* need the sieve. And the operator is named correctly.

$\mathcal{L}^*\mathcal{L} = \omega_n^{-2}\hat{\mathcal{P}}^{\mathcal{S}}_{\varphi_\circ}$ is the compression $P_{\mathcal{S}_n}\hat{\mathcal{A}}^*\hat{\mathcal{A}}|_{\mathcal{S}_n}$, not $\mathcal{R}\big((\hat{\mathcal{A}}^*\hat{\mathcal{A}})^\mu\big) \cap \mathcal{S}_n$; the two are nested in neither direction, and (L2) is the witness that the compression is what the iteration sees, since "the projection is the identity on the iterates" holds *because* $P_{\mathcal{S}_n}$ was absorbed into $\mathcal{L}^*$. Lemma 85(iii) cannot bridge the gap, it is stated at one fixed domain and handles only the scalar rescaling.

(3) (P-a.3) is a **weakening** of the pair, and this is the decisive point. The alternative is to keep (a)+(b) and add a *commutativity* proviso: that $\hat{\mathcal{A}}^*\hat{\mathcal{A}}$ and the compressed scale operator share an eigenbasis on $\mathcal{S}_n$, under which Lemma 84(ii.b)'s device licenses interpolation at every $\mu > 0$. Under that proviso (a)+(b) **imply** (P-a.3). So the hypothesis actually assumed here is strictly weaker than the alternative's, and it needs strictly less proved, assuming the conclusion of a step whose only licence is an unverifiable spectral-alignment condition is the *weaker* assumption, not the stronger one. And nothing is lost that was ever owned. H1's matching is a statement about the *population* operator $\mathcal{A}_{\mathrm{cond}}$ on all of $\mathcal{V}$; (P-a.3) is a statement about a *realised, finite-rank, sieve-compressed* operator at a *random* base point, and Lemma 84(iii) already records that the link condition *cannot* hold on $\mathcal{V}$ because $\hat{\mathcal{A}}$ has finite rank by Proposition 70(ii). H1 therefore never delivers $\mu = s/(2a)$ for the operator (P-a) is about: (P-a) assumes its own matching, and stating it spectrally makes that assumption visible rather than adding one.

(4) What H1 still does, and what a derived μ would cost. H1 is **not** retired. It fixes the scale H^s in which Definition 80, A5(vi′), (P-c) and (P-g) are written; it supplies A2's Picard form; it carries the whole smoothness-budget table. What is withdrawn is only the clause "$\mu = s/(2a)$" from Proposition 28's hypothesis line. *A reader who wants μ derived rather than primitive may have it, at a stated price*, and the price is exactly one proviso: assume in addition that $\hat{\mathcal{A}}^*_{\hat{\varphi}^{(k-1)}}\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}$ and the compressed scale operator commute on $\mathcal{S}_n$; then (a)+(b) at index $s/(2a)$ give (P-a.3) at $\mu = s/(2a)$ with $\mathsf{R}_* \le c\,\mathsf{R}$, c being the interpolation constant, which is a *third* constant Definition 22 would then have to print before the data are seen.

That proviso is not diagnosable and no sufficient condition for it is known: it asserts that a random, rank-$\leq n_{\mathrm{m}}$ operator built from $2n$ design points has its singular system aligned with a spline–Sobolev scale. It is recorded here as the alternative, with its content stated plainly, and it is not assumed. *The honest positioning statement, which belongs in the paper and not in a footnote*: μ is primitive at (P-a.3); what pins it is H1 *plus* a matching proviso, stated and not proved; and that is the same structure the closest antecedent in the literature carries, where a spectral source condition and a link condition are two separate assumptions rather than one derived from the other.

(5) The consequence for the **symbol**: the glyph μ names two numbers, and they are not the same number.

μ_{H1} , *H1's index*, $s/(2a)$. It is the source index of A2, at the **population** operator $\mathcal{A}_{\mathrm{cond}}$ on **all of** $\mathcal{V}$, at $\theta_\dagger$. Its ingredients a (the decay of $\sigma_j(\mathcal{A}_{\mathrm{cond}})$) and s (the order of the *classical* Sobolev space in which $\varphi_\dagger$'s source ball is written, *not* of a $\mathcal{D}$-generated scale) are likewise objects at $\theta_\dagger$, by H1's own sentence.

μ , *(P-a.3)'s index*, and it is **primitive**. It is the exponent of a **spectral** source condition at the **finite-rank sieve compression** (53) of the **realised** operator $\hat{\mathcal{A}}$, at the **anchor** $\theta_\circ$ of Definition 24(b), which Definition 24(b) also states makes a, R_* and $\hat{\delta}_n$ $\theta_\circ$-specific.

Binding reading. **The symbol μ, bare, is (P-a.3)'s primitive index everywhere in Parts H and I**, and in particular in (55), (7), (137), (56), $\kappa = 2\mu/(2\mu+1)$ and every rate display built from them. A sentence that computes μ from (s, a) is a sentence about μ_{H1}, and it transfers to those displays only under the declined proviso of clause (4). *The distinction cuts both ways*: a description of scope in terms of $s > a + \frac{1}{2}$ **excludes** designs the theorem covers, a design with $\mu = 2$ at the compression and $s/(2a) = 0.4$ at the population operator satisfies every hypothesis of Theorem 29 and fails $s > a + \frac{1}{2}$, and it **claims** designs the theorem does not, since $s > a + \frac{1}{2}$ says nothing about the compression's spectral index. *The conclusion of Theorem 29 is untouched*: no proof step computes μ from (s, a).

Remark 87 (Why (P-a)'s three clauses are w.p.→ 1 envelopes and not O_P statements). An O_P reading of (P-a)'s first two clauses is strictly weaker than what (180)'s base case and Lemma 83 consume, and the gap cannot be closed by carrying a constant. The base case of Step 3's induction is $\varepsilon_0 \le \hat{\varepsilon}_0$ *with probability tending to one*; an O_P hypothesis gives instead $\varepsilon_0 \le C_0\hat{\varepsilon}_0$ w.p.→ 1 for a C_0 that is not a number but a family $\{C_{0,\epsilon}\}$ indexed by the tolerance. *Carrying* C_0 *is not available*, and the reason is that c_{14} and c_{15} are **part of the estimator**: Definition 22 prints them before the data are seen, so "choose $\mathsf{c}_{14} \ge \mathsf{c}_{16}\max\{1, C_0^2\}$" either says nothing or makes the estimator, hence (56), indexed by ϵ, at which point the conclusion is no longer $O_P(\delta_n)$ for *one* estimator. So the envelope is declared in the hypothesis, which is how Definition 22 already speaks of $\hat{\varepsilon}_0$.

The conversion device is owned by this block and the cost is a logarithm that is already paid. Proposition 79(iii) is the same manoeuvre one page earlier: an $O_P(a_n)$ statement becomes a w.p.→ 1 envelope at level $a_n \ln n$, at the cost of a divergent factor. Applied here it replaces $\hat{\varepsilon}_0$ by $\hat{\varepsilon}_0 \ln n$, which changes ς_0 by nothing at all, because (181) is a strict inequality between *polynomial* exponents and absorbs any $\ln n$. The same repair closes the second site. Step 3 needs $\mathsf{c}_{15} \ge \mathsf{c}_{17} = 2c_\mu \mathsf{R}_*^{1/(2\mu+1)}$ with c_{15} printed in advance; that is an inequality between numbers exactly because (P-a.3) fixes R_* deterministically and asserts the source condition on an event of probability tending to one, rather than delivering a pre-image of norm $O_P(\mathsf{R}_*)$. *One repair, two sites, and it is Proposition 79(ii)'s point applied to the hypothesis that generates it*: an O_P bound cannot supply a fixed constant to an object printed before the data.

Where the same discipline was already correct, recorded so that it is not re-litigated. A5(vi′) is now also read as a w.p.→ 1 envelope at the *fixed* radius R, $P(\max_{k\le\bar{k}} \|\hat{\varphi}^{(k)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}) \to 1$, because Lemma 83 needs membership of a *deterministic* ball Definition 80 fixes, and R is not a free tuning constant: it appears at every site Definition 80 enumerates, including, through Lemma 81 Step 4, in the x-smoothing bias. *The hypothesis is therefore an existential over*

one R *shared by all of them*, and it is stated that way in A5(vi′) rather than left to be assembled by a reader. Step 4(a.1) and (a.2) are untouched by any of this: they run on the deterministic $\hat{\varepsilon}_k$.

Remark 88 (F-6 / O10: the nonlinear transfer, and why it is declined rather than performed). The registered route for Proposition 28 is nonlinear Landweber–Fridman with a tangential cone condition and the discrepancy principle, after Hanke, Neubauer, and Scherzer (1995) and Engl, Hanke, and Neubauer (2000, Ch. 11), with the operator-noise model of Dunker, Florens, Hohage, Johannes & Mammen (2014). That route does not transfer, and the obstruction is not a technicality. Three things go wrong, of increasing severity, and each is proved or cited rather than reported.

(1) The cone condition is false on $\mathcal{V}$, and on the sieve, where the estimator lives, it is contingent on a product no assumption constrains. Proposition 71 refutes it on every $L^2(\pi)$-ball, for every bounded linearisation. *That is not the same as refuting it for a sieve method*, and the distinction is now printed rather than elided: on $\mathcal{V}_n \cap \mathcal{S}_n$ the spike is not representable at all, the exact first-order *necessary* condition is (109), an "only if", for the reason Remark 72(3) prints, and whether it holds turns on $h_x L_x$, unconstrained by C5′ and A8, and free over a range containing both verdicts inside the block's own admissible window (Remark 72(3)). *So reason (1) is the weakest of the three, not the strongest*, and it is repairable in the ambient space too, by A6b and Proposition 74, at the cost of replacing $\hat{T}$ by $\hat{T}_{h_x}$.

(2) Range invariance, the only route to a cone condition that does not simply assume it, fails. Proposition 76. The strategy memo caps this attempt at one round; the round is spent.

(3) The decisive one, because it is regime-independent: the inherited rate theorem is not available above $\mu = \frac{1}{2}$, and (7) needs $\mu > \frac{1}{2}$. The rate theorem for nonlinear Landweber under a tangential cone condition and the discrepancy principle (Hanke, Neubauer, and Scherzer, 1995; Engl, Hanke, and Neubauer, 2000) delivers $\|x_{\hat{m}} - x^\dagger\| = O(\hat{\delta}^{2\mu/(2\mu+1)})$ for source index $\mu \in (0, \frac{1}{2}]$, and above $\frac{1}{2}$ *the analysis gives no improvement*, the qualification of the method, as

its proof uses operator monotonicity of $z \mapsto z^{2\mu}$, which is the barrier Lemma 84(ii.a) exhibits. Two things must be kept apart. The claim made is that *the theorem is unavailable* above $\frac{1}{2}$; the claim *not* made is that the *rate* fails there, that would be a saturation statement, requiring a lower bound, and no such result is proved or cited in this file. The disposition needs only the first: one cannot inherit a theorem one does not have. At the qualification index the programme then fails, and fails even with a perfect propensity: at $\mu = \frac{1}{2}$ the available rate is $\hat{\delta}_n^{1/2}$, and since $\hat{\delta}_n \geq \hat{\delta}_n^{(T)} = n^{-1/2}\sqrt{\ln n}$ by A5(i)'s floor, *not* by Lemma 77, which bounds the operator error from above and delivers no floor, the best attainable is $\delta_n \asymp n^{-1/4}(\ln n)^{1/4}$, which is *not* $o(n^{-1/4})$; at $\mu < \frac{1}{2}$ it fails polynomially. So the inherited rate and the paper's own feasibility frontier are incompatible, and the incompatibility is exactly at $\mu = \frac{1}{2}$, and it is so on the sieve and off it alike, which is why this and not (1) is the reason to lead with. And the ceiling is not an artefact of the regularisation literature's way of stating things: it is imposed, in the same place and for the same reason, in the closest econometrics treatment. Centorrino, Fève, and Florens (2025) run a nonlinear ill-posed inverse problem of exactly this type and cap their own source index at $\beta \leq 1 - \epsilon$; in the normalisation of this file $\beta = 2\mu$, so their cap *is* $\mu < \frac{1}{2}$, and it is the Heinz range of Lemma 84(ii.a) seen from the applied side. That the same number is reached from operator interpolation here and from an econometric construction there is the strongest available evidence that $\frac{1}{2}$ is a property of the problem and not of the proof.

What is done instead, and at what price. Definition 22 linearises and runs *linear* Landweber–Fridman at each stage, absorbing the nonlinearity into the noise level through Proposition 70(iv)'s exact quadratic remainder. Linear Landweber with the discrepancy principle does not saturate, so the source index is unrestricted from above; what the route needs instead is that the absorbed remainder be dominated by the noise, which is (P-f) and is $\mu > \frac{5}{6}$. The condition that removes the inherited route at $\frac{1}{2}$ is the condition that switches the replacement route on at $\frac{1}{2}$, and the replacement then charges a further $(\frac{1}{2}, \frac{5}{6}]$. *And the replacement's own load-bearing input must be named in the same sentence*: it reaches the same

place **under (P-a.3)**, a random *spectral* source condition at a data-dependent base point, imposed on the finite-rank compression (53) of the realised sample operator and derived from nothing (Q-4). Sold without (P-a.3) the sentence is not true. *This is the substantive resolution of F-6 / O10 and it should be stated in the paper as such, not buried:* the paper does not inherit a nonlinear-regularisation rate, it proves a linearised one under an assumed source condition, and the reason it may is that its own frontier already lives on the side of $\mu = \frac{1}{2}$ where linearisation closes. *And the assumption is stated spectrally rather than as a Hilbert-scale pair, for the reason Remark 86 gives*: the pair does not imply the spectral condition above $2\mu = 1$, so a paper claiming to assume the pair and to apply (F-f) would be claiming an implication Lemma 84(ii.a) refutes.

What is given up. Three things, and they should be printed.

(a) *Global convergence.* Nonlinear Landweber with a cone condition converges from any starting point in the ball where the condition holds. Definition 22 needs a preliminary $\hat{\varphi}^{(0)}$ that is not merely consistent but consistent *at a known polynomial rate* $\hat{\varepsilon}_0$, because the stagewise schedule (37) is built from it. That is L-14's object and is an input, not an output, of Proposition 28. The paper must say where $\hat{\varphi}^{(0)}$ comes from; a fixed number of Landweber steps from 0 on the linearised equation at $\varphi = 0$ will not do without an argument (Q-1).

(b) A sieve, and it is given up twice over. Lemma 84(iii.a) forces any link-type condition onto $\mathcal{S}_n$, and (P-a.3) states its source condition on the compression to $\mathcal{S}_n$ for the same reason, so the estimator is a *sieve* LF and A8's upper end is consumed by the φ-step as well as by the balancing solve. But that is not the only thing the sieve costs, and the second cost was not booked. Even granting a cone condition on $\mathcal{S}_n$, the inherited theorem is stated for *unprojected* nonlinear Landweber, with the cone condition holding on a ball of the *ambient* space containing the solution; Definition 22(b) projects onto $\mathcal{S}_n$, projection does not commute with $\hat{T}_{h_x}$, and $\varphi_\dagger \notin \mathcal{S}_n$. So the sieve restriction also

moves the target to the sieve projection of $\varphi_\dagger$ and would oblige a re-derivation of the cited theorem, not merely its citation. *This is a defect of the inherited route, not of Definition 22*, whose analysis is conducted on $\mathcal{S}_n$ throughout and pays for the projection at (P-c).

(c) The Dunker et al. noise model is adopted and their regulariser is not. The iteratively regularised Gauss–Newton method (Dunker, 2021) is the standard answer to exactly the qualification problem in (3), attaining source indices above $\frac{1}{2}$. *Declining IRGN and then needing $\mu > \frac{5}{6}$ is a live tension in the project's positioning*, and the honest options are two: adopt IRGN, or state, as this file does, that *under* (P-a.3) the linearised route reaches the same place, at a threshold that is the same at $\frac{1}{2}$ and stricter at $\frac{5}{6}$. The second is cheaper and is what is proved; the first should be named in the paper as the alternative and priced, because a methods referee who knows the regularisation literature will ask, and will ask *about* (P-a.3) first. *Two citations must be kept apart*: the IRGN / above-$\frac{1}{2}$ claim belongs to Dunker (2021), while the *operator-noise model* this remark's opening adopts is Dunker, Florens, Hohage, Johannes & Mammen (2014). Both are needed and they are different papers. *And the reason for declining IRGN is not that the linearised route is "cheaper"*: it is that IRGN, as the literature states it, treats $\hat{T}'_\varphi$ and $\hat{T}'^*_{\varphi_\dagger}$ as operators with the same range and *does not model the estimation error arising from using regression residuals in place of the true error*, which is exactly the h_u^{-2} object (176) and (P-f) exist to price. *The route that would remove (P-f) is the route that does not model what (P-f) is about*, and that, not cost, is why it is declined.

Remark 89 (What Proposition 28 does not claim). (1) It is a rate, not a sharpness statement. (56) is an upper bound on $\|\hat{\varphi}^\pi - \varphi_\dagger\|_{L^2(\pi)}$, *the norm being taken at the infeasible reading of Definition 22*(b) *and clause* (7) *below being where that matters.* Nothing here says that δ_n cannot be improved, and nothing here says that $\delta_n = o(n^{-1/4})$ is *necessary. In particular this file does not reinstate the strategy memo §2.5(iii) reading* that "$\delta_n = o(n^{-1/4})$ is sharp

for this moment function": that is a necessity claim about the estimator, Proposition 27 deliberately does not prove it, and Remark 65(1) blocks it. What Theorem 29 may inherit from Part G is the *lower bound* on the second-order form, and what it inherits from here is the *upper bound* (56); the two are different objects and neither is a sharpness claim about $\hat{\varphi}$.

(2) The logarithm is carried in the symbol, not absorbed into a power of n. δ_n in (56) is $(\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)}$, and A5(ii) is that same *rate* by (137); *so the symbol in the clause and the symbol in the conclusion denote one sequence, and that is all this clause settles. It does* ***not*** *follow that the clause is discharged, and the line printed here through 2026-08-27 — "this proposition therefore discharges the clause at the object the block names" — confused the two*: A5(ii) is stated at $\hat{\varphi}$ and (56) at $\hat{\varphi}^{\pi}$, so the *rates* agree and the *objects* do not, which is clause (7). *A downstream result that read A5(ii) as log-free and substituted a power of n would be off by* $(\ln n)^{2\mu/(2\mu+1)}$; Remark 97 carries the disposition. *The logarithm is not absorbed anywhere*: it is genuinely in the rate, Proposition 79(iv) shows the *strictness* of (7) pays for it in $\delta_n = o_P(n^{-1/4})$ and in $\delta_n^2 = o(n^{-1/2})$, and Remark 97(3) says which of the four rate products Theorem 29 needs are paid for that way and which one, A5(iv)'s $\delta_n \epsilon_n$, is not.

(3) It says nothing about $\hat{\ell}$, $\hat{L}$ or $\hat{b}$. A5(iv) prices $\|\hat{\ell} - \ell_\star\|_{L^2(P)}$ and is L-13's business; obligations (O2) and (O3) of Remark 58 are owed by the construction of $\hat{L}$ and are *not* discharged here. *This file discharges* (O4) *and nothing else on that list*, and discharges it with a negative answer on A6, which is a discharge and not an evasion: the obligation was to verify, and the verification has an outcome.

(4) It does not identify $\varphi_\dagger$. A1b(iii) is assumed, as everywhere; (56) is a rate for an estimator of an object assumed identified, and the discrepancy principle stops on the residual, which is informative about φ only to first order.

(5) It is a rate for the *infeasible reading* of the iteration Definition 22 defines — clause (7) being where that qualification is discharged into five parts — and that iteration, under *either* reading, needs four inputs and two constants the analyst must supply. The four inputs are $\hat{\delta}_n$

(hence r and d_Z), the bandwidth h_u (now a function of $\hat{\delta}_n$), the source index μ through κ, and the preliminary's declared rate $\hat{\varepsilon}_0$. *The two constants are* c_{14} *and* c_{15} *of* (37): they are not identified by the data, they must dominate the analytic constants c_{16} of (176) and c_{17} of (178), which depend on the kernel, on the *spectral pre-image* radius R_* of (P-a.3) and on (F-f)'s c_μ, and **not** on the H^s radius R, which is a different object, and if they are set too small the implemented level sits below the true data error, (L3) fails and (F-f) is void. *So the schedule is* ***computable****, which is what Definition 22 needs, and it is not* ***known***; the distinction is stated in Definition 22 and its practical size is what probe CP-13 measures. *A rule that used only the data would not do, and the reason has a sharp part and a cited part*: the realised residual cannot bound the unobservable error from below, because $\|\hat{\mathcal{A}}v_\dagger\| \ll \|v_\dagger\|$ is what ill-posedness *is*; and the general statement, that a purely data-driven choice of regularisation parameter cannot be convergent for an ill-posed problem, is Bakushinskii (1984) (see Engl, Hanke, and Neubauer (2000, §3.3)), which is a worst-case deterministic result and does *not* exclude Lepskii-type balancing in the statistical setting. Definition 22(2a)–(2b) says why the schedule is preferred here rather than forced here. The one input with no supplier in this file is $\hat{\varepsilon}_0$, which is Q-1, and which a balancing stop would soften; that is CP-13's third arm.

(6) It does not derive the source index and does not claim to. μ is **primitive**, at (P-a.3), together with the spectral pre-image radius R_*. H1 does **not** collapse four smoothness parameters to three by delivering $\mu = s/(2a)$, and the paper should not be read as claiming it does. H1 delivers that identity for A2's Picard form and for the *population* operator on $\mathcal{V}$, and the passage from it to the hypothesis (F-f) actually consumes, a *spectral* condition on a finite-rank compression at a random base point, is operator interpolation at exponent 2μ, licensed only at $2\mu \le 1$ and refuted at every $2\mu > 1$ by Lemma 84(ii.a), while (55) forces $2\mu > \frac{5}{3}$. *So a referee may ask what pins μ, and the answer the paper can give is: H1* ***plus*** *a matching proviso, stated and not proved*, Remark 86(4) states the proviso exactly and prices it, and the same two-assumption structure (a spectral source condition and a link condition, assumed separately rather than one derived from the other) is what the closest antecedents

in the nonparametric-IV regularisation literature carry. *What is* ***not*** *conceded*: that the rate is weaker, that H1 is retired, or that any other line of the budget table moves. (56) is unchanged, H1 still fixes the scale and supplies A2's Picard form, and the amendment is a change to the *status* of one identity, not to any conclusion.

(7) *It is a rate for the iteration run at* π*, and Definition 22 runs it at* $\hat{\pi}$*.* (56) is stated in $\|\cdot\|_{L^2(\pi)}$ and its proof runs the recursion through $\hat{\mathcal{A}}^*_\varphi$ as (101) writes it, at the population π. The estimator cannot: π is an unknown feature of P by Definition 5, so the adjoint the analyst forms is (101) at the floored $\hat{\pi}$ that Definition 22 fixes. *That definition accordingly separates the two objects by name from 2026-08-27 —* $\hat{\varphi}^\pi$ *for the reading* (56) *is scoped to,* $\hat{\varphi}$ *for the one Definition 24*(b) *plugs in — and this clause is the account of the passage between them. The gap is in five parts and the division is the point of the clause. Two are settled outright*: (7b) and (7e). *One is settled* ***with a rider it does not discharge***: (7a), which identifies the feasible adjoint and the feasible sieve and then flags that the sieve is *random* while the measurability verification is written at a fixed one. *A part carrying an undischarged rider is not settled outright, and the distinction is not bookkeeping here*: that rider is a hypothesis of another definition, Definition 24(c′.1), whose consumer is Lemma 94's cross-fitting argument — which kills the empirical-process term off $\sigma(\mathcal{I}_{-k})$-measurability and off no rate, so the rider is reached by nothing in (7c) or (7d). *Two are open*: (7c), open and **priced**; (7d), open and unpriced. *So the honest count is two settled, one settled with a rider, two open*, and the line this clause printed through 2026-08-27 — "(7a), (7b) and (7e) settle three of them" — was short by exactly that rider, at the clause whose subject is an accounting. *The parts are lettered* (7a)–(7e) *and not* (a)–(e)*, deliberately*: Definition 22 has clauses of the latter shape and this clause cites them.

(7a) The substitution is not ad hoc. Write $\hat{\mathcal{A}}^*_{\hat{\pi}}$ for (101) with $\hat{\pi}$ in place of π. It is the adjoint of $\hat{\mathcal{A}}_\varphi$ in the $L^2(\hat{\pi})$–$L^2(\Pi)$ pairing: the interchange and the zero-mean step of Proposition 70(iii) read the weight only as a bounded function bounded away from zero, which $\hat{\pi} \in [\varepsilon_\pi, \varepsilon_\pi^{-1}]$ is

at every sample point, and $\int_{\mathcal{X}} \hat{\nu}_{u,t,l}\, dx = 0$ is a statement about the numerator alone. So the feasible recursion is a genuine Landweber iteration for the same forward operator, read on $L^2(\mathcal{X}, \hat{\pi}\, dx)$ rather than on $L^2(\mathcal{X}, \pi\, dx)$. *What moves with the weight is where the adjoint lands*: $\hat{\mathcal{A}}^*_{\hat{\pi}}\Xi$ has zero $\hat{\pi}$-mean and not zero π-mean. So the space the *feasible* construction projects onto in Definition 22(b) is $\mathrm{span}(b_1, \ldots, b_{L_x}) \cap \{h : \int_{\mathcal{X}} h\, d\hat{\pi} = 0\}$, with the projection $L^2(\hat{\pi})$-orthogonal, and **not** the $\mathcal{S}_n$ that definition writes — which the analyst cannot form, $\mathcal{V}$ being defined by π. *One consequence of that is flagged and not discharged, and it is the* ***rider*** *that keeps this part from being settled outright*: the feasible sieve is a **random** finite-dimensional subspace, while the measurability verification in the supplement's discussion of the estimator is written at a fixed $\mathcal{S}_n$. The Carathéodory device that verification already runs for the random functionals (96) is the natural instrument for a random basis as well, *and it is not run for one here. The rider is not a note*: $\sigma(\mathcal{I}_{-k})$-measurability of $\hat{\varphi}^{(-k)}$ is what Definition 24(c′.1) requires and what Lemma 94 consumes, so this part feeds another definition's hypothesis, and *it is reached by neither* (7c) *nor* (7d) — both of those are about rates and operators, and Lemma 94 kills its empirical-process term off measurability and off no rate. *The membership half of the same requirement, $\hat{\varphi}^{(-k)} \in \mathcal{V}$, is a separate matter and* ***is*** (7c), the feasible iterate having zero $\hat{\pi}$-mean where $\mathcal{V}$ asks for zero π-mean.

(7b) The norm in which the rate is read is not the difficulty. Under Definition 22's floor and C1(ii), $\hat{\pi} \geq \varepsilon_\pi \geq (\varepsilon_\pi/\|\pi\|_\infty)\,\pi$ and $\hat{\pi} \leq \varepsilon_\pi^{-1} \leq \varepsilon_\pi^{-1}\|1/\pi\|_\infty\,\pi$ pointwise on $\mathcal{X}$, so

$$\frac{\varepsilon_\pi}{\|\pi\|_\infty}\,\|h\|^2_{L^2(\pi)} \;\leq\; \|h\|^2_{L^2(\hat{\pi})} \;\leq\; \frac{\|1/\pi\|_\infty}{\varepsilon_\pi}\,\|h\|^2_{L^2(\pi)} \qquad \text{surely, for every } h \in L^2(\pi),$$

with both constants deterministic and finite. A rate in either norm is therefore the same rate in the other, and (56) would not have to be restated if its conclusion survived the substitution.

(7c) Open, and quantified: the level of the iterate is pinned by $\hat{\pi}$. By (7a) the feasible sieve is $\mathrm{span}(b_1, \ldots, b_{L_x}) \cap \{h : \int h\, d\hat{\pi} = 0\}$, so $\int_{\mathcal{X}} \hat{\varphi}\, d\hat{\pi} = 0$ while A1b(i) puts $\int_{\mathcal{X}} \varphi_\dagger\, d\pi = 0$. Writing

$h := \hat{\varphi} - \varphi_\dagger$ and using the first identity to subtract,

$$\left| \int_{\mathcal{X}} h \, d\pi \right| \;=\; \left| \int_{\mathcal{X}} \hat{\varphi} \, (\pi - \hat{\pi}) \, dx \right| \;\leq\; \|1/\pi\|_\infty^{1/2} \left\| \hat{\varphi} \right\|_{L^2(\pi)} \left\| \hat{\pi} - \pi \right\|_{L^2(\mathcal{X}, dx)},$$

by Cauchy–Schwarz in $L^2(\mathcal{X}, dx)$ and $\|\hat{\varphi}\|_{L^2(\mathcal{X},dx)} \leq \|1/\pi\|_\infty^{1/2} \|\hat{\varphi}\|_{L^2(\pi)}$. Since π is a probability density, $\|h\|^2_{L^2(\pi)} = \|P_{\mathcal{V}} h\|^2_{L^2(\pi)} + \left(\int h \, d\pi \right)^2$, so the error carries a *constant* component that no argument run inside $\mathcal{V}$ can see, and (56) survives this mechanism as soon as

$$\left\| \hat{\pi} - \pi \right\|_{L^2(\mathcal{X}, dx)} \;=\; O_P(\delta_n).$$

That is a rate condition on a first-stage estimator of exactly the form A4(iii) imposes on $\hat{\varrho}$, and this paper imposes none on $\hat{\pi}$. The mechanism is therefore priced but not discharged: what it needs is one assumption, and the assumption is not made. *It has a **second** consumer, and the two are not closed by the same thing, which is why they are separated here rather than run together*: $\int \hat{\varphi} \, d\hat{\pi} = 0$ with $\int \hat{\varphi} \, d\pi \neq 0$ is exactly the failure of $\hat{\varphi}^{(-k)} \in \mathcal{V}$ that Definition 24(c′.1) requires, so this part is a term in a rate **and** a membership hypothesis of another definition. *The displayed rate condition closes the first and **not** the second*: it makes the constant component $O_P(\delta_n)$, and $O_P(\delta_n)$ is not 0, so exact membership in $\mathcal{V}$ is not restored by any rate. *The repair available for the membership half is a container weakening — read* (c′.1)*'s third entry at $L^2(\pi)$ rather than at $\mathcal{V}$, the move dispatch 14 already made at Lemma 45*(i) *— and it is **not** made here, because it has to be checked at every consumer of that clause and this pass checked none of them.* And neither half touches (7d) or (7a)'s measurability rider, so the rate condition alone does not restore (56) at $\hat{\varphi}$ by any route.

(7d) Open, and not quantified: the operator the source condition is posed at. Since $\hat{\mathcal{A}}^*_{\hat{\pi}} = (\pi/\hat{\pi}) \, \hat{\mathcal{A}}^*_{\varphi}$, the recursion is governed by $\hat{\mathcal{A}}^*_{\hat{\pi}} \hat{\mathcal{A}}_{\varphi} = (\pi/\hat{\pi}) \, \hat{\mathcal{A}}^*_{\varphi} \hat{\mathcal{A}}_{\varphi}$, *the leading factor acting as multiplication by the function $\pi/\hat{\pi}$ and written out rather than given a letter, italic M being retired*: self-adjoint in $L^2(\hat{\pi})$, *not* in $L^2(\pi)$, and a bounded, boundedly invertible, in general

non-constant multiplicative perturbation of the operator (P-a.3) is stated at. *Norm equivalence does not repair this, and that is the trap the clause exists to name*: equivalent renorming preserves rates, and it does not preserve range conditions, since $\mathcal{R}\big((\pi/\hat{\pi})\hat{\mathcal{A}}^*_\varphi\big) = (\pi/\hat{\pi})\,\mathcal{R}\big(\hat{\mathcal{A}}^*_\varphi\big)$ need not be the same subspace unless $\pi/\hat{\pi}$ is constant, and the ranges of the fractional powers $(\mathcal{A}^*\mathcal{A})^\mu$ that (P-a.3) reads are not in general preserved either. So (P-a.3)'s spectral pre-image $\varsigma^{(k)}$ and its radius R_*, and with them (F-f) and Steps 2–3 of the proof, are stated for an operator the feasible iteration does not run.

For (7d) *two routes are available, neither is taken, and that is why* (7d) *is an open item and not a lemma.* (i) Pose the source condition directly at the feasible operator. The cost is that the hypothesis is then a condition on an estimated object rather than on P, which is the standard Remark 58 holds every other nuisance to. (ii) Treat multiplication by $\pi/\hat{\pi}$ as a perturbation of the identity and price it. This needs a *uniform* rate for $\|\hat{\pi}/\pi - 1\|_\infty$ — no condition in this paper supplies one, $\hat{\pi}$ being introduced by Definition 22 and constrained there by nothing beyond the floor — together with a stability result for source conditions under multiplicative perturbation, which this paper does not carry. *Route* (ii) *is the honest one and the work is not done here. The two open parts have one prerequisite in common*: on a bounded $\mathcal{X}$ a uniform rate for $\|\hat{\pi}/\pi - 1\|_\infty$ implies the $L^2(\mathcal{X}, dx)$ rate (7c) asks for, since $\|\hat{\pi} - \pi\|_{L^2(\mathcal{X},dx)} \le |\mathcal{X}|^{1/2}\|\pi\|_\infty\|\hat{\pi}/\pi - 1\|_\infty$, so a single assumption on $\hat{\pi}$ would discharge (7c) outright and open (7d) to route (ii).

A third route would dissolve the item and is declined for a stated reason. The results proved here read π through C1(ii)'s two-sided bounds, through $\bar{c}_\pi$, and through the normalisation of $\mathcal{V}$; no step consumes the identity $\pi = \frac{1}{2}(f_{X_1} + f_{X_2})$ itself. *That last sentence is a reading and not a census: no systematic audit of the corpus for further consumers has been made, and one may exist.* On that reading a *known* π_0 bounded above and below on $\mathcal{X}$ would satisfy C1(ii), would make (101), (35) and (36) computable outright, and would close (7c), (7d) and (7e) below at once, the two pairings of (7e) collapsing to one at $\hat{\pi} = \pi_0 = \pi$. It is declined because

π is the design density the estimator's $\hat{\pi}$ is built to estimate, and because the exchange is not free: the singular system of $\mathcal{A}_{\mathrm{cond}}$, and hence what A2, H1 and (P-a.3) assert, move with the weight, so the smoothness a source condition posed at π_0 asserts is smoothness relative to π_0 and not relative to the design.

(7e) Settled, and it was not among the four parts this clause listed until 2026-08-27: the iteration must be ***relaxation-admissible*** *in the pairing it runs in.* (F-f)'s standing hypothesis $\|\mathcal{L}\| \leq 1$ is a bound on the operator norm *out of the domain Hilbert space whose inner product defines* $\mathcal{L}^*$, Remark 124. By (7a) that space is $L^2(\hat{\pi})$ for the feasible recursion and $L^2(\pi)$ for the one Proposition 28 analyses, so the condition has to be met in whichever is in force. *The other four parts do not reach it, and that is why it is a fifth and not a corollary of one of them*: (7a) settles the adjoint identity and identifies the feasible sieve, (7b) settles the norm the *conclusion* is read in — explicitly a different question — (7c) prices the level of the iterate, (7d) the operator the source condition is posed at. *Nor does* (7b)*'s equivalence do the work, and the trap is the same species it names*: two-sided equivalence with deterministic constants transfers *rates*, and a normalisation is not a rate — it is an inequality against the constant 1, which no equivalence preserves.

What the omission cost, stated at the witness, and stated ***as a gap*** *because that is what it is.* Let $\hat{\pi} \equiv \varepsilon_\pi$, a constant, which the construction convention of Definition 22 admits — it is the function that definition's own non-emptiness argument exhibits, and nothing beyond the two-sided floor constrains $\hat{\pi}$ there. Then $\|h\|^2_{L^2(\hat{\pi})} = \varepsilon_\pi \|h\|^2_{L^2(\mathcal{X},dx)}$ exactly, and (104) gives $\|\hat{\mathcal{A}}_\varphi\|_{L^2(\hat{\pi})\to L^2(\Pi)} \geq (\varepsilon_\pi \|1/\pi\|_\infty)^{-1/2} \|\hat{\mathcal{A}}_\varphi\|_{L^2(\pi)\to L^2(\Pi)}$, whose right-hand side diverges as $\varepsilon_\pi \downarrow 0$ at a fixed ω_n. *Against a scale carrying* $\|1/\pi\|^{1/2}_\infty$ *alone — which is what* (35) *printed through 2026-08-26 —* $\|\mathcal{L}\| \leq 1$ *in* $L^2(\hat{\pi})$ *is then* ***not implied*** *at any* $\varepsilon_\pi < \|1/\pi\|^{-1}_\infty$, that is, throughout the range Definition 22 itself recommends on the separate ground that the clipping there costs nothing: the only bound this corpus carries for that operator norm is (103), and its constant exceeds the old ω_n by the factor $(\varepsilon_\pi \|1/\pi\|_\infty)^{-1/2} > 1$. *Whether the*

inequality was actually ***violated*** *at some sample is not settled here, and the reason it cannot be is itself the point*: that would need a matching *lower* bound on $\|\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\|$, which is **Q-4**'s object and which Lemma 85's closing note records as unavailable. *So the defect is a* ***proof gap*** *and not an exhibited divergence* — (L1) was checked against a bound in the wrong pairing, and the corpus could neither verify it in the right one nor produce a sample at which it failed. *What makes it worth a part of its own rather than a footnote is the failure mode if the inequality does break*: once $\|\mathcal{L}\|^2 > 2$ the map $\mathrm{Id} - \mathcal{L}^*\mathcal{L}$ carries an eigenvalue below -1, the domain being finite-dimensional, and Definition 22(b) diverges geometrically rather than converging slowly — a mode against which a rate theorem offers no protection at all. *The repair below removes the question rather than answering it*: once (35) covers both pairings, how tight Proposition 70(i) is stops bearing on (L1), and Q-4 returns to being about the *rate* and not about convergence.

The repair, and its scope. (35) now carries $\max\{\|1/\pi\|_\infty^{1/2}, \varepsilon_\pi^{-1/2}\}$, the maximum of the two conversion costs Proposition 70(i′) computes from the single primitive estimate (102), so $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi\| \le 1$ holds *surely in both pairings* and (L1) closes whichever the reader runs. *The repair is a constant and not a rate*: ε_π is fixed before the data and $\|1/\pi\|_\infty$ is finite by C1(ii), so the maximum is a fixed positive number free of n, $\omega_n \asymp h_u^{-1}h_x^{-1/2}$ is unchanged, and every asymptotic statement reading ω_n — the power of n in (36), clause (v) of Lemma 85 and its instantiation, Proposition 28 Step 2 (L4)'s stopping-index bound, and the exact cancellation (124), which is stated for *every* $\omega_* > 0$ — is untouched. *What moves is one constant*: the right-hand side of (36) is multiplied by at most $(\varepsilon_\pi\|1/\pi\|_\infty)^{-1}$, a computational budget and not a rate condition. *And the number was already in the corpus, filed one shelf over*: the discussion of the estimator in the supplement offered $\varepsilon_\pi^{-1/2}$ as the analyst's **computable** substitute for an unknown $\|1/\pi\|_\infty^{1/2}$. It is the same number and it was the **stability** constant all along — and, on the range Definition 22 recommends, the maximum *is* $\varepsilon_\pi^{-1/2}$, so the computable choice and the admissible one coincide rather than compete. *What* (7e) *does* ***not*** *close*: (7c) and (7d) are untouched by it and remain open, and (7a)'s random-sieve flag

is likewise unaffected.

One structural observation, which (7e) *settles nothing about and which is recorded because it is what kept the omission from being a false theorem.* (P-a.3) is posed at the *realised* iterates, $\bar{v}_\dagger^{(k)} = P_{\mathcal{S}_n}(\varphi_\dagger - \hat{\varphi}^{(k-1)})$ for every $k \le \bar{k}$, and $\bar{k} \ge 2$ always, Proposition 28 Step 4(b). So on any configuration at which the stage-1 iteration fails to contract, the hypothesis at $k = 2$ demands a bounded spectral pre-image for a divergent $\hat{\varphi}^{(1)}$ and *empties*: the proposition is vacuous there, not false. *That is a thin protection and it should be read as one.* A hypothesis stated at the realised path is checkable only after the estimator has been run, so it silently absorbs defects in the estimator's own tuning — here, a choice of ε_π — and converts what would otherwise be a visible falsehood into an invisible vacuity. *The device is legitimate*, it is what Step 0's conditional route buys and what Remark 86 defends, *and its cost is that the hypothesis set can be emptied by a tuning constant rather than by a feature of P.* (35) as amended removes the one such mechanism identified in this corpus; *no claim is made that it is the only one*, and a systematic audit of which hypotheses can be emptied by an admissible choice of a tuning constant has not been run.

Remark 90 (What the hypotheses of Proposition 28 carry, and why (P-a.3) is load-bearing). (1)

Three tags are narrowed from the blocks "B1", "C1" and "C5′" that stood in the hypothesis list, and each narrowing removes a clause this proposition cannot have. B1(vi) is **retired** and is not available as a hypothesis of any result, so a block citation of B1 charged this proposition with it; C1(i) is consumed at exactly one site in the corpus, Step 2 of Lemma 78, the route this proposition's Step 0 **declines**, while C1(ii) is consumed transitively through Proposition 70 and Lemma 45; and C5′(ii) is a *lower* bound on h_x while this proposition's own (P-g) is an *upper* bound on h_x, so a block citation of C5′ put a floor and a ceiling in one hypothesis set, *and, at the inherited strengthening of C5′(ii), a floor above the ceiling. The two do not meet at every admissible parameter, and the failure is not a corner*: at

$(\mu, s, a, r, d_Z) = (2, 1.2, 0.3, \frac{8}{3}, 3)$, which clears (7) at $\kappa\alpha = 0.256$ and clears (54) at $s > 1$, the strengthened floor is $h_x \gg n^{-0.44}$ and (P-g) is $h_x = O(n^{-0.5143})$, *empty by the exponent gap* 0.0743. The threshold is (63), which reads $s > 1.3182$ at $\alpha = 0.32$ and is failed by $s = 1.2$, and the emptiness band is non-empty at every $\alpha \in (\frac{1}{4}, \approx 0.359)$, which (7) does not exclude. ***What follows from that is vacuity, not falsity***: on the block-cited reading this proposition's hypotheses were unsatisfiable there and it held on an empty hypothesis set, and no conclusion of this corpus is false at that configuration. The narrowing removes the conflict from the hypothesis set; C5′(ii)'s own restatement removes it from the maintained block.

(2) The clause $p_K \geq s - \frac{1}{2}$ is **not** free at every s the budget table admits. Lemma 81 Step 4 consumes it, and $p_K = 8$ being *fixed* by C3 and by C5′(i) it is a hard cap $s \leq \frac{17}{2}$: p_K is not a parameter the analyst may raise against s, so the clause caps s, while (54) is a *floor* on s that diverges as $\alpha \uparrow \frac{1}{2}$. *The two are incompatible at every* $\alpha \geq \frac{68}{145} \approx 0.4690$*, at every* s *and every* a: the floor $9\alpha/\big(8(1-2\alpha)\big)$ reaches $\frac{17}{2}$ there. *Witness, and it is admissible*: $r = 8$, $d_Z = 1$, so $\alpha = \frac{8}{17}$; with $\mu = 2$ the frontier (7) clears at $\kappa\alpha = \frac{8}{17} \cdot \frac{4}{5} \approx 0.3765 > \frac{1}{4}$; (54) demands $s > 9$ and the cap allows $s \leq \frac{17}{2}$; *no* s *exists. The disposition is now a printed restriction, and this remark is where it is derived*: Proposition 28 carries $\alpha < \frac{68}{145}$ in its own hypothesis list, and Theorem 29's group (T-5) inherits it, so on $[\frac{68}{145}, \frac{1}{2})$ the proposition is **excluded rather than vacuous**, it does not speak there, and (T-5) is inapplicable rather than unsatisfiable. *In primitives the restriction is* $9r < 68\, d_Z$, both of which A4(ii) declares, so a reader can check it without leaving the hypothesis list. A4(ii)'s $d_Z \geq 1$ repair delivers only $\alpha < \frac{1}{2}$ and does not reach this range; the cap and the restriction it creates are both recorded as rows of the smoothness-budget table. *And A8 is cited by clause and not as a block*: A8's own lower end $L_x \gg n^{1/(2s)}$ is the balancing route's, A8's own comment already records it as unused on the rate route, and item (8) below prices what carrying it would have cost.

(3) (P-a.1): the measurability half is a requirement on the preliminary and is stated **there** because some hypothesis list must carry it. Definition 22's measurability verification inducts on k *from* it, so the verification that the estimator is a random element consumes it. *The two remaining halves are jointly satisfiable only under* (8)*, and that is not a technicality*: $\hat{\varphi}^{(0)} \in \mathcal{S}_n$ forces $\|\varphi_\dagger - \hat{\varphi}^{(0)}\|_{H^s}$ to be read at a sieve element, so unless $\mathcal{S}_n \subset H^s$ the norm is $+\infty$ surely and the two halves of that one clause are *jointly unsatisfiable*, the probability statement would then hold on the empty event. A8's basis clause supplies (8), with it the well-posedness of (P-a.1), and H1 supplies $\varphi_\dagger \in H^s$.

(4) (P-a.3) *is* ***not*** *the pair, a Hilbert-scale source condition on* $\bar{v}_\dagger^{(k)}$ *together with the link condition of Lemma 84(iii) on* $\mathcal{S}_n$*, and does not follow from it*: the pair delivers (P-a.3) only through operator interpolation at exponent 2μ, which Lemma 84(ii.a) refutes at *every* $2\mu > 1$ while (55) forces $2\mu > \frac{5}{3}$, so the pair never implies what (F-f) consumes. Remark 86 carries the full disposition. *And the operator is named as the compression, which is what* (53) *is for*: the operator (F-f) is applied to in Step 2 of the proof is $\mathcal{L}^*\mathcal{L}$ with $\mathcal{L} = \omega_n^{-1}\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}|_{\mathcal{S}_n}$, and that *is* $\omega_n^{-2}\hat{\mathcal{P}}^{\mathcal{S}}_{\hat{\varphi}^{(k-1)}}$ and not the restriction of $\hat{\mathcal{A}}^*\hat{\mathcal{A}}$ to $\mathcal{S}_n$, two operators whose ranges are nested in neither direction.

(5) What the empty letter (P-b) reserved. An operator-estimation budget $\gamma_n = O(\hat{\delta}_n^{1/(2\mu+1)})$ belonging to the perturbation-of-$\mathcal{A}$ route, which Step 0 of the proof does not take; nothing is imposed at that letter and the list runs (P-a), (P-c)–(P-g). The label is kept empty rather than renumbered so that an existing citation of (P-b) lands on the reservation.

(6) (P-a.2)*'s requirement on* ς_0 *is strictly weaker than the conclusion* (56) *precisely when* $\mu > \frac{5}{6}$*, which is (P-f),* ***the same inequality*** *rather than a second one that agrees with it*, as Step 4(a.1) of the proof shows. *And what (P-f) reads is bandwidth-dependent*: Remark 125 unpacks it, shows that at the unamended bandwidth $h_u \asymp n^{-1/16}$ it reads $\frac{2\mu-1}{2\mu+1} \cdot \frac{r}{2r+d_Z} > \frac{1}{8}$, and shows that at C5′(i) as amended, $h_u \asymp \hat{\delta}_n^{1/p_K}$, the largest bandwidth

whose bias is $O(\hat{\delta}_n)$, it reads $\mu > \frac{5}{6}$, *strictly, with the strictness forced by the logarithms.*

(7) (P-c): the factor h_u is not decoration. The source and link conditions are imposed on $\mathcal{S}_n$, so the proof works with the sieve projection $P_{\mathcal{S}_n}(\varphi_\dagger - \hat{\varphi}^{(k-1)})$ and must carry the residual through $\hat{\mathcal{A}}$, whose norm on the complement is bounded only by $O(h_u^{-1})$. *(P-c) is a* ***lower*** *end for L_x, and it is this proposition's own; it* ***replaces*** *A8's lower end rather than restating it*, being read at $h_u\hat{\delta}_n$ and not at $n^{-1/2}$.

(8) What (54) is numerically, and what carrying A8's lower end as well would have cost. (54) is $s > \frac{27}{16} = 1.6875$ at $\alpha = \frac{3}{8}$ and $s > 1$ at $\alpha = 0.32$. This is a genuine joint line in the smoothness budget and it is recorded there. *Had A8's lower end been carried as well*, the window would instead require $s > \big(2(1-2\alpha)\big)^{-1}$, which is $s > 2$ at $\alpha = \frac{3}{8}$; that is strictly the stronger line at every $\alpha < \frac{4}{9}$, and it is why A8 is cited by clause. A configuration on which *both* routes are to be available at once must clear the stronger one.

(9) What the conclusion discharges, and what it consumes, *and the list is shorter from 2026-08-27 than it was, because the conclusion is scoped.* A5(i) is **discharged, not assumed**, through Lemma 77. *A5(ii) splits*: its second half $\delta_n = o_P(n^{-1/4})$ is discharged, being a statement about the deterministic sequence δ_n alone; its *first* half, the rate itself, is **not**, because that half is stated at $\hat{\varphi}$ — the *feasible* reading of Definition 22(b) — while (56) is stated at the infeasible $\hat{\varphi}^{\pi}$, and the passage between them is clause (7) below, open at (7c) and (7d). A5(iii) is not discharged and no proposition can discharge it, being an inequality among primitives. A5(vi′) is **consumed**, it is a hypothesis of Proposition 28, at Lemma 83(c), and A5(vi) follows from it at $k = \bar{k}$; A5(iv) and A5(v) concern other objects and are untouched. *The line this clause printed through dispatch 27 — "A5(i)–(iii) are discharged, not assumed" — was wrong about A5(iii) before the scoping and about A5(ii)'s first half after it*, and both halves of that correction are recorded here rather than only at the site that consumes them, Theorem 29 Step 0.

(10) Where the threshold (55) comes from, and it is two conditions and not one. The stagewise recursion of Step 4 of the proof is (180), $\hat{\varepsilon}_k = \mathsf{c}_{15}\big(\mathsf{c}_{14}(\hat{\delta}_n \ln n + h_u^{-2}\hat{\varepsilon}_{k-1}^2)\big)^{2\mu/(2\mu+1)}$, a recursion in the *deterministic schedule* of (37) and not in the unobservable error. *Contraction* needs $2\kappa > 1$ with $\kappa = 2\mu/(2\mu+1)$, i.e. $\mu > \frac{1}{2}$; *the fixed point being* (56) needs (P-f), which at the amended bandwidth is $\mu > \frac{5}{6}$. The second implies the first, so (55) is stated at $\frac{5}{6}$ and the $\mu > \frac{1}{2}$ line of the budget table is now slack. *The coincidence worth naming is at* $\frac{1}{2}$*, not at* $\frac{5}{6}$: $\mu > \frac{1}{2}$ is also exactly what A5(iii) proves necessary for (7) to be satisfiable at any r, and it is exactly the index above which the inherited nonlinear route's rate theorem is unavailable (Remark 88(3)). The two arise independently, one from the propensity frontier, one from the linearisation recursion, and they coincide. *What* (55) *charges beyond that is* $(\frac{1}{2}, \frac{5}{6}]$*, and that is a real line in the budget*, listed with the others in the assumption block's table.

(11) And what happens at $\frac{5}{6}$ is **not** a second coincidence. Step 4(a.1) of the proof exhibits a requirement on the preliminary, (181), and shows it is weaker than the rate delivered exactly when $\mu > \frac{5}{6}$. That is (P-f) again, *the same inequality*, written in the same primitives, up to the strictly positive factor $\kappa/(2\kappa - 1)$. *The statement worth printing, and it is not a restatement of (P-f) either*: (P-f) is exactly the condition under which the newly exposed preliminary-rate requirement is *not self-defeating*, so that (181) adds a genuine new demand on $\hat{\varphi}^{(0)}$, open item Q-1, without making the procedure ask its input to be as good as its output.

(12) And the load-bearing hypothesis is (P-a.3), which should be read in the same breath as (55). (P-a.3) is a *random spectral source condition* on a *finite-rank, sieve-compressed* operator at a *data-dependent* base point, assumed to hold with probability tending to one, with no sufficient condition supplied anywhere in this file; it is what makes the conditional-on-sample route legitimate, and it is what carries the source index above $\frac{1}{2}$. *Any statement that "the linearised route reaches the same place under the*

same threshold" is true only with (P-a.3) priced in the same sentence, see Remark 88, Remark 86 and open item Q-4.

Lemma 91 (The moment is valid, its variance is finite, its θ-Jacobian is $-J$, and $J = E[\Delta Z \Delta Z']$)**.** *The moment has mean zero at the truth, a finite second moment, and derivative $-J$ in θ; the finiteness is delivered by the* level *clause of A2′ and not by its derivative clause; J is the second-moment matrix $E[\Delta Z \Delta Z']$, so no rank condition beyond A1b(iv) is needed anywhere; and, with one further clause of C2′, the variance is strictly positive definite, so Theorem 29's studentised conclusion needs no positivity proviso. Suppose A1, A1b(iv), A2′(i),* ***both*** *clauses (i.a) and (i.b), A2′(iii), B1(i), (ii), (iv.a), (iv.d) hold.* Clause (i.b) is in the list and not an "if in addition", for two reasons. *First, every bound below runs through Corollary 49, which is stated at an admissible representer* in the sense of A2b*, and A2b requires $\Lambda^{(0)}, \Lambda^{(1)} < \infty$ and continuous u-differentiability, that is (i.a)* ***and*** *(i.b). Second, A2′(i)'s closing clause fixes $\ell_\star$* pointwise *"from that version", the version satisfying both, so $\ell_\star$* is not even pointwise well defined from (i.a) alone*, and a sup-norm statement about it is not well posed. Theorem 29's group (T-2) lists both clauses, so no result downstream is affected. Then*

(i) $E\|\psi_\star\| < \infty$ *and* $E\big[\psi_\star\big] = 0$*;*

(ii) $E\|\psi_\star\|^2 < \infty$*, so Σ in (40) is a well-defined finite positive semidefinite matrix, with*

$$E\|\psi_\star\|^2 \;\le\; 2\,E\Big\|\Delta Z\,\Delta U\Big\|^2 + 4\,c_{\widetilde{W}}^2\Big\{E\Big\|b_\star(\Delta Z)\Big\|^2 + \Big\|\ell_\star\Big\|_\infty^2\,E\big[\Delta U^2\big]\Big\},$$

the factor $c_{\widetilde{W}}^2$ of Corollary 50 entering through $\|L_\star \widetilde{W}\| \le c_{\widetilde{W}}\|L_\star\|$ and equal to 1 at $\mathsf{L} = 2$*;*

(iii) $\theta \mapsto E[\psi(\theta, \zeta_\star)]$ *is twice differentiable on $\mathbb{R}^p$ with $\partial_\theta E[\psi(\theta,\zeta_\star)]\big|_{\theta=\theta_\dagger} = -J$, $\|J\| < \infty$, and second derivative bounded in norm by $c_{\widetilde{W}}\|\partial_u \ell_\star\|_\infty E\|\Delta Z\|^2$ uniformly in θ , the first derivative using $\Lambda_\star^{(0)}$ and the second using both budgets;*

(iv) the debiasing summand of J vanishes, and the θ-map is exactly affine. Under A1's first *clause,*

$$E\Big[\ell_\star(\Delta U, \Delta Z)\,\widetilde{W}\,\Delta Z'\Big] \;=\; 0, \qquad \textit{so} \qquad J \;=\; E\Big[\Delta Z\,\Delta Z'\Big];$$

consequently J is symmetric positive semidefinite, and J is nonsingular by A1b(iv) alone. Moreover $E\Big[L_\star\Big(\Delta U(\theta, \varphi_\dagger), \Delta Z\Big)\,\widetilde{W}\Big] = 0$ at every $\theta \in \mathbb{R}^p$, so

$$E\Big[\psi(\theta, \zeta_\star)\Big] \;=\; -\,E\Big[\Delta Z\,\Delta Z'\Big]\,\Big(\theta - \theta_\dagger\Big) \qquad \textit{for every } \theta \in \mathbb{R}^p, \tag{125}$$

which is affine with nonsingular gradient: the population *moment has $\theta_\dagger$ as its unique root on all of $\mathbb{R}^p$, and its second derivative is the zero matrix;*

(v) the variance is strictly positive definite, and no hypothesis beyond (ii)*'s list plus C2′(i) is needed for it. If in addition C2′(i) holds, then*

$$\Sigma \;\succ\; 0, \qquad \textit{hence} \qquad J^{-1}\Sigma J^{-1\prime} \;\succ\; 0 \quad \textit{and} \quad \vartheta' J^{-1}\Sigma J^{-1\prime}\vartheta > 0 \;\; \textit{for every } \vartheta \neq 0. \tag{126}$$

Clause (v) converts a **proviso** of Theorem 29 into a **consequence** of its hypotheses. *That theorem's studentised conclusion is stated for directions ϑ with $\vartheta' J^{-1}\Sigma J^{-1\prime}\vartheta > 0$, which reads as a condition the user must check;* (126) *says it holds at every $\vartheta \neq 0$ under hypotheses the theorem already carries in (T-1) and (T-3), namely A1b(iv) and C2′(i). The two ingredients are exactly those and nothing more. By* (127) *below, $\vartheta'\Sigma\vartheta \geq E[(\vartheta'\Delta Z)^2\Delta U^2]$, both summands of Σ being positive semidefinite; A1b(iv) makes $E[\Delta Z \Delta Z']$ nonsingular, so $P(\vartheta'\Delta Z \neq 0) > 0$ for every $\vartheta \neq 0$; and C2′(i) gives ΔY a conditional density given $\mathcal{F}_{XZW}$, hence, since $\Delta U = \Delta Y - \Delta\varphi_\dagger - \Delta Z'\theta_\dagger$ differs from ΔY by an $\mathcal{F}_{XZW}$-measurable shift, gives ΔU one too, so $P(\Delta U = 0 \mid \mathcal{F}_{XZW}) = 0$ almost surely and $P(\Delta U = 0) = 0$. On the positive-probability event $\{\vartheta'\Delta Z \neq 0\}$ the integrand is therefore almost surely strictly positive, and $E[(\vartheta'\Delta Z)^2\Delta U^2] > 0$; finiteness is A1b(iv)'s $E\|\Delta Z \Delta U\|^2 < \infty$.* The second summand of (127) is not used, *so no overlap condition enters: A4(i) is what makes that summand bounded*

below, *and clause* (v) *does not need it. Two things clause* (v) *does not say. It says nothing about the* size *of* Σ*, which is where A4(i) does bind (as* ϱ *approaches a boundary the second summand diverges); and it is a statement at a fixed* P*, not a bound uniform over a class of laws,* $\lambda_{\min}(\Sigma)$ *may be arbitrarily small at nearby* P*, which is the same pointwise-versus-uniform boundary Remark 98*(6) *draws for the theorem itself.*

Three readings of (iv) *must be blocked, and each of the three is a place a reader will over-generalise it.*

(iv.1) *Clause* (iii)*'s second-derivative envelope is a correct bound on the zero matrix, hence carries no information. It is marked here rather than struck, on this file's standing rule that a proved clause is retired in place.* $\|\partial_u \ell_\star\|_\infty$, and hence A2′(i.b)'s budget $\Lambda_\star^{(1)}$, is **not** thereby freed, *for two independent reasons: it is consumed by Proposition 27's inheritance in Lemma 93*(2)*, and A2′(i.b) is in any case required to* license *Corollary 49 at all, which is what makes* $\ell_\star$ *pointwise well defined and* $\|\ell_\star\|_\infty$ *finite, and hence what makes* (40) *a matrix. Only the attribution at Remark 99*(1) *changes.*

(iv.2) *It is a statement about* J *and not about* $\hat{J}$*. A4(i) puts* ϱ *inside the* ε_ϱ*-interior simplex* (11)*, so Definition 24(a)'s truncation is free at the population propensity and the population* $\widetilde{W}$ *is conditionally centred; but* $\widehat{\widetilde{W}}_l^{(-k)} = \mathbb{1}\{W = l\} - \hat{\varrho}_l^{(-k)}(\Delta Z)$ *is* ***not*** *conditionally centred, at any* n*, and the second summand of* (46) *does not vanish. Lemma 95 Step 3 is where its cost* $\|\hat{\ell}^{(-k)}\|_\infty \|\mathsf{r}_k\|_{L^2(P)}$ *is paid, and that step is unchanged.* Replacing $\hat{J}$ by $\mathbb{E}_n[\Delta Z \Delta Z']$ would also be consistent for J; it is not done, *because it would break the exact identity* $\partial_\theta \bar{\psi}_n(\theta) = -\hat{J}(\theta)$ *of Definition 25 on which Theorem 29 Step 1's algebra rests.*

(iv.3) *It does not propagate to the* φ*-direction, and the block would collapse if it did.* Δh *is* $\sigma(X_1, X_2)$*-measurable and* ***not*** $\sigma(\Delta U, \Delta Z)$*-measurable, so* $E[\ell_\star \widetilde{W} \Delta h]$ *is not zero, indeed A2 is precisely the assertion that it equals* $-E[\Delta Z \Delta h]$ *for every* $h \in \mathcal{V}$ *(Proposition 20*(i) *and* (33)*), and were it zero for every* h *one would have* $\nu_{\Delta Z} \equiv 0$*, which fails at any design in which the instrument moves the regressor.* The boundary between the two sides is exactly

the one (P4b)'s warning and (P4c) were split to police, *and* $q = E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2]$ *sits on the non-vanishing side of it.*

(iv.4) *It does not touch* Σ*, and the one thing it* does *say about* Σ *is a decomposition and not a vanishing. The same tower step applied to the cross term of* $\Sigma = E[\psi_\star \psi_\star']$*, legitimate because* $\Delta Z \Delta U \, L_\star'$ *is* $\sigma(\Delta U, \Delta Z)$*-measurable and integrable by clause (ii) and the Cauchy–Schwarz inequality , gives* $E[\Delta Z \Delta U \, \widetilde{W}' L_\star'] = 0$ *and hence, exactly,*

$$\Sigma \;=\; E\Big[\Delta Z \Delta Z' \, \Delta U^2\Big] \;+\; E\Big[L_\star \widetilde{W} \widetilde{W}' \, L_\star'\Big]. \tag{127}$$

Both summands are positive semidefinite, and the second does not vanish, for which one further tag is needed and is **not** on this lemma's list. **And the route to it changes at** $\mathsf{L} \geq 3$**, where the binary one is unavailable rather than merely weaker.** *At* $\mathsf{L} = 2$*,* $\widetilde{W}^2 \geq \varepsilon_\varrho^2$ *surely under A4(i), so* $E[L_\star \widetilde{W} \widetilde{W}' L_\star'] \succeq \varepsilon_\varrho^2 E[L_\star L_\star']$*, which is nonzero whenever* $L_\star$ *is.* At $\mathsf{L} \geq 3$ that argument fails, and not by a constant*:* $\widetilde{W}\widetilde{W}'$ *is a rank-one matrix, so no sure inequality* $\widetilde{W}\widetilde{W}' \succeq cI$ *holds at any* $c > 0$*, and* $L_\star \widetilde{W} \widetilde{W}' L_\star'$ *is singular at every sample point once* $p > 1$*.* The correct route is the conditional one, and (6) was written for it*: under A1,* $\Delta U \perp\!\!\!\perp W \mid \Delta Z$ *gives* $E[\widetilde{W}\widetilde{W}' \mid \Delta U, \Delta Z] = \Omega(\Delta Z)$ *of* (2)*, and A4(i) gives* $\Omega(\Delta Z) \succeq \varepsilon_\Omega \, \mathrm{Id}$ *with* $\varepsilon_\Omega = \varepsilon_\varrho / \mathsf{L}$*, so that* $E[L_\star \widetilde{W} \widetilde{W}' L_\star'] = E[L_\star \Omega(\Delta Z) L_\star'] \succeq \varepsilon_\Omega E[L_\star L_\star']$*, nonzero whenever* $L_\star$ *is. The conditional route uses A1, which is on this lemma's list, and it is valid at every* L *including* 2*, the binary display is retained because its constant* ε_ϱ^2 *is the one earlier rounds printed and, at* $\varepsilon_\varrho < \frac{1}{2}$*, is the weaker of the two.* A4(i) is cited for this sentence and for nothing else here*;* (127) *itself, and every clause* (i)–(iv)*, hold without it.* The reading to take from (127) is the standard debiasing trade and not a defect*: with* $J = E[\Delta Z \Delta Z']$*, the sandwich* $J^{-1} \Sigma J^{-1\prime}$ *is the variance the infeasible plug-in* (45) *would have at a* known $\varphi_\dagger$*, plus a positive semidefinite term; what the debiasing weight buys is not variance but the removal of the first-order bias* $\langle \nu/\pi, h_k \rangle$ *that Lemma 92 Step 4′ shows the plug-in carries at exact order* $\|h_k\|$*.* Neyman orthogonality is untouched throughout*, being a statement about the three*

nuisance *directions of Proposition 20 and not about* ∂_θ. One simplification is available and is **not** taken into the displays*: since* J *is symmetric,* $J^{-1\prime} = J^{-1}$*, so Theorem 29's* $J^{-1}\Sigma J^{-1\prime}$ *and* (48)*'s* $\hat{V}$ *could drop a transpose. They do not, because* $\hat{J}$ *is* not *symmetric by* (iv.2)*, and a variance estimator that dropped the transpose would not be symmetric either.* The transpose is carried in both places so that the population display and the sample display have the same shape*, which is the convention Lemma 95*(ii) *states the limit in.*

A2′(ii) *is not among the hypotheses and is consumed nowhere in Part I. The variance object is* $E\|L_\star \widetilde{W}\|^2$*, which clause (ii) bounds through A2′(iii) and* $\|\ell_\star\|_\infty$*; A2′(ii) bounds* $E\|\ell_\star \widetilde{W}\|^2$*, a different quantity, and that quantity is in any case* implied by A2′(i) in full together with B1(i), (ii), (iv.a), (iv.d)*: Corollary 49 gives* $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty \Lambda_\star^{(0)} < \infty$ *and* $\|\widetilde{W}\| \le c_{\widetilde{W}}$ *by Corollary 50, so* $E\|\ell_\star \widetilde{W}\|^2 \le c_{\widetilde{W}}^2 \|\ell_\star\|_\infty^2 < \infty$ *with no further hypothesis.* The bound on $\|\widetilde{W}\|$ is the Euclidean one and not (P1)*, which is the per-cell statement.* The attribution matters*: the redundancy is* ***not*** *"A2′(ii) is implied by A2′(i.a)", which is not a statement this file can make, since the only route to* $\|\ell_\star\|_\infty < \infty$ *available here is Corollary 49 and that corollary is stated at A2b. The redundancy is real; its ground is A2′(i) in full plus the four B1 clauses. A2′(ii) is* ***retired in place*** *at A2′, on the A3 precedent, a proved clause is retired, its number kept so a stale citation lands somewhere.* The forward Step-8 need is the only ground for retention that survives, and it is a forward obligation rather than a present consumer*; it is recorded at the retirement note.*

Lemma 92 (L-14 with a rate: the plug-in preliminary)**.** *The preliminary inherits the first-stage rate, and it inherits it because* $\theta_\dagger$ *is identified by restriction (B) alone. Suppose A0, A1, A1b(iv), C1(ii) hold, adopt Definition 24, and suppose A5(ii)'s* ***first half, read at each fold****, the first display of* (43). The fold index is cited for the reason Definition 24(d) gives*: A5(ii) prices one* h*, and* (45) *averages over* F *of them.* A5(ii)'s second half $\delta_n = o_P(n^{-1/4})$ is **not** in this list*; it is the hypothesis of clause* (ii) *alone. Then*

(i) the rate, on a list with no threshold in n;

$$\left\|\hat{\theta}^{(0)} - \theta_\dagger\right\| \;=\; O_P\!\left(n^{-1/2}\right) + O_P\!\left(\delta_n\right), \tag{128}$$

and if in addition $\delta_n \to 0$ *then* $\hat{\theta}^{(0)} \xrightarrow{p} \theta_\dagger$*;*

(ii) the threshold, which is the only clause A5(ii)*'s second half enters. If in addition* $\delta_n = o_P(n^{-1/4})$*, then*

$$\left\|\hat{\theta}^{(0)} - \theta_\dagger\right\| \;=\; o_P\!\left(n^{-1/4}\right). \tag{129}$$

Both summands of (i) are needed for this conclusion and neither is redundant to it*: the first is* $o(n^{-1/4})$ *trivially and the second is A5*(ii)*'s second half.*

Why the split, and why it is ***not*** *the* (i)/(i′) *device of Lemma 95. That device states* one *conclusion on two lists; here the two clauses have* different *conclusions, so they are numbered and not primed. The split exists for Corollary 30, whose hypotheses deliberately omit A5(ii)'s second half, doing without* (7)*, hence without that threshold, is what the corollary is* for*, and which consumes this lemma twice: once for the fourth hypothesis of Lemma 95*(i′)*,* $\|\hat{\theta}^{(0)} - \theta_\dagger\| = o_P(1)$*, and once for* $\|\Psi_n^\theta\| = O_P(\|\hat{\theta}^{(0)} - \theta_\dagger\|^2)$*.* Clause (i) supplies both from its statement. If in addition A5(i) and A5(iii) hold*, then* $\delta_n \gg n^{-1/2}$*, because* $\hat{\delta}_n = n^{-\alpha}$ *with* $\alpha = r/(2r + d_Z) < \frac{1}{2}$ *and* $\kappa = 2\mu/(2\mu + 1) < 1$ *give* $\kappa\alpha < \frac{1}{2}$*, so the two summands of* (128) *collapse to* $O_P(\delta_n)$*.* That simplification is cosmetic and is used by neither clause*: Theorem 29 Step 2 works at the uncollapsed rate* (128)*, since* $\sqrt{n}(n^{-1/2} + \delta_n)^2 \le 2(n^{-1/2} + \sqrt{n}\,\delta_n^2)$ *and both summands vanish under A5(ii); which is why A5(i) and A5(iii) are not among the hypotheses above.* No condition on $\hat{\ell}$, on $\hat{L}$, on $\hat{\varrho}$, on L_x or on A2 is used*: the preliminary does not touch the debiased moment.*

Lemma 93 (L-6 and L-7 at the estimated nuisances: the exact bias decomposition)**.** *The conditional bias of the debiased moment at the estimated nuisances is the balancing pairing plus exactly three remainders, one quadratic in the structural error, two products of two estimation*

errors, and no Taylor expansion in the nuisances is taken. Suppose A1, A1b(i), A1b(iv), A2, A2′(i), A2′(iii), N1, B1(i), (ii), (iv), C1(ii) and C2′(i) hold; adopt Definition 24; and suppose hypothesis (R-a) of Theorem 29, so that $\hat{\ell}^{(-k)}$ *is bounded and continuously differentiable in* u *with* $\max_{k\le \mathsf{F}} \|\partial_u \hat{\ell}^{(-k)}\|_\infty = O_P(1)$, an $O_P(1)$ envelope and **not** a bound at every sample point*: (132) carries the envelope on its* right *and is read, as the displays below all are, on the event where it is finite. Put* $g_k := \Delta Z + \hat{\ell}^{(-k)}(\Delta U, \Delta Z)\,\widetilde{W} \in \mathbb{R}^p$*, the contrast standing to the right of the* $p \times (\mathsf{L}-1)$ *estimate as in (38). Then, for every* k*, almost surely*

$$E\Big[\hat{\psi}^{(-k)} \,\Big|\, \sigma(\mathcal{I}_{-k})\Big] \;=\; -\Big\langle \nu_{g_k}/\pi,\; h_k \Big\rangle_{L^2(\pi)} \;-\; \Psi_k^{(1)} \;+\; \Psi_k^{(2)} \;-\; \Psi_k^{(3)}, \tag{130}$$

where, all expectations being conditional on $\sigma(\mathcal{I}_{-k})$*,*

$$\Psi_k^{(1)} := E\Big[\Delta h_k \int_0^1 \Big\{\hat{\ell}^{(-k)}(\Delta U - \lambda \Delta h_k, \Delta Z) - \hat{\ell}^{(-k)}(\Delta U, \Delta Z)\Big\} d\lambda\; \widetilde{W}\Big],$$

$$\Psi_k^{(2)} := E\Big[\Delta h_k \int_0^1 \hat{\ell}^{(-k)}(\Delta U - \lambda \Delta h_k, \Delta Z)\, d\lambda\; \mathsf{r}_k(\Delta Z)\Big], \qquad \Psi_k^{(3)} := E\Big[D_k(\Delta U, \Delta Z)\, \mathsf{r}_k(\Delta Z)\Big],$$

each an $\mathbb{R}^p$-valued object in which the $p \times (\mathsf{L}-1)$ factor stands to the left of the $(\mathsf{L}-1)$-vector it is contracted against and the scalar Δh_k commutes*, and*

$$\Big\|\Big\langle \nu_{g_k}/\pi, h_k \Big\rangle\Big\| \;\le\; 2\, c_{\widetilde{W}}\, \bar{c}_\pi^{1/2} \Big\|\hat{\ell}^{(-k)} - \ell_\star\Big\|_{L^2(P)} \Big\|h_k\Big\|_{L^2(\pi)}, \tag{131}$$

$$\Big\|\Psi_k^{(1)}\Big\| \;\le\; 2\, c_{\widetilde{W}}\, \bar{c}_\pi \Big\|\partial_u \hat{\ell}^{(-k)}\Big\|_\infty \Big\|h_k\Big\|_{L^2(\pi)}^2, \tag{132}$$

$$\Big\|\Psi_k^{(2)}\Big\| \;\le\; 2\, \bar{c}_\pi^{1/2} \Big\|\hat{\ell}^{(-k)}\Big\|_\infty \Big\|\mathsf{r}_k\Big\|_{L^2(P)} \Big\|h_k\Big\|_{L^2(\pi)}, \tag{133}$$

$$\Big\|\Psi_k^{(3)}\Big\| \;\le\; \Big\|\mathsf{r}_k\Big\|_{L^2(P)} \Big\|D_k\Big\|_{L^2(P)}, \qquad \textit{and} \qquad \Psi_k^{(3)} = 0 \;\; \textit{if } D_k \textit{ satisfies N1}', \tag{134}$$

where $\|\mathsf{r}_k\|_{L^2(P)} := \big(E\|\mathsf{r}_k(\Delta Z)\|^2\big)^{1/2}$, the second moment of the norm of the $\mathbb{R}^{\mathsf{L}-1}$-valued propensity error, which is what A4(iii) prices, and not the scalar second moment*, and* $\|D_k\|_{L^2(P)}$, $\|\hat{\ell}^{(-k)}\|_\infty$, $\|\partial_u \hat{\ell}^{(-k)}\|_\infty$ *are Frobenius.* The four left-hand sides are **Euclidean** norms on $\mathbb{R}^p$, and (131)'s is a norm for the same reason the other three are*: the objects*

bounded are the four summands of (130)*, which is an identity between* $\mathbb{R}^p$ *vectors,* ψ *being* $\mathbb{R}^p$*-valued by* (38)*.* That is exactly what the triangle inequality in $\mathbb{R}^p$ needs, and it is the form in which Theorem 29 Step 4 consumes these displays*: applied to* (130) *it gives* $\|E[\hat{\psi}^{(-k)} \mid \sigma(\mathcal{I}_{-k})]\|$ *at most the sum of the four right-hand sides.* (131) and (132) carry $c_{\widetilde{W}} = \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$ of Corollary 50 and (133)–(134) do not, for the reason Lemma 56 gives*: the first two contract a matrix against* $\widetilde{W}$*, the last two against* r_k *or* D_k*, whose norms are already the quantities being priced. At* $\mathsf{L} = 2$*,* $c_{\widetilde{W}} = 1$ *and all four displays carry the constants earlier rounds printed.*

The three remainders are written $\Psi_k^{(j)}$ *and not* $\mathcal{R}_k^{(j)}$*, and the choice is forced.* $\mathcal{R}(\cdot)$ *is the* range *of an operator throughout this file, A2, Lemma 55, Lemma 84, and calligraphic* $\mathcal{C}$*,* $\mathcal{E}$*,* $\mathcal{G}$*,* $\mathcal{H}$*,* $\mathcal{U}$ *are all claimed as well.* Ψ *is the letter this project already binds to* $E[\psi]$ *along a nuisance path (Lemma 56), and these three objects* are *components of* $E[\psi]$ *at the estimated nuisances, so the letter is semantically right;* $\Psi(\tau)$ *always carries an argument and* $\Psi_k^{(j)}$ *never does, which is the same separation device that keeps* $T(\cdot,\cdot)$ *apart from* $\mathcal{T}$*.* Proposition 27's precedent, "the remainder carries no symbol of its own", is followed **where it can be, which is at two objects and not at three***: the Hessian envelope of Theorem 29 Step 1 and the event* $\{\hat{J}$ *nonsingular*$\}$ *are written out, while the* θ*-remainder* ***does*** *carry a symbol,* Ψ_n^{θ}*, because it is referred to at six separate sites across two proofs and writing it out at each would be unreadable.*

Three structural facts are visible in (130) *and each is a claim of the paper.* (1) No second-order term in L and none in ϱ appears, exactly*, because* ψ *is bi-affine in* (L, ϱ)*, this is Lemma 56*(ii) *and it is used here as an* identity *rather than as a derivative statement, which is why* (130) *carries no Taylor remainder in those two directions.* (2) The only square is $\Psi_k^{(1)}$*, and by Proposition 27*(i),(iii) *its order in* $\|h_k\|_{L^2(\pi)}$ *cannot be improved uniformly over an* $L^2(\pi)$*-ball, what is inherited from Part G is the* lower *bound on the leading term, and Remark 65*(1) *is what forbids reading it as a statement about the realised* h_k*.* (3) $\Psi_k^{(3)}$ is the entire cost of N1′

failing at the estimated direction, *which is obligation* (O2) *of Remark 58, and* (134) *prices it.*

Lemma 94 (L-8: cross-fitting removes the empirical-process term)**.** *Once the estimated moment is $L^2(P)$-consistent at a frozen nuisance, the centred empirical average of the estimated moment minus the true one is $o_P(n^{-1/2})$, no entropy condition, no Donsker property, and no stochastic equicontinuity at the estimated point.* What is **not** claimed is that the consistency is free*: (135) is established below from the rate and envelope conditions listed, and the headline is a statement about what* (135) buys *at scale $n^{-1/2}$, not about what* (135) *costs. Suppose A0, C1(ii), A4(ii)–(iii) hold, adopt Definition 24, suppose hypotheses (R-a) and (R-b) of Theorem 29, and suppose A5(ii) and A5(vi) with $s > \frac{1}{2}$* ***read at each fold****, the first two displays of* (43)*.* **A4(ii) is on this line and it is not decoration***: Step 5 concludes $O_P(r_n^{1/2}) = o_P(1)$, which needs $r_n \to 0$, and A4(iii) states only the* form $r_n = n^{-r/(2r+d_Z)}$ *and nothing about r. It is A4(ii)'s $r > 0$ together with $1 \le d_Z \le p-1$ that makes the exponent strictly positive; without it r_n need not vanish, Step 5 delivers $O_P(1)$,* (135) *fails and* (136) *with it.* Of A5(ii) this lemma uses the **first half**, the rate itself, together with $\delta_n \to 0$, and nothing else*: Steps 2 and 3 of the proof read $O_P(\delta_n)$ and conclude $o_P(1)$, and A5*(ii)*'s second half $\delta_n = o_P(n^{-1/4})$ is consumed at no step.* This is the split Lemma 92 states on its own hypothesis line*, and it is stated here for the same reason: an asymmetry between two adjacent lemmas over one clause reads as a gap in the seam rather than as a difference in what each consumes.* The clause is nonetheless left on the list whole rather than narrowed to its first half*, because $\delta_n \to 0$ is what the second half supplies a fortiori and no other hypothesis on this list delivers it.* The fold index is cited rather than left implicit because A5(ii) and A5(vi) are stated at a **single** $h = \hat{\varphi} - \varphi_\dagger$ while this lemma consumes them at F distinct estimators*, one per fold complement; Definition 24(d) is where the reading is declared and Theorem 29 Step 0 is where it is discharged. A1b(iv) is* ***not*** *a hypothesis, on a ground of its own and not on Lemma 77's. No step below cites it, and the two moments any step of this proof or of the clause of Definition 24*(c′.1) *it adopts could want from it, $E\|\Delta Z\|^2$ at Step 2 and $E[\Delta U^2]$ in that clause's derived L^2 membership of $\hat{L}^{(-k)}$, are implied a fortiori by (R-b), which this list*

already carries and which asks for the fourth *moments of both. What this lemma does* ***not*** *use is A1b(iv)'s nonsingularity clause, which is Lemma 95's. Then*

$$\max_{k\le \mathsf{F}} \left\| \hat{\psi}^{(-k)} - \psi_\star \right\|_{L^2(P)} \;=\; o_P(1), \tag{135}$$

where the norm is taken at frozen $\hat{\zeta}^{(-k)}$*, and consequently*

$$\sqrt{n} \sum_{k=1}^{\mathsf{F}} \frac{n_k}{n} \left(\mathbb{E}_{n,k} - E \right) \left[\hat{\psi}^{(-k)} - \psi_\star \right] \;=\; o_P(1). \tag{136}$$

What is and is not conceded, stated because leg 3 of the Chen–Pouzo answer is a positioning commitment. (136) *is proved by conditional Chebyshev at a* frozen *nuisance, so the only input it requires is the* $L^2(P)$ *consistency* (135). This is a different statement from Lemma 81, and the difference is the point*: Lemma 81 concedes a uniform law of large numbers with a rate over a slowly growing ball, at scale* $\hat{\delta}_n \gg n^{-1/2}$*, and that concession is charged in the* φ*-step; the* θ*-step concedes nothing at scale* $n^{-1/2}$*, which is where a penalised sieve minimum-distance treatment of* θ *would need stochastic equicontinuity of the* $\sqrt{n}$*-scaled empirical process at the estimated point.*

Lemma 95 (L-11: the sandwich, and no additional correction for the propensity)**.** *Both factors of the sandwich are consistent, and the sandwich is the* whole *asymptotic variance: no Neyman correction for the propensity appears in it. Suppose the hypotheses of Lemma 94* ***and of Lemma 91****, together with A2, (R-e) of Theorem 29, and* $\|\hat{\theta}^{(0)} - \theta_\dagger\| = o_P(1)$*.* Lemma 91's list is inherited in full and not abbreviated*: Steps 3, 6, 6′ and 7 of the proof cite Lemma 91*(ii)–(iv)*, whose hypotheses run A1, A1b(iv), A2′(i), A2′(iii), B1(i), (ii), (iv.a), (iv.d), the last four being the bridge Corollary 49 needs to make* $\|\ell_\star\|_\infty$ *finite, hence to make* J *of* (40) *a finite matrix.* A3 is **not** on this list*: Step 6 needs* J *nonsingular, Lemma 91*(iv) *delivers* $J = E[\Delta Z \Delta Z']$ *and A1b(iv) delivers its nonsingularity, and both are already on the inherited list. A3 is retired at A3. Then*

(i) $\hat{J}(\theta_\dagger) \xrightarrow{p} J$ *and* $\hat{J} = \hat{J}(\hat{\theta}^{(0)}) \xrightarrow{p} J$*; consequently* $P(\hat{J}$ *nonsingular*$) \to 1$ *and* $\hat{J}^{-1} \xrightarrow{p} J^{-1}$*;*

(i′) clause (i) *uses no rate in* n*, and holds on a strictly weaker list. The conclusions of* (i) *hold under A0, A1, A1b(iv), C1(ii), A2, A2′(i), A2′(iii), B1(i), (ii), (iv.a), (iv.d), A4(iii), Definition 24, (R-a),* $\|\hat{\theta}^{(0)} - \theta_\dagger\| = o_P(1)$ *and the three consistency statements*

$$\max_{k \le \mathsf{F}} \left\| h_k \right\|_{L^2(\pi)} = o_P(1), \qquad \max_{k \le \mathsf{F}} \left\| \hat{\ell}^{(-k)} - \ell_\star \right\|_{L^2(P)} = o_P(1), \qquad \max_{k \le \mathsf{F}} \left\| \mathsf{r}_k \right\|_{L^2(P)} = o_P(1),$$

the last of which A4(ii)–(iii) would supply on their own, and neither is on the list above, because all three statements are assumed **outright** here rather than derived. In particular A4(i), A4(ii), A5(ii), A5(vi), (R-b) and (R-e) are not used, and no product of two rates and no threshold in n appears*;* (135) *is not used either, clause* (i) *being about* $\hat{J}$ *alone. This is the clause Corollary 30 invokes, and it is stated because that corollary's hypotheses contain neither A5(ii) nor a rate product, so an appeal to* (i) *as stated would be an appeal to the* proof *rather than to the statement;*

(ii) if in addition $\hat{\theta} \xrightarrow{p} \theta_\dagger$ *then* $\hat{\Sigma} \xrightarrow{p} \Sigma$ *and* $\hat{V} \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$*.*

A flag, not a side taken. The influence function of Theorem 29 is $J^{-1}\psi_\star$*, in which* ϱ *appears only through* $\widetilde{W}$ *at its true value, and no term correcting for the estimation of* ϱ *is present. That is a* consequence *of Proposition 20*(iii) *under N1: the* ϱ*-derivative of the moment vanishes* exactly*, so there is nothing for a correction to remove.* The project's positioning section states the opposite*, that the sandwich carries an additional correction for* ϱ*; the two statements cannot both be right, and the contradiction is recorded for the writer at §9 item 10 of the strategy memo rather than resolved here.*

Proposition 96 (L-10: the median over partitions)**.** *Averaging or taking medians over a fixed number of independent fold partitions changes neither the influence function nor the limit. Let* $\mathsf{P} \ge 1$ *be fixed and not depending on* n*, let* $\hat{\theta}_1, \dots, \hat{\theta}_\mathsf{P}$ *be the estimators* (47) *computed on* P *partitions drawn independently of the data and indexed by* $\mathsf{p} \in \{1, \dots, \mathsf{P}\}$ *,* sans-serif, the

lowercase of the count P, and **not** q, which is bound to $q = E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2]$, *and suppose the hypotheses of Theorem 29 hold* ***at each partition***. The per-partition reading is part of the hypothesis and not a formality*: the* P *partitions have* PF *distinct fold complements, and (T-5), (R-a), (R-c), (R-e) and* (43) *are conditions on the nuisances computed on a fold complement, so what is required is that they hold at all* PF *of them, with the maxima in* (43) *and (R-a) read over that index set.* Since P and F are both fixed this is finitely many statements and every order is preserved*; it is not a uniformity requirement, and it is the* first *of the three uses of* P *fixed that this proposition's argument makes, enumerated below. Let* $\hat{\theta}^{\mathrm{md}}$ *be their coordinatewise median and* $\hat{\theta}^{\mathrm{av}} := \mathsf{P}^{-1} \sum_{\mathsf{p}} \hat{\theta}_{\mathsf{p}}$. *Then*

$$\sqrt{n}\big(\hat{\theta}_{\mathsf{p}} - \hat{\theta}_{\mathsf{p}'}\big) = o_P(1) \quad \textit{for every } \mathsf{p}, \mathsf{p}', \qquad \sqrt{n}\big(\hat{\theta}^{\mathrm{md}} - \theta_\dagger\big) = \sqrt{n}\big(\hat{\theta}^{\mathrm{av}} - \theta_\dagger\big) + o_P(1) \xrightarrow{d} \textit{ the limit law of } (58).$$

The limit is named by pointing at (58) rather than by printing a symbol, *which is Theorem 29's disposition carried to its third statement site and is also the more informative form: what this proposition asserts* is *that the median and the average have Theorem 29's limit.* The argument is that all P estimators share one influence function, and it uses P fixed three times*: once in the hypothesis, so that "the hypotheses of Theorem 29 at each partition" is finitely many statements; once so that finitely many* $o_P(1)$ *terms may be collected; and once so that the median is a continuous function of finitely many arguments.* It says nothing about $\mathsf{P} \to \infty$*, where the median would have to be handled as a functional of a random measure and the hypothesis itself would become a uniformity requirement over a growing family of fold complements.*

Remark 97 (The logarithm in (56), resolved in A5(ii), and what it costs). *(1) Which object* δ_n *names.* Proposition 28 delivers $\|\hat{\varphi}^{\pi} - \varphi_\dagger\|_{L^2(\pi)} = O_P(\delta_n)$ with $\delta_n = (\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)}$ — *at the infeasible reading* $\hat{\varphi}^{\pi}$ *of Definition 22, which is a separate matter from the one this remark settles and does not bear on it*: this clause is about the **symbol** δ_n and the logarithm inside it, and δ_n is the same deterministic sequence under either reading — and (P-f), Remark 125(0)

and Proposition 79(iv) all read that *log-inflated* object. *A log-free reading of A5(ii) is off by* $(\ln n)^{2\mu/(2\mu+1)}$, and Remark 89(2) is where the warning is discharged.

(2) The resolution taken. A5(ii) is stated so that the symbol is the rate Proposition 28 delivers,

$$\delta_n \;:=\; \big(\hat{\delta}_n \ln n\big)^{2\mu/(2\mu+1)}, \qquad \text{and} \qquad \delta_n = o_P\big(n^{-1/4}\big). \tag{137}$$

The alternative, leave A5(ii) log-free and let Theorem 29's rate conditions carry the logarithm, is rejected, because it leaves (P-f) and A5(ii) reading the same symbol differently and because it would put the logarithm in five places instead of one.

(3) What the resolution costs, term by term. This is the frontier arithmetic and it is not uniform across the four products Theorem 29 needs.

(a) $\delta_n^2 = o(n^{-1/2})$*: free.* (7) is a *strict* inequality between polynomial exponents, so writing $\hat{\delta}_n = n^{-\alpha}$, $\kappa = 2\mu/(2\mu+1)$ and $\kappa\alpha = \frac{1}{4} + \varepsilon$ with $\varepsilon > 0$ gives $\delta_n^2 = n^{-1/2-2\varepsilon}(\ln n)^{2\kappa} = o(n^{-1/2})$. This is Proposition 79(iv) and it is why (7) is **unchanged**. *And the strictness is not decoration*: at $\kappa\alpha = \frac{1}{4}$ exactly, $\delta_n = n^{-1/4}(\ln n)^{\kappa}$ is *not* $o(n^{-1/4})$, so the boundary of (7) fails, which is what makes A5(iii)'s "if and only if" literally correct at the boundary under (137).

(b) $r_n\delta_n = o(n^{-1/2})$: free, and implied by (7) rather than assumed. Since $r_n = n^{-\alpha}$, $r_n\delta_n = n^{-\alpha(1+\kappa)}(\ln n)^{\kappa}$, and (7)'s $\kappa\alpha > \frac{1}{4}$ with $\kappa < 1$ gives $\alpha > 1/(4\kappa)$, hence

$$\alpha(1+\kappa) - \tfrac{1}{2} \;>\; \frac{1+\kappa}{4\kappa} - \frac{1}{2} \;=\; \frac{1-\kappa}{4\kappa} \;>\; 0,$$

a strictly positive polynomial margin that absorbs $(\ln n)^{\kappa}$. *So* (7) *does double duty and no separate condition on* $r_n\delta_n$ *is imposed*; the margin is

$$\frac{1-\kappa}{4\kappa} \;=\; \frac{(2\mu+1)^{-1}}{8\mu/(2\mu+1)} \;=\; \frac{1}{8\mu}, \tag{138}$$

which shrinks as μ grows and is $\frac{3}{20}$ at $\mu = \frac{5}{6}$ and $\frac{1}{16}$ at $\mu = 2$. *The constant is exact at the frontier boundary and therefore cannot be improved from* (7) *alone*: the middle inequality above is an equality at $\alpha = 1/(4\kappa)$, so $1/(8\mu)$ is the infimum of $\alpha(1+\kappa) - \frac{1}{2}$ over the configurations (7) admits.

(c) $\delta_n \epsilon_n = o(n^{-1/2})$: **not** free, and this is the one place the logarithm bites. A5(iv) is a stated condition on ϵ_n, and under (137) it is a strictly stronger demand than under the log-free reading, by the factor $(\ln n)^{\kappa}$. *Nothing in the block pays for it*: it is a demand on the estimator of the representer, is L-13's business, and it is charged here rather than absorbed.

(d) $r_n \epsilon_n^L = o(n^{-1/2})$: not affected by the logarithm, and zero under N1′. (R-c) is where it lives.

(4) What does ***not*** *move.* (55)'s $\mu > \frac{5}{6}$ is unchanged, because Remark 125(2) computes it at the log-inflated δ_n already and derives the *strictness* from precisely the failure of the logarithms to cancel in (P-f). (7) is unchanged by (a). A5(v) becomes *weaker*, its right-hand side being inflated. (54) is unchanged, being a condition on L_x and h_u and not on δ_n.

Remark 98 (The scope: what Theorem 29 covers, and the six things it does not). (1) Mild ill-posedness only, and severe ill-posedness is not a gap in the proof but a place the theorem does not reach. (7) requires $\kappa\alpha > \frac{1}{4}$ with $\kappa = 2\mu/(2\mu+1) < 1$ and $\alpha = r/(2r+d_Z) < \frac{1}{2}$, so it requires δ_n to be a *polynomial* rate faster than $n^{-1/4}$. Under H1's $\sigma_j \asymp j^{-a}$ that is what the source condition delivers. *Under severely ill-posed decay, σ_j decaying exponentially, the rate attainable from a source condition is logarithmic in $\hat{\delta}_n$, hence in n, and $\delta_n = o(n^{-1/4})$ fails at every r*; A5(ii) is then not satisfiable and Theorem 29 is unavailable. *This is plan §2.6's scope and it is restated here without improvement*: mild ill-posedness with $s > a + \frac{1}{2}$, which under H1 reads $\mu > \frac{1}{2} + 1/(4a)$ and which implies (55) when $a \le \frac{3}{4}$ and is an additional restriction otherwise (Remark 125(2)). That last sentence is an **interpretation** and not a scope statement. The two displays it links carry two different objects: $s > a + \frac{1}{2}$ and the

computed $\mu > \frac{1}{2} + 1/(4a)$ are about $\mu_{\mathrm{H1}} = s/(2a)$, the index of A2 at the *population* operator $\mathcal{A}_{\mathrm{cond}}$ on all of $\mathcal{V}$ at $\theta_\dagger$, while (55), (7) and κ carry (P-a.3)'s *primitive* μ, at the finite-rank compression of the realised operator at $\theta_\circ$, and a and s are themselves $\theta_\circ$-specific in (T-5) and $\theta_\dagger$-specific in H1. *The bridge between them is the operator interpolation Remark 86*(1) *shows unavailable at every* $2\mu > 1$*, and Remark 86*(4) *declines the commutativity proviso that would restore it*, so the implication "$s > a + \frac{1}{2} \Rightarrow$ (55)" holds **under that declined proviso and not otherwise**. *Read literally without the qualification, this item both excludes designs the theorem covers and claims designs it does not*; Remark 86(5) states the binding reading and gives the two-sided example. *What survives unqualified is the substance of the item*: severe ill-posedness , exponentially decaying σ_j, makes the rate attainable from *any* source condition logarithmic, so A5(ii) fails at every r whichever index the source condition is stated at, and that argument does not go through $\mu = s/(2a)$ at all.

(2) The rate condition cannot be relaxed by a better remainder bound, and that is a weaker claim than necessity. By Proposition 27(i),(iii) the direct bound on $\Psi_k^{(1)}$ is order-sharp: $\sup_{\|h\| \le \delta} \frac{1}{2}|\langle \mathcal{B}h, h\rangle| \asymp \delta^2$ under O-16(a), so no argument that merely bounds the second-order remainder more carefully will weaken (7). *Three readings must be blocked and Remark 65 blocks them*: the result does *not* say that the realised $\Psi_k^{(1)}$ at $h_k = \hat{\varphi}^{(-k)} - \varphi_\dagger$ fails to be $o_P(\|h_k\|^2)$; it does *not* say that $\delta_n = o(n^{-1/4})$ is necessary for $\sqrt{n}$-estimability of $\theta_\dagger$ by *some* estimator; and it does not say that the moment class is exhausted. *What relaxing* (7) *would require* is a different selection, one with a smaller σ_q and $\mathcal{K}$, which is lever 1 of the strategy memo and is open, or a restriction on the *direction* of the first-stage error, which is an extra hypothesis and not a sharper argument.

(3) It does not derive its own inputs. μ and R_* are primitive at (P-a.3) (Remark 86); the preliminary $\hat{\varphi}^{(0)}$ that Definition 22 needs at the rate (181) has no supplier in this file (open item Q-1); the representer's estimator $\hat{\ell}$ is priced by A5(iv) and constructed nowhere; N1 at $L_\star$ is *assumed* in (T-2) and *arranged* nowhere, the re-anchoring of Remark 52 being available

but not performed, and by Corollary 21 it is $p(\mathsf{L}-1)$ conditions rather than the p a contracted centring supplies; and (79), if (R-e.2) is taken, is assumed. *Theorem 29 is a theorem about an estimator five of whose ingredients are assumed to exist with stated properties*, and saying so is not a hedge: each such ingredient is named in (T-2), (T-5), (R-a) or (R-e) and none is hidden in a regularity condition.

(4) It says nothing about efficiency, about the plug-in, or about what happens when A2 fails. Those are `prp:effbound`, `prp:plugin` and `cor:trend`, Step 8. In particular A5(v) is **not** consumed anywhere in Part I: Definition 24(b) runs the φ-step at the fixed anchor $\theta_\circ$, so no Landweber pass on $\Delta Y - \Delta Z'\hat{\theta}$ occurs and the plug-in cost $\sqrt{\hat{m}}\, n^{-1/2}$ never arises. And when A2 fails, $\ell_\star$ does not exist, $L_\star$ is not defined, $\psi_\star$ is not defined and (40) is not a well-formed display; the object to report is then the Picard diagnostic and the plug-in at its slower rate, and pivotal inference is declined. *What fails under $\neg$A2 is the* existence *of the objects and not a rank condition*: Lemma 91(iv) shows the rank condition never had content beyond A1b(iv), and A1b(iv) holds whether or not A2 does.

(5) It is an asymptotic statement whose asymptotics do not bind at any sample size a user will run, and Corollary 30 and not Theorem 29 is what describes a feasible n. With $\delta_n = (\hat{\delta}_n \ln n)^\kappa$, $\hat{\delta}_n = n^{-\alpha}$ and the notation of item (1), the θ-remainder of Step 2 of the proof is $O_P(n^{-1/2} + \sqrt{n}\,\delta_n^2)$, and its second term is

$$\sqrt{n}\,\delta_n^2 \;=\; n^{-\beta}\,(\ln n)^{2\kappa}, \qquad \beta := 2\kappa\alpha - \tfrac{1}{2} > 0 \;\text{ by (7)}, \tag{139}$$

and $t \mapsto e^{-\beta t} t^{2\kappa}$ is *increasing* for $t < 2\kappa/\beta$. So (139) **rises** in n up to

$$n^* \;=\; \exp\left\{\frac{2\kappa}{2\kappa\alpha - \frac{1}{2}}\right\}$$

and only then falls. *The threshold is astronomically large near the frontier and that is not a pathology of a bad parameterisation but a property of* (7): $\beta \downarrow 0$ as $\kappa\alpha \downarrow \frac{1}{4}$, so $n^* \uparrow \infty$ there.

At $\mu = 2$ (so $\kappa = \frac{4}{5}$) and $\alpha = 0.32$, inside (7), since $\kappa\alpha = 0.256 > \frac{1}{4}$, one gets $\beta = 0.012$ and $n^* = e^{400/3} \approx 10^{58}$. Two things follow and neither weakens (58). First, (58)'s $o_P(1)$ is a statement about the limit and is proved; what (139) says is that the *approach* to that limit is not monotone and that simulation evidence at $n \leq 10^4$ should be read against Corollary 30's (59), in which δ_n^2 appears as a term and not as a remainder. Second, the honest reading of (7) is that it is a *qualitative* frontier: crossing it makes the remainder vanish eventually, and says nothing about where. *Whether the companion term $r_n\delta_n$ turns over as late is a separate computation and is not made here*; Remark 97(3)(b)'s (138) prices its *margin* to $n^{-1/2}$ and is the place to make it.

(6) It is **pointwise in** P, and no uniformity over any class of laws is claimed or proved. Item (5) frames the n^* point as a finite-sample caveat; the more exact statement is that there is no uniformity claim for it to be a caveat about. Every hypothesis of (T-1)–(T-5) and (R-a)–(R-e) is imposed at a *single* data-generating law: μ, R_*, a, s, ε_ϱ, r, $\bar{c}_\pi$, $\Lambda_\star^{(0)}$, $\Lambda_\star^{(1)}$ and the singular system of $\mathcal{A}_{\mathrm{cond}}$ are all P-specific constants, and the conclusion (58) is a convergence in distribution under that P. Three consequences follow and each is a real limitation rather than a formality. First, the $o_P(1)$ in (58) is not uniform over any neighbourhood of P, so the theorem does not deliver confidence sets whose coverage is *honest* over a class, for that one needs the supremum over $P \in \mathcal{P}$ of the absolute coverage error of the studentised interval to vanish, at a fixed nominal level and over a class $\mathcal{P}$, and no such class is even defined here. *The standard reference for that notion in ill-posed models is* Babii (2016), *whose honest sets are built at a linear operator under completeness and so do not cover the equation solved here.* Second, *that is exactly the gap item (5) displays numerically*: $n^* \uparrow \infty$ as $\kappa\alpha \downarrow \frac{1}{4}$, so within any class containing laws arbitrarily close to the frontier the convergence cannot be uniform, and the 10^{58} is the pointwise statement's honest shadow rather than a bad constant. Third, the positive-definiteness (126) is likewise pointwise: $\lambda_{\min}(\Sigma) > 0$ at each P and may be arbitrarily small along a sequence of laws. *What would be needed for a uniform statement*, and it is open work and not a hedge: a class $\mathcal{P}$ on which the constants above are bounded

uniformly, a uniform version of (56), which would require (P-a.3)'s R_* and μ to be uniform over $\mathcal{P}$ at a *random* compressed operator, and a Lindeberg condition uniform in P. None is attempted. *The convention that a limit theorem is read pointwise unless uniformity is asserted is standard and is not being restated as a defect*; what is a defect is asserting a scope remark that enumerates finite-sample caveats and omits the quantifier they all sit inside.

Remark 99 (Where each hypothesis of Theorem 29 is used, by clause). *This is the hypothesis-usage audit.* "Step" refers to the proof of Theorem 29 in `proofs.tex`.

(1) *Consumed, with the site.* A0, fold independence and the i.i.d. central limit theorem (Steps 3, 5), and the compact *interval* $\mathcal{X}$ for the Sobolev embedding (Lemma 94). A1, first clause, $E[\Xi(\Delta U, \Delta Z)\,\widetilde{W}] = 0$ for the $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued generic direction Ξ of Definition 53, hence $\partial_{\tau_2}\Psi \equiv 0$ exactly (Lemma 91(i), Lemma 93 Step 2, through Proposition 20(ii)); second clause, $E[\Delta Z \Delta U] = 0$ (Lemma 91(i), Lemma 92). A1b(i), $\varphi_\dagger \in \mathcal{V}$, so that $h_k \in \mathcal{V}$ and the pairing $\langle \nu_{g_k}/\pi, h_k\rangle$ is the one Lemma 45(iii) computes. A1b(iii), so that the φ-step targets $\varphi_\dagger$ and not a set; consumed through (T-5). A1b(iv), non-singularity of $E[\Delta Z \Delta Z']$ (Lemma 92) and the second-moment clauses (Lemma 91(ii)). A2, existence of $\ell_\star^{\Pi}$, hence of $\nu_{\Delta Z + \ell_\star \widetilde{W}} \equiv 0$ (Lemma 93 (131)). A2′(i.a) **and** (i.b), *jointly*, because both are required for Corollary 49 to apply at all: A2b is what that corollary assumes, and A2′(i)'s closing clause defines $\ell_\star$ pointwise from the version satisfying both. Of the two *bounds* the corollary returns, $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty \Lambda_\star^{(0)}$ uses only the (i.a) budget while $\|\partial_u \ell_\star\|_\infty$ uses both; the latter is consumed by Proposition 27's inheritance in Lemma 93(2). *Lemma 91*(iii) *is* ***not*** *a second consumer of* $\|\partial_u \ell_\star\|_\infty$: by Lemma 91(iv.1) that clause's second-derivative envelope bounds the zero matrix. *A2′(i.b) itself is nonetheless consumed*, it licenses Corollary 49 at all, hence $\ell_\star$ pointwise, and it keeps the Proposition 27 consumer. A2′(iii), $E\|b_\star\|^2 < \infty$ for $\Sigma < \infty$. N1, $E[L_\star \mathsf{r}_k] = 0$ (Lemma 93 Step 2, through Proposition 20(iii)). B1(i), (ii), (iv.a) and (iv.d), Corollary 49; B1(iv) in full, Proposition 20's list, inherited by Lemma 60. C1(ii),

$\bar{c}_\pi$, at every display. C2′(i), Proposition 20(i)'s differentiation, inherited, *and a* ***second and direct*** *consumer*, namely Lemma 91(v), where the conditional density of ΔY given $\mathcal{F}_{XZW}$ is what gives $P(\Delta U = 0) = 0$ and hence $\Sigma \succ 0$. A4(i), *consumed* ***definitionally*** *and at* ***no step*** *of this proof, which is the stronger of the two statements and the one Section 6 needs.* It reaches this theorem through Definition 24(a), which the theorem adopts: A4(i)'s $\varepsilon_\varrho \in (0, 1/\mathsf{L})$ is what makes (11) non-empty, its quantifier over *every* $l \in \{0, \ldots, \mathsf{L}-1\}$, the base cell included, is what puts ϱ itself inside (11), so that truncating $\hat{\varrho}^{(-k)}$ onto that set cannot degrade A4(iii)'s rate, and it is what grounds that clause's per-cell bound $\|\mathsf{r}_k\|_\infty \leq 1 - 2\varepsilon_\varrho$. ***It is not the ground of the sure bound the proof actually uses***: that bound is (183)'s **Euclidean** $\|\mathsf{r}_k\| \leq c_{\widetilde{W}}$, whose derivation states in terms that "no overlap condition is consumed and A4(i) is not on this lemma's list", ϱ lies in the simplex for free and (11) is contained in it at *every* ε_ϱ, with $c_{\widetilde{W}} = \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$ the simplex diameter of Corollary 50, equal to 1 at $\mathsf{L} = 2$. *So no step of the θ-step reads* ε_ϱ *at a value*, and what A4(i) asks of the application here is positivity at every cell and nothing more. A4(ii)–(iii), r_n. A5(i), $\hat{\delta}_n$, through (T-5). A5(ii) as (137), read at each fold per (43), *and the two halves are consumed at different places, which the split of Lemma 92 makes visible*: the *first* half supplies δ_n at Step 0 and is the whole of what Lemma 92(i) needs, while the *second* half, $\delta_n = o_P(n^{-1/4})$, is consumed at Steps 1, 2 and 4 of this proof directly and **not** through Lemma 92, whose clause (ii) this theorem never invokes, Step 2 takes clause (i)'s uncollapsed rate and applies the threshold itself. A5(iii), (7), which delivers (a) and (b) of Remark 97(3), *and which is the one clause of (T-4) that no result in this file delivers.* A5(vi) with $s > \frac{1}{2}$, read at each fold, the L^4 bound in Lemma 94. (R-a), (R-c), (R-e), as stated. (R-b), $E\|\Delta Z\|^4$ at Lemma 94 Step 2 and Lemma 95 Step 9; $E[\Delta U^4]$ at Lemma 94 Step 5.

(2) **Not** listed in (T-1)–(T-5) and not consumed, the five clauses a reader will look for, and why each is absent. A2′(ii), **retired**: the variance needs the *level* and the clause priced the *derivative*, and the clause is in any case implied by A2′(i) in full together

with B1(i), (ii), (iv.a), (iv.d), so it is retired in place at A2′ exactly as A3 was; see Lemma 91's closing paragraph. A5(v), see Remark 98(4). A1b(ii), Lemma 45(ii) is not used; only (i), (iii), (iv) and (v) are, and those need C1(ii) and, for (v), A1. A6, retired (Proposition 71); A6b and Proposition 74 are not consumed, Definition 22 needing no cone condition. A3, retired (A3), being proved from (T-1)'s A1 and A1b(iv) together with (T-2)'s A2, A2′(i), A2′(iii) and (T-3)'s B1(i), (ii), (iv.a), (iv.d), at Lemma 91(iv). *The retirement removes a hypothesis and adds none*; the nonsingularity Lemma 95(i) Step 6 needs is A1b(iv)'s, listed in (T-1).

(3) Inherited as a block through (T-5) and not re-derived. H1, A5(vi′), A8's basis and upper end, $p_K \geq s - \frac{1}{2}$, C3, C5′(i), (P-a.1)–(P-a.3), (P-c)–(P-g), (55), and A7 in the amended form (13). *All are read at* $\theta_\circ$, per Definition 24(b), and each is θ-specific in the sense of Remark 38(iv). *C5′ is inherited* ***by clause****, and a block citation would charge the theorem with C5′(ii).* That clause is a lower bound on h_x and (P-g), inherited on the same line, is an upper bound on h_x; at the strengthened form of C5′(ii) the two have no common solution whenever $s \leq \frac{1}{2} + 18\alpha/(8 - 3\alpha)$, (63), a band that (7) leaves open at every $\alpha \in (\frac{1}{4}, \approx 0.359)$. *The consequence would be that (T-5) is unsatisfiable and this theorem holds* ***vacuously****, not that it is false*; the by-clause citation is what makes (T-5) a satisfiable group at those configurations, and C5′(ii) carries the corresponding repair in the maintained block.

(4) What (181) does and does not discharge, and the answer is negative. A one-step needs a *rate* for its preliminary and L-14 supplies consistency only. (181) *does not close that gap, because it is about a different object*: it is a condition on ς_0, the exponent of $\hat{\varepsilon}_0$, which is the $L^2(\pi)$ rate of the *preliminary structural estimator* $\hat{\varphi}^{(0)}$ inside Definition 22, and it is hypothesis (P-a.2) of the φ-step. The one-step's preliminary is $\hat{\theta}^{(0)}$, an estimator of $\theta_\dagger$; nothing in (181) bears on it. *What closes the gap is Lemma 92*(i): the plug-in (45) has rate $O_P(n^{-1/2} \vee \delta_n)$ on a list containing no threshold, and $\delta_n = o_P(n^{-1/4})$ is A5(ii)'s

second half, applied where it is needed and not before. So the gap is real, the registered instrument is the wrong one, and the discharge required specifying the preliminary, which no file did before Part I.

(5) Why (R-a)–(R-e) are stated on the theorem and **not** promoted to A5. Every clause of A5 constrains an object the *maintained block* introduces, at a single estimator, while (R-a)–(R-e) constrain $\hat{\varphi}^{(-k)}$, $\hat{L}^{(-k)}$, $\hat{\ell}^{(-k)}$, $\hat{b}^{(-k)}$ and D_k, objects that exist only after Definition 24, a definition Part I introduces. A maintained clause cannot quantify over objects the block does not define. In addition (R-a) is a construction claim governed by Remark 58's test, (R-b) is a moment condition on the primitives that would belong beside A1b(iv) rather than in the nuisance-rate block, and (R-c) and (R-e) are **disjunctions**. *The ground is* ***not*** *that A5 prices rates only*, A5(vi) and A5(vi′) being qualitative.

Remark 100 (What the hypothesis groups of Theorem 29 carry, and the six things they do not). Each item below stood as a parenthesis inside the theorem's own hypothesis list. None of it is a hypothesis, and the theorem's content is unchanged by its removal.

(1) (T-2): no local rank condition on $\ell_\star$ is in the group, and neither is A3. The nonsingularity such a moment would ordinarily require follows from (T-1) and the rest of (T-2) at Lemma 91(iv), and the rank condition the theorem needs is A1b(iv), which is in (T-1). A3 stood in this group through dispatch 12 and is retired on that same ground. *Two things about the group's N1 clause are recorded here rather than in the statement.* Its **arity**: N1 at $L_\star$ is $p(\mathsf{L}-1)$ scalar conditions, one per entry, and not a condition on a contraction of the $\mathsf{L}-1$ columns, so it is $p(\mathsf{L}-2)$ conditions *stronger* than a contracted centring at every $\mathsf{L} \geq 3$ and identical to one at $\mathsf{L} = 2$ (Corollary 21). And its **anchor-pinning**: A2′(iii) leaves the anchor $b_\star$ free, but (T-2) and (R-a) together do not — (R-a) forces $\hat{b}^{(-k)} \to b_\star$ and (T-2) requires $L_\star$ centred, so $b_\star$ is the N1 anchor of Remark 52 and any construction of $\hat{L}^{(-k)}$ must target that one.

(2) (T-4) is **not** homogeneous in status, and that is why the group was annotated; *it is* ***less*** *homogeneous from 2026-08-27, one clause having moved from delivered to assumed.* A5(i) is **delivered** by (T-5) rather than required, through Lemma 77, and is listed all the same, because Steps 2 and 4 cite it by name. *A5(ii)'s first half was delivered through dispatch 27 and is now* ***assumed***: the delivery ran through (56), which Proposition 28 is now scoped to state at the infeasible $\hat{\varphi}^{\pi}$, whereas A5(ii) is stated at the $\hat{\varphi}$ that Definition 24(b) plugs in, and Remark 89(7) is open at (7c) and (7d). *The listing is therefore no longer redundant at that clause; it is the charge*, and Theorem 29 is a conditional theorem on it. *The clause is still listed in the form* (137) *for the reason it always was*, which is that the display fixes the object the symbol δ_n denotes and Steps 2 and 4 read that symbol by name. A5(ii)'s *second* half, $\delta_n = o_P(n^{-1/4})$, is still delivered, being about the deterministic δ_n alone. A5(iii) sits at the other end: *no result in this file delivers it and it is therefore assumed*, being a restriction on the primitives (μ, r, d_Z) that Proposition 28 presupposes rather than establishes, that proposition returns (56) *whenever* (7) holds and says nothing when it fails. A5(vi) with $s > \frac{1}{2}$ is a third case: (T-5)'s A5(vi′) delivers the H^s bound at the *final linearisation stage* $\bar{k}$ of Definition 22, *a stage index, not a fold index*, the two being one glyph apart, while $s > \frac{1}{2}$ is a restriction on the primitive s and is delivered by nothing.

(3) (R-a): the *sure* boundedness of $\partial_u \hat{\ell}^{(-k)}$ is **not** required. No first-step construction the block contemplates delivers a sure bound on an estimated object, what a series or kernel estimator of $\ell_\star$ supplies being a bound that is finite with probability tending to one, exactly (R-a)'s $O_P(1)$ display. The sites that read finiteness of the derivative envelope — Lemma 93 Step 4, Lemma 95 Step 5 and Theorem 29 Step 1, where it bounds $\partial^2_{\theta\theta'}\bar{\psi}_n$ and the Taylor remainder Ψ^{θ}_n — carry it on the **right** of an inequality and are read on the event where it is finite, which is where every O_P conclusion in those results is read in any case. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces said "the two sites" and was short by the third, while Definition 25's

own reading named that third and was short by these two — one three-site set described twice, in halves, by two claims neither of which held all of it. *Boundedness of* $\hat{\ell}^{(-k)}$ *itself is retained and is not part of this*: it is what Definition 24(c) already requires, for Definition 19(b) to be well posed. *And Definition 24*(c′.2)*'s* $[0, +\infty]$ *typing is unchanged by (R-a)*: that typing is the *definition*'s, fixing the range of the map $\omega \mapsto \|\cdot\|_\infty$ before any hypothesis is imposed, and it must stay extended-valued because Definition 24 is adopted by results that do not list (R-a); *tightness* of that extended-valued map, and hence finiteness of it with probability tending to one, but **not** at every sample point, is what (R-a) adds to it, at the results that do.

(4) (R-b): a fourth moment of $b_\star$ is **not** needed. $b_\star$ enters Part I at two lines only, both at *second* moment through A2′(iii) (Lemma 91 Step 1), and nothing in Part I needs a fourth moment of $\psi_\star$, Lemma 95 Step 8 applies conditional Markov to squares, its Step 10 needs only $\mathbb{E}_n\|\psi_\star\|^2 = O_P(1)$, and the central limit theorem used is Lindeberg–Lévy. The *fourth*-moment need is on $\hat{b}^{(-k)}$, and (R-a) supplies it.

(5) (R-e.2): (57) is stated on that branch because that branch is the one place both of its ingredients arrive at once. Lemma 60(iii) carries A8 *in full*, so (e.2) imports A8's lower end $L_x \gg n^{1/(2s)}$; (T-5) simultaneously imports (P-d), $L_x = o\big(n^{1-2\alpha}/\ln n\big)$; and (7) forces $\alpha > \frac{1}{4}$, so (P-d) binds and the joint window for L_x is non-empty if and only if (57). *The arithmetic is printed at (P-d) already*, "had A8's lower end been carried as well, the window would instead require $s > (2(1-2\alpha))^{-1}$", and Remark 125(5) records that the reference design clears it. (T-4)'s only s-floor is $s > \frac{1}{2}$, which does not reach it. *Witness, chosen so that nothing else is empty there*: $(\mu, s, a, \alpha) = (2, 1.35, 0.3375, 0.32)$ clears (7) at $\kappa\alpha = 0.256$, clears (54) at $s > 1$ and clears (63) at $s > 1.3182$, while A8's lower end asks $L_x \gg n^{0.3704}$ and (P-d) allows $L_x = o(n^{0.36}/\ln n)$: *empty*, the threshold (57) being $s > 1.3889$. *So without* (57) *the branch is vacuous below that threshold and the theorem is* ***not***: (e.1) is unaffected and carries the conclusion there. *An either/or*

that advertises a choice which does not exist at part of the parameter set is a defect of the enumeration, not of the theorem, and (57) is what makes the choice real wherever it is offered.

(6) (R-d)*'s letter is* reserved and not reused, on the (P-b) precedent, so that a citation to "(R-d)" lands on the reservation and not on the wrong hypothesis.

(7) (R-e.1)*'s fold index is part of the hypothesis and not a reading of it*, for the reason Definition 24(d) gives, and it is what Theorem 29 Step 4(a) and Lemma 95 Step 3 consume.

Remark 101 (How the conclusion of Theorem 29 is read). (1) The quantifier is "every" and not "every ϑ with $\vartheta' J^{-1}\Sigma J^{-1\prime}\vartheta > 0$". By Lemma 91(v) and (126), $\Sigma \succ 0$ under A1b(iv) and C2′(i), both already in (T-1) and (T-3), so $J^{-1}\Sigma J^{-1\prime} \succ 0$ and the positivity guard is *implied by the theorem's own hypotheses*. A proviso that a theorem's hypotheses entail should be stated as a consequence and not as a condition the user checks.

(2) What the guard is **not** replaced by is a uniform lower bound. $\lambda_{\min}(\Sigma) > 0$ at the fixed P of the theorem and may be arbitrarily small at nearby laws, which is Remark 98(6).

(3) *The denominator $\vartheta'\hat{V}\vartheta$ can still be zero or negative in a finite sample*, $\hat{V}$ is a sample sandwich and is set to 0 off $\{\hat{J}$ nonsingular$\}$ by (48), so the studentised display is read, as such displays always are, on the event of probability tending to one where the denominator is strictly positive. That is the reading the statement now records in one clause.

(4) *The contraction direction is ϑ, which is this file's binding letter for exactly this object*: $\vartheta'\psi$ is the moment for $\vartheta'\theta_\dagger$. It is **not** $\varkappa$, which is a local bound variable in (162), nor r, which is the ϱ-direction.

(5) *Both limit laws of Theorem 29, the one in* (58) *and the studentised one, are written out*

and carry no symbol: italic N is a count, a null set or a covering number everywhere else in this corpus, and calligraphic $\mathcal{N}$ is unavailable, being Lemma 41's closed subspace and A2c's null Borel set.

Remark 102 (How much of (59) is in primitives, by branch). The answer differs by branch, which is why it is stated by branch.

(a) *Always in primitives.* $\delta_n = (\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)}$ with $\hat{\delta}_n = n^{-r/(2r+d_Z)}$ and $r_n = n^{-r/(2r+d_Z)}$, so the terms $n^{-1/2}$, δ_n^2 and $r_n\delta_n$ have exponents that are functions of (μ, r, d_Z) alone, through Proposition 28 and A4(iii).

(b) Never in primitives on the branch Corollary 30 states. ϵ_n is priced by A5(iv) and ϵ_n^L by (R-c.2), and *neither has a supplier anywhere in the maintained block*: $\hat{\ell}$ is constructed nowhere (Remark 98(3)) and ϵ_n^L is priced by nothing at all. So on the hypotheses as listed, (R-e.1) with (R-c.2), the terms $\delta_n\epsilon_n$ and $r_n\epsilon_n^L$ carry two exponents that are **assumed and not derived**, and the claim "the rate is in primitives" is false for them. *Note also that s enters no exponent on this branch*, A8's **lower** end being consumed only through (R-e.2); its basis and upper-end clauses reach the theorem through (T-5), and neither carries s into an exponent.

(c) Entirely in primitives on one branch, and that branch is what §19.1's sieve decision buys. Under (R-e.2) **and** (R-c.1), the sieve solve for the first-order bias, N1′ at each D_k for the $\varrho \times L$ term, the two unsupplied symbols both disappear, and

$$\left\|\hat{\theta} - \theta_\dagger\right\| = O_P\Big(n^{-1/2} + n^{-1/2}\delta_n + \mathsf{R}\, L_x^{-s} + \delta_n^2 + r_n\delta_n\Big),$$

every exponent of which is a function of (μ, r, d_Z, s) through Proposition 28, A4(iii), A8's window and the spline direct theorem. That is the sense in which this is a result and not bookkeeping, and it holds on that branch only.

No further index enters. A2c's μ'' supplies a finite *constant* through Lemma 61 and Corollary 49

and appears in no exponent, and the project's second smoothness index is μ'' and no other.

Remark 103 (How Definition 23's two matrices are read). Three clauses, each of which stood inside the definition. None of it is construction, and the objects Definition 23 defines are unchanged by its removal.

(1) *The second summand of* (40) *carries* $\ell_\star = \partial_u L_\star$ *and not* $L_\star$, because θ enters ψ only through $\Delta U(\theta, \varphi)$ and differentiating $L_\star$ in that argument returns its u-derivative. *And under A1 that summand is the zero matrix, so* $J = E[\Delta Z \Delta Z']$ *exactly*, Lemma 91(iv), whose whole content is $E[\widetilde{W} \mid \Delta U, \Delta Z] = 0$, *the zero vector of* $\mathbb{R}^{\mathsf{L}-1}$, applied between the two $\sigma(\Delta U, \Delta Z)$-measurable factors $\ell_\star(\Delta U, \Delta Z)$ and $\Delta Z'$, which is where the contrast sits in $E[\ell_\star \widetilde{W} \Delta Z']$. (40) is nevertheless left in the two-summand form, because J is *defined* as the θ-derivative of (38), which has two summands whatever their values, and because that is the form the estimator (46) takes, where the second summand does **not** vanish.

(2) J *must* ***not*** *be described as "a different matrix" from* $E[\Delta Z \Delta Z']$: wherever (40) is well formed the two are the same matrix, by Lemma 45(v) at the array $\Xi(u, z)$ with entries $\Xi_{jj'l}(u, z) = \ell_{\star,jl}(u, z)\, z_{j'}$, whose last index is the one contracted against $\widetilde{W}$, *which is the array-valued reading that clause states, and at* $\mathsf{L} = 2$ *it is the matrix* $\ell_\star(u, z)z'$ *earlier rounds printed. What* ***is*** *true, and is why identification of* $\theta_\dagger$ *must not be routed through* (40)*, is the conditionality*: $\ell_\star$ exists only under A2, so (40) is a well-formed object only under A2, while identification of $\theta_\dagger$ is A1b(iv)'s and holds whether or not A2 does.

(3) The typing note is about the *form* and not the value. $\ell_\star$ is $\mathbb{R}^{p \times (\mathsf{L}-1)}$-valued by the type declaration of (1), $\widetilde{W}$ is $(\mathsf{L}-1) \times 1$ and $\Delta Z'$ is $1 \times p$, so $E[\ell_\star \widetilde{W} \Delta Z']$ is $p \times p$, *and the three factors multiply in that order and in no other*, and its *form* is neither coordinatewise in the p coordinates of θ nor symmetric, which is what makes the sum with $E[\Delta Z \Delta Z']$ well formed rather than a type error. Its *value* is zero, so J itself is symmetric and positive semidefinite. *The warning is therefore not idle; it transfers to* $\hat{J}$,

whose second summand in (46) is a sample average at $\widehat{\widetilde{W}}^{(-k)}$ and $\hat{\ell}^{(-k)}$, does not vanish, and leaves $\hat{J}$ non-symmetric, which is why (48) writes $\hat{J}^{-1}\hat{\Sigma}\hat{J}^{-1\prime}$ with the transpose and not $\hat{J}^{-1}\hat{\Sigma}\hat{J}^{-1}$.

Remark 104 (How Definition 5's domain declaration is read). Four clauses. The first three stood inside the definition, and none of that is construction: the spaces, the operator, the two inner products, the declared exception and (20) are unchanged by their removal. *The fourth never stood there*: it prices the declaration Definition 5 makes about π, which is construction and stays.

(1) The bound variables of the two inner products. They are h, h' on the domain side and Ξ, Ξ' on the range side, being this paper's letters for a direction in $L^2(\pi)$ and a generic element of a range-type space respectively. Ξ carries **five** value types in this corpus, each declared at its own site, and they must not be conflated, at $\mathsf{L} = 2$ the value types of (a) and (b) are both scalar, and at $\mathsf{L} = 2$ *and* $p = 1$ together three of the first four collapse, which is why the separation is recorded here rather than left to context. (a) In (101) and Proposition 70(iii), Ξ is a *scalar* element of $L^2(\Pi)$, a function of (u, t, l); that is the declaration this clause makes and it does **not** move with L. (b) In Lemma 48(ii) and in Corollary 50, Ξ is $\mathbb{R}^{\mathsf{L}-1}$-valued on $\mathbb{R} \times \mathbb{R}^p$: the first binds it as the generic element of the codomain $L^2(P_{\Delta U, \Delta Z}; \mathbb{R}^{\mathsf{L}-1})$, in which that clause writes its norm, and the second inherits the type because Υ's domain is Lemma 48(ii)'s codomain. (c) In Definition 53, Lemma 45(v) and Remark 99(1), Ξ is the generic *direction*, $\mathbb{R}^{p \times (\mathsf{L}-1)}$-valued, and is contracted against $\widetilde{W}$ from the left. (d) In Remark 103(2) it is array-valued, which is Lemma 45(v)'s stated extension of (c). (e) In Lemma 117 and its proof, Ξ and Ξ' are *generic real random variables* on the underlying probability space, bound in that statement and used nowhere else. *This one is of a different kind from* (a)–(d)*, which are deterministic functions on a range-type index set, and it is the reason the count is five and not four*: it is a random variable on the sample space and not a function of

(u, t, l) or of (u, z), it carries neither L nor p, and it therefore collapses into none of the others at $\mathsf{L} = 2$. *It is listed rather than renamed* because Lemma 117 is a self-contained measure-theoretic lemma whose $(\Xi, \Xi'; \varsigma, \varsigma')$ pairing is itself the product of an earlier collision repair, recorded in its own statement.

(2) Why the restriction to $\mathcal{V}$ costs nothing. The orthogonal decomposition $L^2(\pi) = \mathcal{V} \oplus \{\text{constants}\}$ holds because $\langle c, h \rangle_{L^2(\pi)} = c \int h \, d\pi$, and Δ annihilates constants, so nothing is lost by the restriction while a zero singular value is avoided. In particular the singular system $(\sigma_j; e_j, \tilde{e}_j)_{j \geq 1}$ of Proposition 9, the completeness condition A1c, the source condition A2 and the Hilbert-scale link H1 all refer to $\mathcal{A}_{\text{cond}} : \mathcal{V} \to L^2(\Pi)$ and to that same singular system.

(3) Why $\mathcal{B}$ is the exception, and what governs the form's sharp constant. $\mathcal{B}$ is read on all of $L^2(\pi)$ because $\mathcal{V}$ is *not* invariant under multiplication by a non-constant function and $\mathcal{B}$ contains such a multiplication. The quadratic form of Lemma 56(i) is evaluated only at $h \in \mathcal{V}$, so the form's own sharp constant is governed by the compression $P_{\mathcal{V}} \mathcal{B}_{\text{sym}} P_{\mathcal{V}}|_{\mathcal{V}}$, where $P_{\mathcal{V}}$ is the $L^2(\pi)$-orthogonal projection onto $\mathcal{V}$; Lemma 55(iii) records the relation between the two.

(4) What Definition 5's declaration of π costs, and the one thing it does not settle.

(4a) *π and Π do not have the same status and the shared word "reference" is a trap.* B1's $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ is a set of *choices made by the analyst*, which is why Remark 18(2) can say that B1 survives the passage to a sub-population unchanged and why Lemma 85(vi)–(vii) can rescale its *second* factor in two independent ways with nothing else moving. π is a *functional of P*. Nothing in this paper selects it, and no result below is licensed to treat it as a design lever.

(4b) *Two consequences of the pooled form, both free, and neither of them a weakening of C1(ii).* First, $f_{X_j} \leq f_{X_1} + f_{X_2} = 2\pi$ pointwise for $j = 1, 2$, so $\bar{c}_\pi = \max_j \|f_{X_j}/\pi\|_\infty \leq 2$: the ratio constant that every boundedness statement about $\mathcal{A}_{\text{cond}}$ and about the second-

order form carries has a numerical ceiling it did not have while π was read as an arbitrary reference measure. *The worked example at* $\mathsf{L} = 3$ *above, under* Two constructions for Proposition 17(iv)*, is the check, and it passes*: its marginals are $f_{X_1}(x) = 1 + \frac{\gamma}{3}(\frac{1}{2} - x)$ and $f_{X_2}(x) = 1 + \frac{\gamma}{3}(x - \frac{1}{2})$, whose average is 1 at every x, so that example's declared $\pi \equiv 1$ *is* its pooled design density and was consistent with this reading before the reading was stated; its $\bar{c}_\pi = 1 + \frac{\gamma}{6} < \frac{7}{6}$ at $\gamma \in (0, 1)$ sits inside the ceiling. Second, C1(ii)'s bounds on π follow from its bounds on f_{X_1} and f_{X_2}, since an average of two functions bounded above and below away from zero is one. *C1(ii) is retained verbatim and is not narrowed*: it also bounds f_{X_1,X_2}, f_{X_1} and f_{X_2} themselves, which no pooled identity delivers, and every citation of it stands where it is.

(4c) *Why* π *is fixed at* P *and not re-read at a sub-population.* Remark 18(2) transfers C1(ii)'s *upper* bounds to $Q = P(\cdot \mid W \in \mathsf{S})$ with $\bar{c}_\pi$ inflated to $\bar{c}_\pi / P(W \in \mathsf{S})$, and *both sides of that bound divide by the same* π — the left through $\|f^Q_{X_j}/\pi\|_\infty$ and the right through $\bar{c}_\pi$ itself; the proof of that remark says so in terms. The pooled density of the retained sub-population, $\frac{1}{2}(f^Q_{X_1} + f^Q_{X_2})$, is a different function, and nothing here claims the two agree — which is what the closing sentence of Remark 18(2) already records when it says that nothing is claimed there about π. *A localised implementation that rebuilt* $\hat{\pi}$ *on the retained cells alone would be estimating that other function, and this paper prices no such substitution.*

(4d) *The one thing the declaration does not settle: the adjoint.* π being unknown, every display carrying $1/\pi$ is uncomputable as printed. *Five are worth naming and no claim of exhaustiveness is made*: (1) (101), whose $1/\pi(x)$ stands in front of the whole adjoint; (2) (35), through the entry $\|1/\pi\|_\infty^{1/2}$ of its maximum, *which is the binding entry only above the threshold* $\varepsilon_\pi = \|1/\pi\|_\infty^{-1}$ *and is dominated by the computable* $\varepsilon_\pi^{-1/2}$ *at or below it*; (3) (36), which carries its square on the same reading; (4) Lemma 47(i); and (5) the Cauchy–Schwarz step of the measurability verification in the supplement's

discussion of the estimator. Remark 18(2) lists its own overlapping set and is likewise not a closure claim. *The one that binds is the first*, because Definition 22(b) runs every Landweber step through $\hat{\mathcal{A}}^*_{\varphi}$: the forward operator (100) is free of π and the iteration is not. Definition 22's construction convention makes the substitution of a floored $\hat{\pi}$ well posed and its constants computable; *what no result here establishes is that the substitution leaves* (56) *in place*, and that is stated as an open item at Remark 89(7) rather than assumed away — *in five parts as of 2026-08-27, the fifth being the relaxation condition* (35) *is now sized for. And the statement that would otherwise contradict this sentence is scoped rather than left to be reconciled*: Proposition 28 states (56) at the infeasible reading $\hat{\varphi}^{\pi}$ of Definition 22 and says in its own conclusion that it does not claim the display at $\hat{\varphi}$. *Through 2026-08-27 it claimed exactly that, unhedged*, so this sentence and that proposition were in printed contradiction; the sentence is unchanged and the proposition moved.

Nine conventions are fixed once here and used throughout the proofs that follow, and at several places in the statements above. They are cited by the tags (P1)–(P9) in the way assumption tags are cited: a proof names the tag and does not restate the convention. Four of the nine are results with proofs, (P5), the substitution lemma of (P6), (P7), and the three conversions and the freezing lemma of (P8), and are stated as lemmas; the rest fix versions, bounds and readings. *There is no* (P4a): a third clause of the binary identity was deleted as unused, and the surviving labels (P4b) and (P4c) were not renumbered, so that citations to them did not move.

Lemma 105 (Standing conventions: the free bounds and the mean-zero property)**.** (P1) The per-cell bound. $\left|\widetilde{W}_l\right| = \left|\mathbb{1}\{W = l\} - \varrho_l(\Delta Z)\right| \leq 1$ *almost surely, for every* $l \in \{1, \ldots, \mathsf{L}-1\}$.

(P1b) The vector bound. $\left\|\widetilde{W}\right\| \leq c_{\widetilde{W}}$ *almost surely, with* $\|\cdot\|$ *the Euclidean norm on* $\mathbb{R}^{\mathsf{L}-1}$ *and* $c_{\widetilde{W}} = \min\left\{\sqrt{2}, \sqrt{\mathsf{L}-1}\right\}$ *the constant of Corollary 50.*

(P2) *Suppose B1(i)–(ii) hold. Then* $|w(t'\Phi(z))| \leq \bar{w}(t)$ *for every* $z \in \mathbb{R}^p$ *and every* $t \in \mathcal{T}$*, with* $\bar{w}(t) = e^{c_\Phi \|t\|_1}$.

(P3) $E[\widetilde{W} \mid \Delta Z] = 0$ *almost surely, as an identity in* $\mathbb{R}^{\mathsf{L}-1}$.

Clauses (P1), (P1b) *and* (P3) *use no assumption beyond Definition 4 and Definition 6, and in particular* ***no overlap condition****. Under A4(i) the bound in* (P1) *sharpens, at* every *cell, to* $1 - \varepsilon_\varrho < 1$*; nothing below uses the strict version, and every citation of* (P1) *is a citation of the free per-cell bound.* The two bounds are separate clauses because they are separate statements, and the distinction is not idle*: what the operator norms, the kernel bounds and the Hilbert–Schmidt constant consume is the per-cell bound, which is* 1 *at every* L*; what a contraction of* $\widetilde{W}$ *against a vector consumes is* (P1b)*, whose constant is* $\sqrt{2}$ *at every* $\mathsf{L} \geq 3$*. A derivation routed through the wrong one of the two does not reproduce the printed constants in either direction. At* $\mathsf{L} = 2$ *the contrast index takes the single value* 1*,* $\widetilde{W}_1 = W - \varrho(\Delta Z)$ *and* $c_{\widetilde{W}} = 1$*, so both clauses read* $|\widetilde{W}| \leq 1$*, the single scalar bound earlier rounds printed, with*

no exception and no second constant.

Proof of Lemma 105. (P1). $\mathbb{1}\{W = l\} \in \{0, 1\}$ and $\varrho_l(\Delta Z) = P(W = l \mid \Delta Z) \in [0, 1]$ by Definition 4, so $\mathbb{1}\{W = l\} - \varrho_l(\Delta Z) \in [-1, 1]$ at every sample point and every l. For the sharpening, $\widetilde{W}_l$ takes the two values $1 - \varrho_l$ and $-\varrho_l$, so $|\widetilde{W}_l| \leq \max\{1 - \varrho_l, \varrho_l\}$; under A4(i), $\varrho_l \geq \varepsilon_\varrho$ gives $1 - \varrho_l \leq 1 - \varepsilon_\varrho$, and $\varrho_l = 1 - \sum_{j \neq l} \varrho_j \leq 1 - (\mathsf{L} - 1)\varepsilon_\varrho \leq 1 - \varepsilon_\varrho$, the sum running over the $\mathsf{L} - 1$ cells $j \in \{0, \ldots, \mathsf{L} - 1\} \setminus \{l\}$ and A4(i) bounding each below. At $\mathsf{L} = 2$ this is $\max\{1 - \varrho, \varrho\} \leq 1 - \varepsilon_\varrho$, the printed sharpening. (P1b) is the bound established in the statement of Corollary 50: by (1), $\widetilde{W}$ is the difference of $\big(\mathbb{1}\{W = l\}\big)_{l=1}^{\mathsf{L}-1}$ and ϱ, both points of the simplex $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \geq 0,\ \mathbf{1}'q \leq 1\}$, the first a vertex of it, the origin when $W = 0$, whose diameter is $\sqrt{2}$ at every $\mathsf{L} \geq 3$ and 1 at $\mathsf{L} = 2$; and separately $\|\widetilde{W}\|^2 = \sum_l \widetilde{W}_l^2 \leq \mathsf{L} - 1$ by (P1). The constant is the smaller of the two. *That derivation is written out here in full and deliberately, and it is this clause's own*: what it consumes is (1), (P1), and the fact that ϱ is a vector of conditional probabilities, Definition 4 and Definition 6, exactly as the closing paragraph of the statement records, so a proof that cites (P1b) by tag inherits **no** hypothesis from Corollary 50's hypothesis line, which is Lemma 48's hypotheses together with A1b(iv). *In particular it does not acquire* A1b(iv), which Lemma 48 Step 3 neither has nor needs, and which the tag would import if (P1b) were an inherited clause of the corollary rather than a self-contained one. *What is shared with Corollary 50 is the symbol and the number and not the hypotheses*: $c_{\widetilde{W}}$ is that corollary's symbol, no second constant is introduced, and the two derivations agree because they are the same two-line argument run twice. (P2) is B1(ii) verbatim, B1(i) supplying $\|\Phi\|_\infty \leq c_\Phi$ and hence $|t'\Phi(z)| \leq c_\Phi \|t\|_1$, and $w = \exp$ being increasing. (P3). Coordinatewise, $E[\widetilde{W}_l \mid \Delta Z] = P(W = l \mid \Delta Z) - \varrho_l(\Delta Z) = 0$ by the definition of ϱ_l; each coordinate is bounded by (P1), so the conditional expectations exist, and the $\mathsf{L} - 1$ scalar identities are the vector identity. □

Lemma 106 (Standing convention: the cell-contrast conditional identity)**.**(P4b) At $\sigma(\Delta Z)$. *Let g be a function of W and of variables that are $\sigma(\Delta Z)$-measurable,*

integrable, and write g_j *for a* $\sigma(\Delta Z)$*-measurable version of* $g(j,\cdot)$*,* $j \in \{0,\dots,\mathsf{L}-1\}$*. Then, for every* $l \in \{1,\dots,\mathsf{L}-1\}$*,*

$$E\Big[\widetilde{W}_l\, g(W) \;\Big|\; \Delta Z\Big] \;=\; \varrho_l\Big\{g_l - \sum_{j=0}^{\mathsf{L}-1} \varrho_j\, g_j\Big\} \qquad \textit{almost surely}$$

, cell l *against the* ϱ*-mixture of every cell, and* ***not*** *against any single comparison cell.*

(P4c) The one enlargement A1 licenses. *Suppose A1 and A4(i) hold, and let* $\mathcal{C}$ *be any* σ*-field with* $\sigma(\Delta Z) \subseteq \mathcal{C} \subseteq \sigma(\Delta U, \Delta Z)$*. Then for* ***any*** *integrable* Λ*,* not required to be $\mathcal{C}$-measurable*, and every* $l \in \{1,\dots,\mathsf{L}-1\}$*,*

$$E\Big[\widetilde{W}_l\, \Lambda \;\Big|\; \mathcal{C}\Big] \;=\; \varrho_l\Big\{E[\Lambda \mid \mathcal{C}, W=l] - \sum_{j=0}^{\mathsf{L}-1} \varrho_j\, E[\Lambda \mid \mathcal{C}, W=j]\Big\} \qquad \textit{almost surely,}$$

the L *cell-conditional expectations being well defined by A4(i).*

At $\mathsf{L}=2$ *both displays are the ones earlier rounds printed. There* $l=1$*,* $\varrho_0 = 1-\varrho_1$*, and* $\varrho_1\{g_1 - \varrho_0 g_0 - \varrho_1 g_1\} = \varrho_1\{(1-\varrho_1)g_1 - \varrho_0 g_0\} = \varrho_1\varrho_0\{g_1 - g_0\} = \varrho(1-\varrho)\{g_1 - g_0\}$*, and identically in* (P4c). The mixture form is what (**??**) records as the $\mathsf{L} \geq 3$ content of this identity*, and the clause printed beside that display reads* (P4b) *in this same general form: what separates the two is the conditioning* σ*-field and not the number of cells.*

Proof of Lemma 106. (P4b). Two facts are used, both from Definition 4: that W takes its values in the finite set $\{0,\dots,\mathsf{L}-1\}$, so that $g(W) = \sum_j \mathbb{1}\{W=j\} g_j$ with the sum finite; and that $P(W=j \mid \Delta Z) = \varrho_j$ by definition, for every j. The g_j being $\sigma(\Delta Z)$-measurable, they pull out of the conditional expectation, so $E[\mathbb{1}\{W=l\}g(W) \mid \Delta Z] = \varrho_l g_l$ and $E[g(W) \mid \Delta Z] = \sum_j \varrho_j g_j$. Subtracting ϱ_l times the second from the first, and using that ϱ_l is $\sigma(\Delta Z)$-measurable, gives the display. *The second fact is the one that does not survive enlargement*, which is the content of Remark 107.

(P4c). A1's first clause gives $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, hence $P(W=j \mid \Delta U, \Delta Z) = P(W=j \mid \Delta Z) =$

$\varrho_j(\Delta Z)$ for every j; for $\mathcal{C}$ between the two σ-fields the tower property and $\sigma(\Delta Z)$-measurability of ϱ_j give $P(W=j \mid \mathcal{C}) = E[\varrho_j \mid \mathcal{C}] = \varrho_j$. Therefore $E[\mathbb{1}\{W=l\}\Lambda \mid \mathcal{C}] = \varrho_l\, E[\Lambda \mid \mathcal{C}, W=l]$ and $E[\Lambda \mid \mathcal{C}] = \sum_j \varrho_j\, E[\Lambda \mid \mathcal{C}, W=j]$; subtracting ϱ_l times the second from the first, and using that ϱ_l is $\mathcal{C}$-measurable, gives the display. *A4(i) enters once and only here*, to make every $\{W=j\}$ non-null so that the L cell-conditional expectations are defined. □

Remark 107 (The scope of the cell-contrast identity, and where it fails). (P4b) does **not** extend to σ-fields $\mathcal{C}$ strictly larger than $\sigma(\Delta Z)$, and the split into two clauses exists for that reason. The step uses $P(W=j \mid \mathcal{C}) = \varrho_j(\Delta Z)$ at every cell j, which holds at $\sigma(\Delta Z)$ and, more generally, only when W is conditionally independent of $\mathcal{C}$ given ΔZ, which is exactly what (P4c) assumes and what A1 delivers on the stated range of $\mathcal{C}$.

A counterexample at every $\mathsf{L} \geq 2$, and it is not pathological. Let ΔZ be degenerate, so $\sigma(\Delta Z)$ is trivial; let $V := X_2 - X_1$ have a continuous distribution function F_V; and set $W := \min\Big\{\lfloor \mathsf{L}\, F_V(V) \rfloor,\ \mathsf{L}-1\Big\}$, so that $F_V(V)$ is uniform on $(0,1)$ and $\varrho_j \equiv 1/\mathsf{L}$ for every j, A4(i) holds, at any $\varepsilon_\varrho < 1/\mathsf{L}$. Then W is $\sigma(X_1, X_2)$-measurable, so

$$E\Big[\widetilde{W}_l\, g(W) \mid X_1, X_2\Big] \;=\; \Big(\mathbb{1}\{W=l\} - \tfrac{1}{\mathsf{L}}\Big)\, g(W),$$

which takes the value $\Big(1-\frac{1}{\mathsf{L}}\Big)g_l$ on $\{W=l\}$ and $-\frac{1}{\mathsf{L}}g_j$ on $\{W=j\}$ for each $j \neq l$, in general L distinct values, and is not the constant $\frac{1}{\mathsf{L}}\Big\{g_l - \frac{1}{\mathsf{L}}\sum_j g_j\Big\}$ that (P4b) would give. At $\mathsf{L}=2$, and with (X_1, X_2) exchangeable so that $F_V(0) = \frac{1}{2}$, this is $W = \mathbb{1}\{X_2 > X_1\}$ almost surely with $\varrho = \frac{1}{2}$, the two values $\frac{1}{2}g_1$ and $-\frac{1}{2}g_0$, and the constant $\frac{1}{4}(g_1 - g_0)$, the counterexample earlier rounds printed, recovered verbatim. *This is the case the operator-kernel comparison needs*: there $\mathcal{C} = \sigma(X_1, X_2, \Delta Z)$ and $P(W=j \mid \mathcal{C}) = p_{X,j} \neq \varrho_j$, the weights (**??**) carries, so (P4b) is unavailable and (P4c) does not reach it either, $\mathcal{C}$ is not contained in $\sigma(\Delta U, \Delta Z)$. (P4c) *is therefore not a weakening of this warning but the one enlargement A1 licenses*; it is used only in Proposition 15, and it is **not** available at $\mathcal{F}_{XZW}$.

Lemma 108 (Standing convention: the ν-identity)**.** *Suppose Definition 4 and C1(ii) hold, so that X_1 and X_2 have Lebesgue densities f_{X_1}, f_{X_2} on $\mathcal{X}$.* They are what make (20) an identity between functions rather than a formula in undefined objects*: without a density the two products $E[g \mid X_j = x] f_{X_j}(x)$ have no dx-integral and the right-hand side of* (P5) *is not defined.* The conditional expectations are read at the regular versions Definition 109(P6b) fixes *, a forward reference inside Part 0, the versions being fixed once for the whole file and* (P6b) *naming ν_g expressly as one of the objects they define. Let g be an integrable random variable and let h be measurable on $\mathcal{X}$ with*

$$E\big|g\,h(X_1)\big| < \infty \qquad \text{and} \qquad E\big|g\,h(X_2)\big| < \infty. \tag{140}$$

Then, with ν_g as in (20):

(P5) $E\big[g\,\{h(X_2) - h(X_1)\}\big] = \int_{\mathcal{X}} \nu_g(x)\,h(x)\,dx$*; and, at $h \equiv 1$, for which (140) reduces to $E|g| < \infty$, $\int_{\mathcal{X}} \nu_g\,dx = 0$ for every integrable g.*

The proviso is **term-by-term** integrability and not integrability of the difference*, because the proof splits the difference before conditioning; $E|g\,\Delta h| < \infty$ is strictly weaker and does not license the split. Every citation of* (P5) *discharges* (140) *at its own site.*

Proof of Lemma 108. By (140) the two terms $g\,h(X_2)$ and $g\,h(X_1)$ are separately integrable, so the expectation of the difference is the difference of the expectations. Conditioning on X_2 in the first and on X_1 in the second, $E[g\,h(X_2)] = \int E[g \mid X_2 = x]\,h(x)\,f_{X_2}(x)\,dx$ and $E[g\,h(X_1)] = \int E[g \mid X_1 = x]\,h(x)\,f_{X_1}(x)\,dx$; subtracting and recombining under one integral gives the first display by (20). The second is the first at $h \equiv 1$, for which $\Delta h = 0$, so that the left-hand side vanishes. □

Definition 109 (Standing convention: regular versions fixed once)**.** All spaces here are Polish, so regular conditional distributions exist. The following are fixed once and every later statement is read at them.

(P6a) A regular conditional distribution of $\Delta Y_\theta - \Delta\varphi$ given $\mathcal{F}_{XZW}$, recorded through its distribution function

$$G_\varphi(\omega, v) \;:=\; P\Big(\Delta Y_\theta - \Delta\varphi \le v \;\Big|\; \mathcal{F}_{XZW}\Big)(\omega),$$

the version being chosen so that $v \mapsto G_\varphi(\omega, v)$ is a distribution function for **every** ω and $\omega \mapsto G_\varphi(\omega, v)$ is $\mathcal{F}_{XZW}$-measurable for every v. *The regular conditional distribution itself carries no symbol and is written $dG_\varphi(\omega, \cdot)$, the Lebesgue–Stieltjes measure of that distribution function*, the natural name μ being the **source index** of the Hilbert scale, which appears in every rate display in this project. The argument ω is suppressed wherever no ambiguity arises, so that $G_\varphi(u)$ below means $G_\varphi(\cdot, u)$.

(P6b) Regular conditional distributions $Q_j(x, \cdot)$ of the observation vector given $X_j = x$ for $j = 1, 2$, and $Q_{12}(x, \cdot)$ of the observation vector given $(X_1, X_2) = x = (x_1, x_2)$, through which every conditional expectation entering ν_g, σ_g and q is defined. *Fixing Q_{12} once is what supplies a **jointly measurable** version of q*: for a bounded measurable ς of the observation vector, the map $(x_1, x_2) \mapsto \int \varsigma \, dQ_{12}(x, \cdot)$ is Borel by the definition of a regular conditional distribution, and the monotone class theorem extends this to integrable ς.

The substitution lemma cited throughout as "(P6)'s substitution lemma" and as "(P6a)'s substitution lemma", and the shift identity cited as "the translation statement in (P6)", are Lemma 110, which is stated at the versions fixed here.

Lemma 110 (Standing convention: substitution at the fixed regular version, and the translate identity)**.** *Fix the versions of Definition 109.*

*(i) Substitution, for **integrable** and not merely bounded integrands. Let V be $\mathcal{F}_{XZW}$-measurable, let f be jointly measurable, and suppose $E\big|f(\Delta Y_\theta - \Delta\varphi, V)\big| < \infty$. Then,*

for almost every ω,

$$E\Big[f(\Delta Y_\theta - \Delta\varphi,\, V)\;\Big|\;\mathcal{F}_{XZW}\Big](\omega) \;=\; \int f\Big(v,\, V(\omega)\Big)\, dG_\varphi(\omega, v).$$

(ii) The translate, or shift, identity. *For $h \in \mathcal{V}$, $G_{\varphi+h}(\cdot, v) = G_\varphi(\cdot, v + \Delta h)$ almost surely, for every v; equivalently, $E[\mathbb{1}\{\Delta Y_\theta - \Delta\varphi \le u + \Delta h\} \mid \mathcal{F}_{XZW}] = G_\varphi(u + \Delta h)$ almost surely.*

Proof of Lemma 110. Clause (i). This is the disintegration theorem, Kallenberg (1997, Thm. 5.4), p. 85, read at the σ-field $\mathcal{F}_{XZW}$, at the regular version $dG_\varphi(\omega, \cdot)$ that Definition 109 fixes, and at the $\mathcal{F}_{XZW}$-measurable second argument V; its integrability hypothesis is the one stated here. *The almost-sure existence and the $\mathcal{F}_{XZW}$-measurability of the right-hand integral are part of that theorem's assertion*, which is what this clause needs and what a bare change of variables would not supply.

Clause (ii). Apply clause (i) with $V = \Delta h$ and f bounded. Since $\Delta U(\theta, \varphi + h) = (\Delta Y_\theta - \Delta\varphi) - \Delta h$ with Δh $\mathcal{F}_{XZW}$-measurable, the regular version for $\varphi + h$ is the translate of the version for φ, which is the display. □

Remark 111 (Why the substitution lemma is stated for integrable and not for bounded integrands). The bounded form does not cover the use made of it. Lemma 48 Step 3 applies (P7a) at $f(v, z) = \varpi_u(v) \int_{\mathcal{T}} \ell(v, t)\, w(t'\Phi(z))\, \Pi_t(dt)$, which is jointly measurable and $\mathcal{F}_{XZW}$-compatible but **not bounded**, since ℓ is only in $L^2(\Pi)$. What Step 2 of that proof does deliver is $\mathcal{W}\ell \in L^2(P_{\Delta U, \Delta Z})$, hence in $L^1(P)$ by the Cauchy–Schwarz inequality, which is exactly the hypothesis of Lemma 110(i). Lemma 110(ii) is what licenses the shift identity used in Proposition 8 Step 5 and the exact increment in its Step 1.

Lemma 112 (Standing convention: the range measure has a u-density)**.** *Suppose B1(iv.a) and C2′(i) hold, so that $\Pi_u(du) = \varpi_u(u)\, du$ with ϖ_u bounded and strictly positive everywhere.*

(P7a) *Let V be $\mathcal{F}_{XZW}$-measurable and let ζ be jointly measurable in its two arguments with $E\big|\varpi_u(\Delta U)\,\zeta(\Delta U, V)\big| < \infty$. Then*

$$\int_{\mathbb{R}} f_{\Delta U|\mathcal{F}_{XZW}}(u)\,\zeta\big(u, V\big)\,\Pi_u(du) \;=\; E\Big[\varpi_u(\Delta U)\,\zeta(\Delta U, V)\;\Big|\;\mathcal{F}_{XZW}\Big] \qquad \textit{for almost every } \omega.$$

The second slot is printed because the only live consumer supplies one and because Lemma 110(i), the clause this one delegates to, states the slot in terms*: Lemma 48 Step 3 applies (P7a) once at each contrast index l, at $\zeta(u,z) = \varpi_{\mathsf{w}}(l)\int_{\mathcal{T}} \ell(u,t,l)\,w(t'\Phi(z))\,\Pi_t(dt)$ and $V = \Delta Z$, which is $\mathcal{F}_{XZW}$-measurable.* The one-argument form is the two-argument form at a constant V*, and* the contrast index is not a third slot*: it is fixed before (P7a) is invoked, so the clause is unchanged at every L and reads at $\mathsf{L} = 2$, where $\varpi_{\mathsf{w}}(1) = 1$, exactly as earlier rounds printed it.*

(P7b) *$f_{\Delta U}$ exists and satisfies $\|f_{\Delta U}\|_\infty \le \|\bar f_0\|_\infty < \infty$.*

Proof of Lemma 112. (P7a). By B1(iv.a), $\Pi_u(du) = \varpi_u(u)du$, so the left-hand side is $\int f_{\Delta U|\mathcal{F}_{XZW}}(u)\,\zeta\big(u, V(\omega)\big)\,\varpi_u(u)\,du$. Apply Lemma 110(i), in its **integrable** form, the stated hypothesis being what that clause consumes, and at that clause's $\mathcal{F}_{XZW}$-measurable second argument, which is the V of the statement, to the regular conditional distribution of ΔU given $\mathcal{F}_{XZW}$, whose density is $f_{\Delta U|\mathcal{F}_{XZW}}$ for almost every ω by C2′(i).

(P7b). $f_{\Delta U}(u) = E\big[f_{\Delta U|\mathcal{F}_{XZW}}(u)\big]$ and C2′(i) bounds the integrand by $\bar f_0$ with $\|\bar f_0\|_\infty < \infty$. *The interchange here is Tonelli's theorem and needs no side condition*: the integrand $f_{\Delta U|\mathcal{F}_{XZW}}(u)$ is nonnegative and jointly measurable, so $\int E[\cdot]\,du = E[\int \cdot\, du] = 1$ unconditionally, and no dominating function is required. □

Remark 113 (The two scope points of the density convention). (α) The integrability hypothesis in (P7a) is what Lemma 110(i) consumes. The bounded form of the substitution lemma does not cover Lemma 48 Step 3, where ζ is not bounded; see Remark 111.

(β) "For almost every ω" is exact. C2′(i) supplies the density for almost every ω, not for every ω, so the identification of the conditional law's density with $f_{\Delta U|\mathcal{F}_{XZW}}$ holds off a single null set, fixed once here and not re-fixed at each site.

Without the density the left-hand side of (P7a) *is not a conditional expectation and has no* $(\Delta U, \Delta Z)$*-representative*; that is the whole content of the B1(iv.a) strengthening.

Definition 114 (Standing convention: conditioning on the auxiliary fold)**.** (P8) names four things that do not have the same standing, and they are separated here.

(a) The conditioning convention, which is **Part-H scoped**. Throughout Part H every expectation and every O_P is conditional on $\sigma(\mathcal{I}_a)$ unless said otherwise, so that $\hat{\varrho}$, computed on the auxiliary fold, per A4(iii) and Definition 2, is **frozen** wherever it appears. This is Lemma 60's convention extended to the φ-step, and A0 supplies the independence of the two folds. Definition 24 re-declares it for Part I at $\sigma(\mathcal{I}_{-k})$; it is re-declared and not inherited.

(b) *The three conditional-to-unconditional conversions*, Lemma 115, stated at a generic σ-field and Part-independent.

(c) *The freezing, or substitution, lemma*, Lemma 118, also stated at a generic σ-field. *It is consumed in Parts F, H and I*, which is why the whole of (P8) is stated here and not at the head of Part H.

(d) *The reading of the conditional expectations the freezing lemma produces*, Lemma 117 and Remark 119.

Lemma 115 (Three conditional-to-unconditional conversions, and they are not interchangeable)**.** *Let* $\mathcal{G}$ *be a* σ*-field, let* $a_n > 0$ *be* ***deterministic****, and let* X_n *be real random variables.*

(i) The transfer rule. *Suppose there are* $\mathcal{G}$*-measurable* $\bar{a}_n$ *with* $P\big(|X_n| > \bar{a}_n \mid \mathcal{G}\big) \to 0$ *almost surely, and* $\bar{a}_n/a_n = O_P(1)$ *unconditionally. Then* $X_n = O_P(a_n)$ *unconditionally.* **The moving threshold is written** $\bar{a}_n$ **and carries no letter of its own.** *The bar*

is this project's mark for the enveloping or capping version of the symbol it sits on ($\bar{w}$, $\bar{f}_k$, $\bar{c}_\pi$, $\bar{\tau}$, $\bar{m}_n$, $\bar{v}_\dagger$, $\bar{k}$), and here the reading is exact rather than merely non-colliding: $\bar{a}_n$ is the random *threshold and a_n the deterministic rate it envelopes. It is separated from the* ***reserved*** *degree of ill-posedness a by the bar* and *the subscript, a carrying no subscript anywhere in this project.*

For clauses (ii)–(iv), fix the exponent to be 1 *throughout or* 2 *throughout, and call the* conditional tail bound *the statement that there is a $\mathcal{G}$-measurable $V_n \geq 0$ with*

$$P\Big(|X_n| > \epsilon\, a_n \,\Big|\, \mathcal{G}\Big) \;\leq\; \frac{V_n}{\epsilon} \quad \textit{at exponent one,} \qquad P\Big(|X_n| > \epsilon\, a_n \,\Big|\, \mathcal{G}\Big) \;\leq\; \frac{V_n}{\epsilon^2} \quad \textit{at exponent two,} \tag{141}$$

holding almost surely and simultaneously at every $\epsilon > 0$.

(ii) The O_P companion. *Suppose* (141) *holds at one of the two exponents. If $V_n = O_P(1)$ then $X_n = O_P(a_n)$.*

(iii) The o_P companion. *Suppose* (141) *holds at one of the two exponents. If $V_n = o_P(1)$ then $X_n = o_P(a_n)$.*

(iv) The all-ϵ upgrade, which is what makes (ii)–(iii) citable at their sites. Suppose that, for ***each*** *$\epsilon > 0$ separately, the relevant inequality of* (141) *holds almost surely, a null set per ϵ, which is what conditional Markov and conditional Chebyshev deliver. Then* (141) *holds in the simultaneous form above, with one null set for all ϵ.*

The exponents are written out at the two values this file consumes rather than carried as a symbol , *exponent one at every conditional-Markov site and exponent two at every conditional-Chebyshev site. The same two-line proof gives the conclusion at any fixed positive exponent.*

Proof of Lemma 115. Bare italic K is a tolerance level, bound in place in each of clauses (i) and (ii) and dying in the clause that binds it: at (i) it is the constant a $\bar{a}_n/a_n = O_P(1)$

statement supplies at tolerance ϵ; at (ii) it is the level at which (141) is applied. It is separated by **font** from the **reserved** calligraphic $\mathcal{K}, \tilde{\mathcal{K}}$, the Hilbert–Schmidt operator, and from sans-serif K, Lemma 123(ii)'s mean bound, neither of which occurs in this proof.

Clause (i). Fix $\epsilon > 0$ and choose K with $\limsup_n P(\bar{a}_n > K a_n) < \epsilon/2$, available from $\bar{a}_n / a_n = O_P(1)$. On the $\mathcal{G}$-measurable event $\{\bar{a}_n \le K a_n\}$ one has $\{|X_n| > K a_n\} \subseteq \{|X_n| > \bar{a}_n\}$, so

$$\begin{aligned} P\Big(|X_n| > K a_n\Big) &\le P(\bar{a}_n > K a_n) + E\Big[\mathbb{1}\{\bar{a}_n \le K a_n\}\, P\Big(|X_n| > \bar{a}_n \mid \mathcal{G}\Big)\Big] \\ &\le P(\bar{a}_n > K a_n) + E\Big[P\Big(|X_n| > \bar{a}_n \mid \mathcal{G}\Big)\Big], \end{aligned}$$

and the second term tends to 0 by bounded convergence. Hence $\limsup_n P(|X_n| > K a_n) < \epsilon$. *Both halves of the hypothesis are needed*, and the second is where A4(iii) enters at the one Part-H site that could use this clause: A4(iii) gives $\|\hat{\varrho} - \varrho\|_{L^2} = O_P(r_n)$ **unconditionally**, which is an unconditional $O_P(1)$ statement about the bounding constant.

Clause (ii). Fix $\epsilon > 0$. By $V_n = O_P(1)$ choose η with $\limsup_n P(V_n > \eta) < \epsilon/2$; then choose K with $\eta/K < \epsilon/2$ at exponent one, or $\eta/K^2 < \epsilon/2$ at exponent two. The event $\{V_n \le \eta\}$ lies in $\mathcal{G}$, so, applying (141) at $\epsilon = K$,

$$\begin{aligned} P\Big(|X_n| > K a_n\Big) &\le P(V_n > \eta) + E\Big[\mathbb{1}\{V_n \le \eta\}\, P\Big(|X_n| > K a_n \mid \mathcal{G}\Big)\Big] \\ &\le P(V_n > \eta) + \eta/K \quad \text{(at exponent two, } +\eta/K^2\text{)}, \end{aligned}$$

whence $\limsup_n P(|X_n| > K a_n) < \epsilon$. Since ϵ was arbitrary, $X_n = O_P(a_n)$.

Clause (iii). Fix $\epsilon > 0$ and $\eta > 0$. The same display, with (141) applied at ϵ itself rather than at K, gives $P(|X_n| > \epsilon a_n) \le P(V_n > \eta) + \eta/\epsilon$ (at exponent two, $+\eta/\epsilon^2$). Let $n \to \infty$, so that the first term vanishes by $V_n = o_P(1)$, and then $\eta \to 0$. No appeal to bounded convergence is needed: the second term is bounded by a constant that is then sent to zero.

Clause (iv). For each $\epsilon > 0$ let N_ϵ denote the null set on which the inequality fails at that

ϵ, and put $N := \bigcup N_\epsilon$, the union taken over the **rational** $\epsilon > 0$ only, so that N is null. Fix $\omega \notin N$ and a real $\epsilon > 0$, and take rationals $\epsilon_k \uparrow \epsilon$. Then $\{|X_n| > \epsilon a_n\} \subseteq \{|X_n| > \epsilon_k a_n\}$, so

$$P\Big(|X_n| > \epsilon a_n \mid \mathcal{G}\Big)(\omega) \;\le\; P\Big(|X_n| > \epsilon_k a_n \mid \mathcal{G}\Big)(\omega) \;\le\; \frac{V_n(\omega)}{\epsilon_k} \;\longrightarrow\; \frac{V_n(\omega)}{\epsilon},$$

and likewise with the square in the denominator. Monotonicity of the event in the threshold on the left and continuity of the bound in ϵ on the right are what make the upgrade free. Without this clause the hypothesis of (ii) would be strictly stronger than what conditional Markov and conditional Chebyshev supply, and the sites that assert the simultaneous form would be asserting more than they prove. □

Remark 116 (Which conversion is used where, and what none of the three licenses). *Clause (i) is not a special case of clauses (ii)–(iii), nor they of it, and the distinction is the reason the file carries three statements.* Clause (i) takes a hypothesis at **one moving threshold** $\bar{a}_n$ and needs the conditional probability there to vanish; clauses (ii)–(iii) take a hypothesis at **every** threshold ϵa_n and need only a tightness statement about the numerator. Neither implies the other: (i)'s hypothesis carries no ϵ-uniform envelope, and (ii)–(iii)'s hypothesis never makes any single conditional probability vanish. *At a threshold that is a fixed multiple of a Markov or Chebyshev numerator the conditional probability is bounded by a* ***constant*** *and does not vanish*, which is why no site in this file uses clause (i).

What none of the three licenses. Clause (i) licenses no unconditional statement when the conditional bounding constant is not $O_P(1)$ unconditionally, no site needs that, and Proposition 79(ii) is this block's own record that an O_P hypothesis supplies no *fixed* constant, a different point that clause (i) does not soften. Clause (ii) licenses no conclusion *faster* than a_n: it converts a conditional moment bound whose numerator is tight into an unconditional O_P at the rate the bound was written at, and no better. Clause (iii) licenses no **rate** at all: it says the conclusion is $o_P(a_n)$ and nothing about how fast.

Lemma 117 (Generalised conditional expectations, and the operations that survive)**.** *Let $\mathcal{G}$ be a σ-field and let Ξ, Ξ' be real random variables with $E[|\Xi| \mid \mathcal{G}] < \infty$ and $E[|\Xi'| \mid \mathcal{G}] < \infty$ almost surely,* neither assumed integrable. *Write $E[\Xi \mid \mathcal{G}] := E[\Xi^+ \mid \mathcal{G}] - E[\Xi^- \mid \mathcal{G}]$, the* ***generalised*** *conditional expectation. The generic variables are Ξ, Ξ' and not X, Y, which are the model's regressor and outcome; the generic scalars are ς, ς' and not a, b, which are the* ***reserved*** *degree of ill-posedness and the weight anchor.*

(i) $E[\Xi \mid \mathcal{G}]$ *is well defined and real-valued almost surely, and agrees with the usual conditional expectation when Ξ is integrable.*

(ii) Linearity and the triangle inequality. *For real ς, ς', $E[\varsigma\Xi + \varsigma'\Xi' \mid \mathcal{G}] = \varsigma E[\Xi \mid \mathcal{G}] + \varsigma' E[\Xi' \mid \mathcal{G}]$, and $|E[\Xi \mid \mathcal{G}]| \leq E[|\Xi| \mid \mathcal{G}]$, both almost surely.*

(iii) Tower property and pull-out. *If $\mathcal{G}_0 \subseteq \mathcal{G}$ and $E[|\Xi| \mid \mathcal{G}_0] < \infty$ almost surely, then $E\Big[E[\Xi \mid \mathcal{G}] \mid \mathcal{G}_0\Big] = E[\Xi \mid \mathcal{G}_0]$; and if Λ is $\mathcal{G}$-measurable with $E[|\Lambda\Xi| \mid \mathcal{G}] < \infty$ almost surely, then $E[\Lambda\Xi \mid \mathcal{G}] = \Lambda\, E[\Xi \mid \mathcal{G}]$, both almost surely.*

(iv) Conditional Jensen. *For convex $\phi : \mathbb{R} \to \mathbb{R}$, $E[\phi(\Xi) \mid \mathcal{G}] \geq \phi\Big(E[\Xi \mid \mathcal{G}]\Big)$ almost surely, the left side being defined in $(-\infty, +\infty]$.*

(v) Conditional Cauchy–Schwarz. $E[|\Xi\Xi'| \mid \mathcal{G}] \leq \Big(E[\Xi^2 \mid \mathcal{G}]\Big)^{1/2}\Big(E[\Xi'^2 \mid \mathcal{G}]\Big)^{1/2}$ *almost surely, as an inequality in $[0, +\infty]$ requiring no integrability whatever.*

Proof of Lemma 117. Clause (i). Ξ^+ and Ξ^- are nonnegative, so their conditional expectations are defined in $[0, +\infty]$ with no integrability; both are bounded by $E[|\Xi| \mid \mathcal{G}]$, finite almost surely, so the difference is defined and finite off a null set. When Ξ is integrable the two definitions agree because both are determined by the averaging property, Kallenberg (1997, Thm. 5.1(i)), on $\mathcal{G}$-sets.

Clauses (ii) and (iii). These are Kallenberg (1997, Thm. 5.1), p. 81, which asserts that $E[\,\cdot \mid \mathcal{G}]$ is a *linear* operator and whose clauses (ii), (iv), (v) and (vii) are positivity, conditional monotone convergence, the pull-out of a $\mathcal{G}$-measurable factor and the chain rule;

and which states the last six to hold "whenever the corresponding expressions exist for the absolute values", *which is the regime this lemma is in, and is why the citation reaches a generalised conditional expectation and not only an L^1 one.* What has to be added is one sentence of bookkeeping: apply each to Ξ^+ and Ξ^-, and for the pull-out to Λ^+ and Λ^- against each, and subtract on the almost-sure event where the finitely many envelopes involved are finite, which the hypotheses supply. The triangle inequality is then immediate, $|E[\Xi^+ \mid \mathcal{G}] - E[\Xi^- \mid \mathcal{G}]| \le E[\Xi^+ \mid \mathcal{G}] + E[\Xi^- \mid \mathcal{G}] = E[|\Xi| \mid \mathcal{G}]$.

Clauses (iv) and (v) are not in that theorem and are proved. Kallenberg's conditional Jensen and Hölder inequalities are an unnumbered remark (Kallenberg, 1997, p. 86) and are stated for $\Xi \in L^p$, *under the unconditional integrability this lemma drops*, so neither is available here, and the $\pm$ split that carries clauses (ii) and (iii) reaches neither.

Clause (iv). A finite convex ϕ on $\mathbb{R}$ is the pointwise supremum of a **countable** family of affine functions. Fix one of them and write it $x \mapsto \varsigma' x + \varsigma$, so that $\phi(x) \ge \varsigma' x + \varsigma$ for every x, *the scalars are clause (ii)'s and are used in clause (ii)'s role.* Then $\varsigma'\Xi + \varsigma$ is conditionally integrable, and $\phi(\Xi) - (\varsigma'\Xi + \varsigma) \ge 0$, so the latter's conditional expectation is defined in $[0, +\infty]$ with no integrability. Hence $E[\phi(\Xi) \mid \mathcal{G}]$ is well defined in $(-\infty, +\infty]$ and, by additivity of a nonnegative variable and a conditionally integrable one together with clause (ii),

$$E\Big[\phi(\Xi) \mid \mathcal{G}\Big] \;=\; E\Big[\phi(\Xi) - (\varsigma'\Xi + \varsigma) \mid \mathcal{G}\Big] + \varsigma' E\Big[\Xi \mid \mathcal{G}\Big] + \varsigma \;\ge\; \varsigma' E\Big[\Xi \mid \mathcal{G}\Big] + \varsigma.$$

Take the supremum over the countable family, whose union of null sets is null; the supremum of the right-hand sides is $\phi\Big(E[\Xi \mid \mathcal{G}]\Big)$. *The $\pm$ split of clause (iii) does not prove this clause,* since $\phi(\Xi^+) - \phi(\Xi^-) \ne \phi(\Xi)$ in general; the countable-minorant route is what replaces it, and this is one of the two items the "nothing downstream changes" list asserted without a ground that reached it.

Clause (v). For $\mathcal{G}$-measurable $\varsigma > 0$, *here a random multiplier and not clause (ii)'s real*

constant, declared in place, $|\Xi\Xi'| \le \frac{1}{2}(\varsigma\,\Xi^2 + \varsigma^{-1}\Xi'^2)$ pointwise; take conditional expectations of nonnegative variables and pull out ς by clause (iii), which for a *nonnegative* factor and a *nonnegative* integrand needs no integrability. On the event where $E[\Xi^2 \mid \mathcal{G}]$ and $E[\Xi'^2 \mid \mathcal{G}]$ both lie in $(0,\infty)$, take the inequality at every positive **rational** ς, intersect the countably many almost-sure events, and infimise; the infimum over the rationals equals the infimum over all positive ς because the bound is continuous in ς, and its value is the stated product. On $\{E[\Xi^2 \mid \mathcal{G}] = 0\}$ one has $\Xi = 0$ conditionally almost surely and both sides vanish; on $\{E[\Xi^2 \mid \mathcal{G}] = \infty\}$ the right side is $+\infty$ unless $E[\Xi'^2 \mid \mathcal{G}] = 0$, which is the previous case with the roles exchanged. *This is the second item the list asserted without a reaching ground*: the $\pm$ split alone does not produce the $\mathcal{G}$-measurable multiplier. □

Lemma 118 (Standing convention: the freezing lemma)**.** *Let $\mathcal{G}$ be a σ-field, let ξ be a random element* ***independent of*** *$\mathcal{G}$ with law Q, taking values in a Polish space,* that space and its Borel σ-field are written out rather than named*, the natural calligraphic letters being claimed , and let* N *be a family of objects indexed by the sample point, written* $\mathsf{N}(\omega,\cdot)$*, at which a real function f of such an object and of a point of ξ's space is evaluated.* The frozen nuisance is sans-serif N*; italic N carries, across the three live files, only* counts and null sets*, each bound in the clause that introduces it.* **No measurability, and no container, is imposed on** N **separately.** Six of the nine sites could not supply one*: at each of them* N *is the tuple of estimated nuisances whose* $\hat{\ell}^{(-k)}$ *ingredient Definition 24*(c′.1) *declares through* (41)*, of which that definition says in terms "no equivalence class is formed and no representative is chosen", so there is no space of functions for* N *to be a random element* of. The operative hypothesis is on the **composite** and is a hypothesis of this lemma*:*

$$(\omega, e) \longmapsto f\Big(\mathsf{N}(\omega,\cdot),\, e\Big) \quad \textit{is measurable for the product of } \mathcal{G} \textit{ with the Borel } \sigma\textit{-field of } \xi\textit{'s space.} \tag{142}$$

(F-i) No integrability whatever. *If $f \ge 0$ then, for almost every ω and as an identity in*

$[0, +\infty]$,

$$E\big[f(\mathsf{N}, \xi) \mid \mathcal{G}\big](\omega) \;=\; \int f\Big(\mathsf{N}(\omega, \cdot), e\Big)\, Q(de).$$

(F-ii) *The conditional hypothesis, f of either sign. Suppose*

$$\int \Big|f\Big(\mathsf{N}(\omega, \cdot), e\Big)\Big|\, Q(de) \;<\; \infty \qquad \textit{for almost every } \omega.$$

Then $E[|f(\mathsf{N}, \xi)| \mid \mathcal{G}] < \infty$ almost surely; the generalised conditional expectation $E[f(\mathsf{N}, \xi) \mid \mathcal{G}]$ of Lemma 117 is well defined and real-valued almost surely; and the display of (F-i) *holds.*

The hypothesis of (F-ii) *is* ***conditional*** *and not unconditional. What the consumers prove is that the frozen integral is finite* ***almost surely****, and that is all they can prove: the bounding numerators are $\|\hat{\ell}\|_\infty$, $\|\partial_u \hat{\ell}\|_\infty$, $\|h_k\|_{L^2(\pi)}$ and $\|\hat{\psi} - \psi_\star\|^2_{L^2(P)}$, each O_P or o_P by (R-a), (43) or (135), and an $O_P(1)$ random variable need not be integrable.* Stating (142) on the composite rather than separately on N and on f is what makes the lemma cover the use made of it*, since at every site f involves* ***point evaluation*** *of the nuisance and evaluation is not measurable on a space of equivalence classes.*

Proof of Lemma 118. Neither half is a citation, and the reason is the hypothesis and not the conclusion. The classical statement is Kallenberg (1997, Thm. 5.4) at the law of ξ, the same theorem Lemma 110(i) invokes, and it asks for a measurable space of which the frozen argument is a random element, and for *unconditional* integrability. *This lemma has neither*, by construction: (142) is imposed on the composite precisely because N has no container, and (F-ii)'s hypothesis is conditional. Kallenberg (1997, Lem. 2.11) is not an alternative, being an identity between *unconditional* expectations.

(F-i)*, bounded case.* Consider composites that are the product of the indicator of a set in $\mathcal{G}$ with the indicator of a Borel set in ξ's space, *neither set is given a letter.* At such a composite the left-hand side is the first indicator times the Q-measure of the Borel set, by independence

of ξ from $\mathcal{G}$ together with $\mathcal{G}$-measurability of the first factor, and the right-hand side is the same product. Those rectangles form a π-system generating the product σ-field of (142), and the class of bounded product-measurable composites for which the display holds is a vector space containing the constants and closed under bounded monotone limits, so the functional monotone class theorem gives every **bounded** composite.

(F-i)*, nonnegative unbounded case.* Apply the bounded case to the truncations $f \wedge 1, f \wedge 2, f \wedge 3, \ldots,$ *the truncation level is written out and carries no letter*, n being the sample size and binding throughout. The left-hand sides increase to $E[f(\mathsf{N}, \xi) \mid \mathcal{G}]$ by the **conditional monotone convergence theorem**, which is a statement about nonnegative variables and needs no integrability; the right-hand sides increase to the stated integral by ordinary monotone convergence against Q, at **every** ω. The exceptional sets are indexed by the truncation level, hence countably many, hence their union is null. *This passage from bounded to nonnegative-unbounded is a separate monotone limit and not a corollary of the monotone class theorem*, which by itself delivers only bounded f.

(F-ii). Apply (F-i) to f^+ and to f^-. The hypothesis makes both right-hand sides finite for almost every ω, hence both left-hand sides finite almost surely, hence $E[|f(\mathsf{N}, \xi)| \mid \mathcal{G}] < \infty$ almost surely; the difference of the two identities is then legitimate and is the display. *No conditional dominated convergence is used, and none is available before* (F-i) *has been applied to* $|f|$: the envelope $E[|f| \mid \mathcal{G}] < \infty$ that Lemma 110(i) uses is obtained *from* unconditional integrability, which is exactly the hypothesis (F-ii) drops. □

Remark 119 (What the freezing lemma's conclusion means, and what does not change downstream). *Under* (F-ii) *the conditional expectations this lemma produces are* ***generalised*** *ones* in the sense of Lemma 117, defined as $E[f^+ \mid \mathcal{G}] - E[f^- \mid \mathcal{G}]$ on the almost-sure event where the conditional envelope is finite, *and not conditional expectations of integrable random variables. That reading is declared here once and covers every display the lemma produces*, in particular (130) and the third summand of (186), whose $E[\,\cdot \mid \sigma(\mathcal{I}_{-k})]$ are strictly generalised

for exactly this reason and are not separately annotated at their sites.

Nothing downstream changes, and the ground is now per operation rather than one sentence. Every consumer bounds the *norm* of such a conditional expectation, and each operation the consumers use is a clause of Lemma 117: linearity and the triangle inequality (clause (ii)); the **tower property**, and with it the pull-out of a measurable factor (clause (iii)); conditional Jensen (clause (iv)); conditional Cauchy–Schwarz (clause (v)). *The four grounds are not the same*, and a single ±-split sentence covers only the first three: conditional Jensen needs the countable affine minorants and conditional Cauchy–Schwarz needs the pull-out at a $\mathcal{G}$-measurable multiplier, which is why Lemma 117 states five clauses and proves each.

The tower property is the item two consumers name in terms. Lemma 93 Step 2(b) writes "$E[D\widetilde{W}] = 0 \ldots$ so the tower property applies and pulls D out on the left"; its Step 5 writes $E[D\mathsf{r}] = E[E[D \mid \Delta Z]\,\mathsf{r}(\Delta Z)] = 0$. **Both are generalised**, because what Step 0 supplies is an almost-surely finite *conditional* envelope and not unconditional integrability, $\|\hat{\ell}\|_\infty$ and $E\|\hat{b}\|^2$ being $O_P(1)$ by (R-a) and not bounded by constants. *A list written to say "nothing downstream changes" that omits the one item its two named consumers cite is a defect*, and it is repaired at the tool rather than at the two steps, for the reason the paragraph above gives for declaring the reading once.

The order those two quotations reproduce is the conformable one. D is $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued by Definition 53 while $\widetilde{W}$ and r are $\mathbb{R}^{\mathsf{L}-1}$-valued by (1), so the products that multiply are $D\widetilde{W}$ and $D\mathsf{r}$, both $\mathbb{R}^p$-valued; the reverse orders conform only at $\mathsf{L} = 2$. *Nothing this remark asserts depends on which order is printed*, that the two conditional expectations are generalised, and that the tower property is a clause of Lemma 117, are indifferent to it, and Lemma 45(v) states the conformable order and its ground.

Remark 120 (Where the freezing lemma is consumed, and what supplies its composite hypothesis at each site). *Lemma 118 is used wherever a display of the form "*$E[\,\cdot \mid \sigma(\mathcal{I}_{-k})] = \|\cdot\|_{L^2(P)}$*" or "*$E[\,\cdot \mid \sigma(\mathcal{I}_a)] = E|\cdot|(\Delta Z)$*" appears. There are nine such sites*, and at each

of them $\mathcal{G}$ is $\sigma(\mathcal{I}_{-k})$ or $\sigma(\mathcal{I}_a)$, N is the tuple of estimated nuisances, and ξ is the main-fold observation vector, independent of $\mathcal{G}$ by A0 and, in Part I, by Definition 24. *Every site invokes* (F-ii), and Lemma 77 Step 4 is the one that would also satisfy the unconditional hypothesis, since $|\hat{\varrho}_l - \varrho_l| \le 1$ **surely** there, at *every* cell l, by Definition 2, both quantities lying in $[0,1]$, the truncation being to the ε-interior of the simplex. At $\mathsf{L} = 2$ this is the scalar $|\hat{\varrho} - \varrho| \le 1$ earlier rounds printed. *Lemma 60 Steps 0 and 2 are the archetype*: that proof freezes $\hat{\ell}$ inside ν_g and its Step 2 identity $\nu_g = \nu_{(\hat{\ell}-\ell_\star)\widetilde{W}}$ *is* this lemma at $\mathcal{G} = \sigma(\mathcal{I}_a)$, *the direction to the left of the contrast*, which is the only conformable order once $\hat{\ell} - \ell_\star$ is $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued, and the order Lemma 60(i) prints. At $\mathsf{L} = 2$ the two orders agree.

What supplies (142), and the answer is **not the same at every site**. Without it the right-hand side is an integral of a function not known to be measurable and the left-hand side has no version equal to it.

(a) *Part I, seven sites.* Which clause of Definition 24(c′.1) supplies it depends on which nuisance the composite carries. Where $\hat{\ell}$ enters the composite, Lemma 93 Step 0 and Steps 2–5, Lemma 94 Step 6, Lemma 95 Steps 2, 3 and 8, **six of the seven**, the supplier is (41), product measurability of the estimator's *own pointwise function*, and not the per-realisation "Borel" of Definition 24(c). At the seventh, Lemma 92 Steps 4 and 4′, $\hat{\ell}$ does not appear at all: the composite is $\Delta Z\,\Delta h_k$, the frozen object is h_k, and the supplier is exactly the jointly measurable representative (c′.1) fixes for $\hat{\varphi}^{(-k)}$ in $\mathcal{V}$. At the six sites of the first kind the evaluation point is itself built from $\hat{\varphi}^{(-k)}$, so both clauses of (c′.1) are used, one per ingredient.

(b) *Lemma 77 Step 4, in Part H.* That step is stated hundreds of lines before Definition 24 exists and does not adopt it; what is available there is A4(iii), which now requires joint measurability of $\hat{\varrho}$'s own pointwise function, the (41) shape, one Part earlier, and which supplies this composite by composition with the Borel coordinate maps, and makes A4(iii)'s own O_P display a statement about a measurable function of the sample.

(c) Lemma 60 Steps 0 and 2, in Part F. That lemma's own sample-splitting convention states that $\hat{\ell}$ is a $\sigma(\mathcal{I}_a)$-measurable **random element** of $L^2(P_{\Delta U,\Delta Z})$ with one **jointly measurable representative fixed**. *That is deliberately strictly less than* (41)*, and the scoping is the content*: this site evaluates $\hat{\ell}$ at an **undistorted** $(\Delta U, \Delta Z)$ draw, whose law is the one the container is built on, so the choice of representative is immaterial and (41) is neither needed nor imposed. Importing it here would charge a Part-F lemma with a Part-I hypothesis.

An L^2 representative supplies nothing about $\hat{\ell}$, whose Part-I evaluation points are undominated; it is precisely what supplies the $\hat{\varphi}^{(-k)}$ ingredient, and at Lemma 92 it is the only supplier.

Remark 121 (Standing convention: the kernel read as a function of its scaled argument). (P9) is the reading, not a new object. Wherever a proof below writes Θ_I it means (14) of C3, $\Theta_I(\varsigma) = C_h(h\varsigma, I)$ for whichever of $h = h_u$, $h = h_x$ is in force and for $I \in \mathfrak{I}$ the realised truncation interval, and $\Theta_1 = \Theta_{\mathbb{R}}$ is the interior kernel. The constant is C3(ii)'s $\mathsf{c}_2 = \|\bar{\Theta}\|_\infty$, with $\bar{\Theta}$ the envelope and $c_C = \|\bar{\Theta}\|_{L^1}/(2\mathsf{c}_2)$, so that $2c_C\mathsf{c}_2 = \|\bar{\Theta}\|_{L^1}$.

Every property used below is a clause of C3 and not a convention of this file: vanishing off I and the uniform envelope bound $|\Theta_I| \le \bar{\Theta} \le \mathsf{c}_2$, both C3(ii); the truncated moments $\int_I \varsigma^j \Theta_I = \delta_{j0}$ for $0 \le j \le p_K - 1$, C3(ii) and (15), *in particular* $\int \Theta_I = 1$, *which Lemma 67(i), (iii) and (iv) consume*; $\int |\varsigma|^{p_K} \bar{\Theta} < \infty$, C3(ii); and symmetry with Lipschitz constant c_4 at the interior index, C3(iii). *A family with all of these is exhibited at* $p_K = 8$, Lemma 66, so no proof below rests on an empty hypothesis.

Proof of Proposition 36. The proof has two steps: measurability, then the norm bound.

Step 1: measurability. Fix $l \in \{1, \dots, \mathsf{L}-1\}$. The map $(u, t, \omega) \mapsto \widetilde{W}_l(\omega)\, \mathbb{1}\{\Delta Y_\theta(\omega) - \Delta\varphi(\omega) \le u\}\, w(t'\Phi(\Delta Z(\omega)))$ is jointly Borel measurable, being a product of: a random variable; the indicator of the Borel set $\{(u,\omega) : \Delta Y_\theta(\omega) - \Delta\varphi(\omega) \le u\}$; and a composition of the continuous map $(t, v) \mapsto w(t'v)$ with the Borel map $\Phi \circ \Delta Z$. By (P1)–(P2) its modulus is bounded by

$\bar{w}(t)$, so it is P-integrable for each fixed (u,t) and, on each set $\{\|t\|_1 \le k\}$, is dominated by the constant $\sup_{\|t\|_1\le k}\bar{w}(t) < \infty$. The Fubini–Tonelli theorem applied on each such set makes $(u,t) \mapsto \mathcal{M}(u,t,l;\varphi,\theta)$ Borel measurable there, and letting $k \to \infty$ gives measurability on $\mathbb{R}\times\mathcal{T}$; measurability in l is automatic, the third factor of Π being carried by a finite set, so $\mathcal{M}(\cdot,\cdot,\cdot;\varphi,\theta)$ is measurable on the base set of Π fixed in Definition 5.

Step 2: the norm bound. By (P1), *per cell*, (P2) and $\mathbb{1}\{\cdot\} \le 1$,

$$\Big|\mathcal{M}(u,t,l;\varphi,\theta)\Big| \;\le\; E\Big[|\widetilde{W}_l|\,\Big|w(t'\Phi(\Delta Z))\Big|\Big] \;\le\; \bar{w}(t) \qquad \text{for every } (u,t,l).$$

Hence, by B1(iv.c) and B1(iv.d),

$$\Big\|T(\varphi,\theta)\Big\|^2_{L^2(\Pi)} \;\le\; \sum_{l=1}^{\mathsf{L}-1}\varpi_{\mathsf{w}}(l)\int_{\mathbb{R}}\int_{\mathcal{T}}\bar{w}(t)^2\,\Pi_t(dt)\,\Pi_u(du) \;=\; \Pi_u(\mathbb{R})\int_{\mathcal{T}}\bar{w}(t)^2\,\Pi_t(dt) \;=\; c_\Pi^2,$$

the l-sum contributing 1 because Π_{w} is a probability measure. The bound does not involve (φ,θ) and does not involve L; *at* $\mathsf{L}=2$ *both displays are the ones earlier rounds printed.* □

Proof of Proposition 8. The proof proceeds in five steps: the increment is written exactly by conditioning on the endogenous regressors, which removes the indicator with no interchange of derivative and integral; the conditional distribution function is expanded to second order in integral form; the remainder is bounded in $L^2(\Pi)$, giving (i); boundedness of $\mathcal{A}_\varphi$ is verified; and the same expansion at first order gives (ii).

Throughout, write $\mathcal{F}_{XZW} := \sigma(X_1, X_2, \Delta Z, W)$ and note that $\widetilde{W}$, hence each contrast $\widetilde{W}_l$, $w(t'\Phi(\Delta Z))$ and $\Delta h = h(X_2) - h(X_1)$ are all $\mathcal{F}_{XZW}$-measurable. *Every step below is run at one fixed contrast index* $l \in \{1,\dots,\mathsf{L}-1\}$ *and is one of* $\mathsf{L}-1$ *identical statements*, the index entering only through $\widetilde{W}_l$, which the per-cell bound (P1) treats exactly as the binary contrast was treated; the index is *aggregated* at one place only, the passage from a pointwise bound to an $L^2(\Pi)$ bound in Steps 3 and 5, and that passage is recorded where it occurs. Let $G_\varphi(\omega,v)$ be the regular version fixed in (P6a), so that $v \mapsto G_\varphi(\omega,v)$ is a distribution

function for every ω and $\omega \mapsto G_\varphi(\omega, v)$ is $\mathcal{F}_{XZW}$-measurable for every v; by C2′(i) it admits the density $G'_\varphi(\omega, v) = f_{\Delta Y_\theta - \Delta\varphi | X_1, X_2, \Delta Z, W}(v)$ and is twice continuously differentiable in v with $|G'_\varphi| \le \bar{f}_0$ and $|G''_\varphi| \le \bar{f}_1$. Being measurable in ω and continuous in v, G_φ, G'_φ and G''_φ are jointly measurable in (ω, v) by the Carathéodory criterion; this is used silently below whenever a Fubini argument is invoked. The argument ω is suppressed.

Step 1: an exact expression for the increment. Since $\{\Delta Y_\theta - \Delta(\varphi + h) \le u\} = \{\Delta Y_\theta - \Delta\varphi \le u + \Delta h\}$ and Δh is $\mathcal{F}_{XZW}$-measurable, the substitution lemma of (P6) gives $E[\mathbb{1}\{\Delta Y_\theta - \Delta\varphi \le u + \Delta h\} \mid \mathcal{F}_{XZW}] = G_\varphi(u + \Delta h)$ a.s., and the tower property then gives

$$\mathcal{M}(u, t, l; \varphi + h, \theta) - \mathcal{M}(u, t, l; \varphi, \theta) \;=\; E\Big[\widetilde{W}_l \, w\Big(t'\Phi(\Delta Z)\Big) \Big\{G_\varphi(u + \Delta h) - G_\varphi(u)\Big\}\Big]. \quad (143)$$

This is the step the memo calls L-4, and it is where the strengthened form of `assapp2` earns its place. No derivative is passed through an indicator: the indicator is integrated out by conditioning, and what is required is that the conditional law of ΔY given the *endogenous* regressors (X_1, X_2), together with $(\Delta Z, W)$, admit a density smooth in its first argument. The marginal density $f_{\Delta U}$ named in the draft's `assapp2` is not the relevant object and would not license (143).

Step 2: second-order expansion, in integral form. By Taylor's theorem with the integral form of the remainder, valid pointwise on the underlying probability space because $G_\varphi \in C^2$,

$$G_\varphi(u + \Delta h) - G_\varphi(u) \;=\; G'_\varphi(u)\,\Delta h \;+\; (\Delta h)^2 \int_0^1 (1 - \varsigma)\, G''_\varphi(u + \varsigma\,\Delta h)\, d\varsigma. \quad (144)$$

The integral form is used rather than a Lagrange point $\tilde{u}$ supplied by the mean value theorem: $\tilde{u}$ would be an implicitly defined function of ω with no measurable selection guaranteed, whereas the right-hand side of (144) is jointly measurable in (ω, u) by Fubini and continuity

of G''_φ. Substituting (144) into (143) and using the definition (24) of $\mathcal{A}_\varphi$,

$$\mathcal{M}(u,t,l;\varphi+h,\theta)-\mathcal{M}(u,t,l;\varphi,\theta)-\mathcal{A}_\varphi h(u,t,l) \;=\; E\Big[\widetilde{W}_l\, w\Big(t'\Phi(\Delta Z)\Big)(\Delta h)^2\int_0^1(1-\varsigma)G''_\varphi(u+\varsigma\Delta h)\,d\varsigma\Big] \;=:\; R \tag{145}$$

Step 3: bounding the remainder, giving (i). By (P1), *the per-cell bound* $|\widetilde{W}_l|\le 1$, *which is what the constant is built on*, (P2), $|G''_\varphi|\le\|\bar f_1\|_\infty$ and $\int_0^1(1-\varsigma)d\varsigma=\frac12$,

$$|R_h(u,t,l)| \;\le\; \tfrac12\,\bar w(t)\,\|\bar f_1\|_\infty\,E\Big[(\Delta h)^2\Big] \qquad \text{uniformly in } l.$$

By $(a-b)^2\le 2a^2+2b^2$ and C1(ii),

$$E\Big[(\Delta h)^2\Big] \;\le\; 2\,E\Big[h(X_1)^2\Big]+2\,E\Big[h(X_2)^2\Big] \;=\; 2\int h^2\,(f_{X_1}+f_{X_2})\,dx \;\le\; 4\,\bar c_\pi\int h^2\pi\,dx \;=\; 4\,\bar c_\pi\,\|h\|^2_{L^2(\pi)}, \tag{146}$$

where $\bar c_\pi=\max_j\|f_{X_j}/\pi\|_\infty$. Therefore $|R_h(u,t,l)|\le 2\bar c_\pi\|\bar f_1\|_\infty\bar w(t)\|h\|^2_{L^2(\pi)}$ uniformly in l, and integrating against Π as in Step 2 of the proof of Proposition 36,

$$\|R_h\|^2_{L^2(\Pi)} \;\le\; \Big(2\,\bar c_\pi\,\|\bar f_1\|_\infty\,\|h\|^2_{L^2(\pi)}\Big)^2\sum_{l=1}^{\mathsf{L}-1}\varpi_{\mathsf{w}}(l)\int_{\mathbb{R}\times\mathcal{T}}\bar w(t)^2\,\Pi_u(du)\,\Pi_t(dt) \;=\; \Big(C_{\mathcal{A}}\,\|h\|^2_{L^2(\pi)}\Big)^2,$$

which is (25). *This is the aggregation over l announced above, and it costs nothing*: the l-sum contributes $\sum_l\varpi_{\mathsf{w}}(l)=1$, Π_{w} being a probability measure by B1(iv.d), so $C_{\mathcal{A}}=2\bar c_\pi\|\bar f_1\|_\infty c_\Pi$ is the constant earlier rounds printed, at every L.

Step 4: $\mathcal{A}_\varphi$ is bounded, so the derivative is Fréchet. Both $\mathcal{A}_\varphi h$ and R_h are Borel functions of (u,t,l): at each of the $\mathsf{L}-1$ values of l their integrands are jointly measurable in (u,t,ω), by the joint measurability of G'_φ and G''_φ recorded above, and, for R_h, by Fubini applied to the inner ς-integral in (145), and are dominated on each set $\{\|t\|_1\le k\}$ by an integrable envelope, so the exhaustion argument of Step 1 of the proof of Proposition 36 applies verbatim; measurability in l is automatic, the index set being finite and discrete. Linearity of $h\mapsto\mathcal{A}_\varphi h$ is immediate from (24). By (P1)–(P2), $|G'_\varphi|\le\|\bar f_0\|_\infty$, the Cauchy–Schwarz inequality and

(146),

$$\big|\mathcal{A}_\varphi h(u,t,l)\big| \;\le\; \bar w(t)\,\|\bar f_0\|_\infty\, E\big|\Delta h\big| \;\le\; \bar w(t)\,\|\bar f_0\|_\infty \Big(E[(\Delta h)^2]\Big)^{1/2} \;\le\; 2\,\bar c_\pi^{1/2}\,\|\bar f_0\|_\infty\,\bar w(t)\,\|h\|_{L^2(\pi)},$$

uniformly in l, so $\|\mathcal{A}_\varphi\| \le 2\bar c_\pi^{1/2}\|\bar f_0\|_\infty c_\Pi < \infty$ by the same l-aggregation. Since the remainder in (25) is $O(\|h\|^2) = o(\|h\|)$, $T(\cdot,\theta)$ is Fréchet differentiable at φ with derivative $\mathcal{A}_\varphi$. Evaluating at $\varphi = \varphi_\dagger$, $\theta = \theta_\dagger$ gives $\Delta Y_{\theta_\dagger} - \Delta\varphi_\dagger = \Delta U$ and hence the stated form of $\mathcal{A}_{\text{cond}}$.

Step 5: the derivative map is Lipschitz, giving (ii). Fix $h, h' \in \mathcal{V}$. By the translation statement in (P6), $G_{\varphi+h}(v) = G_\varphi(v + \Delta h)$ a.s. for every v, so $G'_{\varphi+h}(u) = G'_\varphi(u + \Delta h)$ and, by (24),

$$\Big(\mathcal{A}_{\varphi+h} - \mathcal{A}_\varphi\Big)h'(u,t,l) \;=\; E\Big[\widetilde W_l\, w\Big(t'\Phi(\Delta Z)\Big)\Big\{G'_\varphi(u+\Delta h) - G'_\varphi(u)\Big\}\Delta h'\Big].$$

By C2′(i), G'_φ is Lipschitz in v with constant $\|\bar f_1\|_\infty$, so $|G'_\varphi(u+\Delta h) - G'_\varphi(u)| \le \|\bar f_1\|_\infty|\Delta h|$. Using (P1)–(P2), the Cauchy–Schwarz inequality and (146) twice,

$$\Big|\Big(\mathcal{A}_{\varphi+h}-\mathcal{A}_\varphi\Big)h'(u,t,l)\Big| \;\le\; \bar w(t)\,\|\bar f_1\|_\infty\, E\Big[|\Delta h|\,|\Delta h'|\Big] \;\le\; \bar w(t)\,\|\bar f_1\|_\infty \Big(E(\Delta h)^2\Big)^{1/2}\Big(E(\Delta h')^2\Big)^{1/2}$$

$$\le\; 4\,\bar c_\pi\,\|\bar f_1\|_\infty\,\bar w(t)\,\|h\|_{L^2(\pi)}\|h'\|_{L^2(\pi)},$$

uniformly in l. Integrating against Π, the l-sum again contributing $\sum_l \varpi_{\mathsf{w}}(l) = 1$, and taking the supremum over $\|h'\|_{L^2(\pi)} \le 1$,

$$\Big\|\mathcal{A}_{\varphi+h} - \mathcal{A}_\varphi\Big\| \;\le\; 4\,\bar c_\pi\,\|\bar f_1\|_\infty\, c_\Pi\,\|h\|_{L^2(\pi)} \;=\; 2\,C_{\mathcal{A}}\,\|h\|_{L^2(\pi)},$$

which is (26). □

Remark 122 (What the quadratic remainder is, and what it is not). Proposition 8 establishes two distinct facts and they should not be run together. Part (i) is a *quadratic* bound on the first-order remainder of T; part (ii) is the Lipschitz statement, namely Lipschitz

continuity of the map $\varphi \mapsto \mathcal{A}_\varphi$ in operator norm. The two are not independent: integrating (26) along the segment from φ to $\varphi + h$ recovers (25) with exactly the constant $C_{\mathcal{A}}$, since $\int_0^1 \|\mathcal{A}_{\varphi+\varsigma h} - \mathcal{A}_\varphi\| \, \|h\| \, d\varsigma \le 2C_{\mathcal{A}}\|h\|^2 \int_0^1 \varsigma \, d\varsigma = C_{\mathcal{A}}\|h\|^2$.

Two consequences are worth recording now because later results consume them. First, both parts concern the *deterministic* operator T. A tangential cone condition on the *random* operator $\hat{T}$, the retired A6, and the property Definition 73 that replaces it, does not follow from either part: such a condition also constrains the estimation error, which Proposition 8 says nothing about. What (i) and (ii) do supply is the deterministic half of any verification route for a cone condition, with explicit constants. *No result in this file takes one as a hypothesis*; the distinction is retained because Proposition 74 turns on it. Second, the second derivative of the *moment* in φ, the object that generates the quadratic remainder of the whole construction, is a different quantity: it involves $\partial_u \ell_\star$, not $\partial_u f$, and is analysed at Step 4. A quadratic remainder in the operator is standard; the quadratic remainder in the moment is what forces $\delta_n = o_P(n^{-1/4})$.

Proof of Proposition 9. The proof has four steps: the kernel representation; the placement of the kernel slices in $\mathcal{V}$; the Hilbert–Schmidt bound; and the singular system.

Step 0: a jointly measurable version of the kernel. For each (u, t, l) the conditional expectations $E[g_{u,t,l} \mid X_j = x]$ entering $\nu_{g_{u,t,l}}$ are defined only up to f_{X_j}-null sets, so a version jointly measurable in (u, t, l, x) must be constructed rather than assumed; the exhaustion argument of Proposition 36 Step 1 does not supply one, since it concerns integrals over ω at fixed x. Use the regular conditional distributions $Q_j(x, \cdot)$ fixed in (P6b) and *define*

$$E\big[g_{u,t,l} \mid X_j = x\big] \;:=\; \int g_{u,t,l}(\omega') \, Q_j(x, d\omega'), \qquad j \in \{1, 2\}.$$

For each fixed (u, t, l) this is a version of the conditional expectation, by the defining property of a regular conditional distribution. Joint measurability in (u, t, l, x) follows from joint

measurability of $(u,t,\omega') \mapsto g_{u,t,l}(\omega')$ at each of the $\mathsf{L}-1$ values of l, the index set being finite and discrete, so that measurability in l is automatic, together with the domination $|g_{u,t,l}| \le \bar{w}(t)\bar{f}_0$ and $E[\bar{f}_0] < \infty$, by the same exhaustion over $\{\|t\|_1 \le k\}$. All occurrences of ν_g below, and hence k, are understood through this version.

Step 1: kernel representation. Fix $(u,t,l) \in \mathbb{R} \times \mathcal{T} \times \{1,\dots,\mathsf{L}-1\}$ and set $g_{u,t,l} := \widetilde{W}_l\, w(t'\Phi(\Delta Z))\, f_{\Delta U|X_1,X_2,\Delta Z,W}(u)$, which is integrable by (P1), (P2) and $E[\bar{f}_0] < \infty$ from C2′(i). By (24) and the ν_g-identity (P5), whose term-by-term integrability proviso holds here because $E|g_{u,t,l}h(X_j)| \le \bar{w}(t)\|\bar{f}_0\|_\infty (E[h(X_j)^2])^{1/2} < \infty$ for $h \in \mathcal{V}$,

$$\mathcal{A}_{\mathrm{cond}}h(u,t,l) \;=\; E\Big[g_{u,t,l}\,\{h(X_2)-h(X_1)\}\Big] \;=\; \int_{\mathcal{X}} \nu_{g_{u,t,l}}(x)\,h(x)\,dx \;=\; \int_{\mathcal{X}} \mathsf{k}(u,t,l;x)\,h(x)\,\pi(x)\,dx,$$

with $\mathsf{k} = \nu_{g_{u,t,l}}/\pi$, which is (27). Division by π is legitimate because π is bounded below away from zero on $\mathcal{X}$ by C1(ii). *The step is run at one fixed contrast index and is one of* $\mathsf{L}-1$ *identical statements*: only (P1)'s per-cell bound is used, so nothing in it depends on l beyond the symbol.

Step 2: the kernel slices lie in $\mathcal{V}$. By (P5) with $h \equiv 1$, for which $\Delta h = 0$, one has $\int_{\mathcal{X}} \nu_g\,dx = 0$ for every integrable g. Hence

$$\int_{\mathcal{X}} \mathsf{k}(u,t,l;x)\,\pi(x)\,dx \;=\; \int_{\mathcal{X}} \nu_{g_{u,t,l}}(x)\,dx \;=\; 0,$$

so $\mathsf{k}(u,t,l;\cdot) \in \mathcal{V}$ for every (u,t,l), and consequently $\mathcal{A}^*_{\mathrm{cond}}\ell = \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \mathsf{k}(u,t,l;\cdot)\,\ell(u,t,l)\,\Pi_u(du)\,\Pi_t($ $\mathcal{V}$ for every $\ell \in L^2(\Pi)$, a finite sum of elements of a closed subspace, by Definition 5's pairing: no projection onto $\mathcal{V}$ is needed, and $\mathcal{R}(\mathcal{A}^*_{\mathrm{cond}}) \subseteq \mathcal{V}$ by construction. Since $\mathcal{V} = \{\zeta \in L^2(\pi) : \int \zeta\,\pi\,dx = 0\}$, every element of the range satisfies $\int \zeta\,\pi\,dx = 0$. *This is the internal check that the source condition A2 must be imposed on* ν/π *rather than on* ν; the comparison itself requires $E\|\Delta Z\| < \infty$ and is therefore recorded separately, as Remark 37, where A1b(iv) can be cited.

Step 3: the Hilbert–Schmidt bound. By (P1), *the per-cell bound, and not* (P1b), together with (P2) and $|f_{\Delta U|\dots}(u)| \le \|\bar{f}_0\|_\infty$, the conditional expectations entering $\nu_{g_{u,t,l}}$ are bounded in modulus by $\bar{w}(t)\|\bar{f}_0\|_\infty$, so

$$\left|\nu_{g_{u,t,l}}(x)\right| \;\le\; \bar{w}(t)\,\|\bar{f}_0\|_\infty\left(f_{X_1}(x)+f_{X_2}(x)\right), \qquad \text{hence} \qquad \left|\mathsf{k}(u,t,l;x)\right| \;\le\; 2\,\bar{c}_\pi\,\bar{w}(t)\,\|\bar{f}_0\|_\infty,$$

uniformly in l. Since $\int \pi = 1$, and using that the Hilbert–Schmidt norm of an integral operator out of $L^2(\pi)$ is $\iint |\mathsf{k}|^2 \pi\, dx\, d\Pi$,

$$\left\|\mathcal{A}_{\text{cond}}\right\|_{\text{HS}}^2 \;\le\; 4\,\bar{c}_\pi^2\|\bar{f}_0\|_\infty^2\,\Pi_u(\mathbb{R})\int_{\mathcal{T}}\bar{w}(t)^2\,\Pi_t(dt)\sum_{l=1}^{\mathsf{L}-1}\varpi_{\mathsf{w}}(l) \;=\; 4\,\bar{c}_\pi^2\|\bar{f}_0\|_\infty^2\,c_\Pi^2,$$

finite by B1(iv.c), the l-sum contributing the factor 1 because Π_{w} is a *probability* measure by B1(iv.d). *So the constant is the printed one at every* L, and it would not be had the derivation been routed through the Euclidean bound (P1b) or had Π_{w} been allowed a total mass other than one. The bound for the restriction to $\mathcal{V}$ is no larger than the bound for the operator on all of $L^2(\pi)$. An integral operator with square-integrable kernel is Hilbert–Schmidt, and every Hilbert–Schmidt operator between separable Hilbert spaces is compact (Conway, 1990, Prop. II.4.7).

Step 4: singular system. A compact operator between separable Hilbert spaces admits a singular value decomposition $\mathcal{A}_{\text{cond}}h = \sum_{j\in\mathcal{J}}\sigma_j\langle h, e_j\rangle_{L^2(\pi)}\tilde{e}_j$ with $\{e_j\}_{j\in\mathcal{J}}$ an orthonormal basis of $(\ker\mathcal{A}_{\text{cond}})^\perp$ within the domain (Engl, Hanke, and Neubauer, 2000, Ch. 2). Separability of $\mathcal{V}$ and $L^2(\Pi)$ holds because π and Π are σ-finite Borel measures on Polish spaces and $\mathcal{V}$ is a closed subspace of $L^2(\pi)$. By Definition 5 the domain is $\mathcal{V}$, so no zero singular value arising from the constants is carried. The index set $\mathcal{J}$ may be finite, $\mathcal{A}_{\text{cond}}$ of rank one is consistent with every hypothesis here, in which case $\sigma_j \downarrow 0$ is vacuous and the e_j span $(\ker\mathcal{A}_{\text{cond}})^\perp \cap \mathcal{V}$, a proper subspace of $\mathcal{V}$ unless A1c holds. H1 excludes finite rank explicitly and must not be invoked without it. □

Proof of Lemma 10. The proof is in two steps: $\mathcal{I}$ is invariant under translation of θ; and the invariance forces the product structure.

Step 1: $\mathcal{I}$ is closed under translation of θ. Let $(\varphi, \theta) \in \mathcal{I}$ and let $\theta' \in \mathbb{R}^p$ be arbitrary. From (19),

$$\Delta U(\theta', \varphi) \;=\; \Delta U(\theta, \varphi) \;+\; \Delta Z'(\theta - \theta'). \tag{147}$$

Let $\varsigma : \mathbb{R} \to \mathbb{R}$ be bounded and measurable and define $\Lambda(v, z) := \varsigma\Big(v + z'(\theta - \theta')\Big)$, which is bounded and jointly measurable. Because ΔZ is measurable with respect to $\sigma(W, \Delta Z)$, (147) gives

$$E\Big[\varsigma\Big(\Delta U(\theta', \varphi)\Big) \,\Big|\, W, \Delta Z\Big] = E\Big[\Lambda\Big(\Delta U(\theta, \varphi), \Delta Z\Big) \,\Big|\, W, \Delta Z\Big].$$

Conditional independence $\Delta U(\theta, \varphi) \perp\!\!\!\perp W \mid \Delta Z$ implies $E[\Lambda(\Delta U(\theta, \varphi), \Delta Z) \mid W, \Delta Z] = E[\Lambda(\Delta U(\theta, \varphi), \Delta Z) \mid \Delta Z]$ for every bounded jointly measurable Λ: this holds for product functions $\Lambda(v, z) = \varsigma(v)\varkappa(z)$ directly, since $\varkappa(\Delta Z)$ factors out of both sides and conditional independence disposes of ς; and the class of bounded Λ for which it holds is a vector space closed under uniformly bounded monotone limits containing that multiplicative class, so the functional monotone class theorem extends it to all bounded jointly measurable Λ. Hence $E[\varsigma(\Delta U(\theta', \varphi)) \mid W, \Delta Z]$ is $\sigma(\Delta Z)$-measurable for every bounded measurable ς, which is $\Delta U(\theta', \varphi) \perp\!\!\!\perp W \mid \Delta Z$, i.e. $(\varphi, \theta') \in \mathcal{I}$.

Step 2: the product structure. By Step 1, membership of (φ, θ) in $\mathcal{I}$ does not depend on θ; taking $\theta' = 0$, $(\varphi, \theta) \in \mathcal{I}$ if and only if $(\varphi, 0) \in \mathcal{I}$, that is, if and only if $\varphi \in \mathcal{V}_{CI}$ by Definition 7. Therefore $\mathcal{I} = \mathcal{V}_{CI} \times \mathbb{R}^p$, which is (28). Steps 1 and 2 use only Definition 7; no assumption has been invoked.

Step 3: the addendum under A1. Assumption A1 states $(\varphi_\dagger, \theta_\dagger) \in \mathcal{I}$, so $\varphi_\dagger \in \mathcal{V}_{CI}$ and, by Step 1, $(\varphi_\dagger, \theta) \in \mathcal{I}$ for every $\theta \in \mathbb{R}^p$. This is the only place A1 is used, and it is why the lemma is stated in two parts. □

Proof of Lemma 39. The forward implication is immediate; the converse is proved in five

steps.

Step 0: the forward implication. Suppose $E[\varepsilon \mid V] = 0$ a.s. By B1(i)–(ii), $|e^{t'\Phi(V)}| \le \bar{w}(t) < \infty$ for every t, so $E|\varepsilon\, e^{t'\Phi(V)}| \le \bar{w}(t)E|\varepsilon| < \infty$ and the tower property gives $E[\varepsilon\, e^{t'\Phi(V)}] = E\Big[E[\varepsilon \mid V]\, e^{t'\Phi(V)}\Big] = 0$. Boundedness of Φ is doing the work here: without it the product need not be integrable and the step fails absent a moment condition on V.

Step 1: a compactly supported finite signed measure. Define $\Gamma(B) := E\Big[\varepsilon\, \mathbb{1}\{\Phi(V) \in B\}\Big]$ for Borel $B \subseteq \mathbb{R}^p$. Then Γ is a finite signed measure, with total variation $|\Gamma|(\mathbb{R}^p) \le E|\varepsilon| < \infty$, and by B1(i) it is supported in the compact cube $[-c_\Phi, c_\Phi]^p$. By the change-of-variables formula for image measures,

$$\mathsf{L}_\Gamma(t) \;:=\; E\Big[\varepsilon\, e^{t'\Phi(V)}\Big] \;=\; \int_{[-c_\Phi,c_\Phi]^p} e^{t'v}\, \Gamma(dv), \qquad t \in \mathbb{R}^p, \tag{148}$$

the integral being absolutely convergent for every t because $|e^{t'v}| \le e^{c_\Phi \|t\|_1}$ on the cube. *The transform is written* L_Γ *and not* ψ: the letter ψ is reserved for the moment function of the estimator.

Step 2: L_Γ *is real-analytic on* $\mathbb{R}^p$. Boundedness of Φ makes Γ compactly supported, so expanding $e^{t'v}$ under (148) gives a multi-index series whose absolute values sum to at most $|\Gamma|(\mathbb{R}^p)\, e^{c_\Phi\|t\|_1} < \infty$ at every t; term-by-term integration is therefore licensed and L_Γ is an everywhere absolutely convergent power series in t, hence real-analytic on $\mathbb{R}^p$, with coefficient $\int v^{\mathbf{j}}\, \Gamma(dv)/\mathbf{j}!$ at order $\mathbf{j}$, $\mathbf{j}$ running over the tuples of p nonnegative integers, $\mathbf{j}!$ the product of their factorials, and $\partial^{\mathbf{j}}\mathsf{L}_\Gamma(0) = \int v^{\mathbf{j}}\, \Gamma(dv)$. The multi-index is not α, which is **binding** in this project as the propensity rate exponent $r/(2r + d_Z)$. This step is Bierens (1990, proof of Thm. 1) verbatim at the bounded transform $\Phi(V)$, and it is retained as a **named** step, rather than folded into the citation, only because Proposition 15 Step 1 cites it by name at a different ε.

Step 3: L_Γ *vanishes identically.* By hypothesis $\mathsf{L}_\Gamma = 0$ on $\mathcal{T}'$, and $\mathrm{Leb}(\mathcal{T}') > 0$, also by

hypothesis. *This is the only step of the proof that reads the index set at all*, which is why the lemma is stated at a generic Borel $\mathcal{T}'$ and not at B1(iii)'s $\mathcal{T}$; Steps 0–2, 4 and 5 never mention it. A real-analytic function on $\mathbb{R}^p$, which is open and connected, whose zero set has strictly positive Lebesgue measure vanishes identically; this is the identity theorem for real-analytic functions, and it is the step made in Bierens (1990, proof of Thm. 1). Hence $\mathsf{L}_\Gamma \equiv 0$ on $\mathbb{R}^p$. The general statement is on an arbitrary connected open set; it is written at $\mathbb{R}^p$ here, which is the only set the step is applied to.

Step 4: $\Gamma = 0$. By Steps 2 and 3 every moment of Γ vanishes, and a finite signed Borel measure carried by a compact cube is determined by its moments, Stone–Weierstrass, then the Riesz representation theorem on a compact metric space, so $\Gamma = 0$. Taken with Step 3 this is Bierens (1990, Thm. 1) at the bounded random vector $\Phi(V)$, whose conclusion is $E[\varepsilon \mid \Phi(V)] = 0$ almost surely. This is the second place where boundedness of Φ is load-bearing, it is what makes Γ compactly supported and hence moment-determinate, and it is why the theorem cited is stated for a **bounded** vector; for an unbounded transform the moment problem need not be determinate and the step fails.

Step 5: back to V, and this is the half the citation does not carry. Bierens (1990, Thm. 1) is stated at the bounded vector itself and at *every* $t \in \mathbb{R}^p$; what is needed here is the conclusion at $\sigma(V)$ rather than at $\sigma(\Phi(V))$, from a hypothesis quantified over a set of positive Lebesgue measure only. Step 3 supplies the second, and this step supplies the first. $\Gamma = 0$ means $E[\varepsilon\, \mathbb{1}\{\Phi(V) \in B\}] = 0$ for every Borel B, that is, $E[\varepsilon \mid \Phi(V)] = 0$ a.s. By B1(i), Φ is an injective Borel map between Polish spaces, hence bi-measurable by the Lusin–Souslin theorem, so $\Phi(A)$ is Borel for every Borel A and $\sigma(\Phi(V)) = \sigma(V)$. (For the maintained choice Φ is a homeomorphism of $\mathbb{R}^p$ onto an open cube, so bi-measurability is elementary and the general theorem is not needed.) Therefore $E[\varepsilon \mid V] = E[\varepsilon \mid \Phi(V)] = 0$ a.s. □

Proof of Lemma 40. The proof has three steps: existence of a regular version; the cell-contrast identity; and the extension from a countable set of u to all of $\mathbb{R}$.

Step 1: a regular version, fixed for the ***pair*** $(\Delta U, W)$*, which is what lets the two quantifiers meet, and is the only change of substance in this proof.* $(\Delta U(\theta,\varphi), W, \Delta Z)$ takes values in a Polish space, so a regular version $\big(p_0(\cdot \mid z), \dots, p_{\mathsf{L}-1}(\cdot \mid z)\big)$ of $B \mapsto P\big(\Delta U(\theta,\varphi) \in B,\ W = j \mid \Delta Z = z\big)$ exists; fix one, and fix it as the disintegration of the joint law of the *pair* $(\Delta U(\theta,\varphi), W)$ given ΔZ, so that $\sum_{j=0}^{\mathsf{L}-1} p_j(\mathbb{R} \mid z) = 1$ for *every* z and not merely for almost every z. It supplies the statement's three objects at once. Put $\varrho_j(z) := p_j(\mathbb{R} \mid z)$*, so that* $\varrho_0 = 1 - \mathbf{1}'\varrho$ *at every* z*, which is the statement's own convention, by the total mass just fixed*, and let $F_j^{(\theta,\varphi)}(u \mid z) := p_j\big((-\infty, u] \mid z\big)/\varrho_j(z)$ when $\varrho_j(z) > 0$ and an arbitrary fixed distribution function otherwise, $j \in \{0, \dots, \mathsf{L}-1\}$, so that

$$p_j\big((-\infty, u] \mid z\big) \;=\; \varrho_j(z)\, F_j^{(\theta,\varphi)}(u \mid z) \qquad \text{at } \textit{every } (u, z) \text{ and for either branch;}$$

and read the left-hand side of (65) at the function $(u, z) \mapsto p_l\big((-\infty, u] \mid z\big) - \varrho_l(z) \sum_{j=0}^{\mathsf{L}-1} p_j\big((-\infty, u] \mid z\big)$, which is $\widetilde{W}_l \mathbb{1}\{\Delta U(\theta,\varphi) \le u\}$ integrated against that version, the second sum being $P(\Delta U(\theta,\varphi) \le u \mid \Delta Z = z)$ read at it. By construction $u \mapsto F_j^{(\theta,\varphi)}(u \mid z)$ is a distribution function, in particular non-decreasing and right-continuous, for *every* z, not merely for almost every z. *Three properties of that reading of the left-hand side are used below, and all three are immediate from its being an integral against a kernel*: it is jointly measurable in (u, z), each $p_j((-\infty, u] \mid z)$ being measurable in z at fixed u and right-continuous in u at fixed z, which is the **Carathéodory** criterion; it is right-continuous in u at *every* z, being a finite combination of the $p_j((-\infty, u] \mid z)$ with z-only coefficients; and, at each *fixed* u, it is a version of $E[\widetilde{W}_l \mathbb{1}\{\Delta U(\theta,\varphi) \le u\} \mid \Delta Z = \cdot]$, that being what a regular version computes. *It is the third of these that had no designated version through dispatch 25*, and without one the left-hand side of (65) is pinned down only up to a null set free to move with u, which at uncountably many u pins down nothing. At a z at which cell j is null the corresponding $F_j^{(\theta,\varphi)}(\cdot \mid z)$ is an arbitrary version; *this is harmless and no overlap condition is needed*, because F_j enters the right-hand side of (65) only through $\varrho_j(z) F_j(u \mid z)$

inside $\bar{F}$, and through the outer factor $\varrho_l(z)$ when $j = l$, so every occurrence of an arbitrary version is multiplied by a zero. *The set on which* (65) *carries no restriction at all is smaller, and is a different set at* $\mathsf{L} \geq 3$: it is the *degenerate region*

$$\mathcal{Z}_0 \;:=\; \Big\{z:\; \Omega(z) = 0\Big\} \;=\; \Big\{z:\; \max_{0\leq l\leq \mathsf{L}-1} \varrho_l(z) = 1\Big\}, \tag{149}$$

where W is conditionally degenerate, $\widetilde{W} = 0$ almost surely and both sides of (65) vanish whatever the conditional laws are. *A null cell is not by itself degeneracy once* $\mathsf{L} \geq 3$, $\varrho(z) = (q, 0)'$ at $\mathsf{L} = 3$ with $q \in (0, 1)$ has a null cell, has $\det \Omega(z) = 0$, and is not in $\mathcal{Z}_0$, *which is why Step 1 is stated at the null-cell set and* (149) *at the degenerate one*; the two conditions coincide at $\mathsf{L} = 2$, where $\Omega = \varrho(1 - \varrho)$ and $\mathcal{Z}_0 = \{z : \varrho(z)(1 - \varrho(z)) = 0\}$ is the region earlier rounds printed. *The region is written* $\mathcal{Z}_0$ *and not* D, italic D being Part F's perturbation direction in L; $\mathcal{Z}_0$ is calligraphic, matching the convention for sets, and lives on the ΔZ side, which the letter records.

Step 2: the cell-contrast identity, and it has two halves, an identity at ***every*** (u, z), *and an a.s. statement at each* fixed u. *Both are needed and they are proved separately. (2a) At every* (u, z). Fix $l \in \{1, \ldots, \mathsf{L} - 1\}$, fix z, and expand Step 1's reading of the left-hand side by the display fixed there, $p_j((-\infty, u] \mid z) = \varrho_j(z) F_j^{(\theta,\varphi)}(u \mid z)$. Suppressing arguments,

$$p_l\Big((-\infty, u] \mid z\Big) - \varrho_l \sum_{j=0}^{\mathsf{L}-1} p_j\Big((-\infty, u] \mid z\Big) = \varrho_l F_l - \varrho_l \sum_{j=0}^{\mathsf{L}-1} \varrho_j F_j = \varrho_l\Big\{F_l - \bar{F}\Big\},$$

which is (65). *No null set is generated at any point of this display*: it is finitely many algebraic operations on one kernel at one fixed z, so it holds for every $u \in \mathbb{R}$, every l and **every** z, which is the quantifier the statement now prints, and the one the "Consequently" clause needs. *(2b) At each fixed* u, *the reading is a version of the conditional expectation. This is where* (P4b) *is consumed*, and it is an almost-sure statement, as every statement about a conditional expectation must be. Apply (P4b) at that u with $g_j = E[\mathbb{1}\{\Delta U(\theta, \varphi) \leq u\} \mid$

$W = j, \Delta Z] = F_j^{(\theta,\varphi)}(u \mid \Delta Z)$, $j \in \{0, \dots, \mathsf{L}-1\}$, which is legitimate because the conditioning σ-field is exactly $\sigma(\Delta Z)$ and each g_j is a $\sigma(\Delta Z)$-measurable version by Step 1; it gives $E[\widetilde{W}_l \mathbb{1}\{\cdot \le u\} \mid \Delta Z] = \varrho_l\{F_l - \bar{F}\}$ almost surely, which by (2a) is Step 1's reading evaluated at ΔZ. *The two halves are not interchangeable, and running (2a) through (2b) is what failed before*: an a.s. statement made at each u separately carries a null set that may move with u, and there are uncountably many u. *At* $\mathsf{L} = 2$ *this is* $\varrho(1-\varrho)(F_1 - F_0)$*, the display earlier rounds printed*, by the reduction recorded at Lemma 106.

Step 3: from a countable set to $\mathbb{R}$. Let $\mathbb{Q}_0 \subset \mathbb{R}$ be countable and dense and suppose the left-hand side of (65), *read at Step 1's version throughout*, vanishes for every $u \in \mathbb{Q}_0$, every l and a.e. z. For each $u \in \mathbb{Q}_0$ and each l let $N_{u,l}$ be the corresponding null set and put $N := \bigcup_l \bigcup_{u \in \mathbb{Q}_0} N_{u,l}$, a null set because the union is countable, $\mathsf{L} - 1$ being finite and $\mathbb{Q}_0$ countable. Fix $z \notin N$ and l. By (2a), which holds at *that* z and at every u, the right-hand side of (65) vanishes at that z for every $u \in \mathbb{Q}_0$. By Step 1 each $F_j^{(\theta,\varphi)}(\cdot \mid z)$ is right-continuous in u, hence so is $\bar{F}^{(\theta,\varphi)}(\cdot \mid z)$, a finite convex combination of them, and hence so is the whole right-hand side of (65). A right-continuous function of u vanishing on the dense set $\mathbb{Q}_0$ vanishes on all of $\mathbb{R}$: given $u \in \mathbb{R}$, take $u_k \in \mathbb{Q}_0$ with $u_k \downarrow u$ and pass to the limit. So the right-hand side vanishes at that z for every $u \in \mathbb{R}$, and by (2a) once more so does the left-hand side; that is, the left-hand side vanishes for every $u \in \mathbb{R}$, every l and every $z \notin N$, *a single null set, not depending on* u *or on* l*, which is what the statement claims and what (2a)'s "every* z*" is for. No case split on* $\mathcal{Z}_0$ *is made and none is needed*, and in particular no division by a prefactor: the argument bounds nothing and inverts nothing, it only propagates a zero along a right-continuous function. The binary proof divided by $\varrho(1-\varrho)$ to reach $F_1 = F_0$ and then invoked right-continuity of F_1 and F_0 separately, which is why it needed the region; at $\mathsf{L} \ge 3$ that division is not available, there are L conditional laws and $\mathsf{L} - 1$ equations, and it is not wanted. □

Proof of Lemma 11. The strategy is to prove $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (a)$ and then $(c) \Leftrightarrow (d)$.

Step 1: $(a) \Rightarrow (b)$. Let ς be bounded and measurable. Under (a), $\varsigma(\Delta U)$ and W are conditionally independent given ΔZ, so, coordinatewise in l, $E[\widetilde{W}\varsigma(\Delta U) \mid \Delta Z] = E[\widetilde{W} \mid \Delta Z]\, E[\varsigma(\Delta U) \mid \Delta Z]$, which is zero by (P3). *This is an identity in* $\mathbb{R}^{\mathsf{L}-1}$, each $\widetilde{W}_l$ being a bounded function of W alone given ΔZ.

Step 2: $(b) \Rightarrow (c)$. Fix $u \in \mathbb{R}$ and $l \in \{1, \dots, \mathsf{L}-1\}$ and take $\varsigma = \mathbb{1}\{\cdot \le u\}$ in the l-th scalar component of (b), giving $E[\widetilde{W}_l \mathbb{1}\{\Delta U \le u\} \mid \Delta Z] = 0$ a.s. Fix $t \in \mathcal{T}$. The product $\widetilde{W}_l \mathbb{1}\{\Delta U \le u\} w(t'\Phi(\Delta Z))$ is integrable by (P1)–(P2), and $w(t'\Phi(\Delta Z))$ is $\sigma(\Delta Z)$-measurable, so the tower property gives $\mathcal{M}(u,t,l) = E\Big[w(t'\Phi(\Delta Z))\, E[\widetilde{W}_l \mathbb{1}\{\Delta U \le u\} \mid \Delta Z]\Big] = 0$.

Step 3: $(c) \Rightarrow (a)$. Let $\big(p_0(\cdot), \dots, p_{\mathsf{L}-1}(\cdot)\big)$ be a regular version of $B \mapsto P(\Delta U \in B,\, W = j \mid \Delta Z)$, which exists because ΔU takes values in the Polish space $\mathbb{R}$ and W in a finite set; then $p_j(\mathbb{R}) = \varrho_j(\Delta Z)$ and $\sum_{j=0}^{\mathsf{L}-1} p_j(B) = P(\Delta U \in B \mid \Delta Z)$ almost surely, for every Borel B.

Fix $l \in \{1, \dots, \mathsf{L}-1\}$ and, for each $u \in \mathbb{R}$, set $\varepsilon_u := \widetilde{W}_l\, \mathbb{1}\{\Delta U \le u\}$, which satisfies $|\varepsilon_u| \le 1$ by (P1) and hence $E|\varepsilon_u| < \infty$. By (c) and B1(ii), $E[\varepsilon_u\, e^{t'\Phi(\Delta Z)}] = 0$ for every $t \in \mathcal{T}$, so Lemma 39, whose hypotheses are B1(i), B1(ii) and a Borel index set of positive Lebesgue measure, applied at $\mathcal{T}' = \mathcal{T}$ — where the measure is positive by B1(iii), which is on this lemma's own line — and at $V = \Delta Z$ gives $E[\varepsilon_u \mid \Delta Z] = 0$ almost surely, for every $u \in \mathbb{R}$. In the notation just fixed,

$$E\Big[\varepsilon_u \mid \Delta Z\Big] \;=\; p_l\Big((-\infty, u]\Big) \;-\; \varrho_l\, P\Big(\Delta U \le u \mid \Delta Z\Big).$$

Restrict to $u \in \mathbb{Q}$: countably many null sets, and $\mathsf{L}-1$ values of l, so their union N is null. Off N, at each ω, both $s \mapsto p_l((-\infty, s])$ and $s \mapsto \varrho_l P(\Delta U \le s \mid \Delta Z)$ are right-continuous and non-decreasing and they agree on the dense set $\mathbb{Q}$, so they agree at every $s \in \mathbb{R}$. The two set functions $B \mapsto p_l(B)$ and $B \mapsto \varrho_l P(\Delta U \in B \mid \Delta Z)$ are then finite measures agreeing on the π-system $\{(-\infty, s] : s \in \mathbb{R}\} \cup \{\mathbb{R}\}$, hence on $\mathfrak{B}(\mathbb{R})$ by Dynkin's π–λ theorem. So, off N,

$$p_l(B) \;=\; \varrho_l\, P\Big(\Delta U \in B \mid \Delta Z\Big) \qquad \text{for every Borel } B \text{ and every } l \in \{1, \dots, \mathsf{L}-1\}. \quad (150)$$

Summing (150) over $l \geq 1$ and subtracting from $\sum_{j=0}^{\mathsf{L}-1} p_j(B) = P(\Delta U \in B \mid \Delta Z)$, with $\varrho_0 = 1 - \sum_{l \geq 1} \varrho_l$, gives the same identity at $j = 0$. All L cells therefore factorise, which is $\Delta U \perp\!\!\!\perp W \mid \Delta Z$ in the factorisation form of Definition 7; that is (a).

No overlap condition is used, and no case split on the degenerate region (149) is made. The route through (150) does not divide by ϱ_l and does not invert a prefactor, so it needs no lower bound on one: where $\varrho_l = 0$ both sides of (150) vanish, $0 \leq p_l(B) \leq p_l(\mathbb{R}) = \varrho_l$; where $\varrho_l = 1$ both equal $P(\Delta U \in B \mid \Delta Z)$; and neither case is argued separately because neither has to be. *Having no bite on a region is not the same as failing there*, and the point now costs no paragraph.

Step 4: $(c) \Rightarrow (d)$. Immediate, since $\mathcal{M}(u,t,l) = 0$ pointwise implies $T(\varphi_\dagger, \theta_\dagger) = 0$ as an element of $L^2(\Pi)$.

Step 5: $(d) \Rightarrow (c)$. Suppose $T(\varphi_\dagger, \theta_\dagger) = 0$ in $L^2(\Pi)$, i.e. $\mathcal{M}(u,t,l) = 0$ for Π-almost every (u,t,l). By Definition 5, $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ is a product of *three* finite measures, its third factor carried by the finite set $\{1, \ldots, \mathsf{L}-1\}$ with masses $\varpi_{\mathsf{w}}(l)$ that B1(iv.d) makes strictly positive; so "Π-almost every (u,t,l)" already means "for every l, and $\Pi_u \otimes \Pi_t$-almost every (u,t)". Fubini's theorem on $\Pi_u \otimes \Pi_t$, applied at each of the $\mathsf{L}-1$ values of l, supplies Π_t-null sets $N_{\mathcal{T}}^{(l)}$ such that for $t \notin N_{\mathcal{T}}^{(l)}$ one has $\mathcal{M}(u,t,l) = 0$ for Π_u-almost every u; put $N_{\mathcal{T}} := \bigcup_l N_{\mathcal{T}}^{(l)}$, a finite union and hence Π_t-null, which serves every l at once. The contrast index costs nothing here precisely because it is carried by a finite set with positive masses, and it would cost a further argument if Π_{w} were allowed atoms of mass zero, which B1(iv.d) forbids.

Fix such a t and an l, and let $Z_{t,l} := \{u \in \mathbb{R} : \mathcal{M}(u,t,l) = 0\}$, so $\Pi_u(Z_{t,l}^c) = 0$. If $Z_{t,l}$ were not dense in $\mathbb{R}$ there would be a nonempty open O with $O \subseteq Z_{t,l}^c$, whence $\Pi_u(O) = 0$, contradicting the full topological support of Π_u assumed in B1(iv.a). So $Z_{t,l}$ is dense. The map $u \mapsto \mathcal{M}(u,t,l)$ is right-continuous: for $u_k \downarrow u$ one has $\mathbb{1}\{\Delta U \leq u_k\} \downarrow \mathbb{1}\{\Delta U \leq u\}$ pointwise, the integrand is dominated by $\bar{w}(t)$ by (P1)–(P2), and dominated convergence applies. Given any $u \in \mathbb{R}$, density of $Z_{t,l}$ supplies $u_k \in Z_{t,l}$ with $u_k \downarrow u$, and right-continuity

gives $\mathcal{M}(u,t,l)=0$. Hence $\mathcal{M}(\cdot,t,l)\equiv 0$ on $\mathbb{R}$ for every $t\notin N_{\mathcal{T}}$ and every l.

By B1(iv.b) there is a Borel $\mathcal{T}_0\subseteq\mathcal{T}$ with $\mathrm{Leb}|_{\mathcal{T}_0}$ and $\Pi_t|_{\mathcal{T}_0}$ not mutually singular. Suppose $\mathrm{Leb}(\mathcal{T}_0\setminus N_{\mathcal{T}})=0$. Then $\mathrm{Leb}|_{\mathcal{T}_0}$ is concentrated on $N_{\mathcal{T}}$ while $\Pi_t|_{\mathcal{T}_0}$ is concentrated on its complement, since $\Pi_t(N_{\mathcal{T}})=0$; that is, the two measures are mutually singular, a contradiction. Hence $\mathrm{Leb}(\mathcal{T}_0\setminus N_{\mathcal{T}})>0$. *This is where B1(iv.b)'s* ***non-singularity*** *half does the work that cannot be dispensed with*: for a purely atomic Π_t the set $N_{\mathcal{T}}$ could carry full Lebesgue measure and the argument would collapse. *It is not the clause's only reader, on either half, and no claim of exhaustiveness is made.* Step 4 of Lemma 48's proof takes the same nonzero absolutely continuous part to place its index points, and later steps of several proofs read only its consequence $\Pi_t(\mathcal{T})>0$. And the clause's **finiteness** half is used earlier in this same step, at the Fubini on $\Pi_u\otimes\Pi_t$ that produced the $N_{\mathcal{T}}^{(l)}$: that theorem needs the product σ-finite, and Π_u and Π_t are finite by B1(iv.a) and B1(iv.b), which is what Step 0 of Lemma 61's proof states in terms and attributes to those two clauses. The draft this sentence replaces read "this is the only place where B1(iv.b) is used". Now run Step 3 with $\mathcal{T}$ replaced by $\mathcal{T}_0\setminus N_{\mathcal{T}}$, a Borel set of positive Lebesgue measure — $\mathcal{T}_0$ is Borel by B1(iv.b) and $N_{\mathcal{T}}$ is a finite union of Borel Π_t-null sets, the rest of that step being $\mathcal{T}$-free — *which is legitimate because the step's only index-set dependence is its call on Lemma 39, and that lemma is* ***stated*** *at any Borel set of positive Lebesgue measure rather than merely proved at one.* That delivers (a), and (c) on all of $\mathcal{T}$ then follows from Steps 1 and 2. □

Proof of Corollary 12. Fix $\theta\in\mathbb{R}^p$ and $\varphi\in\mathcal{V}$. The proof of Lemma 11 used no property of the pair $(\varphi_\dagger,\theta_\dagger)$ beyond three facts: that $|\widetilde{W}_l|\le 1$ at every cell (P1), which is what makes $\varepsilon_u=\widetilde{W}_l\mathbb{1}\{\cdot\le u\}$ integrable and hence Lemma 39 available; that $E[\widetilde{W}\mid\Delta Z]=0$ (P3); and that the pair (residual, W) admits a regular version of $B\mapsto P(\text{residual}\in B, W=j\mid\Delta Z)$, which Step 3 uses to form (150). *All three hold at arbitrary trial values, and the third holds for* ***any*** *real-valued residual whatever*: $\mathbb{R}$ is Polish and W takes finitely many values by Definition 4, which is the whole hypothesis. *This is where the re-derivation against the lifted*

Step 3 differs from the one earlier rounds printed, which inherited Lemma 40 instead and had to appeal to that lemma's being stated at arbitrary trial values; the lifted route does not pass through Lemma 40 at all, so the inheritance is now the weaker and more transparent one. Substituting $\Delta U(\theta, \varphi)$ for ΔU throughout that proof therefore gives

$$T(\varphi, \theta) = 0 \text{ in } L^2(\Pi) \quad \Longleftrightarrow \quad \Delta U(\theta, \varphi) \perp\!\!\!\perp W \mid \Delta Z \quad \Longleftrightarrow \quad (\varphi, \theta) \in \mathcal{I}.$$

By (28) of Lemma 10, which holds with no assumption, $(\varphi, \theta) \in \mathcal{I}$ if and only if $\varphi \in \mathcal{V}_{CI}$, whatever the value of θ. Hence the solution set is $\mathcal{V}_{CI}$ for every θ, and under A1b(iii) it is $\{\varphi_\dagger\}$. □

Proof of Lemma 41. The proof has three steps, one per claim.

Step 1: claim (i). $\mathcal{G}$ is a closed subspace of the Hilbert space $L^2(P_{\Delta U, \Delta Z})$, so it suffices to show $\mathcal{G}^\perp = \{0\}$. Let $\xi \in \mathcal{G}^\perp$, so that

$$E\left[\xi\, \varsigma(\Delta U)\, e^{t'\Phi(\Delta Z)}\right] \;=\; 0 \qquad \text{for every } \varsigma \in C_b(\mathbb{R}) \text{ and } t \in \mathcal{T}. \tag{151}$$

Fix $\varsigma \in C_b(\mathbb{R})$ and set $\varepsilon := \xi\, \varsigma(\Delta U)$. Then $E|\varepsilon| \le \|\varsigma\|_\infty E|\xi| \le \|\varsigma\|_\infty \|\xi\|_{L^2} < \infty$ by the Cauchy–Schwarz inequality, so Lemma 39 applies with $V = \Delta Z$ and (151) gives

$$E\left[\xi\, \varsigma(\Delta U) \,\middle|\, \Delta Z\right] \;=\; 0 \quad \text{a.s., for every } \varsigma \in C_b(\mathbb{R}). \tag{152}$$

Let $\mathcal{H}_b$ be the set of bounded measurable $\Lambda : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}$ with $E[\xi\, \Lambda(\Delta U, \Delta Z)] = 0$. By (152), $\mathcal{H}_b$ contains every product $\Lambda(v, z) = \varsigma(v)\varkappa(z)$ with $\varsigma \in C_b(\mathbb{R})$ and $\varkappa$ bounded measurable, since multiplying (152) by $\varkappa(\Delta Z)$ and taking expectations is legitimate ($\xi\varsigma\varkappa$ is integrable). That collection is closed under pointwise multiplication and contains the constants (take $\varsigma \equiv 1$, $\varkappa \equiv 1$), and it generates $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathbb{R}^p)$ because C_b separates points of $\mathbb{R}$. $\mathcal{H}_b$ is a vector space closed under uniformly bounded monotone limits, by dominated convergence

with dominating function $|\xi| \sup_k \|\Lambda_k\|_\infty$. The functional monotone class theorem therefore gives that $\mathcal{H}_b$ contains all bounded $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathbb{R}^p)$-measurable functions.

Now take $\Lambda_k := \operatorname{sign}(\xi)\mathbb{1}\{|\xi| \le k\}$, which is bounded and $\sigma(\Delta U, \Delta Z)$-measurable because ξ is. Then $E[|\xi|\mathbb{1}\{|\xi| \le k\}] = 0$ for every k, and monotone convergence gives $E|\xi| = 0$, so $\xi = 0$ almost surely. Hence $\mathcal{G} = L^2(P_{\Delta U, \Delta Z})$.

Step 2: claim (ii). Let $\varsigma_1, \dots, \varsigma_k \in C_b(\mathbb{R})$, $t_1, \dots, t_k \in \mathcal{T}$ and $c_1, \dots, c_k \in \mathbb{R}$, and let γ be the finite signed measure placing mass c_j at t_j. Then $\int \bar{w}\, d|\gamma| = \sum_j |c_j| \bar{w}(t_j) < \infty$, so γ satisfies B1(v), and setting $\lambda(u, t_j) := \varsigma_j(u)$ gives $\int_{\mathcal{T}} \lambda(u,t) w(t'\Phi(z)) \gamma(dt) = \sum_j c_j \varsigma_j(u) w(t_j'\Phi(z))$. Hence $\mathcal{G}_\gamma$ contains every finite linear combination of generators of $\mathcal{G}$, so $\overline{\mathcal{G}_\gamma} \supseteq \mathcal{G} = L^2(P_{\Delta U, \Delta Z})$ by Step 1.

Step 3: claim (iii). The map $P_{N1} L := L - E[L \mid \Delta Z]$ is the orthogonal projection of $L^2(P_{\Delta U, \Delta Z})$ onto the closed subspace $\mathcal{N} := \{L : E[L \mid \Delta Z] = 0\}$, because conditional expectation given $\sigma(\Delta Z)$ is the orthogonal projection onto the closed subspace of $\sigma(\Delta Z)$-measurable elements. Let $L^* \in \mathcal{N}$. By Step 2 there are $L_k \in \mathcal{G}_\gamma$ with $L_k \to L^*$ in L^2; since P_{N1} has operator norm one, $P_{N1} L_k \to P_{N1} L^* = L^*$. As $P_{N1} L_k \in \{L - E[L \mid \Delta Z] : L \in \mathcal{G}_\gamma\}$, the closure of that set contains $\mathcal{N}$. The reverse inclusion holds because $P_{N1} L \in \mathcal{N}$ for every L and $\mathcal{N}$ is closed. □

Proof of Lemma 45. Each clause is short; they are taken in order.

(i). This is (146): $E[(\Delta h)^2] \le 4\bar{c}_\pi \|h\|^2_{L^2(\pi)}$ by $(a-b)^2 \le 2a^2 + 2b^2$ and C1(ii), and taking square roots gives the constant $2\bar{c}_\pi^{1/2}$. *Nothing in that computation uses* $\int h\, d\pi = 0$, which is why the clause is stated for every $h \in L^2(\mathcal{X}, \pi)$ and not only for $h \in \mathcal{V}$: the two displayed steps of (146) are the elementary inequality and $f_{X_1} + f_{X_2} \le 2\bar{c}_\pi \pi$, and the latter is C1(ii). *The same C1(ii) is what makes the statement well posed at a π-equivalence class*: $f_{X_j} \le \bar{c}_\pi \pi$ gives $P_{X_j} \ll \pi$, so two π-versions of h agree almost surely at X_1 and at X_2, and Δh is defined up to a P-null set.

(ii). If h is a.s. constant then $\Delta h = 0$ a.s. Conversely $\Delta h = 0$ a.s. says $h(X_1) = h(X_2)$ a.s., i.e. a function of X_1 equals a function of X_2 almost surely, so A1b(ii) makes it a.s. constant. If moreover $h \in \mathcal{V}$ then $\int h \, d\pi = 0$ *by Definition 5*, and a constant with zero π-mean is zero because $\int \pi = 1$. *The citation is Definition 5 and not A1b(i)*: h here is a generic element of $\mathcal{V}$, and membership of $\mathcal{V}$ carries $\int h \, d\pi = 0$ *by definition.* A1b(i) asserts the same equation for the *truth* $\varphi_\dagger$, which is content; the two statements are typographically identical and semantically disjoint.

(iii). This is convention (P5); its term-by-term integrability proviso is the hypothesis of the lemma. The second equality divides and multiplies by π, legitimate because π is bounded below away from zero on $\mathcal{X}$ by C1(ii).

(iv). Taking $h \equiv 1$ in (iii) gives $\Delta h = 0$ and hence $\int \nu_g \, dx = 0$. For σ_g, integrate (66) term by term: $\int E[g \mid X_j = x] f_{X_j}(x) \, dx = E[g]$ for $j = 1, 2$ by the tower property, and the two terms add rather than cancel, giving $2E[g]$. *This one sign is the whole difference between the first-order and the second-order analysis:* the difference structure that makes $\int \nu_g = 0$ automatic buys nothing at all about $\int \sigma_g$.

(v). Under A1, $\Delta U \perp\!\!\!\perp W \mid \Delta Z$. Work entrywise: for $j \in \{1, \ldots, p\}$ the j-th coordinate of $g = \Xi(\Delta U, \Delta Z)\widetilde{W}$ is $\sum_{l=1}^{\mathsf{L}-1} \Xi_{jl}(\Delta U, \Delta Z)\widetilde{W}_l$, and for each (j, l) the factor $\Xi_{jl}(\Delta U, \Delta Z)$ is a function of $(\Delta U, \Delta Z)$ while $\widetilde{W}_l$ is a bounded function of $(W, \Delta Z)$, so conditional independence and (P3) give

$$E\Big[\Xi_{jl}(\Delta U, \Delta Z)\, \widetilde{W}_l \mid \Delta Z\Big] = E\Big[\Xi_{jl}(\Delta U, \Delta Z) \mid \Delta Z\Big]\, E\Big[\widetilde{W}_l \mid \Delta Z\Big] = 0,$$

whence $E[\Xi(\Delta U, \Delta Z)\widetilde{W} \mid \Delta Z] = E[\Xi(\Delta U, \Delta Z) \mid \Delta Z]\, E[\widetilde{W} \mid \Delta Z] = 0$ as an identity in $\mathbb{R}^p$. *The order is $\Xi\widetilde{W}$ and not $\widetilde{W}\,\Xi$, and at $\mathsf{L} \geq 3$ the two are not interchangeable*: Ξ is $\mathbb{R}^{p \times (\mathsf{L}-1)}$-valued by Definition 53 and $\widetilde{W}$ is $\mathbb{R}^{\mathsf{L}-1}$-valued by (1), so $\Xi\widetilde{W} \in \mathbb{R}^p$, whereas $\widetilde{W}\,\Xi$ fails to conform at all when $p \geq 2$ and, at $p = 1$, conforms but returns the $(\mathsf{L}-1) \times (\mathsf{L}-1)$ outer product of $\widetilde{W}$ with the single row of Ξ. *In neither case is it g*, which is the point: the

reversed order is not a different spelling of the same object. *At* $\mathsf{L} = 2$, $\widetilde{W}$ *is a scalar,* Ξ *is* $\mathbb{R}^p$*-valued and the two orders agree*, which is why earlier rounds could print either. Taking expectations gives $E[g] = 0$, and (iv) then gives $\int \sigma_g \, dx = 0$, both coordinatewise in the p rows. □

Proof of Lemma 46. The proof establishes the envelope (67); Fubini's theorem then applies on the product $\Pi \otimes P$.

Joint measurability of $(u, t, \omega) \mapsto g_{u,t,l}(\omega)$ at each l is Step 0 of the proof of Proposition 9, and ℓ is Π-measurable, so the integrand is $\mathfrak{B}(\mathbb{R} \times \mathcal{T}) \otimes 2^{\{1,\dots,\mathsf{L}-1\}} \otimes \mathcal{F}$-measurable, the third factor of Π being carried by a finite set. By (P1), *per cell*, (P2) and $|f_{\Delta U|\cdots}(u)| \le \|\bar{f}_0\|_\infty$ from C2′(i), $|g_{u,t,l}| \le \bar{w}(t)\|\bar{f}_0\|_\infty$ uniformly in l, so by the Cauchy–Schwarz inequality in P and Lemma 45(i),

$$E\Big[\big|g_{u,t,l}\,\Delta h\big|\Big] \;\le\; \bar{w}(t)\,\|\bar{f}_0\|_\infty\, E\big|\Delta h\big| \;\le\; \bar{w}(t)\,\|\bar{f}_0\|_\infty \Big(E[(\Delta h)^2]\Big)^{1/2} \;\le\; 2\,\bar{c}_\pi^{1/2}\,\|\bar{f}_0\|_\infty\,\bar{w}(t)\,\|h\|_{L^2(\pi)}.$$

Multiplying by $|\ell(u,t,l)|$ and integrating against Π, that is, summing over l against the masses $\varpi_{\mathsf{w}}(l)$ and integrating over $\mathbb{R}\times\mathcal{T}$, which is Definition 5's pairing, the Cauchy–Schwarz inequality in Π gives

$$\sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \bar{w}(t)\,|\ell(u,t,l)|\,\Pi_u(du)\,\Pi_t(dt) \;\le\; \Big(\int \bar{w}(t)^2\,d\Pi\Big)^{1/2} \|\ell\|_{L^2(\Pi)} \;=\; c_\Pi\,\|\ell\|_{L^2(\Pi)},$$

the last equality being B1(iv.c) and the definition of c_Π, the l-sum contributing the factor $\sum_l \varpi_{\mathsf{w}}(l) = 1$ because Π_{w} is a probability measure by B1(iv.d). Combining the two displays gives (67), which is finite. Fubini's theorem for $\Pi \otimes P$, both σ-finite, Π because B1(iv) makes it a product of *three* finite measures, Π_u, Π_t and the finitely supported Π_{w}, and P because it is a *probability* measure by Definition 4, therefore permits the order of integration to be exchanged. *The appeal to A0 made here through round 6 is withdrawn:* σ-finiteness of P needs no sampling assumption, and listing A0 on this lemma would have charged it for a

hypothesis it does not use. □

Proof of Lemma 47. The proof takes the four clauses in order; the only substantive step is (ii), and (iii) follows from it by injectivity rather than by any range computation.

(i). Write $\underline{x} = \inf \mathcal{X} = \min \mathcal{X}$, the minimum existing and the two agreeing because A0's **compactness** clause makes $\mathcal{X}$ a compact subset of $\mathbb{R}$, and $\mathcal{D}^{-1}h'(x) := \int_{[\underline{x},x]\cap\mathcal{X}} h'(v)\,dv$. *The intersection is what frees this step of connectedness, and it is the form clause* (ii) *already uses*; on A0's interval $[\underline{x}, x] \cap \mathcal{X} = [\underline{x}, x]$ and the display is the one earlier rounds printed. By the Cauchy–Schwarz inequality and C1(ii), $|\mathcal{D}^{-1}h'(x)|^2 \le |\mathcal{X}| \int_{\mathcal{X}} |h'|^2 dv \le |\mathcal{X}|\,\|1/\pi\|_\infty \|h'\|^2_{L^2(\pi)}$ for every x, so, using $\int \pi = 1$, $\|\mathcal{D}^{-1}h'\|^2_{L^2(\pi)} \le |\mathcal{X}|\,\|1/\pi\|_\infty \|h'\|^2_{L^2(\pi)}$. Centring is the orthogonal projection onto $\mathcal{V}$ followed by nothing else, and a projection has norm one, so $\|\mathcal{D}_0^{-1}h'\| \le \|\mathcal{D}^{-1}h'\| + |\int \mathcal{D}^{-1}h' d\pi| \le 2\|\mathcal{D}^{-1}h'\|$ by the Cauchy–Schwarz inequality again, giving the stated constant. *For injectivity, and this is the one place the retired spelling was doing work*: suppose $\mathcal{D}_0^{-1}h' = 0$, so that $F := \mathcal{D}^{-1}h'$ equals a constant c at π-almost every point of $\mathcal{X}$, hence at Lebesgue-almost every point of it, π being bounded below away from zero on $\mathcal{X}$ by C1(ii). Since $h' \in L^2(\pi)$ and $|\mathcal{X}| < \infty$, C1(ii) puts h' in $L^1(\mathcal{X}, dx)$, so F is absolutely continuous on $[\underline{x}, \bar{x}]$ with $F' = h'$ a.e. on $\mathcal{X}$ and $F' = 0$ a.e. off it. At almost every point of the measurable set $E := \{F = c\}$ the derivative of F vanishes: almost every point of E is a density point of E, F is differentiable there, and the difference quotient taken along E is identically zero. Since $|\mathcal{X} \setminus E| = 0$, this gives $h' = F' = 0$ a.e. on $\mathcal{X}$. *On A0's interval this is the earlier argument* — F is then continuous and constant a.e. on an interval, hence constant, and Lebesgue differentiation gives the same conclusion — *and neither version reads connectedness anywhere else in this clause.* That $\mathcal{D}$ is unbounded on $L^2(\pi)$ is standard and is exhibited by $h_k(x) = k^{-1}\sin(kx)$, for which $\|h_k\|_{L^2(\pi)} \to 0$ while $\|h_k'\|_{L^2(\pi)} = \|\cos(k\cdot)\|_{L^2(\pi)} \not\to 0$.

(ii). If $\nu_g \equiv 0$ a.e. then $\mathcal{N}_g \equiv 0$. Conversely suppose $\mathcal{N}_g \equiv 0$. Then for every v, $\int_{\{x\ge v\}\cap\mathcal{X}} \nu_g\,dx = 0$, so for $v_1 < v_2$ the difference gives $\int_{[v_1,v_2)\cap\mathcal{X}} \nu_g\,dx = 0$; the collection of such intervals generates $\mathfrak{B}(\mathcal{X})$ and is closed under intersection, so the finite signed mea-

sure $\nu_g\,dx$, finite because ν_g is integrable by hypothesis, vanishes on a generating π-system containing $\mathcal{X}$, hence vanishes identically. Therefore $\nu_g = 0$ a.e.

(iii). What the clause needs is that $(\mathcal{D}_0^{-1})^*$ is **injective on** $\mathcal{V}$, and that is obtained here from clause (ii) and *not* from a range computation. *The route matters, because the obvious one is false off an interval and this lemma no longer assumes one*: $\mathcal{R}(\mathcal{D}_0^{-1})$ does **not** contain every centred C^1 function on a disconnected $\mathcal{X}$, by the witness in the change note below, so the density step cannot be justified that way.

The adjoint, computed rather than named. Let $\zeta \in \mathcal{V}$ and $h' \in L^2(\pi)$. Since ζ is π-centred, the centring inside $\mathcal{D}_0^{-1}$ pairs to zero and $\langle \mathcal{D}_0^{-1}h', \zeta\rangle_{L^2(\pi)} = \langle \mathcal{D}^{-1}h', \zeta\rangle_{L^2(\pi)}$. By the argument of (i), $h' \in L^1(\mathcal{X}, dx)$; and $\int_{\mathcal{X}} |\zeta|\pi\,dx \le \|\zeta\|_{L^2(\pi)} < \infty$ by the Cauchy–Schwarz inequality and $\int \pi = 1$, so $\zeta\pi \in L^1(\mathcal{X}, dx)$. The double integral of $|h'(v)|\,|\zeta(x)|\,\pi(x)$ over $\{(v,x) \in \mathcal{X}\times\mathcal{X} : v \le x\}$ is therefore at most $\|h'\|_{L^1(\mathcal{X},dx)}\|\zeta\|_{L^2(\pi)} < \infty$ and Fubini's theorem gives, at $\nu_g := \zeta\pi$,

$$\left\langle \mathcal{D}_0^{-1}h', \zeta \right\rangle_{L^2(\pi)} = \int_{\mathcal{X}} h'(v)\,\mathcal{N}_g(v)\,dv = \left\langle h',\ \mathcal{N}_g/\pi \right\rangle_{L^2(\pi)},$$

no numbered display is added here, the identity having no consumer outside this clause, with $\mathcal{N}_g$ the tail function of clause (ii) read at that ν_g; so $(\mathcal{D}_0^{-1})^*\zeta = \mathcal{N}_g/\pi$, which lies in $L^2(\pi)$ because $|\mathcal{N}_g| \le \|\zeta\|_{L^2(\pi)}$ and $\|1/\pi\|_\infty < \infty$ by C1(ii). *Clause* (ii) *is available at this* ν_g *although* $\zeta\pi$ *need not be of the form* (20): the clause is stated for an arbitrary integrable function occupying that slot and its proof reads integrability and nothing else.

Injectivity, in three steps, none of which reads connectedness. Suppose $(\mathcal{D}_0^{-1})^*\zeta = 0$ in $L^2(\pi)$. Since π is bounded below on $\mathcal{X}$ by C1(ii), $\mathcal{N}_g = 0$ Lebesgue-almost everywhere on $\mathcal{X}$. (a) $\mathcal{N}_g$ is continuous on $\mathbb{R}$: for $v_1 < v_2$, $|\mathcal{N}_g(v_1) - \mathcal{N}_g(v_2)| = |\int_{[v_1,v_2)\cap\mathcal{X}} \zeta\pi\,dx| \le \int_{[v_1,v_2)} |\zeta|\pi\,dx$, which tends to zero as $v_2 \downarrow v_1$ by absolute continuity of the integral of the fixed L^1 function $|\zeta|\pi$. (b) $\mathcal{N}_g$ vanishes at every $v > \bar{x}$, because $\{x \ge v\} \cap \mathcal{X} = \emptyset$ there, and at every $v \le \underline{x}$, because then $\mathcal{N}_g(v) = \int_{\mathcal{X}} \zeta\pi\,dx = \int \zeta\,d\pi = 0$ — *which is where membership of* $\mathcal{V}$ *is consumed, and it is consumed nowhere else in this clause.* (c) The set $A := \{v \in \mathbb{R} : \mathcal{N}_g(v) \ne 0\}$ is open

by (a) and, by (b), contained in the bounded interval $(\underline{x}, \bar{x}]$. Were it non-empty, let (v_-, v_+) be the connected component of A containing one of its points; v_- is finite and $v_- \notin A$, so $\mathcal{N}_g(v_-) = 0$. Since $\mathcal{N}_g = 0$ almost everywhere on $\mathcal{X}$ while $\mathcal{N}_g \neq 0$ throughout (v_-, v_+), one has $|(v_-, v_+) \cap \mathcal{X}| = 0$, whence for every $v \in (v_-, v_+)$, $\mathcal{N}_g(v) - \mathcal{N}_g(v_-) = -\int_{[v_-, v) \cap \mathcal{X}} \zeta\pi \, dx = 0$ and $\mathcal{N}_g(v) = 0$, contradicting $v \in A$. So $A = \emptyset$ and $\mathcal{N}_g \equiv 0$ on $\mathbb{R}$. Clause (ii) then gives $\zeta\pi = 0$ almost everywhere on $\mathcal{X}$, hence $\zeta = 0$ in $L^2(\pi)$, again because $\pi > 0$ by C1(ii). *So $(\mathcal{D}_0^{-1})^*$ is injective on $\mathcal{V}$; equivalently $\mathcal{R}(\mathcal{D}_0^{-1})$ is dense in $\mathcal{V}$*, the two being the identity $\ker\left((\mathcal{D}_0^{-1})^*\right) = \mathcal{R}(\mathcal{D}_0^{-1})^\perp$ read inside $\mathcal{V}$, which is legitimate because $\mathcal{R}(\mathcal{D}_0^{-1}) \subseteq \mathcal{V}$ by construction. *Density is therefore a **conclusion** here and not a step*, which is the whole of the change: the retired argument used density to get injectivity and justified density by a range computation that a disconnected $\mathcal{X}$ refutes. Now $(\mathcal{A}_{\text{cond}}\mathcal{D}_0^{-1})^* = (\mathcal{D}_0^{-1})^*\mathcal{A}_{\text{cond}}^*$, so $(\mathcal{D}_0^{-1})^*(\nu/\pi) \in \mathcal{R}\left((\mathcal{A}_{\text{cond}}\mathcal{D}_0^{-1})^*\right)$ means $(\mathcal{D}_0^{-1})^*(\nu/\pi) = (\mathcal{D}_0^{-1})^*\mathcal{A}_{\text{cond}}^*\ell$ for some ℓ, and injectivity of $(\mathcal{D}_0^{-1})^*$ gives $\nu/\pi = \mathcal{A}_{\text{cond}}^*\ell$, which is A2. The converse direction is immediate by applying $(\mathcal{D}_0^{-1})^*$ to A2.

(iv). The inequality is (i). For the failure of the converse, the sequence h_k of (i) has $\|h_k\|_{L^2(\pi)} \to 0$ with $\|h_k'\|_{L^2(\pi)}$ bounded away from zero, so no bound of the form $\|h'\| \leq c\|h\|$ holds. *The consequence for the programme is that the rate requirement $\delta_n = o_P(n^{-1/4})$ must be stated in the level norm, where it is the weaker of the two demands:* a derivative-norm rate implies it, by (iv), and it does not imply a derivative-norm rate. Nothing is lost by conducting the second-order analysis in levels, and something would be lost by conducting it in derivatives. □

Proof of Lemma 48. The proof has four steps: the kernel form, which is the adjoint of an integral operator; boundedness of the induction map; its two structural properties; and the closed form, which is a single application of Lemma 46 followed by (P7a), the tower property and (P5).

Step 1: part (i). By Proposition 9, $\mathcal{A}_{\text{cond}}$ is the integral operator on $\mathcal{V}$ with kernel k, and

$\iint |\mathsf{k}|^2 \pi \, dx \, d\Pi < \infty$. For $h \in \mathcal{V}$ and $\ell \in L^2(\Pi)$, Fubini, licensed by that square-integrability and the Cauchy–Schwarz inequality, gives, the $L^2(\Pi)$ pairing being read as Definition 5 fixes it,

$$\langle \mathcal{A}_{\mathrm{cond}} h, \ell \rangle_{L^2(\Pi)} = \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \Big(\int_{\mathcal{X}} \mathsf{k}(u,t,l;x) h(x) \pi(x) dx \Big) \ell(u,t,l) \, \Pi_u(du) \, \Pi_t(dt) = \Big\langle h, \, \mathcal{A}^*_{\mathrm{cond}} \ell \Big\rangle_{L^2(\pi)},$$

which identifies $\mathcal{A}^*_{\mathrm{cond}} \ell$ as the display of part (i). That it lies in $\mathcal{V}$ is Step 2 of the proof of Proposition 9: every kernel slice does, and a finite weighted sum of integrals of elements of a closed subspace against a finite measure stays in it. *The masses $\varpi_{\mathsf{w}}(l)$ enter here through the pairing and not through* k, which is why (27) does not carry them.

Step 2: $\mathcal{W}$ is well defined and bounded, giving the norm bound in part (ii). Fix $\ell \in L^2(\Pi)$ and $l \in \{1, \ldots, \mathsf{L}-1\}$. Joint measurability of $(u,z) \mapsto (\mathcal{W}\ell)_l(u,z)$ follows from joint measurability of $(u,t,z) \mapsto \ell(u,t,l) w(t'\Phi(z))$, a product of a Π-measurable function and a composition of the continuous $(t,v) \mapsto w(t'v)$ with the Borel Φ, together with the domination $|\ell(u,t,l) w(t'\Phi(z))| \le |\ell(u,t,l)| \bar{w}(t)$ of (P2) and the exhaustion over $\{\|t\|_1 \le k\}$ used in Step 1 of the proof of Proposition 36. By the Cauchy–Schwarz inequality in Π_t and (P2), applied inside (68),

$$\big|(\mathcal{W}\ell)_l(u,z)\big|^2 \;\le\; \varpi_{\mathsf{w}}(l)^2 \, \varpi_u(u)^2 \Big(\int_{\mathcal{T}} \bar{w}^2 \, d\Pi_t \Big) \int_{\mathcal{T}} \ell(u,t,l)^2 \, \Pi_t(dt) \qquad \text{for every } (u,z) \text{ and every } l. \tag{153}$$

Summing (153) over l, *which is where the codomain's Euclidean norm is formed*, and it is legitimate coordinate by coordinate because (68) acts diagonally in l, and using $\varpi_{\mathsf{w}}(l)^2 \le \Big(\max_{l'} \varpi_{\mathsf{w}}(l') \Big) \varpi_{\mathsf{w}}(l)$,

$$\big\|(\mathcal{W}\ell)(u,z)\big\|^2 \;\le\; \Big(\max_{l'} \varpi_{\mathsf{w}}(l') \Big) \varpi_u(u)^2 \Big(\int_{\mathcal{T}} \bar{w}^2 \, d\Pi_t \Big) \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathcal{T}} \ell(u,t,l)^2 \, \Pi_t(dt).$$

Taking expectations over $(\Delta U, \Delta Z)$, and using (P7b), $f_{\Delta U}$ exists and is bounded by $\|\bar{f}_0\|_\infty$,

together with $\Pi_u(du) = \varpi_u(u)du$,

$$E\Big[\Big\|(\mathcal{W}\ell)(\Delta U, \Delta Z)\Big\|^2\Big] \;\le\; \Big(\max_{l'} \varpi_{\mathsf{w}}(l')\Big)\|\bar{f}_0\|_\infty \|\varpi_u\|_\infty \Big(\int_{\mathcal{T}} \bar{w}^2 d\Pi_t\Big)\|\ell\|^2_{L^2(\Pi)},$$

where the last step bounds $f_{\Delta U}(u)\varpi_u(u) \le \|\bar{f}_0\|_\infty\|\varpi_u\|_\infty$ and recognises $\varpi_{\mathsf{w}}(l)\,\varpi_u(u)\,du\,\Pi_t(dt)$, summed over l, as Π. Since Π_{w} is a probability measure by B1(iv.d), $\max_{l'}\varpi_{\mathsf{w}}(l') \le 1$ and the factor drops: this is the stated norm bound, and linearity is immediate. The printed constant therefore survives at every L, and is conservative by that factor. *It is the $\varpi_{\mathsf{w}}(l)$ carried inside (68) that makes this work*: without them the same passage would cost $\big(\min_{l'}\varpi_{\mathsf{w}}(l')\big)^{-1/2} \ge 1$, the domain norm carrying the masses and the codomain not, and the printed constant would be too small. At $\mathsf{L} = 2$ there is one column, $\varpi_{\mathsf{w}}(1) = 1$, the codomain is scalar and both displays are the ones earlier rounds printed.

Step 3: the closed form, part (iii). Let $h \in \mathcal{V}$ and $\ell \in L^2(\Pi)$. By Step 1 of the proof of Proposition 9, $\mathcal{A}_{\text{cond}}h(u,t,l) = E[g_{u,t,l}\Delta h]$ with $g_{u,t,l} = \widetilde{W}_l\, w(t'\Phi(\Delta Z))\, f_{\Delta U|\mathcal{F}_{XZW}}(u)$. By Lemma 46,

$$\langle \mathcal{A}_{\text{cond}}h, \ell\rangle_{L^2(\Pi)} = E\Big[\Delta h \sum_{l=1}^{\mathsf{L}-1} \widetilde{W}_l\, \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} w\Big(t'\Phi(\Delta Z)\Big)\, f_{\Delta U|\mathcal{F}_{XZW}}(u)\, \ell(u,t,l)\, \Pi_u(du)\, \Pi_t(dt)\Big]. \tag{154}$$

The passage from Π to the iterated integral $\Pi_t(dt)$ inside $\Pi_u(du)$, at each fixed l, is itself a Fubini step, on the product $\Pi_u \otimes \Pi_t$ of two finite measures, *the third factor of Π is not integrated here but summed, the finite sum having already been written out*, licensed by $\int \Big|w(t'\Phi(\Delta Z)) f_{\Delta U|\mathcal{F}_{XZW}}(u)\ell(u,t,l)\Big|\, d\Pi \le \|\bar{f}_0\|_\infty c_\Pi \|\ell\|_{L^2(\Pi)} < \infty$ from (P2), C2′(i), B1(iv.c) and the Cauchy–Schwarz inequality in Π, together with $\varpi_{\mathsf{w}}(l) > 0$ from B1(iv.d), which makes each of the $\mathsf{L}-1$ non-negative summands of that integral separately finite. *Lemma 46 does not cover it*: that lemma performs the adjacent $\Pi \otimes P$ interchange, not this one.

Because ΔZ is $\mathcal{F}_{XZW}$-measurable, the inner integral at cell l is, by (P7a) applied at $\zeta(u,z) = \varpi_{\mathsf{w}}(l)\int_{\mathcal{T}} \ell(u,t,l)w(t'\Phi(z))\Pi_t(dt)$ and $V = \Delta Z$, equal for almost every

ω to $E[(\mathcal{W}\ell)_l(\Delta U, \Delta Z) \mid \mathcal{F}_{XZW}]$, with $(\mathcal{W}\ell)_l$ as in (68). *The hypothesis (P7a) consumes is* $E|\varpi_u(\Delta U)\zeta(\Delta U, \Delta Z)| = E|(\mathcal{W}\ell)_l(\Delta U, \Delta Z)| < \infty$, *which Step 2 supplies*: $\mathcal{W}\ell \in L^2(P_{\Delta U,\Delta Z}; \mathbb{R}^{\mathsf{L}-1})$, so each component $(\mathcal{W}\ell)_l$ lies in $L^2(P_{\Delta U,\Delta Z}) \subseteq L^1(P)$ by the Cauchy–Schwarz inequality, P being a probability measure. *This is why (P6)'s substitution lemma is stated for integrable rather than bounded integrands*: ζ here is not bounded, since ℓ is only in $L^2(\Pi)$. *This is also the step the strengthened B1*(iv.a) *buys, and it is* ***not*** *the only step that uses it:* without $\Pi_u(du) = \varpi_u(u)du$ the inner integral is an $\mathcal{F}_{XZW}$-measurable object with no representation as a function of $(\Delta U, \Delta Z)$. Step 2 reads the same identity when it takes expectations over $(\Delta U, \Delta Z)$ and recognises $\varpi_{\mathsf{w}}(l)\,\varpi_u(u)\,du\,\Pi_t(dt)$, summed over l, as Π; and Step 4 reads the clause twice more, at the strict positivity of ϖ_u that gives $\bigcup_k \mathcal{U}_k = \mathbb{R}$ and at the finiteness of Π_u that bounds $\mathrm{Leb}(\mathcal{U}_k)$, a distinction that step already prints. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces made one and was short by two steps of its own proof. Substituting into (154) and using that each $\widetilde{W}_l$ and Δh are $\mathcal{F}_{XZW}$-measurable, followed by the tower property and the finiteness of the l-sum,

$$\langle \mathcal{A}_{\mathrm{cond}} h, \ell\rangle_{L^2(\Pi)} = E\Big[\Delta h\; E\Big[\widetilde{W}'\,(\mathcal{W}\ell)(\Delta U, \Delta Z)\,\Big|\,\mathcal{F}_{XZW}\Big]\Big] = E\Big[\widetilde{W}'\,(\mathcal{W}\ell)(\Delta U, \Delta Z)\,\Delta h\Big],$$

$\widetilde{W}'(\mathcal{W}\ell) = \sum_l \widetilde{W}_l(\mathcal{W}\ell)_l$ being the Euclidean inner product (69) abbreviates. Convention (P5) applies to the last expression, its term-by-term integrability proviso holding because $E\big|\widetilde{W}'(\mathcal{W}\ell)(\Delta U, \Delta Z)h(X_j)\big| \le c_{\widetilde{W}}\|\mathcal{W}\ell\|_{L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})}(E[h(X_j)^2])^{1/2} < \infty$ by (P1b), *here, and not* (P1)*: the object contracted is the vector*, Step 2 and C1(ii). Hence $\langle \mathcal{A}_{\mathrm{cond}} h, \ell\rangle_{L^2(\Pi)} = \int \nu_{\widetilde{W}'(\mathcal{W}\ell)} h\,dx = \langle \nu_{\widetilde{W}'(\mathcal{W}\ell)}/\pi,\, h\rangle_{L^2(\pi)}$ for every $h \in \mathcal{V}$, which is (69). The final claim of (iii) reads $\mathcal{A}^*_{\mathrm{cond}}\ell = -\nu/\pi$ as $\nu_{\widetilde{W}'(\mathcal{W}\ell)} = -\nu = -\nu_{\Delta Z}$ almost everywhere, π being strictly positive by C1(ii), and $\nu_{\Delta Z} + \nu_{\widetilde{W}'(\mathcal{W}\ell)} = \nu_{\Delta Z + \widetilde{W}'(\mathcal{W}\ell)}$ by linearity of $g \mapsto \nu_g$. At $\mathsf{L} = 2$ the sum has one term, $c_{\widetilde{W}} = 1$, and every display of this step is the one earlier rounds printed.

Step 4: the two structural claims of part (ii). That $\ker \mathcal{W} \subseteq \ker \mathcal{A}^*_{\mathrm{cond}}$ is immediate from (69):

if $\mathcal{W}\ell = 0$ in $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$ then every component vanishes almost surely, so $\widetilde{W}'(\mathcal{W}\ell) = 0$ a.s., $\nu_{\widetilde{W}'(\mathcal{W}\ell)} = 0$ and $\mathcal{A}^*_{\mathrm{cond}}\ell = 0$.

For density of $\mathcal{R}(\mathcal{W})$ in the $\mathbb{R}^{\mathsf{L}-1}$-valued space, the reduction to one component is made first and is what the diagonal action of (68) buys. Slice l of ℓ generates component l of $\mathcal{W}\ell$ and no other, so, writing $\mathcal{W}_l$ for the scalar map $\ell(\cdot,\cdot,l) \mapsto (\mathcal{W}\ell)_l$, the range $\mathcal{R}(\mathcal{W})$ contains, for each l, the set $\mathcal{R}_l$ of $\mathbb{R}^{\mathsf{L}-1}$-valued functions whose l-th component lies in $\mathcal{R}(\mathcal{W}_l)$ and whose other components vanish identically, take an ℓ supported on slice l, and $\mathcal{R}(\mathcal{W}) \supseteq \sum_l \mathcal{R}_l$ by linearity. If each $\mathcal{R}(\mathcal{W}_l)$ is dense in the scalar space $L^2(P_{\Delta U,\Delta Z})$ then $\sum_l \mathcal{R}_l$ is dense in the product $L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$, approximate each of the $\mathsf{L}-1$ components separately and add, the sum being finite, so it suffices to prove density of $\mathcal{R}(\mathcal{W}_l)$ at one fixed l. *The argument that follows is that scalar statement, and it does not depend on l beyond the symbol*: the mass $\varpi_{\mathsf{w}}(l) > 0$ of B1(iv.d) is a nonzero scalar factor, absorbed into the choice of ℓ_ϵ below.

For that density the argument is in three parts: a reduced generating family; a choice of index points; and the approximation itself.

A reduced generating family. Put $\mathcal{U}_k := \{u \in \mathbb{R} : \varpi_u(u) \geq 1/k\}$ and $\mathcal{T}_1 := \mathcal{T} \cap \operatorname{supp}\Pi_t$, and let

$$\mathcal{G}_* := \Big\{(u,z) \mapsto \varsigma(u)\, w(t_0'\Phi(z)) : \varsigma \text{ bounded measurable with } \operatorname{supp}\varsigma \subseteq \mathcal{U}_k \text{ for some } k,\ t_0 \in \mathcal{T}_1\Big\}.$$

Claim: $\overline{\operatorname{span}}\,\mathcal{G}_* = L^2(P_{\Delta U,\Delta Z})$. Let $\xi \perp \mathcal{G}_*$. Since $\varpi_u > 0$ everywhere by B1(iv.a), $\bigcup_k \mathcal{U}_k = \mathbb{R}$, so for any bounded measurable ς the truncations $\varsigma \mathbb{1}_{\mathcal{U}_k}$ converge to ς pointwise and are dominated by $\|\varsigma\|_\infty$; dominated convergence, with envelope $|\xi|\,\|\varsigma\|_\infty \bar{w}(t_0)$ which is P-integrable by the Cauchy–Schwarz inequality, gives $E[\xi\,\varsigma(\Delta U)e^{t_0'\Phi(\Delta Z)}] = 0$ for every bounded measurable ς and every $t_0 \in \mathcal{T}_1$. Step 1 of the proof of Lemma 41 then applies verbatim with $\mathcal{T}$ replaced by $\mathcal{T}_1$; it reaches Lemma 39, which is *stated* at any Borel index set of positive Lebesgue measure, and is applied here at $\mathcal{T}' = \mathcal{T}_1$, Borel as the intersection of $\mathcal{T}$ with a closed set, with $\mathrm{Leb}(\mathcal{T}_1) > 0$ established in the next paragraph from B1(iv.b). *So this lemma consumes*

no clause of B1(iii), which is why B1(iii) is absent from its hypothesis line and why that absence is now a consequence of Lemma 39's statement rather than of a sentence in its proof. This detour is not decoration: ϖ_u is assumed strictly positive pointwise, not bounded below, so ϖ_u^{-1} need not be integrable against an arbitrary $\varsigma \in C_b$, and the family has to be reduced before ℓ_ϵ below can be placed in $L^2(\Pi)$.

Index points. By B1(iv.b) and the Lebesgue decomposition, Π_t has a nonzero absolutely continuous part $\varpi_t \, d\mathrm{Leb}$ on $\mathcal{T}_0$, the letter matching ϖ_u, both being densities of range-side reference measures, with the caveat that Π_t itself is *not* assumed absolutely continuous and ϖ_t is the density of its absolutely continuous part only. At every Lebesgue density point t_0 of $\{\varpi_t > 0\}$ one has $\Pi_t(B_\epsilon(t_0)) \geq \int_{B_\epsilon(t_0)\cap\{\varpi_t>0\}} \varpi_t > 0$ for every $\epsilon > 0$; the set of such t_0 has positive Lebesgue measure and is contained in $\mathcal{T}_1$, so $\mathrm{Leb}(\mathcal{T}_1) > 0$ as used above.

The approximation. Fix such a t_0, fix ς bounded with $\operatorname{supp}\varsigma \subseteq \mathcal{U}_k$, and set, at the contrast index l fixed above, $\ell_\epsilon(u,t,l') := \mathbb{1}\{l'=l\}\,\varpi_{\mathsf{w}}(l)^{-1}\varpi_u(u)^{-1}\varsigma(u)\,\Pi_t(B_\epsilon(t_0))^{-1}\mathbb{1}\{t \in B_\epsilon(t_0)\}$, *the inverse mass is available because B1(iv.d) makes every* $\varpi_{\mathsf{w}}(l)$ *strictly positive*, and it is exactly what (68)'s own $\varpi_{\mathsf{w}}(l)$ cancels. Then $\|\ell_\epsilon\|^2_{L^2(\Pi)} = \varpi_{\mathsf{w}}(l)^{-1}\Pi_t(B_\epsilon(t_0))^{-1}\int_{\mathcal{U}_k}\varsigma(u)^2\varpi_u(u)^{-1}du \leq k\,\varpi_{\mathsf{w}}(l)^{-1}\Pi_t(B_\epsilon(t_0))^{-1}\|\varsigma\|^2_\infty\,\mathrm{Leb}(\mathcal{U}_k) < \infty$, so $\ell_\epsilon \in L^2(\Pi)$. *Finiteness of* $\mathrm{Leb}(\mathcal{U}_k)$ *is not free and is supplied by* B1(iv.a)*'s finiteness of* Π_u*, not by its positivity clause*: $\mathcal{U}_k = \{\varpi_u \geq 1/k\}$ gives $\mathrm{Leb}(\mathcal{U}_k) \leq k\int_{\mathcal{U}_k}\varpi_u \leq k\,\Pi_u(\mathbb{R}) < \infty$. Moreover $(\mathcal{W}\ell_\epsilon)_{l'} \equiv 0$ for $l' \neq l$, by the diagonal action of (68), while

$$\big(\mathcal{W}\ell_\epsilon\big)_l(u,z) - \varsigma(u)w\big(t_0'\Phi(z)\big) = \varsigma(u)\,\Pi_t(B_\epsilon(t_0))^{-1}\int_{B_\epsilon(t_0)}\Big\{w(t'\Phi(z)) - w(t_0'\Phi(z))\Big\}\Pi_t(dt),$$

and, for $w = \exp$ and $\|\Phi(z)\|_\infty \leq c_\Phi$ by B1(i)–(ii), the mean value theorem applied to $s \mapsto e^s$ on the interval joining $t'\Phi(z)$ and $t_0'\Phi(z)$ gives $|w(t'\Phi(z)) - w(t_0'\Phi(z))| \leq c_\Phi\|t - t_0\|_1\, e^{c_\Phi(\|t_0\|_1+\|t-t_0\|_1)}$ uniformly in z. For $t \in B_\epsilon(t_0)$ one has $\|t-t_0\|_1 \leq \epsilon\sqrt{p}$, so the displayed difference is bounded in modulus by $\|\varsigma\|_\infty\, c_\Phi\,\epsilon\sqrt{p}\,e^{c_\Phi(\|t_0\|_1+\epsilon\sqrt{p})}$, which tends to zero with ϵ uniformly in (u,z) and hence in $L^2(P_{\Delta U,\Delta Z})$. So every element of $\mathcal{G}_*$ lies in $\overline{\mathcal{R}(\mathcal{W}_l)}$, and the

claim of the first paragraph gives $\overline{\mathcal{R}(\mathcal{W}_l)} = L^2(P_{\Delta U,\Delta Z})$; by the reduction opening Step 4, $\overline{\mathcal{R}(\mathcal{W})} = L^2(P_{\Delta U,\Delta Z}; \mathbb{R}^{\mathsf{L}-1})$. At $\mathsf{L} = 2$ the reduction is vacuous, $\varpi_{\mathsf{w}}(1) = 1$, and ℓ_ϵ and the conclusion are the ones earlier rounds printed. □

Proof of Proposition 13. The proof differences out the constant, identifies the increment with a linear functional, and then uses Definition 5, the decomposition $L^2(\pi) = \mathcal{V} \oplus \{\text{constants}\}$, together with $\int_{\mathcal{X}} \nu_g \, dx = 0$ from Lemma 45(iv) to upgrade "orthogonal to $\mathcal{V}$" to "zero". A1b(i) is not cited at any step and $\varphi_\dagger$ occurs nowhere in the proposition. *The first step is taken on increments rather than on the three separate terms of $\Delta U(\theta, \varphi)$, and that is deliberate:* ΔY and ΔZ are not functions of X_1 or X_2, so the proviso (29) does **not** make $E[g\,\Delta Y]$ or $E[g\,\Delta Z']$ separately finite.

Step 0: ν_g is well defined and $\nu_g/\pi \in L^2(\pi)$. $E|g| < \infty$ from (29) makes the conditional expectations in (20) well defined and gives $\int_{\mathcal{X}} |\nu_g| \, dx \le 2E|g| < \infty$, so ν_g/π lies in $L^1(\pi)$ by C1(ii). Next, for $h \in \mathcal{V}$, Jensen's inequality and (29) give $\int |\nu_g h| \, dx \le \sum_{j=1}^{2} E\big|g\, h(X_j)\big| < \infty$, i.e. $(\nu_g/\pi)h \in L^1(\pi)$ for every $h \in \mathcal{V}$; and since $\nu_g/\pi \in L^1(\pi)$ and any $h \in L^2(\pi)$ decomposes as $h_0 + \int h \, d\pi$ with $h_0 \in \mathcal{V}$, in fact $(\nu_g/\pi)h \in L^1(\pi)$ for every $h \in L^2(\pi)$. *No symbol is bound for ν_g/π here, deliberately*: it is a *domain*-side object, an element of $L^2(\pi)$ paired against h in the domain inner product, and Ξ is reserved for the *range* side by Definition 5 and Remark 104(1), whose whole apparatus is that split. Writing it out costs a few glyphs and keeps the two sides apart. *This forces $\nu_g/\pi \in L^2(\pi)$*, by uniform boundedness: the functionals $T_n(h) := \int (\nu_g/\pi)\mathbb{1}\{|\nu_g/\pi| \le n\} h \, d\pi$ are bounded on $L^2(\pi)$ with $\|T_n\| = \|(\nu_g/\pi)\mathbb{1}\{|\nu_g/\pi| \le n\}\|_{L^2(\pi)}$; for each fixed h, $T_n(h) \to \int (\nu_g/\pi) h \, d\pi$ by dominated convergence with dominating function $|(\nu_g/\pi)h| \in L^1(\pi)$, so $\sup_n |T_n(h)| < \infty$ pointwise; the Banach–Steinhaus theorem, Brezis (2011, Thm. 2.2), at the Banach space $L^2(\pi)$, gives $\sup_n \|T_n\| < \infty$; and monotone convergence then gives $\|\nu_g/\pi\|_{L^2(\pi)} = \lim_n \|T_n\| < \infty$. Without this step the phrase "ν_g/π lies in $\mathcal{V}^\perp$" in Step 2 would not be well formed, since $\mathcal{V}^\perp$ is a subset of $L^2(\pi)$. Note also that $h \equiv 1$ is *not* an element of $\mathcal{V}$, so the second clause of (29) does not deliver $E|g| < \infty$; that is

why the first clause is stated separately.

Step 1: the increment. Fix θ and let $\varphi, \varphi_0 \in \mathcal{V}$. *First, $g\,\Delta U(\theta,\varphi)$ is integrable for every $\varphi \in \mathcal{V}$, and clause 3 of (29) is stated only at $\varphi = 0$ because that is all it needs to be:* $\Delta U(\theta,\varphi) = \Delta U(\theta,0) - \Delta\varphi$ by (19), so $E|g\,\Delta U(\theta,\varphi)| \le E|g\,\Delta U(\theta,0)| + E|g\,\Delta\varphi|$, the first term finite by clause 3 and the second by clause 2. (An earlier version quantified clause 3 over all $\varphi \in \mathcal{V}$, which is redundant given clause 2.) By (19) again, $\Delta U(\theta,\varphi) - \Delta U(\theta,\varphi_0) = -\Delta(\varphi - \varphi_0)$, and both products with g are integrable by what was just shown, so their difference may be split and

$$E\big[g\,\Delta U(\theta,\varphi)\big] - E\big[g\,\Delta U(\theta,\varphi_0)\big] \;=\; -\,E\big[g\,\Delta(\varphi-\varphi_0)\big] \;=\; -\int_{\mathcal{X}} \nu_g\,(\varphi-\varphi_0)\,dx, \qquad (155)$$

the last equality by Lemma 45(iii), whose proviso is the second clause of (29). *No term of $\Delta U(\theta,\varphi)$ is evaluated on its own, and θ cancels*, which is the promised "one and hence every θ".

Step 2: constancy is orthogonality of ν_g/π to $\mathcal{V}$. $\mathcal{V}$ is a linear space, so as φ, φ_0 range over it the difference $\varphi - \varphi_0$ ranges over all of $\mathcal{V}$. Hence the map is constant on $\mathcal{V}$ if and only if the right-hand side of (155) vanishes for every $h \in \mathcal{V}$, that is, if and only if $\langle \nu_g/\pi, h\rangle_{L^2(\pi)} = \int \nu_g h\,dx = 0$ for every $h \in \mathcal{V}$, the two forms agreeing by C1(ii). This says $\nu_g/\pi \in \mathcal{V}^{\perp}$, which is meaningful because $\nu_g/\pi \in L^2(\pi)$ by Step 0.

Step 3: the orthogonal complement is the constants, and the constant is zero. By Definition 5, $L^2(\pi) = \mathcal{V} \oplus \{\text{constants}\}$, so $\nu_g/\pi = \mathsf{c}$ for some $\mathsf{c} \in \mathbb{R}$. By Lemma 45(iv) applied with $h \equiv 1$, legitimate because $E|g| < \infty$ by the first clause of (29), one has $\int_{\mathcal{X}} \nu_g\,dx = 0$; and $\int_{\mathcal{X}} (\nu_g/\pi)\,\pi\,dx = \int \nu_g\,dx$, so $\mathsf{c}\int \pi\,dx = \mathsf{c} = 0$. Hence $\nu_g \equiv 0$, because $\pi > 0$ on $\mathcal{X}$ by C1(ii); written out, this is (30). The converse is immediate from (155): $\nu_g \equiv 0$ makes every increment vanish. □

Proof of Corollary 14. The proof has three steps: the normalisation pins the constant of

integration; the moment then collapses to a residual-free weight; and Proposition 13 converts first-order insensitivity into exact constancy.

Step 1: N1 forces $L(u,z) = c(z)u$. Fix z and set $\Lambda_z(u) := L(u,z) - c(z)u$. By hypothesis $L(\cdot, z)$ is differentiable at every point with $\partial_u L(u,z) = c(z)$, so Λ_z is differentiable everywhere with $\Lambda_z' \equiv 0$; by the mean value theorem *applied entrywise* on $[u_1, u_2]$ for arbitrary $u_1 < u_2$, Λ_z takes values in $\mathbb{R}^{p\times(\mathsf{L}-1)}$, so for each of the $p(\mathsf{L}-1)$ entries (j,l) there is a point ξ_{jl} with $\Lambda_{z,jl}(u_2) - \Lambda_{z,jl}(u_1) = \Lambda_{z,jl}'(\xi_{jl})(u_2 - u_1) = 0$, *the point depending on the entry*, Λ_z is constant and $L(u,z) = c(z)u + b(z)$ with $b(z) := \Lambda_z(0)$. ***The word "entrywise" is load-bearing.*** The mean value theorem in its Lagrange form is a statement about *real*-valued functions and is **false** for vector- or matrix-valued ones as soon as there is more than one entry: there is in general no single ξ serving all of them, and at $\mathsf{L} \geq 3$ there are $p(\mathsf{L}-1)$ of them rather than p. Every site in this file that applies it to an $\mathbb{R}^p$- or $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued map wants either an entrywise conclusion, as here, or an *envelope*, for which the valid statement is the **mean value inequality**: if F is continuous on $[u_1, u_2]$ with values in a finite-dimensional Euclidean space, $\mathbb{R}^p$, or $\mathbb{R}^{p\times(\mathsf{L}-1)}$ under the Frobenius norm, the two being the same statement, and differentiable on the interior, then $\|F(u_2) - F(u_1)\| \leq (u_2 - u_1)\sup_\xi \|F'(\xi)\|$, no Lagrange point being produced. At $\mathsf{L} = 2$ there is one column, "entrywise" is the "coordinatewise" earlier rounds printed, and both displays are the printed ones. The scalar issue this paragraph was written for is separate and unchanged: the mean value theorem is the right theorem here and the fundamental theorem of calculus, cited through round 6, is the wrong one. Under the *everywhere-pointwise* reading of the hypothesis, which is the reading now printed in the statement, no absolute continuity is available or needed and the theorem applies directly. Under an *a.e.* reading , which is how derivatives of L-type nuisances are meant elsewhere in the block, in A5(iv) and in Lemma 41, the FTC would be the right citation and *the conclusion would be false without absolute continuity*: the Cantor function has a.e.-derivative zero and is not constant. The hypothesis is therefore stated pointwise. *This does not touch Step 4*, whose contrapositive makes the same move on $\ell_\star$, where Corollary 49 supplies continuous

differentiability in u and a vanishing derivative is an everywhere-vanishing continuous one. Evaluating at the truth and conditioning on ΔZ, $E[L(\Delta U, \Delta Z) \mid \Delta Z] = c(\Delta Z)E[\Delta U \mid \Delta Z] + b(\Delta Z) = b(\Delta Z)$, the first term vanishing by the second clause of A1. N1 sets the left-hand side to zero, so $b \equiv 0$.

Step 2: the weight is $\sigma(W, \Delta Z)$-measurable. Substituting $L(u, z) = c(z)u$ into $\psi = \Delta Z\, \Delta U(\theta, \varphi) + L\big(\Delta U(\theta, \varphi), \Delta Z\big)\, \widetilde{W}$ of (38) gives $\psi = \Delta Z\, \Delta U(\theta, \varphi) + c(\Delta Z)\, \Delta U(\theta, \varphi)\, \widetilde{W} = \{\Delta Z + c(\Delta Z)\, \widetilde{W}\}\, \Delta U(\theta, \varphi) = g\, \Delta U(\theta, \varphi)$, the scalar $\Delta U(\theta, \varphi)$ commuting with the matrix factor. *The order is forced and is not a convention this proof picked*: $c(\Delta Z)$ is $p \times (\mathsf{L} - 1)$ and $\widetilde{W}$ is $(\mathsf{L} - 1) \times 1$, so "$\widetilde{W}c$" does not multiply at $\mathsf{L} \geq 3$ and there is no g to factor out; at $\mathsf{L} = 2$ the two orders are the same scalar-times-vector product, which is why nothing before this dispatch could distinguish them. Since $\widetilde{W}_l = \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ by (1), the weight g is a function of $(W, \Delta Z)$ alone. That g satisfies (29) is the hypothesis of the corollary, and it is a genuine restriction on c rather than a formality: $\big\|c(\Delta Z)\widetilde{W}\big\| \leq c_{\widetilde{W}} \|c(\Delta Z)\|$ by (P1b), *the Euclidean bound on the contrast vector and* ***not*** *the per-cell* (P1), *the two coinciding at* $\mathsf{L} = 2$ *where* $c_{\widetilde{W}} = 1$, and A1b(iv)'s moment clauses cover the ΔZ term, but the clauses involving c must be assumed.

Step 3: the equivalences, with (c) read at the given c. By (155) in the proof of Proposition 13, applied to this g, the increment of $\varphi \mapsto E[\psi]$ between any $\varphi, \varphi_0 \in \mathcal{V}$ is $-\int \nu_g(\varphi - \varphi_0)dx$. So (a), which says $\nu_g \equiv 0$, is exactly the vanishing of every increment, and (a)$\Leftrightarrow$(b) is Proposition 13 applied to this g. For (a)$\Leftrightarrow$(c), linearity of $g \mapsto \nu_g$ gives $\nu_g = \nu + \nu_{c\widetilde{W}}$, so $\nu_g \equiv 0$ holds if and only if $\nu_{c\widetilde{W}} = -\nu$, *with c the function fixed in the hypothesis.* That equality is what clause (c) asserts. *Linearity is read coordinatewise in the p rows and is indifferent to the contraction over l*, both ν and $\nu_{c\widetilde{W}}$ being $\mathbb{R}^p$-valued.

A quantifier that must not be relaxed. Clause (c) is *not* the membership statement $-\nu \in \mathcal{N}_0$ read existentially. $\mathcal{N}_0$ is a set indexed by all admissible $\tilde{c}$, so $-\nu \in \mathcal{N}_0$ asserts only that *some* $\tilde{c}$ realises the equality, whereas (a) and (b) are statements about the *given* c and the L built

from it. The implication (a)⇒($-\nu \in \mathcal{N}_0$) is immediate and is all that the contrapositive below uses; the reverse implication is false, because a different $\tilde{c}$ may realise membership while the given c does not, in which case (a) and (b) fail for the given L and the concluding sentence about the second-order term does not apply to it. An earlier version of this corollary stated (c) existentially and asserted the three-way equivalence; that was an overstatement, and the downstream use, which travels only from (a) to membership, was never affected by it.

Step 4: the second-order term, and the contrapositive. By (155) applied to this g, $\varphi \mapsto E[\psi]$ is *affine* on $\mathcal{V}$, the weight g does not depend on φ, and ψ is linear in $\Delta U(\theta, \varphi)$, which is affine in φ. *Every term of order two or higher therefore vanishes unconditionally*, and not only under (a)–(c); an earlier version wrote "in that case", which attached an unconditional fact to a hypothesis. What (a)–(c) add is the vanishing of the first-order term, i.e. constancy rather than mere affineness. Contrapositively, suppose $-\nu \notin \mathcal{N}_0$ and let $\ell_\star$ satisfy $\nu_{\Delta Z + \ell_\star \widetilde{W}} \equiv 0$. If $\partial_u \ell_\star \equiv 0$ then $\ell_\star(u, z) = \tilde{c}(z)$ for some admissible $\tilde{c}$, and the hypothesis on $\ell_\star$ is statement (a) at that $\tilde{c}$, so $\nu_{\tilde{c}\widetilde{W}} = -\nu$ and therefore $-\nu \in \mathcal{N}_0$, a contradiction. Only the direction (a)⇒ membership is used here, which is why the corrected quantifier in (c) leaves this argument untouched. Hence $\partial_u \ell_\star \not\equiv 0$. □

Proof of Proposition 15. Part (i) is Steps 1–4: (a)⇒(b) by stripping the Bierens index and then the u-index, and (b)⇒(a) by reversing both. Part (ii) is Step 5, the cell-contrast identity (P4c). *Steps 1–4 use neither A1 nor A4(i), and the separation is the point of splitting the statement:* A1 is what makes $P(W = l \mid \Delta U, \Delta Z) = \varrho_l$ available and A4(i) is what makes the L cell-conditional expectations of clause (c) well defined and the per-cell prefactor ϱ_l invertible, and clause (c) is the only place either is needed.

Step 1: (a) implies the pointwise family. Steps 1–4 run at a fixed contrast index $l \in \{1, \ldots, \mathsf{L} - 1\}$ and are repeated at each of them. (24) is scalar-valued at each (u, t, l), so every display in Steps 1–4 is a scalar statement. Clause (a) is one statement about the scalar function $(u, t, l) \mapsto \mathcal{A}_{\text{cond}} h(u, t, l)$ on the product space of Definition 5, and

clause (b) is the $\mathbb{R}^{\mathsf{L}-1}$-valued identity (1) types it as, *that is, the* $\mathsf{L}-1$ *scalar restrictions* $E[\widetilde{W}_l \Delta h \mid \Delta U, \Delta Z] = 0$, *which is the reading the Notes on the proposition print.* The equivalence is therefore proved slicewise and assembled at the end of Steps 3 and 4. Suppose $\mathcal{A}_{\text{cond}} h = 0$ in $L^2(\Pi)$. Since $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ by Definition 5 and $\varpi_{\mathsf{w}}(l) > 0$ at every l by B1(iv.d), this gives $\mathcal{A}_{\text{cond}} h(\cdot, \cdot, l) = 0$ $\Pi_u \otimes \Pi_t$-almost everywhere, at each l separately; fix one. The map $(u, t) \mapsto \mathcal{A}_{\text{cond}} h(u, t, l) = E[g_{u,t,l} \Delta h]$ is continuous in u for each t, by dominated convergence with dominating function $\bar{w}(t) \bar{f}_0 |\Delta h|$, integrable by C2′(i), (P1) read *per cell* as $|\widetilde{W}_l| \leq 1$, (P2) and Lemma 45(i), together with continuity of $u \mapsto f_{\Delta U | \mathcal{F}_{XZW}}(u)$ from C2′(i); and it is real-analytic in t on $\mathbb{R}^p$ by Step 2 of the proof of Lemma 39, applied with $\varepsilon = \widetilde{W}_l f_{\Delta U | \mathcal{F}_{XZW}}(u) \Delta h$, which is integrable by the same bounds. Step 5 of the proof of Lemma 11 therefore applies verbatim, it uses only B1(iv.a) for the u-factor and B1(iv.b) for the t-factor, and upgrades the $\Pi_u \otimes \Pi_t$-a.e. statement to $\mathcal{A}_{\text{cond}} h(u, t, l) = 0$ for *every* $(u, t) \in \mathbb{R} \times \mathcal{T}$. At $\mathsf{L} = 2$ the index takes its single value, Π_{w} is the point mass at 1, and the reduction to the slice is vacuous.

Step 2: stripping the Bierens index. Fix u and set $\varepsilon_u := \widetilde{W}_l \, f_{\Delta U | \mathcal{F}_{XZW}}(u) \, \Delta h$, which satisfies $E|\varepsilon_u| \leq \|\bar{f}_0\|_\infty E|\Delta h| < \infty$. Then $\mathcal{A}_{\text{cond}} h(u, t, l) = E[\varepsilon_u e^{t' \Phi(\Delta Z)}]$, and Step 1 gives that this vanishes for every $t \in \mathcal{T}$. By Lemma 39 with $V = \Delta Z$ at $\mathcal{T}' = \mathcal{T}$, under B1(i) and (ii) with $\text{Leb}(\mathcal{T}) > 0$ from B1(iii), $E[\varepsilon_u \mid \Delta Z] = 0$ almost surely, for each fixed u.

Step 3: stripping the u-index. Let $\varsigma \in C_b(\mathbb{R})$. Multiply the conclusion of Step 2 by $\varsigma(u)$ and integrate over $u \in \mathbb{R}$. *The interchange is licensed by Fubini with the envelope*

$$\begin{aligned} \left| \varsigma(u) \, \widetilde{W}_l \, f_{\Delta U | \mathcal{F}_{XZW}}(u) \, \Delta h \right| &\leq \|\varsigma\|_\infty \, f_{\Delta U | \mathcal{F}_{XZW}}(u) \left| \Delta h \right|, \\ \int\!\!\int_{\mathbb{R}} \|\varsigma\|_\infty f_{\Delta U | \mathcal{F}_{XZW}}(u) |\Delta h| \, dP \, du &= \|\varsigma\|_\infty E \left| \Delta h \right| < \infty, \end{aligned}$$

using (P1) at cell l, $\int_{\mathbb{R}} f_{\Delta U | \mathcal{F}_{XZW}}(u) \, du = 1$ and Lemma 45(i). The constant envelope $\|\varsigma\|_\infty \bar{f}_0 |\Delta h|$ used in an earlier version does not work: $\bar{f}_0$ is a supremum over u, so that bound is constant in u and not integrable over $\mathbb{R}$. The conclusion is unaffected; the justification

was wrong. This gives $E\big[\widetilde{W}_l\,\Delta h\int \varsigma(u) f_{\Delta U|\mathcal{F}_{XZW}}(u)du\,\big|\,\Delta Z\big]=0$. By (P6a)'s substitution lemma the inner integral is $E[\varsigma(\Delta U)\mid \mathcal{F}_{XZW}]$, and since $\widetilde{W}_l\Delta h$ is $\mathcal{F}_{XZW}$-measurable and $\sigma(\Delta Z)\subseteq\mathcal{F}_{XZW}$, the tower property gives

$$E\big[\widetilde{W}_l\,\Delta h\,\varsigma(\Delta U)\,\big|\,\Delta Z\big]\;=\;0\qquad\text{almost surely, for every }\varsigma\in C_b(\mathbb{R}).\tag{156}$$

Multiplying (156) by a bounded measurable $\varkappa(\Delta Z)$ and taking expectations shows that $E[\widetilde{W}_l\Delta h\,\Lambda(\Delta U,\Delta Z)]=0$ for every Λ of product form $\varsigma\otimes\varkappa$ with $\varsigma\in C_b$; that collection is closed under multiplication, contains the constants, and generates $\mathfrak{B}(\mathbb{R})\otimes\mathfrak{B}(\mathbb{R}^p)$ because $C_b(\mathbb{R})$ separates points, so the functional monotone class theorem, with dominating function $|\widetilde{W}_l\Delta h|\sup_k\|\Lambda_k\|_\infty$, integrable by (P1) and Lemma 45(i), extends it to every bounded $\sigma(\Delta U,\Delta Z)$-measurable Λ; taking Λ to be a version of $\mathbb{1}\{E[\widetilde{W}_l\Delta h\mid\Delta U,\Delta Z]>0\}$ and of its complement gives $E[\widetilde{W}_l\Delta h\mid\Delta U,\Delta Z]=0$ almost surely. Ranging over l, that is exactly (b).

Step 4: (b) implies (a). Assume (b). *The argument is run at the unconditional level throughout, because that is the level at which Step 1 established continuity;* an earlier version invoked Step 1's continuity for the ΔZ-conditional map, which Step 1 does not provide. Fix $t\in\mathcal{T}$ and $\varsigma\in C_b(\mathbb{R})$. The function $\varsigma(\Delta U)e^{t'\Phi(\Delta Z)}$ is bounded, by $\|\varsigma\|_\infty\bar{w}(t)$, using (P2), and $\sigma(\Delta U,\Delta Z)$-measurable, so (b) gives, at each l,

$$0=E\Big[\widetilde{W}_l\,\Delta h\,\varsigma(\Delta U)\,e^{t'\Phi(\Delta Z)}\Big]=\int_{\mathbb{R}}\varsigma(u)\,E\Big[\widetilde{W}_l\,\Delta h\,f_{\Delta U|\mathcal{F}_{XZW}}(u)\,e^{t'\Phi(\Delta Z)}\Big]du=\int_{\mathbb{R}}\varsigma(u)\,\mathcal{A}_{\text{cond}}h(u,t,l)\,du,$$

the middle equality by (P6a)'s substitution lemma, the tower property and the Fubini envelope of Step 3. By Step 1 the map $u\mapsto\mathcal{A}_{\text{cond}}h(u,t,l)$ is continuous, and a continuous function that integrates to zero against every $\varsigma\in C_b(\mathbb{R})$ vanishes identically. Hence $\mathcal{A}_{\text{cond}}h(u,t,l)=0$ for every (u,t) and every l, and so in $L^2(\Pi)$.

Step 5: part (ii), (b) is equivalent to (c), and here A1 and A4(i) enter. Write $m_l:=E[\Delta h\mid\Delta U,\Delta Z,W=l]$ for $l\in\{0,\dots,\mathsf{L}-1\}$ and $m:=E[\Delta h\mid\Delta U,\Delta Z]$. Apply (P4c) with

$\mathcal{C} = \sigma(\Delta U, \Delta Z)$ and $\Lambda = \Delta h$, which is integrable by Lemma 45(i). A1's first clause gives $P(W = l \mid \Delta U, \Delta Z) = \varrho_l(\Delta Z)$ at every cell; conditioning on $\sigma(\Delta U, \Delta Z, W)$ and applying the tower property therefore gives $m = \sum_{j=0}^{\mathsf{L}-1} \varrho_j m_j$, *a convex combination over the cells*, while (P4c) gives, for every $l \in \{1, \dots, \mathsf{L}-1\}$,

$$E\big[\widetilde{W}_l \, \Delta h \mid \Delta U, \Delta Z\big] = \varrho_l \Big\{ m_l - \sum_{j=0}^{\mathsf{L}-1} \varrho_j m_j \Big\} = \varrho_l \left(m_l - m \right),$$

the L cell-conditional expectations being well defined because A4(i) makes every W-cell non-null. *The prefactor is the per-cell ϱ_l and the brace is the deviation of cell l from the ϱ-mixture*, which is the comparison clause (c) is stated at. Since A4(i) also bounds $\varrho_l \geq \varepsilon_\varrho > 0$ at every l, the product vanishes almost surely if and only if $m_l = m$ does, for $l = 1, \dots, \mathsf{L}-1$; and the base cell is then free, since $m = \sum_j \varrho_j m_j = \varrho_0 m_0 + (1 - \varrho_0) m$ forces $\varrho_0 (m_0 - m) = 0$ and hence $m_0 = m$, again by $\varrho_0 \geq \varepsilon_\varrho > 0$. Conversely $m_l = m$ at every l makes each $\varrho_l (m_l - m)$ vanish. That is (b)$\Leftrightarrow$(c). *At $\mathsf{L} = 2$ this is the two-cell identity earlier rounds printed*: $l = 1$ and $\varrho_0 = 1 - \varrho_1$ give $\varrho_1 \{ m_1 - \varrho_0 m_0 - \varrho_1 m_1 \} = \varrho_1 \varrho_0 \{ m_1 - m_0 \} = \varrho (1 - \varrho) \{ m_1 - m_0 \}$, whose prefactor A4(i) bounds below by $\varepsilon_\varrho (1 - \varepsilon_\varrho) > 0$, and $m_1 = m$ is $m_1 = m_0$ because $m = \varrho_0 m_0 + \varrho_1 m_1$. □

Proof of Proposition 17. The proof is in six steps: Step 1 makes one null set serve the whole Borel family, Steps 2–5 prove clauses (i)–(iv), and Step 6 discharges the $\mathsf{L} = 2$ reading in the closing sentence of the statement.

Throughout fix $\varphi \in \mathcal{V}$, write $\xi_\varphi := \Delta Y - \Delta \varphi$, and let $\big(p_0(\cdot), \dots, p_{\mathsf{L}-1}(\cdot) \big)$ be a regular version of $B \mapsto P(\xi_\varphi \in B, W = l \mid \Delta Z)$, which exists because ξ_φ takes values in the Polish space $\mathbb{R}$ and W in a finite set. Then $p_l(\mathbb{R}) = \varrho_l(\Delta Z)$ and $\sum_{l=0}^{\mathsf{L}-1} p_l(B) = P(\xi_\varphi \in B \mid \Delta Z)$ almost surely, for every Borel B.

Step 1: one null set serves every Borel set and every pair of cells. Let $\varphi \in \mathcal{V}_{\mathsf{S}}$. For each pair $(l, k) \in \mathsf{S} \times \mathsf{S}$ and each $u \in \mathbb{Q}$ there is a null set off which $\varrho_k p_l((-\infty, u]) = \varrho_l p_k((-\infty, u])$; the

union of these over the at most L^2 pairs and countably many rationals is null. Off that union both sides are, for fixed ω, right-continuous and non-decreasing in u, so the identity holds for every $u \in \mathbb{R}$; and then $B \mapsto \varrho_k p_l(B)$ and $B \mapsto \varrho_l p_k(B)$ are two finite measures agreeing on the π-system $\{(-\infty, u] : u \in \mathbb{R}\} \cup \{\mathbb{R}\}$, hence agreeing on $\mathfrak{B}(\mathbb{R})$ by Dynkin's π–λ theorem. Membership of $\mathcal{V}_{\mathsf{S}}$ may therefore be read with a single null set serving all Borel B and all pairs in S, and it is so read below.

The same step gives the θ-invariance asserted in Definition 16. Replacing ξ_φ by $\Delta U(\theta, \varphi) = \xi_\varphi - \Delta Z'\theta$ replaces $p_l(B)$ by $p_l(B + \Delta Z'\theta)$ on both sides of the defining identity; at each value of ΔZ the map $B \mapsto B + \Delta Z'\theta$ is a bijection of $\mathfrak{B}(\mathbb{R})$, so the family of identities indexed by B is the same family, and the two definitions cut out the same set. *This is used wherever the proof passes between $\Delta Y - \Delta\varphi$ and the residual at $\theta_\dagger$*, which it does in Step 3 and in the proof of Remark 18(1).

Step 2: clause (i). If $\mathsf{S} \subseteq \mathsf{S}'$ the defining family of Definition 16 for S is a sub-family of that for S', so $\mathcal{V}_{\mathsf{S}'} \subseteq \mathcal{V}_{\mathsf{S}}$. If $\#\mathsf{S} \leq 1$ the only admissible pair is (l, l), at which the identity is $\varrho_l p_l(B) = \varrho_l p_l(B)$; so $\mathcal{V}_{\mathsf{S}} = \mathcal{V}$.

For the third assertion, take $\mathsf{S} = \{0, 1, \ldots, \mathsf{L} - 1\}$. If $\varphi \in \mathcal{V}_{CI}$ then by the factorisation form of Definition 7, $p_l(B) = \varrho_l P(\xi_\varphi \in B \mid \Delta Z)$ almost surely for every l and every Borel B; multiplying by ϱ_k gives $\varrho_k p_l(B) = \varrho_k \varrho_l P(\xi_\varphi \in B \mid \Delta Z)$, which is symmetric in (l, k), so $\varphi \in \mathcal{V}_{\{0,\ldots,\mathsf{L}-1\}}$. Conversely let $\varphi \in \mathcal{V}_{\{0,\ldots,\mathsf{L}-1\}}$, fix ω off the null set of Step 1 and a Borel B, and put $A := \{l : \varrho_l > 0\}$. For $l, k \in A$ the defining identity gives $p_l(B)/\varrho_l = p_k(B)/\varrho_k$; call the common value $c(B)$. For $l \notin A$, $0 \leq p_l(B) \leq p_l(\mathbb{R}) = \varrho_l = 0$, so $p_l(B) = 0 = \varrho_l c(B)$. Hence $p_l(B) = \varrho_l c(B)$ for every l, and summing over l with $\sum_l \varrho_l = 1$ gives $c(B) = P(\xi_\varphi \in B \mid \Delta Z)$. That is the factorisation of Definition 7, so $\varphi \in \mathcal{V}_{CI}$. Finally $\mathcal{V}_{CI} = \{\varphi \in \mathcal{V} : T(\varphi, \theta) = 0\}$ for every $\theta \in \mathbb{R}^p$ is Corollary 12, whose hypotheses are B1(i)–(iv). This is the only place in the proof at which B1 is used.

Step 3: clause (ii). Under A1, Lemma 10 gives $\varphi_\dagger \in \mathcal{V}_{CI}$. By Step 2, $\mathcal{V}_{CI} = \mathcal{V}_{\{0,\ldots,\mathsf{L}-1\}} \subseteq \mathcal{V}_{\mathsf{S}}$

for every S, so $\varphi_\dagger \in \mathcal{V}_\mathsf{S}$ and every S is consistent. If some S is not consistent, A1 therefore fails. *No clause of A1b is invoked*: the argument uses A1 and the monotonicity of Step 2 and nothing else.

Step 4: clause (iii). Downward closure of consistency is the monotonicity of Step 2; and $\mathcal{V}_\mathsf{S} = \mathcal{V} \ni 0$ for $\#\mathsf{S} \leq 1$, so those sets are consistent.

Monotonicity of the conflict set. Let $\mathsf{S} \subseteq \mathsf{S}'$ be consistent and let l be in conflict with S. If $l \in \mathsf{S}'$ then $\mathsf{S} \cup \{l\} \subseteq \mathsf{S}'$, so by Step 2 $\mathcal{V}_{\mathsf{S}'} \subseteq \mathcal{V}_{\mathsf{S}\cup\{l\}} = \emptyset$, contradicting consistency of S'; hence $l \notin \mathsf{S}'$. And $\mathcal{V}_{\mathsf{S}'\cup\{l\}} \subseteq \mathcal{V}_{\mathsf{S}\cup\{l\}} = \emptyset$. So l is in conflict with S'.

The value at a maximal consistent set. If $\mathsf{S}^\star$ is maximal among the consistent sets and $l \notin \mathsf{S}^\star$, then $\mathsf{S}^\star \cup \{l\}$ strictly contains $\mathsf{S}^\star$ and so is not consistent; hence the cells in conflict with $\mathsf{S}^\star$ are exactly $\{0, \ldots, \mathsf{L}-1\} \setminus \mathsf{S}^\star$.

Non-emptiness. If l is in conflict with a consistent S then by Step 2 $\mathcal{V}_{CI} = \mathcal{V}_{\{0,\ldots,\mathsf{L}-1\}} \subseteq \mathcal{V}_{\mathsf{S}\cup\{l\}} = \emptyset$. Conversely if $\mathcal{V}_{CI} = \emptyset$ then $\{0, \ldots, \mathsf{L}-1\}$ is not consistent; the consistent sets form a non-empty finite family, since $\emptyset$ belongs to it, so it has a maximal element $\mathsf{S}^\star$, necessarily a proper subset, whose conflict set is its non-empty complement by the previous paragraph.

Uniqueness of the maximal consistent set. If $\mathcal{V}_{CI} \neq \emptyset$ then $\{0, \ldots, \mathsf{L}-1\}$ is consistent and is the unique maximal element. Conversely suppose there is exactly one maximal consistent set $\mathsf{S}^\star$. The family of consistent sets is finite and downward closed, so every consistent set is contained in a maximal one, hence in $\mathsf{S}^\star$; each singleton $\{l\}$ is consistent, so $l \in \mathsf{S}^\star$ for every l, i.e. $\mathsf{S}^\star = \{0, \ldots, \mathsf{L}-1\}$ and $\mathcal{V}_{CI} = \mathcal{V}_{\mathsf{S}^\star} \neq \emptyset$.

No cell is in conflict with every consistent set. Fix l. Then $\{l\}$ is consistent, and by Definition 16 the cells in conflict with $\{l\}$ are drawn from the complement of $\{l\}$, so l is not among them.

Step 5: clause (iv). S_0 is consistent, so the consistent sets containing S_0 form a non-empty finite family; let $\mathsf{S}^\star$ be a maximal element of it. Then $\mathsf{S}^\star$ is maximal among all consistent sets,

any consistent set strictly containing $\mathsf{S}^\star$ would contain S_0 and contradict maximality within that family, and $\#\mathsf{S}^\star \geq \#\mathsf{S}_0 \geq 2$. Since $\mathcal{V}_{CI} = \emptyset$, Step 2 makes $\{0, \ldots, \mathsf{L}-1\}$ inconsistent, so $\mathsf{S}^\star$ is a proper subset and by Step 4 the cells in conflict with it are $\{0, \ldots, \mathsf{L}-1\} \setminus \mathsf{S}^\star$: non-empty, proper, and omitting the at least two cells of $\mathsf{S}^\star$.

For the last sentence, suppose $\mathsf{L} = 2$. Then $\mathsf{S}_0 \subseteq \{0,1\}$ and $\#\mathsf{S}_0 \geq 2$ force $\mathsf{S}_0 = \{0,1\}$, so $\mathcal{V}_{\mathsf{S}_0} = \mathcal{V}_{CI}$ by Step 2, which is empty by hypothesis, contradicting $\mathcal{V}_{\mathsf{S}_0} \neq \emptyset$. The hypothesis of the clause is therefore unsatisfiable at $\mathsf{L} = 2$.

Step 6: the closing sentence of the statement. At $\mathsf{L} = 2$ the subsets of $\{0,1\}$ are $\emptyset, \{0\}, \{1\}, \{0,1\}$. By Step 2 the first three have $\mathcal{V}_{\mathsf{S}} = \mathcal{V}$ and are consistent, and $\mathcal{V}_{\{0,1\}} = \mathcal{V}_{CI}$. If $\mathcal{V}_{CI} \neq \emptyset$ all four are consistent, $\{0,1\}$ is the unique maximal one and, by Step 4, every conflict set is empty. If $\mathcal{V}_{CI} = \emptyset$ the maximal consistent sets are $\{0\}$ and $\{1\}$; by Step 4 the cells in conflict with them are $\{1\}$ and $\{0\}$, while $\emptyset$ has an empty conflict set because $\mathcal{V}_{\{l\}} = \mathcal{V} \neq \emptyset$ for both l. The union over the maximal consistent sets is $\{0,1\}$, every cell. □

Proof of Remark 18. Clauses (1), (2), (6) and (7) carry assertions requiring proof; (3) describes a procedure, (4) cites the Discussion of the assumptions, and (5) is a scope statement. Notation is that of the proof of Proposition 17.

Clause (1), the slice set. Fix $l \in \{1, \ldots, \mathsf{L}-1\}$, $\varphi \in \mathcal{V}$ and $u \in \mathbb{R}$, and put $\varepsilon := \widetilde{W}_l \mathbb{1}\{\Delta U(\theta_\dagger, \varphi) \leq u\}$, which is bounded, so $E|\varepsilon| < \infty$. By B1(ii), $w = \exp$, so (21) reads $\mathcal{M}(u, t, l; \varphi, \theta_\dagger) = E[\varepsilon\, e^{t'\Phi(\Delta Z)}]$, and by Lemma 39, whose hypotheses are B1(i), B1(ii) and a Borel index set of positive Lebesgue measure, applied at $V = \Delta Z$ and at $\mathcal{T}' = \mathcal{T}$, whose measure is positive by B1(iii), this vanishes for every $t \in \mathcal{T}$ if and only if $E[\varepsilon \mid \Delta Z] = 0$ almost surely.

Now $\Delta U(\theta_\dagger, \varphi) = \xi_\varphi - \Delta Z'\theta_\dagger$ by (19), and $\Delta Z'\theta_\dagger$ is ΔZ-measurable, so $\{\Delta U(\theta_\dagger, \varphi) \leq u\} = \{\xi_\varphi \leq u + \Delta Z'\theta_\dagger\}$ and

$$E\big[\varepsilon \mid \Delta Z\big] \;=\; p_l\Big((-\infty,\, u + \Delta Z'\theta_\dagger]\Big) \;-\; \varrho_l\, P\Big(\xi_\varphi \leq u + \Delta Z'\theta_\dagger \mid \Delta Z\Big).$$

Take $u \in \mathbb{Q}$: countably many null sets, whose union is null. Off that union, at each ω the thresholds $\{u + \Delta Z(\omega)'\theta_\dagger : u \in \mathbb{Q}\}$ form a dense subset of $\mathbb{R}$, and both $s \mapsto p_l((-\infty, s])$ and $s \mapsto \varrho_l P(\xi_\varphi \le s \mid \Delta Z)$ are right-continuous and non-decreasing, so the display vanishes at every threshold $s \in \mathbb{R}$. The two set functions $B \mapsto p_l(B)$ and $B \mapsto \varrho_l P(\xi_\varphi \in B \mid \Delta Z)$ are then finite measures agreeing on $\{(-\infty, s] : s \in \mathbb{R}\} \cup \{\mathbb{R}\}$, hence on $\mathfrak{B}(\mathbb{R})$ by the π–λ argument of Step 1. So $p_l(B) = \varrho_l P(\xi_\varphi \in B \mid \Delta Z)$ almost surely, simultaneously for every Borel B. With $g := \mathbb{1}\{\cdot = l\}$ that identity is $P(\xi_\varphi \in B, g(W) = 1 \mid \Delta Z) = P(\xi_\varphi \in B \mid \Delta Z)P(g(W) = 1 \mid \Delta Z)$, and the companion identity at $g(W) = 0$ follows by subtracting it from $\sum_j p_j(B) = P(\xi_\varphi \in B \mid \Delta Z)$. So the slice set is exactly $\mathcal{V}_{CI}(\mathbb{1}\{W = l\})$.

Clause (1)*, the intersection.* If $p_l(B) = \varrho_l P(\xi_\varphi \in B \mid \Delta Z)$ for every $l \in \{1, \dots, \mathsf{L}-1\}$ and every Borel B, then summing over those l and subtracting from $\sum_{j=0}^{\mathsf{L}-1} p_j(B) = P(\xi_\varphi \in B \mid \Delta Z)$, with $\varrho_0 = 1 - \sum_{l \ge 1} \varrho_l$, gives the same identity at $l = 0$. All L cells therefore factorise, which is $\varphi \in \mathcal{V}_{CI}$ by Definition 7. For the reverse inclusion, $\varphi \in \mathcal{V}_{CI}$ gives $p_l(B) = \varrho_l P(\xi_\varphi \in B \mid \Delta Z)$ at every l by Definition 7, which is membership of the l-th slice set by the previous paragraph read backwards. Neither direction uses A4(i) and neither uses A1.

Clause (2). Assume A4(i) and let $\#\mathsf{S} \ge 2$. Write $\varrho_\mathsf{S} := \sum_{k \in \mathsf{S}} \varrho_k$, so $\varrho_\mathsf{S} \ge \varepsilon_\varrho > 0$ almost surely, and let $Q := P(\,\cdot \mid W \in \mathsf{S})$. For bounded measurable ϕ and bounded ΔZ-measurable χ,

$$E_Q\Big[\phi(\xi_\varphi, W)\chi(\Delta Z)\Big] = \frac{E\Big[\chi\, E[\phi\, \mathbb{1}\{W \in \mathsf{S}\} \mid \Delta Z]\Big]}{P(W \in \mathsf{S})} = E_Q\Big[\tfrac{E[\phi\, \mathbb{1}\{W \in \mathsf{S}\} \mid \Delta Z]}{\varrho_\mathsf{S}}\, \chi(\Delta Z)\Big],$$

the second equality because $E_Q[\zeta(\Delta Z)] = E[\zeta \varrho_\mathsf{S}]/P(W \in \mathsf{S})$ for every bounded Borel ζ. So $E_Q[\phi \mid \Delta Z] = E[\phi \mathbb{1}\{W \in \mathsf{S}\} \mid \Delta Z]/\varrho_\mathsf{S}$, and, since $\varrho_\mathsf{S} \ge \varepsilon_\varrho > 0$ almost surely, P and Q have equivalent restrictions to $\sigma(\Delta Z)$, so "almost surely" is unambiguous in what follows. Taking $\phi = \mathbb{1}\{W = l\}$ gives $Q(W = l \mid \Delta Z) = \varrho_l/\varrho_\mathsf{S}$ for $l \in \mathsf{S}$, and the factorisation of Definition 7 under Q therefore reads, for $l \in \mathsf{S}$ and Borel B,

$$\frac{p_l(B)}{\varrho_\mathsf{S}} = \Big(\sum_{k \in \mathsf{S}} \frac{p_k(B)}{\varrho_\mathsf{S}}\Big)\frac{\varrho_l}{\varrho_\mathsf{S}}, \qquad \text{that is} \qquad \varrho_\mathsf{S}\, p_l(B) = \varrho_l \sum_{k \in \mathsf{S}} p_k(B).$$

If $\varphi \in \mathcal{V}_{\mathsf{S}}$, substituting $\varrho_{\mathsf{S}} p_l = \sum_{k\in\mathsf{S}} \varrho_k p_l = \sum_{k\in\mathsf{S}} \varrho_l p_k$ gives the right-hand identity. Conversely, given the right-hand identity for every $l \in \mathsf{S}$, divide by $\varrho_l > 0$, available by A4(i), to find that $p_l(B)/\varrho_l$ does not depend on $l \in \mathsf{S}$, which is the defining identity of Definition 16. A4(i) is used once, for $\varrho_l > 0$, and in the converse direction only.

Clause (2)*, the transfer of the maintained conditions.* A0: the vector $(Y_{i1}, Y_{i2}, X_{i1}, X_{i2}, Z_{i1}, Z_{i2}, W_i)$ is i.i.d. P and $\{W_i \in \mathsf{S}\}$ is an event in the i-th coordinate alone, so the retained observations are i.i.d. draws from Q; the support conditions of A0 are unchanged, $\mathcal{X}$ *fixed and the cell count read at* $\mathsf{L}^Q = \#\mathsf{S}$, the cells outside S carrying no Q-mass, so that A0's carrier $\{0, 1, \ldots, \mathsf{L}-1\}$ becomes S after relabelling and $\mathsf{L}^Q \geq 2$ is the hypothesis $\#\mathsf{S} \geq 2$ already in force. A4(i): the Q-propensities are $\varrho_l/\varrho_{\mathsf{S}}$ by the display above, and $\varrho_l/\varrho_{\mathsf{S}} \geq \varrho_l \geq \varepsilon_\varrho$ because $\varrho_{\mathsf{S}} \leq 1$; the range condition A4(i) imposes at $\#\mathsf{S}$ cells is $\varepsilon_\varrho \in (0, 1/\#\mathsf{S})$, and $\varepsilon_\varrho < 1/\mathsf{L} \leq 1/\#\mathsf{S}$ since $\#\mathsf{S} \leq \mathsf{L}$. *The two sentences must be read at the* ***same*** *cell count, and that is why the sentence above transfers* $\mathsf{L}^Q = \#\mathsf{S}$ *and not* L: A4(i) is a condition at *every* cell of the carrier, so at a carrier still declared to be $\{0, \ldots, \mathsf{L}-1\}$ it would be read at cells outside S, where the Q-propensity $Q(W = l \mid \Delta Z)$ is identically zero and no positive ε_ϱ bounds it below; the transfer would then be false rather than conservative. *No superscripted* ϱ *is bound here*, the Q-propensity being written out at both readings on the write-it-out precedent. B1: every clause of it constrains Φ, w, $\mathcal{T}$ and Π, none of which is a functional of P. C1(ii): since $dQ/dP = \mathbb{1}\{W \in \mathsf{S}\}/P(W \in \mathsf{S})$, the Q-density of X_j is $f^Q_{X_j}(x) = f_{X_j}(x)P(W \in \mathsf{S} \mid X_j = x)/P(W \in \mathsf{S})$, so $f^Q_{X_j} \leq f_{X_j}/P(W \in \mathsf{S})$ and $\|f^Q_{X_j}/\pi\|_\infty \leq \bar{c}_\pi/P(W \in \mathsf{S})$; π is a reference density and does not move. *The lower bound of C1(ii) is NOT obtained*: it needs $\inf_x P(W \in \mathsf{S} \mid X_j = x) > 0$, and A4(i) bounds ϱ_l given ΔZ, from which no bound on $E[\mathbb{1}\{W \in \mathsf{S}\} \mid X_j]$ follows, the two conditionings are not nested and W is not assumed conditionally independent of X_j given ΔZ.

Clause (6). Fix $\varphi \in \mathcal{V}$ and put $h := \varphi_\dagger - \varphi \in \mathcal{V}$, so that $\xi_\varphi = \Delta Y - \Delta\varphi = \Delta h + \Delta U$ with $\Delta h = h(X_2) - h(X_1)$ a Borel function of (X_1, X_2) and hence independent of $(W, \Delta U, \Delta Z)$.

Write ν_h for the law of Δh. Then the conditional law of ξ_φ given $(W = l, \Delta Z)$ is $\nu_h * N(\mu_l, 1)$, which does not depend on ΔZ; and since $\varrho_l \equiv 1/3 > 0$, the defining identity of Definition 16 at a pair (l, k) is, after division by $\varrho_l \varrho_k$, the equality of those two laws.

For the pair $(0, 1)$: $\mu_0 = \mu_1 = 0$, so the two laws are both $\nu_h * N(0, 1)$ and the identity holds, at *every* $\varphi \in \mathcal{V}$. Hence $\mathcal{V}_{\{0,1\}} = \mathcal{V} \neq \emptyset$.

For a pair containing cell 2: the identity would be $\nu_h * N(0, 1) = \nu_h * N(1, 1)$. Writing $\hat{\nu}_h$ for the characteristic function of ν_h and taking characteristic functions of both sides, $\hat{\nu}_h(s)e^{-s^2/2} = \hat{\nu}_h(s)e^{\mathrm{i}s}e^{-s^2/2}$ for every $s \in \mathbb{R}$, i.e. $\hat{\nu}_h(s)\left(1 - e^{\mathrm{i}s}\right) = 0$. So $\hat{\nu}_h(s) = 0$ at every s with $e^{\mathrm{i}s} \neq 1$, in particular for $0 < |s| < 2\pi$. But $\hat{\nu}_h$ is continuous with $\hat{\nu}_h(0) = 1$, so $\hat{\nu}_h(s) \to 1$ as $s \to 0$, a contradiction. Hence $\mathcal{V}_{\{0,2\}} = \mathcal{V}_{\{1,2\}} = \emptyset$, and by the monotonicity of Proposition 17(i), $\mathcal{V}_{CI} = \mathcal{V}_{\{0,1,2\}} = \emptyset$.

The consistent sets are therefore $\emptyset$, $\{0\}$, $\{1\}$, $\{2\}$ and $\{0, 1\}$; the maximal ones are $\{0, 1\}$ and $\{2\}$; and by Proposition 17(iii) the cells in conflict with $\{0, 1\}$ are $\{0, 1, 2\} \setminus \{0, 1\} = \{2\}$. The hypotheses of Proposition 17(iv) hold: B1(i)–(iv) is met by the maintained choices of Φ, w, $\mathcal{T}$ and Π, and $\mathcal{V}_{CI} = \emptyset$ with $\mathcal{V}_{\{0,1\}} \neq \emptyset$ and $\#\{0, 1\} = 2$. A1 fails, as it must: by Proposition 17(ii) it would put $\varphi_\dagger$ in $\mathcal{V}_{CI}$.

At $\mathsf{L} = 4$ with $\varrho_l \equiv 1/4$, $\varepsilon_\varrho = 1/5 < 1/4$, $\mu_0 = \mu_1 = 0$ and $\mu_2 \neq \mu_3$ both non-zero, the same characteristic-function argument run at the difference $\mu_k - \mu_l$ empties every pair other than $\{0, 1\}$, for $\{2, 3\}$ because $\mu_2 \neq \mu_3$, for the four mixed pairs because μ_2 and μ_3 are non-zero. The maximal consistent sets are then $\{0, 1\}$, $\{2\}$ and $\{3\}$, and the cells in conflict with $\{0, 1\}$ are $\{2, 3\}$.

Clause (7)*, in four steps: the two set conclusions, the three relevance claims, the maintained conditions, and the tilt mechanism.* Fix $\varphi \in \mathcal{V}$, put $h := \varphi_\dagger - \varphi \in \mathcal{V}$, and note $\xi_\varphi = \Delta Y - \Delta\varphi = \Delta h + \Delta U$. Write ν_l for the law of Δh under $f^{(l)}$ and $m_l := \int \Delta h \, d\nu_l$. The means exist: $h \in L^2([0, 1]) \subset L^1([0, 1])$ and $f^{(l)} \leq 1 + \gamma \leq 2$, so $E[|\Delta h| \mid W = l] \leq 4\|h\|_{L^1} < \infty$. Because ΔU is independent of $(X_1, X_2, \Delta Z)$ given W, and $(W, X_1, X_2, \Delta U)$ is independent

of ΔZ, the conditional law of ξ_φ given $(W = l, \Delta Z)$ is $\nu_l * N(\mu_l, 1)$ and does not depend on ΔZ; and since $\varrho_l \equiv \frac{1}{3} > 0$, the defining identity of Definition 16 at a pair (l, k) is, after division by $\varrho_l \varrho_k$, equality of those two laws.

Step A: $\varphi_\dagger \in \mathcal{V}_{\{0,1\}}$ and $\mathcal{V}_{\{0,2\}} = \emptyset$. At $h = 0$, $\nu_0 = \nu_1 = \delta_{\{0\}}$ and $\mu_0 = \mu_1$, so the two laws coincide and $\varphi_\dagger \in \mathcal{V}_{\{0,1\}}$. For the pair $(0, 2)$: equality of $\nu_0 * N(\mu_0, 1)$ and $\nu_2 * N(\mu_2, 1)$ forces equality of their first moments, both finite, i.e. $m_0 + \mu_0 = m_2 + \mu_2$. But $f^{(0)} = f^{(2)}$ gives $\nu_0 = \nu_2$ and hence $m_0 = m_2$, so $\mu_0 = \mu_2$, false, $\mu_2 - \mu_0 = 1$. Hence $\mathcal{V}_{\{0,2\}} = \emptyset$, and by Proposition 17(i), $\mathcal{V}_{CI} = \mathcal{V}_{\{0,1,2\}} \subseteq \mathcal{V}_{\{0,2\}} = \emptyset$. Since the only proper extension of $\{0, 1\}$ is $\{0, 1, 2\}$, $\{0, 1\}$ is a maximal consistent set and, by Proposition 17(iii), the cells in conflict with it are $\{0, 1, 2\} \setminus \{0, 1\} = \{2\}$. No characteristic function is needed here, unlike in clause (6): the mean alone decides.

Step B: the three relevance claims. (a) $E[X_2 - X_1 \mid W = 1] = \int_0^1\int_0^1 (x_2 - x_1)\{1 + \gamma(x_2 - x_1)\}\, dx_1 dx_2 = \gamma E[(X_2 - X_1)^2]_{\text{unif}} = \gamma/6$, the linear term integrating to zero and $E[(X_2 - X_1)^2]$ under the uniform law on the square being $\frac{1}{6}$; and $E[X_2 - X_1 \mid W = l] = 0$ for $l \in \{0, 2\}$ by symmetry of $f^{(0)} = f^{(2)}$ in (x_1, x_2). (b) At $h(x) = x - \frac{1}{2}$, which lies in $\mathcal{V}$ since $\int_0^1 (x - \frac{1}{2})dx = 0$, one has $m_0 = m_2 = 0$ and $m_1 = \gamma/6$. Evaluating (24) at $\varphi = \varphi_\dagger$, $\theta = \theta_\dagger$, where the conditional density of ΔU given $(X_1, X_2, \Delta Z, W)$ is $\phi(\cdot - \mu_W)$ and ΔZ is independent of (W, X_1, X_2),

$$\mathcal{A}_{\text{cond}} h(u, t, 1) = E\Big[w(t'\Phi(\Delta Z))\Big] \sum_{l=0}^{2} \varrho_l \Big(\mathbb{1}\{l = 1\} - \varrho_1\Big) \phi(u - \mu_l)\, m_l = \tfrac{2}{9}\, E\Big[w(t'\Phi(\Delta Z))\Big]\, \phi\Big(u + \tfrac{1}{3}\Big)\, \tfrac{\gamma}{6},$$

which is $\frac{\gamma}{27} E[w(t'\Phi(\Delta Z))]\phi(u + \frac{1}{3})$ and is non-zero at every (u, t), $w = \exp > 0$ by B1(ii). The interchange is Fubini's, the integrand being bounded by $\bar{w}(t)\|\phi\|_\infty |\Delta h|$ with $E|\Delta h| < \infty$. This computation does not route through Proposition 15 and so consumes no smoothness condition on the residual law. (c) At the same h, $m_1 - m_0 = \gamma/6 \neq 0$ while $\mu_1 = \mu_0$, so the first moments of $\nu_0 * N(\mu_0, 1)$ and $\nu_1 * N(\mu_1, 1)$ differ and $\varphi_\dagger - h \notin \mathcal{V}_{\{0,1\}}$. Hence $\mathcal{V}_{\{0,1\}} \subsetneq \mathcal{V}$.

Step C: the maintained conditions. A0: $\mathcal{X} = [0, 1]$ is a compact interval and the sample is i.i.d. by construction. A1: $E[\Delta U \mid \Delta Z] = E[\Delta U] = \sum_l \varrho_l \mu_l = \frac{1}{3}(-\frac{1}{3} - \frac{1}{3} + \frac{2}{3}) = 0$; the conditional

independence holds on $\{0,1\}$, where the law is $N(-\frac{1}{3},1)$ whichever of the two cells obtains, and fails at cell 2. A1b(i) by choice of $\varphi_\dagger$; A1b(ii) because $f_{X_1,X_2} \geq 1-\frac{\gamma}{3} > 0$ on all of $[0,1]^2$, so $g(X_1) = k(X_2)$ almost surely forces $g(x_1) = k(x_2)$ for Lebesgue-almost every (x_1,x_2) and hence both constant; A1b(iv) because ΔZ is bounded with $E[\Delta Z \Delta Z']$ nonsingular and $E[\Delta Z \Delta U] = E[\Delta Z]E[\Delta U] = 0$. A4(i): $\varrho_l \equiv \frac{1}{3} \geq \frac{1}{4}$ and $\frac{1}{4} \in (0,\frac{1}{3})$. B1: every clause constrains Φ, w, $\mathcal{T}$ and Π only. C1(i): $f_{\Delta Y,X_1,X_2|\Delta Z,W}(y,x_1,x_2) = f^{(l)}(x_1,x_2)\,\phi\big(y-\varphi_\dagger(x_2)+\varphi_\dagger(x_1)-\mu_l\big)$ is C^∞ and bounded above, $\varphi_\dagger$ being C^∞ and each $f^{(l)}$ a polynomial, and the ΔZ-conditional density is the ϱ-mixture of these. C1(ii): $f_{X_1,X_2} = \sum_l \varrho_l f^{(l)} = 1 + \frac{\gamma}{3}(x_2 - x_1)$, with values in $[1-\frac{\gamma}{3}, 1+\frac{\gamma}{3}]$; $f_{X_1}(x_1) = 1 + \frac{\gamma}{3}(\frac{1}{2} - x_1)$ and $f_{X_2}(x_2) = 1 + \frac{\gamma}{3}(x_2 - \frac{1}{2})$, both in $[1-\frac{\gamma}{6}, 1+\frac{\gamma}{6}]$; $\pi \equiv 1$; so all four are bounded above and below away from zero and $\bar{c}_\pi = 1 + \frac{\gamma}{6}$.

Step D: the tilt form of Step A, which is what generalises. With $\delta_l := f^{(l)}_{X_2} - f^{(l)}_{X_1}$ one has $m_l = E[h(X_2) - h(X_1) \mid W = l] = \int h\,\delta_l$, so Step A's contradiction reads $\int h\,(\delta_0 - \delta_2) = \mu_2 - \mu_0$. If $\delta_0 = \delta_2$ the left side is zero for *every* $h \in \mathcal{V}$, so any $\mu_2 \neq \mu_0$ empties $\mathcal{V}_{\{0,2\}}$, and $f^{(0)} = f^{(2)}$ is sufficient for $\delta_0 = \delta_2$ but not necessary. *If instead $\delta_0 \neq \delta_2$ the argument fails and nothing replaces it here*: scaling h along $\delta_0 - \delta_2$ produces an h with $\int h(\delta_0 - \delta_2) = \mu_2 - \mu_0$, so the first-moment obstruction is genuinely unavailable, and equality of the two laws would then have to be refuted at a higher cumulant, equality forces $\kappa_j(\nu_0) = \kappa_j(\nu_2)$ for every $j \geq 2$ as well. That is not attempted. □

Proof of Corollary 49. The proof differentiates under the Π_t-integral and then applies the product rule. *It is carried out at a generic admissible representer ℓ^Π, that is, under A2b, and not at $\ell^\Pi_\star$*, because no step uses that $\ell^\Pi_\star$ *solves* the adjoint equation: every step consumes only the two budgets $\Lambda^{(0)}, \Lambda^{(1)}$ and the u-differentiability, which is exactly the content of A2b. The specialisation to $\ell^\Pi_\star$ then follows from A2′(i), which asserts A2b at $\ell^\Pi_\star$.

Fix z and a contrast index $l \in \{1,\dots,\mathsf{L}-1\}$; *every display up to and including the product rule is at that fixed l, and the two assemblies over the $\mathsf{L}-1$ columns, of (70) itself and of the two sup-norms, are stated where they occur.* By (P2) and A2b, the integrand $u \mapsto \ell^\Pi(u,t,l)w(t'\Phi(z))$

is continuously differentiable for Π_t-a.e. t, and

$$\sup_{u\in\mathbb{R}}\left\|\partial_u\ell^{\Pi}(u,t,l)\,w(t'\Phi(z))\right\| \;\le\; \left(\sup_{u\in\mathbb{R}}\left\|\partial_u\ell^{\Pi}(u,t,l)\right\|\right)\bar{w}(t) \qquad \text{for every } z,$$

the norm being Euclidean on $\mathbb{R}^p$, which is the norm A2b's budgets are aggregated in, and the right-hand side being Π_t-integrable with integral at most $\Lambda^{(1)}<\infty$ *by the form of A2b, inherited from A2′(i.b), in which the supremum sits inside the integral and the maximum over l sits outside it. The ordering is the whole point*: a clause of the form $\sup_u\int(\cdot)\,d\Pi_t<\infty$ gives $L^1(\Pi_t)$-boundedness in u but not uniform integrability, and Leibniz's rule requires the latter. B1(v) supplies neither, since it dominates $\partial_u\lambda$ in the L-representation while $\mathcal{W}\ell^{\Pi}=\partial_u L$, one derivative further along, and it could not be repaired to supply it, because recovering $\partial_u\ell^{\Pi}$ from a bound on $\partial_u^2\lambda=\partial_u(\varpi_u\ell^{\Pi})$ requires dividing by ϖ_u, which B1(iv.a) does not bound below on any compact set. The standard theorem on differentiation under the integral sign (Leibniz's rule), applied *coordinatewise in the p coordinates of θ*, the envelope above dominating each coordinate because the Euclidean norm dominates each of them, therefore gives that $u\mapsto\int_{\mathcal{T}}\ell^{\Pi}(u,t,l)w(t'\Phi(z))\Pi_t(dt)$ is differentiable with derivative $\int_{\mathcal{T}}\partial_u\ell^{\Pi}(u,t,l)w(t'\Phi(z))\Pi_t(dt)$, uniformly in z.

Continuity of that derivative needs a second dominated-convergence pass, which Leibniz's rule does not supply. Let $u_k\to u$. For Π_t-a.e. t, A2b makes $\partial_u\ell^{\Pi}(\cdot,t,l)$ continuous, so $\partial_u\ell^{\Pi}(u_k,t,l)w(t'\Phi(z))\to\partial_u\ell^{\Pi}(u,t,l)w(t'\Phi(z))$; the norm of the difference is dominated for every k by $2(\sup_v\|\partial_u\ell^{\Pi}(v,t,l)\|)\bar{w}(t)$, which is Π_t-integrable with integral at most $2\Lambda^{(1)}<\infty$ by A2b and (P2). Dominated convergence gives continuity of $u\mapsto\int_{\mathcal{T}}\partial_u\ell^{\Pi}(u,t,l)w(t'\Phi(z))\Pi_t(dt)$, uniformly in z because the dominating function does not depend on z. The same argument with ℓ^{Π} in place of $\partial_u\ell^{\Pi}$ and $\Lambda^{(0)}$ in place of $\Lambda^{(1)}$ gives continuity of the undifferentiated integral.

Multiplying by $\varpi_{\mathsf{w}}(l)\varpi_u$, ϖ_u continuously differentiable by B1(iv.a) and $\varpi_{\mathsf{w}}(l)$ a constant by B1(iv.d), and applying the product rule gives column l of (70); ranging over l gives that

display, and the two continuity statements just proved make each column of $\partial_u \ell$ continuous, hence $\partial_u \ell$ continuous in the Frobenius norm.

The two bounds follow by (P2), the definitions of $\Lambda^{(0)}$ and $\Lambda^{(1)}$, and the two aggregation steps the statement prints. *First, per column, by the triangle inequality for the Π_t-integral of an $\mathbb{R}^p$-valued integrand*:

$$\left\|\ell_l(u,z)\right\| \;\le\; \varpi_{\mathsf{w}}(l)\,\varpi_u(u)\int_{\mathcal{T}}\left\|\ell^{\Pi}(u,t,l)\right\|\bar{w}(t)\,\Pi_t(dt) \;\le\; \varpi_{\mathsf{w}}(l)\,\|\varpi_u\|_\infty\,\Lambda^{(0)},$$

the last step because $\Lambda^{(0)}$ is the *maximum over l* of exactly that integral. *Second, over the columns*: $\|\ell(u,z)\|^2 = \sum_l \|\ell_l(u,z)\|^2 \le \|\varpi_u\|_\infty^2 (\Lambda^{(0)})^2 \sum_l \varpi_{\mathsf{w}}(l)^2$, and $\sum_l \varpi_{\mathsf{w}}(l)^2 \le \left(\sum_l \varpi_{\mathsf{w}}(l)\right)^2 = 1$, the masses being non-negative and summing to one by B1(iv.d); so $\|\ell\|_\infty \le \|\varpi_u\|_\infty \Lambda^{(0)}$ with the Frobenius norm on the $p \times (\mathsf{L}-1)$ array. *No $\sqrt{\mathsf{L}-1}$ is paid, and the reason is that* (68) *carries $\varpi_{\mathsf{w}}(l)$ in column l*: a mass-free induction map would have cost exactly that factor. The same two steps applied to the two terms of (70), the triangle inequality in the column norm first, then the assembly, give $\|\partial_u \ell\|_\infty \le \|\varpi_u'\|_\infty \Lambda^{(0)} + \|\varpi_u\|_\infty \Lambda^{(1)}$, both uniformly in (u,z) and hence on the support of $(\Delta U, \Delta Z)$. Neither constant moves with L or with p, and at $\mathsf{L}=2$ there is one column, $\varpi_{\mathsf{w}}(1)=1$, the Frobenius norm is the Euclidean norm, and both displays are the ones earlier rounds printed.

Specialisation. Under A2′(i.a)–(i.b) the selected solution $\ell_\star^{\Pi}$ satisfies A2b with $\Lambda^{(0)} = \Lambda_\star^{(0)}$ and $\Lambda^{(1)} = \Lambda_\star^{(1)}$, so the displays above hold at $\ell = \ell_\star$ with the starred budgets. *Nothing else about $\ell_\star^{\Pi}$ is used, and in particular not that it solves $\mathcal{A}^*_{\mathrm{cond}} \ell^{\Pi} = -\nu/\pi$*, which is why the generic statement is available and why the four sites that apply this corollary at a representer other than $\ell_\star^{\Pi}$ (Definition 19(a) and (32), Lemma 55(ii) and Proposition 20(i) below) may now cite it by its own hypothesis line rather than restating the bridge. □

Proof of Corollary 50. The proof checks that Υ is well defined and bounded, factors the

adjoint through it, and then reads the two inclusions off density of $\mathcal{R}(\mathcal{W})$ and continuity of Υ.

Step 1: Υ is well defined and bounded. Let $\Xi \in L^2(P_{\Delta U,\Delta Z};\mathbb{R}^{\mathsf{L}-1})$, Lemma 48(ii)'s codomain, which is the corollary's declared domain, and put $g_\Xi := \widetilde{W}'\,\Xi(\Delta U,\Delta Z) = \sum_{l=1}^{\mathsf{L}-1}\widetilde{W}_l\,\Xi_l(\Delta U,\Delta Z)$, the Euclidean inner product of (69) and *the only order that multiplies*, both factors being $(\mathsf{L}-1)$-vectors. The pointwise bound this needs is $|g_\Xi| \le \|\widetilde{W}\|\,\|\Xi\| \le c_{\widetilde{W}}\|\Xi\|$ by the Cauchy–Schwarz inequality in $\mathbb{R}^{\mathsf{L}-1}$ and (P1b), and **not** (P1), which is the per-cell statement $|\widetilde{W}_l| \le 1$ and does not bound the Euclidean norm of the vector. It follows, with a second Cauchy–Schwarz inequality in $L^2(P)$, that $E|g_\Xi| \le c_{\widetilde{W}}E\|\Xi\| \le c_{\widetilde{W}}\|\Xi\| < \infty$ and $E|g_\Xi h(X_j)| \le c_{\widetilde{W}}\|\Xi\|\,(E[h(X_j)^2])^{1/2} < \infty$ for $h \in \mathcal{V}$ by C1(ii), where $\|\Xi\| := (E\|\Xi(\Delta U,\Delta Z)\|^2)^{1/2}$ is the norm of the domain. These are precisely clauses 1 and 2 of (29), so Step 0 of the proof of Proposition 13 applies verbatim to g_Ξ and gives $\nu_{g_\Xi}/\pi \in L^2(\pi)$. That membership is needed before the inner product below can be written at all, and an earlier version of this proof used the inner product to identify the Riesz representer while the membership was still the thing being identified, the same well-formedness gap that Step 0 exists to close one result earlier. By Lemma 45(iii) and then (P1b), the Cauchy–Schwarz inequality and Lemma 45(i),

$$\left|\int_{\mathcal{X}} \nu_{g_\Xi} h\,dx\right| = \left|E[g_\Xi\,\Delta h]\right| \;\le\; c_{\widetilde{W}}\,\|\Xi\|\,\big\|\Delta h\big\|_{L^2(P)} \;\le\; 2\,c_{\widetilde{W}}\,\bar{c}_\pi^{1/2}\,\|\Xi\|\,\|h\|_{L^2(\pi)} \qquad \text{for every } h \in \mathcal{V}.$$

The constant is $c_{\widetilde{W}}$ and not 1*, and this is the step the printed bound $\|\Upsilon\| \le 2c_{\widetilde{W}}\bar{c}_\pi^{1/2}$ rests on*: a version of this proof read the pointwise step off (P1) and produced $2\bar{c}_\pi^{1/2}$, which is **false** at $\mathsf{L} \ge 3$, $\sup\|\widetilde{W}\| = \sqrt{2}$ there, and true at $\mathsf{L} = 2$, where $c_{\widetilde{W}} = 1$. So $h \mapsto \int \nu_{g_\Xi} h\,dx$ is a bounded linear functional on $\mathcal{V}$ of norm at most $2c_{\widetilde{W}}\bar{c}_\pi^{1/2}\|\Xi\|$; since $\nu_{g_\Xi}/\pi \in L^2(\pi)$ the identity $\int \nu_{g_\Xi} h\,dx = \langle \nu_{g_\Xi}/\pi, h\rangle_{L^2(\pi)}$ is well formed, by C1(ii), so the Riesz representer of that functional is the projection of ν_{g_Ξ}/π onto $\mathcal{V}$, and that projection is ν_{g_Ξ}/π itself, because $\int(\nu_{g_\Xi}/\pi)\pi\,dx = \int \nu_{g_\Xi}dx = 0$ by Lemma 45(iv). Hence $\Upsilon\Xi = \nu_{g_\Xi}/\pi \in \mathcal{V}$ with

$\|\Upsilon\Xi\|_{L^2(\pi)} \le 2c_{\widetilde{W}}\bar{c}_\pi^{1/2}\|\Xi\|$, which is the printed bound on $\|\Upsilon\|$; and Υ is linear because $\Xi \mapsto g_\Xi$ and $g \mapsto \nu_g$ are, the first because the inner product is linear in its second argument. At $\mathsf{L} = 2$ the sum has one term, $c_{\widetilde{W}} = 1$, and every display of this step is the one earlier rounds printed.

Step 2: the factorisation and the first inclusion. Lemma 48(iii) reads $\mathcal{A}^*_{\mathrm{cond}}\ell = \nu_{\widetilde{W}'(\mathcal{W}\ell)}/\pi = \Upsilon(\mathcal{W}\ell)$ for every $\ell \in L^2(\Pi)$, i.e. $\mathcal{A}^*_{\mathrm{cond}} = \Upsilon \circ \mathcal{W}$. Therefore $\mathcal{R}(\mathcal{A}^*_{\mathrm{cond}}) = \Upsilon\big(\mathcal{R}(\mathcal{W})\big) \subseteq \mathcal{R}(\Upsilon)$.

Step 3: the second inclusion. Let $\Xi \in L^2(P_{\Delta U,\Delta Z}; \mathbb{R}^{\mathsf{L}-1})$. By Lemma 48(ii), whose density statement is at the $\mathbb{R}^{\mathsf{L}-1}$-valued space and not merely componentwise, there are $\ell_k \in L^2(\Pi)$ with $\mathcal{W}\ell_k \to \Xi$; by Step 1, Υ is continuous, so $\mathcal{A}^*_{\mathrm{cond}}\ell_k = \Upsilon(\mathcal{W}\ell_k) \to \Upsilon\Xi$ in $L^2(\pi)$. Hence $\Upsilon\Xi \in \overline{\mathcal{R}(\mathcal{A}^*_{\mathrm{cond}})}$, which is (71). No property of $\mathcal{R}(\mathcal{W})$ beyond density is used, and in particular the range is not characterised. □

Proof of Lemma 55. The three claims are the affine structure, validity, and the selection functional.

(i). $\mathcal{A}^*_{\mathrm{cond}}$ is linear, so the solution set of an inhomogeneous linear equation is a coset of its kernel; A2 makes it nonempty and $\ell^\Pi_\dagger$ is one member. Applying the linear map $\mathcal{W}$ of Lemma 48(ii) sends the coset to $\mathcal{W}\ell^\Pi_\dagger + \mathcal{W}(\ker \mathcal{A}^*_{\mathrm{cond}})$.

(ii). Let ℓ^Π be an *admissible* selection, $\ell = \mathcal{W}\ell^\Pi$, and L any admissible weight generated by ℓ. Part (i) of Proposition 20, which is stated for an arbitrary admissible representer, applies to ℓ^Π and its right-hand condition holds because ℓ^Π solves the adjoint equation. Parts (ii) and (iii) of Proposition 20 do not involve ℓ at all: (ii) is proved for every admissible direction, and (iii) is an *equivalence* whose right-hand side is N1, so what is inherited by every member of the set is that orthogonality in ϱ holds exactly when N1 does. *Clause (iii) must not be read as asserting orthogonality outright.* N1 is a property that can be arranged on any member of the set by re-anchoring, at the anchor of Remark 52, a change that alters neither $\partial_u L$ nor, by (P3) and A1, the value of $E[\psi]$. *So validity is silent about the selection*, and the design freedom is real.

Why "admissible" cannot be dropped from (ii). The image $\mathcal{W}(\ell^{\Pi}_{\dagger} + \ker \mathcal{A}^{*}_{\mathrm{cond}})$ contains members with no sup-norm bound. For such a member the (P5) proviso still holds, it follows from the Cauchy–Schwarz inequality and C1(ii), since $\mathcal{W}$ maps into $L^2(P_{\Delta U,\Delta Z})$, *but the differentiation under the expectation in Proposition 20(i) does not*: that step is licensed by $\|\ell\|_\infty < \infty$, which is Corollary 49 under **A2b**, i.e. by the *admissibility* property of the member and not by membership of the solution set. *The tag is A2b and not A2′(i.a)–(i.b)*: the quantifier here runs over members of $\mathcal{W}(\ell^{\Pi}_{\dagger} + \ker \mathcal{A}^{*}_{\mathrm{cond}})$, not over $\ell^{\Pi}_{\star}$, and A2′(i) is a maintained assertion about $\ell^{\Pi}_{\star}$ alone. The headline "validity does not select" is therefore true over the admissible members and was stated one class wider than it was proved.

(iii). By Lemma 56(i) the second-order form depends on $\ell_\star$ only through $\partial_u \ell_\star$, and through it only via the two objects q and $\sigma_q = \mathcal{S}\ell_\star$. *Those two are not interchangeable, and the dependence is* ***not*** *"only through σ_q"*: σ_q is the marginalisation (66) of the bivariate q, while $\mathcal{K}$'s kernel is $q\, f_{X_1,X_2}$, which a marginalisation does not determine, two selections may share σ_q and differ in the *symmetric part* of $\mathcal{K}$, which is the only part a quadratic form sees. *The qualifier is load-bearing:* differing in the antisymmetric part alone changes nothing, so a witness must differ symmetrically.

The witness, exhibited inside the paper's own hypotheses. Take $\mathcal{X} = [0,1]$, $\pi \equiv 1$, X_1 and X_2 independent uniform so that $f_{X_1,X_2} \equiv 1$, write $\varsigma(x) := \mathbb{1}\{x > \frac{1}{2}\} - \mathbb{1}\{x \le \frac{1}{2}\}$, and compare

$$q(x_1,x_2) \;=\; -c\,\varsigma(x_1)\varsigma(x_2) \qquad \text{with} \qquad q' \;\equiv\; 0, \qquad c > 0.$$

Since $\int_0^1 \varsigma = 0$, both $\int q\,dx_1$ and $\int q\,dx_2$ vanish identically, so $E[q \mid X_1 = x] f_{X_1}(x) = E[q \mid X_2 = x] f_{X_2}(x) = 0$ and $\sigma_q = \sigma_{q'} \equiv 0$. Yet taking $h = \varsigma$, which lies in $\mathcal{V}$ because $\int \varsigma\, d\pi = 0$,

$$E\big[q\,(\Delta h)^2\big] = \iint q\,(h(x_2) - h(x_1))^2 dx_1 dx_2 = -2 \iint q\,h(x_1)h(x_2)\,dx_1 dx_2 = 2c\Big(\int_0^1 \varsigma h\Big)^2 = 2c \;\neq\; 0,$$

the two squared terms dropping out because $\int q\,dx_j \equiv 0$, while $E[q'(\Delta h)^2] = 0$. Two functions

q with the same σ_q and different quadratic forms, on a continuous $\mathcal{X}$ satisfying Definition 4 and C1(ii).

The transfer to $L^2(\pi)$ must conjugate **both** terms. Clause (i) is stated with the Lebesgue pairing. Fix $\vartheta \in \mathbb{R}^p$ with $\|\vartheta\| = 1$. For $h \in \mathcal{V}$, $\int(\vartheta'\sigma_q)h^2dx = \langle(\vartheta'\sigma_q/\pi)h, h\rangle_{L^2(\pi)}$ and $\langle\vartheta'\mathcal{K}h, h\rangle_{L^2(dx)} = \langle(\vartheta'\mathcal{K}h)/\pi, h\rangle_{L^2(\pi)}$, so on $L^2(\pi)$ the form is $\langle\mathcal{B}h, h\rangle_{L^2(\pi)}$ with $\mathcal{B}$ as in (74): the multiplication is by $\vartheta'\sigma_q/\pi$, not by $\vartheta'\sigma_q$. Writing $\vartheta'\sigma_q \cdot - 2\vartheta'\tilde{\mathcal{K}}$ conjugates the compact part and leaves the multiplication part in the wrong space.

$\mathcal{B}$ is bounded, and that must be said before its spectrum is discussed. The generator of q is $\partial_u\ell_\star\widetilde{W}$, the $p\times(\mathsf{L}-1)$ derivative contracted against the contrast on the *right*, which is the order Lemma 56(i) prints and the only one that multiplies; so $\|\partial_u\ell_\star\widetilde{W}\| \le \|\partial_u\ell_\star\|\,\|\widetilde{W}\| \le c_{\widetilde{W}}\|\partial_u\ell_\star\|_\infty$ with the Frobenius norm on the matrix, by (P1b), *and* ***not*** *by* (P1)*, which bounds each cell and not the Euclidean norm of the vector.* By that bound and conditional Jensen for the norm, $\|q(x_1,x_2)\| \le c_{\widetilde{W}}\|\partial_u\ell_\star\|_\infty$ for a.e. (x_1,x_2); hence $|\vartheta'q| \le \|q\|$ for $\|\vartheta\| = 1$, so by (66) $|\vartheta'\sigma_q(x)| \le c_{\widetilde{W}}\|\partial_u\ell_\star\|_\infty\{f_{X_1}(x) + f_{X_2}(x)\}$ and hence

$$\left\|\vartheta'\sigma_q/\pi\right\|_\infty \;\le\; 2\,c_{\widetilde{W}}\,\bar{c}_\pi\left\|\partial_u\ell_\star\right\|_\infty \;<\; \infty \tag{157}$$

by C1(ii) and Corollary 49 under A2′(i.a)–(i.b), $\|\partial_u\ell_\star\|_\infty$ being that corollary's Frobenius sup-norm. *This is the display clause* (iii) *prints, constant included*; without the factor $c_{\widetilde{W}}$ it would be false at $\mathsf{L} \ge 3$, and at $\mathsf{L} = 2$ $c_{\widetilde{W}} = 1$ and it is the display earlier rounds printed. Without (157) the essential-spectrum statement below would be about an unbounded operator.

$\tilde{\mathcal{K}}$ is Hilbert–Schmidt on $L^2(\pi)$, hence compact. As an operator on $L^2(\pi)$, $\tilde{\mathcal{K}}h = (\mathcal{K}h)/\pi$ has kernel $\tilde{\mathsf{k}}(x_1,x_2) = q(x_1,x_2)f_{X_1,X_2}(x_1,x_2)/\{\pi(x_1)\pi(x_2)\}$ with respect to the measure $\pi(x_2)\,dx_2$, so

$$\left\|\vartheta'\tilde{\mathcal{K}}\right\|_{\mathrm{HS}}^2 = \iint\left|\tilde{\mathsf{k}}\right|^2\pi(x_1)\pi(x_2)\,dx_1dx_2 = \iint\frac{(\vartheta'q)^2 f_{X_1,X_2}^2}{\pi(x_1)\pi(x_2)}\,dx_1dx_2 \;\le\; c_{\widetilde{W}}^2\left\|\partial_u\ell_\star\right\|_\infty^2\left\|1/\pi\right\|_\infty^2\left\|f_{X_1,X_2}\right\|_\infty,$$

using $(\vartheta' q)^2 \le \|q\|^2 \le c_{\widetilde{W}}^2 \|\partial_u \ell_\star\|_\infty^2$ from the previous paragraph and $\iint f_{X_1,X_2} = 1$, and the right side is finite by C1(ii) and Corollary 49, *finiteness is all that is used, so the factor* $c_{\widetilde{W}}^2$ *changes no conclusion here, and it is inserted because the step is otherwise the false one.* A Hilbert–Schmidt operator is compact, the adjoint of a compact operator is compact, and the sum of two compact operators is compact, so $\vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ is compact and self-adjoint. *No compactness of* $\mathcal{X}$ *is used*; only $\iint f_{X_1,X_2} = 1$.

Three further limbs, all bearing on Step 4. *First*, the tilde performs conjugation, not symmetrisation, and $q\, f_{X_1,X_2}$ is not a symmetric kernel, neither q nor f_{X_1,X_2} is symmetric in (x_1, x_2), so $\mathcal{B}$ is *not* self-adjoint. *Second*, for a bounded operator on a real Hilbert space, $\sup_{\|h\|=1} |\langle \mathcal{B}h, h\rangle| = \|\frac{1}{2}(\mathcal{B} + \mathcal{B}^*)\| \le \|\mathcal{B}\|$ with the inequality generally strict, so $\frac{1}{2}\|\mathcal{B}\|$ is an *upper bound* on the sharp constant *for the form on the unit sphere of* $L^2(\pi)$, and that constant is $\frac{1}{2}\|\mathcal{B}_{\mathrm{sym}}\|$ with $\mathcal{B}_{\mathrm{sym}} := \frac{1}{2}(\mathcal{B} + \mathcal{B}^*) = (\vartheta' \sigma_q/\pi)\cdot - \vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$. *Third*, σ_q is $\mathbb{R}^p$-valued, so (74) without a direction is p operators, and the direction is ϑ, *not* ς, which is a reserved local bound variable.

The lower bound, with the two steps that were missing. By (157) the *multiplication part* of $\mathcal{B}$,

$$\mathcal{B}_{\mathrm{mult}} \;:=\; \big(\vartheta' \sigma_q/\pi\big)\cdot, \tag{158}$$

is bounded and self-adjoint on $L^2(\mathcal{X}, \pi\, dx)$, and $\pi\, dx$ *is non-atomic*, being absolutely continuous with respect to Lebesgue measure on an interval; for a multiplication operator on a non-atomic measure space the essential spectrum equals the essential range, $\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{mult}}) = \operatorname{ess\,ran}(\vartheta' \sigma_q/\pi)$. *Non-atomicity is not automatic and is needed*: the *full* spectrum of a multiplication operator is always the essential range, but on an atomic space the multiplier's value at an atom is an eigenvalue of *finite* multiplicity and need not lie in σ_{ess}. *This identity is the one link of the chain that no entry in this bibliography states, and it therefore keeps its proof*; the reason is recorded at Lemma 63(iii). Since $\mathcal{B}_{\mathrm{sym}} - \mathcal{B}_{\mathrm{mult}} = -\vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ is compact by the Hilbert–Schmidt computation, Weyl's theorem, Kato (1995, Thm. IV.5.35),

read at a *bounded* compact perturbation, which is relatively compact in his sense with respect to any closed operator, gives $\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}}) = \sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{mult}})$; and since the norm of a bounded self-adjoint operator dominates the modulus of every point of its spectrum, of which the essential spectrum is a subset, $\left\|\mathcal{B}_{\mathrm{sym}}\right\| \geq \left\|\vartheta'\sigma_q/\pi\right\|_\infty$, *a lower bound and not the constant itself.*

Two things stand between that theorem and this use, and both are discharged here rather than waved through. *First,* ***which*** *essential spectrum*: the cited theorem is stated for a Fredholm-type σ_{ess}, the scalars at which the range of the operator less that scalar fails to be closed, or is closed with nullity and deficiency both infinite, and the source itself warns that the literature diverges and that this is the small version. *For a bounded self-adjoint operator the divergence is void*: the orthogonal complement of the range is the kernel, so nullity and deficiency coincide wherever the range is closed, and the cited condition then holds at exactly the scalars at which the operator, less that scalar, fails to be Fredholm, which for a bounded self-adjoint operator is what σ_{ess} denotes throughout this corpus. The source's own footnote records both that its theorem survives the rival definition and that it is Weyl's, for self-adjoint operators, which is the only case used here. *Second, the* ***field***: that theorem is stated over complex scalars, as a convention declared once for the whole of the book that states it, whereas $L^2(\pi)$ here is **real**. It is therefore read at the complexification. *That transfer is available here and is* ***not*** *available one proposition later, and what differs is the property being transferred and not the field*: a bounded self-adjoint operator on a real space extends to a bounded self-adjoint operator of the same norm, a compact one to a compact one, and at a *real* scalar the kernel and the range complexify coordinatewise, so nullity, deficiency and closedness of the range are unchanged and the two essential spectra agree. Proposition 27 Step 2 meets the neighbouring statement, the numerical radius of a *normal* operator, which the same change of field destroys outright, and refuses its own pin on exactly that ground. *Neither bridge is load-bearing for anything this corpus consumes*: the invariance step is proved a second time at Lemma 63(iii), over $\mathbb{R}$ and from the criterion, in a

form that covers both of its applications here, and the two routes agree.

The symbol is $\mathcal{B}_{\mathrm{mult}}$ *and not* M, which is retired: M named the multiplier, that multiplier is now σ_q, and $\mathcal{B}_{\mathrm{mult}}$ *is not* σ_q *but multiplication by* $\vartheta'\sigma_q/\pi$. The name $\mathcal{B}_{\mathrm{mult}}$ introduces *no new letter*: it joins $\mathcal{B}_{\mathrm{sym}}$ in the word-subscripted $\mathcal{B}$-family, and it says what the object is, the multiplication part of $\mathcal{B}$ in (74), so that $\mathcal{B} = \mathcal{B}_{\mathrm{mult}} - 2\vartheta'\tilde{\mathcal{K}}$ and $\mathcal{B}_{\mathrm{sym}} = \mathcal{B}_{\mathrm{mult}} - \vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ are readable off the notation.

And the space is $L^2(\pi)$, not $\mathcal{V}$, with the compression that transfers the bound. The displays above are statements about operators on all of $L^2(\pi)$, because $\mathcal{V}$ is not invariant under $\mathcal{B}_{\mathrm{mult}}$: multiplication by a non-constant function does not preserve $\int h\,\pi\,dx = 0$. Definition 5 names $\mathcal{B}$ as its one declared exception for exactly this reason. The quadratic form of Lemma 56(i), however, is evaluated only at $h \in \mathcal{V}$, so its own sharp constant, *the one Proposition 27 reports, and a different quantity from the sphere constant of the previous paragraph*, is $\frac{1}{2}\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\|$. *The lower bound transfers.* Since $\mathcal{V}$ has codimension one in $L^2(\pi)$, $\mathrm{Id} - P_{\mathcal{V}}$ has rank one, and $\mathcal{B}_{\mathrm{sym}} - P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}} = (\mathrm{Id} - P_{\mathcal{V}})\mathcal{B}_{\mathrm{sym}} + P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}(\mathrm{Id} - P_{\mathcal{V}})$ has rank at most two, hence is compact; so $\sigma_{\mathrm{ess}}(P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}) = \sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}})$ by Weyl's theorem again, Kato (1995, Thm. IV.5.35) once more, now at a finite-rank and hence compact perturbation, and with the same two bridges, which are stated once above and are not restated here. Writing $L^2(\pi) = \mathcal{V} \oplus \{\text{constants}\}$, the operator $P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}$ is the direct sum of its restriction to $\mathcal{V}$ and the zero operator on a one-dimensional space, whose essential spectrum is empty; hence $\sigma_{\mathrm{ess}}(P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}) = \sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}})$ and $\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\| \geq \|\vartheta'\sigma_q/\pi\|_\infty$. This is the step prp:sharpquad at Step 4 must inherit; without it, that proposition would take a norm on a space Definition 5 forbids.

That $\int \sigma_q\,dx = 0$ for every admissible selection is Lemma 45(v) applied with $\Xi = \partial_u \ell_\star$, which is bounded, hence integrable, by Corollary 49 under A2′(i.a)–(i.b). The constraint $\int \sigma_q = 0$ is therefore free and cannot be used to select; what the selection can move is the shape of σ_q, and, separately, $\mathcal{K}$, never the mean of σ_q. □

Proof of Lemma 56. The proof writes $\Psi(\tau)$ explicitly, verifies that it is finite, and differenti-

ates; the content is that ψ is affine in each of τ_2 and τ_3 separately, bi-affine, with one bilinear term , and that only τ_1 enters nonlinearly.

Step 1: an explicit form for Ψ, and its finiteness. A1b(i) enters here, and only here. $\varphi_\dagger \in \mathcal{V}$ by A1b(i) and $h \in \mathcal{V}$ by hypothesis; $\mathcal{V}$ is a linear subspace of $L^2(\pi)$ by Definition 5, so $\varphi_\dagger + \tau_1 h \in \mathcal{V}$ for every τ_1 and the path $\tau \mapsto \zeta_\tau$ stays inside the nuisance space. *Without A1b(i) the first coordinate of ζ_τ leaves $\mathcal{V}$ at every τ_1 including $\tau_1 = 0$;* Ψ is then the moment at a function outside the parameter space rather than along an admissible path, and Proposition 20's reading of $\partial_{\tau_1}\Psi(0)$ as a Gateaux derivative *within* $\mathcal{V}$, which is what makes "vanishes for every $h \in \mathcal{V}$" the right quantifier there, has no content. The residual identity itself is algebraic and needs nothing: with $\Delta U(\theta_\dagger, \varphi_\dagger + \tau_1 h) = \Delta U - \tau_1 \Delta h$ by (19),

$$\Psi(\tau) = E\Big[\Delta Z\,(\Delta U - \tau_1 \Delta h) + \big\{L_\star + \tau_2 D\big\}\big(\Delta U - \tau_1 \Delta h,\ \Delta Z\big)\big(\widetilde{W} - \tau_3 \mathsf{r}(\Delta Z)\big)\Big], \qquad (159)$$

the contrast standing to the *right* of the $p \times (\mathsf{L}-1)$ weight, which is the order (38) carries and the only one that multiplies; and $\widetilde{W}(\varrho + \tau_3 \mathsf{r}) = \widetilde{W} - \tau_3 \mathsf{r}$ coordinatewise, by (1) read at the perturbed propensity. At $\mathsf{L} = 2$ both factors are scalars against an $\mathbb{R}^p$-valued weight and the display is the one earlier rounds printed. That Ψ is a real number on $[-\bar\tau, \bar\tau] \times \mathbb{R}^2$ is a claim and is proved here, not assumed. The first term is integrable because $E\|\Delta Z(\Delta U - \tau_1 \Delta h)\| \le E\|\Delta Z \Delta U\| + |\tau_1|\,\|\Delta Z\|_{L^2(P)}\|\Delta h\|_{L^2(P)} < \infty$ by A1b(iv), the Cauchy–Schwarz inequality and Lemma 45(i). For the second, $\|\widetilde{W}\| \le c_{\widetilde{W}}$ by (P1b), *the Euclidean bound on the contrast vector, since what multiplies here is a matrix against that vector; the per-cell* (P1) *does not deliver it, and at $\mathsf{L} = 2$ the two coincide*, and r is bounded by Definition 53(a); so, the Euclidean norm of a matrix against a vector being at most the Frobenius norm of the matrix times the Euclidean norm of the vector, it suffices that $E\|L_\star(\Delta U - \tau_1 \Delta h, \Delta Z)\| < \infty$ and $E\|D(\Delta U - \tau_1 \Delta h, \Delta Z)\| < \infty$. The first holds *for every* τ_1 by Remark 52: $\|L_\star(u, z)\| \le \|b_\star(z)\| + \|\ell_\star\|_\infty |u|$ from (31), so $E\|L_\star(\Delta U - \tau_1 \Delta h, \Delta Z)\| \le E\|b_\star(\Delta Z)\| + \|\ell_\star\|_\infty\{E|\Delta U| + |\tau_1| E|\Delta h|\} < \infty$ by A2′(iii),

Corollary 49 and A1b(iv). The second holds for $|\tau_1| \le \bar{\tau}$ by Definition 53(b) and the Cauchy–Schwarz inequality, $E\|D(\Delta U - \tau_1 \Delta h, \Delta Z)\| \le \|D\|_{0,h} < \infty$.

Definition 53(b) is exactly what makes this step work off $\tau_1 = 0$, and nothing available at the true residual replaces it: Remark 54 exhibits a D with $\|D\|_{L^2(P)} = 0$ and $E[D \mid \Delta Z] = 0$ for which $\Psi(1, \tau_2, \tau_3)$ has both parts infinite and is therefore *undefined*.

Which derivatives are exact and which need an interchange, and the envelope for each. Write $\bar{\mathsf{r}} := \|\mathsf{r}\|_\infty$, the supremum of the *Euclidean* norm of the $\mathbb{R}^{\mathsf{L}-1}$-valued direction, as Lemma 56(iii) declares. The (τ_2, τ_3) derivatives need no interchange at all. By (159) the integrand is affine in each of τ_2 and τ_3 *separately*, with the single bilinear term $-\tau_2\tau_3\, D(\Delta U - \tau_1 \Delta h, \Delta Z)\, \mathsf{r}(\Delta Z)$; *it is bi-affine and not jointly affine, and the difference is not pedantic*, joint affineness would force $\partial^2_{\tau_2\tau_3}\Psi \equiv 0$, which is what clause (iii) denies and what Step 4 computes. Step 1 makes each of the four resulting expectations, of $\Delta Z(\Delta U - \tau_1 \Delta h)$, of $L_\star(\cdot)\widetilde{W}$, of $D(\cdot)\widetilde{W}$ and of $\{L_\star, D\}(\cdot)\mathsf{r}$, finite separately on $[-\bar{\tau}, \bar{\tau}] \times \mathbb{R}^2$. So for each fixed τ_1,

$$\Psi(\tau_1, \tau_2, \tau_3) \;=\; \Psi_{00}(\tau_1) + \tau_2\, \Psi_{10}(\tau_1) + \tau_3\, \Psi_{01}(\tau_1) + \tau_2\tau_3\, \Psi_{11}(\tau_1) \tag{160}$$

with Ψ_{jk} the four finite expectations just listed (signs absorbed), the double subscript recording the degree in (τ_2, τ_3): *a polynomial of degree at most one in each of τ_2, τ_3 separately and of total degree two*. Hence $\partial_{\tau_2}\Psi$, $\partial_{\tau_3}\Psi$, $\partial^2_{\tau_2\tau_2}\Psi$, $\partial^2_{\tau_3\tau_3}\Psi$ and $\partial^2_{\tau_2\tau_3}\Psi$ are read off those coefficients exactly, the last being $\Psi_{11}(\tau_1)$. This is why Definition 53's (72) may take its supremum outside the expectation.

The double subscript carries content: Ψ_{jk} is the coefficient of $\tau_2^j \tau_3^k$, so bi-affineness is readable off the notation, and it separates from Ψ_L by subscript *type*, integer digits against a weight symbol.

Only the τ_1-direction requires dominated convergence, and there are exactly three such derivatives.

(a) $\partial^2_{\tau_1\tau_1}\Psi(0)$ differentiates $\tau_1 \mapsto \Psi(\tau_1,0,0) = E[\Delta Z(\Delta U - \tau_1\Delta h) + L_\star(\Delta U - \tau_1\Delta h, \Delta Z)\widetilde{W}]$, in which D *does not appear*. By Definition 19 and Corollary 49 the map $u \mapsto L_\star(u,z)$ is twice continuously differentiable with $\partial_u L_\star = \ell_\star$, $\partial_u^2 L_\star = \partial_u \ell_\star$ and $\|\ell_\star\|_\infty, \|\partial_u \ell_\star\|_\infty < \infty$ in the Frobenius norm, so the *mean value inequality*, the vector- and matrix-valued statement, Corollary 14 Step 1, dominates the first and second difference quotients by $\{\|\Delta Z\| + c_{\widetilde{W}}\|\ell_\star\|_\infty\}\,|\Delta h|$ and by $c_{\widetilde{W}}\|\partial_u \ell_\star\|_\infty (\Delta h)^2$ respectively, the factor $c_{\widetilde{W}}$ from contracting the difference quotient against $\widetilde{W}$, by (P1b), both integrable by A1b(iv), Lemma 45(i) and the Cauchy–Schwarz inequality.

(b) $\partial^2_{\tau_1\tau_3}\Psi(0)$ differentiates the exact τ_3-coefficient $-E[L_\star(\Delta U - \tau_1\Delta h, \Delta Z)\,\mathsf{r}]$, taken at $\tau_2 = 0$, with envelope $\bar{\mathsf{r}}\,\|\ell_\star\|_\infty|\Delta h|$, *no* $c_{\widetilde{W}}$ *here, the contraction being against* r *and not against the contrast*. Again no D appears.

(c) $\partial^2_{\tau_1\tau_2}\Psi(0)$ differentiates the exact τ_2-coefficient $E[D(\Delta U - \tau_1\Delta h, \Delta Z)(\widetilde{W} - \tau_3\mathsf{r})]$ in τ_1, taken at $\tau_3 = 0$. *This is the one derivative that needs an envelope for* $\partial_u D$*, and it must be uniform in* τ_1: by the *mean value inequality* the difference quotients are dominated by $c_{\widetilde{W}} \sup_{|s|\le\bar\tau} \|\partial_u D(\Delta U - s\Delta h, \Delta Z)\|\,|\Delta h|$, whose expectation is at most $c_{\widetilde{W}}\|D\|_{1,h}\,\|\Delta h\|_{L^2(P)} \le 2c_{\widetilde{W}}\bar{c}_\pi^{1/2}\|D\|_{1,h}\|h\|_{L^2(\pi)} < \infty$ by the Cauchy–Schwarz inequality, (73) and Lemma 45(i).

The supremum in (73) *must sit inside the expectation*: the sup-outside condition $\sup_{|s|\le\bar\tau} E\|\partial_u D(\Delta U - s\Delta h, \Delta Z)\|^2 < \infty$ does *not* make the envelope of (c) integrable, because $\sup_s \|\cdot\|_{L^2(P)} < \infty$ does not give $\|\sup_s \cdot\|_{L^2(P)} < \infty$. Remark 54 records the point.

C2′(i) is ***not*** *used in this proof*: the τ_1-derivatives differentiate $L_\star$ along its own first argument, which Corollary 49 makes C^1 with a bounded derivative, so no conditional density of ΔY is required. *The claim does* ***not*** *extend to Proposition 20*: Proposition 20(i) invokes Lemma 48(iii), which consumes C2′(i) through (P7b) and through Proposition 9's bound $|g_{u,t,l}| \le \bar{w}(t)\|\bar{f}_0\|_\infty$, uniform in l. C2′(i) is consumed *directly* by Proposition 8, Proposition 9, Lemma 46, Lemma 48, Proposition 15, where an indicator or a density has to be integrated

out, and *transitively* by Proposition 20; not here.

Step 2: (i), the only nonlinear direction. Differentiating (159) twice in τ_1 at $\tau = 0$ kills the ΔZ term, which is linear in τ_1, and leaves $\partial^2_{\tau_1\tau_1}\Psi(0) = E[\partial_u^2 L_\star(\Delta U, \Delta Z)\,\widetilde{W}\,(\Delta h)^2] = E[\partial_u \ell_\star\,\widetilde{W}\,(\Delta h)^2]$, since $\partial_u L_\star = \ell_\star$ by Definition 19, *the contrast to the right of the derivative, which is the order Lemma 56*(i) *prints, and* $(\Delta h)^2$ *a scalar that commutes.* For the second form, condition on (X_1, X_2) and use the tower property with the jointly measurable version of $q = E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2] \in \mathbb{R}^p$ supplied by (P6b)'s Q_{12}, *a version has to be fixed, because* q *is used below as the kernel of* $\mathcal{K}$ *and inside a sup bound, both of which are statements about a function of* (x_1, x_2) *and not about an equivalence class.* Then $E[\partial_u \ell_\star \widetilde{W}(\Delta h)^2] = E[q\,(\Delta h)^2]$, and expanding $(\Delta h)^2 = h(X_1)^2 - 2h(X_1)h(X_2) + h(X_2)^2$ gives $\int \sigma_q h^2 dx - 2\langle \mathcal{K}h, h\rangle$ by (66) and the definition of $\mathcal{K}$, both read coordinatewise in the p rows. *The term-by-term split of* $E[q(\Delta h)^2]$ *into three expectations needs each term separately integrable*, exactly as (P5)'s scope note requires: each is bounded in norm by $c_{\widetilde{W}}\|\partial_u \ell_\star\|_\infty E|h(X_j)h(X_k)| \le c_{\widetilde{W}}\|\partial_u \ell_\star\|_\infty \bar{c}_\pi \|h\|^2_{L^2(\pi)} < \infty$ by (P1b), the Cauchy–Schwarz inequality and C1(ii), (P1b) *and not* (P1), *the step being* $\|\partial_u \ell_\star \widetilde{W}\| \le \|\partial_u \ell_\star\|\|\widetilde{W}\|$*; finiteness is all that is used here, and at* $\mathsf{L} = 2$ *the factor is* 1. *The two diagonal terms ADD*, which is why the sum functional and not the difference functional governs the second order.

Step 3: (ii), the two exactly-zero directions. In (159) the integrand is affine in τ_2, $L_\star + \tau_2 D$ enters linearly and its argument does not involve τ_2, and affine in τ_3. By Step 1, Ψ is finite at every $\tau \in [-\bar{\tau}, \bar{\tau}] \times \mathbb{R}^2$ and has the form (160) there, so for each fixed (τ_1, τ_3) it is an affine function of τ_2, and for each fixed (τ_1, τ_2) an affine function of τ_3. Only these two one-variable statements are used, and they, unlike joint affineness in (τ_2, τ_3), which is false, are exactly what the two pure second derivatives consume. Hence $\partial^2_{\tau_2\tau_2}\Psi$ and $\partial^2_{\tau_3\tau_3}\Psi$ vanish at every such τ, not only at the truth. *The scope qualifier is not a weakening but a well-posedness requirement*: off $[-\bar{\tau}, \bar{\tau}]$ in τ_1, Definition 53 says nothing and Ψ need not be defined, so "identically in τ" without a scope would be ill-posed rather than false.

Step 4: (iii), the cross term in (L, ϱ), and the exact scope of its vanishing. $\partial^2_{\tau_2\tau_3}$ of the integrand is $-D(\Delta U - \tau_1\Delta h, \Delta Z)\,\mathsf{r}(\Delta Z)$, *the $p \times (\mathsf{L}-1)$ direction against the $\mathbb{R}^{\mathsf{L}-1}$-valued one, the only order that multiplies. At $\tau_1 = 0$, and only there*, its expectation is $-E[D(\Delta U, \Delta Z)\mathsf{r}(\Delta Z)] = -E\Big[E[D \mid \Delta Z]\,\mathsf{r}(\Delta Z)\Big] = 0$ by the tower property and N1′, the $\sigma(\Delta Z)$-measurable factor coming out on the *right*. *The vanishing does not extend to $\tau_1 \neq 0$*, because N1′ is a conditional-mean restriction at the *true* residual ΔU, while at $\tau_1 \neq 0$ the argument is the shifted $\Delta U - \tau_1\Delta h$ and $E[D(\Delta U - \tau_1\Delta h, \Delta Z) \mid \Delta Z]$ is a different object with no reason to vanish. *Part (ii)'s "identically" is a genuinely stronger statement and must not be carried over to (iii)*; an earlier version reused that phrasing here, and a Taylor remainder taken along the path at Step 6 or 7 would have inherited the false version.

Off $\tau_1 = 0$ the correct bound is on the SHIFTED direction, by $\|D\mathsf{r}\| \le \|D\|\,\|\mathsf{r}\|$ with the Frobenius norm on D, *and* ***not*** *by* (P1)*, which an earlier version cited here although no $\widetilde{W}$ occurs in the display: the contraction is against r, so no $c_{\widetilde{W}}$ appears and none should*, exactly as Lemma 56(iii) prints, together with the Cauchy–Schwarz inequality:

$$\Big\|E\Big[D(\Delta U - \tau_1\Delta h, \Delta Z)\,\mathsf{r}(\Delta Z)\Big]\Big\| \;\le\; \|\mathsf{r}\|_\infty \Big\|D(\cdot - \tau_1\Delta h, \cdot)\Big\|_{L^2(P)} \;\le\; \|\mathsf{r}\|_\infty \|D\|_{0,h}, \qquad (161)$$

the outer norm Euclidean on $\mathbb{R}^p$ and $\|D\|_{L^2(P)}$ the Frobenius L^2 norm of (72). The shifted norm is not bounded by $\|D\|_{L^2(P)}$, and a version of this proof asserted that it was. The two are norms of the same function against *different* laws, the law of $(\Delta U, \Delta Z)$ and the law of $(\Delta U - \tau_1\Delta h, \Delta Z)$, and nothing stated at the true residual makes them comparable. This is the very asymmetry the paragraph above was written to catch, reproduced one line later, so the counterexample is printed rather than described. The witness is written on $\mathcal{X} = [0, 1]$ with the uniform design, because C1(ii) is a stated hypothesis of this lemma and a witness must not break one. Take X_1, X_2 independent and uniform on $[0, 1]$, so $f_{X_1,X_2} \equiv 1$,

$f_{X_1} = f_{X_2} \equiv \pi \equiv 1$ and C1(ii) holds with $\bar{c}_\pi = 1$; take

$$h(x) \;=\; \mathbb{1}\{x > \tfrac{1}{2}\} - \tfrac{1}{2}, \qquad \text{so} \qquad \int_{\mathcal{X}} h \, d\pi = \tfrac{1}{2} - \tfrac{1}{2} = 0, \quad h \in \mathcal{V},$$

whence $\Delta h = h(X_2) - h(X_1)$ takes the three values $\{-1, 0, +1\}$ with $P(\Delta h = -1) = P(X_1 > \frac{1}{2})P(X_2 \le \frac{1}{2}) = \frac{1}{4}$, and likewise $P(\Delta h = +1) = \frac{1}{4}$ and $P(\Delta h = 0) = \frac{1}{2}$. Let $\tau_1 = 1$, let $\Delta U \sim \mathrm{Unif}[0,1]$ be independent of $(X_1, X_2, \Delta Z)$, let ΔZ be **non-degenerate** with $E[\Delta Z \Delta Z']$ nonsingular and $E\|\Delta Z\|^2 < \infty$, put $\varrho_l \equiv 1/\mathsf{L}$, fix unit vectors $v \in \mathbb{R}^p$ and $\mathsf{r}_0 \in \mathbb{R}^{\mathsf{L}-1}$, and take $\mathsf{r} \equiv \mathsf{r}_0$ and $D(u, z) = \mathbb{1}\{u \in [1,2]\}\, v\, \mathsf{r}_0' \in \mathbb{R}^{p \times (\mathsf{L}-1)}$, so that $D\mathsf{r}_0 = \mathbb{1}\{u \in [1,2]\}\, v$ and $\|D\| = \mathbb{1}\{u \in [1,2]\}$ in the Frobenius norm and $\|\mathsf{r}\|_\infty = 1$. Then $D(\Delta U, \Delta Z) = 0$ almost surely, $\Delta U \in [0,1]$ meets $[1,2]$ only at the single point 1, so N1′ holds and $\|D\|_{L^2(P)} = 0$. *Yet the shifted expectation does not vanish, and the three branches of* Δh *must each be read*: on $\{\Delta h = -1\}$ the argument is $\Delta U + 1 \in [1,2]$ surely, so the indicator is 1; on $\{\Delta h = 0\}$ it is $\Delta U \in [0,1]$ and the indicator is 0 almost surely; on $\{\Delta h = +1\}$ it is $\Delta U - 1 \in [-1,0]$, disjoint from $[1,2]$, so the indicator is 0. Hence, ΔU being independent of Δh,

$$\Big\| E\Big[D(\Delta U - \tau_1 \Delta h, \Delta Z)\, \mathsf{r} \Big] \Big\| \;=\; P(\Delta h = -1)\, \|v\| \;=\; \tfrac{1}{4} \qquad \text{at } \tau_1 = 1.$$

And the shifted norm is $\frac{1}{2}$*, computed over the whole range of* τ_1 *and not only at* $\tau_1 = 1$: the same three-branch reading at a general τ_1 gives

$$E\Big\| D(\Delta U - \tau_1 \Delta h, \Delta Z) \Big\|^2 = \tfrac{1}{4}\Big| [0,1] \cap [1 - \tau_1,\, 2 - \tau_1] \Big| + \tfrac{1}{4}\Big| [0,1] \cap [1 + \tau_1,\, 2 + \tau_1] \Big|,$$

the two Lebesgue measures coming from the $\Delta h = -1$ and $\Delta h = +1$ branches. Each is at most 1, and equals 1 exactly at $\tau_1 = 1$ and at $\tau_1 = -1$ respectively; the first is positive only for $\tau_1 \in [0,2]$ and the second only for $\tau_1 \in [-2,0]$, and at the one common point $\tau_1 = 0$ both are 0, so *the two are never simultaneously positive* and the maximum of the sum is $\frac{1}{4}$, attained at $\tau_1 = \pm 1$. Hence $\|D\|_{0,h} = (\frac{1}{4})^{1/2} = \frac{1}{2}$ for every $\bar{\tau} \ge 1$, *the shifted norm of* (72)

is a ***square-root*** *of a probability, and it carries the mixing weight* $P(\Delta h = -1)$*, not the conditional value* 1 *that dropping the weight would give.* In particular $\|D\|_{0,h} < \infty$, which is Definition 53(b). The discarded product bound would read $\frac{1}{4} \le C \cdot 0$, false for every C; the bound in (161) reads $\frac{1}{4} \le 1 \cdot \frac{1}{2}$, true. What this witness verifies, and what it does not touch, stated in those terms because the witness it replaces claimed more than it checked. It verifies, at the objects that enter (161): C1(ii) (the uniform design); $h \in \mathcal{V}$, which is what this lemma's own "fix $\bar{\tau} > 0$ and $h \in \mathcal{V}$" asks, together with A1b(i), which is a condition on $\varphi_\dagger$ and holds at $\varphi_\dagger \equiv 0$; A1b(iv), ΔZ being non-degenerate with nonsingular second moment and $E\|\Delta Z \Delta U\|^2 = E\|\Delta Z\|^2 E[\Delta U^2] < \infty$ by independence, ΔU having second moment $\frac{1}{3}$; Definition 53(a), r being bounded and $\varrho + \tau_3 \mathsf{r}_0$ lying in the simplex for all small τ_3 since ϱ is interior at $\varrho_l \equiv 1/\mathsf{L}$; and Definition 53(b), just computed. It places *no* demand on B1, A2 or A2′, and none is claimed: those clauses constrain Φ, w, Π and $\ell^{\Pi}_{\star}$, none of which occurs in (161), whose two sides are functionals of the law of $(\Delta U, \Delta h, \Delta Z)$ and of the pair (D, r) alone. *D is not required to be differentiable*, that being clause (iv)'s hypothesis and not (iii)'s. At $\mathsf{L} = 2$ and $p = 1$, $v = \mathsf{r}_0 = 1$ and this is a scalar bump; the two numbers it produces are $\frac{1}{4}$ and $\frac{1}{2}$ at every L and every p, neither being L-dependent. *Nothing consumes the value*, (161) is the display under test and the counterexample's work is done by $\frac{1}{4} \le C \cdot 0$ being false for every C, which is untouched. *The height of the bump is set to* 1 *without loss*, both sides of (161) being homogeneous of degree one in D; a general height was written M through round 7, in breach of C7's retirement of that letter and one hundred and seventy lines from the M of the lower-bound chain, where the same glyph named a multiplication operator.

What (161) is and, precisely, is not. It is a bound by the *shifted* norm, $\|D\|_{0,h}$, which Definition 53(b) makes finite. *It is not a product bound*, and it must not be quoted as one: a product bound would read $C\|\mathsf{r}\|_\infty \|D\|_{L^2(P)}$ with the *unshifted* norm, and the display above shows that no such bound exists, $\frac{1}{4} \le C \cdot 0$ is false for every C. So the term enters the expansion at order $\|\mathsf{r}\|_\infty \|D\|_{0,h}$, which is *first* order in the L-direction and not a product of two estimation errors, *unless* the estimator constrains $\hat{L}$ so that D satisfies N1′, in which case

the leading cross term vanishes at $\tau_1 = 0$ and the remainder along the path is $O(|\tau_1|)$ times a shifted norm. *Step 6 or 7 must supply that control before expanding along the path, and must not inherit the word "product" from any earlier record of this step.*

Step 5: (iv), the two product terms, and the order in which they are taken. The order used is τ_2 (or τ_3) first, then τ_1, which is the cheap one: by (160) the inner derivative is exact, and only the outer one needs envelope (c). At $\tau_3 = 0$, $\partial_{\tau_2}\Psi(\tau_1, \cdot, 0) = \Psi_{10}(\tau_1) = E[D(\Delta U - \tau_1\Delta h, \Delta Z)\widetilde{W}]$, and differentiating Ψ_{10} at $\tau_1 = 0$ under envelope (c) gives $\partial_{\tau_1}\big[\partial_{\tau_2}\Psi\big](0) = -E[\partial_u D(\Delta U, \Delta Z)\,\widetilde{W}\,\Delta h]$; the same computation at $\tau_2 = 0$ with the exact τ_3-coefficient of (b) gives $\partial_{\tau_1}\big[\partial_{\tau_3}\Psi\big](0) = +E[\ell_\star(\Delta U, \Delta Z)\,\mathsf{r}(\Delta Z)\,\Delta h]$.

The reverse order returns the same two values, and this is verified rather than assumed, because Step 6 or 7 will expand along a path and must know which object it inherits. Under Definition 53's differentiability the map $\tau_1 \mapsto \Psi(\tau)$ is differentiable on $(-\bar\tau, \bar\tau)$ at every fixed (τ_2, τ_3), by the *mean value inequality* with the envelope

$$\Big\{\|\Delta Z\| + \big(c_{\widetilde{W}} + |\tau_3|\bar{\mathsf{r}}\big)\Big(\|\ell_\star\|_\infty + |\tau_2| \sup_{|s|\le\bar\tau}\big\|\partial_u D(\Delta U - s\Delta h, \Delta Z)\big\|\Big)\Big\}\,\big|\Delta h\big|,$$

integrable by A1b(iv), Lemma 45(i), the Cauchy–Schwarz inequality, $\|\ell_\star\|_\infty < \infty$ and $\|D\|_{1,h} < \infty$, *this is the only place the envelope is needed at $\tau_2 \neq 0$, and it is finite for exactly the same reason as envelope (c)*; the factor $\|\widetilde{W} - \tau_3\mathsf{r}\| \le c_{\widetilde{W}} + |\tau_3|\bar{\mathsf{r}}$ is (P1b) and the triangle inequality, and is $1 + |\tau_3|\bar{\mathsf{r}}$ at $\mathsf{L} = 2$. Hence

$$\partial_{\tau_1}\Psi(\tau) = -E\Big[\Delta Z\,\Delta h\Big] - E\Big[\big(\ell_\star + \tau_2\partial_u D\big)\big(\Delta U - \tau_1\Delta h, \Delta Z\big)\big(\widetilde{W} - \tau_3\mathsf{r}\big)\,\Delta h\Big],$$

which is affine in τ_2 at fixed (τ_1, τ_3) and affine in τ_3 at fixed (τ_1, τ_2). *The term-by-term split needed to read that off requires each piece separately integrable*, as (P5)'s scope note demands, and both are: $E[\|\ell_\star\|\,\|\widetilde{W} - \tau_3\mathsf{r}\|\,|\Delta h|] \le (c_{\widetilde{W}} + |\tau_3|\bar{\mathsf{r}})\|\ell_\star\|_\infty E|\Delta h|$ and $|\tau_2|\,E[\|\partial_u D(\Delta U - \tau_1\Delta h, \Delta Z)\|\,\|\widetilde{W} - \tau_3\mathsf{r}\|\,|\Delta h|] \le (c_{\widetilde{W}} + |\tau_3|\bar{\mathsf{r}})|\tau_2|\,\|D\|_{1,h}\,\|\Delta h\|_{L^2(P)}$, both finite. So the second

differentiation is exact and $\partial_{\tau_2}\big[\partial_{\tau_1}\Psi\big](0) = -E[\partial_u D(\Delta U, \Delta Z)\widetilde{W}\Delta h]$ and $\partial_{\tau_3}\big[\partial_{\tau_1}\Psi\big](0) = +E[\ell_\star(\Delta U, \Delta Z)\,\mathsf{r}\,\Delta h]$. Both mixed partials are therefore symmetric at $\tau = 0$ by direct computation, and no appeal to a Clairaut–Schwarz hypothesis is made anywhere.

The bounds, and the norm they are denominated in. By (P1b), the Cauchy–Schwarz inequality and Lemma 45(i),

$$\big\|\partial^2_{\tau_1\tau_2}\Psi(0)\big\| \;\le\; c_{\widetilde{W}}\big\|\partial_u D\big\|_{L^2(P)}\,\|\Delta h\|_{L^2(P)} \;\le\; 2\,c_{\widetilde{W}}\,\bar{c}_\pi^{1/2}\big\|\partial_u D\big\|_{L^2(P)}\|h\|_{L^2(\pi)} \;\le\; 2\,c_{\widetilde{W}}\,\bar{c}_\pi^{1/2}\|D\|_{1,h}\|h\|_{L^2(\pi)},$$

which is (75); the first step is $\|\partial_u D\,\widetilde{W}\| \le \|\partial_u D\|\,\|\widetilde{W}\| \le c_{\widetilde{W}}\|\partial_u D\|$, Frobenius on the matrix, and it is (P1b) and not (P1), a version of this proof read it off the per-cell bound and printed the constant without $c_{\widetilde{W}}$, which contradicts (75) as stated and is false at $\mathsf{L} \ge 3$. For the r-direction, $\|\ell_\star\|_\infty < \infty$ by Corollary 49 under A2′(i.a)–(i.b), so pulling that factor out and applying the Cauchy–Schwarz inequality to the remaining product gives

$$\begin{aligned}\Big\|E\Big[\ell_\star(\Delta U, \Delta Z)\,\mathsf{r}(\Delta Z)\,\Delta h\Big]\Big\| \;&\le\; \|\ell_\star\|_\infty\,E\Big[\|\mathsf{r}(\Delta Z)\|\,|\Delta h|\Big]\\ &\le\; \|\ell_\star\|_\infty\,\|\mathsf{r}\|_{L^2(P)}\,\|\Delta h\|_{L^2(P)}\\ &\le\; 2\bar{c}_\pi^{1/2}\|\ell_\star\|_\infty\|\mathsf{r}\|_{L^2(P)}\|h\|_{L^2(\pi)},\end{aligned}$$

which is (76); the first step is $\|\ell_\star\mathsf{r}\| \le \|\ell_\star\|\,\|\mathsf{r}\|$, and *no* $c_{\widetilde{W}}$ *enters here, the contraction being against* r *and not against the contrast, which is the asymmetry between* (75) *and* (76) *that the statement prints.* The weaker sup-norm form follows from $\|\mathsf{r}\|_{L^2(P)} \le \|\mathsf{r}\|_\infty$, finite by Definition 53(a). In each case a genuine product of the two direction magnitudes, and in each case the operative norm is one the block prices. *The sharp form in the* D*-direction is the one in* $\|\partial_u D\|_{L^2(P)}$: that is the object A5(iv) prices, as ϵ_n, and the last inequality of (75) is a stated weakening to the envelope that licensed the interchange. *Nothing in the assumption block prices* $\|D\|_{0,h}$ *or* $\|D\|_{1,h}$, and they are not rates; they are membership conditions on the direction class. *Nor does anything price* $\|\hat{\varrho} - \varrho\|_\infty$, which is why (76) is denominated in

$\|\mathsf{r}\|_{L^2(P)}$.

What this does and does not give at $D = \hat{L} - L_\star$, $\mathsf{r} = \hat{\varrho} - \varrho$. Read at the estimated directions, the two displays are $O_P(\delta_n \epsilon_n)$ under A5(ii) and A5(iv), and $O_P(\delta_n r_n)$ under A5(ii) and **A4(iii)**, *A4(iii), not A5(iv), is the tag that prices* r_n, through $\|\hat{\varrho} - \varrho\|_{L^2} = O_P(n^{-r/(2r+d_Z)}) =: O_P(r_n)$, with r_n built from the *smoothness index* r of A4(ii), a different object from the direction r; A5(iv) prices ϵ_n and says nothing about r_n. *And the instantiation is licit only because* (76) *is denominated in the* $L^2(P)$ *norm that A4(iii) prices*; against the sup-norm form it would read a sup-norm bound at an L^2 rate. But the rate is not the whole instantiation. Clause (iv) is conditional on $(\hat{L} - L_\star, \hat{\varrho} - \varrho)$ being a *differentiable* admissible direction pair at $(\hat{\varphi} - \varphi_\dagger, \bar{\tau})$, and that membership is neither assumed in A5 nor established here. Remark 57 states the obligation and discharges it under one verifiable property of the estimator, that $\hat{L}$ be an admissible weight generated by a bounded, u-continuous $\hat{\ell}$, via (77). *Step 6 or 7 must cite that verification before quoting either bound at the estimated direction*, exactly as it must supply the N1′ control recorded at the end of Step 4 and as L-21 must deliver the sup-inside form of A2′(i.b). □

Proof of Proposition 20. Part (i) is one differentiation of (32), parts (ii) and (iii) one of (159) at $L = L_\star$; parts (ii) and (iii) are exact because the moment is affine in those directions.

Part (i). Let ℓ^Π be an admissible representer, that is, one satisfying **A2b**, with $\ell = \mathcal{W}\ell^\Pi$ and L an admissible weight generated by ℓ. *Nothing in this part requires* ℓ^Π *to solve anything*; what it requires is Corollary 49, which under B1(i), (ii), (iv.a), (iv.d) and A2b gives $\|\ell\|_\infty < \infty$ and ℓ continuously differentiable in u, and Definition 19, which gives $\partial_u L = \ell$ and $\|L(u,z)\| \le \|b(z)\| + \|\ell\|_\infty |u|$. *The citation is to A2b and not to A2′(i.a)–(i.b)*: Corollary 49 is stated at a generic admissible representer, so this part cites its hypothesis line directly.

Ψ_L is the object defined in (32) and is not imported from Lemma 56. It is finite at every $\tau_1 \in \mathbb{R}$: the first term of (32) by A1b(iv), the Cauchy–Schwarz inequality and Lemma 45(i), the second by $\|L(u,z)\widetilde{W}\| \le c_{\widetilde{W}}\{\|b(z)\| + \|\ell\|_\infty |u|\}$, the displayed bound on $\|L\|$ in the Frobenius

norm, contracted against the contrast by (P1b), which is the estimate Remark 59(3) prints, together with $E\|b(\Delta Z)\|^2 < \infty$ (admissibility), $E|\Delta U| < \infty$ and $E|\Delta h| < \infty$. *No radius $\bar{\tau}$ is needed and no direction pair appears*, because $\tau_2 = \tau_3 = 0$ throughout this part; in particular Definition 53 is not invoked and neither is A2′. Differentiating (32) in τ_1 at $\tau_1 = 0$,

$$\partial_{\tau_1} \Psi_L(0) = -E\Big[\Delta Z\, \Delta h\Big] - E\Big[\ell(\Delta U, \Delta Z)\, \widetilde{W}\, \Delta h\Big] = -E\Big[\Big(\Delta Z + \ell\widetilde{W}\Big)\Delta h\Big],$$

with $\ell\widetilde{W} = \sum_l \widetilde{W}_l\, \ell_{\cdot l}(\Delta U, \Delta Z) \in \mathbb{R}^p$ the contraction Proposition 20(i) declares, the interchange licensed by the *mean value inequality* with envelope $\{\|\Delta Z\| + c_{\widetilde{W}}\|\ell\|_\infty\}\,|\Delta h|$, integrable by A1b(iv), Lemma 45(i) and the Cauchy–Schwarz inequality, *this is envelope* (a) *of Lemma 56 Step 1 with ℓ in place of $\ell_\star$, restated here so that the argument does not borrow a scoping it does not use*; it involves no D. Convention (P5) applies to $g := \Delta Z + \ell\widetilde{W}$, whose proviso (29) holds by A1b(iv), (P1b) and $\|\ell\|_\infty < \infty$, (P1b) *and not* (P1)*, the step being $\|\ell\widetilde{W}\| \le \|\ell\|\,\|\widetilde{W}\|$ with the Frobenius norm on ℓ*; so $\partial_{\tau_1}\Psi_L(0) = -\int \nu_g h\, dx = -\langle \nu_g/\pi, h\rangle_{L^2(\pi)}$ by C1(ii). Vanishing for every $h \in \mathcal{V}$ is, by Steps 2–3 of the proof of Proposition 13, equivalent to $\nu_g \equiv 0$; and $\nu_g = \nu + \nu_{\ell\widetilde{W}}$ by linearity, so $\nu_g \equiv 0$ says $\nu_{\ell\widetilde{W}} = -\nu$, which is $\nu_{\widetilde{W}'(\mathcal{W}\ell^\Pi)} = -\nu$, the two being the same object written with the p rows displayed and suppressed, and which by Lemma 48(iii), which consumes B1(i), (ii), (iv), C1(ii) and C2′(i), is $\mathcal{A}^*_{\mathrm{cond}}\ell^\Pi = -\nu/\pi$. The referent is ν/π and not ν, which is what Proposition 9 Step 2 forces internally and what Remark 51(iv) anchors externally.

Why the quantifier matters. Assuming A2′ and then concluding $\mathcal{A}^*_{\mathrm{cond}}\ell^\Pi_\star = -\nu/\pi$ as the right half of an "if and only if" would assert nothing, because A2′ declares $\ell^\Pi_\star$ to be a solution of exactly that equation. Quantifying instead over admissible *selections* would not repair it, since a selection solves the equation by definition. Quantifying over admissible *representers* does, and it drops A2 and A2′ from this part's hypotheses altogether. Under A2′ the conclusion specialises to $\ell = \ell_\star$, where the left-hand condition holds.

Part (ii). Set $\tau_1 = \tau_3 = 0$ in (159) at $L = L_\star$. The integrand is $\Delta Z \Delta U + L_\star(\Delta U, \Delta Z)\widetilde{W} +$

$\tau_2 D(\Delta U, \Delta Z)\widetilde{W}$, so $\Psi(0, \tau_2, 0) - \Psi(0) = \tau_2 E[D(\Delta U, \Delta Z)\,\widetilde{W}]$, the split being legitimate because $E\|L_\star(\Delta U, \Delta Z)\widetilde{W}\| < \infty$ by Remark 52 and $E\|D\widetilde{W}\| < \infty$ by hypothesis. By A1, $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, so on the event $\{E[\|D\| \mid \Delta Z] < \infty\}$ the conditional expectation factorises, $E[D\widetilde{W} \mid \Delta Z] = E[D \mid \Delta Z]\,E[\widetilde{W} \mid \Delta Z] = 0$ by (P3), *a* $p \times (\mathsf{L}-1)$ *matrix against the zero vector of* $\mathbb{R}^{\mathsf{L}-1}$*, hence* 0 *in* $\mathbb{R}^p$*, and the product exists only in this order. The integrability proviso is why it is stated:* the factorisation of a conditional expectation of a product under conditional independence requires each factor to be conditionally integrable, and $\widetilde{W}$ is bounded by (P1) at every cell while D is not. On the degenerate region $\mathcal{Z}_0$ of (149) the factorisation display need not hold, but there some cell has $\varrho_j = 1$, so $W = j$ almost surely given ΔZ and $\widetilde{W} = 0$ almost surely, and the product vanishes regardless: the conclusion is unaffected. *A null cell is not by itself degeneracy once* $\mathsf{L} \geq 3$, and it is $\mathcal{Z}_0$ and not the null-cell set that is needed here, because it is $\mathcal{Z}_0$ on which $\widetilde{W}$ vanishes; the two coincide at $\mathsf{L} = 2$, where $\mathcal{Z}_0 = \{z : \varrho(z)(1-\varrho(z)) = 0\}$ is the region earlier rounds printed. Taking expectations gives $E[D\widetilde{W}] = 0$, and the difference is zero for *every* τ_2, which is the exactness claim. Note what is used: A1's conditional independence, and nothing about N1 or about the size of D.

Part (iii). Set $\tau_1 = \tau_2 = 0$; A4(i) is what keeps $\varrho + \tau_3 \mathsf{r}$ a propensity, *that is, valued in the simplex* $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \geq 0,\ \mathbf{1}'q \leq 1\}$*, which is the interval* $[0,1]$ *at* $\mathsf{L} = 2$*; the box* $[0,1]^{\mathsf{L}-1}$ *would permit* $\mathbf{1}'(\varrho + \tau_3 \mathsf{r}) > 1$ *and hence a negative base cell*, for small τ_3, so that the perturbed model is in the maintained class. The integrand is $\Delta Z \Delta U + L_\star(\Delta U, \Delta Z)(\widetilde{W} - \tau_3 \mathsf{r}(\Delta Z))$, so $\Psi(0,0,\tau_3) - \Psi(0) = -\tau_3 E[L_\star(\Delta U, \Delta Z)\mathsf{r}(\Delta Z)] = -\tau_3 E\Big[E[L_\star \mid \Delta Z]\,\mathsf{r}(\Delta Z)\Big]$ by the tower property, the $\sigma(\Delta Z)$-measurable factor coming out on the *right*.

Both objects in that display are well defined, and that is not free. By A2′(iii), Corollary 49 and A1b(iv), Remark 52 gives $E\|L_\star(\Delta U, \Delta Z)\| < \infty$ and $E[\|L_\star\| \mid \Delta Z] < \infty$ almost surely, so $E[L_\star \mid \Delta Z]$ exists and is finite a.s. *Without A2′(iii) it need not:* $L_\star$ is an antiderivative of $\ell_\star$ in u, its anchor $b_\star$ is not constrained by the adjoint equation, and N1, which would pin

the anchor, is deliberately withheld in this part.

(⇐) If $E[L_\star \mid \Delta Z] = 0$ a.s. the display vanishes for every r and τ_3. *(⇒)* Note $E[L_\star \mid \Delta Z] \in \mathbb{R}^{p\times(\mathsf{L}-1)}$, fix an entry $(j,l) \in \{1,\dots,p\} \times \{1,\dots,\mathsf{L}-1\}$, and take the perturbation direction $\mathsf{r} = \varepsilon_\varrho \operatorname{sign}\Big(\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big) e_l$, with $e_l \in \mathbb{R}^{\mathsf{L}-1}$ the l-th standard basis vector. It is bounded, and it keeps $\varrho + \tau_3\mathsf{r}$ in the simplex for $|\tau_3| \le 1$: only coordinate l moves, by at most ε_ϱ in modulus, and A4(i) gives both $\varrho_l \ge \varepsilon_\varrho$, so the coordinate stays non-negative, and $1 - \mathbf{1}'\varrho = \varrho_0 \ge \varepsilon_\varrho$, so the sum stays at most one. *One* ***entry*** *is perturbed at a time, with* $p(\mathsf{L}-1)$ *directions of this form*: the contrast index is selected by e_l inside the $\mathbb{R}^{\mathsf{L}-1}$-valued r, and the coordinate of θ is selected afterwards by reading the j-th coordinate of the $\mathbb{R}^p$-valued display, which is not perturbed. *A matrix-valued* r *would not be well typed*, r being a direction in the propensity vector $\varrho \in \mathbb{R}^{\mathsf{L}-1}$ by Definition 53(a). Since $E[L_\star \mid \Delta Z]\,\mathsf{r} = \varepsilon_\varrho \operatorname{sign}\Big(\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big)\big(E[L_\star \mid \Delta Z]\big)_{\cdot l}$, the j-th coordinate of the display reads $-\tau_3\varepsilon_\varrho E\Big|\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big|$, whose vanishing forces $\big(E[L_\star \mid \Delta Z]\big)_{jl} = 0$ a.s.; ranging over (j,l) gives $E[L_\star \mid \Delta Z] \equiv 0$, which is N1. At $\mathsf{L} = 2$, $e_1 = 1$, r is the scalar $\varepsilon_\varrho \operatorname{sign}(E[L_{\star,j} \mid \Delta Z])$, the simplex is $[0,1]$, and there are p directions, the construction earlier rounds printed.

N1 is not among this part's hypotheses, and that is the point. Listing it and then concluding "if and only if $E[L_\star \mid \Delta Z] = 0$, which is N1" would make the equivalence vacuous. The content is that N1 is *necessary and sufficient* for exact orthogonality in the propensity, which is the justification for treating ϱ as a product-rate nuisance rather than a $\sqrt{n}$ one, and it is available only when the proposition does not presuppose it. This is the same repair performed on Lemma 10, whose equality would have been vacuous under A1. □

Proof of Corollary 21. Throughout, $E[L_\star \mid \Delta Z]$ exists, is finite almost surely and is $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued: by A2′(iii), Corollary 49 and A1b(iv), Remark 52 gives $E\|L_\star(\Delta U, \Delta Z)\| < \infty$ and $E[\|L_\star\| \mid \Delta Z] < \infty$ almost surely. This is the same well-posedness Proposition 20(iii) establishes and it is not free; without A2′(iii) the anchor $b_\star$ is unconstrained and the conditional mean need not exist.

Part (i). That N1 is $p(\mathsf{L}-1)$ scalar conditions is the type declaration of (1) read at the zero of $\mathbb{R}^{p\times(\mathsf{L}-1)}$, and $E[L_\star \mid \Delta Z]=0$ trivially gives $E[L_\star \mid \Delta Z]\mathsf{v}=0$ for every v. It remains to exhibit the witness, and to check that it lies in the admissible class rather than outside it, which is the whole content of the clause.

A non-zero $\mathsf{v}^\perp \in \mathbb{R}^{\mathsf{L}-1}$ with $(\mathsf{v}^\perp)'\mathsf{v}=0$ exists if and only if $\{\mathsf{v}\}^\perp \subseteq \mathbb{R}^{\mathsf{L}-1}$ is non-trivial, i.e. if and only if $\mathsf{L}-1\geq 2$: the orthogonal complement of a non-zero vector in $\mathbb{R}^{\mathsf{L}-1}$ has dimension $\mathsf{L}-2$.

Fix such a $\mathsf{v}^\perp$ and an f as in the statement, and put $b := b_{\mathrm{N1}} + \mathsf{f}(\mathsf{v}^\perp)'$, a Borel map $\mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$, the outer product of a p-vector and an $(\mathsf{L}-1)$-vector being $p\times(\mathsf{L}-1)$. *Admissibility.* $\|\mathsf{f}(z)(\mathsf{v}^\perp)'\| = \|\mathsf{f}(z)\|\,\|\mathsf{v}^\perp\|$ with the norm on the left Frobenius and both on the right Euclidean, a rank-one outer product having exactly one non-zero singular value; so $E\|\mathsf{f}(\Delta Z)(\mathsf{v}^\perp)'\|^2 = \|\mathsf{v}^\perp\|^2 E\|\mathsf{f}(\Delta Z)\|^2 < \infty$, and $E\|b_{\mathrm{N1}}(\Delta Z)\|^2<\infty$ by Remark 52 under A1b(iv), A2′(i) and B1(i), (ii), (iv.a), (iv.d), *the four B1 clauses being the bridge and not decoration*: Remark 52's bound is $\|\ell\|_\infty^2 E[\Delta U^2]$, so it is a statement about b_{N1} only once $\|\ell_\star\|_\infty<\infty$, and the only route to that here is Corollary 49, which is stated at A2b. *Naming A2′(i) alone is the attribution Lemma 91's closing paragraph exists to forbid*, whence $E\|b(\Delta Z)\|^2<\infty$ by $(\varsigma+\varsigma')^2\leq 2\varsigma^2+2\varsigma'^2$. So $L := L_{\ell_\star}^{b}$ is admissible in the sense of Definition 19(b). *The derivative is unchanged.* $\partial_u L=\ell_\star$ by (31), the added term $\mathsf{f}(\mathsf{v}^\perp)'$ being free of u; *this is why the witness is a re-anchoring and not a different representer*, and it is why the construction changes nothing that the adjoint equation constrains. *The conditional mean.* By (31) and the defining property of the N1 anchor in Remark 52, $E\big[\int_0^{\Delta U}\ell_\star(v,\Delta Z)\,dv \mid \Delta Z\big] + b_{\mathrm{N1}}(\Delta Z)=0$ almost surely, so

$$E\big[L \mid \Delta Z\big] \;=\; \mathsf{f}(\Delta Z)\,(\mathsf{v}^\perp)' \qquad \text{almost surely.}$$

Hence $E[L\mid\Delta Z]\,\mathsf{v}=\mathsf{f}(\Delta Z)\big((\mathsf{v}^\perp)'\mathsf{v}\big)=0$ identically, so the contracted condition holds; while $\{E[L\mid\Delta Z]\neq 0\}=\{\mathsf{f}(\Delta Z)\neq 0\}$, since $\mathsf{v}^\perp\neq 0$, and that event has positive probability by

hypothesis. So L fails N1 while satisfying the contracted condition, which is the failure of the converse.

Part (ii). The identity is the tower property applied to $E[L_\star(\Delta U, \Delta Z)\,\mathsf{r}(\Delta Z)]$, the $\sigma(\Delta Z)$-measurable factor $\mathsf{r}(\Delta Z)$ coming out *on the right*, which is licensed because $E[\|L_\star\| \mid \Delta Z] < \infty$ almost surely and r is bounded, and which is the order the types force: a $p \times (\mathsf{L}-1)$ matrix against an $(\mathsf{L}-1)$-vector, giving a vector of $\mathbb{R}^p$. It is (159)'s τ_3-term at $\tau_1 = \tau_2 = 0$, so nothing new is computed here.

For the second sentence, suppose $P(E[L_\star \mid \Delta Z] \neq 0) > 0$. Then some entry (j,l) has $P\Big(\big(E[L_\star \mid \Delta Z]\big)_{jl} \neq 0\Big) > 0$. *This is Proposition 20*(iii)*'s own* ($\Rightarrow$) *computation read at that one entry*, and it is reproduced only to the extent the clause needs: the direction $\mathsf{r} = \varepsilon_\varrho \operatorname{sign}\Big(\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big) e_l$ is bounded and keeps $\varrho + \tau_3 \mathsf{r}$ in the simplex for $|\tau_3| \le 1$, only coordinate l moving and A4(i) supplying both $\varrho_l \ge \varepsilon_\varrho$ and $1 - \mathbf{1}'\varrho = \varrho_0 \ge \varepsilon_\varrho$; and the j-th coordinate of the display equals $\varepsilon_\varrho E\Big|\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big|$, which is strictly positive. *Nothing about* v *enters*, which is the point: the contracted condition constrains one $\mathbb{R}^p$-valued combination of the columns and this display reads every entry.

Part (iii). By $\|E[\cdot]\| \le E\|\cdot\|$ applied to the $\mathbb{R}^p$-valued object of part (ii), then $\|Ar\| \le \|A\|\,\|r\|$ for $A \in \mathbb{R}^{p \times (\mathsf{L}-1)}$ and $r \in \mathbb{R}^{\mathsf{L}-1}$ with $\|A\|$ Frobenius and $\|r\|$ Euclidean, then Cauchy–Schwarz on the resulting product of two $L^2(P)$ factors,

$$\Big\|E\Big[E[L_\star \mid \Delta Z]\,\mathsf{r}(\Delta Z)\Big]\Big\| \;\le\; E\Big[\Big\|E[L_\star \mid \Delta Z]\Big\|\,\|\mathsf{r}(\Delta Z)\|\Big] \;\le\; \Big(E\Big\|E[L_\star \mid \Delta Z]\Big\|^2\Big)^{1/2}\Big(E\|\mathsf{r}\|^2\Big)^{1/2},$$

which is (34). *No factor $c_{\widetilde{W}}$ appears, and none should*: the matrix is contracted against r and not against $\widetilde{W}$, so the quantity being priced is already $\|\mathsf{r}\|_{L^2(P)}$ — exactly the distinction (133)–(134) draw against (131)–(132). The exact-order statement is immediate from the lim inf hypothesis together with the upper bound. Finally, at $\mathsf{r} = \hat{\varrho} - \varrho$, A4(iii) prices $\Big(E\|\hat{\varrho} - \varrho\|^2\Big)^{1/2} = \|\hat{\varrho} - \varrho\|_{L^2} = O_P(r_n)$ with $r_n = n^{-\alpha}$ and $\alpha = r/(2r + d_Z)$, the inner norm Euclidean on $\mathbb{R}^{\mathsf{L}-1}$ as A4(iii) states; multiplying by $\sqrt{n}$ gives $n^{1/2-\alpha}$, and $\alpha < \frac{1}{2}$ because

A4(ii) supplies $r > 0$ and $d_Z \geq 1$, so the scaled contribution diverges rather than being $o_P(1)$.

At $\mathsf{L} = 2$ there is one column, $\{\mathsf{v}\}^\perp = \{0\}$ in $\mathbb{R}^1$, and part (i)'s witness cannot be built — which is the collapse the statement records, and the reason the gap was invisible while the corpus was read at two cells. □

Proof of Lemma 60. The proof is the three-term decomposition, a bound on $\|\nu_g/\pi\|$ obtained by reusing the tilted difference map Υ of Corollary 50, and then the two independent routes.

Step 0: what the sample-splitting convention is doing, and why it is a hypothesis. Every statement below is made conditionally on $\sigma(\mathcal{I}_a)$, and on that conditioning $\hat{\ell}$ is a *fixed* function. This matters three times. *First*, (20) defines ν_g for a fixed integrable g; with $g = \Delta Z + \hat{\ell}\,\widetilde{W}$, *the contrast to the* ***right*** *of the* $p \times (\mathsf{L}-1)$ *estimate, which is the order the statement fixes and the only one that multiplies*, the conditional expectations $E[g \mid X_j = x]$ must be taken with $\hat{\ell}$ frozen, since otherwise they would average over estimation noise and Step 2's identity $\nu_g = \nu_{(\hat{\ell}-\ell_\star)\widetilde{W}}$ would fail. *Second*, freezing is legitimate because $\mathcal{I}_a$ and $\mathcal{I}_m$ are independent, which is A0's i.i.d. clause *once the split is declared*, and that declaration is what makes A0 a genuine hypothesis of this lemma rather than an unused tag. *Third*, $\hat{\ell} \in L^2(P_{\Delta U,\Delta Z}; \mathbb{R}^{p\times(\mathsf{L}-1)})$ almost surely is what allows Υ to be applied to each *row* of $\hat{\ell} - \ell_\star$ at all, *membership of the matrix-valued space is membership of each row in* $L^2(P_{\Delta U,\Delta Z}; \mathbb{R}^{\mathsf{L}-1})$*, which is* Υ*'s domain, the two being related by the Frobenius identity Step 2 uses*; A5(iv) gives an O_P rate but not membership, so membership is stated.

Step 1: the decomposition. Add and subtract $\hat{\nu}_g$ and split $h = P_{L_x}h + (\mathrm{Id} - P_{L_x})h$: $\langle \nu_g/\pi, h\rangle = \langle (\nu_g - \hat{\nu}_g)/\pi, h\rangle + \langle \hat{\nu}_g/\pi, P_{L_x}h\rangle + \langle \hat{\nu}_g/\pi, (\mathrm{Id} - P_{L_x})h\rangle$. *This is an identity between vectors of* $\mathbb{R}^p$ *and not between scalars*: $g \in \mathbb{R}^p$ on this lemma, so (20) makes ν_g and $\hat{\nu}_g$ coordinatewise $\mathbb{R}^p$-valued and each of the four pairings is a vector. Take Euclidean norms. The triangle inequality in $\mathbb{R}^p$ bounds the left-hand side by the sum of the three, and on the first and third

$$\left\|\langle F, h\rangle_{L^2(\pi)}\right\| = \left\|\int F\,h\,\pi\,dx\right\| \;\leq\; \int \|F\|\,|h|\,\pi\,dx \;\leq\; \|F\|_{L^2(\pi)}\,\|h\|_{L^2(\pi)},$$

the first inequality being $\|\int \cdot\| \le \int \|\cdot\|$ in $\mathbb{R}^p$ and the second the Cauchy–Schwarz inequality, with $\|F\|_{L^2(\pi)} = \big(\int \|F\|^2 \pi\, dx\big)^{1/2}$ the aggregate the statement declares and Step 2 computes. This gives (78), *whose left-hand side and term* (II) *therefore carry the Euclidean norm on* $\mathbb{R}^p$ *and not a modulus*; the two coincide only at $p = 1$.

Step 2: (i), the size of ν_g *at an estimated weight.* By Proposition 20(i) applied at the admissible selection $\ell_\star^{\Pi}$, whose existence is A2 and whose admissibility, that is, A2b at $\ell_\star^{\Pi}$, is A2′(i), $\nu_{\Delta Z + \ell_\star \widetilde{W}} \equiv 0$, so linearity of $g \mapsto \nu_g$ gives $\nu_g = \nu_{(\hat\ell - \ell_\star)\widetilde{W}}$. *The tilted difference map* Υ *of Corollary 50 is then applied one* ***row*** *at a time, and it has to be*: $\Upsilon\Xi = \nu_{\widetilde{W}'\Xi}/\pi$ is defined on $L^2(P_{\Delta U, \Delta Z}; \mathbb{R}^{\mathsf{L}-1})$, whose argument carries *one* index, whereas $\hat\ell - \ell_\star$ carries two. Write $(\hat\ell - \ell_\star)_{j\cdot}$ for the j-th row, an element of that space; the j-th coordinate of $(\hat\ell - \ell_\star)\widetilde{W} \in \mathbb{R}^p$ is $\sum_l (\hat\ell - \ell_\star)_{jl} \widetilde{W}_l = \widetilde{W}'(\hat\ell - \ell_\star)_{j\cdot}$, so by linearity of $g \mapsto \nu_g$ in each coordinate

$$\big(\nu_g/\pi\big)_j \;=\; \Upsilon\Big((\hat\ell - \ell_\star)_{j\cdot}\Big), \qquad j = 1, \dots, p.$$

Step 1 of Corollary 50's proof bounds $\|\Upsilon\| \le 2\, c_{\widetilde{W}}\, \bar c_\pi^{1/2}$, with $c_{\widetilde{W}} = \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$. Squaring, summing over j, and using that the Frobenius $L^2(P)$ norm of a $p \times (\mathsf{L}-1)$ function is the ℓ^2 aggregate of its rows' $L^2(P_{\Delta U, \Delta Z}; \mathbb{R}^{\mathsf{L}-1})$ norms, an *equality*, so the aggregation costs no constant,

$$\big\|\nu_g/\pi\big\|_{L^2(\pi)}^2 = \sum_{j=1}^{p} \Big\|\Upsilon\Big((\hat\ell - \ell_\star)_{j\cdot}\Big)\Big\|_{L^2(\pi)}^2 \;\le\; \|\Upsilon\|^2 \sum_{j=1}^{p} \big\|(\hat\ell - \ell_\star)_{j\cdot}\big\|^2 \;=\; \|\Upsilon\|^2 \big\|\hat\ell - \ell_\star\big\|_{L^2(P)}^2,$$

which is (i). *The constant is* $2 c_{\widetilde{W}} \bar c_\pi^{1/2}$ *and not* $2\bar c_\pi^{1/2}$, because the pointwise passage $\|(\hat\ell - \ell_\star)\widetilde{W}\| \le \|\hat\ell - \ell_\star\|\, \|\widetilde{W}\|$ inside that step reads the *Euclidean* bound (P1b) on the contrast vector and not the per-cell (P1). At $\mathsf{L} = 2$ each row has one entry, $c_{\widetilde{W}} = 1$, and the display is $\|\nu_g/\pi\| \le 2\bar c_\pi^{1/2} \|\hat\ell - \ell_\star\|$ as earlier rounds printed. *The first-order bias is thus governed by the error in* $\hat\ell$ *and not by its level*, which is what makes the two routes below alternatives rather than a single cumulative requirement.

Step 3: (ii), the rate route. Do not use the decomposition; bound the left-hand side directly. By Step 2 and Step 1's norm passage, $\|\int \cdot\| \le \int \|\cdot\|$ and then the Cauchy–Schwarz inequality, read at $F = \nu_g/\pi$, $\left\|\langle \nu_g/\pi, h\rangle\right\| \le \|\nu_g/\pi\|\,\|h\| \le 2c_{\widetilde{W}}\bar{c}_\pi^{1/2}\|\hat{\ell} - \ell_\star\|\,\|h\| = O_P(\epsilon_n)\,O_P(\delta_n) = o_P(n^{-1/2})$ by A5(iv) and A5(ii), $c_{\widetilde{W}}$ *being a constant free of* n*, the rate is untouched by it, which is what clause* (i) *of the statement records*; *at* $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ *and the constants in the chain are the ones earlier rounds printed.* No projection, no balancing constraint, and no condition on L_x enters, in particular A8's lower end is not used. If instead one routes through (78), the same conclusion follows from $\|(\mathrm{Id} - P_{L_x})h\| \le \|h\|$ and $\|P_{L_x}h\| \le \|h\|$, both orthogonal projections being contractions, together with $\|\hat{\nu}_g/\pi\| \le \|\nu_g/\pi\| + \|(\hat{\nu}_g - \nu_g)/\pi\| = O_P(\epsilon_n) + O_P(n^{-1/2})$ by Step 2 and the first half of (79), *but that routing is optional and the direct bound above uses none of it. Term* (II) *is* ***not*** *bounded by* (III) *via the same projection contraction*, the two involving different projections of h; each is separately bounded by $\|\hat{\nu}_g/\pi\|\,\|h\|$, which is what the argument needs.

The rate route is therefore input-free of (79) entirely. It is the *balancing* route, Step 4, that needs (79), and both halves of it.

Step 4: (iii), the balancing route. If the sieve solve makes $\hat{\nu}_g$ orthogonal to $\mathrm{span}(b_1, \ldots, b_{L_x})$ in $L^2(\pi)$ then (II) $= 0$ exactly. For a B-spline basis of *degree at least* $s - 1$ on a quasi-uniform mesh there is a constant C, depending on the basis degree and the mesh ratio but not on L_x, with

$$\mathcal{E}\big(\zeta, \mathrm{span}(b_1, \ldots, b_{L_x})\big)_{L^2(dx)} := \inf_{\varkappa \in \mathrm{span}(b_1,\ldots,b_{L_x})} \big\|\zeta - \varkappa\big\|_{L^2(dx)} \le C\,L_x^{-s}\,\|\zeta\|_{H^s} \tag{162}$$

the best-approximation error being written $\mathcal{E}$ *and not* E*, which is the expectation*, and the infimum running over $\varkappa$ rather than g, which is the generic moment weight. *The sieve span is written out and carries no symbol*, calligraphic $\mathcal{S}$ being the selection functional $\mathcal{S}\ell = \sigma_{\partial_u \ell \widetilde{W}}$ of Lemma 55(iii), *the contrast to the right of the* $p \times (\mathsf{L} - 1)$ *derivative, the order that lemma prints*; this is also the majority convention already, Lemma 60's statement and Step 4's

opening line both printing $\mathrm{span}(b_1, \ldots, b_{L_x})$ in full. This is the direct, or Jackson, theorem for spline approximation on the interval $\mathcal{X}$ declared by A0 (Schumaker, 2007, Thm. 6.27); see also DeVore and Lorentz (1993, Ch. 12). *What the argument consumes is a best-approximation bound and not a bound on any particular projection*, and that is worth stating because it is what fixes the citation.

The hypothesis above is stated in ***degree****, and the units matter*: "order exceeding s" and "degree exceeding $s - \frac{1}{2}$" are not the same condition, at $s = 2.4$ the second admits degree 2 while the first, read as a degree, would demand degree 3, and *two proof steps consume this display and neither performs a conversion*, Step 4 here and Step 1 of Proposition 28. The arithmetic, and it is what makes the reading safe. Thm. 6.27's smoothness cap is the *order*, which for degree d is $d + 1$, the direct theorem running through the modulus of smoothness of that order; so the rate L_x^{-s} is available for $s \leq d + 1$, i.e. for degree $d \geq s - 1$. That is the hypothesis as now printed, and it is *non-strict*, which is also the form Definition 22's fourth bullet prints ($s \leq d + 1$), so the two printed statements of this hypothesis now agree at $s = d + 1$ as well. Membership, $\mathcal{S}_n \subset H^s$, needs degree $> s - \frac{1}{2}$ and is therefore *strictly the stronger of the two, by half an order. So A8's basis clause implies this display with room and no site has to convert*; the residual check the A8 comment flagged, whether Thm. 6.27's cap is the order or the degree, is the one this paragraph answers, and it is answered in favour of the order.

Since P_{L_x} is the $L^2(\pi)$-orthogonal projection onto $\mathrm{span}(b_1, \ldots, b_{L_x})$, it is the $L^2(\pi)$-nearest point of that span, so for *every* $\varkappa \in \mathrm{span}(b_1, \ldots, b_{L_x})$

$$\left\|(\mathrm{Id} - P_{L_x})\zeta\right\|_{L^2(\pi)} \;\leq\; \left\|\zeta - \varkappa\right\|_{L^2(\pi)} \;\leq\; \|\pi\|_\infty^{1/2} \left\|\zeta - \varkappa\right\|_{L^2(dx)},$$

and taking the infimum over $\varkappa$ gives $\|(\mathrm{Id} - P_{L_x})\zeta\|_{L^2(\pi)} \leq \|\pi\|_\infty^{1/2}\, \mathcal{E}(\zeta, \mathrm{span}(b_1, \ldots, b_{L_x}))_{L^2(dx)}$. *No $L^2(dx)$-orthogonal projection onto the same span appears in the argument*: naming a specific element of the span is incidental, the minimality of P_{L_x} does the work, and the

boundedness of the $L^2(dx)$ projection, which a projection-based route would have to invoke, is not used. Applied to $\zeta = h$ under A5(vi), (III) $= O_P(\|\hat{\nu}_g/\pi\| L_x^{-s})$; with $L_x \gg n^{1/(2s)}$ from A8 and $\|\hat{\nu}_g/\pi\| = O_P(1)$ from (79), this is $o_P(n^{-1/2})$ *with no rate for* $\hat{\ell}$ *at all*, and (I) $= O_P(n^{-1/2}\delta_n)$ by (79) and A5(ii). The window $n^{1/(2s)} \ll L_x \ll n^{1/2}$ is non-empty if and only if $s > 1$; the upper end is the conditioning constraint for a spline basis, and a polynomial basis would require $L_x \ll n^{1/3}$ and hence $s > 3/2$. A8 names splines for exactly this reason. That $s > 1$ cost is charged on this route only, Step 3 pays nothing for it, which is why the smoothness-budget table carries a "Route" column.

A scope note on non-integer s. H1 sets $\mu = s/(2a)$ and A5's table runs $\mu \in \{0.75, 1, 1.5, 2, 3\}$, so s *need not be an integer*. Equation (162) as cited is a statement on the Sobolev scale H^s, which for non-integer s must be read as the interpolation scale between the neighbouring integer orders; on that reading the same L_x^{-s} rate holds, by interpolation between the integer-order direct theorems, with $\|\zeta\|_{H^s}$ the interpolation norm , equivalently, the bound may be stated on the Besov scale $B^s_{2,\infty}(\mathcal{X})$, which is where direct theorems for splines live in their sharpest form. *H1's* H^s *and A5(vi)'s* $\|h\|_{H^s}$ ***are*** *the same space as* (162)*'s, and that is a consequence of H1's* ***declaration*** *rather than a directive from this file*: H1 declares $H^s(\mathcal{X})$ to be the *classical* Sobolev space, read for non-integer s as the interpolation space between the neighbouring integer orders and carrying no boundary condition. ***Declaration and not clause, and the word is load-bearing***: that sentence of H1 fixes what the symbol $H^s(\mathcal{X})$ *names*, so the scale identity above creates no H1 edge — which matters, because H1 is not on this lemma's hypothesis line, that line being Proposition 20(i)'s together with A2, A2′(i) and A2′(iii). H1's *clauses* are its infinite rank, its $\sigma_j \asymp j^{-a}$ decay and (5), and *it is the declaration and none of those three that the scale identity rests on. The claim is about the identity and not about the whole scope note*: the opening sentence of this note reads H1's index $\mu = s/(2a)$, which does involve the decay exponent, and the disposition of the two numbers that glyph names is Remark 86. The register is the assumption block's **three registers** paragraph; through 2026-08-24 this sentence read "H1's clause". The point is kept so that a later author

does not pin (162) to an integer-order theorem while the budget table runs fractional μ.

On the two spline citations, and why the cross-reference is Ch. 12 and not Ch. 13. The load-bearing pin is Schumaker (2007, Thm. 6.27), in a chapter titled "Approximation Power of Splines" whose §§6.4–6.5 are direct theorems, which is the kind of statement (162) is. The DeVore and Lorentz (1993) cross-reference is *Ch. 12, Spline Approximation*, not Ch. 13, *Spline Interpolation and Projections onto Spline Spaces*, because (162) is a statement about the *best approximation* $\mathcal{E}(\zeta, \mathrm{span}(b_1, \ldots, b_{L_x}))$ and not about any projection operator. The distinction is easy to get backwards from the surrounding prose, this proof does end up bounding $\|(\mathrm{Id} - P_{L_x})\zeta\|$, and P_{L_x} *is* a projection onto a spline space, but the projection enters only through its minimality, which is a Hilbert-space fact requiring no spline theory at all. A Ch. 13 pin would cite a theorem about an operator this argument does not use. The rate is L_x^{-s} under either reading, so nothing downstream turns on it; what turns on it is whether the citation names the object.

Two further citations support A8's upper *end only*, Newey (1997) for the sieve conditioning constant and Belloni, Chernozhukov, Chetverikov and Kato for the modern accounting of the sieve conditioning constant, and they are named rather than pinned because nothing in this proof consumes them. *No symbol is attached to that constant here*, since the letter ζ is used in this file for generic functions and the nuisance triple carries a subscript. □

Proof of Lemma 61. The proof is the one-dimensional Sobolev embedding applied slicewise, then integrated in t and maximised over the contrast index; the only non-mechanical points are the choice of version and the measurability of the two suprema, and both are settled before the embedding is used. *Every statement below is written at the $\mathbb{R}^p$-valued arity A2c is stated in*: $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^p$ and $\|\cdot\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}$ is the vector Sobolev scale A2c defines. *And every statement below carries the contrast index $l \in \{1, \ldots, \mathsf{L}-1\}$* ***explicitly***, the base of Π being $\mathbb{R} \times \mathcal{T} \times \{1, \ldots, \mathsf{L}-1\}$ by Definition 5: slices are written $\ell^{\Pi}(\cdot, t, l)$, and the three budgets of (4) and A2′(i.a)–(i.b) are maxima over l of single-contrast integrals. *The*

two indices are separate objects and are discharged by separate devices, and conflating them is the one way to break this proof: the p coordinates of θ sit *inside* a Euclidean norm and are never summed by a step of the argument, while the $\mathsf{L}-1$ contrasts are carried through Steps 0–3 as a free parameter and aggregated by a *maximum* at Step 4. *No constant produced below carries a factor in p or in L, and the reason is structural*: the only sums over the p coordinates anywhere in the argument are the ones that *define* the Euclidean norm and that scale, and the only aggregation over l anywhere is a maximum over a finite set, so no step produces either. Throughout, $\mu'' > \frac{3}{2}$ and $c_{\mu''}$ is (80), finite because $(1+\xi^2)^{1-\mu''} \le \min\{1, |\xi|^{2-2\mu''}\}$ with $2-2\mu'' < -1$; and $c^{(0)}_{\mu''}$, $c^{(1)}_{\mu''}$ are (81), finite by the *same* comparison, since $(1+\xi^2)^{-\mu''}$ and $\xi^2(1+\xi^2)^{-\mu''}$ are both dominated by $(1+\xi^2)^{1-\mu''}$.

Step 0: the representative the argument starts from, and what it does and does not give. The measure-theoretic work is done on the **two**-factor product and the third factor costs nothing, which is why the hypothesis accounting of the statement is unchanged at $\mathsf{L} \ge 3$. By Definition 5 the base of Π is $\mathbb{R} \times \mathcal{T} \times \{1, \dots, \mathsf{L}-1\}$, whose third factor is a finite set carrying its power set; so a function on it is $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathcal{T}) \otimes 2^{\{1,\dots,\mathsf{L}-1\}}$-measurable if and only if each of its $\mathsf{L}-1$ sections is $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathcal{T})$-measurable, the preimage of any set being the finite disjoint union of the sections' preimages. *That criterion is used in both directions here, and it is a **definitional** fact and not a theorem*: Definition 5 puts the **uncompleted** product σ-field on the base, so any representative of an element of $L^2(\Pi)$ is measurable for exactly that σ-field and each of its $\mathsf{L}-1$ sections is $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathcal{T})$-measurable; and conversely, given a $\mathfrak{B}(\mathbb{R}) \otimes \mathfrak{B}(\mathcal{T})$-measurable representative of each section, with the zero function taken at any l of zero Π_{w}-mass, where the section is unconstrained, the same criterion assembles them into a jointly measurable representative on the three-fold base. *The assembly is a finite choice and consumes nothing*, and it is what lets everything from here to Step 3 be a statement about the two-factor product $\Pi_u \otimes \Pi_t$ alone. σ-finiteness supplies **neither** direction, and the attribution is narrowed here to say so. Π_u and Π_t are *finite*, B1(iv.a) and B1(iv.b), hence σ-finite, and $\Pi_u \otimes \Pi_t$ is a product of σ-finite measures; *that is consumed at the two Tonelli*

applications, the one at the close of this step and the one at the close of Step 2, and at neither of the two directions above does a representative have to be selected by a theorem. *And B1(iv.d) still does not enter*: the third factor is reached only through the section criterion, whereas a three-factor route, Π itself read as a product of finite measures, would consume it, B1(iv.d) being the only clause making Π_{w} finite. *Fix the one A2c supplies at* $\ell^{\Pi} = \ell^{\Pi}_{\star}$, *write* ℓ^{Π}_{0} *for it, and let* $\mathcal{T}_{*}$ *be the Borel set of full* Π_t*-measure that A2c attaches to it,* ***one*** *such set, serving every* l*, which is how A2c is stated*, so that ℓ^{Π}_{0} is jointly measurable and $\ell^{\Pi}_{0}(\cdot, t, l) \in H^{\mu''}(\mathbb{R}; \mathbb{R}^p)$ for *every* $t \in \mathcal{T}_{*}$ and *every* l. *The subscript is a version label and nothing else*: ℓ^{Π}_{0} and $\ell^{\Pi}_{\star}$ are the same element of $L^2(\Pi)$, and Steps 1–2 are the business of replacing the first representative by a second one that has pointwise properties. They are kept apart because the whole of Step 2 is a statement relating them, and a single symbol for both is what would make that step cite a property of the wrong function. By Tonelli on $\Pi_u \otimes \Pi_t$, σ-finiteness again, and applied at each l separately, $t \mapsto \int_{\mathbb{R}} \|\ell^{\Pi}_{0}(u, t, l)\|^2 \Pi_u(du)$ is Π_t-measurable, and at any l carrying positive mass $\varpi_{\mathsf{w}}(l)$ it is also Π_t-integrable, that integral being $\varpi_{\mathsf{w}}(l)^{-1}$ times the l-th summand of $\|\ell^{\Pi}_{0}\|^2_{L^2(\Pi)}$; the slice $\ell^{\Pi}_{0}(\cdot, t, l)$ then lies in $L^2(\Pi_u; \mathbb{R}^p)$ for Π_t-almost every t. *Nothing below consumes that conclusion, and the conditional on* $\varpi_{\mathsf{w}}(l)$ *is therefore free*: Remark 62(c) discharges the same conditional from B1(iv.d), but the argument here runs on A2c's hypothesis and not on this sentence, which is printed as the *contrast* that makes the next one legible. *A2c asserts more*, and the gap is real: it asserts that the slices lie in $H^{\mu''}(\mathbb{R}; \mathbb{R}^p) \subset L^2(\mathbb{R}, du; \mathbb{R}^p)$, and $L^2(\Pi_u; \mathbb{R}^p) \not\subseteq L^2(\mathbb{R}, du; \mathbb{R}^p)$ because B1(iv.a) bounds ϖ_u above and only pointwise below. That is the tail-decay content of A2c recorded at Remark 62(c), and it is a hypothesis, not a consequence.

Step 1: the embedding, at a fixed t*, for real* $\mu'' > \frac{3}{2}$*, with the level and the derivative weighed* ***separately***. Let $g \in H^{\mu''}(\mathbb{R}; \mathbb{R}^p)$ and $\mathcal{F}g$ be as in (3), so that $(\mathcal{F}g)(\xi) \in \mathbb{C}^p$ and $\|\cdot\|$ is the norm A2c fixes on $\mathbb{C}^p$. Apply the Cauchy–Schwarz inequality *twice*, once with the weight 1

and once with the weight $|\xi|$, splitting $(1+\xi^2)^{\mu''/2}$ off $\|\mathcal{F}g\|$ in each:

$$\begin{aligned}\int_{\mathbb{R}} \big\|(\mathcal{F}g)(\xi)\big\|\,d\xi \;&\le\; \Big(\int_{\mathbb{R}} \big(1+\xi^2\big)^{-\mu''} d\xi\Big)^{1/2}\Big(\int_{\mathbb{R}} \big(1+\xi^2\big)^{\mu''} \big\|(\mathcal{F}g)(\xi)\big\|^2 d\xi\Big)^{1/2} \;=\; 2\pi\, c^{(0)}_{\mu''}\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)},\\ \int_{\mathbb{R}} |\xi|\big\|(\mathcal{F}g)(\xi)\big\|\,d\xi \;&\le\; \Big(\int_{\mathbb{R}} \xi^2\big(1+\xi^2\big)^{-\mu''} d\xi\Big)^{1/2}\Big(\int_{\mathbb{R}} \big(1+\xi^2\big)^{\mu''} \big\|(\mathcal{F}g)(\xi)\big\|^2 d\xi\Big)^{1/2} \;=\; 2\pi\, c^{(1)}_{\mu''}\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)},\end{aligned} \tag{163}$$

the two equalities by (81) and by $\int(1+\xi^2)^{\mu''}\|\mathcal{F}g\|^2 d\xi = 2\pi\|g\|^2_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}$, which is (3) at A2c's vector scale, *Plancherel holds coordinatewise, and the sum over the p coordinates it produces is the one that* ***defines*** $\|\cdot\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}$ *and not a step of this argument, which is why the two displayed constants are the scalar ones and carry no factor in p. Both right-hand sides are finite, and the binding threshold is the one the hypothesis already supplies*: the first weight integral converges for $\mu'' > \frac{1}{2}$ and the second for $\mu'' > \frac{3}{2}$, so $\mu'' > \frac{3}{2}$ delivers both, and it is the same threshold $c_{\mu''}$ carries, (80)'s integrand $(1+\xi^2)^{1-\mu''}$ being the sum of the two weights. Separating the constants therefore restricts μ'' no further than (80) already did. So $\mathcal{F}g$ and $\xi\,\mathcal{F}g$ are both integrable in norm. Fourier inversion therefore represents g by the everywhere-defined continuous function $u \mapsto (2\pi)^{-1}\int(\mathcal{F}g)(\xi)e^{\mathrm{i}u\xi}d\xi$; differentiating under that integral is licensed by the dominating function $\|\xi\,\mathcal{F}g\| \in L^1$, giving $g \in C^1(\mathbb{R};\mathbb{R}^p)$ with $g'(u) = (2\pi)^{-1}\int \mathrm{i}\xi(\mathcal{F}g)(\xi)e^{\mathrm{i}u\xi}d\xi$; and both g and g' vanish as $|u| \to \infty$ by the Riemann–Lebesgue lemma, applied in each coordinate. Bounding the norm of each inverse transform by $(2\pi)^{-1}$ times the *corresponding* line of (163), by the triangle inequality for the vector-valued integral, $|e^{\mathrm{i}u\xi}| = 1$ leaving the integrand's norm unchanged,

$$\sup_{u\in\mathbb{R}} \big\|g(u)\big\| \;\le\; c^{(0)}_{\mu''}\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}, \qquad \sup_{u\in\mathbb{R}} \big\|g'(u)\big\| \;\le\; c^{(1)}_{\mu''}\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)},$$

which is (83) at that (t,l) once $g = \ell^{\Pi}_0(\cdot,t,l)$, for $t \in \mathcal{T}_*$ and *any* l, *Step 1 sees the contrast index only as a label on the function it is handed, which is why one argument serves all* $\mathsf{L}-1$

of them. Adding the two and applying (82),

$$\sup_{u\in\mathbb{R}}\big\|g(u)\big\|+\sup_{u\in\mathbb{R}}\big\|g'(u)\big\| \;\le\; \big(c^{(0)}_{\mu''}+c^{(1)}_{\mu''}\big)\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} \;\le\; \frac{c_{\mu''}}{\sqrt{2}}\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} \;\le\; c_{\mu''}\big\|g\big\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)},$$

which is (84). The two applications of the Cauchy–Schwarz inequality are kept apart, and that is not presentation. Step 4 maximises over the contrast index *inside* each of the two budgets before adding them, and a bound stated only for the sum $\sup_u\|g(u)\|+\sup_u\|g'(u)\|$ does not survive that order of operations at $\mathsf{L}\ge 3$; only (83) does. *The crude route, one application, preceded by* $1+|\xi|\le 2(1+\xi^2)^{1/2}$*, reaches* $c_{\mu''}$ *directly and loses at least the factor* $\sqrt{2}$ *that* (82) *recovers*, exactly $\sqrt{2}$ at $\mu''=2$ and strictly more elsewhere: the inequality $\max\{1,|\xi|\}\le(1+\xi^2)^{1/2}$ is sharp, but the doubling that follows it is not, and it is the doubling that the separation removes. *This continuous representative is unique*: two continuous functions agreeing Lebesgue-a.e. agree everywhere.

Step 1′: a bound of the same form at $\mu''=2$ *with no transform, recorded because it is what an applied reader will check.* Let $g,g',g''\in L^2(\mathbb{R},du;\mathbb{R}^p)$ in the weak sense, so that $g\in W^{1,2}(\mathbb{R};\mathbb{R}^p)$. *Three classical facts about* $W^{1,2}$ *on an interval are used, each is pinned, and each is read at a single coordinate* g_j: g_j has an absolutely continuous representative satisfying the fundamental-theorem identity (Brezis, 2011, Thm. 8.2); $g_j(u)\to 0$ as $|u|\to\infty$, because $\mathbb{R}$ is unbounded and the integrability exponent is finite (Brezis, 2011, Cor. 8.9); and $g_j^2\in W^{1,2}(\mathbb{R})$ with $(g_j^2)'=2g_jg_j'$ (Brezis, 2011, Cor. 8.10). Summing the last over j, *and that sum is the one that* ***defines*** $\|g(u)\|^2$*, not a step that could cost a factor*, gives $\|g\|^2\in W^{1,2}(\mathbb{R})$ with $(\|g\|^2)'=2\langle g,g'\rangle$, the pairing here and below being the Euclidean inner product on $\mathbb{R}^p$ and not one of this paper's L^2 inner products; which together with the first two gives $\|g(u)\|^2=2\int_{-\infty}^{u}\langle g,g'\rangle\,dv$ and hence, by Cauchy–Schwarz twice, pointwise on $\mathbb{R}^p$ and then in L^2, and $2ab\le a^2+b^2$, $\sup_u\|g(u)\|^2\le 2\|g\|_{L^2}\|g'\|_{L^2}\le\|g\|^2_{H^2(\mathbb{R};\mathbb{R}^p)}$, every L^2 norm here being $\big(\int\|\cdot\|^2du\big)^{1/2}$. Applying the same inequality to g' gives $\sup_u\|g'(u)\|^2\le 2\|g'\|_{L^2}\|g''\|_{L^2}\le\|g\|^2_{H^2(\mathbb{R};\mathbb{R}^p)}$, so $\sup_u\|g(u)\|+\sup_u\|g'(u)\|\le 2\|g\|_{H^2(\mathbb{R};\mathbb{R}^p)}$. *Both dominations are read against*

(86), the H^2 norm of (3) at $\mu''=2$ read at the vector scale, which carries a factor 2 on $\|g'\|_{L^2}^2$: $\|g\|^2+\|g'\|^2$ and $\|g'\|^2+\|g''\|^2$ are each at most $\|g\|^2+2\|g'\|^2+\|g''\|^2$, so the comparison with $c_2=\sqrt{2}$ at Remark 62(d) is like for like.

Step 2: the version, and its joint measurability, clause (i). *Define* $\ell^{\Pi}_{\star}(\cdot,t,l)$, for $t\in\mathcal{T}_*$ and every l, to be the unique continuous representative of $\ell^{\Pi}_0(\cdot,t,l)$ produced by Step 1, and $\ell^{\Pi}_{\star}(\cdot,t,l):=0$ for $t\notin\mathcal{T}_*$. Uniqueness of the continuous representative means that nothing is chosen at any (t,l) with $t\in\mathcal{T}_*$: the prescription determines one function on $\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$. *That this function is a* version *of* ℓ^{Π}_0*, hence a representative of the same element of* $L^2(\Pi)$*, which is what entitles it to the name* $\ell^{\Pi}_{\star}$*, and that it is jointly measurable, are the two things this step proves, and neither is a restatement of Step 0*, which fixed a different representative.

The set on which the argument runs is measurable. $\mathcal{T}_*$ is Borel by A2c (m.1), so $\mathbb{R}\times\mathcal{T}_*\times\{1,\dots,\mathsf{L}-1\}\in\mathfrak{B}(\mathbb{R})\otimes\mathfrak{B}(\mathcal{T})\otimes 2^{\{1,\dots,\mathsf{L}-1\}}$, the third factor being the whole finite set, which is in its own power set, and every assertion below located "on $\mathbb{R}\times\mathcal{T}_*$" is, at each fixed l, an assertion on a product-measurable set. *Nothing here uses that* $\mathcal{T}_*$ *is the largest set carrying the slice condition*, and (m.1) records why that matters: the largest such set need not be Borel.

Joint measurability, by averaging and not by the transform. *Fix* l *for the next three paragraphs*; nothing in them uses anything about l, which is what makes the assembly at the close of this step an assembly and not $\mathsf{L}-1$ separate arguments. For $t\in\mathcal{T}_*$ the slice $\ell^{\Pi}_0(\cdot,t,l)$ lies in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)\subset L^2(\mathbb{R},du;\mathbb{R}^p)$, hence is locally integrable in norm, so

$$\ell^{(N)}(u,t,l)\;:=\;N\int_{u-1/(2N)}^{u+1/(2N)}\ell^{\Pi}_0(v,t,l)\,dv,\qquad N\in\mathbb{N},$$

is well defined on $\mathbb{R}\times\mathcal{T}_*$; and it is jointly measurable there, because the integrand $(v,u,t)\mapsto\ell^{\Pi}_0(v,t,l)\mathbb{1}\{|v-u|\le 1/(2N)\}$ is jointly measurable, *that is Step 0's conclusion, read at the* l*-section of Step 0's representative, which is the only function this display sees*, and is

v-integrable at each (u,t), so Fubini's theorem applies to its positive and negative parts separately.

The average does not see which representative it is computed from, and that is the load-bearing observation of this proof. At each $t \in \mathcal{T}_*$ the slices $\ell_0^\Pi(\cdot,t,l)$ and $\ell_\star^\Pi(\cdot,t,l)$ agree Lebesgue-a.e. in u, being two representatives of one element of $L^2(\mathbb{R},du;\mathbb{R}^p)$; an integral over an interval does not see a Lebesgue-null modification of its integrand; so the interval average of the one equals the interval average of the other *at every* $(u,t) \in \mathbb{R}\times\mathcal{T}_*$. Hence $\ell^{(N)}$, formed above from ℓ_0^Π, is *simultaneously* the corresponding average of $\ell_\star^\Pi$, one function, computable from either representative, jointly measurable because of the first and about to be used to reach the second. This is precisely why averaging works where the transform does not. The mollified family is an invariant of the L^2 class; whereas $\mathcal{F}$ acts on $L^2(\mathbb{R},du;\mathbb{R}^p)$ as an L^2-limit of truncated integrals and not as an integral, so $(\xi,t)\mapsto(\mathcal{F}\ell_0^\Pi(\cdot,t,l))(\xi)$ has no version-invariant pointwise meaning at all, its joint measurability is not a Fubini statement, and recovering it would need an a.e.-convergent subsequence that may depend on t. Averaging needs no transform and no subsequence.

The limit. For $t \in \mathcal{T}_*$ the function $\ell_\star^\Pi(\cdot,t,l)$ is continuous, so the average of it over an interval of length $1/N$ centred at u converges to its value at u, every point is a Lebesgue point of a continuous function, and here continuity alone suffices. By the previous paragraph that average *is* $\ell^{(N)}(u,t,l)$, so $\ell^{(N)}(u,t,l) \to \ell_\star^\Pi(u,t,l)$ for every u and every $t \in \mathcal{T}_*$. A pointwise limit of jointly measurable functions on a measurable set is jointly measurable, so $(u,t)\mapsto\ell_\star^\Pi(u,t,l)$ is jointly measurable on $\mathbb{R}\times\mathcal{T}_*$ and, being identically 0 off it, on $\mathbb{R}\times\mathcal{T}$. The derivative follows by the same device one level up: the difference quotients $N[\ell_\star^\Pi(u+1/N,t,l)-\ell_\star^\Pi(u,t,l)]$ are jointly measurable, being compositions of $\ell_\star^\Pi(\cdot,\cdot,l)$ with a measurable shift, and converge pointwise on $\mathbb{R}\times\mathcal{T}_*$, by Step 1, which puts $\ell_\star^\Pi(\cdot,t,l)$ in $C^1(\mathbb{R};\mathbb{R}^p)$ there, to $\partial_u\ell_\star^\Pi(u,t,l)$, which is therefore jointly measurable on $\mathbb{R}\times\mathcal{T}$.

The assembly over l, which is where clause (i)'s three-fold joint measurability is reached. The

three paragraphs above were run at the l fixed at the head of the first and used nothing about it, since A2c supplies a single $\mathcal{T}_*$ serving every l and $\ell^{\Pi}_{\star}$ is defined slicewise at each; so both conclusions hold at *every* l. Clause (i) asserts joint measurability on $\mathbb{R} \times \mathcal{T} \times \{1, \ldots, \mathsf{L}-1\}$, and the $\mathsf{L}-1$ section statements deliver exactly that, by Step 0's section criterion applied to $\ell^{\Pi}_{\star}$ and to $\partial_u \ell^{\Pi}_{\star}$. *So the* $\mathsf{L}-1$ *contrasts are handled by one argument and not by* $\mathsf{L}-1$ *arguments, and the assembly costs nothing, no constant, no hypothesis, and no interchange. At* $\mathsf{L}=2$ *the disjoint union has one term and this paragraph is vacuous*, which is why earlier rounds could omit it without error.

And it is a version. The set $\{\ell^{\Pi}_{\star} \neq \ell^{\Pi}_{0}\}$ is now known to be measurable in all three arguments, by the same criterion. Its section at each (t, l) with $t \in \mathcal{T}_*$ is Lebesgue-null, hence Π_u-null, *because B1(iv.a) makes* Π_u *absolutely continuous with respect to Lebesgue measure*, a second use of that clause, and like the first it is substantive; and $\Pi_t(\mathcal{T} \setminus \mathcal{T}_*) = 0$ handles the remaining sections, at every l. Tonelli on $\Pi_u \otimes \Pi_t$ therefore gives $\Pi_u \otimes \Pi_t$-measure zero to each of the $\mathsf{L}-1$ sections, and $\Pi(\{\ell^{\Pi}_{\star} \neq \ell^{\Pi}_{0}\})$ is the finite sum of those $\mathsf{L}-1$ zeros against the masses $\varpi_{\mathsf{w}}(l)$, hence zero *whatever those masses are*. So $\ell^{\Pi}_{\star}$ as just defined is a version of the element of $L^2(\Pi)$ that A2c and A2′ speak about.

Boundedness and vanishing at $\pm\infty$ *of both slices* are Step 1. This is what A2′'s "admits a version" clause requires, and it is the point at which the m-3 defect, that neither $\sup_u \|\cdot\|$ nor "continuously differentiable in u" is invariant under Π-null modification, is settled by exhibiting the version rather than asserting one.

Step 3: measurability of the two suprema, first half of clause (iii). For $t \in \mathcal{T}_*$ and every l, both $u \mapsto \ell^{\Pi}_{\star}(u, t, l)$ and $u \mapsto \partial_u \ell^{\Pi}_{\star}(u, t, l)$ are continuous, by Steps 1 and 2; for $t \notin \mathcal{T}_*$ both are identically zero. *The two suprema are of the norms, and the norm is continuous*, so in either case $u \mapsto \|\ell^{\Pi}_{\star}(u, t, l)\|$ and $u \mapsto \|\partial_u \ell^{\Pi}_{\star}(u, t, l)\|$ are continuous and each supremum over $u \in \mathbb{R}$ equals the supremum over $u \in \mathbb{Q}$, a countable set. A countable supremum of measurable functions of t is measurable, and joint measurability from Step 2 makes each $t \mapsto \ell^{\Pi}_{\star}(u, t, l)$

and $t \mapsto \partial_u \ell^{\Pi}_{\star}(u,t,l)$ measurable at fixed u and fixed l, hence so are their norms, the norm being continuous and so Borel. Hence $t \mapsto \sup_u \|\ell^{\Pi}_{\star}(u,t,l)\|$ and $t \mapsto \sup_u \|\partial_u \ell^{\Pi}_{\star}(u,t,l)\|$ are Π_t-measurable *at every* l, which is what clause (iii) asserts, and the single-contrast integrals of Step 4, and hence their maxima $\Lambda^{(0)}_{\star}$ and $\Lambda^{(1)}_{\star}$, are well defined in $[0,\infty]$, a maximum over a finite set of $[0,\infty]$-valued numbers needing no further measurability. This step is not a formality: without it $\Lambda^{(0)}_{\star}$ and $\Lambda^{(1)}_{\star}$ are integrals of functions not known to be measurable, and the same countable-supremum device is the one Definition 53 uses for (73).

Step 4: integration, clause (iii). The two halves of (83) are integrated separately and the maximum over the contrast index is taken **inside** each budget before the two are added. That order is forced at $\mathsf{L} \geq 3$, and Step 1's separation is what makes it available. Fix $l \in \{1,\dots,\mathsf{L}-1\}$. The index has been explicit since Step 0; what is new here is that it is **aggregated by a maximum**. Steps 0, 1 and 3 carried l as a free parameter and used nothing about it beyond A2c's single $\mathcal{T}_*$; Step 4 takes the maximum. *This is not the only step that aggregates* l*, and Step 5 below says so in the plural*: Step 2 aggregates it as the finite sum of $\mathsf{L}-1$ zeros against the masses $\varpi_{\mathsf{w}}(l)$, which is the second of the two forms Step 5 names. Each half of (83) holds at every $t \in \mathcal{T}_*$ by Step 1, and trivially, both sides being zero, at every other t. Multiply each by $\bar{w}(t) \geq 0$, nonnegativity being part of B1(ii)'s definition of $\bar{w}$, and integrate against Π_t over $\mathcal{T}_*$, off which every integrand below vanishes. Write $\Lambda^{(0)}_{l}$, $\Lambda^{(1)}_{l}$ and $\Lambda^{(\mu'')}_{l}$ for the three single-contrast integrals so produced, whose maxima over l are $\Lambda^{(0)}_{\star}$, $\Lambda^{(1)}_{\star}$ and $\Lambda^{(\mu'')}_{\star}$ by A2′(i.a)–(i.b) and (4). Then, at *every* l,

$$\Lambda^{(0)}_{l} \;\leq\; c^{(0)}_{\mu''}\Lambda^{(\mu'')}_{l} \;\leq\; c^{(0)}_{\mu''}\Lambda^{(\mu'')}_{\star}, \qquad \Lambda^{(1)}_{l} \;\leq\; c^{(1)}_{\mu''}\Lambda^{(\mu'')}_{l} \;\leq\; c^{(1)}_{\mu''}\Lambda^{(\mu'')}_{\star}.$$

Maximising each over the finite index set gives $\Lambda^{(0)}_{\star} \leq c^{(0)}_{\mu''}\Lambda^{(\mu'')}_{\star}$ and $\Lambda^{(1)}_{\star} \leq c^{(1)}_{\mu''}\Lambda^{(\mu'')}_{\star}$; adding the two and applying (82) gives (85), whose right-hand side is finite by A2c. Both $\Lambda^{(0)}_{\star} < \infty$ and $\Lambda^{(1)}_{\star} < \infty$ follow because both summands are nonnegative. Integrating the combined bound (84) instead does not reach (85) at $\mathsf{L} \geq 3$, and the failing step is worth naming. It delivers

$\Lambda_l^{(0)} + \Lambda_l^{(1)} \le c_{\mu''}\Lambda_\star^{(\mu'')}$ at each l, hence $\max_l\{\Lambda_l^{(0)} + \Lambda_l^{(1)}\} \le c_{\mu''}\Lambda_\star^{(\mu'')}$; but $\max_l a_l + \max_l b_l \ge \max_l\{a_l + b_l\}$ runs the wrong way, the two budgets being maxima that may be attained at *different* contrasts. *At* $\mathsf{L} = 2$ *the two routes coincide*, the maximum being over a one-element set, and the separated route is then simply sharper by the factor $\sqrt{2}$ of (82). *The supremum was never moved across the* Π_t*-integral*: (83) holds pointwise in t with the supremum already taken, which is what makes A2c's geometry the right one and what meets the target M-1 raised.

Step 5: the consequence for $\ell_\star$. A2′(i.a) and A2′(i.b) now hold at the version of Step 2. *A2b requires one thing more, and it is verified and not inherited*: that $\ell := \mathcal{W}\ell_\star^\Pi$ be defined *pointwise* from that version through (68), *columnwise in* l*, which is the arity that display carries.* Fix (u, z) and l. The integrand $t \mapsto \ell_\star^\Pi(u, t, l)\, w(t'\Phi(z))$ is Π_t-measurable, because Step 2 makes $t \mapsto \ell_\star^\Pi(u, t, l)$ measurable at each fixed u and each fixed l and $t \mapsto w(t'\Phi(z))$ is continuous; and it is dominated in norm by $\big(\sup_{u'} \|\ell_\star^\Pi(u', t, l)\|\big)\bar{w}(t)$, using B1(ii)'s envelope $|w(t'\Phi(z))| \le \bar{w}(t)$, whose Π_t-integral is $\Lambda_l^{(0)}$ and hence *at most* $\Lambda_\star^{(0)}$, that budget being the maximum over the contrast index of exactly this integral, and only finiteness is used here, and so finite by (85). So the integral in (68) converges absolutely at every (u, z) and every l, and $\ell_\star$ is defined pointwise, which is A2b's last clause; *the mass* $\varpi_{\mathsf{w}}(l)$ *that* (68) *carries in front is a bounded factor and does not enter the convergence question.* The same argument at $\partial_u\ell_\star^\Pi$, with $\Lambda_\star^{(1)}$ in place of $\Lambda_\star^{(0)}$, makes the second integral of (70) absolutely convergent at every (u, z) and every l as well; A2b does not require that, but Corollary 49's proof forms the object. *This check is printed because it is what gates the corollary*: the two budgets and the differentiability do not by themselves give the pointwise definition, which needs the measurability Step 2 supplies and the envelope B1(ii) supplies. Hence $\ell_\star^\Pi$ satisfies A2b with $\Lambda^{(0)} = \Lambda_\star^{(0)}$ and $\Lambda^{(1)} = \Lambda_\star^{(1)}$, and Corollary 49 applies at $\ell^\Pi = \ell_\star^\Pi$ under B1(i), (ii), (iv.a), (iv.d). Its two conclusions with (85) substituted give $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty c_{\mu''}\Lambda_\star^{(\mu'')}$ and, using $\Lambda_\star^{(0)} \vee \Lambda_\star^{(1)} \le c_{\mu''}\Lambda_\star^{(\mu'')}$, a weakening of (85), each budget being nonnegative and so at most their sum, $\|\partial_u\ell_\star\|_\infty \le (\|\varpi_u'\|_\infty + \|\varpi_u\|_\infty)c_{\mu''}\Lambda_\star^{(\mu'')}$. *What enters Steps 0–4, exactly.*

A2c, and **B1(iv.a)–(iv.b)** at two points where the use is *substantive and not definitional*: *finiteness* of Π_u and Π_t makes $\Pi_u \otimes \Pi_t$ a product of σ-finite measures, which is what licenses Tonelli, at the close of Step 0 and again at the close of Step 2, *and* ***not*** *what supplies the product-measurable representative, which Definition 5's uncompleted product σ-field gives outright, as Step 0 now records*; and *absolute continuity* of Π_u with respect to Lebesgue measure is what converts a Lebesgue-null modification of a slice into a Π-null modification, which is what makes Step 2's construction a version. B1(ii) enters only as the *definition* of $\bar{w}$, the symbol A2c is written in, its nonnegativity being part of that definition; and B1(i), (ii), (iv.a), (iv.d) enter at this last step, *in* ***two*** *ways of which only one is through Corollary 49*. B1(ii)'s envelope $|w(t'\Phi(z))| \leq \bar{w}(t)$ dominates (68)'s integrand *directly*, which is what makes $\ell_\star := \mathcal{W}\ell_\star^\Pi$ defined pointwise, A2b's last clause, and hence what entitles Corollary 49, stated at an admissible representer, to be applied at all, and the corollary then supplies the two sup-norm bounds under B1(i), (ii), (iv.a) and (iv.d) jointly. B1(i) and B1(iv.d) enter there and nowhere else in this proof; B1(iv.a) enters there and, for the different reason above, at the two earlier points. *Nothing else*: not C1, not A2, and no further clause of B1, *B1(iv.a)–(iv.b) entering at Steps 0 and 2 and nowhere else before Step 5. In particular* ***B1(iv.d) does not enter before Step 5*** — the scope is the preceding clause's and is made explicit here because the statement-side mirror carries it explicitly, at clauses (i)–(iii), and *a quotation may not outlive its source — and that this accounting is unchanged at* $\mathsf{L} \geq 3$ *is a fact and not an oversight*: the third factor of Π is used only through Definition 5's declaration that its base is a *finite set with its power set*, which makes Π-measurability equivalent to measurability of the $\mathsf{L} - 1$ sections and makes every aggregation over l in this proof either a maximum or a finite sum of zeros. *No positivity of ϖ_{w} and no total mass of Π_{w} is consumed anywhere in Steps 0–4*; the one sentence of Step 0 that mentions the masses is printed as a conditional and is consumed nowhere. *The separation is what makes the discharge of* (O1) *clean, and it survives the correction*: the sufficient condition is a statement about the smoothness and tail behaviour of one function, together with two structural properties of the reference measure

that the block assumes anyway, and not about the model. □

Proof of Lemma 63. Clause (i) collects what Lemma 55(iii) proves; clause (ii) is the one construction, and (iii), (iv), (v) follow from it and from the inherited spectral chain. The multiplier $\vartheta'\sigma_q/\pi$ is a real-valued element of $L^\infty(\pi)$ and is *written out throughout.*

(i). The two decompositions are (74) and (158) read together, which is what naming the multiplication part bought. $\mathcal{B}_{\text{mult}}$ is multiplication by a real bounded function, hence self-adjoint on $L^2(\pi)$, and its norm is exactly $\|\vartheta'\sigma_q/\pi\|_{L^\infty(\pi)}$ by Conway (1990, Thm. II.1.5), whose σ-finiteness hypothesis holds here because $\pi\,dx$ is a *probability* measure on $\mathcal{X}$, *and that hypothesis is not decoration*: Conway exhibits its failure, an infinite atom on which the multiplication operator vanishes while the multiplier does not. The bound $\|\vartheta'\sigma_q/\pi\|_\infty \le 2\,c_{\widetilde{W}}\,\bar{c}_\pi\|\partial_u\ell_\star\|_\infty$ is (157), *restated with its constant,* $c_{\widetilde{W}} = \min\{\sqrt{2}, \sqrt{\mathsf{L}-1}\}$, *which is what Lemma 55*(iii) *prints and what* (P1b) *delivers; at* $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ *and this is the display earlier rounds printed*, and the essential supremum with respect to $\pi\,dx$ coincides with the one with respect to Lebesgue measure on $\mathcal{X}$ because C1(ii) bounds π above and below away from zero, so the two measures have the same null sets. Compactness and self-adjointness of $\vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ is the Hilbert–Schmidt computation of Lemma 55(iii) together with the facts that an adjoint of a compact operator is compact and a sum of two compact operators is compact. That $\mathcal{B}$ is bounded but not self-adjoint is Lemma 55(iii), the failure of symmetry being that of the kernel $q\,f_{X_1,X_2}$.

(ii). Let $\varepsilon > 0$, let $\mathcal{X}_0$ be as in the statement, and let $\mathsf{c}_0 = \operatorname{ess\,sup}_{\mathcal{X}_0} |\vartheta'\sigma_q/\pi|$ be as declared there. Such a set exists whenever $\vartheta'\sigma_q \not\equiv 0$, and the order of the two operations matters. Take first $\mathcal{A}_\varepsilon := \{|\vartheta'\sigma_q/\pi| > \|\vartheta'\sigma_q/\pi\|_\infty - \varepsilon\}$, of positive π-measure by definition of the essential supremum, and *then* split it by the sign of the multiplier; at least one of the two halves has positive measure, and $\mathcal{X}_0$ is that half. On it the multiplier has one sign and $|\vartheta'\sigma_q/\pi| > \|\vartheta'\sigma_q/\pi\|_\infty - \varepsilon$, so $\mathsf{c}_0 \ge \|\vartheta'\sigma_q/\pi\|_\infty - \varepsilon$ and *a fortiori* $|\vartheta'\sigma_q/\pi| \ge \mathsf{c}_0 - \varepsilon$ on $\mathcal{X}_0$, since $\mathsf{c}_0 \le \|\vartheta'\sigma_q/\pi\|_\infty$. *Performing the two operations in the other order , fixing a one-signed set*

first and intersecting it with $\mathcal{A}_\varepsilon$ afterwards, can produce the empty set, because the essential supremum may be approached only where the multiplier has the other sign.

Step 1: an orthonormal sequence supported in $\mathcal{X}_0$. The measure $\pi\,dx$ restricted to $\mathcal{X}_0$ is finite, non-atomic and of positive total mass, non-atomic because it is absolutely continuous with respect to Lebesgue measure on an interval, which is *m-1's clause* and is what Lemma 55(iii) already isolates. A finite non-atomic measure of positive mass admits, for every k, a partition of $\mathcal{X}_0$ into 2^k sets of positive measure; so $L^2(\mathcal{X}_0,\pi)$ is infinite dimensional and contains an orthonormal sequence $(v_k)_{k\ge1}$. Extend each v_k by zero to $\mathcal{X}$; orthonormality is preserved and $\operatorname{supp} v_k \subseteq \mathcal{X}_0$.

Step 2: $v_k \rightharpoonup 0$, and the two consequences. An orthonormal sequence in a Hilbert space converges weakly to zero, since Bessel's inequality $\sum_k \langle v_k,\phi\rangle^2 \le \|\phi\|^2$ forces $\langle v_k,\phi\rangle \to 0$ for every ϕ. A compact operator maps a weakly null sequence to a norm null one; applied to the compact $\tilde{\mathcal{K}}$, to $\tilde{\mathcal{K}}^*$ and to the compact $\mathcal{A}_{\mathrm{cond}}$ of Proposition 9, *this is the single use of C2′(i) in the lemma, transitive through that proposition, and it is why the hypothesis line carries C2′(i) separately from Lemma 55's block*, this gives $\|\mathcal{A}_{\mathrm{cond}} v_k\|_{L^2(\Pi)} \to 0$ and

$$\left|\left\langle \vartheta'(\tilde{\mathcal{K}}+\tilde{\mathcal{K}}^*)v_k, v_k\right\rangle_{L^2(\pi)}\right| \;\le\; \left\|\vartheta'(\tilde{\mathcal{K}}+\tilde{\mathcal{K}}^*)v_k\right\|_{L^2(\pi)} \;\longrightarrow\; 0.$$

Step 3: correction into $\mathcal{V}$, which is the compression in its cheapest form. Put $\tilde v_k := P_{\mathcal{V}} v_k = v_k - \langle v_k,1\rangle_{L^2(\pi)}\cdot 1$. Since $1 \in L^2(\pi)$ and $v_k \rightharpoonup 0$, $\langle v_k,1\rangle_{L^2(\pi)} \to 0$; as $\|1\|_{L^2(\pi)} = 1$ because π is a probability density, $\|\tilde v_k - v_k\|_{L^2(\pi)} \to 0$ and $\|\tilde v_k\|_{L^2(\pi)} \to 1$, so $\tilde v_k \neq 0$ for large k. Set $h_k := \tilde v_k/\|\tilde v_k\|_{L^2(\pi)} \in \mathcal{V}$, of unit norm; then $h_k - v_k \to 0$ in $L^2(\pi)$, so $h_k \rightharpoonup 0$, $\|\mathcal{A}_{\mathrm{cond}} h_k\| \to 0$, and, $\mathcal{B}_{\mathrm{sym}}$ being bounded, $\langle \mathcal{B}_{\mathrm{sym}} h_k, h_k\rangle - \langle \mathcal{B}_{\mathrm{sym}} v_k, v_k\rangle \to 0$.

Step 4: the value of the limit. By Step 2 and (i), $\langle \mathcal{B}_{\mathrm{sym}} v_k, v_k\rangle = \int_{\mathcal{X}_0} (\vartheta'\sigma_q/\pi) v_k^2 \pi\,dx + o(1)$, which is (87). On $\mathcal{X}_0$ the multiplier has one sign and modulus at least $\mathsf{c}_0 - \varepsilon$, while $\int_{\mathcal{X}_0} v_k^2 \pi\,dx = 1$; hence $|\int_{\mathcal{X}_0} (\vartheta'\sigma_q/\pi) v_k^2 \pi\,dx| \ge \mathsf{c}_0 - \varepsilon$ for every k, with the sign of $\vartheta'\sigma_q$ on $\mathcal{X}_0$. That sequence

of integrals is bounded, so it has a convergent subsequence, along which the limit in (87) exists and has modulus at least $\mathsf{c}_0 - \varepsilon$; relabelling gives the stated sequence.

(iii). This is the inherited chain, cited and not repeated. Lemma 55(iii) proves: $\pi\,dx$ is non-atomic, so $\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{mult}}) = \operatorname{ess\,ran}(\vartheta'\sigma_q/\pi)$; $\mathcal{B}_{\mathrm{sym}} - \mathcal{B}_{\mathrm{mult}} = -\vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ is compact, so Weyl's theorem, Kato (1995, Thm. IV.5.35), with the definitional and complexification bridges paid at that lemma and not repeated here, gives $\sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}}) = \sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{mult}})$; and $\mathcal{V}$ has codimension one, so $\mathcal{B}_{\mathrm{sym}} - P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}$ has rank at most two, whence $\sigma_{\mathrm{ess}}(P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}) = \sigma_{\mathrm{ess}}(\mathcal{B}_{\mathrm{sym}})$. *The one instance of Weyl's theorem that is consumed is* ***also*** *proved here, from the criterion and over the real field, and the pin now carried at Lemma 55*(iii) *does not make these four lines redundant*: what is pinned there is the *invariance*, over complex scalars, and what is run here is the *criterion*, over $\mathbb{R}$, a different statement on a different field. For self-adjoint S on a real Hilbert space, $\lambda \in \sigma_{\mathrm{ess}}(S)$ if and only if there is an orthonormal sequence (v_k) with $\|(S-\lambda)v_k\| \to 0$. Given that criterion, let C be compact and self-adjoint and let (v_k) be such a sequence for S. Orthonormality gives $v_k \rightharpoonup 0$, so $\|Cv_k\| \to 0$ and $\|(S+C-\lambda)v_k\| \to 0$; hence $\sigma_{\mathrm{ess}}(S) \subseteq \sigma_{\mathrm{ess}}(S+C)$, and the reverse inclusion follows by exchanging S for $S+C$ and C for $-C$. *The lower bound* (88) then holds because the compression is bounded and self-adjoint on $\mathcal{V}$, so its norm is $\max\{|\lambda| : \lambda \in \sigma(\cdot)\}$, which dominates $\sup\{|\lambda| : \lambda \in \sigma_{\mathrm{ess}}(\cdot)\} = \sup\{|\lambda| : \lambda \in \operatorname{ess\,ran}(\vartheta'\sigma_q/\pi)\} = \|\vartheta'\sigma_q/\pi\|_\infty$. *Independently of the chain*, (88) also follows from (ii): by Step 3–4 of that clause, $\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\| \geq \limsup_k |\langle \mathcal{B}_{\mathrm{sym}}h_k, h_k\rangle| \geq \mathsf{c}_0 - \varepsilon$, and taking $\mathcal{X}_0$ as constructed in (ii), for which $\mathsf{c}_0 \geq \|\vartheta'\sigma_q/\pi\|_\infty - \varepsilon$, so that the right-hand side is at least $\|\vartheta'\sigma_q/\pi\|_\infty - 2\varepsilon$, and then letting $\varepsilon \downarrow 0$ gives the same bound with no appeal to any spectral theorem. *The two routes agree, and the second is what makes the first checkable*: it is also the constructive confirmation the strategy memo §3.5 asks for, delivered with orthonormality in place of that memo's $\cos(kx)$, which needs no smoothness of the multiplier.

The nearest published counterpart of the criterion used above is cited as a comparison and ***not*** *as a substitute*, on the same footing as the numerical-radius comparison at Proposition 27

Step 2: Kato (1995, Thm. IV.5.11) characterises an infinite *approximate nullity* by a normalised sequence with no convergent subsequence, on a complex Banach space. *Three things separate it from the line above and each of them matters*: it is stated over $\mathbb{C}$; there is no inner product in its setting, so its sequence is normalised rather than orthonormal; and it prices approximate nullity, reaching σ_{ess} only through two further items of that section. *Four self-contained lines over* $\mathbb{R}$ *are shorter than that bridge would be*, which is why they are run and not looked up.

(iv). If $\vartheta'\sigma_q = 0$ π-a.e. then $\mathcal{B}_{\mathrm{mult}} = 0$ and $\mathcal{B}_{\mathrm{sym}} = -\vartheta'(\tilde{\mathcal{K}} + \tilde{\mathcal{K}}^*)$ is compact by (i); its compression is then compact too, being a product of a compact operator with two bounded projections. Conversely, suppose $\vartheta'\sigma_q \neq 0$ on a set of positive π-measure and let (h_k) be the sequence of (ii) built on such a set with $\varepsilon < \mathsf{c}_0$. A compact self-adjoint C satisfies $\langle Ch_k, h_k\rangle \to 0$ for every weakly null (h_k), by Step 2's argument; but $|\langle \mathcal{B}_{\mathrm{sym}} h_k, h_k\rangle| \geq \mathsf{c}_0 - \varepsilon > 0$ along the subsequence, and the same holds with $\mathcal{B}_{\mathrm{sym}}$ replaced by its compression because $h_k \in \mathcal{V}$. So neither is compact.

(v). $\int_{\mathcal{X}} \sigma_q \, dx = 0$ is Lemma 45(v) applied with $\Xi = \partial_u \ell_\star$, integrable by Corollary 49 under A2′(i.a)–(i.b), as Lemma 55(iii) records. Hence $\int_{\mathcal{X}} (\vartheta'\sigma_q/\pi)\, \pi \, dx = \int_{\mathcal{X}} \vartheta'\sigma_q \, dx = 0$. A real function with zero mean under $\pi\, dx$ that is not π-a.e. zero must be strictly positive on a set of positive measure and strictly negative on another; write $\mathcal{X}_+$ and $\mathcal{X}_-$ for the sign sets $\{\vartheta'\sigma_q/\pi > 0\}$ and $\{\vartheta'\sigma_q/\pi < 0\}$, both therefore of positive measure. *Clause* (ii) *is not applied at* $\mathcal{X}_\pm$ *themselves, and could not be*: they need not meet its premise, because the multiplier may come arbitrarily close to zero on part of a sign set, and there $|\vartheta'\sigma_q/\pi| \geq \mathsf{c}_0 - \varepsilon$ fails for every ε small enough to leave $\mathsf{c}_0 - \varepsilon$ positive. Run instead the construction of (ii)'s proof inside each sign set. On the positive side, let ε be less than half the essential supremum of $|\vartheta'\sigma_q/\pi|$ over $\mathcal{X}_+$, strictly positive, since the multiplier is strictly positive on a subset of $\mathcal{X}_+$ of positive measure, and take for $\mathcal{X}_0$ the near-supremum set of $|\vartheta'\sigma_q/\pi|$ at level ε *relative to* $\mathcal{X}_+$. It has positive measure by definition of the essential supremum; it lies in $\mathcal{X}_+$, so the multiplier is of one sign on it; and, by the computation already made in (ii), its own

c_0 is at most the essential supremum over $\mathcal{X}_+$, so $|\vartheta'\sigma_q/\pi| \geq \mathsf{c}_0 - \varepsilon$ on it, with $\mathsf{c}_0 - \varepsilon$ at least that essential supremum less 2ε and hence strictly positive. *The ordering caveat of* (ii) *is vacuous in this instance*: the near-supremum set is formed *inside* a set that already carries one sign, so no sign split is performed and no intersection can come out empty. The mirror construction inside $\mathcal{X}_-$ does the same on the negative side. Clause (ii) at those two sets produces sequences along which $\langle \mathcal{B}_{\mathrm{sym}} h_k, h_k \rangle$ has a nonzero limit of each sign, so the compression is neither positive nor negative semidefinite; and by (iii) its essential spectrum, being $\operatorname{ess\,ran}(\vartheta'\sigma_q/\pi)$, contains points of both signs, which gives the same conclusion for $\mathcal{B}_{\mathrm{sym}}$. *This is the formal content of the observation that the difference structure buys the mean of* h^2 *and nothing more*: $\int \nu_g \, dx = 0$ constrains a *difference* functional and $\int \sigma_q \, dx = 0$ constrains only the *mean* of the sum functional, leaving its shape, and hence the size and the sign of the form, free. □

Proof of Proposition 27. Clause (i) is three elementary facts about quadratic forms; (ii) collects Lemma 63; (iii) runs on (88) and the numerical-radius identity of Step 2; and (iv) alone runs on Lemma 63(ii)'s sequence, *which is where, and only where, C2′(i) is consumed. Within this proof only*, write $\mathcal{B}_{\mathcal{V}} := P_{\mathcal{V}} \mathcal{B}_{\mathrm{sym}} P_{\mathcal{V}}|_{\mathcal{V}}$; every statement of Lemma 63 and Proposition 27 prints the compression in full. *The subscript type carries the meaning*: a *word* subscript names a part of $\mathcal{B}$ (sym, mult), a *space* subscript names the compression to that space. No new letter and no fourth font of B is introduced.

Step 0: (49) is the object it claims to be. The first equality is Lemma 56(i) contracted in ϑ, which is legitimate coordinatewise because $\partial^2_{\tau_1\tau_1}\Psi(0)$ is the $\mathbb{R}^p$-valued expectation $E[\partial_u \ell_\star \widetilde{W} (\Delta h)^2]$, *the contrast to the* ***right*** *of the* $p \times (\mathsf{L}-1)$ *derivative, which is the order Lemma 56*(i) *prints and the only one that multiplies;* $(\Delta h)^2$ *is a scalar and commutes*, and ϑ is a constant vector. *That lemma is stated at a fixed admissible direction pair, and the pair used here is* $(0,0)$: $\mathsf{r} \equiv 0$ is Borel and bounded, $D \equiv 0$ has $\|D\|_{0,h} = \|D\|_{1,h} = 0$, so the pair is admissible and differentiable at every $(h, \bar{\tau})$ by Definition 53, and $\Psi(\cdot, 0, 0) = \Psi_{L_\star}$

by (32). *So no condition on a direction in L or in ϱ is inherited by this proposition*, which matters because Remark 57's forward obligation (O3) concerns exactly such conditions and does not reach here. The second is the transfer performed in Lemma 55(iii): for $h \in \mathcal{V}$, $\int(\vartheta'\sigma_q)h^2dx = \langle \mathcal{B}_{\text{mult}}h, h\rangle_{L^2(\pi)}$ and $\langle \vartheta'\mathcal{K}h, h\rangle_{L^2(dx)} = \langle \vartheta'\tilde{\mathcal{K}}h, h\rangle_{L^2(\pi)}$, so *both* terms are conjugated and the multiplication is by $\vartheta'\sigma_q/\pi$. The expectation in (49) is finite because $\|\partial_u \ell_\star\|_\infty < \infty$ by Corollary 49 and $E[(\Delta h)^2] \le 4\bar{c}_\pi \|h\|^2_{L^2(\pi)}$ by Lemma 45(i).

Step 1: only the symmetric part is seen, and only its compression. For $h \in \mathcal{V}$, write $\mathcal{B} = \mathcal{B}_{\text{sym}} + \mathcal{B}_{\text{anti}}$ with $\mathcal{B}_{\text{anti}} := \frac{1}{2}(\mathcal{B} - \mathcal{B}^*)$. Over the reals $\langle \mathcal{B}_{\text{anti}}h, h\rangle = \langle h, \mathcal{B}^*_{\text{anti}}h\rangle = -\langle \mathcal{B}_{\text{anti}}h, h\rangle$, so the antisymmetric part contributes zero and $\langle \mathcal{B}h, h\rangle = \langle \mathcal{B}_{\text{sym}}h, h\rangle$. Since $h = P_{\mathcal{V}}h$, $\langle \mathcal{B}_{\text{sym}}h, h\rangle = \langle P_{\mathcal{V}}\mathcal{B}_{\text{sym}}P_{\mathcal{V}}h, h\rangle = \langle \mathcal{B}_{\mathcal{V}}h, h\rangle$, an inner product in $\mathcal{V}$. This is the step that makes the constant the compression's and not $\mathcal{B}_{\text{sym}}$'s, and it is forced by Definition 5: h ranges over $\mathcal{V}$ because that is the parameter space, while $\mathcal{B}$ is read on all of $L^2(\pi)$ because $\mathcal{V}$ is not invariant under a non-constant multiplication.

Step 2: the numerical radius of a self-adjoint operator is its norm. $\mathcal{B}_{\mathcal{V}}$ is bounded and self-adjoint on the real Hilbert space $\mathcal{V}$. Put $\mathsf{c}_1 := \sup_{h\in\mathcal{V},\|h\|=1} |\langle \mathcal{B}_{\mathcal{V}}h, h\rangle|$. Clearly $\mathsf{c}_1 \le \|\mathcal{B}_{\mathcal{V}}\|$. For the converse, symmetry gives, for any $x, y \in \mathcal{V}$, $\langle \mathcal{B}_{\mathcal{V}}(x+y), x+y\rangle - \langle \mathcal{B}_{\mathcal{V}}(x-y), x-y\rangle = 4\langle \mathcal{B}_{\mathcal{V}}x, y\rangle$, so for $\|x\| = \|y\| = 1$

$$4\Big|\big\langle \mathcal{B}_{\mathcal{V}}x, y\big\rangle\Big| \;\le\; \mathsf{c}_1\Big(\|x+y\|^2 + \|x-y\|^2\Big) \;=\; 2\,\mathsf{c}_1\Big(\|x\|^2 + \|y\|^2\Big) \;=\; 4\,\mathsf{c}_1,$$

using the parallelogram law; hence $\|\mathcal{B}_{\mathcal{V}}\| = \sup_{\|x\|=\|y\|=1} |\langle \mathcal{B}_{\mathcal{V}}x, y\rangle| \le \mathsf{c}_1$. *The real field is used twice*, in Step 1 and in the identity above, and over $\mathbb{C}$ the statement would concern the real part of the form. *The complex-field counterparts are Halmos (1982, §§218–219), and they are cited here as a comparison and not as a substitute*: he gives the two-sided equivalence $\frac{1}{2}\|A\| \le w(A) \le \|A\|$ and the identity $w(A) = \|A\|$ for *normal* A, both for a numerical range defined as a set of *complex* numbers, and over $\mathbb{R}$ the normal case is false, a rotation by a quarter turn has $\langle Ah, h\rangle \equiv 0$ with $\|A\| = 1$. *Self-adjointness, not normality, is what survives*

the change of field, which is why the two lines above are run rather than looked up.

Step 3: homogeneity, and clause (i). The map $h \mapsto \langle \mathcal{B}_{\mathcal{V}} h, h\rangle$ is homogeneous of degree two, so

$$\sup_{h\in\mathcal{V},\ \|h\|\le\delta} \left|\left\langle \mathcal{B}h, h\right\rangle\right| = \sup_{0\le c\le\delta}\ c^2 \sup_{h\in\mathcal{V},\|h\|=1} \left|\left\langle \mathcal{B}_{\mathcal{V}} h, h\right\rangle\right| = \delta^2\, \mathsf{c}_1 = \delta^2 \left\|\mathcal{B}_{\mathcal{V}}\right\|,$$

which is (50) after multiplying by $\frac{1}{2}$. *The supremum need not be attained*: it is attained at some h exactly when $\pm\|\mathcal{B}_{\mathcal{V}}\|$ is an eigenvalue of $\mathcal{B}_{\mathcal{V}}$, which by Lemma 63(iii) fails whenever the norm is achieved only on the essential spectrum. So the result is an identity for a supremum and must not be read as exhibiting a worst-case direction. *No hypothesis on q is used in Steps 0–3*: boundedness of $\mathcal{B}$, which is Lemma 55(iii), is all they consume.

Step 4: clause (ii). The first inequality of (51) is (88), which Lemma 63(iii) proves under $\vartheta'\sigma_q \not\equiv 0$, i.e. under O-16(b); when $\vartheta'\sigma_q \equiv 0$ it holds trivially, both sides of (88)'s left member being zero. The second is $\|\mathcal{B}_{\mathcal{V}}\| = \sup_{h\in\mathcal{V},\|h\|=1} \|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}h\| \le \sup_{\|h\|=1} \|\mathcal{B}_{\mathrm{sym}}h\| = \|\mathcal{B}_{\mathrm{sym}}\|$, using $P_{\mathcal{V}}h = h$ for $h \in \mathcal{V}$ and that an orthogonal projection is a contraction. The third is $\|\frac{1}{2}(\mathcal{B}+\mathcal{B}^*)\| \le \frac{1}{2}(\|\mathcal{B}\|+\|\mathcal{B}^*\|) = \|\mathcal{B}\|$. *Both outer inequalities can be strict, and the middle one is strict whenever the extreme point of the spectrum lies in the one-dimensional complement*, which is why the compression, and not $\mathcal{B}_{\mathrm{sym}}$, is the constant.

Step 5: clause (iii). Under O-16(a) there is $h_0 \in \mathcal{V}$ with $\vartheta' E[q(\Delta h_0)^2] \neq 0$, so by (49) $\langle \mathcal{B}_{\mathcal{V}} h_0, h_0\rangle \neq 0$ and therefore $\|\mathcal{B}_{\mathcal{V}}\| > 0$; with Step 3 this gives $\sup_{\|h\|\le\delta} \frac{1}{2}|\langle \mathcal{B}h, h\rangle| = \frac{1}{2}\delta^2\|\mathcal{B}_{\mathcal{V}}\|$, a positive multiple of δ^2 with a factor free of δ, so no function $o(\delta^2)$ can dominate it as $\delta \downarrow 0$. Under O-16(b), Step 4 gives $\|\mathcal{B}_{\mathcal{V}}\| \ge \|\vartheta'\sigma_q/\pi\|_\infty > 0$. And O-16(b) implies O-16(a), by a route that consumes no compactness. Under O-16(b), (88), which Lemma 63(iii) delivers through the inherited spectral chain, needing neither that lemma's clause (ii) nor C2′(i), gives $\|\mathcal{B}_{\mathcal{V}}\| > 0$, so Step 2's identity $\mathsf{c}_1 = \|\mathcal{B}_{\mathcal{V}}\|$ supplies $h \in \mathcal{V}$ of unit norm with $\langle \mathcal{B}_{\mathcal{V}} h, h\rangle \neq 0$, hence with $\vartheta' E[q(\Delta h)^2] \neq 0$ by (49) and Step 1. *Lemma 63*(ii), *at $\varepsilon < \mathsf{c}_0$, reaches the same conclusion, but at the price of C2′*(i), which is why the hypothesis line scopes that assumption to clause (iv) alone and why this step is written the other way round. *The*

converse implication is false, and Lemma 55(iii)'s witness is the counterexample: it exhibits $q \not\equiv 0$ with $\sigma_q \equiv 0$ and $E[q(\Delta h)^2] = 2c \neq 0$. So the two clauses of O-16 are genuinely different hypotheses, and Remark 65(3) records which conclusions survive when only the weaker holds.

Step 6: clause (iv). Suppose (52) held for some finite λ. Take the sequence (h_k) of Lemma 63(ii) built, under O-16(b), on the set $\mathcal{X}_0$ that clause's proof constructs, at any $\varepsilon < \frac{1}{2}\|\vartheta'\sigma_q/\pi\|_\infty$; there $\mathsf{c}_0 - \varepsilon \geq \|\vartheta'\sigma_q/\pi\|_\infty - 2\varepsilon > 0$. Along it $\|\mathcal{A}_{\mathrm{cond}}h_k\|_{L^2(\Pi)} \to 0$ while $\frac{1}{2}|\langle \mathcal{B}h_k, h_k\rangle|$ converges to a limit of modulus at least $\frac{1}{2}(\|\vartheta'\sigma_q/\pi\|_\infty - 2\varepsilon) > 0$, so the right-hand side of (52) tends to zero and the left-hand side does not: contradiction. For the second form, $\mathcal{A}^*_{\mathrm{cond}}$ is bounded, so $\|\mathcal{A}^*_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}}h_k\| \leq \|\mathcal{A}^*_{\mathrm{cond}}\|\|\mathcal{A}_{\mathrm{cond}}h_k\| \to 0$ while $\|h_k\| = 1$, and the same contradiction follows. *Compactness of* $\mathcal{A}_{\mathrm{cond}}$ *is what makes the argument work, and it is proved and not assumed* (Proposition 9); it is also the premise the project's every source-condition and regularisation statement already carries, so the refutation cannot be evaded by weakening it. *This clause is the sole consumer of C2′(i) in the proposition*, transitively through Proposition 9 and Lemma 63(ii), which is what the hypothesis line records. □

Proof of Lemma 66. (a). For $\mathsf{y} \in \mathbb{R}^{p_K} \setminus \{0\}$, $\mathsf{y}'\mathsf{G}(I)\mathsf{y} = \int_I \left(\sum_k \mathsf{y}_k \varsigma^k\right)^2 e^{-\varsigma^2/2} d\varsigma$. The integrand is continuous, nonnegative, and vanishes at no more than $p_K - 1$ points, while I has *positive* length; so the integral is strictly positive. *Positive length is all (a) uses*; the floor $\sup I - \inf I \geq 2$ of $\mathfrak{I}$ is consumed in (c), in (g), and in Lemma 67(iv), whose hypothesis $h_x \leq (\bar{x} - \underline{x})/2$ *is* that floor. *No claim of exhaustiveness is made, and that is deliberate*: every such claim this corpus has printed about this floor has been short by a site. *This is the step a* Θ_1*-weighted Gram–Schmidt cannot take*, its weight being sign-changing.

(b). Θ_I vanishes off I by the indicator in (89), and $\int_I \varsigma^j \Theta_I = \sum_k (\mathsf{G}(I)^{-1}e_0)_k \int_I \varsigma^{j+k} e^{-\varsigma^2/2} = \left(\mathsf{G}(I)\mathsf{G}(I)^{-1}e_0\right)_j = (e_0)_j = \delta_{j0}$.

(c), the topology. (90) is *injective*: tanh is strictly increasing and a closed interval is determined by its two endpoints. Its *image is closed* in $[-1, 0] \times [0, 1]$: if $I_r \in \mathfrak{I}$ and $\left(\tanh \inf I_r, \tanh \sup I_r\right)$ converges in that square, then $\inf I_r$ and $\sup I_r$ converge in

$[-\infty,+\infty]$, artanh being continuous on $[-1,1]$, and the closed interval $I_\circ$ with those two limits as endpoints satisfies $\inf I_\circ \le 0 \le \sup I_\circ$ and $\sup I_\circ - \inf I_\circ \ge 2$, each inequality passing to the limit; so $I_\circ \in \mathfrak{I}$ and the limit lies in the image. The boundary values $\tanh\inf I = -1$ and $\tanh\sup I = 1$ are the admitted half-lines and $\mathbb{R}$. Since $\mathfrak{I}$ carries *by definition* the topology (90) induces, it is homeomorphic to that image, and a closed subset of $[-1,0]\times[0,1]$ is compact and metrisable; *so $\mathfrak{I}$ is compact and metrisable*, which (c)'s own infimum argument below consumes first — a positive continuous function needs a *compact* set to attain a positive infimum on — and (d), (f) and (g) after it. *No claim of exhaustiveness is made here either, and that is deliberate*: the draft this sentence replaces omitted the reader three lines below it.

(c), the infimum. Writing $F_m(\cdot) := \int_{-\infty}^{\cdot} \varsigma^m e^{-\varsigma^2/2} d\varsigma$, we have $\mathsf{G}(I)_{jk} = F_{j+k}(\sup I) - F_{j+k}(\inf I)$, and each F_m is continuous on $[-\infty,+\infty]$ because the Gaussian weight has moments of every order; so G is continuous on $\mathfrak{I}$ for (90), and $\lambda_{\min}\circ\mathsf{G}$ is continuous by Weyl's inequality and strictly positive by (a). A positive continuous function on a compact set attains a positive infimum. *What this step needs of the floor is only that it be positive*: it is what keeps the degenerate point $\inf I = \sup I = 0$, at which $\mathsf{G} = 0$, out of the closure, and $\sup I - \inf I \ge 1$ would serve here. *The floor is therefore an absolute number and not a multiple of c_C*, which it could not be: c_C is defined from $\bar\Theta$, $\bar\Theta$ from the family $\{\Theta_I : I \in \mathfrak{I}\}$, and $\mathfrak{I}$ from the floor. *But the **value** 2 is not free.* (g) *spends it, and so does Lemma 67*(iv)*, whose hypothesis $h_x \le (\bar x - \underline x)/2$ **is** that value and would be a different inequality at any other; so this floor may not be loosened without re-deriving* (93) *and re-stating that hypothesis. Consistently with the disclaimer above, this is not offered as the complete list of the value's consumers.* Step 4 runs its construction **twice**: at ς with a free λ, and again at $\varsigma_\circ = 0$ with $\lambda_0 = 1$, the second needing $\max\{-\inf I, \sup I\} \ge 1$ and so $|I| \ge 2$ outright. Step 6 reads $[-2,0]$ at $|J| = 2$. Step 8 reads it again. *No counterfactual ceiling is quoted here, deliberately*: a smaller floor caps **both** parameters, so it moves $\mathsf{Q}_I(0)$ as well as $\mathsf{Q}_I(\varsigma)$, and a figure computed at one of them is not the ceiling at that floor — a draft of this sentence quoted such a figure. *That the*

live configuration delivers $|I| \geq 10.1$ *is why the floor binds nothing here, and is not a licence to change it.*

(d), the bound. $\mathsf{G}(I) \succ 0$ gives $\|\mathsf{G}(I)^{-1}e_0\|_2 \leq \|\mathsf{G}(I)^{-1}\|_{\mathrm{op}} = \lambda_{\min}(\mathsf{G}(I))^{-1} \leq \underline{g}^{-1}$ by (c), and Cauchy–Schwarz gives $\left|\sum_k (\mathsf{G}(I)^{-1}e_0)_k \varsigma^k\right| \leq \|\mathsf{G}(I)^{-1}e_0\|_2 (\sum_k \varsigma^{2k})^{1/2} \leq \underline{g}^{-1}(\sum_k \varsigma^{2k})^{1/2}$, uniformly in I. Multiplying by $e^{-\varsigma^2/2}$ and bounding the indicator of (89) by 1 gives the second half of (91), for every $I \in \mathfrak{I}$ at once. Its right-hand side is a fixed polynomial times a Gaussian, hence bounded, integrable and of finite mth absolute moment for every m; $\bar{\Theta}$ inherits all three.

(d), continuity. $\Theta_I(\varsigma) = 0$ at every I not containing ς, and $\mathbb{R} \in \mathfrak{I}$ contains every ς, so $\bar{\Theta}(\varsigma) = \sup\{|\Theta_I(\varsigma)| : I \in \mathfrak{I},\ \varsigma \in I\}$, a supremum over a *nonempty* set. By (c), $\mathfrak{I}$ is compact, and $\{(\varsigma, I) \in \mathbb{R} \times \mathfrak{I} : \varsigma \in I\}$ is the closed set $\{\tanh \inf I \leq \tanh \varsigma \leq \tanh \sup I\}$ in the coordinates (90), and on it the indicator of (89) is identically 1, so $(\varsigma, I) \mapsto \Theta_I(\varsigma)$ is jointly continuous there — by (f) for $I \mapsto \mathsf{G}(I)^{-1}e_0$ and by polynomial evaluation for the rest. *The appeal to* (f) *is not circular*: (f) is proved from (a) and (c) alone. The supremum is therefore attained, and continuous in ς in both directions. *Upper semicontinuity*: take $\varsigma_r \to \varsigma$ realising $\limsup_r \bar{\Theta}(\varsigma_r)$, let I_r attain the supremum at ς_r, and pass to a further subsequence with $I_r \to I_\circ$ in the compact parameter set; then $\varsigma \in I_\circ$, since $\tanh \inf I_r \leq \tanh \varsigma_r \leq \tanh \sup I_r$ passes to the limit, and $|\Theta_{I_r}(\varsigma_r)| \to |\Theta_{I_\circ}(\varsigma)| \leq \bar{\Theta}(\varsigma)$. *Lower semicontinuity*: fix $I_\circ = [a_\circ, b_\circ] \in \mathfrak{I}$ with $\varsigma \in I_\circ$ and put $I_r := [a_\circ \wedge \varsigma_r,\ b_\circ \vee \varsigma_r]$, which lies in $\mathfrak{I}$ — its left endpoint is still ≤ 0, its right endpoint still ≥ 0, and its length only grew — and contains ς_r, and $I_r \to I_\circ$ because $a_\circ \leq \varsigma \leq b_\circ$; so $\liminf_r \bar{\Theta}(\varsigma_r) \geq |\Theta_{I_\circ}(\varsigma)|$, and the supremum over $I_\circ$ gives $\liminf_r \bar{\Theta}(\varsigma_r) \geq \bar{\Theta}(\varsigma)$. Hence $\bar{\Theta}$ is continuous, so measurable, *and the supremum never had to be reduced to a countable dense subset* — a reduction (f) would not have licensed, since $I \mapsto \Theta_I(\varsigma)$ **jumps** at each I whose endpoint crosses ς. *Restricting to* $\{I : \varsigma \in I\}$*, which changes nothing, is what removes the jump.*

(d), minimality. If $\bar{\Theta}'$ satisfies C3(ii) then $\bar{\Theta}' \geq |\Theta_I|$ at every $I \in \mathfrak{I}$, hence $\bar{\Theta}' \geq \bar{\Theta}$ pointwise,

hence $\|\bar\Theta'\|_\infty \geq \|\bar\Theta\|_\infty$ and $\|\bar\Theta'\|_{L^1} \geq \|\bar\Theta\|_{L^1}$.

(e). At $I = \mathbb{R}$ the system $\mathsf{G}(\mathbb{R})\mathsf{y} = e_0$ says exactly that $\left(\sum_k \mathsf{y}_k \varsigma^k\right) e^{-\varsigma^2/2}$ integrates to 1 and has vanishing moments of orders $1, \dots, p_K - 1$ — i.e. that it is an order-p_K kernel of the form (polynomial of degree $\leq p_K - 1$) $\times$ Gaussian. The displayed Θ_1 is such a kernel: writing it as $\sum_{j=0}^{3} \frac{(-1)^j}{2^j j!} \mathrm{He}_{2j}(\varsigma)\,(2\pi)^{-1/2} e^{-\varsigma^2/2}$ with He the probabilists' Hermite polynomials — *upright, so that it is not read as $H \cdot e$ against the Sobolev scale H^s and the basis vector e_0* — orthogonality of the He_{2j} against the Gaussian gives $\int \Theta_1 = 1$ and $\int \varsigma^2 \Theta_1 = \int \varsigma^4 \Theta_1 = \int \varsigma^6 \Theta_1 = 0$, while the odd moments vanish by symmetry. *C3(i)'s third clause is the **absolute** moment* $\int_{\mathbb{R}} |\varsigma|^{p_K} |\Theta_1| < \infty$, which holds because Θ_1 is a polynomial times a Gaussian; the signed value $\int \varsigma^8 \Theta_1 = -105$ is a different statement and is recorded only because it is the leading bias constant. And $\mathsf{G}(\mathbb{R}) \succ 0$ by (a) makes the solution *unique*, so $\Theta_{\mathbb{R}} = \Theta_1$ and no compatibility between C3(i) and C3(ii) has to be imposed — it is delivered. Symmetry, smoothness and the Lipschitz bound are read off the closed form, with $\|\Theta_1\|_\infty = 105/(48\sqrt{2\pi}) = 0.873$ and $\|\Theta_1\|_{L^1} = 1.27$.

(f). $\mathsf{G}(-I) = \mathsf{D}\,\mathsf{G}(I)\,\mathsf{D}$ with $\mathsf{D} := \mathrm{diag}((-1)^j)_{j=0}^{p_K - 1}$, and $\mathsf{D}e_0 = e_0$, so $\mathsf{G}(-I)^{-1} e_0 = \mathsf{D}\,\mathsf{G}(I)^{-1} e_0$ and the polynomial at $-I$ is the polynomial at I evaluated at $-\varsigma$; the Gaussian weight is even and $\mathbb{1}\{\varsigma \in -I\} = \mathbb{1}\{-\varsigma \in I\}$. Continuity: G is continuous by (c) and matrix inversion is continuous on the invertible set, which (a) and (c) make all of $\mathfrak{I}$.

(g). One local symbol, bound here and dying here, on the precedent (f)*'s* D *sets. Read* (89)*'s* G*, and the display below, at every interval of **positive length** and not only on* $\mathfrak{I}$: Step 4 evaluates them at intervals of length $\lambda \leq 1$, which $\mathfrak{I}$'s floor excludes, and the extension is legitimate because (a)'s proof consumes positive length and nothing else, so G is positive definite — hence invertible — at every such interval. For such an I and for $\varsigma_\circ \in \mathbb{R}$ write $\mathsf{Q}_I(\varsigma_\circ) := \sum_{j,k} \varsigma_\circ^j \left(\mathsf{G}(I)^{-1}\right)_{jk} \varsigma_\circ^k$. *The letter is* Q *for the quadratic form it is; the obvious K is unavailable, bare italic K being live as Lemma 115's tolerance level — **one** reading, bound in place in each of that lemma's clauses* (i) *and* (ii) *— and $\mathcal{K}$ and* K *both being reserved.* Q *is*

free on the corpus's own sans-serif census, and its font partner $\mathcal{Q}$, the operator $\mathcal{V} \to L^2(\Pi)$, occurs nowhere in this proof — so the two are separated by scope as well as by font.

Step 1: an identity and a Cauchy–Schwarz. $\mathsf{G}(I)$ is symmetric, so $\left(\mathsf{G}(I)^{-1}e_0\right)_k = \left(\mathsf{G}(I)^{-1}\right)_{k0}$ and, by (89), $\Theta_I(\varsigma) = e^{-\varsigma^2/2}\sum_k \left(\mathsf{G}(I)^{-1}\right)_{k0}\varsigma^k \mathbb{1}\{\varsigma \in I\}$. Since $\mathsf{G}(I)^{-1} \succ 0$ by (a), $(\mathsf{u},\mathsf{y}) \mapsto \mathsf{u}'\mathsf{G}(I)^{-1}\mathsf{y}$ is an inner product on $\mathbb{R}^{p_K}$; Cauchy–Schwarz at the monomial vector $(\varsigma^j)_{j=0}^{p_K-1}$ and at $e_0 = (0^j)_{j=0}^{p_K-1}$ gives

$$\left|\Theta_I(\varsigma)\right| \;\le\; e^{-\varsigma^2/2}\,\mathsf{Q}_I(\varsigma)^{1/2}\,\mathsf{Q}_I(0)^{1/2} \qquad (\varsigma \in I),$$

and $\Theta_I(\varsigma) = 0$ off I, so the display holds at every $\varsigma \in \mathbb{R}$. *Note* $\Theta_I(0) = \mathsf{Q}_I(0)$ *exactly*, the Cauchy–Schwarz being an equality there; this is used twice below.

Step 2: a variational form, and the two monotonicities it makes immediate. By the Rayleigh characterisation of $\mathsf{G}(I)^{-1}$ at the monomial vector, and by the display in (a)'s proof identifying $\mathsf{y}'\mathsf{G}(I)\mathsf{y}$ with an integral,

$$\mathsf{Q}_I(\varsigma_\circ) = \sup\left\{\left(\textstyle\sum_k \mathsf{y}_k\varsigma_\circ^k\right)^2 \;:\; \mathsf{y} \in \mathbb{R}^{p_K},\ \textstyle\int_I \left(\sum_k \mathsf{y}_k\varsigma'^k\right)^2 e^{-\varsigma'^2/2}\,d\varsigma' \le 1\right\}.$$

Two consequences, each read straight off the constraint set. *Antitone in I*: if $J \subseteq I$ the constraint at I is the stronger one, so $\mathsf{Q}_I(\varsigma_\circ) \le \mathsf{Q}_J(\varsigma_\circ)$ — *no Loewner inversion step is needed, though the statement is the same one. Weight comparison*: on a bounded J, $e^{-\max_{\varsigma'\in J}\varsigma'^2/2} \le e^{-\varsigma'^2/2} \le 1$, so $\mathsf{Q}_J(\varsigma_\circ)$ lies between the value of the same supremum taken against $d\varsigma'$ alone and $e^{\max_J \varsigma'^2/2}$ times it.

Step 3: the unweighted value at an endpoint. Let J be a bounded interval and $\varsigma_\circ$ an endpoint of it. Then $\sup\{(\sum_k \mathsf{y}_k\varsigma_\circ^k)^2 : \int_J(\sum_k \mathsf{y}_k\varsigma'^k)^2 d\varsigma' \le 1\} = p_K^2/|J|$. Map J affinely onto $[-1,1]$ carrying $\varsigma_\circ$ to 1; this carries the polynomials of degree $\le p_K - 1$ onto themselves and scales $d\varsigma'$ by $|J|/2$. Write the polynomial in the Legendre basis $\{\mathrm{Leg}_j\}$ on $[-1,1]$ — *upright, on the precedent* (e)*'s* He *sets, italic P being probability* — normalised by $\int_{-1}^1 \mathrm{Leg}_j^2 = 2/(2j+1)$ and

$\mathrm{Leg}_j(1) = 1$, with coefficients $\tilde{\mathsf{y}}$. The constraint reads $\sum_j |J|\tilde{\mathsf{y}}_j^2/(2j+1) \le 1$ and the objective is $\big(\sum_j \tilde{\mathsf{y}}_j\big)^2$; Cauchy–Schwarz gives $\big(\sum_j \tilde{\mathsf{y}}_j\big)^2 \le \big(\sum_j |J|\tilde{\mathsf{y}}_j^2/(2j+1)\big)\big(\sum_{j=0}^{p_K-1}(2j+1)/|J|\big) \le p_K^2/|J|$, with equality at $\tilde{\mathsf{y}}_j \propto (2j+1)$, and $\sum_{j=0}^{p_K-1}(2j+1) = p_K^2$.

Step 4: the pointwise bound (92). Fix $I \in \mathfrak{I}$, $\varsigma \in I$ and $0 < \lambda \le 1$. Since $|I| \ge 2$, $\max\{\varsigma - \inf I,\ \sup I - \varsigma\} \ge |I|/2 \ge 1 \ge \lambda$, so one of $[\varsigma-\lambda, \varsigma]$, $[\varsigma, \varsigma+\lambda]$ lies in I; call it J, and note $J \subseteq [\varsigma-\lambda, \varsigma+\lambda]$, whence $\max_J \varsigma'^2 \le (|\varsigma|+\lambda)^2$. Steps 2 and 3 give $\mathsf{Q}_I(\varsigma) \le \mathsf{Q}_J(\varsigma) \le e^{(|\varsigma|+\lambda)^2/2} p_K^2/\lambda$. The same argument at $\varsigma_\circ = 0$ — which lies in I for every $I \in \mathfrak{I}$ — with $\lambda = 1$ gives $\mathsf{Q}_I(0) \le e^{1/2} p_K^2$. Substituting both in Step 1,

$$\big|\Theta_I(\varsigma)\big| \ \le\ \frac{p_K^2 e^{1/4}}{\sqrt{\lambda}} \exp\Big(\frac{\big(|\varsigma|+\lambda\big)^2}{4} - \frac{\varsigma^2}{2}\Big) \ =\ \frac{p_K^2 e^{1/4}}{\sqrt{\lambda}} \exp\Big(\frac{2\lambda^2 - \big(|\varsigma|-\lambda\big)^2}{4}\Big),$$

uniformly in I; taking $\lambda = 2^{-1/2}$ is (92), and taking the supremum over I on the left shows the right-hand side dominates $\bar{\Theta}$. It is a fixed function of Gaussian type $e^{-\varsigma^2/4}$ up to polynomial-free factors, hence bounded, integrable and of finite moment of every order, so it satisfies C3(ii).

Step 5: the upper bounds. The exponent above is maximised at $|\varsigma| = \lambda$, at value $\lambda^2/2$, so $\mathsf{c}_2 \le p_K^2 e^{1/4} \lambda^{-1/2} e^{\lambda^2/2}$; the exponent $-\frac{1}{2}\ln\lambda + \lambda^2/2$ is minimised over $(0,1]$ at $\lambda = 2^{-1/2}$, giving $\mathsf{c}_2 \le 2^{1/4} e^{1/2} p_K^2$. For the L^1 bound, the substitution $r = (\varsigma - \lambda)/2$ gives $\int_{\mathbb{R}} \exp\big(\{2\lambda^2 - (|\varsigma|-\lambda)^2\}/4\big) d\varsigma \ =\ 4e^{\lambda^2/2} \int_{-\lambda/2}^{\infty} e^{-r^2} dr \ =\ 2\sqrt{\pi} e^{\lambda^2/2}\big(1 + \mathrm{erf}(\lambda/2)\big)$, and at $\lambda = 2^{-1/2}$ this is $\|\bar{\Theta}\|_{L^1} \le 2^{5/4}\sqrt{\pi} e^{1/2}\big(1 + \mathrm{erf}(2^{-3/2})\big) p_K^2$. Dividing the two displays gives the pair (16) returns at the right-hand side of (92), in which p_K^2 cancels from c_C.

Step 6: the lower bound, which is the Legendre corroboration made into a theorem. $[-2, 0] \in \mathfrak{I}$, and 0 is an endpoint of it. By Step 1's equality and Step 2's weight comparison in the other direction ($e^{-\varsigma'^2/2} \le 1$), followed by Step 3 at $|J| = 2$, $\mathsf{c}_2 \ge \bar{\Theta}(0) \ge \Theta_{[-2,0]}(0) = \mathsf{Q}_{[-2,0]}(0) \ge p_K^2/2$. *The inequality is one-sided, and the note says why*: the measured value exceeds $p_K^2/2$, and it would exceed it by *exactly* what $e^{-\varsigma'^2/2} \le 1$ on $[-2,0]$ buys **only if** $\big([-2,0], \varsigma = 0\big)$

were the maximiser of (91) — which the note to this lemma names as **unproved**, Step 8 delivering $|I^*| = 2$ only *given* $\varsigma^* = 0$. Nothing below reads the exact form; (93) reads the inequality, and $\mathsf{c}_2 \geq p_K^2/2$ is unaffected.

Step 7: attainment. $\bar{\Theta}$ is continuous by (d), and dominated by the second half of (91), a fixed polynomial times a Gaussian, which tends to 0 as $|\varsigma| \to \infty$; and $\bar{\Theta}(0) \geq p_K^2/2 > 0$ by Step 6. Choose $\mathsf{c} < \infty$ beyond which that majorant is below $p_K^2/2$; then $\|\bar{\Theta}\|_\infty = \max_{[-\mathsf{c},\mathsf{c}]} \bar{\Theta}$, attained at some ς^* by continuity on a compact set. is attained at some $I^* \in \mathfrak{I}$ with $\varsigma^* \in I^*$. *This is where the majorant's decay, and not merely its finiteness, is consumed.*

Step 8: the slice $\varsigma = 0$. $\Theta_I(0) = \mathsf{Q}_I(0)$ is antitone in I by Step 2. Every $I = [a, b] \in \mathfrak{I}$ contains a member of $\mathfrak{I}$ of length exactly 2: the set of admissible left endpoints is $[\max(a, -2),\ \min(b - 2, 0)]$, nonempty because $a \leq 0$, $a \leq b - 2$ and $b \geq 0$. The map $I \mapsto \Theta_I(0)$ is continuous on the compact $\mathfrak{I}$ by (d)'s joint continuity, so the maximum over the closed subfamily $\{|I| = 2\}$ is attained and equals $\bar{\Theta}(0)$. □

Proof of Lemma 67. Write Θ_I as in (14), i.e. (P9), and note $\mathsf{C}_h(v; I) = \int_{-\infty}^{v/h} \Theta_I(\varsigma)\, d\varsigma$ by the change of variable $\varsigma = w/h$ in (9).

(i). $\int_{\mathbb{R}} \bar{\Theta} < \infty$ by C3(ii), so for every $\epsilon > 0$ dominated convergence supplies a finite $c_\Theta(\epsilon)$ with $\int_{|\varsigma| > c_\Theta(\epsilon)} \bar{\Theta} \leq \epsilon$; take the least such, so that c_Θ is well defined. If $v \leq -c_\Theta(\epsilon)h$ then $|\mathsf{C}_h(v)| \leq \int_{-\infty}^{-c_\Theta(\epsilon)} |\Theta_1| \leq \int_{|\varsigma| > c_\Theta(\epsilon)} \bar{\Theta} \leq \epsilon$. If $v \geq c_\Theta(\epsilon)h$ then $|\mathsf{C}_h(v) - 1| = \left| \int_{v/h}^{\infty} \Theta_1 \right| \leq \epsilon$ likewise, using $\int_{\mathbb{R}} \Theta_1 = 1$, which is C3(i)'s normalisation, *printed in C3 rather than read out of the phrase "of order* p_K*".* At every v, $|\mathsf{C}_h(v)| \leq \int_{\mathbb{R}} |\Theta_1| \leq \int_{\mathbb{R}} \bar{\Theta} = 2c_C\mathsf{c}_2$ by (16), so $|\mathsf{C}_h| \leq \mathsf{c}_3 := 1 \vee 2c_C\mathsf{c}_2$. Nonnegativity is not claimed and is not available for a higher-order kernel, and only the two-sided bound is used.

(ii). Θ_1 is continuous (being Lipschitz, C3(iii)) and dominated by the integrable $\bar{\Theta}$, so the integral defining C_h is differentiable in v with $\partial_v \mathsf{C}_h(v) = h^{-1}\Theta_1(v/h) = h^{-1}C_h(v, 1)$, continuous, and bounded by $\mathsf{c}_2 h^{-1}$. For v, v', $|\partial_v \mathsf{C}_h(v) - \partial_v \mathsf{C}_h(v')| = h^{-1}|\Theta_1(v/h) - \Theta_1(v'/h)| \leq$

$h^{-1}\mathsf{c}_4|v-v'|/h$, which is the stated Lipschitz constant. The second-order display is the first-order Taylor formula with integral remainder, $\mathsf{C}_h(v') - \mathsf{C}_h(v) - \partial_v \mathsf{C}_h(v)(v'-v) = \int_0^1 [\partial_v \mathsf{C}_h(v + \varsigma(v'-v)) - \partial_v \mathsf{C}_h(v)](v'-v)\, d\varsigma$, whose modulus is at most $\int_0^1 \mathsf{c}_4 h^{-2} \varsigma (v'-v)^2 d\varsigma = \frac{1}{2}\mathsf{c}_4 h^{-2}(v'-v)^2$. This is the only place a "second derivative" would have been used, and it is not used.

(iii). Two changes from round 11, and the first is that the interchange is **Fubini** and not Tonelli, because the integrand is signed. Let $\gamma_G := dG$, a finite signed Borel measure with $|\gamma_G|(\mathbb{R}) < \infty$ and $G(u) = \gamma_G((-\infty, u])$, both by the hypothesis that G is of bounded variation with $G(-\infty) = 0$. The function $(\varsigma, a) \mapsto \Theta_1(\varsigma) \mathbb{1}\{\varsigma \le (u-a)/h\}$ is jointly measurable and dominated in modulus by the envelope $\bar{\Theta}$, whose integral against $d\varsigma \otimes |\gamma_G|$ is $2 c_C \mathsf{c}_2 |\gamma_G|(\mathbb{R}) < \infty$ by (16). *Fubini's theorem for a finite signed measure therefore applies*, and gives

$$\int_{\mathbb{R}} \mathsf{C}_h(u-a)\, \gamma_G(da) = \int_{\mathbb{R}} \Theta_1(\varsigma) \Big[\int_{\mathbb{R}} \mathbb{1}\{a \le u - h\varsigma\}\, \gamma_G(da) \Big] d\varsigma = \int_{\mathbb{R}} \Theta_1(\varsigma)\, G(u - h\varsigma)\, d\varsigma,$$

the last equality being exactly where $G(-\infty) = 0$ and bounded variation are consumed: without them $\gamma_G((-\infty, u - h\varsigma])$ is not $G(u - h\varsigma)$, and the display fails, at $G \equiv \frac{1}{2}$ the left side is 0 and the right side is $\frac{1}{2}$. Now subtract $G(u) = G(u) \int_{\mathbb{R}} \Theta_1$, which is C3(i)'s normalisation, Taylor G to order p_K about u with Lagrange remainder, and use $\int_{\mathbb{R}} \varsigma^j \Theta_1 = 0$ for $1 \le j \le p_K - 1$; the remainder is bounded by $(p_K!)^{-1} \|G^{(p_K)}\|_\infty h^{p_K} \int_{\mathbb{R}} |\varsigma|^{p_K} |\Theta_1(\varsigma)| d\varsigma$, which is the claim with c_5 as defined. $\mathsf{c}_5 < \infty$ and is uniform in h because C3(i)'s third clause makes $\int_{\mathbb{R}} |\varsigma|^{p_K} |\Theta_1|$ finite, the integral carrying no h. *Two things are **not** used and neither may be*: $g := G' \ge 0$, which would license the name "Tonelli", and $\int g = 1$, which would license replacing $\gamma_G((-\infty, u])$ by $G(u)$. *Every consumer is covered directly, and the two named here are not all of them*: Lemma 81 Step 2 re-runs Lemma 77 Step 3's computation verbatim at $\mathcal{M}(\cdot, t, l; \varphi, \theta_\dagger)$ and carries c_5 into its own aggregation, so it is a third. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces read "both consumers" and was short by that site, as did the clause's own statement, twice. Lemma 77 Step 3's decomposition of $d\mathcal{G}_t$ into $\pm$-parts remains correct and is retained, it is a second, independent route, but it is no

longer *necessary*; Lemma 78 Step 2, which has no ±-device, is now licit as written.

(iv). Substituting $x = x^* + h_x\varsigma$, and with $\mathcal{X}$ an interval by A0,

$$h_x^{-1}\int_{\mathcal{X}} C_{h_x}\big(x - x^*, I(x^*)\big)dx = \int_{(\underline{x}-x^*)/h_x}^{(\bar{x}-x^*)/h_x} \Theta_{I(x^*)}(\varsigma)\,d\varsigma = \int_{I(x^*)} \Theta_{I(x^*)}(\varsigma)\,d\varsigma = 1,$$

the middle equality being the definition (94) of $I(x^*)$ and the last C3(ii)'s *truncated* moment condition (15) at $j = 0$, which applies because $I(x^*) \in \mathfrak{I}$ whenever $h_x \le (\bar{x} - \underline{x})/2$. *One line and one case. This is precisely what a boundary-adjusted generalized kernel is for*, and (15) is the property that makes it one. This and the envelope bound are **not** all of C3(ii) that Part H consumes, under either reading of "part". Lemma 78 Step 2 and Lemma 81 Step 4 read (15) at $1 \le j \le p_K - 1$ and not only at $j = 0$; and that same Step 4 reads the envelope's p_K-th absolute moment, $\int_{\mathbb{R}} |\varsigma|^{p_K}\bar{\Theta} < \infty$, which is neither the $j = 0$ moment nor the sup-norm bound, while clause (i) above reads $\int_{\mathbb{R}} \bar{\Theta} < \infty$. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces made one and was short on both readings at once. □

Proof of Proposition 70. Throughout, the sample is fixed; every statement is conditional on it, and (P8) is not needed for this proposition. Put, for $i \in \mathcal{I}_m$, $\mathsf{d}_i(x) := h_x^{-1}[C_{h_x}(x - X_{i2}, I_{i2}) - C_{h_x}(x - X_{i1}, I_{i1})]$, so that $\Delta_{h_x} v_i = \int_{\mathcal{X}} \mathsf{d}_i v\,dx$ and $\hat{\nu}_{u,t,l} = n_{\mathrm{m}}^{-1}\sum_i \hat{g}_{u,t,l,i}\mathsf{d}_i$. Every object below carries the contrast index $l \in \{1, \dots, \mathsf{L} - 1\}$ that (98), (99), (100) and (101) carry, and $\hat{g}_{\cdot,\cdot,\cdot,i}$ is the three-dot element $(u, t, l) \mapsto \hat{g}_{u,t,l,i}$ of $L^2(\Pi)$ that Definition 69 declares. *The index enters through* $\widehat{\widetilde{W}}_{i,l}$ *and through nothing else*, the kernel and the Bierens factor being l-free, and d_i being a function of x alone; that is what makes every bound below a per-cell bound assembled by $\sum_l \varpi_{\mathsf{w}}(l) = 1$, which is the derivation clauses (i) and (ii) print. At $\mathsf{L} = 2$ the third dot stands for the single value $l = 1$, $\varpi_{\mathsf{w}}(1) = 1$, and every display below is the two-dot one earlier rounds printed. *Sans-serif* d *reads "difference" and is a local of this proof*: within this proof italic d occurs as d_Z, an integer, and as the differentials dx, du, dt and $d\Pi$, and elsewhere in the corpus it carries other jobs still — the spline *degree* in

Lemma 60's direct-theorem paragraph and in Definition 22's reading of $\mathcal{S}_n \subset H^s$, and the cell-versus-base contrast in the assumption block's reading of Ω. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces read "only" and was short both by differentials this very proof prints and by those other jobs. *The* ***upright*** *half of that sentence is left standing, and it was checked rather than assumed*: upright d occurs only in $\mathrm{d}\hat{T}_\varphi[\cdot]$, always adjacent to $\hat{T}$, at every occurrence in the corpus, this sentence's own two aside.

Step 1: two elementary bounds, used four times each.

$$\left\|\hat{g}_{\cdot,\cdot,\cdot,i}\right\|_{L^2(\Pi)} \le \frac{\mathsf{c}_2}{h_u}\Big(\Pi_u(\mathbb{R})\int_{\mathcal{T}}\bar{w}(t)^2\Pi_t(dt)\Big)^{1/2} = \frac{\mathsf{c}_2\, c_\Pi}{h_u}, \tag{164}$$

$$\left\|\mathsf{d}_i\right\|_{L^2(dx)} \le \frac{2}{h_x}\Big(\int_{\mathcal{X}} C_{h_x}(x-X_{i1},I_{i1})^2dx\Big)^{1/2} \le \frac{2}{h_x}\Big(2c_C\mathsf{c}_2^2h_x\Big)^{1/2} = 2\,(2c_C)^{1/2}\,\mathsf{c}_2\,h_x^{-1/2}. \tag{165}$$

(164) uses the *per-cell* bound $|\widehat{\widetilde{W}}_{i,l}| \le 1$ of (95), $|w(t'\Phi(z))| \le \bar{w}(t)$ by (P2), $|\Theta_1| \le \mathsf{c}_2$ by (P9), and B1(iv.c) for c_Π. *The contrast index is aggregated by the pairing and not by a bound*: by Definition 5 and (98), $\|\hat{g}_{\cdot,\cdot,\cdot,i}\|^2_{L^2(\Pi)} = \big(\sum_l \varpi_{\mathsf{w}}(l)\widehat{\widetilde{W}}^2_{i,l}\big)h_u^{-2}\int_{\mathbb{R}} C_{h_u}^2\,d\Pi_u\int_{\mathcal{T}} w^2\,d\Pi_t$, the kernel and the Bierens factor being l-free, and the first factor is at most $\sum_l \varpi_{\mathsf{w}}(l) = 1$ by (95) and B1(iv.d); so the constant does not move with L, which is what clause (i) asserts of it, and would have moved had the Euclidean bound $c_{\widetilde{W}}$ been used instead. (165) uses the triangle inequality, the substitution $x = X_{ij} + h_x\varsigma$, and $\int_{\mathbb{R}}\Theta_I^2 \le \|\bar{\Theta}\|_\infty\|\bar{\Theta}\|_{L^1} = 2c_C\mathsf{c}_2^2$ by (16) — *the same constant the support reading gave, and by the same two factors.*

Step 2: clause (i). By (100), the triangle inequality in $L^2(\Pi)$, (164), Cauchy–Schwarz in $L^2(dx)$, (165) and $\|h\|_{L^2(dx)} \le \|1/\pi\|_\infty^{1/2}\|h\|_{L^2(\pi)}$ (C1(ii)),

$$\left\|\hat{\mathcal{A}}_\varphi h\right\|_{L^2(\Pi)} \le \frac{1}{n_{\mathrm{m}}}\sum_i \left\|\hat{g}_{\cdot,\cdot,\cdot,i}\right\|_{L^2(\Pi)}\left|\Delta_{h_x}h_i\right| \le \frac{\mathsf{c}_2 c_\Pi}{h_u}\cdot 2(2c_C)^{1/2}\mathsf{c}_2h_x^{-1/2}\|1/\pi\|_\infty^{1/2}\|h\|_{L^2(\pi)},$$

which is the stated bound. Linearity is immediate from (100).

Step 2′: clause (i′), which is Step 2 stopped one line earlier and then restarted. The chain just run reaches $\|h\|_{L^2(\mathcal{X},dx)}$ at its penultimate link and converts to $L^2(\pi)$ only at the last; *stopping there is* (102), and no step up to that point names π — (100) integrates against Lebesgue measure, (164) is an $L^2(\Pi)$ bound on $\hat{g}_{\cdot,\cdot,\cdot,i}$, (165) an $L^2(dx)$ bound on d_i, and the Cauchy–Schwarz step pairs the two in $L^2(dx)$. *The estimate therefore holds on all of* $L^2(\mathcal{X},dx)$ *and not merely on* $\mathcal{V}$, no centring having been used. For (103), let $\hat{\pi} \geq \varepsilon_\pi > 0$ Lebesgue-a.e. on $\mathcal{X}$. Then $\|h\|^2_{L^2(\hat{\pi})} = \int h^2 \hat{\pi}\, dx \geq \varepsilon_\pi \|h\|^2_{L^2(\mathcal{X},dx)}$, i.e. $\|h\|_{L^2(\mathcal{X},dx)} \leq \varepsilon_\pi^{-1/2} \|h\|_{L^2(\hat{\pi})}$, and substituting into (102) gives (103). *Only the lower floor is used*: the upper floor $\hat{\pi} \leq \varepsilon_\pi^{-1}$ and the normalisation $\int \hat{\pi} = 1$ are not read, which is why the clause is stated for a weight and not for a density. For (104), take $\hat{\pi} \equiv \varepsilon_\pi$, so that $\|h\|_{L^2(\hat{\pi})} = \varepsilon_\pi^{1/2} \|h\|_{L^2(\mathcal{X},dx)}$ *exactly*, whence $\|\hat{\mathcal{A}}_\varphi\|_{L^2(\hat{\pi})\to L^2(\Pi)} = \varepsilon_\pi^{-1/2} \|\hat{\mathcal{A}}_\varphi\|_{L^2(\mathcal{X},dx)\to L^2(\Pi)}$; and $\|h\|_{L^2(\mathcal{X},dx)} \leq \|1/\pi\|_\infty^{1/2} \|h\|_{L^2(\pi)}$ by C1(ii) gives $\|\hat{\mathcal{A}}_\varphi\|_{L^2(\mathcal{X},dx)\to L^2(\Pi)} \geq \|1/\pi\|_\infty^{-1/2} \|\hat{\mathcal{A}}_\varphi\|_{L^2(\pi)\to L^2(\Pi)}$, the two together being (104). *The supremum defining the middle quantity is over* $L^2(\mathcal{X},dx)$ *in both displays*, so the inequality is between two suprema of the same numerator over the same set, taken against two different denominators, and no interchange is involved.

Step 3: clause (ii). (100) exhibits $\hat{\mathcal{A}}_\varphi$ as $n_{\mathrm{m}}^{-1} \sum_{i \in \mathcal{I}_m} \hat{g}_{\cdot,\cdot,\cdot,i} \otimes \mathsf{d}_i$, a sum of n_{m} rank-one operators, so the rank is at most n_{m} and the operator is Hilbert–Schmidt. *The bound is* n_{m} *and not* $n_{\mathrm{m}}(\mathsf{L}-1)$, *which is clause* (ii)*'s own remark*: each summand is the rank-one map $h \mapsto \hat{g}_{\cdot,\cdot,\cdot,i} \cdot \int \mathsf{d}_i h\, dx$, and the n_{m} functionals $h \mapsto \Delta_{h_x} h_i$ do not depend on l, so the contrast index enlarges the range space and not the number of summands. Its kernel contains $\bigcap_{i \in \mathcal{I}_m} \{h \in \mathcal{V} : \int \mathsf{d}_i h\, dx = 0\}$, an intersection of n_{m} closed hyperplanes of the infinite-dimensional space $\mathcal{V}$, hence of codimension at most n_{m} and therefore infinite-dimensional. $\mathcal{V}$ is infinite-dimensional because $\mathcal{X}$ is a non-degenerate interval by A0 and π is bounded above and below by C1(ii).

Step 4: clause (iii). By Lemma 67(iv), $\int_{\mathcal{X}} \mathsf{d}_i\, dx = 1 - 1 = 0$ for each i, *which is Lemma 67(iv) at* $I_{ij} = I(X_{ij})$ *of* (94), *hence C3(ii)'s truncated moment* (15) *at* $j = 0$, *applied at both*

design points, so $\int_{\mathcal{X}} \hat{\nu}_{u,t,l}(x)\,dx = 0$ for every (u,t,l), *the vanishing is per contrast,* d_i *being* l*-free and* $\hat{g}_{u,t,l,i}$ *entering* (99) *as a scalar factor*, which is the arity clause (iii) states. For $\Xi \in L^2(\Pi)$ the function $(u,t,l,x) \mapsto \hat{\nu}_{u,t,l}(x)\Xi(u,t,l)$ is $\Pi\otimes$Leb-integrable, because $|\hat{\nu}_{u,t,l}(x)| \le \mathsf{c}_2 h_u^{-1}\bar{w}(t)\cdot \sup_i \|\mathsf{d}_i\|_\infty$ at every l, with $\|\mathsf{d}_i\|_\infty \le 2\mathsf{c}_2 h_x^{-1}$, $\mathcal{X}$ is bounded, Π is finite and $\bar{w} \in L^2(\Pi_t)$ by B1(iv.c), so $\Xi\bar{w} \in L^1(\Pi)$ by Cauchy–Schwarz, *the envelope being* l*-free, the* l*-sum contributes the factor* $\sum_l \varpi_{\mathsf{w}}(l) = 1$ *and no* L*-dependence.* Fubini therefore gives both

$$\int_{\mathcal{X}} \big(\hat{\mathcal{A}}_\varphi^* \Xi\big)\pi\,dx = \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \Xi(u,t,l)\Big[\int_{\mathcal{X}} \hat{\nu}_{u,t,l}\,dx\Big]\Pi_u(du)\,\Pi_t(dt) = 0,$$

so $\hat{\mathcal{A}}_\varphi^*\Xi \in \mathcal{V}$, and

$$\big\langle \hat{\mathcal{A}}_\varphi h, \Xi\big\rangle_{L^2(\Pi)} = \int\Big[\int_{\mathcal{X}} \hat{\nu}_{u,t,l}\,h\,dx\Big]\Xi\,d\Pi = \int_{\mathcal{X}} h(x)\Big[\int \hat{\nu}_{u,t,l}(x)\,\Xi\,d\Pi\Big]dx = \big\langle \hat{\mathcal{A}}_\varphi^*\Xi, h\big\rangle_{L^2(\pi)},$$

both outer integrals against $d\Pi$ *being Definition 5's pairing, i.e. the weighted sum over* l *of* $\Pi_u \otimes \Pi_t$*-integrals, written out in the display above and abbreviated here,* the last equality by (101) and cancellation of π. *Where the boundary index is load-bearing:* if the untruncated index $I_{ij} \equiv \mathbb{R}$ were used, then $h_x^{-1}\int_{\mathcal{X}} C_{h_x}(x - X_{ij}, \mathbb{R})dx = \int_{(\underline{x}-X_{ij})/h_x}^{(\bar{x}-X_{ij})/h_x} \Theta_1$ differs from 1 by the tail $\int_{\mathbb{R}\setminus I(X_{ij})} \Theta_1$, the two terms of d_i no longer cancel, and $\hat{\mathcal{A}}_\varphi^*\Xi \notin \mathcal{V}$. *Under an envelope that tail is non-zero at* ***every*** *design point and not only near the boundary*: it is $O(1)$ where the nearer endpoint is within $O(h_x)$ of X_{ij}, and at every interior point it tends to 0 with h_x, by integrability of $\bar{\Theta}$, without being exactly 0 at any of them. Only the $O(1)$ part is confined to an $O(h_x)$ fraction of the sample, which is why the failure stays invisible in a rate; it is nevertheless a type error *at every* i, since def:spaces fixes the codomain of every adjoint at $\mathcal{V}$.

Step 5: clause (iv). Fix φ, v and write $\mathsf{u}_i := u - \Delta Y_{i,\theta} + \Delta_{h_x}\varphi_i$, so that, at each (u,t,l), $\hat{T}_{h_x}(\varphi+v,\theta)(u,t,l) - \hat{T}_{h_x}(\varphi,\theta)(u,t,l) = n_{\mathrm{m}}^{-1}\sum_i \widehat{\widehat{W}}_{i,l}\,w(t'\Phi(\Delta Z_i))\{\mathsf{C}_{h_u}(\mathsf{u}_i + \Delta_{h_x}v_i) - \mathsf{C}_{h_u}(\mathsf{u}_i)\}$ by linearity of $v \mapsto \Delta_{h_x}v_i$, u_i *carries no* l*, so the contrast index sits on the scalar factor alone.* By Lemma 67(ii), each summand differs from $\widehat{\widehat{W}}_{i,l}\,w(t'\Phi(\Delta Z_i))\,h_u^{-1}C_{h_u}(\mathsf{u}_i,\mathbb{R})\Delta_{h_x}v_i =$

$\hat{g}_{u,t,l,i}(\varphi,\theta)\Delta_{h_x}v_i$, note that u_i is exactly the residual appearing in (98), by at most $\frac{1}{2}\mathsf{c}_4 h_u^{-2}(\Delta_{h_x}v_i)^2\bar{w}(t)$ in modulus, using the per-cell $|\widehat{\widetilde{W}}_{i,l}|\le 1$ of (95) and (P2), the majorant being l-free. Summing, taking the $L^2(\Pi)$ norm, whose l-sum contributes $\sum_l \varpi_{\mathsf{w}}(l)=1$ against that l-free majorant, and using B1(iv.c),

$$\left\|\hat{T}_{h_x}(\varphi+v,\theta)-\hat{T}_{h_x}(\varphi,\theta)-\hat{\mathcal{A}}_\varphi v\right\|_{L^2(\Pi)} \;\le\; \frac{\mathsf{c}_4 c_\Pi}{2h_u^2}\cdot\frac{1}{n_{\mathrm{m}}}\sum_{i\in\mathcal{I}_m}\left(\Delta_{h_x}v_i\right)^2,$$

which is (105). Since the right-hand side is $O(\|v\|^2_{L^2(\pi)})$ by (165), $\hat{\mathcal{A}}_\varphi$ is the Fréchet derivative; twice differentiability follows because $v\mapsto\hat{\mathcal{A}}_{\varphi+v}$ is itself differentiable, C_{h_u} being C^1 with Lipschitz derivative, and no third-order statement is made or needed.

Step 6: clause (v), the Gateaux derivative, its formula. For fixed v and scalar τ, $\hat{T}(\varphi+\tau v,\theta)(u,t,l)=n_{\mathrm{m}}^{-1}\sum_i \widehat{\widetilde{W}}_{i,l}\,w_i\mathsf{C}_{h_u}(u-\Delta U_i(\theta,\varphi)+\tau\Delta v_i)$ with $\Delta v_i=v(X_{i2})-v(X_{i1})$, which is a finite sum of C^1 functions of τ; differentiating at $\tau=0$ gives (106), and the convergence of the difference quotient is uniform in (u,t,l) up to $\bar{w}(t)$, *the modulus of continuity being l-free by* (95), by Lemma 67(ii), hence holds in $L^2(\Pi)$.

Step 7: clause (v), unboundedness, and it is the engine of Proposition 71. X is continuously distributed by Definition 4, so almost surely the $2n$ points X_{ij} are pairwise distinct; fix a realisation with this property, and fix any $i^*\in\mathcal{I}_m$. Then $|\widehat{\widetilde{W}}_{i^*,l}|\ge\varepsilon_\varrho>0$ at *every* contrast l by (95), *on that realisation and on every other, the bound is per cell, and it is per cell that it is needed. This is where Definition 2's truncation is consumed, and A4(i) will not serve in its place*: A4(i) constrains ϱ and says nothing about $\hat{\varrho}$. Under (95) there is no exception event at all and the "almost surely" below covers only the distinctness of the X_{ij}. Let $\epsilon_k\downarrow 0$ be smaller than half the minimum interpoint distance and than the distance from X_{i^*2} to $\partial\mathcal{X}$, and let $v_k\in\mathcal{V}$ be the triangular spike of height 1 and base $[X_{i^*2}-\epsilon_k, X_{i^*2}+\epsilon_k]$, less its π-mean. Then

$$\|v_k\|_{L^2(\pi)} \;\le\; 2\|\pi\|_\infty^{1/2}\left(\tfrac{2}{3}\epsilon_k\right)^{1/2}\longrightarrow 0,$$

, the factor 2 because subtracting a π-mean at most doubles an $L^2(\pi)$ norm, and $\frac{2}{3}\epsilon_k$ being the Lebesgue integral of the spike's square, so that its $L^2(\pi)$ norm is at most $\|\pi\|_\infty^{1/2}(\frac{2}{3}\epsilon_k)^{1/2}$, while $\Delta v_{k,i^*} = v_k(X_{i^*2}) - v_k(X_{i^*1}) = 1 - 0 = 1 + O(\epsilon_k)$ and $\Delta v_{k,i} = O(\epsilon_k)$ for $i \neq i^*$, because the spike's support meets no other design point. Hence, by (106), $\mathrm{d}\hat{T}_\varphi[v_k] \to n_{\mathrm{m}}^{-1}\hat{g}_{\cdot,\cdot,\cdot,i^*}$ in $L^2(\Pi)$, whose norm is bounded below by

$$\frac{\left(\sum_l \varpi_{\mathsf{w}}(l)\,\widehat{\widetilde{W}}_{i^*,l}^{\,2}\right)^{1/2}}{n_{\mathrm{m}} h_u}\left(\int_{\mathbb{R}} C_{h_u}\big(u - \Delta U_{i^*}, \mathbb{R}\big)^2 \Pi_u(du) \int_{\mathcal{T}} w\big(t'\Phi(\Delta Z_{i^*})\big)^2 \Pi_t(dt)\right)^{1/2} =: \mathsf{c} > 0,$$

the first factor being Definition 5's pairing applied to the only l-dependent factor of (98), and at least ε_ϱ by (95) and $\sum_l \varpi_{\mathsf{w}}(l) = 1$; strictly positive because $\varpi_u > 0$ everywhere by B1(iv.a), so the u-integral of a nonnegative function that is not Lebesgue-a.e. zero is positive, and because $w = \exp > 0$ with $\Pi_t(\mathcal{T}) > 0$ by B1(iv.b). At $\mathsf{L} = 2$ the sum has one term, $\varpi_{\mathsf{w}}(1) = 1$, and the first factor is $|\widehat{\widetilde{W}}_{i^*}|$ as earlier rounds printed. So $\|v_k\|_{L^2(\pi)} \to 0$ with $\|\mathrm{d}\hat{T}_\varphi[v_k]\|_{L^2(\Pi)} \to \mathsf{c} > 0$: the linear map $v \mapsto \mathrm{d}\hat{T}_\varphi[v]$ is unbounded. A Fréchet derivative is by definition a bounded linear operator, and a Fréchet derivative that exists must agree with the Gateaux derivative; so $\hat{T}(\cdot,\theta)$ has no Fréchet derivative at any $\varphi \in \mathcal{V}$. The obstruction is that $v \mapsto \Delta v_i$ is a difference of point evaluations, and point evaluation is not continuous on L^2. It is not an artefact of the spike: any sequence L^2-convergent to 0 but not uniformly convergent will do. □

Proof of Proposition 71. The strategy is Step 7 of Proposition 70's proof, read against A6's inequality rather than against the definition of a derivative: the spike moves $\hat{T}$ by an $O(1/n_{\mathrm{m}})$ amount and is annihilated in the limit by *any* bounded operator, so A6's left-hand side converges to its own right-hand side divided by η_{TCC}.

Fix a realisation of the sample on which the $2n$ points X_{ij} are pairwise distinct; this set has probability one because X is continuously distributed (Definition 4), and it is the *only* exception event, since $|\widehat{\widetilde{W}}_{i,l}| \geq \varepsilon_\varrho > 0$ for every i and every l surely by (95). *Both conventions*

of Definition 2 are consumed here and neither is cosmetic: the truncation supplies (95), and the fold split supplies the fact that $\hat{\varrho}$ is computed off $\mathcal{I}_m$, without it a first stage that interpolates would give $\hat{\varrho}(\Delta Z_i) = W_i$ for every $i \in \mathcal{I}_m$, whence $\hat{T} \equiv 0$ and the proposition would be *false* with probability one rather than true. Put $i^* := \min \mathcal{I}_m$. Fix $\tilde{\varphi} \in \mathcal{V}$, $\bar{\rho} > 0$ and $\eta_{\mathrm{TCC}} < 1$. Let v_k be the centred triangular spike of Proposition 70 Step 7, at base half-width $\epsilon_k \downarrow 0$ small enough that its support contains no design point other than X_{i^*2} and lies in the interior of $\mathcal{X}$, and put $\varphi_k := \tilde{\varphi} + v_k$. Then $\|v_k\|_{L^2(\pi)} \to 0$, so $\|\varphi_k - \tilde{\varphi}\|_{L^2(\pi)} \le \bar{\rho}$ for all large k. *Note the quantifier order: n is fixed throughout and $k \to \infty$*, which is the right order because A6 is a statement at each n.

Step 1: the operator term vanishes, for every bounded operator, which is clause (i). Let $\mathcal{Q} : \mathcal{V} \to L^2(\Pi)$ be any bounded linear operator. Then

$$\big\|\mathcal{Q} v_k\big\|_{L^2(\Pi)} \;\le\; \|\mathcal{Q}\|\, \|v_k\|_{L^2(\pi)} \;\longrightarrow\; 0.$$

That is the entire content of this step, and it uses nothing about the construction of $\mathcal{Q}$. In particular it holds for $\mathcal{Q} = \hat{\mathcal{A}}_{\tilde{\varphi}}$, which is bounded by Proposition 70(i); the direct verification is $|\Delta_{h_x} v_{k,i}| \le \|\mathsf{d}_i\|_\infty \int_{\mathcal{X}} |v_k|\, dx \le 2\mathsf{c}_2 h_x^{-1}\big(\epsilon_k + \|\pi\|_\infty \epsilon_k(\bar{x} - \underline{x})\big) \to 0$ for every i, whence by (100) and (164) $\|\hat{\mathcal{A}}_{\tilde{\varphi}} v_k\|_{L^2(\Pi)} \le (\mathsf{c}_2 c_\Pi / h_u) \max_i |\Delta_{h_x} v_{k,i}| \to 0$ *at fixed n*, and it is worth seeing because it says *why* the exhibited operator is blind to the spike: it sees v_k only through the smoothed functionals (96). But the refutation does not rest on that mechanism. Boundedness alone suffices, and Steps 2–4 below use no property of $\mathcal{Q}$ whatsoever.

Step 2: the operator-equation term does not. With $\mathsf{u}_i := u - \Delta U_i(\theta, \tilde{\varphi})$ the raw residual, at each (u, t, l),

$$\big\{\hat{T}(\varphi_k, \theta) - \hat{T}(\tilde{\varphi}, \theta)\big\}(u, t, l) = \frac{1}{n_{\mathrm{m}}} \sum_{i \in \mathcal{I}_m} \widehat{\widetilde{W}}_{i,l}\, w\big(t' \Phi(\Delta Z_i)\big) \Big\{ \mathsf{C}_{h_u}\big(\mathsf{u}_i + \Delta v_{k,i}\big) - \mathsf{C}_{h_u}(\mathsf{u}_i) \Big\},$$

and $\Delta v_{k,i^*} = 1 - v_k(X_{i^*1}) \to 1$ while $\Delta v_{k,i} \to 0$ for $i \neq i^*$, because the spike's support meets

no other design point and its π-mean is $O(\epsilon_k)$, $\Delta v_{k,i}$ *carries no* l*, the perturbation being on the domain side.* Since C_{h_u} is Lipschitz and bounded by c_3 (Lemma 67(i)–(ii)), $|\widehat{\widetilde{W}}_{i,l}| \le 1$ by (95), and Π is finite, dominated convergence gives, in $L^2(\Pi)$,

$$\hat{T}(\varphi_k,\theta) - \hat{T}(\tilde{\varphi},\theta) \;\longrightarrow\; \Xi_*(u,t,l) := \frac{\widehat{\widetilde{W}}_{i^*,l}}{n_{\mathrm{m}}}\, w\Big(t'\Phi(\Delta Z_{i^*})\Big)\Big\{\mathsf{C}_{h_u}\Big(\mathsf{u}_{i^*}+1\Big) - \mathsf{C}_{h_u}\Big(\mathsf{u}_{i^*}\Big)\Big\}.$$

Step 3: $\|\Xi_*\|_{L^2(\Pi)} > 0$. Fix $\epsilon < \frac{1}{4}$ and suppose $h_u < 1/\big(2c_\Theta(\epsilon)\big)$, which holds for all large n by C5′(i). By Lemma 67(i), $\mathsf{C}_{h_u}(w+1) - \mathsf{C}_{h_u}(w) \ge (1-\epsilon) - \epsilon = 1 - 2\epsilon > \frac{1}{2}$ for every $w \in [-1 + c_\Theta(\epsilon)h_u,\, -c_\Theta(\epsilon)h_u]$, an interval of length $1 - 2c_\Theta(\epsilon)h_u > 0$. *Step 3 asks for strict positivity and nothing more*, so the ϵ-approximation serves exactly as the exact $1 - 0 = 1$ did. In the variable u this is an interval of the same length, on which ϖ_u is strictly positive by B1(iv.a); so $\int_{\mathbb{R}}\{\mathsf{C}_{h_u}(\mathsf{u}_{i^*}+1) - \mathsf{C}_{h_u}(\mathsf{u}_{i^*})\}^2\Pi_u(du) > 0$. And $\int_{\mathcal{T}} w(t'\Phi(\Delta Z_{i^*}))^2\Pi_t(dt) > 0$ because $w = \exp > 0$ by B1(ii) and $\Pi_t(\mathcal{T}) > 0$ by B1(iv.b). Finally $\sum_l \varpi_{\mathsf{w}}(l)\widehat{\widetilde{W}}^2_{i^*,l} \ge \varepsilon_\varrho^2 > 0$ by (95) and $\sum_l \varpi_{\mathsf{w}}(l) = 1$ (B1(iv.d)). Hence, the three factors being separately positive and Ξ_* a product of them, $\|\Xi_*\|_{L^2(\Pi)} > 0$ by Tonelli on $\Pi_u \otimes \Pi_t$ at each l followed by Definition 5's weighted sum over l. At $\mathsf{L} = 2$ that sum has one term and the step is Tonelli on $\Pi_u \otimes \Pi_t$ alone, as earlier rounds printed.

Step 4: conclusion, and it is clause (i). By Steps 1–3, for every bounded $\mathcal{Q} : \mathcal{V} \to L^2(\Pi)$,

$$\Big\|\hat{T}(\varphi_k,\theta) - \hat{T}(\tilde{\varphi},\theta) - \mathcal{Q}v_k\Big\| \;\longrightarrow\; \|\Xi_*\|, \qquad \eta_{\mathrm{TCC}}\Big\|\hat{T}(\varphi_k,\theta) - \hat{T}(\tilde{\varphi},\theta)\Big\| \;\longrightarrow\; \eta_{\mathrm{TCC}}\|\Xi_*\|,$$

and $\eta_{\mathrm{TCC}}\|\Xi_*\| < \|\Xi_*\|$ because $\eta_{\mathrm{TCC}} < 1$ and $\|\Xi_*\| > 0$. So for all large k the pair $(\varphi_k, \tilde{\varphi})$ violates (107) while lying in the $\bar{\rho}$-ball around $\tilde{\varphi}$. Since $\tilde{\varphi}$, $\bar{\rho}$, $\eta_{\mathrm{TCC}} < 1$ and $\mathcal{Q}$ were arbitrary, and since the realisation was arbitrary in a set of probability one, clause (i) follows; clause (ii) is clause (i) at $\mathcal{Q} = \hat{\mathcal{A}}_{\tilde{\varphi}}$.

Step 5: clause (iii), the unbounded candidate. Take $\mathcal{Q} = \mathrm{d}\hat{T}_{\tilde{\varphi}}[\cdot]$, *which is why the quantified symbol in this proposition is* $\mathcal{Q}$ *and not* $\mathcal{L}$: $\mathcal{L}$ is typed as a *bounded* operator wherever

Part H uses it, in Lemma 85, in Remark 124 and at (L1), and this clause is precisely about a candidate that is not bounded. By (106) and Step 6 of Proposition 70's proof, $\Delta v_{k,i^*} \to 1$ and $\Delta v_{k,i} \to 0$ for $i \neq i^*$, so $\mathcal{Q}v_k \to n_{\mathrm{m}}^{-1}\hat{g}_{\cdot,\cdot,\cdot,i^*}$ in $L^2(\Pi)$, and the left-hand side of (107) converges to $\|\Xi_* - n_{\mathrm{m}}^{-1}\hat{g}_{\cdot,\cdot,\cdot,i^*}\| \geq n_{\mathrm{m}}^{-1}\|\hat{g}_{\cdot,\cdot,\cdot,i^*}\| - \|\Xi_*\|$. Two bounds finish it. *Above*: $|\mathsf{C}_{h_u}(\cdot+1) - \mathsf{C}_{h_u}(\cdot)| \leq 2\mathsf{c}_3$ by Lemma 67(i), the per-cell $|\widehat{\widetilde{W}}_{i^*,l}| \leq 1$ of (95) with $\sum_l \varpi_{\mathsf{w}}(l) = 1$, and B1(iv.c) give $\|\Xi_*\| \leq 2\mathsf{c}_3 c_\Pi n_{\mathrm{m}}^{-1}$. *Below*: by (98) and Definition 5's pairing, the kernel and the Bierens factor being l-free,

$$\left\|\hat{g}_{\cdot,\cdot,\cdot,i^*}\right\|_{L^2(\Pi)} = \frac{\left(\sum_l \varpi_{\mathsf{w}}(l)\,\widehat{\widetilde{W}}^2_{i^*,l}\right)^{1/2}}{h_u}\left(\int_{\mathbb{R}} C_{h_u}\big(u-\Delta U_{i^*},\mathbb{R}\big)^2\Pi_u(du)\right)^{1/2}\left(\int_{\mathcal{T}} w\big(t'\Phi(\Delta Z_{i^*})\big)^2\Pi_t(dt)\right)^{1/2} \geq \frac{\varepsilon_\varrho \mathsf{c}}{h}$$

the last step using $\sum_l \varpi_{\mathsf{w}}(l)\widehat{\widetilde{W}}^2_{i^*,l} \geq \varepsilon_\varrho^2 \sum_l \varpi_{\mathsf{w}}(l) = \varepsilon_\varrho^2$, *which is* (95)*'s* ***lower*** *half read cell by cell; at* $\mathsf{L} = 2$ *the sum has one term and both displays are the ones earlier rounds printed*, where $\mathsf{q}_n := h_u^{-1}\int C_{h_u}(u - \Delta U_{i^*},\mathbb{R})^2\Pi_u(du)$ and $\mathsf{c} > 0$ is the t-integral, positive by B1(ii) and B1(iv.b). *Sans-serif* q *is confined to this proof*, q_n the number above and $\mathsf{q}(\cdot)$ its bandwidth-indexed version, with $\mathsf{q}_n = \mathsf{q}(h_u)$, and is separated by **font** from the binding italic $q = E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2]$, *the contrast to the right of the* $p \times (\mathsf{L}-1)$ *derivative, the order Lemma 56*(i) *prints and the only one that multiplies*, and from the **reserved** σ_q, neither of which occurs in this proof. Under (108), $\liminf_n \mathsf{q}_n > 0$, so $n_{\mathrm{m}}^{-1}\|\hat{g}_{\cdot,\cdot,\cdot,i^*}\| \asymp n_{\mathrm{m}}^{-1}h_u^{-1/2}$ while $\|\Xi_*\| = O(n_{\mathrm{m}}^{-1})$; since $h_u \to 0$ by C5′(i), for all n large enough $n_{\mathrm{m}}^{-1}\|\hat{g}_{\cdot,\cdot,\cdot,i^*}\| > (1+\eta_{\mathrm{TCC}})\|\Xi_*\|$ and (107) holds. *Finally,* (108) *holds with probability one at each fixed* $\tilde{\varphi}$: substituting $u = \Delta U_{i^*} + h\varsigma$ gives $\mathsf{q}(h) = \int \Theta_1(\varsigma)^2\varpi_u(\Delta U_{i^*} + h\varsigma)\,d\varsigma$, which converges to $\varpi_u(\Delta U_{i^*})\int\Theta_1^2$ at every Lebesgue point of a bounded ϖ_u (B1(iv.a)). *The convergence is by a two-piece split and uses no majorant hypothesis at all*: writing $u_0 := \Delta U_{i^*}$, so that $|\mathsf{q}(h) - \varpi_u(u_0)\int\Theta_1^2| \leq \int\Theta_1^2(\varsigma)|\varpi_u(u_0+h\varsigma) - \varpi_u(u_0)|d\varsigma$, fix an arbitrary $\epsilon > 0$ — *a fresh one, Step 3's having died there* — and cut the integral at $|\varsigma| = c_\Theta(\epsilon)$ of Lemma 67(i). *Outside*, since $\Theta_1^2 \leq \|\Theta_1\|_\infty|\Theta_1| \leq \|\Theta_1\|_\infty\bar{\Theta}$ by C3(ii), the piece is at most $2\|\varpi_u\|_\infty\|\Theta_1\|_\infty\epsilon$ — *uniformly in* h, which is the only thing the tail has to be. *Inside*, bounding Θ_1^2 by $\|\Theta_1\|_\infty^2$ and substituting $y = h\varsigma$, the piece is

at most $2c_\Theta(\epsilon)\|\Theta_1\|_\infty^2 \cdot \big(2c_\Theta(\epsilon)h\big)^{-1}\int_{|y|\le c_\Theta(\epsilon)h}|\varpi_u(u_0+y)-\varpi_u(u_0)|dy$, whose second factor tends to 0 as $h\downarrow 0$ *by the definition of a Lebesgue point*, at ϵ and hence $c_\Theta(\epsilon)$ held fixed. Let $h\downarrow 0$ and then $\epsilon\downarrow 0$. *Bounded and integrable is all that is used*, and it is all that C3(ii) gives: ϖ_u bounded by B1(iv.a), Θ_1 bounded and Θ_1^2 integrable by C3(ii) — the limit is strictly positive by B1(iv.a) and because $\int\Theta_1^2>0$, Θ_1 being continuous and not identically zero; and $\Delta U_{i^*}(\theta,\tilde\varphi)$ is absolutely continuous under C2′(i), so it lands in the full-Lebesgue-measure set of Lebesgue points almost surely. *The quantifier here is "at each fixed $\tilde\varphi$, almost surely" and not "almost surely, at every $\tilde\varphi$", and the difference is real*: $\Delta U_{i^*}(\theta,\tilde\varphi)$ shifts with $\tilde\varphi$, so the union over an uncountable family of null sets need not be null.

Three remarks on the scope of what has just been proved. *(a)* A6 requires $\eta_{\mathrm{TCC}}<\frac12$; the proof needs only $\eta_{\mathrm{TCC}}<1$, so it refutes a strictly weaker condition too. *(b)* The proof does not use $\theta=\theta_\dagger$, does not use $\tilde\varphi=\varphi_\dagger$, and does not use A1, A2 or any source condition. *It is a statement about the map $\hat T$, not about the model*, which is why no strengthening of the model's hypotheses can repair it, and, by Step 1, not about the linearisation either, which is why no re-specification of $\hat{\mathcal{A}}$ can. *(c) What it does use*: Definition 4's continuity of X (distinctness), A4(i) with Definition 2 ((95)), A4(iii)'s fold split (non-degeneracy of $\hat T$), B1(iv.a) and C5′(i) (Step 3), and for clause (iii) also C2′(i). The proposition's hypothesis list names all of these; a paraphrase asserting that the refutation uses "no model assumption" would be stronger than what is proved. □

Proof of Proposition 74. (i). Fix $\varphi,\tilde\varphi\in\mathcal{V}_n$, put $v:=\varphi-\tilde\varphi$, $\Xi:=\hat{\mathcal{A}}_{\tilde\varphi}v$ and $\Xi':=\hat T_{h_x}(\varphi,\theta)-\hat T_{h_x}(\tilde\varphi,\theta)$, all norms $L^2(\Pi)$. A6b reads $\|\Xi'-\Xi\|\le\frac{\eta}{1+\eta}\|\Xi\|$. Then $\|\Xi\|\le\|\Xi'\|+\frac{\eta}{1+\eta}\|\Xi\|$, so $\|\Xi\|\le(1+\eta)\|\Xi'\|$, which is the right-hand inequality of (i); substituting it back into A6b gives $\|\Xi'-\Xi\|\le\eta\|\Xi'\|$, which is A6 verbatim at $\eta_{\mathrm{TCC}}=\eta$ with $\hat T$ replaced by $\hat T_{h_x}$. For the left-hand inequality, $\|\Xi'\|\le\|\Xi\|+\frac{\eta}{1+\eta}\|\Xi\|=\frac{1+2\eta}{1+\eta}\|\Xi\|$, and $\frac{1+\eta}{1+2\eta}\ge1-\eta$ because $(1+\eta)-(1-\eta)(1+2\eta)=2\eta^2\ge0$.

(ii). Write $\Delta_{h_x}v=(\Delta_{h_x}v_i)_i\in\mathbb{R}^{n_{\mathrm m}}$. *Fix an arbitrary base point $\tilde\varphi\in\mathcal{V}(\varphi_\circ,\rho_0)$*; every display

below is at that base point, and it is the quantifier over base points that (113) exists to serve. By (100) and the definition (112),

$$\left\|\hat{\mathcal{A}}_{\tilde{\varphi}} v\right\|_{L^2(\Pi)}^2 = \frac{1}{n_{\mathrm{m}}^2} \left(\Delta_{h_x} v\right)' \mathbf{G}_n(\tilde{\varphi}) \left(\Delta_{h_x} v\right) \;\geq\; \frac{\lambda_{\min}(\mathbf{G}_n(\tilde{\varphi}))}{n_{\mathrm{m}}^2} \left\|\Delta_{h_x} v\right\|_2^2 \;\geq\; \frac{\underline{\lambda}_n}{n_{\mathrm{m}}^2} \left\|\Delta_{h_x} v\right\|_2^2, \quad (166)$$

$\mathbf{G}_n(\tilde{\varphi})$ being symmetric positive semi-definite as a Gram matrix, positive definite because $\underline{\lambda}_n > 0$ by (113), and the last step holding because $\tilde{\varphi} \in \mathcal{V}(\varphi_\circ, \rho_0)$. *The first equality is an identity at every* L*, and the contrast index is aggregated* ***inside*** $\mathbf{G}_n$: by (100), $\hat{\mathcal{A}}_{\tilde{\varphi}} v(u,t,l) = n_{\mathrm{m}}^{-1} \sum_i \hat{g}_{u,t,l,i} \Delta_{h_x} v_i$, so expanding the square and pairing in $L^2(\Pi)$, Definition 5's weighted sum over l of $\Pi_u \otimes \Pi_t$-integrals, returns exactly (112)'s entries, which are themselves $L^2(\Pi)$ inner products of the three-dot $\hat{g}_{\cdot,\cdot,\cdot,i}$. *So* $\mathbf{G}_n$ *is* $n_{\mathrm{m}} \times n_{\mathrm{m}}$ *at every* L *and not* $n_{\mathrm{m}}(\mathsf{L}-1)$ *square*, which is the same fact as Proposition 70(ii)'s rank bound and holds for the same reason: the n_{m} functionals $v \mapsto \Delta_{h_x} v_i$ do not depend on l. *This display also proves (iii)*: before the eigenvalue bound it is an equality, and the bound is attained at the eigenvector of $\lambda_{\min}(\mathbf{G}_n(\tilde{\varphi}))$, so $n_{\mathrm{m}}^{-1}\lambda_{\min}(\mathbf{G}_n(\tilde{\varphi}))^{1/2}$ is exactly the infimum of $\|\hat{\mathcal{A}}_{\tilde{\varphi}} v\|/\|\Delta_{h_x} v\|_2$ over v with $\Delta_{h_x} v \neq 0$.

By (105) at base point $\tilde{\varphi}$, the left-hand side of (111) is at most $\frac{1}{2}\mathsf{c}_4 c_\Pi h_u^{-2} n_{\mathrm{m}}^{-1} \|\Delta_{h_x} v\|_2^2$; by (166) its right-hand side is at least $\frac{\eta}{1+\eta} n_{\mathrm{m}}^{-1} \underline{\lambda}_n^{1/2} \|\Delta_{h_x} v\|_2$. So (111) holds at the pair $(\varphi, \tilde{\varphi})$ as soon as

$$\left\|\Delta_{h_x} v\right\|_2 \;\leq\; \frac{2\eta}{1+\eta} \cdot \frac{h_u^2\, \underline{\lambda}_n^{1/2}}{\mathsf{c}_4\, c_\Pi}. \quad (167)$$

By (165) and Cauchy–Schwarz, $|\Delta_{h_x} v_i| \leq 2(2c_C)^{1/2} \mathsf{c}_2 h_x^{-1/2} \|1/\pi\|_\infty^{1/2} \|v\|_{L^2(\pi)}$ for every i, so $\|\Delta_{h_x} v\|_2 \leq 2(2c_C)^{1/2} \mathsf{c}_2 (n_{\mathrm{m}}/h_x)^{1/2} \|1/\pi\|_\infty^{1/2} \|v\|_{L^2(\pi)}$. Substituting into (167) and solving for $\|v\|_{L^2(\pi)}$ gives exactly $\|v\|_{L^2(\pi)} \leq \bar{\rho}_n$ with $\bar{\rho}_n$ of (114).

The two quantifier steps, which a previous draft omitted. (a) The pair, not the centre. If $\varphi, \tilde{\varphi}$ both lie in $\mathcal{V}(\varphi_\circ, \frac{1}{2}\min\{\bar{\rho}_n, \rho_0\})$ then $\|v\|_{L^2(\pi)} = \|\varphi - \tilde{\varphi}\| \leq \min\{\bar{\rho}_n, \rho_0\} \leq \bar{\rho}_n$ by the triangle inequality, which is what the halving buys; the earlier draft placed both points in a ball of

radius $\bar{\rho}_n$ *centred at* $\tilde{\varphi}$, which is not a set the definition quantifies over. (b) The base point, not one base point. $\tilde{\varphi}$ was arbitrary in $\mathcal{V}(\varphi_\circ, \rho_0)$, and $\frac{1}{2}\min\{\bar{\rho}_n, \rho_0\} \le \rho_0$, so (166)'s last step is available at every base point the definition ranges over. *Both steps are needed and neither is implied by the algebra*: (114) is an inequality about one pair at one base point, and (111) is a statement about all pairs at all base points in the ball. A6b at $\left(\eta, \varphi_\circ, \frac{1}{2}\min\{\bar{\rho}_n, \rho_0\}\right)$ follows. The $\ell_\infty \to \ell_2$ step is crude by a factor $n_{\mathrm{m}}^{1/2}$ and can be improved by exploiting that the d_i have effective widths $O(h_x)$, so that at most $O(nh_x)$ of them are non-negligible at any x; the improvement is not claimed because no conclusion uses it, and *under C3(ii) it would now have to be stated with the envelope's tail rather than with a support.* □

Proof of Proposition 76. (i). Write $\mathcal{O} := \mathcal{O}(\varphi, \varphi')$ for the intertwining operator of the statement, which is neither the range nor the contrast covariance: the range is $\mathcal{R}(\cdot)$ throughout and appears in the same display below, and the contrast covariance is Ω, which does not occur in this proof. If $\hat{\mathcal{A}}_\varphi = \mathcal{O}\hat{\mathcal{A}}_{\varphi'}$ then $\mathcal{R}(\hat{\mathcal{A}}_\varphi) \subseteq \mathcal{R}(\hat{\mathcal{A}}_{\varphi'})$ trivially. If in addition $\|\mathcal{O} - \mathrm{Id}\| \le \mathsf{c}\|\varphi - \varphi'\|_{L^2(\pi)} < 1$ then $\mathcal{O}$ is boundedly invertible by the Neumann series, so $\hat{\mathcal{A}}_{\varphi'} = \mathcal{O}^{-1}\hat{\mathcal{A}}_\varphi$ and the reverse inclusion holds too; the two ranges coincide. For a pair at distance $\ge 1/\mathsf{c}$, subdivide the segment $\{(1-\varsigma)\varphi + \varsigma\varphi' : \varsigma \in [0,1]\}$, which lies in $\mathcal{V}(\tilde{\varphi}, \rho)$, that set being convex, into finitely many consecutive pieces of length $< 1/\mathsf{c}$ and compose the equalities. Range equality is transitive, so it holds for all pairs in the ball.

(ii). Two inclusions, and only one of them is free. By (100), $\mathcal{R}(\hat{\mathcal{A}}_\varphi) \subseteq \mathrm{span}\{\hat{g}_{\cdot,\cdot,\cdot,i}(\varphi, \theta) : i \in \mathcal{I}_m\}$ unconditionally, *a span of* n_{m} *elements of* $L^2(\Pi)$*, each a function of* (u, t, l)*, the third dot being Definition 69's contrast index*, and that is the only fact used on the $\tilde{\varphi}$ side. Equality, which is what puts the single function $\hat{g}_{\cdot,\cdot,\cdot,i^*}$ *in* the range on the displaced side, requires the n_{m} functionals $v \mapsto \Delta_{h_x} v_i = \int_{\mathcal{X}} \mathsf{d}_i v \, dx$ to be jointly surjective onto $\mathbb{R}^{n_{\mathrm{m}}}$, equivalently the d_i to be linearly independent in $L^2(\mathcal{X}, dx)$; that is the proviso of clause (ii) and the paragraph following the statement gives both the sufficient condition and its limit. *The constraint* $\int v \, d\pi = 0$ *defining* $\mathcal{V}$ *costs nothing here*: $\int_{\mathcal{X}} \mathsf{d}_i \, dx = 0$ by Lemma 67(iv), so

subtracting from any preimage its π-mean leaves every functional value unchanged and returns a preimage in $\mathcal{V}$. *Distinctness of the X_{ij} is **not** the reason for surjectivity*, the minimum design gap being $O_P(n^{-2})$ and the bump width $h_x \gg n^{-2}$, so the supports overlap. By (98), $\hat{g}_{u,t,l,i}(\varphi,\theta) = \widehat{\widetilde{W}}_{i,l}\, w(t'\Phi(\Delta Z_i))h_u^{-1}\Theta_1\big((u-\mathsf{z}_i(\varphi))/h_u\big)$ with $\mathsf{z}_i(\varphi) := \Delta Y_{i,\theta} - \Delta_{h_x}\varphi_i$; so, in u, the ith spanning function is a **translate** of $\Theta_1(\cdot/h_u)$ centred at $\mathsf{z}_i(\varphi)$. *The centre is the **same** at every l, and this is the fact the whole displacement argument turns on*: by (98) the contrast index enters only through the scalar $\widehat{\widetilde{W}}_{i,l}$, the centre $\mathsf{z}_i(\varphi)$ and the kernel being l-free, so "the ith translate" is well defined without reference to l and no L-dependent bookkeeping enters below.

Linear independence of the translates, which replaces disjointness of supports. If $\sum_i c_i\,\Theta_1\big((u-z_i)/h_u\big) = 0$ for Lebesgue-a.e. u, at distinct centres z_i, then applying the Fourier transform $\mathcal{F}$ of (3) gives $(\mathcal{F}\Theta_1)(h_u\xi)\sum_i c_i e^{-\mathrm{i}\xi z_i} = 0$ for every real ξ. *And $\mathcal{F}\Theta_1$ vanishes at no real frequency. This is one line of Hermite algebra and not a clause of Lemma 66*(e), which gives the closed form of Θ_1 and not its transform. At $p_K = 8$, (e) writes $\Theta_1(\varsigma) = \sum_{j=0}^{3}\frac{(-1)^j}{2^j j!}\mathrm{He}_{2j}(\varsigma)(2\pi)^{-1/2}e^{-\varsigma^2/2}$. Rodrigues' formula makes each summand a derivative of the Gaussian density, $\mathrm{He}_m(\varsigma)(2\pi)^{-1/2}e^{-\varsigma^2/2} = (-1)^m\frac{d^m}{d\varsigma^m}\big[(2\pi)^{-1/2}e^{-\varsigma^2/2}\big]$, and $\mathcal{F}[f'](\xi) = \mathrm{i}\xi(\mathcal{F}f)(\xi)$ with $\mathcal{F}\big[(2\pi)^{-1/2}e^{-\varsigma^2/2}\big](\xi) = e^{-\xi^2/2}$ then give $\mathcal{F}\big[\mathrm{He}_m(2\pi)^{-1/2}e^{-\varsigma^2/2}\big](\xi) = (-\mathrm{i}\xi)^m e^{-\xi^2/2}$. At $m=2j$ that factor is $(-1)^j\xi^{2j}$, which cancels the $(-1)^j$ in the coefficient. Hence

$$(\mathcal{F}\Theta_1)(\xi) = \sum_{j=0}^{3}\frac{\xi^{2j}}{2^j j!}\,e^{-\xi^2/2} = \Big(1+\tfrac{\xi^2}{2}+\tfrac{\xi^4}{8}+\tfrac{\xi^6}{48}\Big)e^{-\xi^2/2} > 0 \qquad \text{for every } \xi\in\mathbb{R},$$

the degree-3 truncation of $e^{\xi^2/2}$ in ξ^2, so every coefficient is positive and strict positivity needs no estimate — the alternating signs of the kernel have all been absorbed. So $\sum_i c_i e^{-\mathrm{i}\xi z_i}\equiv 0$, whence $c\equiv 0$ by linear independence of the characters at distinct frequencies. The scalar factors carry through: $|\widehat{\widetilde{W}}_{i,l}| \ge \varepsilon_\varrho > 0$ by (95) and $w = \exp > 0$ by B1(ii). *So the displacement D_n has only to **separate** $\mathsf{z}_{i^*}(\varphi_{\bar{s}})$ from the other centres, not to clear their whole spread*, which

is a strictly weaker demand on (115) than the disjoint-support route made.

The direction. Put $i^* := \min \mathcal{I}_m$ and

$$b \;:=\; \mathsf{d}_{i^*}/\pi, \qquad \mathsf{d}_i(x) = h_x^{-1}\Big[C_{h_x}(x - X_{i2}, I_{i2}) - C_{h_x}(x - X_{i1}, I_{i1})\Big].$$

Three facts about b, each elementary and each computed rather than assumed. *(1)* $b \in \mathcal{V}$: $\int b\, d\pi = \int_{\mathcal{X}} \mathsf{d}_{i^*}\, dx = 0$ by Lemma 67(iv). *(2) Its own functional value is its squared norm*: $\Delta_{h_x} b_{i^*} = \int \mathsf{d}_{i^*} b\, dx = \int \mathsf{d}_{i^*}^2/\pi\, dx = \|b\|^2_{L^2(\pi)} > 0$. *(3) That norm is of exact order* $h_x^{-1/2}$ *from below*: almost surely $X_{i^*1} \neq X_{i^*2}$, so $|X_{i^*2} - X_{i^*1}|$ is one fixed almost surely positive random variable while $h_x \to 0$, whence $|X_{i^*2} - X_{i^*1}|/h_x \to \infty$ and the cross term between the two translates in d_{i^*} is $o(1)$. *The operative fact is a two-piece tail split and not Cauchy–Schwarz*, which bounds that term but does not make it decay. Substituting $x = X_{i^*1} + h_x\varsigma$ writes the cross term as $\int \Theta_{I_{i^*1}}(\varsigma)\, \Theta_{I_{i^*2}}\big(\varsigma - (X_{i^*2} - X_{i^*1})/h_x\big)\, d\varsigma$; bound both factors by $\bar{\Theta}$ and, for an arbitrary $\epsilon > 0$, cut at $|\varsigma| = c_\Theta(\epsilon)$ of Lemma 67(i). *Outside*, the piece is at most $\mathsf{c}_2 \int_{|\varsigma| > c_\Theta(\epsilon)} \bar{\Theta} \le \mathsf{c}_2\epsilon$. *Inside*, the shifted argument exceeds $c_\Theta(\epsilon)$ in modulus as soon as $|X_{i^*2} - X_{i^*1}|/h_x > 2c_\Theta(\epsilon)$, so that piece is at most $\mathsf{c}_2\epsilon$ as well. The cross term is therefore at most $2\mathsf{c}_2\epsilon$ for every h_x small enough, and $\epsilon > 0$ was arbitrary. Using next that, with $c_\Theta(\frac{1}{2})$ of Lemma 67(i), $\Big|\int_{I \cap [-c_\Theta(1/2),\, c_\Theta(1/2)]} \Theta_I\Big| \ge \frac{1}{2}$ for every $I \in \mathfrak{I}$ by (15) at $j = 0$, so that Cauchy–Schwarz gives, uniformly in I,

$$\int_I \Theta_I^2 \;\ge\; \frac{1}{8\, c_\Theta(\frac{1}{2})} \;=:\; \mathsf{c}_{13} \;>\; 0, \tag{168}$$

we obtain

$$\|b\|^2_{L^2(\pi)} = \int_{\mathcal{X}} \frac{\mathsf{d}_{i^*}^2}{\pi}\, dx \;\ge\; \frac{\|\mathsf{d}_{i^*}\|^2_{L^2(dx)}}{\|\pi\|_\infty} \;\ge\; \frac{2\,\mathsf{c}_{13}\,(1 + o(1))}{\|\pi\|_\infty\, h_x} \;\ge\; \frac{\mathsf{c}_{13}}{\|\pi\|_\infty\, h_x}$$

for all n large enough.

The displacement. For $\varsigma \ge 0$ put $\varphi_\varsigma := \tilde{\varphi} + \varsigma b$, so that $\mathsf{z}_{i^*}(\varphi_\varsigma) = \mathsf{z}_{i^*}(\tilde{\varphi}) - \varsigma\|b\|^2$. Cap the

segment at

$$\bar{\varsigma} \;:=\; \frac{2\,\mathsf{D}_n}{\|b\|^2_{L^2(\pi)}}, \qquad \text{so that} \qquad \bar{\varsigma}\|b\|_{L^2(\pi)} = \frac{2\,\mathsf{D}_n}{\|b\|_{L^2(\pi)}} \;\le\; 2\Big(\|\pi\|_\infty/\mathsf{c}_{13}\Big)^{1/2}\mathsf{D}_n\,h_x^{1/2} \;=\; \mathsf{c}_9\mathsf{D}_n h_x^{1/2} \;<\; \rho,$$

the bar being this project's mark for the capping version of the symbol it sits on, as at $\bar{a}_n$, $\bar{\rho}_n$ *and* $\bar{m}_n$, *and* $\bar{\varsigma}$ *is exactly the largest admitted step; it is* ***not*** ς^*, *which Lemma 66*(g) *binds in this same Part as the maximiser of* (91), *and the two proofs share no line.* By (3) and (115); hence $\varphi_\varsigma \in \mathcal{V}(\tilde{\varphi},\rho)$ for every $\varsigma \in [0,\bar{\varsigma}]$. By construction $\bar{\varsigma}\|b\|^2 = 2\mathsf{D}_n$, which **strictly** exceeds $\mathsf{D}_n = \max_{i,k\in\mathcal{I}_m}|\mathsf{z}_i(\tilde{\varphi}) - \mathsf{z}_k(\tilde{\varphi})|$, the centres being pairwise distinct — which is what the linear-independence step above already requires — so that $\mathsf{D}_n > 0$. Hence the displaced centre $\mathsf{z}_{i^*}(\varphi_{\bar{\varsigma}}) = \mathsf{z}_{i^*}(\tilde{\varphi}) - 2\mathsf{D}_n$ is **distinct** from every $\mathsf{z}_i(\tilde{\varphi})$, $i \in \mathcal{I}_m$ — *at every* l, the centres being l-free. That is all the linear-independence step above requires.

The conclusion. $\hat{g}_{\cdot,\cdot,\cdot,i^*}(\varphi_{\bar{\varsigma}},\theta)$ lies in $\mathcal{R}(\hat{\mathcal{A}}_{\varphi_{\bar{\varsigma}}})$ and is nonzero in $L^2(\Pi)$, by (95) read cell by cell with $\sum_l \varpi_{\mathsf{w}}(l) = 1$, so that $\sum_l \varpi_{\mathsf{w}}(l)\widehat{\widehat{W}}^{\,2}_{i^*,l} \ge \varepsilon_\varrho^2 > 0$; by B1(iv.a) (so a nonnegative function not Lebesgue-a.e. zero has positive Π_u-integral); and by B1(ii), (iv.b) on the t-side. And it is *not* in $\mathcal{R}(\hat{\mathcal{A}}_{\tilde{\varphi}}) \subseteq \operatorname{span}\{\hat{g}_{\cdot,\cdot,\cdot,i}(\tilde{\varphi},\theta)\}$: membership would exhibit it as a combination of translates of $\Theta_1(\cdot/h_u)$ at the n_{m} centres $\mathsf{z}_i(\tilde{\varphi})$, and its own centre is distinct from all of them, so the linear-independence step above forces every coefficient — including its own — to vanish, which contradicts its being nonzero. *One set of centres, common to all* $\mathsf{L}-1$ *contrasts, by the* l*-freeness of the centre and the kernel.* Hence $\mathcal{R}(\hat{\mathcal{A}}_{\varphi_{\bar{\varsigma}}}) \neq \mathcal{R}(\hat{\mathcal{A}}_{\tilde{\varphi}})$ with $\varphi_{\bar{\varsigma}},\tilde{\varphi} \in \mathcal{V}(\tilde{\varphi},\rho)$, contradicting (i). *No genericity of* $(\Delta Y_{i,\theta})_i$ *and no conditioning input is used*: the argument compares one translate against the finitely many others through their centres, rather than against the span of the $n_{\mathrm{m}}-1$ others by a norm estimate. *The one thing it does use is the proviso*, and it uses it in exactly one place, to know that $\hat{g}_{\cdot,\cdot,\cdot,i^*}(\varphi_{\bar{\varsigma}},\theta)$ is in the range and not merely in the span. *That is a qualitative nonsingularity, not a rate*; the $\lambda_{\min}$ information Q-3 records as unavailable is a *lower bound with an order*, and nothing here needs one.

The population statement. By (24) and the shift identity of (P6), at each $(u,t,l) \in \mathbb{R}\times\mathcal{T}\times$

$\{1,\dots,\mathsf{L}-1\}$, $\mathcal{A}_\varphi h(u,t,l) = E[\widetilde{W}_l\, w(t'\Phi(\Delta Z))\, f_{\Delta Y_\theta - \Delta\tilde{\varphi}|\mathcal{F}_{XZW}}(u+\Delta(\varphi-\tilde{\varphi}))\Delta h]$, $\widetilde{W}_l$ *is the* l*-th* ***scalar*** *contrast of* (1)*, which is the arity* (24) *carries; the shift is in* u *and is* l*-free,* so the base point enters as an argument shift *inside* the expectation, with a *random* shift $\Delta(\varphi-\tilde{\varphi})$. A fixed operator $\mathcal{O}$ acting on the range side, that is, on (u,t,l), can reproduce a common shift and cannot reproduce a shift that varies with the integrand, which is why the population route fails for the same reason and not merely by analogy. *This paragraph uses no independence proviso*, the obstruction being the form of the base-point dependence rather than the dimension of a span. The strategy memo's one-round cap on this route is hereby spent, with a proof rather than a report. □

Proof of Lemma 77. Throughout, (P8) is in force: $\hat{\varrho}$ is $\sigma(\mathcal{I}_a)$-measurable and every expectation is over $\mathcal{I}_m$ with $\hat{\varrho}$ frozen.

Step 0: $T(\varphi_\dagger,\theta_\dagger)=0$*, and it is A1 that does it.* By Definition 6, $\mathcal{M}(u,t,l) = E[\widetilde{W}_l \mathbb{1}\{\Delta U \le u\} w(t'\Phi(\Delta Z))]$ for each $l\in\{1,\dots,\mathsf{L}-1\}$. Conditioning on ΔZ and using A1's conditional independence $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, $E[\widetilde{W}_l \mathbb{1}\{\Delta U\le u\}\mid \Delta Z] = E[\widetilde{W}_l \mid \Delta Z]\, P(\Delta U \le u \mid \Delta Z) = 0$ by (P3) read at the l-th coordinate. Multiplying by the $\sigma(\Delta Z)$-measurable $w(t'\Phi(\Delta Z))$ and taking expectations gives $\mathcal{M}\equiv 0$ on $\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$, hence $T(\varphi_\dagger,\theta_\dagger)=0$ in $L^2(\Pi)$. *The argument is run at one contrast and the* $\mathsf{L}-1$ *scalar identities are the statement*; at $\mathsf{L}=2$ the index takes the single value 1, $\widetilde{W}_1 = W - \varrho(\Delta Z)$, and this is the display earlier rounds printed. The integrability needed for the conditioning is (P1), *the per-cell bound, which is all a scalar integrand consumes*, and (P2) with B1(iv.c).

Step 1: the decomposition. Write $\hat{T}(\varphi_\dagger,\theta_\dagger) = \{\hat{T}-\bar{T}\} + \{\bar{T}-E\bar{T}\} + \{E\bar{T} - T(\varphi_\dagger,\theta_\dagger)\}$, the three brackets being (iii), (i) and (ii). This is a decomposition and not an approximation: no cross term is discarded.

Step 2: clause (i). Put $\xi_i := \big[(u,t,l) \mapsto \widetilde{W}_{i,l}\, \mathsf{C}_{h_u}(u-\Delta U_i)\, w(t'\Phi(\Delta Z_i))\big] \in L^2(\Pi)$, so that $\bar{T} = n_{\mathrm{m}}^{-1}\sum_{i\in\mathcal{I}_m}\xi_i$ with ξ_i i.i.d. by A0 and, by (P1), (P2), Lemma 67(i) and B1(iv.c), $\|\xi_i\|_{L^2(\Pi)} \le \mathsf{c}_3\Big(\Pi_u(\mathbb{R})\int \bar{w}^2 d\Pi_t\Big)^{1/2} = \mathsf{c}_3 c_\Pi$ almost surely. *The contrast index is aggregated by*

the pairing and not by a bound, so the constant does not move with L: the kernel and the Bierens factor are l-free, so $\|\xi_i\|^2_{L^2(\Pi)} = \big(\sum_l \varpi_{\mathsf{w}}(l)\widetilde{W}^2_{i,l}\big)\int_{\mathbb{R}} \mathsf{C}^2_{h_u}\, d\Pi_u \int_{\mathcal{T}} w^2\, d\Pi_t$ by Definition 5, and the first factor is at most $\sum_l \varpi_{\mathsf{w}}(l) = 1$ by the *per-cell* (P1) and B1(iv.d), *it would have moved had the Euclidean bound* (P1b) *been used instead. That display is the* ***second*** *half of Bochner integrability and not the whole of it. The first half is* ***strong measurability****, which no inspection of the formula settles for an* $L^2(\Pi)$*-valued map, and which is therefore established here*, on the pattern every other L^2-valued object in this corpus is declared at. *Joint measurability.* The map $(\omega, u, t, l) \mapsto \widetilde{W}_{i,l}(\omega)\, \mathsf{C}_{h_u}(u - \Delta U_i(\omega))\, w(t'\Phi(\Delta Z_i(\omega)))$ is jointly Borel measurable: it is Proposition 36 Step 1's map with the indicator $\mathbb{1}\{\Delta Y_\theta - \Delta\varphi \le u\}$ replaced by $\mathsf{C}_{h_u}(u - \Delta U_i)$, a composition of the Borel map C_{h_u}, continuously differentiable by Lemma 67(ii), with the jointly measurable $(u, \omega) \mapsto u - \Delta U_i(\omega)$, and that step's argument is otherwise unchanged, measurability in l being automatic on a finite set. *Weak measurability.* For $\Xi \in L^2(\Pi)$, *the range-side bound variable of Remark 104*(1)*, at its value type* (a)*, a scalar function of* (u, t, l), the product $(\omega, u, t, l) \mapsto \xi_i(\omega)(u, t, l)\, \Xi(u, t, l)$ is then jointly measurable, and its $P \otimes \Pi$-integral is at most $\mathsf{c}_3 c_\Pi \|\Xi\|_{L^2(\Pi)} < \infty$ by the Cauchy–Schwarz inequality on the base set together with the almost-sure bound $\|\xi_i\|_{L^2(\Pi)} \le \mathsf{c}_3 c_\Pi$ just established, *so no hypothesis beyond that display is read here*; Fubini therefore makes $\omega \mapsto \langle \xi_i(\omega), \Xi\rangle_{L^2(\Pi)}$ measurable for *every* Ξ, which is weak measurability. *Pettis.* $L^2(\Pi)$ is separable, Π being a σ-finite Borel measure on a Polish space, the ground Proposition 9 Step 4 states and uses, so ξ_i is separably valued outright, and the *Pettis measurability theorem* upgrades weak measurability to strong. With $E\|\xi_i\|_{L^2(\Pi)} \le \mathsf{c}_3 c_\Pi < \infty$ from the display above, ξ_i is Bochner integrable; and $E\xi_i = E\bar{T}$, the ξ_i being identically distributed by A0 and $\bar{T}$ their average. Because $L^2(\Pi)$ is a Hilbert space and the $\xi_i - E\xi_i$ are independent with mean zero,

$$E\Big\|\frac{1}{n_{\mathrm{m}}}\sum_i (\xi_i - E\xi_i)\Big\|^2 = \frac{1}{n_{\mathrm{m}}^2}\sum_i E\|\xi_i - E\xi_i\|^2 = \frac{E\|\xi_1 - E\xi_1\|^2}{n_{\mathrm{m}}} \le \frac{\mathsf{c}_3^2 c_\Pi^2}{n_{\mathrm{m}}},$$

the cross terms vanishing because $E\langle \xi_i - E\xi_i, \xi_k - E\xi_k\rangle = \langle E(\xi_i - E\xi_i), E(\xi_k - E\xi_k)\rangle = 0$ for

$i \neq k$ by independence and Fubini for Bochner integrals. Chebyshev's inequality applied to the real random variable $\|\hat{T} - E\hat{T}\|$ gives the second display, and $O_P(n^{-1/2})$ follows. Nothing here is uniform in (u, t, l); the norm does the work that a supremum would otherwise do, and that is why no logarithm appears.

Step 3: clause (ii). Fix t and l and let $\mathcal{G}_{t,l}(a) := \mathcal{M}(a, t, l)$, which by Definition 6 is the difference of the two nondecreasing functions $a \mapsto E[\mathbb{1}\{W = l\}\mathbb{1}\{\Delta U \leq a\}w(t'\Phi(\Delta Z))]$ and $a \mapsto E[\varrho_l(\Delta Z)\mathbb{1}\{\Delta U \leq a\}w(t'\Phi(\Delta Z))]$, each of total variation at most $\bar{w}(t)$ by (P2), *the two multiplicands* $\mathbb{1}\{W = l\}$ *and* $\varrho_l(\Delta Z)$ *taking values in* $[0, 1]$, *which is Definition 4 and is what (P1) is assembled from*, and each tending to 0 as $a \to -\infty$ by dominated convergence with envelope $\bar{w}(t)$; so $\mathcal{G}_{t,l}$ is of *bounded variation with* $\mathcal{G}_{t,l}(-\infty) = 0$, and $d\mathcal{G}_{t,l}$ is a finite signed measure of total variation at most $2\bar{w}(t)$, *at every* l *and with no dependence on* L. *Those are exactly Lemma 67(iii)'s two hypotheses*, and (iii) applies directly; the decomposition into positive and negative parts of $d\mathcal{G}_{t,l}$ below is retained as a second, independent route but is not needed, (iii)'s own proof running on Fubini for a finite signed measure. Either way, $E\bar{T}(u, t, l) = \int_{\mathbb{R}} \Theta_1(\varsigma)\, \mathcal{M}(u - h_u\varsigma, t, l)\, d\varsigma$. By C2′(ii), $u \mapsto \mathcal{M}(u, t, l)$ is p_K times continuously differentiable with $\sup_u |\partial_u^{p_K} \mathcal{M}(u, t, l)| \leq \mathfrak{m}_{p_K}\bar{w}(t)$, $\mathfrak{m}_{p_K}$ *being maximal over* l *as well as over* k, *which is the form the statement declares*, so the Taylor argument of Lemma 67(iii) gives $|E\bar{T}(u, t, l) - \mathcal{M}(u, t, l)| \leq \mathsf{c}_5\mathfrak{m}_{p_K}\bar{w}(t)h_u^{p_K}$ for every (u, t, l). Taking $L^2(\Pi)$ norms and using B1(iv.c) gives the stated bound, *the* l*-aggregation contributing the factor* $\sum_l \varpi_{\mathsf{w}}(l) = 1$ *and no constant*; *under C5′(i)* ***as amended***, $p_K = 8$ and $h_u \asymp \mathrm{sd}(\Delta U)\hat{\delta}_n^{1/p_K}$ make $h_u^{p_K} \asymp \hat{\delta}_n$ and the bound $O(\hat{\delta}_n)$, *which is the reading clause (ii) of the statement calls the only one needed. Under the unamended bandwidth* $h_u \asymp \mathrm{sd}(\Delta U)n^{-1/16}$ *it is* $O(n^{-1/2})$ *instead*, the sharper level the statement records and nothing consumes.

Step 4: clause (iii). At each contrast, $(\hat{T} - \bar{T})(\cdot, \cdot, l) = -n_{\mathrm{m}}^{-1}\sum_i\{\hat{\varrho}_l(\Delta Z_i) - \varrho_l(\Delta Z_i)\}\mathsf{C}_{h_u}(\cdot - \Delta U_i)w(t'\Phi(\Delta Z_i))$, so by the triangle inequality in $L^2(\Pi)$, Lemma 67(i), (P2) and B1(iv.c) its $L^2(\Pi)$ norm is at most $\mathsf{c}_3 c_\Pi n_{\mathrm{m}}^{-1}\sum_i \|\hat{\varrho}(\Delta Z_i) - \varrho(\Delta Z_i)\|$, *the norm being Euclidean on* $\mathbb{R}^{\mathsf{L}-1}$

and the passage from the slicewise bound to it costing nothing: the l-dependence sits in $\hat{\varrho}_l - \varrho_l$ alone, so the Π-pairing returns $\big(\sum_l \varpi_{\mathsf{w}}(l)(\hat{\varrho}_l - \varrho_l)^2\big)^{1/2} \le \max_l |\hat{\varrho}_l - \varrho_l| \le \|\hat{\varrho} - \varrho\|$ by B1(iv.d), and $\mathsf{c}_3 c_\Pi$ is unchanged. At $\mathsf{L} = 2$ the norm is the modulus earlier rounds printed. Under (P8), $E\big[n_{\mathrm{m}}^{-1} \sum_i \|\hat{\varrho} - \varrho\|(\Delta Z_i) \,\big|\, \sigma(\mathcal{I}_a)\big] = E\|\hat{\varrho} - \varrho\|(\Delta Z) \le \|\hat{\varrho} - \varrho\|_{L^2}$ by Jensen's inequality, *at the vector norm on both sides, which is the norm A4(iii) is stated in*, and the right-hand side is $\sigma(\mathcal{I}_a)$-measurable and $O_P(r_n)$ *unconditionally* by A4(iii). Conditional Markov therefore gives, almost surely and for *every* $\epsilon > 0$,

$$P\Big(n_{\mathrm{m}}^{-1} \sum_i \|\hat{\varrho} - \varrho\|(\Delta Z_i) > \epsilon\, r_n \;\Big|\; \sigma(\mathcal{I}_a)\Big) \;\le\; \epsilon^{-1}\, \frac{\|\hat{\varrho} - \varrho\|_{L^2}}{r_n},$$

whose numerator is $\sigma(\mathcal{I}_a)$-measurable and $O_P(1)$ unconditionally by A4(iii); so (P8)'s O_P **companion**, at exponent one and $a_n = r_n$, makes the bound unconditional. (P8)*'s* transfer rule *is* ***not*** *what is used here*: that rule needs $P(|X_n| > \bar{a}_n \mid \sigma(\mathcal{I}_a)) \to 0$ almost surely at a *single* moving threshold, whereas conditional Markov delivers a bound at *every* ϵ and, at a threshold that is a fixed multiple of the numerator, bounds the conditional probability by a *constant* that does not vanish. Lemma 78 Step 3 and Lemma 81 Step 1 cite the *conclusion* of this step and therefore need no separate repair.

Step 5: assembly. Adding the three bounds and using $n_{\mathrm{m}} \asymp n$, which is (10) of Definition 2, $\|\hat{T}(\varphi_\dagger, \theta_\dagger)\| = O_P(n^{-1/2}) + O(h_u^{p_K}) + O_P(r_n)$. Under C5′(i) *as amended* the second is $O(\hat{\delta}_n)$ by Step 3, the first is $O_P(n^{-1/2}) = O_P(\hat{\delta}_n)$ and the third is $O_P(r_n) = O_P(\hat{\delta}_n)$, both by A5(i)'s $\hat{\delta}_n = \max\{n^{-1/2}\sqrt{\ln n}, r_n\}$; so $\hat{\delta}_n$ dominates all three, which is (116). □

Proof of Lemma 78. The operator norm is bounded by the Hilbert–Schmidt norm in the $L^2(\pi)$–$L^2(\Pi)$ pairing, and both operators are integral operators with kernels $\hat{\nu}_{u,t,l}/\pi$ and $\nu_{g_{u,t,l}}/\pi$ by (100) and Proposition 9, so, writing $\int(\cdot)\, d\Pi$ for the three-factor integration of

Definition 5,

$$\left\|\hat{\mathcal{A}}_\varphi - \mathcal{A}_\varphi\right\|^2 \;\le\; \iint_{\mathcal{X}} \left|\frac{\hat{\nu}_{u,t,l}(x) - \nu_{g_{u,t,l}}(x)}{\pi(x)}\right|^2 \pi(x)\,dx\,d\Pi \;\le\; \left\|\frac{1}{\pi}\right\|_\infty \iint_{\mathcal{X}} \left|\hat{\nu}_{u,t,l} - \nu_{g_{u,t,l}}\right|^2 dx\,d\Pi, \tag{169}$$

with $\|1/\pi\|_\infty < \infty$ by C1(ii). It therefore suffices to bound the right-hand side at a fixed φ and then to chain over $\mathcal{V}_n$. Every bound below is taken at a fixed contrast and the index is aggregated by Π_{w}, a probability measure by B1(iv.d), so no constant in this proof moves with L; at $\mathsf{L} = 2$ the index takes the single value 1 and every display is the printed one.

Step 1: variance. Let $\breve{\nu}_{u,t,l}$ be (99) with ϱ in place of $\hat{\varrho}$. ***The breve marks the* infeasible *version of a hatted estimator, the one built at the true propensity ϱ rather than at $\hat{\varrho}$.*** It sits on exactly two stems, $\breve{\nu}$ and $\breve{g}_{u,t,l}$, both living in this proof and in Lemma 78's Step 2. *The reading is what the accent is doing*: $\hat{\nu} - \breve{\nu}$ is the propensity term of Step 3 precisely because the two differ in ϱ alone. Under (P8) the summands of $\breve{\nu}$ are i.i.d., so $\int\!\int E|\breve{\nu} - E\breve{\nu}|^2 dx\,d\Pi \le n_{\mathrm{m}}^{-1} \int\!\int E\big[\hat{g}^2_{u,t,l,1}\mathsf{d}_1(x)^2\big] dx\,d\Pi$ (with ϱ inside $\hat{g}$). Bound the two factors separately and deterministically: $\int_{\mathcal{X}} \mathsf{d}_1(x)^2 dx \le 8 c_C \mathsf{c}_2^2 h_x^{-1}$ by (165), which is l-free, and, using $\Pi_u(du) = \varpi_u(u)du$ from (P7) with $\|\varpi_u\|_\infty < \infty$ by B1(iv.a), together with $\int_{\mathcal{T}} \bar{w}^2\,d\Pi_t < \infty$ by B1(iv.c), *which is the clause the last factor of the display rests on and which this step named nowhere before 2026-08-24, here or at Step 2*,

$$\int \hat{g}^2_{u,t,l,1}\,d\Pi \le \frac{1}{h_u^2}\Big(\int_{\mathbb{R}} C_{h_u}(u - \Delta U_1, \mathbb{R})^2 \varpi_u(u)\,du\Big) \int_{\mathcal{T}} \bar{w}(t)^2 \Pi_t(dt) \le \frac{2 c_C \mathsf{c}_2^2 \|\varpi_u\|_\infty}{h_u} \int_{\mathcal{T}} \bar{w}^2\,d\Pi_t,$$

the middle step by the substitution $u = \Delta U_1 + h_u \varsigma$ and (P9), and the contrast index by $\sum_l \varpi_{\mathsf{w}}(l) \widetilde{\widetilde{W}}^2_{1,l} \le \sum_l \varpi_{\mathsf{w}}(l) = 1$, which is the *per-cell* half of (95) with B1(iv.d) and is *why the displayed constant is the one earlier rounds printed.* Multiplying, $\int\!\int E|\breve{\nu} - E\breve{\nu}|^2 dx\,d\Pi = O\big((n\,h_x\,h_u)^{-1}\big)$, so by Markov the corresponding contribution to (169) is $O_P\big((nh_xh_u)^{-1/2}\big)$.

Step 2: bias, and it has exactly two sources. $E\breve{\nu}_{u,t,l}(x) - \nu_{g_{u,t,l}}(x)$ decomposes as

$$\underbrace{E\breve{\nu}_{u,t,l}(x) - \nu_{\breve{g}_{u,t,l}}(x)}_{x\text{-smoothing}} + \underbrace{\nu_{\breve{g}_{u,t,l}}(x) - \nu_{g_{u,t,l}}(x)}_{u\text{-smoothing}}, \qquad \breve{g}_{u,t,l} := \widetilde{W}_l \, w(t'\Phi(\Delta Z)) \, \tfrac{1}{h_u} C_{h_u}\Big(u - \Delta U(\theta,\varphi), \mathbb{R}\Big),$$

the contrast entering as the scalar $\widetilde{W}_l$ *and through nothing else*, exactly as (98) enters $\widehat{\widetilde{W}}_{i,l}$, the kernel and the Bierens factor being l-free. The first is the classical kernel bias of an order-p_K generalized kernel for the two products $E[\breve{g} \mid X_j = x] f_{X_j}(x)$, which are p_K times continuously differentiable and uniformly bounded by C1(i); it is $O(h_x^{p_K})$ uniformly in (u,t,l,x) up to $\bar{w}(t)$, *by the Taylor-against-the-moments argument of Lemma 81 Step 4, which is the pin for smoothing a* ***function****, whereas Lemma 67(iii) is the statement for integrating against a* ***measure*** *and does* ***not*** *cover this use*, run at order p_K rather than at $\lfloor s - \frac{1}{2} \rfloor$, since C1(i) supplies p_K derivatives here, with the interval index of Lemma 67(iv) removing the endpoint effect and C3(ii)'s truncated moments (15) killing the intermediate terms at truncated design points as well as untruncated ones. The second replaces the conditional density $f_{\Delta Y_\theta - \Delta\varphi | \mathcal{F}_{XZW}}(u)$ inside $g_{u,t,l}$ by its C_{h_u}-smoothed version; by C2′(i) that density has $p_K - 1$ derivatives with integrable envelopes and, after the one u-integration implicit in $\partial_u \mathsf{C}_{h_u}$, the smoothing bias is $O(h_u^{p_K})$ uniformly in (u,t,l,x) up to $\bar{w}(t)$. Integrating $\bar{w}^2$ against Π_t, *finite by* B1(iv.c) — *the clause the last factor of this step rests on, exactly as at Step 1, and the half of the 2026-08-24 repair that was missed: that pass named the clause at Step 1 and left this step naming nothing until 2026-08-26* — summing against Π_{w}, *a probability measure, so the aggregation costs nothing*, and integrating dx over the bounded $\mathcal{X}$ gives an $L^2(\Pi \otimes dx)$ bias of order $h_x^{p_K} + h_u^{p_K}$.

Step 3: propensity, and it carries the bandwidth factor. $\hat{\nu} - \breve{\nu}$ replaces $\widetilde{W}_{i,l}$ by $\widehat{\widetilde{W}}_{i,l}$ inside $\hat{g}$, so

$$\Big\|\hat{\nu} - \breve{\nu}\Big\|_{L^2(\Pi \otimes dx)} \le \frac{1}{n_{\mathrm{m}}} \sum_{i \in \mathcal{I}_m} \Big\|(\hat{\varrho} - \varrho)(\Delta Z_i)\Big\| \cdot \Big\| \tfrac{1}{h_u} C_{h_u}(\cdot - \Delta U_i, \mathbb{R}) \, w(t'\Phi(\Delta Z_i))\Big\|_{L^2(\Pi)} \cdot \Big\|\mathsf{d}_i\Big\|_{L^2(dx)},$$

the first norm Euclidean on $\mathbb{R}^{\mathsf{L}-1}$, by the passage Lemma 77 Step 4 performs: the difference is

l-free in every factor but $\hat{\varrho}_l - \varrho_l$, so the Π_{w}-pairing returns $\big(\sum_l \varpi_{\mathsf{w}}(l)(\hat{\varrho}_l - \varrho_l)^2\big)^{1/2} \le \|\hat{\varrho} - \varrho\|$. By the computation of Step 1, the second factor is $O(h_u^{-1/2})$ and the third is $O(h_x^{-1/2})$ by (165); and $n_{\mathrm{m}}^{-1}\sum_i \|(\hat{\varrho} - \varrho)(\Delta Z_i)\| = O_P(r_n)$ by Lemma 77 Step 4. Hence the contribution to (169) is $O_P\big(r_n(h_x h_u)^{-1/2}\big)$, which is the fourth term of (117). The bandwidth factors do **not** cancel here. The scaling is displayed in the spec `aerr-propensity-scaling`. *What is true, and is why the bound is likely loose*: $E[\hat{\nu} - \breve{\nu}]$ equals, at each contrast, a ν-functional of $(\varrho_l - \hat{\varrho}_l)\,\breve{g}_{u,t,l}$, which Lemma 45 bounds by $O(\|\hat{\varrho} - \varrho\|_{L^2}) = O(r_n)$ with no bandwidth; only the fluctuation about that mean carries $(h_x h_u)^{-1/2}$, and bounding it sharply would need $\|\hat{\varrho} - \varrho\|_\infty$, which no clause of the block prices.

Step 4: uniformity over $\mathcal{V}_n \cap \mathcal{S}_n$, and the restriction is what makes the entropy elementary. The map $\varphi \mapsto \hat{\mathcal{A}}_\varphi$ is Lipschitz in $\|\cdot\|_\infty$: by (98), (P9) and (165), $\|\hat{\mathcal{A}}_\varphi - \hat{\mathcal{A}}_{\varphi'}\| \le \mathsf{c}\, h_u^{-2} h_x^{-1/2}\|\varphi - \varphi'\|_\infty$, the h_u^{-2} coming from the Lipschitz constant of $\partial_v \mathsf{C}_{h_u}$ (Lemma 67(ii)) and the $h_x^{-1/2}$ from (165); the same bound holds for $\varphi \mapsto \mathcal{A}_\varphi$ with h_u^{-2} replaced by $\|\bar{f}_1\|_\infty$ (C2′(i)), which is smaller, and the alignment of the two base points costs $\mathsf{c}\, h_u^{-1}\mathsf{R} h_x^{s-1/2}$, which is $o(1)$ relative to γ_n by hypothesis. *The sup-norm Lipschitz bound at the head of this step is also what makes the supremum in* (118) *a random variable, by the separability paragraph of Lemma 81*(3c) *read at $\mathcal{V}_n \cap \mathcal{S}_n$, which is where that paragraph is stated, once, and it is not restated here*: $\mathcal{V}_n \cap \mathcal{S}_n \subset \mathcal{V}_n$ is $\|\cdot\|_\infty$-separable for the same reason, that bound is the pathwise sup-norm continuity the device asks for, and the *outer* supremum therefore equals the supremum over a fixed countable $\|\cdot\|_\infty$-dense subset, almost surely and with a φ-free exceptional set. *The **inner** supremum, the operator norm itself, taken over the unit ball of $\mathcal{V}$, is uncountable too, and that paragraph does not reach it*; what does is the second paragraph of this lemma's statement, which runs the same device on the unit ball, separability of $\mathcal{V}$ and of $L^2(\Pi)$ being what it consumes. Both suprema are countable before (118) is read. Cover $\mathcal{V}_n \cap \mathcal{S}_n$ by $\|\cdot\|_\infty$-balls of radius $\epsilon_0 := \gamma_n h_u^2 h_x^{1/2}/(2\mathsf{c})$. *The covering number is the volumetric one of Lemma 81(3d) and not a Sobolev-ball entropy*: $\mathcal{V}_n \cap \mathcal{S}_n$ is a bounded subset of $\mathcal{S}_n$, whose dimension is *at most* L_x, so $\log N(\epsilon_0) \le L_x \log(1 + 2\mathsf{c}_{11} L_x^{1/2}\bar{\rho}/\epsilon_0) = O(L_x \ln n)$, since ϵ_0 and

L_x are polynomial in n.

The tail applied at each centre must be exponential; Chebyshev will not serve, and (119) is not calibrated for it. (119) compares $L_x \ln n$ against $nh_xh_u\gamma_n^2$, i.e. it compares a *logarithm* of a covering number against the reciprocal of a variance: that is the calibration of a union bound over an *exponentially* large net with an *exponential* tail. A Chebyshev bound at each centre gives failure probability at most $N(\epsilon_0) \cdot \mathsf{c}'/(nh_xh_u\gamma_n^2)$, which with $N(\epsilon_0) = e^{O(L_x \ln n)}$ requires $e^{O(L_x \ln n)} = o(nh_xh_u\gamma_n^2)$, a *polynomial* bound on an exponential quantity, hence $L_x = O(1)$. So the calibration and the tool did not match, and it is the tool that is wrong. The variance term of Step 1 is an average of i.i.d. summands that are bounded on $\mathcal{S}_n$, by (165) and, for the φ-dependence, by Lemma 81(3d)'s inverse inequality, exactly as in Proposition 28 Step 1 (N2.3), so Lemma 123(ii) applies at each centre and returns an exponential tail $\exp(-\mathsf{c}\, nh_xh_u\gamma_n^2)$; the bias term of Step 2 is deterministic and needs no tail; and the propensity term of Step 3 is φ*-free*, so it is not union-bounded at all. *With the exponential tail the union bound vanishes exactly under* (119)*, which is the condition as printed*, the statement of the lemma is therefore unchanged and only its proof is repaired. *This is the second of the two sites the block's own tool was available at and not used*, the first being Proposition 28 Step 1; Lemma 81(3c)–(3d) is the site that does it right. *The impact here is contained because this lemma has no consumers*, reading (iv) of its statement, and Proposition 28 Step 0, but the defect was in a step, not in a hypothesis, and it is repaired rather than scoped away. Over all of $\mathcal{V}_n$ the same display would need the bracketing entropy of a Sobolev ball, which is item (F-e) and is imported rather than proved here; that is the budget line the restriction removes, and every φ at which this lemma is evaluated lies in $\mathcal{S}_n$ by Definition 22(b).

Step 5: the three readings. (i) The exponents in (117) are read off Step 1: one h_x^{-1} from $\int(\mathsf{d})^2 dx$ and one h_u^{-1} from $\int \hat{g}^2 d\Pi$. C5′(ii)'s $h_x^2 h_u^3$ would be the correct pairing for an estimator of a *derivative* of a density in x and a *second* derivative in u, which is what the cross-sectional companion estimates; $\hat{\mathcal{A}}_\varphi$ estimates neither. C5′(ii) now *states* $(nh_xh_u)^{-1} \ln n \to 0$, which is

what this step consumes. *Its former stronger form implied that condition whenever* $h_x, h_u \le 1$, *but the implication was never a ground for keeping the stronger form*: it is a *lower* bound on h_x, and the corpus imports an *upper* bound on h_x at (P-g), against which the stronger floor is not free, (63). (ii) is arithmetic: γ_n enters any bound only against $\|\varphi_\dagger - \varphi\|$, hence at $\gamma_n\delta_n$. (iii) is Step 4: at a fixed φ no covering is needed and Chebyshev alone gives $O_P((nh_xh_u)^{-1/2})$. *The fourth reading is a scoping statement and needs no proof*: Proposition 28 Step 0 elects the conditional-on-sample route, under which this lemma is not consumed. □

Proof of Proposition 79. (i). By Lemma 77 Steps 2–3, $\|\bar{T}(\varphi_\dagger, \theta_\dagger)\| \le \|\bar{T} - E\bar{T}\| + \mathsf{c}_5\mathfrak{m}_{p_K}c_\Pi h_u^{p_K}$. The two terms are handled at different levels and the statement is at the larger. Under C5′(i) as amended, $h_u^{p_K} \asymp \hat{\delta}_n$, so the second term is at most $\mathsf{c}_5\mathfrak{m}_{p_K}c_\Pi\hat{\delta}_n$ deterministically. For the first, Chebyshev at $\mathsf{s} = \frac{1}{2}\mathsf{c}\,n^{-1/2}\sqrt{\ln n}$ gives $P(\|\bar{T} - E\bar{T}\| \ge \mathsf{s}) \le 4\mathsf{c}_3^2c_\Pi^2 n/(n_{\mathrm{m}}\mathsf{c}^2\ln n) \to 0$ for every fixed $\mathsf{c} > 0$, using $n_{\mathrm{m}} \asymp n$, (10); and $n^{-1/2}\sqrt{\ln n} = \hat{\delta}_n^{(T)} \le \hat{\delta}_n$ by A5(i). Adding, $P(\|\bar{T}\| > \mathsf{c}\,\hat{\delta}_n) \to 0$ at $\mathsf{c} = 4\mathsf{c}_3c_\Pi + \mathsf{c}_5\mathfrak{m}_{p_K}c_\Pi$, which is the display. *Under the unamended bandwidth* $h_u \asymp n^{-1/16}$ *the second term is* $O(n^{-1/2}) = o(\hat{\delta}_n^{(T)})$ *instead, and the same argument gives the sharper conclusion with* $\hat{\delta}_n^{(T)}$ *in place of* $\hat{\delta}_n$; that sharper version is recorded and is not consumed. The logarithm is doing exactly one thing here: it makes a second-moment bound into a vanishing-probability statement at a fixed multiple.

(ii). A4(iii) says $\|\hat{\varrho} - \varrho\|_{L^2} = O_P(r_n)$, i.e. for every $\epsilon > 0$ there is M_ϵ with $\limsup_n P(\|\hat{\varrho} - \varrho\|_{L^2} > \mathsf{M}_\epsilon r_n) < \epsilon$. This is consistent with $P(\|\hat{\varrho} - \varrho\|_{L^2} > \mathsf{c}\,r_n)$ converging to a *strictly positive* limit for every fixed c: take $\hat{\varrho} - \varrho$ equal to $\mathsf{Z}r_n$ times a fixed unit-norm function with Z a standard half-normal variable independent of n; then $\|\hat{\varrho} - \varrho\|_{L^2} = O_P(r_n)$ and $P(\|\hat{\varrho} - \varrho\|_{L^2} > \mathsf{c}\,r_n) = P(\mathsf{Z} > \mathsf{c}) > 0$ for every c, not tending to zero. *The limit carries no symbol*, nothing in the argument using its value. So no fixed-constant conclusion follows from the hypothesis, and Lemma 77(iii) is the best available. This is a statement about what A4(iii) entails and not a claim that the propensity estimator misbehaves. M_ϵ *is the constant an* $O_P(1)$ *statement supplies at tolerance* ϵ, used in this clause and the next.

(iii). Under the strengthened A4(iii), $P(\|\hat{T} - \bar{T}\| > \mathsf{c}_3 c_\Pi \mathsf{c}\, r_n) \to 0$ by Lemma 77 Step 4 with Markov replaced by the assumed bound, and adding (i) gives A7 at $\tau_{\mathrm{DP}} = \max\{4\mathsf{c}_3 c_\Pi,\ \mathsf{c}_3 c_\Pi \mathsf{c}\}$. Under A4(iii) as written, $\|\hat{T}(\varphi_\dagger, \theta_\dagger)\| = O_P(\hat{\delta}_n)$ by Lemma 77. Given $\epsilon > 0$, pick M_ϵ from that O_P statement, so that $\limsup_n P\Big(\|\hat{T}(\varphi_\dagger, \theta_\dagger)\| > \mathsf{M}_\epsilon \hat{\delta}_n\Big) < \epsilon$. Since $\tau_{\mathrm{DP}} \ln n \to \infty$, for all n large enough $\tau_{\mathrm{DP}} \ln n \geq \mathsf{M}_\epsilon$, whence $\Big\{\|\hat{T}(\varphi_\dagger, \theta_\dagger)\| > \tau_{\mathrm{DP}} \hat{\delta}_n \ln n\Big\} \subseteq \Big\{\|\hat{T}(\varphi_\dagger, \theta_\dagger)\| > \mathsf{M}_\epsilon \hat{\delta}_n\Big\}$ and the lim sup of the left-hand probability is below ϵ as well. As $\epsilon > 0$ was arbitrary, that probability tends to zero, which is (13).

(iv), first half, free against (7). Suppose $\frac{2\mu}{2\mu+1} \cdot \frac{r}{2r+d_Z} = \frac{1}{4} + \varepsilon$ with $\varepsilon > 0$, which is (7). Then $(\hat{\delta}_n \ln n)^{2\mu/(2\mu+1)} = n^{-1/4-\varepsilon} (\ln n)^{2\mu/(2\mu+1)}$, and $n^{-\varepsilon} (\ln n)^{2\mu/(2\mu+1)} \to 0$ because any positive power of n dominates any power of $\ln n$. Hence the amended δ_n is still $o(n^{-1/4})$. *The strictness of* (7) *is used and nothing else is*; at equality the argument fails, which is one more reason the frontier is stated strictly.

(iv), second half, not free against (P-f), and the reason is that (P-f) is an exact domination. (7) is an inequality with a strict margin in the *power of* n, and that margin is what absorbs a logarithm. (P-f) is the statement $\delta_n^2 = O\Big(h_u^2 \hat{\delta}_n \ln n\Big)$, which has no margin built into it: with $\delta_n = (\hat{\delta}_n \ln n)^\kappa$ and $\kappa = 2\mu/(2\mu+1)$, its two sides are

$$\delta_n^2 = \hat{\delta}_n^{2\kappa} (\ln n)^{2\kappa}, \qquad h_u^2 \hat{\delta}_n \ln n = h_u^2\, \hat{\delta}_n (\ln n)^1, \qquad 2\kappa = \frac{4\mu}{2\mu+1} \ > \ 1 \iff \mu > \tfrac{1}{2}.$$

So whenever $\mu > \frac{1}{2}$, which (7) makes necessary, the left side carries a *strictly larger* power of $\ln n$ than the right, and the logarithms cannot cancel. Consequence: at any configuration where the powers of n balance exactly, (P-f) fails. The thresholds of Remark 125 must therefore be read strictly: $\alpha(2\kappa - 1) > \frac{1}{8}$ at the unamended bandwidth, and $\mu > \frac{5}{6}$ at the amended one. Under a strict inequality in the powers of n the logarithms are absorbed, by the same argument as in the first half. □

Lemma 123 (Two exponential inequalities, in the forms Part H consumes)**.** *(i)*

Bounded differences. *Let $\zeta_1, \dots, \zeta_N$ be independent random elements of a measurable space and let $F(\zeta_1, \dots, \zeta_N)$ be a real, bounded, measurable function such that changing one argument changes F by at most b. Then, for every $\mathsf{s} > 0$, $P\big(F - EF \geq \mathsf{s}\big) \leq \exp\big(-2\mathsf{s}^2/(N\mathsf{b}^2)\big)$.* Sans-serif b is the bounded-difference constant, *bound by "such that" in this clause, instantiated once at Lemma 81 Step 1, and never carried further. It is separated by* ***font*** *from the roman b-family, the weight anchor $b(\cdot), b_\star$ and the sieve basis $b_1, \dots, b_{L_x}$, and, within that family, by the absence of an argument and of an integer subscript.*

(ii) *Bernstein, for an average of i.i.d. nonnegative bounded summands. Let $\Upsilon_1, \dots, \Upsilon_N$ be i.i.d. real with $0 \leq \Upsilon_1 \leq \mathsf{B}$ almost surely and $E\Upsilon_1 \leq \mathsf{K}$, the* range bound *and the* mean bound, *both bound by "with" here, used in Step 4 of the proof, and instantiated once, at Proposition 28 Step 1, at $\mathsf{B} = (4c_C\mathsf{c}_2\mathsf{c}_{11})^2 L_x$ and at the K of (175). Then, for every $\mathsf{s} > 0$,*

$$P\Big(\frac{1}{N}\sum_{i\leq N}\Upsilon_i \;\geq\; \mathsf{K} + \mathsf{s}\Big) \;\leq\; \exp\Big(-\frac{N\,\mathsf{s}^2}{2\,\mathsf{B}\,(\mathsf{K} + \mathsf{s}/3)}\Big). \tag{170}$$

Why (ii) and not (i) at the site that consumes it. Clause (i) is variance-blind: applied to an average of summands of range B it gives $\exp(-2\mathsf{s}^2 N/\mathsf{B}^2)$, with B squared. *Clause (ii) replaces one factor of B by the mean bound K, and at the site that consumes it, Proposition 28 Step 1, where $\mathsf{B} \asymp L_x$ and $\mathsf{K} = O(1)$, that is the difference between a union bound over $e^{O(L_x)}$ net points surviving under $L_x = o(n^{1/3})$ and surviving under $L_x = o(n^{1/2})$, which is A8's own upper end. The exponent is what decides whether the tool matches the entropy, and clause (i) alone does not match it.*

Proof of Lemma 123. Both clauses are standard and are cited rather than reproved; the only step no citation carries is the second-moment bound of clause (ii), which is three lines and is given.

Clause (i) is the bounded-differences inequality of Boucheron, Lugosi, and Massart (2013,

Thm. 6.2), read at $c_i = \mathsf{b}$ for every $i \le N$, so that their variance factor is $v = N\mathsf{b}^2/4$ and their conclusion is the stated one. Their threshold is strict and this clause's is closed, which costs nothing: the bound is continuous in the threshold, and the closed form is the display their proof ends on (Boucheron, Lugosi, and Massart, 2013, p. 172). Boundedness of F is not consumed by that theorem; it is stated here only because every consumer has it.

Clause (ii)*, the one step that is not a citation.* Since $0 \le \Upsilon_1 \le \mathsf{B}$ almost surely,

$$E\Upsilon_1^2 \;\le\; \mathsf{B}\,E\Upsilon_1 \;\le\; \mathsf{BK}.$$

Nonnegativity is used exactly here and nowhere else, and it is what converts a range bound into a second-moment bound; the inequality is false without it, as $\Upsilon_1 = \pm\mathsf{B}/2$ with $\mathsf{K} = 0$ shows. It is also what puts BK rather than B^2 in the exponent.

The rest is Bennett's inequality. Boucheron, Lugosi, and Massart (2013, Thm. 2.9) at one-sided range bound B and second-moment factor $N\,E\Upsilon_1^2 \le N\mathsf{BK}$, in the Bernstein form of their display (2.10), applied on the containment $\{N^{-1}\sum_i \Upsilon_i \ge \mathsf{K}+\mathsf{s}\} \subseteq \{\sum_i(\Upsilon_i - E\Upsilon_i) \ge N\mathsf{s}\}$, gives (170) at $t = N\mathsf{s}$. *Independence, not identical distribution, is what that theorem consumes*; the statement says i.i.d. because that is what Proposition 28 Step 1 supplies. □

Proof of Lemma 81. Decompose, uniformly in φ, $\hat{T}(\varphi) - T(\varphi) = \{\hat{T}(\varphi) - \bar{T}(\varphi)\} + \{\bar{T}(\varphi) - E\bar{T}(\varphi)\} + \{E\bar{T}(\varphi) - T(\varphi)\}$, the three brackets being handled in Steps 1–3. Throughout, (P8) is in force and $\theta = \theta_\dagger$ is fixed.

Step 1: the propensity term is uniform for free. By the display in Lemma 77 Step 4, $\|\hat{T}(\varphi) - \bar{T}(\varphi)\|_{L^2(\Pi)} \le \mathsf{c}_3 c_\Pi n_{\mathrm{m}}^{-1}\sum_i \|\hat{\varrho} - \varrho\|(\Delta Z_i)$, *the norm Euclidean on* $\mathbb{R}^{\mathsf{L}-1}$*, as that step establishes it*, and the bound does not involve φ at all, because $|\mathsf{C}_{h_u}| \le \mathsf{c}_3$ whatever its argument. So $\sup_{\varphi\in\mathcal{V}_n}\|\hat{T}(\varphi) - \bar{T}(\varphi)\| = O_P(r_n) = O_P(\hat{\delta}_n)$*, the supremum being a random variable by the separability paragraph of* (3c) *below, which states that once for the whole proof, a* φ*-free* ***bound*** *does not by itself make the supremum of what it bounds measurable*, which is

why the point is referred to here rather than passed over.

Step 2: the bias term is uniform, and C2′(i) is what makes it so. Fix $\varphi \in \mathcal{V}_n$ and a contrast l. The computation of Lemma 77 Step 3 applies verbatim with $\mathcal{M}(\cdot,t,l)$ replaced by $\mathcal{M}(\cdot,t,l;\varphi,\theta_\dagger)$, and gives $|E\bar{T}(\varphi)(u,t,l) - \mathcal{M}(u,t,l;\varphi,\theta_\dagger)| \le \mathsf{c}_5 h_u^{p_K} \sup_u |\partial_u^{p_K}\mathcal{M}(u,t,l;\varphi,\theta_\dagger)|$. By (P6)'s substitution lemma and the shift identity, $\mathcal{M}(u,t,l;\varphi,\theta) = E[\widetilde{W}_l\, w(t'\Phi(\Delta Z))\, G_\varphi(u)]$ with $G_\varphi(u) = G_0(u+\Delta\varphi)$, so for $1 \le k \le p_K$

$$\left|\partial_u^k \mathcal{M}(u,t,l;\varphi,\theta)\right| = \left|E\Big[\widetilde{W}_l\, w(t'\Phi(\Delta Z))\, \partial_u^{k-1} f_{\Delta Y_\theta|\mathcal{F}_{XZW}}(u+\Delta\varphi)\Big]\right| \;\le\; \bar{w}(t)\, E\Big[\bar{f}_{k-1}\Big],$$

by (P1), *the per-cell bound, the integrand being scalar at each* l, (P2) and C2′(i)'s envelopes, *and the right-hand side depends on neither* φ *nor* l, because the envelope $\bar{f}_{k-1}$ is a supremum over the first argument. Differentiation under the expectation is licensed by the same envelopes. Hence, the l-aggregation contributing $\sum_l \varpi_{\mathsf{w}}(l) = 1$ by B1(iv.d) and no constant, $\sup_{\varphi\in\mathcal{V}_n} \|E\bar{T}(\varphi) - T(\varphi)\|_{L^2(\Pi)} \le \mathsf{c}_5 c_\Pi \max_{k\le p_K-1} E[\bar{f}_k]\, h_u^{p_K} = O(n^{-1/2})$ under C5′(i). This is a free uniformity and it is worth naming: C2′(i) was introduced for Proposition 8, and it pays a second time here.

Step 3: the centred empirical term, by discretisation and Lemma 123(i). (3a) Lipschitz continuity in the sup-norm. For φ, φ', by Lemma 67(ii), (P1), (P2) and B1(iv.c),

$$\left\|\bar{T}(\varphi) - \bar{T}(\varphi')\right\|_{L^2(\Pi)} \le \frac{\mathsf{c}_2 c_\Pi}{h_u} \max_{i\in\mathcal{I}_m} \left|\Delta\varphi_i - \Delta\varphi_i'\right| \le \frac{2\mathsf{c}_2 c_\Pi}{h_u} \left\|\varphi - \varphi'\right\|_\infty,$$

and the same bound holds for $E\bar{T}$ by Jensen; so $\varphi \mapsto \bar{T}(\varphi) - E\bar{T}(\varphi)$ is $\mathsf{c}_{10} h_u^{-1}$-Lipschitz in $\|\cdot\|_\infty$ with $\mathsf{c}_{10} := 4\mathsf{c}_2 c_\Pi$. *The (P1) citation is the* ***per-cell*** *bound and* c_{10} *does not move with* L: at each i the l-dependence of $\bar{T}$ is the scalar factor $\widetilde{W}_{i,l}$, so the Π-pairing contributes $\big(\sum_l \varpi_{\mathsf{w}}(l)\widetilde{W}_{i,l}^2\big)^{1/2} \le 1$ by (P1) and B1(iv.d).

(3b) The tail at a fixed φ. Put $\mathsf{Z}(\varphi) := \|\bar{T}(\varphi) - E\bar{T}(\varphi)\|_{L^2(\Pi)}$. *The font is the point*: italic Z, ΔZ is the **binding** exogenous control and is live inside this same proof. *In this proof*

Z *never appears without its function argument*, which separates it from Proposition 79(ii)'s bare half-normal Z and from Proposition 96's n-subscripted Z_n, neither of which occurs here. By Lemma 77 Step 2, $E\mathsf{Z}(\varphi) \le (E\mathsf{Z}(\varphi)^2)^{1/2} \le \mathsf{c}_3 c_\Pi n_{\mathrm{m}}^{-1/2}$. Changing one observation in $\mathcal{I}_m$ changes $\bar{T}(\varphi)$ by at most $2\mathsf{c}_3 c_\Pi / n_{\mathrm{m}}$ in $L^2(\Pi)$, hence changes $\mathsf{Z}(\varphi)$ by at most that; so Lemma 123(i) with $N = n_{\mathrm{m}}$ and $\mathsf{b} = 2\mathsf{c}_3 c_\Pi / n_{\mathrm{m}}$ gives, for $\mathsf{s} \ge 2\mathsf{c}_3 c_\Pi n_{\mathrm{m}}^{-1/2}$,

$$P\Big(\mathsf{Z}(\varphi) \ge \mathsf{s}\Big) \le P\Big(\mathsf{Z}(\varphi) - E\mathsf{Z}(\varphi) \ge \mathsf{s}/2\Big) \le \exp\Big(-\frac{n_{\mathrm{m}}\mathsf{s}^2}{8\mathsf{c}_3^2 c_\Pi^2}\Big). \tag{171}$$

(3c) Discretisation and the union bound, clause (a). First, the suprema over $\mathcal{V}_n$ are random variables. This is stated **once**, here, and the other three suprema over $\mathcal{V}_n$ in this block refer to it rather than repeat it. $\mathcal{V}_n$ is separable in $\|\cdot\|_\infty$, and the three inputs are named because the lemma's hypothesis list now carries them. By Definition 80, which chooses R with $\|\varphi_\dagger\|_{H^s} \le \mathsf{R}$, *available because H1's source set is an H^s ball, which is where that choice comes from*, every $\varphi \in \mathcal{V}_n$ has $\|\varphi\|_{H^s} \le 2\mathsf{R}$; so at $s > \frac{1}{2}$, by the one-dimensional *Sobolev embedding* $H^s(\mathcal{X}) \hookrightarrow C(\mathcal{X})$ on A0's compact interval, the same embedding Lemma 94 Step 2 uses, and **not** the $H^s \hookrightarrow C^{0,s-1/2}$ that Step 4 below refuses as vacuous, every element of $\mathcal{V}_n$ has a continuous representative, unique, since C1(ii)'s $\|1/\pi\|_\infty < \infty$ makes π strictly positive on $\mathcal{X}$, so that two continuous representatives of one class agree $\pi\,dx$-a.e. and hence everywhere, and $\|\cdot\|_\infty$ is a function of the class, and $\|\cdot\|_\infty$ is read at it. Hence $\mathcal{V}_n \subset C(\mathcal{X})$, which is $\|\cdot\|_\infty$-separable, and *every* subset of a separable metric space is separable. Fix once a countable $\|\cdot\|_\infty$-dense $\mathcal{V}_n^\circ \subset \mathcal{V}_n$. By (3a), $\varphi \mapsto \mathsf{Z}(\varphi)$ is $\mathsf{c}_{10} h_u^{-1}$-Lipschitz in $\|\cdot\|_\infty$ at every sample point outside one fixed P-null set, the set where (P1)'s $|\widetilde{W}_l| \le 1$ may fail at some l, which carries no φ, (3a)'s bound being otherwise deterministic. (95) is **not** what this appeal or the one two sentences below reads. That display is the *two*-sided per-cell bound $\varepsilon_\varrho \le \big|\widehat{\widetilde{W}}_{i,l}\big| \le 1$ of Definition 68, whose *lower* half rests on A4(i) and Definition 2's truncation: here the object is $\bar{T}$, carrying the population weight $\widetilde{W}_{i,l}$, whose upper bound is (P1) and is free; below it is $\hat{T} - \bar{T}$, carrying $\hat{\varrho} - \varrho$, each coordinate of which is bounded by 1

because ϱ_l and $\hat{\varrho}_l$ both take values in $[0,1]$, Definition 4 and Definition 2's truncation, with no assumption at all. So $\sup_{\varphi\in\mathcal{V}_n}\mathsf{Z}(\varphi)=\sup_{\varphi\in\mathcal{V}_n^{\circ}}\mathsf{Z}(\varphi)$ almost surely, and the right-hand side is a countable supremum of the random variables (3b) applies Lemma 123(i) to. The display below and (122) are read at it. *The same three ingredients, separability of the index set, a sup-norm Lipschitz bound holding pathwise, a countable dense subset, cover the other two suprema of this proof*: Step 1's, because $\varphi\mapsto\hat{T}(\varphi)-\bar{T}(\varphi)$ carries the same Lemma 67(ii) Lipschitz bound in $\|\cdot\|_{\infty}$ that (3a) gives $\bar{T}$, the propensity differences $\hat{\varrho}_l-\varrho_l$ multiplying it being bounded by 1 *at every cell*, so that their Π_{w}-aggregate is bounded by 1 as well, *and not depending on* φ; and Step 4's $\hat{T}_{h_x}$ version, $\varphi\mapsto\hat{T}_{h_x}(\varphi)$ depending on φ only through the *bounded linear* functionals (96), as Definition 68 records. *This is not a formality, and the doctrine is this file's own*: without it $\sup_{\varphi\in\mathcal{V}_n}\mathsf{Z}(\varphi)$ is a function on the sample space not known to be measurable and the display immediately below is not defined, the ground on which Lemma 61 Step 3 refuses the same elision in u, and on which Definition 24(c′.2) refuses it for the inner supremum of (42). *Part H's suprema over* φ *were the one family the device had not been applied to.* Now let $\epsilon_0:=h_u\hat{\delta}_n/(4\mathsf{c}_{10})$ and let $\varphi_1,\dots,\varphi_N$ be an ϵ_0-net of $\mathcal{V}_n$ in $\|\cdot\|_{\infty}$, with $\log N\le\mathsf{c}_6(\mathsf{R}/\epsilon_0)^{1/s}$ by (F-e), for which see Birman and Solomjak (1980, Ch. 2), available because $s>\frac{1}{2}$, and a sup-norm net is a bracketing net, the brackets being $[\varphi_j-\epsilon_0,\varphi_j+\epsilon_0]$. By (3a), $\sup_{\varphi\in\mathcal{V}_n}\mathsf{Z}(\varphi)\le\max_{j\le N}\mathsf{Z}(\varphi_j)+\hat{\delta}_n/4$, and by (171) with $\mathsf{s}=\hat{\delta}_n/2$ and a union bound,

$$P\Big(\sup_{\varphi\in\mathcal{V}_n}\mathsf{Z}(\varphi)\ge\tfrac{3}{4}\hat{\delta}_n\Big)\;\le\;\exp\left(\mathsf{c}_6\Big(\frac{\mathsf{R}}{\epsilon_0}\Big)^{1/s}-\frac{n_{\mathrm{m}}\hat{\delta}_n^2}{32\mathsf{c}_3^2c_{\Pi}^2}\right),$$

which tends to 0 precisely when $(\mathsf{R}/\epsilon_0)^{1/s}=o(n\hat{\delta}_n^2)$; since $\epsilon_0\asymp h_u\hat{\delta}_n$ that is (121) as stated, *in the primitive variables and at no particular bandwidth.* Remark 82 reads it at C5′(i) *as amended*, and at that tuning alone: $n^{9\alpha/(8s)}=o(n^{1-2\alpha})$ at $h_u\asymp\hat{\delta}_n^{1/8}$, which is that remark's reading (1). The unamended $h_u\asymp n^{-1/16}$ would give $n^{(1/16+\alpha)/s}=o(n^{1-2\alpha})$ instead, a slightly stronger condition, the smaller bandwidth making the bracket radius smaller, and Remark 82

does not read it. Combining with Steps 1–2 gives (122). The condition $\mathsf{s} \geq 2\mathsf{c}_3 c_\Pi n_{\mathrm{m}}^{-1/2}$ required by (171) holds because $\hat{\delta}_n \geq n^{-1/2}\sqrt{\ln n}$.

(3d) Clause (b), the sieve version, with no classical entropy bound. $\mathcal{V}_n \cap \mathcal{S}_n$ is contained in an $L^2(\pi)$-ball of radius $\bar{\rho}$ inside $\mathcal{S}_n$, whose dimension is *at most* L_x, *exactly* $L_x - 1$, since $\mathcal{S}_n = \mathrm{span}(b_1, \ldots, b_{L_x}) \cap \mathcal{V}$ and $\zeta \mapsto \int \zeta \, d\pi$ is a nonzero functional on that span, the b_j being nonnegative and π bounded below by C1(ii); *the counts below are stated at* L_x *because every one of them is an* ***upper*** *bound and nothing here is sharp in the dimension*, and the distinction is drawn once, at Definition 22's measurability verification, first step, where a coordinate map is written and where it *does* bind. On a quasi-uniform mesh the B-spline basis of A8 satisfies $\|\zeta\|_\infty \leq \mathsf{c}_{11} L_x^{1/2} \|\zeta\|_{L^2(\pi)}$ for $\zeta \in \mathcal{S}_n$, so the set is contained in a $\|\cdot\|_\infty$-ball of radius $\mathsf{c}_{11} L_x^{1/2} \bar{\rho}$ in a normed space of dimension at most L_x, whose ϵ-covering number is at most $(1 + 2\mathsf{c}_{11} L_x^{1/2} \bar{\rho}/\epsilon)^{L_x}$ by the standard volumetric argument. Hence $\log N(\epsilon_0) \leq L_x \log(1 + 2\mathsf{c}_{11} L_x^{1/2} \bar{\rho}/\epsilon_0) = O(L_x \ln n)$, since ϵ_0 and L_x are polynomial in n. The display of (3c) then vanishes exactly under (123). No Sobolev entropy bound is used in this clause, which is why it and not clause (a) is what Proposition 28 consumes.

Step 4: the $\hat{T}_{h_x}$ *version.* $\hat{T}_{h_x}(\varphi) - \hat{T}(\varphi)$ is bounded, by Lemma 67(ii), (P1), (P2) and B1(iv.c), by $\mathsf{c}_2 c_\Pi h_u^{-1} \max_i |\Delta_{h_x}\varphi_i - \Delta\varphi_i|$; and

$$\left|\Delta_{h_x}\varphi_i - \Delta\varphi_i\right| \;\leq\; 2 \sup_x \left|\varphi_{h_x}(x) - \varphi(x)\right| \;\leq\; \mathsf{c}_{12}\, \mathsf{R}\, h_x^{s-1/2}. \tag{172}$$

The route to (172) is a $\lfloor s - \frac{1}{2} \rfloor$-th order Taylor expansion against C3's moment conditions, and **not** the embedding $H^s \hookrightarrow C^{0,s-1/2}$ that rounds 1–11 named, which is vacuous at every configuration this block admits. The correction is to the justification and not to the rate; (172) stands.

– Why the named embedding is unavailable. On an interval, $C^{0,\beta}$ with $\beta > 1$ contains only the constants: $|f(x) - f(y)| \leq C|x - y|^\beta$ with $\beta > 1$ forces $f' \equiv 0$. So $H^s \hookrightarrow C^{0,s-1/2}$ is vacuous for every $s > \frac{3}{2}$, and (54) requires $s > \frac{27}{16}$ at $\alpha = \frac{3}{8}$ while Remark 125(5)'s

admissible configuration has $s = 3$, where the claim reads "every H^3 function on $\mathcal{X}$ is $\frac{5}{2}$-Hölder", i.e. constant. There is no configuration Proposition 28 admits at which that justification would be non-vacuous.

– *The route that works.* Put $k := \lfloor s - \frac{1}{2} \rfloor \geq 1$ and $\beta := s - \frac{1}{2} - k \in [0, 1)$. The Sobolev embedding on an interval gives $H^s \hookrightarrow C^{k,\beta}$, with $\|\varphi\|_{C^{k,\beta}} \leq \mathsf{c}\,\|\varphi\|_{H^s}$; this is the embedding at the largest integer smoothness the scale supports, and it is non-vacuous at every $s > \frac{1}{2}$ by construction. Now expand: with $\varphi_{h_x}(x) = \int \Theta_{I(x)}(\varsigma)\varphi(x + h_x\varsigma)\, d\varsigma$, the substitution $x' = x + h_x\varsigma$ in the definition of φ_{h_x}, Taylor's theorem to order k about x gives

$$\varphi(x + h_x\varsigma) = \sum_{j=0}^{k} \frac{\varphi^{(j)}(x)}{j!}(h_x\varsigma)^j + \mathcal{E}_k(x, h_x\varsigma), \qquad \left|\mathcal{E}_k(x, h_x\varsigma)\right| \leq \frac{\mathsf{c}\,\|\varphi\|_{H^s}}{k!}\left|h_x\varsigma\right|^{k+\beta},$$

the remainder bound because $\varphi^{(k)}$ is β-Hölder with constant $\mathsf{c}\|\varphi\|_{H^s}$. Integrating against $\Theta_{I(x)}$ and using C3(ii)'s truncated moments (15), $\int_I \Theta_I = 1$ and $\int_I \varsigma^j \Theta_I = 0$ for $1 \leq j \leq p_K - 1$, *kills every term with* $1 \leq j \leq k$ *provided* $k \leq p_K - 1$, and leaves

$$\left|\varphi_{h_x}(x) - \varphi(x)\right| \;\leq\; \frac{\mathsf{c}\,\|\varphi\|_{H^s}}{k!}\, h_x^{k+\beta} \int_{\mathbb{R}} |\varsigma|^{k+\beta} \bar{\Theta}(\varsigma)\, d\varsigma \;\leq\; \tfrac{1}{2}\mathsf{c}_{12}\,\|\varphi\|_{H^s}\, h_x^{s-1/2},$$

since $k + \beta = s - \frac{1}{2} \leq p_K$ and $\int_{\mathbb{R}} |\varsigma|^{p_K} \bar{\Theta} < \infty$ by C3(ii), so the moment integral is finite *uniformly in* I *and* h_x — *which is the clause the envelope supplies and a per-*I *bound could not.* Taking the supremum over x and using $\|\varphi\|_{H^s} \leq \mathsf{R}$ gives (172).

– The unstated condition the route needs, now stated. The step "every term with $1 \leq j \leq k$ is killed" requires $k \leq p_K - 1$, i.e. $\lfloor s - \frac{1}{2} \rfloor \leq p_K - 1$, for which $p_K \geq s - \frac{1}{2}$ suffices. *That condition was nowhere in the block and it is now a hypothesis of Proposition 28*; at $p_K = 8$ it is satisfied at every $s \leq 8.5$, hence at every configuration the budget table admits ($s = 3$ at the admissible one), so it is free in fact and not free in logic. *And it gates two hypotheses*: (P-g), which is (172) read as an upper bound on h_x, and (N1) of

Proposition 28 Step 1, which prices $\hat{T}_{h_x} - \hat{T}$ through this step.

Note which norm is used, because it is not the one (120) *constrains*: the smoothing bias of φ is governed by $\|\varphi\|_{H^s}$ and not by $\|\varphi - \varphi_\dagger\|_{H^s}$, so the display needs $\|\varphi_\dagger\|_{H^s} \le \mathsf{R}$ as well, which H1's source set supplies and which Definition 80 now states. Here $\varphi_{h_x}(x) := h_x^{-1}\int_{\mathcal{X}} C_{h_x}(x'-x, I(x))\varphi(x')dx'$ is the smoothing implicit in (96), so that $\Delta_{h_x}\varphi_i = \Delta(\varphi_{h_x})_i$. *The same displacement is present at $\varphi = \varphi_\dagger$ and is not a discretisation of the parameter but a bias of the x-smoothing*, which is why the hypothesis $\mathsf{R}h_x^{s-1/2} = O\big(h_u\hat{\delta}_n\big)$ is stated separately rather than folded into (121). *The factor h_u is the one the head of this step charges*: the displacement of the two operator families is bounded there in **operator** norm by $\mathsf{c}_2 c_\Pi h_u^{-1}\max_i|\Delta_{h_x}\varphi_i - \Delta\varphi_i|$, so with (172) the composed bound is $\mathsf{c}\,h_u^{-1}\mathsf{R}\,h_x^{s-1/2}$ and it is *that product*, not $\mathsf{R}\,h_x^{s-1/2}$ alone, that must be $O(\hat{\delta}_n)$. □

Proof of Lemma 83. The $L^2(\pi)$ constraint at each stage is Proposition 28 Step 4, which gives $\|\hat{\varphi}^{(k)} - \varphi_\dagger\|_{L^2(\pi)} = O_P(\hat{\varepsilon}_k)$ with $\hat{\varepsilon}_k$ the deterministic schedule (37), nonincreasing in k once $\hat{\varepsilon}_0$ is small enough and with $\hat{\varepsilon}_0 \to 0$; so $\max_{k\le\bar{k}}\|\hat{\varphi}^{(k)} - \varphi_\dagger\|_{L^2(\pi)} \le \bar{\rho}$ with probability tending to one for any fixed $\bar{\rho} > 0$, $\bar{k}$ being finite and not depending on n. *The argument is not circular*: Proposition 28 Step 4's induction assumes containment at stage $k-1$ and derives it at stage k, and the base case is the hypothesis on $\hat{\varphi}^{(0)}$. The sieve constraint is by construction. *The H^s constraint is A5(vi′)*, which is stated at every stage; A5(vi) alone would not serve, since it prices the final iterate only, and it is the *intermediate* iterates that must lie in $\mathcal{V}_n$ for Lemma 81 to be evaluated at them. *Lemma 78 is **not** named here*: Step 0 of Proposition 28 does not take the perturbation route, so nothing in Part H evaluates it anywhere, and reading (iv) of that lemma records why it is retained. Because $\bar{k}$ is finite and does not grow with n, A5(vi′) is a finite union of $O_P(1)$ statements, hence $O_P(1)$, and the H^s half holds with probability tending to one for R fixed and large.

The within-stage bound, recorded because the memo's mechanism gives it and not the containment. Write $\mathcal{L} := \omega_n^{-1}\hat{\mathcal{A}}_{\varphi_\circ}|_{\mathcal{S}_n}$ and $y^\delta := -\omega_n^{-1}\hat{T}_{h_x}(\varphi_\circ, \theta)$ for the normalised pair of

Definition 22, so that $\|\mathcal{L}\| \le 1$ *surely* by (35) and Proposition 70(i)–(i′), *in each of the two domain pairings Proposition 28 Step 2 names, which is why (i′) is cited here beside (i).* With $v_0 = 0$, $\|v_{m+1} - v_m\|_{L^2(\pi)} = \|\mathcal{L}^*(\mathcal{L}v_m - y^\delta)\| \le \|\mathcal{L}v_m - y^\delta\|$, and the Landweber residual is nonincreasing in m for a linear equation under that normalisation (Engl, Hanke, and Neubauer, 2000, Ch. 6), so $\|v_m\| \le m\|y^\delta\| \le 2m\,\omega_n^{-1}\mathsf{c}_3 c_\Pi$, using Lemma 67(i), (P1), (P2) and B1(iv.c). *This bounds the path linearly in the cap $\bar{m}_n$ of (36) and therefore does not by itself place the path in a ball of fixed radius* , the two factors move in opposite directions, ω_n^{-1} shrinking each step and $\bar{m}_n \ge \omega_n^2\hat{\delta}_n^{-2}$ allowing more of them, with the product divergent; it is the stopping rule that places it, through Step 4. □

Remark 124 ((F-f) quoted in full, and the saturation contrast that makes it load-bearing). This is the one external theorem the whole of Proposition 28 rests on. It is displayed here, in this file's own letters, so that Step 2's items (L1)–(L5) read against it line by line.

(F-f) (Engl, Hanke, and Neubauer, 2000, Thm. 6.5). *Let $\mathcal{L}$ be a bounded linear operator between Hilbert spaces with $\|\mathcal{L}\| \le 1$. Let $y := \mathcal{L}\bar{v}_\dagger$ and let y^δ satisfy $\|y^\delta - y\| \le \delta$. Let $v_0 := 0$ and $v_{m+1} := v_m + \mathcal{L}^*(y^\delta - \mathcal{L}v_m)$, and let $\hat{m}$ be the first index m with $\|\mathcal{L}v_m - y^\delta\| \le \tau_{\mathrm{DP}}\delta$, where $\tau_{\mathrm{DP}} > 2$ is fixed. If $\bar{v}_\dagger = (\mathcal{L}^*\mathcal{L})^\mu\varsigma$ for some $\mu > 0$ and some pre-image ς with $\|\varsigma\| \le \mathsf{R}_*$,* the radius is R_*, the **spectral pre-image** norm, which is what (P-a.3) supplies and Lemma 84 measures; it is **not** the H^s radius R of Definition 80, *then $\hat{m}$ is finite and*

$$\hat{m} \;=\; O\Big(\big(\mathsf{R}_*/\delta\big)^{2/(2\mu+1)}\Big), \qquad \big\|v_{\hat{m}} - \bar{v}_\dagger\big\| \;\le\; c_\mu\,\mathsf{R}_*^{1/(2\mu+1)}\,\delta^{2\mu/(2\mu+1)}, \tag{173}$$

with c_μ depending on μ and τ_{DP} only.

One reading of the quotation is load-bearing and is fixed here rather than left to the reader: the ***domain*** *Hilbert space is a single object discharging three roles at once.* It is the space whose inner product defines $\mathcal{L}^*$; it is the space $\|\mathcal{L}\|$ is the operator norm *out of*; and it is the space the source pre-image ς is measured in. *Consequently the pairing must be fixed* ***before*** $\|\mathcal{L}\| \le 1$

is checked, and a bound established out of some other space does not serve: the hypothesis is not "$\mathcal{L}$ is bounded by one in some norm" but a relation between $\mathcal{L}$ and the very inner product through which $\mathcal{L}^*$ is formed, and what it buys is that $\mathrm{Id} - \mathcal{L}^*\mathcal{L}$ is non-expansive. *The failure mode is therefore not a slower rate but a divergence*: $\mathcal{L}^*\mathcal{L}$ is self-adjoint and positive with $\|\mathcal{L}^*\mathcal{L}\| = \|\mathcal{L}\|^2$, so at $\|\mathcal{L}\|^2 > 2$ the operator $\mathrm{Id} - \mathcal{L}^*\mathcal{L}$ carries spectrum below -1 — an eigenvalue, the domain being finite-dimensional in the application — and the error iteration grows geometrically on the corresponding eigenvector from any starting error not orthogonal to it. *Which pairing the estimator of Definition 22 runs in, and why* (35) *is sized to cover both candidates, are Proposition 28 Step 2 and Proposition 70*(i′).

The property that is load-bearing is the **absence of saturation**, and the contrast with the obvious alternative is worth more than the statement. (173) holds at *every* $\mu > 0$: Landweber iteration has infinite qualification, so raising the source index keeps improving the exponent without bound. Tikhonov regularisation under the same stopping rule does not. Its qualification is finite and the discrepancy principle saturates it at $\mu = \frac{1}{2}$: the Tikhonov estimate stopped by the discrepancy principle attains $O(\delta^{2\mu/(2\mu+1)})$ for $\mu \le \frac{1}{2}$ and $O(\delta^{1/2})$ for every $\mu \ge \frac{1}{2}$ (Engl, Hanke, and Neubauer, 2000, Rem. 4.21), and no choice of the regularisation parameter recovers the higher exponent. In this block that is not a matter of taste, and the arithmetic is the same one Remark 88(3) runs at the inherited nonlinear route's qualification index. Part H lives at $\mu > \frac{5}{6}$; a Tikhonov step under the discrepancy principle would return $\delta_n = (\hat{\delta}_n \ln n)^{1/2}$, and since $\hat{\delta}_n \ge n^{-1/2}\sqrt{\ln n}$ by Lemma 77 that is at best $n^{-1/4}(\ln n)^{3/4}$, which is *not* $o(n^{-1/4})$ and so fails A5(ii) and (7) at every configuration. *So the choice of Landweber over Tikhonov is what makes the paper's own feasibility frontier reachable*, and it is the same property, qualification, that removes the inherited nonlinear route at $\mu = \frac{1}{2}$. *What is* ***not*** *claimed*: that Tikhonov's rate *fails* above $\frac{1}{2}$. Saturation as stated here is a statement about the best bound the method's analysis delivers, exactly as in Remark 88(3). *A necessity claim would need a lower bound, and one is now available and is deliberately not invoked*: Engl, Hanke, and Neubauer (2000, Prop. 4.20), attributed there to

Groetsch, shows that if the Tikhonov estimate stopped by the discrepancy principle satisfies $\|x^\delta - x^\dagger\| = o(\delta^{1/2})$ for every admissible y, then $\mathcal{R}(K)$ is finite-dimensional, so on a genuinely ill-posed problem the exponent $\frac{1}{2}$ cannot be beaten. *Nothing below is strengthened on that basis*, because the comparison this block draws needs only the upper bound, and a necessity claim would carry an obligation to check that Engl, Hanke, and Neubauer's "for every y" quantifier matches the configuration Part I actually runs at.

Proof of Lemma 84. (i). Let $\mathcal{P} := \mathcal{A}^*\mathcal{A}$ and $\hat{\mathcal{P}} := \hat{\mathcal{A}}^*\hat{\mathcal{A}}$, both bounded, self-adjoint and positive on $\mathcal{V}$. For every ζ, $\big|\|\mathcal{A}\zeta\| - \|\hat{\mathcal{A}}\zeta\|\big| \le \gamma\|\zeta\|$, i.e. $\langle\mathcal{P}\zeta,\zeta\rangle^{1/2} \le \langle\hat{\mathcal{P}}\zeta,\zeta\rangle^{1/2} + \gamma\|\zeta\|$, so $\mathcal{P} \preceq (\hat{\mathcal{P}}^{1/2} + \gamma\,\mathrm{Id})^2 \preceq 2\hat{\mathcal{P}} + 2\gamma^2\,\mathrm{Id}$ in the operator order. *Heinz's inequality*, if $0 \preceq \mathcal{Q}_1 \preceq \mathcal{Q}_2$ then $\mathcal{Q}_1^{2\mu} \preceq \mathcal{Q}_2^{2\mu}$ *whenever* $2\mu \le 1$, which is operator monotonicity of $z \mapsto z^{2\mu}$, gives $\mathcal{P}^{2\mu} \preceq (2\hat{\mathcal{P}} + 2\gamma^2\,\mathrm{Id})^{2\mu} \preceq 2^{2\mu}(\hat{\mathcal{P}}^{2\mu} + \gamma^{4\mu}\,\mathrm{Id})$. Writing $v = \mathcal{P}^\mu\varsigma$ and using $\|\mathcal{P}^\mu\varsigma\|^2 = \langle\mathcal{P}^{2\mu}\varsigma,\varsigma\rangle$ together with the displayed dominance yields a $\tilde{\varsigma}$ with $\|\tilde{\varsigma}\| \le 2^\mu\mathsf{R}_*$ and $\|v - \hat{\mathcal{P}}^\mu\tilde{\varsigma}\| \le 2^\mu\mathsf{R}_*\gamma^{2\mu}$, by the standard interpolation of a positive operator inequality into a range inclusion with a residual. *The lemma's constant is* $\mathsf{c}_8 := 2^{2\mu}$, the value clause (ii.b) states, and clause (i)'s display follows *a fortiori*: this step delivers the smaller 2^μ, and $2^\mu < 2^{2\mu}$ at every $\mu > 0$. *One symbol, one value*, the sharper constant available on (i)'s range $\mu \in (0, \frac{1}{2}]$ is recorded here and is deliberately not given a second name. The radius is R_*, the **spectral pre-image** norm this lemma's statement declares for clauses (i) and (ii), and **not** the H^s radius R of Definition 80, which appears in clause (iii) alone. The only step that restricts μ is Heinz, and it restricts it to $2\mu \le 1$. *The exponent carries no symbol*, deliberately.

(ii.a): the tool fails above $\mu = \frac{1}{2}$. By Löwner's theorem, $z \mapsto z^p$ is operator monotone on $[0,\infty)$ if and only if $0 \le p \le 1$, Löwner (1934); the modern account, with both the integral representation and the finite-order criterion used below, is Bhatia (1997, Ch. V), item (F-g) of the citation note, so for $2\mu > 1$ the displayed step is not available.

The claim of (ii.a) is at **every** $2\mu > 1$ and in dimension **two**; the explicit matrices below establish it at $\mu = 1$, and the general case is **quoted rather than re-derived**. That

$z \mapsto z^p$ fails to be matrix monotone *of order two* at every $p > 1$ is Löwner's divided-difference criterion applied to $f(z) = z^p$: Löwner (1934) for the criterion and Bhatia (1997, Ch. V) for it, for the integral representation and for the order-n theory in modern form, the same pin the statement of (ii.a) already carries. *Only the order-two conclusion is used*, and it is what the criterion returns by inspection of the 2×2 Löwner matrix at the pair $(1, \epsilon)$ as $\epsilon \downarrow 0$; the threshold $p = 1$ is the same one that delivers Heinz's inequality in clause (i) from the other side.

The witness at $\mu = 1$ is two lines and is worth printing rather than left to the criterion, because $\mu = 1$ is one of this block's two running configurations and because an explicit pair is checkable by hand. Let

$$\mathcal{Q}_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathcal{Q}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then $\mathcal{Q}_1 \succeq 0$ (eigenvalues 0 and 2) and $\mathcal{Q}_2 - \mathcal{Q}_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) \succeq 0$, so $0 \preceq \mathcal{Q}_1 \preceq \mathcal{Q}_2$; but $\mathcal{Q}_2^2 - \mathcal{Q}_1^2 = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 2 \end{smallmatrix}\right) - \left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which has determinant -1 and is therefore not positive semi-definite, explicitly, at $\zeta = (1, -3)'$ the quadratic form equals $3 - 6 + 0 = -3 < 0$. Hence $\mathcal{Q}_1^2 \not\preceq \mathcal{Q}_2^2$.

(ii.b): and no counterexample to the conclusion is offered, because none can be diagonal. Suppose $\mathcal{A}$ and $\hat{\mathcal{A}}$ have a common singular system, so that in that basis $\mathcal{P} = \mathrm{diag}(\sigma_j^2)$ and $\hat{\mathcal{P}} = \mathrm{diag}(\hat{\sigma}_j^2)$ with $|\hat{\sigma}_j - \sigma_j| \le \gamma$. Let $v = \mathcal{P}^\mu \varsigma$, $\|\varsigma\| \le \mathsf{R}_*$, so $v_j = \sigma_j^{2\mu} \varsigma_j$. Split the index set at $\sigma_j \ge 2\gamma$. *On* $\{\sigma_j \ge 2\gamma\}$, $\hat{\sigma}_j \ge \sigma_j - \gamma \ge \sigma_j/2$, so setting $\tilde{\varsigma}_j := v_j / \hat{\sigma}_j^{2\mu} = \varsigma_j (\sigma_j / \hat{\sigma}_j)^{2\mu}$ gives $|\tilde{\varsigma}_j| \le 2^{2\mu} |\varsigma_j|$. *On* $\{\sigma_j < 2\gamma\}$, set $\tilde{\varsigma}_j := 0$; then $|v_j| = \sigma_j^{2\mu} |\varsigma_j| \le (2\gamma)^{2\mu} |\varsigma_j|$. Summing the two, $\|\tilde{\varsigma}\| \le 2^{2\mu} \mathsf{R}_*$ and $\|v - \hat{\mathcal{P}}^\mu \tilde{\varsigma}\| \le (2\gamma)^{2\mu} \mathsf{R}_*$, for *every* $\mu > 0$, *which is the constant* $\mathsf{c}_8 = 2^{2\mu}$ *and the residual* $(2\gamma)^{2\mu} \mathsf{R}_*$ *clause* (ii.b) *states*. So transfer holds at every source index in the commuting case, and the obstruction identified in (ii.a) is one of non-commutativity, not of the size of the perturbation. *In particular no* ***diagonal*** *construction can witness a failure*

of transfer, whether the perturbation is upward or downward. What is claimed here, and it is all that is used, is the unavailability of the device. No result in Part H performs a transfer: (P-a.3) imposes the *spectral* source condition on the compression (53) of the realised $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}$ directly, *which is also why no interpolation step is needed, (ii.a) being what makes one unavailable*, and Remark 88(3) invokes (ii.a) only to say that the inherited nonlinear rate theorem, whose own proof runs through the same operator monotonicity, is unavailable above $\mu = \frac{1}{2}$, not that its conclusion is false there.

(iii). $\|v\|_{H^s} \leq \mathsf{R}$ mentions no operator, so it transfers trivially. That a finite-rank operator cannot satisfy $\|\hat{\mathcal{A}}\zeta\| \asymp \|\zeta\|_{H^{-a}}$ on all of $\mathcal{V}$ is immediate: the left side vanishes on the infinite-dimensional $\ker \hat{\mathcal{A}}$ of Proposition 70(ii), and the right side does not vanish on any nonzero ζ, the H^{-a} norm being a norm. Restricting to $\mathcal{S}_n$, of dimension at most L_x, removes the obstruction whenever $L_x \leq \operatorname{rank} \hat{\mathcal{A}}_{\varphi}$, for which A8's *upper end* $L_x \ll n^{1/2} \leq n_{\mathrm{m}}$ is sufficient in generic position. *The upper end, not A8 in full*: the lower end plays no part here, and Proposition 28 cites A8 by clause for the same reason. □

Proof of Lemma 85. Six clauses: four short computations, (i), (ii), (iv) and (v), of which (iv) is the cancellation, and two longer arguments, the functional-calculus identity of (iii) and the change of reference measure of (vi), which the statement says in terms is **not** covered by (i)–(v). Throughout, $\omega_* > 0$ is fixed and $\mathcal{L}_{\omega_*} = \omega_*^{-1}\mathcal{L}$, $y^{\delta}_{\omega_*} = \omega_*^{-1} y^{\delta}$. The generic scale is ω_* and not bare ω, which is the sample point of the probability space at (P6), (P7a), Lemma 46 and Proposition 9.

(i). The adjoint of $\omega_*^{-1}\mathcal{L}$ is $\omega_*^{-1}\mathcal{L}^*$, so $\mathcal{L}^*_{\omega_*}(y^{\delta}_{\omega_*} - \mathcal{L}_{\omega_*} v) = \omega_*^{-1}\mathcal{L}^*\left(\omega_*^{-1} y^{\delta} - \omega_*^{-1}\mathcal{L} v\right) = \omega_*^{-2}\mathcal{L}^*(y^{\delta} - \mathcal{L} v)$ for every v. Since both recursions start at $v_0 = 0$, induction on m gives identical iterates.

(ii). $\mathcal{L}_{\omega_*} v_m - y^{\delta}_{\omega_*} = \omega_*^{-1}(\mathcal{L} v_m - y^{\delta})$, so the two residual sequences are proportional with the same constant at every m, and $\|\mathcal{L}_{\omega_*} v_m - y^{\delta}_{\omega_*}\| \leq \tau_{\mathrm{DP}}\omega_*^{-1}\delta \iff \|\mathcal{L} v_m - y^{\delta}\| \leq \tau_{\mathrm{DP}}\delta$. The first index at which either holds is therefore the same index.

(iii). $\mathcal{L}_{\omega_*}^* \mathcal{L}_{\omega_*} = \omega_*^{-2} \mathcal{L}^* \mathcal{L}$, which is again bounded, self-adjoint and positive. *Its μth power is defined by the continuous functional calculus for a bounded self-adjoint operator, and the load-bearing fact is that its spectrum is a compact subset of* $[0, \omega_*^{-2}\|\mathcal{L}\|^2]$, so $\lambda \mapsto \lambda^\mu$, continuous on $[0,\infty)$ for every $\mu > 0$, its value at 0 being 0, is continuous on the spectrum and the calculus applies. *Nothing is needed about whether the spectrum touches zero, and no injectivity hypothesis is required*: $\mu > 0$ makes λ^μ extend continuously to 0, injectivity would be needed only for uniqueness of the pre-image (never claimed, Remark 124 says *some* pre-image) or for negative powers (never formed). Then

$$\Big(\omega_*^{-2} \mathcal{L}^* \mathcal{L}\Big)^\mu \;=\; \omega_*^{-2\mu} \Big(\mathcal{L}^* \mathcal{L}\Big)^\mu,$$

by the composition (spectral-mapping) rule $f\Big(g(\mathcal{P})\Big) = (f \circ g)(\mathcal{P})$ *of the continuous functional calculus*, applied with $g(\lambda) = \omega_*^{-2}\lambda$ and $f(\lambda) = \lambda^\mu$, so that $(f \circ g)(\lambda) = \omega_*^{-2\mu}\lambda^\mu$; $\omega_*^{-2} > 0$ is real, which is what makes g map the spectrum into $[0,\infty)$ where f is defined. *This is* ***not*** *"positive homogeneity transferring"*: $z \mapsto z^\mu$ is not linear, so homogeneity of the map is not the mechanism; the mechanism is that composing the two scalar functions before applying the calculus gives the same operator as applying it twice. Hence $\bar{v}_\dagger = (\mathcal{L}^*\mathcal{L})^\mu \varsigma = \Big(\mathcal{L}_{\omega_*}^* \mathcal{L}_{\omega_*}\Big)^\mu \Big(\omega_*^{2\mu}\varsigma\Big)$, and $\|\omega_*^{2\mu}\varsigma\| \le \omega_*^{2\mu}\mathsf{R}_*$. The noise level transforms by (ii)'s proportionality: $\|y_{\omega_*}^\delta - \mathcal{L}_{\omega_*}\bar{v}_\dagger\| = \omega_*^{-1}\|y^\delta - \mathcal{L}\bar{v}_\dagger\| \le \omega_*^{-1}\delta$. Both transformations are exact, not bounds, which is what makes (iv) an identity.

(iv). By (iii), (F-f)'s bound evaluated in the scaled pair is $c_\mu(\omega_*^{2\mu}\mathsf{R}_*)^{1/(2\mu+1)}(\omega_*^{-1}\delta)^{2\mu/(2\mu+1)}$, and the powers of ω_* are $\frac{2\mu}{2\mu+1} - \frac{2\mu}{2\mu+1} = 0$. *The constant c_μ depends on μ and τ_{DP} only, by Remark 124, and on neither ω_* nor the operator*, so nothing else moves. This proves (124). *Note that the conclusion of (F-f) is a bound on $\|v_{\hat{m}} - \bar{v}_\dagger\|$ in the* ***domain*** *norm, which rescaling the equation does not touch*; that is why the invariance is of the bound and not merely of both sides of it.

(v). By the same substitution, $\Big(\omega_*^{2\mu}\mathsf{R}_*/(\omega_*^{-1}\delta)\Big)^{2/(2\mu+1)} = \omega_*^{2(2\mu+1)/(2\mu+1)}(\mathsf{R}_*/\delta)^{2/(2\mu+1)} =$

$\omega_*^2(\mathsf{R}_*/\delta)^{2/(2\mu+1)}$. *The exponents that cancelled in (iv) do not cancel here*, the stopping index scaling with the square of the norm rather than with the source exponent, and that asymmetry is the whole cost of *that* normalisation, but not of the other one, which is (vi).

(vi). Write $\Pi^* := \Pi_u \otimes (c_*\Pi_t) \otimes \Pi_{\mathsf{w}}$, the base measure of Definition 5 with its *second* factor scaled and the other two held fixed, *the statement names all three factors and* ***one*** *of them moves. First,* Π^* *is admissible.* B1(iv.a) constrains ϖ_u alone and is untouched; B1(iv.b) asks for a Borel $\mathcal{T}_0$ on which Leb and Π_t are not mutually singular, a property invariant under multiplication of Π_t by $c_* > 0$; B1(iv.d) constrains the masses $\varpi_{\mathsf{w}}(l)$ alone and is likewise untouched; and $c_{\Pi^*}^2 = \Pi_u(\mathbb{R}) \int_{\mathcal{T}} \bar{w}^2 \, d(c_*\Pi_t) = c_* \, c_\Pi^2 < \infty$, so B1(iv.c) holds, Π_{w} being a probability measure and contributing the factor 1 at every L. So c_* ranges over $(0, \infty)$ inside the maintained block and is a genuine free parameter. *Second, the transformation.* For every measurable ξ, $\|\xi\|^2_{L^2(\Pi^*)} = c_* \, \|\xi\|^2_{L^2(\Pi)}$, so every $L^2(\Pi)$ norm is multiplied by $c_*^{1/2}$; and $c_{\Pi^*} = c_*^{1/2} c_\Pi$ by the display above, so (35), which is linear in c_Π, every other factor in it being Π-free — c_C and c_2 by C3(ii), h_u and h_x by choice, and the factor $\max\{\|1/\pi\|_\infty^{1/2}, \varepsilon_\pi^{-1/2}\}$ because π is a domain-side density and ε_π a constant Definition 22 fixes, *that last factor being the one this list acquired on 2026-08-27 and named here rather than left to "every other"* — gives $\omega_n \mapsto c_*^{1/2}\omega_n$ and $\omega_n^2 \mapsto c_* \, \omega_n^2$. *Third, what does not move.* A5(i)'s $\hat{\delta}_n = \max\{n^{-1/2}\sqrt{\ln n}, r_n\}$ is a functional of n and of P through A4(iii)'s r_n only, and (37) builds $\hat{\delta}_{n,k}^{\text{eff}}$ from $\hat{\delta}_n$, h_u and $\hat{\varepsilon}_{k-1}$; none of the four carries Π. Hence the right-hand side $\omega_n^2 \hat{\delta}_n^{-2}$ of (36) is multiplied by c_* exactly, which is the claim. *And the contrast with (ii) is where the content is*: there the residual and the threshold both carried ω_*^{-1} and the index was preserved; here the residual carries $c_*^{1/2}$ and the threshold carries nothing, so the two scales are not of the same kind and no clause among (i)–(v) covers this one. □

Proof of Proposition 28. The strategy is one sentence: *at each stage, solve a linear ill-posed equation whose data error is the operator noise plus the previous stage's squared error, and iterate the resulting recursion to its fixed point*, which exists and is (56) under (P-f).

Step 0: the route, decided once, because two are available and they are not compatible. The operator of the stage-k equation, $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}$, is random. There are two ways to bring a deterministic regularisation theory to bear on it and *this proof takes the first.*

– *Conditional on the sample.* Freeze the realisation. Then $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}}|_{\mathcal{S}_n}$ is a fixed bounded operator between Hilbert spaces, and (F-f) applies verbatim, with no perturbation argument anywhere. *The price is that the source condition must be assumed for the realised operator*, which is exactly what (P-a.3) says and why (P-a.3) is stated at $\hat{\mathcal{A}}$, indeed at its sieve compression (53), and not at $\mathcal{A}$.

– *Perturbation of $\mathcal{A}$.* Assume the source condition at the population $\mathcal{A}_{\hat{\varphi}^{(k-1)}}$ and transfer it, paying $\|\hat{\mathcal{A}}-\mathcal{A}\| = O_P(\gamma_n)$ by Lemma 78. *This route is not taken, and it is not available*: by Lemma 84(ii.a) the only general device for the transfer, operator monotonicity of $z \mapsto z^{2\mu}$, is unavailable for $2\mu > 1$, which is the whole regime this proposition inhabits.

Three consequences of the choice, all of them simplifications. (a) γ_n, hypothesis (P-b) and Lemma 78 are *not consumed*; (P-b) is struck from the hypothesis list and the $O_P(\gamma_n\varepsilon_{k-1})$ term is struck from (178). (b) The classical theorem is applied at a *fixed* operator, so no statement below is uniform over operators. (c) Everything is on $\mathcal{S}_n$, so the passage from $v_\dagger$ to its sieve projection must be performed explicitly, which Step 2 does.

Step 1: the linearised equation and its exact data error. Fix a stage k and write $\varphi_\circ := \hat{\varphi}^{(k-1)}$, $v_\dagger := \varphi_\dagger - \varphi_\circ$ and $\varepsilon_{k-1} := \|v_\dagger\|_{L^2(\pi)}$, *which is Step 3's ε_k read one stage back*, $\varepsilon_j := \|\varphi_\dagger - \hat{\varphi}^{(j)}\|_{L^2(\pi)}$ being one sequence and not two. Then, in $L^2(\Pi)$,

$$\hat{\mathcal{A}}_{\varphi_\circ} v_\dagger + \hat{T}_{h_x}(\varphi_\circ, \theta_\dagger) = \underbrace{\hat{T}_{h_x}(\varphi_\dagger, \theta_\dagger)}_{\text{(N1)}} - \underbrace{\left\{\hat{T}_{h_x}(\varphi_\dagger, \theta_\dagger) - \hat{T}_{h_x}(\varphi_\circ, \theta_\dagger) - \hat{\mathcal{A}}_{\varphi_\circ} v_\dagger\right\}}_{\text{(N2)}}. \tag{174}$$

This is an identity, not an approximation. (N1) is the operator noise at the truth and is $O_P(\hat{\delta}_n)$ by Lemma 77 together with Lemma 81 Step 4, which prices $\hat{T}_{h_x} - \hat{T}$ at $\mathsf{R}h_x^{s-1/2} = O\big(h_u\hat{\delta}_n\big)$, *hypothesis* (P-g), an upper bound on h_x, and one that C5′ does not carry. *The factor h_u is*

what lets this step close, and it is not decoration: Lemma 81 Step 4 charges the displacement of the two operator families in **operator** norm, at $\mathsf{c}\,h_u^{-1}\mathsf{R}\,h_x^{s-1/2}$, so what must be $O(\hat{\delta}_n)$ is that product and not $\mathsf{R}\,h_x^{s-1/2}$ alone; read at (P-g) as printed, the product is $O(\hat{\delta}_n)$ and (N1) is $O_P(\hat{\delta}_n)$. (N2) is the exact Taylor remainder of $\hat{T}_{h_x}$ and is bounded by (105), hence by $\frac{1}{2}\mathsf{c}_4 c_\Pi h_u^{-2}\cdot n_{\mathrm{m}}^{-1}\sum_i(\Delta_{h_x}v_{\dagger,i})^2$. That empirical second moment must be converted to $\|v_\dagger\|^2_{L^2(\pi)}$ without a bandwidth, and (P-f) presumes exactly that, so the conversion is performed here in full rather than asserted. Three steps, and the third is where the uniformity is bought.

(N2.1) The smoothing operator is bounded on $L^2(dx)$ and on L^∞, with h_x-free constants. Write $v_{h_x}(x) := h_x^{-1}\int_{\mathcal{X}} C_{h_x}(x'-x, I(x))v(x')dx'$, the smoothing implicit in (96), so that $\Delta_{h_x}v_i = \Delta(v_{h_x})_i$. *This is* ***not*** *a convolution*, the boundary index $I(x)$ of Lemma 67(iv) varies with the evaluation point, so Young's inequality does not apply and **Schur's test** is used instead, which needs only the two one-sided kernel integrals and gives the same constant. With $\mathsf{k}_{h_x}(x,x') := h_x^{-1}C_{h_x}(x'-x, I(x))$ and Θ_I of (14),

$$\sup_x\int\left|\mathsf{k}_{h_x}(x,x')\right|dx' = \sup_{I\in\mathfrak{I}}\int_I|\Theta_I| \;\le\; \int_{\mathbb{R}}\bar{\Theta} \;=\; 2c_C\mathsf{c}_2, \qquad \sup_{x'}\int\left|\mathsf{k}_{h_x}(x,x')\right|dx \;\le\; \int_{\mathbb{R}}\bar{\Theta} \;=\; 2c_C\mathsf{c}_2,$$

both by C3(ii) and (16), the second by the substitution $x = x' - h_x\varsigma$ and the same envelope bound. *Both are now the one number* $\|\bar{\Theta}\|_{L^1}$*, read directly*, where the support reading needed a sup times a length. Schur's test therefore gives $\|v_{h_x}\|_{L^2(dx)} \le 2c_C\mathsf{c}_2\|v\|_{L^2(dx)}$, and the same two integrals give $\|v_{h_x}\|_\infty \le 2c_C\mathsf{c}_2\|v\|_\infty$. *Neither constant carries* h_x, and that is the whole reason (P-f) is bandwidth-free in h_x.

(N2.2) The conversion in **expectation** is h_x-free, and it holds on all of $\mathcal{V}$. By (N2.1) and Lemma 45(i),

$$E\left[(\Delta_{h_x}v)^2\right] \;\le\; \mathsf{K}\,\|v\|^2_{L^2(\pi)}, \qquad \mathsf{K} \;:=\; 4\bar{c}_\pi\,(2c_C\mathsf{c}_2)^2\,\|\pi\|_\infty\,\|1/\pi\|_\infty, \tag{175}$$

for *every* $v\in\mathcal{V}$, with K free of n, of h_x and of h_u. The route, and **both** of C1(ii)'s two-sided

bounds are consumed by it, one on each leg of a round trip between the two L^2 norms. Lemma 45(i) at $h = v_{h_x}$ gives $E[(\Delta_{h_x} v)^2] = \|\Delta(v_{h_x})\|^2_{L^2(P)} \le 4\bar{c}_\pi \|v_{h_x}\|^2_{L^2(\pi)}$; (N2.1) is a bound in $L^2(dx)$ and not in $L^2(\pi)$, so reaching it costs $\|v_{h_x}\|^2_{L^2(\pi)} \le \|\pi\|_\infty \|v_{h_x}\|^2_{L^2(dx)}$ on the way out and $\|v\|^2_{L^2(dx)} \le \|1/\pi\|_\infty \|v\|^2_{L^2(\pi)}$ on the way back, both finite by C1(ii), which bounds π above *and* below away from zero. *The factor $\|\pi\|_\infty$ is the outward leg and it is carried here explicitly*: a version of this constant that omits it is short by exactly one bound of C1(ii), and *nothing below reads* K *at its value*, only its finiteness and its positivity, so no conclusion moves. *So the conversion (P-f) presumes is genuinely available, and it is available ambiently*; what is *not* available ambiently is the same statement *uniformly in* v, which is what (N2) needs, and that is (N2.3).

(N2.3) Uniformity over the sieve, by Lemma 123(ii) at the points of a volumetric net, and **not** by a Markov bound, which does not survive the net. Let $\mathcal{S}_n^1 := \{v \in \mathcal{S}_n : \|v\|_{L^2(\pi)} = 1\}$ and, for $v \in \mathcal{S}_n^1$, $\Xi_n(v) := n_{\mathrm{m}}^{-1} \sum_{i \in \mathcal{I}_m} (\Delta_{h_x} v_i)^2$, a positive semi-definite quadratic form in the coefficients of v. Ξ_n is bound here, carries an n subscript and **never** appears without a function argument; the upright Ξ, Ξ' are Remark 104(1)'s range-side bound variables and Ξ_* is Proposition 71's limit, and none of the three occurs in this proof.

– The summands are bounded, and on the sieve the bound is L_x and not h_x^{-1}. By (N2.1)'s sup-norm clause, $|\Delta_{h_x} v_i| \le 2\|v_{h_x}\|_\infty \le 4 c_C \mathsf{c}_2 \|v\|_\infty$, and by the B-spline inverse inequality of Lemma 81(3d), $\|v\|_\infty \le \mathsf{c}_{11} L_x^{1/2}$ on $\mathcal{S}_n^1$. Hence $0 \le (\Delta_{h_x} v_i)^2 \le \mathsf{B} := (4 c_C \mathsf{c}_2 \mathsf{c}_{11})^2 L_x$, uniformly in i and in $v \in \mathcal{S}_n^1$, and free of h_x. *Off the sieve there is no such bound*: the $L^2(\pi)$ unit ball of $\mathcal{V}$ has infinite sup-norm, so the only available summand bound is (165)'s $\mathsf{c}\, h_x^{-1}$, *which is exactly where the spurious h_x^{-1} comes from, and it is a defect of the ambient route and not of (P-f).*

– The tail at a fixed v. The $(\Delta_{h_x} v_i)^2$, $i \in \mathcal{I}_m$, are i.i.d. by A0, nonnegative, bounded by B, with mean at most K by (175). So Lemma 123(ii) with $N = n_{\mathrm{m}}$, at $\mathsf{s} = \mathsf{K}$, gives $P\big(\Xi_n(v) \ge 2\mathsf{K}\big) \le \exp\big(-\mathsf{c}_{18} n_{\mathrm{m}} / L_x\big)$ with $\mathsf{c}_{18} := 3\mathsf{K} \big/ \big(8(4 c_C \mathsf{c}_2 \mathsf{c}_{11})^2\big) > 0$. The

exponent is linear in n_{m}/L_x; Lemma 123(i) would give n_{m}/L_x^2, and Markov would give a reciprocal. c_{18} is declared in place here, is used at four lines of this step, and is **carried** only within this step — but it is **not named** only here. Lemma 66(g)'s note reads it as the one quantity of its accounting that is *invariant* in $(\mathsf{c}_2, \|\bar{\Theta}\|_{L^1})$ and computes its closed form; the note on the constants that follows repeats the ruling; and C3's own notes in the assumption block make the same point about it. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces read "at no other line of any file" and was short by four printed lines, two of which compute the constant.

– The net, and the continuity that links its points to the supremum. Let $v_1, \dots, v_N$ be a $\frac{1}{4}$-net of $\mathcal{S}_n^1$ in $\|\cdot\|_{L^2(\pi)}$; since $\mathcal{S}_n$ has dimension at most L_x, the volumetric bound of Lemma 81(3d) gives $N \le 9^{L_x}$. *The step from net points to the supremum is not an inequality about Ξ_n but the standard one about a symmetric form, and it is supplied rather than presumed*: write $\Xi_n(v, v') := n_{\mathrm{m}}^{-1} \sum_{i \in \mathcal{I}_m} (\Delta_{h_x} v_i)(\Delta_{h_x} v'_i)$ for the symmetric bilinear form the quadratic form is the diagonal of, so that $\Xi_n(v) = \Xi_n(v, v)$. The supremum $\sup_{v \in \mathcal{S}_n^1} \Xi_n(v)$ is finite and is attained at some $v^* \in \mathcal{S}_n^1$, it exists, the sphere being compact in finite dimension. Pick v_j with $\|v^* - v_j\| \le \frac{1}{4}$. By homogeneity $\Xi_n(v) \le \Xi_n(v^*)\|v\|^2$ for every $v \in \mathcal{S}_n$, so the Cauchy–Schwarz inequality for a positive semi-definite form gives $|\Xi_n(v, v')| \le \Xi_n(v)^{1/2} \Xi_n(v')^{1/2} \le \Xi_n(v^*)\|v\|\,\|v'\|$ for all $v, v' \in \mathcal{S}_n$, *the bound variables are v, v' and not a, b, a being the* ***reserved*** *degree of ill-posedness and b the weight anchor and sieve basis*; hence

$$\big|\Xi_n(v^*) - \Xi_n(v_j)\big| = \big|\Xi_n(v^* - v_j, v^*) + \Xi_n(v_j, v^* - v_j)\big| \le \Xi_n(v^*) \cdot \tfrac{1}{4} \cdot \big(\|v^*\| + \|v_j\|\big) = \tfrac{1}{2}\Xi_n(v^*),$$

so $\Xi_n(v_j) \ge \frac{1}{2}\Xi_n(v^*)$ and therefore $\sup_{v \in \mathcal{S}_n^1} \Xi_n(v) = \Xi_n(v^*) \le 2 \max_{j \le N} \Xi_n(v_j)$. *No operator is named here, and none is needed*: everything the step uses is the polarisation of a form this proof has already defined.

– The union bound, and the condition it needs. Combining,

$$P\Big(\sup_{v\in\mathcal{S}_n\setminus\{0\}}\frac{n_{\mathrm{m}}^{-1}\sum_i(\Delta_{h_x}v_i)^2}{\|v\|^2_{L^2(\pi)}}\ \geq\ 4\mathsf{K}\Big)\ \leq\ 9^{L_x}\exp\Big(-\frac{\mathsf{c}_{18}n_{\mathrm{m}}}{L_x}\Big)\ =\ \exp\Big(L_x\ln 9-\frac{\mathsf{c}_{18}n_{\mathrm{m}}}{L_x}\Big),$$

which tends to zero if and only if $L_x^2=o(n_{\mathrm{m}})$, i.e. under (10)'s $n_{\mathrm{m}}\asymp n$, *if and only if* $L_x=o(n^{1/2})$, *which is A8's own upper end*; *that condition carries* ***no*** *assumption tag and is Definition 2's balanced-split* ***construction convention***, (10) *stating exactly what the step reads*; and under the marginally stronger $L_x\ll(n/\ln n)^{1/2}$ it is $O(n^{-\mathsf{c}})$ for every c, so the event may be intersected with the finitely many others of Step 3 at no cost.

Hence $\sup_{v\in\mathcal{S}_n\setminus\{0\}}n_{\mathrm{m}}^{-1}\sum_i(\Delta_{h_x}v_i)^2/\|v\|^2_{L^2(\pi)}=O_P(1)$ and (N2) $=O_P(h_u^{-2}\varepsilon_{k-1}^2)$, *the* h_u^{-2} *being all that survives.*

A Markov bound at each net point will not serve. It has a *reciprocal* tail, the net has $e^{O(L_x\ln n)}$ points, and a union bound over exponentially many events needs an *exponential* tail; that route requires $L_x=O(1)$. Lemma 123(ii)'s variance-sensitive tail is what makes the surviving condition A8's upper end rather than $L_x=o(n^{1/3})$. *And the symmetric-form step above is what links net points to the supremum*, which is why a $\frac{1}{4}$-net suffices with the explicit factor 2. The crude route, $\|\Delta_{h_x}v\|_2^2\leq n_{\mathrm{m}}\max_i(\Delta_{h_x}v_i)^2$ with (165), would give $h_u^{-2}h_x^{-1}\varepsilon_{k-1}^2$, and the spurious h_x^{-1} would make the condition of Remark 125 fail at every configuration of interest. So (P-f) is correct as printed, and the reason is now visible: the conversion it presumes is (175), which is h_x-free on all of $\mathcal{V}$; what needs the sieve is not the conversion but its uniformity, and on $\mathcal{S}_n$ the price of uniformity is L_x, through the inverse inequality, and not h_x^{-1}. A reader who reconstructs the step from (165) alone will conclude that (P-f) is missing an h_x^{-1}; the two displays are the ambient and the sieve conversion constants, and only the second is the one this step runs on. The sieve projection, performed rather than presumed. Put $\bar{v}_\dagger:=P_{\mathcal{S}_n}v_\dagger$, the $L^2(\pi)$-orthogonal projection of $v_\dagger$ on $\mathcal{S}_n$; this is the object the equation of Definition 22(a) can actually reach, since every iterate lies in $\mathcal{S}_n$, and it

is the object (P-a.3)'s source condition is stated for. *By A8's* ***basis*** *clause, B-splines of degree exceeding* $s - \frac{1}{2}$ *on a quasi-uniform mesh, and the spline direct theorem* (162), whose hypothesis is degree at least $s - 1$ and is therefore implied by A8's with half an order to spare, *both are now stated in degree, and this step performs no conversion because none is left to perform* , and *not* by A8's lower end, which this proof does not use and which Proposition 28 does not assume, $\|v_\dagger - \bar{v}_\dagger\|_{L^2(\pi)} \le \mathsf{c}\,\mathsf{R}\,L_x^{-s}$, and that is $O(h_u\hat{\delta}_n)$ by (P-c). *The distinction is not pedantry.* The direct theorem is a statement about the approximation power of the basis at a given L_x; A8's lower end $L_x \gg n^{1/(2s)}$ is a different requirement, calibrated to $n^{-1/2}$ for the balancing route, and (P-c) is this proposition's own replacement for it, calibrated to $h_u\hat{\delta}_n$. *Invoking A8's lower end here as well would cost*: A8's lower end with (P-d) requires $s > \big(2(1-2\alpha)\big)^{-1}$, which at $\alpha = \frac{3}{8}$ is $s > 2$, while the window (P-c) $\wedge$ (P-d) actually used requires only (54), $s > \frac{27}{16}$. Both lines are recorded in the assumption block's budget table. Now, since $\|\omega_n^{-1}\hat{\mathcal{A}}_{\varphi_\circ}|_{\mathcal{S}_n}\| \le 1$ and, by Proposition 70(i) together with Step 1's empirical-moment bound, $\|\hat{\mathcal{A}}_{\varphi_\circ}(v_\dagger - \bar{v}_\dagger)\|_{L^2(\Pi)} \le \mathsf{c}\,h_u^{-1}\|v_\dagger - \bar{v}_\dagger\|_{L^2(\pi)}$, replacing $v_\dagger$ by $\bar{v}_\dagger$ in (174) adds $O_P(h_u^{-1}\mathsf{R}L_x^{-s})$ to the data error. *This is the step (P-c) exists to pay for*, and it is why (P-c) reads $\mathsf{R}L_x^{-s} = O(h_u\hat{\delta}_n)$ rather than $O(\hat{\delta}_n)$.

So $\bar{v}_\dagger$ *solves the linear equation of Definition 22(a) up to a data error of size*

$$\delta_{n,k}^{\mathrm{eff}} \;:=\; \mathsf{c}_{16}\Big(\hat{\delta}_n \ln n \;+\; h_u^{-2}\varepsilon_{k-1}^2\Big) \qquad \text{with } \varepsilon_{k-1} = \Big\|\varphi_\dagger - \hat{\varphi}^{(k-1)}\Big\|_{L^2(\pi)}, \tag{176}$$

$\mathsf{c}_{16} < \infty$ collecting the constants of (N1), (N2) and the projection step. The constant here is c_{16} and not c_{14}, and the two must not be the same symbol. c_{16} is *analytic*: it is delivered by this step, it depends on the kernel constants, on c_Π and on the design, and the analyst does not know it. c_{14} is *chosen*: it is an input to (37), printed before the data are seen. *What (L3) needs is* $\mathsf{c}_{14} \ge \mathsf{c}_{16}$, and Definition 22 says so. *The bandwidth factor* h_u^{-2} *is carried explicitly here, and* ***not*** *only here*: (37), the computable majorant named in the next sentence, carries the same factor, and Step 4's fixed-point recursion reads it at every stage. *No claim*

of exhaustiveness is made, and that is deliberate: the draft this sentence replaces read "and nowhere else" and was short by some thirty printed lines, the pointer in its own next sentence among them. Remark 125 records what it costs and how it is paid. *Note that* (176) *is not observable*, since ε_{k-1} is not; (37) is its computable majorant and Step 2 uses the majorant.

Step 2: linear Landweber–Fridman with the discrepancy principle, applied to that equation, and the five hypotheses of (F-f)*, Remark 124, checked one by one.* Freeze the sample, per Step 0, and write, *in the normalised scale of Definition 22*, $\mathcal{L} := \omega_n^{-1}\hat{\mathcal{A}}_{\varphi_\circ}|_{\mathcal{S}_n}$, $y^\delta := -\omega_n^{-1}\hat{T}_{h_x}(\varphi_\circ, \theta_\dagger)$, with ω_n of (35). *The* ***domain Hilbert space*** *is fixed before anything is checked, because* (F-f) *reads it in three roles at once*, Remark 124: it defines $\mathcal{L}^*$, it is the space $\|\mathcal{L}\|$ is the operator norm out of, and it is the space the source pre-image ς is measured in. *This proof runs the pair* $\big(\mathcal{S}_n \subset L^2(\pi),\, L^2(\Pi)\big)$, which is the infeasible reading, *whose output is the* $\hat{\varphi}^\pi$ *of Definition 22 and is the object* (56) *is scoped to*; the *feasible* construction of Definition 22(b) forms the adjoint at $\hat{\pi}$ and therefore runs the pair whose domain is $\mathrm{span}(b_1, \ldots, b_{L_x}) \cap \{h : \int_{\mathcal{X}} h\, d\hat{\pi} = 0\}$ inside $L^2(\hat{\pi})$, the $\hat{\pi}$-centred sieve of Remark 89(7a), against the same $L^2(\Pi)$. *That set is written out and claims no symbol*, on the write-it-out precedent: it is named here only to identify the pair and is used at no step of this proof, which runs the first. *Two of the five items below are sensitive to which of the two is in force, and they are sensitive in different ways; no claim is made that the other three say the* ***same thing*** *in the two pairings, only that each is true when read consistently inside whichever one is fixed.* (L1) is a bound on $\|\mathcal{L}\|$ *out of* the domain space, so its constant changes with the pairing, and (35) is sized to cover both: that is settled here. (L5) reads a hypothesis that names the domain space through an adjoint and a pre-image norm, so it is stated at one pairing and consumed at the other, and *that is not settled here*: it is the open item Remark 89(7d), and this proof runs (L5) at the infeasible pairing throughout. (L2)–(L4) are statements about iterates, a data error and a stopping index, each computed consistently inside one fixed pair and true in either, with the constants of (L3)'s projection step transferring between them by Remark 89(7b)'s deterministic norm equivalence.

(L1) $\mathcal{L}$ is a bounded linear operator between Hilbert spaces with $\|\mathcal{L}\| \le 1$, *in* ***either*** *of the two domain pairings just named.* Boundedness and linearity are Proposition 70(i); the bound ω_n of (35) is *deterministic* and, by Proposition 70(i′), dominates the conversion cost of *both* $L^2(\pi)$ and $L^2(\hat{\pi})$, its factor being the maximum of the two; so $\|\mathcal{L}\| \le 1$ holds **surely**, at every $\varphi \in \mathcal{V}$ and in particular at the realised $\varphi_\circ$, and it holds whichever pairing the reader runs the iteration in. Neither Lemma 83 nor a probability qualifier is needed here. *The restriction to $\mathcal{S}_n$ is free in both*, an operator norm out of a subspace never exceeding the norm out of the whole space. *That the normalising factor diverges is not a defect*: by Lemma 85 it is a relaxation parameter, the iterates and the stopping index are unchanged, the conclusion (178) is exactly invariant in it, and the one thing that is not invariant, the stopping-index bound, is paid for by the cap (36).

(L2) The iteration analysed is the **infeasible** reading $\hat{\varphi}^{\pi}$ of Definition 22(b) — *and* ***not*** *the feasible iteration that clause instructs*, whose adjoint is formed at $\hat{\pi}$ and whose projection is onto the $\hat{\pi}$-centred sieve of Remark 89(7a) — and the operator it runs on is the **compression**. *The infeasible reading* of Definition 22(b), that is, clause (b) with $\hat{\mathcal{A}}^*_{\varphi_\circ}$ the $L^2(\pi)$–$L^2(\Pi)$ adjoint and the projection $L^2(\pi)$-orthogonal onto $\mathcal{S}_n$, runs $v_{m+1} = v_m - \omega_n^{-2}\hat{\mathcal{A}}^*_{\varphi_\circ}[\hat{\mathcal{A}}_{\varphi_\circ} v_m + \hat{T}_{h_x}]$ from $v_0 = 0$, projected on $\mathcal{S}_n$, and by Lemma 85(i) that is $v_{m+1} = v_m + \mathcal{L}^*(y^\delta - \mathcal{L}v_m)$ in the normalised pair, term by term. Since $\mathcal{L}^*$ already maps into $\mathcal{S}_n$ after the restriction, the projection is the identity on the iterates and the recursion is the classical one. Two consequences, and the second is what (P-a.3) is stated for. First, the projection identity is available *because* $P_{\mathcal{S}_n}$ has been absorbed into $\mathcal{L}^*$: the operator the recursion sees is

$$\mathcal{L}^*\mathcal{L} \;=\; \omega_n^{-2}\, P_{\mathcal{S}_n}\hat{\mathcal{A}}^*_{\varphi_\circ}\hat{\mathcal{A}}_{\varphi_\circ}\Big|_{\mathcal{S}_n} \;=\; \omega_n^{-2}\,\hat{\mathcal{P}}^{\mathcal{S}}_{\varphi_\circ}, \tag{177}$$

the *compression* (53) of $\hat{\mathcal{A}}^*\hat{\mathcal{A}}$ to $\mathcal{S}_n$, and **not** the restriction of $\hat{\mathcal{A}}^*\hat{\mathcal{A}}$ to $\mathcal{S}_n$. Second, *the two are not the same operator and their ranges are nested in neither direction*, so

"$\bar v_\dagger \in \mathcal{R}\big((\hat{\mathcal{A}}^*\hat{\mathcal{A}})^\mu\big) \cap \mathcal{S}_n$" and "$\bar v_\dagger \in \mathcal{R}\big((\hat{\mathcal{P}}^{\mathcal{S}})^\mu\big)$" are different statements: $\hat{\mathcal{A}}^*\hat{\mathcal{A}}$ need not preserve $\mathcal{S}_n$, and $\hat{\mathcal{P}}^{\mathcal{S}}$ is a different operator on $\mathcal{S}_n$ from $\hat{\mathcal{A}}^*\hat{\mathcal{A}}|_{\mathcal{S}_n}$ unless it does. (P-a.3) states the hypothesis at (53), so the two agree, and Lemma 85(iii), stated for a single fixed domain and handling only the scalar rescaling, is not asked to bridge a gap it cannot bridge.

(L3) The data error is bounded by the level used in the rule, and the two constants are what make that an inequality rather than an identity. By Step 1 the exact data error satisfies $\omega_n\|\mathcal{L}\bar v_\dagger - y^\delta\| \le \delta^{\mathrm{eff}}_{n,k}$ w.p.$\to 1$, with $\delta^{\mathrm{eff}}_{n,k}$ of (176); by the induction hypothesis of Step 4, $\varepsilon_{k-1} \le \hat\varepsilon_{k-1}$ w.p.$\to 1$; and $\mathsf{c}_{14} \ge \mathsf{c}_{16}$ by Definition 22. The three together give $\delta^{\mathrm{eff}}_{n,k} \le \hat\delta^{\mathrm{eff}}_{n,k}$ of (37) w.p.$\to 1$. Each of the three is load-bearing. *On the induction.* The induction hypothesis is used at exactly one place, in $h_u^{-2}\varepsilon_{k-1}^2$, and it is used as a *substitution into a bound*, not as a chain of stage-dependent comparisons: the constants $\mathsf{c}_{14}, \mathsf{c}_{16}$ do not depend on k, so nothing accumulates across stages and no quantity of the form "$\varepsilon_{k-1} \le C_k\hat\varepsilon_{k-1}$ with C_k composing" arises. That device, and the finiteness of $\bar k$ that would have been needed to bound $\sup_k C_k$, is not required here. Finiteness of $\bar k$ is still needed, but for the O_P constants of (180) and for A5(vi′), not for this step. *A* ***fixed*** *level $\tau_{\mathrm{DP}}\hat\delta_n \ln n$ would not support this step*: the inequality fails precisely in the regime $h_u^{-2}\varepsilon_{k-1}^2 \ge \hat\delta_n \ln n$, which Step 4(a) is about.

(L4) The discrepancy rule of (F-f) is the rule implemented. Definition 22(c) stops at the first $m \le \bar m_n$ with $\|\mathcal{L}v_m - y^\delta\| \le \tau_{\mathrm{DP}}\omega_n^{-1}\hat\delta^{\mathrm{eff}}_{n,k}$, which by Lemma 85(ii) is the index at which the unscaled rule of Definition 22(c) stops, and $\tau_{\mathrm{DP}} > 2$ by (P-e). The set is non-empty w.p.$\to 1$: by (L3) and (173) the rule stops by index $\mathsf{c}\Big(\omega_n^{2\mu}\mathsf{R}_*\big/(\omega_n^{-1}\hat\delta^{\mathrm{eff}}_{n,k})\Big)^{2/(2\mu+1)} = \mathsf{c}\,\omega_n^2\Big(\mathsf{R}_*/\hat\delta^{\mathrm{eff}}_{n,k}\Big)^{2/(2\mu+1)} \le \omega_n^2\hat\delta_n^{-2} \le \bar m_n$ for all n large, the middle inequality because $2/(2\mu+1) < 2$ for $\mu > \frac12$ and $\hat\delta^{\mathrm{eff}}_{n,k} \ge \mathsf{c}_{14}\hat\delta_n \ln n$. *The factor ω_n^2 is Lemma 85(v) and is exactly why the cap is* (36) *and not $\hat\delta_n^{-2}$.* On the complement the convention $\hat m^{(k)} = \bar m_n$ applies and contributes to an event of vanishing probability. A7 in the form (13) is

what makes the $\hat{\delta}_n \ln n$ part of the level dominate (N1), Proposition 79(iii).

(L5) The source condition holds at the operator used, and it is now **two lines**, because it is assumed where it is consumed. By (P-a.3), on an event of probability tending to one $\bar{v}_\dagger = \big(\hat{\mathcal{P}}^{\mathcal{S}}_{\varphi_\circ}\big)^\mu \varsigma$ with $\|\varsigma\|_{L^2(\pi)} \le \mathsf{R}_*$; by (177), $\hat{\mathcal{P}}^{\mathcal{S}}_{\varphi_\circ} = \omega_n^2\, \mathcal{L}^*\mathcal{L}$; so by Lemma 85(iii) at $\omega = \omega_n^{-1}$, or directly, since $\big(\omega_n^2 \mathcal{L}^*\mathcal{L}\big)^\mu = \omega_n^{2\mu}(\mathcal{L}^*\mathcal{L})^\mu$ by the spectral-mapping rule for the continuous function $\lambda \mapsto \lambda^\mu$ composed with $\lambda \mapsto \omega_n^2\lambda$,

$$\bar{v}_\dagger \;=\; \big(\mathcal{L}^*\mathcal{L}\big)^\mu\big(\omega_n^{2\mu}\varsigma\big), \qquad \big\|\omega_n^{2\mu}\varsigma\big\|_{L^2(\pi)} \;\le\; \omega_n^{2\mu}\mathsf{R}_*,$$

which is exactly (F-f)'s source hypothesis at radius $\omega_n^{2\mu}\mathsf{R}_*$. Three things are absent from this step and each would be a defect. *(a) No interpolation.* Passing from a Hilbert-scale source condition plus a link condition to this display is Hilbert-scale operator interpolation at exponent 2μ, available only at $2\mu \le 1$, while (55) forces $2\mu > \frac{5}{3}$ and this file's own Lemma 84(ii.a) constructs a 2×2 witness refuting the device at *every* exponent above 1. *Finite-dimensionality of $\mathcal{S}_n$ does not rescue it*: range membership there is vacuous, so all of such a step's content is the uniform-in-n pre-image bound, which *is* the interpolation inequality. Remark 86 carries the full disposition. *(b) No naming gap.* The operator is (53), which is what (L2) established and what (177) exhibits. *(c) No O_P radius.* (P-a.3) fixes R_* deterministically on an event of probability tending to one, so c_{17} below is a *number* and "$\mathsf{c}_{15} \ge \mathsf{c}_{17}$" is an inequality between numbers; from a pre-image of norm $O_P(\mathsf{R}_*)$ no such inequality follows, for the reason Remark 87 gives. And no transfer from $\mathcal{A}$ to $\hat{\mathcal{A}}$ is performed, by Step 0.

Therefore (F-f) applies. *It is applied in the normalised scale, so the source radius is $\omega_n^{2\mu}\mathsf{R}_*$ and the noise level $\omega_n^{-1}\hat{\delta}^{\mathrm{eff}}_{n,k}$; the two factors of ω_n cancel exactly, by Lemma 85(iv), and the cancellation is performed here rather than assumed*:

$$\big\|v_{\hat{m}^{(k)}} - \bar{v}_\dagger\big\|_{L^2(\pi)} \;\le\; c_\mu\big(\omega_n^{2\mu}\mathsf{R}_*\big)^{1/(2\mu+1)}\big(\omega_n^{-1}\hat{\delta}^{\mathrm{eff}}_{n,k}\big)^{2\mu/(2\mu+1)} \;=\; c_\mu\,\mathsf{R}_*^{1/(2\mu+1)}\big(\hat{\delta}^{\mathrm{eff}}_{n,k}\big)^{\kappa}, \qquad (178)$$

w.p.$\to 1$, whence, writing $\mathsf{c}_{17} := 2c_\mu \mathsf{R}_*^{1/(2\mu+1)}$,

$$\left\| v_{\hat{m}^{(k)}} - v_\dagger \right\|_{L^2(\pi)} \;\le\; \mathsf{c}_{17}\Big(\hat{\delta}^{\mathrm{eff}}_{n,k}\Big)^{\kappa} \qquad \text{w.p.} \to 1, \tag{179}$$

the passage from (178) to (179) absorbing the sieve approximation error $\|v_\dagger - \bar{v}_\dagger\|_{L^2(\pi)} \le \mathsf{c}\,\mathsf{R} L_x^{-s} = O(h_u \hat{\delta}_n)$ of the projection step, which is $o\Big((\hat{\delta}^{\mathrm{eff}}_{n,k})^{\kappa}\Big)$ by (P-c) and $\kappa < 1$ and is therefore below the halved constant for all n large. (178) *shows* R_*, *and it must*: the radius is the only place ω_n enters upward, so absorbing it into an O_P hides the one computation that makes the divergent normalisation harmless. And the radius shown is R_*, the **spectral pre-image** norm of (P-a.3), not the H^s radius R of Definition 80. *The two radii appear in this proof at disjoint places, and* ***no display carries both***: R only in the projection error $\mathsf{c}\,\mathsf{R} L_x^{-s}$ and in (N1) via Lemma 81 Step 4; R_* in (L4)'s stopping-index bound, throughout (L5), in (178) itself and in c_{17}. (178) *carries no operator-error term*, an $O_P(\gamma_n \varepsilon_{k-1})$ from Lemma 78 belonging to the perturbation route and not to this one. *Linear Landweber under the discrepancy principle does not saturate*: the exponent $2\mu/(2\mu+1)$ is available for every $\mu > 0$, which is the single property of the linear theory that the whole proposition turns on, and it is the property the nonlinear theory does not have. *Where L-19 is consumed, and it is no longer (L1)*: at Step 1, twice, in (N1) through Lemma 81 Step 4, and in the empirical second moment through the volumetric entropy of Lemma 81 (3d), and at Step 5, through Lemma 83's containment. *Both Step-1 uses run through clause (b) of Lemma 81 under* (123), *not* clause (a) and not (121), which is why (P-d) names the former.

Step 3: the recursion, run on the schedule and not on the unobservable error. Read $\varepsilon_k = \|\hat{\varphi}^{(k)} - \varphi_\dagger\|_{L^2(\pi)}$ at the present stage, *the same sequence Step 1 fixed and read there at* $k-1$. Since $\hat{\varphi}^{(k)} = \varphi_\circ + v_{\hat{m}^{(k)}}$ gives $\varepsilon_k = \|v_{\hat{m}^{(k)}} - v_\dagger\|$, (179) reads $\varepsilon_k \le \mathsf{c}_{17}(\hat{\delta}^{\mathrm{eff}}_{n,k})^{\kappa}$ w.p.$\to 1$, and $\mathsf{c}_{15} \ge \mathsf{c}_{17}$ by Definition 22 therefore closes the induction:

$$\varepsilon_k \;\le\; \hat{\varepsilon}_k \quad \text{w.p.} \to 1, \qquad \hat{\varepsilon}_k = \mathsf{c}_{15}\Big(\mathsf{c}_{14}\Big(\hat{\delta}_n \ln n + h_u^{-2}\hat{\varepsilon}_{k-1}^2\Big)\Big)^{\kappa}, \qquad \kappa := \frac{2\mu}{2\mu+1}, \tag{180}$$

which is (37) written out; in particular $\|\hat{\varphi}^{(k)}-\varphi_\dagger\|_{L^2(\pi)} = O_P(\hat{\varepsilon}_k)$. (180) *carries* ***both*** *constants of* (37): writing $\mathsf{c}_{15}(\hat{\delta}_n \ln n + h_u^{-2}\hat{\varepsilon}_{k-1}^2)^\kappa$ would silently fold c_{14}^κ into c_{15}, and the recursion analysed and the schedule implemented must be the same object down to the constants. *This is an induction on* k, its hypothesis is exactly (L3)'s $\varepsilon_{k-1} \le \hat{\varepsilon}_{k-1}$, and *its base case is (P-a.2), which must be a* w.p.$\to 1$ ***envelope*** *and not an* O_P *statement.* The base case needed is $P(\varepsilon_0 \le \hat{\varepsilon}_0) \to 1$; an O_P hypothesis gives $\varepsilon_0 = O_P(\hat{\varepsilon}_0)$, i.e. $\varepsilon_0 \le C_0\hat{\varepsilon}_0$ w.p.$\to 1$ for a C_0 that is a family indexed by tolerance, and the inflation propagates to $k = 1$, where (L3) then needs $\mathsf{c}_{14} \ge \mathsf{c}_{16}\max\{1, C_0^2\}$. *Carrying* C_0 *is not available*, because c_{14} is part of the estimator and is printed before the data are seen, so an ϵ-indexed constant makes (56) itself ϵ-indexed, Remark 87 in full, and Proposition 79(ii) is the block's own proof of the same point in a different slot. *At* $k \ge 2$ *no constant appears*, because $\mathsf{c}_{15} \ge \mathsf{c}_{17}$ absorbs it; the defect was confined to $k = 1$ and it is closed at the hypothesis. *The conversion device is free and is the block's own*: Proposition 79(iii) turns an $O_P(a_n)$ into a w.p.$\to 1$ envelope at $a_n \ln n$, and the resulting $\ln n$ on $\hat{\varepsilon}_0$ is absorbed by the *strictness* of (181), which compares polynomial exponents. Since $\bar{k}$ is finite and does not grow with n, the finitely many "w.p.$\to 1$" events intersect in an event of probability tending to one, which is where finiteness of $\bar{k}$ is used at this step, *and it is used there and not in the constants*, which do not depend on k.

Step 4: the recursion contracts, and the two thresholds it produces are different. Two regimes, both now referring to the *deterministic* sequence $\hat{\varepsilon}_k$. *(a) While* $h_u^{-2}\hat{\varepsilon}_{k-1}^2 \ge \hat{\delta}_n \ln n$, (180) gives $\hat{\varepsilon}_k \le \mathsf{c}_{15}\,(2\mathsf{c}_{14})^\kappa h_u^{-2\kappa}\hat{\varepsilon}_{k-1}^{2\kappa}$, and $2\kappa > 1$ if and only if $4\mu > 2\mu + 1$, i.e. if and only if $\mu > \frac{1}{2}$, which (55) gives with room. *The prefactor is* $\mathsf{c}_{15}(2\mathsf{c}_{14})^\kappa$ *and not* $(2\mathsf{c}_{15})^\kappa$, which is what (180) gives once both of (37)'s constants are carried; only the $O(1)$ term of b_n changes, and nothing below depends on it. Take logarithms, because the recursion is affine in them. Write $\mathsf{a}_k := \log(1/\hat{\varepsilon}_k)$; then

$$\mathsf{a}_k \;\ge\; 2\kappa\,\mathsf{a}_{k-1} - \mathsf{b}_n, \qquad \mathsf{b}_n := \log\left(\mathsf{c}_{15}(2\mathsf{c}_{14})^\kappa\right) + 2\kappa\log(1/h_u) \;=\; \tfrac{\kappa\alpha}{4}\log n + O(1)$$

by C5′(i) as amended, $h_u \asymp n^{-\alpha/8}$. The affine map has the *repelling* fixed point $\mathsf{a}^* := \mathsf{b}_n/(2\kappa-1) = \frac{\kappa\alpha}{4(2\kappa-1)}\log n + O(1)$, and $\mathsf{a}_k - \mathsf{a}^* \geq (2\kappa)^k(\mathsf{a}_0 - \mathsf{a}^*)$. Two things follow and both are substantive.

(a.1) The preliminary must beat the repelling fixed point, and (P-a.2) says so. The iteration improves only if $\mathsf{a}_0 > \mathsf{a}^*$, i.e. only if

$$\varsigma_0 \;>\; \frac{\kappa\,\alpha}{4\,(2\kappa-1)}. \tag{181}$$

This is not vacuous and it is not circular. It is a weaker requirement than the conclusion exactly when $\frac{\kappa\alpha}{4(2\kappa-1)} < \kappa\alpha$, i.e. when $4(2\kappa-1) > 1$, i.e. when $\kappa > \frac{5}{8}$, i.e. when $\mu > \frac{5}{6}$, *which is (P-f).* At $\mu = 1$, $\alpha = \frac{3}{8}$ it asks $\varsigma_0 > \frac{3}{16}$ against a delivered $\kappa\alpha = \frac{1}{4}$. Below $\frac{5}{6}$ the two cross and the procedure would require a preliminary at least as good as its own output, which is why $\mu > \frac{5}{6}$ is stated as a hypothesis and not as a convenience.

And this agreement is an **identity**, not a coincidence. Write $h_u = n^{-\beta}$ and $\hat{\delta}_n = n^{-\alpha}$, leaving β free rather than substituting $\beta = \alpha/8$. Then $\mathsf{b}_n = 2\kappa\beta\log n + O(1)$ and $\mathsf{a}^* = \frac{2\kappa\beta}{2\kappa-1}\log n + O(1)$, so the requirement that (181) be weaker than the delivered exponent $\kappa\alpha$ is

$$\kappa\alpha - \frac{2\kappa\beta}{2\kappa-1} \;>\; 0 \qquad\Longleftrightarrow\qquad \frac{\kappa}{2\kappa-1}\Big[\alpha(2\kappa-1) - 2\beta\Big] \;>\; 0 \qquad\Longleftrightarrow\qquad \alpha(2\kappa-1) \;>\; 2\beta,$$

the last step because $\kappa/(2\kappa-1) > 0$ for $\kappa > \frac{1}{2}$. *And $\alpha(2\kappa-1) \geq 2\beta$ is (P-f) itself*: (P-f) is $\delta_n^2 = O(h_u^2\hat{\delta}_n \ln n)$, which in powers of n with $\delta_n = (\hat{\delta}_n \ln n)^\kappa$ reads $2\kappa\alpha \geq 2\beta + \alpha$. So the two are the same inequality up to a strictly positive factor, the second strict, and the agreement at $\mu > \frac{5}{6}$ is forced rather than fortunate. A third comparison in the block, that branch (a)'s repelling fixed point lies above the regime boundary, is the same inequality again, and it too is forced: for a two-branch map, "the attracting fixed point lies in branch (b)" and "branch (a)'s repelling fixed point lies above it" are equivalent statements about the same map. *The fact worth printing, and it is the reason (a.1) is here at all*: (181) is a **requirement on the**

preliminary estimator, and (P-f) is exactly the condition under which that requirement is not self-defeating. This is what sharpens open item Q-1.

(a.2) The number of stages is finite and depends on $(\mu, \varsigma_0, \alpha)$, not on n. Regime (a) is exited once $\mathsf{a}_k \geq \log(1/h_u) + \frac{1}{2}\log\left(1/(\hat{\delta}_n \ln n)\right) = \left(\frac{\alpha}{8} + \frac{\alpha}{2}\right)\log n + O(\log\ln n)$, so it suffices that $(2\kappa)^k(\mathsf{a}_0 - \mathsf{a}^*) \geq \frac{5\alpha}{8}\log n - \mathsf{a}^*$, i.e. that

$$k \;\geq\; \bar{k}_a \;:=\; \left\lceil\; \log_{2\kappa} \frac{\frac{5\alpha}{8} - \frac{\kappa\alpha}{4(2\kappa-1)}}{\varsigma_0 - \frac{\kappa\alpha}{4(2\kappa-1)}} \;\right\rceil \;\vee\; 1,$$

in which $\log n$ has cancelled from numerator and denominator, both being positive by (181) and by $\kappa > \frac{5}{8}$. Three parameters enter and n does not, and the list is $(\mu, \varsigma_0, \alpha)$. The base 2κ is a function of μ alone; α factors out of the numerator but **not** out of the denominator, because ς_0 is a free exponent and not a multiple of α; and ς_0 enters the denominator directly. So the count is computable from $(\mu, \varsigma_0, \alpha)$, and neither "from μ alone" nor "from μ and ς_0" is right. This is also why (P-a.2) must supply a *rate* for the preliminary and not mere consistency. *If* $\hat{\varepsilon}_0$ *were merely* $o_P(1)$ *the count would not be bounded in* n, and the schedule (37) would not be computable either, the two defects are the same defect, and repairing the estimator repairs both. *(b) Once* $h_u^{-2}\hat{\varepsilon}_{k-1}^2 \leq \hat{\delta}_n \ln n$, (180) gives $\hat{\varepsilon}_k \leq \mathsf{c}_{15}(2\mathsf{c}_{14})^\kappa(\hat{\delta}_n \ln n)^\kappa = O(\delta_n)$, and the regime is self-sustaining: $h_u^{-2}\delta_n^2 = h_u^{-2}(\hat{\delta}_n \ln n)^{2\kappa} \leq \hat{\delta}_n \ln n$ if and only if $(\hat{\delta}_n \ln n)^{2\kappa-1} \leq h_u^2$, which is hypothesis (P-f) and is discussed in Remark 125. Hence the fixed point is $\delta_n = (\hat{\delta}_n \ln n)^\kappa$, it is attained at stage $\bar{k} := \bar{k}_a + 1$, and the schedule (37) the estimator computes tracks it.

Two arithmetic facts about that $\bar{k}$ *are recorded here because a hypothesis of this proposition quantifies over* $k \leq \bar{k}$ *and neither is visible from the statement. First,* $\bar{k} \geq 2$ *at every admissible* $(\mu, \varsigma_0, \alpha)$*, and* $\bar{k} = 1$ *is not attainable*: the ceiling in $\bar{k}_a$ is taken in disjunction with 1, so $\bar{k}_a \geq 1$ and $\bar{k} = \bar{k}_a + 1 \geq 2$. *The disjunction is not idle and its binding case is the favourable one*: when the preliminary already clears the regime-(a) exit level, $\varsigma_0 \geq \frac{5\alpha}{8}$, the ratio inside the logarithm is at most one, the ceiling is at most zero, and $\vee 1$ is what makes $\bar{k}_a$ a stage count. So even a preliminary at or below the stagewise fixed point buys

$\bar{k} = 2$ and not $\bar{k} = 1$; one regime-(b) stage would suffice for the *rate*, and the count this proof certifies is deliberately the conservative one. *Second, and consequently, (P-a.3) is never a one-stage hypothesis*: it is read at $k = 1$, where $\bar{v}_\dagger^{(1)} = P_{\mathcal{S}_n}(\varphi_\dagger - \hat{\varphi}^{(0)})$ involves no iterate of Definition 22(b), **and** at $k = 2$, where $\bar{v}_\dagger^{(2)} = P_{\mathcal{S}_n}(\varphi_\dagger - \hat{\varphi}^{(1)})$ is a functional of the realised first stage. *What that second reading does, and it is the reason the two facts are printed together*: any configuration on which the stage-1 iteration fails to contract drives $\|\hat{\varphi}^{(1)} - \varphi_\dagger\|$ above every deterministic multiple of $\|\hat{\mathcal{P}}^{\mathcal{S}}\|^{\mu}\mathsf{R}_*$, so $\|\varsigma^{(2)}\| \le \mathsf{R}_*$ fails and (P-a.3) empties. On such a configuration this proposition is **vacuous and not false**, which is the same disposition Remark 90(1) records for the block-cited reading of C5′ — and, as there, vacuity is a defect to be repaired at the hypothesis and not a licence. Remark 89(7e) is where the one mechanism of that kind this corpus has identified is priced and closed. If $\mu = \frac{1}{2}$ exactly, $2\kappa = 1$ and the recursion neither contracts nor diverges; if $\mu < \frac{1}{2}$ it diverges. That is the same index at which the inherited nonlinear route's rate theorem ceases to be available, approached from the other side; (55)'s $\frac{5}{6}$ comes from regime (b) and not from regime (a), and it is (P-f) and not the contraction that produces it.

Step 5: containment, and the absence of circularity. Steps 1–4 assume $\hat{\varphi}^{(k-1)} \in \mathcal{V}_n \cap \mathcal{S}_n$, which Lemma 83 supplies by induction on k with base case (P-a.2)'s rate for $\hat{\varphi}^{(0)}$; and Lemma 83 in turn cites Step 4 for stage k given stage $k-1$. The induction is on k and $\bar{k}$ is finite, so the mutual reference is a simultaneous induction and not a circle. The H^s half of Lemma 83 is A5(vi′), a hypothesis of this proposition and not a conclusion of it.

Step 6: the conclusion, and what A5 gains and does not. (56) is Step 4(b) at $k = \bar{k}$, *read at the infeasible pairing this proof has run since Step 2, so the object it bounds is $\hat{\varphi}^{\pi}$ and not $\hat{\varphi}$.* A5(i)'s $O_P(\hat{\delta}_n)$ half is Lemma 77(i)–(iii), its two *branches* being a declaration and so needing no discharge; A5(iii) is the arithmetic already in the block. *A5(ii) is **not** discharged, and the reason is the object and not the arithmetic*: that clause is stated at $\hat{\varphi} = \hat{\varphi}^{(\bar{k})}$ under the *feasible* reading of Definition 22(b), which no step of this proof runs; what this step

establishes is the same display at $\hat{\varphi}^{\pi}$, and the passage between the two is Remark 89(7), open at (7c) and (7d). *The second half of A5(ii)*, $\delta_n = o_P(n^{-1/4})$, is a statement about the deterministic sequence δ_n alone and **is** delivered, by Proposition 79(iv) under (7), at either reading. A5(vi′) is **not** discharged: it is consumed, at Lemma 83(c), and it is listed among this proposition's hypotheses for that reason. A5(vi) follows from A5(vi′) at $k = \bar{k}$ and is likewise not discharged. A5(iv) and A5(v) concern $\hat{\ell}$ and the plug-in and are untouched here. □

Remark 125 ((P-f), why its thresholds are strict, and the re-tuning of h_u). Hypothesis (P-f) is $\delta_n^2 = O\big(h_u^2\,\hat{\delta}_n \ln n\big)$, equivalently $(\hat{\delta}_n \ln n)^{2\kappa-1} = O(h_u^2)$. *A bandwidth appears in it that is not already in C5′, and it is* ***not*** *the only hypothesis of Proposition 28 of which that is true*: (P-c) carries h_u on the right of $\mathsf{R}\,L_x^{-s} = O(h_u\hat{\delta}_n)$, and (P-g) is an upper bound on h_x which, as that proposition prints in the clause itself, C5′ does not carry. *No claim of exhaustiveness is made, and that is deliberate*: the draft this sentence replaces made one and was short by those two. It is stated in Proposition 28 in the rate variables and unpacked only here. Write $\hat{\delta}_n = n^{-\alpha}$, $\alpha = r/(2r + d_Z)$.

(0) The logarithms do not cancel, and they decide the boundary. Read this clause first. *(P-f) is an exact domination, not an inequality with a margin*, and after A7's amendment both sides carry powers of $\ln n$:

$$\delta_n^2 = \hat{\delta}_n^{2\kappa}(\ln n)^{2\kappa}, \qquad h_u^2\hat{\delta}_n \ln n = h_u^2\hat{\delta}_n(\ln n)^1, \qquad 2\kappa = \frac{4\mu}{2\mu+1} > 1 \iff \mu > \tfrac{1}{2}.$$

Since $\mu > \frac{1}{2}$ throughout this block, the left side always carries the *larger* power of $\ln n$. Consequence: at any configuration where the powers of n balance exactly, (P-f) fails by a power of $\ln n$, and every threshold below must therefore be read strictly. Readings (1) and (2) are stated in powers of n and each strict inequality there leaves a polynomial margin that absorbs the logarithms; each equality does not.

(1) At the unamended bandwidth the condition binds, at low μ it is impossible, and at the boundary configuration it fails. With $h_u \asymp n^{-1/16}$, (P-f) reads $\alpha(2\kappa-1) > \frac{1}{8}$. At $\mu = 1$ ($\kappa = \frac{2}{3}$, $2\kappa - 1 = \frac{1}{3}$) this is $\alpha > \frac{3}{8}$, i.e. $r > 4.5$ at $d_Z = 3$, *strictly*, so it is *not* met at the frontier value $r = 4.5$. **At the boundary configuration** $(\mu, d_Z, r) = (1, 3, 4.5)$ **the two sides are** $\delta_n^2 = n^{-1/2}(\ln n)^{4/3}$ **and** $h_u^2 \hat{\delta}_n \ln n = n^{-1/2} \ln n$**, so (P-f) fails by exactly** $(\ln n)^{1/3}$; clause (5) below names that configuration and says why it is an illustration and not a design. At $\mu = 2$ ($2\kappa - 1 = \frac{3}{5}$) the condition is $\alpha > \frac{5}{24} \approx 0.208$, comfortably implied by the frontier's $r > 2.5$ at $d_Z = 3$ ($\alpha > 0.3125$), with a polynomial margin that absorbs the logarithms. At $\mu = \frac{3}{4}$ ($2\kappa - 1 = \frac{1}{5}$) it is $\alpha > \frac{5}{8}$, *impossible* because $\alpha < \frac{1}{2}$ always. So at the unamended bandwidth (P-f) fails for μ near $\frac{1}{2}$, fails at the boundary configuration, and holds with room for $\mu \geq 2$.

(2) And the unamended bandwidth is tuned to the wrong target. It sets $h_u \asymp n^{-1/16}$ by matching the smoothing bias $h_u^{p_K}$ to the *stochastic floor* $n^{-1/2}$. *But the target Step 6 needs the bias to meet is* $\hat{\delta}_n$*, not* $n^{-1/2}$, and $\hat{\delta}_n \geq n^{-1/2}\sqrt{\ln n} \gg n^{-1/2}$ by A5(i)'s floor, in every configuration and not only where the propensity branch binds. Taking the largest admissible bandwidth, $h_u \asymp \hat{\delta}_n^{1/p_K} = n^{-\alpha/8}$, which is C5′(i) as now amended, leaves Lemma 77(ii) intact, the bias being then $O(\hat{\delta}_n)$, which is all (116) needs, and turns (P-f) into $\alpha(2\kappa - 1) > \alpha/4$, a condition free of α altogether:

$$2\kappa - 1 \;>\; \tfrac{1}{4} \qquad \Longleftrightarrow \qquad \kappa > \tfrac{5}{8} \qquad \Longleftrightarrow \qquad \mu \;>\; \tfrac{5}{6}. \tag{182}$$

The inequality is strict, and by clause (0) it must be: at $\mu = \frac{5}{6}$ exactly the powers of n balance and the left side carries $(\ln n)^{2\kappa} = (\ln n)^{5/4}$ against the right side's $(\ln n)^1$, so (P-f) fails by $(\ln n)^{1/4}$. *So re-tuning* h_u *converts a joint restriction on* (μ, α) *into a clean threshold on* μ *alone*, and the threshold is $\frac{5}{6}$, above $\mu > \frac{1}{2}$ and below the $\mu = 1$ row of the frontier table. Under H1 and plan §2.6's scope $s > a + \frac{1}{2}$, which gives $\mu_{\mathrm{H1}} > \frac{1}{2} + \frac{1}{4a}$, (182) is *implied* whenever $a \leq \frac{3}{4}$ and is an additional restriction otherwise. *That last sentence carries the*

same proviso Remark 98(1) *attaches to its copy of it, and it is attached here rather than left to the pointer*: the computed index is $\mu_{\mathrm{H1}} = s/(2a)$, at the population operator at $\theta_\dagger$, while (182)'s μ is (P-a.3)'s primitive index at the sieve compression of the realised operator at $\theta_\circ$, and the bridge between them is the operator interpolation Remark 86(1) shows unavailable at every $2\mu > 1$. The implication therefore holds **under the proviso Remark 86(4) declines and not otherwise**; Remark 86(5) states the binding reading.

(3) What the re-tuning costs. The amended $h_u \asymp \mathrm{sd}(\Delta U)\hat{\delta}_n^{1/p_K}$ *depends on* α*, hence on* r *and* d_Z*, which the analyst must supply*, A4(ii)–(iii) already require both, so no new primitive is introduced, but the unamended bandwidth was α-free and this one is not. *The alternative, keeping the unamended bandwidth and carrying (P-f) as the joint condition* $\alpha(2\kappa - 1) > \frac{1}{8}$*, excludes* μ *below roughly* 0.9 *at* $d_Z = 3$ *and excludes the boundary configuration*; it is a real loss and is not taken. *This is one of the four lines Step 6 charges*, with (123), (P-g) and A5(vi′), and it is the one Theorem 29 must read, because Theorem 29 reads δ_n off A5(ii) and A5(ii) is discharged subject to it.

(4) What $\mu > \frac{5}{6}$ is **also**, and it is **not** a coincidence. Proposition 28 Step 4(a.1) exposes a requirement on the preliminary estimator, (181), and shows that it is weaker than the rate the procedure delivers exactly when $\mu > \frac{5}{6}$. The algebra: with $h_u = n^{-\beta}$ the preliminary requirement being non-self-defeating is $\kappa\alpha - \frac{2\kappa\beta}{2\kappa-1} > 0$, which factors as $\frac{\kappa}{2\kappa-1}\Big[\alpha(2\kappa-1) - 2\beta\Big] > 0$, and $\alpha(2\kappa - 1) \geq 2\beta$ is (P-f) written in powers of n. The two are the same inequality up to a strictly positive factor, the second strict. A third comparison in the block, branch (a)'s repelling fixed point lying above the regime boundary, is the same inequality once more, and for a two-branch map that is forced. *The statement worth printing, and it is not a restatement of (P-f):* (P-f) is exactly the condition under which the newly exposed preliminary-rate requirement is not self-defeating. That requirement is new, it is a real demand on $\hat{\varphi}^{(0)}$, and it is what sharpens open item Q-1. The coincidence worth naming is at $\frac{1}{2}$, not at $\frac{5}{6}$.

(5) The block's two running configurations, named once so that they are not confused. **The**

boundary configuration is $(\mu, d_Z, r) = (1, 3, 4.5)$, equivalently $\alpha = \frac{3}{8}$: it sits *exactly* on the (P-f) boundary in powers of n and fails it by $(\ln n)^{1/3}$, by clause (1), and with A8 in full its sieve window is empty at $s < 2$, by Proposition 28(P-d). It is retained because it is the sharpest illustration of clause (0), and it is not an admissible design. **The frontier-clearing configuration** is $(\mu, s, a, \alpha) = (2, 3, \frac{3}{4}, 0.32)$, equivalently $r = \frac{8}{3}$ at $d_Z = 3$: it clears (7) ($\kappa\alpha = 0.256 > \frac{1}{4}$), clears (P-f) with polynomial margin, clears (54) ($s > 1$) and clears A8 in full ($s > 1.389$). It is the admissible one, it is the one Remark 72(3) already uses, and it is the one a simulation should run.

Proof of Lemma 91. Six steps: the moment bounds; the two centrings; one differentiation under the expectation with the envelope Corollary 49 supplies; then Steps 4 and 5, which apply the *second* of the two centrings a second and a third time, to the Jacobian's debiasing summand, and to the cross term of Σ; and Step 6, clause (v). *Steps 4 and 5 introduce no hypothesis and no device* ***for the clauses they prove***: clause (iv) and the splitting (127) use Step 2's $E[\widetilde{W} \mid \Delta U, \Delta Z] = 0$ and Step 1's integrability, and nothing else. *The one exception is marked where it occurs*: the **lower** bound of reading (iv.4), which is not a clause of the lemma, consumes A4(i) through (6) and the conditional covariance (2), and Step 5 says so at that sentence. *Step 6 introduces exactly one hypothesis, C2′(i), and it is the only step that uses it*, which is why clause (v) carries it as an "if in addition" rather than the lemma carrying it on its list.

Step 1: integrability, and clause (ii). By (P1), $|\widetilde{W}_l| \le 1$ at every cell, and by (P1b) $\|\widetilde{W}\| \le c_{\widetilde{W}}$ in the Euclidean norm on $\mathbb{R}^{\mathsf{L}-1}$, *and it is* (P1b) *and not* (P1) *that this step consumes, the object contracted being the* $p \times (\mathsf{L} - 1)$ *weight against the contrast vector.* By A2′(i), both clauses, which is what makes $\ell^{\Pi}_{\star}$ an admissible representer in the sense of A2b and fixes $\ell_\star$ pointwise, with B1(i), (ii), (iv.a), (iv.d), Corollary 49 gives $\|\ell_\star\|_\infty \le \|\varpi_u\|_\infty \Lambda^{(0)}_\star < \infty$. *Only the budget* $\Lambda^{(0)}_\star$ *enters this bound, but the* ***hypothesis*** *that licenses Corollary 49 is A2b, which is (i.a) and (i.b) together*; the distinction is the one the lemma's statement records.

By Definition 19(b) and (31), $\|L_\star(u,z)\| \le \|b_\star(z)\| + \|\ell_\star\|_\infty |u|$ for every (u,z), *the norm Frobenius, as the type declaration fixes it for* L, ℓ *and* b, so with $(\varsigma+\varsigma')^2 \le 2\varsigma^2 + 2\varsigma'^2$ and $\|L_\star \widetilde{W}\| \le c_{\widetilde{W}} \|L_\star\|$ by (P1b),

$$E\Big\|L_\star(\Delta U, \Delta Z)\,\widetilde{W}\Big\|^2 \;\le\; c_{\widetilde{W}}^2\, E\Big\|L_\star(\Delta U, \Delta Z)\Big\|^2 \;\le\; 2\,c_{\widetilde{W}}^2 \Big\{ E\Big\|b_\star(\Delta Z)\Big\|^2 + \|\ell_\star\|_\infty^2\, E\big[\Delta U^2\big] \Big\} \;<\; \infty$$

by A2′(iii) and A1b(iv). Since $\|\psi_\star\|^2 \le 2\|\Delta Z \Delta U\|^2 + 2\|L_\star \widetilde{W}\|^2$ by (39) and $E\|\Delta Z \Delta U\|^2 < \infty$ by A1b(iv), clause (ii)'s display follows with its factor $4c_{\widetilde{W}}^2$, and $E\|\psi_\star\| \le (E\|\psi_\star\|^2)^{1/2} < \infty$. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$, $\widetilde{W}$ is a scalar, the Frobenius norm is the Euclidean one and $L_\star \widetilde{W} = \widetilde{W} L_\star$, so every display above is the printed object and the factor 4 is unchanged.

Step 2: both centrings, and A1 is used twice for two different things. By A1's second clause, $E[\Delta U \mid \Delta Z] = 0$; since $E\|\Delta Z \Delta U\| < \infty$ by Step 1, the tower property gives $E[\Delta Z \Delta U] = E[\Delta Z\, E[\Delta U \mid \Delta Z]] = 0$. By A1's first clause, $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, so $P(W = l \mid \Delta U, \Delta Z) = P(W = l \mid \Delta Z) = \varrho_l(\Delta Z)$ *for every* $l \in \{1, \dots, \mathsf{L}-1\}$, and hence $E[\widetilde{W}_l \mid \Delta U, \Delta Z] = 0$ almost surely at every l, that is $E[\widetilde{W} \mid \Delta U, \Delta Z] = 0$ as an identity in $\mathbb{R}^{\mathsf{L}-1}$. Since $E\|L_\star(\Delta U, \Delta Z)\,\widetilde{W}\| < \infty$ by Step 1, the tower property gives $E[L_\star(\Delta U, \Delta Z)\,\widetilde{W}] = E\Big[L_\star(\Delta U, \Delta Z)\, E[\widetilde{W} \mid \Delta U, \Delta Z]\Big] = 0$, *the contrast standing to the* ***right*** *of the weight, which is the only order that multiplies and is the order* (38) *declares.* This is the mechanism of Lemma 45(v) at $\Xi = L_\star$, whose type $\mathbb{R}^{p \times (\mathsf{L}-1)}$ is exactly $L_\star$'s, and it is the reason A1 appears in the hypothesis list of a lemma that contains no operator. Adding the two gives $E[\psi_\star] = 0$.

Step 3: clause (iii). By (19), $\Delta U(\theta, \varphi_\dagger) = \Delta U - \Delta Z'(\theta - \theta_\dagger)$, so

$$\psi(\theta, \zeta_\star) = \Delta Z \Big\{ \Delta U - \Delta Z'(\theta - \theta_\dagger) \Big\} + L_\star\Big(\Delta U - \Delta Z'(\theta - \theta_\dagger), \Delta Z \Big)\, \widetilde{W}.$$

The first summand is affine in θ with θ-derivative $-\Delta Z \Delta Z'$, integrable by A1b(iv). For the second, $u \mapsto L_\star(u, z)$ is differentiable at every point with $\partial_u L_\star = \ell_\star$ (Definition 19(b)), so $\theta \mapsto L_\star(\Delta U - \Delta Z'(\theta - \theta_\dagger), \Delta Z)\, \widetilde{W}$ is differentiable at every θ with derivative

$-\ell_\star(\Delta U(\theta,\varphi_\dagger),\Delta Z)\,\widetilde{W}\,\Delta Z'$, a $p\times p$ matrix; by the *mean value inequality*, the map being $\mathbb{R}^p$-valued in $\theta\in\mathbb{R}^p$, so that no Lagrange point is available and none is wanted, Corollary 14 Step 1, every difference quotient is dominated in norm by $c_{\widetilde{W}}\|\ell_\star\|_\infty\|\Delta Z\|$ by (P1b), an envelope free of θ and integrable by A1b(iv) and Step 1. Dominated convergence therefore licenses differentiation under the expectation, and at $\theta=\theta_\dagger$

$$\partial_\theta E\big[\psi(\theta,\zeta_\star)\big]\big|_{\theta_\dagger}=-E\big[\Delta Z\Delta Z'\big]-E\big[\ell_\star(\Delta U,\Delta Z)\,\widetilde{W}\,\Delta Z'\big]=-J,$$

which is (40) *term for term, the second summand being* $(p\times 1)(1\times p)$, with $\|J\|\le E\|\Delta Z\|^2+c_{\widetilde{W}}\|\ell_\star\|_\infty E\|\Delta Z\|<\infty$ by A1b(iv) and the Cauchy–Schwarz inequality. Corollary 49, at the same hypotheses, gives in addition $\|\partial_u\ell_\star\|_\infty\le\|\varpi_u'\|_\infty\Lambda_\star^{(0)}+\|\varpi_u\|_\infty\Lambda_\star^{(1)}<\infty$, *here the second budget* $\Lambda_\star^{(1)}$ *enters the bound and not merely its licence*, and the same argument applied once more, with envelope $c_{\widetilde{W}}\|\partial_u\ell_\star\|_\infty\|\Delta Z\|^2$, integrable by A1b(iv), gives the second derivative and the bound $c_{\widetilde{W}}\|\partial_u\ell_\star\|_\infty E\|\Delta Z\|^2$ clause (iii) states. Both envelopes acquire $c_{\widetilde{W}}$ and both are 1 at $\mathsf{L}=2$. *Step 4 shows that this second bound, though correct, bounds the zero matrix*; the sentence in the lemma's clause (iii) is retained as proved and is annotated there.

Step 4: clause (iv)*, which is Step 2's centring applied a second time and consumes no new hypothesis.* Step 2 established $E[\widetilde{W}\mid\Delta U,\Delta Z]=0$ almost surely, from A1's first clause alone. Fix $j,k\le p$ and write $\ell_{\star j}$ for the j-th *row* of $\ell_\star$, an $\mathbb{R}^{1\times(\mathsf{L}-1)}$-valued map. The random vector $\ell_{\star j}(\Delta U,\Delta Z)\,\Delta Z_k$ is $\sigma(\Delta U,\Delta Z)$-measurable, being a Borel function of $(\Delta U,\Delta Z)$: by (68) $\ell_\star=\mathcal{W}\ell_\star^{\Pi}$ is a fixed Borel map $\mathbb{R}\times\mathbb{R}^p\to\mathbb{R}^{p\times(\mathsf{L}-1)}$, one row per coordinate of θ and one column per contrast, pinned down pointwise by A2′(i)'s closing clause. *That the* ***selection*** *of* $\ell_\star^{\Pi}$ *from* $\ell_\dagger^{\Pi}+\ker\mathcal{A}_{\mathrm{cond}}^*$ *is a functional of the joint law of* $(W,\Delta U,\Delta Z)$ *does not disturb this, and the point is worth stating because it is the natural objection*: a law-dependent choice of a deterministic function is still a deterministic function, and measurability of $\ell_\star(\Delta U,\Delta Z)$ with respect to $\sigma(\Delta U,\Delta Z)$ is a property of the map's *arguments*, of which W is not one. *Nor does N1 enter*: N1 constrains the anchor b of Definition 19(b) and

leaves ℓ untouched, and (40) carries $\ell_\star$ and not $L_\star$. The scalar $\ell_{\star j}\widetilde{W}\Delta Z_k$ is integrable, by $|\ell_{\star j}\widetilde{W}| \le c_{\widetilde{W}}\|\ell_{\star j}\|$ from (P1b) and the Cauchy–Schwarz inequality in $\mathbb{R}^{\mathsf{L}-1}$, $\|\ell_\star\|_\infty < \infty$ (Step 1) and $E\|\Delta Z\| \le (E\|\Delta Z\|^2)^{1/2} < \infty$ (A1b(iv) and the Cauchy–Schwarz inequality), *which is the same bound Step 3 already used to conclude* $\|J\| < \infty$. The tower property therefore gives

$$E\Big[\ell_{\star j}(\Delta U, \Delta Z)\,\widetilde{W}\,\Delta Z_k\Big] = E\Big[\ell_{\star j}(\Delta U, \Delta Z)\,\Delta Z_k\,E\Big[\widetilde{W} \mid \Delta U, \Delta Z\Big]\Big] = 0,$$

the $\sigma(\Delta U, \Delta Z)$*-measurable row passing outside the conditional expectation and the* $\mathbb{R}^{\mathsf{L}-1}$ *zero vector of Step 2 annihilating it*, so the second summand of (40) is the zero matrix and $J = E[\Delta Z \Delta Z']$. Symmetry and positive semidefiniteness are those of a second-moment matrix, and nonsingularity is A1b(iv). *This is Lemma 45*(v) *at* $\Xi(u, z) = \ell_{\star j}(u, z)\,z_k$*, the* $p = 1$ *row of that clause's* $\mathbb{R}^{p\times(\mathsf{L}-1)}$ *type, which its own "coordinatewise in the p rows" sentence licenses, and it is re-derived here rather than cited for the reason Step 2 gives at* $\Xi = L_\star$: that clause is stated under C1(ii), which is not on this lemma's list and which its σ_g half needs.

For the affine display (125), fix θ and put $c := \theta - \theta_\dagger$, so that $\Delta U(\theta, \varphi_\dagger) = \Delta U - \Delta Z' c$ by (19). Then $L_\star(\Delta U - \Delta Z' c, \Delta Z)$ is again a Borel function of $(\Delta U, \Delta Z)$, and it is integrable: $\|L_\star(u, z)\| \le \|b_\star(z)\| + \|\ell_\star\|_\infty |u|$ by (31) gives

$$E\Big\|L_\star\Big(\Delta U - \Delta Z' c, \Delta Z\Big)\Big\| \;\le\; E\Big\|b_\star(\Delta Z)\Big\| + \|\ell_\star\|_\infty\Big\{E|\Delta U| + \|c\|\,E\|\Delta Z\|\Big\} \;<\; \infty$$

by A2′(iii), A1b(iv) and the Cauchy–Schwarz inequality, and so is $L_\star(\Delta U - \Delta Z' c, \Delta Z)\,\widetilde{W}$, at the cost of $c_{\widetilde{W}}$ by (P1b). The same tower step gives $E[L_\star(\Delta U - \Delta Z' c, \Delta Z)\,\widetilde{W}] = 0$; with $E[\Delta Z \Delta U] = 0$ from Step 2, and with $E[\Delta Z\,\Delta Z' c] = E[\Delta Z \Delta Z']\,c$ by linearity of the expectation at the fixed vector c, the summands of $E[\psi(\theta, \zeta_\star)]$ being separately integrable by $E\|\Delta Z \Delta U\| < \infty$ and $E\|\Delta Z\|^2 < \infty$ (A1b(iv)), (125) follows. The second derivative of an affine map is the zero matrix, and uniqueness of the root is nonsingularity of $E[\Delta Z \Delta Z']$,

which is A1b(iv). A3 is used nowhere in this step, which is the point of the clause.

Step 5: the display (127) *of reading* (iv.4)*, recorded because the natural over-generalisation of Step 4 is that the debiasing term drops out of the variance too, and it does not.* The cross term of $\Sigma = E[\psi_\star \psi_\star']$ is $E[\Delta Z \Delta U\, \widetilde{W}' L_\star'] + E[L_\star \widetilde{W}\, \Delta U \Delta Z']$, *the two being transposes of one another, so annihilating either annihilates both*, and, entrywise at $j, k \le p$, $(L_\star \widetilde{W}\, \Delta U \Delta Z')_{jk} = \sum_m \big(L_{\star jm}\, \Delta U\, \Delta Z_k\big) \widetilde{W}_m$ with each $L_{\star jm} \Delta U \Delta Z_k$ $\sigma(\Delta U, \Delta Z)$-measurable and $E\|\Delta Z \Delta U\, L_\star'\| \le (E\|\Delta Z \Delta U\|^2)^{1/2} (E\|L_\star\|^2)^{1/2} < \infty$ by the Cauchy–Schwarz inequality, A1b(iv), and *Step 1's display read before the contraction* $\|L_\star \widetilde{W}\| \le c_{\widetilde{W}} \|L_\star\|$ *is performed*, which bounds $E\|L_\star\|^2$ itself by $2E\|b_\star\|^2 + 2\|\ell_\star\|_\infty^2 E[\Delta U^2]$, without $c_{\widetilde{W}}$. So the same tower step annihilates both cross terms and (127) follows *on this lemma's list, with no further hypothesis.* The **lower** bound in reading (iv.4) is a separate matter and it consumes A4(i), which is **not** on this lemma's list and is cited for that sentence alone. *And the route to it is the* ***conditional*** *one, the sure bound available at* $\mathsf{L} = 2$ *having no analogue above it*: at $\mathsf{L} \ge 3$, $\widetilde{W}\widetilde{W}'$ is rank one, so no sure inequality $\widetilde{W}\widetilde{W}' \succeq cI$ holds at any $c > 0$ and $L_\star \widetilde{W}\widetilde{W}' L_\star'$ is singular at every sample point once $p > 1$. Instead, A1's first clause gives $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, so the conditional law of W given $(\Delta U, \Delta Z)$ is its law given ΔZ and, with $E[\widetilde{W} \mid \Delta Z] = 0$ from (P3), $E[\widetilde{W}\widetilde{W}' \mid \Delta U, \Delta Z] = \mathrm{Var}(\widetilde{W} \mid \Delta Z) = \Omega(\Delta Z)$ of (2); A4(i) gives $\Omega(\Delta Z) \succeq \varepsilon_\Omega \,\mathrm{Id}$ with $\varepsilon_\Omega = \varepsilon_\varrho / \mathsf{L}$ by (6), which that clause carries as a consequence and not as a second hypothesis; and therefore, by the tower property applied entrywise as above,

$$E\Big[L_\star \widetilde{W}\widetilde{W}' L_\star'\Big] = E\Big[L_\star\, \Omega(\Delta Z)\, L_\star'\Big] \;\succeq\; \varepsilon_\Omega\, E\Big[L_\star L_\star'\Big],$$

since $L_\star(\Omega - \varepsilon_\Omega \,\mathrm{Id}) L_\star' \succeq 0$ pointwise; and this is nonzero whenever $L_\star$ is. Without overlap $\Omega(\Delta Z)$ may be singular on a set of positive probability, at $\mathsf{L} = 2$, $E[\widetilde{W}^2 \mid \Delta Z] = \varrho(1 - \varrho)$ may vanish, and no such bound is available. *At* $\mathsf{L} = 2$ *the sure route is also available, and it is* ***weaker*** *rather than sharper, so nothing is lost by taking the conditional one everywhere*: it delivers $\widetilde{W}^2 \ge \varepsilon_\varrho^2$ surely under A4(i), against (6)'s $\varepsilon_\Omega = \varepsilon_\varrho / 2$, and $\varepsilon_\varrho^2 < \varepsilon_\varrho / 2$ whenever $\varepsilon_\varrho < \frac{1}{2}$,

which is A4(i)'s own range $\varepsilon_\varrho \in (0, 1/\mathsf{L})$ read at $\mathsf{L} = 2$. *So the* $\mathsf{L} = 2$ *conclusion is not merely preserved by the passage to the conditional route but improved by it*, and the sharp binary floor $\varrho(1-\varrho) \geq \varepsilon_\varrho(1-\varepsilon_\varrho)$, which (6) does not carry, is larger again. *Nothing in clauses* (i)–(iv) *uses A4(i)*, and in particular $\Sigma < \infty$ (clause (ii)) needs no overlap.

Step 6: clause (v), $\Sigma \succ 0$, *which uses* (127) *and* ***only the first*** *of its two summands.* Fix $\vartheta \in \mathbb{R}^p$ with $\vartheta \neq 0$. By (127) and positive semidefiniteness of $E[L_\star \widetilde{W}\widetilde{W}' L_\star']$, *which is pointwise and needs no hypothesis,* $\vartheta' L_\star \widetilde{W}\widetilde{W}' L_\star' \vartheta = (\widetilde{W}' L_\star' \vartheta)^2 \geq 0$ *at every sample point and every* L,

$$\vartheta'\Sigma\vartheta \;=\; E\Big[(\vartheta'\Delta Z)^2 \Delta U^2\Big] + \vartheta' E\Big[L_\star \widetilde{W}\widetilde{W}' L_\star'\Big]\vartheta \;\geq\; E\Big[(\vartheta'\Delta Z)^2 \Delta U^2\Big],$$

the first summand being finite by A1b(iv)'s $E\|\Delta Z \Delta U\|^2 < \infty$ and the Cauchy–Schwarz inequality. It remains to show that this expectation is strictly positive, and two facts do it. *(a)* $P(\vartheta'\Delta Z \neq 0) > 0$, *by A1b(iv).* If $\vartheta'\Delta Z = 0$ almost surely then $\vartheta' E[\Delta Z \Delta Z']\vartheta = E[(\vartheta'\Delta Z)^2] = 0$, so $E[\Delta Z \Delta Z']$ is singular, contradicting A1b(iv). *(b)* $P(\Delta U = 0) = 0$, *by C2′(i).* That clause gives ΔY a conditional density given $\mathcal{F}_{XZW} = \sigma(X_1, X_2, \Delta Z, W)$; by (18) and (19), $\Delta U = \Delta Y - \Delta\varphi_\dagger - \Delta Z'\theta_\dagger$ differs from ΔY by an $\mathcal{F}_{XZW}$-measurable shift, so the regular conditional distribution of ΔU given $\mathcal{F}_{XZW}$ is the translate of that of ΔY and is likewise absolutely continuous, for almost every ω, *this is (P6)'s substitution lemma and (P7a)'s scope point* (β)*, and "for almost every* ω*" is exact and is all that is needed.* Hence $P(\Delta U = 0 \mid \mathcal{F}_{XZW}) = 0$ almost surely and, taking expectations, $P(\Delta U = 0) = 0$. *Combining.* The set $A := \{\vartheta'\Delta Z \neq 0\}$ has $P(A) > 0$ by (a), and $P(A \cap \{\Delta U \neq 0\}) = P(A) > 0$ by (b), so the nonnegative random variable $(\vartheta'\Delta Z)^2 \Delta U^2$ is strictly positive on a set of positive probability and its expectation is strictly positive. Therefore $\vartheta'\Sigma\vartheta > 0$ for every $\vartheta \neq 0$, which is $\Sigma \succ 0$. Since $J = E[\Delta Z \Delta Z']$ is nonsingular by clause (iv) and A1b(iv), the congruence $J^{-1}\Sigma J^{-1\prime}$ is positive definite as well, which is (126). *Three scope points. First, A4(i) is not used*, in contrast with the lower bound of reading (iv.4): that bound is about the *second*

summand of (127) and this step discards it. *Second, the argument is pointwise in P and gives no lower bound on $\lambda_{\min}(\Sigma)$ uniform over any class of laws*; see Remark 98(6). *Third, C2′(i) is the only hypothesis added to the lemma's list and it is added as an "if in addition"*, because clauses (i)–(iv) do not use it and charging them with it would be the over-attribution this block has corrected four times. □

Proof of Lemma 92. The proof is exact algebra plus two bounds: a Chebyshev rate for the score of restriction (B), and an L^1 bound on the pairing that the plug-in does *not* debias.

Step 1: the normalising matrix. By A1b(iv), $E\|\Delta Z\|^2 < \infty$ and $E[\Delta Z \Delta Z']$ is nonsingular; by A0 and the law of large numbers $\mathbb{E}_n[\Delta Z \Delta Z'] \xrightarrow{p} E[\Delta Z \Delta Z']$, so $P(\mathbb{E}_n[\Delta Z \Delta Z'] \text{ nonsingular}) \to 1$ and, on that event, $\|(\mathbb{E}_n[\Delta Z \Delta Z'])^{-1}\| = O_P(1)$ by continuity of matrix inversion at a nonsingular point.

Step 2: the identity. Write $\Delta h_{k,i} := h_k(X_{i2}) - h_k(X_{i1})$. By (18), $\Delta Y_i - \Delta\hat{\varphi}_i^{(-k)} = \Delta Z_i'\theta_\dagger + \Delta U_i - \Delta h_{k,i}$, so on the event of Step 1

$$\hat{\theta}^{(0)} - \theta_\dagger = \Big(\mathbb{E}_n\big[\Delta Z \Delta Z'\big]\Big)^{-1}\Big[\ \frac{1}{n}\sum_{i=1}^{n} \Delta Z_i \Delta U_i \ - \ \sum_{k=1}^{\mathsf{F}} \frac{n_k}{n}\, \mathbb{E}_{n,k}\big[\Delta Z\, \Delta h_k\big]\ \Big].$$

Step 3: the score of restriction (B). By A1b(iv), $E\|\Delta Z \Delta U\|^2 < \infty$, hence $E\|\Delta Z \Delta U\| < \infty$ by the Cauchy–Schwarz inequality; by A1's second clause $E[\Delta U \mid \Delta Z] = 0$, so the tower property gives $E[\Delta Z \Delta U] = E\big[\Delta Z\, E[\Delta U \mid \Delta Z]\big] = 0$. Chebyshev's inequality and A0 then give $n^{-1}\sum_i \Delta Z_i \Delta U_i = O_P(n^{-1/2})$. *The one-line tower argument is written out rather than cited from Lemma 91*(i)*, and the reason is this lemma's own closing sentence*: that lemma's list runs A1, A1b(iv), A2′(i), A2′(iii), B1(i), (ii), (iv.a), (iv.d), so the citation would import A2′ and B1 into a lemma that advertises using neither, and clause (i) is in any case the statement $E[\psi_\star] = 0$, of which $E[\Delta Z \Delta U] = 0$ is one of two summands, so the appeal would be to that lemma's Step 2 and not to its statement. Both ingredients used here, A1's second clause and $E\|\Delta Z \Delta U\| < \infty$, are on this lemma's own list.

Step 4: the undebiased pairing. Fix k. Conditionally on $\sigma(\mathcal{I}_{-k})$ the function h_k is frozen and the summands of $\mathbb{E}_{n,k}[\Delta Z \Delta h_k]$ are i.i.d. by A0, so

$$\begin{aligned} E\Big[\big\|\mathbb{E}_{n,k}[\Delta Z \Delta h_k]\big\| \;\Big|\; \sigma(\mathcal{I}_{-k})\Big] \;&\le\; E\big[\|\Delta Z\|\,|\Delta h_k| \;\big|\; \sigma(\mathcal{I}_{-k})\big] \\ &\le\; \big(E\|\Delta Z\|^2\big)^{1/2}\big\|\Delta h_k\big\|_{L^2(P)} \\ &\le\; 2\,\bar{c}_\pi^{1/2}\big(E\|\Delta Z\|^2\big)^{1/2}\big\|h_k\big\|_{L^2(\pi)}, \end{aligned}$$

by the Cauchy–Schwarz inequality, A1b(iv), C1(ii) and Lemma 45(i). Conditional Markov therefore gives, almost surely and for *every* $\epsilon > 0$,

$$P\Big(\big\|\mathbb{E}_{n,k}[\Delta Z \Delta h_k]\big\| > \epsilon\,\delta_n \;\Big|\; \sigma(\mathcal{I}_{-k})\Big) \;\le\; \epsilon^{-1}\,\frac{2\,\bar{c}_\pi^{1/2}\big(E\|\Delta Z\|^2\big)^{1/2}\,\big\|h_k\big\|_{L^2(\pi)}}{\delta_n},$$

the constant being written out rather than named, on the precedent Proposition 27 sets for a quantity referred to once. The right-hand numerator is $\sigma(\mathcal{I}_{-k})$-measurable and, by the *first* display of (43), A5(ii)'s first half read at each fold, and the only thing about δ_n this step uses, is $O_P(1)$ *unconditionally*. So (P8)'s O_P **companion**, at exponent one and $a_n = \delta_n$, gives $\mathbb{E}_{n,k}[\Delta Z \Delta h_k] = O_P(\delta_n)$ unconditionally. (P8)*'s* transfer rule *is* ***not*** *the clause invoked*: that rule needs the conditional probability to vanish at a *single* moving threshold, and at a threshold that is a fixed multiple of a Markov numerator it is bounded by a constant instead. Summing the F folds, with F fixed and $\sum_k n_k/n = 1$, gives $O_P(\delta_n)$ for the second bracket of Step 2.

Step 4′: what the conditional mean is, recorded because it is why a one-step is needed at all. The conditional mean of that same average is exactly $E[\Delta Z \Delta h_k \mid \sigma(\mathcal{I}_{-k})] = \langle \nu/\pi, h_k \rangle_{L^2(\pi)}$ by Lemma 45(iii) at $g = \Delta Z$, whose term-by-term integrability proviso holds because $E|\Delta Z_j h_k(X_m)| \le (E[\Delta Z_j^2])^{1/2}\bar{c}_\pi^{1/2}\|h_k\|_{L^2(\pi)} < \infty$ for $m \in \{1, 2\}$. *That pairing is not debiased by anything in* (45), and $\|\nu/\pi\|_{L^2(\pi)} \le 2\bar{c}_\pi^{1/2}(E\|\Delta Z\|^2)^{1/2} < \infty$, by Lemma 45(iii)–(iv), which put ν_j/π in $\mathcal{V}$ and identify $\langle \nu_j/\pi, h \rangle$ with $E[\Delta Z_j \Delta h]$, together with Lemma 45(i) and the

Cauchy–Schwarz inequality, so that the supremum over $\{h \in \mathcal{V} : \|h\| \le 1\}$ is the norm. *No appeal to A2 is made here*: finiteness of $\|\nu/\pi\|$ is a moment statement and range membership is not used. So the plug-in's bias is of exact order $\|h_k\|$, which is why $\hat{\theta}^{(0)}$ is a preliminary and not the estimator.

Step 5: clause (i). Combining Steps 1–4, $\|\hat{\theta}^{(0)} - \theta_\dagger\| = O_P(n^{-1/2}) + O_P(\delta_n)$, which is (128). On the complement of Step 1's event, whose probability tends to zero, $\hat{\theta}^{(0)} := 0$ and the O_P statements are unaffected. If $\delta_n \to 0$ then both summands are $o_P(1)$, so $\hat{\theta}^{(0)} \xrightarrow{p} \theta_\dagger$. *Steps 1–4 use no threshold in n, and that is what clause* (i) *exists to make readable from the statement*: the only fact about δ_n consumed above is the first display of (43), at Step 4; A5(ii)'s second half appears nowhere in Steps 1–4 and is not needed for either conclusion of clause (i).

Step 6: clause (ii)*, and the collapse, which is not clause* (ii). Suppose in addition $\delta_n = o_P(n^{-1/4})$, the second half of A5(ii) in the form (137). The first summand of (128) satisfies $n^{-1/2} = o(n^{-1/4})$ and the second is $o_P(n^{-1/4})$ by that hypothesis, so (129) follows. *The collapse recorded in the statement is a separate assertion and is used by neither clause*: by A5(i)'s floor, and *not* by Lemma 77, whose clause (i) is a Chebyshev tail bound and upper-only, $\hat{\delta}_n \ge \hat{\delta}_n^{(T)} = n^{-1/2}\sqrt{\ln n}$, and writing $\hat{\delta}_n = n^{-\alpha}$, $\kappa = 2\mu/(2\mu+1)$ we have $\kappa < 1$ and $\alpha < \frac{1}{2}$, hence $\kappa\alpha < \frac{1}{2}$ and $\delta_n / n^{-1/2} = n^{1/2-\kappa\alpha}(\ln n)^\kappa \to \infty$; so the second term of (128) dominates and $\|\hat{\theta}^{(0)} - \theta_\dagger\| = O_P(\delta_n)$. □

Proof of Lemma 93. The proof uses three *exact* identities and no Taylor expansion: the fundamental theorem of calculus in the shift, the bi-affineness of ψ in (L, ϱ), and Lemma 45(iii). Fix k and suppress the superscript; every expectation is conditional on $\sigma(\mathcal{I}_{-k})$ and $\hat{\varphi}, \hat{L}, \hat{\ell}, \hat{b}, \hat{\varrho}$ are $\sigma(\mathcal{I}_{-k})$-measurable, hence frozen, by Definition 24 and (P8).

Step 0: everything below is integrable, so every splitting is legitimate. By (19), $\Delta U(\theta_\dagger, \hat{\varphi}) = \Delta U - \Delta h$. By A1b(iv), C1(ii) and Lemma 45(i), $\|\Delta U\|_{L^2(P)} < \infty$ and $\|\Delta h\|_{L^2(P)} \le 2\bar{c}_\pi^{1/2}\|h\|_{L^2(\pi)} < \infty$, so $E\|\Delta Z(\Delta U - \Delta h)\| < \infty$ by the Cauchy–Schwarz inequality. By (R-a) and (31), $\|\hat{L}(u,z)\| \le \|\hat{b}(z)\| + \|\hat{\ell}\|_\infty |u|$, so $E\|\hat{L}(\Delta U - \Delta h, \Delta Z)\| \le (E\|\hat{b}(\Delta Z)\|^2)^{1/2} +$

$\|\hat{\ell}\|_\infty(\|\Delta U\|_{L^2(P)} + \|\Delta h\|_{L^2(P)}) < \infty$ almost surely, and

$$\left\|\widehat{\widetilde{W}}\right\| \le c_{\widetilde{W}} \qquad \text{and} \qquad \left\|\mathsf{r}\right\| \le c_{\widetilde{W}} \qquad \text{surely.} \tag{183}$$

(183) *is* (P1b)*'s two-line derivation read at the* ***estimated*** *propensity, and is* ***not*** (P1b) *itself*, that clause being stated at ϱ: by Definition 24(a), $\hat{\varrho}$ takes values in (11), and ϱ is a vector of conditional probabilities by Definition 4, so both are points of the simplex $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \ge 0,\ \mathbf{1}'q \le 1\}$; $\widehat{\widetilde{W}}$ is the difference of a vertex of that simplex, the origin when $W = 0$, and $\hat{\varrho}$, and $\mathsf{r} = \hat{\varrho} - \varrho$ is a difference of two of its points, so each is bounded in norm by its diameter, which is $\sqrt{2}$ at every $\mathsf{L} \ge 3$ and 1 at $\mathsf{L} = 2$; and separately each is bounded by $\sqrt{\mathsf{L}-1}$, *because every coordinate of* $\hat{\varrho}$ *and of* ϱ *lies in* $[0,1]$, the first by (11), the second by Definition 4, so every coordinate of either difference is at most 1 in modulus and the Euclidean norm of an $(\mathsf{L}-1)$-vector of such coordinates is at most $\sqrt{\mathsf{L}-1}$. The constant is the smaller of the two, which is $c_{\widetilde{W}}$. *No overlap condition is consumed and A4(i) is not on this lemma's list*: the range restriction on $\hat{\varrho}$ is a construction convention of Definition 2 and ϱ lies in the simplex for free. Definition 24(a)'s sharper per-cell bound $\|\mathsf{r}_k\|_\infty \le 1 - 2\varepsilon_\varrho$ would give the same conclusion a fortiori and is **not** read here, because that constant rests on A4(i), as that clause now states, and this lemma's list excludes it. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and (183) is the pair of scalar bounds $|\widehat{\widetilde{W}}| \le 1$, $|\mathsf{r}| \le 1$ that earlier rounds read straight off Definition 24(a). By Remark 52 under A2′(iii), $E\|L_\star(\Delta U, \Delta Z)\| < \infty$ and $E[\|L_\star\| \mid \Delta Z] < \infty$ almost surely. *What the integrability buys, stated once because every norm bound below runs on it*: for an integrand valued in a finite-dimensional normed space and integrable in norm,

$$\left\|E[\,\cdot\,]\right\| \le E\big[\|\cdot\|\big] \qquad \text{and} \qquad \left\|\int_0^1 (\,\cdot\,)\,d\lambda\right\| \le \int_0^1 \|\cdot\|\,d\lambda,$$

which are one inequality read at two measures. The first is the *conditional Jensen inequality*, holding almost surely, in the convention fixed above that E abbreviates $E[\,\cdot \mid \sigma(\mathcal{I}_{-k})]$; the second is the same statement at Lebesgue measure on $[0,1]$ and is what carries the Frobenius

norm inside the λ-integrals of Steps 1, 4 and 5. This is the device Lemma 60 Step 1 writes out in a display, and it is named at every site below at which it is used, not only at the first. *No symbol is coined for the generic integrand*: sans-serif M_ϵ is Proposition 79's O_P constant, italic M is retired by C7, and the corpus's own spelling of this device, at Step 3 below and at Lemma 95 Step 3, is the dotted one used here.

Step 1: the shift identity, exact. Under (R-a), $\hat{L} = L_{\hat{\ell}}^{\hat{b}}$, so by Definition 19(b) $u \mapsto \hat{L}(u, z)$ is differentiable at every point with $\partial_u \hat{L} = \hat{\ell}$, and $\hat{\ell}(\cdot, z)$ is continuous. For every (u, z) and every $a \in \mathbb{R}$ the map $\lambda \mapsto \hat{L}(u - \lambda a, z)$ is therefore C^1 on $[0, 1]$ and the fundamental theorem of calculus gives

$$\hat{L}(u - a, z) - \hat{L}(u, z) \;=\; -\,a \int_0^1 \hat{\ell}(u - \lambda a, z)\, d\lambda. \tag{184}$$

Read at $(u, z) = (\Delta U, \Delta Z)$ and $a = \Delta h$, and noting $\| \int_0^1 \hat{\ell}(\Delta U - \lambda \Delta h, \Delta Z) d\lambda \| \le \|\hat{\ell}\|_\infty$, *Step 0's* ***second*** *display, followed by the sup-norm*, every term is integrable. *The interchange* $E \int_0^1 = \int_0^1 E$ *used below is Fubini's theorem*: the integrand $(\lambda, \omega) \mapsto \hat{\ell}(\Delta U(\omega) - \lambda \Delta h(\omega), \Delta Z(\omega))$ is continuous in λ and measurable in ω, hence jointly measurable, and is dominated by the constant $\|\hat{\ell}\|_\infty$.

Step 2: the three exact vanishings. Write $D = \hat{L} - L_\star$ and $\widehat{\widetilde{W}} = \widetilde{W} - \mathsf{r}$. Then

$$E\big[\hat{\psi}\big] = E[\Delta Z \Delta U] - E[\Delta Z \Delta h] + E\big[\hat{L}(\Delta U - \Delta h, \Delta Z)\,(\widetilde{W} - \mathsf{r})\big].$$

The contrast stands to the ***right*** *of the weight throughout, as* (38) *and* (39) *print it*: $\hat{L}$, $L_\star$, D and $\hat{\ell}$ are $\mathbb{R}^{p \times (\mathsf{L}-1)}$-valued and $\widetilde{W}$, r and $\widehat{\widetilde{W}}$ are $\mathbb{R}^{\mathsf{L}-1}$-valued, so the products that multiply are $\hat{L}\widetilde{W}$, $D\widetilde{W}$, $\hat{L}\mathsf{r}$ and $D\mathsf{r}$, all $\mathbb{R}^p$-valued, and the reversed order is undefined at $\mathsf{L} \ge 3$. *At* $\mathsf{L} = 2$ *every factor above is a scalar or an* $\mathbb{R}^p$*-vector and the two orders agree*, which is Definition 23's closing sentence for $\hat{L}$, $L_\star$ and $\hat{\ell}$ and Proposition 20's for D and r, and it is why earlier rounds

could print either. Insert (184) and then $\hat{L}(\Delta U, \Delta Z) = L_\star(\Delta U, \Delta Z) + D(\Delta U, \Delta Z)$:

$$\begin{aligned} E\Big[\hat{L}(\Delta U - \Delta h, \Delta Z)(\widetilde{W} - \mathsf{r})\Big] &= E\Big[L_\star(\Delta U, \Delta Z)\,\widetilde{W}\Big] + E\Big[D(\Delta U, \Delta Z)\,\widetilde{W}\Big] \\ &\quad - E\Big[L_\star(\Delta U, \Delta Z)\,\mathsf{r}\Big] - E\Big[D(\Delta U, \Delta Z)\,\mathsf{r}\Big] \\ &\quad - E\Big[\Delta h \int_0^1 \hat{\ell}(\Delta U - \lambda \Delta h, \Delta Z) d\lambda\,(\widetilde{W} - \mathsf{r})\Big]. \end{aligned}$$

The first three expectations vanish, for three different reasons, and this is where the moment's structure is consumed:

(a) $E[\Delta Z \Delta U] = 0$ and $E[L_\star \widetilde{W}] = 0$ by Lemma 91(i), under A1, *the two summands of* $E[\psi_\star] = 0$, *the second of which is also the second display of that lemma's clause* (iv) *read at* $\theta = \theta_\dagger$. *No hypothesis is imported by either pointer*: Lemma 91's list is A1, A1b(iv), A2′(i), A2′(iii), B1(i), (ii), (iv.a), (iv.d), every clause of which is on this lemma's own list.

(b) $E[D\widetilde{W}] = 0$ *exactly.* This is Proposition 20(ii): under A1, $E[\widetilde{W} \mid \Delta U, \Delta Z] = 0$ (Step 2 of Lemma 91's proof), $D(\Delta U, \Delta Z)$ is $\sigma(\Delta U, \Delta Z)$-measurable, and $E\|D\widetilde{W}\| < \infty$ by Step 0, so the tower property applies and pulls D out on the left. *That is the direction Proposition 20*(ii)*'s own hypothesis line prints*, which asks for $E[\|D\| \mid \Delta Z] < \infty$ almost surely and $E\|D\widetilde{W}\| < \infty$ and for nothing about $\widetilde{W}D$. *No rate for* $\hat{L}$ *is used and none is available at this point*; the vanishing is a consequence of ψ being *affine* in L, which is Lemma 56(ii)'s first half read as an identity.

(c) $E[L_\star \mathsf{r}] = E\Big[E[L_\star(\Delta U, \Delta Z) \mid \Delta Z]\,\mathsf{r}(\Delta Z)\Big] = 0$ under N1, the inner conditional expectation being finite almost surely by Step 0 and $\mathsf{r}(\Delta Z)$ being $\sigma(\Delta Z)$-measurable, so that the tower property pulls it out on the *right*. This is Proposition 20(iii), and by that proposition's "only if" half it is *equivalent* to N1: without N1 this term is $-E[L_\star \mathsf{r}] \neq 0$ in general and $\hat{\varrho}$ would need a $\sqrt{n}$ rate.

The fourth expectation is $-\Psi^{(3)}$. Splitting $(\widetilde{W} - \mathsf{r})$ in the last line gives $-E[\Delta h \int_0^1 \hat{\ell}(\Delta U -$

$\lambda\Delta h)d\lambda\,\widetilde{W}] + \Psi^{(2)}$, and splitting $\int_0^1 \hat{\ell}(\Delta U - \lambda\Delta h)d\lambda = \hat{\ell}(\Delta U, \Delta Z) + \int_0^1\{\hat{\ell}(\Delta U - \lambda\Delta h) - \hat{\ell}(\Delta U)\}d\lambda$ gives $-E[\Delta h\,\hat{\ell}(\Delta U, \Delta Z)\widetilde{W}] - \Psi^{(1)}$, *the scalar* Δh *commuting and the contrast staying on the right, which is exactly how* (131)–(134)*'s three remainders are defined in the statement.* Collecting with $g = \Delta Z + \hat{\ell}(\Delta U, \Delta Z)\,\widetilde{W}$,

$$E\big[\hat{\psi}\big] = -E\big[g\,\Delta h\big] - \Psi^{(1)} + \Psi^{(2)} - \Psi^{(3)},$$

and $E[g\Delta h] = \langle \nu_g/\pi, h\rangle_{L^2(\pi)}$ by Lemma 45(iii). Its proviso is term-by-term integrability, which holds: $E|g_j h(X_m)| \le (E[g_j^2])^{1/2}(E[h(X_m)^2])^{1/2}$ with $E\|g\|^2 \le 2E\|\Delta Z\|^2 + 2c_{\widetilde{W}}^2\|\hat{\ell}\|_\infty^2 < \infty$ by A1b(iv), (P1b), *the Euclidean contraction* $\|\hat{\ell}\widetilde{W}\| \le c_{\widetilde{W}}\|\hat{\ell}\|$ *and* ***not*** *the per-cell* (P1); at $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and the display is the one earlier rounds printed, and (R-a), and $E[h(X_m)^2] \le \bar{c}_\pi\|h\|_{L^2(\pi)}^2 < \infty$ by C1(ii). This is (130). Note that $h \in \mathcal{V}$, which A1b(i) and Definition 5 supply, is what makes the pairing the one Lemma 45(iii) computes.

Step 3: (131)*, and it is Lemma 60*(i) *with the pairing performed.* The map $g \mapsto \nu_g$ of (20) is linear, so $\nu_g = \nu_{\Delta Z + \ell_\star\widetilde{W}} + \nu_{(\hat{\ell}-\ell_\star)\widetilde{W}}$. Under A2 and A2′(i), $\ell_\star^{\Pi}$ solves $\mathcal{A}_{\mathrm{cond}}^*\ell^{\Pi} = -\nu/\pi$, so by Lemma 48(iii) and (33) of Proposition 20(i), $\nu_{\Delta Z+\ell_\star\widetilde{W}} \equiv 0$ almost everywhere on $\mathcal{X}$, *the subscript being* (33)*'s own object* $\Delta Z + \ell\widetilde{W}$ *read at* $\ell = \ell_\star$*, with the* $\mathsf{L}-1$ *columns of* $\ell_\star$ *contracted against the contrast on the right.* Hence, by Lemma 45(iii) again, then $\|E[\,\cdot\,]\| \le E\|\cdot\|$ in $\mathbb{R}^p$, (P1b), the Cauchy–Schwarz inequality and Lemma 45(i),

$$\Big\|\Big\langle \nu_g/\pi, h\Big\rangle\Big\| = \Big\|E\big[(\hat{\ell}-\ell_\star)(\Delta U, \Delta Z)\,\widetilde{W}\,\Delta h\big]\Big\| \le c_{\widetilde{W}}\Big\|\hat{\ell}-\ell_\star\Big\|_{L^2(P)}\Big\|\Delta h\Big\|_{L^2(P)} \le 2\,c_{\widetilde{W}}\,\bar{c}_\pi^{1/2}\Big\|\hat{\ell}-\ell_\star\Big\|_{L^2(P)}\Big\|h\Big\|_{L^2(\pi)}.$$

The bars are the ***Euclidean*** *norm on* $\mathbb{R}^p$*, as* (131) *prints them, and not a modulus*: $g \in \mathbb{R}^p$, so both the pairing and the expectation are vectors, and the passage to the third member is $\|E[\,\cdot\,]\| \le E\|\cdot\|$, then (P1b) pointwise, then the Cauchy–Schwarz inequality. *This is the form Theorem 29 Step 4 needs*: its triangle inequality over (130) requires a norm on this summand exactly as on the other three, and a modulus would supply one only at $p = 1$. *The*

constant is (131)*'s and the tag is* (P1b) *and not* (P1): what is contracted against $\widetilde{W}$ is the $p \times (\mathsf{L}-1)$ matrix $\hat{\ell} - \ell_\star$, whose Frobenius norm is the quantity ϵ_n prices, so the step is the Euclidean bound and the per-cell bound does not reach it. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and both displayed bounds carry the constants earlier rounds printed.

Step 4: (132)*, the only square.* By (R-a), $\hat{\ell}(\cdot, z)$ is C^1 and $\|\partial_u \hat{\ell}\|_\infty = O_P(1)$, so the *mean value inequality*, $\hat{\ell}$ being $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued, so that no Lagrange point is available and none is wanted, and there are now $p(\mathsf{L}-1)$ entries rather than p; Corollary 14 Step 1, gives $\|\hat{\ell}(u - \lambda a, z) - \hat{\ell}(u, z)\| \le \lambda|a|\,\|\partial_u \hat{\ell}\|_\infty$ and therefore $\|\int_0^1 \{\hat{\ell}(u - \lambda a, z) - \hat{\ell}(u, z)\} d\lambda\| \le \frac{1}{2}|a|\,\|\partial_u \hat{\ell}\|_\infty$, the norms being Frobenius, *the passage from the first of those two bounds to the second is Step 0's* ***second*** *display, followed by* $\int_0^1 \lambda\, d\lambda = \frac{1}{2}$. With $\|\widetilde{W}\| \le c_{\widetilde{W}}$ by (P1b), Lemma 45(i), and *Step 0's* ***first*** *display, the conditional Jensen inequality* $\|E[\,\cdot\,]\| \le E[\|\cdot\|]$*, which is what carries the norm inside the expectation that defines* $\Psi^{(1)}$*, and whose integrability requirement Step 0 supplies*,

$$\left\|\Psi^{(1)}\right\| \le \tfrac{1}{2}\, c_{\widetilde{W}} \left\|\partial_u \hat{\ell}\right\|_\infty E\left[(\Delta h)^2\right] \le 2\, c_{\widetilde{W}}\, \bar{c}_\pi \left\|\partial_u \hat{\ell}\right\|_\infty \|h\|^2_{L^2(\pi)},$$

which is (132), the factor 4 between the two right-hand sides being $\|\Delta h\|^2_{L^2(P)} \le 4\bar{c}_\pi \|h\|^2_{L^2(\pi)}$ from Lemma 45(i) squared. The tag is (P1b) because the object contracted against $\widetilde{W}$ is the $p \times (\mathsf{L}-1)$ integral, and at $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and the display is the printed one. *The envelope is read at (R-a)'s* $O_P(1)$ *display and* ***not*** *at a sure bound*, that clause carrying no sure bound on $\partial_u \hat{\ell}$. *The reading is exact*: the envelope appears only on the *right* of the displayed inequality, where the value $+\infty$ makes the bound vacuously true, and (132) is an O_P statement, read as every O_P statement is on an event whose probability can be made arbitrarily close to one, here the event that the envelope is at most a fixed constant, which (R-a)'s display supplies.

Step 5: (133) *and* (134)*. Both displays begin by moving a norm inside a conditional expectation, and both do it by Step 0's* ***first*** *display, the conditional Jensen inequality* $\|E[\,\cdot\,]\| \le E[\|\cdot\|]$*, integrable at each by Step 0. That step is not the Cauchy–Schwarz inequality, which is what*

the second *inequality of each pair is.* For the first, $\| \int_0^1 \hat{\ell}\| \le \|\hat{\ell}\|_\infty$ in the Frobenius norm, *Step 0's* ***second*** *display*, then the conditional Jensen inequality, and then the Cauchy–Schwarz inequality with Lemma 45(i): $\|\Psi^{(2)}\| \le \|\hat{\ell}\|_\infty E[\|\mathsf{r}\|\,|\Delta h|] \le 2\bar{c}_\pi^{1/2}\|\hat{\ell}\|_\infty \|\mathsf{r}\|_{L^2(P)}\|h\|_{L^2(\pi)}$. For the second, the conditional Jensen inequality and then the Cauchy–Schwarz inequality give $\|\Psi^{(3)}\| \le E[\|D\|\,\|\mathsf{r}\|] \le \|\mathsf{r}\|_{L^2(P)}\|D\|_{L^2(P)}$; and if D satisfies N1′ then $E[D\mathsf{r}] = E[E[D \mid \Delta Z]\,\mathsf{r}(\Delta Z)] = 0$, the inner expectation being finite almost surely because $E\|D\| \le E\|\hat{L}\| + E\|L_\star\| < \infty$ by Step 0, and $\mathsf{r}(\Delta Z)$ pulling out on the *right* because it is $\sigma(\Delta Z)$-measurable. *Neither display carries* $c_{\widetilde{W}}$ *and neither should*, which is the distinction (131)–(134) print in the statement: here the $p \times (\mathsf{L}-1)$ factor is contracted against r and against D, whose norms $\|\mathsf{r}\|_{L^2(P)} = (E\|\mathsf{r}\|^2)^{1/2}$ and $\|D\|_{L^2(P)}$ are already the quantities being priced, by A4(iii) and by (R-c.2) respectively, so a factor $c_{\widetilde{W}}$ inserted here would inflate the bound rather than repair it. *What did change is the modulus*: $\|\mathsf{r}\|$ is the Euclidean norm on $\mathbb{R}^{\mathsf{L}-1}$ and $\|D\|$ the Frobenius norm on $\mathbb{R}^{p\times(\mathsf{L}-1)}$, which is how the statement defines $\|\mathsf{r}_k\|_{L^2(P)}$ and $\|D_k\|_{L^2(P)}$; at $\mathsf{L}=2$ both are scalar moduli and both displays are the ones earlier rounds printed. □

Proof of Lemma 94. Five bounds on the four pieces of $\hat{\psi}^{(-k)} - \psi_\star$, then one conditional Chebyshev inequality. Fix k and suppress the superscript.

Step 1: the decomposition. From (38), (39) and $\Delta U(\theta_\dagger, \hat{\varphi}) = \Delta U - \Delta h$,

$$\hat{\psi}-\psi_\star = \underbrace{-\Delta Z\,\Delta h}_{(1)} + \underbrace{\left\{\hat{L}(\Delta U - \Delta h, \Delta Z) - \hat{L}(\Delta U, \Delta Z)\right\}\widetilde{W}}_{(2)} + \underbrace{D(\Delta U, \Delta Z)\,\widetilde{W}}_{(3)} - \underbrace{\hat{L}(\Delta U - \Delta h, \Delta Z)\,\mathsf{r}}_{(4)},$$

obtained by writing $\widehat{\widetilde{W}} = \widetilde{W} - \mathsf{r}$ in (38) and then inserting $\hat{L}(\Delta U, \Delta Z)$, which splits the $\widetilde{W}$-part into (2) and (3); *every product carries the* $p \times (\mathsf{L}-1)$ *factor on the left, as* (38) *and* (39) *print it, and each of the four terms is* $\mathbb{R}^p$*-valued.* At $\mathsf{L}=2$ the four terms are the ones earlier rounds printed.

Step 2: term (1)*, by a Sobolev embedding and one interpolation.* Since $(\Delta h)^4 \le 8\{h(X_2)^4 + h(X_1)^4\}$ and, by C1(ii), $E[h(X_m)^4] = \int h^4 f_{X_m} dx \le \bar{c}_\pi \int h^4 \pi\, dx \le \bar{c}_\pi \|h\|_\infty^2 \|h\|_{L^2(\pi)}^2$, we get

$E[(\Delta h)^4] \le 16\bar{c}_\pi \|h\|_\infty^2 \|h\|_{L^2(\pi)}^2$. By A0, $\mathcal{X}$ is a compact *interval*, so for $s > \frac{1}{2}$ the Sobolev embedding $H^s(\mathcal{X}) \hookrightarrow C(\mathcal{X})$ holds and $\|h\|_\infty \le C_s \|h\|_{H^s}$ with C_s depending only on s and $|\mathcal{X}|$; A5(vi) gives $\|h\|_{H^s} = O_P(1)$ and A5(ii) gives $\|h\|_{L^2(\pi)} = O_P(\delta_n)$. Hence $(E[(\Delta h)^4])^{1/4} = O_P(\delta_n^{1/2})$ and, by (R-b) and the Cauchy–Schwarz inequality,

$$\big\|\Delta Z \Delta h\big\|_{L^2(P)} \le \Big(E\|\Delta Z\|^4\Big)^{1/4} \Big(E[(\Delta h)^4]\Big)^{1/4} = O_P\Big(\delta_n^{1/2}\Big) = o_P(1).$$

The fourth moment in (R-b) is what this step is for, and it cannot be avoided by Cauchy–Schwarz at exponent two: $E\|\Delta Z\|^2(\Delta h)^2$ is not bounded by $E\|\Delta Z\|^2 \|h\|_{L^2(\pi)}^2$ without a sup-norm on h or a fourth moment on ΔZ.

Step 3: term (2). By (184), $\|\hat{L}(u-a,z) - \hat{L}(u,z)\| \le \|\hat{\ell}\|_\infty |a|$ in the Frobenius norm, so with $\|\widetilde{W}\| \le c_{\widetilde{W}}$ by (P1b) and Lemma 45(i) the $L^2(P)$ norm of (2) is at most $c_{\widetilde{W}} \|\hat{\ell}\|_\infty \|\Delta h\|_{L^2(P)} \le 2 c_{\widetilde{W}} \bar{c}_\pi^{1/2} \|\hat{\ell}\|_\infty \|h\|_{L^2(\pi)} = O_P(\delta_n) = o_P(1)$ by (R-a) and A5(ii).

Step 4: term (3). $\|D\widetilde{W}\|_{L^2(P)} \le c_{\widetilde{W}} \|D\|_{L^2(P)} = o_P(1)$ by (P1b) and the last clause of (R-a). *The tag is* (P1b) *and the constant is not* 1: the per-cell (P1) bounds each $|\widetilde{W}_l|$ and does not bound $\|D\widetilde{W}\|$ by $\|D\|$, which is false at $\mathsf{L} \ge 3$: take $\mathsf{L} = 3$, $D = [d, -d]$ with $d \ne 0$, and $\varrho = (\delta, 1-\delta-\eta)'$ with $W = 1$, so that $\widetilde{W} = (1-\delta, -(1-\delta-\eta))'$ and $\|D\widetilde{W}\| = (2-2\delta-\eta)\|d\|$ against $\|D\|_F = \|D\|_{\mathrm{op}} = \sqrt{2}\|d\|$. The ratio is $(2-2\delta-\eta)/\sqrt{2}$, which *exceeds one* for all small δ, η, so $\|D\widetilde{W}\| \le \|D\|$ is false, and tends to $\sqrt{2} = c_{\widetilde{W}}$, so the repair is sharp and cannot be improved. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and both this step and Step 3 are the displays earlier rounds printed.

Step 5: term (4)*, and this is where the sure bound on* r *earns its place.* By (183), $\|\mathsf{r}\| \le c_{\widetilde{W}}$ surely, so $\|\mathsf{r}\|^2 \le c_{\widetilde{W}} \|\mathsf{r}\|$ pointwise and, by $\|\hat{L}\mathsf{r}\| \le \|\hat{L}\| \, \|\mathsf{r}\|$ and the Cauchy–Schwarz inequality,

$$E\Big[\big\|\mathsf{r}\big\|^2 \big\|\hat{L}(\Delta U - \Delta h, \Delta Z)\big\|^2\Big] \;\le\; c_{\widetilde{W}} \, \big\|\mathsf{r}\big\|_{L^2(P)} \Big(E\big\|\hat{L}(\Delta U - \Delta h, \Delta Z)\big\|^4\Big)^{1/2}.$$

By $\|\hat{L}(u,z)\|^4 \le 8\|\hat{b}(z)\|^4 + 8\|\hat{\ell}\|_\infty^4 u^4$ and $(\Delta U - \Delta h)^4 \le 8\{\Delta U^4 + (\Delta h)^4\}$,

$$E\Big\|\hat{L}(\Delta U - \Delta h, \Delta Z)\Big\|^4 \le 8\,E\|\hat{b}(\Delta Z)\|^4 + 64\,\|\hat{\ell}\|_\infty^4\Big\{E[\Delta U^4] + E[(\Delta h)^4]\Big\} = O_P(1)$$

by (R-a), (R-b) and Step 2. Hence the $L^2(P)$ norm of (4) is $O_P(r_n^{1/2}) = o_P(1)$ by A4(iii), $c_{\widetilde{W}}$ being a fixed constant. *A sup-norm rate for $\hat{\varrho}$ is not used, and none is available*: what is used is the crude sure bound $\|\mathsf{r}\| \le c_{\widetilde{W}}$ of (183), which Definition 24(a)'s truncation supplies, together with the L^2 rate A4(iii) prices. *And the ground of that sure bound is the truncation alone and* ***not*** *A4(i)*, which is not on this lemma's list: (11) is contained in the simplex at every value of ε_ϱ, so the diameter argument runs without an overlap condition. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$, $\|\mathsf{r}\| = |\mathsf{r}| \le 1$ and $\|\mathsf{r}\|^2 \le \|\mathsf{r}\|$ is the pointwise inequality $\mathsf{r}^2 \le |\mathsf{r}|$ earlier rounds printed.

Step 6: (135) and (136). Steps 2–5 give $\|\hat{\psi}^{(-k)} - \psi_\star\|_{L^2(P)} = o_P(1)$ for each k, and the maximum over the fixed number F of folds of finitely many $o_P(1)$ terms is $o_P(1)$; that is (135). For (136): conditionally on $\sigma(\mathcal{I}_{-k})$, the variables $\{\hat{\psi}^{(-k)}(O_i) - \psi_\star(O_i)\}_{i\in\mathcal{I}_k}$ are i.i.d. by A0 with finite conditional second moment, so

$$E\Big[\Big\|(\mathbb{E}_{n,k} - E)\Big[\hat{\psi}^{(-k)} - \psi_\star\Big]\Big\|^2 \;\Big|\; \sigma(\mathcal{I}_{-k})\Big] \;\le\; \frac{1}{n_k}\Big\|\hat{\psi}^{(-k)} - \psi_\star\Big\|_{L^2(P)}^2.$$

The scaled fold term carries no symbol of its own, on the precedent Proposition 27 sets: since $\sqrt{n}\,(n_k/n) = n_k/\sqrt{n}$ and $n_k/n \le 1$, multiplying the display by $n(n_k/n)^2 \le n_k$ gives

$$E\Big[\,n\left(\tfrac{n_k}{n}\right)^2\Big\|(\mathbb{E}_{n,k} - E)\Big[\hat{\psi}^{(-k)} - \psi_\star\Big]\Big\|^2 \;\Big|\; \sigma(\mathcal{I}_{-k})\Big] \;\le\; \Big\|\hat{\psi}^{(-k)} - \psi_\star\Big\|_{L^2(P)}^2,$$

so conditional Chebyshev gives, almost surely and for *every* $\epsilon > 0$,

$$P\Big(\sqrt{n}\,\tfrac{n_k}{n}\Big\|(\mathbb{E}_{n,k} - E)\Big[\hat{\psi}^{(-k)} - \psi_\star\Big]\Big\| > \epsilon \;\Big|\; \sigma(\mathcal{I}_{-k})\Big) \;\le\; \epsilon^{-2}\Big\|\hat{\psi}^{(-k)} - \psi_\star\Big\|_{L^2(P)}^2.$$

The right-hand numerator is $\sigma(\mathcal{I}_{-k})$-measurable and is $o_P(1)$ *unconditionally* by (135), so (P8)'s o_P **companion**, at exponent two and $a_n = 1$, makes the k-th fold term $o_P(1)$. *Neither of* (P8)*'s other two clauses is what is used here and neither would serve*: the *transfer rule* takes an unconditionally $O_P(1)$ bounding constant at a *single moving threshold* and returns $O_P(a_n)$, and the O_P *companion* takes the same display as the o_P companion but with $V_n = O_P(1)$ and likewise returns $O_P(a_n)$, whereas the conclusion needed is o_P and the input available is a conditional second-moment bound with a *vanishing* numerator. Summing the fixed number F of folds gives (136). □

Proof of Lemma 95. Clause (i) in Steps 1–6, clause (i′) in Step 6′, which adds no argument and only records which hypotheses Steps 1–6 consume, and clause (ii) in Steps 7–10. The only device beyond the law of large numbers is that $\hat{\ell}^{(-k)}$ is Lipschitz in u, which is what lets $\hat{\theta}^{(0)}$ be replaced by $\theta_\dagger$ before any conditioning.

Step 1: the first summand of $\hat{J}$. By A1b(iv) and A0, $\mathbb{E}_n[\Delta Z \Delta Z'] \xrightarrow{p} E[\Delta Z \Delta Z']$.

Step 2: the second summand, centred. Fix k and put $A_k := \mathbb{E}_{n,k}\big[\hat{\ell}^{(-k)}(\Delta U - \Delta h_k, \Delta Z)\,\widehat{\widetilde{W}}^{(-k)}\,\Delta Z'\big]$, using $\Delta U_i(\theta_\dagger, \hat{\varphi}^{(-k)}) = \Delta U_i - \Delta h_{k,i}$; *the three factors stand in* (46)*'s order,* $(p \times (\mathsf{L}-1))((\mathsf{L}-1)\times 1)(1 \times p)$*, which is what makes* A_k *a* $p \times p$ *matrix at all.* Conditionally on $\sigma(\mathcal{I}_{-k})$ the summands are i.i.d. with conditional second moment at most $c_{\widetilde{W}}^2 \|\hat{\ell}^{(-k)}\|_\infty^2 E\|\Delta Z\|^2$, by (183), $\|ab'\| = \|a\|\,\|b\|$ at $a = \hat{\ell}^{(-k)}\widehat{\widetilde{W}}^{(-k)}$ and $b = \Delta Z$, and Definition 24(c′)'s reading of the envelope, so their average has conditional variance at most n_k^{-1} times that, and conditional Chebyshev gives, almost surely and for *every* $\epsilon > 0$,

$$P\Big(\big\|A_k - E[A_k \mid \sigma(\mathcal{I}_{-k})]\big\| > \epsilon\, n_k^{-1/2} \;\Big|\; \sigma(\mathcal{I}_{-k})\Big) \;\le\; \epsilon^{-2}\, c_{\widetilde{W}}^2 \,\big\|\hat{\ell}^{(-k)}\big\|_\infty^2 E\|\Delta Z\|^2.$$

The bound is at the ***estimated*** *contrast, so the tag is* (183) *and not* (P1b)*, and it is not* (P1) *either*: Definition 24(a) states its per-cell bound and says in terms that it is not a bound on $\|\widehat{\widetilde{W}}_i^{(-k)}\|$, which is exactly what an outer-product norm needs. *Citing* (183)

from outside Lemma 93 imports no hypothesis of that lemma: the display is derived there from Definition 24(a), Definition 4 and Definition 2's truncation convention alone, and Definition 24 is adopted here. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and both displays are the ones earlier rounds printed. The right-hand numerator is $\sigma(\mathcal{I}_{-k})$-measurable and $O_P(1)$ *unconditionally* by (R-a) and A1b(iv), so (P8)'s O_P **companion**, at exponent two and $a_n = n_k^{-1/2}$, gives $A_k - E[\hat{\ell}^{(-k)}(\Delta U - \Delta h_k, \Delta Z)\,\widehat{\widetilde{W}}^{(-k)}\,\Delta Z' \mid \sigma(\mathcal{I}_{-k})] = O_P(n_k^{-1/2})$. *The transfer rule would* ***not*** *serve here*: it needs the conditional probability to vanish at a *single* moving threshold, which conditional Chebyshev does not deliver.

Step 3: the conditional mean converges. Suppress the superscript $(-k)$, write $\widehat{\widetilde{W}} = \widetilde{W} - \mathsf{r}_k$, and consider

$$E\Big[\hat{\ell}(\Delta U - \Delta h_k, \Delta Z)\,\widehat{\widetilde{W}}\,\Delta Z' \;\Big|\; \sigma(\mathcal{I}_{-k})\Big] \;-\; E\Big[\ell_\star(\Delta U, \Delta Z)\,\widetilde{W}\,\Delta Z'\Big],$$

the second expectation being (40)'s debiasing summand as that display prints it. Separate the r_k-term first, then insert $\hat{\ell}(\Delta U, \Delta Z)$; by (P1b) at the two $\widetilde{W}$-terms, the Cauchy–Schwarz inequality, the *mean value inequality* in u, $\hat{\ell}^{(-k)}$ being $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued, Corollary 14 Step 1, and, at *each* of the three braces, the conditional Jensen inequality

$$\Big\|E\Big[\,\cdot \mid \sigma(\mathcal{I}_{-k})\Big]\Big\| \;\le\; E\Big[\|\cdot\| \;\Big|\; \sigma(\mathcal{I}_{-k})\Big] \qquad \text{almost surely,}$$

valid at any $\mathbb{R}^{p\times p}$-valued integrand of finite first moment, *the device Lemma 60 Step 1 writes out in a display, and it is named here at all three braces and not only at the third*, each brace being reached by carrying a norm inside a conditional expectation; the integrability it asks for is a finite first moment at each brace, which (R-a)'s boundedness of $\hat{\ell}^{(-k)}$, (P1b)'s $\|\widetilde{W}\| \le c_{\widetilde{W}}$ and A1b(iv)'s $E\|\Delta Z\|^2 < \infty$ supply through the Cauchy–Schwarz inequality, and all three are already on this step's line, the norm of the two-term difference displayed at the head of

this step is at most

$$\big(E\|\Delta Z\|^2\big)^{1/2}\Big\{c_{\widetilde{W}}\big\|\partial_u\hat{\ell}\big\|_\infty\big\|\Delta h_k\big\|_{L^2(P)}+c_{\widetilde{W}}\big\|\hat{\ell}-\ell_\star\big\|_{L^2(P)}+\big\|\hat{\ell}\big\|_\infty\big\|\mathsf{r}_k\big\|_{L^2(P)}\Big\}.$$

The first two braces carry $c_{\widetilde{W}}$ *and the third does not*, on the distinction (131)–(134) print: the first two contract a $p\times(\mathsf{L}-1)$ matrix against $\widetilde{W}$, the third against r_k, whose norm A4(iii) already prices. At $\mathsf{L}=2$, $c_{\widetilde{W}}=1$ and the brace is the one earlier rounds printed. No sure bound on r_k is used, and the tag above therefore names none. The third brace is $\|\hat{\ell}\|_\infty\|\mathsf{r}_k\|_{L^2(P)}(E\|\Delta Z\|^2)^{1/2}$, which is $\|E[\hat{\ell}(\Delta U-\Delta h_k,\Delta Z)\,\mathsf{r}_k\,\Delta Z'\mid\sigma(\mathcal{I}_{-k})]\|$ bounded by the conditional Jensen inequality displayed above, the sup-norm of $\hat{\ell}$, $\|ab'\|=\|a\|\,\|b\|$ and the Cauchy–Schwarz inequality alone; (183)'s second half $\|\mathsf{r}\|\le c_{\widetilde{W}}$ would deliver a sure bound, but using it would put $c_{\widetilde{W}}$ into this brace, which the sentence just above records that it does not carry. *No symbol is introduced for the difference*: sans-serif d_i is Part H's kernel-difference function (Definition 68) and every other short candidate is claimed. The three braces are $O_P(\delta_n)$ by (R-a), (43) and Lemma 45(i); $o_P(1)$ by the *third* display of (43) under (R-e.1), or by (R-e.2)'s own $\max_k$ clause, *in either case at* $\hat{\ell}^{(-k)}$ *and not at a single* $\hat{\ell}$, *which is why the fold maximum is part of (R-e) and not a reading of it*; and $O_P(r_n)$ by (R-a) and A4(iii). All three are $o_P(1)$.

Step 4. Steps 1–3 and F fixed give $\hat{J}(\theta_\dagger)\xrightarrow{p}J$.

Step 5: the Lipschitz step in θ. By (19), $\Delta U_i(\theta,\varphi)-\Delta U_i(\theta_\dagger,\varphi)=-\Delta Z_i'(\theta-\theta_\dagger)$, so by (R-a)'s $\max_{k\le\mathsf{F}}\|\partial_u\hat{\ell}^{(-k)}\|_\infty=O_P(1)$, *read at that* O_P *display and* ***not*** *at a sure bound*, the display below having the envelope on its *right* and the conclusion drawn from it being $o_P(1)$, together with (183), which *is* sure,

$$\big\|\hat{J}(\theta)-\hat{J}(\theta_\dagger)\big\|\;\le\;c_{\widetilde{W}}\Big(\max_{k\le\mathsf{F}}\big\|\partial_u\hat{\ell}^{(-k)}\big\|_\infty\Big)\,\mathbb{E}_n\big[\|\Delta Z\|^2\big]\,\big\|\theta-\theta_\dagger\big\|,$$

the factor $c_{\widetilde{W}}$ *coming from contracting the difference quotient of* $\hat{\ell}^{(-k)}$ *against* $\widehat{\widetilde{W}}_i^{(-k)}$ *inside*

(46), *and equal to* 1 *at* $\mathsf{L} = 2$*, where the display is the one earlier rounds printed.* At $\theta = \hat{\theta}^{(0)}$ the right-hand side is $O_P(1) \cdot O_P(1) \cdot o_P(1) = o_P(1)$, using (R-a), A1b(iv) and the hypothesis $\|\hat{\theta}^{(0)} - \theta_\dagger\| = o_P(1)$. Hence $\hat{J} = \hat{J}(\hat{\theta}^{(0)}) \xrightarrow{p} J$.

Step 6. By Lemma 91(iv), $J = E[\Delta Z \Delta Z']$, which A1b(iv) makes nonsingular; det is continuous, so $P(\hat{J}$ nonsingular$) \to 1$, and on that event $\hat{J}^{-1} \xrightarrow{p} J^{-1}$ by continuity of inversion at a nonsingular point. That is clause (i). *A3 is* ***not*** *invoked*, being retired at A3 precisely because Lemma 91(iv) proves it; the two ingredients used instead, A1's first clause, inside that lemma, and A1b(iv), are on this lemma's inherited list. *The limit is nonsingular;* $\hat{J}$ *need not be, at any* n, which is why the event is carried explicitly and why (47) defines $\hat{\theta} := \hat{\theta}^{(0)}$ off it.

Step 6′: clause (i′)*, which is Steps 1–6 read with the rates deleted.* No step above uses a rate. Step 1 uses A0 and A1b(iv); Step 2 uses (R-a), A1b(iv) and conditional Chebyshev, delivering $O_P(n_k^{-1/2}) = o_P(1)$ without reference to any nuisance rate; Step 3 bounds its three braces by $c_{\widetilde{W}} \|\partial_u \hat{\ell}\|_\infty \max_k \|\Delta h_k\|_{L^2(P)}$, $c_{\widetilde{W}} \max_k \|\hat{\ell}^{(-k)} - \ell_\star\|_{L^2(P)}$ and $\|\hat{\ell}\|_\infty \max_k \|\mathsf{r}_k\|_{L^2(P)}$, each of which is $o_P(1)$ under clause (i′)'s three consistency statements together with (R-a), Lemma 45(i) and C1(ii), $c_{\widetilde{W}}$ being a fixed constant, *the first brace needs* $\max_k \|h_k\|_{L^2(\pi)} = o_P(1)$ *and not its rate*; Step 4 uses F fixed; Step 5 uses (R-a), (183), A1b(iv) and $\|\hat{\theta}^{(0)} - \theta_\dagger\| = o_P(1)$; Step 6 uses Lemma 91(iv) and A1b(iv). The limit J is finite by Lemma 91(iii) and equals $E[\Delta Z \Delta Z']$ by Lemma 91(iv), both of which consume A1, A1b(iv), A2′(i), A2′(iii) and B1(i), (ii), (iv.a), (iv.d). That is the list clause (i′) states, and (135) is nowhere invoked. *Clause* (ii)*, by contrast, does use* (135) *and hence A5(vi) and (R-b), so (i′) has no analogue for (ii) and none is claimed.*

Step 7: clause (ii)*, the true part.* By Lemma 91(ii) and A0, $\mathbb{E}_n[\psi_\star \psi_\star'] \xrightarrow{p} \Sigma$.

Step 8: the estimated part at $\theta_\dagger$. Conditionally on $\sigma(\mathcal{I}_{-k})$ the summands of $\mathbb{E}_{n,k}\|\hat{\psi}^{(-k)} - \psi_\star\|^2$ are i.i.d. by A0, nonnegative, with conditional mean exactly $\|\hat{\psi}^{(-k)} - \psi_\star\|^2_{L^2(P)}$, so conditional Markov gives, almost surely and for every $\epsilon > 0$, $P(\mathbb{E}_{n,k}\|\hat{\psi}^{(-k)} - \psi_\star\|^2 > \epsilon \mid \sigma(\mathcal{I}_{-k})) \leq$

$\epsilon^{-1}\|\hat{\psi}^{(-k)} - \psi_\star\|^2_{L^2(P)}$. That numerator is $\sigma(\mathcal{I}_{-k})$-measurable and $o_P(1)$ unconditionally by (135), so (P8)'s o_P **companion**, at exponent one and $a_n = 1$, applies; summing the F folds gives $\mathbb{E}_n\|\hat{\psi}(\theta_\dagger) - \psi_\star\|^2 = o_P(1)$. As at Lemma 94 Step 6, (P8)'s transfer rule and its O_P companion both conclude only O_P and neither is the clause invoked; what distinguishes this site from Step 2 is that the numerator here *vanishes* rather than being merely tight.

Step 9: from $\theta_\dagger$ to $\hat{\theta}$. Writing $\hat{\psi}_i(\theta) := \psi(\theta, \hat{\zeta}^{(-k)})(O_i)$ for $i \in \mathcal{I}_k$, (38), (184) and (183) give $\|\hat{\psi}_i(\theta) - \hat{\psi}_i(\theta_\dagger)\| \le \{\|\Delta Z_i\|^2 + c_{\widetilde{W}}\|\hat{\ell}^{(-k)}\|_\infty\|\Delta Z_i\|\}\|\theta - \theta_\dagger\|$, so $\mathbb{E}_n\|\hat{\psi}(\hat{\theta}) - \hat{\psi}(\theta_\dagger)\|^2 \le 2\|\hat{\theta} - \theta_\dagger\|^2\,\mathbb{E}_n\big[\|\Delta Z\|^4 + c^2_{\widetilde{W}}\max_k\|\hat{\ell}^{(-k)}\|^2_\infty\|\Delta Z\|^2\big] = o_P(1)$ by (R-b), (R-a) and $\hat{\theta} \xrightarrow{p} \theta_\dagger$. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and both displays of this step are the ones earlier rounds printed.

Step 10: assembly. Put $\Delta_i := \hat{\psi}_i(\hat{\theta}) - \psi_{\star i}$, so that $\hat{\psi}_i(\hat{\theta})\hat{\psi}_i(\hat{\theta})' - \psi_{\star i}\psi'_{\star i} = \Delta_i\psi'_{\star i} + \psi_{\star i}\Delta'_i + \Delta_i\Delta'_i$ and hence, by the Cauchy–Schwarz inequality,

$$\big\|\hat{\Sigma} - \mathbb{E}_n[\psi_\star\psi'_\star]\big\| \le 2\big(\mathbb{E}_n\|\Delta\|^2\big)^{1/2}\big(\mathbb{E}_n\|\psi_\star\|^2\big)^{1/2} + \mathbb{E}_n\|\Delta\|^2.$$

Steps 8–9 give $\mathbb{E}_n\|\Delta\|^2 = o_P(1)$ and Step 7 gives $\mathbb{E}_n\|\psi_\star\|^2 = O_P(1)$, so $\hat{\Sigma} \xrightarrow{p} \Sigma$; $\hat{V} \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$ follows from clause (i) and the continuous mapping theorem. □

Proof of Theorem 29. One exact algebraic identity, then four terms. *The order of the steps matters and is not interchangeable*: Lemma 95(i) is applied *before* the limit is established, since it needs only $\hat{\theta}^{(0)} \xrightarrow{p} \theta_\dagger$, and Lemma 95(ii) *after*, since it needs $\hat{\theta} \xrightarrow{p} \theta_\dagger$; there is no circularity.

*Step 0: the φ-step, on each fold, and the one thing it does **not** deliver.* By (T-5), Proposition 28 applies on $\mathcal{I}_{-k}$ at the anchor $\theta_\circ$ of Definition 24(b); since $n_{-k} = n - n_k \asymp n$ and every rate in (56) is a power of n times a power of $\ln n$, the conclusion holds with n in place of n_{-k} up to a constant, so on each fold complement the *infeasible* reading $\hat{\varphi}^\pi$ of Definition 22 satisfies $\|\hat{\varphi}^\pi - \varphi_\dagger\|_{L^2(\pi)} = O_P(\delta_n)$ with δ_n as in (137). *Definition 24*(b) *plugs in the **feasible** $\hat{\varphi}^{(-k)}$, and $h_k := \hat{\varphi}^{(-k)} - \varphi_\dagger$ is defined at that object*, so the display just derived is **not** a statement

about h_k and this step does not produce one: the passage from one reading to the other is Remark 89(7), open at (7c) and (7d). *What supplies* $\|h_k\|_{L^2(\pi)} = O_P(\delta_n)$ *is therefore* (T-4)*'s A5(ii), read at each fold as Definition 24*(d) *directs, and it is a* ***hypothesis*** *of this theorem and not a conclusion of Proposition 28 — which is what makes Theorem 29 a conditional theorem on that one clause, and it is stated here as well as on the hypothesis line because a reader following the proof should meet it at the step that consumes it. Since* F *is fixed, a maximum of finitely many* $O_P(\delta_n)$ *statements is* $O_P(\delta_n)$*, which is the first display of* (43); A5(vi′) is consumed inside Proposition 28 and A5(vi) follows from it, on each fold complement, at the *final linearisation stage* $\bar{k}$ of Definition 22, which with F fixed is the second display of (43) and is what Lemma 94 Step 2 uses. *"*$\bar{k}$*" is a stage index and not a fold index, and the sentence says so in words because the two families are separated by one glyph*: in this paragraph h_k, $\mathcal{I}_{-k}$ and $\hat{\varphi}^{(-k)}$ carry the fold index $k \le \mathsf{F}$, while $\hat{\varphi}^{(0)}, \dots, \hat{\varphi}^{(\bar{k})}$ are the stages of a *single* run of Definition 22 on one fold complement, a nonnegative parenthesised superscript is a stage, a negated one is a fold complement. *Every fold complement is run to its own stage* $\bar{k}$, and $\bar{k}$ is the same finite number on each, not depending on n or on the fold.

What this step does and does not discharge, stated exactly. It **delivers** A5(i), through Lemma 77, which is one of Proposition 28's inputs. It does **not** deliver the *first half* of A5(ii), and the reason is the object and not the arithmetic: Proposition 28 returns (56) at $\hat{\varphi}^{\pi}$ and A5(ii) is stated at $\hat{\varphi}$, the two being the two readings Definition 22 separates. The clause is carried at (T-4) accordingly. *This is not the only place the separation bites, and naming the second is what keeps the theorem's conditional status from being read as one clause*: the $\sigma(\mathcal{I}_{-k})$-measurability and $\mathcal{V}$-membership that Definition 24(c′.1) requires of $\hat{\varphi}^{(-k)}$ are verified at the fixed sieve and pending at the feasible one, and Lemma 94 consumes the measurability at Step 6, where the conditional-on-$\sigma(\mathcal{I}_{-k})$ second-moment bound is taken. *No claim is made that these two exhaust the sites*, and a systematic sweep for further ones has not been run. It does **not** discharge A5(iii) either, and no proposition can: (7) is an inequality among the primitives (μ, r, d_Z), Proposition 28 returns its rate *whenever* that inequality holds, and

A5(ii)'s second half $\delta_n = o_P(n^{-1/4})$ is equivalent to it under (137) by Remark 97(3)(a). *That second half* ***is*** *delivered*, being a statement about the deterministic sequence δ_n alone and free of either reading. So (T-4) lists A5(iii) and A5(ii)'s rate half as assumptions and A5(i) as a delivered symbol, and that is what its annotation says.

Step 1: the exact one-step identity. Under (R-a) each $\hat{L}^{(-k)}$ is twice continuously differentiable in u, with $\partial_u \hat{L}^{(-k)} = \hat{\ell}^{(-k)}$ and $\partial_u^2 \hat{L}^{(-k)} = \partial_u \hat{\ell}^{(-k)}$; since θ enters $\psi(\theta, \hat{\zeta}^{(-k)})$ only through the affine map $\theta \mapsto \Delta U(\theta, \hat{\varphi}^{(-k)})$, $\bar{\psi}_n$ is twice continuously differentiable on $\mathbb{R}^p$ with $\partial_\theta \bar{\psi}_n(\theta) = -\hat{J}(\theta)$ by (46) and, for every θ,

$$\left\| \partial^2_{\theta\theta'} \bar{\psi}_n(\theta) \right\| \;\le\; c_{\widetilde{W}} \Big(\max_{k \le \mathsf{F}} \left\| \partial_u \hat{\ell}^{(-k)} \right\|_\infty \Big) \mathbb{E}_n \big[\|\Delta Z\|^2 \big] \;=\; O_P(1)$$

by (R-a), (183) and A1b(iv), *the right-hand side carries no symbol of its own, because every candidate letter is claimed*; it is written out at the one further place it is used. *The factor* $c_{\widetilde{W}}$ *is where the second* θ*-derivative contracts* $\partial_u \hat{\ell}^{(-k)}$ *against* $\widehat{\widetilde{W}}_i^{(-k)}$: differentiating (46) once more in θ gives the array whose value at (ς, ς) has j-th coordinate $(\partial_u \hat{\ell}^{(-k)} \widehat{\widetilde{W}}_i^{(-k)})_j (\varsigma' \Delta Z_i)^2$, so the estimated contrast is contracted in the Euclidean norm and the per-cell (P1) does not reach it. At $\mathsf{L} = 2$, $c_{\widetilde{W}} = 1$ and both displays here are the ones earlier rounds printed. Taylor's theorem with the integral remainder, applied coordinatewise on the segment joining $\hat{\theta}^{(0)}$ to $\theta_\dagger$, gives

$$\bar{\psi}_n(\theta_\dagger) = \bar{\psi}_n\big(\hat{\theta}^{(0)}\big) - \hat{J}\big(\theta_\dagger - \hat{\theta}^{(0)}\big) + \Psi_n^\theta, \quad \left\| \Psi_n^\theta \right\| \le \tfrac{1}{2}\, c_{\widetilde{W}} \Big(\max_{k \le \mathsf{F}} \left\| \partial_u \hat{\ell}^{(-k)} \right\|_\infty \Big) \mathbb{E}_n \big[\|\Delta Z\|^2 \big] \left\| \hat{\theta}^{(0)} - \theta_\dagger \right\|^2,$$

with $\hat{J} = \hat{J}(\hat{\theta}^{(0)})$. Rearranging and substituting into (47), on the event $\{\hat{J}$ nonsingular$\}$,

$$\hat{\theta} - \theta_\dagger \;=\; \hat{J}^{-1} \Big[\bar{\psi}_n(\theta_\dagger) - \Psi_n^\theta \Big] \qquad \text{exactly.} \tag{185}$$

By Lemma 92(i), whose second half gives $\hat{\theta}^{(0)} \xrightarrow{p} \theta_\dagger$, since A5(ii)'s second half gives $\delta_n \to 0$ a fortiori, and Lemma 95(i), $P(\hat{J}$ nonsingular$) \to 1$ and $\hat{J}^{-1} \xrightarrow{p} J^{-1}$.

Step 2: the θ-remainder, and this is the only place a rate *for $\hat{\theta}^{(0)}$ is used.* By Lemma 92(i), $\|\hat{\theta}^{(0)} - \theta_\dagger\| = O_P(n^{-1/2} + \delta_n)$, and by A5(ii) in the form (137), $\delta_n = o_P(n^{-1/4})$. Hence, by Step 1's bound on $\|\Psi_n^\theta\|$ and $(\varsigma + \varsigma')^2 \le 2\varsigma^2 + 2\varsigma'^2$,

$$\sqrt{n}\left\|\Psi_n^\theta\right\| = O_P\Big(\sqrt{n}\left(n^{-1/2} + \delta_n\right)^2\Big) = O_P\Big(n^{-1/2} + \sqrt{n}\,\delta_n^2\Big) = o_P(1).$$

The uncollapsed rate is used deliberately: Lemma 92 records that the collapse of $O_P(n^{-1/2}) + O_P(\delta_n)$ to $O_P(\delta_n)$ needs A5(i) and A5(iii), and this step needs neither. *Consistency of the preliminary would not suffice*: the threshold is exactly $o_P(n^{-1/4})$, $\sqrt{n}(n^{-1/2} + \delta_n)^2 = o(1)$ if and only if $\delta_n = o(n^{-1/4})$, and it is met because the plug-in inherits δ_n.

Step 3: the moment average at $\theta_\dagger$, split three ways. Since $E[\psi_\star] = 0$ (Lemma 91(i)) and $\sum_k n_k/n = 1$,

$$\bar{\psi}_n(\theta_\dagger) = \mathbb{E}_n\Big[\psi_\star\Big] + \sum_{k=1}^{\mathsf{F}} \frac{n_k}{n}\Big(\mathbb{E}_{n,k} - E\Big)\Big[\hat{\psi}^{(-k)} - \psi_\star\Big] + \sum_{k=1}^{\mathsf{F}} \frac{n_k}{n}\, E\Big[\hat{\psi}^{(-k)} \;\Big|\; \sigma(\mathcal{I}_{-k})\Big], \tag{186}$$

an identity, obtained by adding and subtracting $\mathbb{E}_{n,k}[\psi_\star]$ and $E[\hat{\psi}^{(-k)} \mid \sigma(\mathcal{I}_{-k})]$ inside each fold's average.

Step 4: the bias term. By Lemma 93, for each k, $\|E[\hat{\psi}^{(-k)} \mid \sigma(\mathcal{I}_{-k})]\|$ is at most the sum of the four right-hand sides of (131)–(134). *The passage is the triangle inequality in $\mathbb{R}^p$ applied to* (130), whose four summands, the pairing and the three $\Psi_k^{(j)}$, are exactly the objects (131)–(134) bound, each in the Euclidean norm on $\mathbb{R}^p$, ψ being $\mathbb{R}^p$-valued by (38); *a modulus on* (131) *would license this step only at $p = 1$.* Term by term:

(a) *First-order bias.* Under (R-e.1), (131) with the third display of (43) and Step 0 gives $\max_k \left\|\langle \nu_{g_k}/\pi, h_k\rangle\right\| = O_P(\epsilon_n\delta_n) = o_P(n^{-1/2})$, *the bars being the* ***Euclidean*** *norm on $\mathbb{R}^p$, which is what the triangle inequality just applied consumes and what* (131) *supplies; the fold maximum is exactly what is needed, since the bound is applied at each of the* F *estimators $\hat{\ell}^{(-k)}$ and not at a single $\hat{\ell}$; and δ_n here is* (137)*'s, so A5(iv) is being*

consumed in its log-inflated form, which Remark 97(3)(c) is the record of. Under (R-e.2) the same quantity is $o_P(n^{-1/2})$ by Lemma 60(iii); *that clause's proof transfers verbatim to Definition 24's folds*, because each of the three terms of (78) is a statement at frozen $\hat{\ell}$ and Definition 24 freezes $\hat{\ell}^{(-k)}$ at $\sigma(\mathcal{I}_{-k})$, which is Lemma 60's sample-splitting convention with $(\mathcal{I}_a, \mathcal{I}_m)$ read as $(\mathcal{I}_{-k}, \mathcal{I}_k)$.

(b) *The square.* By (132), (R-a) and Step 0, $\|\Psi_k^{(1)}\| = O_P(\delta_n^2)$, and $\delta_n^2 = o(n^{-1/2})$ by A5(iii): writing $\hat{\delta}_n = n^{-\alpha}$, $\kappa = 2\mu/(2\mu+1)$ and $\kappa\alpha = \frac{1}{4} + \varepsilon$ with $\varepsilon > 0$ by the strict inequality (7), $\delta_n^2 = n^{-1/2-2\varepsilon}(\ln n)^{2\kappa} = o(n^{-1/2})$. This is Proposition 79(iv), and the strictness of (7) is what pays for the logarithm.

(c) The $\varrho \times \varphi$ cross term, whose condition is **implied** by (7). By (133), (R-a), A4(iii) and Step 0, $\|\Psi_k^{(2)}\| = O_P(r_n\delta_n)$. *The substitution that turns this into a rate is* $r_n \le \hat{\delta}_n = n^{-\alpha}$, *and it is printed because it is not the identity it looks like*: the inequality is A5(i), whose $\hat{\delta}_n$ is the *maximum* $\max\{\hat{\delta}_n^{(T)}, \hat{\delta}_n^{(\varrho)}\}$ of which $r_n = \hat{\delta}_n^{(\varrho)}$ is only one branch, and the equality is A5(iii) in the parameterisation clause (b) above has just fixed. With $\delta_n = (\hat{\delta}_n \ln n)^\kappa$ of (137) this gives $r_n\delta_n \le n^{-\alpha(1+\kappa)}(\ln n)^\kappa$ and hence $\|\Psi_k^{(2)}\| = O_P(n^{-\alpha(1+\kappa)}(\ln n)^\kappa)$, *on* ***both*** *branches of that maximum and not only on the* ϱ*-branch*, which is why the step needs an inequality where Remark 97(3)(b), the record of the margin computed next, prints the ϱ-branch equality $r_n = n^{-\alpha}$. From (7), $\kappa\alpha > \frac{1}{4}$ with $0 < \kappa < 1$, so $\alpha > 1/(4\kappa)$ and

$$\alpha(1+\kappa) - \tfrac{1}{2} \;>\; \frac{1+\kappa}{4\kappa} - \frac{1}{2} \;=\; \frac{1-\kappa}{4\kappa} \;=\; \frac{1}{8\mu} \;>\; 0,$$

using $1 - \kappa = (2\mu+1)^{-1}$ and $4\kappa = 8\mu/(2\mu+1)$; a strictly positive polynomial margin, so $r_n\delta_n = o(n^{-1/2})$. *No condition beyond* (7) *is imposed for this term, and that is a fact about the frontier rather than a choice*: $\kappa < 1$ holds at every μ, so the implication is unconditional within the block. *The constant* $1/(8\mu)$ *cannot be improved using* (7) *alone*: at $\alpha = 1/(4\kappa)$, the boundary of (7), the middle inequality holds with equality, so $1/(8\mu)$ is the infimum of $\alpha(1+\kappa) - \frac{1}{2}$ over the frontier and the margin is exact there.

(d) *The $\varrho \times L$ cross term.* Zero under (R-c.1) by (134); under (R-c.2), $O_P(r_n \epsilon_n^L) = o(n^{-1/2})$ by hypothesis.

Summing over the fixed F folds, the third term of (186) is $o_P(n^{-1/2})$.

Step 5: the empirical-process term and the central limit theorem. The second term of (186), multiplied by $\sqrt{n}$, is $o_P(1)$ by Lemma 94. By A0, $\psi_\star(O_i)$ are i.i.d. with mean zero and finite second moment (Lemma 91(i)–(ii)), so the multivariate Lindeberg–Lévy central limit theorem gives the convergence in distribution of $\sqrt{n}\,\mathbb{E}_n[\psi_\star]$ to the centred Gaussian law on $\mathbb{R}^p$ with covariance matrix Σ. *The limit law is written out and carries no symbol*: italic N is a count, a null set or a covering number everywhere else in this corpus, and calligraphic $\mathcal{N}$ is not available either, being Lemma 41's closed subspace and A2c's null Borel set.

Step 6: assembly. Multiply (185) by $\sqrt{n}$ and substitute (186). Steps 2, 4 and 5 give $\sqrt{n}\,\bar{\psi}_n(\theta_\dagger) - \sqrt{n}\,\Psi_n^\theta = \sqrt{n}\,\mathbb{E}_n[\psi_\star] + o_P(1)$, so on $\{\hat{J}$ nonsingular$\}$

$$\sqrt{n}\big(\hat{\theta} - \theta_\dagger\big) = \hat{J}^{-1}\Big\{\sqrt{n}\,\mathbb{E}_n[\psi_\star] + o_P(1)\Big\} = J^{-1}\sqrt{n}\,\mathbb{E}_n[\psi_\star] + o_P(1),$$

the last step by $\hat{J}^{-1} \xrightarrow{p} J^{-1}$ (Step 1) and $\sqrt{n}\mathbb{E}_n[\psi_\star] = O_P(1)$ (Step 5). Since $P(\hat{J}$ nonsingular$) \to 1$ the same holds unconditionally, and Slutsky's theorem gives (58). In particular $\hat{\theta} - \theta_\dagger = O_P(n^{-1/2})$, so $\hat{\theta} \xrightarrow{p} \theta_\dagger$ and Lemma 95(ii) applies, giving $\hat{V} \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$; the studentised statement then follows from the continuous mapping theorem and Slutsky's theorem, since $\vartheta'\hat{V}\vartheta \xrightarrow{p} \vartheta' J^{-1}\Sigma J^{-1\prime}\vartheta$ and that limit is *strictly positive at every unit ϑ*. *The strict positivity is* ***not*** *a proviso*: A1b(iv) is in (T-1) and C2′(i) is in (T-3), so Lemma 91(v) applies and delivers (126), $\Sigma \succ 0$, hence $J^{-1}\Sigma J^{-1\prime} \succ 0$. *Only the limit is claimed positive*; on the event $\{\vartheta'\hat{V}\vartheta \le 0\}$, which has probability tending to zero, the ratio is undefined and the statement is read on the complement, as a studentised limit always is. □

Proof of Corollary 30. Take (185) and collect orders without asserting that any of them is $o(n^{-1/2})$. *Lemma 95*(i′) *is the clause invoked, and not* (i): the hypotheses here contain no rate

product, and (i′)'s three consistency statements are supplied by $\delta_n \to 0$ with (43), by $\epsilon_n \to 0$ with (R-e.1), and by $r_n \to 0$ with A4(ii)–(iii). *Its fourth hypothesis,* $\|\hat{\theta}^{(0)} - \theta_\dagger\| = o_P(1)$*, is supplied by Lemma 92*(i) *and by its* ***statement****, not by its proof*, that clause's list is A0, A1, A1b(iv) (all in (T-1)), C1(ii) (in (T-3)), Definition 24 (adopted, the $\hat{\theta}$ of (59) being Definition 25's one-step estimator and no other object — the reading is that the *estimator* is defined once, not that Definition 24 defines it) and the first display of (43) (delivered by Step 0 of Theorem 29's proof, which (T-5) licenses), while $\delta_n \to 0$ is assumed on the corollary. *A5(ii)'s second half is confined to clause* (ii) *of that lemma and is used at no point in this proof*, which is what the corollary exists to demonstrate. Hence $\hat{J}^{-1} = O_P(1)$ and $P(\hat{J}$ nonsingular$) \to 1$. By Step 3 of Theorem 29's proof and A0, $\mathbb{E}_n[\psi_\star] = O_P(n^{-1/2})$; by Lemma 94, whose hypotheses are met, A5(vi) with $s > \frac{1}{2}$ being listed on the corollary for exactly this reason, the empirical-process term is $o_P(n^{-1/2})$; by Lemma 93 the bias is $O_P(\delta_n\epsilon_n + \delta_n^2 + r_n\delta_n + r_n\epsilon_n^L)$; and by Step 2 of that proof, at Lemma 92(i)'s rate (128) and *not* at its clause-(ii) threshold, $\|\Psi_n^\theta\| = O_P(\|\hat{\theta}^{(0)} - \theta_\dagger\|^2) = O_P(n^{-1} + \delta_n^2)$, absorbed into δ_n^2 and $n^{-1/2}$. Multiplying by $\|\hat{J}^{-1}\| = O_P(1)$ gives (59).

The two branch statements. Under (R-e.2) the first-order bias is instead bounded by the three terms of (78), namely $O_P(n^{-1/2})\|h_k\|_{L^2(\pi)} + 0 + O_P(1)\cdot O_P(\mathsf{R}\, L_x^{-s}) = O_P(n^{-1/2}\delta_n) + O_P(\mathsf{R}\, L_x^{-s})$ by (79) and, for the third, A5(vi) with A8's basis clause and the spline direct theorem (162), both read in *degree*; so no rate for $\hat{\ell}$ enters and ϵ_n disappears from (59). If in addition (R-c.1) holds then $\Psi_k^{(3)} = 0$ for every k by the second half of (134), so ϵ_n^L disappears as well and clause (c) of the corollary's display follows. *Those are the only two symbols in* (59) *without a primitive supplier*, which is what clauses (a)–(c) of the corollary record. □

Proof of Proposition 96. All P estimators share one influence function; the rest is that order statistics are 1-Lipschitz.

The index letter. The P partitions are indexed by $\mathsf{p} \in \{1, \dots, \mathsf{P}\}$ and **not** by q, which is bound project-wide as $q(x_1, x_2) = E[\partial_u \ell_\star \widetilde{W} \mid X_1 = x_1, X_2 = x_2]$, *the contrast to the right of*

the $p \times (\mathsf{L}-1)$ *derivative, which is the order every other site in the corpus prints and the only one that multiplies*, and is live *bare* in Part I at Lemma 91(iv.3). Sans-serif p is the lowercase of the count P this proposition already claims, and it is separated from the **reserved** italic $p = \dim(\Delta Z)$ by font. *And the linear term is* Z_n*, sans-serif*, against the italic exogenous control Z, ΔZ that this very display divides by; the separation from Proposition 79(ii)'s bare Z is the n subscript.

Step 1. The partitions are drawn independently of the data and, by hypothesis, Theorem 29's conditions hold at each of them, *which is what makes the theorem applicable* P *times and not once*, the P partitions having PF distinct fold complements. So Theorem 29 applies to each $\hat{\theta}_{\mathsf{p}}$ and Step 6 of its proof gives $\sqrt{n}(\hat{\theta}_{\mathsf{p}} - \theta_\dagger) = \mathsf{Z}_n + \varepsilon_{\mathsf{p},n}$ with $\mathsf{Z}_n := J^{-1}\sqrt{n}\,\mathbb{E}_n[\psi_\star]$ *not depending on* p and $\varepsilon_{\mathsf{p},n} = o_P(1)$. Since P is fixed, $\max_{\mathsf{p}\le\mathsf{P}} \|\varepsilon_{\mathsf{p},n}\| = o_P(1)$, and subtracting two such expansions gives $\sqrt{n}(\hat{\theta}_{\mathsf{p}} - \hat{\theta}_{\mathsf{p}'}) = o_P(1)$.

Step 2. Fix a coordinate j. The map $x \mapsto \sqrt{n}(x - \theta_{\dagger j})$ is increasing and affine, and order statistics commute with increasing affine maps, so $\sqrt{n}(\hat{\theta}^{\mathrm{md}}_j - \theta_{\dagger j}) = \mathrm{med}_{\mathsf{p}}\{\mathsf{Z}_{n,j} + \varepsilon_{\mathsf{p},n,j}\}$. The median (for even P, the average of the two central order statistics) is 1-Lipschitz from $(\mathbb{R}^{\mathsf{P}}, \|\cdot\|_\infty)$ to $\mathbb{R}$ and fixes constant vectors, so $|\,\mathrm{med}_{\mathsf{p}}\{\mathsf{Z}_{n,j} + \varepsilon_{\mathsf{p},n,j}\} - \mathsf{Z}_{n,j}| \le \max_{\mathsf{p}} |\varepsilon_{\mathsf{p},n,j}| = o_P(1)$. The same bound holds trivially for $\hat{\theta}^{\mathrm{av}}$, whose deviation from $\mathsf{Z}_{n,j}$ is $\mathsf{P}^{-1}\sum_{\mathsf{p}} \varepsilon_{\mathsf{p},n,j}$. Collecting the p coordinates and applying Step 5 of Theorem 29's proof with Slutsky's theorem gives both displays.

Step 3: where P *fixed is used, three times.* Once in the *hypothesis*, so that "Theorem 29 at each partition" is a finite conjunction and the maxima in (43) and (R-a) may be read over PF fold complements at unchanged order; once in Step 1, to pass from "each $\varepsilon_{\mathsf{p},n} = o_P(1)$" to "the maximum is $o_P(1)$"; once in Step 2, for the median to be a Lipschitz function of finitely many arguments. *None survives* $\mathsf{P} \to \infty$: the median would be a functional of a random measure, and the hypothesis would become a uniformity requirement over a growing family. The argument would have to be rebuilt. □

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